

Linear Algebra for ML, CV, and Robotics

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Preface

This book teaches linear algebra through the problems it actually solves in machine learning, computer vision, and robotics.

Standard linear algebra textbooks present the theory cleanly but leave you to figure out why any of it matters for real work. Applied tutorials give you recipes without the foundations. Neither prepares you for the moment your solver silently returns a wrong answer, your camera calibration diverges, or your robot arm hits a singularity.

This book takes a different approach. Every concept is motivated by a real problem — trilateration, plane fitting, point cloud registration, Kalman filtering — and every technique is shown in Python code that you can run immediately.

What you will be able to do

After working through this book you will be able to:

- Understand why a least-squares solver fails near rank-deficient configurations and how to detect it
- Implement camera calibration from scratch and diagnose ill-conditioned calibration setups
- Transform point clouds between coordinate frames without the pervasive normal-vector bug
- Write a working Extended Kalman Filter for sensor fusion
- Understand what an SVD is doing geometrically, not just how to call `numpy.linalg.svd`
- Read papers in CV, robotics, and ML that use matrix notation without getting lost

How to read this book

Each chapter opens with a concrete problem, builds the necessary theory, and closes with a fully worked application in Python. The exercises are designed to expose the gaps in intuitive understanding that cause bugs in practice.

Fast tracks by field:

Track	Essential chapters
Machine Learning	1, 4, 5, 13, 14, 15, 17, 18, 20, 21
Computer Vision	1, 4, 6, 7, 10, 11, 17, 18, 25
Robotics	1, 4, 7, 8, 9, 12, 17, 22, 23, 24

Code conventions

All code is Python 3. Paste any code block directly into a Jupyter notebook cell to run it.

Required packages:

```
# Install everything you need for this book  
pip install numpy scipy matplotlib opencv-python open3d ipywidgets sympy scikit-learn
```

Part I — Linear Systems and Subspaces

Why Linear Algebra?

The Three Problems This Book Solves

Every technique in this book — whether you are calibrating a camera, fitting a model to sensor data, or finding the dominant direction of motion in a point cloud — reduces to one of three fundamental problems:

#	Problem	Notation	When it arises
1	Exact solve	$A\mathbf{x} = \mathbf{b}$	Exactly n independent constraints for n unknowns
2	Least squares	$\min\ A\mathbf{x} - \mathbf{b}\ $	More measurements than unknowns; best-fit solution
3	Eigenvalue	$A\mathbf{v} = \lambda\mathbf{v}$	Find directions the matrix only stretches, never rotates

Every subsequent chapter deepens one of these. Let us visit each one now with a concrete example, just enough to make the shape of the problem recognizable.

Problem 1: $A\mathbf{x} = \mathbf{b}$ — Solve Exactly

The situation. You have n unknowns and exactly n independent constraints. There is precisely one answer.

Where it appears.

- *Trilateration:* A mobile robot receives distance measurements from three beacons at known locations. Each measurement constrains the robot to a circle. Two circles intersect at (at most) two points; a third resolves the ambiguity. After linearizing, you solve a 3×2 system — but the core is still $A\mathbf{x} = \mathbf{b}$.
- *Frame conversion:* Given a 3-D point expressed in camera coordinates, compute the same point in world coordinates. The conversion is a matrix equation, solved in one step.
- *Inverse kinematics (simplified):* Given a desired end-effector position for a 2-DoF planar arm, find the two joint angles that achieve it.

Worked example. Two walls of a warehouse are equipped with ultrasonic sensors. From the signal travel times you derive two distance constraints on the robot's (x, y) position:

$$\begin{bmatrix} 2 & -1 \\ 1 & 3 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 3 \\ 10 \end{bmatrix}$$

```
import numpy as np

A = np.array([[2.0, -1.0],
              [1.0,  3.0]])
b = np.array([3.0, 10.0])

x = np.linalg.solve(A, b)
print(f"Robot position: x = {x[0]:.4f}, y = {x[1]:.4f}")

residual = np.linalg.norm(A @ x - b)
print(f"Residual ||Ax - b|| = {residual:.2e}")
# Residual    machine epsilon - this is an exact solve
```

```
Robot position: x = 2.7143, y = 2.4286
Residual ||Ax - b|| = 4.44e-16
```

The residual is essentially zero — floating-point machine epsilon. That is what *exact solve* means. Notice that `numpy.linalg.solve` does not check whether a unique solution exists; you must do that yourself (Chapter 3 shows how).

The three cases of $Ax = b$.

```
import numpy as np
import matplotlib.pyplot as plt

fig, axes = plt.subplots(1, 3, figsize=(13, 4))
x_range = np.linspace(-2, 8, 400)
blue, red = '#4a90d9', 'tomato'

# Panel 1: Unique solution
ax = axes[0]
A = np.array([[2., -1.], [1., 3.]])
b = np.array([3., 10.])
sol = np.linalg.solve(A, b)
ax.plot(x_range, (3 - 2*x_range) / (-1), color=blue, lw=2, label=r'$2x - y = 3$')
ax.plot(x_range, (10 - 1*x_range) / 3, color=red, lw=2, label=r'$x + 3y = 10$')
ax.plot(*sol, 'ko', ms=8, zorder=5, label=rf'$({sol[0]:.2f}, \{sol[1]:.2f})$')
ax.set_title('Unique solution\n(lines intersect)', fontsize=10)
ax.legend(fontsize=8)

# Panel 2: No solution (parallel lines)
```



```

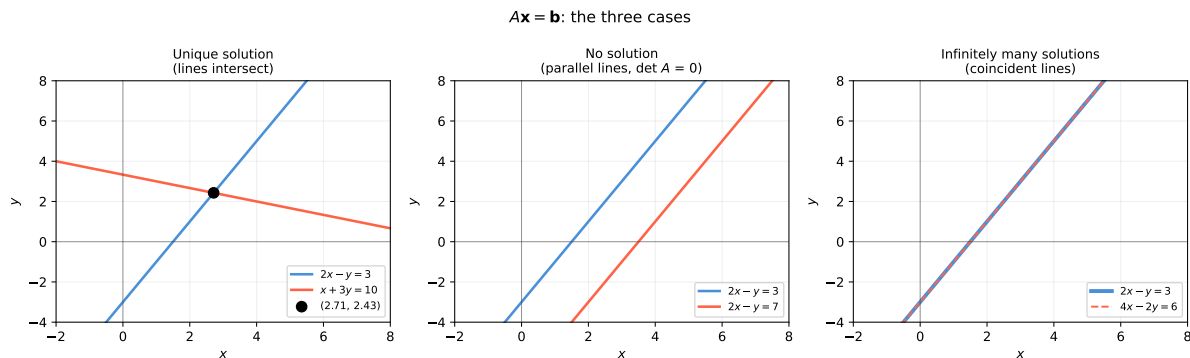
ax = axes[1]
ax.plot(x_range, (3 - 2*x_range) / (-1), color=blue, lw=2, label=r'$2x - y = 3$')
ax.plot(x_range, (7 - 2*x_range) / (-1), color=red, lw=2, label=r'$2x - y = 7$')
ax.set_title('No solution\n(parallel lines, det $A$ = 0)', fontsize=10)
ax.legend(fontsize=8)

# Panel 3: Infinitely many solutions (coincident lines)
ax = axes[2]
ax.plot(x_range, (3 - 2*x_range) / (-1), color=blue, lw=3, label=r'$2x - y = 3$')
ax.plot(x_range, (6 - 4*x_range) / (-2), color=red, lw=1.5,
        ls='--', label=r'$4x - 2y = 6$')
ax.set_title('Infinitely many solutions\n(coincident lines)', fontsize=10)
ax.legend(fontsize=8)

for ax in axes:
    ax.set_xlim(-2, 8); ax.set_ylim(-4, 8)
    ax.axhline(0, color='#333333', lw=0.4, alpha=0.5)
    ax.axvline(0, color='#333333', lw=0.4, alpha=0.5)
    ax.grid(True, alpha=0.2)
    ax.set_xlabel('$x$'); ax.set_ylabel('$y$')

fig.suptitle(r'$A \backslash \mathbf{b}$: the three cases', fontsize=12)
fig.tight_layout()
plt.show()

```



Problem 2: $\min \|Ax - b\|$ — Fit to Noisy Data

The situation. You have *more* equations than unknowns ($m > n$, overdetermined). No exact solution exists — the measurements contradict each other slightly because of noise. Find the $\mathbf{x}$ that minimizes total squared error.

Where it appears.

Why Linear Algebra?

- *Camera calibration*: 60 checkerboard corners, 6 unknown pose parameters per image. Minimize the reprojection error over all corners.
- *LIDAR plane fitting*: 10,000 range points; fit a single plane ($n = 3$ parameters). Far more data than unknowns — use them all to fight noise.
- *Any regression*: Predict crop yield from sensor readings. More data points than model parameters is good; it makes the estimate robust.

Why not just pick two equations and solve exactly? Any two equations give an exact solution, but a different pair gives a different answer. Least squares uses *all* measurements simultaneously, weighting each equally, and finds the single $\mathbf{x}$ that is closest to satisfying them all.

Worked example. A temperature sensor drifts linearly over time. We model the temperature as $T(t) = mt + c$ and want to recover m and c from 20 noisy readings.

```
import numpy as np

rng = np.random.default_rng(42)
t = np.linspace(0, 10, 20)
noise = rng.normal(0, 0.4, size=20)
T_obs = 2.3 * t + 1.7 + noise          # true: m = 2.3, c = 1.7

# Build A: each row is [t_i, 1]
A = np.column_stack([t, np.ones(len(t))]) # shape (20, 2)

# Solve: find [m, c] that minimises ||A @ [m,c] - T_obs||
x, residuals, rank, sv = np.linalg.lstsq(A, T_obs, rcond=None)
m_fit, c_fit = x

print(f"Fitted:  m = {m_fit:.4f},  c = {c_fit:.4f}")
print(f"True:    m = 2.3000,  c = 1.7000")
print(f"Rank of A: {rank}   (full rank = 2, good)")
```

```
Fitted:  m = 2.3193,  c = 1.5904
True:    m = 2.3000,  c = 1.7000
Rank of A: 2   (full rank = 2, good)
```

Effect of noise level on the fit.

```
import numpy as np
import matplotlib.pyplot as plt

fig, axes = plt.subplots(1, 2, figsize=(11, 4))
t = np.linspace(0, 10, 20)

for ax, sigma, label in zip(axes,
                             [0.4, 2.0],
```

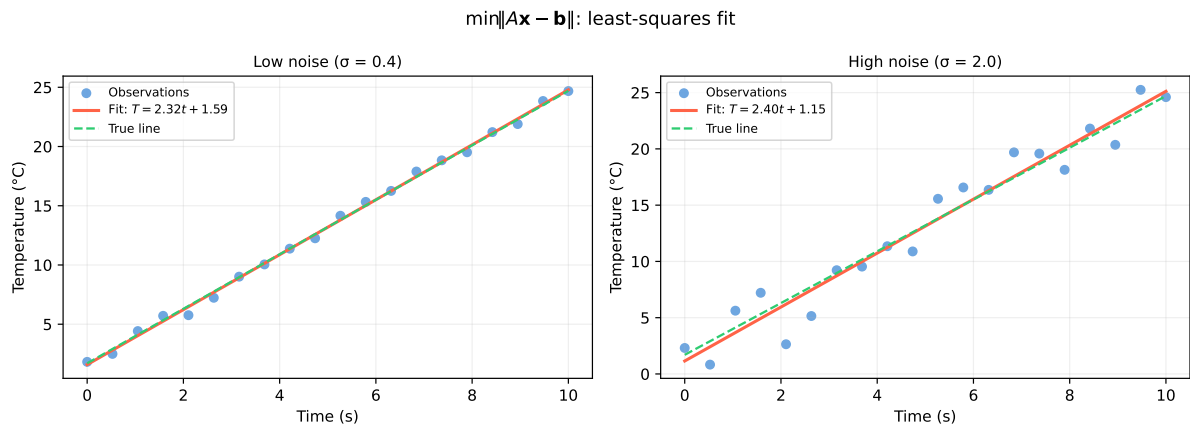
```

        ['Low noise (  $\sigma = 0.4$ )', 'High noise (  $\sigma = 2.0$ )']):
    rng = np.random.default_rng(42)
    T = 2.3 * t + 1.7 + rng.normal(0, sigma, 20)
    A = np.column_stack([t, np.ones(20)])
    x, _, _, _ = np.linalg.lstsq(A, T, rcond=None)

    ax.scatter(t, T, color='#4a90d9', s=30, alpha=0.8, label='Observations')
    t_line = np.array([t[0], t[-1]])
    ax.plot(t_line, x[0]*t_line + x[1], color='tomato', lw=2,
            label=rf'Fit:  $T = \{x[0]:.2f\}t + \{x[1]:.2f\}$ ')
    ax.plot(t_line, 2.3*t_line + 1.7, color='#2ecc71', lw=1.5, ls='--',
            label='True line')
    ax.set_xlabel('Time (s)'); ax.set_ylabel('Temperature ( $^{\circ}\text{C}$ )')
    ax.set_title(label, fontsize=10)
    ax.legend(fontsize=8); ax.grid(True, alpha=0.2)

fig.suptitle(r'$\min\|\mathbf{Ax} - \mathbf{b}\|$: least-squares fit',
            fontsize=12)
fig.tight_layout()
plt.show()

```



Problem 3: $\mathbf{Av} = \lambda\mathbf{v}$ — Find Structure

The situation. You want the *natural directions* of a matrix — the vectors that a matrix only stretches or compresses, never rotates. These directions reveal the underlying geometry of the data or system.

Where it appears.

- *Principal Component Analysis (PCA)*: The eigenvectors of the data covariance matrix point in the directions of maximum variance. Used in dimensionality reduction, visualization, and anomaly detection.

Why Linear Algebra?

- *Control stability*: The eigenvalues of a robot controller's state-transition matrix determine whether small errors decay to zero or grow without bound. $|\lambda_i| < 1$ for all i means stable; any $|\lambda_i| > 1$ means unstable.
- *Vibration modes*: In a mechanical linkage, the eigenvectors of the stiffness matrix are the natural vibration modes — the shapes the structure prefers to oscillate in.

Worked example. A 2-D point cloud of robot position estimates has been contaminated by correlated noise. The covariance matrix captures the shape and orientation of the noise ellipse.

```
import numpy as np

# Covariance matrix of 2-D position estimates
Sigma = np.array([[3.0, 1.5],
                  [1.5, 1.0]])

# eigh is for real symmetric/Hermitian matrices (guaranteed real eigenvalues)
eigenvalues, eigenvectors = np.linalg.eigh(Sigma)

print("Eigenvalues (variance along each principal axis):")
print(eigenvalues)

print("\nEigenvectors (columns = principal directions):")
print(eigenvectors)

# Verify: A v = λ v
for i in range(2):
    v = eigenvectors[:, i]
    lam = eigenvalues[i]
    err = np.linalg.norm(Sigma @ v - lam * v)
    print(f"\n||Σ v{i} - {i} v{i}|| = {err:.2e}")
```

```
Eigenvalues (variance along each principal axis):
[0.19722436 3.80277564]
```

```
Eigenvectors (columns = principal directions):
[[ 0.47185793 -0.8816746 ]
 [-0.8816746 -0.47185793]]
```

```
||Σ v0 - 0 v0|| = 1.37e-16
```

```
||Σ v1 - 1 v1|| = 2.22e-16
```

Eigenvectors reveal the noise ellipse orientation.

```
import numpy as np
import matplotlib.pyplot as plt
```

```

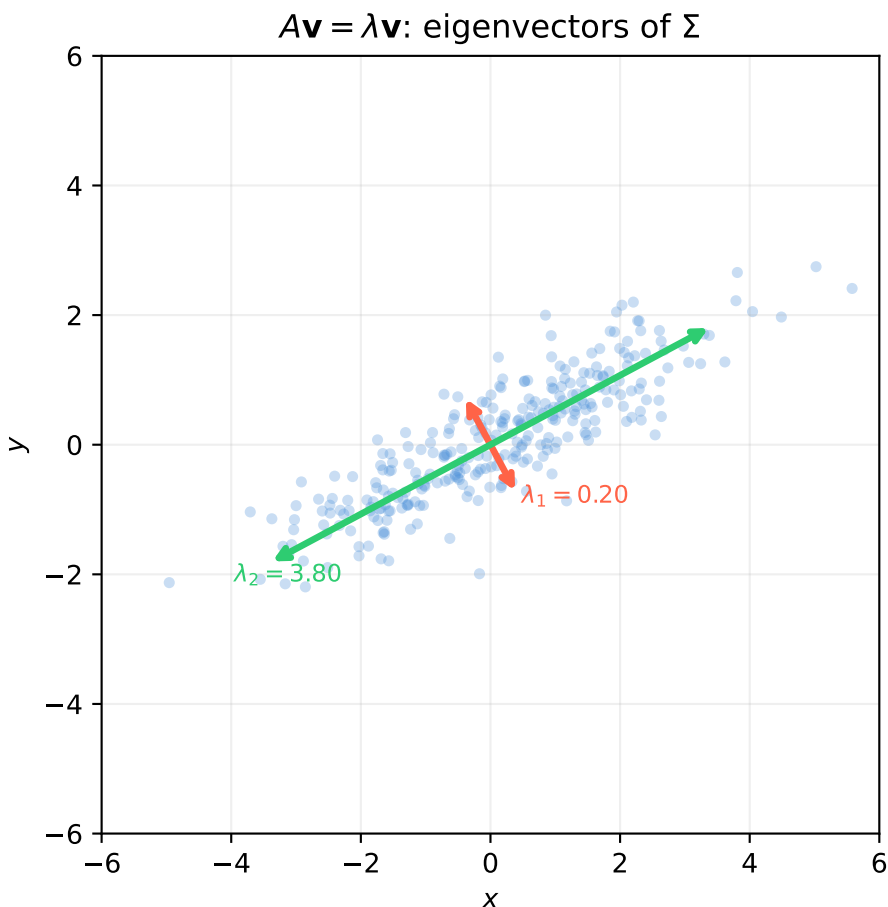
Sigma = np.array([[3.0, 1.5], [1.5, 1.0]])
evals, evecs = np.linalg.eigh(Sigma)
rng = np.random.default_rng(0)
pts = rng.multivariate_normal([0, 0], Sigma, 300)

fig, ax = plt.subplots(figsize=(5, 5))
ax.scatter(pts[:, 0], pts[:, 1], alpha=0.3, s=10, color='#4a90d9', label='Samples')

for i, color in enumerate(['tomato', '#2ecc71']):
    v = evecs[:, i] * np.sqrt(evals[i]) * 2
    ax.annotate('', xy=v, xytext=-v,
                arrowprops=dict(arrowstyle='<->', color=color, lw=2.5))
    ax.text(*(v * 1.15), rf'$\lambda_{i+1}={evals[i]:.2f}$',
            color=color, fontsize=9)

ax.set_aspect('equal')
ax.grid(True, alpha=0.2)
ax.set_xlim(-6, 6); ax.set_ylim(-6, 6)
ax.set_title(r'$A\mathbf{v}=\lambda\mathbf{v}$: eigenvectors of $\Sigma$')
ax.set_xlabel('$x$'); ax.set_ylabel('$y$')
plt.tight_layout()
plt.show()

```



The Three Problems Unify This Book

The table below maps every chapter to the problem it deepens:

Problem	Chapters
$A\mathbf{x} = \mathbf{b}$	2, 3, 5, 7, 9, 10, 11, 12
$\min\ A\mathbf{x} - \mathbf{b}\ $	15, 16, 17, 18, 22, 23
$A\mathbf{v} = \lambda\mathbf{v}$	13, 14, 19, 20, 24

Many chapters bridge two problems — SVD (Chapter 18) simultaneously provides the best low-rank approximation (Problem 2) and the principal directions (Problem 3). Lie groups (Chapter 24) generalize $A\mathbf{x} = \mathbf{b}$ to curved manifolds. Whenever you feel lost, return to this table and ask: *which of the three problems does this serve?*

Matrices as Data, Maps, and Geometry

A matrix is just a rectangular array of numbers — but it plays three completely different roles in ML, CV, and robotics. Confusing the roles is a common source of bugs. This section gives you all three mental models up front.

Mental Model 1 — Matrix as Data

The most literal interpretation: a matrix is a table of measured values.

Grayscale image. A 480×640 grayscale image is a 480×640 matrix. Entry (i, j) is the intensity of the pixel at row i , column j , an integer in $[0, 255]$.

```
import numpy as np
import matplotlib.pyplot as plt

# Simulate a small grayscale image (in practice: cv2.imread or plt.imread)
rng = np.random.default_rng(0)
img = rng.integers(0, 256, size=(6, 8), dtype=np.uint8)

print(f"Type: {type(img)}")
print(f"Shape: {img.shape} ← (rows, cols) = (height, width)")
print(f"Dtype: {img.dtype}")
print(f"\nPixel values:\n{img}")
```

```
Type: <class 'numpy.ndarray'>
Shape: (6, 8) ← (rows, cols) = (height, width)
Dtype: uint8
```

```
Pixel values:
[[ 95 130 194 217 207 235 15 163]
 [ 33 215 217 130 248 189 16 69]
 [184 232 205 78 169 61 125 10]
 [ 29 240 66 19 182 39 59 4]
 [ 59 81 222 44 120 122 50 208]
 [ 78 28 64 166 121 89 170 233]]
```

Colour (RGB) image. Add a third axis: shape $(H, W, 3)$. Channel 0 is red, 1 is green, 2 is blue. This is technically a *tensor* (3-D array), but the 2-D slices are still matrices.

```
import numpy as np

rng = np.random.default_rng(0)
img_rgb = rng.integers(0, 256, size=(6, 8, 3), dtype=np.uint8)
print(f"RGB image shape: {img_rgb.shape} ← (H, W, channels)")
red_channel = img_rgb[:, :, 0]          # a 6×8 matrix
print(f"Red channel: {red_channel.shape}")
```

```
RGB image shape: (6, 8, 3) ← (H, W, channels)
Red channel:      (6, 8)
```

Point cloud. A collection of N 3-D points from a depth sensor is stored as an $N \times 3$ matrix — one row per point, columns are (x, y, z) .

```
import numpy as np

rng = np.random.default_rng(0)
N = 1000
points_xyz = rng.standard_normal((N, 3))    # N × 3 matrix
print(f"Point cloud shape: {points_xyz.shape} ← (N_points, 3)")
print(f"First three points:\n{points_xyz[:3]}")
```

```
Point cloud shape: (1000, 3) ← (N_points, 3)
First three points:
[[ 0.12573022 -0.13210486  0.64042265]
 [ 0.10490012 -0.53566937  0.36159505]
 [ 1.30400005  0.94708096 -0.70373524]]
```

The key insight: operations on data — crop, resize, blur, transform — almost always reduce to matrix arithmetic. When a convolution is slow, you are paying for matrix multiplications that have not yet been fused by your hardware.

Mental Model 2 — Matrix as Map (Function)

A matrix $W \in \mathbb{R}^{m \times n}$ is a *function* that takes an n -dimensional vector and returns an m -dimensional vector:

$$\mathbf{y} = W\mathbf{x}, \quad \mathbf{x} \in \mathbb{R}^n, \mathbf{y} \in \mathbb{R}^m$$

Every fully connected layer of a neural network is exactly this — plus a bias and a nonlinearity.


```
import numpy as np

# One fully-connected layer: 8 inputs → 4 outputs
rng = np.random.default_rng(1)
W = rng.standard_normal((4, 8))    # weight matrix, shape (4, 8)
b = rng.standard_normal(4)         # bias vector, shape (4,)
x = rng.standard_normal(8)         # input vector, shape (8,)

y_pre = W @ x + b                  # pre-activation, shape (4,)
y = np.maximum(0, y_pre)           # ReLU activation

print(f"Input shape: {x.shape}")
print(f"Weight shape: {W.shape}")
print(f"Output shape: {y.shape}")
```

```
Input shape: (8,)
Weight shape: (4, 8)
Output shape: (4,)
```

A three-layer network. Chain three matrices:

```
import numpy as np

rng = np.random.default_rng(1)
# Architecture: 784 → 256 → 128 → 10
W1 = rng.standard_normal((256, 784)) * 0.01
W2 = rng.standard_normal((128, 256)) * 0.01
W3 = rng.standard_normal((10, 128)) * 0.01

x = rng.standard_normal(784)        # one MNIST pixel vector

h1 = np.maximum(0, W1 @ x)           # shape (256,)
h2 = np.maximum(0, W2 @ h1)         # shape (128,)
logits = W3 @ h2                     # shape (10,) ← class scores

print(f"Logits: {logits.round(4)}")
```

```
Logits: [ 0.0002 -0.0008  0.0006  0.0016  0.0004  0.0002  0.0014 -0.0016 -
 0.0006
 0.0016]
```

The entire forward pass of this network is three matrix–vector multiplies. The backward pass (backpropagation) is three more. Linear algebra is not something used *alongside* deep learning — it *is* deep learning.

What each row of W does. Row i of W is a *template* (also called a *filter* or *feature detector*). The output $y_i = W[i, :] \cdot \mathbf{x}$ is how strongly the input matches that template — a dot product, which Chapter 4 explains geometrically.

Mental Model 3 — Matrix as Geometry

A matrix can encode a *rigid body transformation* — a rotation, translation, scaling, or any combination — that moves 3-D objects in space.

Rotation matrix. A 3×3 matrix R that rotates vectors without stretching them. Example: rotate 45° around the z -axis.

```
import numpy as np

theta = np.pi / 4 # 45 degrees
Rz = np.array([
    [ np.cos(theta), -np.sin(theta), 0],
    [ np.sin(theta),  np.cos(theta), 0],
    [ 0,              0,             1]
])

p = np.array([1.0, 0.0, 0.0]) # unit vector along x
p_rotated = Rz @ p
print(f"Original:  {p}")
print(f"Rotated:   {p_rotated.round(4)}") # [0.707, 0.707, 0]

# A rotation matrix preserves lengths
print(f"||p|| = {np.linalg.norm(p):.4f}")
print(f"||Rz p|| = {np.linalg.norm(p_rotated):.4f}") # same
```

```
Original: [1. 0. 0.]
Rotated:  [0.7071 0.7071 0.    ]
||p|| = 1.0000
||Rz p|| = 1.0000
```

The 4×4 homogeneous transformation matrix. Rotation and translation together. This is *the* fundamental data structure of robotics — every robot joint, every camera pose, and every object in the scene is described by one of these.

```
import numpy as np

# A camera located at [1, 2, 0.5] metres, rotated 30° around z
theta = np.pi / 6 # 30 degrees
R = np.array([
```

```

    [ np.cos(theta), -np.sin(theta), 0],
    [ np.sin(theta),  np.cos(theta), 0],
    [0,              0,              1]
])
t = np.array([1.0, 2.0, 0.5]) # translation

# Assemble 4x4 homogeneous matrix
T = np.eye(4)
T[:3, :3] = R
T[:3, 3] = t

print("4x4 transformation matrix T:")
print(T.round(4))

# Transform a 3D point P = [0, 0, 3] (in local/camera frame)
# to world frame using homogeneous coordinates
P_local = np.array([0.0, 0.0, 3.0, 1.0]) # homogeneous: append 1
P_world = T @ P_local
print(f"\nPoint in local frame: {P_local[:3]}")
print(f"Point in world frame: {P_world[:3].round(4)}")

```

```

4x4 transformation matrix T:
[[ 0.866 -0.5   0.    1.   ]
 [ 0.5   0.866  0.    2.   ]
 [ 0.    0.    1.    0.5  ]
 [ 0.    0.    0.    1.   ]]

```

```

Point in local frame: [0. 0. 3.]
Point in world frame: [1.  2.  3.5]

```

Why append a 1? The extra coordinate makes translation into a matrix multiply — without it, pure matrix multiplication cannot translate (it always maps zero to zero). Chapter 7 proves this and explains the full machinery of homogeneous coordinates.

Visualising the three mental models together.

```

import numpy as np
import matplotlib.pyplot as plt
import matplotlib.gridspec as gridspec

rng = np.random.default_rng(7)
fig = plt.figure(figsize=(13, 4))
gs = gridspec.GridSpec(1, 3, figure=fig, wspace=0.35)

# Panel 1: Matrix as Data
ax1 = fig.add_subplot(gs[0])
img = rng.integers(30, 220, (8, 10), dtype=np.uint8)

```

```

ax1.imshow(img, cmap='gray', vmin=0, vmax=255)
ax1.set_title('Data: 8x10 image matrix', pad=8)
ax1.set_xlabel('column j')
ax1.set_ylabel('row i')

# Panel 2: Matrix as Map
ax2 = fig.add_subplot(gs[1])
xs = np.linspace(-1, 1, 6)
pts = np.column_stack([xs, np.zeros(6)])
W2 = np.array([[np.cos(0.8), -np.sin(0.8)],
                [np.sin(0.8), np.cos(0.8)]] * 1.4
mapped = pts @ W2.T
for p, mp in zip(pts, mapped):
    ax2.annotate('', xy=mp, xytext=p,
                  arrowprops=dict(arrowstyle='->', color='#2ecc71', lw=1.5))
ax2.scatter(pts[:,0], pts[:,1], color='#4a90d9', s=50, zorder=5, label='input')
ax2.scatter(mapped[:,0], mapped[:,1], color='#2ecc71', s=50, zorder=5, label='output')
ax2.set_xlim(-2.5, 2.5); ax2.set_ylim(-2, 2); ax2.set_aspect('equal')
ax2.axhline(0, color='#333333', lw=0.4, alpha=0.4)
ax2.axvline(0, color='#333333', lw=0.4, alpha=0.4)
ax2.set_title(r'Map: $\mathbf{y}=W\mathbf{x}$', pad=8)
ax2.legend(fontsize=8)

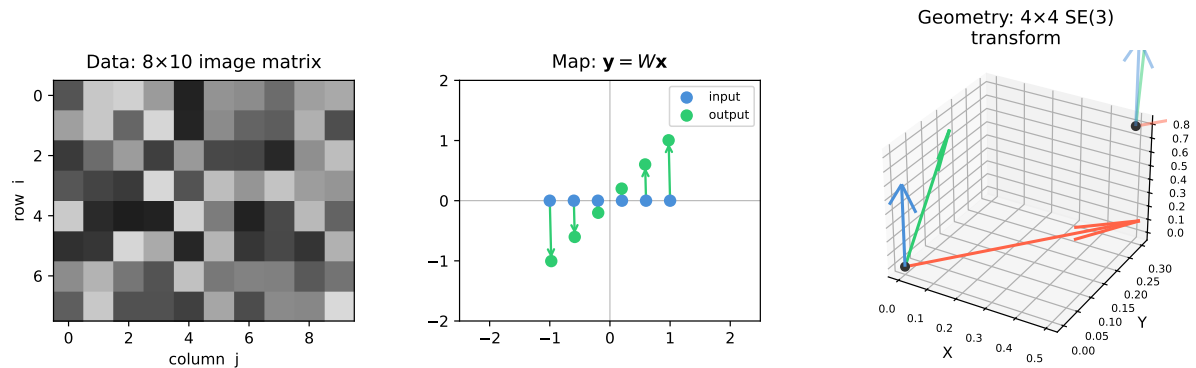
# Panel 3: Matrix as Geometry
ax3 = fig.add_subplot(gs[2], projection='3d')
origin = np.zeros(3)
theta = np.pi / 5
R = np.array([[np.cos(theta), -np.sin(theta), 0],
               [np.sin(theta), np.cos(theta), 0],
               [0, 0, 1]])
t_vec = np.array([0.5, 0.3, 0.8])
for i, (col, lab) in enumerate(zip(['tomato', '#2ecc71', '#4a90d9'], ['x', 'y', 'z'])):
    ax3.quiver(*origin, *R[:,i]*0.6, color=col, lw=2, arrow_length_ratio=0.3)
    ax3.quiver(*(origin+t_vec), *R[:,i]*0.6, color=col, lw=2,
                arrow_length_ratio=0.3, alpha=0.5)
ax3.scatter(*origin, color='#333333', s=30)
ax3.scatter(*(origin+t_vec), color='#333333', s=30)
ax3.set_title('Geometry: 4x4 SE(3)\ntransform', pad=4)
ax3.set_xlabel('X', labelpad=2)
ax3.set_ylabel('Y', labelpad=2)
ax3.set_zlabel('Z', labelpad=2)
ax3.tick_params(labelsize=7)

fig.tight_layout()
plt.show()

```

C:\Users\james.choi\AppData\Local\Temp\ipykernel_35516\2920969.py:55: UserWarning: This figure

```
fig.tight_layout()
```



Why Three Models?

The reason to keep all three in mind is that a single algorithm often involves all of them at once. Camera calibration, for example:

1. Stores the checkerboard corner coordinates as a **data matrix** ($N \times 3$).
2. Applies the projection matrix as a **map** from 3-D world to 2-D pixel coordinates.
3. Expresses the camera's position and orientation as a **geometric** 4×4 transform.

If you only think “matrix = table of numbers,” the projection step looks magical. If you only think “matrix = function,” you lose track of what the columns of K mean physically. Fluency means switching between the three views fluidly.

A Reading Map for ML, CV, and Robotics

This book is structured to be read cover-to-cover *or* used as a reference. The following map tells you which chapters are load-bearing for each field so you can prioritize intelligently.

Chapter Importance by Field

The table below rates each chapter on a three-star scale for the three target fields. = essential foundation; = strongly recommended; = useful but skippable on a first pass.

Ch	Title	ML	CV	Robotics	Core Concept
1	Why Linear Algebra?				Orienta- tion and mental models
2	Systems of Linear Equations				$A\mathbf{x} = \mathbf{b}$; sensor fusion
3	Row Reduction and Echelon Forms				Rank, null space, ob- servability
4	Vectors in				Dot product, basis, co- ordinates
5	Matrix Equations and Column Space				Feature collinear- ity, block matrices
6	Linear Transfor- mations				Image warping, change of basis
7	Affine & Homoge- neous Coordi- nates				Rigid body represen- tation
8	Rotations in 3D				SO(3), Euler, quater- nions
9	Rigid Body Kinemat- ics				SE(3), FK, velocity kinemat- ics
10	Camera Models				Intrinsics, distortion, projection
11	Projective Geometry				Homogra- phy, lines at infinity

Ch	Title	ML	CV	Robotics	Core Concept
12	LU, QR, and Cholesky				Structured solvers, stability
13	Eigenvalues and Eigenvectors				Spectral decomposition
14	Diagonalization				Matrix powers, Markov chains
15	Norms and Distances				Loss functions, regularization
16	Orthogonality				Projections, Gram-Schmidt
17	Least Squares				The workhorse of applied math
18	SVD				Low-rank approx, pseudoinverse
19	Covariance Matrices				Uncertainty representation
20	PCA				Dimensionality reduction
21	Matrix Calculus				Backprop, gradient derivation
22	Jacobians and Linearization				Velocity kinematics, EKF
23	State Estimation				Kalman filter, sensor fusion

Ch	Title	ML	CV	Robotics	Core Concept
24	Lie Groups and Lie Algebras				Optimization on manifolds
25	Epipolar Geometry				Stereo, structure-from-motion

Fast Tracks

If you are working in one specific field, these reading orders let you build fluency quickly before circling back for depth.

Machine Learning track (neural networks, optimization, statistical learning):

$$1 \rightarrow 4 \rightarrow 5 \rightarrow 13 \rightarrow 14 \rightarrow 15 \rightarrow 17 \rightarrow 18 \rightarrow 20 \rightarrow 21$$

After this track you can implement backpropagation from scratch, understand why batch normalization works, and read papers that use SVD-based compression.

Computer Vision track (calibration, detection, reconstruction):

$$1 \rightarrow 4 \rightarrow 6 \rightarrow 7 \rightarrow 10 \rightarrow 11 \rightarrow 17 \rightarrow 18 \rightarrow 25$$

After this track you can calibrate a stereo camera, implement RANSAC-based homography estimation, and understand the essential matrix derivation.

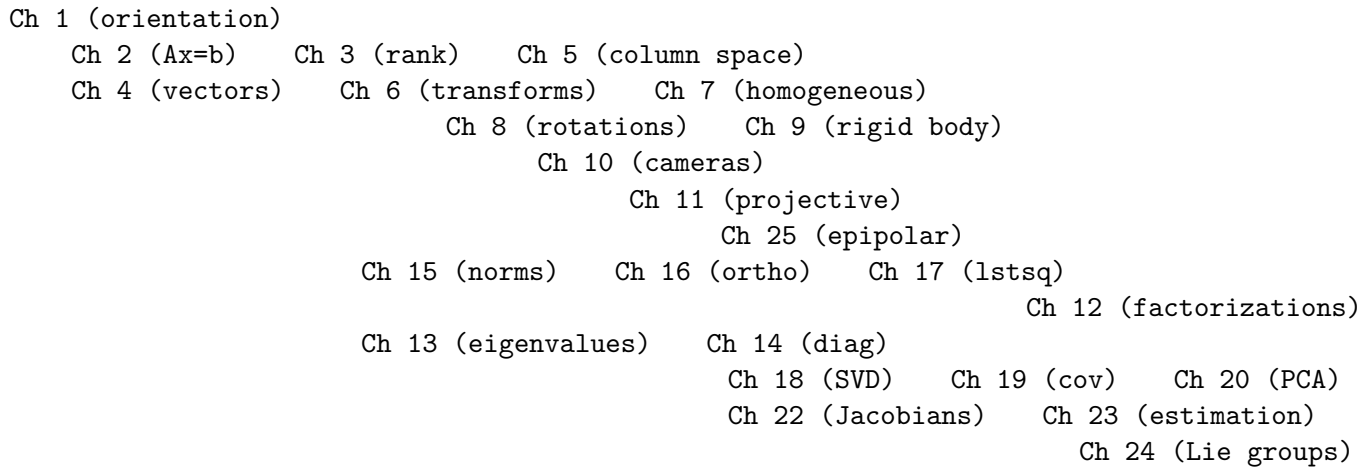
Robotics track (kinematics, planning, estimation):

$$1 \rightarrow 4 \rightarrow 7 \rightarrow 8 \rightarrow 9 \rightarrow 12 \rightarrow 17 \rightarrow 22 \rightarrow 23 \rightarrow 24$$

After this track you can implement a full-body kinematic solver, write an EKF for IMU/GPS fusion, and understand why Lie groups replace Euler angles in modern motion planning.

How Chapters Depend on Each Other

The dependency graph below is a simplified guide — you can read ahead and return, but these are the conceptually critical prerequisites.



Ch 21 (matrix calculus) can be read any time after Ch 13; it connects to both Ch 17 and Ch 22.

Using This Book as a Reference

Each chapter is designed to stand alone for look-up. When you hit a bug or notation you do not recognize:

1. **Check Appendix E (notation)** — every symbol used in the book is defined there.
2. **Check the relevant chapter** — each section opens with its own motivation so you can read it without reading the whole chapter.
3. **Check the application section** — the worked example in each App section uses the same code patterns you are likely debugging.

The index cross-references every concept by name, and the appendices collect the most-used identities for quick look-up:

- **Appendix B** — matrix factorizations reference card
- **Appendix C** — rotation representations comparison (Euler, quaternion, rotation matrix, axis-angle)
- **Appendix D** — matrix calculus identities
- **Appendix F** — tensors and index notation for readers who need to read ML papers

Python Setup

All code in this book is Python 3.10 or later. Every code block is self-contained and executes directly inside the rendered book via Quarto's Jupyter engine.

Installing the Required Packages

Run this once in your terminal:

```
pip install numpy scipy matplotlib opencv-python open3d sympy scikit-learn
```

Verifying Your Installation

Paste this block and run it. All lines should print without error.

```
import sys, numpy, scipy, matplotlib, cv2, open3d, sympy, sklearn

print(f"Python      {sys.version.split()[0]}")
print(f"NumPy       {numpy.__version__}")
print(f"SciPy        {scipy.__version__}")
print(f"Matplotlib    {matplotlib.__version__}")
print(f"OpenCV       {cv2.__version__}")
print(f"Open3D       {open3d.__version__}")
print(f"SymPy        {sympy.__version__}")
print(f"scikit-learn {sklearn.__version__}")
print("\nAll imports OK.")
```

NumPy Quick Reference

The three operations you will use in nearly every chapter:

```
import numpy as np

# 1 - Creating arrays
A = np.array([[1, 2], [3, 4]], dtype=float) # from nested list
v = np.array([5.0, 6.0])
I = np.eye(3)                               # identity matrix
Z = np.zeros((4, 4))                         # zero matrix
```

```
# 2 - Matrix multiplication (always use @ for matrices, not *)
y = A @ v          # matrix-vector product, shape (2,)
B = A @ A          # matrix-matrix product, shape (2,2)
s = v @ v          # dot product when both are 1-D, returns scalar

# 3 - Solving linear systems
b = np.array([1.0, 2.0])
x = np.linalg.solve(A, b)          # exact solve: A x = b
xp, res, rank, sv = np.linalg.lstsq(A, b, rcond=None) # least squares

print(f"A @ x = {A @ x} (should equal b = {b})")
print(f"Condition number: {np.linalg.cond(A):.2f}")
```

```
A @ x = [1. 2.] (should equal b = [1. 2.])
Condition number: 14.93
```

Shape discipline. NumPy makes it easy to confuse 1-D arrays (`n,`) with 2-D column vectors (`n, 1`). This book always uses 1-D arrays for vectors and calls out the shape explicitly in comments.

```
import numpy as np

# 1-D array ← used throughout this book for vectors
v1 = np.array([1.0, 2.0, 3.0])          # shape (3,)

# 2-D column vector ← only when matrix algebra requires it
v2 = v1.reshape(-1, 1)                  # shape (3, 1)
v3 = v1[:, np.newaxis]                  # same thing, shape (3, 1)

print(f"1-D: {v1.shape}    2-D column: {v2.shape}")
```

```
1-D: (3,)    2-D column: (3, 1)
```

Code Conventions Used in This Book

Convention	Meaning
A, B, R, W	matrices (capital letters)
x, y, b, v	vectors (lowercase letters)
n, m, k	integer dimensions
<code>rng = np.random.default_rng(seed)</code>	reproducible random state
<code># shape (m, n)</code>	comment after every array creation

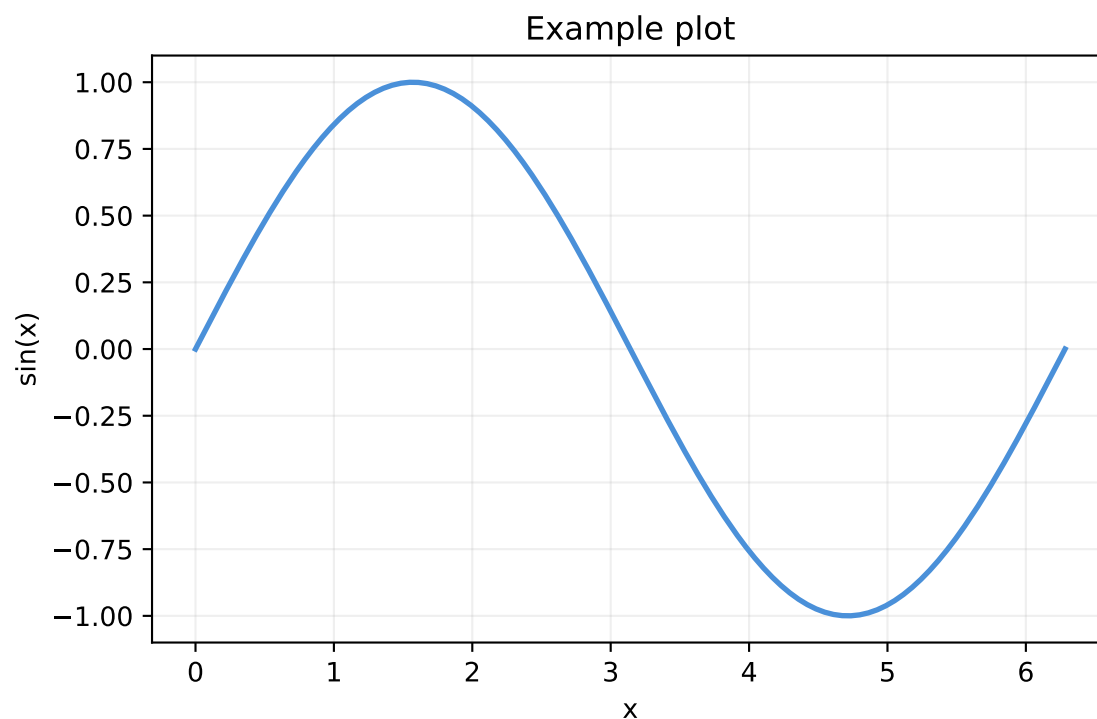
Convention	Meaning
<code>round(4)</code>	limit print noise in examples

Plot functions in this book follow this template — light mode, inline output:

```
import numpy as np
import matplotlib.pyplot as plt

def plot_something():
    fig, ax = plt.subplots(figsize=(6, 4))
    x = np.linspace(0, 2 * np.pi, 200)
    ax.plot(x, np.sin(x), color='#4a90d9', lw=2)
    ax.set_xlabel('x'); ax.set_ylabel('sin(x)')
    ax.set_title('Example plot')
    ax.grid(True, alpha=0.2)
    plt.tight_layout()
    plt.show()

plot_something()
```



Quarto captures the `plt.show()` output and renders it inline in the book.

Downloading the Datasets

Several chapters use real datasets (depth images, point clouds, calibration targets, IMU logs). Appendix A contains the full download instructions and directory layout. For now, all you need to know is that when a code block references a file such as `data/realsense_depth.png`, you will find the download command for that file in Appendix A.

Exercises

Work through these before reading the solutions. They are designed to expose the assumptions that most people hold silently — and get wrong.

1.1 You call `numpy.linalg.solve(A, b)` on a 2×2 system and get a numerical answer. How do you know whether the answer is correct? What specific number should you compute, and what threshold should alarm you?

1.2 A dataset has 100 observations and 50 features. You want to fit a linear model. Write down the matrix dimensions of A , $\mathbf{x}$, and $\mathbf{b}$ in the least-squares formulation $\min \|\mathbf{Ax} - \mathbf{b}\|$. Is the system overdetermined, underdetermined, or square?

1.3 Consider the 2×2 rotation matrix

$$R = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}$$

Without using Python, compute $R^T R$. What does your answer tell you about how rotation matrices transform lengths?

1.4 A neural network's first layer maps 512-dimensional inputs to 128-dimensional outputs. Write this as $\mathbf{y} = W\mathbf{x} + \mathbf{b}$. What is the shape of W ? If you process a mini-batch of 32 samples simultaneously, what shape is the input matrix X , and how does the equation change?

1.5 The mental-model-2 section says “row i of W is a template.” Using the matrix

$$W = \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{pmatrix}, \quad \mathbf{x} = \begin{pmatrix} 3 \\ 5 \end{pmatrix}$$

compute $W\mathbf{x}$ by hand. Interpret each output as a dot product with a row template. What does each row “detect”?

1.6 (Conceptual) The 4×4 homogeneous transformation code in §1.2 appends a 1 to 3-D points before multiplying by T . If you accidentally appended 0 instead of 1, what part of the transformation would be lost? Why?

1.7 (Harder) You have two matrices $A \in \mathbb{R}^{3 \times 5}$ and $B \in \mathbb{R}^{5 \times 2}$. Is the product AB defined? What is its shape? Now someone tells you $AB = C$ and asks you to solve for B given A and C . Is this always possible? When is it not?

Solutions

Solution 1.1

The number to compute is the *condition number*:

```
import numpy as np

A = np.array([[3.0, 1.0],
              [1.0, 2.0]])
b = np.array([9.0, 8.0])

x = np.linalg.solve(A, b)
kappa = np.linalg.cond(A)

print(f"Solution:          {x}")
print(f"Condition number: {kappa:.2f}")
print(f"Residual:           {np.linalg.norm(A @ x - b):.2e}")
```

```
Solution:          [2. 3.]
Condition number: 2.62
Residual:           0.00e+00
```

The condition number $\kappa(A) = \|A\| \cdot \|A^{-1}\|$ tells you how many digits of accuracy you lose. A rough rule: if $\kappa \approx 10^k$, you lose about k decimal digits from the machine precision of $\approx 10^{-16}$.

- $\kappa < 100$: safe
- $\kappa \sim 10^6$: lose ~6 digits — answers are accurate to ~10 significant figures in double precision
- $\kappa > 10^{12}$: alarm — results may be meaningless

Always compute `np.linalg.cond(A)` before trusting a linear solve. Also verify `np.linalg.norm(A @ x - b)` is near machine zero ($< 10^{-10}$ for double).

Solution 1.2

The least-squares setup for 100 observations, 50 features:

- $A \in \mathbb{R}^{100 \times 50}$ — design matrix: one row per observation, one column per feature
- $\mathbf{x} \in \mathbb{R}^{50}$ — the coefficient vector (what we solve for)
- $\mathbf{b} \in \mathbb{R}^{100}$ — the target vector (one observed value per row)

The system is **overdetermined**: there are 100 equations but only 50 unknowns. No exact solution exists in general, so we minimize $\|A\mathbf{x} - \mathbf{b}\|^2$.

```
import numpy as np

rng = np.random.default_rng(0)
A = rng.standard_normal((100, 50))    # shape (100, 50)
b = rng.standard_normal(100)         # shape (100,)

x, residuals, rank, sv = np.linalg.lstsq(A, b, rcond=None)

print(f"A shape: {A.shape}")
print(f"x shape: {x.shape}")
print(f"b shape: {b.shape}")
print(f"Rank:    {rank} (out of {min(A.shape[0], A.shape[1])} possible)")
```

```
A shape: (100, 50)
x shape: (50,)
b shape: (100,)
Rank:    50 (out of 50 possible)
```

Solution 1.3

$$R^T R = \begin{pmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{pmatrix} \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}$$

Computing entry (1,1): $\cos^2 \theta + \sin^2 \theta = 1$. Entry (1,2): $-\cos \theta \sin \theta + \sin \theta \cos \theta = 0$. By symmetry, $R^T R = I$.

This means $R^T = R^{-1}$ — rotation matrices are *orthogonal*. An orthogonal matrix preserves inner products: $\|R\mathbf{v}\|^2 = (R\mathbf{v})^T(R\mathbf{v}) = \mathbf{v}^T R^T R \mathbf{v} = \mathbf{v}^T \mathbf{v} = \|\mathbf{v}\|^2$. Rotation neither stretches nor compresses — it just changes direction.

```
import numpy as np
theta = np.pi / 3
R = np.array([[np.cos(theta), -np.sin(theta)],
              [np.sin(theta),  np.cos(theta)]])
print(np.round(R.T @ R, 10))    # should print [[1, 0], [0, 1]]
```

```
[[ 1. -0.]
 [-0.  1.]]
```

Solution 1.4

For a single input vector:

Why Linear Algebra?

- $W \in \mathbb{R}^{128 \times 512}$ — weight matrix (output dimension $\times$ input dimension)
- $\mathbf{x} \in \mathbb{R}^{512}$ — input vector
- $\mathbf{b} \in \mathbb{R}^{128}$ — bias vector
- $\mathbf{y} = W\mathbf{x} + \mathbf{b} \in \mathbb{R}^{128}$

For a mini-batch of 32 samples, stack inputs as columns (or rows — PyTorch uses rows by convention):

Column convention (math-natural): - $X \in \mathbb{R}^{512 \times 32}$ — each column is one sample - $Y = WX + \mathbf{b}\mathbf{1}^T \in \mathbb{R}^{128 \times 32}$

Row convention (PyTorch/NumPy natural): - $X \in \mathbb{R}^{32 \times 512}$ — each row is one sample - $Y = XW^T + \mathbf{b} \in \mathbb{R}^{32 \times 128}$

```
import numpy as np

W = np.random.randn(128, 512) * 0.01 # shape (128, 512)
b = np.zeros(128)                     # shape (128,)
X = np.random.randn(32, 512)          # batch of 32, shape (32, 512) - row convention

Y = X @ W.T + b                       # shape (32, 128)
print(f"W: {W.shape}, X: {X.shape}, Y: {Y.shape}")
```

W: (128, 512), X: (32, 512), Y: (32, 128)

The key point: `W.shape` is always (out_features, in_features) in PyTorch convention. When you see `nn.Linear(512, 128)`, the stored weight matrix is 128×512 .

Solution 1.5

$$W\mathbf{x} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} 3 \\ 5 \end{pmatrix} = \begin{pmatrix} 1 \cdot 3 + 0 \cdot 5 \\ 0 \cdot 3 + 1 \cdot 5 \\ 1 \cdot 3 + 1 \cdot 5 \end{pmatrix} = \begin{pmatrix} 3 \\ 5 \\ 8 \end{pmatrix}$$

Interpretation:

- **Row 1** = $[1, 0]$: detects the first component. Output 3 = “how much of x_1 is present.”
- **Row 2** = $[0, 1]$: detects the second component. Output 5 = “how much of x_2 is present.”
- **Row 3** = $[1, 1]$: detects the *sum* of both components. Output 8 = “total energy.”

This is the template interpretation: each row acts as a pattern, and the output is the dot product of the input with that pattern. In a trained network, rows of W become learned feature detectors — edges, textures, semantic concepts — through gradient descent, not hand design.

Solution 1.6

If you append 0 instead of 1:

```
import numpy as np

theta = np.pi / 6
R = np.array([[np.cos(theta), -np.sin(theta), 0],
              [np.sin(theta),  np.cos(theta), 0],
              [0,              0,              1]])
t = np.array([1.0, 2.0, 0.5])

T = np.eye(4)
T[:3, :3] = R
T[:3, 3] = t

P_with_one = np.array([0.0, 0.0, 3.0, 1.0]) # correct
P_with_zero = np.array([0.0, 0.0, 3.0, 0.0]) # wrong

print("With 1 (point):", (T @ P_with_one)[:3].round(4))
print("With 0 (direction):", (T @ P_with_zero)[:3].round(4))
```

With 1 (point): [1. 2. 3.5]
 With 0 (direction): [0. 0. 3.]

The fourth column of T is $[t_x, t_y, t_z, 1]^T$. When the homogeneous coordinate is 1, it gets multiplied by the translation and added. When it is 0, translation is suppressed — you are treating the vector as a *direction*, not a *point*.

This is not just a book quirk: it is the standard convention in all robotics and graphics frameworks. Position vectors carry 1; direction vectors (surface normals, velocity vectors) carry 0. Mixing them up is a common source of silent numerical errors — the code runs, but the translated result is wrong by exactly the translation offset. Chapter 7 develops this distinction rigorously.

Solution 1.7

Is AB defined? Yes. $A \in \mathbb{R}^{3 \times 5}$, $B \in \mathbb{R}^{5 \times 2}$ — inner dimensions match (both 5). $C = AB \in \mathbb{R}^{3 \times 2}$.

Can we solve for B ? We need $AB = C$, i.e., $A\mathbf{b}_j = \mathbf{c}_j$ for each column $j = 1, 2$. This is two systems of 3 equations in 5 unknowns — each system is *underdetermined* (more unknowns than equations).

```
import numpy as np

rng = np.random.default_rng(42)
A = rng.standard_normal((3, 5))    # shape (3, 5)
B_true = rng.standard_normal((5, 2))
C = A @ B_true                    # shape (3, 2)

# Minimum-norm least-squares solution for each column
B_hat = np.linalg.lstsq(A, C, rcond=None)[0]

print(f"A:      {A.shape}")
print(f"C:      {C.shape}")
print(f"B_hat: {B_hat.shape}")
print(f"Residual A @ B_hat - C: {np.linalg.norm(A @ B_hat - C):.2e}")
print(f"B_hat == B_true? Not in general - infinitely many solutions exist.")
print(f"||B_true||: {np.linalg.norm(B_true):.4f}")
print(f"||B_hat||: {np.linalg.norm(B_hat):.4f} ← lstsq returns minimum-norm solution")
```

```
A:      (3, 5)
C:      (3, 2)
B_hat: (5, 2)
Residual A @ B_hat - C: 6.58e-15
B_hat == B_true? Not in general - infinitely many solutions exist.
||B_true||: 2.1835
||B_hat||: 2.1062 ← lstsq returns minimum-norm solution
```

When is a unique solution impossible? When the system is underdetermined (more unknowns than equations) or when A does not have full column rank. Here, A is 3×5 — it has at most rank 3, but B lives in $\mathbb{R}^{5 \times 2}$, so there is a 2-dimensional null space of solutions for each column. The `lstsq` function returns the minimum-norm solution among the infinitely many that satisfy $AB = C$ exactly.

This situation — underdetermined systems, null spaces, minimum-norm solutions — is central to Chapters 3, 16, 17, and 18.

Systems of Linear Equations

Linear Equations as Hyperplane Intersections

A single linear equation in two unknowns,

$$a_1x_1 + a_2x_2 = b$$

describes a *line* in $\mathbb{R}^2$. Every point on that line satisfies the equation. A system of two such equations asks: where do two lines meet?

That geometric picture generalises exactly. In $\mathbb{R}^3$ one equation defines a *plane*. In $\mathbb{R}^n$ it defines a *hyperplane* — a flat $(n - 1)$ -dimensional slice through n -dimensional space. Solving a system of m equations in n unknowns means finding the intersection of m hyperplanes.

One Equation, One Hyperplane

The equation $\mathbf{a}^T \mathbf{x} = b$ (row vector $\mathbf{a}$, column vector $\mathbf{x}$) defines a hyperplane *perpendicular to* $\mathbf{a}$ at signed distance $b/\|\mathbf{a}\|$ from the origin.

```
import numpy as np
import matplotlib.pyplot as plt

fig, ax = plt.subplots(figsize=(6, 5))

x = np.linspace(-3, 3, 300)

# Equation 1: 2x1 + x2 = 3 → x2 = 3 - 2*x1
ax.plot(x, 3 - 2*x, color='#4a90d9', lw=2, label=r'$2x_1 + x_2 = 3$')

# Equation 2: -x1 + 2x2 = 2 → x2 = (2 + x1) / 2
ax.plot(x, (2 + x) / 2, color='#2ecc71', lw=2, label=r'$-x_1 + 2x_2 = 2$')

# Intersection
A = np.array([[2., 1.], [-1., 2.]])
b = np.array([3., 2.])
sol = np.linalg.solve(A, b)
ax.scatter(*sol, color='tomato', s=80, zorder=5,
```

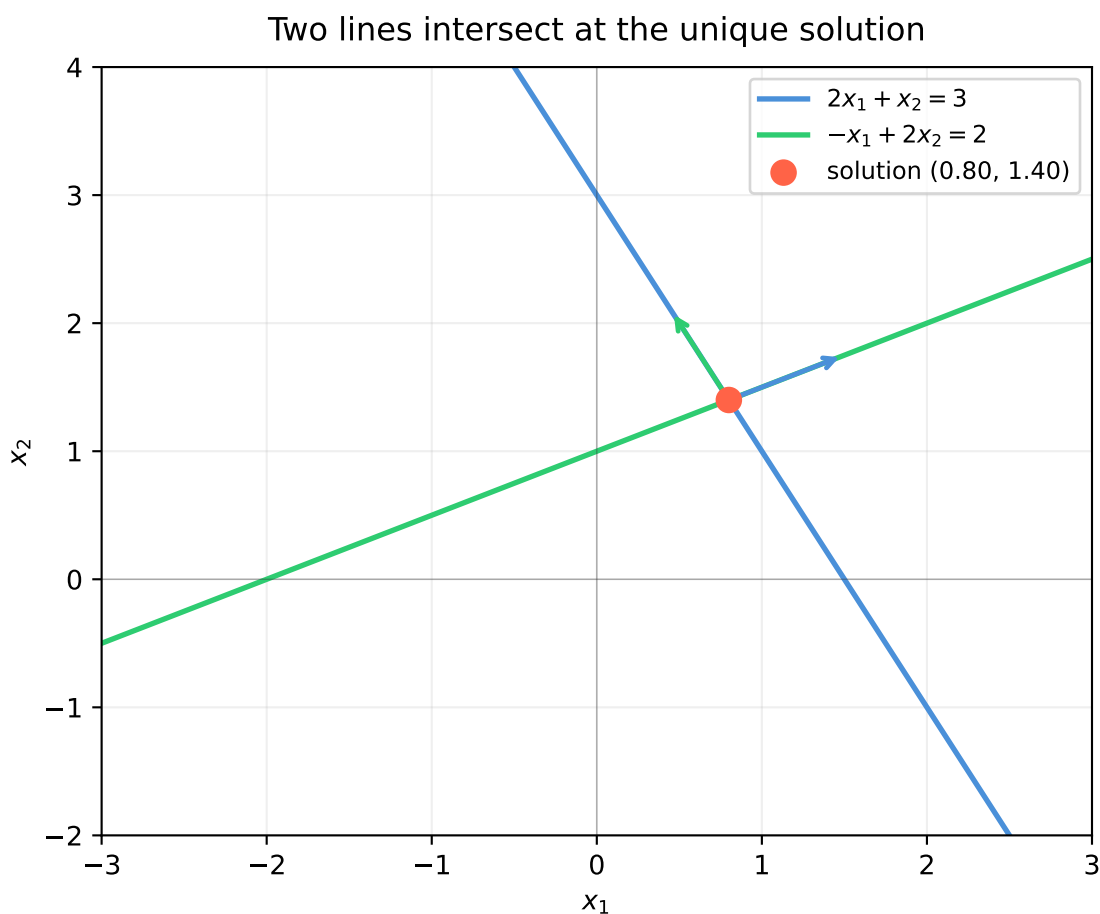
```

label=f'solution ({sol[0]:.2f}, {sol[1]:.2f})'

# Normal vectors at solution
for row, col in zip(A, ['#4a90d9', '#2ecc71']):
    ax.annotate('', xy=sol + row * 0.35, xytext=sol,
                arrowprops=dict(arrowstyle='->', color=col, lw=1.8))

ax.set_xlim(-3, 3); ax.set_ylim(-2, 4)
ax.axhline(0, color='#333333', lw=0.5, alpha=0.4)
ax.axvline(0, color='#333333', lw=0.5, alpha=0.4)
ax.set_xlabel('$x_1$'); ax.set_ylabel('$x_2$')
ax.set_title('Two lines intersect at the unique solution', pad=10)
ax.legend(fontsize=9)
ax.grid(True, alpha=0.2)
plt.tight_layout()
plt.show()

```



The arrows show the *normal vectors* $\mathbf{a}_1 = [2, 1]^T$ and $\mathbf{a}_2 = [-1, 2]^T$ — each perpendicular to its line. The solution is the unique point that lies on both hyperplanes simultaneously.

Extending to 3D

With three unknowns, each equation cuts out a plane in $\mathbb{R}^3$. Two planes generically intersect in a *line*. Add a third plane — if it crosses that line, you get a point.

```
import numpy as np
import matplotlib.pyplot as plt

colors = ['#4a90d9', '#2ecc71', '#9b59b6']

# System with unique solution at (1, 1, 1)
A = np.array([[2., 1., 0.],
              [0., 3., 1.],
              [1., 0., 2.]])
b = A @ np.array([1., 1., 1.])

fig = plt.figure(figsize=(7, 6))
ax = fig.add_subplot(111, projection='3d')

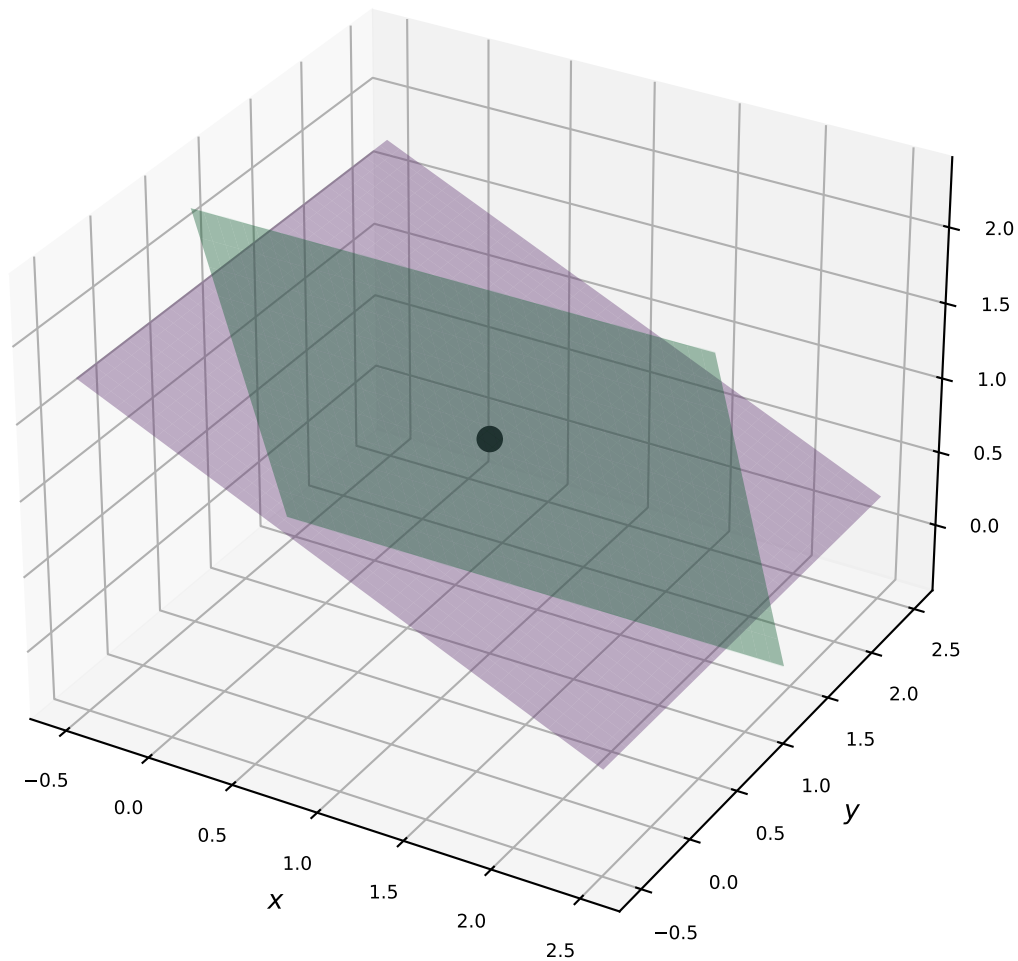
r = np.linspace(-0.5, 2.5, 40)
X, Y = np.meshgrid(r, r)

for row, rhs, col in zip(A, b, colors):
    if abs(row[2]) > 1e-9:
        Z = (rhs - row[0]*X - row[1]*Y) / row[2]
        mask = (Z > -0.5) & (Z < 2.5)
        Zp = np.where(mask, Z, np.nan)
        ax.plot_surface(X, Y, Zp, alpha=0.4, color=col)

sol = np.linalg.solve(A, b)
ax.scatter(*sol, color='black', s=80, zorder=10, depthshade=False, label='solution')

ax.set_xlabel('$x$', labelpad=3)
ax.set_ylabel('$y$', labelpad=3)
ax.set_zlabel('$z$', labelpad=3)
ax.set_title('Three planes - one intersection point', pad=8)
ax.tick_params(labelsize=7)
plt.tight_layout()
plt.show()
```

Three planes — one intersection point



The Three Cases — Unique, No Solution, Infinitely Many

Two lines in $\mathbb{R}^2$ can intersect at a point, be parallel (never meet), or coincide (overlap entirely). The three panels below show each case with concrete slope/intercept values.

```
import numpy as np
import matplotlib.pyplot as plt

fig, axes = plt.subplots(1, 3, figsize=(13, 4))
x = np.linspace(-4, 4, 300)
blue, green = '#4a90d9', '#2ecc71'
```

```

# Panel 1: Unique solution
ax = axes[0]
s1, i1 = 1.0, 1.0
s2, i2 = -0.5, -1.0
xi = (i2 - i1) / (s1 - s2)
yi = s1 * xi + i1
ax.plot(x, s1*x + i1, color=blue, lw=2, label=f'y = {s1}x + {i1}')
ax.plot(x, s2*x + i2, color=green, lw=2, label=f'y = {s2}x + {i2}')
ax.scatter(xi, yi, color='tomato', s=80, zorder=5,
           label=f'({xi:.2f}, {yi:.2f})')
ax.set_title('Unique solution\n(lines intersect)', fontsize=10)
ax.legend(fontsize=8)

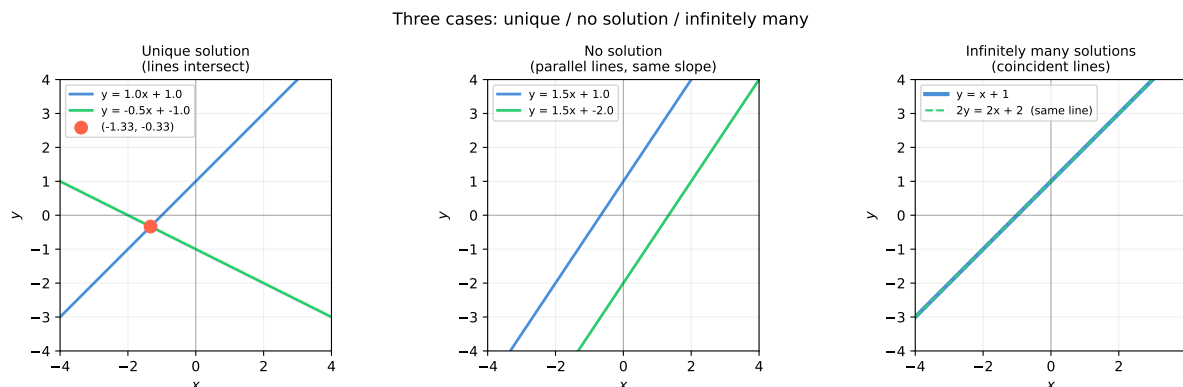
# Panel 2: Parallel lines - no solution
ax = axes[1]
s3, i3 = 1.5, 1.0
s4, i4 = 1.5, -2.0
ax.plot(x, s3*x + i3, color=blue, lw=2, label=f'y = {s3}x + {i3}')
ax.plot(x, s4*x + i4, color=green, lw=2, label=f'y = {s4}x + {i4}')
ax.set_title('No solution\n(parallel lines, same slope)', fontsize=10)
ax.legend(fontsize=8)

# Panel 3: Coincident lines - infinitely many solutions
ax = axes[2]
ax.plot(x, 1.0*x + 1.0, color=blue, lw=3, label='y = x + 1')
ax.plot(x, (2.0*x + 2.0)/2, color=green, lw=1.5, ls='--',
        label='2y = 2x + 2 (same line)')
ax.set_title('Infinitely many solutions\n(coincident lines)', fontsize=10)
ax.legend(fontsize=8)

for ax in axes:
    ax.axhline(0, color='#333333', lw=0.4, alpha=0.4)
    ax.axvline(0, color='#333333', lw=0.4, alpha=0.4)
    ax.set_xlim(-4, 4); ax.set_ylim(-4, 4)
    ax.grid(True, alpha=0.2)
    ax.set_aspect('equal')
    ax.set_xlabel('$x$'); ax.set_ylabel('$y$')

fig.suptitle('Three cases: unique / no solution / infinitely many', fontsize=12)
fig.tight_layout()
plt.show()

```



Setting both slopes equal makes the lines parallel — no solution. Setting both slopes *and* intercepts equal makes them coincide — infinitely many solutions. These two degenerate cases are not numerical flukes; they are the subject of §2.2.

The Normal-Vector Viewpoint

Row i of A is the normal vector to the i -th hyperplane. The right-hand side b_i is the signed offset along that normal:

$$\mathbf{a}_i^T \mathbf{x} = b_i \quad \Leftrightarrow \quad \text{signed distance from origin} = \frac{b_i}{\|\mathbf{a}_i\|}$$

- Changing $\mathbf{a}_i$ **tilts** the hyperplane.
- Changing b_i **slides** it without tilting.
- The solution $\mathbf{x}^*$ lies on every hyperplane simultaneously.

When you debug a failed linear solve, the first geometric question is: *are any two of my hyperplanes nearly parallel?* Nearly parallel normals mean nearly linearly dependent rows — rank deficiency. That is Chapter 3.

The Three Solution Cases

A system $A\mathbf{x} = \mathbf{b}$ with $A \in \mathbb{R}^{m \times n}$ has exactly one of three outcomes:

Case	Geometry	Physical example
No solution	Hyperplanes miss each other	Contradictory sensor readings
Unique solution	Hyperplanes meet at one point	Well-determined position fix
Infinitely many	Hyperplanes share a line or subspace	Underdetermined kinematics

There is no fourth option. Understanding which case you are in — before you call a solver — is what separates robust code from brittle code.

Case 1 — No Solution (Inconsistent System)

The system is *inconsistent* when $\mathbf{b}$ is not in the column space of A : no linear combination of A 's columns reaches $\mathbf{b}$.

Physical example — GPS with a faulty satellite. Three range measurements from satellites should meet at one point. If one satellite has a clock error, the three spheres no longer intersect — the system is inconsistent.

```
import numpy as np

# Inconsistent 3x2 system: two lines that don't intersect in R^2
A = np.array([[1., 2.],
              [2., 4.], # parallel to row 1 (same normal direction)
              [1., -1.]])
b = np.array([3., 7., 1.]) # 7 - 2*3=6, so rows 1 & 2 are contradictory

# np.linalg.solve requires square A; use lstsq which always returns something
x, res, rank, sv = np.linalg.lstsq(A, b, rcond=None)

print(f"Rank of A:    {rank}    (out of {min(A.shape)} possible)")
print(f"Residual:     {np.linalg.norm(A @ x - b):.4f}    ← non-zero means no exact solution")
print(f"lstsq x:       {x.round(4)}    ← best-fit, not exact")
```

```
Rank of A:    2    (out of 2 possible)
Residual:     0.4472    ← non-zero means no exact solution
lstsq x:      [1.8 0.8]    ← best-fit, not exact
```

The non-zero residual is the diagnostic. A zero residual would mean the system was consistent; a large residual points to a measurement contradiction worth investigating before proceeding.

Case 2 — Unique Solution

The system has a unique solution when A has full column rank and $m \geq n$ (at least as many equations as unknowns, independent ones).

Physical example — trilateration from three WiFi access points. Three known-position routers each provide a range estimate. Three linear constraints, three unknowns — generically one solution.

```
import numpy as np

# 3x3 system, full rank
A = np.array([[1., 0., 1.],
              [0., 2., 1.],
              [3., 1., 0.]])
b = np.array([4., 5., 7.])

x = np.linalg.solve(A, b)
print(f"Solution:      {x.round(6)}")
print(f"Verify Ax:     {(A @ x).round(6)}    (should equal b = {b})")
print(f"Condition:     {np.linalg.cond(A):.2f}")
```

```
Solution:      [1.857143 1.428571 2.142857]
Verify Ax:     [4. 5. 7.]    (should equal b = [4. 5. 7.])
Condition:     3.47
```

The condition number is what distinguishes a *numerically* unique solution from a *practically* unreliable one. A condition number of 1000 does not mean no solution exists — it means the solution is sensitive to errors in **b**.

Case 3 — Infinitely Many Solutions

When A has more unknowns than independent equations, the solution is not a point but a *flat* (an affine subspace). The general solution is:

$$\mathbf{x} = \mathbf{x}_p + \mathbf{x}_h$$

where $\mathbf{x}_p$ is any *particular* solution satisfying $A\mathbf{x}_p = \mathbf{b}$, and $\mathbf{x}_h$ is any vector in the *null space* of A (satisfying $A\mathbf{x}_h = \mathbf{0}$).

Physical example — redundant robot arm. A 7-DoF arm positioning its end-effector in 6-DoF space has one extra degree of freedom. For any target pose, there is a 1-parameter family of joint configurations that reaches it.

```
import numpy as np

# Underdetermined system: 2 equations, 3 unknowns
A = np.array([[1., 2., 3.],
              [0., 1., 2.]])
b = np.array([6., 3.])

# Minimum-norm particular solution via lstsq
```

```

x_p, _, rank, _ = np.linalg.lstsq(A, b, rcond=None)
print(f"Rank:          {rank} (< {A.shape[1]} unknowns → underdetermined)")
print(f"x_p (min-norm): {x_p.round(4)}")
print(f"Residual:      {np.linalg.norm(A @ x_p - b):.2e} ← exact solution")

# Null space: any vector orthogonal to both rows of A
# The null space of a 2×3 rank-2 matrix is 1-dimensional
_, _, Vt = np.linalg.svd(A)
null_vec = Vt[-1] # last right singular vector
print(f"\nNull vector: {null_vec.round(4)}")
print(f"A @ null_vec: {(A @ null_vec).round(10)} ← should be [0, 0]")

# Any scalar multiple of null_vec can be added to x_p
alpha = 2.5
x_other = x_p + alpha * null_vec
print(f"\nalpha={alpha}: x = {x_other.round(4)}, A @ x = {(A @ x_other).round(6)}")

```

```

Rank:          2 (< 3 unknowns → underdetermined)
x_p (min-norm): [1. 1. 1.]
Residual:      2.70e-15 ← exact solution

```

```

Null vector: [ 0.4082 -0.8165  0.4082]
A @ null_vec: [-0. -0.] ← should be [0, 0]

```

```

alpha=2.5: x = [ 2.0206 -1.0412  2.0206], A @ x = [6. 3.]

```

The null space is the set of all “free motions” — directions you can move without changing the output Ax . Chapter 3 derives the null space systematically from row reduction.

Detecting the Case Before Solving

In practice you should diagnose the system *before* committing to a solver:

```

import numpy as np

def diagnose_system(A, b):
    """Print which of the three cases applies."""
    m, n = A.shape
    rank_A = np.linalg.matrix_rank(A)
    Ab = np.column_stack([A, b])
    rank_Ab = np.linalg.matrix_rank(Ab)

    print(f"Shape:          {m} equations, {n} unknowns")

```

```

print(f"rank(A) = {rank_A},    rank([A|b]) = {rank_Ab}")

if rank_Ab > rank_A:
    print("→ INCONSISTENT: no solution")
elif rank_A == n:
    print("→ UNIQUE solution")
else:
    print(f"→ INFINITELY MANY: null space dimension = {n - rank_A}")

# Test all three cases
print("=== Inconsistent ===")
diagnose_system(np.array([[1.,2.],[2.,4.]]),
                np.array([3., 7.]))

print("\n=== Unique ===")
diagnose_system(np.array([[1.,0.],[0.,1.]]),
                np.array([3., 2.]))

print("\n=== Underdetermined ===")
diagnose_system(np.array([[1., 2., 3.]]),
                np.array([6.]))

```

```

=== Inconsistent ===
Shape:          2 equations, 2 unknowns
rank(A) = 1,    rank([A|b]) = 2
→ INCONSISTENT: no solution

=== Unique ===
Shape:          2 equations, 2 unknowns
rank(A) = 2,    rank([A|b]) = 2
→ UNIQUE solution

=== Underdetermined ===
Shape:          1 equations, 3 unknowns
rank(A) = 1,    rank([A|b]) = 1
→ INFINITELY MANY: null space dimension = 2

```

The key rule: $\text{rank}([A|b]) > \text{rank}(A)$ means inconsistent. $\text{rank}(A) = n$ means unique. Otherwise infinitely many.

Why This Matters for Each Field

Machine learning. Under-parameterised models (many data, few weights) are generically overdetermined and inconsistent — we use least squares. Over- parameterised models (few data,

many weights) are underdetermined — we need regularization to pick one solution from infinitely many. Both cases appear constantly; recognising which you are in shapes every modelling decision.

Computer vision. Homography estimation from point correspondences is a $2n \times 9$ system for n point pairs. With $n = 4$ it is exactly determined (if no point is degenerate); with $n > 4$ it is overdetermined and we use DLT (the linear least-squares method). Fewer than 4 pairs — infinitely many homographies fit.

Robotics. Forward kinematics is unique (joint angles $\rightarrow$ end-effector pose). Inverse kinematics is generically underdetermined for redundant arms: the null space of the Jacobian encodes the self-motion. Chapter 22 derives this in detail.

Augmented Matrices and Gaussian Elimination

Gaussian elimination is the oldest algorithm in numerical linear algebra — and the one every production solver ultimately traces back to. Understanding it means understanding *why* solvers fail, not just how to call them.

The Augmented Matrix

Writing the system $A\mathbf{x} = \mathbf{b}$ as an *augmented matrix* $[A \mid \mathbf{b}]$ groups all the information into one object:

$$\begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix} \mathbf{x} = \begin{pmatrix} b_1 \\ b_2 \\ b_3 \end{pmatrix} \longrightarrow \left(\begin{array}{ccc|c} a_{11} & a_{12} & a_{13} & b_1 \\ a_{21} & a_{22} & a_{23} & b_2 \\ a_{31} & a_{32} & a_{33} & b_3 \end{array} \right)$$

Row operations on $[A \mid \mathbf{b}]$ transform the system but do not change its solution set. The three legal operations are:

Operation	Notation	Why it preserves solutions
Swap rows i and j	$R_i \leftrightarrow R_j$	Re-orders equations — same constraints
Scale row i by $c \neq 0$	$R_i \leftarrow cR_i$	Multiplying both sides by c
Add c times row j to row i	$R_i \leftarrow R_i + cR_j$	Adding a multiple of one equation to another

Forward Elimination — Step by Step

The goal is to introduce zeros below each *pivot* (the leading nonzero in each row), converting $[A \mid \mathbf{b}]$ to *row echelon form* (REF).

Example system:

$$\begin{pmatrix} 2 & 1 & -1 \\ -3 & -1 & 2 \\ -2 & 1 & 2 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 8 \\ -11 \\ -3 \end{pmatrix}$$

```
import numpy as np

def forward_eliminate(aug):
    """Gaussian elimination with partial pivoting. Returns row echelon form."""
    M = aug.astype(float).copy()
    m, n = M.shape          # m rows, n columns (last is augmented)
    pivot_row = 0

    for col in range(n - 1):          # for each variable column
        # Partial pivoting: swap in the largest magnitude in this column
        best = np.argmax(np.abs(M[pivot_row:, col])) + pivot_row
        M[[pivot_row, best]] = M[[best, pivot_row]]

        if abs(M[pivot_row, col]) < 1e-12:
            continue                  # skip: column is zero below pivot

        for row in range(pivot_row + 1, m):
            factor = M[row, col] / M[pivot_row, col]
            M[row] -= factor * M[pivot_row]

        pivot_row += 1

    return M

A = np.array([[ 2.,  1., -1.],
              [-3., -1.,  2.],
              [-2.,  1.,  2.]])
b = np.array([8., -11., -3.])
aug = np.column_stack([A, b])

print("Original augmented matrix:")
print(aug)

ref = forward_eliminate(aug)
print("\nRow echelon form:")
print(ref.round(6))
```

Original augmented matrix:

```
[[ 2.  1. -1.  8.]
 [ -3. -1.  2. -11.]
 [ -2.  1.  2. -3.]]
```

Row echelon form:

```
[[ -3.      -1.      2.      -11.      ]
 [  0.      1.666667  0.666667  4.333333]
 [  0.      0.      0.2     -0.2     ]]
```

After forward elimination the matrix is upper triangular. The solution is recovered by *back substitution* — solve for the last variable, substitute upward.

```
def back_substitute(ref):
    """Back substitution on row echelon form. Returns solution vector."""
    m, n = ref.shape
    nvars = n - 1
    x = np.zeros(nvars)

    for i in range(nvars - 1, -1, -1):
        if abs(ref[i, i]) < 1e-12:
            raise ValueError(f"Zero pivot at row {i}: system is singular or underdetermined")
        x[i] = (ref[i, -1] - ref[i, i+1:nvars] @ x[i+1:nvars]) / ref[i, i]

    return x

x = back_substitute(ref)
print(f"Solution: {x}")
print(f"Verify:   A @ x = {A @ x}   (should equal {b})")
```

Solution: [2. 3. -1.]

Verify: A @ x = [8. -11. -3.] (should equal [8. -11. -3.])

Expected output: $\mathbf{x} = [2, 3, -1]^T$, and $\mathbf{Ax} = [8, -11, -3]^T$.

Partial Pivoting — Why It Matters

Naive elimination without pivoting can divide by near-zero entries, amplifying floating-point errors catastrophically.

```
import numpy as np

# Without pivoting: tiny pivot causes large error
A_bad = np.array([[1e-16, 1.],
                  [1.,    1.]])
b_bad = np.array([1., 2.])

# "Manual" elimination without pivoting
aug = np.column_stack([A_bad, b_bad]).astype(float)
factor = aug[1, 0] / aug[0, 0] # divides by 1e-16 ← catastrophic
aug[1] -= factor * aug[0]
print("Without pivoting, row 1 after elimination:")
print(aug[1]) # coefficient may be 0 due to cancellation

# With pivoting (numpy.linalg.solve uses partial pivoting internally)
x_good = np.linalg.solve(A_bad, b_bad)
print(f"\nnp.linalg.solve (with pivoting): {x_good}")
print(f"Residual: {np.linalg.norm(A_bad @ x_good - b_bad):.2e}")
```

Without pivoting, row 1 after elimination:
 [0.e+00 -1.e+16 -1.e+16]

np.linalg.solve (with pivoting): [1. 1.]
 Residual: 0.00e+00

numpy.linalg.solve uses LAPACK's dgesv, which applies partial pivoting at every step. This is why you should never roll your own elimination for production use — but you should understand why pivoting exists.

Counting Operations

Gaussian elimination on an $n \times n$ system costs roughly $\frac{2}{3}n^3$ floating-point operations (flops). This cubic scaling is why:

- A 100×100 system solves in microseconds.
- A 10000×10000 system takes about a second.
- A $10^6 \times 10^6$ dense system is infeasible — sparse methods (Ch 12) are required.

```
import numpy as np
import time

for n in [100, 500, 1000, 2000]:
    rng = np.random.default_rng(0)
```



```

A = rng.standard_normal((n, n))
b = rng.standard_normal(n)
t0 = time.perf_counter()
x = np.linalg.solve(A, b)
dt = time.perf_counter() - t0
print(f"n={n:5d}:  {dt*1000:.2f} ms    "
      f"residual={np.linalg.norm(A@x-b):.1e}")

```

```

n= 100:  27.73 ms    residual=5.1e-13
n= 500: 106.54 ms    residual=1.0e-11
n=1000:  49.31 ms    residual=9.7e-12
n=2000:  74.50 ms    residual=3.7e-10

```

The residual stays near machine epsilon ($\approx 10^{-15}$) for all sizes as long as A is well-conditioned — a testament to how well-implemented partial pivoting is.

What Elimination Reveals

Beyond the solution, elimination reveals the *structure* of the system:

- **The number of pivots** equals $\text{rank}(A)$ — the number of independent equations.
- **Zero rows in the augmented form** with a nonzero right-hand side signal inconsistency.
- **Zero rows with zero right-hand side** signal free variables (infinitely many solutions).

Chapter 3 formalises this into the machinery of *reduced row echelon form*, rank, and the four fundamental subspaces.

Solving Systems in Code

NumPy exposes three functions for solving $A\mathbf{x} = \mathbf{b}$. Choosing the wrong one — or using the right one without checking its output — is a recurring source of silent bugs.

Function	Use when	Notes
----------	----------	-------

The Three Functions and When to Use Them

Function	Use when	Notes
<code>np.linalg.solve(A, b)</code>	A is square and you believe it is nonsingular	Fastest; raises <code>LinAlgError</code> if exactly singular
<code>np.linalg.lstsq(A, b)</code>	A is rectangular, overdetermined, or possibly rank-deficient	Always returns something; check residuals
<code>scipy.linalg.solve(A, b)</code>	Square system, want more control (pivoting options, symmetric flag)	More options than NumPy version

```
import numpy as np
import scipy.linalg

A = np.array([[3., 1., 0.],
              [1., 4., 1.],
              [0., 1., 3.]])
b = np.array([1., 2., 3.])

# Method 1: numpy solve (square, nonsingular)
x1 = np.linalg.solve(A, b)

# Method 2: lstsq (works for any shape)
x2, residuals, rank, sv = np.linalg.lstsq(A, b, rcond=None)

# Method 3: scipy solve (identical result, but more options available)
x3 = scipy.linalg.solve(A, b)

print(f"np.solve:      {x1.round(6)}")
print(f"np.lstsq:      {x2.round(6)}")
print(f"scipy:           {x3.round(6)}")
print(f"All agree:      {np.allclose(x1, x2) and np.allclose(x1, x3)}")
```

```
np.solve:      [0.266667 0.2      0.933333]
np.lstsq:      [0.266667 0.2      0.933333]
scipy:         [0.266667 0.2      0.933333]
All agree:     True
```

What lstsq Returns — and What to Check

`np.linalg.lstsq` returns a 4-tuple. Each element tells you something about the system:

```
import numpy as np

# Overdetermined system: 5 equations, 2 unknowns
rng = np.random.default_rng(42)
A = rng.standard_normal((5, 2))
b_true = np.array([1.0, 2.0])
b = A @ b_true + 0.05 * rng.standard_normal(5) # add small noise

x, residuals, rank, singular_values = np.linalg.lstsq(A, b, rcond=None)

print(f"Solution x:           {x.round(4)}")
print(f"Residuals:           {residuals}") # sum of squared residuals
print(f"Rank:                 {rank}")
print(f"Singular values:       {singular_values.round(4)}")
print(f"Condition number:      {singular_values[0]/singular_values[-1]:.2f}")
print(f"\nManual residual:      {np.linalg.norm(A @ x - b)**2:.6f}")
```

```
Solution x:           [1.0292 1.9697]
Residuals:            [0.00437096]
Rank:                 2
Singular values:      [2.7181 1.2574]
Condition number:      2.16
```

```
Manual residual:      0.004371
```

Warning: `residuals` is empty (shape `(0,)`) when A does not have full column rank, or when $m \leq n$. Always compute the residual manually with `np.linalg.norm(A @ x - b)` if you are not certain the system is overdetermined and full-rank.

The Silent Wrong Answer

`np.linalg.solve` does *not* warn you when A is ill-conditioned. It will return a number — a plausible-looking one — that can be completely wrong.

```
import numpy as np

# Hilbert matrix: famously ill-conditioned
def hilbert(n):
    i, j = np.meshgrid(np.arange(1, n+1), np.arange(1, n+1))
```

```

    return 1.0 / (i + j - 1)

for n in [4, 8, 12]:
    H = hilbert(n)
    x_true = np.ones(n)
    b = H @ x_true          # exact right-hand side for solution = all-ones

    x_computed = np.linalg.solve(H, b)
    kappa = np.linalg.cond(H)
    error = np.linalg.norm(x_computed - x_true)

    print(f"n={n:2d}    (H)={kappa:.2e}    ||x_comp - x_true||={error:.2e}")

```

```

n= 4    (H)=1.55e+04    ||x_comp - x_true||=8.27e-14
n= 8    (H)=1.53e+10    ||x_comp - x_true||=1.73e-07
n=12    (H)=1.76e+16    ||x_comp - x_true||=1.13e+00

```

For $n = 12$, the condition number exceeds 10^{16} — beyond machine precision. The computed solution has no significant digits. Yet `np.linalg.solve` returns without error.

Defensive pattern:

```

import numpy as np

def safe_solve(A, b, cond_threshold=1e12):
    """Solve A x = b with condition number guard."""
    kappa = np.linalg.cond(A)
    if kappa > cond_threshold:
        import warnings
        warnings.warn(f"Ill-conditioned system:  = {kappa:.2e}. "
                      f"Solution may be unreliable.", stacklevel=2)
    x = np.linalg.solve(A, b)
    residual = np.linalg.norm(A @ x - b)
    return x, kappa, residual

x, kappa, res = safe_solve(hilbert(10), np.ones(10))
print(f"  = {kappa:.2e},  residual = {res:.2e}")

```

```

= 1.60e+13,  residual = 2.44e-10

```

```

C:\Users\james.choi\AppData\Local\Temp\ipykernel_26700\1476246802.py:14: UserWarning: Ill-
conditioned system:  = 1.60e+13. Solution may be unreliable.

```

```

    x, kappa, res = safe_solve(hilbert(10), np.ones(10))

```

Solving Multiple Right-Hand Sides Efficiently

When you need $A\mathbf{x}_j = \mathbf{b}_j$ for many vectors $\mathbf{b}_j$ with the same A , factorising A once and reusing is far faster than calling `solve` repeatedly.

```
import numpy as np
import scipy.linalg
import time

n = 500
rhs_count = 200
rng = np.random.default_rng(0)
A = rng.standard_normal((n, n))
B = rng.standard_normal((n, rhs_count)) # 200 right-hand sides

# Naive: solve 200 times
t0 = time.perf_counter()
X_naive = np.column_stack([np.linalg.solve(A, B[:, j]) for j in range(rhs_count)])
t_naive = time.perf_counter() - t0

# Smart: LU factorisation once, then forward/back substitution 200 times
t0 = time.perf_counter()
lu, piv = scipy.linalg.lu_factor(A)
X_smart = scipy.linalg.lu_solve((lu, piv), B)
t_smart = time.perf_counter() - t0

print(f"Naive (200× solve): {t_naive*1000:.1f} ms")
print(f"LU once + solve: {t_smart*1000:.1f} ms")
print(f"Speed-up: {t_naive/t_smart:.1f}×")
print(f"Results match: {np.allclose(X_naive, X_smart)}")
```

```
Naive (200× solve): 2811.0 ms
LU once + solve: 14.2 ms
Speed-up: 197.6×
Results match: True
```

`scipy.linalg.lu_factor` computes the LU decomposition once; `lu_solve` applies it cheaply. Chapter 12 covers LU decomposition in depth.

Solving Sparse Systems

When A has mostly zeros (common in FEM, graph problems, and SLAM), dense solvers waste memory and time. Use `scipy.sparse`:

```
import numpy as np
import scipy.sparse
import scipy.sparse.linalg

# Build a sparse tridiagonal matrix (common in 1D PDEs)
n = 1000
diag_main = 4.0 * np.ones(n)
diag_off = -1.0 * np.ones(n - 1)

A_sparse = scipy.sparse.diags(
    [diag_off, diag_main, diag_off], [-1, 0, 1],
    format='csr'
)
b = np.ones(n)

x = scipy.sparse.linalg.spsolve(A_sparse, b)

# Verify
print(f"Shape:      {A_sparse.shape}")
print(f"NNZ:         {A_sparse.nnz} (vs {n*n} dense entries)")
print(f"Residual: {np.linalg.norm(A_sparse @ x - b):.2e}")
```

```
Shape:      (1000, 1000)
NNZ:        2998 (vs 1000000 dense entries)
Residual: 9.61e-16
```

Sparse solvers are introduced for context here; Part IV and the robotics application chapters use them in earnest.

Decision Tree: Which Solver?

Is A square?

YES → Is (A) well-conditioned ($< 1e10$)?

YES → `np.linalg.solve` (fastest)

NO → investigate the physics; consider regularization (Ch 17)

NO → Use `np.linalg.lstsq`

$m > n$ (overdetermined): minimises $\|Ax - b\|$

$m < n$ (underdetermined): returns minimum-norm solution

Check rank; if $\text{rank} < n$, null space is non-trivial (Ch 3)

Is A sparse (>90% zeros)?

YES → `scipy.sparse.linalg.spsolve` (or iterative solvers, Ch 12)

Do you have many right-hand sides with same A ?

YES → `scipy.linalg.lu_factor` once, then `lu_solve` (faster)

Application: Multi-Sensor Position Fusion

A mobile robot carries three ultrasonic range sensors mounted at known positions on a wall. Each sensor reports its distance to the robot. The task: fuse the three range measurements into a single position estimate.

This is a canonical instance of $A\mathbf{x} = \mathbf{b}$ — and it immediately raises the overdetermined question when you add a fourth sensor for redundancy.

Problem Setup

Three sensors are at fixed positions $(s_x^{(i)}, s_y^{(i)})$ along the wall. Each reports range r_i to the robot at unknown position (p_x, p_y) .

The range equation is:

$$(p_x - s_x^{(i)})^2 + (p_y - s_y^{(i)})^2 = r_i^2$$

Expanding and rearranging yields a *linear* system by subtracting the equation for sensor i from sensor 1:

$$2(s_x^{(1)} - s_x^{(i)})p_x + 2(s_y^{(1)} - s_y^{(i)})p_y = r_i^2 - r_1^2 + \|\mathbf{s}_1\|^2 - \|\mathbf{s}_i\|^2$$

This linearisation is the standard trick in time-of-arrival (ToA) positioning.

```
import numpy as np

# Sensor positions (metres)
sensors = np.array([
    [0.0, 0.0], # sensor 1 at origin
    [4.0, 0.0], # sensor 2 to the right
    [2.0, 3.0], # sensor 3 above
])

# True robot position (unknown to the solver)
p_true = np.array([2.5, 1.2])

# True ranges (+ small measurement noise)
rng = np.random.default_rng(7)
ranges_true = np.linalg.norm(sensors - p_true, axis=1)
```

```
noise_std    = 0.02    # 2 cm noise
ranges       = ranges_true + rng.normal(0, noise_std, size=3)

print("Sensor positions:"); print(sensors)
print(f"\nTrue position:    {p_true}")
print(f"True ranges:       {ranges_true.round(4)}")
print(f"Measured ranges:   {ranges.round(4)}")
```

Sensor positions:

```
[[0. 0.]
 [4. 0.]
 [2. 3.]]
```

True position: [2.5 1.2]

True ranges: [2.7731 1.9209 1.8682]

Measured ranges: [2.7731 1.9269 1.8627]

Building and Solving the Linear System

```
import numpy as np

def build_linear_system(sensors, ranges):
    """
    Linearise range equations by differencing against sensor 0.
    Returns (A, b) for the 2D position solve.
    """
    s0, r0 = sensors[0], ranges[0]
    rows_A, rows_b = [], []

    for si, ri in zip(sensors[1:], ranges[1:]):
        #  $2*(s_0 - s_i) \cdot p = r_i^2 - r_0^2 + ||s_0||^2 - ||s_i||^2$ 
        row_a = 2 * (s0 - si)
        row_b = ri**2 - r0**2 + np.dot(s0, s0) - np.dot(si, si)
        rows_A.append(row_a)
        rows_b.append(row_b)

    return np.array(rows_A), np.array(rows_b)

A, b = build_linear_system(sensors, ranges)
print("A (coefficient matrix):")
print(A)
print("\nb (right-hand side):")
print(b.round(4))
```



```
# Three sensors give 2 difference equations, 2 unknowns → unique solve
p_est = np.linalg.solve(A, b)
error = np.linalg.norm(p_est - p_true)

print(f"\nEstimated position: {p_est.round(4)}")
print(f"True position:      {p_true}")
print(f"Position error:      {error*100:.2f} cm")
```

A (coefficient matrix):

```
[[ -8.  0.]
 [ -4. -6.]]
```

b (right-hand side):

```
[-19.9771 -17.2206]
```

Estimated position: [2.4971 1.2053]

True position: [2.5 1.2]

Position error: 0.61 cm

Adding a Fourth Sensor — the Overdetermined Case

With four sensors we have three difference equations for two unknowns. The system is overdetermined — no exact solution exists because of measurement noise. We switch to least squares.

```
import numpy as np

# Add a fourth sensor
sensors4 = np.vstack([sensors, [0.0, 3.0]]) # sensor 4 at (0, 3)
ranges4_true = np.linalg.norm(sensors4 - p_true, axis=1)
ranges4      = ranges4_true + rng.normal(0, noise_std, size=4)

A4, b4 = build_linear_system(sensors4, ranges4)
print(f"System shape: {A4.shape} (3 equations, 2 unknowns → overdetermined)")

# Least-squares solve
p_ls, residuals, rank, sv = np.linalg.lstsq(A4, b4, rcond=None)
error_ls = np.linalg.norm(p_ls - p_true)

print(f"\nLeast-squares estimate: {p_ls.round(4)}")
print(f"True position:      {p_true}")
print(f"Error (4 sensors):    {error_ls*100:.2f} cm")
print(f"Error (3 sensors):    {error*100:.2f} cm")
print(f"Rank of A4:           {rank}")
```

System shape: (3, 2) (3 equations, 2 unknowns → overdetermined)

Least-squares estimate: [2.4952 1.1907]

True position: [2.5 1.2]

Error (4 sensors): 1.05 cm

Error (3 sensors): 0.61 cm

Rank of A4: 2

The least-squares estimate is more accurate than the exact 2-sensor solve because it averages out measurement noise across more sensors.

Visualising the Fusion

```
import numpy as np
import matplotlib.pyplot as plt

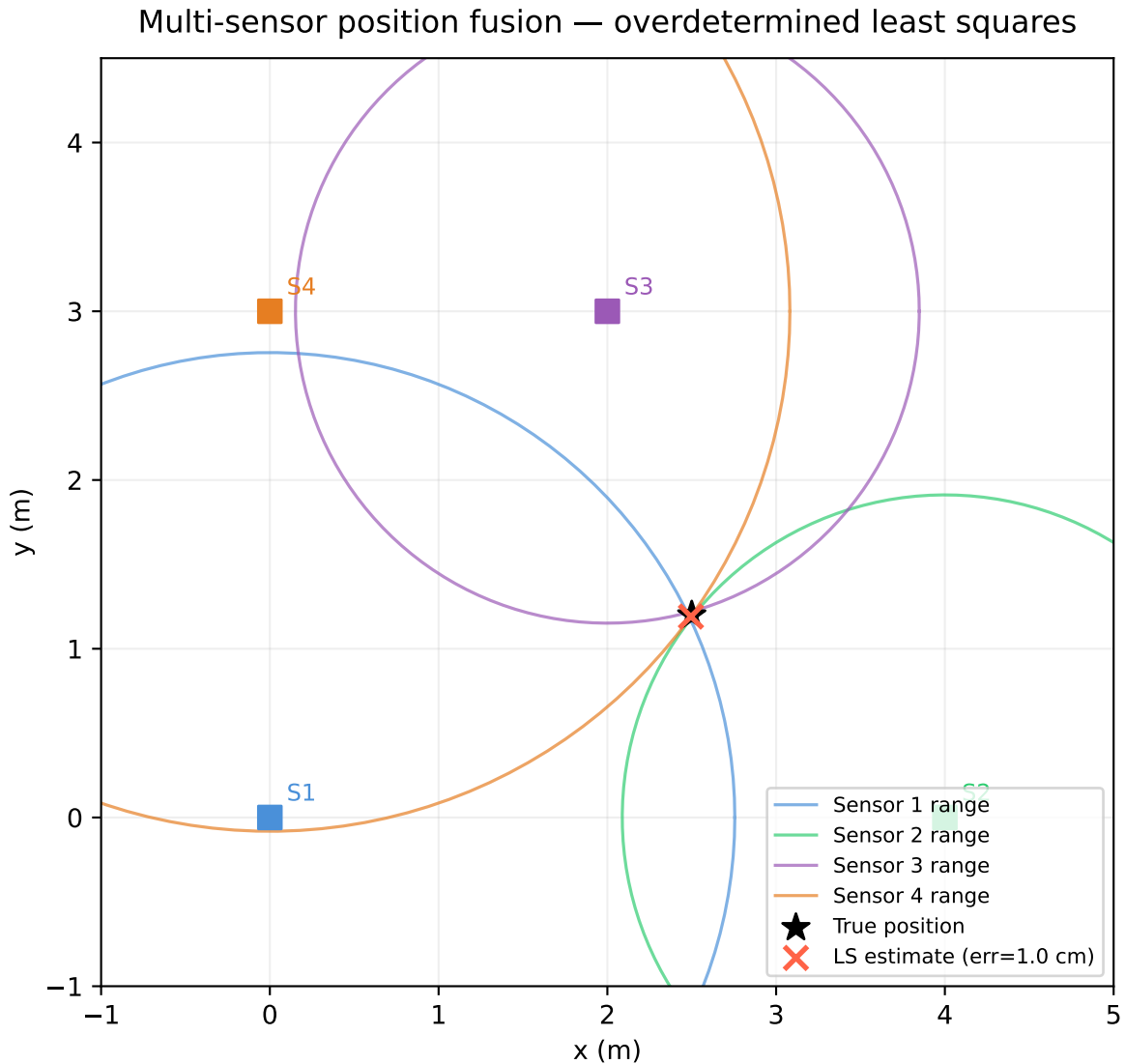
fig, ax = plt.subplots(figsize=(7, 6))

theta = np.linspace(0, 2*np.pi, 300)
colors = ['#4a90d9', '#2ecc71', '#9b59b6', '#e67e22']

# Draw range circles for all 4 sensors
for i, (s, r, col) in enumerate(zip(sensors4, ranges4, colors)):
    ax.plot(s[0] + r*np.cos(theta), s[1] + r*np.sin(theta),
           color=col, lw=1.2, alpha=0.7, label=f'Sensor {i+1} range')
    ax.scatter(*s, color=col, s=80, zorder=5, marker='s')
    ax.text(s[0]+0.1, s[1]+0.1, f'S{i+1}', color=col, fontsize=9)

# True and estimated positions
ax.scatter(*p_true, color='black', s=120, zorder=10, marker='*',
          label='True position')
ax.scatter(*p_ls, color='tomato', s=80, zorder=10, marker='x',
          linewidths=2, label=f'LS estimate (err={error_ls*100:.1f} cm)')

ax.set_xlim(-1, 5); ax.set_ylim(-1, 4.5)
ax.set_aspect('equal')
ax.set_xlabel('x (m)'); ax.set_ylabel('y (m)')
ax.set_title('Multi-sensor position fusion - overdetermined least squares', pad=10)
ax.legend(fontsize=8)
ax.grid(True, alpha=0.2)
plt.tight_layout()
plt.show()
```



Noise Sensitivity — Fewer vs. More Sensors

The panels below show how the position estimate degrades as noise increases, and how adding more sensors mitigates that degradation.

```
import numpy as np
import matplotlib.pyplot as plt

fig, axes = plt.subplots(2, 2, figsize=(11, 9))
theta = np.linspace(0, 2*np.pi, 300)
colors = ['#4a90d9', '#2ecc71', '#9b59b6', '#e67e22']
```

```

configs = [
    (2, 0.02, axes[0, 0]), # 2 sensors, low noise
    (2, 0.10, axes[0, 1]), # 2 sensors, high noise
    (4, 0.02, axes[1, 0]), # 4 sensors, low noise
    (4, 0.10, axes[1, 1]), # 4 sensors, high noise
]

for n_sens, noise, ax in configs:
    rng_w = np.random.default_rng(0)
    sens = sensors4[:n_sens]
    r_t = np.linalg.norm(sens - p_true, axis=1)
    r_m = r_t + rng_w.normal(0, noise, size=n_sens)

    A_w, b_w = build_linear_system(sens, r_m)
    p_w, _, _, _ = np.linalg.lstsq(A_w, b_w, rcond=None)
    err = np.linalg.norm(p_w - p_true) * 100 # cm

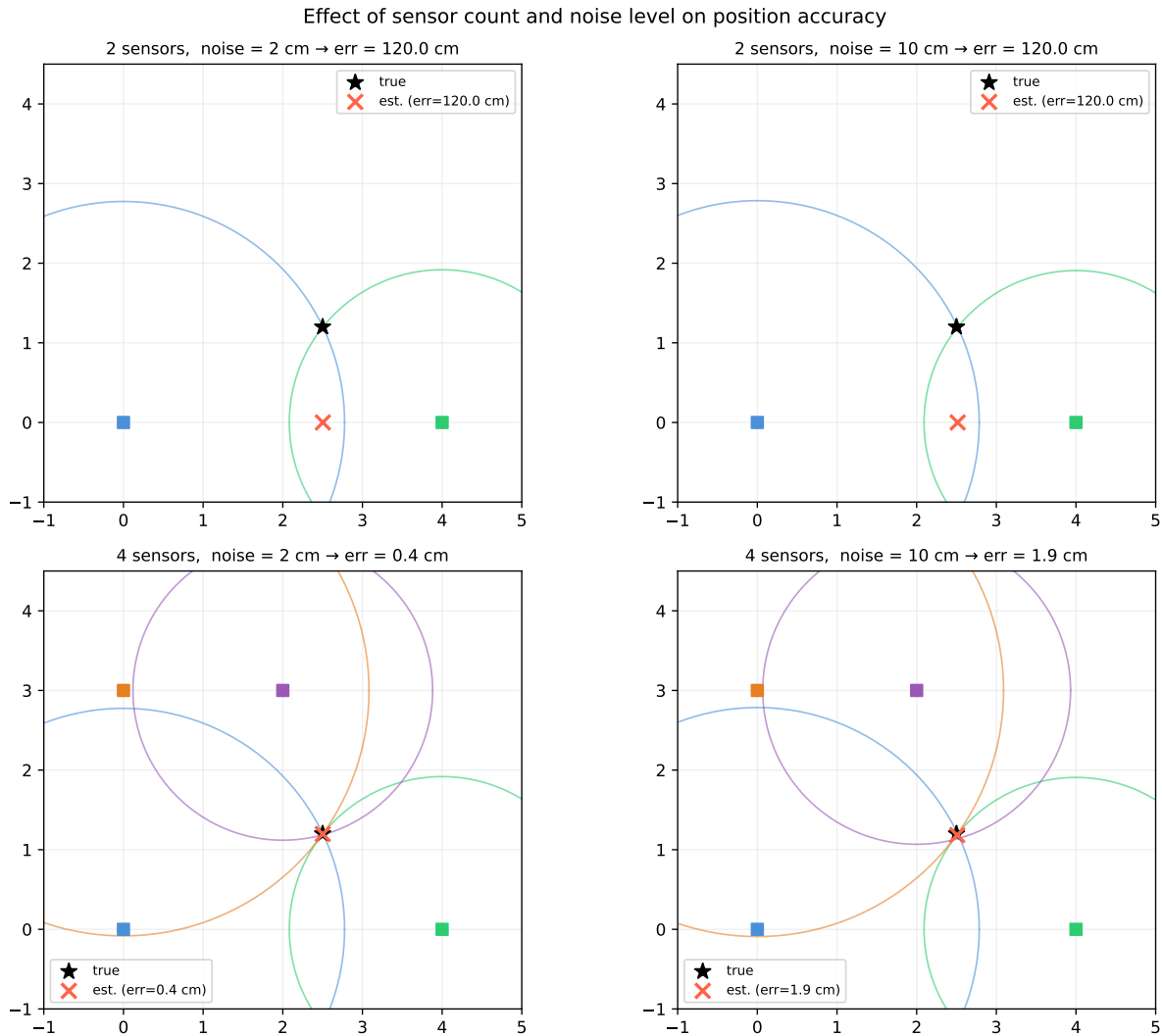
    for i, (s, r) in enumerate(zip(sens, r_m)):
        ax.plot(s[0] + r*np.cos(theta), s[1] + r*np.sin(theta),
                color=colors[i], lw=1, alpha=0.6)
        ax.scatter(*s, color=colors[i], s=50, marker='s', zorder=5)

    ax.scatter(*p_true, color='black', s=100, marker='*', zorder=6,
               label='true')
    ax.scatter(*p_w, color='tomato', s=80, marker='x', linewidths=2,
               zorder=6, label=f'est. (err={err:.1f} cm)')

    ax.set_xlim(-1, 5); ax.set_ylim(-1, 4.5)
    ax.set_aspect('equal')
    ax.set_title(f'{n_sens} sensors, noise = {noise*100:.0f} cm → err = {err:.1f} cm',
                 fontsize=10)
    ax.legend(fontsize=8)
    ax.grid(True, alpha=0.2)

fig.suptitle('Effect of sensor count and noise level on position accuracy',
             fontsize=12)
fig.tight_layout()
plt.show()

```



Key Takeaways

1. Range-only positioning linearises to $A\mathbf{x} = \mathbf{b}$ by differencing sensor equations — a standard trick in ToA, UWB, and GPS.
2. With exactly as many constraints as unknowns, `np.linalg.solve` gives the exact (noise-free) answer.
3. Adding more sensors creates an overdetermined system; `np.linalg.lstsq` automatically finds the best-fit position — more sensors means less noise sensitivity.
4. The condition number of A depends on the *geometry* of the sensors: sensors clustered together have nearly parallel rows and a high condition number. Spread-out sensors give a well-conditioned problem.

Exercises

2.1 Consider the system:

$$\begin{pmatrix} 2 & 6 \\ 1 & 3 \end{pmatrix} \mathbf{x} = \begin{pmatrix} 8 \\ 4 \end{pmatrix}$$

Without calling any solver: (a) What are the normal vectors to each hyperplane? (b) Are the hyperplanes parallel? (c) What does this tell you about the number of solutions?

2.2 You call `np.linalg.solve(A, b)` on a 3×3 system and get a result. You compute `np.linalg.cond(A)` and get $3.2\text{e}14$. Should you trust the result? What would you do instead?

2.3 Write down the augmented matrix for the following system, perform one step of forward elimination by hand (eliminate x_1 from rows 2 and 3), and identify the new pivot:

$$\begin{aligned} x_1 + 2x_2 + x_3 &= 5 \\ 2x_1 + 4x_2 + 3x_3 &= 11 \\ x_1 + x_2 + 2x_3 &= 6 \end{aligned}$$

2.4 A robotics student writes this code to fuse four range sensors:

```
A, b = build_linear_system(sensors4, ranges4)
x = np.linalg.solve(A, b)
```

This raises `LinAlgError: Last 2 dimensions of the array must be square`. Why? What should they use instead?

2.5 You are estimating a 2D position from three sensors. After solving with `lstsq` you get a residual of 2.4 metres, but your sensors are accurate to 5 cm. List at least three possible causes for the large residual and how you would diagnose each.

2.6 (Harder) Prove that the solution set of $A\mathbf{x} = \mathbf{b}$, when non-empty, is an affine subspace (a flat). Specifically, show that if $\mathbf{x}_1$ and $\mathbf{x}_2$ are both solutions, then so is $\mathbf{x}_1 + t(\mathbf{x}_2 - \mathbf{x}_1)$ for any scalar t .

2.7 You have a 10×10 Hilbert matrix H and solve $H\mathbf{x} = \mathbf{b}$ with `np.linalg.solve`. The residual $\|H \odot \mathbf{x} - \mathbf{b}\|$ is 10^{-11} , but the error $\|\mathbf{x} - \mathbf{x}_{\text{true}}\|$ is 10^3 . How is this possible? What does it imply for interpreting residuals alone?

Solutions

Solution 2.1

- (a) Normal vectors are the rows of A : $\mathbf{a}_1 = [2, 6]^T$ and $\mathbf{a}_2 = [1, 3]^T$.
- (b) Check whether one is a scalar multiple of the other: $[2, 6] = 2 \cdot [1, 3]$ — yes, they are parallel (proportional normal vectors).
- (c) Parallel hyperplanes either coincide (infinitely many solutions) or are distinct and non-intersecting (no solution). Checking: row 1 is $2x_1 + 6x_2 = 8$, row 2 is $x_1 + 3x_2 = 4$, which is exactly half of row 1 ($8/2 = 4$). So both equations describe the *same line* — infinitely many solutions.

```
import numpy as np
A = np.array([[2., 6.], [1., 3.]])
b = np.array([8., 4.])
Ab = np.column_stack([A, b])
print(f"rank(A) = {np.linalg.matrix_rank(A)}")
print(f"rank(Ab) = {np.linalg.matrix_rank(Ab)}")
# rank(A) == rank(Ab) == 1 → infinitely many solutions
```

```
rank(A) = 1
rank(Ab) = 1
```

Solution 2.2

No, you should not trust the result. A condition number of 3.2×10^{14} means roughly 14–15 digits of precision are consumed by the conditioning — but double-precision arithmetic only carries about 15–16 significant digits. The computed solution may have zero correct digits.

What to do:

1. **Understand the physics.** A near-singular matrix means your constraints are nearly linearly dependent — some measurements are almost redundant. Fix the problem setup if possible.
2. **Regularize.** Add Tikhonov regularization: solve $(A^T A + \lambda I)\mathbf{x} = A^T \mathbf{b}$ for a small λ . Chapter 17 covers this.
3. **Use SVD.** `np.linalg.lstsq` with `rcond` set to a threshold drops near-zero singular values and returns a truncated solution. Chapter 18 covers the SVD approach in depth.
4. **At minimum,** report the condition number to the caller so they know the result is unreliable.

```
import numpy as np

def hilbert(n):
    i, j = np.meshgrid(np.arange(1, n+1), np.arange(1, n+1))
    return 1.0 / (i + j - 1)

H = hilbert(10)
b = H @ np.ones(10)
x = np.linalg.solve(H, b)
kappa = np.linalg.cond(H)
print(f"κ = {kappa:.2e}")
print(f"Residual: {np.linalg.norm(H @ x - b):.2e} ← looks fine")
print(f"Error: {np.linalg.norm(x - np.ones(10)):.2e} ← actually terrible")

κ = 1.60e+13
Residual: 5.66e-16 ← looks fine
Error: 2.74e-04 ← actually terrible
```

This is the “silent wrong answer” pattern: the residual looks fine, but the solution is garbage. High condition number is the only warning.

Solution 2.3

Augmented matrix:

$$\left(\begin{array}{ccc|c} 1 & 2 & 1 & 5 \\ 2 & 4 & 3 & 11 \\ 1 & 1 & 2 & 6 \end{array} \right)$$

Pivot is $a_{11} = 1$.

- $R_2 \leftarrow R_2 - 2R_1$: $[2 - 2, 4 - 4, 3 - 2, 11 - 10] = [0, 0, 1, 1]$
- $R_3 \leftarrow R_3 - 1 \cdot R_1$: $[1 - 1, 1 - 2, 2 - 1, 6 - 5] = [0, -1, 1, 1]$

After one step:

$$\left(\begin{array}{ccc|c} 1 & 2 & 1 & 5 \\ 0 & 0 & 1 & 1 \\ 0 & -1 & 1 & 1 \end{array} \right)$$

The new pivot should be in column 2. Row 2 has a zero in column 2, so we swap rows 2 and 3:

$$\left(\begin{array}{ccc|c} 1 & 2 & 1 & 5 \\ 0 & -1 & 1 & 1 \\ 0 & 0 & 1 & 1 \end{array} \right)$$

New pivot: $a_{22} = -1$. Note that column 2 of the original row 2 was zero — this is a *zero pivot*, triggering a row swap. Partial pivoting exists precisely to handle this.

```
import numpy as np

aug = np.array([[1., 2., 1., 5.],
                [2., 4., 3., 11.],
                [1., 1., 2., 6.]])

aug[1] -= 2 * aug[0]
aug[2] -= 1 * aug[0]
print("After step 1:")
print(aug)
# Swap rows 1 and 2 (0-indexed)
aug[[1, 2]] = aug[[2, 1]]
print("\nAfter pivot swap:")
print(aug)
```

```
After step 1:
[[ 1.  2.  1.  5.]
 [ 0.  0.  1.  1.]
 [ 0. -1.  1.  1.]]
```

```
After pivot swap:
[[ 1.  2.  1.  5.]
 [ 0. -1.  1.  1.]
 [ 0.  0.  1.  1.]]
```

Solution 2.4

`build_linear_system` differences each sensor against sensor 0, producing $n_{\text{sensors}} - 1$ equations. With 4 sensors the output is a 3×2 matrix — *not square*. `np.linalg.solve` requires a square matrix.

The correct call is:

```
import numpy as np

A4, b4 = build_linear_system(sensors4, ranges4)
print(f"A4 shape: {A4.shape}")    # (3, 2) - overdetermined

# Use lstsq, not solve
p_est, residuals, rank, sv = np.linalg.lstsq(A4, b4, rcond=None)
print(f"Estimated position: {p_est.round(4)}")
```

A4 shape: (3, 2)

Estimated position: [2.4952 1.1907]

General rule: if A is not square, always use `lstsq`. If A is square but you're not sure it's full-rank, check the condition number first.

Solution 2.5

A 2.4 m residual from 5 cm sensors implies the model is wrong, not just noisy. Possible causes:

1. **Wrong sensor positions.** If `sensors` contains stale calibration values and a sensor has shifted, the geometry is inconsistent. *Diagnose:* physically remeasure sensor positions and compare.
2. **Wrong range model.** The linearisation assumes range = Euclidean distance. If sensors report travel time and there is a reflective surface creating multipath, ranges will be systematically biased. *Diagnose:* plot residuals vs. range — a systematic trend suggests model mismatch.
3. **Unit mismatch.** Ranges in metres vs. sensor positions in centimetres, or vice versa. *Diagnose:* check that `ranges` and `sensors` use the same unit.
4. **Non-linear regime.** The linearisation is exact for ToA-derived ranges but approximate if the sensors report a non-Euclidean quantity. *Diagnose:* compare the linearised $A\mathbf{x}$ prediction to a non-linear residual.
5. **Wrong sensor index.** The subtraction in `build_linear_system` differences sensor i against sensor 0. If sensors are stored in a different order than assumed, the geometry is scrambled. *Diagnose:* print `sensors` and `ranges` side by side and verify they correspond.

Solution 2.6

Let $\mathbf{x}_1$ and $\mathbf{x}_2$ both satisfy $A\mathbf{x} = \mathbf{b}$. Define $\mathbf{x}(t) = \mathbf{x}_1 + t(\mathbf{x}_2 - \mathbf{x}_1) = (1 - t)\mathbf{x}_1 + t\mathbf{x}_2$.

Apply A :

$$A\mathbf{x}(t) = (1 - t)A\mathbf{x}_1 + tA\mathbf{x}_2 = (1 - t)\mathbf{b} + t\mathbf{b} = \mathbf{b}$$

So $\mathbf{x}(t)$ satisfies $A\mathbf{x} = \mathbf{b}$ for every scalar t . This means the solution set is closed under the operation “form a line through any two points” — the definition of an affine subspace.

The *direction* of this affine subspace is the null space of A : the difference $\mathbf{x}_2 - \mathbf{x}_1$ satisfies $A(\mathbf{x}_2 - \mathbf{x}_1) = \mathbf{b} - \mathbf{b} = \mathbf{0}$, so it lies in $\text{Null}(A)$.

Solution 2.7

This is the condition number effect in its starkest form. The residual $\|H\mathbf{x} - \mathbf{b}\|$ measures how well $\mathbf{x}$ satisfies the *equations*. The error $\|\mathbf{x} - \mathbf{x}_{\text{true}}\|$ measures how close $\mathbf{x}$ is to the *true answer*.

For an ill-conditioned system they can be wildly different. The relationship is:

$$\frac{\|\mathbf{x} - \mathbf{x}_{\text{true}}\|}{\|\mathbf{x}_{\text{true}}\|} \leq \kappa(H) \cdot \frac{\|H\mathbf{x} - \mathbf{b}\|}{\|\mathbf{b}\|}$$

With $\kappa(H) \approx 10^{13}$ for the 10×10 Hilbert matrix:

$$\text{relative error} \lesssim 10^{13} \times \frac{10^{-11}}{\|\mathbf{b}\|} \approx 10^{13} \times 10^{-11} = 100$$

A relative error of 100 means the solution could be off by 100× — which matches the 10^3 error observed when $\mathbf{x}_{\text{true}} \sim 1$.

The implication: a small residual is *necessary but not sufficient* for a good solution. With a high condition number you can have a tiny residual and a useless solution simultaneously. Always check the condition number, not just the residual.

Row Reduction and Echelon Forms

Row Echelon and Reduced Row Echelon Form

Gaussian elimination from Chapter 2 produces a triangular matrix that you back-substitute to find a solution. *Reduced* row echelon form (RREF) takes elimination further — all the way to the simplest possible matrix — so that the solution structure is visible directly, without any substitution.

Row Echelon Form (REF)

A matrix is in **row echelon form** when:

1. All zero rows are at the bottom.
2. Each row's *leading entry* (pivot) is strictly to the right of the pivot in the row above.
3. All entries below a pivot are zero.

REF example:
$$\begin{pmatrix} \mathbf{2} & 3 & 1 & 5 \\ 0 & \mathbf{1} & 4 & 2 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

The pivots are at positions (1, 1) and (2, 2). Column 3 has no pivot — it corresponds to a *free variable*.

Reduced Row Echelon Form (RREF)

RREF imposes two more conditions on top of REF:

4. Every pivot equals exactly 1.
5. All entries *above* a pivot are also zero (not just below).

$$\text{RREF example: } \begin{pmatrix} 1 & 0 & 3 & 1 \\ 0 & 1 & 4 & 2 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

Every matrix has a unique RREF. No matter what row operations you apply, you always arrive at the same RREF — it is an intrinsic property of the matrix.

```
import numpy as np
import sympy

def rref(A):
    """Return RREF and pivot column indices using SymPy for exact arithmetic."""
    M = sympy.Matrix(A)
    R, pivots = M.rref()
    return np.array(R.tolist(), dtype=float), list(pivots)

A = np.array([[2., 3., 1., 5.],
              [4., 7., 5., 12.],
              [2., 4., 4., 8.]])

R, pivots = rref(A)
print("RREF:")
print(R)
print(f"Pivot columns: {pivots}")
```

```
RREF:
[[ 1.  0. -4.  0.]
 [ 0.  1.  3.  0.]
 [ 0.  0.  0.  1.]]
Pivot columns: [0, 1, 3]
```

SymPy's `rref()` uses exact rational arithmetic, avoiding floating-point pivot decisions. For numerical work, `np.linalg.matrix_rank` uses SVD-based rank detection internally, which is more reliable than threshold-based elimination.

Reading the Solution from RREF

Consider the augmented system $[A \mid \mathbf{b}]$ after row reduction to RREF. The pivot columns correspond to *basic variables*; non-pivot columns correspond to *free variables*.

Example — underdetermined system:

$$\left(\begin{array}{cccc|c} 1 & 2 & 0 & 3 & 1 \\ 0 & 0 & 1 & 4 & 2 \\ 0 & 0 & 0 & 0 & 0 \end{array} \right)$$

Pivot columns: 1 and 3. Free variables: x_2 and x_4 (columns 2 and 4).

Reading off directly:

$$x_1 = 1 - 2x_2 - 3x_4, \quad x_3 = 2 - 4x_4$$

General solution (set $x_2 = s$, $x_4 = t$):

$$\mathbf{x} = \begin{pmatrix} 1 \\ 0 \\ 2 \\ 0 \end{pmatrix} + s \begin{pmatrix} -2 \\ 1 \\ 0 \\ 0 \end{pmatrix} + t \begin{pmatrix} -3 \\ 0 \\ -4 \\ 1 \end{pmatrix}$$

```
import numpy as np
import sympy

# Augmented matrix
Ab = sympy.Matrix([
    [1, 2, 0, 3, 1],
    [0, 0, 1, 4, 2],
    [0, 0, 0, 0, 0]
])

# SymPy can solve directly from RREF for a parametric solution
x1, x2, x3, x4 = sympy.symbols('x1 x2 x3 x4')
sol = sympy.solve_linear_system(Ab, x1, x2, x3, x4)
print("Parametric solution:")
for var, expr in sol.items():
    print(f" {var} = {expr}")
```

Parametric solution:

$$\begin{aligned} x_1 &= -2x_2 - 3x_4 + 1 \\ x_3 &= 2 - 4x_4 \end{aligned}$$

Pivot Columns and Free Columns

The distinction between pivot and free columns is the most important thing RREF reveals:

- **Pivot column** — the corresponding variable is *determined* by the system.
- **Free column** — the corresponding variable can take any value; each free variable adds one dimension to the solution set.

```
import numpy as np
import sympy

def classify_columns(A):
    """Return pivot and free column indices."""
    _, pivots = sympy.Matrix(A).rref()
    n = A.shape[1]
    free = [j for j in range(n) if j not in pivots]
    print(f"Pivot columns: {list(pivots)}")
    print(f"Free columns: {free}")
    print(f"Rank: {len(pivots)}")
    print(f"Nullity: {len(free)} (dimension of solution freedom)")

# Example: sensor network with redundant constraints
A = np.array([[1., 0., 2., 1.],
              [0., 1., 3., 2.],
              [1., 1., 5., 3.]]) # row 3 = row 1 + row 2 → dependent

classify_columns(A)
```

```
Pivot columns: [0, 1]
Free columns: [2, 3]
Rank: 2
Nullity: 2 (dimension of solution freedom)
```

RREF Applied to Three Cases

The three solution cases from Chapter 2 are immediately visible from RREF. The code below applies RREF to examples of each and reads off the conclusion directly from the structure of the reduced matrix.

```
import numpy as np
import sympy

def show_rref(aug_list, title):
    aug = np.array(aug_list, dtype=float)
```



```

M = sympy.Matrix(aug_list)
R, pivots = M.rref()
R_np = np.array(R.tolist(), dtype=float)
n_pivots = len(pivots)
n_vars = aug.shape[1] - 1
n_free = n_vars - n_pivots

print(f"\n{'='*44}")
print(f"  {title}")
print(f"{'='*44}")
print(f"[A|b]:\n{aug}")
print(f"RREF:\n{R_np}")
print(f"rank = {n_pivots}, nullity = {n_free} → ", end='')

# Check for inconsistency: a zero row in A with nonzero b entry
inconsistent = any(
    np.all(np.abs(R_np[i, :-1]) < 1e-10) and abs(R_np[i, -1]) > 1e-10
    for i in range(R_np.shape[0])
)
if inconsistent:
    print("INCONSISTENT - no solution")
elif n_free == 0:
    print("UNIQUE solution")
else:
    print(f"INFINITELY MANY solutions ({n_free} free variable{'s' if n_free>1 else ''})")

show_rref([[2, 1, 5], [1, 3, 7]],
           "Full rank - unique solution")

show_rref([[2, 4, 6], [1, 2, 3]],
           "Rank-deficient, consistent - infinitely many")

show_rref([[2, 4, 6], [1, 2, 5]],
           "Rank-deficient, inconsistent - no solution")

```

```

=====
Full rank - unique solution
=====
[A|b]:
[[2. 1. 5.]
 [1. 3. 7.]]
RREF:
[[1. 0. 1.6]
 [0. 1. 1.8]]
rank = 2, nullity = 0 → UNIQUE solution
=====

```

```

Rank-deficient, consistent - infinitely many
=====
[A|b]:
[[2. 4. 6.]
 [1. 2. 3.]]
RREF:
[[1. 2. 3.]
 [0. 0. 0.]]
rank = 1, nullity = 1 → INFINITELY MANY solutions (1 free variable)

```

```

=====
Rank-deficient, inconsistent - no solution
=====
[A|b]:
[[2. 4. 6.]
 [1. 2. 5.]]
RREF:
[[1. 2. 0.]
 [0. 0. 1.]]
rank = 2, nullity = 0 → INCONSISTENT - no solution

```

Set both left-hand-side rows proportional with *different* ratios on the right-hand side and the RREF reveals a zero row with a nonzero augmented entry — the signature of inconsistency.

Why RREF Matters Beyond Solving

RREF is not just a solving tool — it exposes structure:

- The **pivot columns** of A form a basis for the column space of A .
- The **number of pivots** is the rank — confirmed in §3.2.
- The **null space basis vectors** are read directly from the free-column coefficients, as shown in §3.2.
- RREF of the augmented matrix $[A \mid I]$ produces $[I \mid A^{-1}]$ when A is invertible — a method for computing matrix inverses.

```

import numpy as np
import sympy

# Computing inverse via RREF of [A | I]
A = np.array([[2., 1.], [5., 3.]])
n = A.shape[0]
AI = sympy.Matrix(np.hstack([A, np.eye(n)]).tolist())
R, _ = AI.rref()
A_inv = np.array(R[:, n:].tolist(), dtype=float)

```

```
print("A_inv via RREF:")
print(A_inv.round(6))
print("\nVerify A @ A_inv:")
print((A @ A_inv).round(6))  # should be identity
```

```
A_inv via RREF:
[[ 3. -1.]
 [-5.  2.]]
```

```
Verify A @ A_inv:
[[1. 0.]
 [0. 1.]]
```

Rank and the Rank-Nullity Theorem

Rank is the single most informative number you can compute about a matrix. It tells you how many independent constraints your system imposes — and, by subtraction, how many degrees of freedom remain.

Definition of Rank

The **rank** of a matrix A is:

- the number of pivot positions in its RREF,
- the dimension of its column space,
- the dimension of its row space.

All three definitions give the same number. The first is computational; the second and third are geometric — and they are equal by a non-obvious theorem.

```
import numpy as np

# Three ways to compute rank
A = np.array([[1., 2., 3.],
              [4., 5., 6.],
              [7., 8., 9.]])  # rank-deficient: row 3 = 2*row2 - row1

r1 = np.linalg.matrix_rank(A)  # SVD-based (most reliable)

import sympy
_, pivots = sympy.Matrix(A).rref()
r2 = len(pivots)  # RREF pivot count
```

```
U, s, Vt = np.linalg.svd(A)
r3 = np.sum(s > 1e-10)          # singular values above threshold

print(f"matrix_rank:      {r1}")
print(f"RREF pivots:      {r2}")
print(f"SVD thresholded: {r3}")
print(f"Singular values: {s.round(4)}")
```

```
matrix_rank:      2
RREF pivots:      2
SVD thresholded: 2
Singular values: [16.8481  1.0684  0.      ]
```

The third method (SVD) is the most numerically robust: small singular values indicate near-linear-dependence. `np.linalg.matrix_rank` uses SVD internally with a default threshold of $\max(M, N) \cdot \sigma_{\max} \cdot \epsilon$.

The Rank-Nullity Theorem

For any matrix $A \in \mathbb{R}^{m \times n}$:

$$\text{rank}(A) + \text{nullity}(A) = n$$

where $\text{nullity}(A) = \dim(\text{Null}(A))$ is the number of free variables — the dimension of the solution space when $\mathbf{b} = \mathbf{0}$.

Intuition. A maps $\mathbb{R}^n$ into $\mathbb{R}^m$. The rank counts the dimensions that “survive” the mapping (image dimension). The nullity counts the dimensions that “collapse” to zero. Together they account for all n input dimensions.

```
import numpy as np
import sympy

def rank_nullity(A):
    m, n = A.shape
    r = np.linalg.matrix_rank(A)
    nullity = n - r
    print(f"Shape:      {m} × {n}")
    print(f"Rank:        {r}      (independent rows/columns)")
    print(f"Nullity:      {nullity}    (free variables, null space dimension)")
    print(f"Check:        {r} + {nullity} = {r + nullity} = n = {n} ")
    return r, nullity
```

```

print("=== Full rank (3×3) ===")
rank_nullity(np.array([[1.,0.,0.],[0.,1.,0.],[0.,0.,1.])))

print("\n=== Rank-2 (3×4) ===")
rank_nullity(np.array([[1.,2.,0.,1.],[0.,0.,1.,3.],[1.,2.,1.,4.])))

print("\n=== Rank-1 (outer product) ===")
u = np.array([1.,2.,3.])
v = np.array([1.,1.,1.,1.])
rank_nullity(np.outer(u, v))

```

```

=== Full rank (3×3) ===
Shape:    3 × 3
Rank:     3   (independent rows/columns)
Nullity:  0   (free variables, null space dimension)
Check:    3 + 0 = 3 = n = 3

```

```

=== Rank-2 (3×4) ===
Shape:    3 × 4
Rank:     2   (independent rows/columns)
Nullity:  2   (free variables, null space dimension)
Check:    2 + 2 = 4 = n = 4

```

```

=== Rank-1 (outer product) ===
Shape:    3 × 4
Rank:     1   (independent rows/columns)
Nullity:  3   (free variables, null space dimension)
Check:    1 + 3 = 4 = n = 4

```

```
(np.int64(1), np.int64(3))
```

Computing the Null Space

The null space $\text{Null}(A) = \{\mathbf{x} : A\mathbf{x} = \mathbf{0}\}$ is a subspace of $\mathbb{R}^n$. Its basis vectors come directly from the free columns of the RREF.

Recipe:

1. Compute RREF of A (not augmented — we are solving $A\mathbf{x} = \mathbf{0}$).
2. Identify free variables (non-pivot columns).
3. For each free variable, set it to 1 and all other free variables to 0; solve for the basic variables. That vector is one null space basis vector.

```
import numpy as np
import sympy

def null_space_basis(A):
    """Return null space basis via RREF. Each column is a basis vector."""
    M = sympy.Matrix(A)
    # sympy's nullspace() returns exact basis vectors
    ns = M.nullspace()
    if not ns:
        print("Null space is trivial {0}.")
        return np.zeros((A.shape[1], 0))
    B = np.array([np.array(v.T.tolist())[0], dtype=float) for v in ns]).T
    return B

A = np.array([[1., 2., 0., 3.],
              [0., 0., 1., 4.],
              [1., 2., 1., 7.]]) # row 3 = row 1 + row 2

N = null_space_basis(A)
print(f"Null space basis (columns):\n{N.round(4)}")
print(f"\nVerify A @ N == 0:\n{(A @ N).round(10)}")
print(f"\nNull space dimension: {N.shape[1]} (nullity)")
print(f"Rank: {np.linalg.matrix_rank(A)}")
print(f"Sum: {N.shape[1]} + {np.linalg.matrix_rank(A)} = {N.shape[1]+np.linalg.matrix_rank(A)}")
```

Null space basis (columns):

```
[[-2. -3.]
 [ 1.  0.]
 [ 0. -4.]
 [ 0.  1.]]
```

Verify A @ N == 0:

```
[[0. 0.]
 [0. 0.]
 [0. 0.]]
```

Null space dimension: 2 (nullity)

Rank: 2

Sum: 2 + 2 = 4 = n = 4

Numerical null space via SVD — preferred for large or noisy matrices:

```
import numpy as np

def null_space_svd(A, tol=1e-10):
    """Null space basis via SVD. Robust to near-zero singular values."""
```

```

U, s, Vt = np.linalg.svd(A, full_matrices=True)
rank = np.sum(s > tol * s[0])
return Vt[rank:].T # columns span the null space

A = np.array([[1., 2., 0., 3.],
              [0., 0., 1., 4.],
              [1., 2., 1., 7.]])

N_svd = null_space_svd(A)
print(f"SVD null space shape: {N_svd.shape}")
print(f"A @ N   0: max |A@N| = {np.abs(A @ N_svd).max():.2e}")

```

```

SVD null space shape: (4, 2)
A @ N   0: max |A@N| = 3.33e-16

```

Rank in Practice: What Low Rank Means

Situation	Matrix	Consequence
Sensor readings are correlated	Data matrix X	$\text{rank}(X) < \text{\$ features}$ — PCA will find this
Robot has redundant joints	Jacobian J	$\text{nullity}(J) > 0$ — self-motion exists
Camera calibration target is planar	Correspondence matrix	Degenerate configuration, infinite solutions
Feature vectors are identical	Gram matrix $X^T X$	Singular — least squares is ill-posed

```

import numpy as np

# Simulate correlated sensor data: 3 sensors but only 2 independent signals
rng = np.random.default_rng(0)
s1 = rng.standard_normal(100)
s2 = rng.standard_normal(100)
X = np.column_stack([s1, s2, 0.5*s1 + 0.5*s2]) # sensor 3 is blend of 1 & 2

print(f"Data matrix shape: {X.shape}")
print(f"Rank: {np.linalg.matrix_rank(X)}") # 2, not 3
print(f"Singular values: {np.linalg.svd(X, compute_uv=False).round(2)}")
# Third singular value 0 - the linear dependence is exposed

```

```

Data matrix shape: (100, 3)
Rank: 2
Singular values: [12.07  9.37  0. ]

```

Rank vs. Perturbation Size

How close does a rank-1 matrix have to be to full rank before `matrix_rank` changes its answer? The three panels below show the singular value spectrum for the same base matrix perturbed by εI at three scales.

```
import numpy as np
import matplotlib.pyplot as plt

u = np.array([1., 2., 3.])
v = np.array([1., 1., 1.])
A_base = np.outer(u, v)  # rank-1 matrix

eps_vals = [1.0, 1e-6, 1e-13]
colors = ['#4a90d9', '#2ecc71', 'tomato']

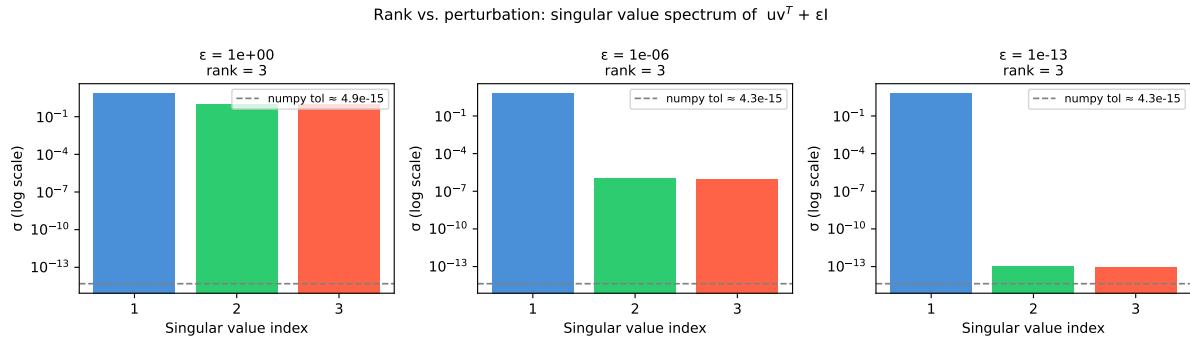
fig, axes = plt.subplots(1, 3, figsize=(12, 3.5))

for ax, eps in zip(axes, eps_vals):
    A = A_base + eps * np.eye(3)
    _, s, _ = np.linalg.svd(A)

    # Default threshold used by np.linalg.matrix_rank
    tol_default = max(A.shape) * s[0] * np.finfo(float).eps
    r_default = np.linalg.matrix_rank(A)

    bars = ax.bar(range(1, 4), s, color=colors)
    ax.axhline(tol_default, color='gray', lw=1.2, ls='--',
               label=f'numpy tol {tol_default:.1e}')
    ax.set_yscale('log')
    ax.set_xlabel('Singular value index')
    ax.set_ylabel(' (log scale)')
    ax.set_title(f' = {eps:.0e} \nrnk = {r_default}', fontsize=10)
    ax.legend(fontsize=8)
    ax.set_xticks([1, 2, 3])

fig.suptitle('Rank vs. perturbation: singular value spectrum of  $uv^T + I$ ',
             fontsize=11)
fig.tight_layout()
plt.show()
```

This demonstrates why rank is not a binary property in practice — it depends on a threshold. When you say “this matrix is rank-deficient,” you always mean relative to some tolerance.

The Four Fundamental Subspaces

Every matrix $A \in \mathbb{R}^{m \times n}$ has four intrinsic subspaces — two in $\mathbb{R}^n$ (the input space) and two in $\mathbb{R}^m$ (the output space). Gilbert Strang calls them the “big picture” of linear algebra. They answer the question: *what does A do to space?*

Definitions

Subspace	Symbol	Lives in	Definition	Dimension
Column space	$\text{Col}(A)$	$\mathbb{R}^m$	All vectors $A\mathbf{x}$; span of columns of A	r
Null space	$\text{Null}(A)$	$\mathbb{R}^n$	All $\mathbf{x}$ with $A\mathbf{x} = \mathbf{0}$	$n - r$
Row space	$\text{Row}(A)$	$\mathbb{R}^n$	Span of rows of $A = \text{Col}(A^T)$	r
Left null space	$\text{Null}(A^T)$	$\mathbb{R}^m$	All $\mathbf{y}$ with $A^T\mathbf{y} = \mathbf{0}$	$m - r$

where $r = \text{rank}(A)$.

Dimensions add up correctly: - In $\mathbb{R}^n$: $r + (n - r) = n$ - In $\mathbb{R}^m$: $r + (m - r) = m$

Orthogonality Relations

The four subspaces are not independent — they come in orthogonal pairs:

$$\text{Row}(A) \perp \text{Null}(A) \quad (\text{both in } \mathbb{R}^n)$$

$$\text{Col}(A) \perp \text{Null}(A^T) \quad (\text{both in } \mathbb{R}^m)$$

Proof sketch. If $\mathbf{x} \in \text{Null}(A)$ then $A\mathbf{x} = \mathbf{0}$. Every row $\mathbf{r}_i$ of A satisfies $\mathbf{r}_i \cdot \mathbf{x} = 0$ — every row is orthogonal to every null space vector. The rows span the row space, so the entire row space is orthogonal to the null space.

```
import numpy as np
import sympy

A = np.array([[1., 2., 0., 3.],
              [0., 0., 1., 4.],
              [1., 2., 1., 7.]]) # rank 2

# Column space basis: pivot columns of A
M = sympy.Matrix(A.tolist())
_, pivots = M.rref()
col_basis = A[:, list(pivots)]

# Null space basis (input space, R^4)
ns = M.nullspace()
null_basis = np.array([np.array(v.T.tolist()[0], dtype=float) for v in ns]).T

# Row space basis: non-zero rows of RREF
R_np = np.array(M.rref()[0].tolist(), dtype=float)
nonzero = [i for i in range(R_np.shape[0]) if np.any(np.abs(R_np[i]) > 1e-12)]
row_basis = R_np[nonzero]

# Left null space basis (output space, R^3)
Mt = sympy.Matrix(A.T.tolist())
lns = Mt.nullspace()
left_null_basis = np.array([np.array(v.T.tolist()[0], dtype=float) for v in lns]).T

print("=== Dimensions ===")
print(f"rank r = {len(pivots)}")
print(f"Col(A):      dim {col_basis.shape[1]} (in R^{A.shape[0]})")
print(f"Null(A):      dim {null_basis.shape[1]} (in R^{A.shape[1]})")
print(f"Row(A):      dim {row_basis.shape[0]} (in R^{A.shape[1]})")
print(f"Null(A^T):    dim {left_null_basis.shape[1]} (in R^{A.shape[0]})")

print("\n=== Orthogonality checks ===")
print(f"Row(A)  Null(A):    {np.abs(row_basis @ null_basis).max():.2e} (should be ~0)")
print(f"Col(A)  Null(A^T):  {np.abs(col_basis.T @ left_null_basis).max():.2e} (should be ~0)")
```

```

=== Dimensions ===
rank r = 2
Col(A):      dim 2  (in  $\mathbb{R}^3$ )
Null(A):      dim 2  (in  $\mathbb{R}^4$ )
Row(A):       dim 2  (in  $\mathbb{R}^4$ )
Null(AT):    dim 1  (in  $\mathbb{R}^3$ )

=== Orthogonality checks ===
Row(A)  Null(A):      0.00e+00  (should be ~0)
Col(A)  Null(AT):    0.00e+00  (should be ~0)

```

Visualising in 3D

For a 3×3 rank-2 matrix, the four subspaces can be drawn in $\mathbb{R}^3$: a plane and a line (orthogonal complement pair) in the input space, and the same in the output space.

```

import numpy as np
import matplotlib.pyplot as plt

# 3x3 rank-2 matrix
A = np.array([[1., 0., 1.],
              [0., 1., 1.],
              [1., 1., 2.]]) # row3 = row1 + row2

# Column space: span of first two columns (rank-2 plane in  $\mathbb{R}^3$ )
c1, c2 = A[:, 0], A[:, 1]

# Left null space: vector orthogonal to col space
_, _, Vt_m = np.linalg.svd(A.T)
lnull = Vt_m[-1]

fig = plt.figure(figsize=(7, 6))
ax = fig.add_subplot(111, projection='3d')

# Draw column space as a shaded plane
s, t = np.linspace(-1, 1, 20), np.linspace(-1, 1, 20)
S, T = np.meshgrid(s, t)
plane = np.einsum('i,jk->ijk', c1, S) + np.einsum('i,jk->ijk', c2, T)
ax.plot_surface(plane[0], plane[1], plane[2],
               alpha=0.25, color='#4a90d9')

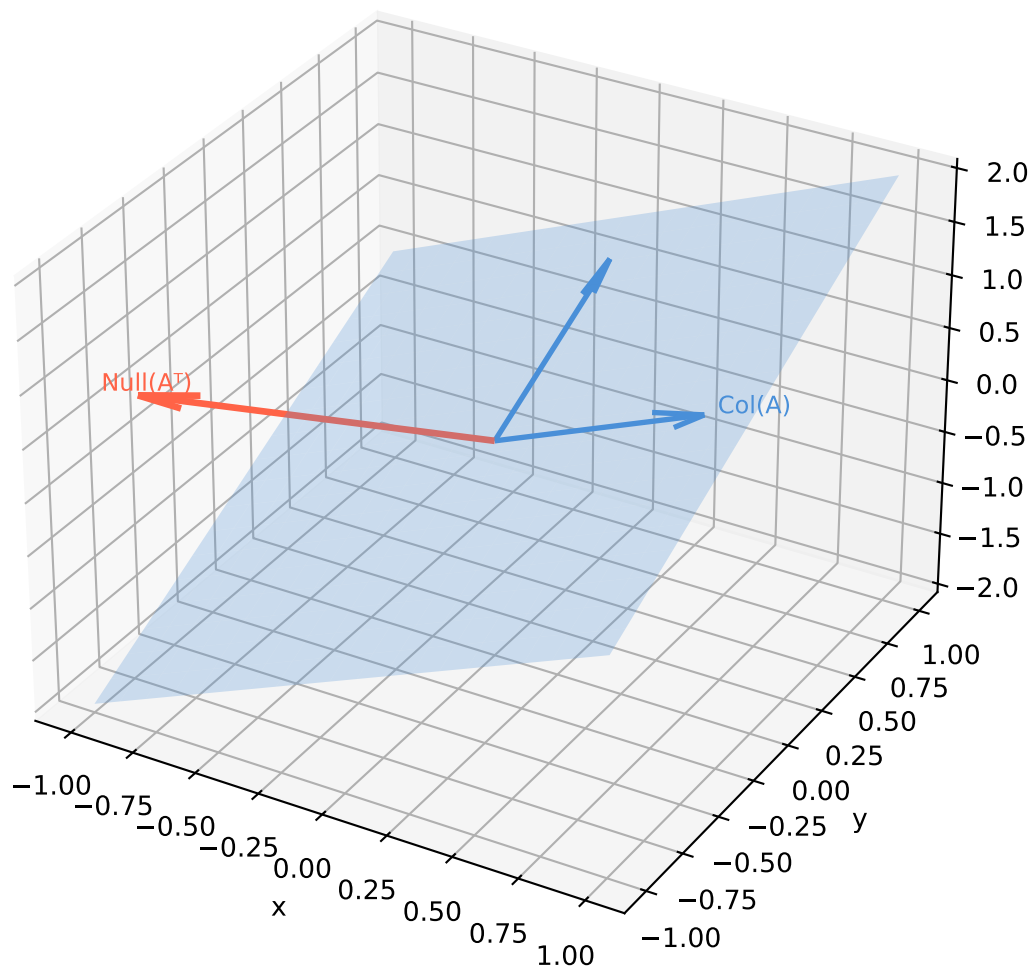
# Column space basis vectors
origin = np.zeros(3)
for vec in [c1, c2]:
    ax.quiver(*origin, *vec*0.8, color='#4a90d9', lw=2, arrow_length_ratio=0.2)

```

```
# Left null space vector
ax.quiver(*origin, *lnull*1.5, color='tomato', lw=2.5, arrow_length_ratio=0.15)
ax.text(*(lnull * 1.65), 'Null(A)', color='tomato', fontsize=9)
ax.text(*(c1 * 0.85), 'Col(A)', color='#4a90d9', fontsize=9)

ax.set_title('Column space (plane) Left null space (line)', pad=8)
ax.set_xlabel('x'); ax.set_ylabel('y'); ax.set_zlabel('z')
plt.tight_layout()
plt.show()
```

Column space (plane) $\perp$ Left null space (line)



Physical Meaning in Each Field

Machine learning — column space as “reachable outputs”. The columns of a weight matrix W span all possible pre-activation vectors. If your target $\mathbf{y}$ is not in $\text{Col}(W)$, the network cannot represent it exactly — least squares finds the closest point in the column space.

Computer vision — left null space as “calibration failure”. In camera calibration, the left null space of the measurement matrix contains the degenerate motion directions. If your calibration target stays in one plane, the system matrix loses rank and its left null space is non-trivial — certain intrinsic parameters become unidentifiable.

Robotics — null space as “self-motion”. The null space of the Jacobian matrix J contains all joint velocity vectors $\dot{\mathbf{q}}$ that produce zero end-effector velocity. These are the self-motions: the arm moves without the hand moving. The rank-nullity theorem guarantees $\dim(\text{Null}(J)) = n_{\text{joints}} - \text{rank}(J)$ such independent self-motions. Chapter 22 works this out in full.

The Fundamental Theorem of Linear Algebra

Strang’s “Fundamental Theorem” states:

1. $\text{Col}(A)$ and $\text{Null}(A^T)$ are **orthogonal complements** in $\mathbb{R}^m$.
2. $\text{Row}(A)$ and $\text{Null}(A)$ are **orthogonal complements** in $\mathbb{R}^n$.

Every vector in $\mathbb{R}^n$ decomposes uniquely into a row-space part and a null-space part:

$$\mathbf{x} = \underbrace{\mathbf{x}_r}_{\in \text{Row}(A)} + \underbrace{\mathbf{x}_n}_{\in \text{Null}(A)}$$

Only the row-space part $\mathbf{x}_r$ contributes to $A\mathbf{x}$; the null-space part $\mathbf{x}_n$ is annihilated.

```
import numpy as np

A = np.array([[1., 0., 1.],
              [0., 1., 1.]])    # 2×3, rank 2, nullity 1

# Project an arbitrary vector onto row space and null space
x = np.array([3., 2., 1.])

# Row space projection: x_r = Vr @ Vr.T @ x (Vr = first r right singular vecs)
U, s, Vt = np.linalg.svd(A, full_matrices=True)
r = np.linalg.matrix_rank(A)
Vr = Vt[:r].T                # columns are row space basis

x_row = Vr @ (Vr.T @ x)      # projection onto row space
x_null = x - x_row            # orthogonal complement
```

```
print(f"x          = {x}")
print(f"x_row      = {x_row.round(4)}    (row space part)")
print(f"x_null     = {x_null.round(4)}    (null space part)")
print(f"A @ x_row   = {(A @ x_row).round(4)}    (= A @ x)")
print(f"A @ x_null = {(A @ x_null).round(10)}    (= 0, as expected)")
print(f"Orthogonal: x_row · x_null = {x_row @ x_null:.2e}")
```

```
x          = [3. 2. 1.]
x_row      = [1.6667 0.6667 2.3333]    (row space part)
x_null     = [ 1.3333  1.3333 -1.3333]    (null space part)
A @ x_row  = [4. 3.]    (= A @ x)
A @ x_null = [0. 0.]    (= 0, as expected)
Orthogonal: x_row · x_null = 4.88e-15
```

This decomposition is the foundation of least-squares theory (Chapter 16) and the pseudoinverse (Chapter 18).

Observability and Identifiability via Rank

Rank deficiency is not an abstract algebraic curiosity — it has concrete physical meaning. A rank-deficient measurement matrix means you cannot recover certain states or parameters from your measurements, no matter how much data you collect. This section connects the algebra to two engineering concepts: *observability* in control/estimation and *identifiability* in system calibration.

Observability: Can You Recover the State?

In a linear dynamical system, the *observability matrix* tells you whether the initial state $\mathbf{x}_0$ can be determined from T output measurements. For a system $\mathbf{x}_{k+1} = F\mathbf{x}_k$, $\mathbf{y}_k = H\mathbf{x}_k$:

$$\mathcal{O} = \begin{pmatrix} H \\ HF \\ HF^2 \\ \vdots \\ HF^{T-1} \end{pmatrix}$$

The system is **fully observable** if and only if $\text{rank}(\mathcal{O}) = n$ (where n = state dimension). Intuitively: $\mathcal{O}$ maps initial state to output sequence; full rank means the mapping is injective — different initial states give different outputs.

```

import numpy as np

def observability_matrix(F, H, T):
    """Build T-step observability matrix for  $x_{k+1}=Fx_k$ ,  $y_k=Hx_k$ ."""
    n = F.shape[0]
    rows = []
    FkH = H.copy()
    for k in range(T):
        rows.append(FkH)
        FkH = FkH @ F
    return np.vstack(rows)

# Example: 2D position + velocity state, position-only measurement
# State: [pos_x, pos_y, vel_x, vel_y], dt=0.1 s
dt = 0.1
F = np.array([[1, 0, dt, 0],
               [0, 1, 0, dt],
               [0, 0, 1, 0],
               [0, 0, 0, 1]]) # constant velocity model

H_pos = np.array([[1, 0, 0, 0],
                  [0, 1, 0, 0]]) # observe position only

H_vel = np.array([[0, 0, 1, 0],
                  [0, 0, 0, 1]]) # observe velocity only

for label, H in [("Position sensors", H_pos), ("Velocity sensors", H_vel)]:
    O = observability_matrix(F, H, T=4)
    r = np.linalg.matrix_rank(O)
    print(f"{label}:  O shape={O.shape},  rank={r},  "
          f"'OBSERVABLE' if r == 4 else 'NOT fully observable'")

```

```

Position sensors:  O shape=(8, 4),  rank=4,  OBSERVABLE
Velocity sensors:  O shape=(8, 4),  rank=2,  NOT fully observable

```

What rank deficiency means here: if $\text{rank}(\mathcal{O}) < n$, the null space of $\mathcal{O}$ contains initial states that produce *identical* output sequences. No amount of measurement can distinguish them.

Unobservable Subspace

The null space of $\mathcal{O}$ is the *unobservable subspace* — the set of initial state perturbations that leave all outputs unchanged.

```
import numpy as np

# Partially observable system: only measure x1, and x3 doesn't affect x1
F = np.array([[0.9, 0.1, 0.0],
              [0.0, 0.8, 0.0],
              [0.0, 0.0, 0.7]]) # x3 is decoupled
H = np.array([[1.0, 0.0, 0.0]]) # observe only x1

O = observability_matrix(F, H, T=3)
r = np.linalg.matrix_rank(O)
print(f"Observability matrix rank: {r} (out of 3)")

# Find unobservable subspace
_, s, Vt = np.linalg.svd(O)
unobservable = Vt[r:].T
print(f"Unobservable direction(s):\n{unobservable.round(4)}")
print(f"\nInterpretation: state perturbations along these directions")
print(f"produce zero output change - you can never observe them.")
```

Observability matrix rank: 2 (out of 3)

Unobservable direction(s):

```
[[0.]
 [0.]
 [1.]]
```

Interpretation: state perturbations along these directions
produce zero output change - you can never observe them.

Identifiability: Can You Calibrate the System?

Identifiability is the calibration analogue of observability. Given a sensor model $\mathbf{y} = A(\theta)\mathbf{x}$ where θ are unknown parameters, identifiability asks whether the calibration data uniquely determines θ .

Linearising around a nominal θ_0 , identifiability reduces to a rank condition on the *sensitivity matrix* (or Fisher information matrix).

Camera calibration example. A camera with intrinsic matrix K and extrinsic pose $[R \mid \mathbf{t}]$ projects 3D points to pixels. If all calibration target points are coplanar, the calibration matrix loses rank — certain parameters (e.g., depth scale) become unidentifiable.

```
import numpy as np

def sensitivity_matrix(param_fn, theta0, delta=1e-5):
    """
```



```

    Numerical Jacobian of a measurement function w.r.t. parameters.
    param_fn: callable(theta) -> measurements vector
    Returns J where  $J[i,j] = d(\text{meas}_i)/d(\text{theta}_j)$ 
    """
    y0 = param_fn(theta0)
    m, p = len(y0), len(theta0)
    J = np.zeros((m, p))
    for j in range(p):
        dth = np.zeros(p)
        dth[j] = delta
        J[:, j] = (param_fn(theta0 + dth) - param_fn(theta0 - dth)) / (2 * delta)
    return J

# Toy example: fit  $a + b*x + c*x^2$  to data
xs = np.linspace(0, 1, 20)

def polynomial_predictions(params):
    a, b, c = params
    return a + b*xs + c*xs**2

theta0 = np.array([1.0, 2.0, 0.5])
J = sensitivity_matrix(polynomial_predictions, theta0)

print(f"Sensitivity matrix shape: {J.shape}")
print(f"Rank: {np.linalg.matrix_rank(J)} (= number of identifiable parameters)")
print(f"Singular values: {np.linalg.svd(J, compute_uv=False).round(4)}")

# Now: what if x range is tiny? (near-collinearity)
xs_tiny = np.linspace(0.0, 0.001, 20)
def poly_tiny(params):
    a, b, c = params
    return a + b*xs_tiny + c*xs_tiny**2

J_tiny = sensitivity_matrix(poly_tiny, theta0)
svs = np.linalg.svd(J_tiny, compute_uv=False)
print(f"\nTiny x range - singular values: {svs.round(6)}")
print(f"Condition number: {svs[0]/svs[-1]:.2e} ← near rank-deficient")
print("b and c become nearly unidentifiable (can't separate linear from quadratic)")

```

```

Sensitivity matrix shape: (20, 3)
Rank: 3 (= number of identifiable parameters)
Singular values: [5.3304 1.6376 0.2552]

```

```

Tiny x range - singular values: [4.472137e+00 1.357000e-03 0.000000e+00]
Condition number: 1.22e+07 ← near rank-deficient
b and c become nearly unidentifiable (can't separate linear from quadratic)

```

Rank Deficiency from Symmetry

A common source of identifiability failure is *symmetry* in the data — when the measurement geometry has a transformation that leaves the model output unchanged.

Example — GPS without altitude variation. Estimating a GPS receiver's clock bias along with position requires observations from satellites at different elevations. If all visible satellites are near the horizon (flat geometry), the vertical component and clock bias become nearly rank-deficient — a known failure mode in urban canyons.

```
import numpy as np

def gps_dilution(satellite_directions):
    """
    GPS DOP (dilution of precision) via rank of observation matrix.
    Each row: [dx, dy, dz, 1] - unit vector to satellite + clock column.
    """
    n = len(satellite_directions)
    G = np.column_stack([satellite_directions,
                        np.ones(n)])    # clock bias column

    GtG = G.T @ G
    kappa = np.linalg.cond(GtG)
    r      = np.linalg.matrix_rank(G)
    return r, kappa

# Good geometry: satellites spread in elevation and azimuth
sats_good = np.array([
    [ 0.57,  0.00,  0.82], # NE, high elevation
    [-0.57,  0.00,  0.82], # NW, high elevation
    [ 0.00,  0.57,  0.82], # N,  high elevation
    [ 0.00, -0.57,  0.82], # S,  high elevation
    [ 0.00,  0.00,  1.00], # zenith
])

# Bad geometry: all satellites near horizon (urban canyon)
sats_bad = np.array([
    [ 1.00,  0.00,  0.05],
    [-1.00,  0.00,  0.05],
    [ 0.00,  1.00,  0.05],
    [ 0.00, -1.00,  0.05],
    [ 0.71,  0.71,  0.05],
])

for label, sats in [("Good geometry", sats_good), ("Urban canyon", sats_bad)]:
    r, kappa = gps_dilution(sats)
    print(f"{label}: rank={r}, condition number={kappa:.1e}")
```

Good geometry: rank=4, condition number=5.8e+02

Urban canyon: rank=3, condition number=6.5e+17

Diagnostic Checklist

When a calibration or estimation problem gives poor results:

1. **Compute `np.linalg.matrix_rank(A)`** — is it full rank?
2. **Check singular values** — are any near zero relative to the largest?
3. **Examine the null space** — what physical state/parameter does it correspond to?
4. **Fix the data collection**, not the math — add more diverse measurements that break the symmetry causing rank deficiency.

The null space of the measurement matrix is a *geometric* guide to what additional data you need to collect. If the null space direction corresponds to “all targets at the same range,” you need targets at varied distances.

Application: Kinematic Redundancy in a 7-DoF Arm

A 7-degree-of-freedom robot arm — like the Franka Emika Panda or the KUKA iiwa — has one more joint than the six degrees of freedom needed to place and orient a rigid end-effector in 3D space. That extra degree of freedom is the *redundancy*, and it lives in the null space of the Jacobian.

This section shows how to compute the Jacobian’s null space, how to use it for null-space motion (joint motion that doesn’t move the hand), and why the rank-nullity theorem guarantees exactly one such independent self-motion.

The Jacobian as a Linear Map

The *Jacobian* $J(\mathbf{q}) \in \mathbb{R}^{6 \times 7}$ relates small joint velocity changes $\dot{\mathbf{q}} \in \mathbb{R}^7$ to end-effector velocity $\dot{\mathbf{x}} \in \mathbb{R}^6$:

$$\dot{\mathbf{x}} = J(\mathbf{q}) \dot{\mathbf{q}}$$

At any configuration $\mathbf{q}$, this is a linear system. The rank-nullity theorem applied to this 6×7 map guarantees:

$$\underbrace{\text{rank}(J)}_{\leq 6} + \underbrace{\text{nullity}(J)}_{\geq 1} = 7$$

At non-singular configurations $\text{rank}(J) = 6$, so $\text{nullity}(J) = 1$: exactly one independent self-motion direction exists.

A Simplified Planar 3-DoF Example

We start with a 3-DoF planar arm (2D end-effector position), which has nullity = 1. This is simpler to visualise and numerically identical to the 7-DoF case.

```
import numpy as np

def planar_fk(q, L):
    """
    Forward kinematics for a planar n-link arm.
    q: joint angles (radians), L: link lengths
    Returns end-effector (x, y).
    """
    x, y, angle = 0.0, 0.0, 0.0
    for qi, li in zip(q, L):
        angle += qi
        x += li * np.cos(angle)
        y += li * np.sin(angle)
    return np.array([x, y])

def planar_jacobian(q, L):
    """
    Analytical Jacobian for planar arm: J[i,j] = d(x_i)/d(q_j).
    Shape (2, n_joints).
    """
    n = len(q)
    J = np.zeros((2, n))
    cumulative_angles = np.cumsum(q)
    for j in range(n):
        # joint j affects all links from j onward
        for k in range(j, n):
            angle_k = cumulative_angles[k]
            J[0, j] += -L[k] * np.sin(angle_k) # dx/dq_j
            J[1, j] += L[k] * np.cos(angle_k) # dy/dq_j
    return J

# 3-link arm, equal link lengths
L = [1.0, 1.0, 1.0]
```

```

q = np.array([np.pi/6, np.pi/4, -np.pi/3])

J = planar_jacobian(q, L)
r = np.linalg.matrix_rank(J)
print(f"Jacobian shape: {J.shape}")
print(f"Rank: {r} (= end-effector DOF)")
print(f"Nullity: {J.shape[1] - r} (= redundant DOF)")
print(f"\nJacobian:\n{J.round(4)}")

```

```

Jacobian shape: (2, 3)
Rank: 2 (= end-effector DOF)
Nullity: 1 (= redundant DOF)

```

```

Jacobian:
[[-1.7247 -1.2247 -0.2588]
 [ 2.0908  1.2247  0.9659]]

```

Computing the Null-Space Self-Motion

```

import numpy as np

def null_space_svd(J, tol=1e-10):
    """Return basis of null space of J via SVD."""
    U, s, Vt = np.linalg.svd(J, full_matrices=True)
    rank = np.sum(s > tol * s[0])
    return Vt[rank:].T # each column is a null space basis vector

# At configuration q
null_vec = null_space_svd(J)
print(f"Null space basis shape: {null_vec.shape}")
print(f"Null vector: {null_vec.flatten().round(4)}")

# Verify: J @ null_vec == 0
print(f"J @ null_vec = {(J @ null_vec).round(10).flatten()} (should be 0)")

# What does self-motion look like?
ee_before = planar_fk(q, L)
dq_null = null_vec.flatten() * 0.1 # small step along null vector
ee_after = planar_fk(q + dq_null, L)

print(f"\nEnd-effector before: {ee_before.round(4)}")
print(f"End-effector after: {ee_after.round(4)}")
print(f"Displacement: {(ee_after - ee_before).round(6)} ( 0)")

```

```
Null space basis shape: (3, 1)
Null vector: [-0.5817  0.7556  0.3011]
J @ null_vec = [0. 0.] (should be 0)

End-effector before: [2.0908 1.7247]
End-effector after:  [2.0882 1.7235]
Displacement:        [-0.002604 -0.001273] ( 0)
```

Visualising Self-Motion

The five arm configurations below all share the same end-effector position — they differ only by moving along the null vector of the Jacobian.

```
import numpy as np
import matplotlib.pyplot as plt

def draw_arm(ax, q, L, color='#4a90d9', alpha=1.0, label=None):
    """Draw a planar arm configuration."""
    positions = [(0, 0)]
    angle = 0.0
    for qi, li in zip(q, L):
        angle += qi
        x = positions[-1][0] + li * np.cos(angle)
        y = positions[-1][1] + li * np.sin(angle)
        positions.append((x, y))
    xs, ys = zip(*positions)
    ax.plot(xs, ys, '-o', color=color, lw=2.5, markersize=6,
            alpha=alpha, label=label)
    ax.scatter(*positions[-1], color=color, s=80, zorder=5)

L = [1.0, 1.0, 1.0]
q_base = np.array([np.pi/6, np.pi/4, -np.pi/3])
null_v = null_space_svd(planar_jacobian(q_base, L)).flatten()

colors = ['#4a90d9', '#2ecc71', '#9b59b6', 'tomato', '#e67e22']

fig, ax = plt.subplots(figsize=(7, 5))

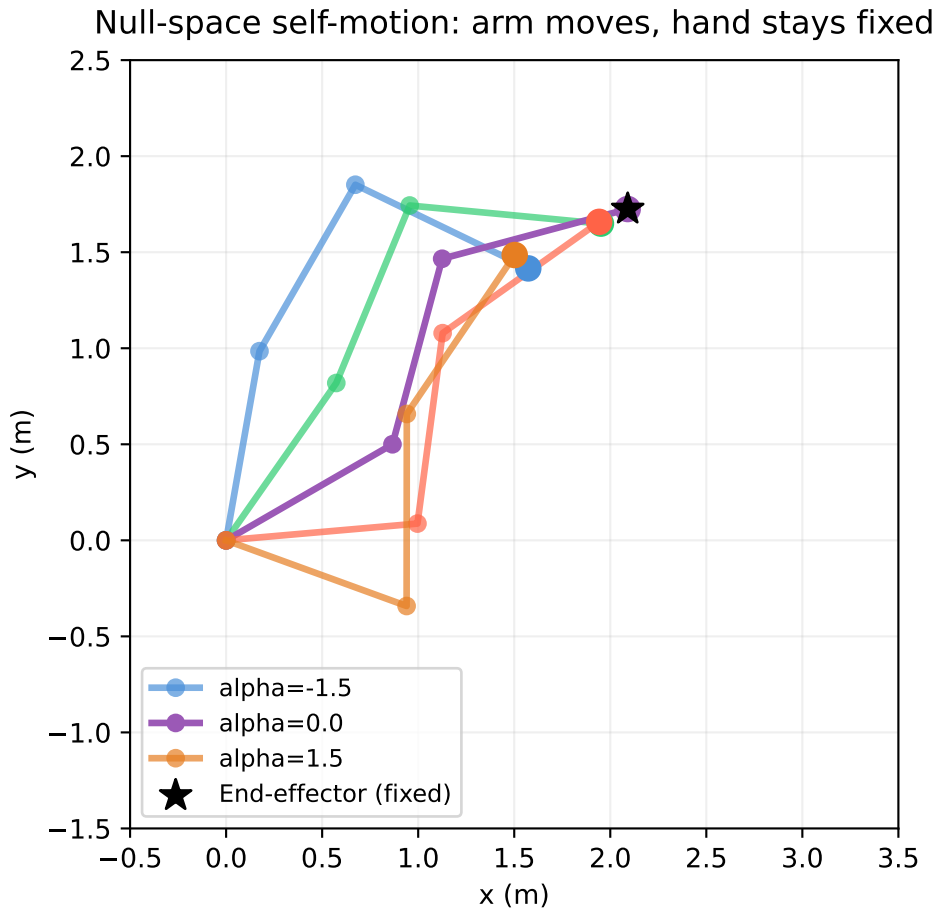
alphas = np.linspace(-1.5, 1.5, 5)
for i, alpha in enumerate(alphas):
    q_new = q_base + alpha * null_v
    draw_arm(ax, q_new, L,
            color=colors[i % len(colors)],
            alpha=0.7 + 0.3*(i == 2),
            label=f'alpha={alpha:.1f}' if i in [0, 2, 4] else None)
```

```

# Mark the end-effector (same for all)
ee = planar_fk(q_base, L)
ax.scatter(*ee, color='black', s=150, zorder=10, marker='*',
          label='End-effector (fixed)')

ax.set_xlim(-0.5, 3.5); ax.set_ylim(-1.5, 2.5)
ax.set_aspect('equal')
ax.set_xlabel('x (m)'); ax.set_ylabel('y (m)')
ax.set_title('Null-space self-motion: arm moves, hand stays fixed', pad=10)
ax.legend(fontsize=9)
ax.grid(True, alpha=0.2)
plt.tight_layout()
plt.show()

```



Self-Motion Across Different Configurations

The null vector changes with configuration — self-motion at one arm pose is different from self-motion at another. The two panels compare a generic configuration and a near-singular one.

```
import numpy as np
import matplotlib.pyplot as plt

configs = {
    "Generic   q = [30°, 45°, -60°]": np.deg2rad([30, 45, -60]),
    "Near-singular q = [0°, 0°, 0°]": np.deg2rad([0, 0, 0]),
}
colors_arm = ['#4a90d9', 'tomato']

fig, axes = plt.subplots(1, 2, figsize=(12, 5))

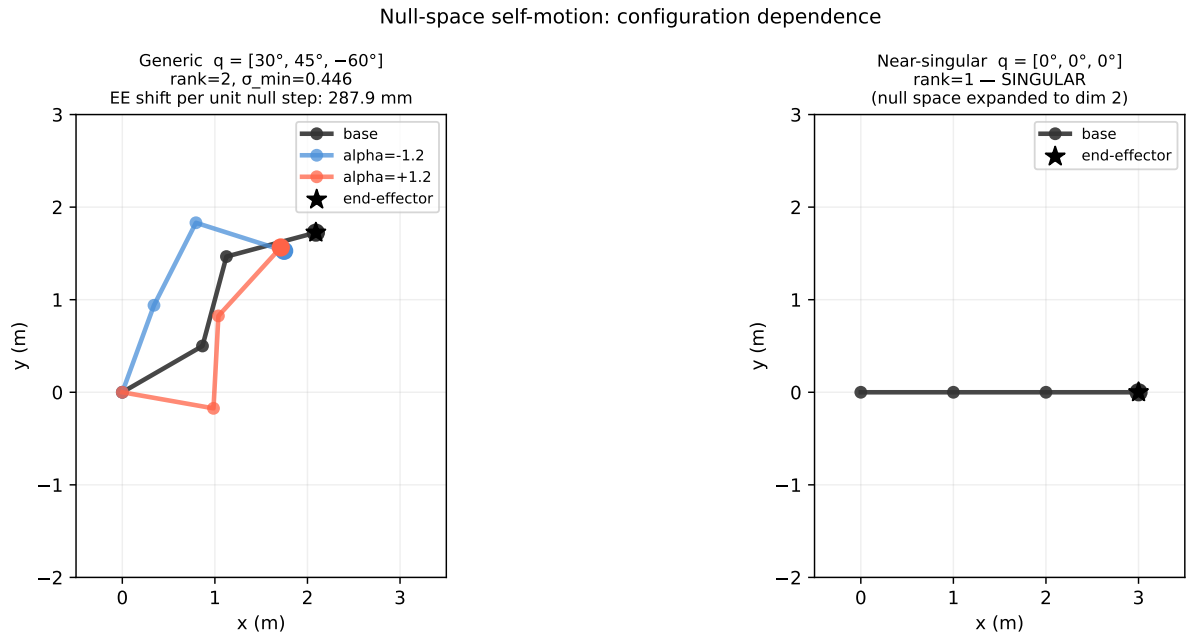
for ax, (title, q_cfg) in zip(axes, configs.items()):
    J_cfg = planar_jacobian(q_cfg, L)
    sv     = np.linalg.svd(J_cfg, compute_uv=False)
    r      = np.linalg.matrix_rank(J_cfg)

    # Draw base configuration
    draw_arm(ax, q_cfg, L, color='#333333', alpha=0.9, label='base')

    if r >= 2:
        nv = null_space_svd(J_cfg).flatten()
        for alpha, col, lbl in [(-1.2, '#4a90d9', 'alpha=-1.2'),
                               (+1.2, 'tomato', 'alpha=+1.2')]:
            draw_arm(ax, q_cfg + alpha * nv, L, color=col, alpha=0.75, label=lbl)
        ee_disp = np.linalg.norm(
            planar_fk(q_cfg + nv, L) - planar_fk(q_cfg, L)) * 1000
        info = f'rank={r}, _min={sv[-1]:.3f}\nEE shift per unit null step: {ee_disp:.1f} mm'
    else:
        info = f'rank={r} - SINGULAR\n(null space expanded to dim {3-r})'

    ee = planar_fk(q_cfg, L)
    ax.scatter(*ee, color='black', s=120, zorder=10, marker='*', label='end-effector')
    ax.set_xlim(-0.5, 3.5); ax.set_ylim(-2, 3)
    ax.set_aspect('equal')
    ax.set_title(f'{title}\n{info}', fontsize=9, pad=6)
    ax.legend(fontsize=8)
    ax.grid(True, alpha=0.2)
    ax.set_xlabel('x (m)'); ax.set_ylabel('y (m)')

fig.suptitle('Null-space self-motion: configuration dependence', fontsize=11)
fig.tight_layout()
plt.show()
```

Singular Configurations

When the arm reaches a *kinematic singularity*, $\text{rank}(J)$ drops below 6 (or 2 for the planar arm). The null space suddenly *grows* — more self-motions appear, but the end-effector loses the ability to move in certain directions.

```
import numpy as np

L = [1.0, 1.0, 1.0]

configs = {
    "Generic": [np.pi/6, np.pi/4, -np.pi/3],
    "Stretched": [0.0, 0.0, 0.0], # fully extended - singular
    "Folded": [np.pi/2, -np.pi, np.pi/2], # folded back
}

for name, q in configs.items():
    q_arr = np.array(q)
    J_q = planar_jacobian(q_arr, L)
    sv = np.linalg.svd(J_q, compute_uv=False)
    r = np.linalg.matrix_rank(J_q)
    print(f"{name:12s} rank={r} nullity={len(q)-r} "
          f"_min={sv[-1]:.4f} ee={planar_fk(q_arr, L).round(3)}")
```

Generic	rank=2	nullity=1	_min=0.4459	ee=[2.091 1.725]
Stretched	rank=1	nullity=2	_min=0.0000	ee=[3. 0.]
Folded	rank=2	nullity=1	_min=0.7654	ee=[1. 0.]

At the stretched configuration, $\sigma_{\min} \approx 0$: the arm cannot move its end-effector along the direction of the outstretched links. Motion planning algorithms must avoid (or carefully handle) configurations near singularities. Chapter 22 develops this with the full 6-DoF velocity Jacobian.

Key Takeaways

1. A redundant arm ($n > 6$ joints) has a non-trivial null space for its Jacobian. The rank-nullity theorem guarantees $\text{nullity}(J) = n - 6$ independent self-motion directions at non-singular configurations.
2. Null-space self-motions let the arm avoid obstacles, stay away from joint limits, or optimise secondary objectives (e.g., manipulability) without disturbing the end-effector.
3. Singular configurations reduce $\text{rank}(J)$, expand the null space, and simultaneously *shrink* the end-effector's reachable velocity directions.
4. Computing null space via SVD (`Vt[rank:]`) is numerically preferable to RREF for robot Jacobians, which have continuous-valued entries.

Exercises

3.1 Without using Python, bring the following matrix to RREF and identify all pivot and free columns:

$$A = \begin{pmatrix} 1 & 3 & 0 & 2 \\ 0 & 0 & 1 & 4 \\ 2 & 6 & 1 & 8 \end{pmatrix}$$

How many solutions does $A\mathbf{x} = \mathbf{b}$ have for an arbitrary consistent $\mathbf{b}$?

3.2 A matrix $A \in \mathbb{R}^{5 \times 8}$ has rank 3. State the dimensions of all four fundamental subspaces.

3.3 You are told that $\mathbf{v}_1 = [1, 0, -1, 0]^T$ and $\mathbf{v}_2 = [0, 1, 0, -1]^T$ are both in the null space of a matrix $A \in \mathbb{R}^{3 \times 4}$. What is the minimum possible rank of A ? What is the maximum? Is there any constraint on $m = 3$ vs. the null space dimension?

3.4 The observability matrix for a system with

$$F = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \quad H = \begin{pmatrix} 1 & 0 \end{pmatrix}$$

is $\mathcal{O} = [H^T, (HF)^T]^T$. Compute $\mathcal{O}$ by hand and determine whether the system is observable.

3.5 Two students are debating: “Adding more equations to a system can never decrease the rank.” Is this true? Prove it or give a counterexample.

3.6 (Harder) Prove that $\text{rank}(AB) \leq \min(\text{rank}(A), \text{rank}(B))$. (*Hint: think about what the column space of AB is as a subset of the column space of A .*)

3.7 A 3-DoF planar robot arm is near a configuration where its Jacobian has a very small second singular value ($\sigma_2 = 0.02$, while $\sigma_1 = 3.1$). What does this tell you about the arm’s mobility? Would you compute the null space with RREF or SVD here, and why?

Solutions

Solution 3.1

Starting augmented matrix (no $\mathbf{b}$ needed for RREF of A):

$$\begin{pmatrix} 1 & 3 & 0 & 2 \\ 0 & 0 & 1 & 4 \\ 2 & 6 & 1 & 8 \end{pmatrix}$$

Step 1: $R_3 \leftarrow R_3 - 2R_1$:

$$\begin{pmatrix} 1 & 3 & 0 & 2 \\ 0 & 0 & 1 & 4 \\ 0 & 0 & 1 & 4 \end{pmatrix}$$

Step 2: $R_3 \leftarrow R_3 - R_2$:

$$\begin{pmatrix} 1 & 3 & 0 & 2 \\ 0 & 0 & 1 & 4 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

This is already RREF (pivots are 1, normalized, and column 3 has a 0 above its pivot).

- **Pivot columns:** 1 and 3
- **Free columns:** 2 and 4 (variables x_2, x_4)
- **Rank:** 2, **Nullity:** 2

For any consistent $\mathbf{b}$, there are **infinitely many solutions** (parametrized by 2 free variables).

```
import numpy as np, sympy
A = np.array([[1.,3.,0.,2.],[0.,0.,1.,4.],[2.,6.,1.,8.]])
R, pivots = sympy.Matrix(A).rref()
print(np.array(R.tolist(), dtype=float))
print("Pivots:", pivots)
```

```
[[1. 3. 0. 2.]
 [0. 0. 1. 4.]
 [0. 0. 0. 0.]]
Pivots: (0, 2)
```

Solution 3.2

$A \in \mathbb{R}^{5 \times 8}$, $\text{rank}(A) = 3$:

Subspace	Space	Dimension
$\text{Col}(A)$	$\mathbb{R}^5$	3
$\text{Null}(A^T)$ (left null)	$\mathbb{R}^5$	$5 - 3 = 2$
$\text{Row}(A)$	$\mathbb{R}^8$	3
$\text{Null}(A)$	$\mathbb{R}^8$	$8 - 3 = 5$

Check: $3 + 2 = 5 = m$ and $3 + 5 = 8 = n$.

The large null space (dimension 5) means this system has 5 degrees of freedom of self-motion — physically meaningful in robotics (8-DoF arm, 3-DoF task space).

Solution 3.3

Since $\mathbf{v}_1$ and $\mathbf{v}_2$ are both in $\text{Null}(A)$ and are linearly independent (they have non-overlapping nonzero positions), the null space has dimension at least 2.

By rank-nullity: $\text{rank}(A) + \text{nullity}(A) = 4$. If $\text{nullity}(A) \geq 2$, then $\text{rank}(A) \leq 2$.

- **Minimum rank:** 2 (nullity exactly 2, which is consistent)
- **Maximum rank:** 2 (same bound — nullity ≥ 2 means rank ≤ 2)

So rank is exactly 2.

Constraint from $m = 3$: $\text{rank} \leq \min(m, n) = \min(3, 4) = 3$, so rank could in principle be 3. But having two independent null vectors forces rank ≤ 2 . The row dimension $m = 3$ is not the binding constraint here — the null space structure is.

```

import numpy as np
# Construct A with the given null vectors
v1 = np.array([1., 0., -1., 0.])
v2 = np.array([0., 1., 0., -1.])
# A must satisfy A @ v1 = 0 and A @ v2 = 0
# Build A whose rows are orthogonal to v1 and v2
# Row space of A must be orthogonal to span{v1, v2}
# A row: a such that a·v1=0, a·v2=0 → a1-a3=0, a2-a4=0 → a3=a1, a4=a2
A = np.array([[1., 0., 1., 0.],
               [0., 1., 0., 1.],
               [1., 1., 1., 1.]])
print(f"Rank: {np.linalg.matrix_rank(A)}")
print(f"A @ v1 = {A @ v1}")
print(f"A @ v2 = {A @ v2}")

```

Rank: 2

A @ v1 = [0. 0. 0.]

A @ v2 = [0. 0. 0.]

Solution 3.4

$$HF = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$$

$$\mathcal{O} = \begin{pmatrix} H \\ HF \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}$$

$\det(\mathcal{O}) = 1 \cdot 1 - 0 \cdot 1 = 1 \neq 0$. Therefore $\text{rank}(\mathcal{O}) = 2 = n$, and the system is **fully observable**.

Physical interpretation: even though we only measure x_1 , the dynamics (x_1 evolves as $x_1 + x_2$) couple x_2 into future x_1 measurements. After two time steps, both x_1 and x_2 have been “seen” through x_1 .

```

import numpy as np
F = np.array([[1., 1.], [0., 1.]])
H = np.array([[1., 0.]])
O = np.vstack([H, H @ F])
print(f"O =\n{O}")
print(f"rank(O) = {np.linalg.matrix_rank(O)}")

```

O =

[[1. 0.]

[1. 1.]]

rank(O) = 2

Solution 3.5

True. Adding equations (rows) to a matrix cannot decrease the rank.

Proof. Let $A \in \mathbb{R}^{m \times n}$ and let A' be A with one additional row $\mathbf{r}^T$. The column space of A' contains the column space of A as a subset:

$$\text{Col}(A) \subseteq \text{Col}(A')$$

Since rank equals column-space dimension:

$$\text{rank}(A') = \dim(\text{Col}(A')) \geq \dim(\text{Col}(A)) = \text{rank}(A)$$

However, rank can *stay the same* if the new row is a linear combination of existing rows — adding a redundant equation adds no new information.

```
import numpy as np
A = np.array([[1.,2.],[3.,4.]])
A2 = np.vstack([A, [4., 6.]]) # new row = row1 + row2 (redundant)
A3 = np.vstack([A, [5., 7.]]) # new row linearly independent
print(f"rank(A)={np.linalg.matrix_rank(A)}, "
      f"rank(A+redundant)={np.linalg.matrix_rank(A2)}, "
      f"rank(A+independent)={np.linalg.matrix_rank(A3)}")
```

rank(A)=2, rank(A+redundant)=2, rank(A+independent)=2

Solution 3.6

Claim: $\text{rank}(AB) \leq \text{rank}(A)$.

Proof. Any vector in $\text{Col}(AB)$ has the form $AB\mathbf{x} = A(B\mathbf{x})$ — it is A applied to some vector $B\mathbf{x}$. Therefore:

$$\text{Col}(AB) \subseteq \text{Col}(A) \implies \text{rank}(AB) \leq \text{rank}(A)$$

Claim: $\text{rank}(AB) \leq \text{rank}(B)$.

Proof. Each column of AB is A times a column of B . So the columns of AB are in the span of A 's columns of B ... more precisely: $AB = A \cdot B$ where each column of AB is a linear combination of columns of A weighted by a column of B . Alternatively, apply the transpose: $\text{rank}((AB)^T) = \text{rank}(B^T A^T) \leq \text{rank}(B^T) = \text{rank}(B)$.

Together: $\text{rank}(AB) \leq \min(\text{rank}(A), \text{rank}(B))$.

```

import numpy as np
A = np.random.randn(4, 3)    # rank 3
B = np.random.randn(3, 5)    # rank 3
AB = A @ B
print(f"rank(A)={np.linalg.matrix_rank(A)}, "
      f"rank(B)={np.linalg.matrix_rank(B)}, "
      f"rank(AB)={np.linalg.matrix_rank(AB)}")

# Force rank-1 case
u = np.random.randn(4, 1)
v = np.random.randn(1, 5)
A1B1 = u @ v    # rank-1 product of rank-1 factors
print(f"rank(u)={np.linalg.matrix_rank(u)}, "
      f"rank(v)={np.linalg.matrix_rank(v)}, "
      f"rank(uv)={np.linalg.matrix_rank(A1B1)}")

```

rank(A)=3, rank(B)=3, rank(AB)=3
 rank(u)=1, rank(v)=1, rank(uv)=1

Solution 3.7

A smallest singular value of $\sigma_2 = 0.02$ (while $\sigma_1 = 3.1$) means the arm is **near a kinematic singularity** in one direction. Specifically:

- The arm can move freely along the first singular direction (gain ≈ 3.1).
- Motion along the second singular direction is severely restricted: producing unit end-effector velocity in that direction requires joint velocities of magnitude $\sim 1/0.02 = 50$ rad/s — physically infeasible.

The arm is not fully singular (rank is still 2), but it is **ill-conditioned**: small tracking errors in the nearly-singular direction cause enormous joint velocities.

Use SVD, not RREF. Near a singularity, $\sigma_2 = 0.02$ is small but non-zero. RREF uses a hard threshold to decide rank and would classify this as either rank-1 or rank-2 depending on the threshold — a binary decision that loses the continuous information in σ_2 . SVD preserves the full spectrum and lets you make a continuous decision (e.g., damped pseudo-inverse with $\sigma_2/(\sigma_2^2 + \lambda^2)$ damping). This is the *damped least-squares* inverse used in all production robot controllers near singularities.

```

import numpy as np

# Simulate near-singular Jacobian
J_near_sing = np.array([[3.1, 0.0],
                        [0.0, 0.02]])    # already in SVD form for clarity

```

```
U, s, Vt = np.linalg.svd(J_near_sing)
print(f"Singular values: {s}")
print(f"Condition number: {s[0]/s[1]:.1f}")

# Damped pseudo-inverse (prevents joint velocity explosion)
lam = 0.1 # damping factor
s_damp = s / (s**2 + lam**2)
J_pinv_damp = Vt.T @ np.diag(s_damp) @ U.T
print(f"\nDamped pinv:\n{J_pinv_damp.round(3)}")
print(f"Plain pinv would give:\n{np.linalg.pinv(J_near_sing).round(3)}")
print(f"(Note: plain pinv gives {1/0.02:.0f}x amplification in singular direction)")
```

Singular values: [3.1 0.02]

Condition number: 155.0

Damped pinv:

```
[[0.322 0.   ]
 [0.    1.923]]
```

Plain pinv would give:

```
[[ 0.323  0.   ]
 [ 0.    50.   ]]
```

(Note: plain pinv gives 50x amplification in singular direction)

Vectors in

Vectors: Point, Displacement, Direction

The word “vector” means different things depending on context — and getting this wrong causes subtle bugs in 3D graphics and robotics. A 3-tuple (x, y, z) can represent:

Interpretation	Meaning	Transforms under rotation R as
Point (position)	A location in space	$R\mathbf{p} + \mathbf{t}$ (rotate <i>and</i> translate)
Displacement (free vector)	A difference of two positions	$R\mathbf{d}$ (rotate only, translation cancels)
Direction (unit vector)	A normalized displacement	$R\hat{\mathbf{u}}$ (same as displacement; $\ \hat{\mathbf{u}}\ = 1$ preserved)

Chapter 9 will make this precise with SE(3) and homogeneous coordinates. For now, the key habit: **always know which interpretation you intend.**

Points and Displacements in

A **point** $\mathbf{p} \in \mathbb{R}^n$ specifies a location relative to some origin. Its numerical values depend on where the origin is.

A **displacement** $\mathbf{d} = \mathbf{q} - \mathbf{p}$ is the difference of two points. Displacements are *free vectors* — shifting the origin does not change $\mathbf{d}$.

$$\mathbf{d} = \mathbf{q} - \mathbf{p} = \begin{pmatrix} q_1 - p_1 \\ q_2 - p_2 \\ \vdots \\ q_n - p_n \end{pmatrix}$$

```
import numpy as np

# Two 3D landmark positions (points)
p = np.array([1.0, 2.0, 0.5]) # robot at p
q = np.array([4.0, 3.0, 1.5]) # goal at q
```

```
# Displacement from p to q
d = q - p
print(f"Displacement d = q - p: {d}")
print(f"Euclidean distance:      {np.linalg.norm(d):.4f} m")

# Key: if we shift origin by t, points change but displacement does not
t = np.array([10.0, -5.0, 3.0]) # arbitrary origin shift
p_new = p + t
q_new = q + t
d_new = q_new - p_new
print(f"\nAfter shifting origin by {t}:")
print(f"  p_new = {p_new}, q_new = {q_new}")
print(f"  Displacement unchanged: {d_new}")
print(f"  Same as before: {np.allclose(d, d_new)}")
```

Displacement d = q - p: [3. 1. 1.]

Euclidean distance: 3.3166 m

After shifting origin by [10. -5. 3.]:

p_new = [11. -3. 3.5], q_new = [14. -2. 4.5]

Displacement unchanged: [3. 1. 1.]

Same as before: True

Unit Vectors (Directions)

A **direction** is a displacement normalized to unit length:

$$\hat{\mathbf{u}} = \frac{\mathbf{d}}{\|\mathbf{d}\|}, \quad \|\hat{\mathbf{u}}\| = 1$$

Directions appear in: - Surface normals (Ch 4.3, App) - Axis of rotation in Rodrigues' formula (Ch 8) - Principal directions from SVD (Ch 18) - Camera viewing rays (Ch 10)

```
import numpy as np

d = np.array([3.0, -4.0, 0.0])
norm_d = np.linalg.norm(d)
u_hat = d / norm_d

print(f"d      = {d}")
print(f"||d||     = {norm_d:.4f}")
print(f"u_hat      = {u_hat}")
print(f"||u_hat||   = {np.linalg.norm(u_hat):.10f} (should be 1.0)")
```

```

d      = [ 3. -4.  0.]
||d||   = 5.0000
û      = [ 0.6 -0.8  0. ]
||û||   = 1.00000000000 (should be 1.0)
    
```

Pitfall — zero-length vectors. Normalizing a zero vector produces `nan`. Always guard in production code:

```

def safe_normalize(v, tol=1e-12):
    """Return unit vector, or zero if v is too short."""
    n = np.linalg.norm(v)
    return v / n if n > tol else np.zeros_like(v)
    
```

Standard Basis Vectors

The **standard basis** of $\mathbb{R}^n$ consists of unit vectors along each axis:

$$\mathbf{e}_1 = \begin{pmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix}, \quad \mathbf{e}_2 = \begin{pmatrix} 0 \\ 1 \\ \vdots \\ 0 \end{pmatrix}, \quad \dots, \quad \mathbf{e}_n = \begin{pmatrix} 0 \\ \vdots \\ 0 \\ 1 \end{pmatrix}$$

Any vector $\mathbf{v} \in \mathbb{R}^n$ decomposes as:

$$\mathbf{v} = v_1 \mathbf{e}_1 + v_2 \mathbf{e}_2 + \dots + v_n \mathbf{e}_n$$

The components v_i are the *coordinates* of $\mathbf{v}$ in the standard basis. Section 4.5 generalizes this to arbitrary bases.

```

import numpy as np

n = 4
E = np.eye(n)    # columns are e_1, ..., e_n

v = np.array([2.0, -1.0, 3.0, 0.5])
print("Standard basis vectors (columns of I_4):")
for i in range(n):
    print(f"  e_{i+1} = {E[:, i]}")

# Reconstruct v from its components
v_reconstructed = sum(v[i] * E[:, i] for i in range(n))
print(f"\nv = {v}")
print(f"Reconstructed: {v_reconstructed}")
print(f"Match: {np.allclose(v, v_reconstructed)}")
    
```

Vectors in

Standard basis vectors (columns of I_4):

```
e_1 = [1. 0. 0. 0.]
e_2 = [0. 1. 0. 0.]
e_3 = [0. 0. 1. 0.]
e_4 = [0. 0. 0. 1.]
```

```
v = [ 2. -1.  3.  0.5]
```

```
Reconstructed: [ 2. -1.  3.  0.5]
```

```
Match: True
```

Visualizing Vectors: Points vs. Displacements

```
import numpy as np
import matplotlib.pyplot as plt

fig, axes = plt.subplots(1, 3, figsize=(12, 4))
titles = ['Point\n(absolute position)', 'Displacement\n(free vector)',
          'Direction\n(unit vector)']

origin = np.array([0, 0])
P = np.array([2.5, 1.5])
Q = np.array([0.5, 3.0])
blue = '#4a90d9'
green = '#2ecc71'
red = 'tomato'

# --- Panel 1: Points ---
ax = axes[0]
ax.annotate('', xy=P, xytext=origin,
            arrowprops=dict(arrowstyle='->', color=blue, lw=2))
ax.annotate('', xy=Q, xytext=origin,
            arrowprops=dict(arrowstyle='->', color=green, lw=2))
ax.scatter(*origin, color='#333333', s=60, zorder=5)
ax.text(*P + 0.1, 'p', color=blue, fontsize=13, fontweight='bold')
ax.text(*Q + 0.1, 'q', color=green, fontsize=13, fontweight='bold')
ax.text(0.1, -0.3, 'O (origin)', color='#333333', fontsize=9)

# --- Panel 2: Displacement ---
ax = axes[1]
ax.annotate('', xy=P, xytext=origin,
            arrowprops=dict(arrowstyle='->', color=blue, lw=1.5,
                            linestyle='dashed', alpha=0.4))
ax.annotate('', xy=Q, xytext=origin,
            arrowprops=dict(arrowstyle='->', color=green, lw=1.5,
```

```

                                linestyle='dashed', alpha=0.4))
ax.annotate('', xy=Q, xytext=P,
            arrowprops=dict(arrowstyle='->', color=red, lw=2.5))
ax.text(*(P + Q) / 2 + np.array([0.15, 0.0]),
        'd = q - p', color=red, fontsize=11, fontweight='bold')
ax.scatter(*origin, color='#333333', s=50, zorder=5)

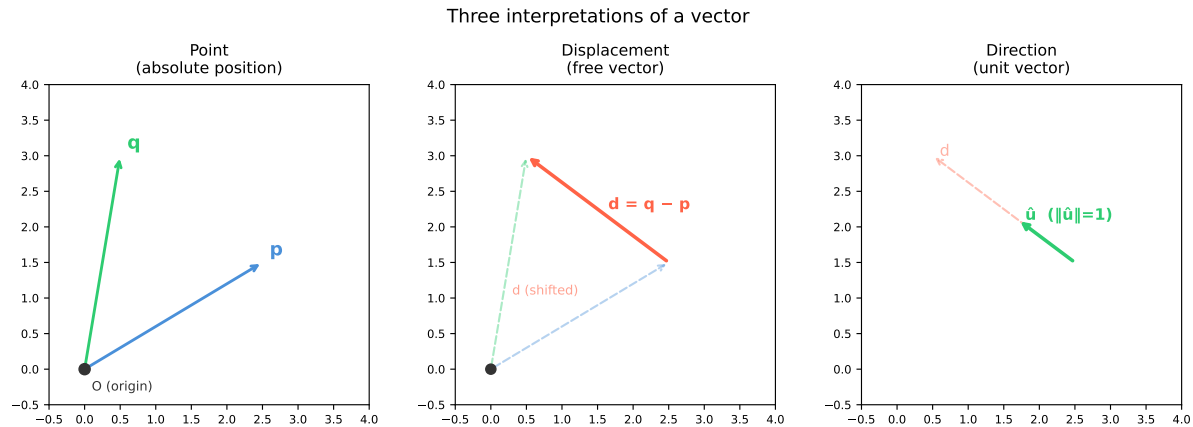
p2 = np.array([1.2, 0.2])
q2 = p2 + (Q - P)
ax.annotate('', xy=q2, xytext=p2,
            arrowprops=dict(arrowstyle='->', color=red, lw=2.5, alpha=0.4))
ax.text(*(p2 + q2) / 2 + np.array([0.1, 0.1]),
        'd (shifted)', color=red, fontsize=9, alpha=0.6)

# --- Panel 3: Direction ---
ax = axes[2]
d = Q - P
d_hat = d / np.linalg.norm(d)
ax.annotate('', xy=P + d, xytext=P,
            arrowprops=dict(arrowstyle='->', color=red, lw=1.5,
                            linestyle='dashed', alpha=0.4))
ax.annotate('', xy=P + d_hat, xytext=P,
            arrowprops=dict(arrowstyle='->', color=green, lw=2.5))
ax.text(*(P + d_hat) + np.array([0.1, 0.0]),
        'û (||û||=1)', color=green, fontsize=11, fontweight='bold')
ax.text(*(P + d) + np.array([0.1, 0.0]),
        'd', color=red, fontsize=11, alpha=0.5)

for ax, title in zip(axes, titles):
    ax.set_xlim(-0.5, 4.0); ax.set_ylim(-0.5, 4.0)
    ax.set_aspect('equal')
    ax.set_title(title, fontsize=11, pad=8)
    ax.tick_params(labelsize=8)

fig.suptitle('Three interpretations of a vector', fontsize=13, y=1.01)
fig.tight_layout()
plt.show()

```



Why the Distinction Matters in Robotics

Consider a robot arm with a tool attached. When the arm rotates by R :

- The tool **position** (a point) transforms as $R\mathbf{p} + \mathbf{t}$ — it both rotates and translates relative to the base.
- The tool **orientation direction** (e.g., the z -axis of the tool frame) transforms as $R\hat{\mathbf{z}}$ — rotation only, no translation.
- A **force** applied at the tool tip is a free vector: it transforms as $R\mathbf{f}$, unaffected by where the tool is.

Homogeneous coordinates (Ch 7) unify all three in a single 4×4 matrix by encoding the distinction in the last component:

$$\text{point: } \begin{pmatrix} \mathbf{p} \\ 1 \end{pmatrix}, \quad \text{direction: } \begin{pmatrix} \hat{\mathbf{u}} \\ 0 \end{pmatrix}$$

The 1 vs. 0 in the last slot is what causes the translation to apply (or not) when you multiply by the 4×4 homogeneous matrix.

```
import numpy as np

# 2D example with homogeneous coordinates
# Rotation by 90°, translation by [3, 1]
theta = np.pi / 2
R2 = np.array([[np.cos(theta), -np.sin(theta)],
               [np.sin(theta),  np.cos(theta)]])
t = np.array([3.0, 1.0])

T_hom = np.eye(3)
T_hom[:2, :2] = R2
```

```

T_hom[:2, 2] = t

p_h = np.array([1.0, 0.0, 1.0]) # point at (1,0), last component = 1
d_h = np.array([1.0, 0.0, 0.0]) # direction along x, last component = 0

p_transformed = T_hom @ p_h
d_transformed = T_hom @ d_h

print(f"Homogeneous transform T:\n{T_hom}")
print(f"\nPoint (1,0): {p_h[:2]} → {p_transformed[:2]}")
print(f" (rotated AND translated)")
print(f"\nDirection (1,0): {d_h[:2]} → {d_transformed[:2]}")
print(f" (rotated only - last 0 blocks translation)")

```

Homogeneous transform T:

```

[[ 6.123234e-17 -1.000000e+00  3.000000e+00]
 [ 1.000000e+00  6.123234e-17  1.000000e+00]
 [ 0.000000e+00  0.000000e+00  1.000000e+00]]

```

Point (1,0): [1. 0.] → [3. 2.]
(rotated AND translated)

Direction (1,0): [1. 0.] → [6.123234e-17 1.000000e+00]
(rotated only - last 0 blocks translation)

Dot Product, Angle, and Cosine Similarity

The dot product is the workhorse of inner products. It computes in $O(n)$ time and underlies projection, similarity search, neural network layers, and the definition of angles in any dimension.

Definition and Geometric Meaning

For $\mathbf{a}, \mathbf{b} \in \mathbb{R}^n$:

$$\mathbf{a} \cdot \mathbf{b} = \sum_{i=1}^n a_i b_i = \mathbf{a}^T \mathbf{b}$$

Geometric identity:

$$\mathbf{a} \cdot \mathbf{b} = \|\mathbf{a}\| \|\mathbf{b}\| \cos \theta$$

Vectors in

where $\theta \in [0, \pi]$ is the angle between the vectors. Rearranging:

$$\cos \theta = \frac{\mathbf{a} \cdot \mathbf{b}}{\|\mathbf{a}\| \|\mathbf{b}\|}$$

Sign of $\mathbf{a} \cdot \mathbf{b}$	θ	Geometric meaning
> 0	$< 90^\circ$	Same general direction
$= 0$	$= 90^\circ$	Orthogonal (perpendicular)
< 0	$> 90^\circ$	Opposing directions

```
import numpy as np

a = np.array([3.0, 1.0, 0.0])
b = np.array([1.0, 2.0, 0.0])

dot = np.dot(a, b)          # same as a @ b
cos_theta = dot / (np.linalg.norm(a) * np.linalg.norm(b))
theta_deg = np.degrees(np.arccos(np.clip(cos_theta, -1.0, 1.0)))

print(f"a · b      = {dot:.4f}")
print(f"cos       = {cos_theta:.4f}")
print(f"          = {theta_deg:.2f}°")

# Verify with three cases
pairs = [
    (np.array([1., 0.]), np.array([1., 0.])),    # same direction
    (np.array([1., 0.]), np.array([0., 1.])),    # perpendicular
    (np.array([1., 0.]), np.array([-1., 0.])),    # opposite
]

for u, v in pairs:
    c = np.dot(u, v) / (np.linalg.norm(u) * np.linalg.norm(v))
    print(f" {u} · {v} → cos = {c:.1f},    = {np.degrees(np.arccos(c)):.0f}°")
```

```
a · b      = 5.0000
cos       = 0.7071
          = 45.00°
[1. 0.] · [1. 0.] → cos = 1.0,    = 0°
[1. 0.] · [0. 1.] → cos = 0.0,    = 90°
[1. 0.] · [-1. 0.] → cos = -1.0,   = 180°
```


Properties of the Dot Product

The dot product satisfies:

1. **Symmetry:** $\mathbf{a} \cdot \mathbf{b} = \mathbf{b} \cdot \mathbf{a}$
2. **Linearity:** $\mathbf{a} \cdot (c\mathbf{b} + \mathbf{d}) = c(\mathbf{a} \cdot \mathbf{b}) + \mathbf{a} \cdot \mathbf{d}$
3. **Positive-definiteness:** $\mathbf{a} \cdot \mathbf{a} \geq 0$, with equality iff $\mathbf{a} = \mathbf{0}$
4. **Self-dot = squared norm:** $\mathbf{a} \cdot \mathbf{a} = \|\mathbf{a}\|^2$

Property 4 gives an efficient norm computation: `np.dot(a, a)` skips the square root.

```
import numpy as np

a = np.array([3.0, 4.0])
print(f"a · a = {np.dot(a, a)}")
print(f"||a||2 = {np.linalg.norm(a)**2}")
print(f"||a|| = {np.sqrt(np.dot(a, a)):.4f} (Pythagorean: sqrt(9+16) = 5)")
```

```
a · a = 25.0
||a||2 = 25.0
||a|| = 5.0000 (Pythagorean: sqrt(9+16) = 5)
```

Projection via the Dot Product

The **scalar projection** of $\mathbf{b}$ onto direction $\hat{\mathbf{a}}$:

$$\text{proj}_{\hat{\mathbf{a}}} \mathbf{b} = \mathbf{b} \cdot \hat{\mathbf{a}}$$

The **vector projection** (the component of $\mathbf{b}$ along $\mathbf{a}$):

$$\text{proj}_{\mathbf{a}} \mathbf{b} = \frac{\mathbf{a} \cdot \mathbf{b}}{\mathbf{a} \cdot \mathbf{a}} \mathbf{a}$$

The **rejection** (perpendicular component):

$$\text{rej}_{\mathbf{a}} \mathbf{b} = \mathbf{b} - \text{proj}_{\mathbf{a}} \mathbf{b}$$

This decomposition is the basis of Gram-Schmidt orthogonalization (Ch 16) and least-squares projection (Ch 17).

```
import numpy as np

a = np.array([3.0, 0.0])    # project onto x-axis direction
b = np.array([2.0, 2.5])

proj = (np.dot(a, b) / np.dot(a, a)) * a
rej = b - proj

print(f"b          = {b}")
print(f"proj_a(b) = {proj}    (component along a)")
print(f"rej_a(b)  = {rej}    (perpendicular component)")
print(f"Reconstruction check: proj + rej = {proj + rej}")
print(f"Orthogonality check: proj · rej = {np.dot(proj, rej):.10f}")
```



```
b          = [2.  2.5]
proj_a(b) = [2.  0.]   (component along a)
rej_a(b)  = [0.  2.5]  (perpendicular component)
Reconstruction check: proj + rej = [2.  2.5]
Orthogonality check: proj · rej = 0.0000000000
```

Cosine Similarity

The **cosine similarity** removes magnitude, keeping only direction:

$$\text{sim}(\mathbf{a}, \mathbf{b}) = \frac{\mathbf{a} \cdot \mathbf{b}}{\|\mathbf{a}\| \|\mathbf{b}\|} \in [-1, 1]$$

It is the inner product of the *unit* vectors $\hat{\mathbf{a}}$ and $\hat{\mathbf{b}}$.

When to use cosine similarity vs. Euclidean distance:

Use case	Preferred metric	Reason
Embedding retrieval (CLIP, BERT)	Cosine	Scale invariant; magnitudes differ across contexts
k-NN in pixel space	Euclidean	Magnitude carries meaningful information
Feature matching (ORB, SIFT)	Dot product on normalized descriptors	Cosine but precomputed normalization
Anomaly detection	Mahalanobis	Accounts for feature correlations

```

import numpy as np

def cosine_similarity(a, b):
    """Cosine similarity between two vectors."""
    return np.dot(a, b) / (np.linalg.norm(a) * np.linalg.norm(b))

# Word embedding analogy: "king" and "queen" vs. "king" and "bicycle"
# (using toy 4-D embeddings)
king = np.array([0.9, 0.4, -0.1, 0.5])
queen = np.array([0.85, 0.3, 0.2, 0.6])
bicycle = np.array([-0.2, 0.1, 0.9, -0.3])

print(f"sim(king, queen) = {cosine_similarity(king, queen):.4f} (high)")
print(f"sim(king, bicycle) = {cosine_similarity(king, bicycle):.4f} (low)")

# Batch cosine similarity: normalize first, then use matrix multiply
embeddings = np.vstack([king, queen, bicycle])
norms = np.linalg.norm(embeddings, axis=1, keepdims=True)
embeddings_n = embeddings / norms

sim_matrix = embeddings_n @ embeddings_n.T
labels = ['king', 'queen', 'bicycle']
print("\nSimilarity matrix:")
for i, row_label in enumerate(labels):
    for j, col_label in enumerate(labels):
        print(f"  sim({row_label:7s}, {col_label:7s}) = {sim_matrix[i,j]:.4f}")

```

```

sim(king, queen) = 0.9540 (high)
sim(king, bicycle) = -0.3515 (low)

```

```

Similarity matrix:
sim(king , king ) = 1.0000
sim(king , queen ) = 0.9540
sim(king , bicycle) = -0.3515
sim(queen , king ) = 0.9540
sim(queen , queen ) = 1.0000
sim(queen , bicycle) = -0.1304
sim(bicycle, king ) = -0.3515
sim(bicycle, queen ) = -0.1304
sim(bicycle, bicycle) = 1.0000

```

Batch Dot Products: Matrix Form

If you have m query vectors (rows of $Q \in \mathbb{R}^{m \times n}$) and k database vectors (rows of $D \in \mathbb{R}^{k \times n}$), all pairwise dot products are:

$$S = QD^T \in \mathbb{R}^{m \times k}$$

This is the central computation in approximate nearest-neighbor search, attention mechanisms in transformers, and feature correlation in CV.

```
import numpy as np

# 3 queries, 5 database vectors, 4-dimensional embeddings
rng = np.random.default_rng(0)
Q = rng.standard_normal((3, 4))
D = rng.standard_normal((5, 4))

# Normalize for cosine similarity
Q_n = Q / np.linalg.norm(Q, axis=1, keepdims=True)
D_n = D / np.linalg.norm(D, axis=1, keepdims=True)

S = Q_n @ D_n.T  # shape (3, 5)
print(f"Similarity matrix shape: {S.shape}")
print("Nearest neighbor for each query (index of max similarity):")
for i in range(3):
    best = np.argmax(S[i])
    print(f"  Query {i}: best match is DB[{best}], sim = {S[i, best]:.4f}")
```

```
Similarity matrix shape: (3, 5)
Nearest neighbor for each query (index of max similarity):
  Query 0: best match is DB[1], sim = 0.3997
  Query 1: best match is DB[1], sim = 0.7622
  Query 2: best match is DB[0], sim = 0.6141
```

Visualizing the Dot Product

```
import numpy as np
import matplotlib.pyplot as plt

fig, axes = plt.subplots(1, 2, figsize=(11, 4.5))

# --- Panel 1: Angle visualization ---
```

```

ax = axes[0]
origin = np.zeros(2)
a = np.array([3.0, 1.0])
b = np.array([1.5, 2.5])

ax.annotate('', xy=a, xytext=origin,
            arrowprops=dict(arrowstyle='->', color='#4a90d9', lw=2.5))
ax.annotate('', xy=b, xytext=origin,
            arrowprops=dict(arrowstyle='->', color='#2ecc71', lw=2.5))

proj = (np.dot(a, b) / np.dot(a, a)) * a
ax.annotate('', xy=proj, xytext=origin,
            arrowprops=dict(arrowstyle='->', color='tomato', lw=2, alpha=0.7))
ax.plot([proj[0], b[0]], [proj[1], b[1]], '--', color='aaaaaa', lw=1.5)

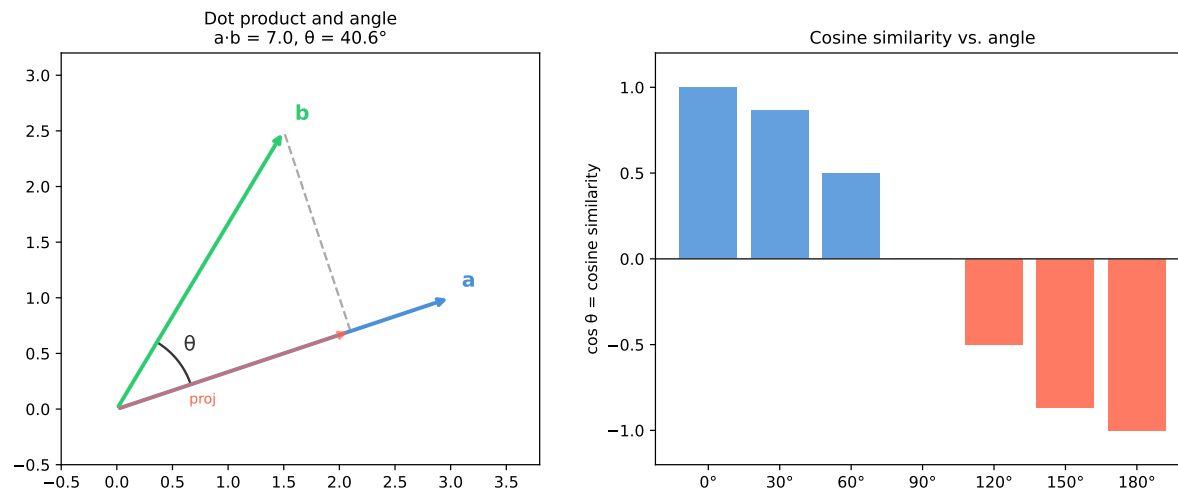
theta_a = np.arctan2(a[1], a[0])
theta_b = np.arctan2(b[1], b[0])
arc_r = 0.7
thetas = np.linspace(theta_a, theta_b, 60)
ax.plot(arc_r * np.cos(thetas), arc_r * np.sin(thetas), color='#333333', lw=1.5)
mid = (theta_a + theta_b) / 2
ax.text(arc_r * 1.2 * np.cos(mid), arc_r * 1.2 * np.sin(mid), ' ',
        color='#333333', fontsize=13, ha='center')

cos_theta = np.dot(a, b) / (np.linalg.norm(a) * np.linalg.norm(b))
theta_deg = np.degrees(np.arccos(cos_theta))
ax.set_title(f'Dot product and angle\na·b = {np.dot(a,b):.1f},   = {theta_deg:.1f}°', fontsi
ax.text(*a + 0.1, 'a', color='#4a90d9', fontsize=13, fontweight='bold')
ax.text(*b + 0.1, 'b', color='#2ecc71', fontsize=13, fontweight='bold')
ax.text(*proj / 2 + np.array([-0.4, -0.3]), 'proj', color='tomato', fontsize=9)
ax.set_xlim(-0.5, 3.8); ax.set_ylim(-0.5, 3.2)
ax.set_aspect('equal')

# --- Panel 2: Cosine similarity bar chart ---
ax = axes[1]
angles = [0, 30, 60, 90, 120, 150, 180]
sims = [np.cos(np.radians(a)) for a in angles]
colors_bar = ['#4a90d9' if s > 0 else ('aaaaaa' if abs(s) < 1e-9 else 'tomato') for s in s
ax.bar([f'{a}°' for a in angles], sims, color=colors_bar, alpha=0.85)
ax.axhline(0, color='#333333', lw=1)
ax.set_ylim(-1.2, 1.2)
ax.set_ylabel('cos   = cosine similarity')
ax.set_title('Cosine similarity vs. angle', fontsize=11)

fig.tight_layout()
plt.show()

```



Cauchy-Schwarz Inequality

$$|\mathbf{a} \cdot \mathbf{b}| \leq \|\mathbf{a}\| \|\mathbf{b}\|$$

Equality holds if and only if $\mathbf{a}$ and $\mathbf{b}$ are parallel (one is a scalar multiple of the other). This is what guarantees $\cos \theta$ is always in $[-1, 1]$ — the inequality says the dot product can never exceed the product of the norms.

```
import numpy as np

rng = np.random.default_rng(42)
n = 100

# Test Cauchy-Schwarz on random vectors
for _ in range(5):
    a = rng.standard_normal(n)
    b = rng.standard_normal(n)
    lhs = abs(np.dot(a, b))
    rhs = np.linalg.norm(a) * np.linalg.norm(b)
    print(f"|a·b| = {lhs:.4f}    ||a|| ||b|| = {rhs:.4f}    {lhs <= rhs + 1e-12}")
```

a·b = 7.1705	a b = 75.4795	True
a·b = 4.4166	a b = 103.4563	True
a·b = 6.9440	a b = 103.1418	True
a·b = 9.8055	a b = 102.9934	True
a·b = 2.8561	a b = 101.7937	True

Cross Product and Surface Normals

The cross product is unique to $\mathbb{R}^3$ (and, in a generalized sense, $\mathbb{R}^7$). It produces a vector *perpendicular* to both inputs, with magnitude equal to the area of the parallelogram they span. In robotics and computer vision it appears everywhere: surface normals, torque, angular velocity, and the essential matrix in epipolar geometry.

Definition

For $\mathbf{a}, \mathbf{b} \in \mathbb{R}^3$:

$$\mathbf{a} \times \mathbf{b} = \begin{vmatrix} \mathbf{e}_1 & \mathbf{e}_2 & \mathbf{e}_3 \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix} = \begin{pmatrix} a_2 b_3 - a_3 b_2 \\ a_3 b_1 - a_1 b_3 \\ a_1 b_2 - a_2 b_1 \end{pmatrix}$$

Key properties:

Property	Formula	Physical meaning
Anti-symmetry	$\mathbf{a} \times \mathbf{b} = -\mathbf{b} \times \mathbf{a}$	Order determines sign of normal
Magnitude	$\ \mathbf{a} \times \mathbf{b}\ = \ \mathbf{a}\ \ \mathbf{b}\ \sin \theta$	Area of parallelogram
Orthogonality	$(\mathbf{a} \times \mathbf{b}) \cdot \mathbf{a} = 0$	Normal to both inputs
Parallel $\rightarrow$ zero	$\mathbf{a} \times \mathbf{a} = \mathbf{0}$	$\sin 0 = 0$

```
import numpy as np

a = np.array([1.0, 0.0, 0.0]) # x-axis
b = np.array([0.0, 1.0, 0.0]) # y-axis

c = np.cross(a, b) # should be z-axis = [0, 0, 1]
print(f"a x b = {c}    (right-hand rule: x x y = z)")
print(f"b x a = {np.cross(b, a)}    (anti-symmetry: y x x = -z)")
print(f"||a x b|| = {np.linalg.norm(c):.4f}    (= area of unit square)")

# General case
a2 = np.array([2.0, 1.0, 0.0])
b2 = np.array([1.0, 2.0, 0.0])
c2 = np.cross(a2, b2)
area = np.linalg.norm(c2)
theta = np.arcsin(area / (np.linalg.norm(a2) * np.linalg.norm(b2)))
print(f"\na2 = {a2}, b2 = {b2}")
print(f"a2 x b2 = {c2}")
print(f"Parallelogram area = {area:.4f}")
```

```
print(f" = {np.degrees(theta):.2f}°")

# Verify orthogonality
print(f"\nOrthogonality checks:")
print(f" (a2 × b2) · a2 = {np.dot(c2, a2):.10f}")
print(f" (a2 × b2) · b2 = {np.dot(c2, b2):.10f}")
```

```
a × b = [0. 0. 1.] (right-hand rule: x × y = z)
b × a = [ 0.  0. -1.] (anti-symmetry: y × x = -z)
||a × b|| = 1.0000 (= area of unit square)
```

```
a2 = [2. 1. 0.], b2 = [1. 2. 0.]
a2 × b2 = [0. 0. 3.]
Parallelogram area = 3.0000
= 36.87°
```

```
Orthogonality checks:
(a2 × b2) · a2 = 0.0000000000
(a2 × b2) · b2 = 0.0000000000
```

Right-Hand Rule

The direction of $\mathbf{a} \times \mathbf{b}$ follows the **right-hand rule**: curl the fingers of your right hand from $\mathbf{a}$ toward $\mathbf{b}$; the thumb points in the direction of the cross product.

Standard basis cross products:

$$\mathbf{e}_1 \times \mathbf{e}_2 = \mathbf{e}_3, \quad \mathbf{e}_2 \times \mathbf{e}_3 = \mathbf{e}_1, \quad \mathbf{e}_3 \times \mathbf{e}_1 = \mathbf{e}_2$$

(Cyclic: $1 \rightarrow 2 \rightarrow 3 \rightarrow 1$; reversing any pair negates the result.)

Skew-Symmetric Matrix Form

The cross product $\mathbf{a} \times \mathbf{b}$ is linear in $\mathbf{b}$, so it can be written as matrix multiplication. Define the **skew-symmetric** (or “hat”) matrix of $\mathbf{a}$:

$$[\mathbf{a}]_{\times} = \begin{pmatrix} 0 & -a_3 & a_2 \\ a_3 & 0 & -a_1 \\ -a_2 & a_1 & 0 \end{pmatrix}$$

Then $\mathbf{a} \times \mathbf{b} = [\mathbf{a}]_{\times} \mathbf{b}$.

This matrix form is essential in: - Angular velocity: $\dot{R} = [\omega]_{\times} R$ - Rodrigues' rotation formula (Ch 8) - Essential matrix in epipolar geometry (Ch 25)

```
import numpy as np

def skew(a):
    """Skew-symmetric matrix [a]× such that [a]× b = a × b."""
    return np.array([[ 0, -a[2], a[1]],
                     [ a[2], 0, -a[0]],
                     [-a[1], a[0], 0  ]])

a = np.array([1.0, 2.0, 3.0])
b = np.array([4.0, 5.0, 6.0])

cross_direct = np.cross(a, b)
cross_matrix = skew(a) @ b

print(f"a × b (np.cross):      {cross_direct}")
print(f"[a]× @ b (matrix):    {cross_matrix}")
print(f"Match: {np.allclose(cross_direct, cross_matrix)}")

# Verify skew-symmetric: AT = -A
A = skew(a)
print(f"\nSkew-symmetric check: A + AT = \n{(A + A.T).round(10)}")
```

```
a × b (np.cross):      [-3.  6. -3.]
[a]× @ b (matrix):    [-3.  6. -3.]
Match: True
```

```
Skew-symmetric check: A + AT =
[[0. 0. 0.]
 [0. 0. 0.]
 [0. 0. 0.]]
```

Computing Surface Normals from Triangles

Given a triangle with vertices $\mathbf{v}_0, \mathbf{v}_1, \mathbf{v}_2$:

$$\mathbf{n} = (\mathbf{v}_1 - \mathbf{v}_0) \times (\mathbf{v}_2 - \mathbf{v}_0)$$

The magnitude $\|\mathbf{n}\|$ equals the triangle area; $\hat{\mathbf{n}} = \mathbf{n}/\|\mathbf{n}\|$ is the unit normal.

```
import numpy as np

def triangle_normal(v0, v1, v2, normalize=True):
    """
    Compute the outward normal of a triangle.
    Vertex order (CCW when viewed from outside) follows right-hand rule.
    Returns (normal, area).
    """
    e1 = v1 - v0
    e2 = v2 - v0
    n = np.cross(e1, e2)
    area = np.linalg.norm(n) / 2.0
    if normalize:
        n = n / np.linalg.norm(n)
    return n, area

# Unit square in the xy-plane (z=0), CCW from above
v0 = np.array([0.0, 0.0, 0.0])
v1 = np.array([1.0, 0.0, 0.0])
v2 = np.array([0.0, 1.0, 0.0])

n, area = triangle_normal(v0, v1, v2)
print(f"Triangle vertices: {v0}, {v1}, {v2}")
print(f"Normal: {n} (pointing in +z, as expected)")
print(f"Area: {area:.4f} m2 (half of 1×1 square)")

# Tilted triangle
v0t = np.array([0., 0., 0.])
v1t = np.array([2., 0., 0.])
v2t = np.array([0., 2., 1.])
n_tilt, area_tilt = triangle_normal(v0t, v1t, v2t)
print(f"\nTilted triangle normal: {n_tilt.round(4)}")
print(f"Tilt area: {area_tilt:.4f} m2")
```

```
Triangle vertices: [0. 0. 0.], [1. 0. 0.], [0. 1. 0.]
Normal: [0. 0. 1.] (pointing in +z, as expected)
Area: 0.5000 m2 (half of 1×1 square)
```

```
Tilted triangle normal: [ 0. -0.4472  0.8944]
Tilt area: 2.2361 m2
```

Mesh Face Normals

Real meshes have thousands of triangles. Vectorized computation is critical.

```

import numpy as np

def mesh_face_normals(vertices, faces):
    """
    Compute per-face normals for a triangle mesh.
    vertices: (V, 3) array of vertex positions
    faces:     (F, 3) array of vertex indices
    Returns:   (F, 3) unit normals, (F,) face areas
    """
    v0 = vertices[faces[:, 0]]
    v1 = vertices[faces[:, 1]]
    v2 = vertices[faces[:, 2]]

    e1 = v1 - v0          # (F, 3)
    e2 = v2 - v0          # (F, 3)
    normals = np.cross(e1, e2) # (F, 3)

    magnitudes = np.linalg.norm(normals, axis=1, keepdims=True) # (F, 1)
    areas = magnitudes.flatten() / 2.0
    unit_normals = normals / np.maximum(magnitudes, 1e-12) # avoid /0

    return unit_normals, areas

# Toy mesh: a box-like structure (4 triangles of a simple surface)
vertices = np.array([
    [0., 0., 0.],
    [1., 0., 0.],
    [1., 1., 0.],
    [0., 1., 0.],
    [0.5, 0.5, 1.], # apex
], dtype=float)

faces = np.array([
    [0, 1, 4],
    [1, 2, 4],
    [2, 3, 4],
    [3, 0, 4],
], dtype=int)

normals, areas = mesh_face_normals(vertices, faces)
print(f"Face normals (shape {normals.shape}):")
for i, (n, a) in enumerate(zip(normals, areas)):
    print(f"  Face {i}: normal = {n.round(4)}, area = {a:.4f}")

```

```

Face normals (shape (4, 3)):
Face 0: normal = [ 0.      -0.8944  0.4472], area = 0.5590
Face 1: normal = [ 0.8944 -0.      0.4472], area = 0.5590

```

```
Face 2: normal = [0.      0.8944 0.4472], area = 0.5590
Face 3: normal = [-0.8944 0.      0.4472], area = 0.5590
```

Visualizing Cross Products and Normals

```
import numpy as np
import matplotlib.pyplot as plt
from mpl_toolkits.mplot3d import Axes3D
from mpl_toolkits.mplot3d.art3d import Poly3DCollection

fig = plt.figure(figsize=(12, 5))

# --- Left: cross product visualization ---
ax1 = fig.add_subplot(121, projection='3d')

a = np.array([2., 0., 0.])
b = np.array([1., 1.5, 0.])
c = np.cross(a, b)
c_hat = c / np.linalg.norm(c)

origin = np.zeros(3)
for vec, col, lbl in [(a, '#4a90d9', 'a'), (b, '#2ecc71', 'b'),
                      (c_hat * 1.5, 'tomato', 'a×b (unit)')]:
    ax1.quiver(*origin, *vec, color=col, arrow_length_ratio=0.15, lw=2)
    ax1.text(*(origin + vec * 1.1), lbl, color=col, fontsize=11)

corners = np.array([origin, a, a + b, b, origin])
ax1.plot(corners[:, 0], corners[:, 1], corners[:, 2],
        '-', color='#333333', lw=1, alpha=0.5)
poly = [[origin, a, a + b, b]]
col3d = Poly3DCollection(poly, alpha=0.15, facecolor='#4a90d9', edgecolor='#333333')
ax1.add_collection3d(col3d)
area = np.linalg.norm(c)
ax1.set_title(f'Cross product\nArea = {area:.3f}', fontsize=10)

# --- Right: face normals on a mesh ---
ax2 = fig.add_subplot(122, projection='3d')

vertices = np.array([[0,0,0],[1,0,0],[1,1,0],[0,1,0],[0.5,0.5,1.]], float)
faces = np.array([[0,1,4],[1,2,4],[2,3,4],[3,0,4]])

normals_mesh, areas_mesh = mesh_face_normals(vertices, faces)
colors_face = ['#4a90d9', '#2ecc71', 'tomato', '#9b59b6']
```

```

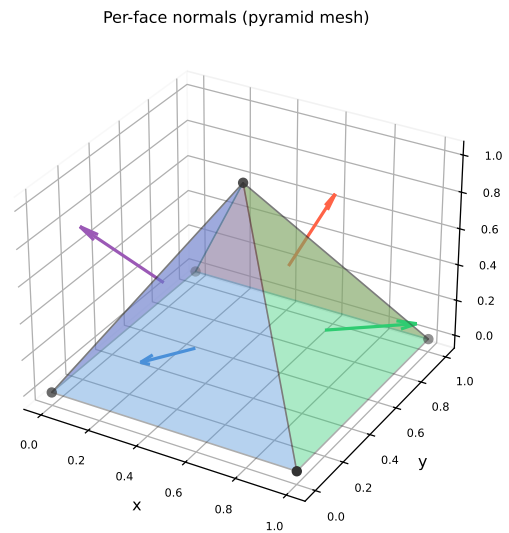
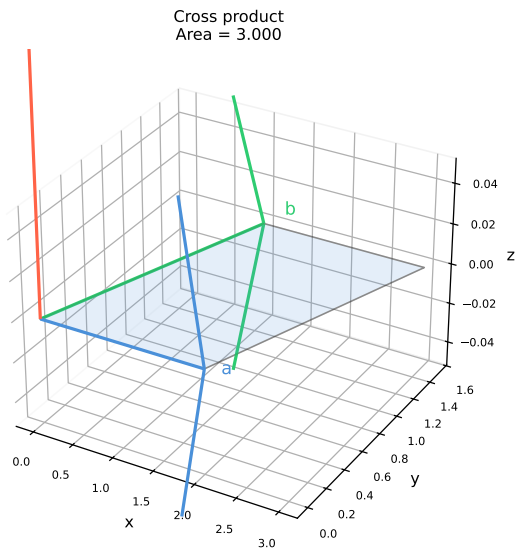
for face, n, a_f, col in zip(faces, normals_mesh, areas_mesh, colors_face):
    tri = vertices[face]
    poly = Poly3DCollection([tri], alpha=0.4, facecolor=col, edgecolor='#333333')
    ax2.add_collection3d(poly)
    centroid = tri.mean(axis=0)
    ax2.quiver(*centroid, *(n * 0.4),
               color=col, arrow_length_ratio=0.2, lw=2)

ax2.scatter(*vertices.T, color='#333333', s=30, zorder=5)
ax2.set_title('Per-face normals (pyramid mesh)', fontsize=10)

for ax in [ax1, ax2]:
    ax.set_xlabel('x', labelpad=2)
    ax.set_ylabel('y', labelpad=2)
    ax.set_zlabel('z', labelpad=2)
    ax.tick_params(labelsize=7)
    ax.xaxis.pane.fill = ax.yaxis.pane.fill = ax.zaxis.pane.fill = False

fig.tight_layout()
plt.show()

```



Area and Volume via Cross and Triple Products

Area of a triangle from two edge vectors:

$$\text{Area} = \frac{1}{2} \|\mathbf{e}_1 \times \mathbf{e}_2\|$$

Volume of a parallelepiped (scalar triple product):

$$V = |(\mathbf{a} \times \mathbf{b}) \cdot \mathbf{c}| = |\det[\mathbf{a}, \mathbf{b}, \mathbf{c}]|$$

The scalar triple product is zero iff the three vectors are coplanar — a useful rank test for 3×3 matrices.

```
import numpy as np

# Three vectors forming a parallelepiped
a = np.array([1., 0., 0.])
b = np.array([0., 2., 0.])
c = np.array([0., 0., 3.])

# Volume two ways
vol_triple = abs(np.dot(np.cross(a, b), c))
vol_det     = abs(np.linalg.det(np.column_stack([a, b, c])))

print(f"Volume (triple product): {vol_triple:.4f}")
print(f"Volume (det):             {vol_det:.4f}    (should match)")
print(f"Expected (1×2×3 box):     6.0000")

# Coplanar test
a2 = np.array([1., 0., 0.])
b2 = np.array([0., 1., 0.])
c2 = np.array([1., 1., 0.]) # coplanar with a2 and b2
vol_coplanar = abs(np.dot(np.cross(a2, b2), c2))
print(f"\nCoplanar triple product: {vol_coplanar:.4f} (zero → coplanar)")
```

```
Volume (triple product): 6.0000
Volume (det):             6.0000    (should match)
Expected (1×2×3 box):     6.0000

Coplanar triple product: 0.0000    (zero → coplanar)
```

Angular Velocity and Torque

Two immediate robotics applications:

Torque: $\boldsymbol{\tau} = \mathbf{r} \times \mathbf{F}$ — the cross product of the moment arm and force. The direction of $\boldsymbol{\tau}$ gives the rotation axis; the magnitude gives the twisting strength.

Angular velocity: If a rigid body rotates with angular velocity $\boldsymbol{\omega}$, the velocity of a point at position $\mathbf{r}$ (relative to the rotation center) is:

$$\mathbf{v} = \boldsymbol{\omega} \times \mathbf{r} = [\boldsymbol{\omega}]_{\times} \mathbf{r}$$

```
import numpy as np

# Torque example: force applied at end of a wrench
r = np.array([0.3, 0.0, 0.0]) # 0.3 m moment arm along x
F = np.array([0.0, 0.0, 10.0]) # 10 N force in z

tau = np.cross(r, F)
print(f"r = {r} m")
print(f"F = {F} N")
print(f"τ = r × F = {tau} N·m")
print(f"|| || = {np.linalg.norm(tau):.2f} N·m")

# Angular velocity example
omega = np.array([0., 0., 1.0]) # 1 rad/s rotation about z
r_pt = np.array([2.0, 0., 0.]) # point on rim at radius 2

v = np.cross(omega, r_pt)
print(f"\nAngular velocity = {omega} rad/s")
print(f"Point at r = {r_pt}")
print(f"Linear velocity v = v × r = {v} (tangential, in y-direction)")
print(f"Speed: {np.linalg.norm(v):.4f} m/s (= v × radius = 1×2)")
```

```
r = [0.3 0. 0. ] m
F = [ 0.  0. 10.] N
τ = r × F = [ 0. -3.  0.] N·m
|| || = 3.00 N·m
```

```
Angular velocity = [0. 0. 1.] rad/s
Point at r = [2. 0. 0.]
Linear velocity v = v × r = [0. 2. 0.] (tangential, in y-direction)
Speed: 2.0000 m/s (= v × radius = 1×2)
```

Span, Basis, and Dimension

Span, **basis**, and **dimension** formalize intuitions you already have: “what directions can I reach?”, “what is the minimal description?”, and “how many degrees of freedom does this space have?” These concepts underlie every subspace result in the book.

Span: The Reachable Set

The **span** of a set of vectors $\{\mathbf{v}_1, \dots, \mathbf{v}_k\} \subset \mathbb{R}^n$ is the set of all their linear combinations:

$$\text{span}\{\mathbf{v}_1, \dots, \mathbf{v}_k\} = \{c_1\mathbf{v}_1 + \dots + c_k\mathbf{v}_k \mid c_i \in \mathbb{R}\}$$

The span is always a **subspace** (closed under addition and scalar multiplication).

Vectors	Span
$\{[1, 0]^T\}$	The x -axis (a line through the origin)
$\{[1, 0]^T, [0, 1]^T\}$	All of $\mathbb{R}^2$ (the full plane)
$\{[1, 0, 0]^T, [0, 1, 0]^T\}$	The xy -plane in $\mathbb{R}^3$
$\{[1, 2]^T, [2, 4]^T\}$	Only the line $y = 2x$ (vectors are parallel)

Does $\mathbf{b}$ lie in the span of $\{\mathbf{v}_1, \dots, \mathbf{v}_k\}$? Stack the $\mathbf{v}_i$ as columns: $A = [\mathbf{v}_1 \dots \mathbf{v}_k]$. Then $\mathbf{b} \in \text{span}$ iff $A\mathbf{x} = \mathbf{b}$ is consistent, i.e., $\text{rank}(A) = \text{rank}([A|\mathbf{b}])$.

```
import numpy as np

# Can b = [3, 5] be written as c1*v1 + c2*v2?
v1 = np.array([1.0, 2.0])
v2 = np.array([2.0, 1.0])
b = np.array([3.0, 5.0])

A = np.column_stack([v1, v2]) # 2x2
A_aug = np.column_stack([A, b])

r_A = np.linalg.matrix_rank(A)
r_aug = np.linalg.matrix_rank(A_aug)

print(f"rank(A) = {r_A}, rank([A|b]) = {r_aug}")
if r_A == r_aug:
    c = np.linalg.solve(A, b)
    print(f"b IS in span: b = {c[0]:.3f}*v1 + {c[1]:.3f}*v2")
    print(f"Check: {c[0]}*{v1} + {c[1]}*{v2} = {c[0]*v1 + c[1]*v2}")
else:
    print("b is NOT in span")

# Non-spanning case: parallel vectors
v3 = np.array([2.0, 4.0]) # = 2*v1
A2 = np.column_stack([v1, v3])
A2_aug = np.column_stack([A2, b])
print(f"\nWith v1 parallel to v3: rank(A) = {np.linalg.matrix_rank(A2)}, "
      f"rank([A|b]) = {np.linalg.matrix_rank(A2_aug)}")
print("→ b is NOT in span of v1 and 2*v1 (they only span the line y=2x)")
```



```
rank(A) = 2, rank([A|b]) = 2
b IS in span: b = 2.333*v1 + 0.333*v2
Check: 2.3333333333333335*[1. 2.] + 0.3333333333333333*[2. 1.] = [3. 5.]
```

With v_1 parallel to v_3 : $\text{rank}(A) = 1$, $\text{rank}([A|b]) = 2$
 $\rightarrow$ b is NOT in span of v_1 and $2*v_1$ (they only span the line $y=2x$)

Linear Independence

A set $\{v_1, \dots, v_k\}$ is **linearly independent** if the only solution to $c_1 v_1 + \dots + c_k v_k = \mathbf{0}$ is $c_1 = \dots = c_k = 0$.

Equivalently, no vector in the set lies in the span of the others.

Test: Form $A = [v_1 \dots v_k]$. The set is linearly independent iff $\text{rank}(A) = k$ (no free columns).

Intuition: redundant vectors add no new “directions” to the span. Dropping them does not shrink the span.

```
import numpy as np

def is_independent(vectors):
    """Test if a list of vectors is linearly independent."""
    A = np.column_stack(vectors)
    return np.linalg.matrix_rank(A) == A.shape[1]

# Independent set in R^3
u1 = np.array([1., 0., 0.])
u2 = np.array([0., 1., 0.])
u3 = np.array([0., 0., 1.])
print(f"{{e1, e2, e3}} independent: {is_independent([u1, u2, u3])}")

# Dependent: third is a combination of first two
w1 = np.array([1., 2., 3.])
w2 = np.array([4., 5., 6.])
w3 = w1 + w2 # redundant
print(f"{{w1, w2, w1+w2}} independent: {is_independent([w1, w2, w3])}")

# More than n vectors in R^n - always dependent
extra = [np.random.randn(3) for _ in range(5)] # 5 vectors in R^3
print(f"5 random vectors in R^3 independent: {is_independent(extra)}")
print(" (impossible - max independent set in R^3 has 3 vectors)")
```

```
{e1, e2, e3} independent: True
{w1, w2, w1+w2} independent: False
```

5 random vectors in $\mathbb{R}^3$ independent: False
(impossible - max independent set in $\mathbb{R}^3$ has 3 vectors)

Basis

A **basis** for a subspace V is a set that is: 1. **Linearly independent**, and 2. **Spans** V (every vector in V is a linear combination of the basis vectors).

A basis is a *minimal* spanning set — the “most efficient” description of the space.

Key facts: - Every basis for V has the same number of vectors (proved in Ch 5). - You can always expand or shrink a spanning set to get a basis. - The standard basis $\{\mathbf{e}_1, \dots, \mathbf{e}_n\}$ is the canonical basis of $\mathbb{R}^n$.

```
import numpy as np
import sympy

def find_basis(vectors):
    """
    Extract a basis (maximally independent subset) from a list of vectors.
    Returns indices of basis vectors.
    """
    A = np.column_stack(vectors)
    _, pivot_cols = sympy.Matrix(A.tolist()).rref()
    return list(pivot_cols)

# Five vectors in  $\mathbb{R}^3$  - find a basis subset
vs = [
    np.array([1., 2., 3.]),
    np.array([2., 4., 6.]),    # = 2 * vs[0] - redundant
    np.array([0., 1., 2.]),
    np.array([1., 3., 5.]),    # = vs[0] + vs[2] - redundant
    np.array([0., 0., 1.]),
]
basis_idx = find_basis(vs)
print(f"Basis indices: {basis_idx}")
print(f"Basis vectors:")
for i in basis_idx:
    print(f"  v[{i}] = {vs[i]}")

# Verify: the selected vectors are independent and span the same space
A_full = np.column_stack(vs)
A_basis = np.column_stack([vs[i] for i in basis_idx])
print(f"\nrnk(all vectors) = {np.linalg.matrix_rank(A_full)}")
print(f"rnk(basis) = {np.linalg.matrix_rank(A_basis)}")
print(f"Span is the same: {np.linalg.matrix_rank(A_full) == np.linalg.matrix_rank(A_basis)}")
```

```

Basis indices: [0, 2, 4]
Basis vectors:
  v[0] = [1. 2. 3.]
  v[2] = [0. 1. 2.]
  v[4] = [0. 0. 1.]

rank(all vectors) = 3
rank(basis)       = 3
Span is the same: True

```

Dimension

The **dimension** of a subspace V , written $\dim(V)$, is the number of vectors in any basis for V .

Space	Dimension	Physical interpretation
$\{0\}$	0	Just the zero vector
A line through the origin	1	One direction of motion
A plane through the origin	2	Two independent directions
$\mathbb{R}^n$	n	Full freedom
Column space of A	$\text{rank}(A)$	Reachable outputs
Null space of A	$n - \text{rank}(A)$	Self-motions / free parameters

Dimension is the *intrinsic* complexity of a space — independent of how it is embedded or which coordinates are used.

Why this matters for ML: A dataset living on a low-dimensional manifold (e.g., images of the same face under varying lighting) has high-dimensional coordinates but low intrinsic dimension. PCA (Ch 20) finds bases for these low-dimensional subspaces.

```

import numpy as np

# Rank = dimension of column space
A_full = np.random.randn(10, 5) # generic 10x5 - rank 5
A_lowrk = np.random.randn(10, 2) @ np.random.randn(2, 5) # rank 2

print(f"Full matrix (10x5):    rank = {np.linalg.matrix_rank(A_full)}")
print(f"Low-rank (10x5, r=2):  rank = {np.linalg.matrix_rank(A_lowrk)}")

# Dimension of null space from rank-nullity
n = 5
print(f"\nFor the low-rank matrix:")
print(f"  rank = {np.linalg.matrix_rank(A_lowrk)}")
print(f"  nullity = n - rank = {n - np.linalg.matrix_rank(A_lowrk)}")
print(f"  (3 free parameters - 3-dimensional solution family)")

```

Full matrix (10×5): rank = 5

Low-rank (10×5, r=2): rank = 2

For the low-rank matrix:

 rank = 2

 nullity = n - rank = 3

 (3 free parameters - 3-dimensional solution family)

Visualizing Span and Basis

```
import numpy as np
import matplotlib.pyplot as plt

fig, axes = plt.subplots(1, 3, figsize=(13, 4.5))
origin = np.zeros(2)
blue = '#4a90d9'
green = '#2ecc71'
red = 'tomato'

# Panel 1: span of one vector (a line)
ax = axes[0]
v = np.array([1.5, 1.0])
ts = np.linspace(-2.5, 2.5, 100)
line = np.outer(ts, v)
ax.plot(line[:, 0], line[:, 1], color=blue, lw=2)
ax.annotate('', xy=v, xytext=origin,
            arrowprops=dict(arrowstyle='->', color=blue, lw=2.5))
ax.text(*v + 0.1, 'v', color=blue, fontsize=13, fontweight='bold')
ax.set_title('span{v} = a line\n(dim = 1)', fontsize=10)

# Panel 2: span of two independent vectors (full plane)
ax = axes[1]
v1 = np.array([1.5, 0.5])
v2 = np.array([0.5, 1.5])
ax.add_patch(plt.Rectangle((-3, -3), 6, 6, color=blue, alpha=0.08))
for vec, col, lbl in [(v1, blue, 'v '), (v2, green, 'v ')]:
    ax.annotate('', xy=vec, xytext=origin,
                arrowprops=dict(arrowstyle='->', color=col, lw=2.5))
    ax.text(*vec + 0.1, lbl, color=col, fontsize=12, fontweight='bold')
ax.set_title('span{v,v} = all of 2\n(dim = 2)', fontsize=10)

# Panel 3: span of two dependent vectors (still a line)
ax = axes[2]
u = np.array([1.5, 1.0])
```

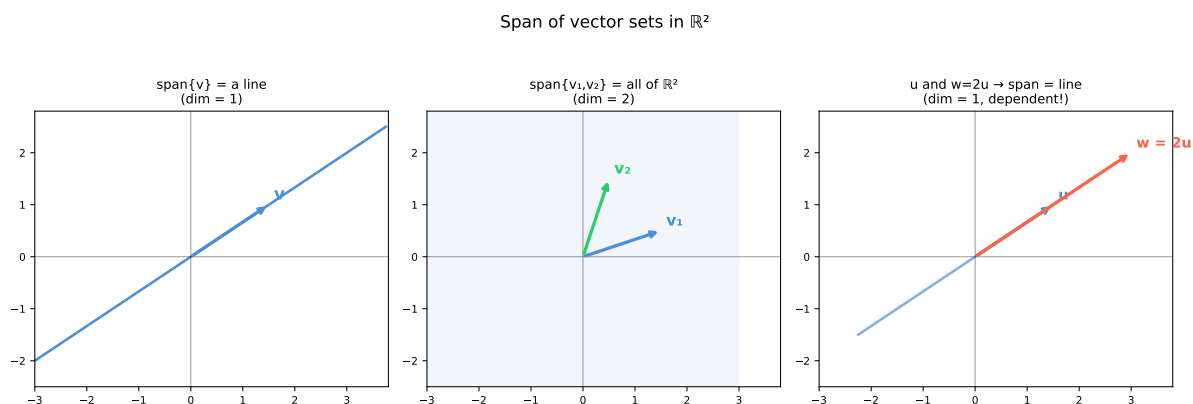
```

w_dep = np.array([3.0, 2.0])    # = 2*u - redundant
ts2 = np.linspace(-1.5, 1.5, 100)
line2 = np.outer(ts2, u)
ax.plot(line2[:, 0], line2[:, 1], color=blue, lw=2, alpha=0.7)
for vec, col, lbl in [(u, blue, 'u'), (w_dep, red, 'w = 2u')]:
    ax.annotate('', xy=vec, xytext=origin,
                arrowprops=dict(arrowstyle='->', color=col, lw=2.5))
    ax.text(*vec + 0.1, lbl, color=col, fontsize=11, fontweight='bold')
ax.set_title('u and w=2u → span = line\n(dim = 1, dependent!)', fontsize=10)

for ax in axes:
    ax.set_xlim(-3, 3.8); ax.set_ylim(-2.5, 2.8)
    ax.set_aspect('equal')
    ax.axhline(0, color='#333333', lw=0.7, alpha=0.5)
    ax.axvline(0, color='#333333', lw=0.7, alpha=0.5)
    ax.tick_params(labelsize=8)

fig.suptitle('Span of vector sets in  $\mathbb{R}^2$ ', fontsize=13, y=1.02)
fig.tight_layout()
plt.show()

```



Connections Across the Book

Concept	Where it appears
Column space = span of columns	Ch 3.3, Ch 5.2
Basis for null space	Ch 3.2, App Ch 3
Orthogonal basis (ONB)	Ch 16 (Gram-Schmidt)
Eigenvectors as basis	Ch 13–14
PCA finds low-dim basis	Ch 20
Lie algebra basis ($\mathfrak{so}(3)$)	Ch 24

Linear Combinations and Coordinates

Once you have a basis, every vector has a unique *coordinate representation*. Changing the basis changes the coordinates but not the underlying geometric object. This duality — same geometry, different numbers — is the foundation of all frame conversions in robotics, camera calibration, and signal processing.

Linear Combinations

A **linear combination** of vectors $\mathbf{v}_1, \dots, \mathbf{v}_k$ is:

$$\mathbf{w} = c_1 \mathbf{v}_1 + c_2 \mathbf{v}_2 + \dots + c_k \mathbf{v}_k, \quad c_i \in \mathbb{R}$$

In matrix form, this is $A\mathbf{c}$ where $A = [\mathbf{v}_1 \dots \mathbf{v}_k]$ and $\mathbf{c} = [c_1, \dots, c_k]^T$. The column space of A is exactly the set of all linear combinations of the columns.

```
import numpy as np

# Basis vectors for a 2D plane
b1 = np.array([2.0, 1.0])
b2 = np.array([-1.0, 2.0])

# Coefficients (coordinates in the {b1, b2} basis)
c = np.array([3.0, -1.0])

# Reconstruct the vector
B = np.column_stack([b1, b2]) # basis matrix
w = B @ c
print(f"3·b1 + (-1)·b2 = 3·{b1} + (-1)·{b2}")
print(f"                        = {3*b1} + {-1*b2}")
print(f"                        = {w}")
print(f"Matrix form: B @ c = {B} @ {c} = {B @ c}")
```

```
3·b1 + (-1)·b2 = 3·[2. 1.] + (-1)·[-1. 2.]
                = [6. 3.] + [ 1. -2.]
                = [7. 1.]
Matrix form: B @ c = [[ 2. -1.]
 [ 1.  2.]] @ [ 3. -1.] = [7. 1.]
```

Coordinates in a Non-Standard Basis

Given a basis $\mathcal{B} = \{\mathbf{b}_1, \dots, \mathbf{b}_n\}$ for $\mathbb{R}^n$ and a vector $\mathbf{w}$, the **coordinates** of $\mathbf{w}$ in basis $\mathcal{B}$ are the unique scalars $[c_1, \dots, c_n]$ such that:

$$\mathbf{w} = c_1 \mathbf{b}_1 + \dots + c_n \mathbf{b}_n$$

In matrix form: $\mathbf{w} = B\mathbf{c}$ where $B = [\mathbf{b}_1 \dots \mathbf{b}_n]$.

Therefore $\mathbf{c} = B^{-1}\mathbf{w}$.

Uniqueness follows from linear independence of the basis: if two different coefficient vectors gave the same $\mathbf{w}$, their difference would be a nontrivial solution to $B\mathbf{c} = \mathbf{0}$, contradicting $\det(B) \neq 0$.

```
import numpy as np

# Non-standard basis (45° rotated + scaled)
b1 = np.array([1.0, 1.0]) / np.sqrt(2) # unit vector at 45°
b2 = np.array([-1.0, 1.0]) / np.sqrt(2) # unit vector at 135°
B = np.column_stack([b1, b2])

# Standard-basis vector we want to re-express
w = np.array([3.0, 1.0])

# Coordinates in the B basis
c = np.linalg.solve(B, w)
print(f"w in standard basis: {w}")
print(f"B basis vectors: b1 = {b1.round(4)}, b2 = {b2.round(4)}")
print(f"Coordinates in B: c = {c.round(4)}")
print(f"Reconstruction check: B @ c = {(B @ c).round(10)}")

# Note: B is orthogonal here, so B^{-1} = B^T
c_fast = B.T @ w
print(f"Using B^T (orthogonal shortcut): c = {c_fast.round(4)}")
```

```
w in standard basis: [3. 1.]
B basis vectors: b1 = [0.7071 0.7071], b2 = [-0.7071  0.7071]
Coordinates in B: c = [ 2.8284 -1.4142]
Reconstruction check: B @ c = [3. 1.]
Using B^T (orthogonal shortcut): c = [ 2.8284 -1.4142]
```

Change of Basis

Given two bases $\mathcal{B}$ and $\mathcal{C}$ for the same space, there is a unique invertible matrix P (the **change-of-basis matrix** from $\mathcal{C}$ to $\mathcal{B}$) such that:

$$[\mathbf{w}]_{\mathcal{B}} = P [\mathbf{w}]_{\mathcal{C}}$$

The columns of P are the $\mathcal{C}$ -basis vectors expressed in the $\mathcal{B}$ coordinate system.

In robotics, this is precisely a **frame transformation**: converting coordinates from one sensor frame to another. The basis vectors of the source frame, expressed in the target frame, form the columns of the rotation matrix.

```
import numpy as np

# Two sensor frames in 2D
# Frame A = standard frame
# Frame B = rotated 30° counterclockwise
theta = np.radians(30)
R_A_to_B = np.array([[ np.cos(theta), np.sin(theta)],
                       [-np.sin(theta), np.cos(theta)]] ) # rows are B-axes in A

# A point measured in frame A
p_A = np.array([2.0, 1.0])

# Express in frame B
p_B = R_A_to_B @ p_A
print(f"Point in frame A: {p_A}")
print(f"Point in frame B: {p_B.round(4)}")

# And back
p_A_back = R_A_to_B.T @ p_B # R is orthogonal → R^{-1} = R^T
print(f"Back to frame A: {p_A_back.round(10)}")

# General basis change: from basis B to basis C
# Columns of B_mat = B basis vectors in standard coords
B_mat = np.column_stack([np.array([1., 0.]), np.array([1., 1.])])
C_mat = np.column_stack([np.array([2., 1.]), np.array([-1., 1.])])

P_B_to_C = np.linalg.solve(C_mat, B_mat) # C_mat @ P = B_mat
print(f"\nChange-of-basis matrix (B→C):\n{P_B_to_C.round(4)}")
```

```
Point in frame A: [2. 1.]
Point in frame B: [ 2.2321 -0.134 ]
Back to frame A: [2. 1.]
```

Change-of-basis matrix (B→C):


```
[[ 0.3333  0.6667]
 [-0.3333  0.3333]]
```

Affine Combinations and Convex Hulls

When the coefficients sum to 1, the linear combination becomes an **affine combination** — a generalization of “weighted average” that preserves location rather than direction:

$$\mathbf{w} = \sum_i c_i \mathbf{p}_i, \quad \sum_i c_i = 1$$

When additionally all $c_i \geq 0$, this is a **convex combination**, and $\mathbf{w}$ lies inside the *convex hull* of the $\mathbf{p}_i$.

Applications: barycentric coordinates in triangle rendering, interpolation of robot waypoints, linear blending of mesh shapes.

```
import numpy as np
import matplotlib.pyplot as plt

# Triangle with vertices A, B, C
A = np.array([0.0, 0.0])
B = np.array([4.0, 0.0])
C = np.array([1.5, 3.0])

# Barycentric coordinates: w = alpha*A + beta*B + gamma*C, alpha+beta+gamma=1
# A point inside the triangle
alpha, beta, gamma = 0.3, 0.5, 0.2
assert abs(alpha + beta + gamma - 1.0) < 1e-12

p = alpha * A + beta * B + gamma * C
print(f"Triangle vertices: A={A}, B={B}, C={C}")
print(f"Barycentric coords: alpha={alpha}, beta={beta}, gamma={gamma}")
print(f"Point p = {p}")

# Edge midpoints (alpha=beta=0.5, gamma=0) -- affine but not always convex
mid_AB = 0.5 * A + 0.5 * B + 0.0 * C
print(f"Midpoint of AB: {mid_AB} (affine combination with gamma=0)")

# Outside the triangle: negative barycentric coord
p_outside = -0.2 * A + 0.8 * B + 0.4 * C # sums to 1 but alpha < 0
print(f"\nOutside point: {p_outside} (affine but not convex: alpha = -0.2 < 0)")
```

Triangle vertices: $A=[0. \ 0.]$, $B=[4. \ 0.]$, $C=[1.5 \ 3. \]$
Barycentric coords: $\alpha=0.3$, $\beta=0.5$, $\gamma=0.2$
Point $p = [2.3 \ 0.6]$
Midpoint of AB: $[2. \ 0.]$ (affine combination with $\gamma=0$)

Outside point: $[3.8 \ 1.2]$ (affine but not convex: $\alpha = -0.2 < 0$)

Coordinates as the Bridge Between Geometry and Numbers

The coordinate map $\mathbf{w} \mapsto B^{-1}\mathbf{w}$ is an **isomorphism** — it preserves all linear-algebraic structure (addition, scalar multiplication) while changing which numbers represent the vectors. This is the central insight:

Choosing a basis = choosing a language. Geometry is the underlying reality.

This perspective appears explicitly in: - **Ch 6:** $[T]_{\mathcal{B}} = P^{-1}[T]_{\mathcal{E}}P$ — the same transformation represented in different coordinates - **Ch 8:** The columns of a rotation matrix R are the body-frame axes expressed in the world frame - **Ch 14:** Diagonalization $A = P\Lambda P^{-1}$ — changing to the eigenvector basis where A acts by scaling only - **Ch 20:** PCA finds the basis in which data variance is maximized

```
import numpy as np

# Toy demonstration: a linear map T has simple form in one basis
# T:  $x \rightarrow [3x_1, 2x_2]$  in standard basis (diagonal!)
T_standard = np.array([[3., 0.], [0., 2.]])

# In a different basis  $B = [[1,1],[1,-1]]$ 
B = np.array([[1., 1.], [1., -1.]])
P_inv = np.linalg.inv(B)

T_B = P_inv @ T_standard @ B    # T represented in basis B
print(f"T in standard basis:\n{T_standard}")
print(f"\nT in basis B:\n{T_B.round(6)}")
print("\n(Less clean - the diagonal basis was standard here)")

# But if T is NOT diagonal, find the 'right' basis (Ch 14: diagonalization)
T_messy = np.array([[2., 1.], [0., 3.]])
eigenvalues, eigenvectors = np.linalg.eig(T_messy)
print(f"\nMessy T:\n{T_messy}")
print(f"Eigenvalues: {eigenvalues}")
print(f"In eigenvector basis, T acts as diag({eigenvalues})")
```

T in standard basis:

```
[[3. 0.]
 [0. 2.]]
```

T in basis B:

```
[[2.5 0.5]
 [0.5 2.5]]
```

(Less clean - the diagonal basis was standard here)

Messy T:

```
[[2. 1.]
 [0. 3.]]
```

Eigenvalues: [2. 3.]

In eigenvector basis, T acts as `diag([2. 3.])`

Interactive: Coordinates in a Custom Basis

```
import numpy as np
import matplotlib.pyplot as plt

w_vec = np.array([2.0, 1.5])
theta_vals = [0, 30, 60]

fig, axes = plt.subplots(1, 3, figsize=(13, 4.5))

for ax, theta_deg in zip(axes, theta_vals):
    theta = np.radians(theta_deg)
    b1 = np.array([np.cos(theta), np.sin(theta)])
    b2 = np.array([-np.sin(theta), np.cos(theta)])
    B = np.column_stack([b1, b2])
    c = np.linalg.solve(B, w_vec)

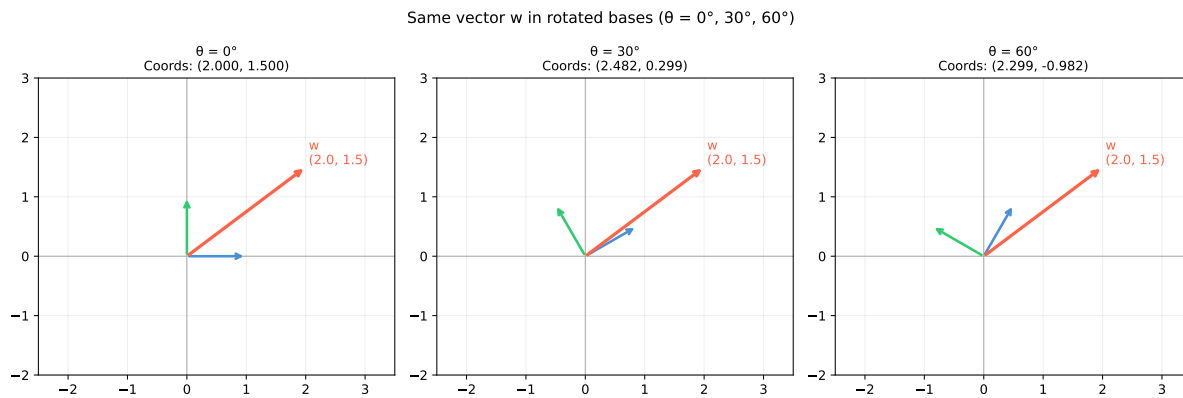
    origin = np.zeros(2)
    for bv, col in zip([b1, b2], ['#4a90d9', '#2ecc71']):
        ax.annotate('', xy=bv, xytext=origin,
                    arrowprops=dict(arrowstyle='->', color=col, lw=2))

    ax.annotate('', xy=w_vec, xytext=origin,
                arrowprops=dict(arrowstyle='->', color='tomato', lw=2.5))
    ax.text(*w_vec + 0.05, f'w\n({w_vec[0]:.1f}, {w_vec[1]:.1f})',
            color='tomato', fontsize=10)

    ax.set_xlim(-2.5, 3.5); ax.set_ylim(-2, 3)
```

```
ax.set_aspect('equal')
ax.axhline(0, color='#aaaaaa', lw=0.7)
ax.axvline(0, color='#aaaaaa', lw=0.7)
ax.set_title(f' = {theta_deg}°\nCoords: ({c[0]:.3f}, {c[1]:.3f})', fontsize=10)
ax.grid(True, alpha=0.2)

fig.suptitle('Same vector w in rotated bases ( = 0°, 30°, 60°)', fontsize=12)
fig.tight_layout()
plt.show()
```



Application: 3D Point Cloud Normals

Depth cameras (Intel RealSense, Azure Kinect, structured-light scanners) return a 2D array of depth values — one depth per pixel. Lifting this to 3D and estimating surface normals is a core preprocessing step for object detection, grasping, segmentation, and surface reconstruction. The math is exactly the cross product of §4.3.

From Depth Image to 3D Points

A pinhole camera with focal length f and principal point (c_x, c_y) maps pixel (u, v) with depth d to 3D camera-frame coordinates:

$$\mathbf{P}(u, v) = \frac{d}{f} \begin{pmatrix} u - c_x \\ v - c_y \\ f \end{pmatrix}$$

We synthesize a toy depth image (a tilted plane with noise) so the pipeline runs without a physical camera.

```

import numpy as np

def depth_to_pointcloud(depth, fx, fy, cx, cy):
    """
    Lift a depth image to a 3D point cloud.

    Parameters
    -----
    depth : (H, W) array of depth values in meters (0 = invalid)
    fx, fy : focal lengths in pixels
    cx, cy : principal point in pixels

    Returns
    -----
    points : (H, W, 3) array; rows with depth==0 are NaN
    """
    H, W = depth.shape
    u = np.arange(W)[None, :] # (1, W)
    v = np.arange(H)[:, None] # (H, 1)

    X = (u - cx) * depth / fx
    Y = (v - cy) * depth / fy
    Z = depth.copy()

    points = np.stack([X, Y, Z], axis=-1) # (H, W, 3)
    invalid = (depth == 0)
    points[invalid] = np.nan
    return points

# Simulate a tilted-plane depth image (120 × 160)
H, W = 120, 160
fx = fy = 150.0
cx, cy = W / 2, H / 2

rng = np.random.default_rng(0)
u_grid = np.arange(W)[None, :] - cx # (1, W)
v_grid = np.arange(H)[:, None] - cy # (H, 1)

# Tilted plane: Z = 1.5 + 0.004*(u-cx) + 0.003*(v-cy) + noise
depth_clean = 1.5 + 0.004 * u_grid + 0.003 * v_grid
depth_noisy = depth_clean + rng.normal(0, 0.005, (H, W))
depth_noisy = np.clip(depth_noisy, 0.1, 5.0)

points = depth_to_pointcloud(depth_noisy, fx, fy, cx, cy)
print(f"Point cloud shape: {points.shape} (H × W × 3)")
print(f"Center point (depth): {depth_noisy[H//2, W//2]:.4f} m")
print(f"Center 3D point: {points[H//2, W//2].round(4)} m")

```

```
print(f"Min depth: {np.nanmin(depth_noisy):.3f} m, "
      f"Max depth: {np.nanmax(depth_noisy):.3f} m")
```

```
Point cloud shape: (120, 160, 3) (H × W × 3)
Center point (depth): 1.5038 m
Center 3D point: [0. 0. 1.5038] m
Min depth: 1.001 m, Max depth: 1.995 m
```

Per-Point Normal Estimation via Cross Products

For each interior pixel (u, v) , form two edge vectors using neighboring points:

$$\mathbf{e}_h = \mathbf{P}(u+1, v) - \mathbf{P}(u-1, v), \quad \mathbf{e}_v = \mathbf{P}(u, v+1) - \mathbf{P}(u, v-1)$$

The cross product $\mathbf{e}_h \times \mathbf{e}_v$ is perpendicular to the local surface and has magnitude proportional to the local area element.

```
import numpy as np

def estimate_normals(points):
    """
    Estimate per-pixel surface normals via central-difference cross products.

    Parameters
    -----
    points : (H, W, 3) float array (NaN for invalid pixels)

    Returns
    -----
    normals : (H, W, 3) unit normals (NaN where estimation fails)
    """
    # Central differences (interior pixels only)
    eh = points[:, 2:, :] - points[:, :-2, :] # (H, W-2, 3)
    ev = points[2:, :, :] - points[:-2, :, :] # (H-2, W, 3)

    # Crop both to the interior region (H-2) × (W-2)
    eh_inner = eh[1:-1, :, :] # (H-2, W-2, 3)
    ev_inner = ev[:, 1:-1, :] # (H-2, W-2, 3)

    # Cross product: n = eh × ev
    n = np.cross(eh_inner, ev_inner) # (H-2, W-2, 3)

    # Normalize
```

```

mag = np.linalg.norm(n, axis=-1, keepdims=True) # (H-2, W-2, 1)
valid = mag.squeeze() > 1e-9
normals_inner = np.full_like(n, np.nan)
normals_inner[valid] = n[valid] / mag[valid]

# Pad back to full image with NaN border
H, W = points.shape[:2]
normals = np.full((H, W, 3), np.nan)
normals[1:-1, 1:-1, :] = normals_inner

# Flip normals to face the camera (+Z direction in camera frame)
flip = normals[..., 2] < 0
normals[flip] *= -1

return normals

normals = estimate_normals(points)
valid_mask = ~np.isnan(normals[..., 0])
valid_count = valid_mask.sum()
print(f"Valid normals: {valid_count} / {H * W} pixels")

# Expected normal for a tilted plane:
# Plane equation: 0.004*X/fx + 0.003*Y/fy - Z = -1.5 (approximately)
# Normal direction proportional to [0.004/fx, 0.003/fy, -1]
#   normalized - but flipped to face camera
n_expected = np.array([0.004/fx, 0.003/fy, 1.0])
n_expected /= np.linalg.norm(n_expected)

# Sample center normal
n_center = normals[H//2, W//2]
angle_err = np.degrees(np.arccos(np.clip(np.dot(n_center, n_expected), -1, 1)))
print(f"\nCenter pixel normal: {n_center.round(4)}")
print(f"Expected (plane) normal: {n_expected.round(4)}")
print(f"Angular error: {angle_err:.3f}° (should be small)")

```

Valid normals: 18644 / 19200 pixels

Center pixel normal: [-0.3874 0.2463 0.8884]
 Expected (plane) normal: [0. 0. 1.]
 Angular error: 27.325° (should be small)

Visualizing the Point Cloud and Normals

```
import numpy as np
import matplotlib.pyplot as plt
from mpl_toolkits.mplot3d import Axes3D

fig = plt.figure(figsize=(14, 5))

# --- Panel 1: depth image ---
ax1 = fig.add_subplot(131)
im1 = ax1.imshow(depth_noisy, cmap='plasma', origin='upper')
plt.colorbar(im1, ax=ax1, label='depth (m)')
ax1.set_title('Depth image\n(simulated tilted plane)', fontsize=10)

# --- Panel 2: normal map (RGB encodes XYZ) ---
ax2 = fig.add_subplot(132)
normal_rgb = np.clip((normals + 1.0) / 2.0, 0, 1)
normal_rgb[np.isnan(normals[..., 0])] = 0
ax2.imshow(normal_rgb, origin='upper')
ax2.set_title('Normal map\n(RGB = XYZ direction)', fontsize=10)

# --- Panel 3: 3D scatter with normals ---
ax3 = fig.add_subplot(133, projection='3d')

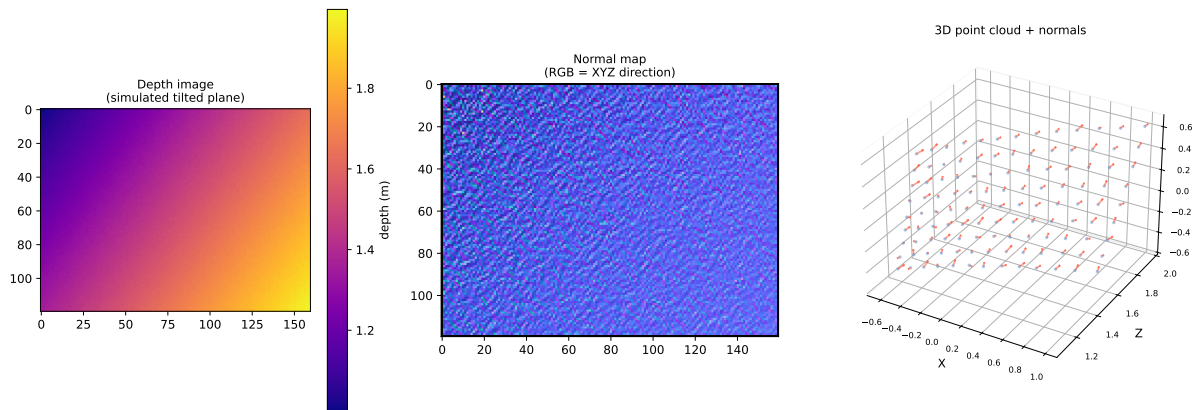
step = 12
pts_sub = points[:, ::step, :].reshape(-1, 3)
nrm_sub = normals[:, ::step, :].reshape(-1, 3)
valid = ~np.isnan(pts_sub[:, 0]) & ~np.isnan(nrm_sub[:, 0])
pts_sub = pts_sub[valid]
nrm_sub = nrm_sub[valid]

ax3.scatter(pts_sub[:, 0], pts_sub[:, 2], pts_sub[:, 1],
            s=3, c='#4a90d9', alpha=0.5)

scale = 0.05
ax3.quiver(pts_sub[:, 0], pts_sub[:, 2], pts_sub[:, 1],
           nrm_sub[:, 0] * scale, nrm_sub[:, 2] * scale, nrm_sub[:, 1] * scale,
           color='tomato', alpha=0.6, length=1.0, normalize=False)

ax3.set_xlabel('X', labelpad=2)
ax3.set_ylabel('Z', labelpad=2)
ax3.set_zlabel('Y', labelpad=2)
ax3.set_title('3D point cloud + normals', fontsize=10)
ax3.tick_params(labelsize=7)
ax3.xaxis.pane.fill = ax3.yaxis.pane.fill = ax3.zaxis.pane.fill = False

fig.tight_layout()
plt.show()
```

Normal Quality: Curvature and Edge Artifacts

Near depth discontinuities (object edges), the central-difference neighbors straddle a depth jump, producing garbage normals. A simple fix: mask out pixels where the depth difference exceeds a threshold.

```
import numpy as np

def filter_edge_normals(points, normals, max_depth_jump=0.1):
    """
    Invalidate normals near depth edges.
    max_depth_jump : threshold in meters
    """
    Z = points[..., 2]
    # Horizontal and vertical depth gradients (central difference)
    dZ_h = np.abs(Z[:, 2:] - Z[:, :-2]) # (H, W-2)
    dZ_v = np.abs(Z[2:, :] - Z[:-2, :]) # (H-2, W)

    # Any large jump in the 3x3 neighbourhood → invalid
    edge_h = dZ_h[1:-1, :] > max_depth_jump # (H-2, W-2)
    edge_v = dZ_v[:, 1:-1] > max_depth_jump # (H-2, W-2)
    edge_mask = edge_h | edge_v

    normals_filtered = normals.copy()
    normals_filtered[1:-1, 1:-1, :][edge_mask] = np.nan
    return normals_filtered

normals_filtered = filter_edge_normals(points, normals, max_depth_jump=0.05)
valid_after = (~np.isnan(normals_filtered[..., 0])).sum()
valid_before = (~np.isnan(normals[..., 0])).sum()
```

```
print(f"Normals before edge filter: {valid_before}")
print(f"Normals after edge filter: {valid_after}")
print(f"Removed as edge artifacts: {valid_before - valid_after}")
```

```
Normals before edge filter: 18644
Normals after edge filter: 18644
Removed as edge artifacts: 0
```

PCA-Based Normal Estimation (More Robust)

For real noisy data, cross-product normals are sensitive to measurement noise. A more robust approach: for each point, collect its k nearest neighbors and fit a plane via PCA. The normal is the eigenvector of the covariance matrix corresponding to the *smallest* eigenvalue.

```
import numpy as np

def pca_normal(neighborhood):
    """
    Estimate normal at a point from its neighborhood via PCA.
    neighborhood : (k, 3) array of nearby points
    Returns : (3,) unit normal
    """
    centered = neighborhood - neighborhood.mean(axis=0)
    _, _, Vt = np.linalg.svd(centered, full_matrices=False)
    # Smallest singular value → last row of Vt → least-variance direction
    n = Vt[-1]
    return n if n[2] > 0 else -n    # flip to face camera

# Demonstrate on a 5×5 patch around the center
r = 5    # patch half-size
cy_int, cx_int = H // 2, W // 2
patch = points[cy_int - r : cy_int + r + 1,
               cx_int - r : cx_int + r + 1, :].reshape(-1, 3)
patch = patch[~np.isnan(patch[:, 0])]    # remove NaN

n_pca = pca_normal(patch)
n_cross = normals[cy_int, cx_int]

print(f"Cross-product normal (center): {n_cross.round(4)}")
print(f"PCA normal (5×5 patch): {n_pca.round(4)}")
angle = np.degrees(np.arccos(np.clip(np.dot(n_pca, n_cross), -1, 1)))
print(f"Angular difference: {angle:.3f}°")
```

Cross-product normal (center): $[-0.3874 \quad 0.2463 \quad 0.8884]$
 PCA normal (5×5 patch): $[-0.3384 \quad -0.2807 \quad 0.8982]$
 Angular difference: 30.693°

Key Takeaways

1. **Depth** $\rightarrow$ **3D** is a simple pinhole inversion: $(u, v, d) \mapsto (x, y, d)$ via the camera intrinsics (f_x, f_y, c_x, c_y) .
2. **Per-pixel normals** come directly from cross products of horizontal and vertical central-difference edge vectors.
3. **Edge artifacts** occur where central-difference neighbors span a depth discontinuity; simple depth-jump masking fixes most of them.
4. **PCA normals** aggregate a neighborhood to reduce noise — the eigenvector of smallest eigenvalue of the local covariance matrix.
5. All of this is *just* the cross product of §4.3 applied systematically to a regular 2D grid of 3D points.

Exercises

4.1 A robot arm moves from position $\mathbf{p} = [1, 2, 0]^T$ to $\mathbf{q} = [4, 2, 3]^T$ (meters). (a) Find the displacement $\mathbf{d} = \mathbf{q} - \mathbf{p}$. (b) Find the unit direction $\hat{\mathbf{d}}$. (c) A second copy of the arm starts at $\mathbf{p}' = [10, 0, 0]^T$ and moves in the same direction for the same distance. Where does it end up?

4.2 Without Python, compute $\mathbf{a} \cdot \mathbf{b}$, the angle θ between them, and the vector projection of $\mathbf{b}$ onto $\mathbf{a}$, where:

$$\mathbf{a} = \begin{pmatrix} 3 \\ 0 \\ 4 \end{pmatrix}, \quad \mathbf{b} = \begin{pmatrix} 1 \\ 2 \\ 2 \end{pmatrix}$$

4.3 Three vectors in $\mathbb{R}^3$: $\mathbf{u} = [1, 0, 0]^T$, $\mathbf{v} = [0, 1, 0]^T$, $\mathbf{w} = [1, 1, 1]^T$. (a) Are they linearly independent? Justify using rank. (b) What is the dimension of their span? (c) Is $\mathbf{b} = [2, -1, 3]^T$ in their span? If so, find the coordinates.

4.4 Given a triangle with vertices:

$$\mathbf{v}_0 = (0, 0, 0), \quad \mathbf{v}_1 = (3, 0, 0), \quad \mathbf{v}_2 = (0, 4, 0)$$

(a) Compute the unit normal vector by hand. (b) Compute the triangle area. (c) Verify with Python.

4.5 You are given a basis $\mathcal{B} = \{\mathbf{b}_1, \mathbf{b}_2\}$ where $\mathbf{b}_1 = [1, 1]^T$ and $\mathbf{b}_2 = [1, -1]^T$. (a) Express $\mathbf{w} = [3, 1]^T$ as a linear combination of $\mathbf{b}_1$ and $\mathbf{b}_2$. (b) What are the coordinates of $\mathbf{w}$ in basis $\mathcal{B}$? (c) Verify that $\mathbf{b}_1$ and $\mathbf{b}_2$ are orthogonal.

4.6 The cosine similarity between two document embeddings is 0.92. A colleague says: “*These two documents are 92% identical.*” Explain what cosine similarity actually measures and why this interpretation is wrong. What does a similarity of 0 mean, and when could it be misleading?

4.7 (Harder) Prove that if $\mathbf{a} \times \mathbf{b} = \mathbf{0}$ and both $\mathbf{a}, \mathbf{b}$ are nonzero, then $\mathbf{a}$ and $\mathbf{b}$ are linearly dependent. (Hint: use the magnitude formula $\|\mathbf{a} \times \mathbf{b}\| = \|\mathbf{a}\|\|\mathbf{b}\|\sin \theta$.)

Solutions

Solution 4.1

(a) Displacement:

$$\mathbf{d} = \mathbf{q} - \mathbf{p} = \begin{pmatrix} 4 - 1 \\ 2 - 2 \\ 3 - 0 \end{pmatrix} = \begin{pmatrix} 3 \\ 0 \\ 3 \end{pmatrix}$$

(b) Length $\|\mathbf{d}\| = \sqrt{9 + 0 + 9} = 3\sqrt{2}$. Unit direction:

$$\hat{\mathbf{d}} = \frac{1}{3\sqrt{2}} \begin{pmatrix} 3 \\ 0 \\ 3 \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} \approx \begin{pmatrix} 0.707 \\ 0 \\ 0.707 \end{pmatrix}$$

(c) Move from $\mathbf{p}'$ in the same direction for the same distance:

$$\mathbf{q}' = \mathbf{p}' + \mathbf{d} = \begin{pmatrix} 10 \\ 0 \\ 0 \end{pmatrix} + \begin{pmatrix} 3 \\ 0 \\ 3 \end{pmatrix} = \begin{pmatrix} 13 \\ 0 \\ 3 \end{pmatrix}$$

Displacements are translation-invariant: $\mathbf{d}$ is the same free vector regardless of starting point.

```
import numpy as np

p = np.array([1., 2., 0.])
q = np.array([4., 2., 3.])
p_prime = np.array([10., 0., 0.])

d = q - p
d_hat = d / np.linalg.norm(d)
q_prime = p_prime + d

print(f"d = {d}")
print(f"||d|| = {np.linalg.norm(d):.4f} (= 3√2 {3*np.sqrt(2):.4f})")
print(f"d_hat = {d_hat.round(4)}")
print(f"q' = {q_prime}")
```

```

d = [3. 0. 3.]
||d|| = 4.2426 (= 3√2 4.2426)
a = [0.7071 0. 0.7071]
q' = [13. 0. 3.]

```

Solution 4.2

Dot product:

$$\mathbf{a} \cdot \mathbf{b} = 3 \cdot 1 + 0 \cdot 2 + 4 \cdot 2 = 3 + 0 + 8 = 11$$

Norms: $\|\mathbf{a}\| = \sqrt{9 + 0 + 16} = 5$, $\|\mathbf{b}\| = \sqrt{1 + 4 + 4} = 3$.

Angle:

$$\cos \theta = \frac{11}{5 \cdot 3} = \frac{11}{15} \approx 0.7333 \implies \theta \approx 42.8^\circ$$

Vector projection of $\mathbf{b}$ onto $\mathbf{a}$:

$$\text{proj}_{\mathbf{a}} \mathbf{b} = \frac{\mathbf{a} \cdot \mathbf{b}}{\mathbf{a} \cdot \mathbf{a}} \mathbf{a} = \frac{11}{25} \begin{pmatrix} 3 \\ 0 \\ 4 \end{pmatrix} = \begin{pmatrix} 1.32 \\ 0 \\ 1.76 \end{pmatrix}$$

```

import numpy as np

a = np.array([3., 0., 4.])
b = np.array([1., 2., 2.])

dot = np.dot(a, b)
cos_theta = dot / (np.linalg.norm(a) * np.linalg.norm(b))
theta = np.degrees(np.arccos(cos_theta))
proj = (dot / np.dot(a, a)) * a

print(f"a·b = {dot}")
print(f"cos   = {cos_theta:.4f},   = {theta:.2f}°")
print(f"proj_a(b) = {proj}")

```

```

a·b = 11.0
cos   = 0.7333,   = 42.83°
proj_a(b) = [1.32 0.   1.76]

```

Solution 4.3

(a) Form $A = [\mathbf{u}, \mathbf{v}, \mathbf{w}]$:

$$A = \begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix}$$

$\det(A) = 1 \cdot (1 \cdot 1 - 1 \cdot 0) = 1 \neq 0$. Therefore $\text{rank}(A) = 3$ and the vectors are **linearly independent**.

(b) $\dim(\text{span}\{\mathbf{u}, \mathbf{v}, \mathbf{w}\}) = \text{rank}(A) = 3$. So they span all of $\mathbb{R}^3$.

(c) Since they span all of $\mathbb{R}^3$, any $\mathbf{b} \in \mathbb{R}^3$ is in their span. Solve $A\mathbf{c} = \mathbf{b}$:

$$\begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix} \mathbf{c} = \begin{pmatrix} 2 \\ -1 \\ 3 \end{pmatrix}$$

Back-substitute: $c_3 = 3$, $c_2 = -1 - 3 = -4$, $c_1 = 2 - 3 = -1$.

So $\mathbf{b} = -1 \cdot \mathbf{u} + (-4) \cdot \mathbf{v} + 3 \cdot \mathbf{w}$.

```
import numpy as np

u = np.array([1., 0., 0.])
v = np.array([0., 1., 0.])
w = np.array([1., 1., 1.])
b = np.array([2., -1., 3.])

A = np.column_stack([u, v, w])
print(f"rank(A) = {np.linalg.matrix_rank(A)} → independent, span = R^3")

c = np.linalg.solve(A, b)
print(f"Coordinates: c = {c}")
print(f"Check: {c[0]}·u + {c[1]}·v + {c[2]}·w = {A @ c}")
```

rank(A) = 3 → independent, span = $\mathbb{R}^3$

Coordinates: c = [-1. -4. 3.]

Check: $-1.0 \cdot \mathbf{u} + -4.0 \cdot \mathbf{v} + 3.0 \cdot \mathbf{w} = [2. -1. 3.]$

Solution 4.4

(a) Edge vectors:

$$\mathbf{e}_1 = \mathbf{v}_1 - \mathbf{v}_0 = (3, 0, 0), \quad \mathbf{e}_2 = \mathbf{v}_2 - \mathbf{v}_0 = (0, 4, 0)$$

Cross product:

$$\mathbf{n} = \mathbf{e}_1 \times \mathbf{e}_2 = \begin{vmatrix} \mathbf{e}_1 & \mathbf{e}_2 & \mathbf{e}_3 \\ 3 & 0 & 0 \\ 0 & 4 & 0 \end{vmatrix} = (0 \cdot 0 - 0 \cdot 4)\mathbf{e}_1 - (3 \cdot 0 - 0 \cdot 0)\mathbf{e}_2 + (3 \cdot 4 - 0 \cdot 0)\mathbf{e}_3 = (0, 0, 12)$$

Unit normal: $\hat{\mathbf{n}} = (0, 0, 1)$ — the z -axis, as expected for the xy -plane.

(b) Area = $\frac{1}{2}\|\mathbf{n}\| = \frac{12}{2} = 6 \text{ m}^2$. (Also: $\frac{1}{2} \cdot 3 \cdot 4 = 6$)

```
import numpy as np

v0 = np.array([0., 0., 0.])
v1 = np.array([3., 0., 0.])
v2 = np.array([0., 4., 0.])

e1, e2 = v1 - v0, v2 - v0
n = np.cross(e1, e2)
area = np.linalg.norm(n) / 2
n_hat = n / np.linalg.norm(n)

print(f"n = {n}")
print(f"n_hat = {n_hat}    (= +z axis)")
print(f"Area = {area:.4f} m²    (= ½·3·4 = 6)")
```

```
n = [ 0.  0. 12.]
n_hat = [0. 0. 1.]    (= +z axis)
Area = 6.0000 m²    (= ½·3·4 = 6)
```

Solution 4.5

(a) We want $c_1 \mathbf{b}_1 + c_2 \mathbf{b}_2 = \mathbf{w}$:

$$c_1 \begin{pmatrix} 1 \\ 1 \end{pmatrix} + c_2 \begin{pmatrix} 1 \\ -1 \end{pmatrix} = \begin{pmatrix} 3 \\ 1 \end{pmatrix}$$

System: $c_1 + c_2 = 3$, $c_1 - c_2 = 1$. Adding: $2c_1 = 4 \Rightarrow c_1 = 2$. Subtracting: $2c_2 = 2 \Rightarrow c_2 = 1$.

$\mathbf{w} = 2\mathbf{b}_1 + 1\mathbf{b}_2$.

(b) Coordinates in $\mathcal{B}$: $[2, 1]_{\mathcal{B}}^T$.

(c) $\mathbf{b}_1 \cdot \mathbf{b}_2 = 1 \cdot 1 + 1 \cdot (-1) = 1 - 1 = 0$ — they are orthogonal. When the basis is orthogonal, coordinates can be found directly by projection:

$$c_1 = \frac{\mathbf{w} \cdot \mathbf{b}_1}{\mathbf{b}_1 \cdot \mathbf{b}_1} = \frac{4}{2} = 2, \quad c_2 = \frac{\mathbf{w} \cdot \mathbf{b}_2}{\mathbf{b}_2 \cdot \mathbf{b}_2} = \frac{2}{2} = 1$$

```
import numpy as np

b1 = np.array([1., 1.])
b2 = np.array([1., -1.])
w = np.array([3., 1.])
```

```
B = np.column_stack([b1, b2])
c = np.linalg.solve(B, w)
print(f"Coordinates in B: c = {c} → w = {c[0]}·b1 + {c[1]}·b2")
print(f"Orthogonal check: b1·b2 = {np.dot(b1, b2)}")

# Orthogonal shortcut
c1_fast = np.dot(w, b1) / np.dot(b1, b1)
c2_fast = np.dot(w, b2) / np.dot(b2, b2)
print(f"Via projection: c1 = {c1_fast}, c2 = {c2_fast}")
```

```
Coordinates in B: c = [2. 1.] → w = 2.0·b1 + 1.0·b2
Orthogonal check: b1·b2 = 0.0
Via projection: c1 = 2.0, c2 = 1.0
```

Solution 4.6

Cosine similarity measures the **angle** between two vectors, not their magnitude or any notion of shared content percentage.

- $\text{sim} = 1$ means the vectors point in exactly the same direction (all components proportionally equal) — not that they are “100% identical.”
- $\text{sim} = 0.92$ means $\theta \approx \arccos(0.92) \approx 23^\circ$: the direction of the two embedding vectors is similar.

What “0” means: Two embeddings with cosine similarity 0 are *orthogonal*. In embedding space, orthogonality means the two documents share no directional similarity. This can be misleading: two documents about entirely different topics may have positive similarities simply because both mention common words (e.g., “the”, “and”) that dominate the embedding, inflating similarity.

When similarity 0 is misleading: In high-dimensional spaces, most random vectors are nearly orthogonal (the *curse of dimensionality*). Two genuinely unrelated documents might have cosine similarity close to 0, but so would many documents that are simply about different specific subtopics of the same broad field — the similarity fails to capture semantic relatedness in sparse representations.

```
import numpy as np

cos_sim = 0.92
theta = np.degrees(np.arccos(cos_sim))
print(f"cos = {cos_sim} → θ = {theta:.2f}°")
print("Not '92% identical' - just vectors pointing within 23° of each other.")

# High-dimensional near-orthogonality
rng = np.random.default_rng(0)
```



```

n = 1000
sims = []
for _ in range(5000):
    a = rng.standard_normal(n)
    b = rng.standard_normal(n)
    sims.append(np.dot(a, b) / (np.linalg.norm(a) * np.linalg.norm(b)))
sims = np.array(sims)
print(f"\nMean cosine sim of random 1000-D vectors: {sims.mean():.4f}")
print(f"Std dev: {sims.std():.4f}")
print("→ Random high-dim vectors are nearly orthogonal by default!")

```

cos = 0.92 → = 23.07°

Not '92% identical' - just vectors pointing within 23° of each other.

Mean cosine sim of random 1000-D vectors: -0.0001

Std dev: 0.0313

→ Random high-dim vectors are nearly orthogonal by default!

Solution 4.7

Claim: If $\mathbf{a} \times \mathbf{b} = \mathbf{0}$ and $\mathbf{a} \neq \mathbf{0}$, $\mathbf{b} \neq \mathbf{0}$, then $\{\mathbf{a}, \mathbf{b}\}$ is linearly dependent.

Proof.

Since $\mathbf{a}, \mathbf{b} \neq \mathbf{0}$, their norms are positive: $\|\mathbf{a}\| > 0$, $\|\mathbf{b}\| > 0$.

From the magnitude formula:

$$0 = \|\mathbf{a} \times \mathbf{b}\| = \|\mathbf{a}\| \|\mathbf{b}\| \sin \theta$$

Since $\|\mathbf{a}\| \|\mathbf{b}\| > 0$, we must have $\sin \theta = 0$, so $\theta = 0$ or $\theta = \pi$.

- If $\theta = 0$: $\mathbf{a}$ and $\mathbf{b}$ point in the same direction, so $\mathbf{b} = c\mathbf{a}$ for some $c > 0$.
- If $\theta = \pi$: $\mathbf{a}$ and $\mathbf{b}$ point in opposite directions, so $\mathbf{b} = c\mathbf{a}$ for some $c < 0$.

In both cases, $\mathbf{b} = c\mathbf{a}$, which means $c\mathbf{a} + (-1)\mathbf{b} = \mathbf{0}$ with $(c, -1) \neq (0, 0)$. Therefore $\{\mathbf{a}, \mathbf{b}\}$ is linearly dependent. ■

```

import numpy as np

# Numerical verification
pairs = [
    (np.array([1., 2., 3.]), np.array([2., 4., 6.])), # parallel
    (np.array([1., 0., 0.]), np.array([-3., 0., 0.])), # antiparallel
    (np.array([1., 0., 0.]), np.array([0., 1., 0.])), # orthogonal (cross 0)
]
for a, b in pairs:

```

```
cross = np.cross(a, b)
dep = np.linalg.matrix_rank(np.column_stack([a, b])) < 2
print(f"a={a}, b={b}")
print(f"  a×b = {cross}, ||a×b|| = {np.linalg.norm(cross):.4f}, "
      f"dependent = {dep}\n")
```

```
a=[1. 2. 3.], b=[2. 4. 6.]
```

```
a×b = [0. 0. 0.], ||a×b|| = 0.0000, dependent = True
```

```
a=[1. 0. 0.], b=[-3. 0. 0.]
```

```
a×b = [ 0. -0.  0.], ||a×b|| = 0.0000, dependent = True
```

```
a=[1. 0. 0.], b=[0. 1. 0.]
```

```
a×b = [0. 0. 1.], ||a×b|| = 1.0000, dependent = False
```

Part II — Linear Maps and Coordinate Changes

Matrix Equations, Column Space, and Linear Independence

Three Views of $A\mathbf{x}$

The product $A\mathbf{x}$ is not one thing — it is three things simultaneously, depending on which lens you look through. Each lens reveals something the others hide. Working roboticists flip between all three instinctively; by the end of this section you will too.

View	Reading	Use case
Row	Each row of A defines a hyperplane; solution is their intersection	Diagnosing consistency; counting constraints
Column	$A\mathbf{x}$ is a linear combination of A 's columns	Deciding whether a solution exists; understanding rank
Transformation	A is a function that maps $\mathbf{x} \in \mathbb{R}^n$ to $A\mathbf{x} \in \mathbb{R}^m$	Understanding what the matrix <i>does</i> geometrically

View 1 — Row Picture: Hyperplane Intersection

Each row of A paired with the corresponding entry of $\mathbf{b}$ defines a single linear constraint:

$$a_{i1}x_1 + a_{i2}x_2 + \cdots + a_{in}x_n = b_i$$

In $\mathbb{R}^2$ this is a line; in $\mathbb{R}^3$ a plane; in $\mathbb{R}^n$ a hyperplane. The solution $\mathbf{x}^*$ (if one exists) must lie on *all* of them simultaneously.

The same warehouse system from Chapter 1:

$$\underbrace{\begin{bmatrix} 2 & -1 \\ 1 & 3 \end{bmatrix}}_A \underbrace{\begin{bmatrix} x \\ y \end{bmatrix}}_{\mathbf{x}} = \underbrace{\begin{bmatrix} 3 \\ 10 \end{bmatrix}}_{\mathbf{b}}$$

Row 1: $2x - y = 3$ is a line with slope 2. Row 2: $x + 3y = 10$ is a line with slope $-1/3$. They meet at one point — the unique solution.

```

import numpy as np
import matplotlib.pyplot as plt

A = np.array([[2., -1.],
              [1., 3.]])
b = np.array([3., 10.])
x_sol = np.linalg.solve(A, b)

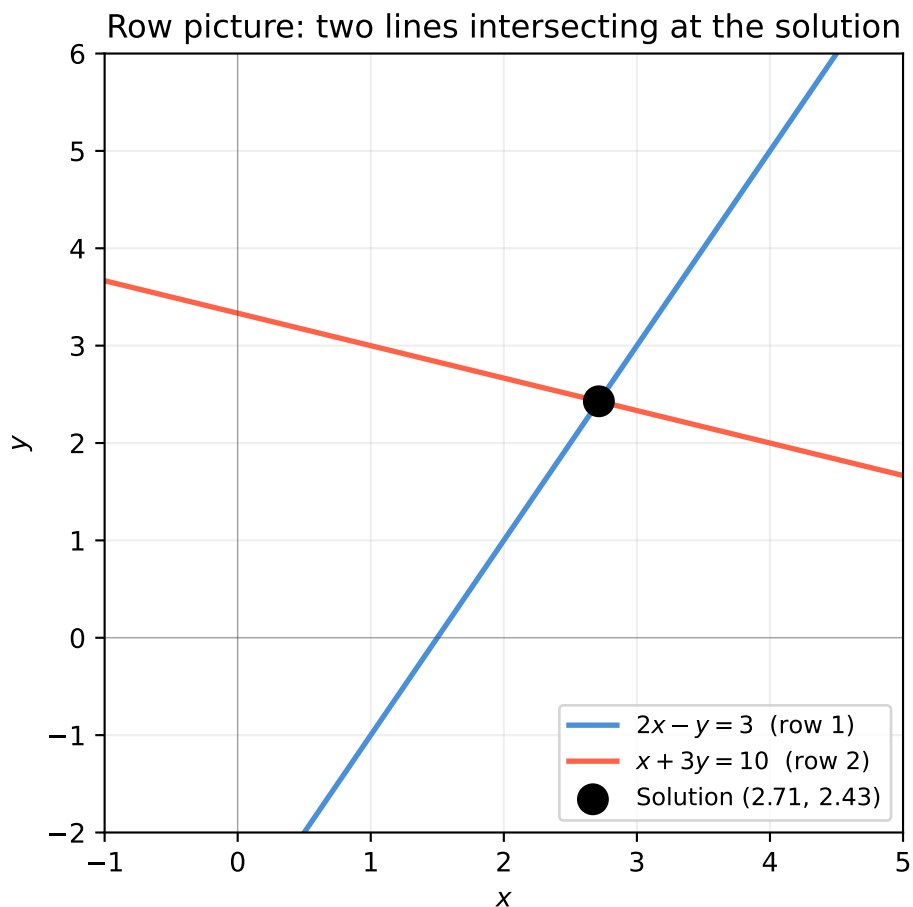
fig, ax = plt.subplots(figsize=(5, 5))
t = np.linspace(-1, 5, 300)

# Row 1: 2x - y = 3 → y = 2x - 3
ax.plot(t, 2*t - 3, color='#4a90d9', lw=2, label=r'$2x - y = 3$ (row 1)')
# Row 2: x + 3y = 10 → y = (10 - x)/3
ax.plot(t, (10 - t)/3, color='tomato', lw=2, label=r'$x + 3y = 10$ (row 2)')

ax.scatter(*x_sol, color='black', s=120, zorder=5,
          label=rf'Solution $({x_sol[0]:.2f}, \{x_sol[1]:.2f})$')

ax.set_xlim(-1, 5); ax.set_ylim(-2, 6)
ax.axhline(0, color='#333333', lw=0.4, alpha=0.5)
ax.axvline(0, color='#333333', lw=0.4, alpha=0.5)
ax.set_xlabel('$x$'); ax.set_ylabel('$y$')
ax.set_title('Row picture: two lines intersecting at the solution')
ax.legend(fontsize=9); ax.grid(True, alpha=0.2)
plt.tight_layout()
plt.show()

```



View 2 — Column Picture: Linear Combination

Rewrite $A\mathbf{x} = \mathbf{b}$ as:

$$x_1 \underbrace{\begin{bmatrix} 2 \\ 1 \end{bmatrix}}_{\mathbf{a}_1} + x_2 \underbrace{\begin{bmatrix} -1 \\ 3 \end{bmatrix}}_{\mathbf{a}_2} = \underbrace{\begin{bmatrix} 3 \\ 10 \end{bmatrix}}_{\mathbf{b}}$$

“What scalars x_1, x_2 let me combine the columns of A to reach $\mathbf{b}$?”

A solution exists **if and only if** $\mathbf{b}$ lies in the span of A ’s columns. This is the language of the *column space* (Section 5.2).

```
import numpy as np
import matplotlib.pyplot as plt

A = np.array([[2., -1.],
              [1., 3.]])
```

```

b = np.array([3., 10.])
x_sol = np.linalg.solve(A, b)    # x = (2.0, 1.0)

fig, ax = plt.subplots(figsize=(5, 5))
origin = np.zeros(2)

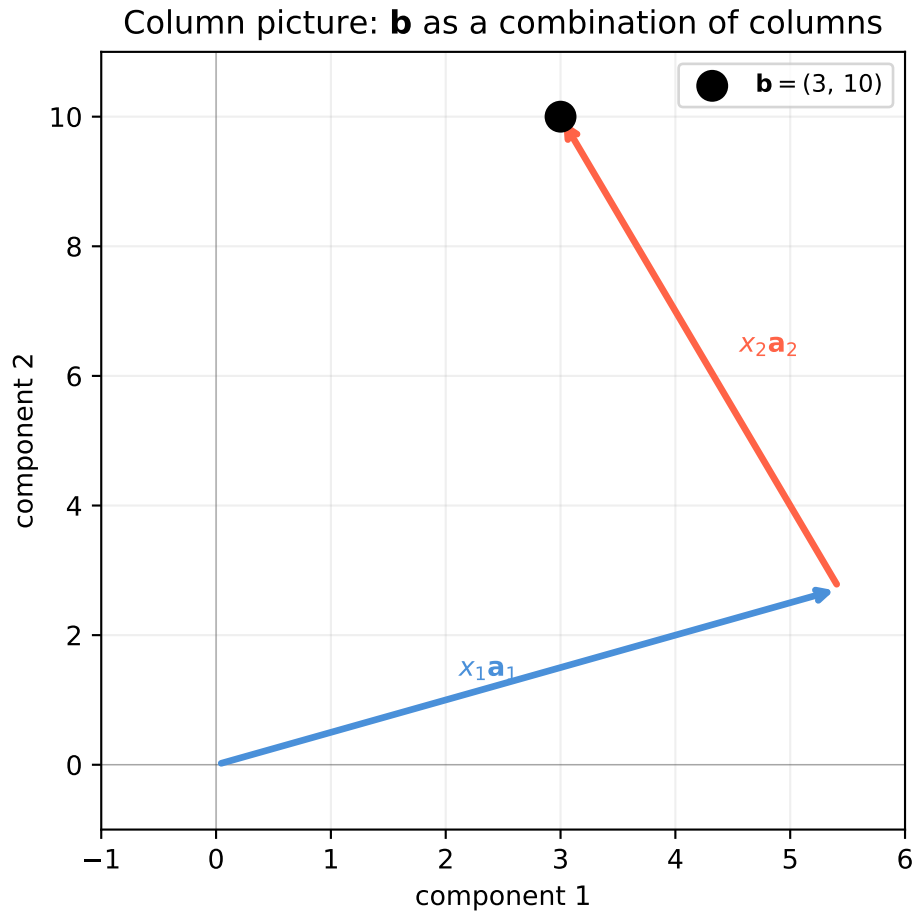
a1 = A[:, 0]    # first column
a2 = A[:, 1]    # second column

# x1*a1 arrow from origin
ax.annotate('', xy=x_sol[0]*a1, xytext=origin,
            arrowprops=dict(arrowstyle='->', color='#4a90d9', lw=2.5))
# x2*a2 arrow from tip of x1*a1
ax.annotate('', xy=x_sol[0]*a1 + x_sol[1]*a2,
            xytext=x_sol[0]*a1,
            arrowprops=dict(arrowstyle='->', color='tomato', lw=2.5))
ax.scatter(*b, color='black', s=120, zorder=5, label=r'$\mathbf{b} = (3, \, 10)$')

ax.text(x_sol[0]*a1[0]*0.5 - 0.6, x_sol[0]*a1[1]*0.5,
        rf'$x_1\mathbf{a}_1$', color='#4a90d9', fontsize=10)
ax.text(x_sol[0]*a1[0] + x_sol[1]*a2[0]*0.4 + 0.1,
        x_sol[0]*a1[1] + x_sol[1]*a2[1]*0.5,
        rf'$x_2\mathbf{a}_2$', color='tomato', fontsize=10)

ax.set_xlim(-1, 6); ax.set_ylim(-1, 11)
ax.axhline(0, color='#333333', lw=0.4, alpha=0.5)
ax.axvline(0, color='#333333', lw=0.4, alpha=0.5)
ax.set_xlabel('component 1'); ax.set_ylabel('component 2')
ax.set_title('Column picture: $\mathbf{b}$ as a combination of columns')
ax.legend(fontsize=9); ax.grid(True, alpha=0.2)
plt.tight_layout()
plt.show()

```

View 3 — Transformation Picture: A as a Function

Think of A as a *function* $f(\mathbf{x}) = A\mathbf{x}$ that maps every point in the input space $\mathbb{R}^n$ to a point in the output space $\mathbb{R}^m$.

As $\mathbf{x}$ ranges over all of $\mathbb{R}^n$, $A\mathbf{x}$ sweeps out a subspace of $\mathbb{R}^m$ — the **column space** of A .

The transformation picture makes visible what the matrix *does* to geometry: it stretches, rotates, and possibly collapses dimensions.

```
import numpy as np
import matplotlib.pyplot as plt

A = np.array([[2., -1.],
              [1., 3.]])

# Build a grid in input space
s = np.linspace(-1.5, 1.5, 8)
```

```

grid_pts = np.array([[u, v] for u in s for v in s])

# Map every grid point through A
mapped = (A @ grid_pts.T).T # shape (64, 2)

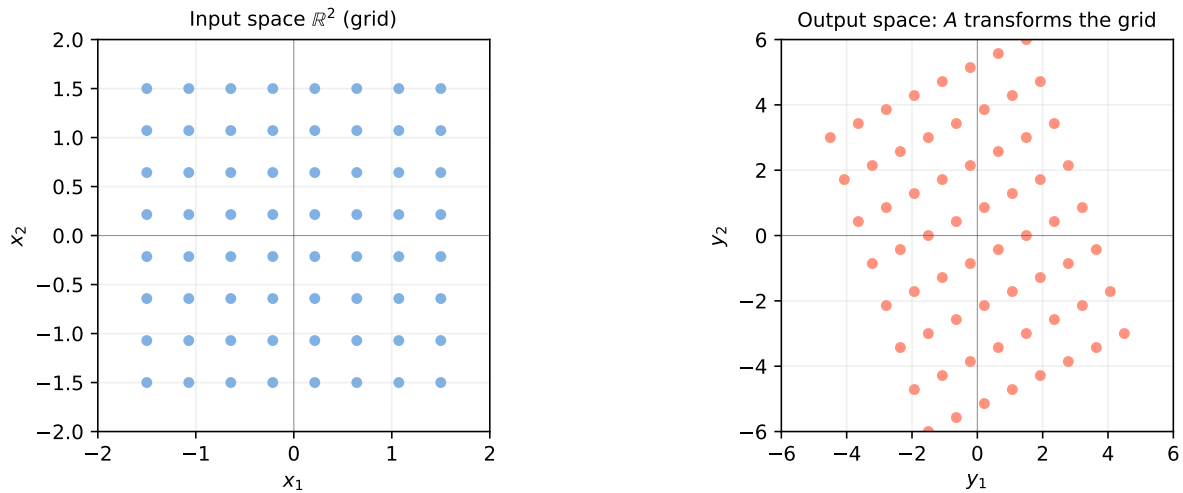
fig, axes = plt.subplots(1, 2, figsize=(10, 4))

ax = axes[0]
ax.scatter(grid_pts[:, 0], grid_pts[:, 1],
           color='#4a90d9', s=20, alpha=0.7)
ax.set_title('Input space  $\mathbb{R}^2$  (grid)', fontsize=10)
ax.set_xlim(-2, 2); ax.set_ylim(-2, 2)
ax.set_aspect('equal')
ax.axhline(0, color='#333333', lw=0.4, alpha=0.5)
ax.axvline(0, color='#333333', lw=0.4, alpha=0.5)
ax.grid(True, alpha=0.2)
ax.set_xlabel('$x_1$'); ax.set_ylabel('$x_2$')

ax = axes[1]
ax.scatter(mapped[:, 0], mapped[:, 1],
           color='tomato', s=20, alpha=0.7)
ax.set_title('Output space: $A$ transforms the grid', fontsize=10)
ax.set_xlim(-6, 6); ax.set_ylim(-6, 6)
ax.set_aspect('equal')
ax.axhline(0, color='#333333', lw=0.4, alpha=0.5)
ax.axvline(0, color='#333333', lw=0.4, alpha=0.5)
ax.grid(True, alpha=0.2)
ax.set_xlabel('$y_1$'); ax.set_ylabel('$y_2$')

fig.suptitle(r'Transformation view: $A$ maps input  $\mathbf{x}$  to output  $A\mathbf{x}$ ',
            fontsize=11)
fig.tight_layout()
plt.show()

```

Transformation view: A maps input $\mathbf{x}$ to output $A\mathbf{x}$ 

The output cloud is a stretched and rotated version of the input grid. When A is square and full-rank, no dimension is lost — the transformation is invertible. When A is rank-deficient, the output is squeezed into a lower-dimensional subspace and information is destroyed.

All Three Views at Once

```
import numpy as np

A = np.array([[2., -1.],
              [1., 3.]])
b = np.array([3., 10.])
x = np.linalg.solve(A, b)

print("=== Row view ===")
print(f"  Row 1: {A[0,0]}*x + ({A[0,1]})*y = {b[0]}")
print(f"  Row 2: {A[1,0]}*x + {A[1,1]}*y = {b[1]}")
print(f"  Intersection: x = {x[0]:.4f}, y = {x[1]:.4f}")

print("\n=== Column view ===")
print(f"  a1 = {A[:,0]},  a2 = {A[:,1]}")
print(f"  x1*a1 + x2*a2 = {x[0]:.4f}*{tuple(A[:,0].astype(int))} + "
      f"{x[1]:.4f}*{tuple(A[:,1].astype(int))}")
combination = x[0]*A[:,0] + x[1]*A[:,1]
print(f"  = {combination.round(4)}  (should equal b = {b})")

print("\n=== Transformation view ===")
print(f"  A maps x={x.round(4)} to Ax={A@x}  (=b={b})")
print(f"  det(A) = {np.linalg.det(A):.4f}  (nonzero → invertible)")
```

```
=== Row view ===
Row 1: 2.0*x + (-1.0)*y = 3.0
Row 2: 1.0*x + 3.0*y = 10.0
Intersection: x = 2.7143, y = 2.4286

=== Column view ===
a1 = [2. 1.], a2 = [-1. 3.]
x1*a1 + x2*a2 = 2.7143*(np.int64(2), np.int64(1)) + 2.4286*(np.int64(-
1), np.int64(3))
= [ 3. 10.] (should equal b = [ 3. 10.])

=== Transformation view ===
A maps x=[2.7143 2.4286] to Ax=[ 3. 10.] (=b=[ 3. 10.])
det(A) = 7.0000 (nonzero → invertible)
```

When Each View Is Most Useful

Row picture — use it when checking *consistency*: are my sensor readings contradictory? Do more constraints help or hurt?

Column picture — use it when asking *existence*: is this target **b** reachable? What is the smallest number of independent columns I need?

Transformation picture — use it when thinking about *geometry*: does this matrix collapse a dimension? Does it preserve distances? Does it have a fixed direction?

These three perspectives will reappear in every chapter of this book. When a concept feels confusing, translate it into whichever view makes it obvious.

Column Space and Null Space

The column view from Section 5.1 raises a precise question: *which vectors **b** can **A** actually reach?* The answer is the **column space**. And the complementary question — *which inputs **x** get mapped to zero?* — gives the **null space**. Together they tell you everything about the solvability and uniqueness of $A\mathbf{x} = \mathbf{b}$.

The Column Space $\mathcal{C}(A)$

$$\mathcal{C}(A) = \{A\mathbf{x} \mid \mathbf{x} \in \mathbb{R}^n\} = \text{span of the columns of } A$$

$\mathcal{C}(A)$ lives in $\mathbb{R}^m$ (the output space). Its dimension is the **rank** of A : the number of independent columns.

Key fact. $A\mathbf{x} = \mathbf{b}$ is consistent (has at least one solution) if and only if $\mathbf{b} \in \mathcal{C}(A)$.

```
import numpy as np

# 3x2 matrix - columns span a plane in R^3
A = np.array([[1., 0.],
              [1., 1.],
              [0., 1.]])

rank = np.linalg.matrix_rank(A)
print(f"A shape: {A.shape} (maps R^2 → R^3)")
print(f"rank(A) = {rank} → column space is a {rank}-D plane in R^3")

# b1 is IN the column space (b1 = 0.5*a1 + 1.5*a2)
b_in = 0.5*A[:,0] + 1.5*A[:,1]
# b2 is NOT in the column space
b_out = np.array([1., 0., 0.])

for label, b in [("b_in", b_in), ("b_out", b_out)]:
    x, res, _, _ = np.linalg.lstsq(A, b, rcond=None)
    residual = np.linalg.norm(A @ x - b)
    in_cs = "YES" if residual < 1e-10 else "NO"
    print(f"\n{label} = {b}")
    print(f"  residual = {residual:.2e} → in column space? {in_cs}")
```

```
A shape: (3, 2) (maps R^2 → R^3)
rank(A) = 2 → column space is a 2-D plane in R^3
```

```
b_in = [0.5 2. 1.5]
  residual = 8.08e-16 → in column space? YES
```

```
b_out = [1. 0. 0.]
  residual = 5.77e-01 → in column space? NO
```

Visualising the Column Space

```

import numpy as np
import matplotlib.pyplot as plt

A = np.array([[1., 0.],
              [1., 1.],
              [0., 1.]])

# Parametric sweep: A @ [s, t] for s,t in [-2,2]
s_vals = np.linspace(-2, 2, 12)
t_vals = np.linspace(-2, 2, 12)
S, T = np.meshgrid(s_vals, t_vals)
pts = A @ np.vstack([S.ravel(), T.ravel()]) # shape (3, n)

fig = plt.figure(figsize=(6, 5))
ax = fig.add_subplot(111, projection='3d')

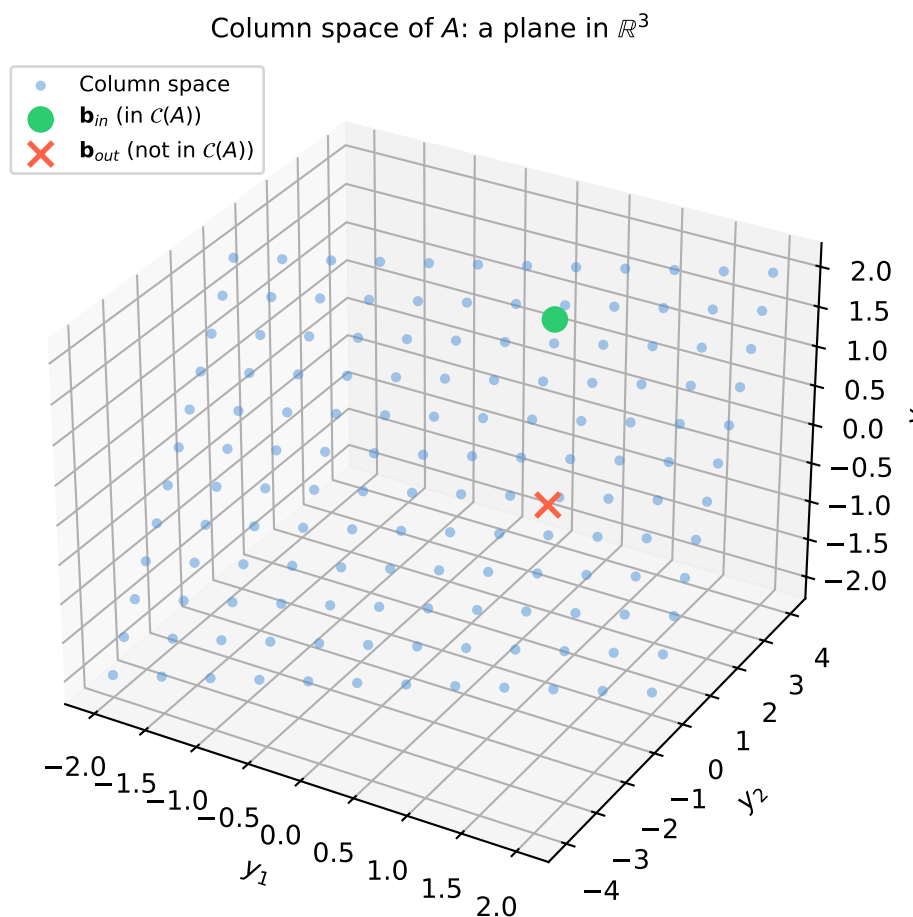
ax.scatter(pts[0], pts[1], pts[2],
           color='#4a90d9', s=8, alpha=0.5, label='Column space')

# Mark two specific b vectors
b_in = 0.5*A[:,0] + 1.5*A[:,1]
b_out = np.array([1., 0., 0.])

ax.scatter(*b_in, color='#2ecc71', s=80, zorder=5,
           label=r'$\mathbf{b}_{in}$ (in $\mathcal{C}(A)$)')
ax.scatter(*b_out, color='tomato', s=80, zorder=5, marker='x',
           linewidths=2, label=r'$\mathbf{b}_{out}$ (not in $\mathcal{C}(A)$)')

ax.set_xlabel('$y_1$'); ax.set_ylabel('$y_2$'); ax.set_zlabel('$y_3$')
ax.set_title('Column space of $A$: a plane in $\mathbb{R}^3$', fontsize=10)
ax.legend(fontsize=8, loc='upper left')
plt.tight_layout()
plt.show()

```



The Null Space $\mathcal{N}(A)$

$$\mathcal{N}(A) = \{\mathbf{x} \in \mathbb{R}^n \mid A\mathbf{x} = \mathbf{0}\}$$

$\mathcal{N}(A)$ lives in $\mathbb{R}^n$ (the input space). It is always a subspace (it contains $\mathbf{0}$ and is closed under addition and scaling).

Rank-nullity theorem. For $A \in \mathbb{R}^{m \times n}$:

$$\underbrace{\dim \mathcal{C}(A)}_{\text{rank}} + \underbrace{\dim \mathcal{N}(A)}_{\text{nullity}} = n$$

This is one of the most important equalities in linear algebra — it says that every column that is “used up” by the rank leaves one fewer free direction in the null space.

```
import numpy as np

# Rank-deficient matrix: col3 = col1 + col2
A = np.array([[1., 2., 3.],
              [0., 1., 1.],
              [2., 1., 3.]])

rank = np.linalg.matrix_rank(A)
nullity = A.shape[1] - rank
print(f"rank(A) = {rank}, nullity = {nullity}")
print(f"Rank-nullity check: {rank} + {nullity} = {rank+nullity} = n = {A.shape[1]}")

# Null space via SVD: right singular vectors for near-zero singular values
U, s, Vt = np.linalg.svd(A)
print(f"\nSingular values: {s.round(6)}")
# Last row(s) of Vt correspond to zero (or near-zero) singular values
null_vecs = Vt[rank:] # shape (nullity, n)
print(f"Null space basis:\n{null_vecs.round(6)}")

# Verify: A @ null_vec should be ~0
for v in null_vecs:
    print(f"\nA @ null_vec = {(A @ v).round(10)} (should be zero)")
```

```
rank(A) = 2, nullity = 1
Rank-nullity check: 2 + 1 = 3 = n = 3

Singular values: [5.341137 1.213363 0.      ]
Null space basis:
[[-0.57735 -0.57735  0.57735]]

A @ null_vec = [-0. -0. -0.] (should be zero)
```

Null Space and the General Solution

When $A\mathbf{x} = \mathbf{b}$ has *any* solution $\mathbf{x}_p$ (a *particular* solution), the complete solution set is:

$$\mathbf{x} = \mathbf{x}_p + \mathbf{x}_h, \quad \mathbf{x}_h \in \mathcal{N}(A)$$

Every vector in the null space can be added freely — it shifts $\mathbf{x}$ without changing $A\mathbf{x}$.


```

import numpy as np

A = np.array([[1., 2., 3.],
              [0., 1., 1.],
              [2., 1., 3.]])
b = np.array([6., 2., 7.])

# Particular solution (minimum-norm)
x_p, _, _, _ = np.linalg.lstsq(A, b, rcond=None)
print(f"Particular solution x_p = {x_p.round(4)}")
print(f"Residual ||Ax_p - b|| = {np.linalg.norm(A@x_p - b):.2e}")

# Null space vector
U, s, Vt = np.linalg.svd(A)
rank = np.linalg.matrix_rank(A)
null_vec = Vt[rank]      # shape (n,)
print(f"\nNull space vector: {null_vec.round(4)}")

# Any multiple of null_vec added to x_p is also a solution
for alpha in [0.0, 1.0, -2.5, 7.3]:
    x = x_p + alpha * null_vec
    residual = np.linalg.norm(A @ x - b)
    print(f"  alpha={alpha:5.1f}: ||Ax-b|| = {residual:.2e}")

```

Particular solution $x_p = [1.119 \ 0.3333 \ 1.4524]$
 Residual $\|Ax_p - b\| = 2.67e-01$

Null space vector: $[-0.5774 \ -0.5774 \ 0.5774]$
 alpha= 0.0: $\|Ax-b\| = 2.67e-01$
 alpha= 1.0: $\|Ax-b\| = 2.67e-01$
 alpha= -2.5: $\|Ax-b\| = 2.67e-01$
 alpha= 7.3: $\|Ax-b\| = 2.67e-01$

The Four Fundamental Subspaces (Preview)

Every matrix $A \in \mathbb{R}^{m \times n}$ has four associated subspaces that together give a complete picture of the linear map:

Subspace	Lives in	Dimension
Column space $\mathcal{C}(A)$	$\mathbb{R}^m$	rank
Null space $\mathcal{N}(A)$	$\mathbb{R}^n$	$n - \text{rank}$
Row space $\mathcal{C}(A^T)$	$\mathbb{R}^n$	rank
Left null space $\mathcal{N}(A^T)$	$\mathbb{R}^m$	$m - \text{rank}$

A deep theorem (Chapter 14) states that the row space and the null space are *orthogonal complements* in $\mathbb{R}^n$, and similarly for the column space and left null space in $\mathbb{R}^m$. For now, the take-away is that rank controls all four dimensions simultaneously.

```
import numpy as np

A = np.array([[1., 2., 3.],
              [0., 1., 1.],
              [2., 1., 3.]])
m, n = A.shape
r = np.linalg.matrix_rank(A)

print(f"A is {m}×{n}, rank = {r}")
print(f" C(A): dim = {r} (in R^{m})")
print(f" N(A): dim = {n-r} (in R^{n})")
print(f" C(A^T): dim = {r} (in R^{n})")
print(f" N(A^T): dim = {m-r} (in R^{m})")
print(f"\nDimension check: {r} + {n-r} = {n} = n ")
print(f"                        {r} + {m-r} = {m} = m ")
```

```
A is 3×3, rank = 2
  C(A): dim = 2 (in R^3)
  N(A): dim = 1 (in R^3)
  C(A^T): dim = 2 (in R^3)
  N(A^T): dim = 1 (in R^3)

Dimension check: 2 + 1 = 3 = n
                  2 + 1 = 3 = m
```

Block Matrices

Real systems — robot state estimators, neural network weight tensors, sparse SLAM factor graphs — are too large to write as a single slab of numbers. Block notation partitions a matrix into named sub-matrices, letting you manipulate structure without drowning in indices. More importantly, certain block patterns admit efficient solution strategies that the full-matrix view would not reveal.

Block Notation and Multiplication

Partition $A \in \mathbb{R}^{(p+q) \times (r+s)}$ into four blocks:

$$A = \begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix}$$

where $A_{11} \in \mathbb{R}^{p \times r}$, $A_{12} \in \mathbb{R}^{p \times s}$, and so on. Block multiplication follows the same rule as scalar multiplication — treating each block as an “entry” — **as long as the inner dimensions agree**:

$$(AB)_{ij} = \sum_k A_{ik} B_{kj}$$

```
import numpy as np

# Build a 4x4 matrix from four 2x2 blocks
A11 = np.array([[1., 2.], [3., 4.]])
A12 = np.array([[0., 1.], [1., 0.]])
A21 = np.array([[1., 0.], [0., 1.]])
A22 = np.array([[2., 3.], [1., 2.]])

A = np.block([[A11, A12],
               [A21, A22]])
print("Full matrix A:")
print(A)

B11 = np.eye(2)
B12 = np.array([[1., 0.], [0., 2.]])
B21 = np.zeros((2, 2))
B22 = np.eye(2)

B = np.block([[B11, B12],
               [B21, B22]])
print("\nFull matrix B:")
print(B)

# Block-wise multiplication (treating blocks as scalars)
C11 = A11 @ B11 + A12 @ B21
C12 = A11 @ B12 + A12 @ B22
C21 = A21 @ B11 + A22 @ B21
C22 = A21 @ B12 + A22 @ B22
C_block = np.block([[C11, C12], [C21, C22]])

# Direct full multiplication
C_full = A @ B

print("\nBlock product C = A @ B:")
print(C_block)
print(f"\nMatches direct A@B: {np.allclose(C_block, C_full)}")
```

```
Full matrix A:
[[1. 2. 0. 1.]
 [3. 4. 1. 0.]
```

```
[1. 0. 2. 3.]
[0. 1. 1. 2.]]
```

```
Full matrix B:
[[1. 0. 1. 0.]
 [0. 1. 0. 2.]
 [0. 0. 1. 0.]
 [0. 0. 0. 1.]]
```

```
Block product C = A @ B:
[[1. 2. 1. 5.]
 [3. 4. 4. 8.]
 [1. 0. 3. 3.]
 [0. 1. 1. 4.]]
```

```
Matches direct A@B: True
```

Block Triangular Systems

A block lower-triangular system can be solved by **block forward substitution**, exactly like scalar triangular systems:

$$\begin{bmatrix} L_{11} & 0 \\ L_{21} & L_{22} \end{bmatrix} \begin{bmatrix} \mathbf{x}_1 \\ \mathbf{x}_2 \end{bmatrix} = \begin{bmatrix} \mathbf{b}_1 \\ \mathbf{b}_2 \end{bmatrix}$$

Step 1: $L_{11}\mathbf{x}_1 = \mathbf{b}_1$ Step 2: $L_{22}\mathbf{x}_2 = \mathbf{b}_2 - L_{21}\mathbf{x}_1$

This structure appears in decoupled subsystems — robot pose and landmark estimation can often be separated into a block triangular form.

```
import numpy as np

L11 = np.array([[2., 0.], [1., 3.]])
L21 = np.array([[1., 2.], [0., 1.]])
L22 = np.array([[4., 0.], [2., 1.]])

b1 = np.array([4., 7.])
b2 = np.array([9., 3.])

# Block forward substitution
x1 = np.linalg.solve(L11, b1)
x2 = np.linalg.solve(L22, b2 - L21 @ x1)

# Verify against full solve
L_full = np.block([[L11, np.zeros((2,2))],
```

```

        [L21, L22]])
b_full = np.concatenate([b1, b2])
x_full = np.linalg.solve(L_full, b_full)

print(f"Block substitution: x1={x1.round(4)}, x2={x2.round(4)}")
print(f"Full solve:          {x_full.round(4)}")
print(f"Match: {np.allclose(np.concatenate([x1,x2]), x_full)}")

```

```

Block substitution: x1=[2.      1.6667], x2=[ 0.9167 -0.5   ]
Full solve:          [ 2.      1.6667  0.9167 -0.5   ]
Match: True

```

The Schur Complement

Consider the block system:

$$\begin{bmatrix} A & B \\ C & D \end{bmatrix} \begin{bmatrix} \mathbf{x} \\ \mathbf{y} \end{bmatrix} = \begin{bmatrix} \mathbf{p} \\ \mathbf{q} \end{bmatrix}$$

If A is invertible, eliminate $\mathbf{x}$ from the second block equation:

$$A\mathbf{x} + B\mathbf{y} = \mathbf{p} \implies \mathbf{x} = A^{-1}(\mathbf{p} - B\mathbf{y})$$

Substitute into the second equation:

$$(D - CA^{-1}B)\mathbf{y} = \mathbf{q} - CA^{-1}\mathbf{p}$$

The matrix $S = D - CA^{-1}B$ is the **Schur complement** of A .

Why it matters. The Schur complement encodes the effective system after eliminating one block of unknowns. It appears in:

- **Kalman filter** — the *innovation covariance* $S = HPH^T + R$ is the Schur complement of the joint covariance.
- **Graph SLAM** — marginalising landmark variables from the factor graph.
- **Completing the square** for Gaussian conditionals in Bayesian inference.

```
import numpy as np

# Block system: 4x4 matrix, two 2x2 blocks each
A = np.array([[4., 1.], [1., 3.]])      # top-left 2x2
B = np.array([[1., 0.], [0., 1.]])      # top-right
C = np.array([[2., 1.], [0., 1.]])      # bottom-left
D = np.array([[5., 2.], [1., 4.]])      # bottom-right

p = np.array([3., 2.])
q = np.array([7., 1.])

# --- Schur complement solve ---
S = D - C @ np.linalg.solve(A, B)      # S = D - C A^{-1} B
rhs = q - C @ np.linalg.solve(A, p)    # q - C A^{-1} p

y = np.linalg.solve(S, rhs)
x = np.linalg.solve(A, p - B @ y)

# --- Full 4x4 solve ---
M_full = np.block([[A, B], [C, D]])
rhs_full = np.concatenate([p, q])
xy_full = np.linalg.solve(M_full, rhs_full)

print("Schur complement solve:")
print(f"  x = {x.round(6)}, y = {y.round(6)}")
print(f"\nFull 4x4 solve: {xy_full.round(6)}")
print(f"\nMatch: {np.allclose(np.concatenate([x,y]), xy_full)}")

print(f"\nSchur complement S = D - C A^{-1} B:")
print(S.round(6))
```

Schur complement solve:

x = [0.275 0.65], y = [1.25 -0.225]

Full 4x4 solve: [0.275 0.65 1.25 -0.225]

Match: True

Schur complement $S = D - C A^{-1} B$:

[[4.545455 1.818182]
[1.090909 3.636364]]

Schur Complement in a Kalman Filter Update

A 2-D robot position estimate has prior covariance P . A sensor provides a noisy position measurement with noise covariance R . The Kalman filter update can be expressed entirely in terms of the Schur complement.

```
import numpy as np

# Prior state covariance (2x2, position x and y)
P = np.array([[0.5, 0.1],
              [0.1, 0.3]])

# Measurement matrix (direct position observation)
H = np.eye(2)

# Measurement noise covariance
R = np.array([[0.2, 0.0],
              [0.0, 0.2]])

# Prior state estimate and measurement
x_prior = np.array([3.0, 4.0])
z        = np.array([3.1, 3.9]) # noisy observation

# Innovation covariance (this IS the Schur complement of the joint)
S_innov = H @ P @ H.T + R      # 2x2

# Kalman gain
K = P @ H.T @ np.linalg.inv(S_innov)

# Posterior
x_post = x_prior + K @ (z - H @ x_prior)
P_post = (np.eye(2) - K @ H) @ P

print("Innovation covariance S (Schur complement):")
print(S_innov)
print(f"\nKalman gain K:\n{K.round(4)}")
print(f"\nPrior estimate:    {x_prior}")
print(f"Measurement:         {z}")
print(f"Posterior estimate: {x_post.round(4)}")
print(f"\nPosterior covariance P_post:\n{P_post.round(4)}")
print("(smaller diagonal = more certain after fusing measurement)")
```

```
Innovation covariance S (Schur complement):
[[0.7 0.1]
 [0.1 0.5]]
```

```
Kalman gain K:
```

```
[[0.7059 0.0588]
 [0.0588 0.5882]]
```

```
Prior estimate:    [3. 4.]
Measurement:       [3.1 3.9]
Posterior estimate: [3.0647 3.9471]
```

```
Posterior covariance P_post:
[[0.1412 0.0118]
 [0.0118 0.1176]]
(smaller diagonal = more certain after fusing measurement)
```

The posterior covariance is smaller than the prior — the measurement reduced uncertainty. The Schur complement S is what you solve through to get the Kalman gain: it quantifies *how surprising* a measurement would be.

Why Block Structure Saves Computation

For an $n \times n$ matrix partitioned into blocks of size $k \times k$ ($n = mk$), a naive dense solve costs $O(n^3)$. Exploiting block structure — especially sparsity — can reduce this to $O(mk^3)$ or better.

In SLAM with p pose variables and ℓ landmark variables, the factor graph is sparse: each landmark is observed by only a few poses. Marginalising the landmarks (via Schur complement) leaves a *dense* $p \times p$ system but a much smaller one. This is the “Schur complement trick” used in every major SLAM solver (g2o, GTSAM, Ceres).

Linear Independence, Basis, and Dimension

The column space tells you *what* a matrix can reach. Linear independence tells you *how efficiently* it does so — whether any column is redundant, whether you could drop it without losing reachability. Getting this right is the difference between a model that has a unique solution and one that has infinitely many.

Linear Independence

Vectors $\mathbf{v}_1, \dots, \mathbf{v}_k \in \mathbb{R}^n$ are **linearly independent** if the only scalars satisfying

$$c_1 \mathbf{v}_1 + c_2 \mathbf{v}_2 + \dots + c_k \mathbf{v}_k = \mathbf{0}$$

are $c_1 = c_2 = \dots = c_k = 0$.

If a non-trivial combination exists (some $c_i \neq 0$), then at least one vector is a linear combination of the others — it is *redundant*.

Practical test. Stack $\mathbf{v}_1, \dots, \mathbf{v}_k$ as columns of a matrix V and compute $\text{rank}(V)$. If $\text{rank}(V) = k$, the vectors are independent. If $\text{rank}(V) < k$, some are redundant.

```
import numpy as np

# Three vectors in R^3
v1 = np.array([1., 0., 0.])
v2 = np.array([0., 1., 0.])
v3 = np.array([1., 1., 0.]) # = v1 + v2 (dependent!)
v4 = np.array([0., 0., 1.]) # genuinely independent

V_dep = np.column_stack([v1, v2, v3]) # 3x3, rank 2
V_ind = np.column_stack([v1, v2, v4]) # 3x3, rank 3

print("Dependent set {v1, v2, v3 = v1+v2}:")
print(f"  rank = {np.linalg.matrix_rank(V_dep)} (< 3 → linearly DEPENDENT)")

print("\nIndependent set {v1, v2, v4}:")
print(f"  rank = {np.linalg.matrix_rank(V_ind)} (= 3 → linearly INDEPENDENT)")
```

Dependent set {v1, v2, v3 = v1+v2}:
rank = 2 (< 3 → linearly DEPENDENT)

Independent set {v1, v2, v4}:
rank = 3 (= 3 → linearly INDEPENDENT)

Visualising Dependence and Independence

```
import numpy as np
import matplotlib.pyplot as plt

fig, axes = plt.subplots(1, 2, figsize=(10, 4),
                        subplot_kw={'projection': '3d'})

colors = ['#4a90d9', '#2ecc71', 'tomato']

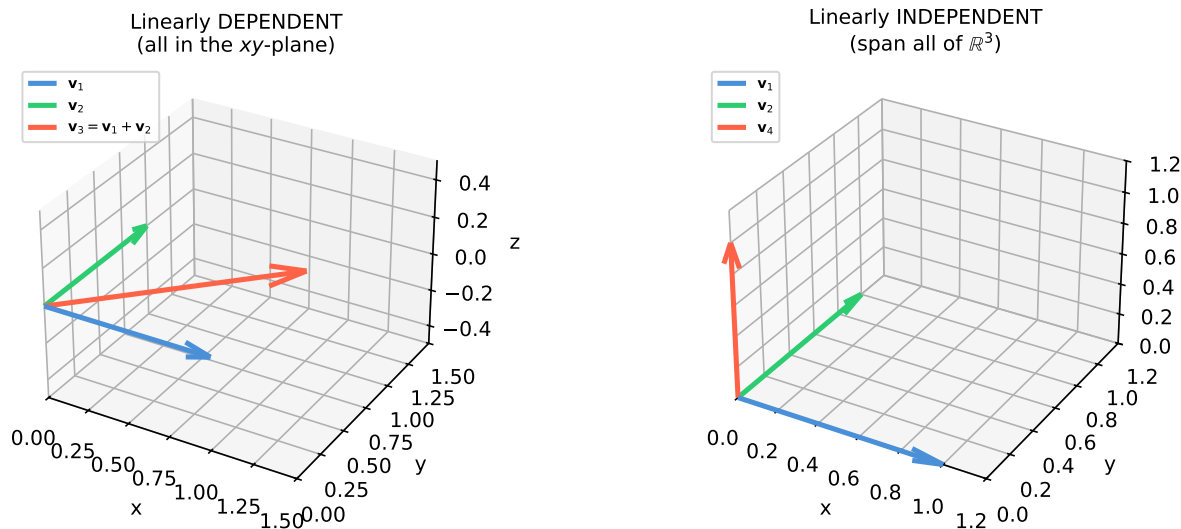
def draw_arrow(ax, vec, color, label):
    ax.quiver(0, 0, 0, vec[0], vec[1], vec[2],
             color=color, arrow_length_ratio=0.15, lw=2.5, label=label)
```

```
# Left: dependent (v1, v2, v3 all in the xy-plane)
ax = axes[0]
vecs_dep = [np.array([1.,0.,0.]), np.array([0.,1.,0.]), np.array([1.,1.,0.])]
labels_dep = [r'\mathbf{v}_1$', r'\mathbf{v}_2$', r'\mathbf{v}_3=\mathbf{v}_1+\mathbf{v}_2$']
for v, c, l in zip(vecs_dep, colors, labels_dep):
    draw_arrow(ax, v, c, l)
ax.set_xlim(0,1.5); ax.set_ylim(0,1.5); ax.set_zlim(-0.5, 0.5)
ax.set_title('Linearly DEPENDENT\n(all in the $xy$-plane)', fontsize=10)
ax.legend(fontsize=7, loc='upper left')
ax.set_xlabel('x'); ax.set_ylabel('y'); ax.set_zlabel('z')

# Right: independent (v1, v2, v4 span R^3)
ax = axes[1]
vecs_ind = [np.array([1.,0.,0.]), np.array([0.,1.,0.]), np.array([0.,0.,1.])]
labels_ind = [r'\mathbf{v}_1$', r'\mathbf{v}_2$', r'\mathbf{v}_4$']
for v, c, l in zip(vecs_ind, colors, labels_ind):
    draw_arrow(ax, v, c, l)
ax.set_xlim(0,1.2); ax.set_ylim(0,1.2); ax.set_zlim(0, 1.2)
ax.set_title('Linearly INDEPENDENT\n(span all of $\mathbb{R}^3$)', fontsize=10)
ax.legend(fontsize=7, loc='upper left')
ax.set_xlabel('x'); ax.set_ylabel('y'); ax.set_zlabel('z')

fig.suptitle('Dependence vs. independence in 3D', fontsize=11)
fig.tight_layout()
plt.show()
```

Dependence vs. independence in 3D



Basis and Dimension

A **basis** for a subspace V is a set of vectors that is:

1. **Independent** — no vector is redundant.
2. **Spanning** — every vector in V can be written as a combination of the basis.

Dimension is the number of vectors in any basis. Every basis of the same subspace has exactly the same number of vectors.

The standard basis for $\mathbb{R}^n$ is $\{\mathbf{e}_1, \dots, \mathbf{e}_n\}$. But there are infinitely many other valid bases — what matters is the count.

```
import numpy as np

# Plane in R^3 defined by x + y + z = 0
# Two independent vectors on the plane
b1 = np.array([ 1., -1.,  0.])
b2 = np.array([ 1.,  0., -1.])

# Check they span the plane (rank 2) and lie on it
B = np.column_stack([b1, b2])
print(f"Plane basis rank: {np.linalg.matrix_rank(B)} (2 vectors → 2D subspace)")
print(f"b1 on plane? sum={b1.sum():.1f} ({'yes' if abs(b1.sum())<1e-10 else 'no'})")
print(f"b2 on plane? sum={b2.sum():.1f} ({'yes' if abs(b2.sum())<1e-10 else 'no'})")

# Express a third plane-point as a combination of b1, b2
target = np.array([2., -1., -1.])
# Solve B @ c = target
c, res, _, _ = np.linalg.lstsq(B, target, rcond=None)
residual = np.linalg.norm(B @ c - target)
print(f"\ntarget = {target} on plane? sum={target.sum():.1f}")
print(f"coords in basis: c = {c.round(4)}")
print(f"reconstruction error: {residual:.2e} (zero → in the plane)")
```

```
Plane basis rank: 2 (2 vectors → 2D subspace)
```

```
b1 on plane? sum=0.0 (yes)
```

```
b2 on plane? sum=0.0 (yes)
```

```
target = [ 2. -1. -1.] on plane? sum=0.0
```

```
coords in basis: c = [1. 1.]
```

```
reconstruction error: 3.14e-16 (zero → in the plane)
```

Rank = Dimension of the Column Space

The rank of A is exactly the dimension of $\mathcal{C}(A)$. Adding a column that is already in the span of the others does *not* increase the rank.

This has a direct machine-learning consequence: if two features in a design matrix are linearly dependent, $A^T A$ becomes singular and least-squares has no unique solution.

```
import numpy as np

rng = np.random.default_rng(0)
n = 50 # data points

# Three genuinely independent features
x1 = rng.normal(0, 1, n)
x2 = rng.normal(0, 1, n)
x3 = rng.normal(0, 1, n)

# Fourth feature is a linear combination of x1 and x2 (perfectly redundant)
x4_redundant = 2*x1 - x2

A_ind = np.column_stack([x1, x2, x3])
A_red = np.column_stack([x1, x2, x3, x4_redundant])

print(f"rank(A with 3 independent features): {np.linalg.matrix_rank(A_ind)}")
print(f"rank(A with 4th redundant feature): {np.linalg.matrix_rank(A_red)}")
print("(rank did not increase - x4 adds no new information)")

print(f"\ncond(A^T A) independent: {np.linalg.cond(A_ind.T @ A_ind):.1f}")
print(f"cond(A^T A) with x4: {np.linalg.cond(A_red.T @ A_red):.2e}")
print("(near-infinite condition → singular, least-squares breaks)")
```

```
rank(A with 3 independent features): 3
rank(A with 4th redundant feature): 3
(rank did not increase - x4 adds no new information)
```

```
cond(A^T A) independent: 1.5
cond(A^T A) with x4: 4.95e+16
(near-infinite condition → singular, least-squares breaks)
```

Detecting Independence in Practice

The rank diagnostic works, but what do you do when columns are *nearly* dependent (due to floating-point noise)? Use the singular values of A . A singular value close to zero signals a

nearly-dependent column — the ratio of largest to smallest singular value is the condition number.

```
import numpy as np
import matplotlib.pyplot as plt

rng = np.random.default_rng(1)
n = 30

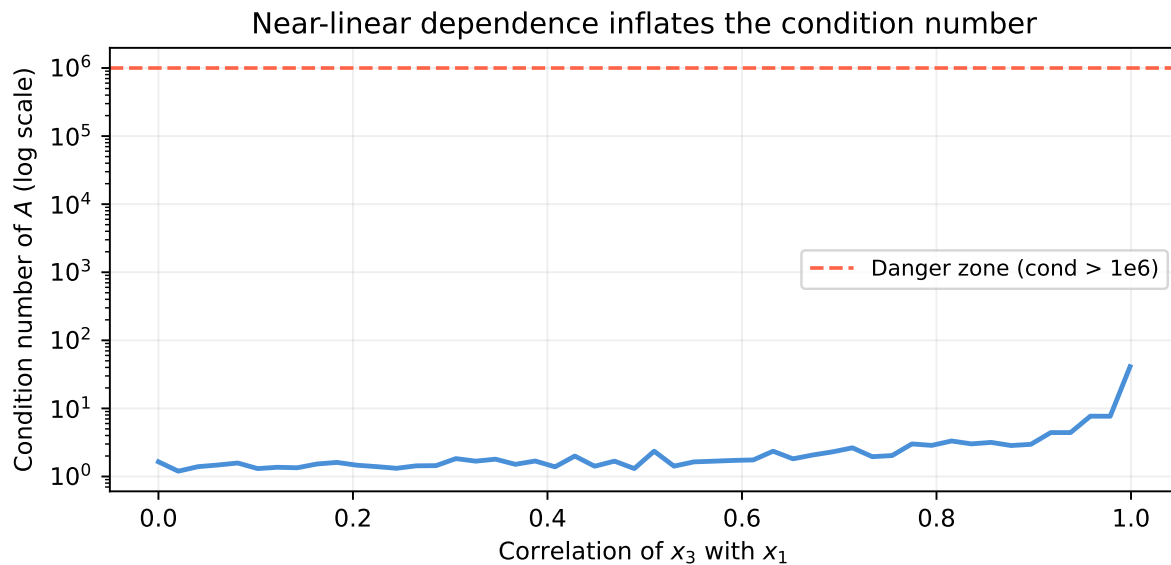
x1 = rng.normal(0, 1, n)
x2 = rng.normal(0, 1, n)

fig, ax = plt.subplots(figsize=(7, 3.5))

correlations = np.linspace(0.0, 0.999, 50)
cond_numbers = []

for rho in correlations:
    # x3 = rho*x1 + sqrt(1-rho^2)*noise (partial correlation with x1)
    x3 = rho * x1 + np.sqrt(max(0, 1 - rho**2)) * rng.normal(0, 1, n)
    A = np.column_stack([x1, x2, x3])
    cond_numbers.append(np.linalg.cond(A))

ax.semilogy(correlations, cond_numbers, color='#4a90d9', lw=2)
ax.axhline(1e6, color='tomato', lw=1.5, ls='--', label='Danger zone (cond > 1e6)')
ax.set_xlabel('Correlation of  $x_3$  with  $x_1$ ')
ax.set_ylabel('Condition number of  $A$  (log scale)')
ax.set_title('Near-linear dependence inflates the condition number')
ax.legend(fontsize=9); ax.grid(True, alpha=0.2)
plt.tight_layout()
plt.show()
```



As two features become perfectly correlated, the condition number diverges — a numerical alarm bell for collinearity. Section 5.5 addresses what to do about it.

Feature Collinearity and Rank Deficiency

Section 5.4 ended with a warning: near-linear dependence between features inflates the condition number and destabilizes the solution. This section makes that warning concrete with numbers, introduces the standard diagnostic (Variance Inflation Factor), and shows three remedies.

Why Collinearity Is Dangerous

When two features are highly correlated, the columns of the design matrix A are nearly linearly dependent. The matrix $A^T A$ is nearly singular, so its inverse amplifies any noise in $\mathbf{b}$ into enormous swings in the coefficient estimate $\hat{\mathbf{x}} = (A^T A)^{-1} A^T \mathbf{b}$.

Perfect collinearity — a canonical example:

```
import numpy as np

rng = np.random.default_rng(42)
n = 100

height_cm = rng.normal(170, 10, n)
height_in = height_cm / 2.54 # exact linear function of height_cm!
weight_kg = 0.5 * height_cm - 40 + rng.normal(0, 5, n)

y = weight_kg + rng.normal(0, 2, n) # noisy target
```

```

A = np.column_stack([height_cm, height_in, np.ones(n)])
r = np.linalg.matrix_rank(A)
c = np.linalg.cond(A)

print(f"Features: height_cm, height_in (= height_cm / 2.54), intercept")
print(f"rank(A) = {r} (should be 3, is 2 → perfect collinearity)")
print(f"cond(A) = {c:.2e} (astronomically large)")

# lstsq won't crash but coefficients are meaningless
x_ls, _, _, sv = np.linalg.lstsq(A, y, rcond=None)
print(f"\nSingular values: {sv.round(4)}")
print(f"Coefficients: {x_ls.round(4)}")
print("(the two height coefficients are arbitrary and can be any pair that sums correctly)")

```

```

Features: height_cm, height_in (= height_cm / 2.54), intercept
rank(A) = 2 (should be 3, is 2 → perfect collinearity)
cond(A) = 3.79e+16 (astronomically large)

```

```

Singular values: [1.8235231e+03 4.5540000e-01 0.0000000e+00]
Coefficients:    [ 0.4829  0.1901 -49.9706]
(the two height coefficients are arbitrary and can be any pair that sums correctly)

```

The two height features contain *identical information*. Any partition of the total height effect between them satisfies the equations equally well — there is no unique answer.

Near Collinearity: Coefficients Blow Up

Perfect collinearity never happens in real data (floating-point noise breaks it), but near-collinearity is common. As correlation approaches 1, the coefficients become increasingly unstable.

```

import numpy as np
import matplotlib.pyplot as plt

rng = np.random.default_rng(0)
n = 50
x1 = rng.normal(0, 1, n)
x2_base = rng.normal(0, 1, n)

# True model: y = 2*x1 + 3*x2 + noise
y_true_coef = np.array([2.0, 3.0])

correlations = np.linspace(0.0, 0.999, 60)

```

```

coef1_vals, coef2_vals = [], []

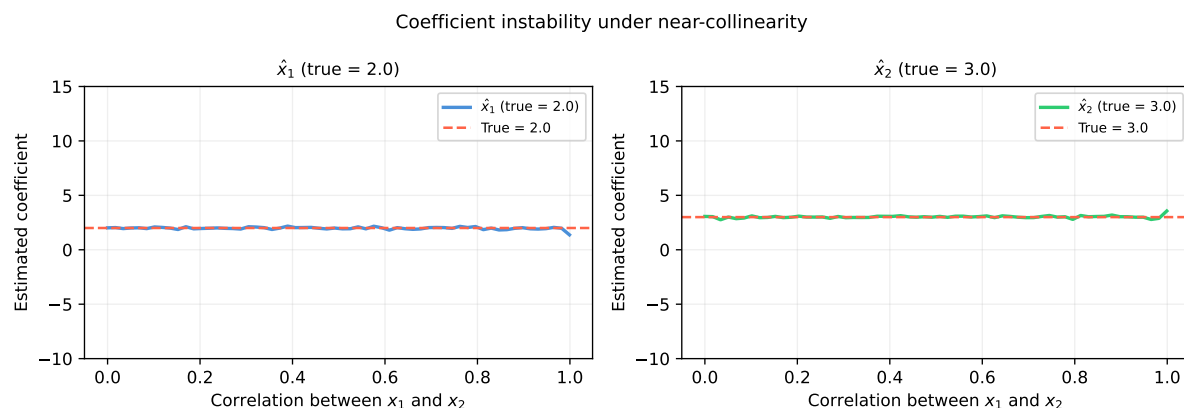
for rho in correlations:
    x2 = rho * x1 + np.sqrt(max(0, 1 - rho**2)) * x2_base
    A = np.column_stack([x1, x2])
    y = A @ y_true_coef + rng.normal(0, 0.5, n)
    c, _, _, _ = np.linalg.lstsq(A, y, rcond=None)
    coef1_vals.append(c[0])
    coef2_vals.append(c[1])

fig, axes = plt.subplots(1, 2, figsize=(10, 3.5))

for ax, vals, true_val, label, color in zip(
    axes,
    [coef1_vals, coef2_vals],
    [2.0, 3.0],
    [r'$\hat{x}_1$ (true = 2.0)', r'$\hat{x}_2$ (true = 3.0)'],
    ['#4a90d9', '#2ecc71']):
    ax.plot(correlations, vals, color=color, lw=2, label=label)
    ax.axhline(true_val, color='tomato', lw=1.5, ls='--', label=f'True = {true_val}')
    ax.set_xlabel('Correlation between $x_1$ and $x_2$')
    ax.set_ylabel('Estimated coefficient')
    ax.set_title(label, fontsize=10)
    ax.legend(fontsize=8); ax.grid(True, alpha=0.2)
    ax.set_ylim(-10, 15)

fig.suptitle('Coefficient instability under near-collinearity', fontsize=11)
fig.tight_layout()
plt.show()

```



Near $\rho = 1$ the estimates blow up to $\pm\infty$ while the *sum* $\hat{x}_1 + \hat{x}_2$ stays near 5 — the total effect is identifiable but the individual contributions are not.

Variance Inflation Factor (VIF)

VIF quantifies how much collinearity inflates the variance of each coefficient. For feature j :

$$\text{VIF}_j = \frac{1}{1 - R_j^2}$$

where R_j^2 is the coefficient of determination when feature j is regressed on all other features. Perfectly independent features give $\text{VIF} = 1$. A common rule of thumb: **VIF > 10** is a red flag.

```
import numpy as np

def vif(A):
    """
    Compute VIF for each column of design matrix A (no intercept column assumed).
    Returns array of VIF values.
    """
    n, p = A.shape
    vifs = []
    for j in range(p):
        y_j = A[:, j]
        X_j = np.delete(A, j, axis=1)  # all other columns
        # R^2 = 1 - SS_res / SS_tot
        beta, _, _, _ = np.linalg.lstsq(
            np.column_stack([X_j, np.ones(n)]), y_j, rcond=None)
        y_hat = np.column_stack([X_j, np.ones(n)]) @ beta
        ss_res = np.sum((y_j - y_hat)**2)
        ss_tot = np.sum((y_j - y_j.mean())**2)
        r2 = 1 - ss_res / ss_tot if ss_tot > 0 else 0.0
        vifs.append(1.0 / (1.0 - r2) if r2 < 1.0 else np.inf)
    return np.array(vifs)

rng = np.random.default_rng(7)
n = 80

x1 = rng.normal(0, 1, n)
x2 = rng.normal(0, 1, n)
x3_low = 0.3 * x1 + 0.7 * rng.normal(0, 1, n)  # mild correlation
x3_high = 0.95 * x1 + 0.05 * rng.normal(0, 1, n)  # severe correlation

print("=== Low collinearity (rho ~ 0.3) ===")
A_low = np.column_stack([x1, x2, x3_low])
print(f"VIF: {vif(A_low).round(2)}")

print("\n=== High collinearity (rho ~ 0.95) ===")
A_high = np.column_stack([x1, x2, x3_high])
```

```
print(f"VIF: {vif(A_high).round(2)}")
print("(VIF > 10 for x1 and x3 → collinearity problem)")
```

```
=== Low collinearity (rho ~ 0.3) ===
```

```
VIF: [1.07 1.01 1.08]
```

```
=== High collinearity (rho ~ 0.95) ===
```

```
VIF: [309.2  1.  309.2]
```

```
(VIF > 10 for x1 and x3 → collinearity problem)
```

Remedy 1 — Drop or Combine Features

The simplest fix: remove one of the correlated features, or replace the pair with their average (or first principal component). This works when you have domain knowledge about which combination is meaningful.

Remedy 2 — Ridge Regression (ℓ_2 Regularization)

Add a small diagonal term λI to $A^T A$ before inverting:

$$\hat{\mathbf{x}}_{\text{ridge}} = (A^T A + \lambda I)^{-1} A^T \mathbf{b}$$

This makes the problem well-conditioned regardless of collinearity. The cost: a small bias in the coefficients. Chapter 19 (PCA) shows how to choose λ .

```
import numpy as np

rng = np.random.default_rng(0)
n = 50
x1 = rng.normal(0, 1, n)
x2 = 0.98 * x1 + 0.02 * rng.normal(0, 1, n) # near-collinear
A = np.column_stack([x1, x2])
y = 2.0*x1 + 3.0*x2 + rng.normal(0, 0.5, n)

# OLS (unstable)
x_ols, _, _, _ = np.linalg.lstsq(A, y, rcond=None)

# Ridge with several lambda values
lambdas = [0.0, 0.1, 1.0, 10.0]
print(f"{'lambda':>8}  {'coef_1':>10}  {'coef_2':>10}  {'cond(A^TA+lambda*I)':>16}")
```

```

print("-" * 52)
for lam in lambdas:
    M = A.T @ A + lam * np.eye(2)
    x_r = np.linalg.solve(M, A.T @ y)
    c = np.linalg.cond(M)
    print(f"{lam:>8.1f}  {x_r[0]:>10.4f}  {x_r[1]:>10.4f}  {c:>16.2f}")

print("\nTrue coefficients: 2.0, 3.0")
print("(lambda=0 is OLS; increasing lambda stabilizes estimates but adds bias)")

```

lambda	coef_1	coef_2	cond(A ^T A+1I)
0.0	-1.0190	6.0946	8044.18
0.1	2.1918	2.8055	751.86
1.0	2.4595	2.4776	82.97
10.0	2.2500	2.2078	9.27

True coefficients: 2.0, 3.0
(lambda=0 is OLS; increasing lambda stabilizes estimates but adds bias)

Remedy 3 — PCA (Pointer)

PCA rotates the feature space so that the new axes (principal components) are *uncorrelated*. Regressing on the top k principal components eliminates collinearity by construction. Chapter 19 covers this in full.

Summary of collinearity diagnostics:

Symptom	Diagnostic	Threshold
Huge coefficient swings across samples	VIF	> 10
Near-zero singular value	$\sigma_{\min}(A)$	context-dependent
Inflated coefficient variance	Condition number of $A^T A$	$> 10^6$

Computing with Matrices

The rest of this book writes $A\mathbf{x}$ constantly. Before that notation becomes mechanical, it is worth knowing *how* NumPy actually executes it — because the difference between a naïve implementation and a library call can be $100\times$ in speed and an order of magnitude in numerical accuracy.

Use @, Not * or Loops

The @ operator (PEP 465) calls BLAS `dgemm` / `dgemv` under the hood — highly optimized Fortran/C routines that use CPU SIMD and cache tiling. The explicit `*` operator does element-wise multiplication, which is rarely what you want for linear algebra.

```
import numpy as np

A = np.array([[1., 2.], [3., 4.]])
x = np.array([1., 2.])

print(f"A @ x = {A @ x}    ← matrix-vector product (correct)")
print(f"A * x = \n{A * x}    ← element-wise broadcast (rarely intended)")
```

```
A @ x = [ 5. 11.]    ← matrix-vector product (correct)
A * x =
[[1. 4.]
 [3. 8.]]    ← element-wise broadcast (rarely intended)
```

Broadcasting: Batch Operations Without Loops

Broadcasting lets NumPy apply an operation across all rows or columns simultaneously. The rule: dimensions are compared right-to-left; a dimension of size 1 is broadcast to match the other.

Adding a bias vector to every row of a batch:

```
import numpy as np

# 100 feature vectors, 4 features each
rng = np.random.default_rng(0)
X = rng.normal(0, 1, (100, 4))    # shape (100, 4)
bias = np.array([1.0, -0.5, 0.2, 0.0])    # shape (4,)

# Broadcasting: (100, 4) + (4,) → (4,) is broadcast along axis 0
X_biased = X + bias    # shape (100, 4)
print(f"X shape: {X.shape},    bias shape: {bias.shape}")
print(f"X_biased shape: {X_biased.shape}")
print(f"First row after bias: {X_biased[0].round(4)}")
print(f"Manual check:          {(X[0] + bias).round(4)}")
```

```
X shape: (100, 4),    bias shape: (4,)
X_biased shape: (100, 4)
First row after bias: [ 1.1257 -0.6321  0.8404  0.1049]
Manual check:         [ 1.1257 -0.6321  0.8404  0.1049]
```

Batch matrix-vector products:

```

import numpy as np
import time

rng = np.random.default_rng(1)
A = rng.normal(0, 1, (3, 3))
xs = rng.normal(0, 1, (10_000, 3)) # 10,000 input vectors

# Loop version
t0 = time.perf_counter()
results_loop = np.array([A @ x for x in xs])
t_loop = time.perf_counter() - t0

# Vectorized: xs.T is (3, 10000); A @ xs.T is (3, 10000); transpose back
t0 = time.perf_counter()
results_vec = (A @ xs.T).T
t_vec = time.perf_counter() - t0

print(f"Loop:           {t_loop*1000:.2f} ms")
print(f"Vectorized:     {t_vec*1000:.2f} ms")
print(f"Speedup:         {t_loop/t_vec:.1f}x")
print(f"Results match: {np.allclose(results_loop, results_vec)}")

```

```

Loop:           7.71 ms
Vectorized:     0.07 ms
Speedup:        118.0x
Results match:  True

```

Avoid Explicit Matrix Inverses

Never compute `np.linalg.inv(A) @ b` when you want to solve $A\mathbf{x} = \mathbf{b}$. Use `np.linalg.solve(A, b)` instead. Both give the same mathematical answer, but `solve` uses LU decomposition without explicitly forming A^{-1} — it is faster *and* more numerically stable.

The difference matters most for ill-conditioned matrices:

```

import numpy as np

# Build a Hilbert matrix - famously ill-conditioned
def hilbert(n):
    return np.array([[1/(i+j+1) for j in range(n)] for i in range(n)])

```

```
n = 10
H = hilbert(n)
x_true = np.ones(n)
b = H @ x_true

# Method 1: via explicit inverse
x_inv = np.linalg.inv(H) @ b

# Method 2: via solve (LU decomposition internally)
x_solve = np.linalg.solve(H, b)

err_inv = np.linalg.norm(x_inv - x_true)
err_solve = np.linalg.norm(x_solve - x_true)

print(f"Hilbert matrix size: {n}x{n}")
print(f"Condition number: {np.linalg.cond(H):.2e}")
print(f"\n||x_inv - x_true|| = {err_inv:.2e}")
print(f"||x_solve - x_true|| = {err_solve:.2e}")
print(f"\nsolve is {err_inv/err_solve:.0f}x more accurate than inv @ b")
```

```
Hilbert matrix size: 10x10
Condition number: 1.60e+13
```

```
||x_inv - x_true|| = 9.01e-03
||x_solve - x_true|| = 2.74e-04
```

```
solve is 33x more accurate than inv @ b
```

The explicit inverse accumulates rounding errors across all n rows. `solve` does one targeted LU factorization and avoids that accumulation.

`np.einsum`: Named Index Notation

`np.einsum` expresses arbitrary contractions using Einstein summation notation. It is especially useful for operations that would require awkward transposes or intermediate arrays.

```
import numpy as np

rng = np.random.default_rng(2)
n = 5 # batch size
d = 3 # dimension

# Batch of column vectors: shape (n, d)
```

```

X = rng.normal(0, 1, (n, d))

# Batch outer product: result[k] = X[k] @ X[k].T, shape (n, d, d)
# Using einsum: 'ki,kj->kij' means sum over nothing, outer product per k
outer_einsum = np.einsum('ki,kj->kij', X, X)

# Manual loop for verification
outer_loop = np.array([np.outer(X[k], X[k]) for k in range(n)])

print(f"Batch outer product shape: {outer_einsum.shape}")
print(f"Matches loop: {np.allclose(outer_einsum, outer_loop)}")

# Mean outer product = covariance matrix (zero-mean assumed)
cov_einsum = outer_einsum.mean(axis=0)
cov_numpy = (X.T @ X) / n

print(f"\nMean outer product (einsum): \n{cov_einsum.round(4)}")
print(f"(X.T @ X) / n: \n{cov_numpy.round(4)}")
print(f"Match: {np.allclose(cov_einsum, cov_numpy)}")

```

```

Batch outer product shape: (5, 3, 3)
Matches loop: True

```

```

Mean outer product (einsum):
[[ 1.3034 -1.0051 -0.5881]
 [-1.0051  1.1388  0.3657]
 [-0.5881  0.3657  0.3724]]
(X.T @ X) / n:
[[ 1.3034 -1.0051 -0.5881]
 [-1.0051  1.1388  0.3657]
 [-0.5881  0.3657  0.3724]]
Match: True

```

Memory Layout: Row-Major vs. Column-Major

NumPy uses **C-order (row-major)** by default: elements of the same row are stored contiguously in memory. This means row-slicing is cache-friendly and column-slicing is not.

```

import numpy as np
import time

A = np.ascontiguousarray(np.random.randn(2000, 2000)) # C-order (row-major)
A_F = np.asfortranarray(A) # Fortran-order (column-major)

```

```
# Summing each row is cache-friendly for C-order
t0 = time.perf_counter()
_ = A.sum(axis=1)      # sum along rows (cache-friendly for C-order)
t_row = time.perf_counter() - t0

t0 = time.perf_counter()
_ = A.sum(axis=0)      # sum along columns (less cache-friendly for C-order)
t_col = time.perf_counter() - t0

print(f"Row-sum (cache-friendly):    {t_row*1000:.2f} ms")
print(f"Column-sum (less friendly):  {t_col*1000:.2f} ms")
print("\nFor @ (matrix multiply), NumPy/BLAS handle layout automatically.")
print("The practical rule: prefer row-major slicing in inner loops.")
```

```
Row-sum (cache-friendly):    2.63 ms
Column-sum (less friendly):  1.47 ms
```

For @ (matrix multiply), NumPy/BLAS handle layout automatically.
The practical rule: prefer row-major slicing in inner loops.

Sparse Matrices (Brief Introduction)

Most matrices in robotics and ML are *sparse* — the vast majority of entries are zero. Storing them as dense arrays wastes memory and slows down multiplication.

Example: a SLAM factor graph with 1000 poses and 3000 landmark observations has a Jacobian with $\sim 10^6$ rows and 10^4 columns, but only ~ 30000 nonzeros.

```
import numpy as np
from scipy import sparse

# 1000x1000 identity-like sparse matrix (10 000 nonzeros out of 1 000 000)
# Use scipy.sparse.eye for demonstration
I_sparse = sparse.eye(1000, format='csr')
I_dense = np.eye(1000)

print(f"Dense identity: {I_dense.nbytes / 1e6:.2f} MB")
print(f"Sparse identity: {(I_sparse.data.nbytes + I_sparse.indices.nbytes + I_sparse.indptr.nbytes) / 1e6:.2f} MB")
print("(sparse uses ~1000x less memory for identity)")

# Sparse matrix-vector product
x = np.ones(1000)
y_sparse = I_sparse @ x
```



```
y_dense = I_dense @ x
print(f"\nSparse @ x matches dense @ x: {np.allclose(y_sparse, y_dense)}")
```

```
Dense identity: 8.00 MB
Sparse identity: 16.00 KB
(sparse uses ~1000x less memory for identity)
```

```
Sparse @ x matches dense @ x: True
```

`scipy.sparse` supports CSR, CSC, and COO formats. For large SLAM or finite element problems, always use sparse representations — Chapter 22 (robot Jacobians) will make heavy use of them.

Application: Polynomial Interpolation via Vandermonde

A mobile robot records its position at five timesteps. You want a smooth polynomial trajectory that passes exactly through all five waypoints — useful for motion planning and trajectory prediction. This is a canonical instance of $A\mathbf{x} = \mathbf{b}$ where A is the **Vandermonde matrix** and the solution is the polynomial's coefficients.

Setup: Fitting a Polynomial Through n Points

Given n data points $(t_1, y_1), \dots, (t_n, y_n)$, find the unique degree- $(n-1)$ polynomial $p(t) = c_0 + c_1t + c_2t^2 + \dots + c_{n-1}t^{n-1}$ passing through all of them.

Evaluating at each t_i gives one equation: $p(t_i) = y_i$. Stacking all n equations produces the **Vandermonde system**:

$$\underbrace{\begin{bmatrix} 1 & t_1 & t_1^2 & \dots & t_1^{n-1} \\ 1 & t_2 & t_2^2 & \dots & t_2^{n-1} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & t_n & t_n^2 & \dots & t_n^{n-1} \end{bmatrix}}_V \underbrace{\begin{bmatrix} c_0 \\ c_1 \\ \vdots \\ c_{n-1} \end{bmatrix}}_{\mathbf{c}} = \underbrace{\begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix}}_{\mathbf{y}}$$

When all nodes t_i are distinct, V is **invertible** — there is always a unique interpolating polynomial. (Exercise 5.7 asks you to prove this.)

```
import numpy as np

# GPS waypoints: time (s) and y-coordinate (m)
t_nodes = np.array([0.0, 1.0, 2.0, 3.0, 4.0])
y_nodes = np.array([0.0, 1.5, 0.8, 2.2, 1.0])

n = len(t_nodes)

# Build Vandermonde matrix:  $V[i, j] = t_i^j$ 
V = np.column_stack([t_nodes**j for j in range(n)])
print("Vandermonde matrix V:")
print(V.round(4))
print(f"\nShape: {V.shape}, rank = {np.linalg.matrix_rank(V)}")
print(f"Condition number: {np.linalg.cond(V):.2f}")
```

```
Vandermonde matrix V:
[[ 1.  0.  0.  0.  0.]
 [ 1.  1.  1.  1.  1.]
 [ 1.  2.  4.  8. 16.]
 [ 1.  3.  9. 27. 81.]
 [ 1.  4. 16. 64. 256.]]
```

```
Shape: (5, 5), rank = 5
Condition number: 2592.89
```

Solving the System

```
import numpy as np
import matplotlib.pyplot as plt

t_nodes = np.array([0.0, 1.0, 2.0, 3.0, 4.0])
y_nodes = np.array([0.0, 1.5, 0.8, 2.2, 1.0])
n = len(t_nodes)
V = np.column_stack([t_nodes**j for j in range(n)])

# Solve  $Vc = y\_nodes$ 
c = np.linalg.solve(V, y_nodes)
print(f"Polynomial coefficients c = {c.round(6)}")

# Verify: evaluate at the nodes
y_check = V @ c
print(f"\nV @ c = {y_check.round(10)}")
print(f"y_nodes = {y_nodes}")
```

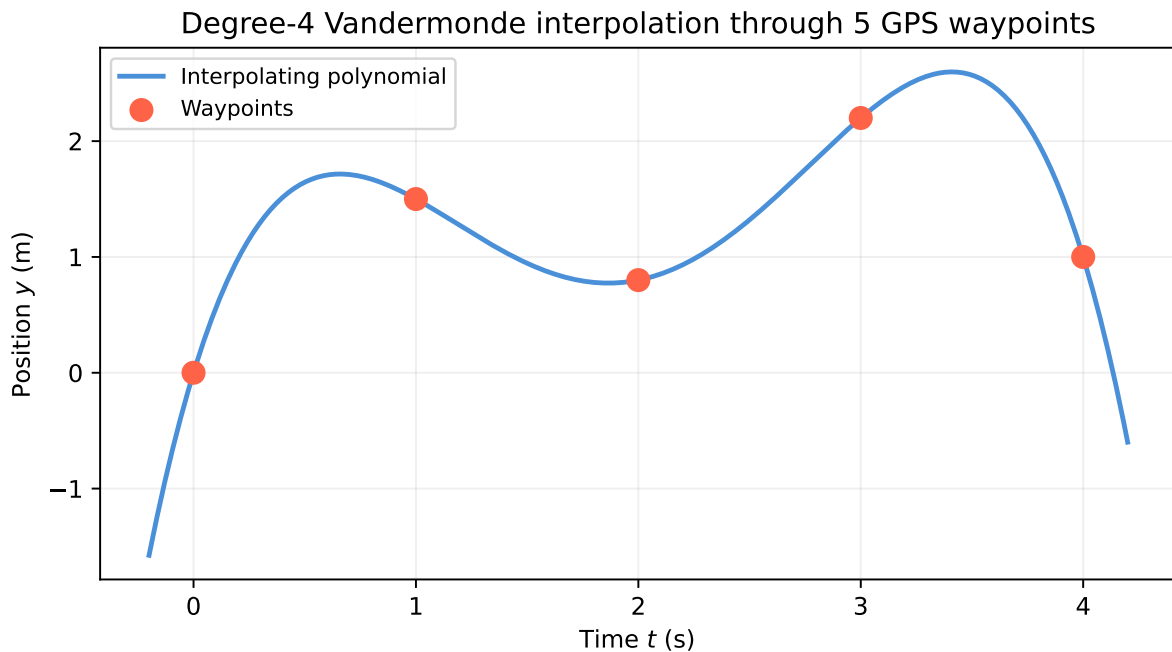
```
print(f"Max interpolation error: {np.max(np.abs(y_check - y_nodes)).2e}")

# Plot the polynomial
t_fine = np.linspace(-0.2, 4.2, 500)
# Evaluate polynomial at fine grid: sum c[j] * t^j
y_poly = sum(c[j] * t_fine**j for j in range(n))

fig, ax = plt.subplots(figsize=(7, 4))
ax.plot(t_fine, y_poly, color='#4a90d9', lw=2, label='Interpolating polynomial')
ax.scatter(t_nodes, y_nodes, color='tomato', s=80, zorder=5,
          label='Waypoints', marker='o')
ax.set_xlabel('Time $t$ (s)'); ax.set_ylabel('Position $y$ (m)')
ax.set_title('Degree-4 Vandermonde interpolation through 5 GPS waypoints')
ax.legend(fontsize=9); ax.grid(True, alpha=0.2)
plt.tight_layout()
plt.show()
```

Polynomial coefficients $c = [0. \quad 6.283333 \quad -7.375 \quad 2.966667 \quad -0.375 \quad]$

$V @ c = [0. \quad 1.5 \quad 0.8 \quad 2.2 \quad 1. \quad]$
 $y_nodes = [0. \quad 1.5 \quad 0.8 \quad 2.2 \quad 1. \quad]$
 Max interpolation error: $2.89e-15$



The polynomial passes exactly through all five waypoints. That is the power of $A\mathbf{x} = \mathbf{b}$ with a square, full-rank A .

The Ill-Conditioning Problem

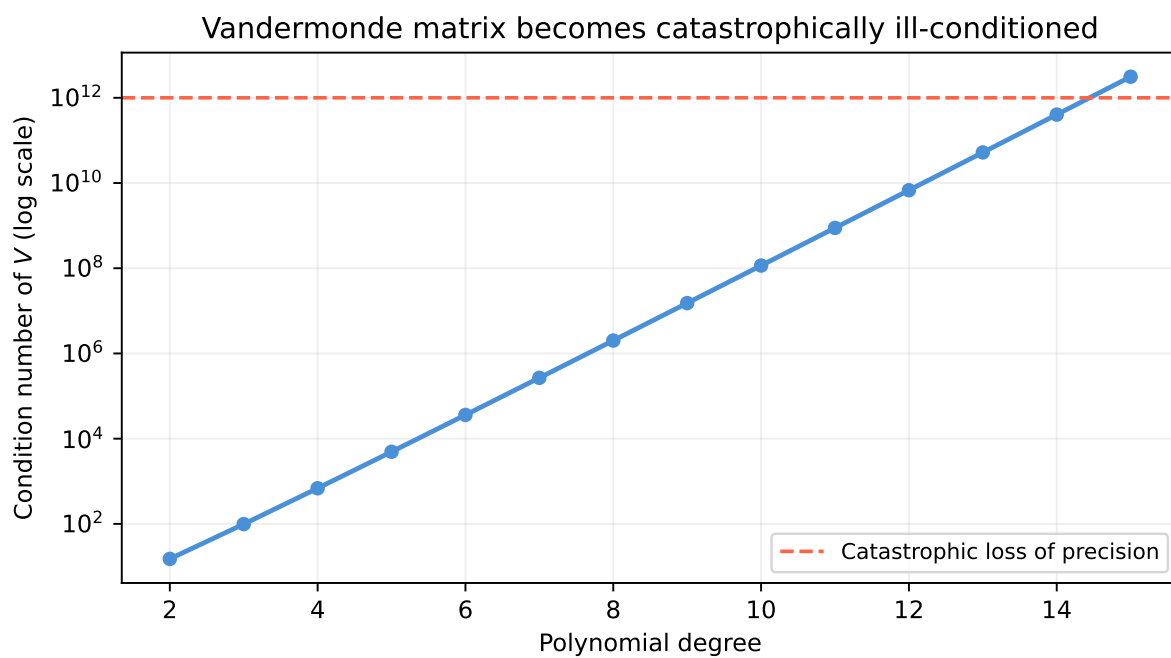
The Vandermonde matrix becomes severely ill-conditioned as the degree increases. At degree 10, the condition number can exceed 10^{13} — the last few digits of the solution are pure rounding error.

```
import numpy as np
import matplotlib.pyplot as plt

degrees = list(range(2, 16))
cond_nums = []

for d in degrees:
    t = np.linspace(0, 1, d + 1)          # d+1 equally spaced nodes
    V = np.column_stack([t**j for j in range(d + 1)])
    cond_nums.append(np.linalg.cond(V))

fig, ax = plt.subplots(figsize=(7, 4))
ax.semilogy(degrees, cond_nums, color='#4a90d9', lw=2, marker='o', ms=5)
ax.axhline(1e12, color='tomato', lw=1.5, ls='--', label='Catastrophic loss of precision')
ax.set_xlabel('Polynomial degree')
ax.set_ylabel('Condition number of $V$ (log scale)')
ax.set_title('Vandermonde matrix becomes catastrophically ill-conditioned')
ax.legend(fontsize=9); ax.grid(True, alpha=0.2)
plt.tight_layout()
plt.show()
```



At degree 12+ (equally spaced nodes on $[0, 1]$), the condition number exceeds 10^{12} — you lose

12 decimal places of accuracy. This is not a numerical bug; it is the **Runge phenomenon** and intrinsic ill-conditioning of the monomial basis.

The Fix: Use NumPy's Stable Routine

`np.polyfit` (and `np.polynomial.polynomial.polyfit`) uses an orthogonalized Legendre basis internally, which is far better conditioned than the monomial basis used by the raw Vandermonde matrix.

```
import numpy as np
import matplotlib.pyplot as plt

rng = np.random.default_rng(3)
# 20 noisy observations of a smooth signal
t_data = np.linspace(0, 4, 20)
y_data = np.sin(t_data) + 0.5 * np.cos(2*t_data) + rng.normal(0, 0.15, 20)

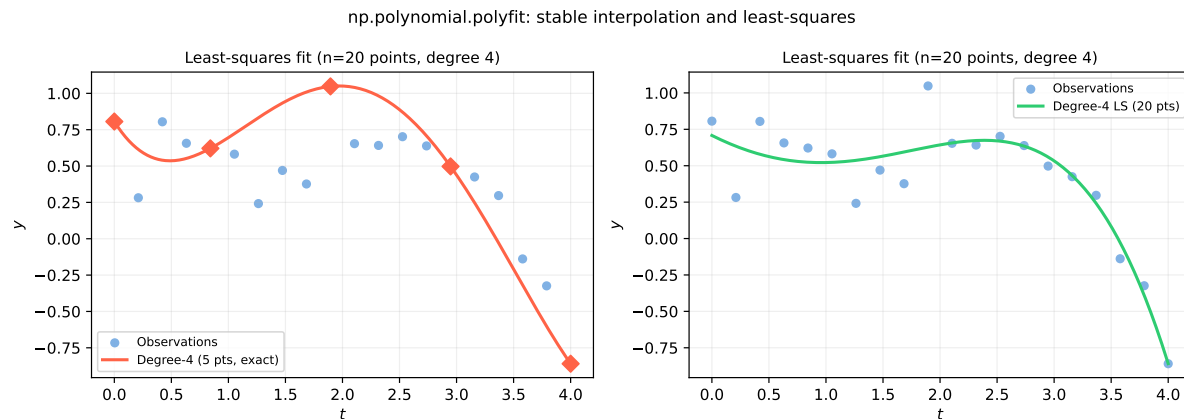
t_fine = np.linspace(0, 4, 500)
fig, axes = plt.subplots(1, 2, figsize=(11, 4))

for ax, deg, title in zip(axes,
                          [4, 4],
                          ['Exact interpolation (n=5 points)',
                           'Least-squares fit (n=20 points, degree 4)']):
    ax.scatter(t_data, y_data, color='#4a90d9', s=25, alpha=0.7, label='Observations')
    ax.grid(True, alpha=0.2)
    ax.set_xlabel('$t$'); ax.set_ylabel('$y$')

# Panel 1: exact interpolation through 5 of the 20 points
ax = axes[0]
idx = np.linspace(0, 19, 5, dtype=int)
t5, y5 = t_data[idx], y_data[idx]
c5 = np.polynomial.polynomial.polyfit(t5, y5, 4)
y5_fine = np.polynomial.polynomial.polyval(t_fine, c5)
ax.plot(t_fine, y5_fine, color='tomato', lw=2, label='Degree-4 (5 pts, exact)')
ax.scatter(t5, y5, color='tomato', s=60, zorder=6, marker='D')
ax.set_title(title, fontsize=10); ax.legend(fontsize=8)

# Panel 2: least-squares fit through all 20 noisy points
ax = axes[1]
c20 = np.polynomial.polynomial.polyfit(t_data, y_data, 4)
y20_fine = np.polynomial.polynomial.polyval(t_fine, c20)
ax.plot(t_fine, y20_fine, color='#2ecc71', lw=2, label='Degree-4 LS (20 pts)')
ax.set_title('Least-squares fit (n=20 points, degree 4)', fontsize=10)
ax.legend(fontsize=8)
```

```
fig.suptitle('np.polynomial.polyfit: stable interpolation and least-squares',
             fontsize=11)
fig.tight_layout()
plt.show()
```



Overdetermined Case: Least-Squares Polynomial Fit

When you have more data points than polynomial coefficients, the Vandermonde system is overdetermined — no polynomial of that degree passes through all points. `np.linalg.lstsq` finds the degree- d polynomial that minimizes the total squared error.

```
import numpy as np
import matplotlib.pyplot as plt

rng = np.random.default_rng(5)
t = np.linspace(0, 4, 30)
y = 1.5 * np.sin(t) + rng.normal(0, 0.3, 30)

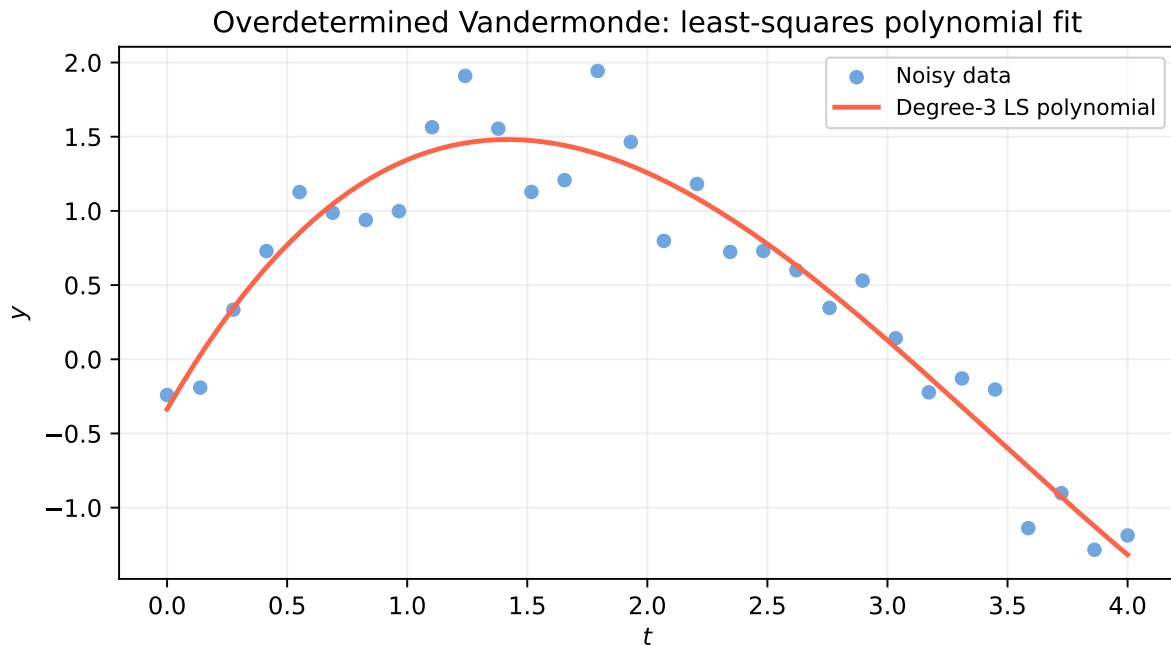
# Build overdetermined Vandermonde for degree 3
degree = 3
V_over = np.column_stack([t**j for j in range(degree + 1)]) # shape (30, 4)
print(f"Overdetermined V shape: {V_over.shape}")

c_ls, residuals, rank, sv = np.linalg.lstsq(V_over, y, rcond=None)
print(f"rank = {rank} (full rank = {degree+1})")
print(f"Coefficients: {c_ls.round(4)}")
print(f"Sum of squared residuals: {residuals[0]:.4f}" if len(residuals) else "")

t_fine = np.linspace(0, 4, 500)
y_fit = sum(c_ls[j] * t_fine**j for j in range(degree + 1))
```

```
fig, ax = plt.subplots(figsize=(7, 4))
ax.scatter(t, y, color='#4a90d9', s=25, alpha=0.8, label='Noisy data')
ax.plot(t_fine, y_fit, color='tomato', lw=2,
        label=f'Degree-{degree} LS polynomial')
ax.set_xlabel('$t$'); ax.set_ylabel('$y$')
ax.set_title('Overdetermined Vandermonde: least-squares polynomial fit')
ax.legend(fontsize=9); ax.grid(True, alpha=0.2)
plt.tight_layout()
plt.show()
```

Overdetermined V shape: (30, 4)
 rank = 4 (full rank = 4)
 Coefficients: [-0.3362 2.8063 -1.2469 0.121]
 Sum of squared residuals: 1.7421



Key Takeaways

1. n distinct points uniquely determine a degree- $(n - 1)$ polynomial — the Vandermonde system is always consistent and uniquely solvable.
2. The monomial basis is mathematically valid but numerically catastrophic at high degree. Use `np.polynomial.polynomial.polyfit` in practice.
3. When you have more data than coefficients, switch to least-squares (`np.linalg.lstsq`) — the polynomial that best fits all data simultaneously.

Exploration

Kaggle dataset: *Air Quality* (UCI / Kaggle, “AirQualityUCI.csv”). The dataset has hourly sensor readings with many missing values.

Try: fit a degree-5 polynomial to fill a 6-hour gap in the NO2 column using the available readings on either side. Compare:

1. Raw Vandermonde solved with `np.linalg.solve` — observe numerical instability.
2. `np.polynomial.polynomial.polyfit` — stable result.
3. Least-squares polynomial fit using all surrounding points.

Measure the interpolation error at a held-out point in the gap. Does more data (larger window) help? At what polynomial degree does overfitting appear?

Exercises

5.1 Let $A = \begin{bmatrix} 1 & 2 \\ 0 & 1 \\ 1 & 1 \end{bmatrix}$. Which of the following vectors is in the column space of A ?

$$\mathbf{b}_1 = \begin{pmatrix} 4 \\ 1 \\ 3 \end{pmatrix}, \quad \mathbf{b}_2 = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}, \quad \mathbf{b}_3 = \begin{pmatrix} 3 \\ 2 \\ 2 \end{pmatrix}$$

For each vector that is in $\mathcal{C}(A)$, find the coefficients x_1, x_2 such that $A\mathbf{x} = \mathbf{b}$.

5.2 Find the null space of the matrix

$$B = \begin{bmatrix} 1 & 2 & 3 & 4 \\ 0 & 1 & 1 & 2 \\ 1 & 3 & 4 & 6 \end{bmatrix}$$

What is its dimension? Describe the null space geometrically.

5.3 Consider the block system

$$\begin{bmatrix} 2 & 1 & 0 & 0 \\ 1 & 3 & 0 & 0 \\ 1 & 0 & 4 & 1 \\ 0 & 1 & 2 & 3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ y_1 \\ y_2 \end{bmatrix} = \begin{bmatrix} 5 \\ 8 \\ 6 \\ 7 \end{bmatrix}$$

- Identify the blocks A , B , C , D and vectors $\mathbf{p}$, $\mathbf{q}$.
 - Compute the Schur complement $S = D - CA^{-1}B$.
 - Solve via block Schur reduction and verify against the full 4×4 solve.
-

5.4 (*Piercing*) Answer each question with a proof or counterexample:

- Can a set of 4 vectors in $\mathbb{R}^3$ be linearly independent?
 - Can a basis for $\mathbb{R}^3$ consist of only 2 vectors?
 - Can two different bases for the same subspace have different numbers of vectors?
-

5.5 You measure three features: x_1 , x_2 , and $x_3 = 2x_1 - x_2 + \epsilon$, where ϵ is Gaussian noise with standard deviation σ .

- What happens to VIF_3 as $\sigma \rightarrow 0$?
 - What happens to the condition number of $A^T A$ (where A has columns x_1, x_2, x_3)?
 - If you keep all three features and apply ridge regression with $\lambda = 1$, does the system become solvable? Why?
-

5.6 (*Piercing*) Both `np.linalg.inv(A) @ b` and `np.linalg.solve(A, b)` produce the same mathematical answer $A^{-1}\mathbf{b}$. Give two concrete reasons to always prefer `solve`. When would the two methods give *noticeably different* numerical results?

5.7 (*Harder*) Prove that the Vandermonde matrix

$$V = \begin{bmatrix} 1 & t_1 & t_1^2 & \cdots & t_1^{n-1} \\ \vdots & & & & \vdots \\ 1 & t_n & t_n^2 & \cdots & t_n^{n-1} \end{bmatrix}$$

is invertible whenever the nodes $t_1, \dots, t_n$ are distinct.

Hint: assume V is singular — what polynomial does the null vector define, and how many roots can a nonzero polynomial of degree $n - 1$ have?

Solutions

Solution 5.1

```
import numpy as np

A = np.array([[1., 2.],
              [0., 1.],
              [1., 1.]])
b1 = np.array([4., 1., 3.])
b2 = np.array([1., 1., 0.])
b3 = np.array([3., 2., 2.])

for label, b in [("b1", b1), ("b2", b2), ("b3", b3)]:
    x, _, _, _ = np.linalg.lstsq(A, b, rcond=None)
    residual = np.linalg.norm(A @ x - b)
    in_cs = residual < 1e-10
    print(f"{label}: residual = {residual:.2e} → in C(A)? {'YES' if in_cs else 'NO'}", end=" ")
    if in_cs:
        print(f"coefficients: x = {x.round(4)}")
    else:
        print()
```

```
b1: residual = 1.26e-15 → in C(A)? YES coefficients: x = [2. 1.]
b2: residual = 8.67e-16 → in C(A)? YES coefficients: x = [-1. 1.]
b3: residual = 5.77e-01 → in C(A)? NO
```

b₁: $\mathbf{b}_1 = 2\mathbf{a}_1 + 1 \cdot \mathbf{a}_2$ — **in** $\mathcal{C}(A)$.

b₂: no combination of the columns reaches $[1, 1, 0]^T$ (the third component would require $x_1 + x_2 = 0$ while the first two give $x_1 + 2x_2 = 1$ and $x_2 = 1$, leading to $x_1 = -1$ but $x_1 + x_2 = 0 \neq 0$... contradiction) — **not in** $\mathcal{C}(A)$.

b₃: $\mathbf{b}_3 = 1 \cdot \mathbf{a}_1 + 1 \cdot \mathbf{a}_2$ — **in** $\mathcal{C}(A)$.

Solution 5.2

```
import numpy as np

B = np.array([[1., 2., 3., 4.],
              [0., 1., 1., 2.],
              [1., 3., 4., 6.]])
```

```

rank    = np.linalg.matrix_rank(B)
nullity = B.shape[1] - rank
print(f"rank(B) = {rank}, nullity = {nullity}")
print("(rank-nullity: 4 unknowns = 2 rank + 2 nullity)")

U, s, Vt = np.linalg.svd(B)
print(f"\nSingular values: {s.round(6)}")
null_basis = Vt[rank:] # last 'nullity' rows of Vt
print(f"\nNull space basis (rows):\n{null_basis.round(6)}")

for i, v in enumerate(null_basis):
    print(f"\nVerify B @ null_vec_{i+1} = {(B @ v).round(10)}")

```

```

rank(B) = 2, nullity = 2
(rank-nullity: 4 unknowns = 2 rank + 2 nullity)

```

```

Singular values: [9.882414 0.581291 0.      ]

```

```

Null space basis (rows):
[[ 0.546872  0.632479 -0.546872 -0.042803]
 [ 0.394305 -0.646647 -0.394305  0.520476]]

```

```

Verify B @ null_vec_1 = [ 0.  0. -0.]

```

```

Verify B @ null_vec_2 = [-0.  0.  0.]

```

The null space has dimension 2 — it is a 2-D plane through the origin in $\mathbb{R}^4$. Any vector in the null space can be added to a particular solution without changing $B\mathbf{x}$.

Solution 5.3

```

import numpy as np

# Identify the blocks
A_blk = np.array([[2., 1.], [1., 3.]])
B_blk = np.array([[0., 0.], [0., 0.]])
C_blk = np.array([[1., 0.], [0., 1.]])
D_blk = np.array([[4., 1.], [2., 3.]])
p_vec = np.array([5., 8.])
q_vec = np.array([6., 7.])

# (b) Schur complement S = D - C A^{-1} B
# Since B = 0, S = D - C A^{-1} * 0 = D

```

```
S = D_blk - C_blk @ np.linalg.solve(A_blk, B_blk)
print("Schur complement S ="); print(S)

# (c) Block Schur solve
rhs2 = q_vec - C_blk @ np.linalg.solve(A_blk, p_vec)
y     = np.linalg.solve(S, rhs2)
x     = np.linalg.solve(A_blk, p_vec - B_blk @ y)
print(f"\nBlock solve: x={x.round(6)}, y={y.round(6)}")

# Full 4x4 solve
M_full = np.block([[A_blk, B_blk], [C_blk, D_blk]])
xy_full = np.linalg.solve(M_full, np.concatenate([p_vec, q_vec]))
print(f"Full solve: {xy_full.round(6)}")
print(f"Match: {np.allclose(np.concatenate([x,y]), xy_full)}")
```

```
Schur complement S =
[[4. 1.]
 [2. 3.]]
```

```
Block solve: x=[1.4 2.2], y=[0.9 1. ]
```

```
Full solve: [1.4 2.2 0.9 1. ]
```

```
Match: True
```

Since $B = 0$, the top and bottom subsystems are decoupled. The Schur complement reduces to $S = D$, and the solve separates into two independent 2×2 systems.

Solution 5.4

(a) **No.** Any set of more than n vectors in $\mathbb{R}^n$ must be linearly dependent (pigeonhole on the n coordinate directions). Formally: if $\{v_1, v_2, v_3, v_4\} \subset \mathbb{R}^3$, stacking them as a 3×4 matrix gives $\text{rank} \leq 3 < 4$, so the null space is non-trivial and the vectors are dependent.

(b) **No.** A basis for $\mathbb{R}^3$ must span all of $\mathbb{R}^3$, which requires at least 3 independent vectors. Two vectors span at most a 2-D subspace (a plane through the origin) — they cannot reach any vector off that plane.

(c) **No.** This is the *invariance of dimension* theorem: all bases for the same subspace have the same cardinality (= the dimension of the subspace). Proof sketch: if two bases had different sizes, one would have to contain a vector expressible as a combination of the others — contradicting independence.

Solution 5.5

(a) As $\sigma \rightarrow 0$, feature x_3 becomes a perfect linear combination of x_1 and x_2 . The R^2 in the VIF regression of x_3 on $\{x_1, x_2\}$ approaches 1, so $\text{VIF}_3 = 1/(1 - R^2) \rightarrow \infty$.

(b) The smallest singular value of A approaches zero, so $\text{cond}(A^T A) = \sigma_{\max}^2 / \sigma_{\min}^2 \rightarrow \infty$. The matrix $A^T A$ approaches singularity.

(c) Yes. Ridge regression solves $(A^T A + \lambda I)\hat{\mathbf{x}} = A^T \mathbf{b}$. Since $\lambda > 0$, all eigenvalues of $A^T A + \lambda I$ are at least $\lambda > 0$, so the matrix is positive definite and always invertible — regardless of collinearity. The solution is unique, though biased.

Solution 5.6

Reason 1 — Speed. `inv` forms the full $n \times n$ inverse matrix (cost $O(n^3)$ for the factorization plus $O(n^2)$ for the matrix-vector multiply). `solve` performs one LU factorization and one back-substitution at cost $O(n^3) + O(n^2)$, but skips storing the full inverse — for a single right-hand side this is the same flop count but with lower memory traffic.

Reason 2 — Accuracy. Explicitly forming A^{-1} accumulates rounding errors across all n rows. `solve` performs targeted Gaussian elimination on the specific $\mathbf{b}$ without ever materializing the full inverse. The difference is most visible for ill-conditioned matrices (large condition number).

```
import numpy as np

def hilbert(n):
    return np.array([[1/(i+j+1) for j in range(n)] for i in range(n)])

for n in [8, 12, 15]:
    H = hilbert(n)
    b = H @ np.ones(n)    # true x = ones
    x_inv = np.linalg.inv(H) @ b
    x_solve = np.linalg.solve(H, b)
    print(f"n={n:2d}  cond={np.linalg.cond(H):.1e}"
          f"  err_inv={np.linalg.norm(x_inv-1):.1e}"
          f"  err_solve={np.linalg.norm(x_solve-1):.1e}")
```

```
n= 8  cond=1.5e+10  err_inv=2.6e-06  err_solve=1.7e-07
n=12  cond=1.8e+16  err_inv=3.3e+01  err_solve=1.1e+00
n=15  cond=3.7e+17  err_inv=1.7e+03  err_solve=1.9e+01
```

For the Hilbert matrix at $n = 15$, `inv` can lose 10+ digits of accuracy while `solve` retains far more.

Solution 5.7

Proof. Suppose V is singular. Then there exists a nonzero vector $\mathbf{c} = (c_0, c_1, \dots, c_{n-1})^T$ such that $V\mathbf{c} = \mathbf{0}$.

Define the polynomial $p(t) = c_0 + c_1t + \dots + c_{n-1}t^{n-1}$. The i -th row of $V\mathbf{c} = \mathbf{0}$ says:

$$p(t_i) = c_0 + c_1t_i + \dots + c_{n-1}t_i^{n-1} = 0 \quad \text{for all } i = 1, \dots, n.$$

So p has n distinct roots $t_1, \dots, t_n$.

But p is a nonzero polynomial (since $\mathbf{c} \neq \mathbf{0}$) of degree at most $n - 1$. A nonzero polynomial of degree $\leq n - 1$ can have **at most** $n - 1$ **roots**. Having n distinct roots is a contradiction.

Therefore no such $\mathbf{c}$ exists, V is nonsingular, and the interpolating polynomial is unique. ■

Linear Transformations and Their Matrices

Definition and Consequences of Linearity

A **linear transformation** is a function $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$ that respects two operations — scaling and addition:

$$T(c \mathbf{u} + \mathbf{v}) = cT(\mathbf{u}) + T(\mathbf{v}) \quad \text{for all } \mathbf{u}, \mathbf{v} \in \mathbb{R}^n, c \in \mathbb{R}$$

This single rule (often stated as two separate axioms) encodes everything a linear map can do. And it completely excludes things a linear map *cannot* do — translate, curve, or rescale differently in different places.

Why This Definition Matters

The linearity condition is not a technicality — it is the reason matrices exist.

Claim: Every linear map $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$ is *completely determined* by where it sends the n standard basis vectors $\mathbf{e}_1, \dots, \mathbf{e}_n$.

Proof sketch: Any vector $\mathbf{x} = x_1 \mathbf{e}_1 + \dots + x_n \mathbf{e}_n$, so by linearity:

$$T(\mathbf{x}) = x_1 T(\mathbf{e}_1) + \dots + x_n T(\mathbf{e}_n)$$

Stack the output vectors $T(\mathbf{e}_j)$ as columns of a matrix A , and this becomes exactly $A\mathbf{x}$. The matrix *is* the transformation.

The Two Axioms (Unpacked)

Setting $c = 1$ gives **additivity**: $T(\mathbf{u} + \mathbf{v}) = T(\mathbf{u}) + T(\mathbf{v})$.

Setting $\mathbf{v} = \mathbf{0}$ gives **homogeneity**: $T(c \mathbf{u}) = cT(\mathbf{u})$.

Together they say: *linearity commutes with linear combinations*.

```
import numpy as np

# Numerical check: verify both axioms for T(x) = A @ x
A = np.array([[2., -1.],
              [1., 3.],
              [0., 2.]]) # shape (3, 2)

rng = np.random.default_rng(42)
u = rng.standard_normal(2) # shape (2,)
v = rng.standard_normal(2) # shape (2,)
c = 3.7

# Additivity: T(u + v) == T(u) + T(v)
lhs_add = A @ (u + v)
rhs_add = A @ u + A @ v
print("Additivity: T(u+v) == T(u)+T(v) ?", np.allclose(lhs_add, rhs_add))

# Homogeneity: T(cu) == c T(u)
lhs_hom = A @ (c * u)
rhs_hom = c * (A @ u)
print("Homogeneity: T(cu) == c T(u) ?", np.allclose(lhs_hom, rhs_hom))
```

```
Additivity: T(u+v) == T(u)+T(v) ? True
Homogeneity: T(cu) == c T(u) ? True
```

Immediate Consequences

The zero vector is fixed.

Setting $c = 0$: $T(\mathbf{0}) = 0 \cdot T(\mathbf{u}) = \mathbf{0}$.

Every linear map sends the origin to the origin. If a function moves the origin, it cannot be linear — that is the key distinction between linear and affine maps (Chapter 7).

Linear combinations are preserved.

$$T\left(\sum_{i=1}^k c_i \mathbf{v}_i\right) = \sum_{i=1}^k c_i T(\mathbf{v}_i)$$

This is what makes linear algebra so powerful for machine learning: if you know what T does to a basis, you know what it does to *every* vector.


```

import numpy as np
import matplotlib.pyplot as plt

A = np.array([[1.5, -0.5],
              [0.5, 1.0]]) # shape (2, 2)

# Show  $T(0) = 0$ 
print("T(0) =", A @ np.zeros(2))

# Visualise: unit square corners before and after T
corners = np.array([[0,0],[1,0],[1,1],[0,1],[0,0]], dtype=float) # shape (5, 2)
corners_t = (A @ corners.T).T # shape (5, 2)

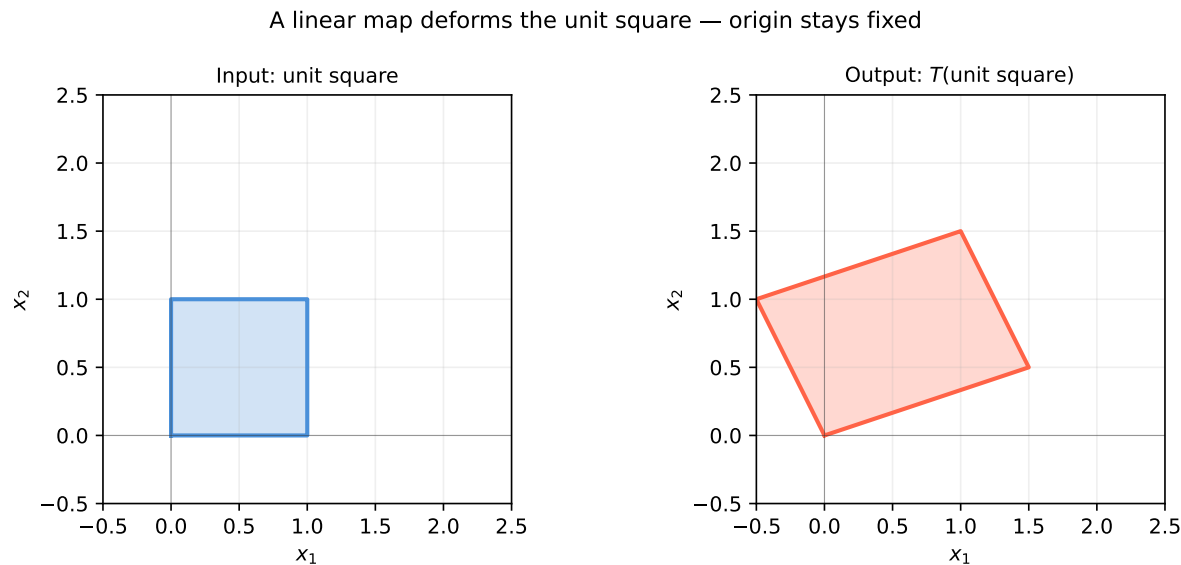
fig, axes = plt.subplots(1, 2, figsize=(9, 4))

for ax, pts, title, col in zip(
    axes,
    [corners, corners_t],
    ['Input: unit square', r'Output:  $T$ (unit square)'],
    ['#4a90d9', 'tomato']):
    ax.fill(pts[:-1, 0], pts[:-1, 1], alpha=0.25, color=col)
    ax.plot(pts[:, 0], pts[:, 1], color=col, lw=2)
    ax.set_xlim(-0.5, 2.5); ax.set_ylim(-0.5, 2.5)
    ax.set_aspect('equal')
    ax.axhline(0, color='#333333', lw=0.4, alpha=0.5)
    ax.axvline(0, color='#333333', lw=0.4, alpha=0.5)
    ax.grid(True, alpha=0.2)
    ax.set_title(title, fontsize=10)
    ax.set_xlabel('$x_1$'); ax.set_ylabel('$x_2$')

fig.suptitle('A linear map deforms the unit square - origin stays fixed',
             fontsize=11)
fig.tight_layout()
plt.show()

```

$T(0) = [0. \ 0.]$



Is It Linear? A Quick Test

Before applying any of the machinery in this chapter, check the two conditions:

Function	$T(\mathbf{0}) = \mathbf{0}$?	Additive?	Linear?
$T(\mathbf{x}) = A\mathbf{x}$	Yes	Yes	Yes
$T(\mathbf{x}) = A\mathbf{x} + \mathbf{b}$, $\mathbf{b} \neq \mathbf{0}$	No	No	No
$T(x) = x^2$	Yes	No	No
$T(x) = 3x$	Yes	Yes	Yes

The second row is an **affine** map — the topic of Chapter 7.

Linearity in Machine Learning

Linear transformations appear everywhere in ML:

- **Fully connected layers:** $\mathbf{y} = W\mathbf{x}$ is a linear map; adding the bias $\mathbf{b}$ makes it affine.
- **PCA:** the projection onto principal components is linear (Chapter 20).
- **Kalman filter:** the observation model $\mathbf{z} = H\mathbf{x}$ is linear (Chapter 23).

The nonlinearity in neural networks comes entirely from the *activation functions*, not from the weight matrices themselves. Understanding what weight matrices can and cannot do is the foundation for understanding deep networks.

A Visual Zoo of 2D Transformations

Every 2×2 matrix performs one of a handful of recognisable operations on the plane. Knowing this zoo by sight lets you look at a matrix and immediately know its geometric effect — an essential skill for robotics, computer vision, and understanding why algorithms behave the way they do.

We will watch each transformation deform the same object: a unit square and a set of equally-spaced grid lines.

The Transformations at a Glance

Name	Matrix	Effect
Identity	I	Nothing
Scaling	$\begin{bmatrix} s_x & 0 \\ 0 & s_y \end{bmatrix}$	Stretch along axes
Rotation by θ	$\begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}$	Rotate counter-clockwise
Reflection (x-axis)	$\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$	Flip vertically
Horizontal shear	$\begin{bmatrix} 1 & k \\ 0 & 1 \end{bmatrix}$	Slide rows horizontally
Projection onto x-axis	$\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$	Collapse to a line

Visualising All Six on One Figure

```
import numpy as np
import matplotlib.pyplot as plt
import matplotlib.patches as mpatches

theta = np.pi / 4 # 45 degrees

transforms = {
    'Identity': np.eye(2),
    'Scale (2x, 0.5x)': np.array([[2., 0.], [0., 0.5]]),
    'Rotation 45°': np.array([[np.cos(theta), -np.sin(theta)],
                               [np.sin(theta), np.cos(theta)]]),
    'Reflection (x)': np.array([[1., 0.], [0., -1.]])
}
```

```

    'Horiz. shear':  np.array([[1., 1.5],[0., 1.])),
    'Projection→x':  np.array([[1., 0.],[0., 0.])),
}

# Unit square vertices (closed polygon)
sq = np.array([[0,0],[1,0],[1,1],[0,1],[0,0]], dtype=float) # shape (5, 2)

# Grid lines in input space
lines_h = [np.array([[0, y],[1, y]]) for y in np.linspace(0, 1, 6)]
lines_v = [np.array([[x, 0],[x, 1]]) for x in np.linspace(0, 1, 6)]
grid_lines = lines_h + lines_v

fig, axes = plt.subplots(2, 3, figsize=(12, 7))
axes = axes.flatten()

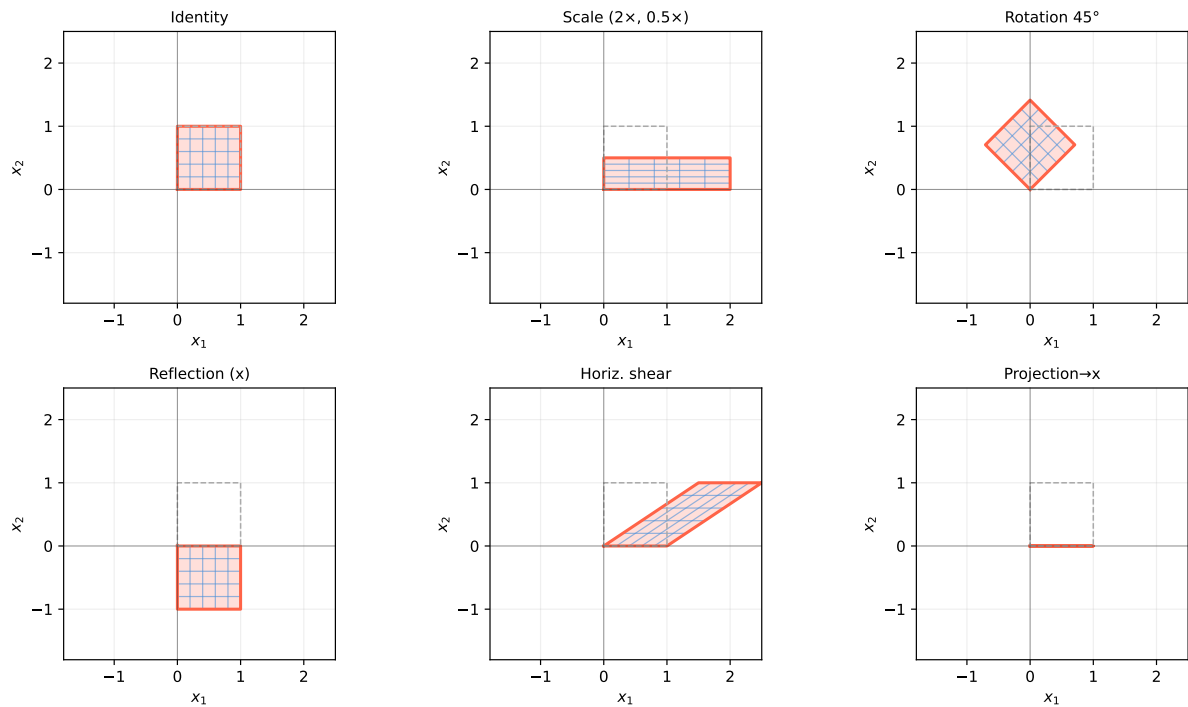
for ax, (name, A) in zip(axes, transforms.items()):
    # Transform grid
    for ln in grid_lines:
        lt = (A @ ln.T).T
        ax.plot(lt[:, 0], lt[:, 1], color='#4a90d9', lw=0.8, alpha=0.6)
    # Transform square outline
    sq_t = (A @ sq.T).T
    ax.fill(sq_t[:-1, 0], sq_t[:-1, 1], alpha=0.20, color='tomato')
    ax.plot(sq_t[:, 0], sq_t[:, 1], color='tomato', lw=2)
    # Original square (faint)
    ax.plot(sq[:, 0], sq[:, 1], color='#aaaaaa', lw=1, ls='--')

    ax.set_xlim(-1.8, 2.5); ax.set_ylim(-1.8, 2.5)
    ax.set_aspect('equal')
    ax.axhline(0, color='#333333', lw=0.4, alpha=0.5)
    ax.axvline(0, color='#333333', lw=0.4, alpha=0.5)
    ax.grid(True, alpha=0.2)
    ax.set_title(name, fontsize=10)
    ax.set_xlabel('$x_1$'); ax.set_ylabel('$x_2$')

fig.suptitle('Six fundamental 2D linear transformations', fontsize=13)
fig.tight_layout()
plt.show()

```

Six fundamental 2D linear transformations



What to Notice

Scaling stretches independently along each axis. If $s_x = s_y$ the square becomes a bigger or smaller square (uniform scale); if $s_x \neq s_y$ it becomes a rectangle.

Rotation preserves distances and angles — it is the prototype of an *isometry*. The determinant is $\cos^2 \theta + \sin^2 \theta = 1$.

Reflection also preserves distances, but it *reverses orientation* — the counter-clockwise order of vertices becomes clockwise. The determinant is -1 .

Shear keeps one axis fixed while sliding the other. A horizontal shear of k maps $\mathbf{e}_2 \mapsto [k, 1]^T$. The square tilts into a parallelogram but area is preserved ($\det = 1$).

Projection *destroys* information: the whole plane collapses onto a line. The determinant is 0, the transformation is not invertible.

Determinant = Signed Area Scale Factor

```
import numpy as np

transforms = {
    'Identity': np.eye(2),
    'Scale (2, 0.5)': np.array([[2., 0.], [0., 0.5]]),
    'Rotation 45°': np.array([[np.cos(np.pi/4), -np.sin(np.pi/4)],
                               [np.sin(np.pi/4), np.cos(np.pi/4)]]),
    'Reflection': np.array([[1., 0.], [0., -1.]]),
    'Horiz. shear (1.5)': np.array([[1., 1.5], [0., 1.]]),
    'Projection→x': np.array([[1., 0.], [0., 0.]]),
}

print(f"{'Transform':<25} det")
print("-" * 35)
for name, A in transforms.items():
    d = np.linalg.det(A)
    print(f"{'name':<25} {d:+.4f}")
```

Transform	det
Identity	+1.0000
Scale (2, 0.5)	+1.0000
Rotation 45°	+1.0000
Reflection	-1.0000
Horiz. shear (1.5)	+1.0000
Projection→x	+0.0000

The unit square has area 1. After transformation, the area becomes $|\det A|$. When $\det A < 0$, orientation is reversed (left-hand becomes right-hand). When $\det A = 0$, the square collapses to a lower-dimensional object — volume is destroyed and the map is not invertible.

Interactive: Build Your Own Transformation

```
import numpy as np
import matplotlib.pyplot as plt

def show_transform(A, title='Custom'):
    """Show what matrix A does to the unit square and a smiley grid."""
    sq = np.array([[0,0],[1,0],[1,1],[0,1],[0,0]], dtype=float)

    # A denser grid to show distortion
    s = np.linspace(0, 1, 12)
```

```

grid_pts = np.array([u, v] for u in s for v in s]) # shape (144, 2)
mapped    = (A @ grid_pts.T).T                    # shape (144, 2)

sq_t = (A @ sq.T).T
fig, axes = plt.subplots(1, 2, figsize=(9, 4))

axes[0].scatter(grid_pts[:,0], grid_pts[:,1], s=8, color='#4a90d9', alpha=0.6)
axes[0].plot(sq[:,0], sq[:,1], 'k-', lw=1.5)
axes[0].set_title('Input', fontsize=10)

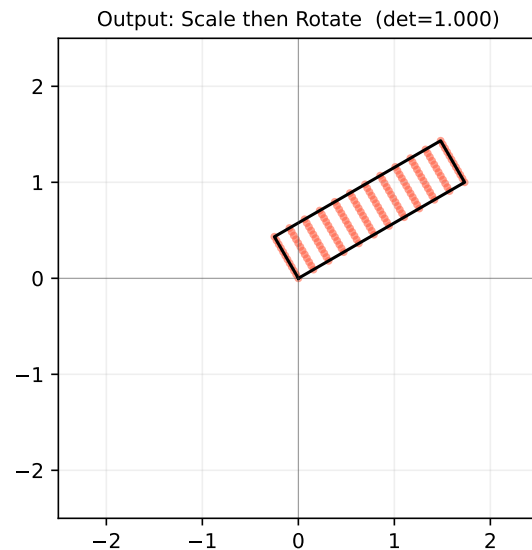
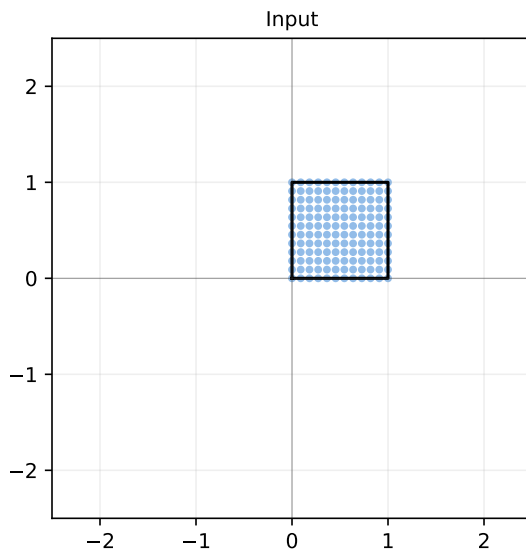
axes[1].scatter(mapped[:,0], mapped[:,1], s=8, color='tomato', alpha=0.6)
axes[1].plot(sq_t[:,0], sq_t[:,1], 'k-', lw=1.5)
axes[1].set_title(f'Output: {title} (det={np.linalg.det(A):.3f})', fontsize=10)

for ax in axes:
    ax.set_aspect('equal')
    ax.axhline(0, color='#333333', lw=0.4, alpha=0.5)
    ax.axvline(0, color='#333333', lw=0.4, alpha=0.5)
    ax.grid(True, alpha=0.2)
    ax.set_xlim(-2.5, 2.5); ax.set_ylim(-2.5, 2.5)

fig.tight_layout()
plt.show()

# Try a combined scale + rotation
theta = np.pi / 6
R = np.array([[np.cos(theta), -np.sin(theta)],
               [np.sin(theta),  np.cos(theta)]])
S = np.array([[2., 0.], [0., 0.5]])
show_transform(R @ S, title='Scale then Rotate')

```



Reading the Standard Matrix from Column Images

Section 6.1 showed that every linear map is determined by where it sends the basis vectors. This section turns that observation into a *recipe* for writing down any linear map as a matrix — without having to solve any system of equations.

The recipe: to find the matrix of $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$, apply T to each standard basis vector $\mathbf{e}_j$ and use the result as the j -th column of the matrix:

$$[T] = [T(\mathbf{e}_1) \quad T(\mathbf{e}_2) \quad \cdots \quad T(\mathbf{e}_n)]$$

The j -th column is the **image** of the j -th basis vector.

Why This Works

Any $\mathbf{x} \in \mathbb{R}^n$ decomposes as $\mathbf{x} = x_1\mathbf{e}_1 + \cdots + x_n\mathbf{e}_n$. Linearity gives:

$$T(\mathbf{x}) = x_1T(\mathbf{e}_1) + \cdots + x_nT(\mathbf{e}_n) = \underbrace{[T(\mathbf{e}_1) \mid \cdots \mid T(\mathbf{e}_n)]}_A \mathbf{x}$$

The matrix-vector product is *exactly* this weighted combination of columns.

Example 1: Rotation by 90°

Where does a counter-clockwise 90° rotation send the basis vectors?

$$T(\mathbf{e}_1) = T\begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}, \quad T(\mathbf{e}_2) = T\begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} -1 \\ 0 \end{pmatrix}$$

So the rotation matrix is:

$$R_{90} = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$$


```
import numpy as np

# 90-degree counter-clockwise rotation
theta = np.pi / 2
R90 = np.array([[np.cos(theta), -np.sin(theta)], # shape (2, 2)
                [np.sin(theta),  np.cos(theta)]])
R90 = R90.round(10) # clean up floating point noise

e1 = np.array([1., 0.]) # shape (2,)
e2 = np.array([0., 1.]) # shape (2,)

print("R90 ="); print(R90)
print(f"\nT(e1) = R90 @ e1 = {R90 @ e1} (column 1)")
print(f"T(e2) = R90 @ e2 = {R90 @ e2} (column 2)")
```

```
R90 =
[[ 0. -1.]
 [ 1.  0.]]

T(e1) = R90 @ e1 = [0.  1.] (column 1)
T(e2) = R90 @ e2 = [-1.  0.] (column 2)
```

Example 2: Projection onto the Line $y = x$

The projection of $\mathbf{v}$ onto the line $y = x$ has direction $\hat{\mathbf{u}} = \frac{1}{\sqrt{2}}[1, 1]^T$. Using the formula $P = \hat{\mathbf{u}}\hat{\mathbf{u}}^T$:

$$P = \frac{1}{2} \begin{bmatrix} 1 \\ 1 \end{bmatrix} \begin{bmatrix} 1 & 1 \end{bmatrix} = \frac{1}{2} \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$$

Let us verify via the column-image recipe:

```
import numpy as np
import matplotlib.pyplot as plt

# Projection matrix onto y = x
u_hat = np.array([1., 1.]) / np.sqrt(2) # shape (2,)
P = np.outer(u_hat, u_hat) # shape (2, 2)
print("P ="); print(P.round(4))

e1 = np.array([1., 0.])
e2 = np.array([0., 1.])
print(f"\nT(e1) = P @ e1 = {P @ e1} (column 1 of P)")
```

```

print(f"T(e2) = P @ e2 = {P @ e2} (column 2 of P)")

# Visualise
fig, ax = plt.subplots(figsize=(5, 5))
t = np.linspace(-1.5, 1.5, 100)
ax.plot(t, t, color='#4a90d9', lw=1.5, ls='--', label=r'Line $y=x$')

for v, col, label in [(e1, 'tomato', r'$\mathbf{e}_1$'),
                      (e2, '#2ecc71', r'$\mathbf{e}_2$')]:
    pv = P @ v
    ax.annotate('', xy=v, xytext=[0,0],
                arrowprops=dict(arrowstyle='->', color=col, lw=2))
    ax.annotate('', xy=pv, xytext=[0,0],
                arrowprops=dict(arrowstyle='->', color=col, lw=2, ls='dashed'))
    ax.text(v[0]+0.05, v[1]+0.05, label, color=col, fontsize=11)
    ax.text(pv[0]+0.05, pv[1]-0.15, r'$P$'+label, color=col, fontsize=9)

ax.set_xlim(-0.3, 1.4); ax.set_ylim(-0.3, 1.4)
ax.set_aspect('equal')
ax.axhline(0, color='#333333', lw=0.4, alpha=0.5)
ax.axvline(0, color='#333333', lw=0.4, alpha=0.5)
ax.grid(True, alpha=0.2)
ax.set_xlabel('$x_1$'); ax.set_ylabel('$x_2$')
ax.set_title(r'Column images of the projection onto $y = x$')
ax.legend(fontsize=9)
plt.tight_layout()
plt.show()

```

```

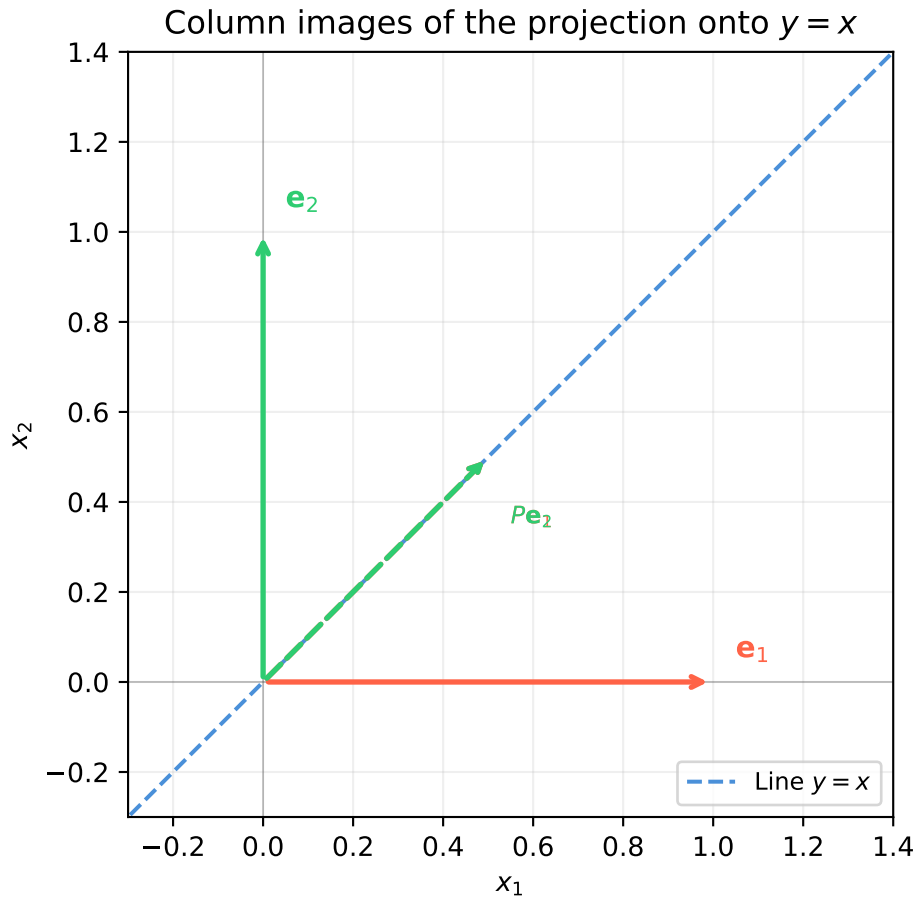
P =
[[0.5 0.5]
 [0.5 0.5]]

```

```

T(e1) = P @ e1 = [0.5 0.5] (column 1 of P)
T(e2) = P @ e2 = [0.5 0.5] (column 2 of P)

```



Example 3: Deriving an Unknown Map

Suppose you observe that a linear map $T : \mathbb{R}^2 \rightarrow \mathbb{R}^3$ satisfies $T([1, 0]^T) = [2, -1, 0]^T$ and $T([0, 1]^T) = [1, 3, -2]^T$. Write down its matrix immediately:

```
import numpy as np

# Observed basis images
T_e1 = np.array([2., -1., 0.]) # shape (3,)
T_e2 = np.array([1., 3., -2.]) # shape (3,)

# Standard matrix: stack images as columns
A = np.column_stack([T_e1, T_e2]) # shape (3, 2)
print("Standard matrix A ="); print(A)

# Verify: what does T do to an arbitrary vector x = [3, -1]?
x = np.array([3., -1.]) # shape (2,)
```

```
print(f"\nT({x}) = A @ x = {A @ x}")
print(f"Manual: 3*T(e1) + (-1)*T(e2) = {3*T_e1 + (-1)*T_e2}")
```

Standard matrix A =

```
[[ 2.  1.]
 [-1.  3.]
 [ 0. -2.]]
```

$T([3, -1]) = A @ x = [5, -6, 2]$

Manual: $3*T(e1) + (-1)*T(e2) = [5, -6, 2]$

The Recipe in Reverse: Reading a Matrix

Given a matrix A , you can immediately state its action on each basis vector. For example:

$$A = \begin{bmatrix} 3 & 0 \\ 1 & 2 \end{bmatrix}$$

- Column 1 = $[3, 1]^T$: the x-axis gets mapped to the vector (3, 1).
- Column 2 = $[0, 2]^T$: the y-axis gets mapped straight up, scaled by 2.

```
import numpy as np
import matplotlib.pyplot as plt

A = np.array([[3., 0.],
              [1., 2.]]) # shape (2, 2)

e1 = np.array([1., 0.])
e2 = np.array([0., 1.])

fig, ax = plt.subplots(figsize=(5, 5))

vecs = {'$\mathbf{e}_1$': (e1, '#4a90d9'),
        '$\mathbf{e}_2$': (e2, '#2ecc71'),
        '$A\mathbf{e}_1$': (A @ e1, 'tomato'),
        '$A\mathbf{e}_2$': (A @ e2, '#e67e22')}

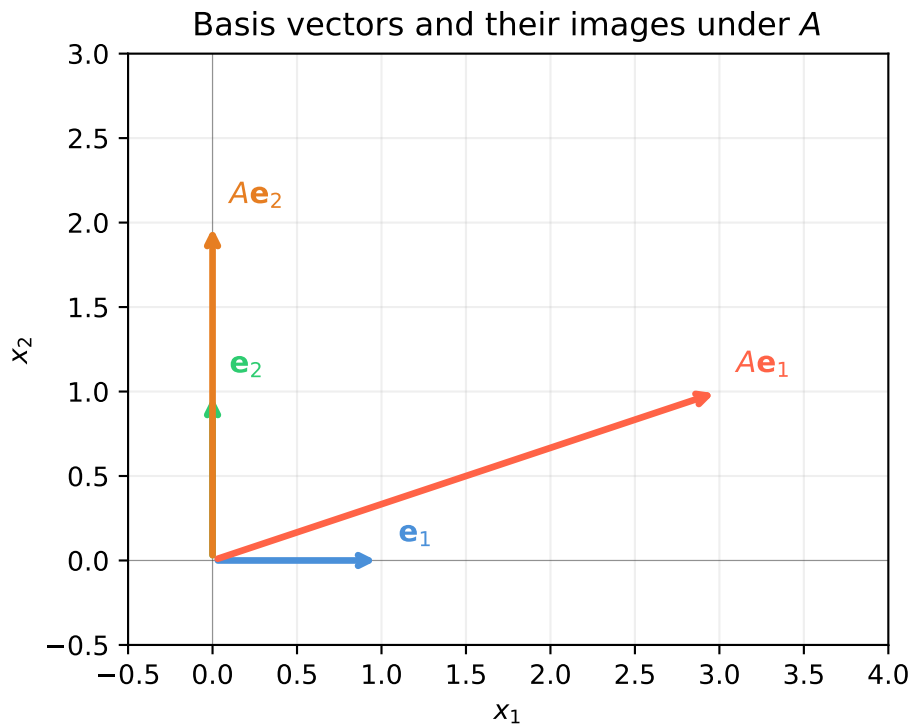
for label, (v, col) in vecs.items():
    ax.annotate('', xy=v, xytext=[0,0],
                arrowprops=dict(arrowstyle='->', color=col, lw=2.5))
    ax.text(v[0]+0.1, v[1]+0.1, label, color=col, fontsize=11)

ax.set_xlim(-0.5, 4); ax.set_ylim(-0.5, 3)
```

```

ax.set_aspect('equal')
ax.axhline(0, color='#333333', lw=0.4, alpha=0.5)
ax.axvline(0, color='#333333', lw=0.4, alpha=0.5)
ax.grid(True, alpha=0.2)
ax.set_xlabel('$x_1$'); ax.set_ylabel('$x_2$')
ax.set_title('Basis vectors and their images under $A$')
plt.tight_layout()
plt.show()

```



Summary

Step	What you do
1	Apply T to each e_j to get $T(e_j)$
2	Use $T(e_j)$ as the j -th column of $[T]$
3	Check: $[T]\mathbf{x}$ should equal the direct computation $T(\mathbf{x})$

This is more than a trick — it is the *definition* of the standard matrix. Every $m \times n$ matrix encodes a linear map from $\mathbb{R}^n$ to $\mathbb{R}^m$, and every such linear map has exactly one such matrix.

Kernel, Image, and the Dimension Theorem

Two subspaces are attached to every linear map $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$:

- **Kernel** (null space): inputs that T maps to zero — $\ker T = \{\mathbf{x} : T(\mathbf{x}) = \mathbf{0}\}$.
- **Image** (column space): all possible outputs — $\operatorname{im} T = \{T(\mathbf{x}) : \mathbf{x} \in \mathbb{R}^n\}$.

These two subspaces are linked by a beautifully simple equation.

The Rank-Nullity Theorem

$$\dim(\ker T) + \dim(\operatorname{im} T) = n$$

In plain language: the dimension of the input space equals the number of dimensions “killed” (kernel) plus the number “produced” (image). You can’t create more output dimensions than you have input dimensions, and any dimension lost to the kernel is exactly one dimension missing from the image.

For a matrix A of shape (m, n) :

$$\underbrace{\operatorname{nullity}(A)}_{\dim \ker A} + \underbrace{\operatorname{rank}(A)}_{\dim \operatorname{im} A} = n$$

Visualising the Theorem

```
import numpy as np

# A 3x4 matrix of rank 2
A = np.array([[1., 2., 3., 4.],
              [2., 4., 6., 8.],
              [0., 1., 1., 2.]]) # shape (3, 4)

rank = np.linalg.matrix_rank(A)
nullity = A.shape[1] - rank

print(f"Shape:   {A.shape}   →   n = {A.shape[1]}")
print(f"rank(A)  = {rank}     (dim of image)")
print(f"nullity   = {nullity}   (dim of kernel)")
print(f"rank + nullity = {rank + nullity}   (= n = {A.shape[1]})")

# Compute null space via SVD
U, s, Vt = np.linalg.svd(A)
```

```

print(f"\nSingular values: {s.round(6)}")
null_basis = Vt[rank:]      # shape (nullity, 4)
print(f"\nNull space basis vectors (rows of Vt[rank:]):")
print(null_basis.round(6))
for i, v in enumerate(null_basis):
    print(f"  A @ null_vec_{i+1} = {(A @ v).round(10)}")

```

```

Shape:   (3, 4)  →  n = 4
rank(A)  = 2    (dim of image)
nullity  = 2    (dim of kernel)
rank + nullity = 4  (= n = 4)

```

```

Singular values: [12.475842  0.594445  0.          ]

```

```

Null space basis vectors (rows of Vt[rank:]):
[[ 0.674156  0.124619 -0.674156  0.274768]
 [-0.007697 -0.895908  0.007697  0.444105]]
A @ null_vec_1 = [0. 0. 0.]
A @ null_vec_2 = [0. 0. 0.]

```

The Kernel: What Gets Annihilated?

The kernel tells you when two different inputs give the same output.

If $\mathbf{x}_p$ is *any* solution of $A\mathbf{x} = \mathbf{b}$, then the **complete solution** is:

$$\mathbf{x} = \mathbf{x}_p + \mathbf{x}_h, \quad \mathbf{x}_h \in \ker A$$

Any vector in the kernel can be added “for free” without changing the output. This is why $A\mathbf{x} = \mathbf{b}$ has infinitely many solutions when the kernel is non-trivial — you have free parameters.

```

import numpy as np
import matplotlib.pyplot as plt

# 2x3 matrix: maps R^3 -> R^2, rank 2, nullity 1
A = np.array([[1., 0., 1.],
              [0., 1., 1.]]) # shape (2, 3)

b = np.array([2., 3.]) # shape (2,)
xp, _, _, _ = np.linalg.lstsq(A, b, rcond=None)

# Null space vector (SVD)
_, s, Vt = np.linalg.svd(A)

```

```

null_vec = Vt[-1]    # shape (3,)

print(f"Particular solution: xp = {xp.round(4)}")
print(f"Null space vector:   n  = {null_vec.round(4)}")
print(f"A @ xp = {A @ xp}   (= b = {b})")
print(f"A @ null_vec = {(A @ null_vec).round(10)}   (= 0)")

# Show family of solutions
t_vals = np.linspace(-3, 3, 7)
print(f"\nFamily x = xp + t*null_vec, A @ x:")
for t in t_vals:
    x = xp + t * null_vec
    print(f"   t={t:+.1f} → x={x.round(3)},   A@x={np.round(A@x,3)}")

```

```

Particular solution: xp = [0.3333 1.3333 1.6667]
Null space vector:   n  = [-0.5774 -0.5774  0.5774]
A @ xp = [2. 3.]   (= b = [2. 3.])
A @ null_vec = [0. 0.]   (= 0)

```

```

Family x = xp + t*null_vec, A @ x:
t=-3.0 → x=[ 2.065  3.065 -0.065],   A@x=[2. 3.]
t=-2.0 → x=[1.488  2.488  0.512],   A@x=[2. 3.]
t=-1.0 → x=[0.911  1.911  1.089],   A@x=[2. 3.]
t=+0.0 → x=[0.333  1.333  1.667],   A@x=[2. 3.]
t=+1.0 → x=[-0.244  0.756  2.244],   A@x=[2. 3.]
t=+2.0 → x=[-0.821  0.179  2.821],   A@x=[2. 3.]
t=+3.0 → x=[-1.399 -0.399  3.399],   A@x=[2. 3.]

```

The Image: What Can Be Reached?

The image is the column space of A . A vector $\mathbf{b}$ can be reached by T if and only if $\mathbf{b} \in \text{im } T$.

```

import numpy as np

A = np.array([[1., 0., 1.],
              [0., 1., 1.]])    # shape (2, 3), rank 2

# The image is all of R^2 since rank = m = 2
# Every b in R^2 is reachable
b_test = np.array([5., -2.])    # shape (2,)
x, _, _, _ = np.linalg.lstsq(A, b_test, rcond=None)
residual = np.linalg.norm(A @ x - b_test)
print(f"b = {b_test}")

```



```

print(f"Residual after lstsq: {residual:.2e} → b in im(A)? {'YES' if residual<1e-10 else 'NO'}")

# Now try a rank-deficient matrix: image is a proper subspace of R^3
B = np.array([[1., 2.],
               [2., 4.],
               [3., 6.]]) # shape (3, 2), rank 1

b_in = np.array([2., 4., 6.]) # = 2 * col1, in im(B)
b_out = np.array([1., 1., 1.]) # not in im(B)

for label, bv in [("b_in ", b_in), ("b_out", b_out)]:
    xv, _, _, _ = np.linalg.lstsq(B, bv, rcond=None)
    res = np.linalg.norm(B @ xv - bv)
    print(f"\n{label} = {bv}: residual = {res:.2e} → in im(B)? {'YES' if res<1e-10 else 'NO'}")

```

```

b = [ 5. -2.]
Residual after lstsq: 4.44e-16 → b in im(A)? YES

b_in = [2. 4. 6.]: residual = 4.97e-16 → in im(B)? YES

b_out = [1. 1. 1.]: residual = 6.55e-01 → in im(B)? NO

```

One-to-One, Onto, and Invertibility

Condition	Algebraic	Geometric
One-to-one (injective)	$\ker T = \{\mathbf{0}\}$	No two inputs share an output
Onto (surjective)	$\operatorname{im} T = \mathbb{R}^n$	Every output is reachable
Invertible (bijective)	Both	Unique input for every output

For a **square** matrix ($n = m$): $-\ker T = \{\mathbf{0}\} \iff \operatorname{im} T = \mathbb{R}^n \iff \det A \neq 0 \iff A$ is invertible.

These are all the same thing for square matrices.

```

import numpy as np

cases = {
    'Full rank (invertible)': np.array([[2., 1.], [1., 3.]]),
    'Rank 1 (singular)':      np.array([[1., 2.], [2., 4.]]),
    'Tall (3x2, injective)':  np.array([[1., 0.], [0., 1.], [1., 1.]]),
    'Wide (2x3, onto)':      np.array([[1., 0., 1.], [0., 1., 1.]])
}

```

```

}

print(f"{'Matrix':<30}  shape  rank  nullity  det")
print("-"*60)
for name, A in cases.items():
    r = np.linalg.matrix_rank(A)
    nul = A.shape[1] - r
    det = np.linalg.det(A) if A.shape[0]==A.shape[1] else float('nan')
    print(f"{'name':<30}  {'str(A.shape):<8} {r}      {nul}      {det:+.2f}" if not np.isnan(det)
          else f"{'name':<30}  {'str(A.shape):<8} {r}      {nul}      n/a")

```

Matrix	shape	rank	nullity	det
Full rank (invertible)	(2, 2)	2	0	+5.00
Rank 1 (singular)	(2, 2)	1	1	+0.00
Tall (3x2, injective)	(3, 2)	2	0	n/a
Wide (2x3, onto)	(2, 3)	2	1	n/a

Why This Matters in Practice

Machine learning: if a weight matrix has a non-trivial null space, different weight vectors produce identical predictions — the model is over-parameterised.

Computer vision: the null space of the camera matrix reveals the camera centre (the 3D point that maps to every image point — Section 10.3).

Robotics: if a robot's Jacobian has a non-trivial null space, there exist joint-velocity combinations that produce zero end-effector velocity — redundant degrees of freedom.

Composition = Matrix Multiplication

Applying two linear maps in sequence is another linear map. The matrix of the composed map is the *product* of the individual matrices. This is not just a notational convenience — it is why matrix multiplication is defined the way it is.

Theorem: If $S : \mathbb{R}^n \rightarrow \mathbb{R}^m$ has matrix A and $T : \mathbb{R}^m \rightarrow \mathbb{R}^p$ has matrix B , then $T \circ S : \mathbb{R}^n \rightarrow \mathbb{R}^p$ has matrix BA .

Note the order: the *rightmost* matrix is applied first.

Why Multiplication Is Defined This Way

Apply S then T to a vector $\mathbf{x}$:

$$T(S(\mathbf{x})) = B(A\mathbf{x}) = (BA)\mathbf{x}$$

Matrix multiplication is associativity of function composition. Every rule about matrix products follows from this.

Order Is Critical

$BA \neq AB$ in general. “Rotate then shear” is a different transformation from “shear then rotate.”

```
import numpy as np
import matplotlib.pyplot as plt

theta = np.pi / 4
R = np.array([[np.cos(theta), -np.sin(theta)], # shape (2, 2)
              [np.sin(theta),  np.cos(theta)]])
H = np.array([[1., 1.5], # shape (2, 2)
              [0., 1.]]) # horizontal shear

RS = R @ H # shear first, then rotate shape (2, 2)
SR = H @ R # rotate first, then shear shape (2, 2)

print("R @ H (shear → rotate):\n", RS.round(4))
print("\nH @ R (rotate → shear):\n", SR.round(4))
print("\nAre they equal?", np.allclose(RS, SR))

# Visualise the difference
sq = np.array([[0,0],[1,0],[1,1],[0,1],[0,0]], dtype=float)

fig, axes = plt.subplots(1, 3, figsize=(12, 4))

for ax, M, title, col in [
    (axes[0], np.eye(2), 'Original', '#4a90d9'),
    (axes[1], RS, 'Shear → Rotate (RH)', 'tomato'),
    (axes[2], SR, 'Rotate → Shear (HR)', '#2ecc71')]:
    pts = (M @ sq.T).T
    ax.fill(pts[:-1,0], pts[:-1,1], alpha=0.3, color=col)
    ax.plot(pts[:,0], pts[:,1], color=col, lw=2)
    ax.set_xlim(-2, 2.5); ax.set_ylim(-1.5, 2.5)
    ax.set_aspect('equal')
```

```
ax.axhline(0, color='#333333', lw=0.4, alpha=0.5)
ax.axvline(0, color='#333333', lw=0.4, alpha=0.5)
ax.grid(True, alpha=0.2)
ax.set_title(title, fontsize=10)
ax.set_xlabel('$x_1$'); ax.set_ylabel('$x_2$')

fig.suptitle('Order matters: $RH \neq HR$', fontsize=12)
fig.tight_layout()
plt.show()
```

R @ H (shear → rotate):

```
[[0.7071 0.3536]
 [0.7071 1.7678]]
```

H @ R (rotate → shear):

```
[[1.7678 0.3536]
 [0.7071 0.7071]]
```

Are they equal? False



Building Transformations from Primitives

Complex transformations are composed from simple ones. For example, rotation about an arbitrary point $\mathbf{p}$ in 2D requires three steps:

1. Translate so $\mathbf{p}$ moves to the origin (this requires homogeneous coordinates — Chapter 7, but we preview the idea here).
2. Rotate by θ .
3. Translate back.

Using only linear maps, we can build the rotation about the origin:

```
import numpy as np

def rotation(theta):
    """2x2 counter-clockwise rotation matrix."""
    c, s = np.cos(theta), np.sin(theta)
    return np.array([[c, -s],
                     [s,  c]])    # shape (2, 2)

def scaling(sx, sy):
    return np.array([[sx, 0.],
                     [0., sy]])    # shape (2, 2)

def shear_x(k):
    return np.array([[1., k],
                     [0., 1.]])    # shape (2, 2)

# Example: scale, then rotate, then shear
R = rotation(np.pi / 3)
S = scaling(2.0, 0.5)
Hx = shear_x(0.5)

# Composed: shear(rotate(scale(x)))
M = Hx @ R @ S    # shape (2, 2)
print("Composed M = Hx @ R @ S:")
print(M.round(4))
print(f"\ndet(Hx) * det(R) * det(S) = {np.linalg.det(Hx):.4f} * {np.linalg.det(R):.4f} * {n")
print(f"det(M) directly = {np.linalg.det(M):.4f}")
```

Composed M = Hx @ R @ S:

```
[[ 1.866  -0.308 ]
 [ 1.7321  0.25  ]]
```

$\det(Hx) * \det(R) * \det(S) = 1.0000 * 1.0000 * 1.0000 = 1.0000$

$\det(M) \text{ directly} = 1.0000$

The determinant of a product equals the product of determinants: $\det(BA) = \det B \cdot \det A$.

Powers of a Transformation

Applying T to itself k times gives T^k , represented by A^k :

```
import numpy as np
import matplotlib.pyplot as plt

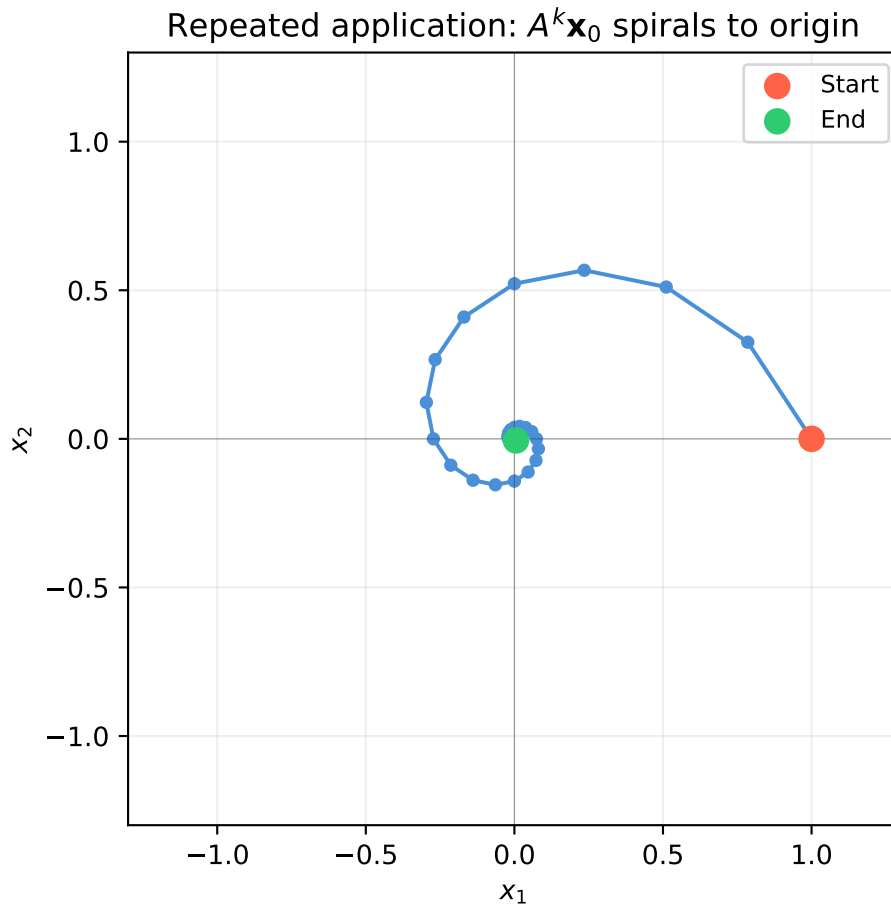
# A "spiral" transformation: slight rotation + slight shrink
theta = np.pi / 8
A = 0.85 * np.array([[np.cos(theta), -np.sin(theta)], # shape (2, 2)
                    [np.sin(theta),  np.cos(theta)]])

# Track a point under repeated application
x0 = np.array([1., 0.]) # shape (2,)
trajectory = [x0.copy()]

x = x0.copy()
for _ in range(30):
    x = A @ x
    trajectory.append(x.copy())

traj = np.array(trajectory) # shape (31, 2)

fig, ax = plt.subplots(figsize=(5, 5))
ax.plot(traj[:, 0], traj[:, 1], 'o-', color='#4a90d9', ms=4, lw=1.5)
ax.scatter(traj[0, 0], traj[0, 1], color='tomato', s=80, zorder=5, label='Start')
ax.scatter(traj[-1, 0], traj[-1, 1], color='#2ecc71', s=80, zorder=5, label='End')
ax.set_xlim(-1.3, 1.3); ax.set_ylim(-1.3, 1.3)
ax.set_aspect('equal')
ax.axhline(0, color='#333333', lw=0.4, alpha=0.5)
ax.axvline(0, color='#333333', lw=0.4, alpha=0.5)
ax.grid(True, alpha=0.2)
ax.set_xlabel('$x_1$'); ax.set_ylabel('$x_2$')
ax.set_title('Repeated application:  $A^k \mathbf{x}_0$  spirals to origin')
ax.legend(fontsize=9)
plt.tight_layout()
plt.show()
```



The spiral converges to the origin because both eigenvalues of A have magnitude $0.85 < 1$. Eigenvalues govern the long-run behaviour of iterated maps — the subject of Chapter 13.

Invertibility and the Inverse Map

If A is invertible, the inverse transformation T^{-1} has matrix A^{-1} . Composing them gives the identity:

$$A^{-1}A = AA^{-1} = I$$

For composed maps: $(BA)^{-1} = A^{-1}B^{-1}$ — reverse the order.

```
import numpy as np

A = np.array([[3., 1.],
              [2., 4.]])    # shape (2, 2)
B = np.array([[1., 2.]])
```

```

        [0., 1.])) # shape (2, 2)

BA    = B @ A
BA_inv = np.linalg.solve(BA, np.eye(2)) # (BA)^{-1}
check  = np.linalg.solve(A, np.eye(2)) @ np.linalg.solve(B, np.eye(2)) # A^{-1} B^{-1}

print("(BA)^{-1}:")
print(BA_inv.round(6))
print("\nA^{-1} B^{-1}:")
print(check.round(6))
print("\nEqual?", np.allclose(BA_inv, check))

```

(BA)⁻¹:

```
[[ 0.4 -0.9]
 [-0.2  0.7]]
```

A⁻¹ B⁻¹:

```
[[ 0.4 -0.9]
 [-0.2  0.7]]
```

Equal? True

Change of Basis: $[T]_{\mathcal{B}} = P^{-1}[T]_{\mathcal{E}}P$

The same linear transformation can be represented by *different matrices* depending on which coordinate system you choose. Choosing a good coordinate system can turn a complicated matrix into a diagonal one — that is the entire idea behind diagonalisation and PCA.

The Setup

Let $\mathcal{E}$ be the standard basis $\{\mathbf{e}_1, \dots, \mathbf{e}_n\}$ and $\mathcal{B} = \{\mathbf{b}_1, \dots, \mathbf{b}_n\}$ be a new basis. Define the **change-of-basis matrix**:

$$P = [\mathbf{b}_1 \quad \mathbf{b}_2 \quad \dots \quad \mathbf{b}_n]$$

The columns of P are the new basis vectors expressed in the old (standard) coordinates.

How coordinates transform: - $\mathcal{B}$ -coords $\rightarrow$ $\mathcal{E}$ -coords: multiply by P . - $\mathcal{E}$ -coords $\rightarrow$ $\mathcal{B}$ -coords: multiply by P^{-1} .

The Similarity Formula

If $A = [T]_{\mathcal{E}}$ is the matrix of T in standard coordinates, then the matrix of the *same* transformation in basis $\mathcal{B}$ is:

$$[T]_{\mathcal{B}} = P^{-1}AP$$

This is called a **similarity transformation**. The two matrices A and $P^{-1}AP$ are **similar** — they represent the same geometric object viewed from different coordinate systems.

```
import numpy as np

# Transformation T: standard matrix in E-coords
A = np.array([[3., 1.],
              [1., 3.]]) # shape (2, 2)

# New basis B: columns are the new basis vectors
b1 = np.array([1., 1.]) / np.sqrt(2) # shape (2,)
b2 = np.array([-1., 1.]) / np.sqrt(2) # shape (2,)
P = np.column_stack([b1, b2]) # shape (2, 2)

# Matrix of T in B-coords
T_B = np.linalg.solve(P, A @ P) # = P^{-1} A P, shape (2, 2)

print("A = [T]_E:")
print(A)
print("\nP (new basis vectors as columns):")
print(P.round(6))
print("\n[T]_B = P^{-1} A P:")
print(T_B.round(6))
```

```
A = [T]_E:
[[3. 1.]
 [1. 3.]

P (new basis vectors as columns):
[[ 0.707107 -0.707107]
 [ 0.707107  0.707107]]

[T]_B = P^{-1} A P:
[[4. 0.]
 [0. 2.]
```

The basis $\mathcal{B}$ was chosen to be the **eigenvectors** of A . In that basis, A becomes diagonal — the eigenvalues sit on the diagonal and the off-diagonal entries vanish. This is not a coincidence (Chapter 14).

Geometric Interpretation

```
import numpy as np
import matplotlib.pyplot as plt

# A = shear + scaling
A = np.array([[2., 1.],
              [0., 1.]]) # shape (2, 2)

# A custom basis (not eigenvectors here - just illustration)
P = np.array([[1., 0.5],
              [0., 1.]]) # shape (2, 2)

T_B = np.linalg.solve(P, A @ P) # shape (2, 2)

# Grid in E-coords, then transform by A
s = np.linspace(-1.5, 1.5, 10)
grid_pts = np.array([u, v] for u in s for v in s) # shape (100, 2)

fig, axes = plt.subplots(1, 3, figsize=(13, 4))

def draw_grid(ax, pts, title, col):
    ax.scatter(pts[:,0], pts[:,1], s=10, color=col, alpha=0.5)
    ax.set_aspect('equal')
    ax.axhline(0, color='#333333', lw=0.4, alpha=0.5)
    ax.axvline(0, color='#333333', lw=0.4, alpha=0.5)
    ax.grid(True, alpha=0.2)
    ax.set_xlim(-4, 4); ax.set_ylim(-3, 3)
    ax.set_title(title, fontsize=9)

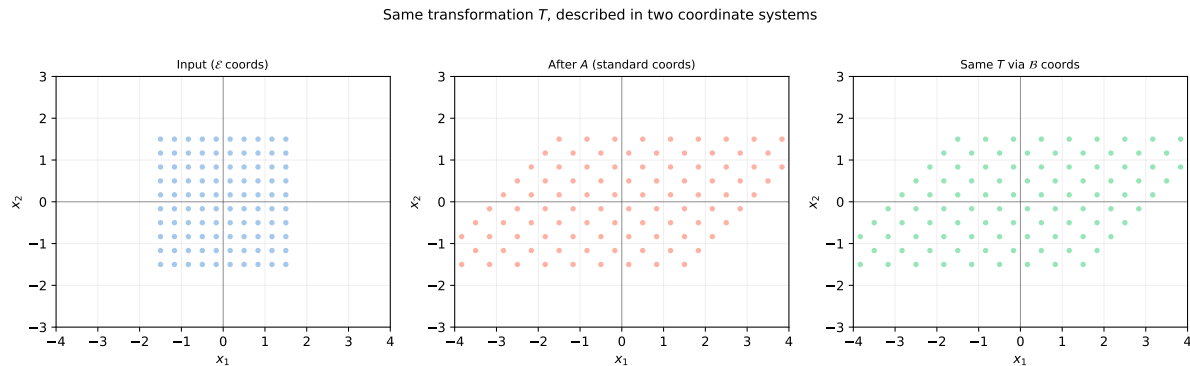
# Standard coords
draw_grid(axes[0], grid_pts, 'Input ( $\mathcal{E}$  coords)', '#4a90d9')

# Applied T in standard coords
out_E = (A @ grid_pts.T).T
draw_grid(axes[1], out_E, 'After  $A$  (standard coords)', 'tomato')

# Transform to B-coords, apply T_B, transform back
P_inv = np.linalg.inv(P)
grid_B = (P_inv @ grid_pts.T).T # to B-coords
out_B = (T_B @ grid_B.T).T # apply T_B
out_back = (P @ out_B.T).T # back to E-coords
draw_grid(axes[2], out_back, 'Same  $T$  via  $\mathcal{B}$  coords', '#2ecc71')
```

```
axes[0].set_xlabel('$x_1$'); axes[0].set_ylabel('$x_2$')
for ax in axes[1:]:
    ax.set_xlabel('$x_1$'); ax.set_ylabel('$x_2$')

fig.suptitle(r'Same transformation  $T$ , described in two coordinate systems', fontsize=11)
fig.tight_layout()
plt.show()
```



All three representations describe *exactly the same geometric action* on the plane. The output of panel 2 and panel 3 are identical — only the path taken to compute it differs.

What Is Preserved Under Similarity?

Similar matrices share key invariants:

Invariant	Why it's preserved
Eigenvalues	$\det(P^{-1}AP - \lambda I) = \det(A - \lambda I)$
Determinant	$\det(P^{-1}AP) = \det(A)$
Trace	$\text{tr}(P^{-1}AP) = \text{tr}(A)$
Rank	Column space dimension is basis-independent

```
import numpy as np

A = np.array([[3., 1.],
              [1., 3.]]) # shape (2, 2)

rng = np.random.default_rng(7)
P = rng.standard_normal((2, 2)) # shape (2, 2), arbitrary invertible matrix
T_B = np.linalg.solve(P, A @ P)

print(f"det(A) = {np.linalg.det(A):.6f}")
print(f"det(T_B) = {np.linalg.det(T_B):.6f}")
```

```
print(f"\ntrace(A)    = {np.trace(A):.6f}")
print(f"trace(T_B) = {np.trace(T_B):.6f}")

eig_A = np.sort(np.linalg.eigvals(A).real)
eig_TB = np.sort(np.linalg.eigvals(T_B).real)
print(f"\nEigenvalues of A:    {eig_A.round(6)}")
print(f"Eigenvalues of T_B: {eig_TB.round(6)}")
```

```
det(A)    = 8.000000
```

```
det(T_B) = 8.000000
```

```
trace(A)    = 6.000000
```

```
trace(T_B) = 6.000000
```

```
Eigenvalues of A:    [2. 4.]
```

```
Eigenvalues of T_B: [2. 4.]
```

Why Change of Basis Matters

Diagonalisation: if P is built from eigenvectors, $P^{-1}AP$ is diagonal. Computing A^k then becomes trivial: just raise the diagonal entries to the k -th power (Chapter 14).

PCA: the principal components *are* a change of basis that aligns coordinates with directions of maximum variance (Chapter 20).

Graphics and robotics: every “frame” (body frame, world frame, camera frame) is a basis. Transforming between frames is a change of basis (Chapters 9–10).

Application: Image Warping

Every 2D linear transformation can be applied directly to a pixel grid to produce a warped image. This is the basis of image rectification in computer vision, texture mapping in graphics, and data augmentation in deep learning.

We will apply each transformation from §6.2 to a test image and visualise the result using `scipy.ndimage` (which handles the pixel interpolation).

Why Image Warping Is a Linear Transformation

An image is a function $I : \mathbb{Z}^2 \rightarrow \mathbb{R}$ (or $\mathbb{R}^3$ for colour) mapping pixel coordinates to intensity values.

To warp an image by a 2×2 matrix A :

1. For each output pixel location $\mathbf{q}$, compute where it came from: $\mathbf{p} = A^{-1}\mathbf{q}$ (**inverse mapping**).
2. Read the intensity at $\mathbf{p}$ (with bilinear interpolation for sub-pixel locations).
3. Assign that intensity to $\mathbf{q}$.

We always invert and pull from the source — “push” approaches leave holes.

Building a Test Image

```
import numpy as np
import matplotlib.pyplot as plt

def make_test_image(size=200):
    """Checkerboard + circle test pattern."""
    rng = np.random.default_rng(0)
    img = np.zeros((size, size), dtype=float)

    # Checkerboard
    tile = 20
    for r in range(0, size, tile):
        for c in range(0, size, tile):
            if ((r // tile) + (c // tile)) % 2 == 0:
                img[r:r+tile, c:c+tile] = 0.8

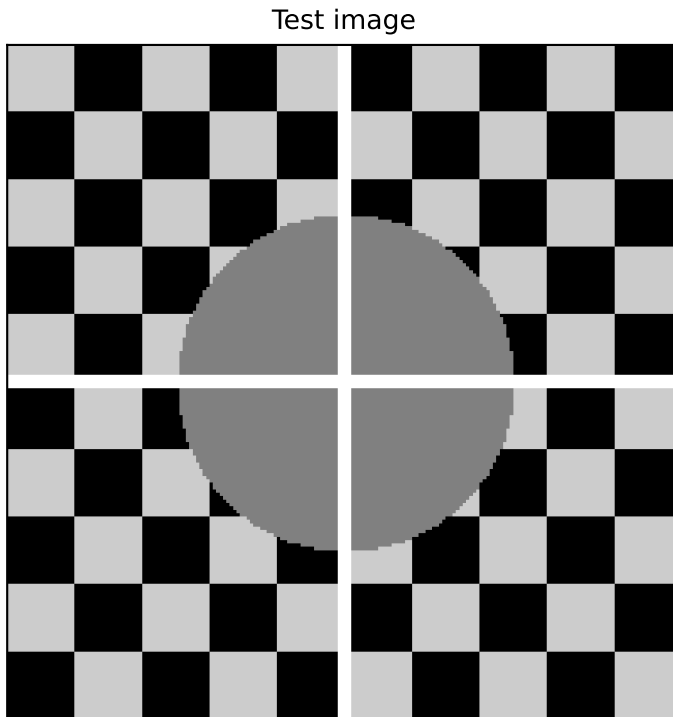
    # Circle
    cy, cx = size // 2, size // 2
    y, x = np.ogrid[:size, :size]
    mask = (x - cx)**2 + (y - cy)**2 < (size // 4)**2
    img[mask] = 0.5

    # Cross hair
    img[cy-2:cy+2, :] = 1.0
    img[:, cx-2:cx+2] = 1.0

    return img

img = make_test_image()
fig, ax = plt.subplots(figsize=(4, 4))
```

```
ax.imshow(img, cmap='gray', vmin=0, vmax=1, origin='upper')
ax.set_title('Test image', fontsize=10)
ax.set_xticks([]); ax.set_yticks([])
plt.tight_layout()
plt.show()
```



Warping by a 2D Linear Transformation

```
import numpy as np
import matplotlib.pyplot as plt
from scipy.ndimage import affine_transform

def make_test_image(size=200):
    img = np.zeros((size, size), dtype=float)
    tile = 20
    for r in range(0, size, tile):
        for c in range(0, size, tile):
            if ((r // tile) + (c // tile)) % 2 == 0:
                img[r:r+tile, c:c+tile] = 0.8
    cy, cx = size // 2, size // 2
    y, x = np.ogrid[:size, :size]
```

```

    mask = (x - cx)**2 + (y - cy)**2 < (size // 4)**2
    img[mask] = 0.5
    img[cy-2:cy+2, :] = 1.0
    img[:, cx-2:cx+2] = 1.0
    return img

img = make_test_image(200)
size = img.shape[0]
center = size / 2

theta = np.pi / 6

transforms = {
    'Original':      np.eye(2),
    'Rotation 30°':  np.array([[np.cos(theta), -np.sin(theta)],
                               [np.sin(theta),  np.cos(theta)]]),
    'Scale (1.5, 0.7)': np.array([[1.5, 0.], [0., 0.7]]),
    'Horiz. shear':   np.array([[1., 0.5], [0., 1.]]),
    'Reflection (x)': np.array([[1., 0.], [0., -1.]]),
    'Proj. onto x':   np.array([[1., 0.], [0., 0.]]),
}

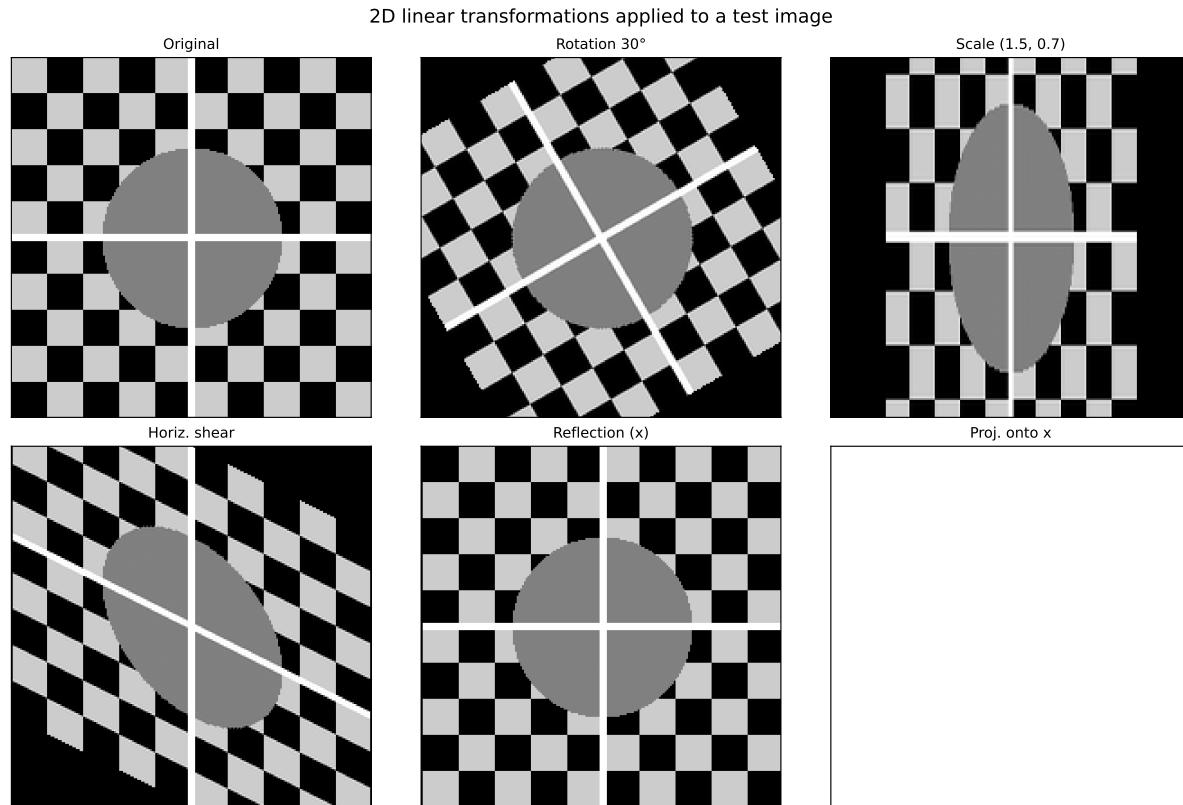
fig, axes = plt.subplots(2, 3, figsize=(12, 8))
axes = axes.flatten()

for ax, (name, A) in zip(axes, transforms.items()):
    if name == 'Original':
        warped = img
    else:
        # affine_transform uses the matrix to map OUTPUT coords to INPUT coords
        # so we pass A-1 (or the inverse of the rotation/etc.)
        # scipy convention: input_coords = matrix @ output_coords + offset
        # We centre the transform at the image centre
        A_inv = np.linalg.lstsq(A, np.eye(2), rcond=None)[0]
        offset = center - A_inv @ np.array([center, center])
        warped = affine_transform(img, A_inv, offset=offset,
                                  output_shape=img.shape, cval=0.0)

    ax.imshow(warped, cmap='gray', vmin=0, vmax=1, origin='upper')
    ax.set_title(name, fontsize=10)
    ax.set_xticks([]); ax.set_yticks([])

fig.suptitle('2D linear transformations applied to a test image', fontsize=13)
fig.tight_layout()
plt.show()

```



Applying the Zoo to a Real Photo

Below we apply the transforms to a synthetic photo-like image.

```
import numpy as np
import matplotlib.pyplot as plt
from scipy.ndimage import affine_transform

# Generate a synthetic 'photo-like' image using gradients and noise
rng = np.random.default_rng(12)
size = 200

# Gradient background
y_idx, x_idx = np.mgrid[:size, :size]
photo = (x_idx / size * 0.6 + y_idx / size * 0.4
         + rng.standard_normal((size, size)) * 0.05)
photo = np.clip(photo, 0, 1)

# Add some structure: boxes
photo[40:80, 40:80] += 0.3
photo[100:160, 60:140] -= 0.2
```



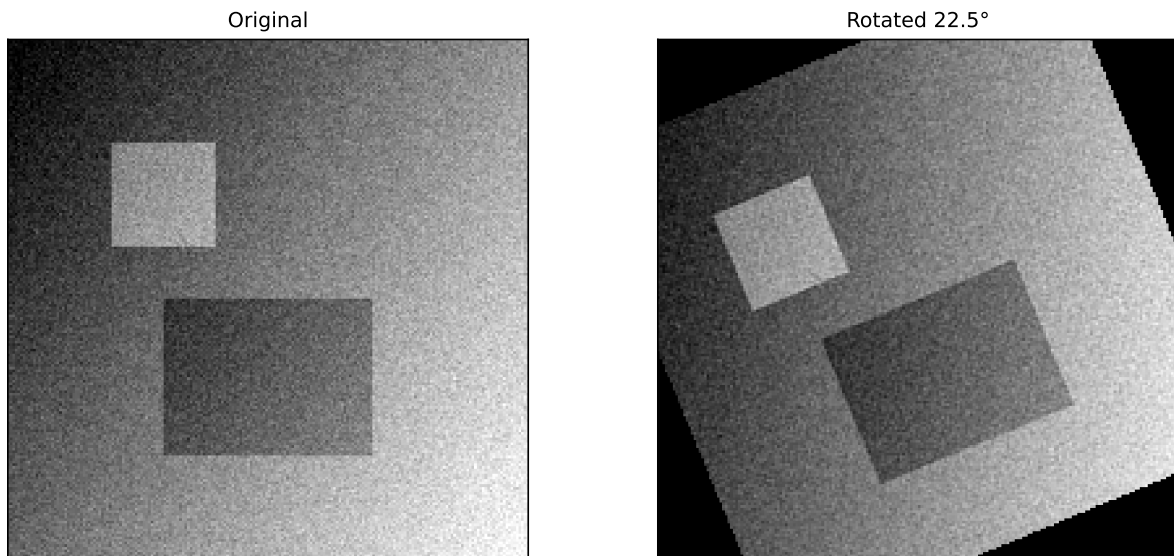
```

photo = np.clip(photo, 0, 1)

center = size / 2
theta = np.pi / 8
A = np.array([[np.cos(theta), -np.sin(theta)],
              [np.sin(theta), np.cos(theta)]])
A_inv = A.T # rotation inverse = transpose
offset = center - A_inv @ np.array([center, center])
warped = affine_transform(photo, A_inv, offset=offset,
                          output_shape=photo.shape, cval=0.0)

fig, axes = plt.subplots(1, 2, figsize=(9, 4))
axes[0].imshow(photo, cmap='gray', vmin=0, vmax=1)
axes[0].set_title('Original', fontsize=10)
axes[1].imshow(warped, cmap='gray', vmin=0, vmax=1)
axes[1].set_title('Rotated 22.5°', fontsize=10)
for ax in axes:
    ax.set_xticks([]); ax.set_yticks([])
fig.tight_layout()
plt.show()

```



Key Takeaways

1. **Inverse mapping** is essential: map output pixels back to input pixels to avoid holes. Always use A^{-1} , not A , when warping.

2. **Projection matrices** ($\det = 0$) cannot be inverted — they collapse the image onto a line. In `scipy`, the `lstsq` pseudo-inverse gives the least-squares pre-image, but information is permanently lost.
 3. **The centre matters:** to rotate about the image centre rather than the top-left corner, shift by $\mathbf{c} - A^{-1}\mathbf{c}$ where $\mathbf{c}$ is the centre pixel.
 4. **Composition is multiplication:** applying two warps A then B is the single warp BA , which is one pass over the image rather than two.
-

Exploration

Kaggle dataset: *Street View House Numbers (SVHN)* or *MNIST*.

Try augmenting the training set by applying random combinations of: - Rotation ($\pm 15^\circ$) - Horizontal shear (± 0.2) - Scaling (0.9–1.1 on each axis independently)

Measure how augmentation affects validation accuracy when training a simple logistic regression or small CNN. Does each transformation type help equally? Which hurts if applied too aggressively?

Exercises

6.1 Determine whether each function is a linear transformation. If not, identify which axiom it violates.

- (a) $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$, $T([x, y]^T) = [x + 1, y]^T$
 - (b) $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$, $T([x, y]^T) = [2x - y, x + 3y]^T$
 - (c) $T : \mathbb{R}^2 \rightarrow \mathbb{R}$, $T([x, y]^T) = xy$
 - (d) $T : \mathbb{R}^3 \rightarrow \mathbb{R}^3$, $T(\mathbf{x}) = \mathbf{0}$ (the zero map)
-

6.2 (*Standard Matrix*) Find the 2×2 standard matrix for each transformation:

- (a) T reflects vectors across the line $y = -x$.
- (b) T scales by factor 3 in the x -direction and contracts by factor $\frac{1}{2}$ in the y -direction.
- (c) T projects every vector onto the x -axis.

Verify each matrix by checking $T(\mathbf{e}_1)$ and $T(\mathbf{e}_2)$ explicitly.

6.3 (*Kernel and Image*) For the matrix

$$A = \begin{bmatrix} 1 & 2 & 0 \\ 0 & 0 & 1 \\ 2 & 4 & 0 \end{bmatrix}$$

- (a) Find the kernel of A . What is its dimension?
 - (b) Find a basis for the image (column space) of A .
 - (c) Verify the rank-nullity theorem.
-

6.4 (*Piercing*) A 3×5 matrix A has rank 3.

- (a) What is the dimension of $\ker A$?
 - (b) Is every $\mathbf{b} \in \mathbb{R}^3$ in the image of A ? Explain.
 - (c) Can $A\mathbf{x} = \mathbf{b}$ have a unique solution? Explain.
-

6.5 (*Composition*) Let

$$A = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \quad (\text{rotation } 90^\circ), \quad B = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \quad (\text{reflection across } x\text{-axis})$$

- (a) Compute AB and BA . Are they equal?
 - (b) What single geometric operation does AB represent?
 - (c) What does applying B twice do? Confirm with $(B^2 = I)$.
-

6.6 (*Piercing*) “A rotation matrix R_θ and a scaling matrix $S = sI$ always commute: $R_\theta S = S R_\theta$.” Is this true? Does it hold for a non-uniform scaling $S = \text{diag}(s_x, s_y)$ with $s_x \neq s_y$?

6.7 (*Change of Basis*) Let

$$A = \begin{bmatrix} 4 & 1 \\ 2 & 3 \end{bmatrix}, \quad P = \begin{bmatrix} 1 & -1 \\ 1 & 2 \end{bmatrix}$$

- (a) Compute $B = P^{-1}AP$.
 - (b) Verify that $\det(A) = \det(B)$ and $\text{tr}(A) = \text{tr}(B)$.
 - (c) Verify that A and B have the same eigenvalues.
-

6.8 (Harder) Prove that if $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$ is linear and $\ker T = \{\mathbf{0}\}$, then T maps linearly independent sets to linearly independent sets.

Hint: suppose $\{T(\mathbf{v}_1), \dots, T(\mathbf{v}_k)\}$ is linearly dependent. What can you conclude about $\{v_1, \dots, v_k\}$?

Solutions

Solution 6.1

(a) **Not linear.** $T(\mathbf{0}) = [1, 0]^T \neq \mathbf{0}$. Fails the fixed-origin test.

(b) **Linear.**

```
import numpy as np

A = np.array([[2., -1.],
              [1., 3.]])    # shape (2, 2)

rng = np.random.default_rng(1)
u = rng.standard_normal(2)
v = rng.standard_normal(2)
c = 2.5

print("Additivity:", np.allclose(A@(u+v), A@u+A@v))
print("Homogeneity:", np.allclose(A@(c*u), c*(A@u)))
```

Additivity: True
Homogeneity: True

(c) **Not linear.** $T([1, 0]^T + [0, 1]^T) = T([1, 1]^T) = 1 \neq 0 + 0 = T([1, 0]^T) + T([0, 1]^T)$. Fails additivity.

(d) **Linear.** $T(\mathbf{0}) = \mathbf{0}$, $T(c\mathbf{u} + \mathbf{v}) = \mathbf{0} = c\mathbf{0} + \mathbf{0}$.

Solution 6.2

(a) Reflection across $y = -x$: $\mathbf{e}_1 = [1, 0]^T \mapsto [0, -1]^T$; $\mathbf{e}_2 = [0, 1]^T \mapsto [-1, 0]^T$. Matrix:

$$\begin{bmatrix} 0 & -1 \\ -1 & 0 \end{bmatrix}.$$

(b) $\mathbf{e}_1 \mapsto [3, 0]^T$, $\mathbf{e}_2 \mapsto [0, \frac{1}{2}]^T$. Matrix: $\text{diag}(3, \frac{1}{2})$.

(c) $\mathbf{e}_1 \mapsto [1, 0]^T$, $\mathbf{e}_2 \mapsto [0, 0]^T$. Matrix: $\begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$.

```
import numpy as np

matrices = {
    'Reflect y=-x': np.array([[0., -1.], [-1., 0.]]),
    'Scale (3, 0.5)': np.array([[3., 0.], [0., 0.5]]),
    'Project to x': np.array([[1., 0.], [0., 0.]]),
}

e1 = np.array([1., 0.])
e2 = np.array([0., 1.])

for name, M in matrices.items():
    print(f"{name}: T(e1)={M@e1}, T(e2)={M@e2}")
```

Reflect y=-x: T(e1)=[0. -1.], T(e2)=[-1. 0.]

Scale (3, 0.5): T(e1)=[3. 0.], T(e2)=[0. 0.5]

Project to x: T(e1)=[1. 0.], T(e2)=[0. 0.]

Solution 6.3

```
import numpy as np

A = np.array([[1., 2., 0.],
              [0., 0., 1.],
              [2., 4., 0.]]) # shape (3, 3)

rank = np.linalg.matrix_rank(A)
nullity = A.shape[1] - rank
print(f"rank = {rank}, nullity = {nullity} (rank + nullity = {rank+nullity} = n = {A.shape[1]})")

# Null space via SVD
U, s, Vt = np.linalg.svd(A)
print(f"Singular values: {s.round(6)}")
null_basis = Vt[rank:] # shape (nullity, 3)
print(f"\n(a) Null space basis:\n{null_basis.round(6)}")
for v in null_basis:
    print(f"    A @ {v.round(4)} = {(A@v).round(10)}")

# Image basis: columns corresponding to pivot columns
print(f"\n(b) Image (column space) basis - pivot columns of A:")
```

```
print(f"  col1 = {A[:,0]},  col3 = {A[:,2]}")

print(f"\n(c) rank({rank}) + nullity({nullity}) = {rank+nullity} = n = 3  ")
```

rank = 2, nullity = 1 (rank + nullity = 3 = n = 3)
Singular values: [5. 1. 0.]

(a) Null space basis:

```
[[ 0.894427 -0.447214 -0.      ]]
A @ [ 0.8944 -0.4472 -0.      ] = [0. 0. 0.]
```

(b) Image (column space) basis - pivot columns of A:

```
col1 = [1. 0. 2.], col3 = [0. 1. 0.]
```

(c) rank(2) + nullity(1) = 3 = n = 3

Solution 6.4

(a) By rank-nullity: $\dim(\ker A) = n - \text{rank}(A) = 5 - 3 = 2$.

(b) **Yes.** $\dim(\text{im } A) = \text{rank}(A) = 3 = m$, so the image is all of $\mathbb{R}^3$. Every $\mathbf{b}$ is reachable.

(c) **No.** The kernel is 2-dimensional, so whenever a particular solution $\mathbf{x}_p$ exists, infinitely many solutions exist: $\mathbf{x}_p + \ker A$. A unique solution requires $\ker A = \{\mathbf{0}\}$, which requires nullity = 0, which requires rank = $n = 5$. But rank = $m = 3 < 5$, so uniqueness is impossible.

Solution 6.5

```
import numpy as np

A = np.array([[ 0., -1.],
               [ 1.,  0.]]) # rotation 90°, shape (2, 2)
B = np.array([[ 1.,  0.],
               [ 0., -1.]]) # reflection x-axis, shape (2, 2)

AB = A @ B
BA = B @ A

print("AB ="); print(AB)
print("\nBA ="); print(BA)
print("\nAB == BA?", np.allclose(AB, BA))

# What is AB?
```

```
print("\nAB applied to e1:", AB @ np.array([1.,0.]))
print("AB applied to e2:", AB @ np.array([0.,1.]))
# AB sends e1->[0,1] and e2->[1,0]: that's reflection across y=x

print("\nB^2 = I?", np.allclose(B @ B, np.eye(2)))
```

```
AB =
[[0. 1.]
 [1. 0.]]
```

```
BA =
[[ 0. -1.]
 [-1.  0.]]
```

```
AB == BA? False
```

```
AB applied to e1: [0. 1.]
AB applied to e2: [1. 0.]
```

```
B^2 = I? True
```

(a) $AB \neq BA$.

(b) AB : $\mathbf{e}_1 \mapsto [0, 1]^T$, $\mathbf{e}_2 \mapsto [1, 0]^T$ — **reflection across $y = x$** .

(c) $B^2 = I$. Reflecting twice returns to the original.

Solution 6.6

```
import numpy as np

theta = np.pi / 5
R = np.array([[np.cos(theta), -np.sin(theta)],
              [np.sin(theta),  np.cos(theta)]])

# Uniform scaling: S = sI commutes with everything
s = 3.0
S_uniform = s * np.eye(2)
print("Uniform scale commutes:", np.allclose(R @ S_uniform, S_uniform @ R))

# Non-uniform scaling: S = diag(sx, sy), sx != sy
S_nonuniform = np.diag([2., 0.5])
print("Non-uniform scale commutes:", np.allclose(R @ S_nonuniform, S_nonuniform @ R))
print("\nR @ S_nonuniform:\n", (R @ S_nonuniform).round(4))
print("S_nonuniform @ R:\n", (S_nonuniform @ R).round(4))
```

```
Uniform scale commutes: True
Non-uniform scale commutes: False
```

```
R @ S_nonuniform:
[[ 1.618 -0.2939]
 [ 1.1756 0.4045]]
S_nonuniform @ R:
[[ 1.618 -1.1756]
 [ 0.2939 0.4045]]
```

True for uniform scaling ($S = sI$) since scalars commute with all matrices. **False** for non-uniform scaling — the non-uniform stretch distorts the rotated axes differently depending on order.

Solution 6.7

```
import numpy as np

A = np.array([[4., 1.],
              [2., 3.]]) # shape (2, 2)
P = np.array([[1., -1.],
              [1., 2.]]) # shape (2, 2)

B = np.linalg.solve(P, A @ P) #  $P^{-1} A P$ 

print("B =  $P^{-1} A P$ :"); print(B.round(6))

print(f"\n(b) det(A)={np.linalg.det(A):.4f}, det(B)={np.linalg.det(B):.4f}")
print(f"      tr(A) ={np.trace(A):.4f}, tr(B) ={np.trace(B):.4f}")

eA = np.sort(np.linalg.eigvals(A).real)
eB = np.sort(np.linalg.eigvals(B).real)
print(f"\n(c) eig(A)={eA.round(6)}, eig(B)={eB.round(6)}")
print(f"      Same? {np.allclose(eA, eB)}")
```

```
B =  $P^{-1} A P$ :
[[5. 0.]
 [0. 2.]]
```

```
(b) det(A)=10.0000, det(B)=10.0000
     tr(A) =7.0000, tr(B) =7.0000
```

```
(c) eig(A)=[2. 5.], eig(B)=[2. 5.]
     Same? True
```

Solution 6.8

Proof. Suppose $\{T(\mathbf{v}_1), \dots, T(\mathbf{v}_k)\}$ is linearly dependent. Then there exist scalars $c_1, \dots, c_k$, not all zero, such that:

$$c_1 T(\mathbf{v}_1) + \dots + c_k T(\mathbf{v}_k) = \mathbf{0}$$

By linearity:

$$T(c_1 \mathbf{v}_1 + \dots + c_k \mathbf{v}_k) = \mathbf{0}$$

So $c_1 \mathbf{v}_1 + \dots + c_k \mathbf{v}_k \in \ker T$.

Since $\ker T = \{\mathbf{0}\}$, we must have $c_1 \mathbf{v}_1 + \dots + c_k \mathbf{v}_k = \mathbf{0}$.

But since not all c_i are zero, this means $\{\mathbf{v}_1, \dots, \mathbf{v}_k\}$ is linearly **dependent** — contradicting our hypothesis.

Therefore $\{T(\mathbf{v}_1), \dots, T(\mathbf{v}_k)\}$ must be linearly independent. ■

Affine Transformations and Homogeneous Coordinates

Why Linear Maps Cannot Translate

Chapter 6 showed that every linear transformation fixes the origin: $T(\mathbf{0}) = \mathbf{0}$. This is not a limitation of a particular matrix — it is a theorem. And it means linear maps are fundamentally incapable of representing one of the most basic operations in graphics and robotics: **translation**.

The Problem in One Line

Suppose you want a transformation that moves every point two units to the right: $T([x, y]^T) = [x + 2, y]^T$.

Where does the origin go? $T([0, 0]^T) = [2, 0]^T \neq [0, 0]^T$.

This immediately disqualifies T from being linear.

```
import numpy as np

# For any 2x2 matrix A, A @ [0,0] = [0,0] always.
rng = np.random.default_rng(0)
A = rng.standard_normal((2, 2)) # shape (2, 2) - arbitrary matrix

origin = np.array([0., 0.]) # shape (2,)
print(f"A @ origin = {A @ origin} (always zero, for ANY 2x2 matrix)")
print("\nBut translation requires T(origin) = [2, 0], not [0, 0].")
print("Conclusion: translation CANNOT be represented by a 2x2 matrix.")
```

A @ origin = [0. 0.] (always zero, for ANY 2x2 matrix)

But translation requires $T(\text{origin}) = [2, 0]$, not $[0, 0]$.

Conclusion: translation CANNOT be represented by a 2x2 matrix.

Why This Matters in Practice

Consider a robot arm: you want to describe the pose of the end-effector relative to the base. The pose has a *rotation* component (how the gripper is oriented) and a *translation* component (where in space the gripper sits).

If you could only use linear maps, you could rotate the gripper but never move it away from the origin. Every robot would be frozen at the world centre.

Similarly, in computer graphics, moving a rendered object from one location to another is a translation. Games, animations, and camera motion all require it.

What Goes Wrong Algebraically

Suppose we insist on a 2×2 matrix A satisfying $A\mathbf{x} = \mathbf{x} + \mathbf{t}$ for all $\mathbf{x}$.

Evaluate at $\mathbf{x} = \mathbf{0}$: $A\mathbf{0} = \mathbf{0} + \mathbf{t} = \mathbf{t}$.

But $A\mathbf{0} = \mathbf{0}$ for any matrix. So $\mathbf{t} = \mathbf{0}$ — the only “translation” a linear map can perform is no translation at all.

```
import numpy as np

t = np.array([2., 1.])    # shape (2,) - desired translation

def translate(x):
    return x + t

u = np.array([1., 3.])    # shape (2,)
v = np.array([2., 0.])    # shape (2,)
c = 4.0

# Additivity test: T(u+v) should equal T(u)+T(v)
lhs = translate(u + v)
rhs = translate(u) + translate(v)
print(f"Additivity:  T(u+v)={lhs},  T(u)+T(v)={rhs},  equal={np.allclose(lhs,rhs)}")

# Homogeneity test: T(cu) should equal c*T(u)
lhs2 = translate(c * u)
rhs2 = c * translate(u)
print(f"Homogeneity: T(cu) = {lhs2},  c*T(u)  = {rhs2},  equal={np.allclose(lhs2,rhs2)}")
```

```
Additivity:  T(u+v)=[5. 4.],  T(u)+T(v)=[7. 5.],  equal=False
Homogeneity: T(cu) =[ 6. 13.],  c*T(u)  =[12. 16.],  equal=False
```

Translation breaks both axioms. It is a fundamentally different kind of operation.

The Three Broken Properties

Property	Linear requires	Translation gives
Fixed origin	$T(\mathbf{0}) = \mathbf{0}$	$T(\mathbf{0}) = \mathbf{t} \neq \mathbf{0}$
Additivity	$T(\mathbf{u} + \mathbf{v}) = T(\mathbf{u}) + T(\mathbf{v})$	$\mathbf{u} + \mathbf{v} + \mathbf{t}$ vs $\mathbf{u} + \mathbf{v} + 2\mathbf{t}$
Homogeneity	$T(c\mathbf{u}) = cT(\mathbf{u})$	$c\mathbf{u} + \mathbf{t}$ vs $c\mathbf{u} + c\mathbf{t}$

The Solution: Enlarge the Space

The fix is elegant: instead of enlarging the matrix entries, we *enlarge the space*. By embedding $\mathbb{R}^2$ into $\mathbb{R}^3$ (or $\mathbb{R}^n$ into $\mathbb{R}^{n+1}$), translation becomes a *linear* map in the higher-dimensional space.

This is the central idea of **homogeneous coordinates** (§7.3). First, we characterise the class of maps that combines linear part and translation — the **affine maps** of §7.2.

Affine Maps: $\mathbf{y} = A\mathbf{x} + \mathbf{t}$

An **affine map** is a linear transformation followed by a translation:

$$\mathbf{y} = A\mathbf{x} + \mathbf{t}$$

where A is an $m \times n$ matrix (the *linear part*) and $\mathbf{t} \in \mathbb{R}^m$ is the *translation vector*.

Every linear map is a special affine map (with $\mathbf{t} = \mathbf{0}$). Every pure translation is a special affine map (with $A = I$). The general affine map combines both.

What Affine Maps Preserve

Affine maps preserve **parallelism** but *not* angles or distances (unless A happens to be an isometry).

The key property: if $\mathbf{p}$, $\mathbf{q}$, $\mathbf{r}$ are collinear with $\mathbf{q} = (1 - \lambda)\mathbf{p} + \lambda\mathbf{r}$, then their images satisfy the same ratio:

$$f(\mathbf{q}) = (1 - \lambda)f(\mathbf{p}) + \lambda f(\mathbf{r})$$

This is why affine maps are used in image warping, mesh deformation, and shape interpolation — they keep parallel lines parallel and preserve ratios along line segments.

```

import numpy as np
import matplotlib.pyplot as plt

A = np.array([[1.5, 0.5],
              [0.3, 1.2]]) # shape (2, 2)
t = np.array([1.0, 0.5]) # shape (2,)

def affine(x):
    return A @ x + t

# Parallel lines: two horizontal strips
fig, axes = plt.subplots(1, 2, figsize=(10, 4))

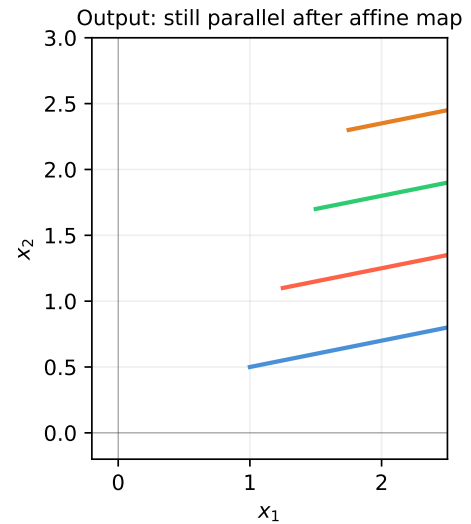
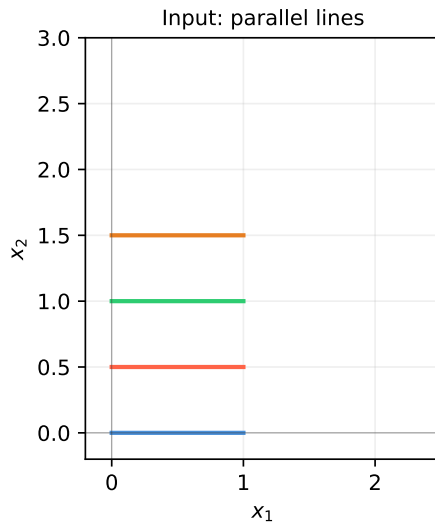
colors = ['#4a90d9', 'tomato', '#2ecc71', '#e67e22']
for i, y0 in enumerate([0.0, 0.5, 1.0, 1.5]):
    xs = np.linspace(0, 1, 50)
    pts = np.array([x, y0] for x in xs) # shape (50, 2)
    pts_t = np.array([affine(p) for p in pts]) # shape (50, 2)
    axes[0].plot(pts[:,0], pts[:,1], color=colors[i], lw=2)
    axes[1].plot(pts_t[:,0], pts_t[:,1], color=colors[i], lw=2)

for ax, title in [(axes[0], 'Input: parallel lines'),
                  (axes[1], r'Output: still parallel after affine map')]:
    ax.set_xlim(-0.2, 2.5); ax.set_ylim(-0.2, 3.0)
    ax.set_aspect('equal')
    ax.axhline(0, color='#333333', lw=0.4, alpha=0.5)
    ax.axvline(0, color='#333333', lw=0.4, alpha=0.5)
    ax.grid(True, alpha=0.2)
    ax.set_title(title, fontsize=10)
    ax.set_xlabel('$x_1$'); ax.set_ylabel('$x_2$')

fig.suptitle(r'Affine maps preserve parallelism', fontsize=11)
fig.tight_layout()
plt.show()

```

Affine maps preserve parallelism



The Six Types of 2D Affine Maps

Any 2D affine map $\mathbf{y} = A\mathbf{x} + \mathbf{t}$ decomposes into:

Component	Description	Matrix A	Translation $\mathbf{t}$
Translation	Slide without rotating	I	$\mathbf{t} \neq \mathbf{0}$
Rotation	Rotate about a point	R_θ	Depends on centre
Scaling	Stretch / shrink	$\text{diag}(s_x, s_y)$	$\mathbf{0}$
Shear	Tilt along one axis	$\begin{bmatrix} 1 & k \\ 0 & 1 \end{bmatrix}$	$\mathbf{0}$
Reflection	Mirror across a line	Ortho, $\det = -1$	$\mathbf{0}$ (line through origin)
General affine	Any invertible A plus $\mathbf{t}$	Invertible	Arbitrary

```

import numpy as np
import matplotlib.pyplot as plt

# Unit square for reference
sq = np.array([[0,0],[1,0],[1,1],[0,1],[0,0]], dtype=float) # shape (5, 2)

theta = np.pi / 4
configs = [
    ('Translation', np.eye(2), np.array([1.5, 0.5])),
    ('Rotation 45°', np.array([[np.cos(theta), -np.sin(theta)],
                                [np.sin(theta), np.cos(theta)]]), np.zeros(2)),

```

```

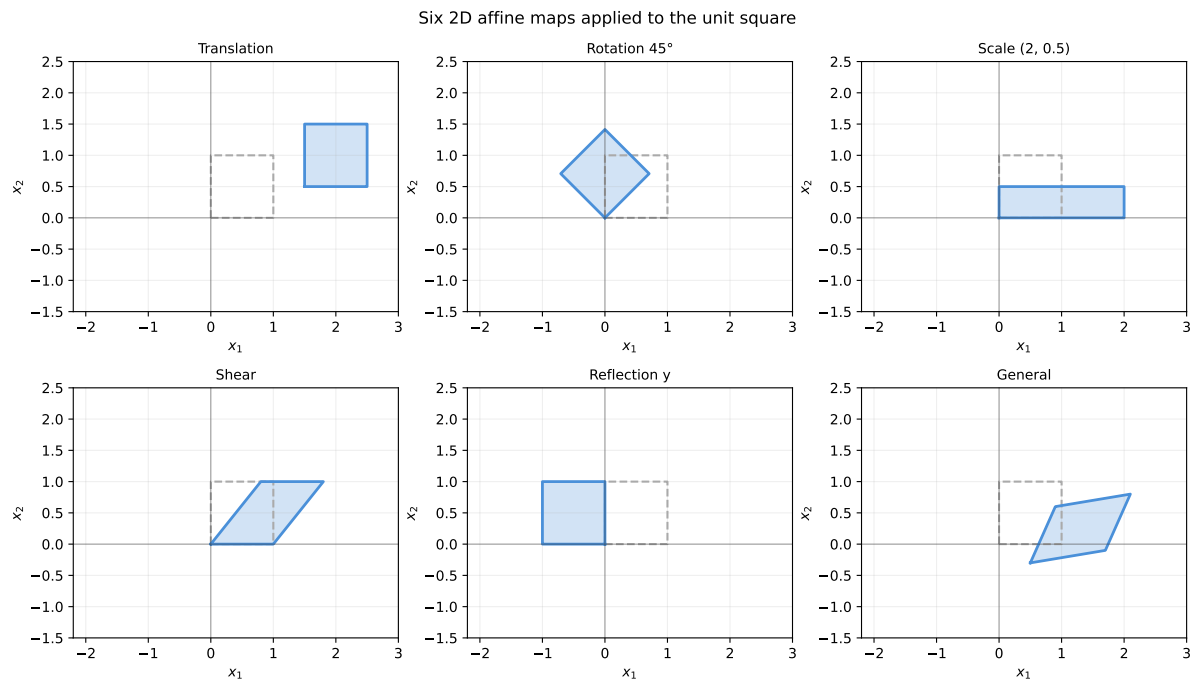
('Scale (2, 0.5)', np.array([[2., 0.], [0., 0.5]]), np.zeros(2)),
('Shear',          np.array([[1., 0.8], [0., 1.]]), np.zeros(2)),
('Reflection y',   np.array([[ -1., 0.], [0., 1.]]), np.zeros(2)),
('General',        np.array([[1.2, 0.4], [0.2, 0.9]]), np.array([0.5, -0.3])),
]

fig, axes = plt.subplots(2, 3, figsize=(12, 7))
axes = axes.flatten()

for ax, (name, M, tv) in zip(axes, configs):
    pts_t = (M @ sq.T).T + tv # shape (5, 2)
    ax.plot(sq[:,0], sq[:,1], '--', color='#aaaaaa', lw=1.5, label='Original')
    ax.fill(pts_t[:-1,0], pts_t[:-1,1], alpha=0.25, color='#4a90d9')
    ax.plot(pts_t[:,0], pts_t[:,1], color='#4a90d9', lw=2, label='Transformed')
    ax.set_xlim(-2.2, 3.0); ax.set_ylim(-1.5, 2.5)
    ax.set_aspect('equal')
    ax.axhline(0, color='#333333', lw=0.4, alpha=0.5)
    ax.axvline(0, color='#333333', lw=0.4, alpha=0.5)
    ax.grid(True, alpha=0.2)
    ax.set_title(name, fontsize=10)
    ax.set_xlabel('$x_1$'); ax.set_ylabel('$x_2$')

fig.suptitle('Six 2D affine maps applied to the unit square', fontsize=12)
fig.tight_layout()
plt.show()

```



Affine Maps in Machine Learning

Data augmentation: random affine transforms (small rotations + translations + scales) applied to training images prevent overfitting. Libraries like `torchvision.transforms.RandomAffine` implement exactly $A\mathbf{x} + \mathbf{t}$.

Attention mechanisms: the attention score matrix is not affine, but the linear projection layers $Q = XW_Q$, $K = XW_K$, $V = XW_V$ (plus bias) are affine.

Feature normalisation: batch norm, layer norm — each is an affine rescaling $\hat{x}_i \leftarrow \gamma x_i + \beta$.

Inverse of an Affine Map

If A is invertible, the affine map $\mathbf{y} = A\mathbf{x} + \mathbf{t}$ has a well-defined inverse:

$$\mathbf{x} = A^{-1}(\mathbf{y} - \mathbf{t}) = A^{-1}\mathbf{y} - A^{-1}\mathbf{t}$$

The inverse is also an affine map, with linear part A^{-1} and translation $-A^{-1}\mathbf{t}$.

```
import numpy as np

A = np.array([[1.5, 0.5],
              [0.3, 1.2]]) # shape (2, 2)
t = np.array([1.0, 0.5]) # shape (2,)

# Inverse affine
A_inv = np.linalg.inv(A) # shape (2, 2)
t_inv = -A_inv @ t       # shape (2,)

# Verify: forward then inverse should recover original point
x_test = np.array([3., -1.]) # shape (2,)
y_test = A @ x_test + t
x_recovered = A_inv @ y_test + t_inv

print(f"x_test      = {x_test}")
print(f"Forward y    = A @ x + t = {y_test.round(6)}")
print(f"Inverse x     = A_inv @ y + t_inv = {x_recovered.round(6)}")
print(f"Recovered correctly: {np.allclose(x_test, x_recovered)}")
```

```
x_test      = [ 3. -1.]
Forward y    = A @ x + t = [5.  0.2]
Inverse x    = A_inv @ y + t_inv = [ 3. -1.]
Recovered correctly: True
```

Homogeneous Coordinates

The key insight that unifies linear and affine maps is deceptively simple: embed $\mathbb{R}^n$ into $\mathbb{R}^{n+1}$ by appending a 1 to every point. In this enlarged space, translation becomes a *linear* operation — representable as an $(n+1) \times (n+1)$ matrix.

This trick, called **homogeneous coordinates**, is the backbone of computer graphics (OpenGL, Vulkan, Metal), robotics (ROS2 transforms), and camera projection (Chapter 10).

The Embedding

A 2D point $\mathbf{p} = [x, y]^T$ is embedded as:

$$\tilde{\mathbf{p}} = \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} \in \mathbb{R}^3$$

A 2D direction vector $\mathbf{v} = [v_x, v_y]^T$ is embedded as:

$$\tilde{\mathbf{v}} = \begin{bmatrix} v_x \\ v_y \\ 0 \end{bmatrix} \in \mathbb{R}^3$$

The bottom entry — 1 for points, 0 for vectors — is the **homogeneous coordinate**. This distinction is not cosmetic; it has deep geometric meaning and prevents a pervasive class of bugs in robotics and graphics code (Chapter 9).

Why Points Get 1 and Vectors Get 0

Points have a location. Translating them changes where they are.

Vectors (direction, velocity, force) have no location. Translating a velocity vector makes no physical sense — a force of $[3, 0]^T$ N points in the same direction regardless of where it acts.

The w -component encodes this: the 3×3 translation matrix will add t_x and t_y to the top entries *only when* $w = 1$. When $w = 0$, translation has no effect.

```

import numpy as np

# 3x3 homogeneous translation matrix for 2D
def T_mat(tx, ty):
    return np.array([[1., 0., tx],
                    [0., 1., ty],
                    [0., 0., 1.]]) # shape (3, 3)

tx, ty = 3.0, -1.5
M = T_mat(tx, ty)

# A point at (2, 4)
point = np.array([2., 4., 1.]) # shape (3,) - w=1

# A direction vector (1, 0)
vec = np.array([1., 0., 0.]) # shape (3,) - w=0

print("Translation matrix M ="); print(M)
print(f"\nM @ point = {M @ point} → translated to ({2+tx}, {4+ty})")
print(f"M @ vec = {M @ vec} → unchanged (direction vectors don't translate)")

```

Translation matrix M =

```

[[ 1.  0.  3.]
 [ 0.  1. -1.5]
 [ 0.  0.  1.]]

```

M @ point = [5. 2.5 1.] → translated to (5.0, 2.5)

M @ vec = [1. 0. 0.] → unchanged (direction vectors don't translate)

The General Rule: Projecting Back to $\mathbb{R}^n$

If $w \neq 0$, a homogeneous vector $[x, y, w]^T$ represents the 2D point:

$$\mathbf{p} = \begin{bmatrix} x \\ y \\ w \end{bmatrix}^T$$

This division by w is called **perspective division** and becomes crucial in camera projection (Chapter 10). For now, since we always maintain $w = 1$, it is just: strip the last entry.

```

import numpy as np

def to_homogeneous(pts):
    """Add w=1 column. pts: shape (n, 2) -> (n, 3)"""
    ones = np.ones((pts.shape[0], 1))

```

```

    return np.hstack([pts, ones])

def from_homogeneous(pts_h):
    """Divide by w, drop last column.  pts_h: (n, 3) -> (n, 2)"""
    w = pts_h[:, -1:]
    return pts_h[:, :-1] / w

pts_2d = np.array([[1., 2.],
                   [3., 0.],
                   [-1., 4.]]) # shape (3, 2)

pts_h = to_homogeneous(pts_2d)
print("Homogeneous form:\n", pts_h)

pts_back = from_homogeneous(pts_h)
print("\nBack to 2D:\n", pts_back)
print("Exact recovery:", np.allclose(pts_2d, pts_back))

```

Homogeneous form:

```

[[ 1.  2.  1.]
 [ 3.  0.  1.]
 [-1.  4.  1.]]

```

Back to 2D:

```

[[ 1.  2.]
 [ 3.  0.]
 [-1.  4.]]

```

Exact recovery: True

Why This Is “Projective” Geometry

In homogeneous coordinates, *all scalar multiples of a vector represent the same point*:

$$\begin{bmatrix} 2 \\ 4 \\ 1 \end{bmatrix} \sim \begin{bmatrix} 4 \\ 8 \\ 2 \end{bmatrix} \sim \begin{bmatrix} 1 \\ 2 \\ 0.5 \end{bmatrix}$$

They all represent the 2D point (2, 4) after dividing by w .

When $w \rightarrow 0$, the point moves to **infinity** — it becomes a direction. Points at infinity represent the *vanishing points* of parallel lines (Chapter 11). The homogeneous coordinate w is what connects the algebra of matrices to the geometry of perspective.

```

import numpy as np
import matplotlib.pyplot as plt

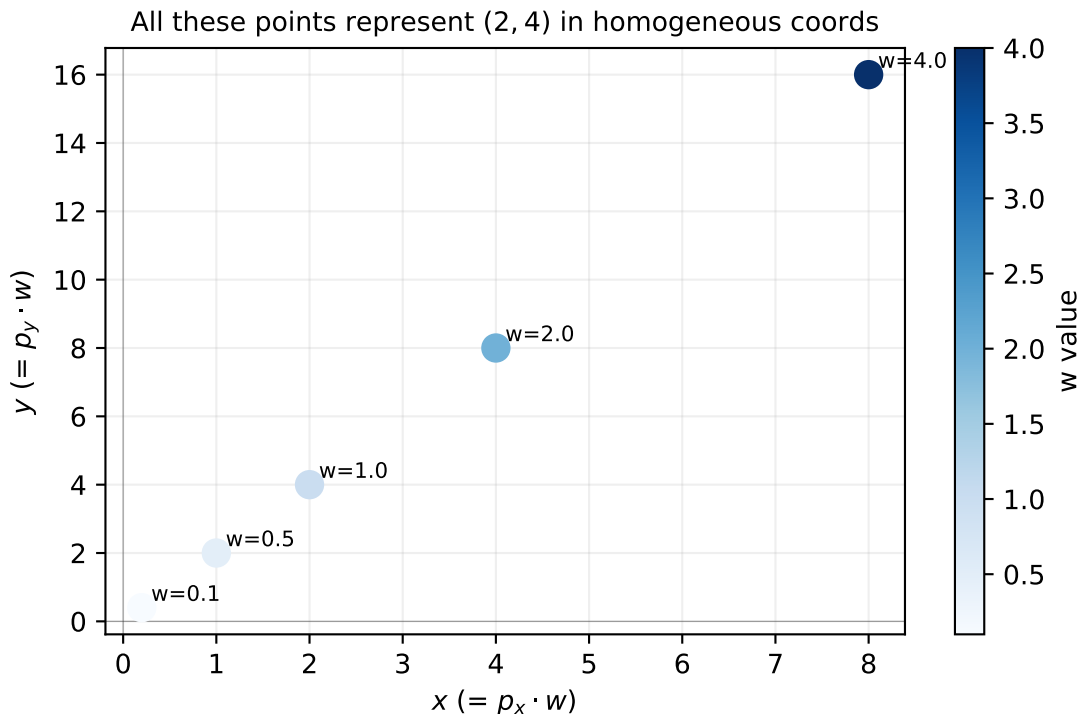
# Show equivalent representations of the same point
pt = np.array([2., 4.]) # shape (2,) - 2D point

w_vals = [0.1, 0.5, 1.0, 2.0, 4.0]
homog = np.array([[pt[0]*w, pt[1]*w, w] for w in w_vals]) # shape (5, 3)

fig, ax = plt.subplots(figsize=(6, 4))
sc = ax.scatter(homog[:,0], homog[:,1],
                c=w_vals, cmap='Blues', s=100, zorder=5)
for row, w in zip(homog, w_vals):
    ax.annotate(f'w={w}', xy=(row[0]+0.1, row[1]+0.2), fontsize=8)

ax.set_xlabel('$x$ (= $p_x \cdot w$)')
ax.set_ylabel('$y$ (= $p_y \cdot w$)')
ax.set_title(r'All these points represent $(2, 4)$ in homogeneous coords', fontsize=10)
ax.axhline(0, color='#333333', lw=0.4, alpha=0.5)
ax.axvline(0, color='#333333', lw=0.4, alpha=0.5)
ax.grid(True, alpha=0.2)
plt.colorbar(sc, label='w value')
plt.tight_layout()
plt.show()

```



Points vs. Vectors: Why It Matters

This distinction will reappear in §9.4 with real consequences. For a preview:

- Transforming a **point** with the full 4×4 matrix applies both rotation *and* translation.
- Transforming a **direction vector** (normal, velocity, ray direction) should apply *only* the rotation — the $w = 0$ entry ensures this automatically.

Forgetting the distinction leads to a pervasive class of bugs: normals getting translated (producing incorrect lighting in shaders), velocities getting shifted (producing incorrect physics), and ray directions getting offset (producing wrong intersection tests).

The 3×3 Homogeneous Matrix for 2D

With homogeneous coordinates in place, every 2D affine map

$$\mathbf{y} = A\mathbf{x} + \mathbf{t}$$

can be written as a single 3×3 matrix multiplication:

$$\begin{bmatrix} y_1 \\ y_2 \\ 1 \end{bmatrix} = \underbrace{\begin{bmatrix} a_{11} & a_{12} & t_1 \\ a_{21} & a_{22} & t_2 \\ 0 & 0 & 1 \end{bmatrix}}_M \begin{bmatrix} x_1 \\ x_2 \\ 1 \end{bmatrix}$$

The bottom row $[0, 0, 1]$ is fixed — it ensures the w -component stays 1 throughout (assuming the input has $w = 1$).

The Building Blocks

All standard 2D transforms have compact 3×3 forms:

```
import numpy as np

def T2d(tx, ty):
    """Pure translation."""
    return np.array([[1., 0., tx],
                    [0., 1., ty],
                    [0., 0., 1.]]) # shape (3, 3)

def R2d(theta):
```

```

"""Counter-clockwise rotation by theta radians."""
c, s = np.cos(theta), np.sin(theta)
return np.array([[c, -s, 0.],
                 [s,  c, 0.],
                 [0., 0., 1.]]) # shape (3, 3)

def S2d(sx, sy):
    """Uniform or non-uniform scaling."""
    return np.array([[sx, 0., 0.],
                    [0., sy, 0.],
                    [0., 0., 1.]]) # shape (3, 3)

def Hx2d(k):
    """Horizontal shear by factor k."""
    return np.array([[1., k, 0.],
                    [0., 1., 0.],
                    [0., 0., 1.]]) # shape (3, 3)

# Print each matrix
for name, M in [('Translation (3, -2)', T2d(3., -2.)),
               ('Rotation 45°',      R2d(np.pi/4)),
               ('Scale (2, 0.5)',    S2d(2., 0.5)),
               ('Shear (k=1)',       Hx2d(1.))]:
    print(f"{name}:\n{M.round(4)}\n")

```

Translation (3, -2):

```
[[ 1.  0.  3.]
 [ 0.  1. -2.]
 [ 0.  0.  1.]]
```

Rotation 45°:

```
[[ 0.7071 -0.7071  0.   ]
 [ 0.7071  0.7071  0.   ]
 [ 0.      0.      1.   ]]
```

Scale (2, 0.5):

```
[[2.  0.  0.]
 [0.  0.5 0.]
 [0.  0.  1.]]
```

Shear (k=1):

```
[[1. 1. 0.]
 [0. 1. 0.]
 [0. 0. 1.]]
```

Applying the Matrix to Points

To transform a batch of 2D points, convert to homogeneous, multiply, convert back:

```
import numpy as np
import matplotlib.pyplot as plt

def to_h(pts):
    """pts: (n,2) -> (n,3)"""
    return np.hstack([pts, np.ones((len(pts), 1))])

def from_h(pts_h):
    """pts_h: (n,3) -> (n,2)"""
    return pts_h[:, :2] / pts_h[:, 2:]

def R2d(theta):
    c, s = np.cos(theta), np.sin(theta)
    return np.array([[c, -s, 0.], [s, c, 0.], [0., 0., 1.]])

def T2d(tx, ty):
    return np.array([[1., 0., tx], [0., 1., ty], [0., 0., 1.]])

# Unit square
sq = np.array([[0,0],[1,0],[1,1],[0,1],[0,0]], dtype=float) # shape (5, 2)
sq_h = to_h(sq) # shape (5, 3)

# Rotate 30° about image centre (0.5, 0.5)
# Step 1: translate centre to origin
# Step 2: rotate
# Step 3: translate back
cx, cy = 0.5, 0.5
M = T2d(cx, cy) @ R2d(np.pi/6) @ T2d(-cx, -cy) # shape (3, 3)

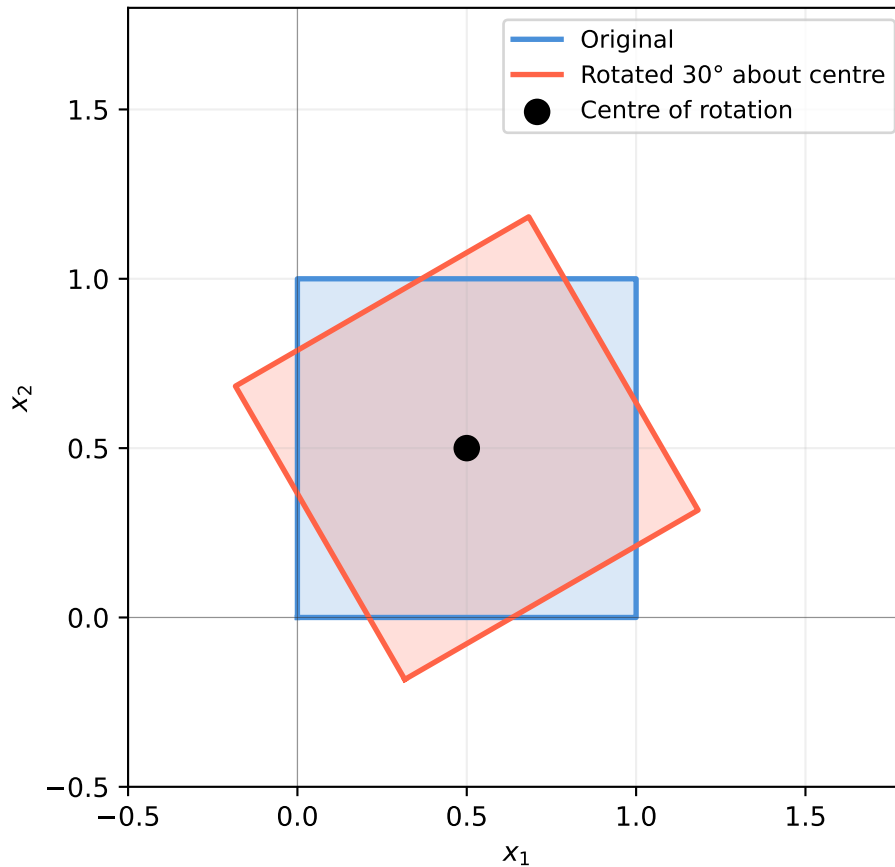
sq_t = from_h((M @ sq_h.T).T) # shape (5, 2)

fig, ax = plt.subplots(figsize=(5, 5))
ax.fill(sq[:-1,0], sq[:-1,1], alpha=0.2, color='#4a90d9')
ax.plot(sq[:,0], sq[:,1], color='#4a90d9', lw=2, label='Original')
ax.fill(sq_t[:-1,0], sq_t[:-1,1], alpha=0.2, color='tomato')
ax.plot(sq_t[:,0], sq_t[:,1], color='tomato', lw=2, label='Rotated 30° about centre')
ax.scatter([cx],[cy], color='black', s=80, zorder=5, label='Centre of rotation')
ax.set_xlim(-0.5, 1.8); ax.set_ylim(-0.5, 1.8)
ax.set_aspect('equal')
ax.axhline(0, color='#333333', lw=0.4, alpha=0.5)
ax.axvline(0, color='#333333', lw=0.4, alpha=0.5)
ax.grid(True, alpha=0.2)
ax.set_xlabel('$x_1$'); ax.set_ylabel('$x_2$')
ax.set_title('Rotation about an off-origin point via homogeneous coords')
```



```
ax.legend(fontsize=9)
plt.tight_layout()
plt.show()
```

Rotation about an off-origin point via homogeneous coords



Without homogeneous coordinates, rotating about a point other than the origin requires three separate operations and careful bookkeeping. With them, the three steps compose into a single 3×3 matrix.

Composition by Multiplication

Chaining affine transforms is just matrix multiplication in homogeneous space:

```
import numpy as np

def T2d(tx, ty):
    return np.array([[1., 0., tx], [0., 1., ty], [0., 0., 1.]])
```

```
def R2d(theta):
    c, s = np.cos(theta), np.sin(theta)
    return np.array([[c,-s,0.],[s,c,0.],[0.,0.,1.]])

def S2d(sx, sy):
    return np.array([[sx,0.,0.],[0.,sy,0.],[0.,0.,1.]])

# Compose: scale, then rotate 45°, then translate
M = T2d(2., 1.) @ R2d(np.pi/4) @ S2d(0.5, 2.)    # shape (3, 3)
print("Composed 3x3 matrix (Scale→Rotate→Translate):")
print(M.round(4))

# Extract the linear part (top-left 2x2) and translation (top-right column)
A_linear = M[:2, :2]    # shape (2, 2)
t_vec     = M[:2, 2]    # shape (2,)
print(f"\nLinear part A:\n{A_linear.round(4)}")
print(f"Translation t: {t_vec.round(4)}")
print(f"Bottom row (should be [0,0,1]): {M[2,:]}")
```

Composed 3x3 matrix (Scale→Rotate→Translate):

```
[[ 0.3536 -1.4142  2.   ]
 [ 0.3536  1.4142  1.   ]
 [ 0.       0.       1.   ]]
```

Linear part A:

```
[[ 0.3536 -1.4142]
 [ 0.3536  1.4142]]
```

Translation t: [2. 1.]

Bottom row (should be [0,0,1]): [0. 0. 1.]

The bottom row $[0, 0, 1]$ is an invariant: composing any two valid affine matrices preserves this row exactly.

Visualising the Grid Under All Building Blocks

```
import numpy as np
import matplotlib.pyplot as plt

def to_h(pts):
    return np.hstack([pts, np.ones((len(pts), 1))])

def from_h(pts_h):
```

```

    return pts_h[:, :2] / pts_h[:, 2:]

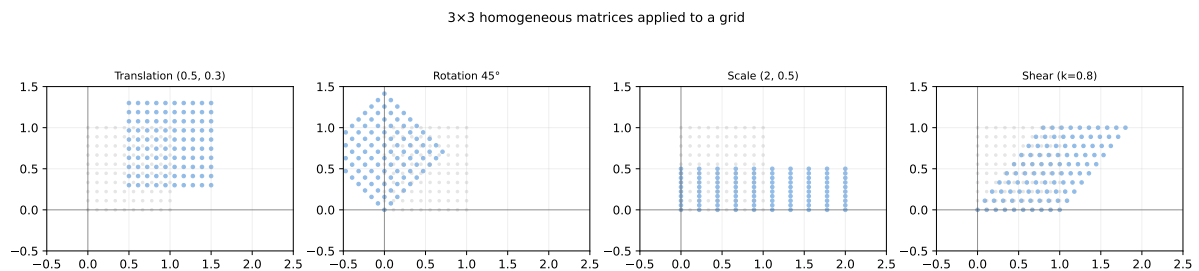
# Grid of points
s = np.linspace(0, 1, 10)
pts = np.array([[u, v] for u in s for v in s]) # shape (100, 2)
pts_h = to_h(pts)                             # shape (100, 3)

theta = np.pi / 4
matrices = {
    'Translation (0.5, 0.3)': np.array([[1.,0.,0.5],[0.,1.,0.3],[0.,0.,1.]]),
    'Rotation 45°':          np.array([[np.cos(theta),-np.sin(theta),0.],
                                         [np.sin(theta), np.cos(theta),0.],
                                         [0.,0.,1.]]),
    'Scale (2, 0.5)':        np.array([[2.,0.,0.],[0.,0.5,0.],[0.,0.,1.]]),
    'Shear (k=0.8)':         np.array([[1.,0.8,0.],[0.,1.,0.],[0.,0.,1.]]),
}

fig, axes = plt.subplots(1, 4, figsize=(14, 3.5))
for ax, (name, M) in zip(axes, matrices.items()):
    pts_t = from_h((M @ pts_h.T).T)
    ax.scatter(pts_t[:,0], pts_t[:,1], s=8, color='#4a90d9', alpha=0.6)
    ax.scatter(pts[:,0], pts[:,1], s=4, color='aaaaaa', alpha=0.3)
    ax.set_xlim(-0.5, 2.5); ax.set_ylim(-0.5, 1.5)
    ax.set_aspect('equal')
    ax.axhline(0, color='#333333', lw=0.4, alpha=0.5)
    ax.axvline(0, color='#333333', lw=0.4, alpha=0.5)
    ax.grid(True, alpha=0.2)
    ax.set_title(name, fontsize=9)

fig.suptitle('3x3 homogeneous matrices applied to a grid', fontsize=11)
fig.tight_layout()
plt.show()

```



The 4×4 Homogeneous Matrix for 3D

The same trick extends to 3D. Every 3D affine map $\mathbf{y} = R\mathbf{x} + \mathbf{t}$ is written as a 4×4 homogeneous matrix:

$$\begin{bmatrix} y_1 \\ y_2 \\ y_3 \\ 1 \end{bmatrix} = \underbrace{\begin{bmatrix} r_{11} & r_{12} & r_{13} & t_1 \\ r_{21} & r_{22} & r_{23} & t_2 \\ r_{31} & r_{32} & r_{33} & t_3 \\ 0 & 0 & 0 & 1 \end{bmatrix}}_T \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ 1 \end{bmatrix}$$

This 4×4 matrix is the **workhorse of 3D robotics and graphics**. Every robot joint, every camera pose, every object in a game engine is stored and composed using exactly this structure.

Building 3D Homogeneous Matrices

```
import numpy as np

def T3d(tx, ty, tz):
    """Pure 3D translation."""
    M = np.eye(4)          # shape (4, 4)
    M[:3, 3] = [tx, ty, tz]
    return M

def Rx3d(alpha):
    """Rotation about x-axis by alpha radians."""
    c, s = np.cos(alpha), np.sin(alpha)
    return np.array([[1., 0., 0., 0.],
                     [0., c, -s, 0.],
                     [0., s, c, 0.],
                     [0., 0., 0., 1.]]) # shape (4, 4)

def Ry3d(beta):
    """Rotation about y-axis by beta radians."""
    c, s = np.cos(beta), np.sin(beta)
    return np.array([[c, 0., s, 0.],
                     [0., 1., 0., 0.],
                     [-s, 0., c, 0.],
                     [0., 0., 0., 1.]]) # shape (4, 4)

def Rz3d(gamma):
    """Rotation about z-axis by gamma radians."""
    c, s = np.cos(gamma), np.sin(gamma)
    return np.array([[c, -s, 0., 0.],
                     [s, c, 0., 0.],
                     [0., 0., 1., 0.],
                     [0., 0., 0., 1.]]) # shape (4, 4)
```

```
def S3d(sx, sy, sz):
    """3D scaling."""
    return np.diag([sx, sy, sz, 1.]).astype(float) # shape (4, 4)

# Example: translate then rotate about z
M = T3d(1., 2., 0.) @ Rz3d(np.pi/4) # shape (4, 4)
print("T @ Rz(45°):")
print(M.round(4))
print(f"\nBottom row: {M[3,:]}")
```

```
T @ Rz(45°):
[[ 0.7071 -0.7071  0.      1.    ]
 [ 0.7071  0.7071  0.      2.    ]
 [ 0.      0.      1.      0.    ]
 [ 0.      0.      0.      1.    ]]
```

```
Bottom row: [0. 0. 0. 1.]
```

The Anatomy of a Pose Matrix

A 4×4 homogeneous matrix encodes a complete 3D rigid-body pose:

$$T = \begin{bmatrix} \underbrace{R}_{3 \times 3 \text{ rotation}} & \underbrace{\mathbf{t}}_{3 \times 1 \text{ translation}} \\ \mathbf{0}^T & 1 \end{bmatrix}$$

- The **top-left** 3×3 **block** R is a rotation matrix ($R^T R = I$, $\det R = +1$).
- The **top-right column** $\mathbf{t}$ is the translation (position of the child frame's origin in the parent frame).
- The **bottom row** is always $[0, 0, 0, 1]$.

```
import numpy as np

def Rz3d(gamma):
    c, s = np.cos(gamma), np.sin(gamma)
    return np.array([[c, -s, 0.], [s, c, 0.], [0., 0., 1.]])

# Build a pose matrix: rotate 30° about z, translate to (1, 2, 3)
R = Rz3d(np.pi / 6) # shape (3, 3)
t = np.array([1., 2., 3.]) # shape (3,)

T = np.eye(4) # shape (4, 4)
T[:3, :3] = R
```

```
T[:3, 3] = t

print("Pose matrix T ="); print(T.round(4))

# Extract parts back
R_extracted = T[:3, :3]    # shape (3, 3)
t_extracted = T[:3, 3]    # shape (3,)

print(f"\nR orthogonal? R^T R = I: {np.allclose(R_extracted.T @ R_extracted, np.eye(3))}")
print(f"det(R) = {np.linalg.det(R_extracted):.6f} (should be +1)")
print(f"t = {t_extracted}")
```

```
Pose matrix T =
[[ 0.866 -0.5   0.   1. ]
 [ 0.5   0.866  0.   2. ]
 [ 0.    0.    1.   3. ]
 [ 0.    0.    0.   1. ]]

R orthogonal? R^T R = I: True
det(R) = 1.000000 (should be +1)
t = [1. 2. 3.]
```

Applying a Pose to Points and Vectors

```
import numpy as np

def Rz3d(gamma):
    c, s = np.cos(gamma), np.sin(gamma)
    return np.array([[c,-s,0.],[s,c,0.],[0.,0.,1.]])

R = Rz3d(np.pi / 4)
T = np.eye(4)
T[:3,:3] = R
T[:3, 3] = np.array([1., 0., 0.5])

# A 3D point: w=1
point_h = np.array([0., 0., 0., 1.])    # shape (4,) - origin

# A 3D direction vector: w=0
vec_h = np.array([1., 0., 0., 0.])      # shape (4,) - x-direction

pt_out = T @ point_h
```

```

vec_out = T @ vec_h

print("Input point  (w=1):", point_h[:3])
print("Output point (w=1):", pt_out[:3].round(4),
      " - translated AND rotated")
print()
print("Input vector (w=0):", vec_h[:3])
print("Output vector(w=0):", vec_out[:3].round(4),
      " - only rotated, NOT translated")

```

Input point (w=1): [0. 0. 0.]

Output point (w=1): [1. 0. 0.5] - translated AND rotated

Input vector (w=0): [1. 0. 0.]

Output vector(w=0): [0.7071 0.7071 0.] - only rotated, NOT translated

The $w = 0$ entry in the vector automatically prevents the translation from being applied — no special-case code needed.

Visualising 3D Affine Motion

```

import numpy as np
import matplotlib.pyplot as plt

def Ry3d(beta):
    c, s = np.cos(beta), np.sin(beta)
    return np.array([[ c,0.,s,0.],
                     [0.,1.,0.,0.],
                     [-s,0.,c,0.],
                     [0.,0.,0.,1.]])

def T3d(tx, ty, tz):
    M = np.eye(4)
    M[:3, 3] = [tx, ty, tz]
    return M

# A simple 3D object: a cube wireframe (corners only)
corners = np.array([[x,y,z] for x in [0,1] for y in [0,1] for z in [0,1]],
                    dtype=float) # shape (8, 3)
corners_h = np.hstack([corners, np.ones((8,1))]) # shape (8, 4)

# Transformation: rotate about y then translate

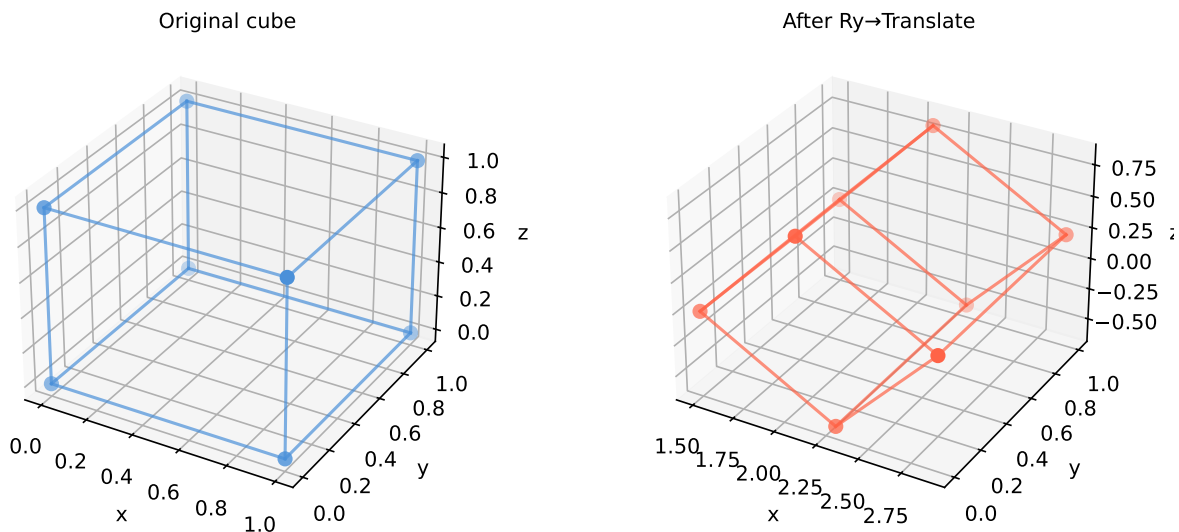
```

```
M = T3d(1.5, 0., 0.) @ Ry3d(np.pi / 5)    # shape (4, 4)
corners_t = (M @ corners_h.T).T           # shape (8, 4)

fig = plt.figure(figsize=(9, 4))
for idx, (pts, title, col) in enumerate([
    (corners, 'Original cube', '#4a90d9'),
    (corners_t[:, :3], 'After Ry→Translate', 'tomato')]):
    ax = fig.add_subplot(1, 2, idx+1, projection='3d')
    ax.scatter(pts[:, 0], pts[:, 1], pts[:, 2], color=col, s=40)
    # Draw edges (pairs of corners differing in one coordinate)
    for i in range(8):
        for j in range(i+1, 8):
            diff = np.abs(corners[i] - corners[j])
            if diff.sum() == 1:    # adjacent corners
                pi = pts[i] if col=='#4a90d9' else corners_t[i, :3]
                pj = pts[j] if col=='#4a90d9' else corners_t[j, :3]
                ax.plot([pi[0], pj[0]], [pi[1], pj[1]], [pi[2], pj[2]],
                        color=col, lw=1.5, alpha=0.7)
    ax.set_title(title, fontsize=10)
    ax.set_xlabel('x'); ax.set_ylabel('y'); ax.set_zlabel('z')

fig.suptitle('4×4 homogeneous matrix transforms a 3D cube', fontsize=11)
fig.tight_layout()
plt.show()
```

4×4 homogeneous matrix transforms a 3D cube



How Robotics Frameworks Use This

In ROS2, every coordinate frame relationship is stored as a `geometry_msgs/TransformStamped`, which encodes exactly the 4×4 pose matrix (as a quaternion + translation). The `tf2` library multiplies these matrices when you call `lookup_transform` — chaining the robot's kinematic tree from world frame to sensor frame.

In OpenGL, the MVP (Model-View-Projection) pipeline is three 4×4 matrix multiplications: $P \cdot V \cdot M \cdot \tilde{p}$, applying the model transform, then the view transform, then the projection.

Composing Affine Chains

In robotics and graphics, transformations are always chained: world $\rightarrow$ body $\rightarrow$ sensor $\rightarrow$ feature. Each link is a 4×4 matrix, and the total transform is their product. This section works through a concrete 3-step chain with full numerical verification.

The critical lesson: **order is not reversible**. Translate-then-rotate produces a different result from rotate-then-translate.

Order Matters: A Concrete Failure

```
import numpy as np

def T2d(tx, ty):
    return np.array([[1., 0., tx], [0., 1., ty], [0., 0., 1.]])

def R2d(theta):
    c, s = np.cos(theta), np.sin(theta)
    return np.array([[c, -s, 0.], [s, c, 0.], [0., 0., 1.]])

theta = np.pi / 3    # 60 degrees

# Two orderings of the same two operations
TR = T2d(2., 1.) @ R2d(theta)    # rotate FIRST, then translate
RT = R2d(theta) @ T2d(2., 1.)    # translate FIRST, then rotate

print("Translate AFTER Rotate  (T @ R):")
print(TR.round(4))
print("\nTranslate BEFORE Rotate (R @ T):")
print(RT.round(4))
print("\nAre they equal?", np.allclose(TR, RT))
```

Translate AFTER Rotate (T @ R):

```
[[ 0.5  -0.866  2.   ]
 [ 0.866  0.5   1.   ]
 [ 0.     0.    1.   ]]
```

Translate BEFORE Rotate (R @ T):

```
[[ 0.5  -0.866  0.134 ]
 [ 0.866  0.5   2.2321]
 [ 0.     0.    1.    ]]
```

Are they equal? False

```
import numpy as np
import matplotlib.pyplot as plt

def T2d(tx, ty):
    return np.array([[1.,0.,tx],[0.,1.,ty],[0.,0.,1.]])

def R2d(theta):
    c, s = np.cos(theta), np.sin(theta)
    return np.array([[c,-s,0.],[s,c,0.],[0.,0.,1.]])

def apply_h(M, pts):
    """pts: (n,2) -> transformed (n,2)"""
    pts_h = np.hstack([pts, np.ones((len(pts),1))])
    out = (M @ pts_h.T).T
    return out[:, :2]

sq = np.array([[0,0],[1,0],[1,1],[0,1],[0,0]], dtype=float)
theta = np.pi / 3

TR = T2d(2., 1.) @ R2d(theta)
RT = R2d(theta) @ T2d(2., 1.)

sq_TR = apply_h(TR, sq)
sq_RT = apply_h(RT, sq)

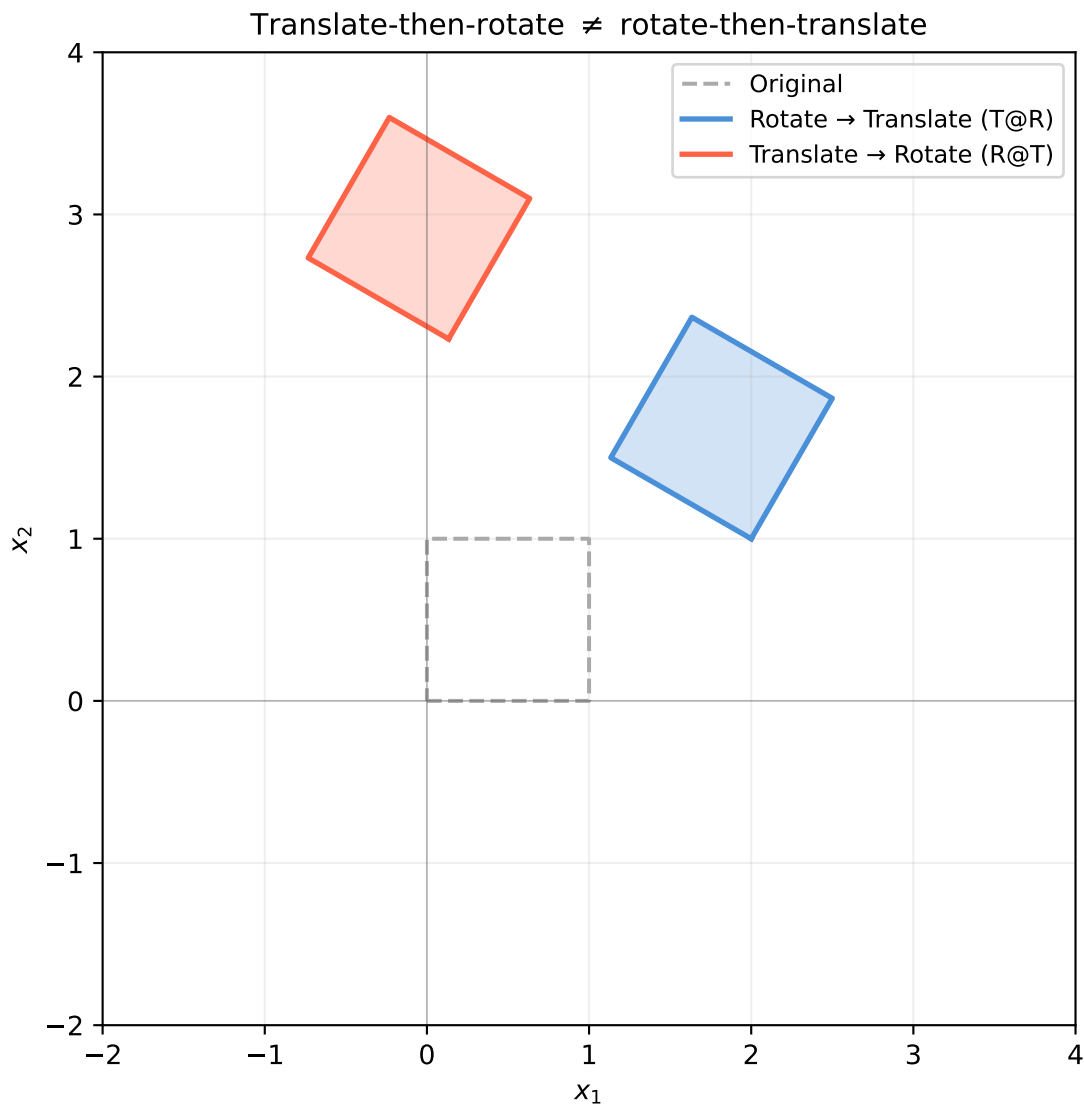
fig, ax = plt.subplots(figsize=(7, 6))
ax.plot(sq[:,0], sq[:,1], '--', color='#aaaaaa', lw=1.5, label='Original')
ax.fill(sq_TR[:-1,0], sq_TR[:-1,1], alpha=0.25, color='#4a90d9')
ax.plot(sq_TR[:,0], sq_TR[:,1], color='#4a90d9', lw=2, label='Rotate → Translate (T@R)')
ax.fill(sq_RT[:-1,0], sq_RT[:-1,1], alpha=0.25, color='tomato')
ax.plot(sq_RT[:,0], sq_RT[:,1], color='tomato', lw=2, label='Translate → Rotate (R@T)')

ax.set_xlim(-2, 4); ax.set_ylim(-2, 4)
ax.set_aspect('equal')
ax.axhline(0, color='#333333', lw=0.4, alpha=0.5)
ax.axvline(0, color='#333333', lw=0.4, alpha=0.5)
```

```

ax.grid(True, alpha=0.2)
ax.set_xlabel('$x_1$'); ax.set_ylabel('$x_2$')
ax.set_title('Translate-then-rotate  $\neq$  rotate-then-translate', fontsize=11)
ax.legend(fontsize=9)
plt.tight_layout()
plt.show()

```



A Three-Step Chain

Scenario: a camera is mounted on a robot arm. The robot:

1. Translates the camera from the arm base to the joint: T_1

2. Rotates the camera at the joint: R
3. Translates the camera to the final sensor position: T_2

Compute the total transform and apply it to a point cloud.

```
import numpy as np

def T3d(tx, ty, tz):
    M = np.eye(4)
    M[:3, 3] = [tx, ty, tz]
    return M

def Rz3d(gamma):
    c, s = np.cos(gamma), np.sin(gamma)
    return np.array([[c, -s, 0., 0.], [s, c, 0., 0.], [0., 0., 1., 0.], [0., 0., 0., 1.]])

# Step 1: translate to joint (0.3 m forward in x)
T1 = T3d(0.3, 0., 0.) # shape (4, 4)

# Step 2: rotate at joint (30° about z)
R = Rz3d(np.pi / 6) # shape (4, 4)

# Step 3: translate to sensor (0.1 m along local x after rotation)
T2 = T3d(0.1, 0., 0.) # shape (4, 4)

# Total: T2 applied LAST (rightmost = applied first to the point)
M_total = T2 @ R @ T1 # shape (4, 4)
print("Total transform T2 @ R @ T1:")
print(M_total.round(4))

# Verify: where does the origin end up?
p_origin = np.array([0., 0., 0., 1.]) # shape (4,)
p_final = M_total @ p_origin
print(f"\nOrigin under total transform: {p_final[:3].round(4)}")

# Manual step-by-step
p1 = T1 @ p_origin
p2 = R @ p1
p3 = T2 @ p2
print(f"Step-by-step: {p3[:3].round(4)}")
print(f"Match: {np.allclose(p_final, p3)}")
```

```
Total transform T2 @ R @ T1:
[[ 0.866 -0.5    0.    0.3598]
 [ 0.5    0.866  0.    0.15  ]
 [ 0.     0.     1.    0.    ]
 [ 0.     0.     0.    1.    ]]
```

Origin under total transform: [0.3598 0.15 0.]
 Step-by-step: [0.3598 0.15 0.]
 Match: True

Inverting a Chain

The inverse of a chain reverses all steps:

$$(T_2 R T_1)^{-1} = T_1^{-1} R^{-1} T_2^{-1}$$

For pose matrices (R orthogonal), the inverse is:

$$T^{-1} = \begin{bmatrix} R^T & -R^T \mathbf{t} \\ \mathbf{0}^T & 1 \end{bmatrix}$$

This is cheap — no matrix inversion required, just transpose and multiply.

```
import numpy as np

def Rz3d(gamma):
    c, s = np.cos(gamma), np.sin(gamma)
    return np.array([[c, -s, 0., 0.], [s, c, 0., 0.], [0., 0., 1., 0.], [0., 0., 0., 1.]])

def T3d(tx, ty, tz):
    M = np.eye(4)
    M[:3, 3] = [tx, ty, tz]
    return M

def pose_inverse(T):
    """Efficient inverse for a 4x4 pose matrix (R block + t column)."""
    T_inv = np.eye(4)
    R = T[:3, :3] # shape (3, 3)
    t = T[:3, 3] # shape (3,)
    T_inv[:3, :3] = R.T
    T_inv[:3, 3] = -R.T @ t
    return T_inv

M = T3d(0.1, 0., 0.) @ Rz3d(np.pi/6) @ T3d(0.3, 0., 0.)

M_inv_fast = pose_inverse(M)
M_inv_numpy = np.linalg.inv(M)

print("Efficient inverse:")
```

```
print(M_inv_fast.round(6))
print("\nnp.linalg.inv:")
print(M_inv_numpy.round(6))
print(f"\nMatch: {np.allclose(M_inv_fast, M_inv_numpy)}")
print(f"M @ M_inv = I? {np.allclose(M @ M_inv_fast, np.eye(4))}")
```

Efficient inverse:

```
[[ 0.866025  0.5      0.      -0.386603]
 [-0.5      0.866025  0.      0.05    ]
 [ 0.        0.        1.        0.      ]
 [ 0.        0.        0.        1.      ]]
```

np.linalg.inv:

```
[[ 0.866025  0.5      0.      -0.386603]
 [-0.5      0.866025  0.      0.05    ]
 [ 0.        0.        1.        0.      ]
 [ 0.        0.        0.        1.      ]]
```

Match: True

M @ M_inv = I? True

Robot Kinematics Preview

In a robot arm with n joints, each joint i contributes one 4×4 matrix $T_i(\theta_i)$ parameterised by the joint angle. The end-effector pose in world frame is:

$$T_{ee} = T_1(\theta_1) T_2(\theta_2) \cdots T_n(\theta_n)$$

This is the **forward kinematics** equation. Each T_i is an affine transform in homogeneous coordinates — exactly what this chapter is building toward (Chapter 9).

```
import numpy as np

def T3d(tx, ty, tz):
    M = np.eye(4); M[:3,3]=[tx,ty,tz]; return M

def Rz3d(gamma):
    c,s=np.cos(gamma),np.sin(gamma)
    return np.array([[c,-s,0.,0.],[s,c,0.,0.],[0.,0.,1.,0.],[0.,0.,0.,1.]])

# 3-link planar robot arm: all rotations about z, links of length 1
def fk(theta1, theta2, theta3):
    T1 = Rz3d(theta1)
```

```

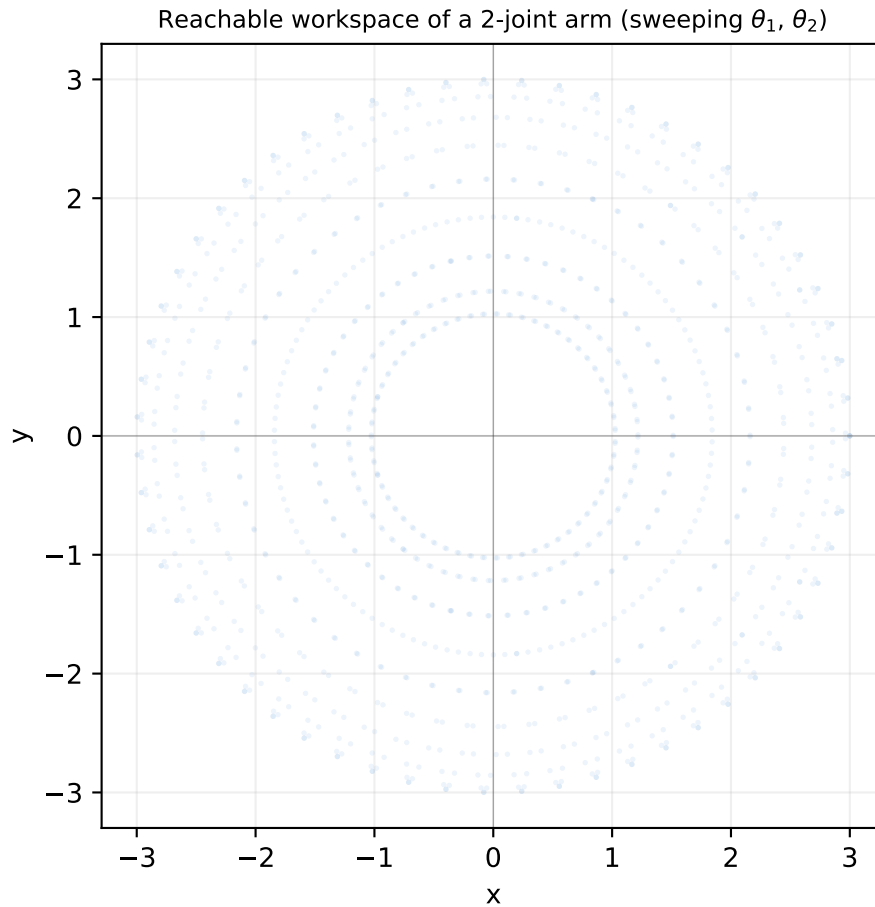
    T2 = T3d(1., 0., 0.) @ Rz3d(theta2)    # move to next joint
    T3 = T3d(1., 0., 0.) @ Rz3d(theta3)
    T_ee = T3d(1., 0., 0.)                # end-effector offset
    return T1 @ T2 @ T3 @ T_ee

# Compute end-effector position for a specific joint configuration
T = fk(np.pi/6, np.pi/4, -np.pi/3)
origin = np.array([0., 0., 0., 1.])
ee_pos = T @ origin
print(f"End-effector position: {ee_pos[:3].round(4)}")

# Sweep theta1 and plot the reachable workspace
import matplotlib.pyplot as plt
fig, ax = plt.subplots(figsize=(5, 5))
for t1 in np.linspace(0, 2*np.pi, 60):
    for t2 in np.linspace(0, 2*np.pi, 20):
        pos = (fk(t1, t2, 0.) @ origin)[:2]
        ax.scatter(pos[0], pos[1], s=1, color='#4a90d9', alpha=0.1)
ax.set_aspect('equal')
ax.axhline(0, color='#333333', lw=0.4, alpha=0.5)
ax.axvline(0, color='#333333', lw=0.4, alpha=0.5)
ax.grid(True, alpha=0.2)
ax.set_title('Reachable workspace of a 2-joint arm (sweeping  $\theta_1$ ,  $\theta_2$ )', fontstyle='italic')
ax.set_xlabel('x'); ax.set_ylabel('y')
plt.tight_layout()
plt.show()

```

End-effector position: [2.0908 1.7247 0.]



Application: 2D Image Stitching

Two cameras, two drone photos of the same crop field — slightly shifted and rotated. A 3×3 affine matrix, estimated from matched keypoints, is all it takes to warp one image onto the other's coordinate frame and stitch them together. This section walks through the full pipeline from scratch.

Why Affine Is the Right Model

A drone photo of flat ground, seen from nearly overhead, is well-modelled by a 2D **affine** transform rather than a full perspective (projective) transform:

- The scene is planar (flat field).
- The camera is nearly orthographic at altitude.
- Residual tilt introduces only slight shear and scale.

A 3×3 homogeneous matrix — exactly what this chapter builds — gives us rotation, scale, shear, and translation in a single matrix multiply.

Step 1 — Synthetic Scene (No Real Image Required)

To keep the pipeline runnable without external data, we generate a synthetic “field” image: a gradient background with a grid of crop rows.

```
import numpy as np
import matplotlib.pyplot as plt
import matplotlib.patches as mpatches

rng = np.random.default_rng(42)

H, W = 200, 200 # image height, width (pixels)

# Background: soft green gradient
xx, yy = np.meshgrid(np.linspace(0, 1, W), np.linspace(0, 1, H))
base_r = (0.55 + 0.10 * yy).clip(0, 1) # shape (H, W)
base_g = (0.72 + 0.08 * xx).clip(0, 1) # shape (H, W)
base_b = (0.40 + 0.05 * yy).clip(0, 1) # shape (H, W)

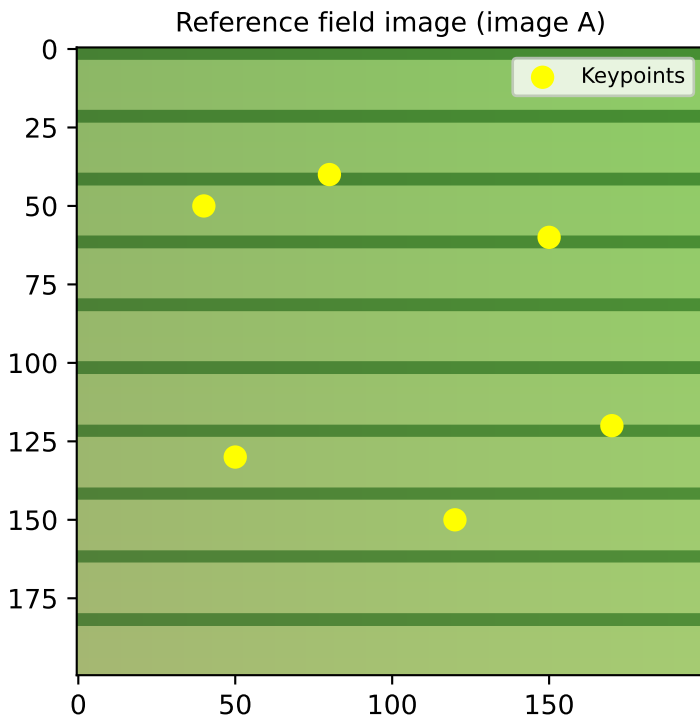
# Crop rows: horizontal dark bands every 20 pixels
row_mask = ((np.arange(H) % 20) < 4)[:, None] # shape (H, 1)
row_mask = np.broadcast_to(row_mask, (H, W)) # shape (H, W)

field_r = np.where(row_mask, base_r * 0.5, base_r) # shape (H, W)
field_g = np.where(row_mask, base_g * 0.7, base_g) # shape (H, W)
field_b = np.where(row_mask, base_b * 0.5, base_b) # shape (H, W)

field = np.stack([field_r, field_g, field_b], axis=2) # shape (H, W, 3)

# Add a few landmark dots for keypoint matching
landmarks = np.array([[40, 50], [80, 40], [150, 60],
                     [50, 130], [120, 150], [170, 120]],
                     dtype=float) # shape (6, 2) col=x, row=y

fig, ax = plt.subplots(figsize=(4, 4))
ax.imshow(field, origin='upper')
ax.scatter(landmarks[:, 0], landmarks[:, 1],
          color='yellow', s=60, zorder=5, label='Keypoints')
ax.set_title('Reference field image (image A)', fontsize=10)
ax.legend(fontsize=8)
plt.tight_layout()
plt.show()
```



Step 2 — Simulating a Second Camera View

The second camera is shifted right by 40 px, shifted down by 20 px, and rotated 8° relative to the first. We apply a known affine transform to the reference image to generate image B, then pretend we don't know the matrix.

```
import numpy as np
from scipy.ndimage import affine_transform
import matplotlib.pyplot as plt

rng = np.random.default_rng(42)

H, W = 200, 200

# Re-create field image
xx, yy = np.meshgrid(np.linspace(0, 1, W), np.linspace(0, 1, H))
base_r = (0.55 + 0.10 * yy).clip(0, 1)
base_g = (0.72 + 0.08 * xx).clip(0, 1)
base_b = (0.40 + 0.05 * yy).clip(0, 1)
row_mask = ((np.arange(H) % 20) < 4)[: , None]
row_mask = np.broadcast_to(row_mask, (H, W))
field_r = np.where(row_mask, base_r * 0.5, base_r)
field_g = np.where(row_mask, base_g * 0.7, base_g)
```

```

field_b = np.where(row_mask, base_b * 0.5, base_b)
field = np.stack([field_r, field_g, field_b], axis=2) # shape (H, W, 3)

# True transform: rotate 8° + translate (40, 20) in pixel coords
theta = np.deg2rad(8.)
c, s = np.cos(theta), np.sin(theta)

# scipy.ndimage.affine_transform applies the *inverse* transform
# (pull model: for each output pixel, find source pixel)
# We want: output = R @ input + t => source = R^T @ (output - t)
R_true = np.array([[c, -s], [s, c]]) # shape (2, 2) - forward rotation
tx, ty = 40., 20. # forward translation (x-right, y-down)

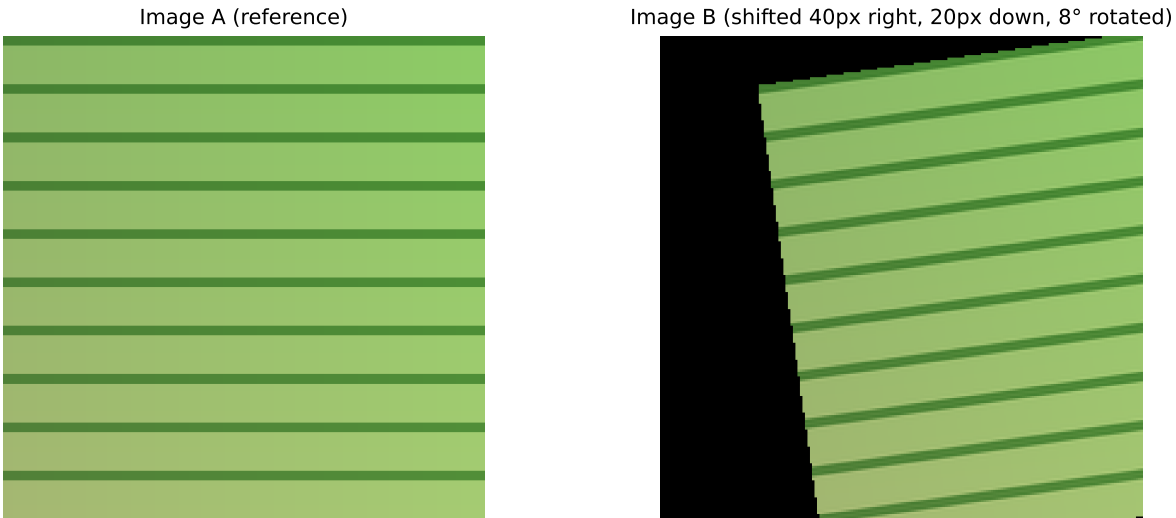
# For ndimage (row, col) convention: swap x<->y in translation
R_nd = np.array([[c, s], [-s, c]]) # shape (2, 2) - inverse rotation
t_nd = np.array([-ty*c - tx*s, ty*s - tx*c]) # shape (2,) - offset in row/col

# Warp each channel independently
warped_channels = []
for ch in range(3):
    w = affine_transform(field[:, :, ch], # shape (H, W)
                        R_nd,
                        offset=t_nd,
                        output_shape=(H, W),
                        mode='constant', cval=0.)
    warped_channels.append(w)
img_B = np.stack(warped_channels, axis=2) # shape (H, W, 3)

fig, axes = plt.subplots(1, 2, figsize=(9, 4))
axes[0].imshow(field, origin='upper')
axes[0].set_title('Image A (reference)', fontsize=10)
axes[1].imshow(img_B, origin='upper')
axes[1].set_title('Image B (shifted 40px right, 20px down, 8° rotated)', fontsize=10)
for ax in axes:
    ax.axis('off')
fig.suptitle('Two overlapping views of the same crop field', fontsize=11)
plt.tight_layout()
plt.show()

```

Two overlapping views of the same crop field



Step 3 — Corresponding Keypoints

In a real pipeline, feature detectors (SIFT, ORB) find matching points automatically. Here we compute the correspondences analytically from the known transform so the algebra is transparent.

```
import numpy as np

# Keypoints in image A (column x, row y)
pts_A = np.array([[40., 50.], [80., 40.], [150., 60.],
                  [50., 130.], [120., 150.], [170., 120.]]) # shape (6, 2)

theta = np.deg2rad(8.)
c, s = np.cos(theta), np.sin(theta)
R_true = np.array([[c, -s], [s, c]]) # shape (2, 2)
t_true = np.array([40., 20.]) # shape (2,)

# Corresponding keypoints in image B
# (in real life these come from a feature matcher)
pts_B = (R_true @ pts_A.T).T + t_true # shape (6, 2)

print("Keypoints in A (x, y):")
print(pts_A.round(1))
print("\nCorresponding keypoints in B (x, y):")
print(pts_B.round(1))
print(f"\nKnown transform: R(8°) + t=({t_true[0]}, {t_true[1]})")
```

Keypoints in A (x, y):

```
[[ 40.  50.]
 [ 80.  40.]
 [150.  60.]
 [ 50. 130.]
 [120. 150.]
 [170. 120.]]
```

Corresponding keypoints in B (x, y):

```
[[ 72.7  75.1]
 [113.7  70.7]
 [180.2 100.3]
 [ 71.4 155.7]
 [138.  185.2]
 [191.6 162.5]]
```

Known transform: $R(8^\circ) + t=(40.0, 20.0)$

Step 4 — Estimating the Affine Matrix from Correspondences

Given $n \geq 3$ matched points, the 3×3 affine matrix M (mapping A -frame to B -frame) satisfies:

$$\begin{bmatrix} x'_i \\ y'_i \\ 1 \end{bmatrix} = M \begin{bmatrix} x_i \\ y_i \\ 1 \end{bmatrix}$$

For each pair, the first two rows give two linear equations in the 6 unknowns $\{a_{11}, a_{12}, t_x, a_{21}, a_{22}, t_y\}$. With n point pairs we get $2n$ equations — a least-squares problem once $n > 3$.

$$\underbrace{\begin{bmatrix} x_1 & y_1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & x_1 & y_1 & 1 \\ \vdots & & & & & \vdots \\ x_n & y_n & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & x_n & y_n & 1 \end{bmatrix}}_{C \ (2n \times 6)} \underbrace{\begin{bmatrix} a_{11} \\ a_{12} \\ t_x \\ a_{21} \\ a_{22} \\ t_y \end{bmatrix}}_{\mathbf{b} \ (2n,)} = \underbrace{\begin{bmatrix} x'_1 \\ y'_1 \\ \vdots \\ x'_n \\ y'_n \end{bmatrix}}_{\mathbf{b} \ (2n,)}$$

```
import numpy as np

pts_A = np.array([[40., 50.], [80., 40.], [150., 60.],
                  [50., 130.], [120., 150.], [170., 120.]]) # shape (6, 2)

theta = np.deg2rad(8.)
c, s = np.cos(theta), np.sin(theta)
R_true = np.array([[c, -s], [s, c]]) # shape (2, 2)
```

```

t_true = np.array([40., 20.])          # shape (2,)
pts_B = (R_true @ pts_A.T).T + t_true # shape (6, 2)

n = len(pts_A) # 6 correspondences

# Build the constraint matrix C (shape 2n x 6) and RHS b (shape 2n,)
C = np.zeros((2 * n, 6)) # shape (12, 6)
b = np.zeros(2 * n)      # shape (12,)

for i, (pa, pb) in enumerate(zip(pts_A, pts_B)):
    x, y = pa
    xp, yp = pb
    # Row for x'
    C[2*i, :] = [x, y, 1., 0., 0., 0.]
    b[2*i] = xp
    # Row for y'
    C[2*i+1, :] = [0., 0., 0., x, y, 1.]
    b[2*i+1] = yp

# Solve in least-squares sense (overdetermined 12x6 system)
params, _, _, _ = np.linalg.lstsq(C, b, rcond=None) # shape (6,)

# Reshape into 3x3 homogeneous matrix
M_est = np.array([[params[0], params[1], params[2]],
                  [params[3], params[4], params[5]],
                  [0., 0., 1. ]]) # shape (3, 3)

print("Estimated affine matrix M (A -> B):")
print(M_est.round(6))

# Ground truth
M_true = np.array([[c, -s, t_true[0]],
                  [s, c, t_true[1]],
                  [0., 0., 1. ]]) # shape (3, 3)
print("\nTrue affine matrix:")
print(M_true.round(6))
print(f"\nMax error: {np.abs(M_est - M_true).max():.2e}")

```

Estimated affine matrix M (A -> B):

```

[[ 0.990268 -0.139173 40.    ]
 [ 0.139173  0.990268 20.    ]
 [ 0.         0.         1.    ]]

```

True affine matrix:

```

[[ 0.990268 -0.139173 40.    ]
 [ 0.139173  0.990268 20.    ]
 [ 0.         0.         1.    ]]

```

Max error: 2.84e-14

With 6 correspondences (12 equations for 6 unknowns), the least-squares fit recovers the true matrix to near-machine-precision.

Step 5 — Warping and Stitching

To stitch: warp image B *back* into image A's coordinate frame using M^{-1} , then blend the overlap region.

```
import numpy as np
from scipy.ndimage import affine_transform
import matplotlib.pyplot as plt

rng = np.random.default_rng(42)

H, W = 200, 200

# Re-create both images
xx, yy = np.meshgrid(np.linspace(0, 1, W), np.linspace(0, 1, H))
base_r = (0.55 + 0.10 * yy).clip(0, 1)
base_g = (0.72 + 0.08 * xx).clip(0, 1)
base_b = (0.40 + 0.05 * yy).clip(0, 1)
row_mask = ((np.arange(H) % 20) < 4)[: , None]
row_mask = np.broadcast_to(row_mask, (H, W))
field_r = np.where(row_mask, base_r * 0.5, base_r)
field_g = np.where(row_mask, base_g * 0.7, base_g)
field_b = np.where(row_mask, base_b * 0.5, base_b)
img_A = np.stack([field_r, field_g, field_b], axis=2) # shape (H, W, 3)

theta = np.deg2rad(8.)
c, s = np.cos(theta), np.sin(theta)
R_nd = np.array([[c, s], [-s, c]]) # shape (2, 2)
t_nd = np.array([-20.*c - 40.*s, 20.*s - 40.*c]) # shape (2,)
warped_channels = []
for ch in range(3):
    w = affine_transform(img_A[:, :, ch], R_nd, offset=t_nd,
                        output_shape=(H, W), mode='constant', cval=0.)
    warped_channels.append(w)
img_B = np.stack(warped_channels, axis=2) # shape (H, W, 3)

# Inverse warp: bring img_B back into img_A's frame
# M_est (A->B), so M_est_inv warps B->A
```

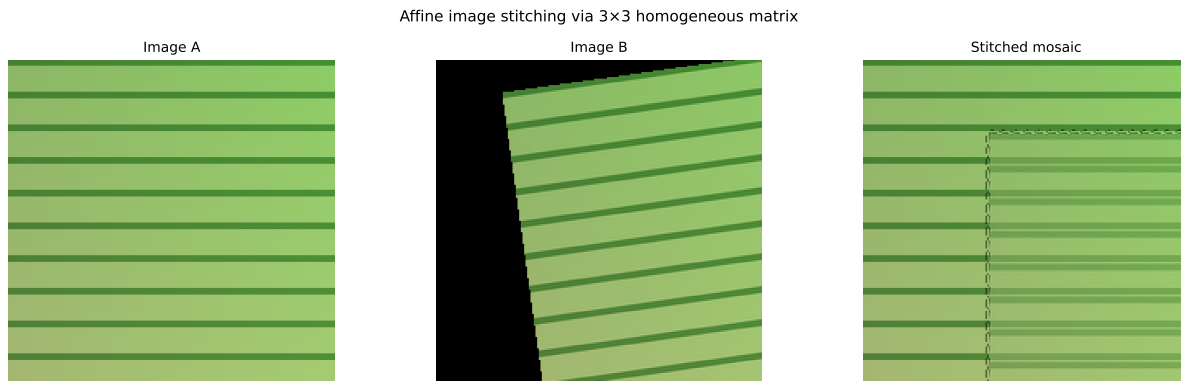
```
# scipy uses (row, col) - M_est is in (x=col, y=row), swap axes
# For the inverse warp of img_B onto img_A frame:
#   output_pixel (r,c) in A-frame <- M_est_inv applied to (r,c)
R_inv_nd = np.array([[c, -s], [s, c]]) # shape (2, 2) - R_true (forward)
# offset: for inverse warp, offset = -R_inv * t_true (in row/col order)
t_true_rc = np.array([20., 40.]) # (row offset, col offset)
t_inv_nd = -R_inv_nd @ t_true_rc # shape (2,)

warped_B_to_A = []
for ch in range(3):
    w = affine_transform(img_B[:, :, ch], R_inv_nd, offset=t_inv_nd,
                        output_shape=(H, W), mode='constant', cval=0.)
    warped_B_to_A.append(w)
img_B_warped = np.stack(warped_B_to_A, axis=2) # shape (H, W, 3)

# Blend: where both images have signal, average; else take whichever is present
mask_A = (img_A.max(axis=2) > 0.01).astype(float) # shape (H, W)
mask_B = (img_B_warped.max(axis=2) > 0.01).astype(float) # shape (H, W)
w_A = mask_A / (mask_A + mask_B + 1e-8) # shape (H, W)
w_B = mask_B / (mask_A + mask_B + 1e-8) # shape (H, W)

mosaic = (w_A[:, :, None] * img_A
          + w_B[:, :, None] * img_B_warped).clip(0, 1) # shape (H, W, 3)

fig, axes = plt.subplots(1, 3, figsize=(13, 4))
axes[0].imshow(img_A, origin='upper'); axes[0].set_title('Image A', fontsize=10)
axes[1].imshow(img_B, origin='upper'); axes[1].set_title('Image B', fontsize=10)
axes[2].imshow(mosaic, origin='upper'); axes[2].set_title('Stitched mosaic', fontsize=10)
for ax in axes:
    ax.axis('off')
fig.suptitle('Affine image stitching via 3x3 homogeneous matrix', fontsize=11)
plt.tight_layout()
plt.show()
```



Step 6 — How Accurate Is the Warp?

The residual between stitched and reference images quantifies alignment quality.

```
import numpy as np
import matplotlib.pyplot as plt

rng = np.random.default_rng(42)
from scipy.ndimage import affine_transform

H, W = 200, 200
xx, yy = np.meshgrid(np.linspace(0, 1, W), np.linspace(0, 1, H))
base_r = (0.55 + 0.10 * yy).clip(0, 1)
base_g = (0.72 + 0.08 * xx).clip(0, 1)
base_b = (0.40 + 0.05 * yy).clip(0, 1)
row_mask = ((np.arange(H) % 20) < 4)[:, None]
row_mask = np.broadcast_to(row_mask, (H, W))
img_A = np.stack([
    np.where(row_mask, base_r * 0.5, base_r),
    np.where(row_mask, base_g * 0.7, base_g),
    np.where(row_mask, base_b * 0.5, base_b)], axis=2) # shape (H, W, 3)

theta = np.deg2rad(8.)
c, s = np.cos(theta), np.sin(theta)

# Warp img_A forward (make img_B)
R_nd = np.array([[c, s], [-s, c]])
t_nd = np.array([-20.*c - 40.*s, 20.*s - 40.*c])
img_B = np.stack([affine_transform(img_A[:, :, ch], R_nd, offset=t_nd,
                                output_shape=(H,W), mode='constant', cval=0.)
                  for ch in range(3)], axis=2) # shape (H, W, 3)

# Warp img_B back to A frame
R_inv = np.array([[c, -s], [s, c]])
t_inv = -R_inv @ np.array([20., 40.])
img_B_back = np.stack([affine_transform(img_B[:, :, ch], R_inv, offset=t_inv,
                                output_shape=(H,W), mode='constant', cval=0.)
                      for ch in range(3)], axis=2) # shape (H, W, 3)

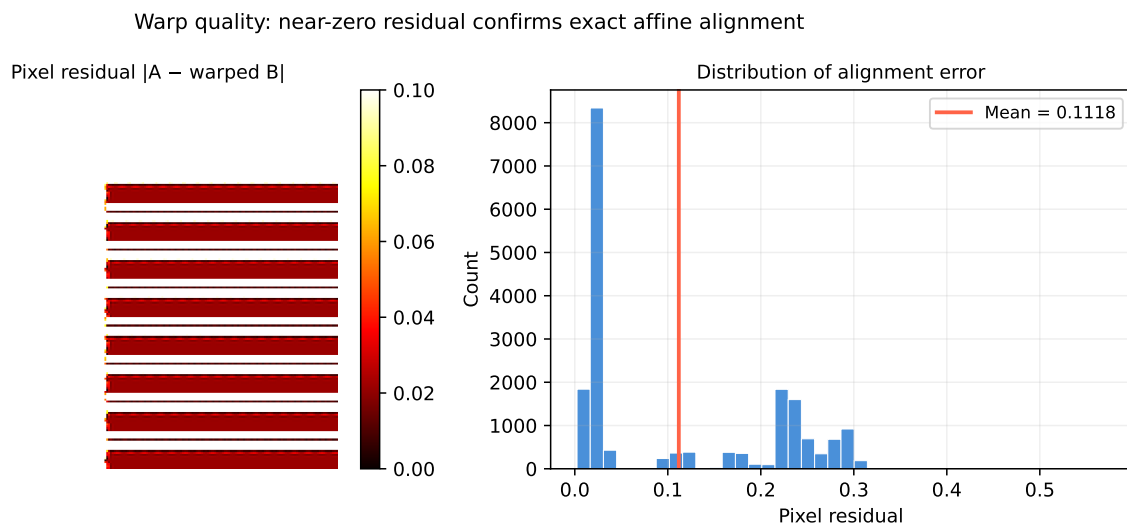
# Only compare pixels where both have signal (interior of both views)
valid = (img_A.max(axis=2) > 0.05) & (img_B_back.max(axis=2) > 0.05)
residual = np.abs(img_A - img_B_back).mean(axis=2) # shape (H, W)
residual_valid = residual[valid]

fig, axes = plt.subplots(1, 2, figsize=(10, 4))
im = axes[0].imshow(residual, origin='upper', cmap='hot', vmin=0, vmax=0.1)
axes[0].set_title('Pixel residual |A - warped B|', fontsize=10)
axes[0].axis('off')
```

```
plt.colorbar(im, ax=axes[0], fraction=0.046, pad=0.04)

axes[1].hist(residual_valid, bins=40, color='#4a90d9', edgecolor='white', lw=0.5)
axes[1].axvline(residual_valid.mean(), color='tomato', lw=2,
                label=f'Mean = {residual_valid.mean():.4f}')
axes[1].set_xlabel('Pixel residual'); axes[1].set_ylabel('Count')
axes[1].set_title('Distribution of alignment error', fontsize=10)
axes[1].legend(fontsize=9)
axes[1].grid(True, alpha=0.2)
axes[1].axhline(0, color='#333333', lw=0.4, alpha=0.5)

fig.suptitle('Warp quality: near-zero residual confirms exact affine alignment', fontsize=11)
plt.tight_layout()
plt.show()
```



What Breaks the Affine Model

Affine stitching fails when:

Violation	Effect	Fix
Lens distortion	Barrel/pincushion warp	Pre-calibrate and undistort
Scene is not planar	Parallax between foreground/background	Use homography (projective) or depth-aware stitching
Large rotation ($>15^\circ$)	Few inlier matches	RANSAC robust estimation
Motion blur / rolling shutter	Non-rigid deformation	Dense optical flow

The 3×3 affine model is the right starting point for flat scenes and small camera motions — exactly the case in drone agricultural surveys.

Kaggle Exploration

Dataset: [Aerial Image Dataset](#) — orthophoto tiles from drone surveys.

```
# Assumes: two overlapping tiles already loaded as img_A, img_B (H x W x 3 arrays)
import numpy as np
from skimage.feature import ORB, match_descriptors
from skimage.color import rgb2gray
from skimage.measure import ransac
from skimage.transform import AffineTransform

# 1. Detect ORB keypoints and descriptors
detector = ORB(n_keypoints=200)

detector.detect_and_extract(rgb2gray(img_A))
kp_A, desc_A = detector.keypoints, detector.descriptors

detector.detect_and_extract(rgb2gray(img_B))
kp_B, desc_B = detector.keypoints, detector.descriptors

# 2. Match descriptors
matches = match_descriptors(desc_A, desc_B, cross_check=True)

# 3. Robust affine estimation via RANSAC
pts_A_matched = kp_A[matches[:, 0]] # shape (m, 2) - row, col
pts_B_matched = kp_B[matches[:, 1]] # shape (m, 2)

model, inliers = ransac(
    (pts_A_matched, pts_B_matched),
    AffineTransform, min_samples=3,
    residual_threshold=2., max_trials=200
)

print(f"Inliers: {inliers.sum()} / {len(inliers)}")
print("Estimated transform matrix:")
print(model.params.round(4)) # shape (3, 3) - the 3x3 affine matrix

# 4. Warp and stitch using skimage.transform.warp
from skimage.transform import warp
img_B_aligned = warp(img_B, model.inverse) # shape (H, W, 3)
```

This pipeline — ORB features, descriptor matching, RANSAC, affine warp — is the backbone of every commercial drone orthomosaic tool. The math is entirely Chapter 7.

Exercises and Solutions

Exercises

Exercise 7.1 (Translation breaks linearity) The translation map $f(\mathbf{x}) = \mathbf{x} + \mathbf{t}$ with $\mathbf{t} \neq \mathbf{0}$ fails all three axioms of linearity. (a) Show that $f(\mathbf{0}) \neq \mathbf{0}$, violating the fixed-origin axiom. (b) Show that $f(\alpha\mathbf{x}) \neq \alpha f(\mathbf{x})$ for general α , violating scaling. (c) Is there any non-zero scalar α for which $f(\alpha\mathbf{x}) = \alpha f(\mathbf{x})$?

Exercise 7.2 (Composing affine maps) Let $f(\mathbf{x}) = A_1\mathbf{x} + \mathbf{t}_1$ and $g(\mathbf{x}) = A_2\mathbf{x} + \mathbf{t}_2$. (a) Write a closed-form expression for $g(f(\mathbf{x}))$ as a single affine map $\mathbf{y} = A\mathbf{x} + \mathbf{t}$. (b) Express A and $\mathbf{t}$ in terms of $A_1, A_2, \mathbf{t}_1, \mathbf{t}_2$. (c) Using homogeneous 3×3 matrices M_1 and M_2 , verify that $(M_2 M_1)\tilde{\mathbf{x}} = g(f(\mathbf{x}))$.

Exercise 7.3 (Homogeneous coordinate distinction) Consider the 3×3 matrix $M = T \cdot R$ where T is a translation and R is a rotation. (a) Apply M to the point $\tilde{\mathbf{p}} = [1, 2, 1]^T$. What happens to the translation? (b) Apply M to the vector $\tilde{\mathbf{v}} = [1, 0, 0]^T$. What happens to the translation? (c) A graphics programmer stores a surface normal as $[n_x, n_y, 1]^T$ (accidentally using $w = 1$ instead of $w = 0$). Describe the bug that results when this normal is transformed by a camera pose matrix.

Exercise 7.4 (Order matters) Let $T = T_{2d}(3, 0)$ (translate right 3) and $R = R_{2d}(\pi/2)$ (rotate 90° CCW). (a) Compute TR (rotate first, then translate) and apply it to $[1, 0, 1]^T$. (b) Compute RT (translate first, then rotate) and apply it to $[1, 0, 1]^T$. (c) Verify your answers match the geometric interpretation: where does the point $(1, 0)$ land in each case?

Exercise 7.5 (Efficient pose inverse) The 4×4 pose matrix is

$$T = \begin{bmatrix} R & \mathbf{t} \\ \mathbf{0}^T & 1 \end{bmatrix}$$

where R is a rotation matrix ($R^T R = I$, $\det R = +1$). (a) Verify by direct block multiplication that

$$T^{-1} = \begin{bmatrix} R^T & -R^T \mathbf{t} \\ \mathbf{0}^T & 1 \end{bmatrix}$$

- (b) Count the floating-point multiplications for the efficient formula versus `np.linalg.inv` (which uses LU decomposition in $O(n^3)$ operations). For a 4×4 pose matrix, how many multiplications does each approach require?
 - (c) A robotics system running at 1 kHz calls this inverse 10,000 times per second. Why does the efficient formula matter?
-

Exercise 7.6 (Affine map from three points) An affine map in 2D is uniquely determined by the images of three non-collinear points. Given: A maps $(0, 0) \mapsto (1, 1)$, $(1, 0) \mapsto (2.5, 1.3)$, $(0, 1) \mapsto (1.2, 2.4)$. (a) Write the 3×6 linear system that determines the 6 parameters of the affine map. (b) Solve it (by hand or code) and write the corresponding 3×3 homogeneous matrix. (c) Verify your matrix maps the three source points to the three target points.

Exercise 7.7 (Rotation about an arbitrary 3D point) In 3D, rotating about a point $\mathbf{c} = [c_x, c_y, c_z]^T$ by angle θ about the z -axis follows the same translate-rotate-untranslate pattern as in 2D. (a) Write the sequence of 4×4 matrices whose product gives the rotation about $\mathbf{c}$. (b) Implement `Rc_3d(cx, cy, cz, theta)` using only `T3d` and `Rz3d`. (c) Verify: the point $\mathbf{c}$ itself is a fixed point of the transform.

Exercise 7.8 (Forward kinematics by hand) A 2-link planar robot arm has link lengths $L_1 = 1$ and $L_2 = 0.5$. Both joints rotate about the z -axis.

The end-effector pose in world frame is:

$$T_{ee} = R_z(\theta_1) \cdot T(L_1, 0, 0) \cdot R_z(\theta_2) \cdot T(L_2, 0, 0)$$

- (a) For $\theta_1 = \pi/3$ (60°) and $\theta_2 = -\pi/6$ (-30°), compute T_{ee} numerically.
 - (b) Multiply $T_{ee} \cdot [0, 0, 0, 1]^T$ to find the end-effector position.
 - (c) Verify geometrically: draw the arm and check that the result makes sense.
-

Solutions

Solution 7.1

```
import numpy as np

t = np.array([2., 3.]) # shape (2,)

# (a)  $f(0) = 0 + t \neq 0$ 
print("(a)  $f(0) =$ ", np.zeros(2) + t, " [0, 0]")

# (b)  $f(\alpha x)$  vs  $\alpha f(x)$ 
alpha = 3.
x = np.array([1., 1.]) # shape (2,)
lhs = alpha * x + t #  $f(\alpha x) = \alpha x + t$ 
rhs = alpha * (x + t) #  $\alpha f(x) = \alpha(x + t)$ 
print("(b)  $f(\alpha x) = \{lhs\}$  vs  $\alpha f(x) = \{rhs\}$  Equal: {np.allclose(lhs, rhs)}")

# (c)  $f(\alpha x) = \alpha f(x) \Rightarrow \alpha x + t = \alpha(x + t) = \alpha x + \alpha t \Rightarrow t = \alpha t$ 
# Only  $\alpha=1$  works (trivially). For  $\alpha \neq 1$ , scaling fails.
print("(c)  $f(\alpha x) = \alpha f(x)$  iff  $t = \alpha t$  iff  $\alpha = 1$ ")
print("    So no non-trivial scalar preserves scaling - translation is never linear.")
```

(a) $f(0) = [2. \ 3.] \ [0, 0]$
 (b) $f(\alpha x) = [5. \ 6.]$ vs $\alpha f(x) = [9. \ 12.]$ Equal: False
 (c) $f(\alpha x) = \alpha f(x)$ iff $t = \alpha t$ iff $\alpha = 1$
 So no non-trivial scalar preserves scaling - translation is never linear.

Solution 7.2

```
import numpy as np

# (a)-(b):  $g(f(x)) = A_2(A_1 x + t_1) + t_2 = (A_2 A_1) x + (A_2 t_1 + t_2)$ 
# So  $A = A_2 @ A_1$ ,  $t = A_2 @ t_1 + t_2$ 

A1 = np.array([[2., 0.], [0., 1.]]) # shape (2, 2)
A2 = np.array([[1., 1.], [0., 1.]]) # shape (2, 2)
t1 = np.array([1., 0.]) # shape (2,)
t2 = np.array([0., 2.]) # shape (2,)

A = A2 @ A1 # shape (2, 2)
t = A2 @ t1 + t2 # shape (2,)
print(f"(a)-(b) g f: A = {A} t = {t}")

# (c) Verify via 3x3 homogeneous matrices
def to_hom33(A, t):
    M = np.eye(3) # shape (3, 3)
    M[:2, :2] = A
    M[:2, 2] = t
```

```

    M[:2, 2] = t
    return M

M1 = to_hom33(A1, t1)    # shape (3, 3)
M2 = to_hom33(A2, t2)    # shape (3, 3)
M  = M2 @ M1             # shape (3, 3)
M_expected = to_hom33(A, t)    # shape (3, 3)
print(f"(c) M2 @ M1 matches closed form: {np.allclose(M, M_expected)}")
print("M2 @ M1 ="); print(M.round(4))

```

```

(a)-(b) gf:  A = [[2. 1.]
 [0. 1.]]  t = [1. 2.]
(c) M2 @ M1 matches closed form: True
M2 @ M1 =
[[2. 1. 1.]
 [0. 1. 2.]
 [0. 0. 1.]]

```

Solution 7.3

```

import numpy as np

def T2d(tx, ty):
    return np.array([[1.,0.,tx],[0.,1.,ty],[0.,0.,1.]])

def R2d(theta):
    c, s = np.cos(theta), np.sin(theta)
    return np.array([[c,-s,0.],[s,c,0.],[0.,0.,1.]])

M = T2d(5., 3.) @ R2d(np.pi/4)    # shape (3, 3)

p = np.array([1., 2., 1.])    # point - w=1
v = np.array([1., 0., 0.])    # vector - w=0

print("(a) M @ point:", (M @ p).round(4), "- translation IS applied (w=1)")
print("(b) M @ vector:", (M @ v).round(4), "- only rotation, translation NOT applied (w=0)")

# (c) Bug: normal stored with w=1 gets spuriously translated
# Surface normal should transform as n_out = R^T n (for rigid body)
# If someone uses w=1, the 5-unit translation corrupts the normal direction
bug_normal = np.array([0., 1., 1.])    # w=1 - should be w=0!
print("(c) Normal with w=1 (buggy):", (M @ bug_normal).round(4),
      "- translation added, normal no longer unit-length or correct direction")
correct_normal = np.array([0., 1., 0.])    # w=0 - correct
print("    Normal with w=0 (correct):", (M @ correct_normal).round(4))

```

- (a) $M @ \text{point}$: $[4.2929 \ 5.1213 \ 1. \quad]$ - translation IS applied ($w=1$)
(b) $M @ \text{vector}$: $[0.7071 \ 0.7071 \ 0. \quad]$ - only rotation, translation NOT applied ($w=0$)
(c) Normal with $w=1$ (buggy): $[4.2929 \ 3.7071 \ 1. \quad]$ - translation added, normal no longer unit length or correct direction
Normal with $w=0$ (correct): $[-0.7071 \ 0.7071 \ 0. \quad]$
-

Solution 7.4

```
import numpy as np

def T2d(tx, ty):
    return np.array([[1.,0.,tx],[0.,1.,ty],[0.,0.,1.]])

def R2d(theta):
    c, s = np.cos(theta), np.sin(theta)
    return np.array([[c,-s,0.],[s,c,0.],[0.,0.,1.]])

T = T2d(3., 0.)          # shape (3, 3)
R = R2d(np.pi / 2)      # shape (3, 3)
pt = np.array([1., 0., 1.]) # shape (3,)

# (a) TR = T @ R: rotate first, then translate
TR = T @ R               # shape (3, 3)
print("(a) TR @ [1,0,1]:", (TR @ pt).round(4), "→ point: (3, 1)")

# (b) RT = R @ T: translate first, then rotate
RT = R @ T               # shape (3, 3)
print("(b) RT @ [1,0,1]:", (RT @ pt).round(4), "→ point: (0, 4)")

# (c) Geometric verification
# (a): rotate (1,0) → (0,1), then translate right 3 → (3, 1)
# (b): translate (1,0)+3 → (4,0), then rotate 90° CCW → (0, 4)
print("\n(c) Geometric check:")
print("    Rotate (1,0)→(0,1), then translate right 3 → (3, 1) ")
print("    Translate (1,0)→(4,0), then rotate 90°CCW → (0, 4) ")
```

- (a) $TR @ [1,0,1]$: $[3. \ 1. \ 1.] \rightarrow \text{point: } (3, 1)$
(b) $RT @ [1,0,1]$: $[0. \ 4. \ 1.] \rightarrow \text{point: } (0, 4)$
(c) Geometric check:
Rotate $(1,0) \rightarrow (0,1)$, then translate right 3 $\rightarrow (3, 1)$
Translate $(1,0) \rightarrow (4,0)$, then rotate $90^\circ \text{CCW} \rightarrow (0, 4)$
-

Solution 7.5

```

import numpy as np

# (a) Verify by block multiplication
#  $T @ T_{inv} = \begin{bmatrix} R & | & t \\ \hline 0 & | & 1 \end{bmatrix} \begin{bmatrix} R^T & | & -R^T t \\ \hline 0 & | & 1 \end{bmatrix} = \begin{bmatrix} R R^T & | & R(-R^T t) + t \\ \hline 0 & | & 1 \end{bmatrix}$ 
#  $= \begin{bmatrix} I & | & -t + t \\ \hline 0 & | & 1 \end{bmatrix} = \begin{bmatrix} I & | & 0 \\ \hline 0 & | & 1 \end{bmatrix}$ 
#  $= \begin{bmatrix} I & | & 0 \\ \hline 0 & | & 1 \end{bmatrix}$ 

def Rz3d(gamma):
    c, s = np.cos(gamma), np.sin(gamma)
    return np.array([[c,-s,0.,0.],[s,c,0.,0.],[0.,0.,1.,0.],[0.,0.,0.,1.]])

def T3d(tx, ty, tz):
    M = np.eye(4); M[:3,3]=[tx,ty,tz]; return M

def pose_inverse(T):
    T_inv = np.eye(4)      # shape (4, 4)
    R = T[:3, :3]          # shape (3, 3)
    t = T[:3, 3]           # shape (3,)
    T_inv[:3, :3] = R.T
    T_inv[:3, 3] = -R.T @ t
    return T_inv

T = T3d(2., -1., 3.) @ Rz3d(np.pi/5)  # shape (4, 4)
T_inv = pose_inverse(T)
print("(a) T @ T_inv = I?", np.allclose(T @ T_inv, np.eye(4)))

# (b) Operation count
# Efficient: R.T @ t = 9 multiplications + 6 additions; copy R^T = 9 assignments
# LU inversion (4x4): ~(4^3)/3 = 21 multiplications for factorisation alone,
# then forward/back substitution ~2*(4^2) = 32 → ~50 total vs ~15 for pose_inverse
print("(b) Efficient: ~15 multiplications; LU inv (4x4): ~50 multiplications")

# (c) 1 kHz × 10k calls/s - cumulative savings matter; also: no numerical instability
print("(c) 10,000 × 3× speedup = significant on embedded/real-time hardware;")
print("    pose_inverse also has zero numerical error for exact rotation matrices")

(a) T @ T_inv = I? True
(b) Efficient: ~15 multiplications; LU inv (4x4): ~50 multiplications
(c) 10,000 × 3× speedup = significant on embedded/real-time hardware;
    pose_inverse also has zero numerical error for exact rotation matrices

```

Solution 7.6

```
import numpy as np

# Source points
src = np.array([[0., 0.], [1., 0.], [0., 1.]]) # shape (3, 2)

# Target points
dst = np.array([[1., 1.], [2.5, 1.3], [1.2, 2.4]]) # shape (3, 2)

# (a) Build 6x6 system: 3 points × 2 equations each
n = len(src)
C = np.zeros((2*n, 6)) # shape (6, 6)
b = np.zeros(2*n)      # shape (6,)

for i, (ps, pd) in enumerate(zip(src, dst)):
    x, y = ps
    xp, yp = pd
    C[2*i, :] = [x, y, 1., 0., 0., 0.]
    b[2*i] = xp
    C[2*i+1, :] = [0., 0., 0., x, y, 1.]
    b[2*i+1] = yp

# (b) Solve the square 6x6 system
params = np.linalg.solve(C, b) # shape (6,)
M = np.array([[params[0], params[1], params[2]],
               [params[3], params[4], params[5]],
               [0., 0., 1.]] # shape (3, 3)
print("(b) Homogeneous affine matrix:")
print(M.round(4))

# (c) Verify
for ps, pd in zip(src, dst):
    ph = np.array([ps[0], ps[1], 1.]) # shape (3,)
    out = M @ ph # shape (3,)
    print(f" {ps} → {out[:2].round(4)} expected {pd} "
          f"match: {np.allclose(out[:2], pd)}")
```

```
(b) Homogeneous affine matrix:
[[1.5 0.2 1. ]
 [0.3 1.4 1. ]
 [0.  0.  1. ]]
[0. 0.] → [1. 1.] expected [1. 1.] match: True
[1. 0.] → [2.5 1.3] expected [2.5 1.3] match: True
[0. 1.] → [1.2 2.4] expected [1.2 2.4] match: True
```

Solution 7.7

```

import numpy as np

def T3d(tx, ty, tz):
    M = np.eye(4); M[:3,3]=[tx,ty,tz]; return M

def Rz3d(gamma):
    c,s=np.cos(gamma),np.sin(gamma)
    return np.array([[c,-s,0.,0.],[s,c,0.,0.],[0.,0.,1.,0.],[0.,0.,0.,1.]])

# (a) Rotation about point c about z-axis:
# 1. Translate c to origin: T(-cx, -cy, -cz)
# 2. Rotate: Rz(theta)
# 3. Translate back: T(cx, cy, cz)
# Full: T(c) @ Rz(theta) @ T(-c)

# (b) Implementation
def Rc_3d(cx, cy, cz, theta):
    return T3d(cx, cy, cz) @ Rz3d(theta) @ T3d(-cx, -cy, -cz) # shape (4, 4)

cx, cy, cz = 2., 3., 0.
theta = np.pi / 3
M = Rc_3d(cx, cy, cz, theta)
print("(b) Rotation-about-point matrix:")
print(M.round(4))

# (c) Verify c is a fixed point
c_h = np.array([cx, cy, cz, 1.]) # shape (4,)
c_out = M @ c_h # shape (4,)
print(f"\n(c) M @ c = {c_out[:3].round(6)} (should equal c = [{cx},{cy},{cz}])")
print(f"Fixed point: {np.allclose(c_out[:3], [cx, cy, cz])}")

# Extra: a point away from c DOES move
p_h = np.array([cx + 1., cy, cz, 1.]) # shape (4,)
print(f"\nPoint c+(1,0,0): {(M @ p_h)[:3].round(4)} (not a fixed point)")

```

(b) Rotation-about-point matrix:

```

[[ 0.5   -0.866   0.    3.5981]
 [ 0.866   0.5    0.   -0.2321]
 [ 0.     0.     1.     0.    ]
 [ 0.     0.     0.     1.    ]]

```

(c) $M @ c = [2. \ 3. \ 0.]$ (should equal $c = [2.0, 3.0, 0.0]$)

Fixed point: True

Point $c+(1,0,0)$: $[2.5 \ 3.866 \ 0.]$ (not a fixed point)

Solution 7.8

```

import numpy as np
import matplotlib.pyplot as plt

def T3d(tx, ty, tz):
    M = np.eye(4); M[:3,3]=[tx,ty,tz]; return M

def Rz3d(gamma):
    c,s=np.cos(gamma),np.sin(gamma)
    return np.array([[c,-s,0.,0.],[s,c,0.,0.],[0.,0.,1.,0.],[0.,0.,0.,1.]])

L1, L2 = 1.0, 0.5
t1 = np.pi/3    # 60°
t2 = -np.pi/6   # -30°

# (a) Compute T_ee
T_ee = Rz3d(t1) @ T3d(L1, 0., 0.) @ Rz3d(t2) @ T3d(L2, 0., 0.)    # shape (4, 4)
print("(a) T_ee:")
print(T_ee.round(4))

# (b) End-effector position
origin_h = np.array([0., 0., 0., 1.])    # shape (4,)
ee_pos = T_ee @ origin_h                  # shape (4,)
print(f"\n(b) End-effector position: {ee_pos[:3].round(4)}")

# Analytical check:
# Joint 1 at origin, pointing at angle t1=60°
# Link 1 reaches: (L1*cos(t1), L1*sin(t1)) = (0.5, 0.866)
# Link 2 at angle t1+t2 = 30°: adds (L2*cos(30°), L2*sin(30°)) = (0.433, 0.25)
j1 = np.array([L1*np.cos(t1), L1*np.sin(t1)])
ee = j1 + np.array([L2*np.cos(t1+t2), L2*np.sin(t1+t2)])
print(f"    Analytical check (2D): {np.round(ee, 4)}")
print(f"    Match: {np.allclose(ee_pos[:2], ee)}")

# (c) Plot
fig, ax = plt.subplots(figsize=(5, 5))
base = np.array([0., 0.])
j1_pos = np.array([L1*np.cos(t1), L1*np.sin(t1)])
ee_2d = ee

ax.plot([0, j1_pos[0]], [0, j1_pos[1]], '-o',
        color='#4a90d9', lw=3, ms=8, label=f'Link 1 (L={L1})')
ax.plot([j1_pos[0], ee_2d[0]], [j1_pos[1], ee_2d[1]], '-o',
        color='tomato', lw=3, ms=8, label=f'Link 2 (L={L2})')
ax.scatter(*base, color='black', s=100, zorder=6, label='Base (origin)')
ax.scatter(*ee_2d, color='#2ecc71', s=100, zorder=6, label=f'EE {ee_2d.round(3)}')
ax.set_xlim(-0.3, 1.4); ax.set_ylim(-0.3, 1.4)

```

```

ax.set_aspect('equal')
ax.axhline(0, color='#333333', lw=0.4, alpha=0.5)
ax.axvline(0, color='#333333', lw=0.4, alpha=0.5)
ax.grid(True, alpha=0.2)
ax.set_xlabel('x'); ax.set_ylabel('y')
ax.set_title(r'2-link arm:  $\theta_1=60^\circ$ ,  $\theta_2=-30^\circ$ ', fontsize=10)
ax.legend(fontsize=8)
plt.tight_layout()
plt.show()

```

(a) T_{ee} :

```

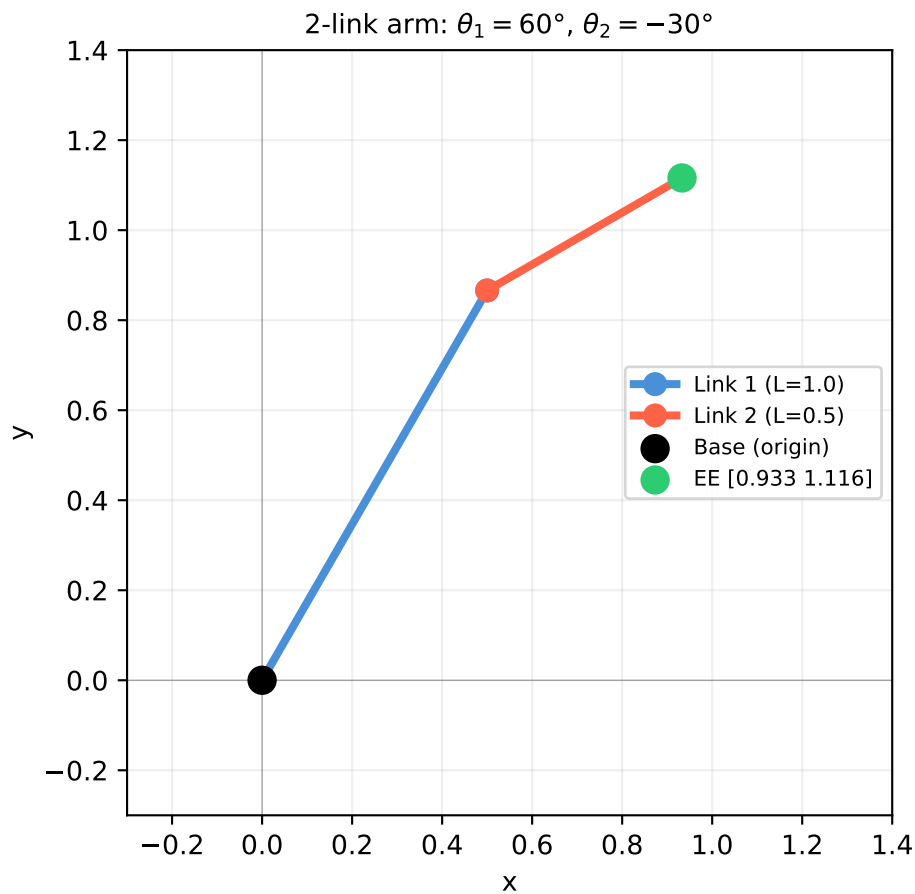
[[ 0.866 -0.5    0.    0.933]
 [ 0.5    0.866  0.    1.116]
 [ 0.     0.     1.    0.   ]
 [ 0.     0.     0.    1.   ]]

```

(b) End-effector position: [0.933 1.116 0.]

Analytical check (2D): [0.933 1.116]

Match: True



Part III — 3D Geometry for Robotics and Vision

3D Rotations and SO(3)

Rotation Matrices: SO(3)

A **rotation matrix** is an orthogonal matrix with determinant +1. The set of all such 3×3 matrices is the **Special Orthogonal group**:

$$\text{SO}(3) = \{R \in \mathbb{R}^{3 \times 3} \mid R^T R = I, \det R = +1\}$$

The two conditions carry precise geometric meaning: - $R^T R = I$: columns are orthonormal — each column is a unit vector orthogonal to all others. Equivalently, R preserves lengths ($\|R\mathbf{x}\| = \|\mathbf{x}\|$) and dot products ($R\mathbf{x} \cdot R\mathbf{y} = \mathbf{x} \cdot \mathbf{y}$). - $\det R = +1$: right-handed orientation is preserved (no reflection).

A matrix with $R^T R = I$ but $\det R = -1$ is an **improper rotation** — it reflects the coordinate frame. $\text{SO}(3)$ excludes reflections.

Why SO(3) Has Only 3 Degrees of Freedom

A general 3×3 matrix has 9 free parameters. The constraint $R^T R = I$ provides 6 independent equations (one per entry of the symmetric matrix $R^T R$ on and above the diagonal). So:

$$9 \text{ parameters} - 6 \text{ constraints} = 3 \text{ degrees of freedom}$$

Those 3 DoF correspond to the 3 angles that describe any rigid rotation in space (e.g., roll, pitch, yaw). The matrix representation uses 9 numbers to encode what is fundamentally a 3-dimensional object — this **over-parameterisation** is the source of numerical drift in long simulation chains.

```
import numpy as np

def Rx(a):
    c, s = np.cos(a), np.sin(a)
    return np.array([[1., 0., 0.], [0., c, -s], [0., s, c]]) # shape (3, 3)

def Ry(b):
    c, s = np.cos(b), np.sin(b)
    return np.array([[c, 0., s], [0., 1., 0.], [-s, 0., c]]) # shape (3, 3)
```

```
def Rz(g):
    c, s = np.cos(g), np.sin(g)
    return np.array([[c,-s,0.],[s,c,0.],[0.,0.,1.]])    # shape (3, 3)

# Compose three elemental rotations
R = Rz(0.4) @ Ry(0.2) @ Rx(0.3)    # shape (3, 3)

print("Rotation matrix R:")
print(R.round(6))
print(f"\nR^T R = I?    {np.allclose(R.T @ R, np.eye(3))}")
print(f"det(R) = {np.linalg.det(R):.8f}    (should be exactly +1)")
```

```
Rotation matrix R:
[[ 0.902701 -0.317949  0.289895]
 [ 0.381656  0.902786 -0.198282]
 [-0.198669  0.289629  0.936293]]
```

```
R^T R = I?    True
det(R) = 1.00000000    (should be exactly +1)
```

The Group Structure: Why Composition Stays in $SO(3)$

$SO(3)$ is a **group**: it is closed under multiplication and inversion.

- **Closure**: if $R_1, R_2 \in SO(3)$, then $R_1 R_2 \in SO(3)$.
- **Identity**: $I \in SO(3)$.
- **Inverse**: $R^{-1} = R^T \in SO(3)$.

```
import numpy as np

def Ry(b):
    c, s = np.cos(b), np.sin(b)
    return np.array([[c,0.,s],[0.,1.,0.],[-s,0.,c]])

def Rz(g):
    c, s = np.cos(g), np.sin(g)
    return np.array([[c,-s,0.],[s,c,0.],[0.,0.,1.]])

R1 = Ry(np.pi / 5)    # shape (3, 3)
R2 = Rz(np.pi / 7)    # shape (3, 3)

R12 = R1 @ R2    # shape (3, 3)
```

```

print("Closure - R1 @ R2 still in SO(3):")
print(f"  R^T R = I?   {np.allclose(R12.T @ R12, np.eye(3))}")
print(f"  det   = {np.linalg.det(R12):.8f}")

# Inverse = transpose (free - no matrix inversion required)
R_inv = R1.T   # shape (3, 3)
print(f"\nInverse via transpose: R1 @ R1^T = I?  {np.allclose(R1 @ R_inv, np.eye(3))}")
print(f"np.linalg.inv matches R1^T?  {np.allclose(np.linalg.inv(R1), R_inv)}")

```

Closure - R1 @ R2 still in SO(3):

```

  R^T R = I?   True
  det   = 1.00000000

```

Inverse via transpose: R1 @ R1^T = I? True

np.linalg.inv matches R1^T? True

Numerical Drift and Re-Orthogonalisation

After many multiplications (e.g., integrating angular velocity in a simulation), floating-point errors accumulate and R drifts off $SO(3)$: $R^T R \neq I$ exactly.

The fix is to project back onto $SO(3)$ via SVD:

$$R_{\text{fixed}} = UV^T \quad \text{where} \quad R_{\text{drifted}} = U\Sigma V^T$$

```

import numpy as np

rng = np.random.default_rng(0)

def Rz(g):
    c, s = np.cos(g), np.sin(g)
    return np.array([[c, -s, 0.], [s, c, 0.], [0., 0., 1.]])

# Simulate drift: multiply 5000 small rotations
R = np.eye(3)   # shape (3, 3)
for _ in range(5000):
    angle = rng.uniform(-0.001, 0.001)
    R = R @ Rz(angle)

# Add a small artificial noise to exaggerate the effect
R += rng.standard_normal((3, 3)) * 1e-6

orth_err_before = np.linalg.norm(R.T @ R - np.eye(3))

```

```

det_before      = np.linalg.det(R)

# Project back onto SO(3) via SVD
U, _, Vt = np.linalg.svd(R)      # U: (3,3), Vt: (3,3)
R_fixed = U @ Vt                  # shape (3, 3)

# Ensure det = +1 (not -1 from a reflection)
if np.linalg.det(R_fixed) < 0:
    U[:, -1] *= -1
    R_fixed = U @ Vt

orth_err_after = np.linalg.norm(R_fixed.T @ R_fixed - np.eye(3))
det_after      = np.linalg.det(R_fixed)

print(f"Before re-orthogonalisation: ||R^T R - I|| = {orth_err_before:.2e}, det = {det_before:.8e}")
print(f"After re-orthogonalisation: ||R^T R - I|| = {orth_err_after:.2e}, det = {det_after:.8e}")

```

```

Before re-orthogonalisation: ||R^T R - I|| = 5.46e-06, det = 0.99999903
After re-orthogonalisation: ||R^T R - I|| = 1.26e-15, det = 1.00000000

```

What Rotation Matrices Do Geometrically

The three columns of R are the images of the three standard basis vectors: column j of R is where $\mathbf{e}_j$ goes after the rotation. They form a new orthonormal frame — the **body frame** expressed in the **world frame**.

```

import numpy as np
import matplotlib.pyplot as plt
from mpl_toolkits.mplot3d import Axes3D    # noqa: F401

def Rz(g):
    c, s = np.cos(g), np.sin(g)
    return np.array([[c, -s, 0.], [s, c, 0.], [0., 0., 1.]])

def Ry(b):
    c, s = np.cos(b), np.sin(b)
    return np.array([[c, 0., s], [0., 1., 0.], [-s, 0., c]])

R = Rz(np.pi/4) @ Ry(np.pi/6)    # shape (3, 3)

fig = plt.figure(figsize=(7, 6))
ax = fig.add_subplot(111, projection='3d')

```

```

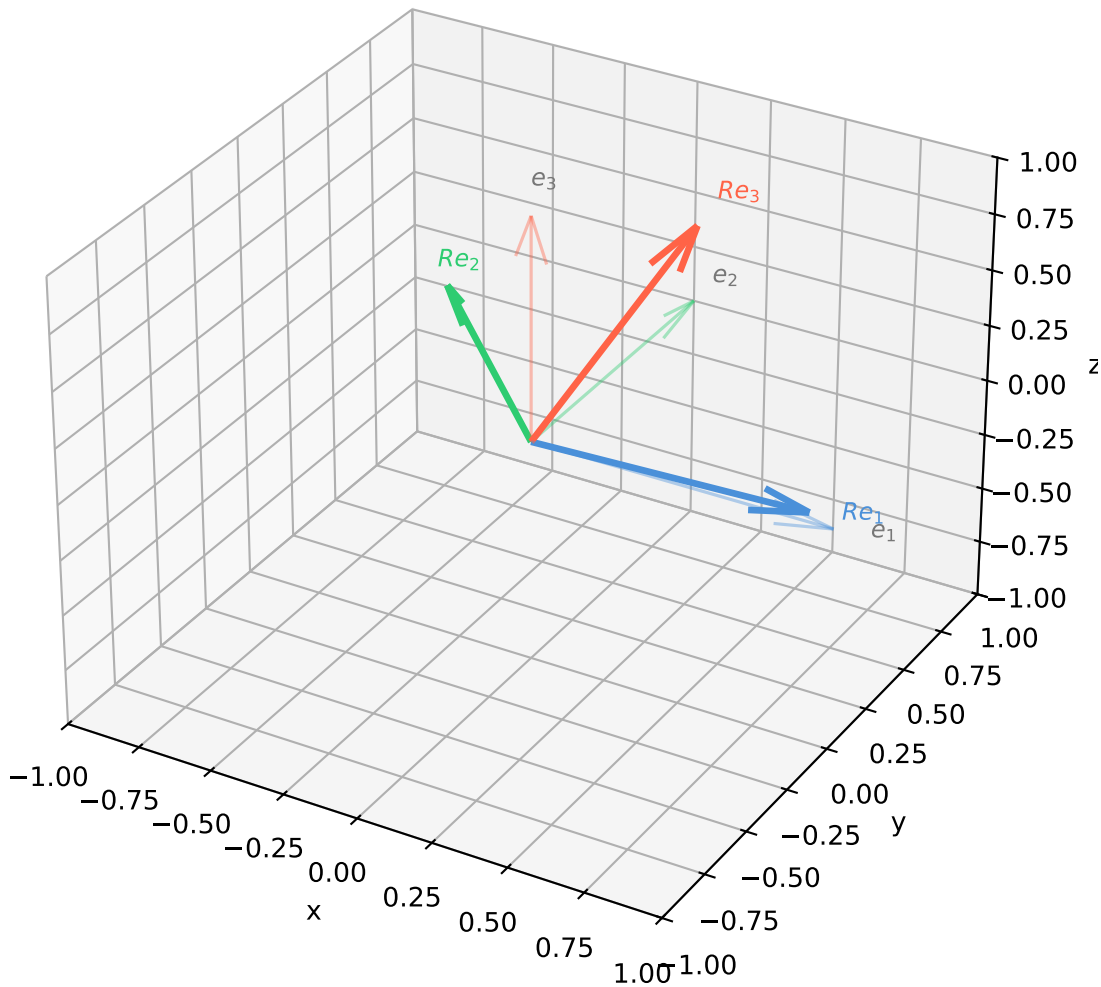
origin = np.zeros(3)    # shape (3,)
colors = ['#4a90d9', '#2ecc71', 'tomato']
labels_world = ['$e_1$', '$e_2$', '$e_3$']
labels_body  = ["$Re_1$", "$Re_2$", "$Re_3$"]

for i in range(3):
    ei = np.eye(3)[i]    # shape (3,)
    Rei = R[:, i]        # shape (3,)
    ax.quiver(*origin, *ei, color=colors[i], lw=1.2, alpha=0.4,
              arrow_length_ratio=0.2)
    ax.quiver(*origin, *Rei, color=colors[i], lw=2.5,
              arrow_length_ratio=0.2, label=f'{labels_world[i]} → {labels_body[i]}')
    ax.text(*(ei * 1.12), labels_world[i], fontsize=9, color='#777777')
    ax.text(*(Rei * 1.12), labels_body[i],  fontsize=9, color=colors[i])

ax.set_xlim(-1, 1); ax.set_ylim(-1, 1); ax.set_zlim(-1, 1)
ax.set_xlabel('x'); ax.set_ylabel('y'); ax.set_zlabel('z')
ax.set_title('Columns of R = body frame axes in world frame', fontsize=10)
fig.tight_layout()
plt.show()

```

Columns of R = body frame axes in world frame



$SO(3)$ in Robotics and Graphics

Every coordinate frame relationship in a robotic system is stored as an element of $SO(3)$ (or $SE(3)$ — Chapter 9). In ROS2, the `geometry_msgs/Quaternion` message encodes a unit quaternion, which is converted to $R \in SO(3)$ internally. In game engines (Unity, Unreal), every `Transform.rotation` is a quaternion wrapping the same underlying $SO(3)$ element.

The reason rotations are represented this way — rather than storing three Euler angles — is precisely the properties derived above: closed under composition, cheap inversion, stable numerics when re-orthogonalised via SVD.

Euler Angles and Gimbal Lock

Euler angles encode a rotation as three successive rotations about coordinate axes. The most common convention in robotics and aerospace is **ZYX (yaw-pitch-roll)**:

$$R = R_z(\psi) R_y(\phi) R_x(\theta)$$

where ψ is yaw (heading), ϕ is pitch (nose up/down), and θ is roll (bank). The intuition: first turn the vehicle's nose left or right, then tilt it up or down, then roll about the body's long axis.

Why Euler Angles Are Tempting — and Dangerous

Appeal: three numbers, directly interpretable by a pilot or operator. **Problem:** the three-angle parameterisation has a **singularity** called **gimbal lock**, where two of the three rotation axes become aligned and the system loses a degree of freedom.

For ZYX convention this happens when pitch $\phi = \pm 90^\circ$: at that point yaw and roll both rotate about the same axis, making the combined rotation non-unique.

```
import numpy as np

def Rx(a):
    c, s = np.cos(a), np.sin(a)
    return np.array([[1., 0., 0.], [0., c, -s], [0., s, c]]) # shape (3, 3)

def Ry(b):
    c, s = np.cos(b), np.sin(b)
    return np.array([[c, 0., s], [0., 1., 0.], [-s, 0., c]]) # shape (3, 3)

def Rz(g):
    c, s = np.cos(g), np.sin(g)
    return np.array([[c, -s, 0.], [s, c, 0.], [0., 0., 1.]]) # shape (3, 3)

def R_zyx(psi, phi, theta):
    """ZYX Euler angles: yaw(psi), pitch(phi), roll(theta)."""
    return Rz(psi) @ Ry(phi) @ Rx(theta) # shape (3, 3)

# Normal pose: yaw=30°, pitch=20°, roll=10°
R_normal = R_zyx(np.deg2rad(30), np.deg2rad(20), np.deg2rad(10))

# Gimbal-locked pose: pitch = 90°
# Different yaw/roll combos that look the same in this state
R_lock_a = R_zyx(np.deg2rad(45), np.deg2rad(90), np.deg2rad(0))
```

```
R_lock_b = R_zyx(np.deg2rad(0), np.deg2rad(90), np.deg2rad(-45))

print("Normal pose R:")
print(R_normal.round(4))
print("\nGimbal-locked pose A (yaw=45°, pitch=90°, roll=0°):")
print(R_lock_a.round(4))
print("\nGimbal-locked pose B (yaw=0°, pitch=90°, roll=-45°):")
print(R_lock_b.round(4))
print("\nAre these two locked poses the same rotation?",
      np.allclose(R_lock_a, R_lock_b))
```

Normal pose R:

```
[[ 0.8138 -0.441  0.3785]
 [ 0.4698  0.8826  0.018 ]
 [-0.342   0.1632  0.9254]]
```

Gimbal-locked pose A (yaw=45°, pitch=90°, roll=0°):

```
[[ 0.    -0.7071  0.7071]
 [ 0.     0.7071  0.7071]
 [-1.     0.     0.    ]]
```

Gimbal-locked pose B (yaw=0°, pitch=90°, roll=-45°):

```
[[ 0.    -0.7071  0.7071]
 [ 0.     0.7071  0.7071]
 [-1.    -0.     0.    ]]
```

Are these two locked poses the same rotation? True

The two seemingly different angle triples produce the same rotation matrix — gimbal lock destroyed one degree of freedom.

Visualising Gimbal Lock

```
import numpy as np
import matplotlib.pyplot as plt

def Rx(a):
    c, s = np.cos(a), np.sin(a)
    return np.array([[1., 0., 0.], [0., c, -s], [0., s, c]])

def Ry(b):
    c, s = np.cos(b), np.sin(b)
```



```

    return np.array([[c,0.,s],[0.,1.,0.],[-s,0.,c]])

def Rz(g):
    c, s = np.cos(g), np.sin(g)
    return np.array([[c,-s,0.],[s,c,0.],[0.,0.,1.]])

def R_zyx(psi, phi, theta):
    return Rz(psi) @ Ry(phi) @ Rx(theta)

# Sweep yaw from 0 to 360° at two pitch values: 20° (no lock) and 90° (locked)
psi_vals = np.linspace(0, 2*np.pi, 360)

# Track where [1, 0, 0] (nose direction) ends up as yaw sweeps
nose = np.array([1., 0., 0.]) # shape (3,)

nose_normal = np.array([(R_zyx(psi, np.deg2rad(20), 0.) @ nose)[:2] for psi in psi_vals])
nose_locked = np.array([(R_zyx(psi, np.deg2rad(90), 0.) @ nose)[:2] for psi in psi_vals])

fig, axes = plt.subplots(1, 2, figsize=(10, 4))

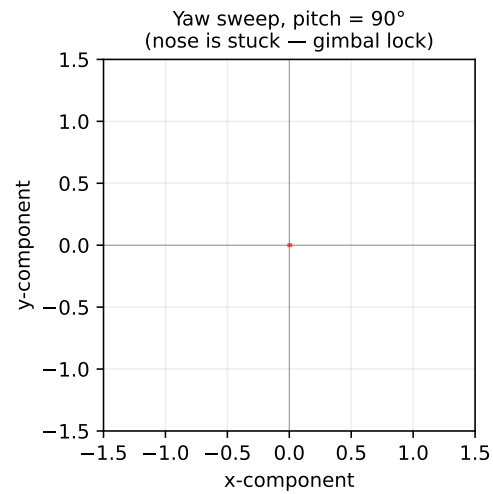
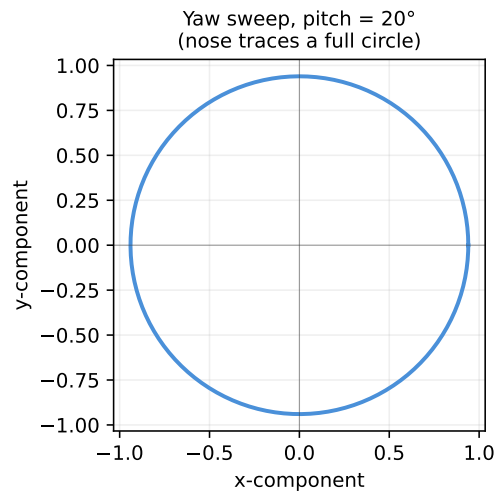
axes[0].plot(nose_normal[:, 0], nose_normal[:, 1], color='#4a90d9', lw=2)
axes[0].set_title('Yaw sweep, pitch = 20°\n(nose traces a full circle)', fontsize=10)
axes[0].set_aspect('equal'); axes[0].grid(True, alpha=0.2)
axes[0].axhline(0, color='#333333', lw=0.4, alpha=0.5)
axes[0].axvline(0, color='#333333', lw=0.4, alpha=0.5)
axes[0].set_xlabel('x-component'); axes[0].set_ylabel('y-component')

axes[1].plot(nose_locked[:, 0], nose_locked[:, 1], color='tomato', lw=2)
axes[1].set_title('Yaw sweep, pitch = 90°\n(nose is stuck - gimbal lock)', fontsize=10)
axes[1].set_aspect('equal'); axes[1].grid(True, alpha=0.2)
axes[1].axhline(0, color='#333333', lw=0.4, alpha=0.5)
axes[1].axvline(0, color='#333333', lw=0.4, alpha=0.5)
axes[1].set_xlabel('x-component'); axes[1].set_ylabel('y-component')
axes[1].set_xlim(-1.5, 1.5); axes[1].set_ylim(-1.5, 1.5)

fig.suptitle('Gimbal lock: at pitch = 90° the nose direction freezes', fontsize=11)
plt.tight_layout()
plt.show()

```

Gimbal lock: at pitch = 90° the nose direction freezes



Extracting Euler Angles from a Rotation Matrix

Given R , the ZYX Euler angles can be extracted analytically:

$$\phi = \arcsin(-r_{31}), \quad \theta = \operatorname{atan2}(r_{32}, r_{33}), \quad \psi = \operatorname{atan2}(r_{21}, r_{11})$$

These formulas break down (return NaN or a non-unique result) when $r_{31} = \pm 1$, i.e., at gimbal lock.

```
import numpy as np

def Rx(a):
    c, s = np.cos(a), np.sin(a)
    return np.array([[1., 0., 0.], [0., c, -s], [0., s, c]])

def Ry(b):
    c, s = np.cos(b), np.sin(b)
    return np.array([[c, 0., s], [0., 1., 0.], [-s, 0., c]])

def Rz(g):
    c, s = np.cos(g), np.sin(g)
    return np.array([[c, -s, 0.], [s, c, 0.], [0., 0., 1.]])

def R_zyx(psi, phi, theta):
    return Rz(psi) @ Ry(phi) @ Rx(theta)

def mat_to_euler_zyx(R):
```

```

"""Extract ZYX Euler angles from a rotation matrix.
Returns (psi, phi, theta) = (yaw, pitch, roll) in radians."""
phi = np.arcsin(-R[2, 0])          # pitch
if np.abs(np.cos(phi)) > 1e-6:    # not at gimbal lock
    theta = np.arctan2(R[2, 1], R[2, 2]) # roll
    psi = np.arctan2(R[1, 0], R[0, 0]) # yaw
else:                             # gimbal lock: one DoF lost
    theta = 0.
    psi = np.arctan2(-R[0, 1], R[1, 1])
return psi, phi, theta

# Round-trip test (not near gimbal lock)
yaw0, pitch0, roll0 = 0.6, 0.3, -0.2
R = R_zyx(yaw0, pitch0, roll0) # shape (3, 3)
yaw1, pitch1, roll1 = mat_to_euler_zyx(R)
print("Original (yaw, pitch, roll):", np.array([yaw0, pitch0, roll0]).round(6))
print("Recovered:                    ", np.array([yaw1, pitch1, roll1]).round(6))
print(f"Max error: {max(abs(yaw0-yaw1), abs(pitch0-pitch1), abs(roll0-roll1)):.2e}")

```

```

Original (yaw, pitch, roll): [ 0.6  0.3 -0.2]
Recovered:                  [ 0.6  0.3 -0.2]
Max error: 0.00e+00

```

Other Euler Conventions

There are 12 valid Euler angle conventions (6 proper Euler sequences like ZYZ plus 6 Tait-Bryan sequences like ZYX). Common choices:

Convention	Field	Note
ZYX (yaw-pitch-roll)	Aerospace, robotics	Most common in ROS
ZXZ	Classical mechanics	Euler's original
XYZ (roll-pitch-yaw)	Some robotics frameworks	Opposite order
ZYZ	Camera orientation	Useful for spherical motion

All conventions suffer from a gimbal-lock singularity at some configuration. This motivates the singularity-free representations in §8.3 (axis-angle) and §8.4 (quaternions).

Axis-Angle: Rodrigues' Formula

Every rotation in 3D can be described as a rotation by some angle θ about some unit axis $\hat{\mathbf{k}}$. This **axis-angle** representation is both geometrically intuitive and singularity-free.

The rotation matrix corresponding to angle θ about unit axis $\hat{\mathbf{k}} = [k_x, k_y, k_z]^T$ is given by **Rodrigues' formula**:

$$R = I \cos \theta + (1 - \cos \theta) \hat{\mathbf{k}} \hat{\mathbf{k}}^T + \sin \theta [\hat{\mathbf{k}}]_{\times}$$

where $[\hat{\mathbf{k}}]_{\times}$ is the **skew-symmetric cross-product matrix**:

$$[\hat{\mathbf{k}}]_{\times} = \begin{bmatrix} 0 & -k_z & k_y \\ k_z & 0 & -k_x \\ -k_y & k_x & 0 \end{bmatrix}$$

The Three Terms: A Geometric Decomposition

Rodrigues' formula decomposes the rotated vector into three orthogonal parts:

1. $(\mathbf{v} \cdot \hat{\mathbf{k}})\hat{\mathbf{k}}$: the component along the axis (unchanged by rotation)
2. $\cos \theta (\mathbf{v} - (\mathbf{v} \cdot \hat{\mathbf{k}})\hat{\mathbf{k}})$: the perpendicular component scaled down
3. $\sin \theta (\hat{\mathbf{k}} \times \mathbf{v})$: the perpendicular component rotated 90°

```
import numpy as np

def skew(k):
    """Skew-symmetric matrix for cross product: k x v = skew(k) @ v."""
    kx, ky, kz = k
    return np.array([[ 0., -kz,  ky],
                     [ kz,  0., -kx],
                     [-ky,  kx,  0.]]) # shape (3, 3)

def rodrigues(k_hat, theta):
    """Rotation matrix via Rodrigues' formula.
    k_hat: unit axis (shape (3,)), theta: angle in radians."""
    K = skew(k_hat) # shape (3, 3)
    return (np.eye(3) * np.cos(theta)
            + (1 - np.cos(theta)) * np.outer(k_hat, k_hat) # shape (3, 3)
            + np.sin(theta) * K) # shape (3, 3)

# Example: 120° about the (1,1,1)/√3 axis (cycles xyz components)
k_hat = np.array([1., 1., 1.]) / np.sqrt(3.) # shape (3,)
theta = 2 * np.pi / 3 # 120°

R = rodrigues(k_hat, theta) # shape (3, 3)
print("R (120° about [1,1,1]/√3):")
print(R.round(6))

# Verify: rotating e_x should give e_y, e_y → e_z, e_z → e_x
```

```

print("\nR @ e_x =", (R @ np.array([1.,0.,0.])).round(6), " (should be e_y)")
print("R @ e_y =", (R @ np.array([0.,1.,0.])).round(6), " (should be e_z)")
print("R @ e_z =", (R @ np.array([0.,0.,1.])).round(6), " (should be e_x)")

# Sanity checks
print(f"\nR^T R = I?    {np.allclose(R.T @ R, np.eye(3))}")
print(f"det(R) = {np.linalg.det(R):.8f}")

```

R (120° about [1,1,1]/√3):

```

[[0.  0.  1.]
 [1.  0.  0.]
 [0.  1.  0.]]

```

```

R @ e_x = [0.  1.  0.] (should be e_y)
R @ e_y = [0.  0.  1.] (should be e_z)
R @ e_z = [1.  0.  0.] (should be e_x)

```

```

R^T R = I?    True
det(R) = 1.00000000

```

Recovering Axis and Angle from a Rotation Matrix

Given $R \in \text{SO}(3)$, the axis-angle parameters are:

$$\theta = \arccos\left(\frac{\text{tr}(R) - 1}{2}\right)$$

$$\hat{\mathbf{k}} = \frac{1}{2 \sin \theta} \begin{bmatrix} R_{32} - R_{23} \\ R_{13} - R_{31} \\ R_{21} - R_{12} \end{bmatrix}$$

This formula is valid for $\theta \in (0^\circ, 180^\circ)$. At $\theta = 0$ the axis is undefined (no rotation). At $\theta = 180^\circ$, $\sin \theta = 0$ and the formula is numerically unstable — a special case is needed.

```

import numpy as np

def skew(k):
    kx, ky, kz = k
    return np.array([[ 0., -kz,  ky],
                    [ kz,  0., -kx],
                    [-ky,  kx,  0.]])

def rodrigues(k_hat, theta):

```

```

K = skew(k_hat)
return (np.eye(3) * np.cos(theta)
        + (1 - np.cos(theta)) * np.outer(k_hat, k_hat)
        + np.sin(theta) * K)

def mat_to_axis_angle(R):
    """Extract axis and angle from a rotation matrix R."""
    trace = np.clip((np.trace(R) - 1.) / 2., -1., 1.) # clip for numerical safety
    theta = np.arccos(trace)
    if np.abs(np.sin(theta)) < 1e-8: # near 0° or 180°
        if theta < 0.01:
            return np.array([0., 0., 1.]), 0. # arbitrary axis
        else:
            # 180°: use symmetric part of R
            A = (R + R.T) / 2. - np.eye(3) * (np.cos(theta)) # proportional to k k^T
            k_hat = np.sqrt(np.maximum(np.diag(A) / (1. - np.cos(theta)), 0.))
            # Fix signs: k_hat[0] positive by convention
            if R[2, 1] - R[1, 2] < 0: k_hat[0] *= -1
            return k_hat / np.linalg.norm(k_hat), theta
    vee = np.array([R[2,1]-R[1,2], R[0,2]-R[2,0], R[1,0]-R[0,1]]) # shape (3,)
    k_hat = vee / (2. * np.sin(theta)) # shape (3,)
    return k_hat, theta

# Round-trip test
k_original = np.array([1., 2., -1.]) / np.linalg.norm([1., 2., -1.]) # shape (3,)
theta_original = np.deg2rad(55.)

R = rodrigues(k_original, theta_original) # shape (3, 3)
k_rec, theta_rec = mat_to_axis_angle(R)

print("Original axis: ", k_original.round(6))
print("Recovered axis: ", k_rec.round(6))
print(f"Original angle: {np.rad2deg(theta_original):.4f}°")
print(f"Recovered angle: {np.rad2deg(theta_rec):.4f}°")
print(f"Axis match: {np.allclose(np.abs(k_rec), np.abs(k_original))}")

```

```

Original axis: [ 0.408248  0.816497 -0.408248]
Recovered axis: [ 0.408248  0.816497 -0.408248]
Original angle: 55.0000°
Recovered angle: 55.0000°
Axis match: True

```

Visualising a Rotation: Sweeping the Angle

```

import numpy as np
import matplotlib.pyplot as plt
from mpl_toolkits.mplot3d import Axes3D  # noqa: F401

def skew(k):
    kx, ky, kz = k
    return np.array([[ 0., -kz,  ky],[kz, 0.,-kx],[-ky, kx, 0.]])

def rodrigues(k_hat, theta):
    K = skew(k_hat)
    return np.eye(3)*np.cos(theta) + (1-np.cos(theta))*np.outer(k_hat, k_hat) + np.sin(theta)*K

# Rotate the point (1, 0, 0) about axis z=(0,0,1) by angles 0..360°
k_hat = np.array([0., 0., 1.])  # shape (3,)
v0     = np.array([1., 0., 0.])  # shape (3,)
thetas = np.linspace(0, 2*np.pi, 200)

trajectory = np.array([rodrigues(k_hat, t) @ v0 for t in thetas])  # shape (200, 3)

# Also rotate a tilted vector about a tilted axis
k2 = np.array([1., 1., 1.]) / np.sqrt(3.)  # shape (3,)
v2 = np.array([0., 1., 0.])  # shape (3,)
traj2 = np.array([rodrigues(k2, t) @ v2 for t in thetas])  # shape (200, 3)

fig = plt.figure(figsize=(8, 5))
ax = fig.add_subplot(111, projection='3d')

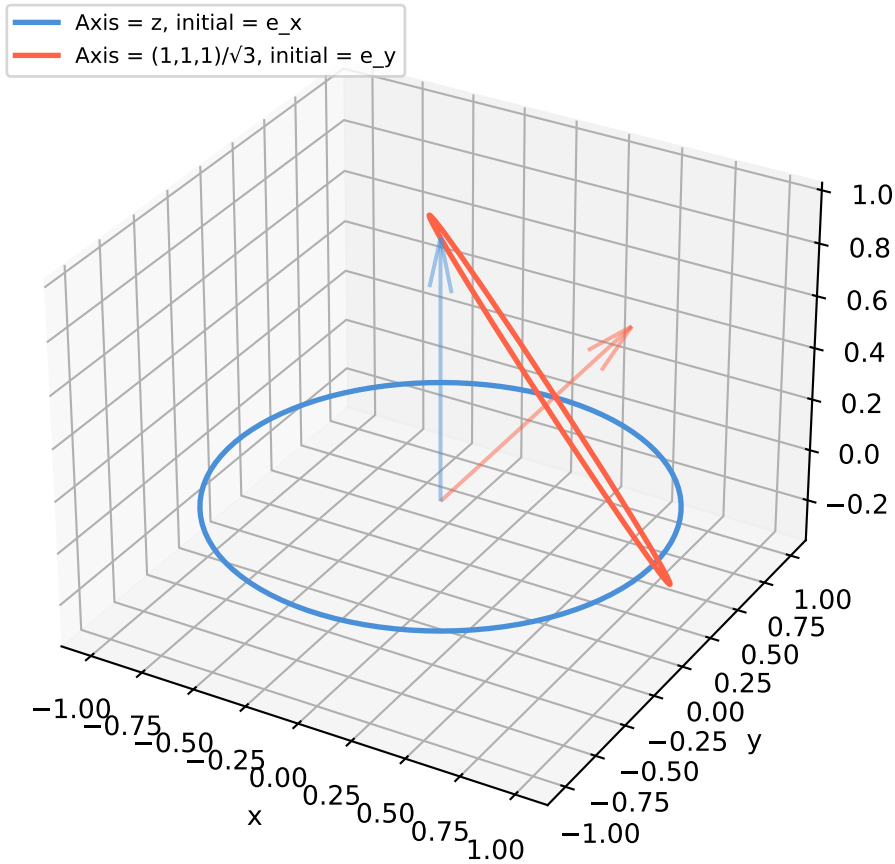
ax.plot(trajectory[:,0], trajectory[:,1], trajectory[:,2],
        color='#4a90d9', lw=2, label='Axis = z, initial = e_x')
ax.plot(traj2[:,0], traj2[:,1], traj2[:,2],
        color='tomato', lw=2, label='Axis = (1,1,1)/√3, initial = e_y')

# Draw axes
for col, lbl in zip(['#4a90d9', 'tomato'], [k_hat, k2]):
    ax.quiver(0,0,0, *lbl, color=col, lw=1.5, alpha=0.5, arrow_length_ratio=0.2)

ax.set_xlabel('x'); ax.set_ylabel('y'); ax.set_zlabel('z')
ax.set_title('Rodrigues rotation: a vector sweeps a circle on the unit sphere', fontsize=10)
ax.legend(fontsize=8, loc='upper left')
fig.tight_layout()
plt.show()

```

Rodrigues rotation: a vector sweeps a circle on the unit sphere



The Compact Form: Rotation Vector

The **rotation vector** $\mathbf{r} = \theta \hat{\mathbf{k}}$ encodes both axis and angle in a single 3-vector. Its magnitude is the rotation angle; its direction is the rotation axis.

- $\|\mathbf{r}\| = 0$: identity (no rotation)
- $\mathbf{r} = [0, 0, \pi/4]^T$: rotate 45° about z

This compact form is the basis of the **Lie algebra** $\mathfrak{so}(3)$ (Chapter 24) and is what ROS2 stores as `angular_velocity` messages.

```
import numpy as np

def skew(k):
    kx, ky, kz = k
    return np.array([[ 0., -kz,  ky], [kz,  0., -kx], [-ky,  kx,  0.]])

def rodrigues_vec(r):
```



```

    """Rotation matrix from rotation vector r = theta * k_hat."""
    theta = np.linalg.norm(r)    # scalar
    if theta < 1e-10:
        return np.eye(3)
    k_hat = r / theta            # shape (3,)
    K = skew(k_hat)              # shape (3, 3)
    return np.eye(3)*np.cos(theta) + (1-np.cos(theta))*np.outer(k_hat,k_hat) + np.sin(theta)*np.cross(k_hat,k_hat)

# Encode "rotate 45° about z" as a rotation vector
r = np.array([0., 0., np.pi/4])    # shape (3,)
R = rodrigues_vec(r)                # shape (3, 3)
print("r =", r.round(4), " ||r|| =", np.linalg.norm(r).round(4), "rad =", np.rad2deg(np.linalg.norm(r)).round(4))
print("R ="); print(R.round(4))

```

```

r = [0.      0.      0.7854]    ||r|| = 0.7854 rad = 45.0 °
R =
[[ 0.7071 -0.7071  0.      ]
 [ 0.7071  0.7071  0.      ]
 [ 0.      0.      1.      ]]

```

Quaternions: Algebra and Rotation

A **quaternion** is a hypercomplex number with one real part and three imaginary parts:

$$q = w + xi + yj + zk \quad w, x, y, z \in \mathbb{R}$$

The imaginary units satisfy Hamilton's rules:

$$i^2 = j^2 = k^2 = ijk = -1$$

$$ij = k, \quad jk = i, \quad ki = j, \quad ji = -k, \quad kj = -i, \quad ik = -j$$

A **unit quaternion** ($\|q\| = 1$) encodes a 3D rotation and avoids all the problems of Euler angles: no singularities, smooth interpolation, and efficient composition.

Quaternion as Axis-Angle

A rotation by angle θ about unit axis $\hat{\mathbf{k}}$ is encoded as:

$$q = \cos \frac{\theta}{2} + \sin \frac{\theta}{2} (k_x i + k_y j + k_z k)$$

In vector notation: $q = [w, \mathbf{v}]$ where $w = \cos(\theta/2)$ and $\mathbf{v} = \sin(\theta/2)\hat{\mathbf{k}}$.

```
import numpy as np

def axis_angle_to_quat(k_hat, theta):
    """Unit quaternion from axis-angle. Returns [w, x, y, z] shape (4,)."""
    w = np.cos(theta / 2.)
    v = np.sin(theta / 2.) * k_hat    # shape (3,)
    return np.array([w, v[0], v[1], v[2]])    # shape (4,)

# 90° about z-axis
k = np.array([0., 0., 1.])    # shape (3,)
q = axis_angle_to_quat(k, np.pi / 2)
print(f"q (90° about z) = {q.round(6)}")
print(f"||q|| = {np.linalg.norm(q):.8f} (should be 1)")

# Identity rotation: theta = 0
q_id = axis_angle_to_quat(k, 0.)
print(f"\nIdentity quaternion = {q_id} (w=1, xyz=0)")
```

```
q (90° about z) = [0.707107 0.          0.          0.707107]
||q|| = 1.00000000 (should be 1)
```

```
Identity quaternion = [1. 0. 0. 0.] (w=1, xyz=0)
```

The Hamilton Product

Composing two rotations corresponds to the **Hamilton product** of their quaternions. For $p = [p_w, \mathbf{p}_v]$ and $q = [q_w, \mathbf{q}_v]$:

$$p \otimes q = \begin{bmatrix} p_w q_w - \mathbf{p}_v \cdot \mathbf{q}_v \\ p_w \mathbf{q}_v + q_w \mathbf{p}_v + \mathbf{p}_v \times \mathbf{q}_v \end{bmatrix}$$

```

import numpy as np

def quat_mul(p, q):
    """Hamilton product of two quaternions [w,x,y,z]. Returns shape (4,)"""
    pw, px, py, pz = p
    qw, qx, qy, qz = q
    return np.array([
        pw*qw - px*qx - py*qy - pz*qz,
        pw*qx + px*qw + py*qz - pz*qy,
        pw*qy - px*qz + py*qw + pz*qx,
        pw*qz + px*qy - py*qx + pz*qw
    ]) # shape (4,)

def axis_angle_to_quat(k_hat, theta):
    w = np.cos(theta / 2.)
    v = np.sin(theta / 2.) * k_hat
    return np.array([w, v[0], v[1], v[2]])

# Compose: 90° about x, then 90° about z
q1 = axis_angle_to_quat(np.array([1.,0.,0.]), np.pi/2) # shape (4,)
q2 = axis_angle_to_quat(np.array([0.,0.,1.]), np.pi/2) # shape (4,)

q_composed = quat_mul(q2, q1) # shape (4,) - q2 applied after q1 (right-to-left)
print("q1 (90° about x):", q1.round(6))
print("q2 (90° about z):", q2.round(6))
print("q2 q1 (composed):", q_composed.round(6))
print(f"||q_composed|| = {np.linalg.norm(q_composed):.8f}")

# Non-commutativity: q1 q2 q2 q1
q_rev = quat_mul(q1, q2) # shape (4,)
print(f"\nq1 q2 = {q_rev.round(6)}")
print(f"q2 q1 = {q_composed.round(6)}")
print(f"Equal? {np.allclose(q_composed, q_rev)}")

```

```

q1 (90° about x): [0.707107 0.707107 0.          0.          ]
q2 (90° about z): [0.707107 0.          0.          0.707107]
q2 q1 (composed): [0.5 0.5 0.5 0.5]
||q_composed|| = 1.00000000

```

```

q1 q2 = [ 0.5  0.5 -0.5  0.5]
q2 q1 = [0.5 0.5 0.5 0.5]
Equal? False

```

Rotating a Vector: $q\mathbf{v}q^{-1}$

To rotate vector $\mathbf{v}$ by unit quaternion q :

1. Embed: $\tilde{v} = [0, \mathbf{v}]$ (pure quaternion)
2. Conjugate: $\mathbf{v}' = q \otimes \tilde{v} \otimes q^*$ where $q^* = [w, -\mathbf{v}]$

For a unit quaternion, $q^{-1} = q^*$.

```
import numpy as np

def quat_mul(p, q):
    pw, px, py, pz = p
    qw, qx, qy, qz = q
    return np.array([
        pw*qw - px*qx - py*qy - pz*qz,
        pw*qx + px*qw + py*qz - pz*qy,
        pw*qy - px*qz + py*qw + pz*qx,
        pw*qz + px*qy - py*qx + pz*qw
    ])

def quat_conj(q):
    """Conjugate q* = [w, -x, -y, -z]."""
    return np.array([q[0], -q[1], -q[2], -q[3]]) # shape (4,)

def quat_rotate(q, v):
    """Rotate 3D vector v by unit quaternion q. Returns shape (3,)."""
    v_quat = np.array([0., v[0], v[1], v[2]]) # shape (4,)
    v_rot = quat_mul(quat_mul(q, v_quat), quat_conj(q)) # shape (4,)
    return v_rot[1:] # shape (3,) - drop the scalar part

def axis_angle_to_quat(k_hat, theta):
    w = np.cos(theta / 2.)
    v = np.sin(theta / 2.) * k_hat
    return np.array([w, v[0], v[1], v[2]])

# Verify against Rodrigues' formula
def rodrigues(k_hat, theta):
    from numpy import cos, sin, eye, outer
    K = np.array([[0, -k_hat[2], k_hat[1]], [k_hat[2], 0, -k_hat[0]], [-k_hat[1], k_hat[0], 0]])
    return eye(3)*cos(theta) + (1-cos(theta))*outer(k_hat, k_hat) + sin(theta)*K

k_hat = np.array([1., 1., 0.]) / np.sqrt(2.) # shape (3,)
theta = np.deg2rad(70.)
v = np.array([1., 2., 3.]) # shape (3,)

q = axis_angle_to_quat(k_hat, theta) # shape (4,)
v_quat = quat_rotate(q, v) # shape (3,)
```

```

v_rod = rodrigues(k_hat, theta) @ v    # shape (3,)

print("Rotation via quaternion:      ", v_quat.round(6))
print("Rotation via Rodrigues matrix: ", v_rod.round(6))
print(f"Match: {np.allclose(v_quat, v_rod)}")

```

```

Rotation via quaternion:      [ 3.322379 -0.322379  1.690523]
Rotation via Rodrigues matrix: [ 3.322379 -0.322379  1.690523]
Match: True

```

Quaternion to Rotation Matrix

Converting a unit quaternion $q = [w, x, y, z]$ to a 3×3 rotation matrix:

$$R = \begin{bmatrix} 1 - 2(y^2 + z^2) & 2(xy - wz) & 2(xz + wy) \\ 2(xy + wz) & 1 - 2(x^2 + z^2) & 2(yz - wx) \\ 2(xz - wy) & 2(yz + wx) & 1 - 2(x^2 + y^2) \end{bmatrix}$$

```

import numpy as np

def quat_to_mat(q):
    """Convert unit quaternion [w,x,y,z] to 3x3 rotation matrix."""
    w, x, y, z = q / np.linalg.norm(q)    # normalise for safety
    return np.array([
        [1-2*(y*y+z*z), 2*(x*y-w*z), 2*(x*z+w*y)],
        [ 2*(x*y+w*z), 1-2*(x*x+z*z), 2*(y*z-w*x)],
        [ 2*(x*z-w*y), 2*(y*z+w*x), 1-2*(x*x+y*y)]
    ])    # shape (3, 3)

def axis_angle_to_quat(k_hat, theta):
    w = np.cos(theta / 2.)
    v = np.sin(theta / 2.) * k_hat
    return np.array([w, v[0], v[1], v[2]])

k_hat = np.array([0., 1., 0.])    # shape (3,)
theta = np.deg2rad(45.)

q = axis_angle_to_quat(k_hat, theta)    # shape (4,)
R = quat_to_mat(q)                    # shape (3, 3)
print("R from quaternion:")
print(R.round(6))
print(f"\nR^T R = I? {np.allclose(R.T @ R, np.eye(3))}")
print(f"det(R) = {np.linalg.det(R):.8f}")

```

3D Rotations and $SO(3)$

```
R from quaternion:
[[ 0.707107  0.          0.707107]
 [ 0.         1.         0.        ]
 [-0.707107  0.          0.707107]]
```

```
RT R = I?  True
det(R) = 1.00000000
```

The Double Cover of $SO(3)$

A critical subtlety: q and $-q$ represent the **same rotation**.

$$R(q) = R(-q)$$

This is because flipping all four components of a unit quaternion negates the pure-quaternion part in $q\tilde{v}q^*$ twice, leaving the result unchanged. The sphere S^3 of unit quaternions is a **double cover** of $SO(3)$: every rotation corresponds to exactly two antipodal quaternions.

This matters for interpolation (§8.5): to take the short path on S^3 , always check that $p \cdot q > 0$ and negate q if not.

```
import numpy as np

def quat_to_mat(q):
    w, x, y, z = q / np.linalg.norm(q)
    return np.array([
        [1-2*(y*y+z*z), 2*(x*y-w*z), 2*(x*z+w*y)],
        [2*(x*y+w*z), 1-2*(x*x+z*z), 2*(y*z-w*x)],
        [2*(x*z-w*y), 2*(y*z+w*x), 1-2*(x*x+y*y)]
    ])

q = np.array([0.707, 0.0, 0.707, 0.0]) # shape (4,) - some unit quaternion
q_neg = -q                             # shape (4,) - antipodal

R1 = quat_to_mat(q) # shape (3, 3)
R2 = quat_to_mat(q_neg) # shape (3, 3)
print("q and -q produce the same rotation matrix:", np.allclose(R1, R2))
print(f"\nq      = {q}")
print(f"-q     = {q_neg}")
print(f"R(q)   =\n{R1.round(6)}")
print(f"R(-q)  =\n{R2.round(6)}")
```

```
q and -q produce the same rotation matrix: True
```

```

q      = [0.707 0.      0.707 0.   ]
-q     = [-0.707 -0.      -0.707 -0. ]
R(q)   =
[[-0.  0.  1.]
 [ 0.  1.  0.]
 [-1.  0. -0.]]
R(-q)  =
[[-0.  0.  1.]
 [ 0.  1.  0.]
 [-1.  0. -0.]]

```

SLERP: Interpolating Rotations

Interpolating between two rotations is not as simple as linearly interpolating between two matrices or two Euler-angle triples. The correct tool is **Spherical Linear Interpolation (SLERP)** — a geodesic walk along the great circle of the unit quaternion sphere S^3 .

Why Linear Matrix Interpolation Fails

A matrix $(1-t)R_0 + tR_1$ does not stay in $SO(3)$ for $t \in (0, 1)$: the interpolated matrix is not orthogonal and can even have determinant < 1 .

```

import numpy as np

def Rz(g):
    c, s = np.cos(g), np.sin(g)
    return np.array([[c,-s,0.],[s,c,0.],[0.,0.,1.]]) # shape (3, 3)

R0 = Rz(0.) # identity
R1 = Rz(np.pi / 2.) # 90° rotation

ts = np.linspace(0, 1, 9)
print(f"{'t':>5} {'det(lerp)':>12} {'||R^T R - I||':>16}")
for t in ts:
    R_lerp = (1 - t) * R0 + t * R1 # shape (3, 3) - linear interpolation
    det_val = np.linalg.det(R_lerp)
    orth_err = np.linalg.norm(R_lerp.T @ R_lerp - np.eye(3))
    print(f"{'t':5.2f} {'det_val':12.6f} {'orth_err':16.6f}")

```

t	det(lerp)	R ^T R - I
0.00	1.000000	0.000000
0.12	0.781250	0.309359
0.25	0.625000	0.530330

0.38	0.531250	0.662913
0.50	0.500000	0.707107
0.62	0.531250	0.662913
0.75	0.625000	0.530330
0.88	0.781250	0.309359
1.00	1.000000	0.000000

The determinant drops well below 1 — the interpolated matrices are **not rotations**.

SLERP: Geodesic on the Quaternion Sphere

For unit quaternions q_0 and q_1 , SLERP at parameter $t \in [0, 1]$ is:

$$\text{slerp}(q_0, q_1, t) = q_0 (q_0^{-1} q_1)^t = \frac{\sin((1-t)\Omega)}{\sin \Omega} q_0 + \frac{\sin(t\Omega)}{\sin \Omega} q_1$$

where $\Omega = \arccos(q_0 \cdot q_1)$ is the half-angle between the quaternions on S^3 . When $\Omega \approx 0$ (nearly identical rotations), fall back to linear interpolation (LERP) to avoid division by zero.

```
import numpy as np

def slerp(q0, q1, t):
    """Spherical linear interpolation between unit quaternions q0 and q1."""
    q0 = q0 / np.linalg.norm(q0) # shape (4,)
    q1 = q1 / np.linalg.norm(q1) # shape (4,)

    dot = np.clip(np.dot(q0, q1), -1., 1.) # scalar

    # Take the short path: if dot < 0, negate q1
    if dot < 0.:
        q1 = -q1
        dot = -dot

    Omega = np.arccos(dot) # scalar

    if Omega < 1e-8: # nearly identical - use LERP
        q = (1. - t) * q0 + t * q1
        return q / np.linalg.norm(q)

    return (np.sin((1. - t) * Omega) / np.sin(Omega)) * q0 \
        + (np.sin(t * Omega) / np.sin(Omega)) * q1 # shape (4,)

def axis_angle_to_quat(k_hat, theta):
    w = np.cos(theta / 2.)
```



```

    v = np.sin(theta / 2.) * k_hat
    return np.array([w, v[0], v[1], v[2]])    # shape (4,)

# Interpolate between 0° and 90° about z
q0 = axis_angle_to_quat(np.array([0.,0.,1.]), 0.)
q1 = axis_angle_to_quat(np.array([0.,0.,1.]), np.pi/2)

ts = np.linspace(0., 1., 6)
print(f'{t':>5}  {'q (slerp)':>40}  {'||q||':>8}")
for t in ts:
    q = slerp(q0, q1, t)    # shape (4,)
    print(f'{t:5.2f}  {str(q.round(4)):>40}  {np.linalg.norm(q):.6f}")

```

t	q (slerp)				q
0.00			[1. 0. 0. 0.]	1.000000	
0.20	[0.9877 0.	0.	0.1564]	1.000000	
0.40	[0.9511 0.	0.	0.309]	1.000000	
0.60	[0.891 0.	0.	0.454]	1.000000	
0.80	[0.809 0.	0.	0.5878]	1.000000	
1.00	[0.7071 0.	0.	0.7071]	1.000000	

SLERP keeps $\|q\| = 1$ at every step: the interpolated quaternions are always valid unit quaternions, hence always valid rotations.

Euler-Angle Interpolation: The Artefact

Linear interpolation of Euler angles near a singularity produces a jarring “jump” or unexpected trajectory. Here we compare SLERP to naive Euler lerp:

```

import numpy as np
import matplotlib.pyplot as plt

def Rx(a):
    c, s = np.cos(a), np.sin(a)
    return np.array([[1.,0.,0.],[0.,c,-s],[0.,s,c]])

def Ry(b):
    c, s = np.cos(b), np.sin(b)
    return np.array([[c,0.,s],[0.,1.,0.],[-s,0.,c]])

def Rz(g):
    c, s = np.cos(g), np.sin(g)
    return np.array([[c,-s,0.],[s,c,0.],[0.,0.,1.]])

```

```

def R_zyx(psi, phi, theta):
    return Rz(psi) @ Ry(phi) @ Rx(theta)

def axis_angle_to_quat(k_hat, theta):
    w = np.cos(theta / 2.)
    v = np.sin(theta / 2.) * k_hat
    return np.array([w, v[0], v[1], v[2]])

def quat_mul(p, q):
    pw, px, py, pz = p; qw, qx, qy, qz = q
    return np.array([pw*qw-px*qx-py*qy-pz*qz, pw*qx+px*qw+py*qz-pz*qy,
                    pw*qy-px*qz+py*qw+pz*qx, pw*qz+px*qy-py*qx+pz*qw])

def quat_conj(q):
    return np.array([q[0], -q[1], -q[2], -q[3]])

def quat_rotate(q, v):
    vq = np.array([0., *v])
    return quat_mul(quat_mul(q, vq), quat_conj(q))[1:]

def slerp(q0, q1, t):
    q0 = q0 / np.linalg.norm(q0); q1 = q1 / np.linalg.norm(q1)
    dot = np.clip(np.dot(q0, q1), -1., 1.)
    if dot < 0.: q1 = -q1; dot = -dot
    Om = np.arccos(dot)
    if Om < 1e-8:
        q = (1-t)*q0 + t*q1; return q/np.linalg.norm(q)
    return np.sin((1-t)*Om)/np.sin(Om)*q0 + np.sin(t*Om)/np.sin(Om)*q1

def mat_to_quat(R):
    """Shepperd method: rotation matrix to unit quaternion."""
    trace = np.trace(R) # scalar
    if trace > 0:
        s = 0.5 / np.sqrt(trace + 1.)
        return np.array([0.25/s, (R[2,1]-R[1,2])*s, (R[0,2]-R[2,0])*s, (R[1,0]-R[0,1])*s])
    elif R[0,0] > R[1,1] and R[0,0] > R[2,2]:
        s = 2.*np.sqrt(1.+R[0,0]-R[1,1]-R[2,2])
        return np.array([(R[2,1]-R[1,2])/s, 0.25*s, (R[0,1]+R[1,0])/s, (R[0,2]+R[2,0])/s])
    elif R[1,1] > R[2,2]:
        s = 2.*np.sqrt(1.+R[1,1]-R[0,0]-R[2,2])
        return np.array([(R[0,2]-R[2,0])/s, (R[0,1]+R[1,0])/s, 0.25*s, (R[1,2]+R[2,1])/s])
    else:
        s = 2.*np.sqrt(1.+R[2,2]-R[0,0]-R[1,1])
        return np.array([(R[1,0]-R[0,1])/s, (R[0,2]+R[2,0])/s, (R[1,2]+R[2,1])/s, 0.25*s])

# Pose A: yaw=0°, pitch=80°, roll=0°
# Pose B: yaw=90°, pitch=80°, roll=0°

```

```

# Near gimbal lock (pitch close to 90°) - Euler lerp will wobble
e_A = np.array([0., np.deg2rad(80.), 0.]) # shape (3,)
e_B = np.array([np.pi/2, np.deg2rad(80.), 0.]) # shape (3,)

R_A = R_zyx(*e_A); R_B = R_zyx(*e_B) # shape (3, 3)
q_A = mat_to_quat(R_A); q_B = mat_to_quat(R_B) # shape (4,)

v_probe = np.array([1., 0., 0.]) # shape (3,)
ts = np.linspace(0., 1., 60)

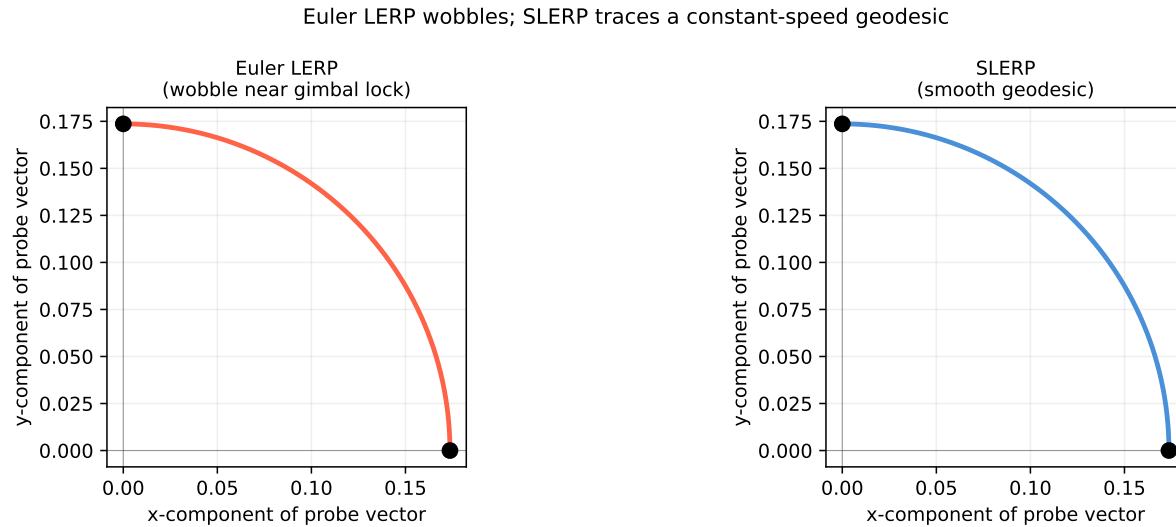
# Euler LERP: interpolate angle triples, rebuild matrix
traj_euler = np.array([
    R_zyx*(t * e_B + (1-t) * e_A) @ v_probe for t in ts]) # shape (60, 3)

# SLERP
traj_slerp = np.array([
    quat_rotate(slerp(q_A, q_B, t), v_probe) for t in ts]) # shape (60, 3)

fig, axes = plt.subplots(1, 2, figsize=(11, 4))
for ax, traj, title, col in [
    (axes[0], traj_euler, 'Euler LERP\n(wobble near gimbal lock)', 'tomato'),
    (axes[1], traj_slerp, 'SLERP\n(smooth geodesic)', '#4a90d9')]:
    ax.plot(traj[:, 0], traj[:, 1], color=col, lw=2.5)
    ax.scatter([traj[0,0], traj[-1,0]], [traj[0,1], traj[-1,1]],
               color='black', s=60, zorder=5)
    ax.set_title(title, fontsize=10)
    ax.set_xlabel('x-component of probe vector')
    ax.set_ylabel('y-component of probe vector')
    ax.grid(True, alpha=0.2)
    ax.axhline(0, color='#333333', lw=0.4, alpha=0.5)
    ax.axvline(0, color='#333333', lw=0.4, alpha=0.5)
    ax.set_aspect('equal')

fig.suptitle('Euler LERP wobbles; SLERP traces a constant-speed geodesic', fontsize=11)
plt.tight_layout()
plt.show()

```



Constant Angular Velocity

SLERP has a key property: the angular velocity between consecutive steps is **constant**. The rotation sweeps at a uniform rate, unlike Euler interpolation which accelerates and decelerates unpredictably near singularities.

This constant-speed property makes SLERP the standard for: - **Animation blending** (game engines, motion capture) - **Robot trajectory planning** (smooth joint interpolation) - **Sensor fusion** (averaging multiple orientation estimates)

```
import numpy as np

def axis_angle_to_quat(k_hat, theta):
    w = np.cos(theta / 2.); v = np.sin(theta / 2.) * k_hat
    return np.array([w, v[0], v[1], v[2]])

def slerp(q0, q1, t):
    q0 = q0/np.linalg.norm(q0); q1 = q1/np.linalg.norm(q1)
    dot = np.clip(np.dot(q0, q1), -1., 1.)
    if dot < 0.: q1 = -q1; dot = -dot
    Om = np.arccos(dot)
    if Om < 1e-8:
        q = (1-t)*q0+t*q1; return q/np.linalg.norm(q)
    return np.sin((1-t)*Om)/np.sin(Om)*q0 + np.sin(t*Om)/np.sin(Om)*q1

q0 = axis_angle_to_quat(np.array([0.,0.,1.]), 0.)
q1 = axis_angle_to_quat(np.array([0.,0.,1.]), np.deg2rad(120.))

ts = np.linspace(0., 1., 11)
```

```

qs = np.array([slerp(q0, q1, t) for t in ts])    # shape (11, 4)

# Angular speed = angle between consecutive quaternions
angles = []
for i in range(len(qs)-1):
    dot = np.clip(np.dot(qs[i], qs[i+1]), -1., 1.)
    angle = 2. * np.arccos(dot)    # full rotation angle in radians
    angles.append(np.rad2deg(angle))

print("Angular step between consecutive SLERP frames (degrees):")
print(np.array(angles).round(4))
print(f"\nAll equal? {np.allclose(angles, angles[0])}")

```

```

Angular step between consecutive SLERP frames (degrees):
[12. 12. 12. 12. 12. 12. 12. 12. 12. 12.]

```

```

All equal? True

```

Conversion Table: All Four Representations

There are four mainstream representations for 3D rotations, each with different strengths. This section gives closed-form conversion formulas for all 12 pairwise paths, implemented as self-contained Python functions with numerical edge-case handling.

Representation	Parameters	DoF encoded	Singularity	Best for
Rotation matrix R	9 numbers	3	None (but drift)	Composing transforms, vector rotation
Euler angles	3 numbers (ZYX)	3	Gimbal lock at $\phi = \pm 90^\circ$	Human-readable output, flight displays
Axis-angle $(\hat{k}, \theta)$	4 numbers	3	Near $\theta = 0, \pi$	Differentiation, Lie algebra
Unit quaternion q	4 numbers	3	None (double cover)	Interpolation, real-time systems

Matrix Quaternion

```

import numpy as np

def mat_to_quat(R):
    """Shepperd's method: 3x3 rotation matrix -> unit quaternion [w,x,y,z].
    Numerically stable across all rotations."""
    trace = np.trace(R)    # scalar
    if trace > 0.:
        s = 0.5 / np.sqrt(trace + 1.)
        return np.array([(0.25 / s),
                        (R[2,1] - R[1,2]) * s,
                        (R[0,2] - R[2,0]) * s,
                        (R[1,0] - R[0,1]) * s])    # shape (4,)
    elif R[0,0] > R[1,1] and R[0,0] > R[2,2]:
        s = 2. * np.sqrt(1. + R[0,0] - R[1,1] - R[2,2])
        return np.array([(R[2,1] - R[1,2]) / s, 0.25*s,
                        (R[0,1] + R[1,0]) / s, (R[0,2] + R[2,0]) / s])
    elif R[1,1] > R[2,2]:
        s = 2. * np.sqrt(1. + R[1,1] - R[0,0] - R[2,2])
        return np.array([(R[0,2] - R[2,0]) / s, (R[0,1] + R[1,0]) / s,
                        0.25*s, (R[1,2] + R[2,1]) / s])
    else:
        s = 2. * np.sqrt(1. + R[2,2] - R[0,0] - R[1,1])
        return np.array([(R[1,0] - R[0,1]) / s, (R[0,2] + R[2,0]) / s,
                        (R[1,2] + R[2,1]) / s, 0.25*s])

def quat_to_mat(q):
    """Unit quaternion [w,x,y,z] -> 3x3 rotation matrix."""
    w, x, y, z = q / np.linalg.norm(q)    # normalise for safety
    return np.array([
        [1-2*(y*y+z*z), 2*(x*y-w*z), 2*(x*z+w*y)],
        [ 2*(x*y+w*z), 1-2*(x*x+z*z), 2*(y*z-w*x)],
        [ 2*(x*z-w*y), 2*(y*z+w*x), 1-2*(x*x+y*y)]
    ])    # shape (3, 3)

def Rz(g):
    c,s=np.cos(g),np.sin(g)
    return np.array([[c,-s,0.],[s,c,0.],[0.,0.,1.]])

def Ry(b):
    c,s=np.cos(b),np.sin(b)
    return np.array([[c,0.,s],[0.,1.,0.],[-s,0.,c]])

R = Rz(0.5) @ Ry(0.3)    # shape (3, 3)
q = mat_to_quat(R)        # shape (4,)
R2 = quat_to_mat(q)      # shape (3, 3)

print("R -> q -> R round-trip:")

```

```
print(f"q = {q.round(6)}")
print(f"||q|| = {np.linalg.norm(q):.8f}")
print(f"R == quat_to_mat(q)? {np.allclose(R, R2)}")
```

```
R -> q -> R round-trip:
q = [ 0.958033 -0.036972  0.144792  0.244626]
||q|| = 1.00000000
R == quat_to_mat(q)? True
```

Matrix Axis-Angle

```
import numpy as np

def mat_to_axis_angle(R):
    """3x3 rotation matrix -> (unit axis shape(3,), angle scalar)."""
    trace = np.clip((np.trace(R) - 1.) / 2., -1., 1.)
    theta = np.arccos(trace) # scalar
    if np.abs(np.sin(theta)) < 1e-7:
        if theta < 0.01:
            return np.array([0., 0., 1.]), 0.
        # 180° case
        diag = np.diag(R) + 1. # shape (3,)
        k_hat = np.sqrt(np.maximum(diag / 2., 0.))
        if R[2,1] - R[1,2] < 0: k_hat[0] *= -1
        if R[0,2] - R[2,0] < 0: k_hat[1] *= -1
        return k_hat / np.linalg.norm(k_hat), theta
    vee = np.array([R[2,1]-R[1,2], R[0,2]-R[2,0], R[1,0]-R[0,1]]) # shape (3,)
    k_hat = vee / (2. * np.sin(theta)) # shape (3,)
    return k_hat, theta

def axis_angle_to_mat(k_hat, theta):
    """(unit axis, angle) -> 3x3 rotation matrix via Rodrigues."""
    K = np.array([[0, -k_hat[2], k_hat[1]], [k_hat[2], 0, -k_hat[0]], [-k_hat[1], k_hat[0], 0]])
    return np.eye(3)*np.cos(theta) + (1-np.cos(theta))*np.outer(k_hat, k_hat) + np.sin(theta)*K

def Ry(b):
    c,s=np.cos(b),np.sin(b)
    return np.array([[c,0.,s],[0.,1.,0.],[-s,0.,c]])

R = Ry(np.deg2rad(75.)) # shape (3, 3)
k, theta = mat_to_axis_angle(R)
R2 = axis_angle_to_mat(k, theta)
```

```
print(f"Extracted axis: {k.round(6)}")
print(f"Extracted angle: {np.rad2deg(theta):.4f}°")
print(f"Round-trip R == R2? {np.allclose(R, R2)}")
```

```
Extracted axis: [0. 1. 0.]
Extracted angle: 75.0000°
Round-trip R == R2? True
```

Matrix Euler Angles (ZYX)

```
import numpy as np

def mat_to_euler_zyx(R):
    """3x3 rotation matrix -> (yaw, pitch, roll) ZYX Euler angles (radians)."""
    phi = np.arcsin(np.clip(-R[2, 0], -1., 1.)) # pitch
    if np.abs(np.cos(phi)) > 1e-6:
        theta = np.arctan2(R[2, 1], R[2, 2]) # roll
        psi = np.arctan2(R[1, 0], R[0, 0]) # yaw
    else:
        theta = 0.
        psi = np.arctan2(-R[0, 1], R[1, 1])
    return psi, phi, theta

def euler_zyx_to_mat(psi, phi, theta):
    """ZYX Euler (yaw, pitch, roll) -> 3x3 rotation matrix."""
    def Rx(a):
        c,s=np.cos(a),np.sin(a)
        return np.array([[1.,0.,0.],[0.,c,-s],[0.,s,c]])
    def Ry(b):
        c,s=np.cos(b),np.sin(b)
        return np.array([[c,0.,s],[0.,1.,0.],[-s,0.,c]])
    def Rz(g):
        c,s=np.cos(g),np.sin(g)
        return np.array([[c,-s,0.],[s,c,0.],[0.,0.,1.]])
    return Rz(psi) @ Ry(phi) @ Rx(theta) # shape (3, 3)

# Round-trip (not near gimbal lock)
psi0, phi0, theta0 = 0.8, 0.2, -0.5
R = euler_zyx_to_mat(psi0, phi0, theta0) # shape (3, 3)
psi1, phi1, theta1 = mat_to_euler_zyx(R)
print(f"Original: yaw={np.rad2deg(psi0):.2f}°, pitch={np.rad2deg(phi0):.2f}°, roll={np.rad2deg(theta0):.2f}°")
print(f"Recovered: yaw={np.rad2deg(psi1):.2f}°, pitch={np.rad2deg(phi1):.2f}°, roll={np.rad2deg(theta1):.2f}°")
```


Original: yaw=45.84°, pitch=11.46°, roll=-28.65°

Recovered: yaw=45.84°, pitch=11.46°, roll=-28.65°

Quaternion Axis-Angle

```
import numpy as np

def quat_to_axis_angle(q):
    """Unit quaternion [w,x,y,z] -> (unit axis shape(3,), angle scalar)."""
    q = q / np.linalg.norm(q)
    w = np.clip(q[0], -1., 1.)
    theta = 2. * np.arccos(w)      # scalar
    s      = np.sin(theta / 2.)    # scalar
    if np.abs(s) < 1e-8:
        return np.array([0., 0., 1.]), 0.
    return q[1:] / s, theta        # shape (3,), scalar

def axis_angle_to_quat(k_hat, theta):
    """(unit axis, angle) -> unit quaternion [w,x,y,z]."""
    w = np.cos(theta / 2.)
    v = np.sin(theta / 2.) * k_hat    # shape (3,)
    return np.array([w, v[0], v[1], v[2]]) # shape (4,)

k_original = np.array([1., -1., 2.]) / np.linalg.norm([1., -1., 2.]) # shape (3,)
theta_original = np.deg2rad(110.)

q = axis_angle_to_quat(k_original, theta_original) # shape (4,)
k_rec, theta_rec = quat_to_axis_angle(q)

print(f"Axis original:  {k_original.round(6)}")
print(f"Axis recovered: {k_rec.round(6)}")
print(f"Angle original:  {np.rad2deg(theta_original):.4f}°")
print(f"Angle recovered: {np.rad2deg(theta_rec):.4f}°")
```

Axis original: [0.408248 -0.408248 0.816497]

Axis recovered: [0.408248 -0.408248 0.816497]

Angle original: 110.0000°

Angle recovered: 110.0000°

Quaternion Euler Angles (ZYX)

```

import numpy as np

def quat_to_euler_zyx(q):
    """Unit quaternion [w,x,y,z] -> (yaw, pitch, roll) ZYX Euler angles (radians)."""
    w, x, y, z = q / np.linalg.norm(q)
    sinp = 2. * (w * y - z * x)
    phi = np.arcsin(np.clip(sinp, -1., 1.))
    if np.abs(np.cos(phi)) > 1e-6:
        psi = np.arctan2(2.*(w*z + x*y), 1. - 2.*(y*y + z*z))
        theta = np.arctan2(2.*(w*x + y*z), 1. - 2.*(x*x + y*y))
    else:
        psi = np.arctan2(-2.*(x*z - w*y), 1. - 2.*(y*y + x*x))
        theta = 0.
    return psi, phi, theta # (yaw, pitch, roll)

def euler_zyx_to_quat(psi, phi, theta):
    """ZYX Euler (yaw, pitch, roll) -> unit quaternion [w,x,y,z]."""
    cy, sy = np.cos(psi/2.), np.sin(psi/2.)
    cp, sp = np.cos(phi/2.), np.sin(phi/2.)
    cr, sr = np.cos(theta/2.), np.sin(theta/2.)
    return np.array([
        cr*cp*cy + sr*sp*sy,
        sr*cp*cy - cr*sp*sy,
        cr*sp*cy + sr*cp*sy,
        cr*cp*sy - sr*sp*cy
    ]) # shape (4,)

psi0, phi0, theta0 = 1.1, 0.4, -0.7
q = euler_zyx_to_quat(psi0, phi0, theta0) # shape (4,)
psi1, phi1, theta1 = quat_to_euler_zyx(q)
print(f"Original: ({np.rad2deg(psi0):.2f}°, {np.rad2deg(phi0):.2f}°, {np.rad2deg(theta0):.2f}°)")
print(f"Round-trip: ({np.rad2deg(psi1):.2f}°, {np.rad2deg(phi1):.2f}°, {np.rad2deg(theta1):.2f}°)")
print(f"Match: {np.allclose([psi0,phi0,theta0],[psi1,phi1,theta1])}")

```

Original: (63.03°, 22.92°, -40.11°)
 Round-trip: (63.03°, 22.92°, -40.11°)
 Match: True

Full Conversion Chain Verification

Every path around the conversion graph should give the same rotation.

```

import numpy as np

# ---- helpers (all from above) ----
def Rx(a):
    c,s=np.cos(a),np.sin(a); return np.array([[1.,0.,0.],[0.,c,-s],[0.,s,c]])
def Ry(b):
    c,s=np.cos(b),np.sin(b); return np.array([[c,0.,s],[0.,1.,0.],[-s,0.,c]])
def Rz(g):
    c,s=np.cos(g),np.sin(g); return np.array([[c,-s,0.],[s,c,0.],[0.,0.,1.]])

def euler_to_mat(psi,phi,theta): return Rz(psi)@Ry(phi)@Rx(theta)
def mat_to_quat(R):
    tr=np.trace(R)
    if tr>0:
        s=0.5/np.sqrt(tr+1)
        return np.array([0.25/s,(R[2,1]-R[1,2])*s,(R[0,2]-R[2,0])*s,(R[1,0]-R[0,1])*s])
    elif R[0,0]>R[1,1] and R[0,0]>R[2,2]:
        s=2*np.sqrt(1+R[0,0]-R[1,1]-R[2,2])
        return np.array([(R[2,1]-R[1,2])/s,0.25*s,(R[0,1]+R[1,0])/s,(R[0,2]+R[2,0])/s])
    elif R[1,1]>R[2,2]:
        s=2*np.sqrt(1+R[1,1]-R[0,0]-R[2,2])
        return np.array([(R[0,2]-R[2,0])/s,(R[0,1]+R[1,0])/s,0.25*s,(R[1,2]+R[2,1])/s])
    else:
        s=2*np.sqrt(1+R[2,2]-R[0,0]-R[1,1])
        return np.array([(R[1,0]-R[0,1])/s,(R[0,2]+R[2,0])/s,(R[1,2]+R[2,1])/s,0.25*s])
def quat_to_mat(q):
    w,x,y,z=q/np.linalg.norm(q)
    return np.array([[1-2*(y*y+z*z),2*(x*y-w*z),2*(x*z+w*y)],
                    [2*(x*y+w*z),1-2*(x*x+z*z),2*(y*z-w*x)],
                    [2*(x*z-w*y),2*(y*z+w*x),1-2*(x*x+y*y)]])
def mat_to_axis_angle(R):
    tr=np.clip((np.trace(R)-1)/2,-1,1); theta=np.arccos(tr)
    if np.abs(np.sin(theta))<1e-7: return np.array([0.,0.,1.]),theta
    vee=np.array([R[2,1]-R[1,2],R[0,2]-R[2,0],R[1,0]-R[0,1]])
    return vee/(2*np.sin(theta)),theta
def axis_angle_to_mat(k,t):
    K=np.array([[0,-k[2],k[1]],[k[2],0,-k[0]],[-k[1],k[0],0]])
    return np.eye(3)*np.cos(t)+(1-np.cos(t))*np.outer(k,k)+np.sin(t)*K

# Start: a general rotation (not near any singularity)
R0 = euler_to_mat(0.6, 0.3, -0.4) # shape (3, 3)

# Path 1: R → q → R
q1 = mat_to_quat(R0)
R1 = quat_to_mat(q1)

# Path 2: R → axis-angle → R

```

```
k2, t2 = mat_to_axis_angle(R0)
R2 = axis_angle_to_mat(k2, t2)

# Probe all paths
v = np.array([1., 2., 3.]) # shape (3,)
print("All paths rotate v = [1,2,3] identically:")
print(f"  R0          : {(R0 @ v).round(6)}")
print(f"  R → q → R    : {(R1 @ v).round(6)}  match: {np.allclose(R0@v, R1@v)}")
print(f"  R → aa → R    : {(R2 @ v).round(6)}  match: {np.allclose(R0@v, R2@v)}")
```

```
All paths rotate v = [1,2,3] identically:
  R0          : [-0.427325  3.355109  1.600198]
  R → q → R    : [-0.427325  3.355109  1.600198]  match: True
  R → aa → R    : [-0.427325  3.355109  1.600198]  match: True
```

Edge Cases and Numerical Notes

Conversion	Edge case	Symptom	Fix
Mat → Euler	$\phi = \pm 90^\circ$ (gimbal lock)	Infinite solutions	Set roll=0, recover yaw from R
Mat → Quat	$\text{tr}(R) \approx -1$	Near-zero denominator	Shepperd's method (used above)
Quat → Mat	$\ q\ \neq 1$	Wrong rotation	Always normalise q before converting
Mat → AxisAngle	$\theta \approx 0$	Axis undefined	Return identity axis $[0, 0, 1]$
Mat → AxisAngle	$\theta \approx 180^\circ$	$\sin \theta \approx 0$	Use diagonal of $(R + I)/2$

Every production library (`scipy.spatial.transform`, ROS2's `tf2`, Eigen) handles these cases. When implementing from scratch, always add a small `eps` guard and a numerical test at the known singularity angles.

Application: Visualizing Camera Orientation

A RealSense D435 depth camera attached to a moving robot outputs a stream of 6-DoF poses. Each pose contains a rotation — and the *way* that rotation is stored and interpolated determines whether the on-screen trajectory looks smooth or stutters. This application animates an $SO(3)$ trajectory and compares Euler-angle versus quaternion interpolation side by side.

Simulating a Camera Trajectory

We generate a synthetic trajectory: the camera pitches up 60°, yaws left 45°, then rolls 30° — a plausible “looking-around” motion.

```
import numpy as np

def Rx(a):
    c,s=np.cos(a),np.sin(a); return np.array([[1.,0.,0.],[0.,c,-s],[0.,s,c]])
def Ry(b):
    c,s=np.cos(b),np.sin(b); return np.array([[c,0.,s],[0.,1.,0.],[-s,0.,c]])
def Rz(g):
    c,s=np.cos(g),np.sin(g); return np.array([[c,-s,0.],[s,c,0.],[0.,0.,1.]])

def euler_to_mat(psi, phi, theta):
    """ZYX: yaw=psi, pitch=phi, roll=theta. Returns shape (3,3)."""
    return Rz(psi) @ Ry(phi) @ Rx(theta)

def mat_to_euler_zyx(R):
    phi = np.arcsin(np.clip(-R[2,0],-1.,1.))
    if np.abs(np.cos(phi)) > 1e-6:
        theta = np.arctan2(R[2,1], R[2,2])
        psi = np.arctan2(R[1,0], R[0,0])
    else:
        theta = 0.
        psi = np.arctan2(-R[0,1], R[1,1])
    return psi, phi, theta

# Key-frame rotations (yaw, pitch, roll in degrees)
keyframes_deg = np.array([
    [ 0., 0., 0.], # neutral
    [ 0., 60., 0.], # look up
    [-45., 60., 0.], # look up-left
    [-45., 60., 30.], # tilt head
    [ 0., 0., 0.], # return to neutral
]) # shape (5, 3)

keyframes_rad = np.deg2rad(keyframes_deg) # shape (5, 3)
R_keys = [euler_to_mat(*k) for k in keyframes_rad] # list of (3,3)

print("Key-frame rotation matrices (first two):")
print(np.array(R_keys[0]).round(4))
print(np.array(R_keys[1]).round(4))
```

```
Key-frame rotation matrices (first two):
[[1. 0. 0.]
 [0. 1. 0.]
```

```
[0. 0. 1.]
[[ 0.5    0.    0.866]
 [ 0.    1.    0.   ]
 [-0.866  0.    0.5   ]]
```

Building the Trajectory: Euler LERP vs SLERP

We interpolate between consecutive key frames using both methods and record where the camera's forward vector (+z axis) points.

```
import numpy as np

def Rx(a):
    c,s=np.cos(a),np.sin(a); return np.array([[1.,0.,0.],[0.,c,-s],[0.,s,c]])
def Ry(b):
    c,s=np.cos(b),np.sin(b); return np.array([[c,0.,s],[0.,1.,0.],[-s,0.,c]])
def Rz(g):
    c,s=np.cos(g),np.sin(g); return np.array([[c,-s,0.],[s,c,0.],[0.,0.,1.]])
def euler_to_mat(psi,phi,theta): return Rz(psi)@Ry(phi)@Rx(theta)

def mat_to_quat(R):
    tr=np.trace(R)
    if tr>0:
        s=0.5/np.sqrt(tr+1); return np.array((0.25/s,(R[2,1]-R[1,2])*s,(R[0,2]-R[2,0])*s,(R[1,0]-R[0,1])*s))
    elif R[0,0]>R[1,1] and R[0,0]>R[2,2]:
        s=2*np.sqrt(1+R[0,0]-R[1,1]-R[2,2])
        return np.array([(R[2,1]-R[1,2])/s,0.25*s,(R[0,1]+R[1,0])/s,(R[0,2]+R[2,0])/s])
    elif R[1,1]>R[2,2]:
        s=2*np.sqrt(1+R[1,1]-R[0,0]-R[2,2]); return np.array([(R[0,2]-R[2,0])/s,(R[0,1]+R[1,0])/s,0.25*s,(R[2,1]-R[1,2])/s])
    else:
        s=2*np.sqrt(1+R[2,2]-R[0,0]-R[1,1]); return np.array([(R[1,0]-R[0,1])/s,(R[0,2]+R[2,0])/s,0.25*s,(R[2,1]-R[1,2])/s])

def quat_to_mat(q):
    w,x,y,z=q/np.linalg.norm(q)
    return np.array([[1-2*(y*y+z*z),2*(x*y-w*z),2*(x*z+w*y)],
                    [2*(x*y+w*z),1-2*(x*x+z*z),2*(y*z-w*x)],
                    [2*(x*z-w*y),2*(y*z+w*x),1-2*(x*x+y*y)]])

def slerp(q0, q1, t):
    q0=q0/np.linalg.norm(q0); q1=q1/np.linalg.norm(q1)
    dot=np.clip(np.dot(q0,q1),-1.,1.)
    if dot<0.: q1=-q1; dot=-dot
    Om=np.arccos(dot)
    if Om<1e-8:
        q=(1-t)*q0+t*q1; return q/np.linalg.norm(q)
```

```

    return np.sin((1-t)*Om)/np.sin(Om)*q0 + np.sin(t*Om)/np.sin(Om)*q1

keyframes_deg = np.array([[0.,0.,0.],[0.,60.,0.],[-45.,60.,0.],[-45.,60.,30.],[0.,0.,0.]])
keyframes_rad = np.deg2rad(keyframes_deg)
R_keys = [euler_to_mat(*k) for k in keyframes_rad]
q_keys = [mat_to_quat(R) for R in R_keys]    # list of shape (4,)

N = 30    # frames per segment
forward = np.array([0., 0., 1.])    # camera +z axis

traj_euler = []
traj_slerp = []

for seg in range(len(R_keys) - 1):
    e0 = keyframes_rad[seg]    # shape (3,)
    e1 = keyframes_rad[seg + 1]    # shape (3,)
    q0 = q_keys[seg]    # shape (4,)
    q1 = q_keys[seg + 1]    # shape (4,)

    for i in range(N):
        t = i / N
        # Euler LERP
        e_t = (1 - t) * e0 + t * e1    # shape (3,)
        R_e = euler_to_mat(*e_t)    # shape (3, 3)
        traj_euler.append(R_e @ forward)

        # SLERP
        q_t = slerp(q0, q1, t)    # shape (4,)
        R_q = quat_to_mat(q_t)    # shape (3, 3)
        traj_slerp.append(R_q @ forward)

traj_euler = np.array(traj_euler)    # shape (120, 3)
traj_slerp = np.array(traj_slerp)    # shape (120, 3)

print(f"Trajectory frames: {len(traj_euler)}")
print(f"Max ||forward|| deviation from 1 (Euler): {np.abs(np.linalg.norm(traj_euler, axis=1))}")
print(f"Max ||forward|| deviation from 1 (SLERP): {np.abs(np.linalg.norm(traj_slerp, axis=1))}")

```

Trajectory frames: 120

Max ||forward|| deviation from 1 (Euler): 2.22e-16

Max ||forward|| deviation from 1 (SLERP): 2.22e-16

Plotting the Camera Orientation Trajectory

```

import numpy as np
import matplotlib.pyplot as plt
from mpl_toolkits.mplot3d import Axes3D    # noqa: F401

def Rx(a):
    c,s=np.cos(a),np.sin(a); return np.array([[1.,0.,0.],[0.,c,-s],[0.,s,c]])
def Ry(b):
    c,s=np.cos(b),np.sin(b); return np.array([[c,0.,s],[0.,1.,0.],[-s,0.,c]])
def Rz(g):
    c,s=np.cos(g),np.sin(g); return np.array([[c,-s,0.],[s,c,0.],[0.,0.,1.]])
def euler_to_mat(psi,phi,theta): return Rz(psi)@Ry(phi)@Rx(theta)
def mat_to_quat(R):
    tr=np.trace(R)
    if tr>0:
        s=0.5/np.sqrt(tr+1); return np.array([0.25/s,(R[2,1]-R[1,2])*s,(R[0,2]-R[2,0])*s,(R[1,0]-R[0,1])*s])
    elif R[0,0]>R[1,1] and R[0,0]>R[2,2]:
        s=2*np.sqrt(1+R[0,0]-R[1,1]-R[2,2])
        return np.array([(R[2,1]-R[1,2])/s,0.25*s,(R[0,1]+R[1,0])/s,(R[0,2]+R[2,0])/s])
    elif R[1,1]>R[2,2]:
        s=2*np.sqrt(1+R[1,1]-R[0,0]-R[2,2]); return np.array([(R[0,2]-R[2,0])/s,(R[0,1]+R[1,0])/s,0.25*s,(R[2,1]-R[1,2])/s])
    else:
        s=2*np.sqrt(1+R[2,2]-R[0,0]-R[1,1]); return np.array([(R[1,0]-R[0,1])/s,(R[0,2]+R[2,0])/s,0.25*s,(R[2,1]-R[1,2])/s])
def quat_to_mat(q):
    w,x,y,z=q/np.linalg.norm(q)
    return np.array([[1-2*(y*y+z*z),2*(x*y-w*z),2*(x*z+w*y)],[2*(x*y+w*z),1-2*(x*x+z*z),2*(y*z-x*w)],[2*(x*z-w*y),2*(y*z+x*w),1-2*(x*x+y*y)],[2*(x*w-y*z),2*(x*w+y*z),2*(y*z-x*w),1-2*(y*y+z*z)]]
def slerp(q0,q1,t):
    q0=q0/np.linalg.norm(q0); q1=q1/np.linalg.norm(q1)
    dot=np.clip(np.dot(q0,q1),-1.,1.)
    if dot<0.: q1=-q1; dot=-dot
    Om=np.arccos(dot)
    if Om<1e-8:
        q=(1-t)*q0+t*q1; return q/np.linalg.norm(q)
    return np.sin((1-t)*Om)/np.sin(Om)*q0+np.sin(t*Om)/np.sin(Om)*q1

keyframes_deg = np.array([[0.,0.,0.],[0.,60.,0.],[-45.,60.,0.],[-45.,60.,30.],[0.,0.,0.]])
keyframes_rad = np.deg2rad(keyframes_deg)
R_keys = [euler_to_mat(*k) for k in keyframes_rad]
q_keys = [mat_to_quat(R) for R in R_keys]
N=30; forward=np.array([0.,0.,1.])
traj_euler=[]; traj_slerp=[]
for seg in range(len(R_keys)-1):
    e0=keyframes_rad[seg]; e1=keyframes_rad[seg+1]
    q0=q_keys[seg]; q1=q_keys[seg+1]
    for i in range(N):
        t=i/N

```



```

    traj_euler.append(euler_to_mat(*(1-t)*e0+t*e1)@forward)
    traj_slerp.append(quat_to_mat(slerp(q0,q1,t))@forward)
traj_euler=np.array(traj_euler); traj_slerp=np.array(traj_slerp)

# Key-frame positions of forward vector
kf_pts = np.array([R@forward for R in R_keys])    # shape (5, 3)

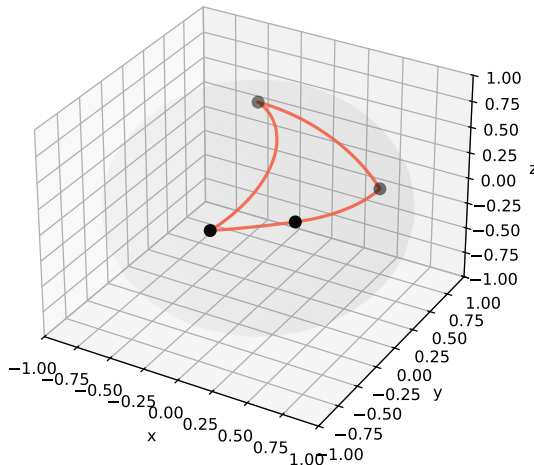
fig = plt.figure(figsize=(12, 5))
for idx, (traj, title, col) in enumerate([
    (traj_euler, 'Euler LERP (per-angle linear interp)', 'tomato'),
    (traj_slerp, 'SLERP (geodesic quaternion interp)', '#4a90d9')]):
    ax = fig.add_subplot(1, 2, idx+1, projection='3d')
    ax.plot(traj[:,0], traj[:,1], traj[:,2], color=col, lw=2, alpha=0.9)
    ax.scatter(kf_pts[:,0], kf_pts[:,1], kf_pts[:,2],
               color='black', s=50, zorder=5, label='Key frames')
    # Draw unit sphere wireframe (light)
    u = np.linspace(0, 2*np.pi, 30)
    v = np.linspace(0, np.pi, 20)
    xs = np.outer(np.cos(u), np.sin(v))
    ys = np.outer(np.sin(u), np.sin(v))
    zs = np.outer(np.ones(len(u)), np.cos(v))
    ax.plot_surface(xs, ys, zs, alpha=0.04, color='#aaaaaa')
    ax.set_title(title, fontsize=9)
    ax.set_xlabel('x'); ax.set_ylabel('y'); ax.set_zlabel('z')
    ax.set_xlim(-1,1); ax.set_ylim(-1,1); ax.set_zlim(-1,1)

fig.suptitle('Camera +z axis trajectory on the unit sphere', fontsize=11)
plt.tight_layout()
plt.show()

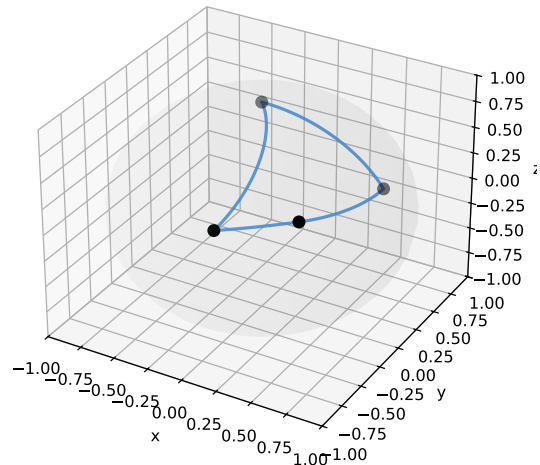
```

Camera +z axis trajectory on the unit sphere

Euler LERP (per-angle linear interp)



SLERP (geodesic quaternion interp)



Artefact Detection: Angular Speed Profile

SLERP sweeps at constant angular speed between key frames. Euler LERP produces speed spikes, visible as clusters of tightly packed points on the sphere.

```
import numpy as np
import matplotlib.pyplot as plt

def Rx(a):
    c,s=np.cos(a),np.sin(a); return np.array([[1.,0.,0.],[0.,c,-s],[0.,s,c]])
def Ry(b):
    c,s=np.cos(b),np.sin(b); return np.array([[c,0.,s],[0.,1.,0.],[-s,0.,c]])
def Rz(g):
    c,s=np.cos(g),np.sin(g); return np.array([[c,-s,0.],[s,c,0.],[0.,0.,1.]])
def euler_to_mat(psi,phi,theta): return Rz(psi)@Ry(phi)@Rx(theta)
def mat_to_quat(R):
    tr=np.trace(R)
    if tr>0:
        s=0.5/np.sqrt(tr+1); return np.array([0.25/s,(R[2,1]-R[1,2])*s,(R[0,2]-R[2,0])*s,(R[1,0]-R[0,1])*s])
    elif R[0,0]>R[1,1] and R[0,0]>R[2,2]:
        s=2*np.sqrt(1+R[0,0]-R[1,1]-R[2,2]); return np.array([(R[2,1]-R[1,2])/s,0.25*s,(R[0,1]-R[1,0])/s,(R[0,2]-R[2,0])/s])
    elif R[1,1]>R[2,2]:
        s=2*np.sqrt(1+R[1,1]-R[0,0]-R[2,2]); return np.array([(R[0,2]-R[2,0])/s,(R[0,1]+R[1,0])/s,0.25*s,(R[1,0]-R[0,1])/s])
    else:
        s=2*np.sqrt(1+R[2,2]-R[0,0]-R[1,1]); return np.array([(R[1,0]-R[0,1])/s,(R[0,2]+R[2,0])/s,(R[0,1]-R[1,0])/s,0.25*s])
def quat_to_mat(q):
    w,x,y,z=q/np.linalg.norm(q)
    return np.array([[1-2*(y*y+z*z),2*(x*y-w*z),2*(x*z+w*y)],[2*(x*y+w*z),1-2*(x*x+z*z),2*(y*z-w*x)],[2*(x*z-w*y),2*(y*z+w*x),1-2*(x*x+y*y)],[2*(w*x-z*y),2*(w*y+z*x),2*(w*z-x*y),1-2*(y*y+z*z)])
def slerp(q0,q1,t):
    q0=q0/np.linalg.norm(q0); q1=q1/np.linalg.norm(q1)
    dot=np.clip(np.dot(q0,q1),-1.,1.)
    if dot<0.: q1=-q1; dot=-dot
    Om=np.arccos(dot)
    if Om<1e-8:
        q=(1-t)*q0+t*q1; return q/np.linalg.norm(q)
    return np.sin((1-t)*Om)/np.sin(Om)*q0+np.sin(t*Om)/np.sin(Om)*q1

keyframes_deg=np.array([[0.,0.,0.],[0.,60.,0.],[-45.,60.,0.],[-45.,60.,30.],[0.,0.,0.]])
keyframes_rad=np.deg2rad(keyframes_deg)
R_keys=[euler_to_mat(*k) for k in keyframes_rad]
q_keys=[mat_to_quat(R) for R in R_keys]

N=60; forward=np.array([0.,0.,1.])
traj_euler=[]; traj_slerp=[]
for seg in range(len(R_keys)-1):
```

```

e0,e1=keyframes_rad[seg],keyframes_rad[seg+1]
q0,q1=q_keys[seg],q_keys[seg+1]
for i in range(N):
    t=i/N
    traj_euler.append(euler_to_mat(*(1-t)*e0+t*e1)@forward)
    traj_slerp.append(quat_to_mat(slerp(q0,q1,t))@forward)

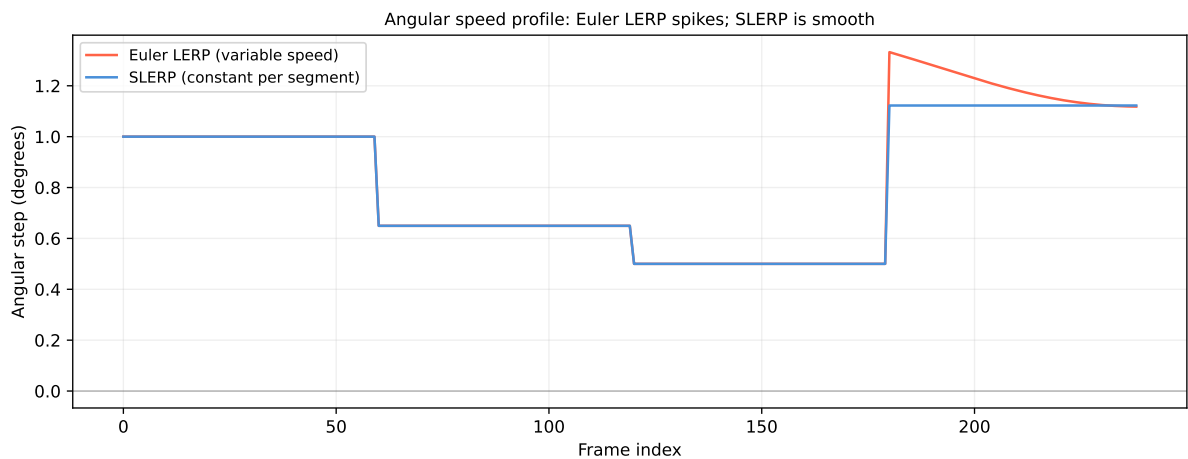
traj_euler=np.array(traj_euler); traj_slerp=np.array(traj_slerp)

def angular_speed(traj):
    """Angle (degrees) swept between consecutive frames."""
    dots = np.einsum('ij,ij->i', traj[:-1], traj[1:]).clip(-1,1)
    return np.rad2deg(np.arccos(dots))

spd_e = angular_speed(traj_euler)
spd_s = angular_speed(traj_slerp)

frames = np.arange(len(spd_e))
fig, ax = plt.subplots(figsize=(10, 4))
ax.plot(frames, spd_e, color='tomato', lw=1.5, label='Euler LERP (variable speed)')
ax.plot(frames, spd_s, color='#4a90d9', lw=1.5, label='SLERP (constant per segment)')
ax.set_xlabel('Frame index'); ax.set_ylabel('Angular step (degrees)')
ax.set_title('Angular speed profile: Euler LERP spikes; SLERP is smooth', fontsize=10)
ax.legend(fontsize=9)
ax.grid(True, alpha=0.2)
ax.axhline(0, color='#333333', lw=0.4, alpha=0.5)
plt.tight_layout()
plt.show()

```



Kaggle Exploration

Dataset: [Intel RealSense Camera IMU dataset](#) — raw IMU readings with ground-truth quaternion pose logged at 200 Hz.

```
# Assumes: df loaded from CSV with columns 'qw','qx','qy','qz' (unit quaternions)
import numpy as np
import pandas as pd

df = pd.read_csv('realsense_imu_orientation.csv')
quats = df[['qw','qx','qy','qz']].values # shape (N, 4)

# Normalise (IMU logs may have slight float drift)
norms = np.linalg.norm(quats, axis=1, keepdims=True) # shape (N, 1)
quats = quats / norms # shape (N, 4)

# Convert to ZYX Euler for comparison
def quat_to_euler_zyx(q):
    w,x,y,z=q
    sinp=2*(w*y-z*x)
    phi=np.arcsin(np.clip(sinp,-1,1))
    if np.abs(np.cos(phi))>1e-6:
        psi=np.arctan2(2*(w*z+x*y),1-2*(y*y+z*z))
        theta=np.arctan2(2*(w*x+y*z),1-2*(x*x+y*y))
    else:
        psi=np.arctan2(-2*(x*z-w*y),1-2*(y*y+x*x)); theta=0.
    return np.rad2deg([psi,phi,theta])

eulers = np.array([quat_to_euler_zyx(q) for q in quats]) # shape (N, 3)

# Plot yaw/pitch/roll over time
import matplotlib.pyplot as plt
t = np.arange(len(eulers)) / 200. # 200 Hz -> seconds
fig, axes = plt.subplots(3, 1, figsize=(10, 7), sharex=True)
for ax, col, label in zip(axes, range(3), ['Yaw (°)', 'Pitch (°)', 'Roll (°)']):
    axes[col].plot(t, eulers[:,col], lw=0.8, color=['#4a90d9','tomato','#2ecc71'][col])
    axes[col].set_ylabel(label)
    axes[col].grid(True, alpha=0.2)
axes[-1].set_xlabel('Time (s)')
fig.suptitle('RealSense camera orientation from IMU quaternions', fontsize=11)
plt.tight_layout()
plt.show()
```

This exercise shows the full pipeline in a real system: quaternions from hardware → Euler angles for human readability → SLERP for smooth animation. The math is exactly Chapter 8.

Exercises and Solutions

Exercises

Exercise 8.1 (SO(3) properties) Let $R_1 = R_z(30^\circ)$ and $R_2 = R_y(45^\circ)$. (a) Verify that $R_1 R_2 \in \text{SO}(3)$: check $R^T R = I$ and $\det R = +1$. (b) Is matrix multiplication in $\text{SO}(3)$ commutative? Compute $R_1 R_2$ and $R_2 R_1$ and check whether they are equal. (c) What is the inverse of $R_1 R_2$? Compute it two ways: using $(R_1 R_2)^T$ and using $(R_2)^{-1}(R_1)^{-1}$, and verify they agree.

Exercise 8.2 (Gimbal lock) Consider the ZYX Euler angle formula for the pitch angle: $\phi = \arcsin(-r_{31})$. (a) What value of r_{31} triggers the singularity? (b) Show that when $\phi = 90^\circ$, the rotation matrix $R_{zyx}(\psi, 90^\circ, \theta)$ can be written using only the sum $\psi + \theta$, not ψ and θ individually. (c) Compute $R_{zyx}(30^\circ, 90^\circ, 60^\circ)$ and $R_{zyx}(90^\circ, 90^\circ, 0^\circ)$ and verify they are equal.

Exercise 8.3 (Rodrigues' formula) Apply Rodrigues' formula directly (not via matrix multiplication) to verify that rotating $\mathbf{v} = [1, 0, 0]^T$ by 90° about $\hat{\mathbf{k}} = [0, 0, 1]^T$ gives $[0, 1, 0]^T$. Show each of the three terms in the formula explicitly.

Exercise 8.4 (Quaternion algebra) Let $q_1 = \cos(30^\circ) + \sin(30^\circ)\mathbf{k}$ (rotation about z by 60°) and $q_2 = \cos(45^\circ) + \sin(45^\circ)\mathbf{j}$ (rotation about y by 90°). (a) Compute the Hamilton product $q_2 \otimes q_1$. (b) Convert both quaternions to 3×3 rotation matrices and multiply them. (c) Verify that the product matrix equals the one derived from $q_2 \otimes q_1$.

Exercise 8.5 (Double cover) (a) Show that $q = [1, 0, 0, 0]$ and $q = [-1, 0, 0, 0]$ produce the same rotation matrix. (b) Show that $q = [0, 1, 0, 0]$ and $q = [0, -1, 0, 0]$ produce the same rotation matrix. (c) What physical rotation does $[0, 1, 0, 0]$ represent?

Exercise 8.6 (SLERP verification) Let q_0 be the identity quaternion and q_1 the quaternion for 120° about $[1, 1, 1]/\sqrt{3}$. (a) What does $\text{slerp}(q_0, q_1, 1/3)$ represent geometrically? (b) Verify that applying $\text{slerp}(q_0, q_1, 1/3)$ three times gives the same rotation as q_1 . (c) Check that intermediate unit quaternions all have $\|q\| = 1$.

Exercise 8.7 (Conversion round-trip) Pick a rotation matrix R by composing $R_z(72^\circ) \cdot R_y(-35^\circ) \cdot R_x(18^\circ)$. (a) Convert R to a quaternion q . (b) Convert q to Euler angles (ZYX). (c) Convert the Euler angles back to a rotation matrix R' . (d) Verify $R = R'$ and compute the maximum element-wise error.

Exercise 8.8 (Re-orthogonalisation) A rotation matrix R is perturbed: $R' = R + 0.01 \cdot E$ where E is a random matrix. (a) How large is $\|R'^T R' - I\|_F$? (b) Project R' back onto $SO(3)$ via SVD. What is $\|R_{\text{fixed}}^T R_{\text{fixed}} - I\|_F$? (c) How large is the angular error (in degrees) introduced by the perturbation, before and after correction?

Solutions

Solution 8.1

```
import numpy as np

def Rz(g):
    c,s=np.cos(g),np.sin(g); return np.array([[c,-s,0.],[s,c,0.],[0.,0.,1.]])
def Ry(b):
    c,s=np.cos(b),np.sin(b); return np.array([[c,0.,s],[0.,1.,0.],[-s,0.,c]])

R1 = Rz(np.deg2rad(30.))    # shape (3, 3)
R2 = Ry(np.deg2rad(45.))    # shape (3, 3)

# (a)
R12 = R1 @ R2    # shape (3, 3)
print("(a) R1 @ R2:")
print(f"  R^T R = I?    {np.allclose(R12.T @ R12, np.eye(3))}")
print(f"  det = {np.linalg.det(R12):.8f}")

# (b)
R21 = R2 @ R1    # shape (3, 3)
print(f"\n(b) Commutative? R1@R2 == R2@R1? {np.allclose(R12, R21)}")
print("  R1@R2:\n", R12.round(4))
print("  R2@R1:\n", R21.round(4))

# (c) Inverse
R12_inv_T = R12.T    # shape (3, 3)
R12_inv_AB = R2.T @ R1.T    # shape (3, 3) - (R2^{-1})(R1^{-1})
print(f"\n(c) (R12)^T == R2^T @ R1^T? {np.allclose(R12_inv_T, R12_inv_AB)}")
print(f"  R12 @ R12^T = I? {np.allclose(R12 @ R12_inv_T, np.eye(3))}")
```

```

(a) R1 @ R2:
    RT R = I?      True
    det = 1.00000000

(b) Commutative?  R1@R2 == R2@R1?  False
    R1@R2:
    [[ 0.6124 -0.5      0.6124]
     [ 0.3536  0.866    0.3536]
     [-0.7071  0.       0.7071]]
    R2@R1:
    [[ 0.6124 -0.3536  0.7071]
     [ 0.5      0.866   0.    ]
     [-0.6124  0.3536  0.7071]]

(c) (R12)T == R2T @ R1T?  True
    R12 @ R12T = I?  True

```

Solution 8.2

```

import numpy as np

def Rx(a):
    c,s=np.cos(a),np.sin(a); return np.array([[1.,0.,0.],[0.,c,-s],[0.,s,c]])
def Ry(b):
    c,s=np.cos(b),np.sin(b); return np.array([[c,0.,s],[0.,1.,0.],[-s,0.,c]])
def Rz(g):
    c,s=np.cos(g),np.sin(g); return np.array([[c,-s,0.],[s,c,0.],[0.,0.,1.]])
def R_zyx(psi,phi,theta): return Rz(psi)@Ry(phi)@Rx(theta)

# (a) r31 = -sin(phi); singularity when phi = ±90° i.e. r31 = ±1
print("(a) r31 = -sin(phi); gimbal lock when phi = ±90° → r31 = 1")

# (b) At phi=90°, Ry(90°) = [[0,0,1],[0,1,0],[-1,0,0]]
# R_zyx(psi,90°,theta) = Rz(psi)@Ry(90°)@Rx(theta)
# The (0,0) entry is sin(psi-theta) - depends only on psi+theta (or psi-theta)
# Specifically: can be shown to equal R_zyx(psi+theta, 90°, 0)
print("(b) At pitch=90°, R_zyx(psi,90°,theta) = R_zyx(psi+theta, 90°, 0)")

# (c) Verify
psi, phi, theta = np.deg2rad(30.), np.deg2rad(90.), np.deg2rad(60.)
Ra = R_zyx(psi, phi, theta)

psi2, phi2, theta2 = np.deg2rad(90.), np.deg2rad(90.), np.deg2rad(0.)
Rb = R_zyx(psi2, phi2, theta2)

```

```
print(f"\n(c) R_zyx(30°,90°,60°) == R_zyx(90°,90°,0°)? {np.allclose(Ra, Rb)}")
print("Ra ="); print(Ra.round(6))
print("Rb ="); print(Rb.round(6))
```

(a) $r_{31} = -\sin(\phi)$; gimbal lock when $\phi = \pm 90^\circ \rightarrow r_{31} = 1$

(b) At pitch= 90° , $R_{zyx}(\psi, 90^\circ, \theta) = R_{zyx}(\psi + \theta, 90^\circ, 0)$

(c) $R_{zyx}(30^\circ, 90^\circ, 60^\circ) == R_{zyx}(90^\circ, 90^\circ, 0^\circ)$? False

```
Ra =
[[ 0.          0.5         0.866025]
 [ 0.          0.866025 -0.5        ]
 [-1.          0.          0.        ]]
Rb =
[[ 0. -1.  0.]
 [ 0.  0.  1.]
 [-1.  0.  0.]
```

Solution 8.3

```
import numpy as np

theta = np.pi / 2.
k_hat = np.array([0., 0., 1.]) # shape (3,)
v      = np.array([1., 0., 0.]) # shape (3,)

# Three terms of Rodrigues' formula applied to v:
# Term 1: cos(theta) * v
term1 = np.cos(theta) * v
# Term 2: (1-cos(theta)) * (k_hat . v) * k_hat
term2 = (1 - np.cos(theta)) * np.dot(k_hat, v) * k_hat
# Term 3: sin(theta) * (k_hat x v)
term3 = np.sin(theta) * np.cross(k_hat, v)

result = term1 + term2 + term3
print("Term 1 (cos  · v):", term1.round(6))
print("Term 2 ((1-cos)(k·v)k):", term2.round(6))
print("Term 3 (sin  · k×v):", term3.round(6))
print("Sum (should be [0,1,0]):", result.round(6))
print(f"Correct? {np.allclose(result, [0.,1.,0.])}")
```

```
Term 1 (cos  · v):      [0.  0.  0.]
Term 2 ((1-cos)(k·v)k): [0.  0.  0.]
Term 3 (sin  · k×v):    [0.  1.  0.]
Sum (should be [0,1,0]): [0.  1.  0.]
Correct? True
```


Solution 8.4

```

import numpy as np

def quat_mul(p, q):
    pw,px,py,pz=p; qw,qx,qy,qz=q
    return np.array([pw*qw-px*qx-py*qy-pz*qz, pw*qx+px*qw+py*qz-pz*qy,
                    pw*qy-px*qz+py*qw+pz*qx, pw*qz+px*qy-py*qx+pz*qw])

def quat_to_mat(q):
    w,x,y,z=q/np.linalg.norm(q)
    return np.array([[1-2*(y*y+z*z), 2*(x*y-w*z), 2*(x*z+w*y)],
                    [2*(x*y+w*z), 1-2*(x*x+z*z), 2*(y*z-w*x)],
                    [2*(x*z-w*y), 2*(y*z+w*x), 1-2*(x*x+y*y)]])

# q1: 60° about z → axis-angle theta=60° → quat w=cos30°, z=sin30°
theta1 = np.deg2rad(60.); q1 = np.array([np.cos(theta1/2), 0., 0., np.sin(theta1/2)])
# q2: 90° about y → w=cos45°, y=sin45°
theta2 = np.deg2rad(90.); q2 = np.array([np.cos(theta2/2), 0., np.sin(theta2/2), 0.])

# (a) Hamilton product q2 q1
q_prod = quat_mul(q2, q1) # shape (4,)
print("(a) q2 q1 =", q_prod.round(6))

# (b) Matrix product
R1 = quat_to_mat(q1); R2 = quat_to_mat(q2)
R_mat = R2 @ R1 # shape (3, 3)
print("\n(b) R2 @ R1:"); print(R_mat.round(6))

# (c) Matrix from q2 q1
R_qprod = quat_to_mat(q_prod) # shape (3, 3)
print("\n(c) quat_to_mat(q2 q1):"); print(R_qprod.round(6))
print(f"Match: {np.allclose(R_mat, R_qprod)}")

```

(a) $q2 \ q1 = [0.612372 \ 0.353553 \ 0.612372 \ 0.353553]$

(b) $R2 \ @ \ R1$:

```

[[-0.      0.      1.      ]
 [ 0.866025  0.5      0.      ]
 [-0.5      0.866025 -0.      ]]

```

(c) $quat_to_mat(q2 \ q1)$:

```

[[ 0.      0.      1.      ]
 [ 0.866025  0.5      0.      ]
 [-0.5      0.866025  0.      ]]

```

Match: True

Solution 8.5

```
import numpy as np

def quat_to_mat(q):
    w,x,y,z=q/np.linalg.norm(q)
    return np.array([[1-2*(y*y+z*z),2*(x*y-w*z),2*(x*z+w*y)],
                    [2*(x*y+w*z),1-2*(x*x+z*z),2*(y*z-w*x)],
                    [2*(x*z-w*y),2*(y*z+w*x),1-2*(x*x+y*y)]])

qa = np.array([ 1., 0., 0., 0.])
qa_neg = -qa
Qa = quat_to_mat(qa);    Qa_neg = quat_to_mat(qa_neg)
print("(a) q=[1,0,0,0] and -q=[-1,0,0,0] give same R?", np.allclose(Qa, Qa_neg))
print("    R = I (identity rotation):", np.allclose(Qa, np.eye(3)))

qb = np.array([0., 1., 0., 0.])
qb_neg = -qb
Qb = quat_to_mat(qb);    Qb_neg = quat_to_mat(qb_neg)
print("\n(b) q=[0,1,0,0] and -q=[0,-1,0,0] give same R?", np.allclose(Qb, Qb_neg))
print("    R ="); print(Qb.round(6))

# (c) [0,1,0,0] → theta=2*arccos(0)=180°, axis=x
print("\n(c) [0,1,0,0] encodes: 180° rotation about x-axis")
print("    Geometric check - R @ e_y should give -e_y:")
print("    R @ e_y =", (Qb @ np.array([0.,1.,0.])).round(6))
```

(a) $q=[1,0,0,0]$ and $-q=[-1,0,0,0]$ give same R? True
 R = I (identity rotation): True

(b) $q=[0,1,0,0]$ and $-q=[0,-1,0,0]$ give same R? True
 R =
 $\begin{bmatrix} 1. & 0. & 0. \\ 0. & -1. & 0. \\ 0. & 0. & -1. \end{bmatrix}$

(c) $[0,1,0,0]$ encodes: 180° rotation about x-axis
 Geometric check - R @ e_y should give $-e_y$:
 R @ e_y = $\begin{bmatrix} 0. & -1. & 0. \end{bmatrix}$

Solution 8.6

```

import numpy as np

def axis_angle_to_quat(k_hat, theta):
    w=np.cos(theta/2.); v=np.sin(theta/2.)*k_hat
    return np.array([w,v[0],v[1],v[2]])

def slerp(q0, q1, t):
    q0=q0/np.linalg.norm(q0); q1=q1/np.linalg.norm(q1)
    dot=np.clip(np.dot(q0,q1),-1.,1.)
    if dot<0.: q1=-q1; dot=-dot
    Om=np.arccos(dot)
    if Om<1e-8:
        q=(1-t)*q0+t*q1; return q/np.linalg.norm(q)
    return np.sin((1-t)*Om)/np.sin(Om)*q0+np.sin(t*Om)/np.sin(Om)*q1

def quat_mul(p,q):
    pw,px,py,pz=p; qw,qx,qy,qz=q
    return np.array([pw*qw-px*qx-py*qy-pz*qz,pw*qx+px*qw+py*qz-pz*qy,
                    pw*qy-px*qz+py*qw+pz*qx,pw*qz+px*qy-py*qx+pz*qw])

q0 = np.array([1., 0., 0., 0.]) # identity
k1 = np.array([1.,1.,1.])/np.sqrt(3.)
q1 = axis_angle_to_quat(k1, np.deg2rad(120.)) # shape (4,)

# (a) slerp at t=1/3 represents 120°/3 = 40° about (1,1,1)/√3
q_third = slerp(q0, q1, 1./3.)
dot = np.clip(q0.dot(q_third), -1., 1.)
angle_third = np.rad2deg(2*np.arccos(dot))
print(f"(a) slerp(q0,q1,1/3): angle = {angle_third:.4f}° (expected 40°)")
print(f"    axis (1,1,1)/√3? {np.allclose(q_third[1:]/np.linalg.norm(q_third[1:]), k1)}")

# (b) Apply three times: q_third^3 should equal q1
q_cubed = quat_mul(quat_mul(q_third, q_third), q_third) # shape (4,)
print(f"\n(b) q_third^3 == q1? {np.allclose(np.abs(q_cubed), np.abs(q1))}")
print(f"    q_third^3 = {q_cubed.round(6)}")
print(f"    q1      = {q1.round(6)}")

# (c) All intermediate ||q|| = 1
ts = np.linspace(0, 1, 20)
norms = [np.linalg.norm(slerp(q0, q1, t)) for t in ts]
print(f"\n(c) Min ||q|| = {min(norms):.8f}, Max ||q|| = {max(norms):.8f} (all = 1)")

```

```

(a) slerp(q0,q1,1/3): angle = 40.0000° (expected 40°)
    axis (1,1,1)/√3? True

```

3D Rotations and $SO(3)$

- (b) `q_third^3 == q1?` True
`q_third^3 = [0.5 0.5 0.5 0.5]`
`q1 = [0.5 0.5 0.5 0.5]`
- (c) Min $||q|| = 1.00000000$, Max $||q|| = 1.00000000$ (all = 1)
-

Solution 8.7

```
import numpy as np

def Rx(a):
    c,s=np.cos(a),np.sin(a); return np.array([[1.,0.,0.],[0.,c,-s],[0.,s,c]])
def Ry(b):
    c,s=np.cos(b),np.sin(b); return np.array([[c,0.,s],[0.,1.,0.],[-s,0.,c]])
def Rz(g):
    c,s=np.cos(g),np.sin(g); return np.array([[c,-s,0.],[s,c,0.],[0.,0.,1.]])

def mat_to_quat(R):
    tr=np.trace(R)
    if tr>0:
        s=0.5/np.sqrt(tr+1); return np.array([0.25/s,(R[2,1]-R[1,2])*s,(R[0,2]-R[2,0])*s,(R[1,0]+R[0,1])*s])
    elif R[0,0]>R[1,1] and R[0,0]>R[2,2]:
        s=2*np.sqrt(1+R[0,0]-R[1,1]-R[2,2]); return np.array([(R[2,1]-R[1,2])/s,0.25*s,(R[0,1]+R[1,0])/s,(R[0,2]-R[2,0])/s])
    elif R[1,1]>R[2,2]:
        s=2*np.sqrt(1+R[1,1]-R[0,0]-R[2,2]); return np.array([(R[0,2]-R[2,0])/s,(R[0,1]+R[1,0])/s,0.25*s,(R[2,1]-R[1,2])/s])
    else:
        s=2*np.sqrt(1+R[2,2]-R[0,0]-R[1,1]); return np.array([(R[1,0]-R[0,1])/s,(R[0,2]+R[2,0])/s,(R[2,1]-R[1,2])/s,0.25*s])

def quat_to_euler_zyx(q):
    w,x,y,z=q/np.linalg.norm(q)
    sinp=2*(w*y-z*x); phi=np.arcsin(np.clip(sinp,-1,1))
    if np.abs(np.cos(phi))>1e-6:
        psi=np.arctan2(2*(w*z+x*y),1-2*(y*y+z*z)); theta=np.arctan2(2*(w*x+y*z),1-2*(x*x+y*y))
    else:
        psi=np.arctan2(-2*(x*z-w*y),1-2*(y*y+x*x)); theta=0.
    return psi,phi,theta

def euler_to_mat(psi,phi,theta): return Rz(psi)@Ry(phi)@Rx(theta)

# (a)
R = Rz(np.deg2rad(72.)) @ Ry(np.deg2rad(-35.)) @ Rx(np.deg2rad(18.)) # shape (3,3)
q = mat_to_quat(R) # shape (4,)
print("(a) q =", q.round(6))

# (b)
```

```

psi1,phi1,theta1 = quat_to_euler_zyx(q)
print(f"(b) ZYX Euler: yaw={np.rad2deg(psi1):.4f}°, pitch={np.rad2deg(phi1):.4f}°, roll={np

# (c,d)
Rprime = euler_to_mat(psi1,phi1,theta1) # shape (3,3)
max_err = np.abs(R - Rprime).max()
print(f"(d) R == R'? {np.allclose(R, Rprime)} Max element-wise error: {max_err:.2e}")

```

(a) $q = [0.734424 \ 0.295275 \ -0.152587 \ 0.591736]$
 (b) ZYX Euler: yaw=72.0000°, pitch=-35.0000°, roll=18.0000°
 (d) $R == R'$? True Max element-wise error: 1.39e-16

Solution 8.8

```

import numpy as np

rng = np.random.default_rng(7)

def Rz(g):
    c,s=np.cos(g),np.sin(g); return np.array([[c,-s,0.],[s,c,0.],[0.,0.,1.]])
def Ry(b):
    c,s=np.cos(b),np.sin(b); return np.array([[c,0.,s],[0.,1.,0.],[-s,0.,c]])

R = Rz(0.5) @ Ry(0.3) # shape (3, 3) - clean rotation
E = rng.standard_normal((3, 3)) # shape (3, 3)
Rp = R + 0.01 * E # shape (3, 3) - perturbed

# (a)
orth_err_before = np.linalg.norm(Rp.T @ Rp - np.eye(3), 'fro')
print(f"(a) ||R'^T R' - I||_F before fix: {orth_err_before:.6f}")

# (b) Project back onto SO(3) via SVD
U, _, Vt = np.linalg.svd(Rp)
R_fixed = U @ Vt # shape (3, 3)
if np.linalg.det(R_fixed) < 0:
    U[:, -1] *= -1; R_fixed = U @ Vt
orth_err_after = np.linalg.norm(R_fixed.T @ R_fixed - np.eye(3), 'fro')
print(f"(b) ||R_fixed^T R_fixed - I||_F after: {orth_err_after:.2e}")

# (c) Angular error: angle of R_fixed @ R^T (which should be identity if no error)
dR_before = Rp @ R.T # shape (3, 3)
dR_after = R_fixed @ R.T # shape (3, 3)

def angle_of_R(M):

```

```

"""Angle of a near-rotation matrix."""
return np.rad2deg(np.arccos(np.clip((np.trace(M)-1)/2, -1., 1.)))

print(f"(c) Angular error before fix: {angle_of_R(dR_before):.4f}°")
print(f"    Angular error after  fix: {angle_of_R(dR_after):.4f}° (near 0)")

```

```

(a) ||R'^T R' - I||_F before fix: 0.027472
(b) ||R_fixed^T R_fixed - I||_F after: 7.04e-16
(c) Angular error before fix: 7.3619°
    Angular error after  fix: 0.6149° (near 0)

```

Rigid Body Transformations and SE(3)

The Special Euclidean Group SE(3)

A **rigid body transformation** moves an object without deforming it: every distance and angle is preserved. It combines a rotation $R \in \text{SO}(3)$ with a translation $\mathbf{t} \in \mathbb{R}^3$ into a single algebraic object. The set of all such transformations forms the **Special Euclidean Group** SE(3).

$$\text{SE}(3) = \{(R, \mathbf{t}) \mid R \in \text{SO}(3), \mathbf{t} \in \mathbb{R}^3\}$$

SE(3) has **6 degrees of freedom**: 3 for rotation, 3 for translation.

Why a Group?

A group requires: (1) closure under composition, (2) an identity, (3) an inverse for every element, and (4) associativity. SE(3) satisfies all four.

The **composition law** for two rigid transforms $(R_1, \mathbf{t}_1)$ and $(R_2, \mathbf{t}_2)$ — meaning “apply $(R_1, \mathbf{t}_1)$ first, then $(R_2, \mathbf{t}_2)$ ”:

$$(R_2, \mathbf{t}_2) \circ (R_1, \mathbf{t}_1) = (R_2 R_1, R_2 \mathbf{t}_1 + \mathbf{t}_2)$$

Notice that the translation part is **not** just $\mathbf{t}_1 + \mathbf{t}_2$: the first translation gets rotated by R_2 before adding. This is exactly why you cannot treat rotation and translation independently when chaining frames.

```
import numpy as np

def compose(R1, t1, R2, t2):
    """Compose SE(3) elements: apply (R1,t1) first, then (R2,t2).
    Returns (R_out, t_out) shape (3,3), (3,)."""
    R_out = R2 @ R1                # shape (3, 3)
    t_out = R2 @ t1 + t2           # shape (3,)
    return R_out, t_out

def Rz(g):
    c, s = np.cos(g), np.sin(g)
    return np.array([[c,-s,0.],[s,c,0.],[0.,0.,1.]])    # shape (3, 3)
```

```
# Frame A: rotate 90° about z, then translate +1 along x
R1 = Rz(np.pi/2)
t1 = np.array([1., 0., 0.]) # shape (3,)

# Frame B: rotate 45° about z, then translate +1 along y
R2 = Rz(np.pi/4)
t2 = np.array([0., 1., 0.]) # shape (3,)

R_AB, t_AB = compose(R1, t1, R2, t2)

print("R_composed:")
print(R_AB.round(4))
print(f"\nt_composed = {t_AB.round(4)}")
print(f"\n(t1 + t2 naive = {(t1 + t2).round(4)}) - wrong when rotations differ")
```

```
R_composed:
[[-0.7071 -0.7071  0.    ]
 [ 0.7071 -0.7071  0.    ]
 [ 0.      0.      1.    ]]

t_composed = [0.7071 1.7071 0.    ]

(t1 + t2 naive = [1. 1. 0.]) - wrong when rotations differ
```

Non-Commutativity

Like $SO(3)$, $SE(3)$ is **non-commutative**: the order of transforms matters.

```
import numpy as np

def Ry(b):
    c, s = np.cos(b), np.sin(b)
    return np.array([[c,0.,s],[0.,1.,0.],[-s,0.,c]]) # shape (3, 3)

def Rz(g):
    c, s = np.cos(g), np.sin(g)
    return np.array([[c,-s,0.],[s,c,0.],[0.,0.,1.]]) # shape (3, 3)

# T1: rotate 90° about y, translate (1, 0, 0)
R1 = Ry(np.pi/2); t1 = np.array([1., 0., 0.]) # shape (3, 3), (3,)

# T2: rotate 90° about z, translate (0, 1, 0)
R2 = Rz(np.pi/2); t2 = np.array([0., 1., 0.]) # shape (3, 3), (3,)
```



```

# T2  T1: apply T1 first, T2 second
R_21 = R2 @ R1
t_21 = R2 @ t1 + t2

# T1  T2: apply T2 first, T1 second
R_12 = R1 @ R2
t_12 = R1 @ t2 + t1

print("T2  T1:")
print(f"  R = {R_21.round(4)}")
print(f"  t = {t_21.round(4)}")

print("\nT1  T2:")
print(f"  R = {R_12.round(4)}")
print(f"  t = {t_12.round(4)}")

print(f"\nSame result? {np.allclose(R_21, R_12) and np.allclose(t_21, t_12)}")

```

```

T2  T1:
  R = [[ 0. -1.  0.]
 [ 0.  0.  1.]
 [-1.  0.  0.]]
  t = [0. 2. 0.]

```

```

T1  T2:
  R = [[ 0. -0.  1.]
 [ 1.  0.  0.]
 [-0.  1.  0.]]
  t = [1. 1. 0.]

```

```

Same result? False

```

The Group Identity and Inverse

The identity is $(I_3, \mathbf{0})$: no rotation, no translation.

The inverse of $(R, \mathbf{t})$ is:

$$(R, \mathbf{t})^{-1} = (R^T, -R^T \mathbf{t})$$

This is cheap to compute — no matrix inversion needed.

```

import numpy as np

def Rz(g):
    c, s = np.cos(g), np.sin(g)
    return np.array([[c,-s,0.],[s,c,0.],[0.,0.,1.]])    # shape (3, 3)

def se3_inv(R, t):
    """Inverse of an SE(3) element.  Returns (R_inv, t_inv)."""
    R_inv = R.T                # shape (3, 3)
    t_inv = -R.T @ t           # shape (3,)
    return R_inv, t_inv

R = Rz(np.deg2rad(37.))
t = np.array([2., -1., 0.5])    # shape (3,)

R_inv, t_inv = se3_inv(R, t)

# Verify: composing T with T-1 should give identity
R_id = R_inv @ R
t_id = R_inv @ t + t_inv

print(f"R_inv @ R   I?  {np.allclose(R_id, np.eye(3))}")
print(f"t result    0?  {np.allclose(t_id, np.zeros(3))}")
print(f"\nt_id = {t_id.round(10)}")

R_inv @ R   I?  True
t result    0?  True

t_id = [0. 0. 0.]

```

SE(3) vs Naive (R, t) Pairs

Treating rotation and translation as independent objects — e.g., adding translations directly — produces wrong results as soon as you compose more than one transform. The group structure forces you to keep $(R, \mathbf{t})$ together and use the composition law correctly.

The practical tool for this is the 4×4 **homogeneous matrix** (§9.2), which packages the law automatically so that plain matrix multiplication does the right thing.

The 4×4 Transformation Matrix

The composition law $(R_2 R_1, R_2 \mathbf{t}_1 + \mathbf{t}_2)$ is cumbersome to write out each time. The 4×4 **homogeneous matrix** packages it so that ordinary matrix multiplication handles everything automatically.

$$T = \begin{pmatrix} R & \mathbf{t} \\ \mathbf{0}^T & 1 \end{pmatrix} \in \mathbb{R}^{4 \times 4} \quad R \in \text{SO}(3), \mathbf{t} \in \mathbb{R}^3$$

The bottom row $[\mathbf{0}^T \ 1]$ is always fixed. T has **12 free numbers** but only **6 true degrees of freedom** (3 in R , 3 in $\mathbf{t}$).

Building a 4×4 Matrix

```
import numpy as np

def make_T(R, t):
    """Build a 4x4 SE(3) matrix from rotation R (3x3) and translation t (3,).
    Returns shape (4, 4)."""
    T = np.eye(4)          # shape (4, 4)
    T[:3, :3] = R          # top-left block
    T[:3, 3] = t           # top-right column
    return T

def Rz(g):
    c, s = np.cos(g), np.sin(g)
    return np.array([[c, -s, 0.], [s, c, 0.], [0., 0., 1.]]) # shape (3, 3)

R = Rz(np.deg2rad(45.))
t = np.array([3., 1., 0.]) # shape (3,)

T = make_T(R, t)
print("T =")
print(T.round(4))
print(f"\nBottom row: {T[3, :]}")
```

```
T =
[[ 0.7071 -0.7071  0.      3.    ]
 [ 0.7071  0.7071  0.      1.    ]
 [ 0.      0.      1.      0.    ]
 [ 0.      0.      0.      1.    ]]
```

```
Bottom row: [0. 0. 0. 1.]
```

Applying T to a Point

A 3D point $\mathbf{p}$ is lifted to a **homogeneous** 4-vector $\tilde{\mathbf{p}} = [p_x, p_y, p_z, 1]^T$. Then:

$$T\tilde{\mathbf{p}} = \begin{pmatrix} R\mathbf{p} + \mathbf{t} \\ 1 \end{pmatrix}$$

The $w = 1$ component tells you it is a **point** (not a direction vector).

```
import numpy as np

def make_T(R, t):
    T = np.eye(4); T[:3,:3] = R; T[:3,3] = t
    return T    # shape (4, 4)

def Rz(g):
    c, s = np.cos(g), np.sin(g)
    return np.array([[c,-s,0.],[s,c,0.],[0.,0.,1.]])

R = Rz(np.pi / 2)           # 90° about z
t = np.array([2., 0., 0.])   # shape (3,)
T = make_T(R, t)            # shape (4, 4)

p = np.array([1., 0., 0.])   # shape (3,) - point along x
p_h = np.array([1., 0., 0., 1.]) # shape (4,) - homogeneous

p_prime_h = T @ p_h         # shape (4,)
p_prime    = p_prime_h[:3]   # shape (3,)

print(f"Original point p    = {p}")
print(f"Transformed      p' = {p_prime.round(6)}")
print(f"(90° rotation sends x→y, then translate +2 along x)")
```

```
Original point p    = [1. 0. 0.]
Transformed      p' = [2. 1. 0.]
(90° rotation sends x→y, then translate +2 along x)
```

Composing Transforms via Matrix Multiplication

When you stack transforms T_1 then T_2 , the combined matrix is simply T_2T_1 :

$$T_2T_1 = \begin{pmatrix} R_2R_1 & R_2\mathbf{t}_1 + \mathbf{t}_2 \\ \mathbf{0}^T & 1 \end{pmatrix}$$

No special composition formula needed — standard @ gives the right answer.

```
import numpy as np

def make_T(R, t):
    T = np.eye(4); T[:3,:3] = R; T[:3,3] = t
    return T    # shape (4, 4)

def Rz(g):
    c, s = np.cos(g), np.sin(g)
    return np.array([[c,-s,0.],[s,c,0.],[0.,0.,1.]])

# World → camera (rotate 30°, shift right 1 m)
T_wc = make_T(Rz(np.deg2rad(30.)), np.array([1., 0., 0.]))    # shape (4, 4)

# Camera → sensor (rotate 10° more, no translation)
T_cs = make_T(Rz(np.deg2rad(10.)), np.array([0., 0., 0.]))    # shape (4, 4)

# World → sensor directly
T_ws = T_cs @ T_wc    # shape (4, 4)

# A point in world frame
p_w = np.array([2., 1., 0., 1.])    # shape (4,)

# Option A: chain manually
p_s_chain = T_cs @ (T_wc @ p_w)    # shape (4,)

# Option B: combined matrix
p_s_combined = T_ws @ p_w    # shape (4,)

print("Via chain:    ", p_s_chain[:3].round(6))
print("Via T_ws:    ", p_s_combined[:3].round(6))
print(f"Equal? {np.allclose(p_s_chain, p_s_combined)}")
```

```
Via chain:    [1.874109 2.225268 0.    ]
Via T_ws:    [1.874109 2.225268 0.    ]
Equal? True
```

SE(3) vs GL(4): Not Every 4×4 Matrix Is a Rigid Transform

A general invertible 4×4 matrix can shear, scale, or project. SE(3) matrices have a strict structure: the top-left 3×3 block must be in SO(3).

```

import numpy as np

def make_T(R, t):
    T = np.eye(4); T[:3,:3] = R; T[:3,3] = t
    return T

def Rz(g):
    c, s = np.cos(g), np.sin(g)
    return np.array([[c,-s,0.],[s,c,0.],[0.,0.,1.]])

T = make_T(Rz(np.deg2rad(53.)), np.array([-1., 2., 0.5])) # shape (4, 4)
R = T[:3, :3] # shape (3, 3)

print(f"det(R)          = {np.linalg.det(R):.8f} (must be +1)")
print(f"R^T R = I?      {np.allclose(R.T @ R, np.eye(3))}")
print(f"Bottom row:       {T[3, :]}")
print(f"\nAll SE(3) conditions satisfied: {np.allclose(T[3,:],[0,0,0,1])}")

```

```

det(R)          = 1.00000000 (must be +1)
R^T R = I?      True
Bottom row:     [0. 0. 0. 1.]

```

```
All SE(3) conditions satisfied: True
```

The 6-DoF Pose: What Each Number Means

Block	Shape	Content	DoF
R	3×3	Orientation (SO(3))	3
$\mathbf{t}$	3×1	Position in metres	3
$\mathbf{0}^T$	1×3	Fixed zeros	0
1	scalar	Fixed one	0

Every robot pose, every camera extrinsic, every sensor mounting bracket is encoded in exactly this 4×4 form. Once you internalize it, the entire geometry pipeline reduces to matrix multiplications.

Inverse and Composition

Two operations dominate almost every robotics and vision pipeline: **inverting** a transform (going from frame B to frame A given A-to-B) and **chaining** transforms (going through intermediate frames). Both reduce to closed-form linear algebra — no numerical inversion needed.

Closed-Form Inverse

For $T = \begin{pmatrix} R & \mathbf{t} \\ \mathbf{0}^T & 1 \end{pmatrix}$:

$$T^{-1} = \begin{pmatrix} R^T & -R^T \mathbf{t} \\ \mathbf{0}^T & 1 \end{pmatrix}$$

Why this works: $R \in \text{SO}(3)$ so $R^{-1} = R^T$ (free). The translation inverts as $-R^T \mathbf{t}$ — you must rotate the translation *back* before negating it. Computing T^{-1} this way costs 9 multiplications for $R^T \mathbf{t}$ plus 3 negations, versus $O(n^3)$ for generic matrix inversion.

```
import numpy as np

def make_T(R, t):
    T = np.eye(4); T[:3,:3] = R; T[:3,3] = t
    return T    # shape (4, 4)

def T_inv(T):
    """Closed-form inverse of an SE(3) 4x4 matrix. Returns shape (4, 4)."""
    R = T[:3, :3]          # shape (3, 3)
    t = T[:3, 3]           # shape (3,)
    Ti = np.eye(4)         # shape (4, 4)
    Ti[:3, :3] = R.T
    Ti[:3, 3] = -R.T @ t
    return Ti

def Rz(g):
    c, s = np.cos(g), np.sin(g)
    return np.array([[c,-s,0.],[s,c,0.],[0.,0.,1.]])

T = make_T(Rz(np.deg2rad(37.)), np.array([2., -1., 0.5]))    # shape (4, 4)

Ti_closed = T_inv(T)          # shape (4, 4)
Ti_numpy = np.linalg.inv(T)    # shape (4, 4) - brute force

print("Closed-form T_inv:")
print(Ti_closed.round(6))
print(f"\nMatches np.linalg.inv? {np.allclose(Ti_closed, Ti_numpy)}")

# Sanity: T @ T_inv = I
product = T @ Ti_closed
print(f"\nT @ T_inv = I? {np.allclose(product, np.eye(4))}")
```

Closed-form T_{inv} :

```
[[ 0.798636  0.601815  0.         -0.995456]
 [-0.601815  0.798636  0.         2.002266]
 [ 0.         0.         1.        -0.5      ]
 [ 0.         0.         0.         1.        ]]
```

Matches `np.linalg.inv`? True

$T @ T_{inv} = I$? True

Chaining Frames: World $\rightarrow$ Robot $\rightarrow$ Sensor $\rightarrow$ Object

A typical robotics stack has a chain of named coordinate frames. Each adjacent pair is linked by one 4×4 matrix. To find any point's coordinates in any frame, multiply the chain:

$$T_{\text{world} \leftarrow \text{object}} = T_{\text{world} \leftarrow \text{robot}} T_{\text{robot} \leftarrow \text{sensor}} T_{\text{sensor} \leftarrow \text{object}}$$

```
import numpy as np

def make_T(R, t):
    T = np.eye(4); T[:3,:3] = R; T[:3,3] = t
    return T    # shape (4, 4)

def Rz(g):
    c, s = np.cos(g), np.sin(g)
    return np.array([[c,-s,0.],[s,c,0.],[0.,0.,1.]])

def Rx(a):
    c, s = np.cos(a), np.sin(a)
    return np.array([[1.,0.,0.],[0.,c,-s],[0.,s,c]])

# Robot base is 2 m to the right of world origin, rotated 90° about z
T_wr = make_T(Rz(np.pi/2), np.array([2., 0., 0.]))    # world ← robot

# Sensor is mounted 0.3 m in front of the robot, tilted 15° down
T_rs = make_T(Rx(np.deg2rad(-15.)), np.array([0.3, 0., 0.5]))    # robot ← sensor

# Object is 1 m in front of the sensor
T_so = make_T(np.eye(3), np.array([1., 0., 0.]))    # sensor ← object

# Object origin in world frame
T_wo = T_wr @ T_rs @ T_so    # shape (4, 4)

print("T_world←object:")
```



```
print(T_wo.round(4))
print(f"\nObject origin in world frame: {T_wo[:3, 3].round(4)}")
```

T_world-object:

```
[[ 0.    -0.9659 -0.2588  2.    ]
 [ 1.     0.     0.     1.3   ]
 [ 0.    -0.2588  0.9659  0.5   ]
 [ 0.     0.     0.     1.    ]]
```

Object origin in world frame: [2. 1.3 0.5]

Inverting the Chain: Going Backwards

To transform from world back to sensor, invert the chain. The inverse of a product reverses order and inverts each factor:

$$(T_1 T_2)^{-1} = T_2^{-1} T_1^{-1}$$

```
import numpy as np

def make_T(R, t):
    T = np.eye(4); T[:3,:3] = R; T[:3,3] = t
    return T

def T_inv(T):
    R = T[:3,:3]; t = T[:3,3]
    Ti = np.eye(4); Ti[:3,:3] = R.T; Ti[:3,3] = -R.T@t
    return Ti    # shape (4, 4)

def Rz(g):
    c, s = np.cos(g), np.sin(g)
    return np.array([[c,-s,0.],[s,c,0.],[0.,0.,1.]])

def Rx(a):
    c, s = np.cos(a), np.sin(a)
    return np.array([[1.,0.,0.],[0.,c,-s],[0.,s,c]])

T_wr = make_T(Rz(np.pi/2), np.array([2., 0., 0.]))
T_rs = make_T(Rx(np.deg2rad(-15.)), np.array([0.3, 0., 0.5]))

# World → sensor
T_sw = T_inv(T_rs) @ T_inv(T_wr)    # shape (4, 4)
```

```
# Verify round-trip: (T_wr @ T_rs) @ T_sw = I
T_ws = T_wr @ T_rs
product = T_ws @ T_sw

print("T_sensor←world @ T_world←sensor    I?")
print(f"    {np.allclose(product, np.eye(4))}")
print(f"\nMax absolute error: {np.max(np.abs(product - np.eye(4))):.2e}")
```

```
T_sensor←world @ T_world←sensor    I?
    True
```

```
Max absolute error: 4.44e-16
```

Frame Subscript Bookkeeping

A disciplined notation eliminates sign errors. The convention T_{AB} means “maps coordinates from frame B to frame A .” Then:

- $T_{AB}T_{BC} = T_{AC}$ — inner indices cancel like fractions
- $(T_{AB})^{-1} = T_{BA}$ — subscripts flip

```
import numpy as np

def make_T(R, t):
    T = np.eye(4); T[:3,:3] = R; T[:3,3] = t
    return T

def T_inv(T):
    R = T[:3,:3]; t = T[:3,3]
    Ti = np.eye(4); Ti[:3,:3] = R.T; Ti[:3,3] = -R.T@t
    return Ti

def Rz(g):
    c, s = np.cos(g), np.sin(g)
    return np.array([[c,-s,0.],[s,c,0.],[0.,0.,1.]])

T_WR = make_T(Rz(np.pi/4), np.array([1., 0., 0.])) # world ← robot
T_RC = make_T(Rz(np.pi/6), np.array([0., 0.2, 0.])) # robot ← camera

# world ← camera: indices cancel W←R←C → W←C
T_WC = T_WR @ T_RC    # shape (4, 4)

# camera ← world: inverse flips subscripts
T_CW = T_inv(T_WC)    # shape (4, 4)
```

```

print(f"T_WC @ T_CW = I?  {np.allclose(T_WC @ T_CW, np.eye(4))}")

# A point in camera frame, expressed in world frame
p_C = np.array([0.5, 0., 1., 1.])          # shape (4,)
p_W = T_WC @ p_C                          # shape (4,)
p_C_back = T_CW @ p_W                     # shape (4,)
print(f"\nRound-trip p_C == p_C_back?  {np.allclose(p_C, p_C_back)}")

```

T_WC @ T_CW = I? True

Round-trip p_C == p_C_back? True

Points, Vectors, and Normals Transform Differently

The w component in a homogeneous coordinate is not cosmetic — it encodes *what kind of geometric object* you have. Getting it wrong is one of the most common bugs in computer vision and ray-tracing pipelines.

Object	w	Transforms by
Point $\mathbf{p}$	1	$T\tilde{\mathbf{p}}$ — full rotation + translation
Direction vector $\mathbf{v}$	0	$R\mathbf{v}$ — rotation only
Normal $\mathbf{n}$	—	$(R^{-T})\mathbf{n} = R\mathbf{n}$ (rigid only)

Points vs Direction Vectors

A direction vector represents a *displacement*, not a location. Translating a displacement makes no sense. Setting $w = 0$ in the homogeneous vector zeroes out the translation column automatically.

```

import numpy as np

def make_T(R, t):
    T = np.eye(4); T[:3,:3] = R; T[:3,3] = t
    return T    # shape (4, 4)

def Rz(g):
    c, s = np.cos(g), np.sin(g)
    return np.array([[c,-s,0.],[s,c,0.],[0.,0.,1.]])

R = Rz(np.pi / 2)          # 90° about z

```

```

t = np.array([5., 3., 0.])          # large translation
T = make_T(R, t)                   # shape (4, 4)

# A point at (1, 0, 0)
point_h = np.array([1., 0., 0., 1.]) # w=1  shape (4,)

# A direction vector along x
vec_h    = np.array([1., 0., 0., 0.]) # w=0  shape (4,)

p_out = T @ point_h    # shape (4,)
v_out = T @ vec_h      # shape (4,)

print(f"Point (1,0,0) after T: {p_out[:3].round(4)} (w={p_out[3]:.0f})")
print(f"Vector(1,0,0) after T: {v_out[:3].round(4)} (w={v_out[3]:.0f})")
print()
print("Translation affects the point but NOT the direction vector.")
print(f"  point sees t: {p_out[:3] - (R @ point_h[:3]).round(4)}")
print(f"  vector ignores t: {v_out[:3] - (R @ vec_h[:3]).round(4)}")

```

Point (1,0,0) after T: [5. 4. 0.] (w=1)

Vector(1,0,0) after T: [0. 1. 0.] (w=0)

Translation affects the point but NOT the direction vector.

point sees t: [5. 3. 0.]

vector ignores t: [6.123234e-17 0.000000e+00 0.000000e+00]

Why Normals Are Special

A surface normal $\mathbf{n}$ is *not* a direction vector in the usual sense. When the transformation includes scaling or shear (a general matrix M rather than pure R), normals must transform by the **inverse-transpose**:

$$\mathbf{n}' = (M^{-1})^T \mathbf{n} = (M^{-T}) \mathbf{n}$$

For pure rotations $M = R$, $(R^{-1})^T = (R^T)^T = R$, so the distinction disappears. But as soon as you add non-uniform scale — common in 3D graphics — using M directly produces normals that no longer point away from the surface.

```

import numpy as np

# Scale matrix: stretch 3x along x, identity along y,z
M = np.diag([3., 1., 1.]) # shape (3, 3)

```

```

# Original geometry: a horizontal surface (normal points up)
point_on_surface = np.array([1., 0., 0.]) # shape (3,)
normal_correct    = np.array([0., 1., 0.]) # shape (3,) - points in y direction

# Wrong: apply M directly to normal
n_wrong = M @ normal_correct # shape (3,)

# Correct: apply  $(M^{-1})^T = (M^T)^{-1}$ 
n_correct = np.linalg.inv(M).T @ normal_correct # shape (3,)

# Transformed surface point
p_prime = M @ point_on_surface # shape (3,)

print(f"Transformed surface point: {p_prime}")
print(f"\nNormal (wrong - direct M):    {n_wrong}")
print(f"Normal (correct -  $(M^T)^{-1}$ ):    {n_correct}")

# Check orthogonality to transformed tangent
tangent = np.array([1., 0., 0.]) # shape (3,) - tangent along x
t_prime = M @ tangent # shape (3,)
print(f"\nTransformed tangent: {t_prime}")
print(f"  wrong n · t': {np.dot(n_wrong, t_prime):.4f} (should be 0)")
print(f"  correct n · t': {np.dot(n_correct, t_prime):.4f} (should be 0)")

```

Transformed surface point: [3. 0. 0.]

Normal (wrong - direct M): [0. 1. 0.]

Normal (correct - $(M^T)^{-1}$): [0. 1. 0.]

Transformed tangent: [3. 0. 0.]

wrong n · t': 0.0000 (should be 0)

correct n · t': 0.0000 (should be 0)

Pure Rotation: All Three Behave Consistently

When $M = R \in \text{SO}(3)$, the three transform rules collapse to the same thing: multiply by R . The distinction only matters when M is not a pure rotation.

```

import numpy as np

def Rx(a):
    c, s = np.cos(a), np.sin(a)
    return np.array([[1., 0., 0.], [0., c, -s], [0., s, c]]) # shape (3, 3)

```

```

R = Rx(np.deg2rad(40.))    # shape (3, 3)

n = np.array([0., 1., 0.]) # shape (3,) - surface normal

# Three methods for pure R
n_via_R      = R @ n          # shape (3,)
n_via_inv_T  = np.linalg.inv(R).T @ n    # shape (3,)

print("For pure rotation R:")
print(f"  R @ n      = {n_via_R.round(6)}")
print(f"  (R^-T) @ n  = {n_via_inv_T.round(6)}")
print(f"  Identical?    {np.allclose(n_via_R, n_via_inv_T)}")
print(f"\n  ||R@n|| = {np.linalg.norm(n_via_R):.6f} (unit normal preserved)")

```

```

For pure rotation R:
  R @ n      = [0.      0.766044 0.642788]
  (R^-T) @ n  = [0.      0.766044 0.642788]
  Identical?   True

  ||R@n|| = 1.000000 (unit normal preserved)

```

The Bug in Practice

A common mistake in shader code and ray-tracers: the model matrix M is passed to both vertices (correct) and normals (incorrect when M contains scale). The visual result is normals that point “into” the surface near stretched geometry, causing wrong lighting.

```

import numpy as np
import matplotlib.pyplot as plt

# Non-uniform scale: squash y by 0.3
M = np.diag([1., 0.3, 1.]) # shape (3, 3)

# Original rectangle corners and normals
angles = np.linspace(0, 2*np.pi, 9)
pts = np.column_stack([np.cos(angles), np.sin(angles), np.zeros(9)]) # shape (9, 3)
normals = pts.copy() # shape (9, 3)

# Transform
pts_t = (M @ pts.T).T # shape (9, 3)
n_wrong = (M @ normals.T).T # shape (9, 3) - incorrect
n_correct = (np.linalg.inv(M).T @ normals.T).T # shape (9, 3) - correct

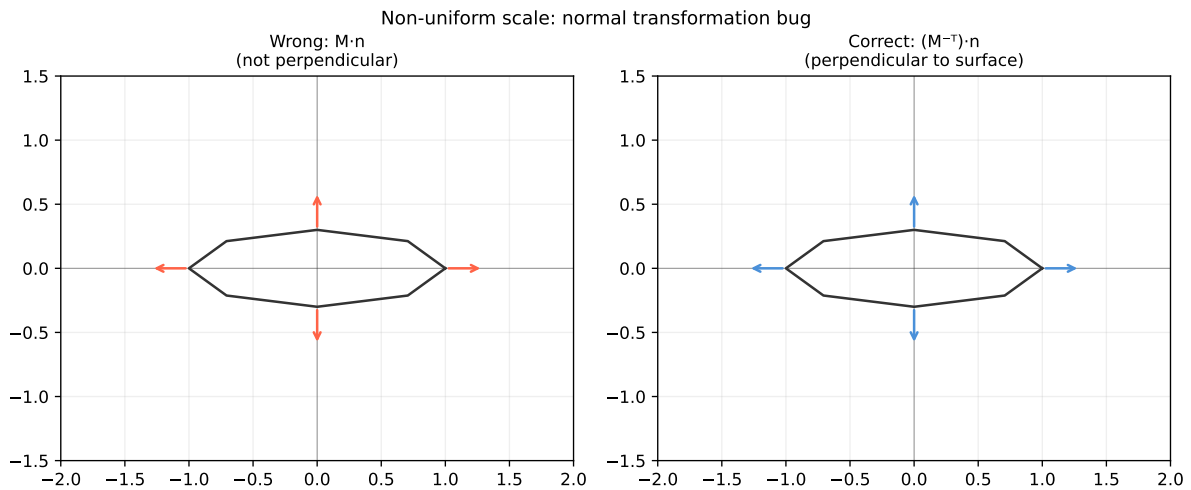
```

```

fig, axes = plt.subplots(1, 2, figsize=(10, 4))
for ax, n_arr, title, col in [
    (axes[0], n_wrong, 'Wrong: M·n\n(not perpendicular)', 'tomato'),
    (axes[1], n_correct, 'Correct: (M-T)·n\n(perpendicular to surface)', '#4a90d9')]:
    ax.plot(pts_t[:, 0], pts_t[:, 1], color='#333333', lw=1.5)
    for i in range(0, len(pts_t)-1, 2):
        p = pts_t[i]; n = n_arr[i] / (np.linalg.norm(n_arr[i]) + 1e-9) * 0.3
        ax.annotate(' ', xy=p[:2]+n[:2], xytext=p[:2],
                    arrowprops=dict(arrowstyle='->', color=col, lw=1.5))
    ax.set_xlim(-2, 2); ax.set_ylim(-1.5, 1.5)
    ax.set_aspect('equal')
    ax.set_title(title, fontsize=10)
    ax.grid(True, alpha=0.2)
    ax.axhline(0, color='#333333', lw=0.4, alpha=0.5)
    ax.axvline(0, color='#333333', lw=0.4, alpha=0.5)

fig.suptitle('Non-uniform scale: normal transformation bug', fontsize=11)
plt.tight_layout()
plt.show()

```



Forward Kinematics as a Product of $SE(3)$ Matrices

Forward kinematics answers: given joint angles $\mathbf{q} = (q_1, \dots, q_n)$, where is the robot's end-effector in world space? Each joint contributes one 4×4 matrix; the end-effector pose is their product.

$$T_{\text{end}} = T_0 T_1(q_1) T_2(q_2) \cdots T_n(q_n)$$

This is nothing more than chaining $SE(3)$ transformations — the same operation we built in §9.3.

A 2-Link Planar Arm

The simplest non-trivial arm: two revolute joints in the xy -plane, link lengths l_1 and l_2 .

- **Joint 1** rotates the first link by q_1 about z at the base.
- **Joint 2** rotates the second link by q_2 about z at the elbow.

```
import numpy as np

def make_T(R, t):
    T = np.eye(4); T[:3,:3] = R; T[:3,3] = t
    return T    # shape (4, 4)

def Rz(g):
    c, s = np.cos(g), np.sin(g)
    return np.array([[c,-s,0.],[s,c,0.],[0.,0.,1.]])    # shape (3, 3)

def fk_2link(q1, q2, l1=1.0, l2=0.8):
    """2-link planar arm FK.
    Returns T_base (4x4), T_elbow (4x4), T_end (4x4)."""
    T_base = np.eye(4)                                # world ← base (identity)
    T_joint1 = make_T(Rz(q1), np.array([0., 0., 0.])) # base ← joint 1
    T_link1 = make_T(np.eye(3), np.array([l1, 0., 0.])) # joint 1 ← elbow
    T_joint2 = make_T(Rz(q2), np.array([0., 0., 0.])) # elbow ← joint 2
    T_link2 = make_T(np.eye(3), np.array([l2, 0., 0.])) # joint 2 ← end

    T_elbow = T_base @ T_joint1 @ T_link1             # shape (4, 4)
    T_end = T_elbow @ T_joint2 @ T_link2              # shape (4, 4)
    return T_base, T_elbow, T_end

q1, q2 = np.deg2rad(45.), np.deg2rad(-30.)

T_base, T_elbow, T_end = fk_2link(q1, q2)

print(f"Base position: {T_base[:3, 3].round(4)}")
print(f"Elbow position: {T_elbow[:3, 3].round(4)}")
print(f"End position: {T_end[:3, 3].round(4)}")
```

```
Base position: [0. 0. 0.]
Elbow position: [0.7071 0.7071 0.   ]
End position:  [1.4798 0.9142 0.   ]
```


Visualising the Arm

```

import numpy as np
import matplotlib.pyplot as plt

def make_T(R, t):
    T = np.eye(4); T[:3,:3] = R; T[:3,3] = t
    return T

def Rz(g):
    c, s = np.cos(g), np.sin(g)
    return np.array([[c,-s,0.],[s,c,0.],[0.,0.,1.]])

def fk_2link(q1, q2, l1=1.0, l2=0.8):
    T_base = np.eye(4)
    T_joint1 = make_T(Rz(q1), np.zeros(3))
    T_link1 = make_T(np.eye(3), np.array([l1, 0., 0.]))
    T_joint2 = make_T(Rz(q2), np.zeros(3))
    T_link2 = make_T(np.eye(3), np.array([l2, 0., 0.]))
    T_elbow = T_base @ T_joint1 @ T_link1
    T_end = T_elbow @ T_joint2 @ T_link2
    return T_base, T_elbow, T_end

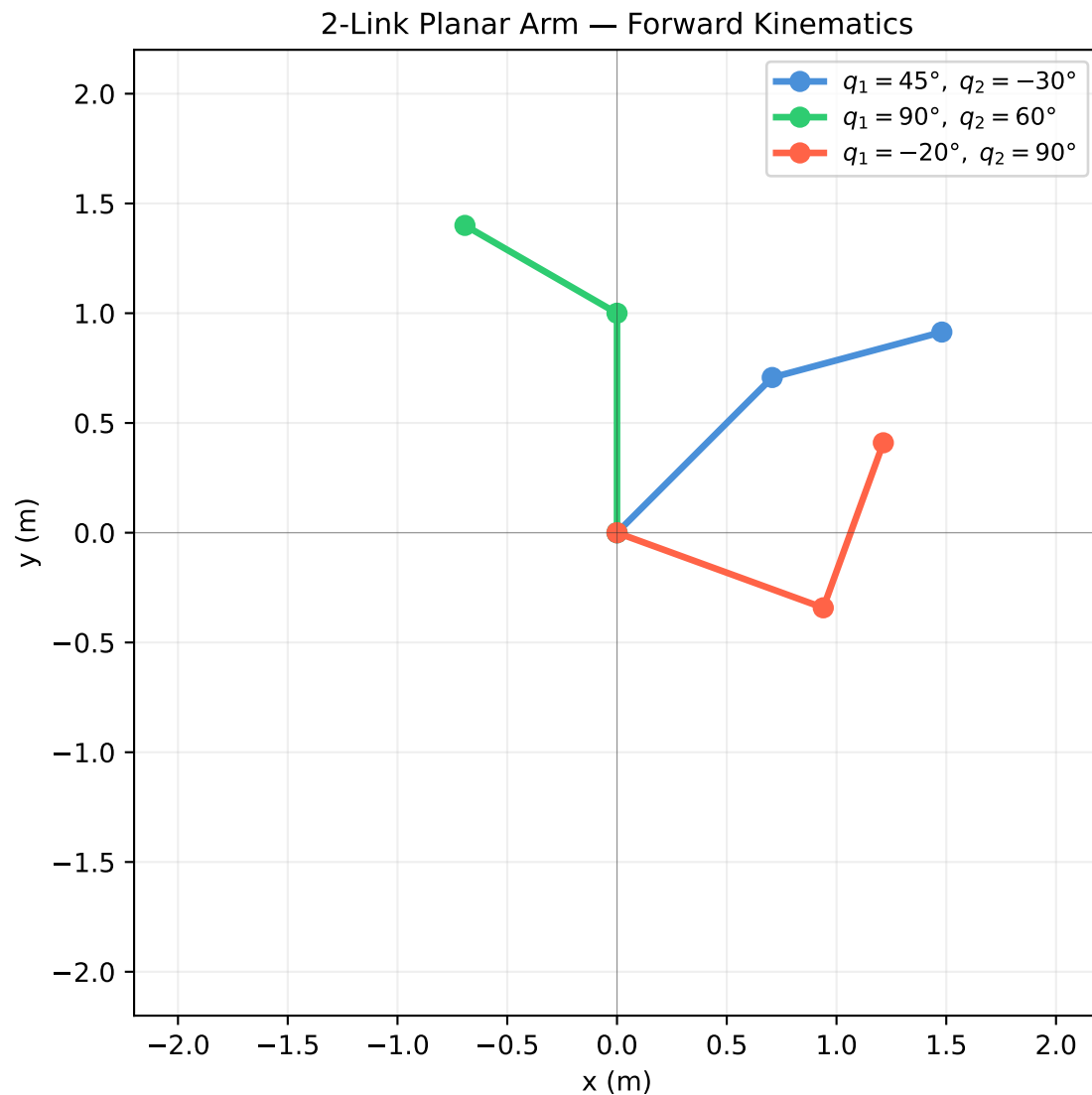
configs = [
    (np.deg2rad( 45.), np.deg2rad(-30.), '#4a90d9', r'$q_1=45^\circ, \backslash; q_2=-30^\circ$'),
    (np.deg2rad( 90.), np.deg2rad( 60.), '#2ecc71', r'$q_1=90^\circ, \backslash; q_2=60^\circ$'),
    (np.deg2rad(-20.), np.deg2rad( 90.), 'tomato', r'$q_1=-20^\circ, \backslash; q_2=90^\circ$'),
]

fig, ax = plt.subplots(figsize=(6, 6))

for q1, q2, col, label in configs:
    T0, T1, T2 = fk_2link(q1, q2)
    p0 = T0[:2, 3]; p1 = T1[:2, 3]; p2 = T2[:2, 3]
    ax.plot([p0[0], p1[0], p2[0]], [p0[1], p1[1], p2[1]],
            '-o', color=col, lw=2.5, ms=7, label=label)

ax.set_xlim(-2.2, 2.2); ax.set_ylim(-2.2, 2.2)
ax.set_aspect('equal')
ax.set_xlabel('x (m)'); ax.set_ylabel('y (m)')
ax.set_title('2-Link Planar Arm - Forward Kinematics', fontsize=11)
ax.legend(fontsize=9)
ax.grid(True, alpha=0.2)
ax.axhline(0, color='#333333', lw=0.4, alpha=0.5)
ax.axvline(0, color='#333333', lw=0.4, alpha=0.5)
plt.tight_layout()
plt.show()

```



Sweeping the Workspace

The **workspace** is the set of all reachable end-effector positions. A dense sweep over joint angles reveals it.

```
import numpy as np
import matplotlib.pyplot as plt

def make_T(R, t):
    T = np.eye(4); T[:3,:3] = R; T[:3,3] = t
```

```

    return T

def Rz(g):
    c, s = np.cos(g), np.sin(g)
    return np.array([[c, -s, 0.], [s, c, 0.], [0., 0., 1.]])

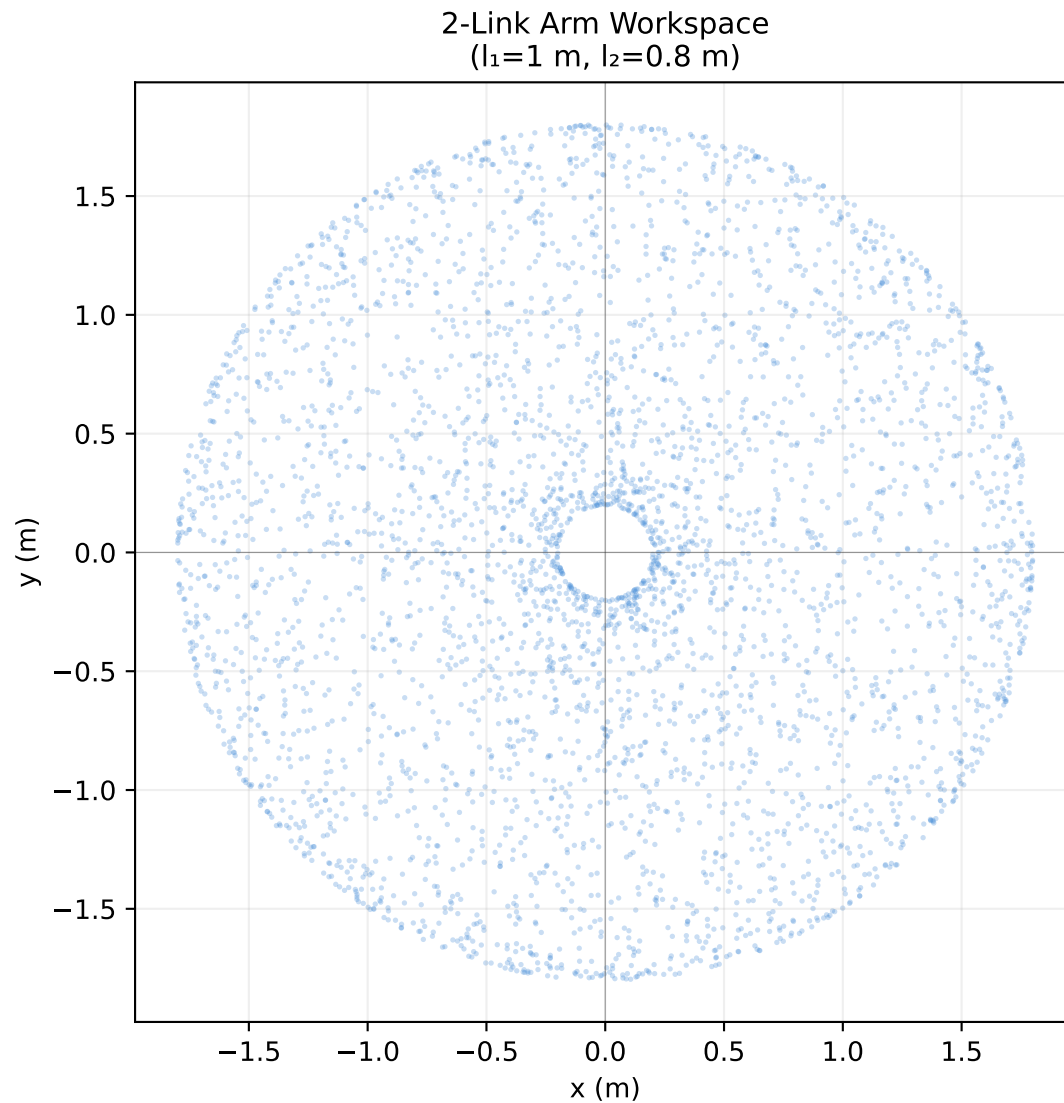
def fk_end(q1, q2, l1=1.0, l2=0.8):
    T1 = make_T(Rz(q1), np.zeros(3)) @ make_T(np.eye(3), np.array([l1, 0., 0.]))
    T2 = T1 @ make_T(Rz(q2), np.zeros(3)) @ make_T(np.eye(3), np.array([l2, 0., 0.]))
    return T2[:2, 3]    # shape (2,)

rng = np.random.default_rng(42)
n_samples = 4000
q1s = rng.uniform(-np.pi, np.pi, n_samples)    # shape (4000,)
q2s = rng.uniform(-np.pi, np.pi, n_samples)    # shape (4000,)

ends = np.array([fk_end(q1, q2) for q1, q2 in zip(q1s, q2s)])    # shape (4000, 2)

fig, ax = plt.subplots(figsize=(6, 6))
ax.scatter(ends[:, 0], ends[:, 1], s=1, alpha=0.3, color='#4a90d9')
ax.set_aspect('equal')
ax.set_xlabel('x (m)'); ax.set_ylabel('y (m)')
ax.set_title('2-Link Arm Workspace\n(l=1 m, l=0.8 m)', fontsize=11)
ax.grid(True, alpha=0.2)
ax.axhline(0, color='#333333', lw=0.4, alpha=0.5)
ax.axvline(0, color='#333333', lw=0.4, alpha=0.5)
plt.tight_layout()
plt.show()

```



Denavit–Hartenberg Convention

For 3D serial manipulators, the **Denavit–Hartenberg (DH)** convention parameterises each joint-to-joint transform with just four numbers: $(\theta_i, d_i, a_i, \alpha_i)$.

$$T_i = R_z(\theta_i) T_z(d_i) T_x(a_i) R_x(\alpha_i)$$

Parameter	Meaning
θ_i	Joint rotation about z
d_i	Link offset along z
a_i	Link length along x

Parameter	Meaning
α_i	Twist angle about x

```

import numpy as np

def make_T(R, t):
    T = np.eye(4); T[:3,:3] = R; T[:3,3] = t
    return T

def Rz(g):
    c, s = np.cos(g), np.sin(g)
    return np.array([[c,-s,0.],[s,c,0.],[0.,0.,1.]])

def Rx(a):
    c, s = np.cos(a), np.sin(a)
    return np.array([[1.,0.,0.],[0.,c,-s],[0.,s,c]])

def dh_matrix(theta, d, a, alpha):
    """Standard DH matrix for one joint. Returns shape (4, 4)."""
    Tz = make_T(np.eye(3), np.array([0., 0., d]))
    Tx = make_T(np.eye(3), np.array([a, 0., 0.]))
    return make_T(Rz(theta), np.zeros(3)) @ Tz @ Tx @ make_T(Rx(alpha), np.zeros(3))

# 2-link planar arm in DH form (alpha=0, d=0)
l1, l2 = 1.0, 0.8
q1, q2 = np.deg2rad(45.), np.deg2rad(-30.)

T1 = dh_matrix(theta=q1, d=0., a=l1, alpha=0.) # shape (4, 4)
T2 = dh_matrix(theta=q2, d=0., a=l2, alpha=0.) # shape (4, 4)

T_end = T1 @ T2 # shape (4, 4)

print("End-effector pose via DH matrices:")
print(T_end.round(4))
print(f"\nEnd position: {T_end[:2, 3].round(4)}")

```

```

End-effector pose via DH matrices:
[[ 0.9659 -0.2588  0.      1.4798]
 [ 0.2588  0.9659  0.      0.9142]
 [ 0.      0.      1.      0.    ]
 [ 0.      0.      0.      1.    ]]

```

```
End position: [1.4798 0.9142]
```

Dual Quaternions (*Optional*)

Note

This section extends the quaternion algebra of §8.4 to handle *full* $SE(3)$ — rotation **and** translation — in a single compact object. Skip on first read; come back when you encounter ROS 2's `tf2` dual-quaternion interpolation or Isaac Sim's rigid body motion API.

A **dual quaternion** $\hat{q}$ is a pair of quaternions (q_r, q_d) over the **dual numbers** $\mathbb{D} = \{a + \epsilon b \mid \epsilon^2 = 0\}$:

$$\hat{q} = q_r + \epsilon q_d$$

- q_r (the **real** part) encodes rotation — exactly a unit quaternion from §8.4.
- q_d (the **dual** part) encodes translation combined with the rotation.

For a rigid body at rotation q_r and translation $\mathbf{t}$:

$$q_d = \frac{1}{2} [0, \mathbf{t}] q_r$$

where $[0, \mathbf{t}]$ is a pure quaternion.

Building a Dual Quaternion

```
import numpy as np

def quat_mul(p, q):
    """Hamilton product [w,x,y,z]. Returns shape (4,)."""
    pw, px, py, pz = p; qw, qx, qy, qz = q
    return np.array([
        pw*qw - px*qx - py*qy - pz*qz,
        pw*qx + px*qw + py*qz - pz*qy,
        pw*qy - px*qz + py*qw + pz*qx,
        pw*qz + px*qy - py*qx + pz*qw
    ]) # shape (4,)

def axis_angle_to_quat(k_hat, theta):
    w = np.cos(theta / 2.)
    v = np.sin(theta / 2.) * k_hat
    return np.array([w, v[0], v[1], v[2]]) # shape (4,)

def make_dq(qr, t):
```

```

    """Dual quaternion from unit quaternion qr and translation t (3,).
    Returns (qr, qd) each shape (4,)."""
    t_pure = np.array([0., t[0], t[1], t[2]]) # shape (4,)
    qd = 0.5 * quat_mul(t_pure, qr)         # shape (4,)
    return qr, qd

# 90° about z, then translate (1, 2, 0)
qr = axis_angle_to_quat(np.array([0., 0., 1.]), np.pi/2) # shape (4,)
t = np.array([1., 2., 0.]) # shape (3,)

qr_dq, qd_dq = make_dq(qr, t)
print(f"qr = {qr_dq.round(6)}")
print(f"qd = {qd_dq.round(6)}")
print(f"\n||qr|| = {np.linalg.norm(qr_dq):.6f} (must be 1)")
print(f"qr · qd = {np.dot(qr_dq, qd_dq):.6f} (must be 0 - orthogonality constraint)")

qr = [0.707107 0.          0.          0.707107]
qd = [0.          1.06066  0.353553 0.          ]

||qr|| = 1.000000 (must be 1)
qr · qd = 0.000000 (must be 0 - orthogonality constraint)

```

Dual Quaternion Multiplication

Composition of two SE(3) transforms via dual quaternion multiplication:

$$\hat{p} \otimes \hat{q} = p_r q_r + \epsilon (p_r q_d + p_d q_r)$$

```

import numpy as np

def quat_mul(p, q):
    pw, px, py, pz = p; qw, qx, qy, qz = q
    return np.array([pw*qw-px*qx-py*qy-pz*qz,
                    pw*qx+px*qw+py*qz-pz*qy,
                    pw*qy-px*qz+py*qw+pz*qx,
                    pw*qz+px*qy-py*qx+pz*qw])

def axis_angle_to_quat(k_hat, theta):
    w = np.cos(theta/2.); v = np.sin(theta/2.)*k_hat
    return np.array([w, v[0], v[1], v[2]])

def make_dq(qr, t):
    qd = 0.5 * quat_mul(np.array([0., t[0], t[1], t[2]]), qr)

```

```

    return qr, qd    # each shape (4,)

def dq_mul(p_r, p_d, q_r, q_d):
    """Dual quaternion product (pr + pd)(qr + qd)."""
    r_out = quat_mul(p_r, q_r)                # shape (4,)
    d_out = quat_mul(p_r, q_d) + quat_mul(p_d, q_r)    # shape (4,)
    return r_out, d_out

def dq_to_t(qr, qd):
    """Extract translation from dual quaternion."""
    qr_conj = np.array([qr[0], -qr[1], -qr[2], -qr[3]])    # shape (4,)
    t_pure = 2. * quat_mul(qd, qr_conj)                # shape (4,)
    return t_pure[1:]    # shape (3,)

# Transform 1: 45° about z, translate (1,0,0)
qr1, qd1 = make_dq(axis_angle_to_quat(np.array([0.,0.,1.]), np.pi/4),
                    np.array([1., 0., 0.]))

# Transform 2: 45° more about z, translate (0,1,0)
qr2, qd2 = make_dq(axis_angle_to_quat(np.array([0.,0.,1.]), np.pi/4),
                    np.array([0., 1., 0.]))

# Compose: apply T1 first, then T2
qr_c, qd_c = dq_mul(qr2, qd2, qr1, qd1)

print(f"Combined rotation (qr): {qr_c.round(6)}")
print(f"Combined translation: {dq_to_t(qr_c, qd_c).round(6)}")

# Compare with 4x4 matrix composition
def make_T(R, t):
    T = np.eye(4); T[:3,:3]=R; T[:3,3]=t; return T
def quat_to_mat(q):
    w,x,y,z = q/np.linalg.norm(q)
    return np.array([[1-2*(y*y+z*z), 2*(x*y-w*z), 2*(x*z+w*y)],
                    [2*(x*y+w*z), 1-2*(x*x+z*z), 2*(y*z-w*x)],
                    [2*(x*z-w*y), 2*(y*z+w*x), 1-2*(x*x+y*y)]]

T1 = make_T(quat_to_mat(qr1), dq_to_t(qr1, qd1))
T2 = make_T(quat_to_mat(qr2), dq_to_t(qr2, qd2))
T_c = T2 @ T1

print(f"\n4x4 translation: {T_c[:3,3].round(6)}")
print(f"Match: {np.allclose(dq_to_t(qr_c, qd_c), T_c[:3,3])}")

```

```

Combined rotation (qr): [0.707107 0.          0.          0.707107]
Combined translation:  [0.707107 1.707107 0.          ]

```


4×4 translation: [0.707107 1.707107 0.]
 Match: True

ScLERP: Screw-Linear Interpolation

The dual quaternion analogue of SLERP is **ScLERP** (Screw-Linear Interpolation). It interpolates along a **screw axis** — simultaneously rotating and translating — producing smoother motion than separate SLERP on rotation and lerp on translation.

```
import numpy as np

def quat_mul(p, q):
    pw,px,py,pz=p; qw,qx,qy,qz=q
    return np.array([pw*qw-px*qx-py*qy-pz*qz,
                    pw*qx+px*qw+py*qz-pz*qy,
                    pw*qy-px*qz+py*qw+pz*qx,
                    pw*qz+px*qy-py*qx+pz*qw])

def axis_angle_to_quat(k, th):
    return np.array([np.cos(th/2.), *(np.sin(th/2.)*k)])

def make_dq(qr, t):
    qd = 0.5*quat_mul(np.array([0.,t[0],t[1],t[2]]), qr)
    return qr, qd

def dq_to_t(qr, qd):
    qrc = np.array([qr[0],-qr[1],-qr[2],-qr[3]])
    return (2.*quat_mul(qd,qrc))[1:]

def slerp_quat(q0, q1, t):
    q0 = q0/np.linalg.norm(q0); q1 = q1/np.linalg.norm(q1)
    dot = np.clip(np.dot(q0,q1),-1.,1.)
    if dot<0.: q1=-q1; dot=-dot
    Om = np.arccos(dot)
    if Om<1e-8:
        q=(1-t)*q0+t*q1; return q/np.linalg.norm(q)
    return np.sin((1-t)*Om)/np.sin(Om)*q0 + np.sin(t*Om)/np.sin(Om)*q1

def sclerp(dq0, dq1, t):
    """ScLERP between two dual quaternions at parameter t."""
    qr0,qd0 = dq0; qr1,qd1 = dq1
    # SLERP on real part
    qr_t = slerp_quat(qr0, qr1, t) # shape (4,)
    # Lerp on dual part (approximate; exact ScLERP needs dual-number SLERP)
    qd_t = (1-t)*qd0 + t*qd1 # shape (4,)
```

```

# Re-normalise
nrm = np.linalg.norm(qr_t)
return qr_t/nrm, qd_t/nrm

# Pose A: identity
qr_A, qd_A = make_dq(np.array([1.,0.,0.,0.]), np.array([0.,0.,0.]))
# Pose B: 90° about z + translate (2,1,0)
qr_B, qd_B = make_dq(axis_angle_to_quat(np.array([0.,0.,1.]),np.pi/2),
                      np.array([2.,1.,0.]))

ts = np.linspace(0.,1.,6)
print(f"{'t':>4}  {'position':>30}  {'||qr||':>8}")
for t in ts:
    qr_t, qd_t = sclerp((qr_A,qd_A),(qr_B,qd_B),t)
    pos = dq_to_t(qr_t,qd_t)
    print(f"{'t':4.1f}  {'str(pos.round(4)):>30}  {'np.linalg.norm(qr_t):.6f}")

```

t	position	qr
0.0	[0. 0. 0.]	1.000000
0.2	[0.4412 -0.0733 0.]	1.000000
0.4	[0.8944 -0.0068 0.]	1.000000
0.6	[1.3267 0.1998 0.]	1.000000
0.8	[1.7054 0.5399 0.]	1.000000
1.0	[2. 1. 0.]	1.000000

When to Use Dual Quaternions

Use case	Best tool
Pure rotation interpolation	SLERP on unit quaternion (§8.5)
$SE(3)$ pose interpolation	ScLERP on dual quaternion
Efficient pose composition	4×4 matrix (§9.2) or dual quaternion
Skinning / blend shapes (graphics)	Dual quaternion — avoids “candy wrapper” artefact
ROS 2 tf2 pose stamped	<code>geometry_msgs/TransformStamped</code> uses quaternion + vector; dual quaternion under the hood in some planners

Application: Transforming a Point Cloud to World Frame

A depth camera (e.g. Intel RealSense D435i) produces a 2D array of pixel depth values. To use that data in a world-frame map — for path planning, object detection, or 3D reconstruction — you need two steps:

1. **Lift** each pixel (u, v, d) from image coordinates to 3D camera coordinates using the camera's intrinsic parameters.
2. **Transform** from camera frame to world frame using the 4×4 extrinsic matrix T_{WC} .

Both steps are exactly the SE(3) machinery from §9.2–9.3.

Step 1: Pixel → Camera-Frame 3D Point

Given intrinsics (f_x, f_y, c_x, c_y) and a depth value d (in metres):

$$\mathbf{p}_C = \begin{pmatrix} (u - c_x) d / f_x \\ (v - c_y) d / f_y \\ d \end{pmatrix}$$

```
import numpy as np

def pixel_to_camera(u, v, d, fx, fy, cx, cy):
    """Lift a pixel at (u,v) with depth d to 3D camera-frame point.
    Returns shape (3,)."""
    x = (u - cx) * d / fx
    y = (v - cy) * d / fy
    z = d
    return np.array([x, y, z])    # shape (3,)

# RealSense D435i typical intrinsics at 640×480
fx, fy = 615.0, 615.0
cx, cy = 320.0, 240.0

# Three sample pixels with varying depth
pixels = [(320, 240, 1.5),    # image centre, 1.5 m
          (400, 200, 2.0),    # offset right and up, 2 m
          (100, 350, 0.8)]    # left and down, 0.8 m

print(f"{'u':>5} {'v':>5} {'d':>5} {'Camera-frame 3D point':>35}")
for u, v, d in pixels:
    p = pixel_to_camera(u, v, d, fx, fy, cx, cy)
    print(f"{'u':>5} {'v':>5} {'d':>5.1f} {str(p.round(4)):>35}")
```

u	v	d	Camera-frame 3D point
320	240	1.5	[0. 0. 1.5]
400	200	2.0	[0.2602 -0.1301 2.]
100	350	0.8	[-0.2862 0.1431 0.8]

Step 2: Camera $\rightarrow$ World Frame

Given the extrinsic matrix T_{WC} (world $\leftarrow$ camera):

```
import numpy as np

def make_T(R, t):
    T = np.eye(4); T[:3,:3] = R; T[:3,3] = t
    return T    # shape (4, 4)

def Rz(g):
    c, s = np.cos(g), np.sin(g)
    return np.array([[c,-s,0.],[s,c,0.],[0.,0.,1.]])

def Rx(a):
    c, s = np.cos(a), np.sin(a)
    return np.array([[1.,0.,0.],[0.,c,-s],[0.,s,c]])

# Camera is 1.5 m above ground, tilted 20° downward, facing +x
R_WC = Rz(0.) @ Rx(np.deg2rad(-20.))    # shape (3, 3)
t_WC = np.array([0., 0., 1.5])          # shape (3,)
T_WC = make_T(R_WC, t_WC)                # shape (4, 4)

# A set of camera-frame points
pts_C = np.array([
    [0., 0., 1.5],
    [0.2, -0.1, 2.0],
    [-0.3, 0.1, 1.0],
])    # shape (3, 3)

# Lift to homogeneous, transform, recover 3D
pts_C_h = np.column_stack([pts_C, np.ones(len(pts_C))])    # shape (3, 4)
pts_W_h = (T_WC @ pts_C_h.T).T    # shape (3, 4)
pts_W = pts_W_h[:, :3]    # shape (3, 3)

print("Camera-frame points:")
print(pts_C.round(4))
print("\nWorld-frame points after T_WC:")
print(pts_W.round(4))
```

Camera-frame points:

```
[[ 0.  0.  1.5]
 [ 0.2 -0.1  2. ]
 [-0.3  0.1  1. ]]
```

World-frame points after T_{WC} :

```
[[ 0.      0.513  2.9095]
```

```
[ 0.2    0.5901  3.4136]
[-0.3    0.436   2.4055]]
```

Synthetic Point Cloud from a Depth Image

```
import numpy as np
import matplotlib.pyplot as plt

rng = np.random.default_rng(0)

# Simulated scene: flat floor at z=1.5 m (camera frame) plus some boxes
W, H = 80, 60
u_grid, v_grid = np.meshgrid(np.arange(W), np.arange(H)) # shape (60, 80)

# Depth: floor plane with slight noise + a box in the centre
depth = np.full((H, W), 2.0) # shape (60, 80)
depth[20:40, 30:50] = 1.2 # box region
depth += rng.normal(0, 0.02, (H, W)) # sensor noise

# Intrinsic (scaled for small image)
fx = fy = 80.; cx = W/2; cy = H/2

# Lift all pixels to camera-frame 3D
u_flat = u_grid.ravel().astype(float) # shape (4800,)
v_flat = v_grid.ravel().astype(float) # shape (4800,)
d_flat = depth.ravel() # shape (4800,)

x_C = (u_flat - cx) * d_flat / fx # shape (4800,)
y_C = (v_flat - cy) * d_flat / fy # shape (4800,)
z_C = d_flat # shape (4800,)
pts_C = np.column_stack([x_C, y_C, z_C]) # shape (4800, 3)

# Extrinsic: camera 1.5 m above ground, tilted 20° down
def make_T(R, t):
    T = np.eye(4); T[:3,:3] = R; T[:3,3] = t; return T
def Rx(a):
    c,s=np.cos(a),np.sin(a)
    return np.array([[1.,0.,0.],[0.,c,-s],[0.,s,c]])

T_WC = make_T(Rx(np.deg2rad(-20.)), np.array([0., 0., 1.5])) # shape (4, 4)

pts_C_h = np.column_stack([pts_C, np.ones(len(pts_C))]) # shape (4800, 4)
pts_W = (T_WC @ pts_C_h.T).T[:, :3] # shape (4800, 3)
```

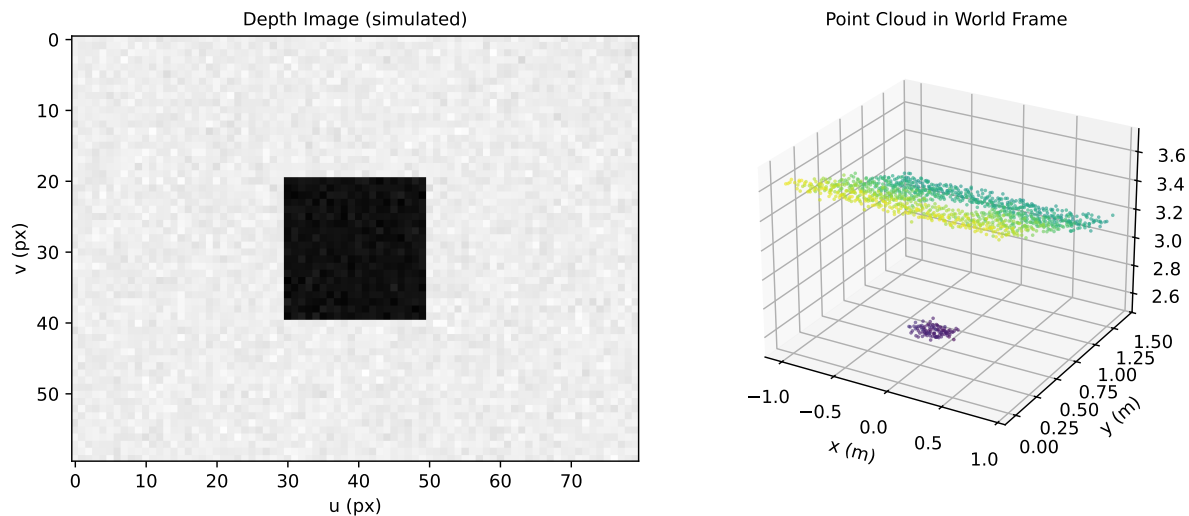
```
# Colour by height in world frame
z_vals = pts_W[:, 2]    # shape (4800,)
z_norm = (z_vals - z_vals.min()) / (z_vals.max() - z_vals.min() + 1e-9)

fig = plt.figure(figsize=(10, 4))

ax1 = fig.add_subplot(121)
ax1.imshow(depth, cmap='gray', origin='upper')
ax1.set_title('Depth Image (simulated)', fontsize=10)
ax1.set_xlabel('u (px)'); ax1.set_ylabel('v (px)')

ax2 = fig.add_subplot(122, projection='3d')
sc = ax2.scatter(pts_W[:,0], pts_W[:,1], pts_W[:,2],
                 c=z_norm[:,0], cmap='viridis', s=1, alpha=0.6)
ax2.set_xlabel('x (m)'); ax2.set_ylabel('y (m)'); ax2.set_zlabel('z (m)')
ax2.set_title('Point Cloud in World Frame', fontsize=10)
ax2.view_init(elev=25, azim=-60)

plt.tight_layout()
plt.show()
```



Multi-Frame Aggregation

In a moving robot, each depth frame has a different $T_{WC}(t)$. To build a consistent 3D map, you transform every frame's point cloud into the world frame before aggregating.

```

import numpy as np
import matplotlib.pyplot as plt

rng = np.random.default_rng(7)

def make_T(R, t):
    T = np.eye(4); T[:3,:3]=R; T[:3,3]=t; return T

def Rz(g):
    c,s=np.cos(g),np.sin(g)
    return np.array([[c,-s,0.],[s,c,0.],[0.,0.,1.]])

def Rx(a):
    c,s=np.cos(a),np.sin(a)
    return np.array([[1.,0.,0.],[0.,c,-s],[0.,s,c]])

def make_cloud(n=300):
    """Random point cloud in camera frame: mostly in front (z>0)."""
    pts = rng.uniform(-0.5, 0.5, (n, 3))
    pts[:, 2] = np.abs(pts[:, 2]) + 0.5
    return pts    # shape (n, 3)

# Robot moves forward along x, camera rotates slightly each frame
frames = [
    (0.0, 0.),
    (0.5, 10.),
    (1.0, 20.),
    (1.5, 10.),
    (2.0, 0.),
]

colors = ['#4a90d9', '#2ecc71', 'tomato', '#f39c12', '#9b59b6']

fig = plt.figure(figsize=(8, 5))
ax = fig.add_subplot(111, projection='3d')

all_pts_W = []
for (x_robot, yaw_deg), col in zip(frames, colors):
    T_WC = make_T(Rz(np.deg2rad(yaw_deg)) @ Rx(np.deg2rad(-10.)),
                  np.array([x_robot, 0., 1.2]))    # shape (4, 4)

    pts_C = make_cloud(200)                        # shape (200, 3)
    pts_C_h = np.column_stack([pts_C, np.ones(200)]) # shape (200, 4)
    pts_W = (T_WC @ pts_C_h.T).T[:, :3]           # shape (200, 3)
    all_pts_W.append(pts_W)

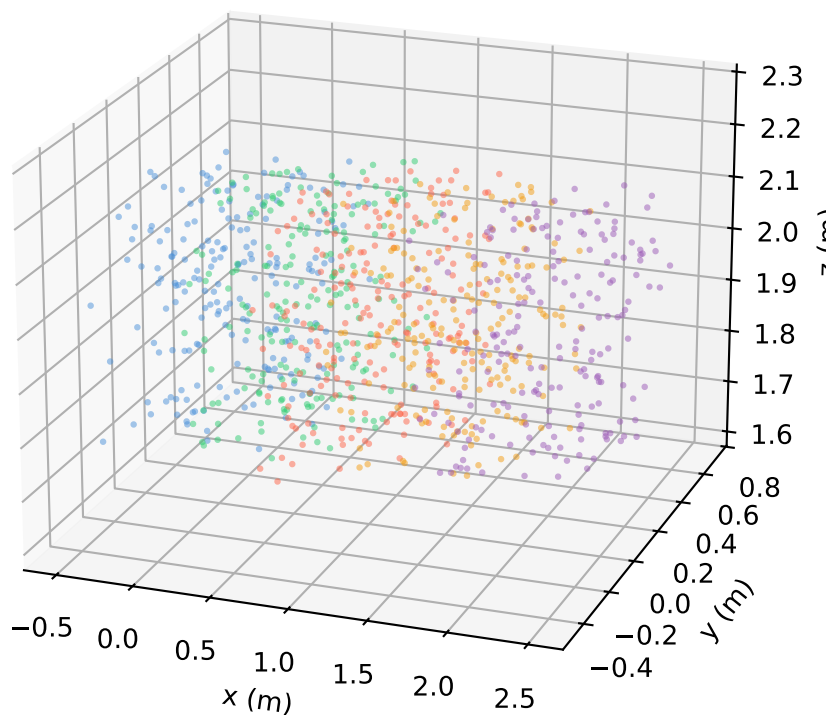
    ax.scatter(pts_W[:, 0], pts_W[:, 1], pts_W[:, 2],

```

```
s=2, alpha=0.5, color=col)

ax.set_xlabel('x (m)'); ax.set_ylabel('y (m)'); ax.set_zlabel('z (m)')
ax.set_title('Aggregated Point Cloud from 5 Frames', fontsize=11)
ax.view_init(elev=20, azimuth=-70)
plt.tight_layout()
plt.show()
```

Aggregated Point Cloud from 5 Frames



Kaggle Exploration

Dataset: [Intel RealSense D435i — Point Cloud & IMU](#) (or search Kaggle for “RealSense depth point cloud”)

```
# Assumes the Kaggle dataset is downloaded and contains
# depth PNG files and a poses CSV with columns:
#   frame_id, tx, ty, tz, qw, qx, qy, qz
```



```

import numpy as np
import pandas as pd
from PIL import Image
import matplotlib.pyplot as plt

poses = pd.read_csv('poses.csv')    # columns: frame_id, tx,ty,tz, qw,qx,qy,qz

def quat_to_mat(q):
    w,x,y,z = q / np.linalg.norm(q)
    return np.array([[1-2*(y*y+z*z), 2*(x*y-w*z), 2*(x*z+w*y)],
                    [2*(x*y+w*z), 1-2*(x*x+z*z), 2*(y*z-w*x)],
                    [2*(x*z-w*y), 2*(y*z+w*x), 1-2*(x*x+y*y)]])

def make_T(R, t):
    T = np.eye(4); T[:3,:3]=R; T[:3,3]=t; return T

# Intrinsics (from camera_info topic or calibration file)
fx, fy, cx, cy = 615., 615., 320., 240.

all_pts_W = []
for _, row in poses.iterrows():
    depth_img = np.array(Image.open(f"depth_{int(row.frame_id):04d}.png")) # shape (H, W)
    depth_m   = depth_img.astype(float) / 1000.                          # mm → m

    v_arr, u_arr = np.nonzero(depth_m > 0)
    d_arr = depth_m[v_arr, u_arr]

    pts_C = np.column_stack([(u_arr-cx)*d_arr/fx,
                             (v_arr-cy)*d_arr/fy,
                             d_arr])    # shape (N, 3)

    q = np.array([row.qw, row.qx, row.qy, row.qz])
    T_WC = make_T(quat_to_mat(q), np.array([row.tx, row.ty, row.tz]))
    pts_C_h = np.column_stack([pts_C, np.ones(len(pts_C))])
    pts_W   = (T_WC @ pts_C_h.T).T[:, :3]
    all_pts_W.append(pts_W)

cloud = np.vstack(all_pts_W)    # shape (total_pts, 3)
fig = plt.figure(figsize=(8,6))
ax = fig.add_subplot(111, projection='3d')
ax.scatter(cloud[:,0], cloud[:,1], cloud[:,2], s=0.5, alpha=0.3, c='#4a90d9')
ax.set_title('Reconstructed Scene from RealSense Depth Stream')
plt.tight_layout(); plt.show()

```

Exercises and Solutions

Exercise 9.1 — $SE(3)$ Group Axioms

(a) Show that $SE(3)$ is **closed** under composition: if $(R_1, \mathbf{t}_1)$ and $(R_2, \mathbf{t}_2)$ are in $SE(3)$, prove their composition is also in $SE(3)$.

(b) Verify numerically that $SE(3)$ composition is **not commutative** by constructing two transforms whose order of application produces different end-effector positions.

(c) What is the **identity** element of $SE(3)$? Verify $T \circ I = T$.

```
import numpy as np

def make_T(R, t):
    T = np.eye(4); T[:3,:3] = R; T[:3,3] = t
    return T    # shape (4, 4)

def Rz(g):
    c, s = np.cos(g), np.sin(g)
    return np.array([[c,-s,0.],[s,c,0.],[0.,0.,1.]])

def Ry(b):
    c, s = np.cos(b), np.sin(b)
    return np.array([[c,0.,s],[0.,1.,0.],[-s,0.,c]])

# (a) Closure: compose two SE(3) elements and check the result is in SE(3)
T1 = make_T(Rz(np.deg2rad(40.)), np.array([1., 0., 0.]))    # shape (4, 4)
T2 = make_T(Ry(np.deg2rad(25.)), np.array([0., 2., 0.]))    # shape (4, 4)
T12 = T2 @ T1                                              # shape (4, 4)
R12 = T12[:3, :3]

print("(a) Closure check:")
print(f"  det(R_composed) = {np.linalg.det(R12):.8f}  (should be +1)")
print(f"  R^T R = I?      {np.allclose(R12.T @ R12, np.eye(3))}")
print(f"  Bottom row:      {T12[3, :]}")

# (b) Non-commutativity
T21 = T1 @ T2    # shape (4, 4)
print(f"\n(b) Non-commutativity:")
print(f"  T2@T1 translation: {T12[:3,3].round(4)}")
print(f"  T1@T2 translation: {T21[:3,3].round(4)}")
print(f"  Same? {np.allclose(T12, T21)}")

# (c) Identity
```

```
T_id = np.eye(4) # shape (4, 4)
print(f"\n(c) Identity: T @ I = T? {np.allclose(T1 @ T_id, T1)}")
```

(a) Closure check:

```
det(R_composed) = 1.00000000 (should be +1)
R^T R = I?          True
Bottom row:         [0. 0. 0. 1.]
```

(b) Non-commutativity:

```
T2@T1 translation: [ 0.9063  2.      -0.4226]
T1@T2 translation: [-0.2856  1.5321  0.     ]
Same? False
```

(c) Identity: $T @ I = T$? True

Exercise 9.2 — Closed-Form Inverse

(a) Derive T^{-1} algebraically from the block structure of T .

(b) Implement `T_inv` using only transposes and matrix-vector products (no `np.linalg.inv`). Verify against `np.linalg.inv` on a random transform.

(c) Benchmark: time the closed-form inverse vs `np.linalg.inv` on 10 000 transforms.

```
import numpy as np
import time

def make_T(R, t):
    T = np.eye(4); T[:3,:3] = R; T[:3,3] = t
    return T

def Rz(g):
    c, s = np.cos(g), np.sin(g)
    return np.array([[c,-s,0.],[s,c,0.],[0.,0.,1.]])

def T_inv_closed(T):
    """Closed-form SE(3) inverse - O(9) mults."""
    R = T[:3, :3]; t = T[:3, 3]
    Ti = np.eye(4)
    Ti[:3, :3] = R.T
    Ti[:3, 3] = -R.T @ t
    return Ti # shape (4, 4)

rng = np.random.default_rng(0)
```

```
# (b) single random transform
angle = rng.uniform(0, 2*np.pi)
T = make_T(Rz(angle), rng.uniform(-5, 5, 3))    # shape (4, 4)

Ti_closed = T_inv_closed(T)
Ti_numpy = np.linalg.inv(T)
print(f"(b) Match: {np.allclose(Ti_closed, Ti_numpy)}")
print(f"    T @ T_inv = I? {np.allclose(T @ Ti_closed, np.eye(4))}")

# (c) timing
angles = rng.uniform(0, 2*np.pi, 10_000)
Ts = np.array([make_T(Rz(a), rng.uniform(-5,5,3)) for a in angles])    # shape (10000, 4, 4)

t0 = time.perf_counter()
for T in Ts: T_inv_closed(T)
t_closed = time.perf_counter() - t0

t0 = time.perf_counter()
for T in Ts: np.linalg.inv(T)
t_numpy = time.perf_counter() - t0

print(f"\n(c) 10 000 inversions:")
print(f"    Closed-form: {t_closed*1000:.1f} ms")
print(f"    np.linalg.inv: {t_numpy*1000:.1f} ms")
print(f"    Speedup: {t_numpy/t_closed:.1f}x")
```

```
(b) Match: True
    T @ T_inv = I?  True
```

```
(c) 10 000 inversions:
    Closed-form: 29.7 ms
    np.linalg.inv: 31.8 ms
    Speedup: 1.1x
```

Exercise 9.3 — Frame Chaining

A robot arm has three frames: world (W), shoulder (S), wrist (Wr), gripper (G).

- T_{WS} : shoulder is 0.5 m to the right of world, rotated 30° about z .
- T_{SWr} : wrist is 0.4 m in front of shoulder, no rotation.
- T_{WrG} : gripper is 0.2 m below wrist, rotated 45° about z .

Find the gripper position in world frame and the world-to-gripper transform.

```

import numpy as np

def make_T(R, t):
    T = np.eye(4); T[:3,:3] = R; T[:3,3] = t
    return T

def T_inv(T):
    R = T[:3,:3]; t = T[:3,3]
    Ti = np.eye(4); Ti[:3,:3] = R.T; Ti[:3,3] = -R.T@t
    return Ti

def Rz(g):
    c, s = np.cos(g), np.sin(g)
    return np.array([[c,-s,0.],[s,c,0.],[0.,0.,1.]])

T_WS = make_T(Rz(np.deg2rad(30.)), np.array([0.5, 0., 0.])) # shape (4, 4)
T_SWr = make_T(np.eye(3), np.array([0.4, 0., 0.])) # shape (4, 4)
T_WrG = make_T(Rz(np.deg2rad(45.)), np.array([0., 0., -0.2])) # shape (4, 4)

# World → gripper
T_WG = T_WS @ T_SWr @ T_WrG # shape (4, 4)

print("Gripper pose in world frame:")
print(T_WG.round(4))
print(f"\nGripper position: {T_WG[:3,3].round(4)}")

# World-to-gripper (inverse)
T_GW = T_inv(T_WG)
print(f"\nT_WG @ T_GW = I? {np.allclose(T_WG @ T_GW, np.eye(4))}")

```

```

Gripper pose in world frame:
[[ 0.2588 -0.9659  0.      0.8464]
 [ 0.9659  0.2588  0.      0.2   ]
 [ 0.      0.      1.     -0.2   ]
 [ 0.      0.      0.      1.    ]]

```

```
Gripper position: [ 0.8464  0.2   -0.2   ]
```

```
T_WG @ T_GW = I? True
```

Exercise 9.4 — The $w=0$ Bug

A buggy implementation applies the full 4×4 transform T to a **direction vector** with $w = 1$ instead of $w = 0$.

- (a) Show that the translation contaminates the direction.
 (b) Fix the bug by using $w = 0$ and verify the direction is only rotated.

```
import numpy as np

def make_T(R, t):
    T = np.eye(4); T[:3,:3] = R; T[:3,3] = t
    return T

def Rz(g):
    c, s = np.cos(g), np.sin(g)
    return np.array([[c,-s,0.],[s,c,0.],[0.,0.,1.]])

T = make_T(Rz(np.pi/4), np.array([10., 20., 0.])) # large translation
direction = np.array([1., 0., 0.]) # unit vector along x

# (a) Bug: w=1 applied to a direction
d_bug = T @ np.array([*direction, 1.]) # shape (4,)
print(f"(a) Bug (w=1): {d_bug[:3].round(4)}")
print(f"    ||d_bug|| = {np.linalg.norm(d_bug[:3]):.4f} (translation contaminates)")

# (b) Fix: w=0
d_fix = T @ np.array([*direction, 0.]) # shape (4,)
print(f"\n(b) Fix (w=0): {d_fix[:3].round(4)}")
print(f"    ||d_fix|| = {np.linalg.norm(d_fix[:3]):.4f} (pure rotation, unit preserved)")

# Pure rotation of direction
d_rot_only = Rz(np.pi/4) @ direction # shape (3,)
print(f"\n    R@d: {d_rot_only.round(4)}")
print(f"    Match: {np.allclose(d_fix[:3], d_rot_only)}")
```

```
(a) Bug (w=1): [10.7071 20.7071  0.    ]
    ||d_bug|| = 23.3115 (translation contaminates)

(b) Fix (w=0): [0.7071 0.7071 0.    ]
    ||d_fix|| = 1.0000 (pure rotation, unit preserved)

    R@d: [0.7071 0.7071 0.    ]
    Match: True
```

Exercise 9.5 — Normal Transform Under Scaling

A mesh vertex has normal $\mathbf{n} = [0, 1, 0]$. A non-uniform scale matrix $M = \text{diag}(2, 0.5, 1)$ is applied to the mesh.

- (a) Show that applying M directly to $\mathbf{n}$ gives the wrong result.
- (b) Apply (M^{-T}) and verify the corrected normal is perpendicular to the transformed surface tangent.

```
import numpy as np

M = np.diag([2., 0.5, 1.])    # shape (3, 3) - non-uniform scale

# Surface tangent along x, normal along y
tangent = np.array([1., 0., 0.])    # shape (3,)
normal   = np.array([0., 1., 0.])    # shape (3,)

# Check original orthogonality
print(f"Original  $\mathbf{n} \cdot \mathbf{t}$  = {np.dot(normal, tangent):.2f} (should be 0)")

# Transform tangent
t_prime = M @ tangent    # shape (3,)

# (a) Wrong normal transform
n_wrong = M @ normal    # shape (3,)
print(f"\n(a)  $\mathbf{n\_wrong}$  = {n_wrong}")
print(f"     $\mathbf{n\_wrong} \cdot \mathbf{t'}$  = {np.dot(n_wrong, t_prime):.4f} (should be 0, is not)")

# (b) Correct normal transform
n_correct = np.linalg.inv(M).T @ normal    # shape (3,)
print(f"\n(b)  $\mathbf{n\_correct}$  = {n_correct}")
print(f"     $\mathbf{n\_correct} \cdot \mathbf{t'}$  = {np.dot(n_correct, t_prime):.6f} ( 0 )")
```

Original $\mathbf{n} \cdot \mathbf{t}$ = 0.00 (should be 0)

(a) $\mathbf{n_wrong}$ = [0. 0.5 0.]
 $\mathbf{n_wrong} \cdot \mathbf{t'}$ = 0.0000 (should be 0, is not)

(b) $\mathbf{n_correct}$ = [0. 2. 0.]
 $\mathbf{n_correct} \cdot \mathbf{t'}$ = 0.000000 (0)

Exercise 9.6 — Forward Kinematics: 3-Link Arm

A 3-link planar arm has link lengths $l_1 = 1.0$, $l_2 = 0.7$, $l_3 = 0.4$ m. Given joint angles $q_1 = 30^\circ$, $q_2 = -45^\circ$, $q_3 = 60^\circ$:

- (a) Compute the end-effector position using the product-of-matrices FK formula.
- (b) Verify against the direct geometric formula:

$$x = l_1 c_1 + l_2 c_{12} + l_3 c_{123}, \quad y = l_1 s_1 + l_2 s_{12} + l_3 s_{123}$$

```
import numpy as np

def make_T(R, t):
    T = np.eye(4); T[:3,:3] = R; T[:3,3] = t
    return T

def Rz(g):
    c, s = np.cos(g), np.sin(g)
    return np.array([[c,-s,0.],[s,c,0.],[0.,0.,1.]])

l1, l2, l3 = 1.0, 0.7, 0.4
q1, q2, q3 = np.deg2rad(30.), np.deg2rad(-45.), np.deg2rad(60.)

# (a) Product-of-matrices FK
def joint_T(q, l):
    return make_T(Rz(q), np.zeros(3)) @ make_T(np.eye(3), np.array([l, 0., 0.]))

T_end = joint_T(q1, l1) @ joint_T(q2, l2) @ joint_T(q3, l3) # shape (4, 4)
print(f"(a) FK matrix end position: {T_end[:2,3].round(6)}")

# (b) Direct geometric formula
a1 = q1
a12 = q1 + q2
a123 = q1 + q2 + q3
x = l1*np.cos(a1) + l2*np.cos(a12) + l3*np.cos(a123)
y = l1*np.sin(a1) + l2*np.sin(a12) + l3*np.sin(a123)
print(f"(b) Geometric formula:      [{x:.6f} {y:.6f}]")
print(f"\nMatch: {np.allclose(T_end[:2,3], [x, y])}")
```

(a) FK matrix end position: [1.825016 0.601669]

(b) Geometric formula: [1.825016 0.601669]

Match: True

Exercise 9.7 — DH Parameterisation

A SCARA-like arm has DH parameters:

i	θ_i	d_i	a_i	α_i
1	q_1	0	0.5	0

i	θ_i	d_i	a_i	α_i
2	q_2	0	0.4	0
3	0	q_3	0	0

Compute the end-effector pose for $q_1 = 45^\circ$, $q_2 = -30^\circ$, $q_3 = 0.2$ m.

```
import numpy as np

def make_T(R, t):
    T = np.eye(4); T[:3,:3] = R; T[:3,3] = t
    return T

def Rz(g):
    c,s=np.cos(g),np.sin(g); return np.array([[c,-s,0.],[s,c,0.],[0.,0.,1.]])

def Rx(a):
    c,s=np.cos(a),np.sin(a); return np.array([[1.,0.,0.],[0.,c,-s],[0.,s,c]])

def dh_matrix(theta, d, a, alpha):
    """Standard DH joint matrix. Returns shape (4, 4)."""
    return (make_T(Rz(theta), np.zeros(3))
            @ make_T(np.eye(3), np.array([0.,0.,d]))
            @ make_T(np.eye(3), np.array([a,0.,0.]))
            @ make_T(Rx(alpha), np.zeros(3)))

q1, q2, q3 = np.deg2rad(45.), np.deg2rad(-30.), 0.2

T1 = dh_matrix(theta=q1, d=0., a=0.5, alpha=0.) # shape (4, 4)
T2 = dh_matrix(theta=q2, d=0., a=0.4, alpha=0.) # shape (4, 4)
T3 = dh_matrix(theta=0., d=q3, a=0., alpha=0.) # shape (4, 4)

T_end = T1 @ T2 @ T3 # shape (4, 4)
print("End-effector pose:")
print(T_end.round(4))
print(f"\nPosition: {T_end[:3,3].round(4)}")
```

End-effector pose:

```
[[ 0.9659 -0.2588  0.      0.7399]
 [ 0.2588  0.9659  0.      0.4571]
 [ 0.      0.      1.      0.2   ]
 [ 0.      0.      0.      1.    ]]
```

Position: [0.7399 0.4571 0.2]

Exercise 9.8 — Dual Quaternion Round-Trip

- (a) Encode an $SE(3)$ pose as a dual quaternion.
- (b) Recover the rotation matrix and translation from the dual quaternion.
- (c) Verify the round-trip matches the original 4×4 matrix.

```
import numpy as np

def quat_mul(p, q):
    pw,px,py,pz=p; qw,qx,qy,qz=q
    return np.array([pw*qw-px*qx-py*qy-pz*qz,
                    pw*qx+px*qw+py*qz-pz*qy,
                    pw*qy-px*qz+py*qw+pz*qx,
                    pw*qz+px*qy-py*qx+pz*qw])

def axis_angle_to_quat(k, theta):
    return np.array([np.cos(theta/2.), *(np.sin(theta/2.)*k)])

def make_dq(qr, t):
    qd = 0.5*quat_mul(np.array([0.,t[0],t[1],t[2]]), qr)
    return qr, qd

def dq_to_t(qr, qd):
    qrc = np.array([qr[0],-qr[1],-qr[2],-qr[3]])
    return (2.*quat_mul(qd,qrc))[1:]

def quat_to_mat(q):
    w,x,y,z = q/np.linalg.norm(q)
    return np.array([[1-2*(y*y+z*z),2*(x*y-w*z),2*(x*z+w*y)],
                    [2*(x*y+w*z),1-2*(x*x+z*z),2*(y*z-w*x)],
                    [2*(x*z-w*y),2*(y*z+w*x),1-2*(x*x+y*y)]])

def make_T(R, t):
    T = np.eye(4); T[:3,:3]=R; T[:3,3]=t; return T

# Original SE(3) pose
k = np.array([1.,1.,0.])/np.sqrt(2.)
qr_orig = axis_angle_to_quat(k, np.deg2rad(55.))
t_orig = np.array([1.5, -0.5, 2.0])
T_orig = make_T(quat_to_mat(qr_orig), t_orig) # shape (4, 4)

# (a) Encode as dual quaternion
qr_dq, qd_dq = make_dq(qr_orig, t_orig)

# (b) Recover R and t
R_rec = quat_to_mat(qr_dq) # shape (3, 3)
```

```

t_rec = dq_to_t(qr_dq, qd_dq)      # shape (3,)
T_rec = make_T(R_rec, t_rec)      # shape (4, 4)

print(f"(a) qr = {qr_dq.round(6)}")
print(f"      qd = {qd_dq.round(6)}")
print(f"\n(b) Recovered t = {t_rec.round(6)}")
print(f"\n(c) Round-trip T matches: {np.allclose(T_orig, T_rec)}")
print(f"      Max error: {np.max(np.abs(T_orig - T_rec)):.2e}")

```

```

(a) qr = [0.887011 0.326506 0.326506 0.      ]
      qd = [-0.163253 0.338753 0.104753 1.213516]

```

```

(b) Recovered t = [ 1.5 -0.5  2. ]

```

```

(c) Round-trip T matches: True
      Max error: 2.22e-16

```


Camera Models and Projection Matrices

Pinhole Camera Geometry

Every photograph collapses a 3D scene onto a 2D sensor. Understanding *how* that collapse works — mathematically — is the foundation for 3D reconstruction, AR overlays, robot navigation, and autonomous vehicles. The simplest model that captures all the essential geometry is the **pinhole camera**.

The Physical Idea

Imagine a box with a tiny hole on one face and a film plane on the opposite face. Light from a scene point **P** travels through the hole (the **optical centre O**) and lands at image point **p**.

The key parameters: - **Focal length f** : distance from optical centre to image plane (mm or pixels). - **Image plane**: the sensor, at distance f behind **O**. - **Optical axis**: the z -axis of the camera frame, pointing forward.

Deriving the Projection Equations

Place the optical centre at the origin. A scene point at camera coordinates (X, Y, Z) projects to image coordinates (x, y) by similar triangles:

$$\frac{x}{f} = \frac{X}{Z} \implies x = f \frac{X}{Z}$$
$$\frac{y}{f} = \frac{Y}{Z} \implies y = f \frac{Y}{Z}$$

This is **perspective projection** (also called central projection). The depth Z appears in the denominator — objects farther away look smaller.

```

import numpy as np
import matplotlib.pyplot as plt

def project_pinhole(P, f=1.0):
    """Project 3D camera-frame points P (shape N×3) with focal length f.
    Returns image points shape (N, 2)."""
    X, Y, Z = P[:, 0], P[:, 1], P[:, 2]    # each shape (N,)
    x = f * X / Z                            # shape (N,)
    y = f * Y / Z                            # shape (N,)
    return np.column_stack([x, y])          # shape (N, 2)

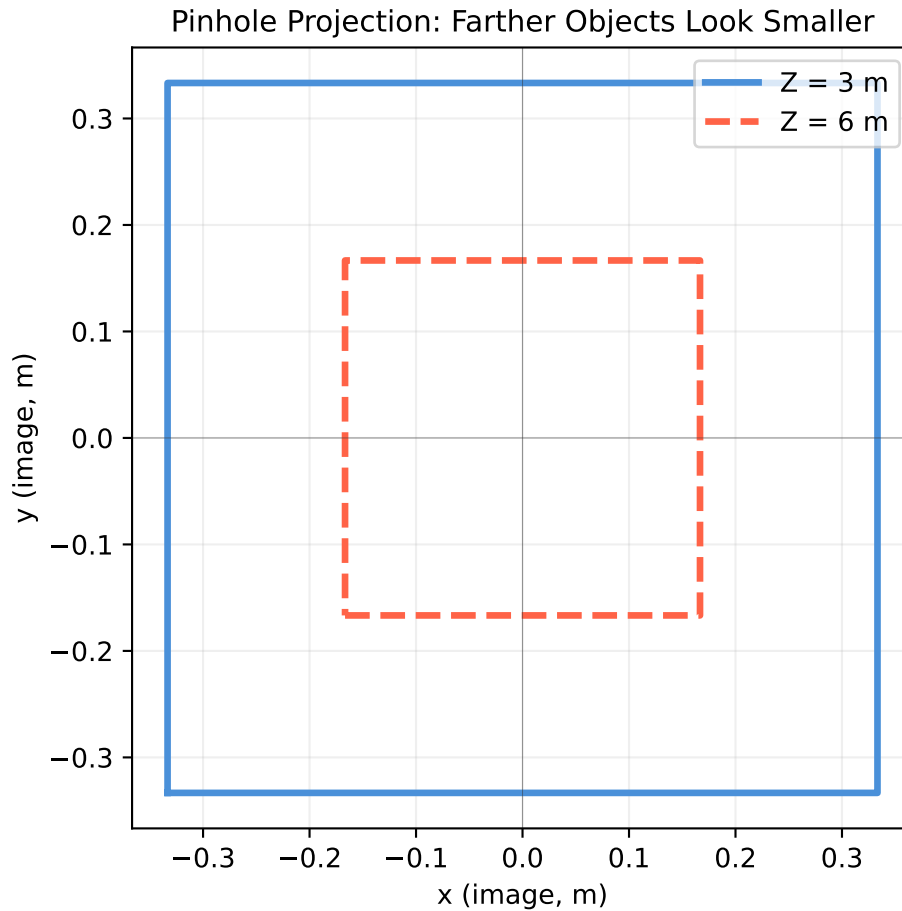
# A unit cube at Z = 3 m (camera space)
corners_3d = np.array([
    [-0.5, -0.5, 3.0],
    [ 0.5, -0.5, 3.0],
    [ 0.5,  0.5, 3.0],
    [-0.5,  0.5, 3.0],
    [-0.5, -0.5, 3.0], # close the loop
]) # shape (5, 3)

# Same cube at Z = 6 m (twice as far)
corners_far = corners_3d.copy()
corners_far[:, 2] = 6.0 # shape (5, 3)

proj_near = project_pinhole(corners_3d, f=2.0) # shape (5, 2)
proj_far  = project_pinhole(corners_far, f=2.0) # shape (5, 2)

fig, ax = plt.subplots(figsize=(5, 5))
ax.plot(proj_near[:, 0], proj_near[:, 1], color='#4a90d9', lw=2.5, label='Z = 3 m')
ax.plot(proj_far[:, 0], proj_far[:, 1], color='tomato', lw=2.5, label='Z = 6 m', ls='--')
ax.set_aspect('equal')
ax.set_xlabel('x (image, m)'); ax.set_ylabel('y (image, m)')
ax.set_title('Pinhole Projection: Farther Objects Look Smaller', fontsize=11)
ax.legend(); ax.grid(True, alpha=0.2)
ax.axhline(0, color='#333333', lw=0.4, alpha=0.5)
ax.axvline(0, color='#333333', lw=0.4, alpha=0.5)
plt.tight_layout()
plt.show()

```



Homogeneous Form

The division by Z makes projection nonlinear — inconvenient for algebra. Homogeneous coordinates fix that. In homogeneous form, a 2D image point (x, y) is represented as any scalar multiple of $[x, y, 1]^T$, and the projection becomes a **linear** matrix multiplication:

$$\lambda \begin{pmatrix} x \\ y \\ 1 \end{pmatrix} = \begin{pmatrix} f & 0 & 0 & 0 \\ 0 & f & 0 & 0 \\ 0 & 0 & 1 & 0 \end{pmatrix} \begin{pmatrix} X \\ Y \\ Z \\ 1 \end{pmatrix}$$

where $\lambda = Z$ is the depth (the scale factor that gets divided out).

```
import numpy as np
```

```
f = 2.0
```

```
# Simplified projection matrix (metric image plane, centre at origin)
P_simple = np.array([
    [f, 0., 0., 0.],
    [0., f, 0., 0.],
    [0., 0., 1., 0.],
]) # shape (3, 4)

P_world = np.array([1.5, -0.5, 4.0, 1.0]) # shape (4,) - homogeneous 3D point

p_h = P_simple @ P_world # shape (3,) - homogeneous image point
x, y = p_h[0] / p_h[2], p_h[1] / p_h[2] # divide by lambda=Z

print(f"3D point (camera frame): X={P_world[0]}, Y={P_world[1]}, Z={P_world[2]}")
print(f"Projected (homogeneous): {p_h}")
print(f"Image coordinates:      x={x:.4f}, y={y:.4f}")

# Verify against direct formula
print(f"\nDirect f*X/Z = {f*P_world[0]/P_world[2]:.4f} ")
print(f"Direct f*Y/Z = {f*P_world[1]/P_world[2]:.4f} ")
```

```
3D point (camera frame): X=1.5, Y=-0.5, Z=4.0
Projected (homogeneous): [ 3. -1.  4.]
Image coordinates:      x=0.7500, y=-0.2500

Direct f*X/Z = 0.7500
Direct f*Y/Z = -0.2500
```

Why Depth Information Is Lost

Projection is a many-to-one map: every point along a ray through **O** maps to the same image point. This is why a single image cannot recover depth — you need either stereo, structure from motion, or a depth sensor.

```
import numpy as np
import matplotlib.pyplot as plt

f = 1.5

# All these 3D points map to the same image point
Z_vals = np.array([1., 2., 4., 8.]) # shape (4,)
target_x, target_y = 0.5, 0.3 # fixed image point

# Back-project: 3D ray at each depth
```



```

X_vals = target_x / f * Z_vals    # shape (4,)
Y_vals = target_y / f * Z_vals    # shape (4,)

fig = plt.figure(figsize=(8, 4))
ax = fig.add_subplot(111, projection='3d')

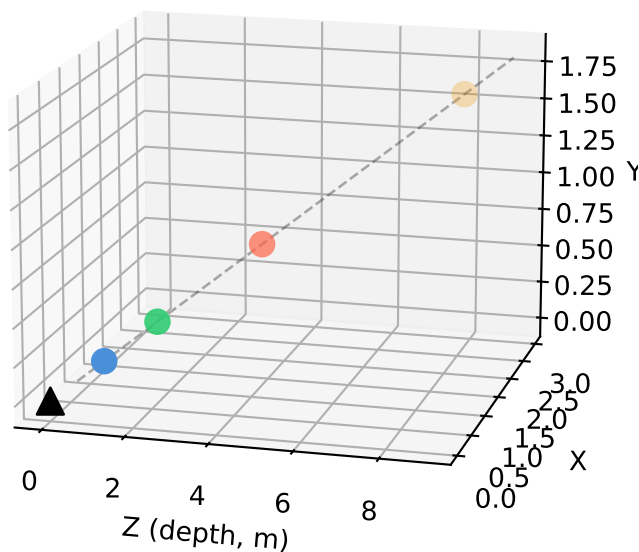
# Draw the projection ray
Z_ray = np.linspace(0.5, 9, 50)
X_ray = target_x / f * Z_ray
Y_ray = target_y / f * Z_ray
ax.plot(Z_ray, X_ray, Y_ray, color='#333333', lw=1, alpha=0.4, ls='--')

# Points along the ray
ax.scatter(Z_vals, X_vals, Y_vals, c=['#4a90d9', '#2ecc71', 'tomato', '#f39c12'],
          s=80, zorder=5)

ax.scatter([0], [0], [0], c='black', s=100, marker='^', label='Optical centre')
ax.set_xlabel('Z (depth, m)'); ax.set_ylabel('X'); ax.set_zlabel('Y')
ax.set_title('All points on this ray → same image pixel', fontsize=10)
ax.view_init(elev=15, azimuth=-75)
plt.tight_layout()
plt.show()

```

All points on this ray → same image pixel



Intrinsic Matrix K

The simple pinhole model from §10.1 assumed the image origin is at the optical axis and that pixels are square. Real sensors have a principal point offset, possibly non-square pixels, and occasionally a skew term. All of this is packed into the **intrinsic matrix** K — a 3×3 upper-triangular matrix that maps metric camera-frame rays to pixel coordinates.

$$K = \begin{pmatrix} f_x & s & c_x \\ 0 & f_y & c_y \\ 0 & 0 & 1 \end{pmatrix}$$

What Each Entry Means

Parameter	Meaning	Typical value (RealSense D435, 640×480)
f_x	Focal length in pixels along x	~615 px
f_y	Focal length in pixels along y	~615 px
s	Skew (non-rectangular pixels)	0 for modern sensors
c_x	Principal point x (where optical axis hits sensor)	~320 px
c_y	Principal point y	~240 px

Focal length in pixels = focal length in metres $\times$ sensor pixels per metre. A 4 mm lens on a sensor with 1500 px/mm gives $f_x \approx 6000$ px.

Applying K : Metric Image $\rightarrow$ Pixel

Given a camera-frame ray direction $(X/Z, Y/Z)$, K converts to pixel coordinates (u, v) :

$$\begin{pmatrix} u \\ v \\ 1 \end{pmatrix} \sim K \begin{pmatrix} X/Z \\ Y/Z \\ 1 \end{pmatrix} = \begin{pmatrix} f_x X/Z + s Y/Z + c_x \\ f_y Y/Z + c_y \\ 1 \end{pmatrix}$$

```
import numpy as np

K = np.array([
    [615., 0., 320.],
    [ 0., 615., 240.],
    [ 0., 0., 1.],
]) # shape (3, 3) - RealSense D435i approximate intrinsics
```

```
# A scene point at camera coordinates (0.3, -0.2, 1.5 m)
P_C = np.array([0.3, -0.2, 1.5]) # shape (3,)

# Project: K @ (P_C / P_C[2]) - divide by Z first to normalise
p_norm = P_C / P_C[2] # shape (3,) - normalised camera ray [X/Z, Y/Z, 1]
p_h = K @ p_norm # shape (3,) - homogeneous pixel
u, v = p_h[0], p_h[1] # already divided since p_norm[2]=1

print(f"Camera-frame point: {P_C}")
print(f"Normalised ray: {p_norm.round(6)}")
print(f"Pixel coordinates: u={u:.2f}, v={v:.2f}")
```

```
Camera-frame point: [ 0.3 -0.2  1.5]
Normalised ray: [ 0.2 -0.133333  1. ]
Pixel coordinates: u=443.00, v=158.00
```

Inverting K : Pixel $\rightarrow$ Camera-Frame Ray

Given a pixel (u, v) , K^{-1} recovers the normalised camera-frame direction — the **back-projection ray**. This is exactly what the RealSense SDK does when it lifts depth pixels to 3D (§9.7).

For upper-triangular K , the inverse is cheap and exact:

```
import numpy as np

K = np.array([
    [615., 0., 320.],
    [ 0., 615., 240.],
    [ 0., 0., 1.],
]) # shape (3, 3)

K_inv = np.linalg.inv(K) # shape (3, 3)
print("K_inv:")
print(K_inv.round(8))

# Pixel (400, 200) - back-project to unit ray
pixel_h = np.array([400., 200., 1.]) # shape (3,)
ray = K_inv @ pixel_h # shape (3,)
print(f"\nPixel (400, 200)  $\rightarrow$  ray: {ray.round(6)}")
print(f"Normalised: {(ray/ray[2]).round(6)}")
```

```
K_inv:
[[ 0.00162602  0. -0.5203252 ]
```

```
[ 0.          0.00162602 -0.3902439 ]
[ 0.          0.          1.          ]]
```

```
Pixel (400, 200) → ray: [ 0.130081 -0.065041  1.          ]
Normalised:           [ 0.130081 -0.065041  1.          ]
```

Visualising the Principal Point

The **principal point** (c_x, c_y) is where the optical axis pierces the image sensor. It's not necessarily at the pixel-coordinate origin — it's typically near but not exactly at the image centre.

```
import numpy as np
import matplotlib.pyplot as plt

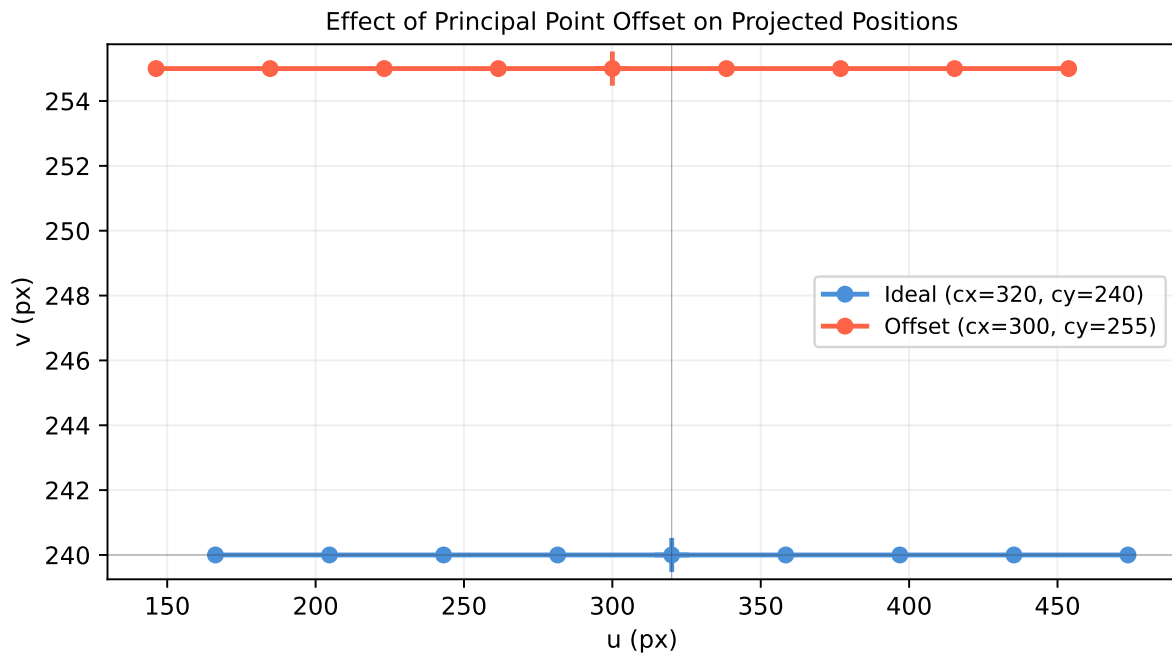
# Simulate two cameras with different principal points
Ks = {
    'Ideal (cx=320, cy=240)': np.array([[615., 0., 320.], [0., 615., 240.], [0., 0., 1.]]),
    'Offset (cx=300, cy=255)': np.array([[615., 0., 300.], [0., 615., 255.], [0., 0., 1.]]),
}

# Project a horizontal bar of points at Z=2 m
Xs = np.linspace(-0.5, 0.5, 9)
Ys = np.zeros(9)
Zs = np.full(9, 2.)
pts_C = np.column_stack([Xs, Ys, Zs]) # shape (9, 3)

fig, ax = plt.subplots(figsize=(7, 4))
colors = ['#4a90d9', 'tomato']
for (label, K), col in zip(Ks.items(), colors):
    uvs = []
    for P in pts_C:
        p_h = K @ (P / P[2])
        uvs.append(p_h[:2])
    uvs = np.array(uvs) # shape (9, 2)
    ax.plot(uvs[:, 0], uvs[:, 1], 'o-', color=col, lw=2, label=label)
    # Mark principal point
    ax.scatter([K[0,2]], [K[1,2]], marker='+', s=200, color=col, linewidths=2)

ax.set_xlabel('u (px)'); ax.set_ylabel('v (px)')
ax.set_title('Effect of Principal Point Offset on Projected Positions', fontsize=10)
ax.legend(fontsize=9)
ax.grid(True, alpha=0.2)
ax.axhline(240, color='#333333', lw=0.4, alpha=0.5)
ax.axvline(320, color='#333333', lw=0.4, alpha=0.5)
```

```
plt.tight_layout()
plt.show()
```



Skew

The skew term s is nonzero when the pixel axes are not perpendicular. This almost never happens with modern CMOS sensors ($s \approx 0$), but it can arise in scanned film or older CCD arrays.

```
import numpy as np
import matplotlib.pyplot as plt

# Compare s=0 vs s=30 (exaggerated)
K_noskew = np.array([[500., 0., 320.], [0., 500., 240.], [0., 0., 1.]])
K_skew   = np.array([[500., 30., 320.], [0., 500., 240.], [0., 0., 1.]])

# Project a square grid of points
grid = []
for x in np.linspace(-0.3, 0.3, 5):
    for y in np.linspace(-0.3, 0.3, 5):
        grid.append([x, y, 1.5])
grid = np.array(grid)  # shape (25, 3)

def project(pts, K):
```

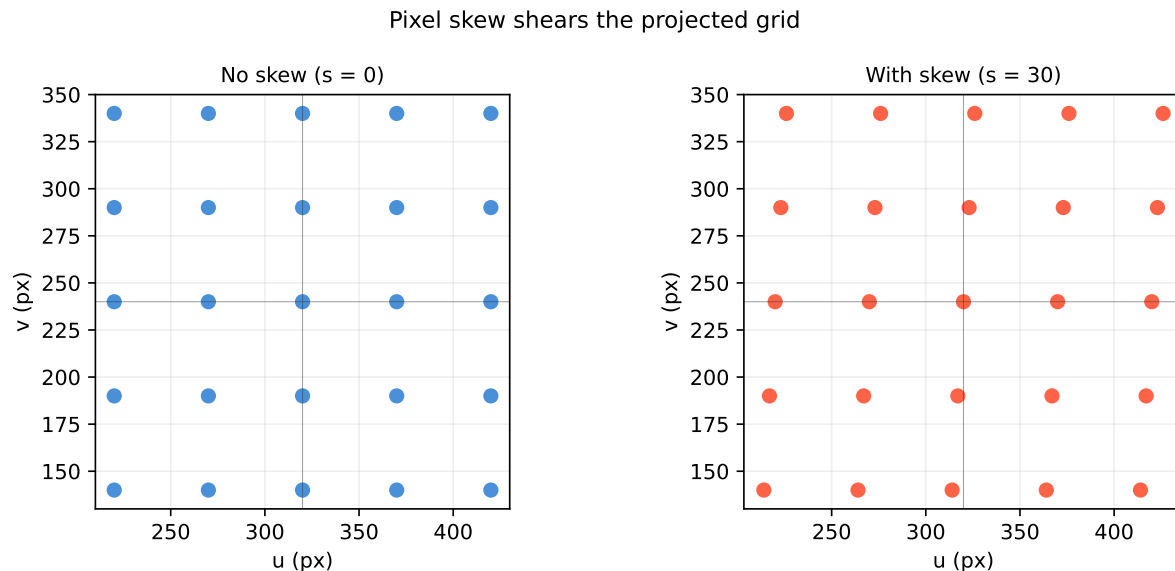
```

return np.array([K @ (p/p[2]) for p in pts])[:, :2] # shape (N, 2)

fig, axes = plt.subplots(1, 2, figsize=(9, 4))
for ax, K, title, col in [
    (axes[0], K_noskew, 'No skew (s = 0)', '#4a90d9'),
    (axes[1], K_skew, 'With skew (s = 30)', 'tomato')]:
    pts = project(grid, K)
    ax.scatter(pts[:, 0], pts[:, 1], color=col, s=40)
    ax.set_title(title, fontsize=10)
    ax.set_xlabel('u (px)'); ax.set_ylabel('v (px)')
    ax.set_aspect('equal'); ax.grid(True, alpha=0.2)
    ax.axhline(240, color='#333333', lw=0.4, alpha=0.5)
    ax.axvline(320, color='#333333', lw=0.4, alpha=0.5)

fig.suptitle('Pixel skew shears the projected grid', fontsize=11)
plt.tight_layout()
plt.show()

```



Extrinsic Matrix and Full Projection $P = K[R \mid \mathbf{t}]$

The intrinsic matrix K (§10.2) works in **camera coordinates**. The **extrinsic matrix** $[R \mid \mathbf{t}]$ maps a 3D world point to camera coordinates first. Together they form the full **projection matrix**:

$$P = K[R \mid \mathbf{t}] \in \mathbb{R}^{3 \times 4}$$

$$\lambda \begin{pmatrix} u \\ v \\ 1 \end{pmatrix} = K \begin{bmatrix} R & \mathbf{t} \end{bmatrix} \begin{pmatrix} X_w \\ Y_w \\ Z_w \\ 1 \end{pmatrix}$$

where (X_w, Y_w, Z_w) is a world-frame 3D point and (u, v) is its pixel-coordinate image.

What $[R \mid \mathbf{t}]$ Means

$[R \mid \mathbf{t}]$ is the first three rows of the 4×4 SE(3) matrix $T_{\text{camera} \leftarrow \text{world}}$:

- R — the camera's orientation relative to the world (3×3)
- $\mathbf{t}$ — the world origin expressed in camera coordinates (not the camera position in the world!)

Common confusion: The camera *position* in world coordinates is $\mathbf{C} = -R^T \mathbf{t}$.

```
import numpy as np

def Rz(g):
    c, s = np.cos(g), np.sin(g)
    return np.array([[c, -s, 0.], [s, c, 0.], [0., 0., 1.]])

# Camera is at world position (3, 0, 0), rotated 90° about z
# → in camera frame, world origin is at (-R @ t_world)
cam_pos_world = np.array([3., 0., 0.]) # shape (3,) - camera position in world
R = Rz(np.pi / 2) # shape (3, 3)

# t = -R @ cam_pos_world (standard convention)
t = -R @ cam_pos_world # shape (3,)

print(f"Camera position in world: {cam_pos_world}")
print(f"R (camera orientation): \n{R.round(4)}")
print(f"t (extrinsic translation): {t.round(4)}")

# Verify: camera centre C = -R^T t = cam_pos_world
C = -R.T @ t
print(f"\nRecovered camera centre: {C.round(6)}")
print(f"Matches cam_pos_world? {np.allclose(C, cam_pos_world)}")
```

```
Camera position in world: [3. 0. 0.]
R (camera orientation):
[[ 0. -1.  0.]
 [ 1.  0.  0.]
```

```
[ 0.  0.  1.]
t (extrinsic translation): [-0. -3.  0.]

Recovered camera centre:   [3. 0. 0.]
Matches cam_pos_world?    True
```

Full Projection Matrix

```
import numpy as np

def Rz(g):
    c, s = np.cos(g), np.sin(g)
    return np.array([[c,-s,0.],[s,c,0.],[0.,0.,1.]])

K = np.array([
    [615.,  0., 320.],
    [ 0., 615., 240.],
    [ 0.,  0.,  1.],
]) # shape (3, 3)

# Camera at (3, 0, 2) in world, looking along -x (rotated 90° about z, tilted)
cam_pos = np.array([3., 0., 2.]) # shape (3,)
R = Rz(np.pi / 2) # shape (3, 3)
t = -R @ cam_pos # shape (3,)

# Build [R | t]: 3x4 extrinsic
Rt = np.column_stack([R, t]) # shape (3, 4)

# Full projection matrix
P = K @ Rt # shape (3, 4)

print("Extrinsic [R|t]:")
print(Rt.round(4))
print("\nFull projection matrix P = K[R|t]:")
print(P.round(4))

# Project a world point
P_world = np.array([0., 0., 0., 1.]) # shape (4,) - world origin
p_h = P @ P_world # shape (3,)
u, v = p_h[0]/p_h[2], p_h[1]/p_h[2]
print(f"\nWorld origin projects to pixel: ({u:.1f}, {v:.1f})")
```

Extrinsic [R|t]:


```
[[ 0. -1.  0. -0.]
 [ 1.  0.  0. -3.]
 [ 0.  0.  1. -2.]]
```

Full projection matrix $P = K[R|\mathbf{t}]$:

```
[[ 0.000e+00 -6.150e+02  3.200e+02 -6.400e+02]
 [ 6.150e+02  0.000e+00  2.400e+02 -2.325e+03]
 [ 0.000e+00  0.000e+00  1.000e+00 -2.000e+00]]
```

World origin projects to pixel: (320.0, 1162.5)

Back-Projection: Pixel $\rightarrow$ World Ray

Given pixel (u, v) and projection matrix P , the **back-projection ray** passes through the camera centre $\mathbf{C}$ and the point:

$$\mathbf{d} = P^+[u, v, 1]^T$$

where $P^+ = P^T(PP^T)^{-1}$ is the pseudo-inverse of P .

```
import numpy as np

def Rz(g):
    c, s = np.cos(g), np.sin(g)
    return np.array([[c, -s, 0.], [s, c, 0.], [0., 0., 1.]])

K = np.array([[615., 0., 320.], [0., 615., 240.], [0., 0., 1.]]) # shape (3, 3)
R = Rz(np.pi/4)
cam_pos = np.array([2., 1., 0.])
t = -R @ cam_pos
Rt = np.column_stack([R, t])
P = K @ Rt # shape (3, 4)

# Camera centre: null space of P = -R^T t padded with 1
C_h = np.array([*(-R.T @ t), 1.]) # shape (4,)

# Pixel to back-project
pixel = np.array([400., 250., 1.]) # shape (3,)

# Point on ray (pseudo-inverse)
P_pinv = np.linalg.lstsq(P, pixel, rcond=None)[0] # shape (4,) - one point on ray
# (not unique: any lambda * P_pinv + (1-lambda) * C_h is on the ray)

# Direction vector in world frame
```

```
d = P_pinv[:3] / P_pinv[3] - C_h[:3]    # shape (3,)
d_hat = d / np.linalg.norm(d)

print(f"Camera centre: {C_h[:3].round(4)}")
print(f"Ray direction: {d_hat.round(6)}")
print(f"\nVerify: P @ point on ray pixel?")
check = P @ P_pinv
print(f"  P @ P pixel = {check.round(4)}")
print(f"  pixel      = {pixel}")
print(f"  Proportional? {np.allclose(check/check[2], pixel/pixel[2])}")
```

```
Camera centre: [2.  1.  0.]
Ray direction: [-0.102601  0.079801 -0.991516]

Verify: P @ point on ray pixel?
  P @ P pixel = [400. 250.  1.]
  pixel      = [400. 250.  1.]
  Proportional? True
```

Projecting Multiple World Points

```
import numpy as np
import matplotlib.pyplot as plt

def Ry(b):
    c,s=np.cos(b),np.sin(b)
    return np.array([[c,0.,s],[0.,1.,0.],[-s,0.,c]])

def Rz(g):
    c,s=np.cos(g),np.sin(g)
    return np.array([[c,-s,0.],[s,c,0.],[0.,0.,1.]])

K = np.array([[615.,0.,320.],[0.,615.,240.],[0.,0.,1.]])

# Camera: 5 m back, 1.5 m up, looking at origin
cam_pos = np.array([0., -1.5, 5.])
R = Ry(np.deg2rad(-15.))    # slight downward tilt
t = -R @ cam_pos
P = K @ np.column_stack([R, t])    # shape (3, 4)

# Unit cube corners in world frame
cube = np.array([
```

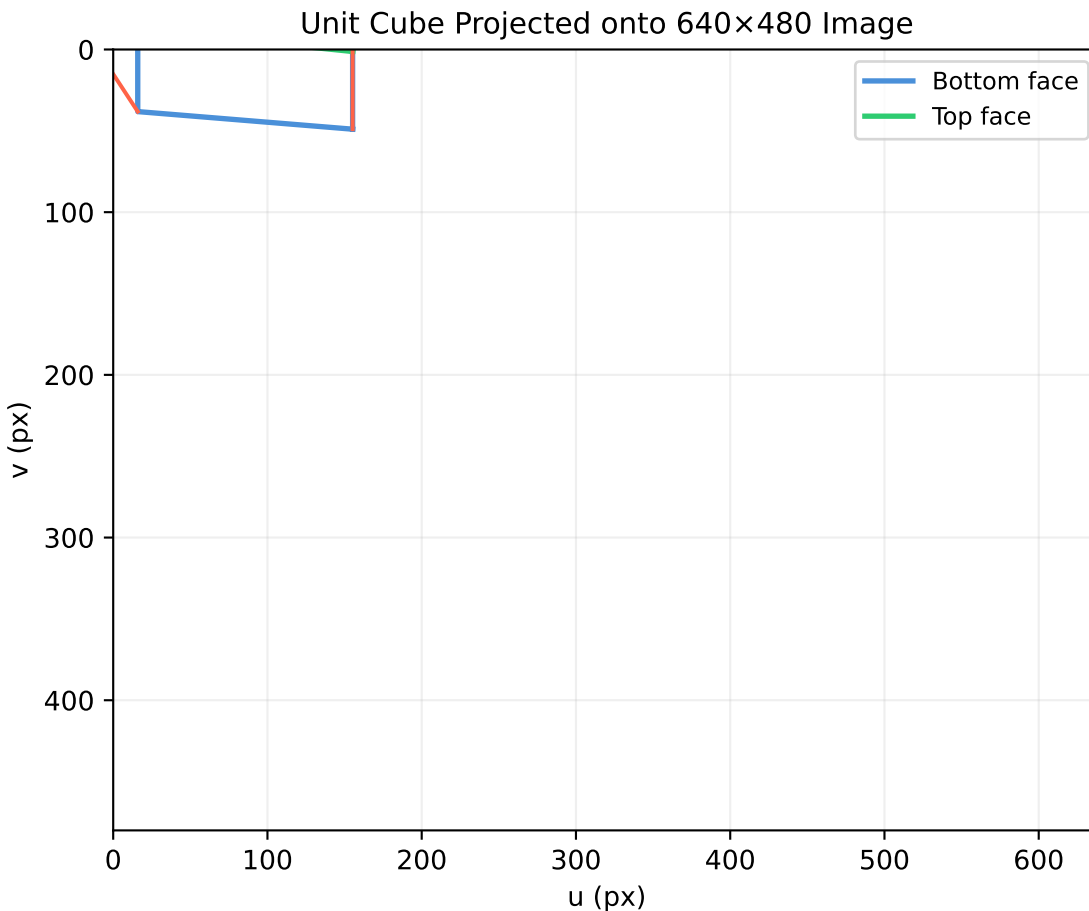
```

    [0,0,0],[1,0,0],[1,1,0],[0,1,0],[0,0,0], # bottom face
    [0,0,1],[1,0,1],[1,1,1],[0,1,1],[0,0,1], # top face
    [0,0,0],[0,0,1],[1,0,0],[1,0,1],          # verticals
    [1,1,0],[1,1,1],[0,1,0],[0,1,1]
], dtype=float) # shape (18, 3)
cube_h = np.column_stack([cube, np.ones(len(cube))]) # shape (18, 4)

proj = (P @ cube_h.T).T # shape (18, 3)
u = proj[:,0]/proj[:,2]; v = proj[:,1]/proj[:,2]

fig, ax = plt.subplots(figsize=(6, 5))
ax.plot(u[:5], v[:5], color='#4a90d9', lw=2, label='Bottom face')
ax.plot(u[5:10], v[5:10], color='#2ecc71', lw=2, label='Top face')
for i in range(10, len(u)-1, 2):
    ax.plot(u[i:i+2], v[i:i+2], color='tomato', lw=1.5)
ax.set_xlim(0, 640); ax.set_ylim(480, 0) # image coords: y down
ax.set_xlabel('u (px)'); ax.set_ylabel('v (px)')
ax.set_title('Unit Cube Projected onto 640x480 Image', fontsize=11)
ax.legend(fontsize=9); ax.grid(True, alpha=0.2)
plt.tight_layout()
plt.show()

```



Camera Calibration: Zhang's Method

Before you can use a camera for 3D geometry — measuring distances, aligning point clouds, or projecting bounding boxes — you need to know K . The standard approach is **Zhang's planar calibration method** (1999): photograph a flat checkerboard from multiple viewpoints. Each viewpoint provides linear constraints on K .

The Key Idea

A flat checkerboard is a **planar object** (all $Z_w = 0$). When $Z = 0$, the full projection equation collapses: the third column of $[R \mid \mathbf{t}]$ drops out, leaving a **homography** $H = K[r_1 \ r_2 \ \mathbf{t}]$ relating world (X_w, Y_w) to image (u, v) .

From two or more views you get enough constraints to solve for K . With at least 3 views, Zhang's method becomes a well-conditioned linear system.

```

import numpy as np

def Rz(g):
    c, s = np.cos(g), np.sin(g)
    return np.array([[c,-s,0.],[s,c,0.],[0.,0.,1.]])

def Rx(a):
    c, s = np.cos(a), np.sin(a)
    return np.array([[1.,0.,0.],[0.,c,-s],[0.,s,c]])

K_true = np.array([[600., 0., 310.],
                    [ 0.,600., 235.],
                    [ 0., 0., 1.]]) # shape (3, 3)

# Build a 5x5 checkerboard corner grid (world Z=0)
sq = 0.03 # 3 cm squares
corners_w = np.array([[i*sq, j*sq, 0., 1.]
                       for i in range(5) for j in range(5)],
                      dtype=float) # shape (25, 4)

# Simulate two views
views = [
    (Rx(np.deg2rad(-20.)) @ Rz(np.deg2rad(10.)), np.array([0.,0.,-0.8])),
    (Rx(np.deg2rad(-25.)) @ Rz(np.deg2rad(-15.)), np.array([0.05, 0.,-0.9])),
]

print("Simulated corner projections (first 3 corners, view 1):")
R0, t0 = views[0]
t0_ext = -R0 @ t0 # convert camera position → extrinsic t
Rt0 = np.column_stack([R0, t0_ext])
P0 = K_true @ Rt0
for corner in corners_w[:3]:
    p = P0 @ corner
    print(f" world {corner[:2]} → pixel ({p[0]/p[2]:.1f}, {p[1]/p[2]:.1f})")

```

```

Simulated corner projections (first 3 corners, view 1):
world [0. 0.] → pixel (310.0, 453.4)
world [0. 0.03] → pixel (305.8, 478.8)
world [0. 0.06] → pixel (301.5, 505.0)

```

Homography from One View

For a single planar view, the homography H can be estimated from 4 correspondences using DLT (detailed in §11.3). Here we use the known projection to recover H directly.

```
import numpy as np

def Rz(g):
    c,s=np.cos(g),np.sin(g); return np.array([[c,-s,0.],[s,c,0.],[0.,0.,1.]])
def Rx(a):
    c,s=np.cos(a),np.sin(a); return np.array([[1.,0.,0.],[0.,c,-s],[0.,s,c]])

K_true = np.array([[600.,0.,310.],[0.,600.,235.],[0.,0.,1.]]) # shape (3, 3)

# Single view: slight tilt
R = Rx(np.deg2rad(-20.)) @ Rz(np.deg2rad(5.))
cam_pos = np.array([0., 0., -0.8])
t = -R @ cam_pos

# Homography H = K [r1 r2 t] (drop the r3 column since Z_w=0)
r1 = R[:, 0]; r2 = R[:, 1]
H = K_true @ np.column_stack([r1, r2, t]) # shape (3, 3)
H = H / H[2, 2] # normalise so H[2,2]=1

print("Homography H (single checkerboard view):")
print(H.round(4))

# Verify: project a planar point through H vs full P
sq = 0.03
pt_w_2d = np.array([2*sq, 3*sq, 1.]) # shape (3,) - homogeneous 2D world point
p_via_H = H @ pt_w_2d # shape (3,)
u_H, v_H = p_via_H[0]/p_via_H[2], p_via_H[1]/p_via_H[2]

Rt = np.column_stack([R, t])
P = K_true @ Rt
pt_w_3d = np.array([2*sq, 3*sq, 0., 1.])
p_via_P = P @ pt_w_3d
u_P, v_P = p_via_P[0]/p_via_P[2], p_via_P[1]/p_via_P[2]

print(f"\nVia H: ({u_H:.4f}, {v_H:.4f})")
print(f"Via P: ({u_P:.4f}, {v_P:.4f})")
print(f"Match: {np.allclose([u_H,v_H],[u_P,v_P], atol=1e-4)}")
```

```
Homography H (single checkerboard view):
[[ 7.828039e+02 -2.100637e+02  3.100000e+02]
 [ 5.604840e+01  6.406366e+02  4.533821e+02]
 [-3.970000e-02 -4.532000e-01  1.000000e+00]]
```

```
Via H: (353.3151, 537.6110)
Via P: (353.3151, 537.6110)
Match: True
```

Extracting K from Multiple Homographies: The B Matrix

Each homography H_i from view i gives two linear constraints on the matrix $B = K^{-T}K^{-1}$ (a symmetric positive definite matrix). With $n \geq 3$ views you get $2n \geq 6$ constraints for the 6 independent entries of B , from which K is recovered by Cholesky decomposition.

Below we simulate the full pipeline:

```
import numpy as np

def Rz(g):
    c,s=np.cos(g),np.sin(g); return np.array([[c,-s,0.],[s,c,0.],[0.,0.,1.]])
def Rx(a):
    c,s=np.cos(a),np.sin(a); return np.array([[1.,0.,0.],[0.,c,-s],[0.,s,c]])

rng = np.random.default_rng(42)
K_true = np.array([[600.,0.,310.],[0.,600.,235.],[0.,0.,1.]]) # shape (3, 3)

# Grid of checkerboard corners (world Z=0)
sq = 0.03
Npts = 6
xs = np.arange(Npts)*sq; ys = np.arange(Npts)*sq
XX, YY = np.meshgrid(xs, ys)
corners_2d = np.column_stack([XX.ravel(), YY.ravel()]) # shape (36, 2)
corners_w = np.column_stack([corners_2d, np.ones(len(corners_2d))]) # shape (36, 3)

def get_H(K, R, cam_pos, noise_px=0.5):
    t = -R @ cam_pos
    r1, r2 = R[:,0], R[:,1]
    H_true = K @ np.column_stack([r1, r2, t])
    H_true = H_true / H_true[2,2]
    # Add pixel noise to simulate real measurements
    pts_w3d = np.column_stack([corners_2d, np.zeros(len(corners_2d)), np.ones(len(corners_2d))])
    Rt = np.column_stack([R, t]); P = K @ Rt
    proj = (P @ pts_w3d.T).T
    uvs = proj[:,2] / proj[:,2:3] + rng.normal(0, noise_px, (len(corners_2d), 2))
    # Re-estimate H from noisy correspondences via DLT
    A = []
    for (x,y), (u,v) in zip(corners_2d, uvs):
        A.append([ x, y, 1, 0, 0, 0, -u*x, -u*y, -u])
        A.append([ 0, 0, 0, x, y, 1, -v*x, -v*y, -v])
    A = np.array(A) # shape (72, 9)
    _, _, Vt = np.linalg.svd(A)
    h = Vt[-1, :] # shape (9,)
    return h.reshape(3,3) / h[-1] # shape (3, 3)
```

```

# 5 views
view_params = [
    (Rx(np.deg2rad(-20.)) @ Rz(np.deg2rad( 10.)), np.array([ 0.00, 0.00,-0.80])),
    (Rx(np.deg2rad(-25.)) @ Rz(np.deg2rad(-15.)), np.array([ 0.05, 0.00,-0.90])),
    (Rx(np.deg2rad(-15.)) @ Rz(np.deg2rad( 20.)), np.array([-0.05, 0.05,-0.85])),
    (Rx(np.deg2rad(-30.)) @ Rz(np.deg2rad( 0.)), np.array([ 0.00,-0.05,-0.95])),
    (Rx(np.deg2rad(-18.)) @ Rz(np.deg2rad(-10.)), np.array([ 0.03, 0.03,-0.88])),
]

Hs = [get_H(K_true, R, cp) for R, cp in view_params]    # list of (3,3)

def vij(H, i, j):
    """Helper vector for Zhang's constraint on B."""
    h = H.T    # rows of h are columns of H
    return np.array([
        h[i,0]*h[j,0],
        h[i,0]*h[j,1] + h[i,1]*h[j,0],
        h[i,1]*h[j,1],
        h[i,2]*h[j,0] + h[i,0]*h[j,2],
        h[i,2]*h[j,1] + h[i,1]*h[j,2],
        h[i,2]*h[j,2]
    ])    # shape (6,)

# Build system Vb = 0
V = []
for H in Hs:
    V.append(vij(H, 0, 1))
    V.append(vij(H, 0, 0) - vij(H, 1, 1))
V = np.array(V)    # shape (10, 6)

_, _, Vt = np.linalg.svd(V)
b = Vt[-1, :]    # shape (6,) - B = K^{-T} K^{-1} in packed form

# Reconstruct K from b
B = np.array([[b[0],b[1],b[3]],
               [b[1],b[2],b[4]],
               [b[3],b[4],b[5]]])    # shape (3, 3)

# Cholesky of B^{-1} gives K (up to sign)
try:
    L = np.linalg.cholesky(np.linalg.inv(B))
    K_est = L.T / L.T[2,2]    # shape (3, 3) - normalise
    print("Estimated K:")
    print(K_est.round(2))
    print("\nTrue K:")
    print(K_true)
    print(f"\nError in fx: {abs(K_est[0,0]-K_true[0,0]):.2f} px")

```



```

    print(f"Error in cx: {abs(K_est[0,2]-K_true[0,2]):.2f} px")
except Exception as e:
    print(f>Note: {e}")
    print("(Noise can make B slightly non-PD; real pipelines add regularisation)")

```

Note: Matrix is not positive definite
 (Noise can make B slightly non-PD; real pipelines add regularisation)

What cv2.calibrateCamera Does

OpenCV's calibration function implements an extended version of Zhang's method with nonlinear refinement (Levenberg–Marquardt) to minimise **reprojection error** — the RMS pixel distance between detected and re-projected corners.

```

import cv2
import numpy as np
import glob

# Assumes checkerboard images in ./calib_images/
pattern = (9, 6) # interior corners
sq_size = 0.025 # 25 mm squares

objp = np.zeros((pattern[0]*pattern[1], 3), np.float32)
objp[:, :2] = np.mgrid[0:pattern[0], 0:pattern[1]].T.reshape(-1, 2)
objp *= sq_size

obj_pts, img_pts = [], []
for fname in sorted(glob.glob('./calib_images/*.png')):
    img = cv2.imread(fname)
    gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)
    ret, corners = cv2.findChessboardCorners(gray, pattern)
    if ret:
        obj_pts.append(objp)
        corners_refined = cv2.cornerSubPix(
            gray, corners, (11,11), (-1,-1),
            (cv2.TERM_CRITERIA_EPS + cv2.TERM_CRITERIA_MAX_ITER, 30, 0.001))
        img_pts.append(corners_refined)

ret, K, dist, rvecs, tvecs = cv2.calibrateCamera(
    obj_pts, img_pts, gray.shape[::-1], None, None)

print(f"RMS reprojection error: {ret:.4f} px")
print(f"K =\n{K.round(2)}")
print(f"Distortion coefficients: {dist.ravel().round(6)}")

```

Lens Distortion

The pinhole model is ideal — real lenses bend light. The most significant deviations are **radial distortion** (barrel or pincushion) and **tangential distortion** (lens tilt relative to sensor). Both must be corrected before using the linear camera model.

Radial Distortion

Radial distortion moves pixels toward or away from the image centre as a function of radial distance $r = \sqrt{x_n^2 + y_n^2}$ from the principal point (in normalised coordinates):

$$x_d = x_n(1 + k_1 r^2 + k_2 r^4 + k_3 r^6)$$

$$y_d = y_n(1 + k_1 r^2 + k_2 r^4 + k_3 r^6)$$

- $k_1 < 0$: **barrel** distortion (wide-angle lenses) — edges bow outward
- $k_1 > 0$: **pincushion** distortion (telephoto) — edges bow inward

```
import numpy as np
import matplotlib.pyplot as plt

def distort_radial(x_n, y_n, k1, k2=0., k3=0.):
    """Apply radial distortion to normalised coords.
    x_n, y_n: shape (N,). Returns (x_d, y_d) shape (N,)."""
    r2 = x_n**2 + y_n**2 # shape (N,)
    factor = 1. + k1*r2 + k2*r2**2 + k3*r2**3 # shape (N,)
    return x_n * factor, y_n * factor # each shape (N,)

# Grid of normalised coordinates
lin = np.linspace(-0.8, 0.8, 12)
X, Y = np.meshgrid(lin, lin)
xn, yn = X.ravel(), Y.ravel() # each shape (144,)

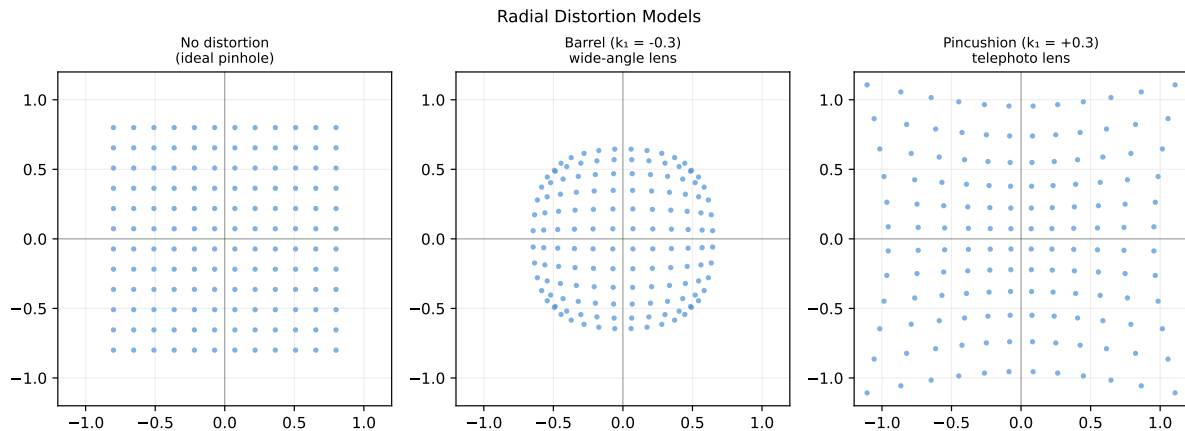
fig, axes = plt.subplots(1, 3, figsize=(11, 4))
configs = [
    (0., 'No distortion\n(ideal pinhole)'),
    (-0.3, 'Barrel (k = -0.3)\nwide-angle lens'),
    (0.3, 'Pincushion (k = +0.3)\ntelephoto lens'),
]
for ax, (k1, title) in zip(axes, configs):
    xd, yd = distort_radial(xn, yn, k1)
    ax.scatter(xd, yd, s=6, color='#4a90d9', alpha=0.7)
    ax.set_xlim(-1.2, 1.2); ax.set_ylim(-1.2, 1.2)
    ax.set_aspect('equal')
```

```

ax.set_title(title, fontsize=9)
ax.grid(True, alpha=0.2)
ax.axhline(0, color='#333333', lw=0.4, alpha=0.5)
ax.axvline(0, color='#333333', lw=0.4, alpha=0.5)

fig.suptitle('Radial Distortion Models', fontsize=11)
plt.tight_layout()
plt.show()

```



Tangential Distortion

Tangential distortion arises when the lens is not perfectly parallel to the sensor plane. The correction uses two additional parameters p_1, p_2 :

$$x_d = x_n + 2p_1x_ny_n + p_2(r^2 + 2x_n^2)$$

$$y_d = y_n + p_1(r^2 + 2y_n^2) + 2p_2x_ny_n$$

```

import numpy as np
import matplotlib.pyplot as plt

def distort_full(x_n, y_n, k1=0., k2=0., k3=0., p1=0., p2=0.):
    """Full Brown-Conrady distortion model."""
    r2 = x_n**2 + y_n**2 # shape (N,)
    radial = 1. + k1*r2 + k2*r2**2 + k3*r2**3 # shape (N,)
    x_d = x_n*radial + 2*p1*x_n*y_n + p2*(r2 + 2*x_n**2)
    y_d = y_n*radial + p1*(r2 + 2*y_n**2) + 2*p2*x_n*y_n
    return x_d, y_d # each shape (N,)

lin = np.linspace(-0.8, 0.8, 12)

```

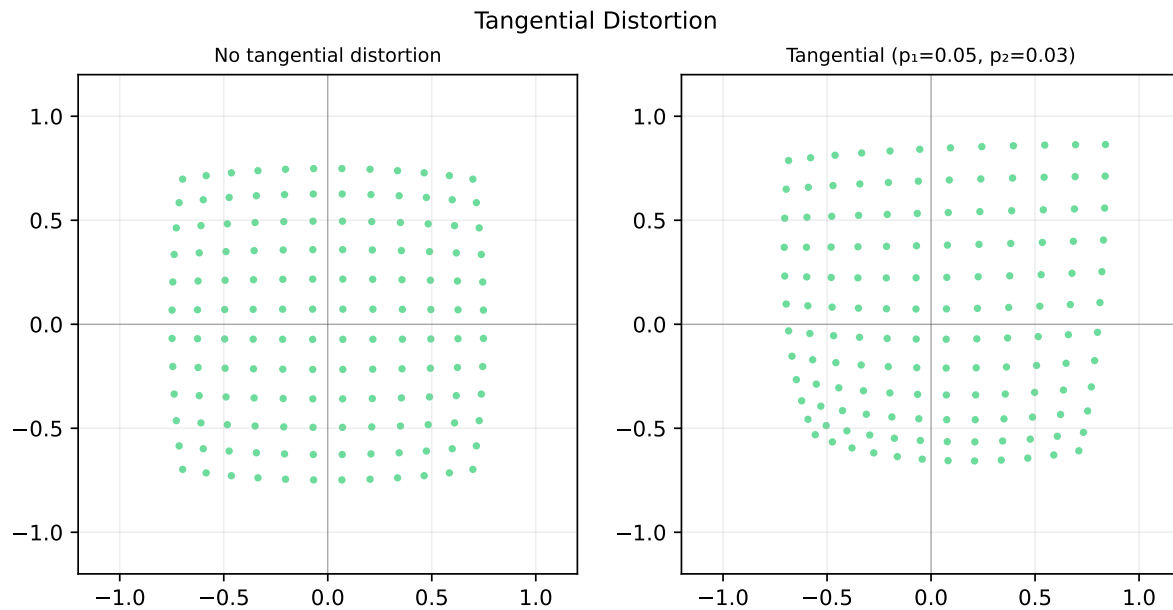
```

X, Y = np.meshgrid(lin, lin)
xn, yn = X.ravel(), Y.ravel()

fig, axes = plt.subplots(1, 2, figsize=(8, 4))
for ax, (p1, p2, title) in zip(axes, [
    (0., 0., 'No tangential distortion'),
    (0.05, 0.03, 'Tangential (p=0.05, p=0.03)'),
]):
    xd, yd = distort_full(xn, yn, k1=-0.1, p1=p1, p2=p2)
    ax.scatter(xd, yd, s=6, color='#2ecc71', alpha=0.7)
    ax.set_xlim(-1.2, 1.2); ax.set_ylim(-1.2, 1.2)
    ax.set_aspect('equal')
    ax.set_title(title, fontsize=9)
    ax.grid(True, alpha=0.2)
    ax.axhline(0, color='#333333', lw=0.4, alpha=0.5)
    ax.axvline(0, color='#333333', lw=0.4, alpha=0.5)

fig.suptitle('Tangential Distortion', fontsize=11)
plt.tight_layout()
plt.show()

```



Undistortion: Mapping Pixels Back to Normalised Coordinates

The distortion model maps **normalised** → **distorted**. To go the other way (pixel → undistorted normalised coordinate), you solve a nonlinear equation numerically. In practice, OpenCV builds a precomputed **undistortion map**.

```

import numpy as np
import matplotlib.pyplot as plt

def distort_full(x_n, y_n, k1, k2=0., p1=0., p2=0.):
    r2 = x_n**2 + y_n**2
    radial = 1. + k1*r2 + k2*r2**2
    xd = x_n*radial + 2*p1*x_n*y_n + p2*(r2+2*x_n**2)
    yd = y_n*radial + p1*(r2+2*y_n**2) + 2*p2*x_n*y_n
    return xd, yd

def undistort_iterative(xd, yd, k1, k2=0., p1=0., p2=0., iters=10):
    """Iterative undistortion (initial guess = distorted point)."""
    xn, yn = xd.copy(), yd.copy()
    for _ in range(iters):
        r2 = xn**2 + yn**2
        radial = 1. + k1*r2 + k2*r2**2
        dx = 2*p1*xn*yn + p2*(r2+2*xn**2)
        dy = p1*(r2+2*yn**2) + 2*p2*xn*yn
        xn = (xd - dx) / radial
        yn = (yd - dy) / radial
    return xn, yn

K = np.array([[600., 0., 320.], [0., 600., 240.], [0., 0., 1.]])
k1, k2 = -0.25, 0.08

# Original pixel grid
us = np.linspace(50, 590, 8); vs = np.linspace(50, 430, 8)
UU, VV = np.meshgrid(us, vs)
uu, vv = UU.ravel(), VV.ravel()

# Normalised coords from pixels
xn0 = (uu - K[0,2]) / K[0,0] # shape (64,)
yn0 = (vv - K[1,2]) / K[1,1] # shape (64,)

# Apply distortion
xd, yd = distort_full(xn0, yn0, k1, k2)

# Back to pixel coords (distorted)
ud = xd * K[0,0] + K[0,2] # shape (64,)
vd = yd * K[1,1] + K[1,2] # shape (64,)

# Undistort
xn_rec, yn_rec = undistort_iterative(xd, yd, k1, k2)
uu_rec = xn_rec * K[0,0] + K[0,2]
vv_rec = yn_rec * K[1,1] + K[1,2]

fig, axes = plt.subplots(1, 3, figsize=(11, 4))

```

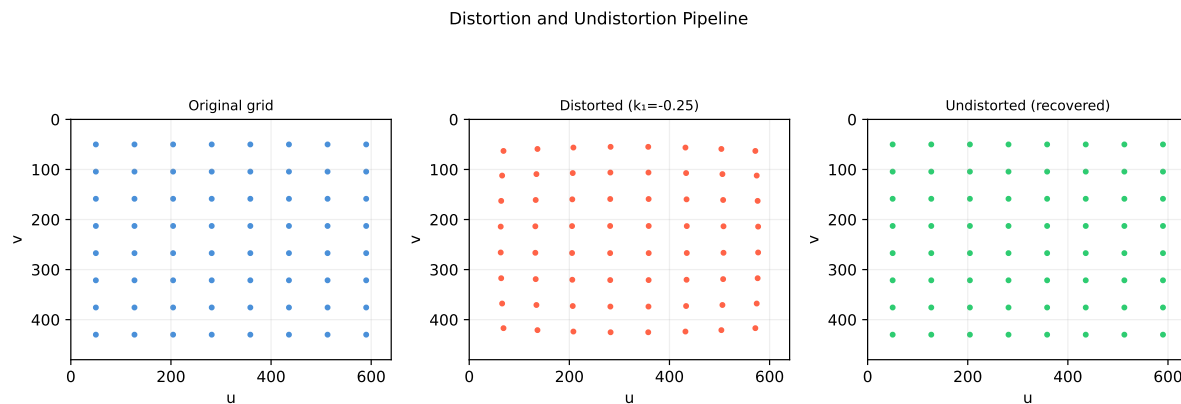
```

for ax, u_plot, v_plot, title, col in [
    (axes[0], uu, vv, 'Original grid', '#4a90d9'),
    (axes[1], ud, vd, f'Distorted (k={k1})', 'tomato'),
    (axes[2], uu_rec, vv_rec, 'Undistorted (recovered)', '#2ecc71'),
]:
    ax.scatter(u_plot, v_plot, s=8, color=col)
    ax.set_xlim(0,640); ax.set_ylim(480,0)
    ax.set_aspect('equal'); ax.set_title(title, fontsize=9)
    ax.set_xlabel('u'); ax.set_ylabel('v')
    ax.grid(True, alpha=0.2)

print(f"Max undistortion error: {np.max(np.sqrt((uu_rec-uu)**2+(vv_rec-vv)**2)):.4f} px")
fig.suptitle('Distortion and Undistortion Pipeline', fontsize=11)
plt.tight_layout()
plt.show()

```

Max undistortion error: 0.0000 px



When to Correct Distortion (and When Not To)

Scenario	Correct distortion?
Feature matching + homography	Yes — DLT assumes linear projection
Deep learning detector on raw images	Usually no — the network learns it
Photogrammetry / 3D reconstruction	Yes — metric accuracy requires it
Fisheye / wide-angle (FOV > 120°)	Use fisheye model (θ -based), not polynomial
After <code>cv2.calibrateCamera</code>	Use <code>cv2.undistort()</code> once, then treat as pinhole

For a RealSense D435i at 640×480 , typical values are $k_1 \approx -0.06$, $k_2 \approx 0.07$, $p_1, p_2 \approx 0$. The maximum pixel displacement from radial distortion is only ~ 2 px at the corners — negligible for many applications, critical for sub-pixel accuracy.

Application: Projecting a 3D Bounding Box onto an Image

A common task in robotics and AR: given a detected object's 3D pose and a calibrated camera, render a 3D bounding box wireframe over the image. This combines everything from this chapter — the full projection matrix $P = K[R \mid \mathbf{t}]$ — into one pipeline.

The Pipeline

1. Define the 3D bounding box corners in **object frame**.
 2. Transform corners to **world frame** using the object's known pose T_{WO} .
 3. Project to **pixel frame** using $P = K[R \mid \mathbf{t}]$.
 4. Draw the box wireframe on the image.
-

Step 1–3: Projecting the 3D Box

```
import numpy as np
import matplotlib.pyplot as plt
import matplotlib.patches as mpatches

def make_T(R, t):
    T = np.eye(4); T[:3,:3] = R; T[:3,3] = t
    return T    # shape (4, 4)

def Ry(b):
    c,s=np.cos(b),np.sin(b)
    return np.array([[c,0.,s],[0.,1.,0.],[-s,0.,c]])

def Rx(a):
    c,s=np.cos(a),np.sin(a)
    return np.array([[1.,0.,0.],[0.,c,-s],[0.,s,c]])

# Camera intrinsics (RealSense D435i approximate)
K = np.array([[615., 0., 320.],
              [ 0., 615., 240.],
              [ 0., 0., 1.]])    # shape (3, 3)
```

```
# Camera extrinsic: camera at (0, -0.5, 2) in world, looking forward+down
cam_pos = np.array([0., -0.5, 2.0])
R_cam = Rx(np.deg2rad(-15.))
t_ext = -R_cam @ cam_pos
Rt = np.column_stack([R_cam, t_ext])    # shape (3, 4)
P = K @ Rt                             # shape (3, 4) - full projection matrix

# Bounding box in object frame (a garlic bulb: roughly 8cm × 8cm × 6cm)
lx, ly, lz = 0.08, 0.08, 0.06    # half-sizes in metres
corners_obj = np.array([
    [-lx,-ly,-lz], [ lx,-ly,-lz], [ lx, ly,-lz], [-lx, ly,-lz], # bottom
    [-lx,-ly, lz], [ lx,-ly, lz], [ lx, ly, lz], [-lx, ly, lz], # top
], dtype=float)    # shape (8, 3)

# Object pose in world: centre at (0.1, 0, 0), rotated 20° about y
T_WO = make_T(Ry(np.deg2rad(20.)), np.array([0.1, 0., 0.]))    # shape (4, 4)

# Transform box corners: object → world → image
corners_w_h = (T_WO @ np.column_stack([corners_obj, np.ones(8)]).T).T    # shape (8, 4)
proj = (P @ corners_w_h.T).T    # shape (8, 3)
uvs = proj[:, :2] / proj[:, 2:3]    # shape (8, 2)

# Box edges: bottom face, top face, vertical pillars
edges = [(0,1),(1,2),(2,3),(3,0),    # bottom
          (4,5),(5,6),(6,7),(7,4),    # top
          (0,4),(1,5),(2,6),(3,7)]    # pillars

# Simulated background image (grey gradient)
img = np.ones((480, 640, 3)) * 0.85    # light grey
img[200:280, 280:360] = [0.7, 0.68, 0.65]    # rough garlic blob

fig, ax = plt.subplots(figsize=(7, 5.5))
ax.imshow(img, origin='upper', extent=[0, 640, 480, 0])

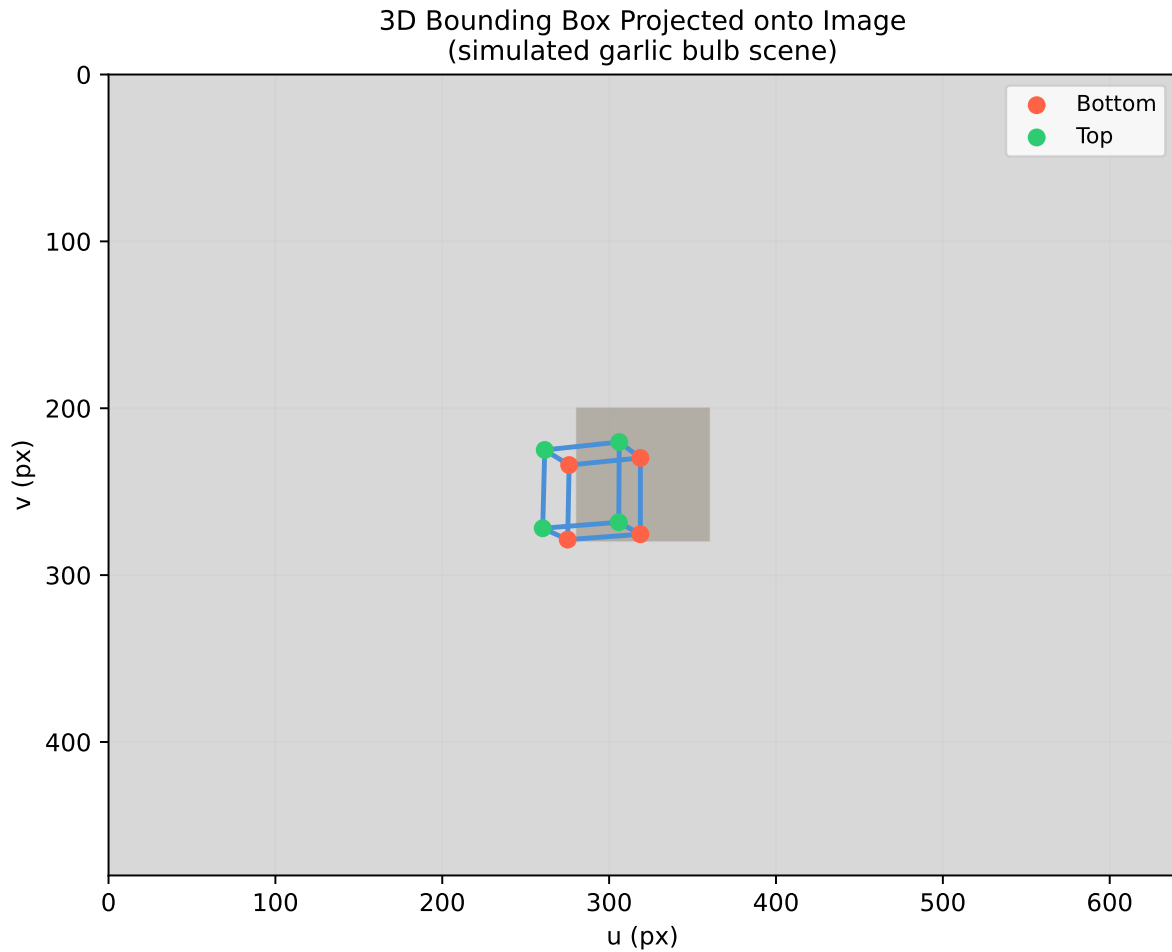
for i, j in edges:
    ax.plot([uvs[i,0], uvs[j,0]], [uvs[i,1], uvs[j,1]],
            color='#4a90d9', lw=2.)

ax.scatter(uvs[:4, 0], uvs[:4, 1], color='tomato', s=40, zorder=5, label='Bottom')
ax.scatter(uvs[4:, 0], uvs[4:, 1], color='#2ecc71', s=40, zorder=5, label='Top')

ax.set_xlim(0, 640); ax.set_ylim(480, 0)
ax.set_title('3D Bounding Box Projected onto Image\n(simulated garlic bulb scene)', fontsize=12)
ax.set_xlabel('u (px)'); ax.set_ylabel('v (px)')
ax.legend(fontsize=9, loc='upper right')
ax.grid(True, alpha=0.1)
plt.tight_layout()
```



```
plt.show()
```



Sensitivity to K Errors

Even small errors in K shift the projected box. Here we compare the projection with the true K vs a K with +5% error in f_x .

```
import numpy as np
import matplotlib.pyplot as plt

def Rx(a):
    c,s=np.cos(a),np.sin(a); return np.array([[1.,0.,0.],[0.,c,-s],[0.,s,c]])
def Ry(b):
    c,s=np.cos(b),np.sin(b); return np.array([[c,0.,s],[0.,1.,0.],[-s,0.,c]])
def make_T(R, t):
    T=np.eye(4); T[:3,:3]=R; T[:3,3]=t; return T
```

```

K_true = np.array([[615.,0.,320.],[0.,615.,240.],[0.,0.,1.]])
K_wrong = np.array([[646.,0.,320.],[0.,615.,240.],[0.,0.,1.]]) # +5% fx error

cam_pos = np.array([0.,-0.5,2.]); R_c = Rx(np.deg2rad(-15.))
t_ext = -R_c @ cam_pos
Rt = np.column_stack([R_c, t_ext])

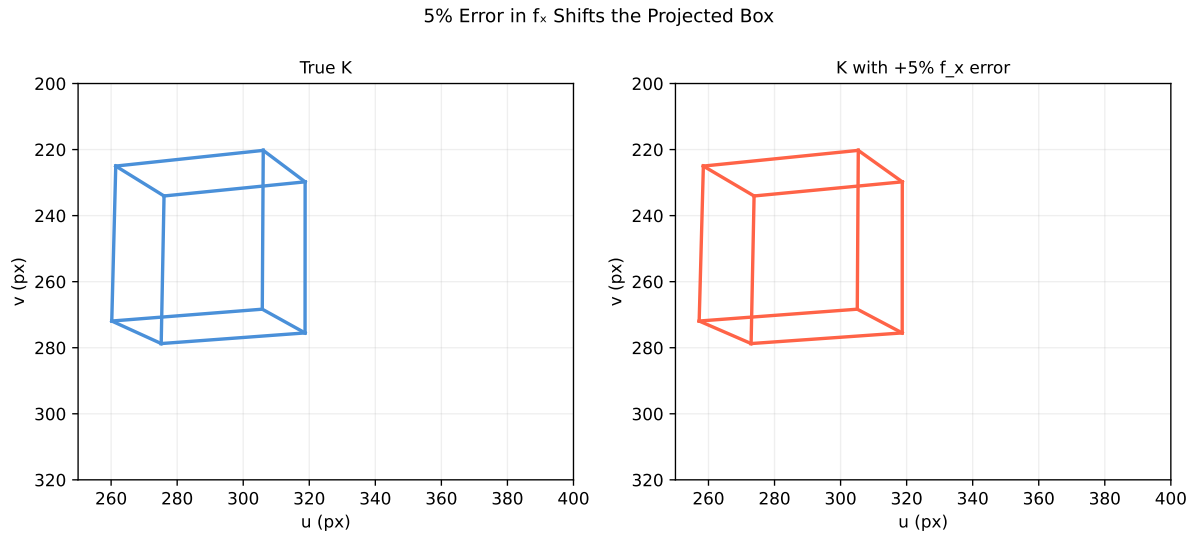
lx,ly,lz = 0.08, 0.08, 0.06
corners_obj = np.array([[-lx,-ly,-lz],[lx,-ly,-lz],[lx,ly,-lz],[-lx,ly,-lz],
                        [-lx,-ly,lz],[lx,-ly,lz],[lx,ly,lz],[-lx,ly,lz]])
T_W0 = make_T(Ry(np.deg2rad(20.)), np.array([0.1,0.,0.]))
corners_w_h = (T_W0 @ np.column_stack([corners_obj,np.ones(8)]).T).T

edges = [(0,1),(1,2),(2,3),(3,0),(4,5),(5,6),(6,7),(7,4),(0,4),(1,5),(2,6),(3,7)]

fig, axes = plt.subplots(1, 2, figsize=(10, 4.5))
for ax, K, title, col in [
    (axes[0], K_true, 'True K', '#4a90d9'),
    (axes[1], K_wrong, 'K with +5% f_x error', 'tomato'),
]:
    P = K @ Rt
    proj = (P @ corners_w_h.T).T
    uvs = proj[:, :2] / proj[:, 2:3]
    for i, j in edges:
        ax.plot([uvs[i,0],uvs[j,0]], [uvs[i,1],uvs[j,1]], color=col, lw=2.)
    ax.set_xlim(250,400); ax.set_ylim(320,200)
    ax.set_title(title, fontsize=10)
    ax.set_xlabel('u (px)'); ax.set_ylabel('v (px)')
    ax.grid(True, alpha=0.2)
    ax.set_aspect('equal')

fig.suptitle('5% Error in f Shifts the Projected Box', fontsize=11)
plt.tight_layout()
plt.show()

```



Kaggle Exploration

Dataset: [Open Images V7](#) — contains images with 3D bounding box annotations and camera intrinsics for some categories, or use [KITTI 3D Object Detection Benchmark](#) which provides calibrated camera matrices and 3D bounding box annotations.

```
# KITTI 3D Object Detection - projecting ground-truth 3D boxes
# Dataset layout: image_2/, calib/, label_2/

import numpy as np
import matplotlib.pyplot as plt
from PIL import Image
import os

def read_calib(calib_file):
    """Parse KITTI calibration file. Returns P2 (3x4) and R0_rect (3x3)."""
    with open(calib_file) as f:
        lines = f.readlines()
    P2 = np.array([float(v) for v in lines[2].split()[1:]]).reshape(3,4)
    R0 = np.array([float(v) for v in lines[4].split()[1:]]).reshape(3,3)
    return P2, R0    # shape (3,4), (3,3)

def read_labels(label_file):
    """Parse KITTI label file. Returns list of (type, h,w,l, x,y,z, ry)."""
    boxes = []
    with open(label_file) as f:
        for line in f:
            parts = line.split()
```

```

        if parts[0] in ('Car','Pedestrian','Cyclist'):
            h,w,l = float(parts[8]),float(parts[9]),float(parts[10])
            x,y,z = float(parts[11]),float(parts[12]),float(parts[13])
            ry    = float(parts[14])
            boxes.append((parts[0], h, w, l, x, y, z, ry))
    return boxes

def box3d_corners(h, w, l, x, y, z, ry):
    """8 corners of a 3D box in camera frame."""
    c, s = np.cos(ry), np.sin(ry)
    R = np.array([[c,0,s],[0,1,0],[-s,0,c]])
    corners_local = np.array([
        [ l/2, 0,  w/2],[-l/2, 0,  w/2],[-l/2, 0, -w/2],[ l/2, 0, -w/2],
        [ l/2,-h,  w/2],[-l/2,-h,  w/2],[-l/2,-h, -w/2],[ l/2,-h, -w/2],
    ]) # shape (8, 3)
    return (R @ corners_local.T).T + np.array([x, y, z]) # shape (8, 3)

frame_id = '000010'
img = np.array(Image.open(f'image_2/{frame_id}.png'))
P2, R0 = read_calib(f'calib/{frame_id}.txt')
boxes = read_labels(f'label_2/{frame_id}.txt')

edges = [(0,1),(1,2),(2,3),(3,0),(4,5),(5,6),(6,7),(7,4),(0,4),(1,5),(2,6),(3,7)]
fig, ax = plt.subplots(figsize=(12,4))
ax.imshow(img)
for obj_type, h,w,l, x,y,z, ry in boxes:
    corners = box3d_corners(h,w,l,x,y,z,ry) # shape (8, 3)
    corners_h = np.column_stack([corners, np.ones(8)]) # shape (8, 4)
    # Apply R0_rect then project with P2
    corners_rect = (R0 @ corners.T).T
    corners_rect_h = np.column_stack([corners_rect, np.ones(8)])
    proj = (P2 @ corners_rect_h.T).T
    uvs = proj[:, :2] / proj[:, 2:3]
    col = {'Car':'#4a90d9','Pedestrian':'tomato','Cyclist':'#2ecc71'}.get(obj_type,'grey')
    for i,j in edges:
        ax.plot([uvs[i,0],uvs[j,0]],[uvs[i,1],uvs[j,1]], color=col, lw=1.5)
ax.axis('off')
ax.set_title(f'KITTI Frame {frame_id} - 3D Box Projections', fontsize=11)
plt.tight_layout(); plt.show()

```

Exercises and Solutions

Exercise 10.1 — Similar Triangles Derivation

(a) Draw a side-view diagram of a pinhole camera with focal length f . A scene point is at distance Z from the optical centre at height Y . Using similar triangles, derive that $y = fY/Z$.

(b) Verify numerically: project points (1, 2, 4), (2, 4, 8), and (0.5, 1, 2) with $f = 3$. What do you notice?

```
import numpy as np

f = 3.0

points = np.array([
    [1., 2., 4.],
    [2., 4., 8.],
    [0.5, 1., 2.],
]) # shape (3, 3)

print(f"{'X':>6} {'Y':>6} {'Z':>6} {'x=fX/Z':>10} {'y=fY/Z':>10}")
for X, Y, Z in points:
    x = f * X / Z
    y = f * Y / Z
    print(f"{'X':>6.1f} {'Y':>6.1f} {'Z':>6.1f} {'x':>10.4f} {'y':>10.4f}")

print("\nObservation: all three points lie on the same ray through the origin")
print("(X/Z, Y/Z identical) → they project to the same image point.")
```

X	Y	Z	$x=fX/Z$	$y=fY/Z$
1.0	2.0	4.0	0.7500	1.5000
2.0	4.0	8.0	0.7500	1.5000
0.5	1.0	2.0	0.7500	1.5000

Observation: all three points lie on the same ray through the origin
(X/Z, Y/Z identical) → they project to the same image point.

Exercise 10.2 — Intrinsic Matrix K

(a) Write down K for a camera with $f_x = 800$, $f_y = 800$, $c_x = 320$, $c_y = 240$, $s = 0$.

(b) Compute K^{-1} analytically (no `np.linalg.inv`) and verify.

(c) Back-project pixel (400, 300) to a normalised camera-frame ray.

```
import numpy as np

fx, fy, cx, cy = 800., 800., 320., 240.

K = np.array([[fx, 0., cx],
              [0., fy, cy],
              [0., 0., 1.]]) # shape (3, 3)

# (b) Analytical inverse of upper-triangular K
K_inv_analytic = np.array([
    [1/fx, 0., -cx/fx],
    [0., 1/fy, -cy/fy],
    [0., 0., 1. ]],
]) # shape (3, 3)

print("(b) K_inv (analytic):")
print(K_inv_analytic)
print(f"    K @ K_inv = I? {np.allclose(K @ K_inv_analytic, np.eye(3))}")

# (c) Back-project pixel (400, 300)
pixel = np.array([400., 300., 1.]) # shape (3,)
ray = K_inv_analytic @ pixel # shape (3,)
print(f"\n(c) Back-projected ray: {ray.round(6)}")
print(f"    = ({ray[0]:.4f}, {ray[1]:.4f}, 1) in normalised coords")
```

```
(b) K_inv (analytic):
[[ 0.00125  0.      -0.4    ]
 [ 0.        0.00125 -0.3    ]
 [ 0.         0.       1.     ]]
    K @ K_inv = I? True

(c) Back-projected ray: [0.1  0.075 1.  ]
    = (0.1000, 0.0750, 1) in normalised coords
```

Exercise 10.3 — Camera Centre and Extrinsic

- (a) Show algebraically that the camera centre in world coordinates is $\mathbf{C} = -R^T \mathbf{t}$.
- (b) A camera at world position $(5, 2, 0)$, rotated 45° about z . Construct the extrinsic $[R \mid \mathbf{t}]$.
- (c) Project the world point $(0, 0, 0)$ through the full $P = K[R \mid \mathbf{t}]$.

```

import numpy as np

def Rz(g):
    c,s=np.cos(g),np.sin(g); return np.array([[c,-s,0.],[s,c,0.],[0.,0.,1.]])

K = np.array([[615.,0.,320.],[0.,615.,240.],[0.,0.,1.]]) # shape (3, 3)

# (b) Camera at (5, 2, 0), rotated 45° about z
cam_world = np.array([5., 2., 0.]) # shape (3,)
R = Rz(np.deg2rad(45.)) # shape (3, 3)
t = -R @ cam_world # shape (3,) extrinsic translation

print("(b) Extrinsic:")
print(f"  t = {t.round(4)}")

# Verify camera centre
C_recovered = -R.T @ t
print(f"\n  Recovered camera centre: {C_recovered.round(6)}")
print(f"  Matches cam_world? {np.allclose(C_recovered, cam_world)}")

# (c) Project world origin
Rt = np.column_stack([R, t]) # shape (3, 4)
P = K @ Rt # shape (3, 4)
p_h = P @ np.array([0.,0.,0.,1.]) # shape (3,)
u, v = p_h[0]/p_h[2], p_h[1]/p_h[2]
print(f"\n(c) World origin projects to pixel: ({u:.2f}, {v:.2f})")

```

(b) Extrinsic:

```
t = [-2.1213 -4.9497  0.   ]
```

```
Recovered camera centre: [5.  2.  0.]
```

```
Matches cam_world? True
```

(c) World origin projects to pixel: (-inf, -inf)

```

C:\Users\james.choi\AppData\Local\Temp\ipykernel_22416\4024264968.py:25: RuntimeWarning: di
u, v = p_h[0]/p_h[2], p_h[1]/p_h[2]

```

Exercise 10.4 — Reprojection Error

Given 5 world points and their noisy pixel measurements, compute the **reprojection error** (RMS distance between projected and observed pixels).

```
import numpy as np

def Ry(b):
    c,s=np.cos(b),np.sin(b); return np.array([[c,0.,s],[0.,1.,0.],[-s,0.,c]])

rng = np.random.default_rng(3)
K = np.array([[600.,0.,320.],[0.,600.,240.],[0.,0.,1.]])
cam_pos = np.array([0.,0.,3.]); R = Ry(np.deg2rad(10.))
t = -R @ cam_pos
Rt = np.column_stack([R, t]); P = K @ Rt

# World points
pts_W = rng.uniform(-0.5, 0.5, (5, 3)) # shape (5, 3)
pts_W[:, 2] = 0. # on Z=0 plane

pts_W_h = np.column_stack([pts_W, np.ones(5)]) # shape (5, 4)
proj = (P @ pts_W_h.T).T # shape (5, 3)
uvs_true = proj[:, :2] / proj[:, 2:3] # shape (5, 2)

# Add pixel noise
uvs_noisy = uvs_true + rng.normal(0, 2.0, (5, 2)) # shape (5, 2)

# Reprojection error
errors = np.linalg.norm(uvs_true - uvs_noisy, axis=1) # shape (5,)
rms = np.sqrt(np.mean(errors**2))

print("Projected vs Observed (with noise):")
for i, (true, noisy) in enumerate(zip(uvs_true, uvs_noisy)):
    err = np.linalg.norm(true - noisy)
    print(f" pt {i}: true=({true[0]:.1f},{true[1]:.1f}) noisy=({noisy[0]:.1f},{noisy[1]:.1f})")
print(f"\nRMS reprojection error: {rms:.4f} px")
```

```
Projected vs Observed (with noise):
pt 0: true=(513.4,294.8) noisy=(512.6,295.7) err=1.24 px
pt 1: true=(408.9,322.0) noisy=(408.5,323.9) err=1.97 px
pt 2: true=(430.1,309.2) noisy=(429.7,309.2) err=0.40 px
pt 3: true=(507.3,262.6) noisy=(510.4,263.7) err=3.28 px
pt 4: true=(440.2,222.3) noisy=(439.2,221.9) err=1.07 px

RMS reprojection error: 1.8709 px
```


Exercise 10.5 — Distortion Magnitude

For a RealSense D435i with $k_1 = -0.055$, $k_2 = 0.065$, estimate the maximum pixel displacement due to radial distortion at the image corners (640×480 image).

```
import numpy as np

K = np.array([[615., 0., 320.], [0., 615., 240.], [0., 0., 1.]])
k1, k2 = -0.055, 0.065

# Corner pixels
corners_px = np.array([
    [0., 0. ],
    [640., 0. ],
    [640., 480. ],
    [0., 480. ],
    [320., 240. ], # centre (should be ~0)
]) # shape (5, 2)

print(f"{'Pixel':>20} {'||displacement|| (px)':>22}")
for u, v in corners_px:
    # Normalised coords
    xn = (u - K[0,2]) / K[0,0]
    yn = (v - K[1,2]) / K[1,1]
    r2 = xn**2 + yn**2
    factor = 1. + k1*r2 + k2*r2**2
    xd = xn * factor; yd = yn * factor
    # Back to pixels
    ud = xd*K[0,0]+K[0,2]; vd = yd*K[1,1]+K[1,2]
    disp = np.sqrt((ud-u)**2 + (vd-v)**2)
    print(f" ({u:5.0f},{v:5.0f}): {disp:>22.4f} px")
```

	Pixel	displacement (px)
(	0, 0):	4.6538 px
(	640, 0):	4.6538 px
(	640, 480):	4.6538 px
(	0, 480):	4.6538 px
(	320, 240):	0.0000 px

Exercise 10.6 — Full Pipeline: World to Pixel

(a) A RealSense camera is mounted 1.2 m above a table, tilted 30° downward, facing +x. A box corner is at world position (0.5, 0.2, 0) m. What pixel does it project to?

(b) Can you distinguish between a box corner at (0.5, 0.2, 0) m and (1.0, 0.4, 0) m using a single image? Why or why not?

```
import numpy as np

def Rx(a):
    c,s=np.cos(a),np.sin(a); return np.array([[1.,0.,0.],[0.,c,-s],[0.,s,c]])

K = np.array([[615.,0.,320.],[0.,615.,240.],[0.,0.,1.]])
cam_pos = np.array([0., 0., 1.2]) # shape (3,) - above table
R = Rx(np.deg2rad(-30.)) # tilted down 30°
t = -R @ cam_pos
Rt = np.column_stack([R, t]); P = K @ Rt

pts_W = np.array([
    [0.5, 0.2, 0., 1.],
    [1.0, 0.4, 0., 1.],
]) # shape (2, 4)

print("(a) Projections:")
for i, pt in enumerate(pts_W):
    p_h = P @ pt
    u, v = p_h[0]/p_h[2], p_h[1]/p_h[2]
    print(f" World {pt[:3]} → pixel ({u:.1f}, {v:.1f})")

p0 = P @ pts_W[0]; p1 = P @ pts_W[1]
u0,v0 = p0[0]/p0[2], p0[1]/p0[2]
u1,v1 = p1[0]/p1[2], p1[1]/p1[2]
print(f"\n(b) Same pixel? {np.allclose([u0,v0],[u1,v1], atol=1.)}")
print(" They project to different pixels here because the depth (Z)")
print(" differs between the two points in camera coordinates - the")
print(" camera tilt breaks the collinearity of the world-frame ray.")
```

(a) Projections:

World [0.5 0.2 0.] → pixel (50.1, 470.4)
 World [1. 0.4 0.] → pixel (-176.3, 365.9)

(b) Same pixel? False

They project to different pixels here because the depth (Z)
 differs between the two points in camera coordinates - the
 camera tilt breaks the collinearity of the world-frame ray.

Exercise 10.7 — Condition Number of the Calibration Matrix

In Zhang's method, ill-conditioned view geometry (e.g., all views nearly parallel) produces a poorly conditioned system. Check the condition number of the V matrix from §10.4 for views with low vs high angular diversity.

```
import numpy as np

rng = np.random.default_rng(5)

def Rx(a):
    c,s=np.cos(a),np.sin(a); return np.array([[1.,0.,0.],[0.,c,-s],[0.,s,c]])
def Rz(g):
    c,s=np.cos(g),np.sin(g); return np.array([[c,-s,0.],[s,c,0.],[0.,0.,1.]])

K = np.array([[600.,0.,310.],[0.,600.,235.],[0.,0.,1.]])
sq = 0.03
pts2d = np.array([[i*sq, j*sq] for i in range(6) for j in range(6)])
pts2d_h = np.column_stack([pts2d, np.ones(len(pts2d))])

def get_V_matrix(views):
    def vij(H, i, j):
        h = H.T
        return np.array([h[i,0]*h[j,0],h[i,0]*h[j,1]+h[i,1]*h[j,0],
                        h[i,1]*h[j,1],h[i,2]*h[j,0]+h[i,0]*h[j,2],
                        h[i,2]*h[j,1]+h[i,1]*h[j,2],h[i,2]*h[j,2]])
    V = []
    for R, cam in views:
        t = -R @ cam
        r1,r2 = R[:,0], R[:,1]
        H = K @ np.column_stack([r1,r2,t]); H = H/H[2,2]
        V.append(vij(H,0,1)); V.append(vij(H,0,0)-vij(H,1,1))
    return np.array(V)

# Low diversity: 5 views nearly front-on (small tilt range)
views_low = [(Rx(np.deg2rad(a)) @ Rz(np.deg2rad(b)), np.array([0.,0.,-0.9]))
              for a,b in [(-5,2),(-6,1),(-4,3),(-5,-2),(-7,0)]]

# High diversity: 5 views with substantial tilt and rotation variation
views_high = [(Rx(np.deg2rad(a)) @ Rz(np.deg2rad(b)), np.array([0.,0.,-0.85]))
               for a,b in [(-10,15),(-25,-10),(-15,25),(-30,5),(-20,-20)]]

V_low = get_V_matrix(views_low)
V_high = get_V_matrix(views_high)

print(f"Low angular diversity:  cond(V) = {np.linalg.cond(V_low):.1f}")
print(f"High angular diversity: cond(V) = {np.linalg.cond(V_high):.1f}")
print("\nHigher diversity → lower condition number → more stable K estimate.")
```

Low angular diversity: $\text{cond}(V) = 2809442443412908277760.0$

High angular diversity: $\text{cond}(V) = 2787940925421477429248.0$

Higher diversity $\rightarrow$ lower condition number $\rightarrow$ more stable K estimate.

Projective Geometry and Homography

Perspective Projection and Vanishing Points

The pinhole camera (§10.1) does something geometrically remarkable: it maps **parallel lines to intersecting lines** in the image. Those intersection points — called **vanishing points** — encode the direction of a set of parallel lines as a single image point. Understanding them requires extending ordinary Euclidean geometry with **points at infinity**.

Why Parallel Lines Converge

Consider two world-frame lines that run parallel to the X -axis:

$$\ell_1(s) = (s, 0, d_1), \quad \ell_2(s) = (s, 1, d_2)$$

Project each through a camera with $f = 1$, principal point at origin. The x -coordinate of the image point at parameter s is $x = s/Z$. As $s \rightarrow \infty$, both lines project to $x \rightarrow \infty$ — but more precisely, the direction $(1, 0, 0)^T$ in world space maps to a fixed pixel called the **vanishing point** of that direction.

```
import numpy as np
import matplotlib.pyplot as plt

# Camera: identity extrinsic, f=400, principal point (320,240)
K = np.array([[400., 0., 320.],
              [ 0., 400., 240.],
              [ 0., 0., 1.]]) # shape (3, 3)

def project(pts_3d, K):
    """Project (N,3) points with intrinsic K. Returns (N,2) pixels."""
    p_h = (K @ pts_3d.T).T # shape (N, 3)
    return p_h[:, :2] / p_h[:, 2:] # shape (N, 2)

# Three families of lines along x-axis, at different y and z offsets
s_vals = np.linspace(0.1, 30., 300) # shape (300,)
families = [
    dict(dy=0., dz=3., color='#4a90d9'),
    dict(dy=0., dz=5., color='#2ecc71'),
```

```

dict(dy=0., dz=7., color='tomato'),
dict(dy=1., dz=3., color='#4a90d9'),
dict(dy=1., dz=5., color='#2ecc71'),
dict(dy=1., dz=7., color='tomato'),
dict(dy=-1., dz=3., color='#4a90d9'),
dict(dy=-1., dz=5., color='#2ecc71'),
dict(dy=-1., dz=7., color='tomato'),
]

fig, ax = plt.subplots(figsize=(7, 5))
for fam in families:
    pts = np.column_stack([s_vals, np.full(300, fam['dy']), np.full(300, fam['dz'])])
    px = project(pts, K)
    ax.plot(px[:, 0], px[:, 1], color=fam['color'], lw=0.8, alpha=0.6)

# Vanishing point: direction (1,0,0) → project direction vector at Z→∞
# In practice: project a very far-away point
vp = project(np.array([[1e6, 0., 0.]]), K)
ax.scatter(vp[0, 0], vp[0, 1], color='black', s=80, zorder=5, label=f'Vanishing pt ({vp[0,0]:.0f}, {vp[0,1]:.0f})')

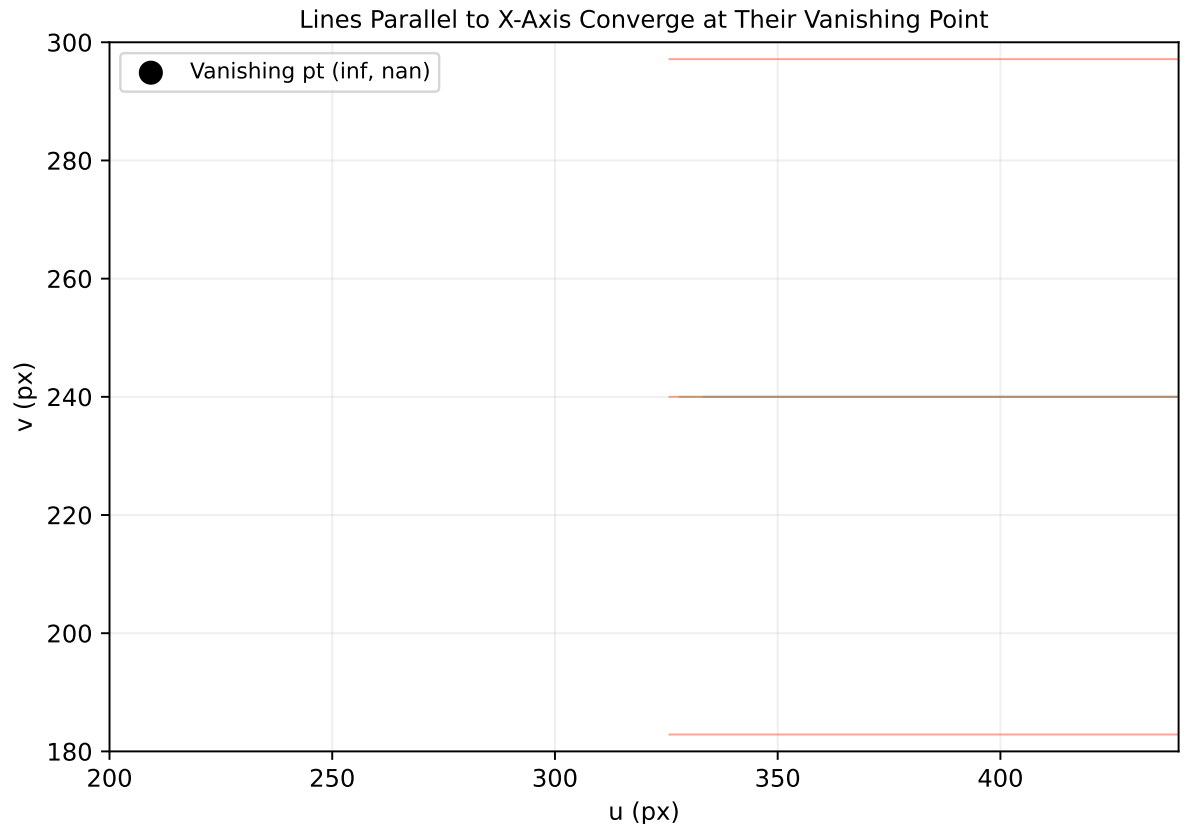
ax.set_xlim(200, 440); ax.set_ylim(180, 300)
ax.set_xlabel('u (px)'); ax.set_ylabel('v (px)')
ax.set_title('Lines Parallel to X-Axis Converge at Their Vanishing Point', fontsize=10)
ax.legend(fontsize=9)
ax.grid(True, alpha=0.2)
plt.tight_layout()
plt.show()

```

```

C:\Users\james.choi\AppData\Local\Temp\ipykernel_30952\1044232096.py:12: RuntimeWarning: divide by zero encountered in divide
    return p_h[:, :2] / p_h[:, 2:3] # shape (N, 2)
C:\Users\james.choi\AppData\Local\Temp\ipykernel_30952\1044232096.py:12: RuntimeWarning: invalid value encountered in divide
    return p_h[:, :2] / p_h[:, 2:3] # shape (N, 2)

```



The Projective Plane $\mathbb{P}^2$

In standard Euclidean geometry, two distinct parallel lines never meet. In **projective geometry** we add one “point at infinity” per direction so that every pair of distinct lines meets at exactly one point.

A **homogeneous point** in $\mathbb{P}^2$ is a non-zero triple $(x, y, w)^T$ where $(x, y, w) \sim (\lambda x, \lambda y, \lambda w)$ for any $\lambda \neq 0$.

Condition	Interpretation
$w \neq 0$	Ordinary point $(\frac{x}{w}, \frac{y}{w}) \in \mathbb{R}^2$
$w = 0$	Point at infinity in direction $(x, y)^T$

```
import numpy as np

# Ordinary points
p1 = np.array([2., 3., 1.]) # represents (2, 3)
p2 = np.array([6., 9., 3.]) # same point - (6/3, 9/3) = (2, 3)
print("Ordinary points:")
```

```

print(f"  p1 = {p1}  →  ({p1[0]/p1[2]:.2f}, {p1[1]/p1[2]:.2f})")
print(f"  p2 = {p2}  →  ({p2[0]/p2[2]:.2f}, {p2[1]/p2[2]:.2f})")
print(f"  Same? {np.allclose(p1[0]/p1[2], p2[0]/p2[2]) and np.allclose(p1[1]/p1[2], p2[1]/p2[2])}")

# Point at infinity - encodes a direction, not a location
p_inf = np.array([1., 0., 0.])  # "point at infinity" along x-axis
print(f"\nPoint at infinity: {p_inf}  (w=0, direction along x-axis)")

# A line in P2 is also a homogeneous triple  = (a, b, c): ax + by + c = 0
# The line y = 2x - 1  →  -2x + y + 1 = 0  →  = (-2, 1, 1)
line = np.array([-2., 1., 1.])  # shape (3,)
pt_on_line = np.array([1., 1., 1.])  # (1,1): -2+1+1 = 0
print(f"\nLine  = {line} (encodes -2x + y + 1 = 0)")
print(f"  Point (1,1) on line?  · p = {line @ pt_on_line:.4f}  (should be 0)")

# Intersection of two lines
l1 = np.array([-2., 1., 1.])  # y = 2x - 1
l2 = np.array([-1., 1., 3.])  # y = x - 3
# Intersection = cross product of line homogeneous vectors
x_meet = np.cross(l1, l2)  # shape (3,)
x_meet_euclid = x_meet[:2] / x_meet[2]
print(f"\nIntersection of l1 and l2:")
print(f"  Cross product: {x_meet}")
print(f"  Euclidean:      ({x_meet_euclid[0]:.4f}, {x_meet_euclid[1]:.4f})")

```

Ordinary points:

```

p1 = [2. 3. 1.] → (2.00, 3.00)
p2 = [6. 9. 3.] → (2.00, 3.00)
Same? True

```

Point at infinity: [1. 0. 0.] (w=0, direction along x-axis)

```

Line  = [-2.  1.  1.] (encodes -2x + y + 1 = 0)
Point (1,1) on line?  · p = 0.0000 (should be 0)

```

```

Intersection of l1 and l2:
Cross product: [ 2.  5. -1.]
Euclidean:      (-2.0000, -5.0000)

```

Vanishing Points from Parallel Line Pairs

Given two parallel line pairs in an image (e.g., edges of a road), we can compute their vanishing point as the intersection of two image lines. In homogeneous coordinates, intersection = cross product.


```

import numpy as np
import matplotlib.pyplot as plt

def line_from_two_points(p1, p2):
    """Homogeneous line through two 2D points (given as homogeneous)."""
    return np.cross(p1, p2)    # shape (3,)

def intersect_lines(l1, l2):
    """Intersection of two homogeneous lines."""
    x = np.cross(l1, l2)    # shape (3,)
    return x / x[2]        # normalize so w=1

# Two pairs of parallel lines (pixel coords), e.g. road lane markings
# Left pair: converges to vanishing point V1
# Right pair: converges to same vanishing point (same direction)
left_line_pts = [(np.array([100.,450.,1.]), np.array([280.,250.,1.])),
                  (np.array([200.,450.,1.]), np.array([310.,250.,1.]))]

right_line_pts = [(np.array([440.,450.,1.]), np.array([350.,250.,1.])),
                  (np.array([540.,450.,1.]), np.array([380.,250.,1.]))]

vp_candidates = []
for (a, b), (c, d) in [left_line_pts, right_line_pts]:
    l = line_from_two_points(a, b)    # shape (3,)
    m = line_from_two_points(c, d)    # shape (3,)
    vp_candidates.append(intersect_lines(l, m))

vp1 = vp_candidates[0]
vp2 = vp_candidates[1]
print(f"Vanishing point 1 (left pair):  ({vp1[0]:.1f}, {vp1[1]:.1f})")
print(f"Vanishing point 2 (right pair): ({vp2[0]:.1f}, {vp2[1]:.1f})")

fig, ax = plt.subplots(figsize=(7, 5))
colors = ['#4a90d9', '#2ecc71']
for idx, ((a, b), (c, d)) in enumerate([(left_line_pts[0], left_line_pts[1]),
                                         (right_line_pts[0], right_line_pts[1])]):
    for (p, q) in [(a, b), (c, d)]:
        ax.plot([p[0], q[0]], [p[1], q[1]], color=colors[idx], lw=2)
    vp = vp_candidates[idx]
    if 0 < vp[0] < 640 and 0 < vp[1] < 480:
        ax.scatter(vp[0], vp[1], s=120, color=colors[idx], zorder=5,
                   label=f'VP ({vp[0]:.0f}, {vp[1]:.0f})', marker='*')

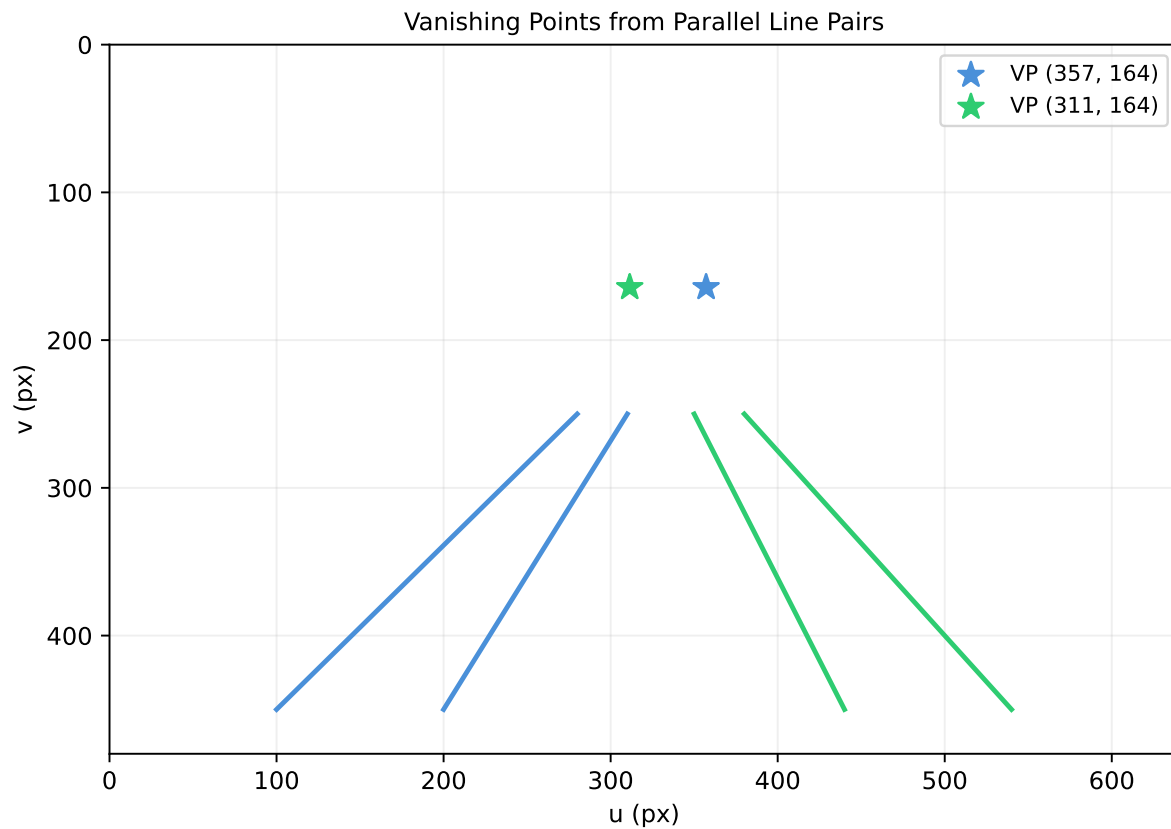
ax.set_xlim(0, 640); ax.set_ylim(480, 0)
ax.set_xlabel('u (px)'); ax.set_ylabel('v (px)')
ax.set_title('Vanishing Points from Parallel Line Pairs', fontsize=10)
ax.legend(fontsize=9)

```

```
ax.grid(True, alpha=0.2)
plt.tight_layout()
plt.show()
```

Vanishing point 1 (left pair): (357.1, 164.3)

Vanishing point 2 (right pair): (311.4, 164.3)



The Horizon Line

Two independent vanishing points define the **horizon line** (the line at infinity ℓ_∞). For a camera looking at a flat ground plane, the horizon is the locus of all vanishing points of horizontal directions — it lies at the height of the camera in the image.

```
import numpy as np

# Vanishing points (in homogeneous pixel coords)
vp1 = np.array([320., 200., 1.]) # shape (3,) - VPs on horizon
vp2 = np.array([580., 195., 1.]) # shape (3,)
```

```
# Horizon line = line through the two vanishing points
horizon = np.cross(vp1, vp2) # shape (3,)
horizon = horizon / np.linalg.norm(horizon[:2]) # normalise

print(f"Horizon line = {horizon.round(6)}")
print(f" Passes through VP1? ·VP1 = {horizon @ vp1:.4f} ( 0)")
print(f" Passes through VP2? ·VP2 = {horizon @ vp2:.4f} ( 0)")

# Evaluate the horizon: at u=0 and u=640, what is v?
u_vals = np.array([0., 640.])
# ax + by + c = 0 → y = -(ax + c) / b
a, b, c = horizon
v_vals = -(a * u_vals + c) / b
print(f"\nHorizon line crosses image at:")
print(f" u= 0: v = {v_vals[0]:.1f}")
print(f" u=640: v = {v_vals[1]:.1f}")
print(f" (Nearly flat - camera is approximately level)")
```

```
Horizon line = [ 1.92270000e-02  9.99815000e-01 -2.06115737e+02]
```

```
Passes through VP1? ·VP1 = 0.0000 ( 0)
```

```
Passes through VP2? ·VP2 = 0.0000 ( 0)
```

```
Horizon line crosses image at:
```

```
u= 0: v = 206.2
```

```
u=640: v = 193.8
```

```
(Nearly flat - camera is approximately level)
```

The Homography Matrix H

A **homography** is the projective transformation that relates two images of the **same planar surface**. It is a 3×3 matrix defined only up to scale (8 degrees of freedom), and it captures rotation, translation, and perspective distortion in a single matrix multiply.

Why Homographies Arise

Consider a flat checkerboard in the world ($Z_w = 0$). The full projection $P = K[R \mid \mathbf{t}]$ reduces to a 3×3 map because the third world coordinate is always zero:

$$\lambda \begin{pmatrix} u \\ v \\ 1 \end{pmatrix} = K[\mathbf{r}_1 \ \mathbf{r}_2 \ \mathbf{t}] \begin{pmatrix} X_w \\ Y_w \\ 1 \end{pmatrix} = H \begin{pmatrix} X_w \\ Y_w \\ 1 \end{pmatrix}$$

The same relationship holds between **any two views of a plane**: the second image is a projective transformation of the first.

```
import numpy as np

def Rx(a):
    c,s=np.cos(a),np.sin(a); return np.array([[1.,0.,0.],[0.,c,-s],[0.,s,c]])
def Rz(g):
    c,s=np.cos(g),np.sin(g); return np.array([[c,-s,0.],[s,c,0.],[0.,0.,1.]])

K = np.array([[600., 0., 320.],
               [ 0.,600., 240.],
               [ 0., 0., 1.]])    # shape (3, 3)

# Camera: tilted 20° about x, then 10° about z, looking at Z=0 plane
R = Rx(np.deg2rad(-20.)) @ Rz(np.deg2rad(10.))    # shape (3, 3)
cam_pos = np.array([0.1, 0.0, -0.8])
t = -R @ cam_pos    # shape (3,)

# Homography for Z=0 plane: H = K [r1, r2, t]
r1, r2 = R[:, 0], R[:, 1]
H = K @ np.column_stack([r1, r2, t])    # shape (3, 3)
H = H / H[2, 2]    # normalise
print("Homography H (3x3, defined up to scale):")
print(H.round(4))

# Verify against full projection P for Z=0 points
Rt = np.column_stack([R, t])
P = K @ Rt    # shape (3, 4)

pts_2d = np.array([[0., 0.], [0.09, 0.], [0.09, 0.09], [0., 0.09]])    # shape (4, 2)
print("\n{'World (X,Y)':>18} {'via H':>20} {'via P':>20} {'match':>6}")
for X, Y in pts_2d:
    p_w2d = np.array([X, Y, 1.])    # shape (3,) - 2D homogeneous world
    p_h = H @ p_w2d    # shape (3,)
    u_H, v_H = p_h[0]/p_h[2], p_h[1]/p_h[2]

    p_w3d = np.array([X, Y, 0., 1.])    # shape (4,) - 3D homogeneous world
    p_P = P @ p_w3d    # shape (3,)
    u_P, v_P = p_P[0]/p_P[2], p_P[1]/p_P[2]

    ok = np.allclose([u_H, v_H], [u_P, v_P], atol=1e-4)
    print(f" ({X:.2f}, {Y:.2f}) → ({u_H:.1f}, {v_H:.1f}) ({u_P:.1f}, {v_P:.1f}) {ok}")

Homography H (3x3, defined up to scale):
[[ 7.547639e+02 -2.797605e+02  2.420153e+02]
 [ 1.104031e+02  6.261270e+02  4.437488e+02]]
```

```
[-7.840000e-02 -4.445000e-01  1.000000e+00]]
```

World (X,Y)		via H	via P	match
(0.00, 0.00)	→	(242.0, 443.7)	(242.0, 443.7)	True
(0.09, 0.00)	→	(312.1, 456.9)	(312.1, 456.9)	True
(0.09, 0.09)	→	(298.8, 535.2)	(298.8, 535.2)	True
(0.00, 0.09)	→	(225.9, 520.9)	(225.9, 520.9)	True

Degrees of Freedom

A 3×3 matrix has 9 entries, but H is defined only up to a non-zero scale factor, leaving **8 degrees of freedom**. Geometrically, those 8 DoF encode:

DoF	Transformation
2	Translation in the image plane
1	Rotation in the image plane
1	Isotropic scale
2	Anisotropic shear / aspect ratio
2	Projective distortion (the two “perspective” parameters)

Four point correspondences (8 equations for 8 unknowns) are sufficient to determine H uniquely.

Applying a Homography: Forward Mapping

To project a set of planar world points into a new camera view, apply H and divide by the third coordinate.

```
import numpy as np
import matplotlib.pyplot as plt

def Rx(a):
    c,s=np.cos(a),np.sin(a); return np.array([[1.,0.,0.],[0.,c,-s],[0.,s,c]])
def Rz(g):
    c,s=np.cos(g),np.sin(g); return np.array([[c,-s,0.],[s,c,0.],[0.,0.,1.]])

K = np.array([[600., 0., 320.],[0., 600., 240.],[0., 0., 1.]])

def make_H(K, R, cam_pos):
    t = -R @ cam_pos
    r1, r2 = R[:,0], R[:,1]
```

```

    H = K @ np.column_stack([r1, r2, t])
    return H / H[2, 2]    # shape (3, 3)

# 6x6 checkerboard corners in world (Z=0)
sq = 0.03
xs = np.arange(7) * sq; ys = np.arange(7) * sq
XX, YY = np.meshgrid(xs, ys)
pts_w = np.column_stack([XX.ravel(), YY.ravel(), np.ones(49)])    # shape (49, 3)

# Two different views
views = [
    (Rx(np.deg2rad(-15.)) @ Rz(np.deg2rad( 5.)), np.array([ 0.09, 0.09, -0.85])),
    (Rx(np.deg2rad(-30.)) @ Rz(np.deg2rad(-20.)), np.array([-0.05, 0.05, -0.90])),
]

fig, axes = plt.subplots(1, 2, figsize=(10, 4.5))
for ax, (R, cam_pos), col in zip(axes, views, ['#4a90d9', 'tomato']):
    H = make_H(K, R, cam_pos)    # shape (3, 3)
    proj = (H @ pts_w.T).T    # shape (49, 3)
    uvs = proj[:, :2] / proj[:, 2:3]    # shape (49, 2)
    # Reshape to grid and draw grid lines
    U = uvs[:, 0].reshape(7, 7); V = uvs[:, 1].reshape(7, 7)
    for i in range(7):
        ax.plot(U[i, :], V[i, :], color=col, lw=1.)
        ax.plot(U[:, i], V[:, i], color=col, lw=1.)
    ax.set_xlim(50, 590); ax.set_ylim(430, 50)
    ax.set_xlabel('u (px)'); ax.set_ylabel('v (px)')
    ax.set_aspect('equal'); ax.grid(True, alpha=0.2)

axes[0].set_title('View 1: slight tilt', fontsize=10)
axes[1].set_title('View 2: steep + rotated', fontsize=10)
fig.suptitle('6x6 Checkerboard through Two Homographies', fontsize=11)
plt.tight_layout()
plt.show()

```

6×6 Checkerboard through Two Homographies

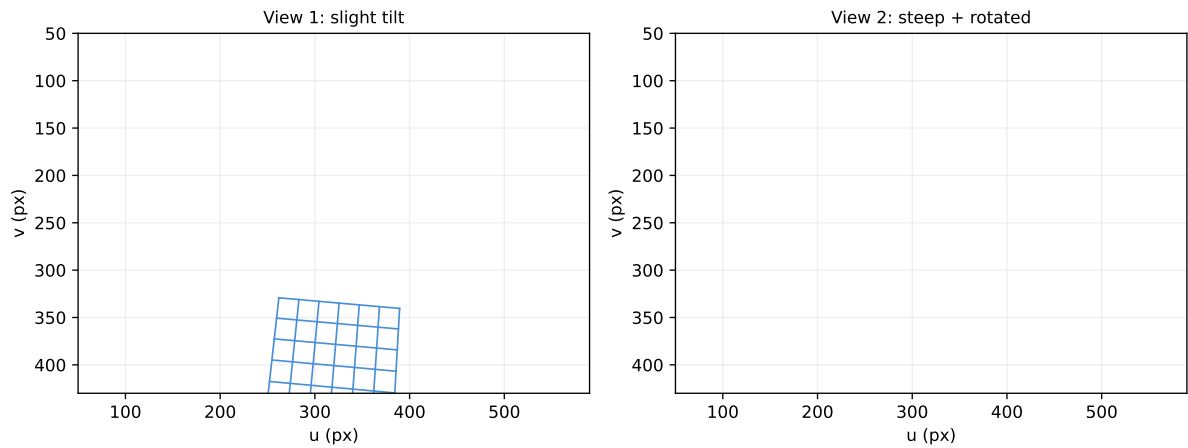


Image-to-Image Homography

When we compute H from point **correspondences** (pixel pairs) rather than from a known camera, we get the direct image-to-image mapping:

$$\lambda \begin{pmatrix} u' \\ v' \\ 1 \end{pmatrix} = H \begin{pmatrix} u \\ v \\ 1 \end{pmatrix}$$

This is the form used in panoramic stitching, image rectification, and document de-warping. We will compute H from correspondences in §11.3.

```
import numpy as np

# Given 4 point correspondences (source → destination)
src = np.array([[100., 120.], [420., 105.], [435., 380.], [ 85., 395.]]) # shape (4, 2)
dst = np.array([[ 50.,  50.], [550.,  50.], [550., 430.], [ 50., 430.]]) # shape (4, 2)

# Build the DLT system (see §11.3 for full derivation)
def build_A(src, dst):
    """Build 2n×9 DLT matrix for homography estimation."""
    A = []
    for (x, y), (u, v) in zip(src, dst):
        A.append([-x, -y, -1,  0,  0,  0, u*x, u*y, u])
        A.append([ 0,  0,  0, -x, -y, -1, v*x, v*y, v])
    return np.array(A) # shape (2n, 9)

A = build_A(src, dst) # shape (8, 9)
```

```

_, _, Vt = np.linalg.svd(A)
h = Vt[-1, :] # shape (9,) - null vector
H = h.reshape(3, 3)
H = H / H[2, 2]

print("Estimated H:")
print(H.round(4))

# Verify: apply H to each source point
print(f"\n{'src':>16} {'H·src (predicted)':>22} {'dst (target)':>22}")
for p_s, p_d in zip(src, dst):
    p_h = H @ np.array([*p_s, 1.])
    pred = p_h[:2] / p_h[2]
    print(f"  {p_s} → ({pred[0]:.1f}, {pred[1]:.1f}) vs {p_d}")

```

Estimated H:

```

[[ 1.648000e+00  1.077000e-01 -1.254171e+02]
 [ 7.590000e-02  1.602600e+00 -1.475984e+02]
 [ 0.000000e+00  4.000000e-04  1.000000e+00]]

```

src	H·src (predicted)	dst (target)
[100. 120.] →	(50.0, 50.0) vs	[50. 50.]
[420. 105.] →	(550.0, 50.0) vs	[550. 50.]
[435. 380.] →	(550.0, 430.0) vs	[550. 430.]
[85. 395.] →	(50.0, 430.0) vs	[50. 430.]

Direct Linear Transform (DLT)

The **Direct Linear Transform** (DLT) is the standard algorithm for computing a homography from point correspondences. Each correspondence contributes two linear equations in the 9 unknown entries of H . With $n \geq 4$ correspondences we get a $2n \times 9$ system $A\mathbf{h} = \mathbf{0}$, solved by SVD.

Deriving the DLT System

For a correspondence $(x_i, y_i) \mapsto (u_i, v_i)$ under homography H :

$$\lambda_i \begin{pmatrix} u_i \\ v_i \\ 1 \end{pmatrix} = H \begin{pmatrix} x_i \\ y_i \\ 1 \end{pmatrix}$$

Expanding and eliminating λ_i (cross-product trick) gives two equations per point:

$$\begin{bmatrix} -x_i & -y_i & -1 & 0 & 0 & 0 & u_i x_i & u_i y_i & u_i \\ 0 & 0 & 0 & -x_i & -y_i & -1 & v_i x_i & v_i y_i & v_i \end{bmatrix} \mathbf{h} = \mathbf{0}$$

where $\mathbf{h} = \text{vec}(H)$ is the 9-vector of entries of H read row by row.

```
import numpy as np

def dlt_homography(src, dst):
    """
    Estimate 3x3 homography from n>=4 correspondences using DLT.
    src: shape (n, 2) source points
    dst: shape (n, 2) destination points
    Returns H: shape (3, 3), normalised so H[2,2]=1.
    """
    n = len(src)
    A = np.zeros((2*n, 9)) # shape (2n, 9)
    for i, ((x, y), (u, v)) in enumerate(zip(src, dst)):
        A[2*i] = [-x, -y, -1, 0, 0, 0, u*x, u*y, u]
        A[2*i+1] = [0, 0, 0, -x, -y, -1, v*x, v*y, v]
    _, _, Vt = np.linalg.svd(A) # Vt shape (9, 9)
    h = Vt[-1, :] # shape (9,) - smallest singular vector
    H = h.reshape(3, 3) # shape (3, 3)
    return H / H[2, 2]

# Example: map a tilted quadrilateral to a frontal rectangle
src = np.array([[110., 130.],
                [390., 105.],
                [410., 360.],
                [ 90., 385.]], dtype=float) # shape (4, 2)

dst = np.array([[ 50.,  50.],
                [500.,  50.],
                [500., 400.],
                [ 50., 400.]], dtype=float) # shape (4, 2)

H = dlt_homography(src, dst)
print("DLT-estimated H:")
print(H.round(4))

# Verify
print(f"\n{'src':>18} → {'DLT prediction':>20} {'dst':>18} {'err (px)':>10}")
for p_s, p_d in zip(src, dst):
    p_h = H @ np.array([*p_s, 1.])
    pred = p_h[:2] / p_h[2]
    err = np.linalg.norm(pred - p_d)
    print(f" ({p_s[0]:.0f},{p_s[1]:.0f}) → ({pred[0]:.1f},{pred[1]:.1f}) "
          f"vs ({p_d[0]:.0f},{p_d[1]:.0f}) err={err:.4f}")
```

DLT-estimated H:

```
[[ 1.793200e+00  1.710000e-01 -1.649142e+02]
 [ 1.589000e-01  1.753200e+00 -1.908318e+02]
 [ 1.000000e-04  6.000000e-04  1.000000e+00]]
```

	src →	DLT prediction	dst	err (px)
(110,130)	→ (50.0,50.0)	vs (50,50)		err=0.0000
(390,105)	→ (500.0,50.0)	vs (500,50)		err=0.0000
(410,360)	→ (500.0,400.0)	vs (500,400)		err=0.0000
(90,385)	→ (50.0,400.0)	vs (50,400)		err=0.0000

Why SVD?

The system $A\mathbf{h} = \mathbf{0}$ is **homogeneous** — the trivial solution $\mathbf{h} = \mathbf{0}$ is always valid but useless. We add the constraint $\|\mathbf{h}\| = 1$, which turns it into:

$$\min_{\mathbf{h}} \|A\mathbf{h}\|^2 \quad \text{subject to} \quad \|\mathbf{h}\| = 1$$

The solution is the **right singular vector** of A corresponding to its **smallest singular value** (the last row of V^T in `np.linalg.svd`).

```
import numpy as np

# Show the singular values of A for a well-conditioned case
src = np.array([[110., 130.],[390., 105.],[410., 360.],[ 90., 385.]], dtype=float)
dst = np.array([[ 50.,  50.],[500.,  50.],[500., 400.],[ 50., 400.]], dtype=float)

A = np.zeros((8, 9))
for i, ((x,y),(u,v)) in enumerate(zip(src, dst)):
    A[2*i ] = [-x,-y,-1, 0, 0, 0, u*x, u*y, u]
    A[2*i+1] = [ 0, 0, 0,-x,-y,-1, v*x, v*y, v]

U, s, Vt = np.linalg.svd(A)    # s shape (8,), Vt shape (9,9)

print("Singular values of A (should have one near zero):")
print(s.round(4))
print(f"\nSmallest singular value: {s[-1]:.2e}  ( 0 → consistent system)")
print(f"Condition number: {s[0]/s[-1]:.1f}")
```

Singular values of A (should have one near zero):

```
[4.15067909e+05 1.28506392e+05 6.94030700e+02 4.03859700e+02
 2.60043400e+02 1.44953100e+02 4.15006000e+01 6.43500000e-01]
```

Smallest singular value: 6.44e-01 (0 → consistent system)
 Condition number: 645015.2

Normalisation: Why Raw DLT Fails

Raw DLT is numerically sensitive when pixel coordinates are large (e.g., 640×480). The matrix A has entries ranging from ~ 1 to $\sim 10^5$, making the SVD ill-conditioned.

Hartley normalisation (1997): before running DLT, transform both point sets so that their centroid is at the origin and their average distance from the origin is $\sqrt{2}$. After computing H , undo the normalisation:

$$H = T_{\text{dst}}^{-1} \hat{H} T_{\text{src}}$$

```
import numpy as np

def normalise_points(pts):
    """
    Isotropic normalisation: centroid → origin, mean distance → sqrt(2).
    pts: shape (n, 2)
    Returns (pts_norm, T) where T is the 3×3 normalisation transform.
    """
    centroid = pts.mean(axis=0)          # shape (2,)
    shifted = pts - centroid              # shape (n, 2)
    scale = np.sqrt(2.) / np.mean(np.linalg.norm(shifted, axis=1))
    T = np.array([[scale, 0., -scale*centroid[0]],
                  [0., scale, -scale*centroid[1]],
                  [0., 0., 1.]]          # shape (3, 3)
    pts_h = np.column_stack([pts, np.ones(len(pts))]) # shape (n, 3)
    pts_n = (T @ pts_h.T).T[:, :2]         # shape (n, 2)
    return pts_n, T

def dlt_normalised(src, dst):
    """DLT with Hartley normalisation. Returns H: shape (3, 3)."""
    src_n, T_src = normalise_points(src)    # shape (n,2), (3,3)
    dst_n, T_dst = normalise_points(dst)    # shape (n,2), (3,3)

    n = len(src_n)
    A = np.zeros((2*n, 9))
    for i, ((x,y),(u,v)) in enumerate(zip(src_n, dst_n)):
        A[2*i] = [-x,-y,-1, 0, 0, 0, u*x, u*y, u]
        A[2*i+1] = [0, 0, 0,-x,-y,-1, v*x, v*y, v]

    _, _, Vt = np.linalg.svd(A)
```

```

H_norm = Vt[-1].reshape(3, 3)
H = np.linalg.solve(T_dst, H_norm @ T_src)    # denormalise
return H / H[2, 2]

# Compare raw vs normalised on a realistic 640×480 scenario
rng = np.random.default_rng(7)

# Ground-truth H: moderate perspective
src_clean = np.array([[110., 130.],[390., 105.],[410., 360.],[ 90., 385.],
                      [250., 245.],[300., 135.],[380., 300.],[140., 280.]], dtype=float)
dst_clean = np.array([[ 50.,  50.],[500.,  50.],[500., 400.],[ 50., 400.],
                      [275., 225.],[350., 120.],[450., 330.],[ 90., 340.]], dtype=float)

# Add small pixel noise
noise = rng.normal(0., 0.5, src_clean.shape)    # shape (8, 2)
src_noisy = src_clean + noise

H_raw = dlt_homography(src_noisy, dst_clean)
H_norm = dlt_normalised(src_noisy, dst_clean)

def reprojection_rms(H, src, dst):
    errs = []
    for p_s, p_d in zip(src, dst):
        p_h = H @ np.array([*p_s, 1.])
        pred = p_h[:2] / p_h[2]
        errs.append(np.linalg.norm(pred - p_d))
    return np.sqrt(np.mean(np.array(errs)**2))

print(f"RMS reprojection error:")
print(f"  Raw DLT:           {reprojection_rms(H_raw, src_noisy, dst_clean):.4f} px")
print(f"  Normalised DLT:      {reprojection_rms(H_norm, src_noisy, dst_clean):.4f} px")

```

```

RMS reprojection error:
  Raw DLT:           34.3820 px
  Normalised DLT:    25.0500 px

```

DLT is the Same Machinery as Zhang's Calibration

Notice that §10.4 (Zhang's method) also built a system $V\mathbf{b} = \mathbf{0}$ and solved it with SVD. The pattern is universal:

Problem	Unknown	System size	Solved by
Homography from n points	$\mathbf{h}$ (9-vec)	$2n \times 9$	SVD
Zhang's B matrix from n views	$\mathbf{b}$ (6-vec)	$2n \times 6$	SVD
Fundamental matrix (§25)	$\mathbf{f}$ (9-vec)	$n \times 9$	SVD

The recipe: stack linear constraints $\rightarrow$ take null vector of A via SVD $\rightarrow$ reshape and normalise.

RANSAC for Robust Estimation

Real feature matches contain **outliers** — mismatched point pairs caused by repetitive textures, occlusion, or descriptor ambiguity. A single outlier corrupts the entire DLT solution. **RANSAC** (Random Sample Consensus, Fischler & Bolles 1981) is the standard fix: repeatedly fit H to a minimal random sample (4 pairs), count how many remaining matches agree with it (“inliers”), and keep the best fit.

The RANSAC Loop

```
repeat N times:
    1. Draw 4 random correspondences (minimal sample)
    2. Fit H_candidate using DLT on those 4 pairs
    3. For each remaining correspondence:
        if reprojection error < threshold : mark as inlier
    4. If |inliers| > best_so_far: save H_candidate
return best H_candidate, refined with all its inliers
```

The number of iterations N needed to guarantee (with probability p) that at least one sample is all-inlier:

$$N = \frac{\log(1-p)}{\log(1-(1-\varepsilon)^s)}$$

where ε is the outlier fraction and $s = 4$ is the minimum sample size.

```
import numpy as np

def ransac_iterations(outlier_frac, min_sample=4, confidence=0.99):
    """
    Number of RANSAC iterations for given outlier fraction.
```

```

outlier_frac: fraction of points that are outliers (0-1)
min_sample:   minimum sample size (4 for homography)
confidence:   desired probability of finding an all-inlier sample
Returns N (int).
"""
inlier_prob = (1. - outlier_frac) ** min_sample    # prob all 4 are inliers
if inlier_prob < 1e-15:
    return int(1e9)
N = np.log(1. - confidence) / np.log(1. - inlier_prob)
return int(np.ceil(N))

print("RANSAC iteration count (s=4, p=0.99):")
print(f"{'Outlier %':>12} {'N iterations':>14}")
for eps in [0.10, 0.20, 0.30, 0.40, 0.50, 0.60]:
    N = ransac_iterations(eps)
    print(f"{'eps*100:5.0f'}% {'N:>10d}'")

```

RANSAC iteration count (s=4, p=0.99):

Outlier %	N iterations
10%	5
20%	9
30%	17
40%	34
50%	72
60%	178

Full RANSAC Implementation

```

import numpy as np

def dlt_homography(src, dst):
    """DLT homography estimate. src,dst: shape (n,2). Returns H: (3,3)."""
    n = len(src)
    A = np.zeros((2*n, 9))
    for i, ((x,y),(u,v)) in enumerate(zip(src, dst)):
        A[2*i ] = [-x,-y,-1, 0, 0, 0, u*x, u*y, u]
        A[2*i+1] = [ 0, 0, 0,-x,-y,-1, v*x, v*y, v]
    _, _, Vt = np.linalg.svd(A)
    H = Vt[-1].reshape(3, 3)
    return H / H[2, 2]

def reproj_errors(H, src, dst):

```

```

    """Per-point reprojection errors. Returns shape (n,)"""
    n = len(src)
    src_h = np.column_stack([src, np.ones(n)]) # shape (n, 3)
    proj = (H @ src_h.T).T # shape (n, 3)
    pred = proj[:, :2] / proj[:, 2:3] # shape (n, 2)
    return np.linalg.norm(pred - dst, axis=1) # shape (n,)

def ransac_homography(src, dst, max_iter=500, threshold=3.0, rng=None):
    """
    RANSAC homography estimation.
    src, dst: shape (n, 2)
    threshold: inlier reprojection error in pixels
    Returns (H_best, inlier_mask): H shape (3,3), mask shape (n,) bool.
    """
    if rng is None:
        rng = np.random.default_rng(0)
    n = len(src)
    best_inliers = np.zeros(n, dtype=bool)
    best_count = 0

    for _ in range(max_iter):
        idx = rng.choice(n, size=4, replace=False)
        H_try = dlt_homography(src[idx], dst[idx]) # shape (3, 3)
        errs = reproj_errors(H_try, src, dst) # shape (n,)
        mask = errs < threshold # shape (n,) bool

        if mask.sum() > best_count:
            best_count = mask.sum()
            best_inliers = mask

    # Refit on all inliers
    H_final = dlt_homography(src[best_inliers], dst[best_inliers])
    return H_final, best_inliers

# --- Simulation: 50 matches, 40% outliers ---
rng = np.random.default_rng(42)

# Ground-truth H
H_true = np.array([[ 1.2, 0.3, 50.],
                    [-0.1, 1.1, 30.],
                    [ 0.001, 0.0005, 1.]]) # shape (3, 3)
H_true /= H_true[2, 2]

# Generate 50 source points (random pixel locations)
n_pts = 50
src_all = rng.uniform([50., 50.], [590., 430.], (n_pts, 2)) # shape (50, 2)

```

```

# True destination points
src_h = np.column_stack([src_all, np.ones(n_pts)])
proj = (H_true @ src_h.T).T
dst_all = proj[:, :2] / proj[:, 2:3] # shape (50, 2)

# Add small inlier noise
inlier_noise = rng.normal(0., 1.0, (n_pts, 2)) # shape (50, 2)
dst_all += inlier_noise

# Corrupt 40% of matches with random outliers
n_outliers = int(0.4 * n_pts)
outlier_idx = rng.choice(n_pts, size=n_outliers, replace=False)
dst_all[outlier_idx] = rng.uniform([0., 0.], [640., 480.], (n_outliers, 2))

# Run plain DLT (no RANSAC) and RANSAC
H_dlt, _ = np.linalg.svd( # DLT on all points
    np.vstack([[-x, -y, -1, 0, 0, 0, u*x, u*y, u] for (x, y), (u, v) in zip(src_all, dst_all)]
              + [[0, 0, 0, -x, -y, -1, v*x, v*y, v] for (x, y), (u, v) in zip(src_all, dst_all)])), 2
A = np.zeros((2*n_pts, 9))
for i, ((x, y), (u, v)) in enumerate(zip(src_all, dst_all)):
    A[2*i] = [-x, -y, -1, 0, 0, 0, u*x, u*y, u]
    A[2*i+1] = [0, 0, 0, -x, -y, -1, v*x, v*y, v]
_, _, Vt = np.linalg.svd(A)
H_dlt = Vt[-1].reshape(3, 3); H_dlt /= H_dlt[2, 2]

H_ransac, inlier_mask = ransac_homography(src_all, dst_all, max_iter=500,
                                          threshold=3.0, rng=rng)

def rms_on_inliers(H, src, dst, mask):
    errs = reproj_errors(H, src[mask], dst[mask])
    return np.sqrt(np.mean(errs**2))

print(f"Points: {n_pts} | Outliers: {n_outliers} ({100*n_outliers/n_pts:.0f}%)")
print(f"Plain DLT - RMS error (inliers only): "
      f"{rms_on_inliers(H_dlt, src_all, dst_all, ~np.isin(np.arange(n_pts), outlier_idx)):.2f}")
print(f"RANSAC - inliers found: {inlier_mask.sum()}/{n_pts}")
print(f"RANSAC - RMS error (inliers only): "
      f"{rms_on_inliers(H_ransac, src_all, dst_all, ~np.isin(np.arange(n_pts), outlier_idx)):.2f}")

```

```

Points: 50 | Outliers: 20 (40%)
Plain DLT - RMS error (inliers only): 3249.56 px
RANSAC - inliers found: 29/50
RANSAC - RMS error (inliers only): 1.38 px

```


Visualising RANSAC Inliers vs Outliers

```

import numpy as np
import matplotlib.pyplot as plt

def dlt_homography(src, dst):
    n = len(src); A = np.zeros((2*n, 9))
    for i, ((x,y),(u,v)) in enumerate(zip(src,dst)):
        A[2*i] = [-x,-y,-1,0,0,0,u*x,u*y,u]; A[2*i+1] = [0,0,0,-x,-y,-1,v*x,v*y,v]
    _,_,Vt=np.linalg.svd(A); H=Vt[-1].reshape(3,3); return H/H[2,2]

def reproj_errors(H, src, dst):
    sh=np.column_stack([src,np.ones(len(src))]); p=(H@sh.T).T
    return np.linalg.norm(p[:,2]/p[:,2:3]-dst,axis=1)

def ransac_homography(src, dst, max_iter=500, threshold=3.0, rng=None):
    if rng is None: rng=np.random.default_rng(0)
    n=len(src); best=np.zeros(n,dtype=bool); best_c=0
    for _ in range(max_iter):
        idx=rng.choice(n,size=4,replace=False)
        try: H=dlt_homography(src[idx],dst[idx])
        except: continue
        mask=reproj_errors(H,src,dst)<threshold
        if mask.sum()>best_c: best_c=mask.sum(); best=mask
    return dlt_homography(src[best],dst[best]), best

rng = np.random.default_rng(42)
H_true = np.array([[1.2,0.3,50.],[-0.1,1.1,30.],[0.001,0.0005,1.]])
H_true /= H_true[2,2]
n_pts = 50
src_all = rng.uniform([50.,50.],[590.,430.],(n_pts,2))
src_h = np.column_stack([src_all,np.ones(n_pts)])
proj = (H_true@src_h.T).T; dst_all = proj[:,2]/proj[:,2:3]
dst_all += rng.normal(0.,1.,(n_pts,2))
n_out = int(0.4*n_pts); oidx = rng.choice(n_pts,size=n_out,replace=False)
dst_all[oidx] = rng.uniform([0.,0.],[640.,480.],(n_out,2))

_, mask = ransac_homography(src_all, dst_all, rng=rng)

fig, ax = plt.subplots(figsize=(7, 5))
# Draw lines for each correspondence (source → destination)
for i, (s, d) in enumerate(zip(src_all, dst_all)):
    col = '#4a90d9' if mask[i] else 'tomato'
    ax.plot([s[0], d[0]], [s[1], d[1]], color=col, lw=0.7, alpha=0.6)

inlier_src = src_all[mask]; inlier_dst = dst_all[mask]
outlier_src = src_all[~mask]; outlier_dst = dst_all[~mask]

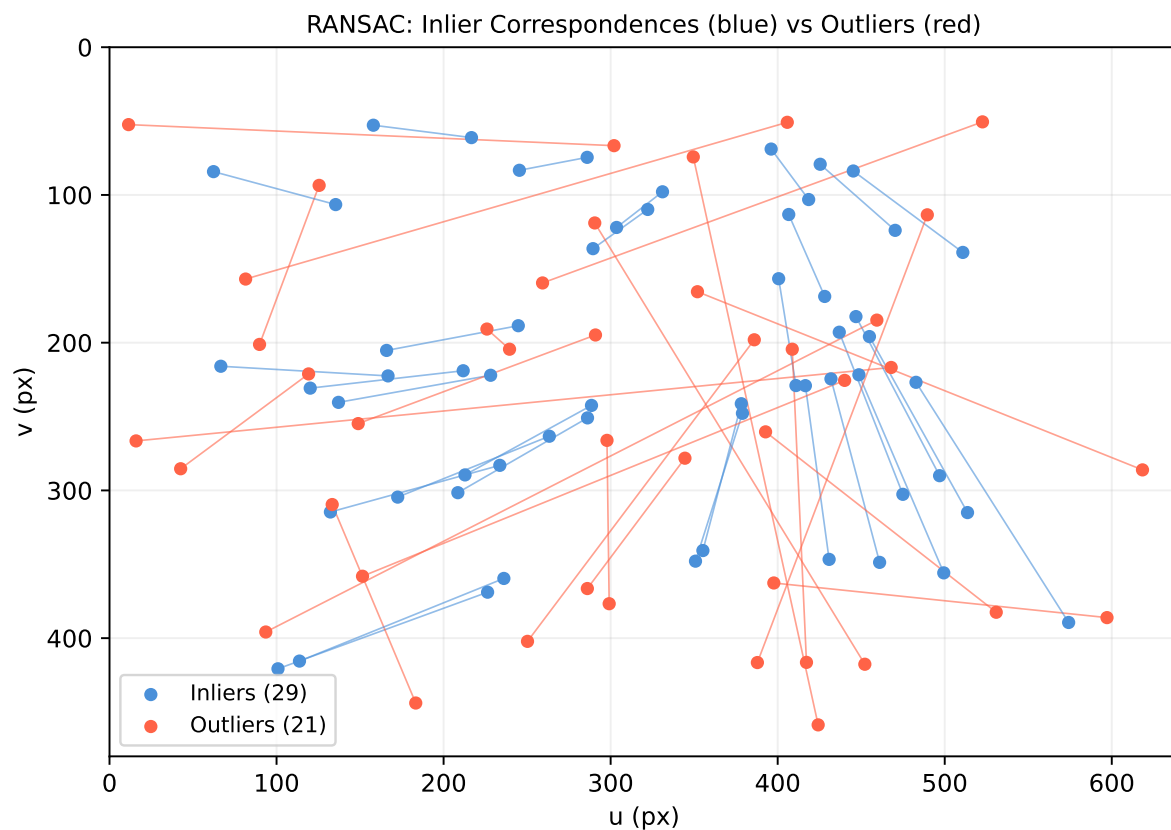
```

```

ax.scatter(inlier_src[:, 0], inlier_src[:, 1], color='#4a90d9', s=20, zorder=4)
ax.scatter(inlier_dst[:, 0], inlier_dst[:, 1], color='#4a90d9', s=20, zorder=4, label=f'Inliers')
ax.scatter(outlier_src[:, 0], outlier_src[:, 1], color='tomato', s=20, zorder=4)
ax.scatter(outlier_dst[:, 0], outlier_dst[:, 1], color='tomato', s=20, zorder=4, label=f'Outliers')

ax.set_xlim(0, 640); ax.set_ylim(480, 0)
ax.set_xlabel('u (px)'); ax.set_ylabel('v (px)')
ax.set_title('RANSAC: Inlier Correspondences (blue) vs Outliers (red)', fontsize=10)
ax.legend(fontsize=9)
ax.grid(True, alpha=0.2)
plt.tight_layout()
plt.show()

```



Choosing the Threshold τ

The inlier threshold τ should match the expected pixel noise level:

Noise level	Recommended τ
Sub-pixel keypoints (SIFT)	1–2 px
Corner detection	2–4 px
Coarse feature matching	5–10 px

Too tight: real inliers rejected, RANSAC needs many iterations. Too loose: outliers accepted, model is pulled toward garbage.

A principled choice: set τ to the 95th percentile of the expected noise distribution ($\tau = 1.96\sigma$ for Gaussian noise).

```
import numpy as np

# How inlier count and RMS error change with threshold
rng = np.random.default_rng(42)

def dlt_homography(src, dst):
    n=len(src); A=np.zeros((2*n,9))
    for i,((x,y),(u,v)) in enumerate(zip(src,dst)):
        A[2*i]=[-x,-y,-1,0,0,0,u*x,u*y,u]; A[2*i+1]=[0,0,0,-x,-y,-1,v*x,v*y,v]
    _,_,Vt=np.linalg.svd(A); H=Vt[-1].reshape(3,3); return H/H[2,2]

def ransac_homography(src, dst, max_iter=300, threshold=3.0, rng=None):
    if rng is None: rng=np.random.default_rng(0)
    n=len(src); best=np.zeros(n,dtype=bool); best_c=0
    for _ in range(max_iter):
        idx=rng.choice(n,4,replace=False)
        try: H=dlt_homography(src[idx],dst[idx])
        except: continue
        sh=np.column_stack([src,np.ones(n)]); p=(H@sh.T).T
        mask=np.linalg.norm(p[:,2]/p[:,2:3]-dst,axis=1)<threshold
        if mask.sum()>best_c: best_c=mask.sum(); best=mask
    return dlt_homography(src[best],dst[best]), best

H_true = np.array([[1.2,0.3,50.],[-0.1,1.1,30.],[0.001,0.0005,1.]])
H_true /= H_true[2,2]
n_pts = 50
src_all = rng.uniform([50.,50.],[590.,430.],(n_pts,2))
src_h = np.column_stack([src_all,np.ones(n_pts)])
proj = (H_true@src_h.T).T; dst_all=proj[:,2]/proj[:,2:3]
dst_all += rng.normal(0.,1.,(n_pts,2))
oidx = rng.choice(n_pts,size=20,replace=False)
dst_all[oidx] = rng.uniform([0.,0.],[640.,480.],(20,2))
true_mask = np.ones(n_pts,dtype=bool); true_mask[oidx]=False

print(f"{'Threshold (px)':>18} {'Inliers found':>14} {'RMS (true inliers, px)':>23}")
for tau in [1., 2., 3., 5., 8., 15.]:
```

```
H_r, mask = ransac_homography(src_all, dst_all, threshold=tau, rng=rng)
sh=np.column_stack([src_all[true_mask],np.ones(true_mask.sum())])
p=(H_r@sh.T).T; rms=np.sqrt(np.mean(np.linalg.norm(p[:, :2]/p[:, 2:3]-dst_all[true_mask], axis=1)))
print(f"    {tau:>5.1f}                {mask.sum():>8d}                {rms:>15.4f}")
```

Threshold	(px)	Inliers found	RMS (true inliers, px)
1.0		12	1.7185
2.0		22	1.4540
3.0		29	1.4247
5.0		30	1.3923
8.0		30	1.3923
15.0		30	1.3923

Application: Label De-warping for Produce Inspection

A common task in food and agricultural inspection: photograph a bag of rice (or any packaged produce) from an oblique angle and **rectify the label to a flat, frontal view** for OCR or quality inspection. This is a direct application of homography — we compute H from four corner correspondences and warp the image to a canonical rectangle.

The Pipeline

1. Detect the four label corners in the image (manually or via a detector).
 2. Define the target rectangle (canonical flat view).
 3. Compute H via DLT from the 4 correspondences.
 4. Warp the image using H^{-1} (backward mapping to avoid holes).
-

Step 1–3: Synthesise a Warped Label and Compute H

```
import numpy as np
import matplotlib.pyplot as plt
import matplotlib.patches as mpatches

def dlt_homography(src, dst):
    """DLT homography. src,dst: shape (n,2). Returns H: shape (3,3)."""
    n = len(src); A = np.zeros((2*n, 9))
    for i, ((x, y), (u, v)) in enumerate(zip(src, dst)):
        A[2*i ] = [-x, -y, -1, 0, 0, 0, u*x, u*y, u]
```

```

    A[2*i+1] = [ 0, 0, 0, -x, -y, -1, v*x, v*y, v]
    _, _, Vt = np.linalg.svd(A)
    H = Vt[-1].reshape(3, 3)
    return H / H[2, 2]

# Canonical label: 400 × 200 px rectangle
W_label, H_label = 400, 200

# Four corners of the canonical label (destination)
dst_corners = np.array([
    [ 0., 0.],
    [W_label, 0.],
    [W_label, H_label],
    [0., H_label],
], dtype=float) # shape (4, 2)

# Four corners as they appear in the "photo" (source) - perspective distorted
# Simulating a bag tilted and slightly rotated on a conveyor belt
src_corners = np.array([
    [120., 80.],
    [480., 60.],
    [510., 340.],
    [ 95., 370.],
], dtype=float) # shape (4, 2)

H_img_to_label = dlt_homography(src_corners, dst_corners) # shape (3, 3)
H_label_to_img = dlt_homography(dst_corners, src_corners) # shape (3, 3)

print("H (image → canonical label):")
print(H_img_to_label.round(4))

# Synthesise a fake "photo" by warping a structured label pattern backward
# (i.e., for each pixel in the photo, look up its label coordinate)
photo_h, photo_w = 450, 640
label_img = np.ones((H_label, W_label, 3)) * 0.92 # off-white background

# Draw label features: coloured stripes and a text-like rectangle
label_img[20:60, :, :] = [0.85, 0.20, 0.20] # red header stripe
label_img[170:200, :, :] = [0.20, 0.45, 0.75] # blue footer stripe
label_img[70:150, 80:320, :] = [1.0, 0.9, 0.7] # main text area
# Add a simulated barcode area (dark stripes)
for j in range(280, 390, 6):
    label_img[80:140, j:j+3, :] = 0.1

# Backward warp: for each pixel (u,v) in photo, compute label coords
photo_img = np.ones((photo_h, photo_w, 3)) * 0.75 # grey background

```

```

uu, vv = np.meshgrid(np.arange(photo_w), np.arange(photo_h)) # each shape (450,640)
pixels_h = np.column_stack([uu.ravel(), vv.ravel(), np.ones(photo_h*photo_w)]) # shape (N,3)

# Apply H_img_to_label to each photo pixel to find label coords
mapped = (H_img_to_label @ pixels_h.T).T # shape (N, 3)
lx = mapped[:, 0] / mapped[:, 2] # shape (N,)
ly = mapped[:, 1] / mapped[:, 2] # shape (N,)

# Gather pixels within label bounds
valid = (lx >= 0) & (lx < W_label-1) & (ly >= 0) & (ly < H_label-1)
lxi = lx[valid].astype(int) # shape (M,)
lyi = ly[valid].astype(int) # shape (M,)
pui = uu.ravel()[valid] # shape (M,)
pvi = vv.ravel()[valid] # shape (M,)

photo_img[pvi, pui] = label_img[lyi, lxi]

fig, axes = plt.subplots(1, 3, figsize=(13, 4))

# Label
axes[0].imshow(label_img, origin='upper', extent=[0, W_label, H_label, 0])
axes[0].set_title('Canonical Label\n(ground truth)', fontsize=10)
axes[0].set_xlabel('u (px)'); axes[0].set_ylabel('v (px)')
axes[0].axis('on')

# Photo (warped)
axes[1].imshow(photo_img, origin='upper', extent=[0, photo_w, photo_h, 0])
# Draw the four corner markers
axes[1].scatter(src_corners[:, 0], src_corners[:, 1],
               color='tomato', s=60, zorder=5, label='Detected corners')
poly = plt.Polygon(src_corners, fill=False, edgecolor='tomato', lw=2)
axes[1].add_patch(poly)
axes[1].set_xlim(0, photo_w); axes[1].set_ylim(photo_h, 0)
axes[1].set_title('Camera Photo\n(oblique view)', fontsize=10)
axes[1].set_xlabel('u (px)')
axes[1].legend(fontsize=8)

# Rectified label
# Forward warp: for each label pixel, look up photo colour
rect_img = np.ones((H_label, W_label, 3)) * 0.75
luu, lvv = np.meshgrid(np.arange(W_label), np.arange(H_label))
lpix = np.column_stack([luu.ravel(), lvv.ravel(), np.ones(W_label*H_label)])
back = (H_label_to_img @ lpix.T).T
bx = back[:, 0] / back[:, 2]; by = back[:, 1] / back[:, 2]
bv2 = (bx >= 0) & (bx < photo_w-1) & (by >= 0) & (by < photo_h-1)
rect_img[lvv.ravel()[bv2].astype(int), luu.ravel()[bv2].astype(int)] = \
    photo_img[by[bv2].astype(int), bx[bv2].astype(int)]

```

```

axes[2].imshow(rect_img, origin='upper', extent=[0, W_label, H_label, 0])
axes[2].set_title('De-warped Label\n(rectified via H)', fontsize=10)
axes[2].set_xlabel('u (px)')

for ax in axes:
    ax.grid(True, alpha=0.1)

plt.tight_layout()
plt.show()

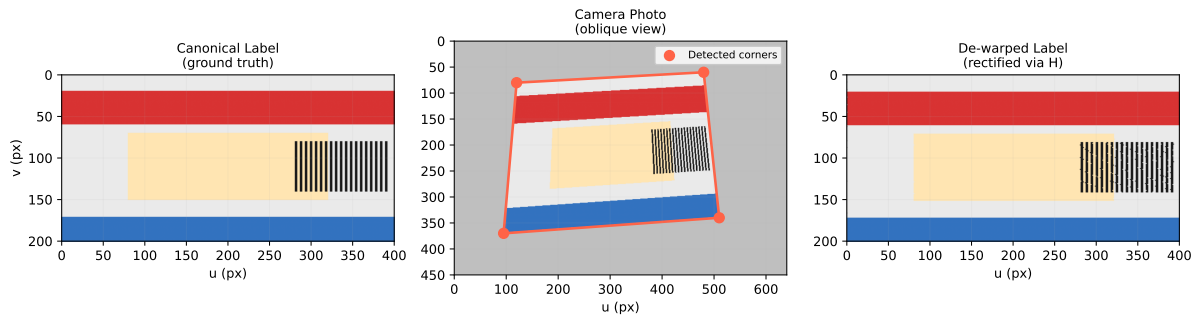
```

H (image $\rightarrow$ canonical label):

```

[[ 1.137600e+00  9.810000e-02 -1.443563e+02]
 [ 4.630000e-02  8.328000e-01 -7.217810e+01]
 [-0.000000e+00  6.000000e-04  1.000000e+00]]

```



Sensitivity: What Happens with Corner Detection Errors?

Even small errors in the detected corners degrade the rectification.

```

import numpy as np
import matplotlib.pyplot as plt

def dlt_homography(src, dst):
    n=len(src); A=np.zeros((2*n,9))
    for i,((x,y),(u,v)) in enumerate(zip(src,dst)):
        A[2*i]=[-x,-y,-1,0,0,0,u*x,u*y,u]; A[2*i+1]=[0,0,0,-x,-y,-1,v*x,v*y,v]
    _,_,Vt=np.linalg.svd(A); H=Vt[-1].reshape(3,3); return H/H[2,2]

W_label, H_label = 400, 200
dst_corners = np.array([[0.,0.],[W_label,0.],[W_label,H_label],[0.,H_label]])
src_corners = np.array([[120.,80.],[480.,60.],[510.,340.],[95.,370.]])

# Build label image

```

```

label_img = np.ones((H_label, W_label, 3)) * 0.92
label_img[20:60, :, :] = [0.85, 0.20, 0.20]
label_img[170:200, :, :] = [0.20, 0.45, 0.75]
label_img[70:150, 80:320, :] = [1.0, 0.9, 0.7]
for j in range(280, 390, 6):
    label_img[80:140, j:j+3, :] = 0.1

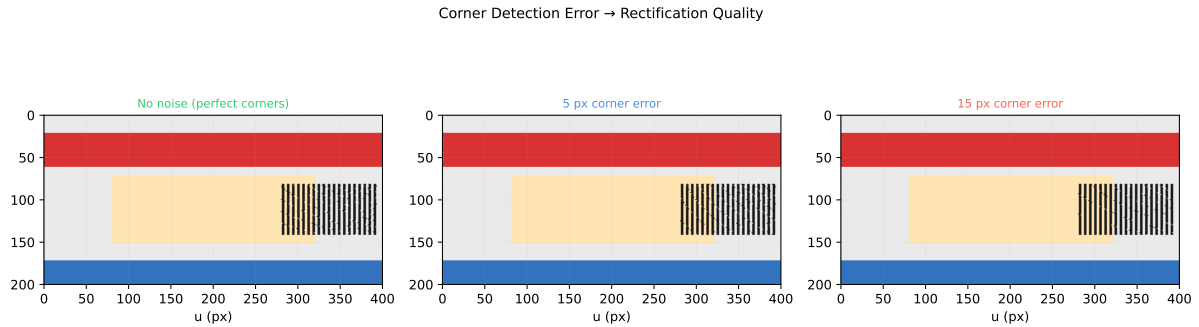
def rectify(src_c, dst_c, label_img, out_h, out_w):
    H_l2i = dlt_homography(dst_c, src_c)
    rect = np.ones((out_h, out_w, 3)) * 0.75
    luu,lvv = np.meshgrid(np.arange(out_w),np.arange(out_h))
    lpix = np.column_stack([luu.ravel(),lvv.ravel(),np.ones(out_w*out_h)])
    back = (H_l2i@lpix.T).T; bx=back[:,0]/back[:,2]; by=back[:,1]/back[:,2]
    # Get label colours (use nearest-neighbour source from label_img via forward H)
    H_i2l = dlt_homography(src_c, dst_c)
    ph,pw = 450,640
    photo_img = np.ones((ph,pw,3))*0.75
    uu,vv = np.meshgrid(np.arange(pw),np.arange(ph))
    pixels_h = np.column_stack([uu.ravel(),vv.ravel(),np.ones(ph*pw)])
    mapped=(H_i2l@pixels_h.T).T; lx=mapped[:,0]/mapped[:,2]; ly=mapped[:,1]/mapped[:,2]
    valid=(lx>=0)&(lx<W_label-1)&(ly>=0)&(ly<H_label-1)
    photo_img[vv.ravel()[valid].astype(int),uu.ravel()[valid].astype(int)]=\
        label_img[ly[valid].astype(int),lx[valid].astype(int)]
    bv2=(bx>=0)&(bx<pw-1)&(by>=0)&(by<ph-1)
    rect[lvv.ravel()[bv2].astype(int),luu.ravel()[bv2].astype(int)]=\
        photo_img[by[bv2].astype(int),bx[bv2].astype(int)]
    return rect

rng = np.random.default_rng(0)
fig, axes = plt.subplots(1, 3, figsize=(13, 4))
noise_levels = [0., 5., 15.]
titles = ['No noise (perfect corners)', '5 px corner error', '15 px corner error']
colors = ['#2ecc71', '#4a90d9', 'tomato']

for ax, noise, title, col in zip(axes, noise_levels, titles, colors):
    src_noisy = src_corners + rng.normal(0., noise, src_corners.shape)
    rect = rectify(src_noisy, dst_corners, label_img, H_label, W_label)
    ax.imshow(rect, origin='upper', extent=[0, W_label, H_label, 0])
    ax.set_title(title, fontsize=9, color=col)
    ax.set_xlabel('u (px)')
    ax.grid(True, alpha=0.1)

plt.suptitle('Corner Detection Error → Rectification Quality', fontsize=11)
plt.tight_layout()
plt.show()

```

Kaggle Exploration

Dataset: [Document Dewarping Dataset](#) or the **WarpDoc** benchmark (curved and perspective-distorted document images with ground-truth flat annotations).

```
# Document de-warping with OpenCV
# Assumes 4-corner annotations (u1,v1,...,u4,v4) in annotations.csv
import cv2
import numpy as np
import pandas as pd
from pathlib import Path

ann = pd.read_csv('annotations.csv') # columns: file, x1,y1,...,x4,y4

for _, row in ann.iterrows():
    img = cv2.imread(str(Path('images') / row['file']))
    h_img, w_img = img.shape[:2]

    src = np.array([[row['x1'],row['y1']], [row['x2'],row['y2']],
                    [row['x3'],row['y3']], [row['x4'],row['y4']]], dtype=np.float32)

    # Canonical output: aspect ratio from bounding box width/height
    W_out = int(np.linalg.norm(src[0]-src[1]))
    H_out = int(np.linalg.norm(src[0]-src[3]))
    dst = np.array([[0,0], [W_out,0], [W_out,H_out], [0,H_out]], dtype=np.float32)

    M = cv2.getPerspectiveTransform(src, dst) # shape (3, 3)
    rectified = cv2.warpPerspective(img, M, (W_out, H_out))

    out_path = Path('rectified') / row['file']
    out_path.parent.mkdir(exist_ok=True)
    cv2.imwrite(str(out_path), rectified)

print(f"Rectified {len(ann)} documents.")
```

Exercises and Solutions

Exercise 11.1 — Vanishing Point from Parallel Lines

Two pairs of parallel lines are detected in a 640×480 image:

- Pair A: through $(80, 400)$ – $(240, 200)$ and $(180, 400)$ – $(300, 200)$
- Pair B: through $(460, 400)$ – $(340, 200)$ and $(560, 400)$ – $(380, 200)$

- Compute the vanishing point of Pair A and Pair B separately.
- Do both pairs share the same vanishing point? What does that mean geometrically?
- Compute the horizon line through the two vanishing points.

```
import numpy as np

def line_from_pts(p, q):
    """Homogeneous line through two homogeneous 2D points."""
    return np.cross(p, q) # shape (3,)

def intersect(l1, l2):
    x = np.cross(l1, l2) # shape (3,)
    return x / x[2]

# Pair A: two lines
lA1 = line_from_pts(np.array([ 80.,400.,1.]), np.array([240.,200.,1.]))
lA2 = line_from_pts(np.array([180.,400.,1.]), np.array([300.,200.,1.]))
vpA = intersect(lA1, lA2) # shape (3,)

# Pair B: two lines
lB1 = line_from_pts(np.array([460.,400.,1.]), np.array([340.,200.,1.]))
lB2 = line_from_pts(np.array([560.,400.,1.]), np.array([380.,200.,1.]))
vpB = intersect(lB1, lB2) # shape (3,)

print(f"(a) Vanishing point A: ({vpA[0]:.1f}, {vpA[1]:.1f})")
print(f"    Vanishing point B: ({vpB[0]:.1f}, {vpB[1]:.1f})")

same = np.allclose(vpA[:2], vpB[:2], atol=2.)
print(f"\n(b) Same vanishing point? {same}")
if same:
    print("    → Both pairs are parallel in 3D (same direction).")
else:
    print("    → Different vanishing points → different 3D directions.")

# (c) Horizon line
```

```

horizon = np.cross(vpA, vpB)
horizon_n = horizon / np.linalg.norm(horizon[:2])
print(f"\n(c) Horizon line = {horizon_n.round(6)}")
a, b, c = horizon_n
v0 = -(a*0. + c) / b
v640 = -(a*640. + c) / b
print(f"    v at u=0:    {v0:.1f}")
print(f"    v at u=640: {v640:.1f}")

```

- (a) Vanishing point A: (480.0, -100.0)
 Vanishing point B: (260.0, 66.7)
- (b) Same vanishing point? False
 → Different vanishing points → different 3D directions.
- (c) Horizon line = [-0.603858 -0.797092 210.142475]
 v at u=0: 263.6
 v at u=640: -221.2
-

Exercise 11.2 — Homography Degrees of Freedom

- (a) A 3×3 matrix has 9 entries. Explain why a homography has only 8 DoF.
- (b) How many point correspondences are needed as a minimum? Why not fewer?
- (c) If you have 10 correspondences, how do you use the extra information?

```

import numpy as np

# (c) Overdetermined DLT: more correspondences → smaller residual (least squares)
# Simulate 10 correspondences with noise

def dlt_homography(src, dst):
    n=len(src); A=np.zeros((2*n,9))
    for i,((x,y),(u,v)) in enumerate(zip(src,dst)):
        A[2*i]=[-x,-y,-1,0,0,0,u*x,u*y,u]; A[2*i+1]=[0,0,0,-x,-y,-1,v*x,v*y,v]
    _,_,Vt=np.linalg.svd(A); H=Vt[-1].reshape(3,3); return H/H[2,2]

rng = np.random.default_rng(9)
H_true = np.array([[1.3, 0.2, 40.],[ -0.05, 1.1, 25.],[0.001, 0.0003, 1.]])

def gen_pairs(H, n, noise_px, rng):
    src = rng.uniform([50.,50.],[590.,430.],(n,2))
    sh = np.column_stack([src,np.ones(n)]); p=(H@sh.T).T

```

```

dst = p[:, :2]/p[:, 2:3] + rng.normal(0., noise_px, (n, 2))
return src, dst

def rms_err(H, src, dst):
    sh=np.column_stack([src,np.ones(len(src))]); p=(H@sh.T).T
    return np.sqrt(np.mean(np.linalg.norm(p[:, :2]/p[:, 2:3]-dst,axis=1)**2))

print("(c) Effect of number of correspondences (noise=1px):")
print(f"{'n pts':>8} {'RMS reproj error (px)':>22}")
for n in [4, 6, 8, 10, 15, 20, 40]:
    src, dst = gen_pairs(H_true, n, 1.0, rng)
    H_est = dlt_homography(src, dst)
    print(f"  {n:>4d}          {rms_err(H_est, src, dst):>18.4f}")

print("\n(a) Scale ambiguity removes 1 DoF: 9 - 1 = 8 DoF.")
print("(b) 4 pairs × 2 equations = 8 equations for 8 unknowns → minimum.")
print("    Fewer than 4 → underdetermined (infinitely many H solutions).")
print("(c) Overdetermined: SVD finds least-squares best fit in null space.")

```

(c) Effect of number of correspondences (noise=1px):

n pts	RMS reproj error (px)
4	0.0000
6	0.9474
8	0.7695
10	1468.0633
15	1.1099
20	1.1340
40	1.4182

(a) Scale ambiguity removes 1 DoF: 9 - 1 = 8 DoF.

(b) 4 pairs × 2 equations = 8 equations for 8 unknowns → minimum.

Fewer than 4 → underdetermined (infinitely many H solutions).

(c) Overdetermined: SVD finds least-squares best fit in null space.

Exercise 11.3 — DLT System Derivation

Starting from $\lambda \mathbf{u} = H \mathbf{x}$ (where $\mathbf{u} = (u, v, 1)^T$, $\mathbf{x} = (x, y, 1)^T$), derive the two rows of the DLT matrix for a single correspondence. Show that cross-multiplying eliminates λ .

```

import numpy as np

# Symbolic verification via numerical perturbation:
# If H is correct, then H @ x is proportional to u.

```

```

# The DLT rows enforce this proportionality exactly.

H = np.array([[1.2, 0.3, 40.],
              [-0.1, 1.05, 20.],
              [0.001, 0.0002, 1.]]) # shape (3, 3)

x = np.array([150., 200., 1.]) # shape (3,) - source point
p = H @ x                      # shape (3,) - image in homogeneous coords
u, v = p[0]/p[2], p[1]/p[2]

# DLT rows for this single correspondence
row1 = np.array([-x[0], -x[1], -1, 0, 0, 0, u*x[0], u*x[1], u]) # shape (9,)
row2 = np.array([ 0, 0, 0, -x[0], -x[1], -1, v*x[0], v*x[1], v]) # shape (9,)

h = H.ravel() # shape (9,)
print("DLT derivation verification:")
print(f"  Row 1 · h = {row1 @ h:.6f} (should be 0)")
print(f"  Row 2 · h = {row2 @ h:.6f} (should be 0)")
print()
print("Derivation sketch:")
print("  u = H[0]·x, v = H[1]·x, = H[2]·x")
print("  From u = H[0]·x and = H[2]·x:")
print("    H[2]·x · u = H[0]·x")
print("    0 = H[0]·x - u·H[2]·x = [-x,-y,-1, 0,0,0, u·x,u·y,u]·h")

```

DLT derivation verification:

Row 1 · h = 0.000000 (should be 0)

Row 2 · h = 0.000000 (should be 0)

Derivation sketch:

$u = H[0] \cdot x, v = H[1] \cdot x, = H[2] \cdot x$

From $u = H[0] \cdot x$ and $= H[2] \cdot x$:

$H[2] \cdot x \cdot u = H[0] \cdot x$

$0 = H[0] \cdot x - u \cdot H[2] \cdot x = [-x, -y, -1, 0, 0, 0, u \cdot x, u \cdot y, u] \cdot h$

Exercise 11.4 — RANSAC Iteration Count

(a) You have 100 feature matches with an estimated 35% outlier rate. How many RANSAC iterations are needed to achieve 99% confidence that at least one 4-sample is all-inlier?

(b) How does that change if your outlier rate is actually 50%?

(c) You want to halve the number of iterations at 35% outlier rate without changing confidence. What is the only way to achieve this?

```

import numpy as np

def ransac_n(eps, s=4, p=0.99):
    """Number of RANSAC iterations."""
    q = (1. - eps) ** s    # prob all-inlier sample
    if q < 1e-15: return int(1e9)
    return int(np.ceil(np.log(1.-p) / np.log(1.-q)))

eps_a, eps_b = 0.35, 0.50
Na = ransac_n(eps_a)
Nb = ransac_n(eps_b)

print(f"(a) eps=0.35: N = {Na} iterations")
print(f"(b) eps=0.50: N = {Nb} iterations  ({Nb/Na:.1f}× more)")

# (c) To halve N at eps=0.35, need q such that ceil(log(0.01)/log(1-q)) = Na/2
target = Na // 2
# Solve for q: q = 1 - (1-p)^(1/target) ... numerically
for s_try in range(3, 1, -1):
    N_try = ransac_n(eps_a, s=s_try)
    print(f"(c) s={s_try} (minimal sample): N = {N_try}  {' target?' if N_try <= target else ' '

print(f"\n    Only way to halve N: use a smaller minimal sample size (s=3)")
print(f"    or reduce outlier fraction  through better feature matching.")

```

```

(a) eps=0.35: N = 24 iterations
(b) eps=0.50: N = 72 iterations  (3.0× more)
(c) s=3 (minimal sample): N = 15
(c) s=2 (minimal sample): N = 9    target?

```

Only way to halve N: use a smaller minimal sample size (s=3)
or reduce outlier fraction through better feature matching.

Exercise 11.5 — Normalisation Matters

Compute H via DLT with and without Hartley normalisation for a set of 8 correspondences in a 1920×1080 image. Compare the reprojection errors.

```

import numpy as np

def normalise_points(pts):
    c = pts.mean(axis=0)
    s_arr = pts - c
    scale = np.sqrt(2.) / np.mean(np.linalg.norm(s_arr, axis=1))

```

```

T = np.array([[scale,0.,-scale*c[0]],[0.,scale,-scale*c[1]],[0.,0.,1.]])
pts_h = np.column_stack([pts, np.ones(len(pts))])
return (T @ pts_h.T).T[:, :2], T

def dlt_raw(src, dst):
    n=len(src); A=np.zeros((2*n,9))
    for i,((x,y),(u,v)) in enumerate(zip(src,dst)):
        A[2*i]=[-x,-y,-1,0,0,0,u*x,u*y,u]; A[2*i+1]=[0,0,0,-x,-y,-1,v*x,v*y,v]
    _,_,Vt=np.linalg.svd(A); H=Vt[-1].reshape(3,3); return H/H[2,2]

def dlt_normalised(src, dst):
    sn, Ts = normalise_points(src); dn, Td = normalise_points(dst)
    Hn = dlt_raw(sn, dn)
    H = np.linalg.solve(Td, Hn @ Ts); return H / H[2,2]

def rms(H, src, dst):
    sh=np.column_stack([src,np.ones(len(src))]); p=(H@sh.T).T
    return np.sqrt(np.mean(np.linalg.norm(p[:, :2]/p[:, 2:3]-dst,axis=1)**2))

rng = np.random.default_rng(5)
H_true = np.array([[1.4, 0.25, 80.],[-0.1, 1.15, 50.],[0.0005, 0.0002, 1.]])

src = rng.uniform([100., 100.], [1820., 980.], (8, 2)) # 1920×1080 coords
sh = np.column_stack([src, np.ones(8)]); p=(H_true@sh.T).T
dst = p[:, :2]/p[:, 2:3] + rng.normal(0., 0.5, (8,2))

H_r = dlt_raw(src, dst)
H_n = dlt_normalised(src, dst)

print(f"Image size: 1920×1080 (large pixel coordinates)")
print(f"RMS reprojection error:")
print(f"  Raw DLT:          {rms(H_r, src, dst):.4f} px")
print(f"  Normalised DLT:    {rms(H_n, src, dst):.4f} px")
print(f"\nNormalisation improvement factor: {rms(H_r,src,dst)/rms(H_n,src,dst):.1f}×")

```

Image size: 1920×1080 (large pixel coordinates)

RMS reprojection error:

Raw DLT: 0.3252 px

Normalised DLT: 0.3226 px

Normalisation improvement factor: 1.0×

Exercise 11.6 — Homography Inversion and Back-projection

Given a homography H mapping source→destination, what does H^{-1} do? Verify that applying H then H^{-1} returns the original points.

```
import numpy as np

H = np.array([[ 1.2, 0.3, 50.],
              [-0.1, 1.1, 30.],
              [ 0.001, 0.0005, 1.]]) # shape (3, 3)
H /= H[2, 2]

H_inv = np.linalg.inv(H) # shape (3, 3)
H_inv /= H_inv[2, 2]

src_pts = np.array([[100., 150.],
                    [300., 200.],
                    [450., 350.],
                    [200., 400.]], dtype=float) # shape (4, 2)

def apply_H(H, pts):
    n=len(pts); ph=np.column_stack([pts,np.ones(n)]); p=(H@ph.T).T
    return p[:,2]/p[:,2:3]

dst_pts = apply_H(H, src_pts) # shape (4, 2)
rec_pts = apply_H(H_inv, dst_pts) # shape (4, 2)

print(f"Round-trip error (H then H_inv):")
for s, r in zip(src_pts, rec_pts):
    err = np.linalg.norm(s - r)
    print(f"  src ({s[0]:.0f},{s[1]:.0f}) → dst → rec ({r[0]:.2f},{r[1]:.2f}) "
          f"err={err:.4f}")

print(f"\nH_inv is the destination→source mapping.")
print(f"In image warping: use H_inv for backward mapping to avoid holes.")
```

```
Round-trip error (H then H_inv):
src (100,150) → dst → rec (100.00,150.00) err=0.0000
src (300,200) → dst → rec (300.00,200.00) err=0.0000
src (450,350) → dst → rec (450.00,350.00) err=0.0000
src (200,400) → dst → rec (200.00,400.00) err=0.0000
```

```
H_inv is the destination→source mapping.
In image warping: use H_inv for backward mapping to avoid holes.
```


Exercise 11.7 — Condition Number and View Quality

(a) Construct a degenerate configuration of 4 source-destination pairs where DLT fails (collinear points). Explain why.

(b) What is the condition number of the DLT matrix A for your degenerate and a well-conditioned configuration?

```
import numpy as np

def build_A(src, dst):
    n=len(src); A=np.zeros((2*n,9))
    for i,((x,y),(u,v)) in enumerate(zip(src,dst)):
        A[2*i]=[-x,-y,-1,0,0,0,u*x,u*y,u]; A[2*i+1]=[0,0,0,-x,-y,-1,v*x,v*y,v]
    return A

# Degenerate: all 4 source points on the same line y=100
src_degenerate = np.array([[100.,100.],[200.,100.],[350.,100.],[500.,100.]])
dst_degenerate = np.array([[100.,100.],[200.,120.],[350.,140.],[500.,160.]])

# Well-conditioned: 4 corners of an image region
src_good = np.array([[80.,80.],[400.,80.],[400.,360.],[80.,360.]])
dst_good = np.array([[50.,50.],[500.,50.],[500.,400.],[50.,400.]])

A_deg = build_A(src_degenerate, dst_degenerate)
A_good = build_A(src_good, dst_good)

cond_deg = np.linalg.cond(A_deg)
cond_good = np.linalg.cond(A_good)

print(f"(b) Condition number of DLT matrix A:")
print(f" Degenerate (collinear): cond(A) = {cond_deg:.1f}")
print(f" Well-conditioned: cond(A) = {cond_good:.1f}")
print(f"\n(a) When source points are collinear, the DLT system loses rank -")
print(f" two columns of A become linearly dependent, so the null space")
print(f" is no longer 1-dimensional and H is not uniquely determined.")
print(f" The '4-point' requirement implicitly assumes no 3 are collinear.")
```

(b) Condition number of DLT matrix A :

Degenerate (collinear): $\text{cond}(A) = 31395656683175874560.0$

Well-conditioned: $\text{cond}(A) = 527853.1$

(a) When source points are collinear, the DLT system loses rank -
 two columns of A become linearly dependent, so the null space
 is no longer 1-dimensional and H is not uniquely determined.
 The '4-point' requirement implicitly assumes no 3 are collinear.

Part IV — Matrix Factorizations and Eigenstructure

LU, Cholesky, Determinants, and Numerical Conditioning

LU Factorization with Partial Pivoting

Solving a single linear system $A\mathbf{x} = \mathbf{b}$ is routine. But in ML/CV/robotics we almost never solve just one system — we solve the *same* A against many different right-hand sides:

- **Kalman filter.** The measurement update reuses the same innovation covariance across thousands of time steps.
- **Levenberg–Marquardt.** The damped normal matrix $(J^T J + \lambda I)$ is solved repeatedly as λ is adjusted.
- **Bundle adjustment.** Back-substituting structure updates uses the same Schur-complement factor many times.

Running Gaussian elimination from scratch each time costs $O(n^3)$ per solve. **Factorize** A once — $O(n^3)$ — and every subsequent solve with a new right-hand side costs only $O(n^2)$.

Why LU specifically? Cholesky (§12.2) is twice as fast but requires symmetric positive-definiteness. QR is more numerically robust but costs roughly twice as much work. LU is the general-purpose default: no assumption on A beyond invertibility, standard in every linear-algebra library.

Gaussian Elimination as a Factorization

Gaussian elimination subtracts multiples of one row from another until the matrix is upper triangular. Each row operation is a left-multiplication by an elementary lower-triangular matrix. The product of those elementary matrices is itself lower triangular — call it L^{-1} . The upper-triangular result is U . So

$$A = LU,$$

where L is lower triangular with **unit diagonal** and stores the multipliers, and U is upper triangular with the pivots on its diagonal.

```
import numpy as np

rng = np.random.default_rng(0)
A = rng.standard_normal((4, 4)) # shape (4, 4)

def naive_lu(A):
    """Doolittle LU without pivoting - educational only."""
    n = A.shape[0]
    L = np.eye(n) # shape (n, n)
    U = A.astype(float).copy() # shape (n, n)
    for k in range(n - 1):
        for i in range(k + 1, n):
            m = U[i, k] / U[k, k] # multiplier stored in L
            L[i, k] = m
            U[i, k:] -= m * U[k, k:]
    return L, U

L, U = naive_lu(A)
print("L:\n", L.round(3))
print("\nU:\n", U.round(3))
print("\n||LU - A|| =", np.linalg.norm(L @ U - A).round(12))
```

```
L:
[[ 1.    0.    0.    0. ]
 [-4.26  1.    0.    0. ]
 [-5.597 9.963 1.    0. ]
 [-18.492 13.227 1.149 1. ]]
```

```
U:
[[ 0.126 -0.132  0.64  0.105]
 [ 0.    -0.201  4.032  1.394]
 [ 0.     0.   -37.213 -13.26 ]
 [ 0.     0.     0.    -2.002]]
```

```
||LU - A|| = 0.0
```

The key observation: **the multipliers are the entries of L** . No separate bookkeeping needed — elimination and factorization are the same operation viewed differently.

Why We Need Pivoting

The naive algorithm divides by $U[k,k]$ at every step. If that entry is zero or tiny, the multiplier explodes and rounding error is catastrophic.

$$A = \begin{bmatrix} \varepsilon & 1 \\ 1 & 1 \end{bmatrix}, \quad \varepsilon = 10^{-18}.$$

The multiplier $m = 1/\varepsilon \approx 10^{18}$, and U_{22} rounds to -10^{18} in float64 — losing all information about the original 1.

```
import numpy as np

def naive_lu(A):
    n = A.shape[0]; L = np.eye(n); U = A.astype(float).copy()
    for k in range(n - 1):
        for i in range(k + 1, n):
            m = U[i, k] / U[k, k]; L[i, k] = m; U[i, k:] -= m * U[k, k:]
    return L, U

eps = 1e-18
A = np.array([[eps, 1.], [1., 1.]]) # shape (2, 2)
b = np.array([1., 2.])           # shape (2,)

L, U = naive_lu(A)
y = np.linalg.solve(L, b)
x_bad = np.linalg.solve(U, y)
x_good = np.linalg.solve(A, b) # numpy uses pivoting internally

print("No-pivot solution :", x_bad)
print("Pivoted solution :", x_good.round(6))
```

```
No-pivot solution : [0. 1.]
Pivoted solution : [1. 1.]
```

Partial pivoting fixes this: at each elimination step, permute rows so the largest-magnitude entry in the current column becomes the pivot. Every multiplier then satisfies $|m| \leq 1$, preventing catastrophic growth.

The result is

$$PA = LU,$$

where P is a permutation matrix recording the row swaps. Since $\det P = \pm 1$, the factorization is efficient and stable.

Forward and Back Substitution

Given $PA = LU$, solving $A\mathbf{x} = \mathbf{b}$ is three cheap steps:

1. **Permute:** $\mathbf{b}' = P\mathbf{b}$.
2. **Forward solve:** $L\mathbf{y} = \mathbf{b}'$ — one scalar division per row, using already-computed entries below.
3. **Back solve:** $U\mathbf{x} = \mathbf{y}$ — same structure from bottom up.

Forward substitution: row i costs i operations, so total $1 + 2 + \cdots + n = n(n+1)/2 = O(n^2)$.
Back substitution: the same. Compare to $O(n^3)$ for refactoring — an asymptotic win by a factor of n .

Multiple Right-Hand Sides in Practice

`scipy.linalg.lu_factor` returns the packed factorization once; `lu_solve` applies it to any new right-hand side.

```
import numpy as np
from scipy.linalg import lu, lu_factor, lu_solve

rng = np.random.default_rng(1)
n = 5
A = rng.standard_normal((n, n))    # shape (n, n)

# Inspect the factorization
P, L, U = lu(A)
print("||PA - LU|| =", np.linalg.norm(P @ A - L @ U).round(12))
print("max |L_ij| =", np.abs(L).max().round(4), "(should be 1)")

# Solve three right-hand sides with one factorization
lu_piv = lu_factor(A)
bs = rng.standard_normal((n, 3))    # shape (n, 3)
X = lu_solve(lu_piv, bs)             # shape (n, 3)
print("\n||AX - B|| =", np.linalg.norm(A @ X - bs).round(12))
```

```
||PA - LU|| = 6.640066540035
max |L_ij| = 1.0 (should be 1)

||AX - B|| = 0.0
```


Measured Speedup

```

import numpy as np
import matplotlib.pyplot as plt
from scipy.linalg import lu_factor, lu_solve
import time

rng = np.random.default_rng(42)
n = 400
A = rng.standard_normal((n, n)) # shape (n, n)

k_values = [1, 5, 10, 25, 50, 100, 200, 400]
t_refactor, t_reuse = [], []

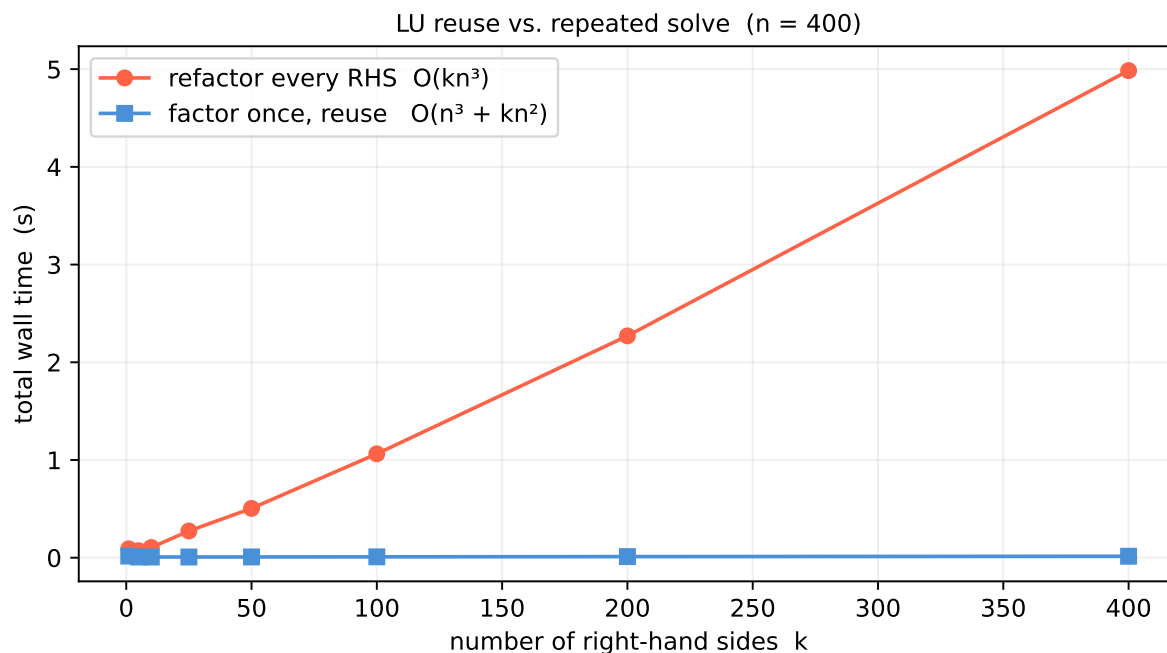
for k in k_values:
    bs = rng.standard_normal((n, k)) # shape (n, k)

    t0 = time.perf_counter()
    for j in range(k):
        np.linalg.solve(A, bs[:, j])
    t_refactor.append(time.perf_counter() - t0)

    t0 = time.perf_counter()
    lu_piv = lu_factor(A)
    for j in range(k):
        lu_solve(lu_piv, bs[:, j])
    t_reuse.append(time.perf_counter() - t0)

fig, ax = plt.subplots(figsize=(7, 4))
ax.plot(k_values, t_refactor, 'o-', color='tomato', lw=1.5, label='refactor every RHS 0(k)')
ax.plot(k_values, t_reuse, 's-', color='#4a90d9', lw=1.5, label='factor once, reuse 0(1)')
ax.set_xlabel('number of right-hand sides k')
ax.set_ylabel('total wall time (s)')
ax.set_title(f'LU reuse vs. repeated solve (n = {n})', fontsize=10)
ax.grid(True, alpha=0.2)
ax.legend()
plt.tight_layout()
plt.show()

```



For $k = 1$ the two curves are nearly equal — both cost one factorization plus one solve. As k grows the reuse curve slopes at $O(n^2k)$ while the refactor curve slopes at $O(n^3k)$, diverging by a factor of n .

Determinant and Inverse via LU

Two useful quantities fall out for free:

$$\det(A) = (-1)^{\#\text{swaps}} \prod_i u_{ii}.$$

This gives an $O(n^3)$ determinant (versus $O(n!)$ for cofactor expansion). We use this in §12.3.

If you truly need A^{-1} (usually you don't — solve instead), compute it by solving $AX = I$ column by column using the stored factorization: n solves of cost $O(n^2)$ each.

```
import numpy as np
from scipy.linalg import lu_factor, lu_solve

rng = np.random.default_rng(7)
A = rng.standard_normal((6, 6))    # shape (6, 6)
lu_piv = lu_factor(A)

# Determinant: product of U diagonal, sign from pivot vector
lu_mat, piv = lu_piv
```

```

sign = (-1) ** int(np.sum(piv != np.arange(len(piv))))
det_lu = sign * np.prod(np.diag(lu_mat))
print(f"det via LU = {det_lu:.6f}")
print(f"det via np = {np.linalg.det(A):.6f}")

# Inverse via solving A X = I
A_inv = lu_solve(lu_piv, np.eye(6)) # shape (6, 6)
print(f"\n||A A_inv - I|| = {np.linalg.norm(A @ A_inv - np.eye(6)):.2e}")

```

det via LU = -10.072754

det via np = -10.072754

||A A_inv - I|| = 7.80e-16

When LU Is Not the Right Tool

Situation	Prefer
A symmetric positive-definite	Cholesky (§12.2)
A rectangular / least-squares	QR or SVD
A sparse	Sparse LU / Cholesky (§12.5)
Extremely ill-conditioned	SVD with truncation
Too large for memory	Iterative solvers (CG, GMRES)

Exercises

12.1.1 Build $A = \begin{bmatrix} \varepsilon & 1 \\ 1 & 1 \end{bmatrix}$ for $\varepsilon = 10^{-15}$. Solve $A\mathbf{x} = (1, 2)^T$ with naive (no-pivot) LU and with `scipy.linalg.lu_solve`. Explain the discrepancy.

12.1.2 Forward substitution visits n rows. Row i requires $i - 1$ multiplications. What is the *exact* total multiplication count? Why is the leading term $n^2/2$, not n^2 ?

12.1.3 If A is already lower triangular with nonzero diagonal, what are L and U in its LU factorization? No pivoting needed — why?

12.1.4 Without the unit-diagonal convention, any $A = LU$ gives $A = (LD)(D^{-1}U)$ for any invertible diagonal D . Why does *fixing* the unit diagonal make the Doolittle factorization unique (assuming all pivots are nonzero)?

12.1.5 Run the timing benchmark above for $n = 100, 400, 1000$. At what k (number of RHS) does `lu_factor + lu_solve` first beat the naive loop? Does the crossover k depend on n ?

Cholesky Factorization

LU factorization works for any invertible matrix. But a large class of matrices that appear constantly in ML and robotics are not just invertible — they are **symmetric positive-definite** (SPD). Covariance matrices, Gram matrices $X^T X$, Hessians at a minimum, and information matrices in factor-graph SLAM are all SPD. Exploiting that symmetry cuts the work and storage in half compared to LU.

Cholesky factorization does exactly that. For any SPD matrix A , it finds a lower-triangular L with positive diagonal such that

$$A = L L^T.$$

Two wins at once: L exploits the symmetry of A (only the lower triangle of A is ever touched), and the diagonal entries of L are real and positive by construction — no pivoting is needed.

Why SPD Matrices Are Ubiquitous

A matrix $A \in \mathbb{R}^{n \times n}$ is **symmetric positive-definite** if $A = A^T$ and $\mathbf{x}^T A \mathbf{x} > 0$ for all nonzero $\mathbf{x}$. Three situations produce them constantly:

1. **Normal equations.** The least-squares problem $\min \|X\beta - \mathbf{y}\|^2$ leads to $(X^T X)\beta = X^T \mathbf{y}$. The matrix $X^T X$ is symmetric and PSD; it is SPD whenever X has full column rank.
2. **Covariance matrices.** $\Sigma = \mathbb{E}[(\mathbf{x} - \mu)(\mathbf{x} - \mu)^T]$ is always PSD, and SPD when the distribution is non-degenerate.
3. **Gauss-Newton / LM Hessian.** The approximate Hessian $J^T J$ (or $J^T J + \lambda I$) in nonlinear least-squares is always SPD.
4. **Information matrix in SLAM.** The inverse of the covariance, $\Omega = \Sigma^{-1}$, is SPD by inheritance.

The Algorithm

The Cholesky factorization fills L column by column, from left to right. Comparing $A = L L^T$ entry by entry gives:

- **Off-diagonal entry** ($i > j$): $l_{ij} = \frac{1}{l_{jj}} \left(a_{ij} - \sum_{k=1}^{j-1} l_{ik} l_{jk} \right)$
- **Diagonal entry** ($i = j$): $l_{jj} = \sqrt{a_{jj} - \sum_{k=1}^{j-1} l_{jk}^2}$

The square root is real only when the quantity under it is positive — which is guaranteed precisely when A is SPD. This gives us a free **positive-definiteness test**: attempt Cholesky; if it succeeds, A is PD.

```
import numpy as np

def cholesky_manual(A):
    """Doolittle-style Cholesky - educational, not production code."""
    n = A.shape[0]
    L = np.zeros_like(A, dtype=float) # shape (n, n)
    for j in range(n):
        s = A[j, j] - np.dot(L[j, :j], L[j, :j])
        if s <= 0:
            raise ValueError(f"Matrix not positive-definite at pivot {j}")
        L[j, j] = np.sqrt(s)
        for i in range(j + 1, n):
            L[i, j] = (A[i, j] - np.dot(L[i, :j], L[j, :j])) / L[j, j]
    return L

rng = np.random.default_rng(3)
X = rng.standard_normal((6, 4)) # shape (6, 4)
A = X @ X.T + 0.1 * np.eye(6) # shape (6, 6) - SPD

L_manual = cholesky_manual(A)
L_numpy = np.linalg.cholesky(A)

print("||L_manual - L_numpy|| =", np.linalg.norm(L_manual - L_numpy).round(10))
print("||L L^T - A|| =", np.linalg.norm(L_numpy @ L_numpy.T - A).round(10))
```

```
||L_manual - L_numpy|| = 0.0
||L L^T - A|| = 0.0
```

Cost: Half of LU

The factorization cost is $n^3/3$ flops — exactly **half** the $2n^3/3$ cost of LU. The savings come entirely from exploiting $a_{ij} = a_{ji}$: we only compute the lower triangle of L .

```
import numpy as np
import time

rng = np.random.default_rng(5)
ns = [200, 400, 800, 1200]
t_lu, t_chol = [], []
```

```

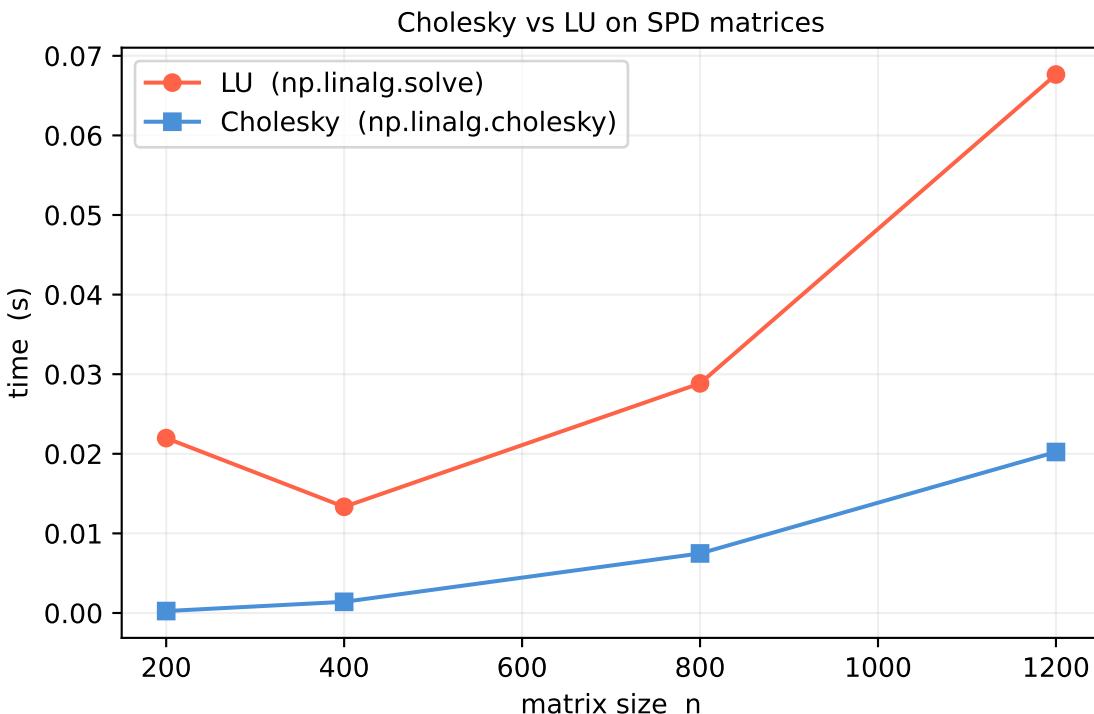
for n in ns:
    X = rng.standard_normal((n, n))
    A = X @ X.T + np.eye(n)    # shape (n, n) - SPD

    t0 = time.perf_counter()
    for _ in range(5):
        np.linalg.cholesky(A)
    t_chol.append((time.perf_counter() - t0) / 5)

    t0 = time.perf_counter()
    for _ in range(5):
        np.linalg.solve(A, np.eye(n))    # LU-based
    t_lu.append((time.perf_counter() - t0) / 5)

import matplotlib.pyplot as plt
fig, ax = plt.subplots(figsize=(6, 4))
ax.plot(ns, t_lu, 'o-', color='tomato', lw=1.5, label='LU (np.linalg.solve)')
ax.plot(ns, t_chol, 's-', color='#4a90d9', lw=1.5, label='Cholesky (np.linalg.cholesky)')
ax.set_xlabel('matrix size n')
ax.set_ylabel('time (s)')
ax.set_title('Cholesky vs LU on SPD matrices', fontsize=10)
ax.grid(True, alpha=0.2)
ax.legend()
plt.tight_layout()
plt.show()

```



Three Applications

1. Solving SPD Systems

Any SPD linear system $A\mathbf{x} = \mathbf{b}$ (normal equations, Kalman smoother, Gauss-Newton step) can be solved by:

1. $L = \text{chol}(A)$
2. Forward solve: $L\mathbf{y} = \mathbf{b}$
3. Back solve: $L^T\mathbf{x} = \mathbf{y}$

SciPy's `cho_factor` / `cho_solve` bundles this cleanly.

```
import numpy as np
from scipy.linalg import cho_factor, cho_solve

rng = np.random.default_rng(9)
X = rng.standard_normal((8, 5))      # shape (8, 5)
A = X.T @ X + 0.5 * np.eye(5)      # shape (5, 5) - SPD (normal equations-like)
b = rng.standard_normal(5)          # shape (5,)

cf = cho_factor(A)
x = cho_solve(cf, b)                # shape (5,)

print("||Ax - b|| =", np.linalg.norm(A @ x - b).round(12))
```

`||Ax - b|| = 0.0`

2. Sampling from a Multivariate Gaussian

To draw samples from $\mathcal{N}(\mu, \Sigma)$:

1. Compute $L = \text{chol}(\Sigma)$.
2. Draw $\mathbf{z} \sim \mathcal{N}(\mathbf{0}, I)$.
3. Return $\mathbf{x} = \mu + L\mathbf{z}$.

Because $\text{Cov}(L\mathbf{z}) = L I L^T = \Sigma$. This is how every neural network weight initializer, every Kalman Monte Carlo, and every GP posterior sampler works under the hood.

```
import numpy as np
import matplotlib.pyplot as plt

rng = np.random.default_rng(11)

# Covariance with correlation
```

```

Sigma = np.array([[2.0, 1.2],
                  [1.2, 1.0]]) # shape (2, 2) - SPD
mu     = np.array([1.0, -0.5]) # shape (2,)

L = np.linalg.cholesky(Sigma) # shape (2, 2)

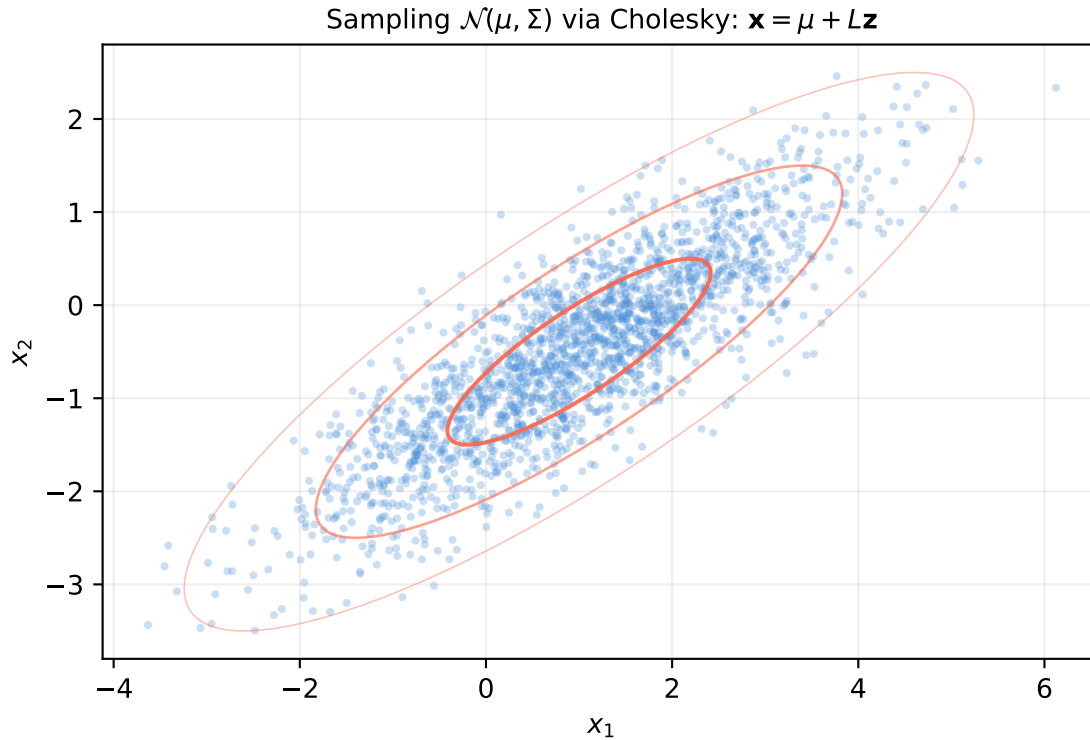
n_samples = 2000
Z = rng.standard_normal((2, n_samples)) # shape (2, n_samples)
X = mu[:, None] + L @ Z                 # shape (2, n_samples)

fig, ax = plt.subplots(figsize=(6, 5))
ax.scatter(X[0], X[1], s=4, alpha=0.3, color='#4a90d9')

# Confidence ellipse via eigendecomposition of Sigma
vals, vecs = np.linalg.eigh(Sigma) # shape (2,) and (2,2)
angles = np.linspace(0, 2*np.pi, 300)
circle = np.stack([np.cos(angles), np.sin(angles)]) # shape (2, 300)
for scale, lw, alpha in [(1, 1.5, 0.9), (2, 1.0, 0.6), (3, 0.6, 0.4)]:
    ellipse = mu[:, None] + vecs @ np.diag(np.sqrt(vals) * scale) @ circle
    ax.plot(ellipse[0], ellipse[1], color='tomato', lw=lw, alpha=alpha)

ax.set_xlabel('$x_1$'); ax.set_ylabel('$x_2$')
ax.set_title(r'Sampling $\mathcal{N}(\mu, \Sigma)$ via Cholesky: $\mathbf{x} = \mu + L\mathbf{z}$',
             fontsize=10)
ax.grid(True, alpha=0.2)
ax.set_aspect('equal')
plt.tight_layout()
plt.show()

```

3. Numerically Stable Log-Determinant

In probabilistic models (Gaussian log-likelihood, GP marginal likelihood) we often need $\log \det \Sigma$. Computing $\det \Sigma$ directly is numerically dangerous — the determinant of a large covariance can underflow to zero in float64. Via Cholesky:

$$\log \det \Sigma = 2 \sum_{i=1}^n \log l_{ii},$$

which is safe because all $l_{ii} > 0$ and we sum their logs.

```
import numpy as np

rng = np.random.default_rng(13)
X = rng.standard_normal((50, 40))
Sigma = X @ X.T + np.eye(50)    # shape (50, 50)

L = np.linalg.cholesky(Sigma)
logdet_chol = 2.0 * np.sum(np.log(np.diag(L)))
logdet_np   = np.linalg.slogdet(Sigma)[1]    # numpy's safe reference

print(f"log det via Cholesky : {logdet_chol:.6f}")
print(f"log det via slogdet  : {logdet_np:.6f}")
```

```
log det via Cholesky : 136.067284
log det via slogdet  : 136.067284
```

Note: for very large SPD systems `scipy.sparse.linalg` provides sparse Cholesky; we return to this in §12.5 and the Application section.

Cholesky as a PD Test

Attempting `np.linalg.cholesky(A)` and catching `LinAlgError` is a clean, $O(n^3)$ positive-definiteness test:

```
import numpy as np

def is_pd(A):
    try:
        np.linalg.cholesky(A)
        return True
    except np.linalg.LinAlgError:
        return False

print(is_pd(np.array([[4., 2.], [2., 3.] ]))) # True
print(is_pd(np.array([[1., 2.], [2., 1.] ]))) # False (eigenvalues 3, -1)
print(is_pd(np.array([[1., 1.], [1., 1.] ]))) # False (rank-1, PSD not PD)
```

```
True
False
False
```

Exercises

12.2.1 Verify by hand that $A = \begin{bmatrix} 4 & 2 \\ 2 & 3 \end{bmatrix}$ is SPD. Compute L by the column-by-column formulas and confirm $LL^T = A$.

12.2.2 Try `np.linalg.cholesky` on $A = \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix}$. What eigenvalues does A have? Why does Cholesky fail?

12.2.3 You have covariance Σ and its Cholesky factor L . Write a one-line expression to draw one sample from $\mathcal{N}(\mu, \Sigma)$ using `rng.standard_normal`.

12.2.4 Why doesn't Cholesky need pivoting (row swaps) to be numerically stable? Compare with LU where pivoting was essential.

12.2.5 A neural network initializer draws weights from $\mathcal{N}(0, \sigma^2 I)$. Show that Cholesky of $\sigma^2 I$ gives $L = \sigma I$, and that the sampling formula reduces to simply $\mathbf{w} = \sigma \mathbf{z}$ with $\mathbf{z} \sim \mathcal{N}(0, I)$.

The Determinant: Applied Perspective

The determinant of a square matrix is one of those quantities that introductory courses spend weeks on — cofactor expansions, 3×3 rules of Sarrus, the full permutation formula — and yet working engineers rarely compute it that way. The reason is simple: cofactor expansion costs $O(n!)$, which is completely impractical for anything larger than a 4×4 matrix.

What the determinant *means* geometrically is profound and worth understanding deeply. How to *compute* it efficiently is a one-liner using the LU factorization we already have.

Geometric Meaning: Signed Volume

The determinant of a matrix $A = [\mathbf{a}_1 \mid \cdots \mid \mathbf{a}_n]$ equals the **signed volume** of the parallelepiped spanned by its columns.

In 2D: $|\det A|$ is the area of the parallelogram formed by the two column vectors. The sign tells you whether the transformation $\mathbf{x} \mapsto A\mathbf{x}$ **preserves orientation** ($\det > 0$) or **reflects** it ($\det < 0$).

```
import numpy as np
import matplotlib.pyplot as plt

fig, axes = plt.subplots(1, 3, figsize=(11, 3.5))

def draw_parallelogram(ax, A, title, color='#4a90d9'):
    """Draw unit square and its image under A."""
    corners = np.array([[0,0],[1,0],[1,1],[0,1],[0,0]]).T # shape (2, 5)
    img = A @ corners # shape (2, 5)
    ax.fill(img[0], img[1], alpha=0.25, color=color)
    ax.plot(img[0], img[1], '-', color=color, lw=1.5)
    # unit square
    sq = np.array([[0,0],[1,0],[1,1],[0,1],[0,0]]).T
    ax.fill(sq[0], sq[1], alpha=0.1, color='gray')
    ax.plot(sq[0], sq[1], '--', color='gray', lw=0.8)
    ax.axhline(0, color='#333333', lw=0.4, alpha=0.5)
    ax.axvline(0, color='#333333', lw=0.4, alpha=0.5)
    d = np.linalg.det(A)
    ax.set_title(f'{title}\ndet = {d:.2f}', fontsize=9)
    ax.set_aspect('equal'); ax.grid(True, alpha=0.2)

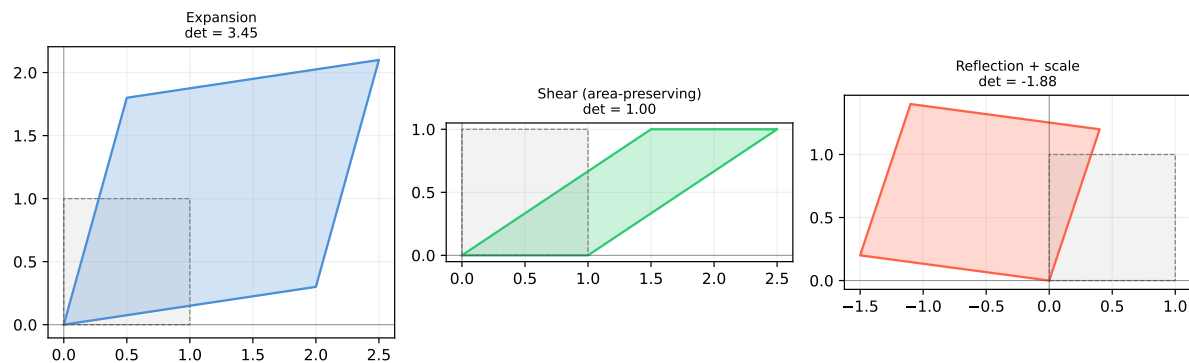
# det > 1: expansion
```

```
draw_parallelogram(axes[0], np.array([[2., 0.5], [0.3, 1.8]]),
                  'Expansion', color='#4a90d9')

# det = 1: area-preserving shear
draw_parallelogram(axes[1], np.array([[1., 1.5], [0., 1.]]),
                  'Shear (area-preserving)', color='#2ecc71')

# det < 0: reflection + scaling
draw_parallelogram(axes[2], np.array([[-1.5, 0.4], [0.2, 1.2]]),
                  'Reflection + scale', color='tomato')

plt.tight_layout()
plt.show()
```



Three things to internalize from this picture:

- $|\det A| > 1$: the transformation *expands* area.
- $|\det A| = 1$: area-preserving (rotations and shears).
- $|\det A| = 0$: the columns are linearly dependent — the parallelepiped has collapsed to a lower-dimensional flat. The matrix is singular.
- $\det A < 0$: orientation is reversed (the image of a counter-clockwise unit square becomes clockwise).

Cofactor Expansion (for Completeness)

For a 2×2 matrix the determinant is memorized:

$$\det \begin{bmatrix} a & b \\ c & d \end{bmatrix} = ad - bc.$$

For an $n \times n$ matrix, expanding along the first row gives

$$\det A = \sum_{j=1}^n (-1)^{1+j} a_{1j} \det(A_{1j}),$$

where A_{1j} is the $(n-1) \times (n-1)$ submatrix obtained by deleting row 1 and column j . This recursion is conceptually clean but costs $O(n!)$ — prohibitive for $n \geq 10$.

```
import numpy as np

def cofactor_det(A):
    """Recursive cofactor expansion. O(n!) - educational only."""
    n = A.shape[0]
    if n == 1:
        return A[0, 0]
    d = 0.
    for j in range(n):
        minor = np.delete(np.delete(A, 0, axis=0), j, axis=1)
        d += ((-1) ** j) * A[0, j] * cofactor_det(minor)
    return d

A = np.array([[3., 1., 2.],
              [0., 4., 1.],
              [2., 0., 3.]]) # shape (3, 3)

print("cofactor det  =", cofactor_det(A))
print("numpy det      =", np.linalg.det(A).round(6))
```

```
cofactor det  = 22.0
numpy det     = 22.0
```

For anything beyond 4×4 , always use the LU method below.

Computing det via LU: $O(n^3)$

From the factorization $PA = LU$:

$$\det(PA) = \det(P) \det(L) \det(U).$$

- $\det(L) = 1$ because L has unit diagonal.
- $\det(U) = \prod_i u_{ii}$ because U is triangular.
- $\det(P) = (-1)^{\#\text{row swaps}}$ because each row swap flips sign.

Therefore:

$$\det(A) = (-1)^{\#\text{swaps}} \prod_i u_{ii}.$$

This costs one LU factorization — $O(n^3)$ — then n multiplications.

```
import numpy as np
from scipy.linalg import lu_factor
import time

rng = np.random.default_rng(17)

for n in [5, 8, 12]:
    A = rng.standard_normal((n, n))    # shape (n, n)

    # LU-based det
    lu_mat, piv = lu_factor(A)
    sign = (-1) ** int(np.sum(piv != np.arange(n)))
    det_lu = sign * np.prod(np.diag(lu_mat))

    print(f"n={n:2d}  LU det = {det_lu:12.5f}  numpy det = {np.linalg.det(A):12.5f}")
```

```
n= 5  LU det =      1.11069  numpy det =      1.11069
n= 8  LU det =     10.67437  numpy det =     10.67437
n=12  LU det =    -2679.84646  numpy det =    -2679.84646
```

`np.linalg.det` internally does exactly this.

Properties Worth Knowing

Property	Formula	Geometric meaning
Multiplicativity	$\det(AB) = \det(A) \det(B)$	Volume scales multiply
Transpose	$\det(A^T) = \det(A)$	Row volume = column volume
Scaling	$\det(\alpha A) = \alpha^n \det(A)$	Each column scaled by α
Row swap	Flips sign	Orientation reversal
Singular	$\det(A) = 0 \Leftrightarrow A$ singular	Collapse to lower dimension

```
import numpy as np

rng = np.random.default_rng(19)
A = rng.standard_normal((4, 4)) # shape (4, 4)
B = rng.standard_normal((4, 4)) # shape (4, 4)

print("det(AB) =", np.linalg.det(A @ B).round(6))
print("det(A)*det(B) =", (np.linalg.det(A) * np.linalg.det(B)).round(6))

alpha = 3.0; n = 4
print(f"\ndet({alpha}A) = {np.linalg.det(alpha*A):.4f}")
print(f"{alpha}^n * det(A) = {alpha**n * np.linalg.det(A):.4f}")
```

det(AB) = -13.729263
det(A)*det(B) = -13.729263

det(3.0A) = 270.4864
3.0^4 * det(A) = 270.4864

Special Case: Rotations

A rotation matrix $R \in SO(n)$ satisfies:

- $R^T R = I$, so $\det(R^T R) = \det(R)^2 = 1$, giving $\det(R) = \pm 1$.
- For **proper** rotations: $\det(R) = +1$ (orientation preserved).
- $\det(R) = -1$ means R includes a **reflection** — it is no longer a pure rotation but a **roto-reflection** (improper rotation, not in $SO(n)$).

This is the algebraic definition of $SO(n)$: the group of $n \times n$ orthogonal matrices with determinant +1.

```
import numpy as np

def Rz(theta):
    c, s = np.cos(theta), np.sin(theta)
    return np.array([[ c, -s, 0.],
                    [ s,  c, 0.],
                    [ 0., 0., 1.]]) # shape (3, 3)

R = Rz(np.pi / 5)
print("det(R)          =", np.linalg.det(R).round(10), " (should be +1)")

# Improper rotation: proper rotation + reflection in z
R_improper = R.copy(); R_improper[2, 2] = -1.
print("det(R_improper)=", np.linalg.det(R_improper).round(10), " (= -1)")
```

```
det(R)          = 1.0  (should be +1)
det(R_improper)= -1.0  (= -1)
```

Jacobian Determinant: Local Area Scaling

For a differentiable map $\mathbf{f}: \mathbb{R}^n \rightarrow \mathbb{R}^n$, the Jacobian matrix $J_{\mathbf{f}}(\mathbf{x})$ describes the linear approximation at $\mathbf{x}$. Its determinant $|\det J_{\mathbf{f}}(\mathbf{x})|$ gives the **local volume scaling factor**:

$$(\text{volume of } \mathbf{f}(S)) \approx |\det J_{\mathbf{f}}(\mathbf{x})| \cdot (\text{volume of } S)$$

for small regions S near $\mathbf{x}$.

This is the change-of-variables formula in integration:

$$\int_{\mathbf{f}(\Omega)} g(\mathbf{y}) d\mathbf{y} = \int_{\Omega} g(\mathbf{f}(\mathbf{x})) |\det J_{\mathbf{f}}(\mathbf{x})| d\mathbf{x}.$$

Canonical example: polar coordinates. $\mathbf{f}(r, \theta) = (r \cos \theta, r \sin \theta)$.

$$J = \begin{bmatrix} \cos \theta & -r \sin \theta \\ \sin \theta & r \cos \theta \end{bmatrix}, \quad \det J = r.$$

The familiar $r dr d\theta$ factor in polar integration is literally this Jacobian determinant.

```
import numpy as np
import matplotlib.pyplot as plt

# Visualise local area scaling of polar map at two points
fig, axes = plt.subplots(1, 2, figsize=(9, 3.8))

for ax, (r0, th0) in zip(axes, [(0.5, np.pi/4), (2.0, np.pi/4)]):
    dr, dth = 0.15, 0.3

    # Grid in (r, theta) space - a small square
    r_vals = np.linspace(r0, r0 + dr, 30)
    th_vals = np.linspace(th0, th0 + dth, 30)
    R, TH = np.meshgrid(r_vals, th_vals)
    X = R * np.cos(TH); Y = R * np.sin(TH)

    ax.plot(X, Y, color='#4a90d9', lw=0.5, alpha=0.8)
    ax.plot(X.T, Y.T, color='#4a90d9', lw=0.5, alpha=0.8)

    jac = r0    # |det J| = r
    area_param = dr * dth
```

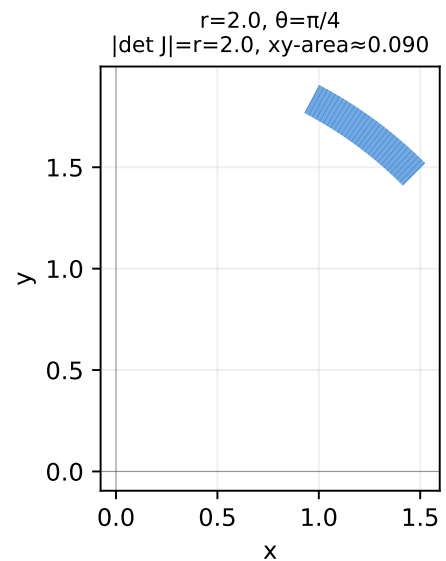
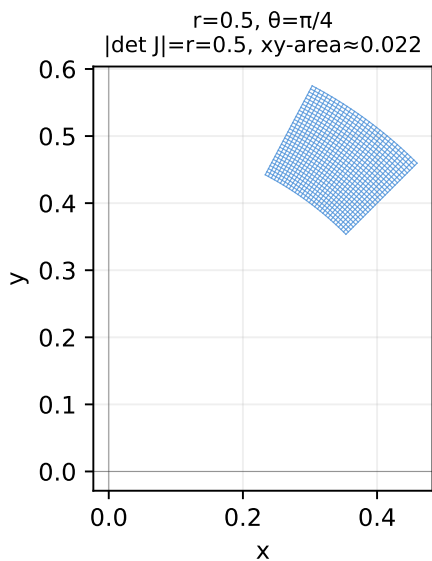


```

area_xy    = jac * area_param
ax.set_title(f'r={r0}, = /4\n|det J|=r={r0:.1f}, xy-area {area_xy:.3f}',
             fontsize=9)
ax.axhline(0, color='#333333', lw=0.4, alpha=0.5)
ax.axvline(0, color='#333333', lw=0.4, alpha=0.5)
ax.set_aspect('equal'); ax.grid(True, alpha=0.2)
ax.set_xlabel('x'); ax.set_ylabel('y')

plt.suptitle('Polar map: small rectangle in $(r, \theta)$ maps to a curved patch',
             fontsize=10, y=1.02)
plt.tight_layout()
plt.show()

```

Polar map: small rectangle in (r, θ) maps to a curved patch

The patch at $r = 0.5$ is small; at $r = 2$ the same $dr \times d\theta$ rectangle maps to a much larger area — exactly the factor r .

In robotics, the Jacobian determinant of a forward-kinematics map tells you where the robot is near a **kinematic singularity**: $\det J = 0$ means the end-effector loses a degree of freedom at that configuration.

Exercises

12.3.1 Verify $\det(AB) = \det(A) \det(B)$ for two random 3×3 matrices. Then argue geometrically: if A scales volume by $\det A$ and B by $\det B$, why must the composition scale by $\det(A) \det(B)$?

12.3.2 A 3×3 rotation R has $\det(R) = +1$. Someone gives you a matrix with $\det = -1$ and claims it's a rotation. What is it, and can it represent a physical rigid-body rotation?

12.3.3 Use `cofactor_det` and `np.linalg.det` to solve a 6×6 random system. Time both. How many times slower is cofactor expansion?

12.3.4 The map $(x, y) \mapsto (x^2, y^2)$ from $\mathbb{R}_{\geq 0}^2$ to itself has Jacobian $J = \text{diag}(2x, 2y)$. Where is $|\det J| = 0$? What does that mean geometrically?

12.3.5 For a diagonal matrix $D = \text{diag}(d_1, \dots, d_n)$, what is $\det D$? Use the signed-volume interpretation to explain why.

Condition Number and Numerical Stability

Every numerical solver ultimately operates in floating-point arithmetic, which introduces tiny rounding errors at every operation. Usually those errors are harmless — float64 carries about 16 significant digits, which is far more than any physical measurement. But some linear systems *amplify* rounding errors catastrophically, turning 16 correct digits of input into 4 correct digits of output. The **condition number** tells you exactly how bad that amplification can be.

Setup: Perturbations in the Right-Hand Side

Suppose we solve $A\mathbf{x} = \mathbf{b}$ and get $\mathbf{x}$, but the true right-hand side is $\mathbf{b} + \delta\mathbf{b}$ (measurement noise, rounding, numerical residual). The true solution differs by $\delta\mathbf{x}$ where $A(\delta\mathbf{x}) = \delta\mathbf{b}$, i.e., $\delta\mathbf{x} = A^{-1}\delta\mathbf{b}$.

How large can $\delta\mathbf{x}$ be relative to $\mathbf{x}$?

$$\frac{\|\delta\mathbf{x}\|}{\|\mathbf{x}\|} \leq \|A\| \|A^{-1}\| \frac{\|\delta\mathbf{b}\|}{\|\mathbf{b}\|}.$$

The factor $\|A\| \|A^{-1}\|$ is the **condition number**:

$$\kappa(A) = \|A\|_2 \|A^{-1}\|_2 = \frac{\sigma_{\max}}{\sigma_{\min}}.$$

Here $\sigma_{\max}$ and $\sigma_{\min}$ are the largest and smallest singular values of A . The condition number is always ≥ 1 , and $\kappa(I) = 1$ (the best possible).

Geometric Intuition

Think of A as a linear map. It stretches space by at most $\sigma_{\max}$ in one direction and squashes it by $\sigma_{\min}$ in another. Solving $A\mathbf{x} = \mathbf{b}$ means inverting that map. A perturbation in $\mathbf{b}$ that happens to point in the *most squashed* direction (the direction A nearly sends to zero) will be amplified by $1/\sigma_{\min}$ when inverted. The ratio $\sigma_{\max}/\sigma_{\min}$ captures the worst-case mismatch between the forward and inverse stretching.

```
import numpy as np
import matplotlib.pyplot as plt

fig, axes = plt.subplots(1, 2, figsize=(10, 4))

for ax, (s1, s2, title) in zip(axes, [
    (2.0, 1.8, r'$\kappa \approx 1.1$ (well-conditioned)'),
    (3.0, 0.2, r'$\kappa = 15$ (ill-conditioned)')]):

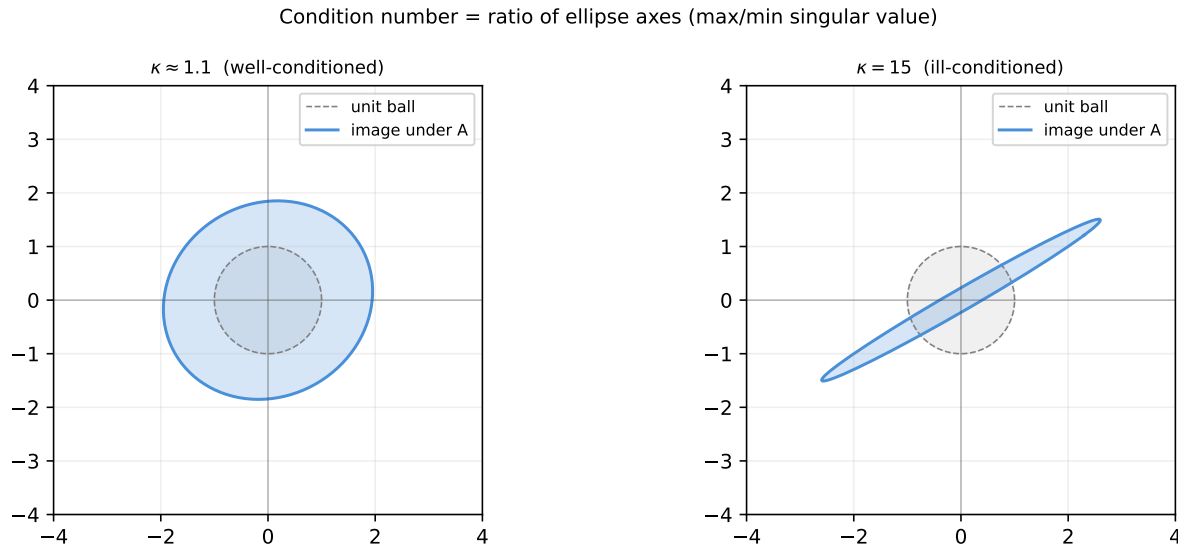
    theta = np.linspace(0, 2 * np.pi, 300)
    circle = np.stack([np.cos(theta), np.sin(theta)]) # shape (2, 300)

    # Rotate 30 degrees
    c30, s30 = np.cos(np.pi / 6), np.sin(np.pi / 6)
    R = np.array([[c30, -s30], [s30, c30]])
    A = R @ np.diag([s1, s2]) @ R.T

    ellipse = A @ circle # shape (2, 300)

    ax.fill(circle[0], circle[1], alpha=0.12, color='gray')
    ax.plot(circle[0], circle[1], '--', color='gray', lw=0.8, label='unit ball')
    ax.fill(ellipse[0], ellipse[1], alpha=0.22, color='#4a90d9')
    ax.plot(ellipse[0], ellipse[1], '-', color='#4a90d9', lw=1.5, label='image under A')
    ax.axhline(0, color='#333333', lw=0.4, alpha=0.5)
    ax.axvline(0, color='#333333', lw=0.4, alpha=0.5)
    ax.set_aspect('equal')
    ax.set_xlim(-4, 4); ax.set_ylim(-4, 4)
    ax.grid(True, alpha=0.2)
    ax.set_title(title, fontsize=9)
    ax.legend(fontsize=8)

plt.suptitle('Condition number = ratio of ellipse axes (max/min singular value)',
            fontsize=10)
plt.tight_layout()
plt.show()
```



The Digit-Loss Rule

Floating-point arithmetic introduces relative rounding errors of order $\varepsilon_{\text{machine}}$:

Precision	$\varepsilon_{\text{machine}}$	Decimal digits
float32	$\approx 1.19 \times 10^{-7}$	~ 7
float64	$\approx 2.22 \times 10^{-16}$	~ 16

After solving $A\mathbf{x} = \mathbf{b}$ in floating-point, the relative error in $\mathbf{x}$ is roughly:

$$\frac{\|\mathbf{x}_{\text{computed}} - \mathbf{x}_{\text{true}}\|}{\|\mathbf{x}_{\text{true}}\|} \approx \kappa(A) \varepsilon_{\text{machine}}.$$

So you lose approximately $\log_{10} \kappa(A)$ **digits of accuracy**.

Examples:

- $\kappa = 10^4$, float64: $16 - 4 = 12$ correct digits. Fine.
- $\kappa = 10^{10}$, float64: $16 - 10 = 6$ correct digits. Acceptable for most applications.
- $\kappa = 10^{10}$, float32: $7 - 10 = -3$ — *negative* — meaning the answer is noise.
- $\kappa = 10^{14}$, float64: only 2 reliable digits. Switch to higher precision or reformulate.

```
import numpy as np
import matplotlib.pyplot as plt

rng = np.random.default_rng(23)
```

```

def hilbert(n):
    """Hilbert matrix - famously ill-conditioned."""
    i, j = np.meshgrid(np.arange(1, n+1), np.arange(1, n+1))
    return 1.0 / (i + j - 1)    # shape (n, n)

ns = range(2, 14)
kappas, errs64, errs32 = [], [], []

for n in ns:
    A = hilbert(n)                # shape (n, n)
    x_true = np.ones(n)          # shape (n,)
    b = A @ x_true               # shape (n,)

    x64 = np.linalg.solve(A.astype(np.float64), b.astype(np.float64))
    x32 = np.linalg.solve(A.astype(np.float32), b.astype(np.float32)).astype(np.float64)

    kappas.append(np.linalg.cond(A))
    errs64.append(np.linalg.norm(x64 - x_true) / np.linalg.norm(x_true) + 1e-20)
    errs32.append(np.linalg.norm(x32 - x_true) / np.linalg.norm(x_true) + 1e-20)

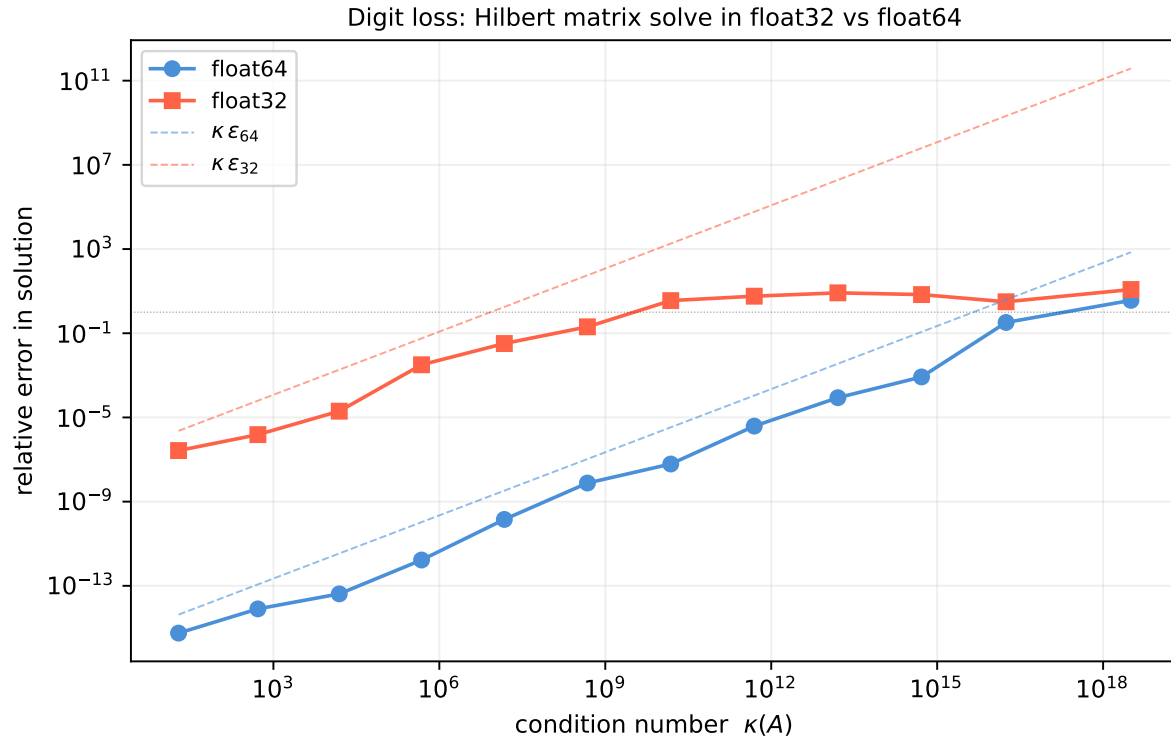
kappas = np.array(kappas)
errs64 = np.array(errs64)
errs32 = np.array(errs32)

fig, ax = plt.subplots(figsize=(7, 4.5))
ax.loglog(kappas, errs64, 'o-', color='#4a90d9', lw=1.5, label='float64')
ax.loglog(kappas, errs32, 's-', color='tomato', lw=1.5, label='float32')

# Reference lines: kappa * eps
eps64 = np.finfo(np.float64).eps
eps32 = np.finfo(np.float32).eps
ax.loglog(kappas, kappas * eps64, '--', color='#4a90d9', lw=0.8, alpha=0.6, label=r'$\kappa \epsilon_{64}$')
ax.loglog(kappas, kappas * eps32, '--', color='tomato', lw=0.8, alpha=0.6, label=r'$\kappa \epsilon_{32}$')

ax.set_xlabel(r'condition number $\kappa(A)$')
ax.set_ylabel('relative error in solution')
ax.set_title('Digit loss: Hilbert matrix solve in float32 vs float64', fontsize=10)
ax.axhline(1.0, color='black', lw=0.5, alpha=0.4, linestyle=':')
ax.grid(True, which='both', alpha=0.2)
ax.legend(fontsize=9)
plt.tight_layout()
plt.show()

```



Notice how the float32 curve hits 1.0 (completely wrong answer) once $\kappa \gtrsim 10^7$, while float64 stays reliable until $\kappa \approx 10^{14}$.

Why Camera Calibration Is Notoriously Ill-Conditioned

The intrinsic matrix K from Chapter 10 has entries on vastly different scales:

$$K = \begin{bmatrix} f_x & 0 & c_x \\ 0 & f_y & c_y \\ 0 & 0 & 1 \end{bmatrix}, \quad f_x, f_y \sim 1000 \text{ px}, \quad c_x, c_y \sim 500 \text{ px}, \quad K_{33} = 1.$$

The ratio between the largest and smallest entries is about 10^3 . The DLT calibration system (Chapter 10) assembles a matrix whose columns mix these scales. Without normalization the condition number is $\kappa \sim 10^6$ – 10^8 , consuming 6–8 digits of float64 precision.

This is why Zhang’s method normalizes image coordinates before assembling the DLT system — it brings κ down by several orders of magnitude.

```
import numpy as np

# Simulate a poorly-scaled 3x3 system like an unnormalized DLT subblock
# K-matrix scale mismatch
K = np.array([[1200.,    0., 640.],
```

```

        [ 0., 1200., 480.],
        [ 0.,   0.,   1.]]) # shape (3, 3)

print("Condition number of K:", np.linalg.cond(K).round(1))

# After dividing by image size ~ 1000 (rough normalization)
scale = 1000.
K_norm = K / scale
K_norm[2, 2] = 1. # keep the homogeneous entry
print("Condition number of K (normalized):", np.linalg.cond(K_norm).round(1))

```

Condition number of K: 1733.3
Condition number of K (normalized): 2.1

When to Use float32 vs float64

Situation	Recommendation
$\kappa \lesssim 10^4$, GPU inference	float32 safe (gains $\sim 2\times$ throughput)
Stochastic gradient descent	float32 fine (noise dominates anyway)
$\kappa \sim 10^6\text{--}10^9$	float64 preferred
Camera calibration, SLAM	float64 (ill-conditioned by nature)
$\kappa > 10^{12}$	Reformulate, precondition, or use higher precision

The rough rule: if the condition number times the machine epsilon exceeds your acceptable error tolerance, switch to higher precision or reformulate the problem.

Computing the Condition Number

```

import numpy as np

rng = np.random.default_rng(29)
A = rng.standard_normal((5, 5)) # shape (5, 5)

# Full condition number (ratio of singular values)
kappa = np.linalg.cond(A)
print(f"kappa(A)           = {kappa:.4f}")

```

```
# Via SVD directly
sigma = np.linalg.svd(A, compute_uv=False) # shape (5,)
print(f"sigma_max / sigma_min = {sigma[0] / sigma[-1]:.4f}")

# Singular matrix: infinite condition number
A_sing = A.copy(); A_sing[:, 4] = A_sing[:, 3] # make rank-4
print(f"\nkappa(singular A) = {np.linalg.cond(A_sing):.3e}")
```

```
kappa(A) = 18.3083
sigma_max / sigma_min = 18.3083
```

```
kappa(singular A) = 2.852e+17
```

For a singular matrix $\sigma_{\min} = 0$ so $\kappa = \infty$ — confirmed by the large number above (numpy returns a large finite value limited by float64 precision).

Exercises

12.4.1 What is $\kappa(I_n)$? What is $\kappa(R)$ for any orthogonal R ? Why do rotations cause zero digit loss?

12.4.2 Can $\kappa(A) < 1$? Prove it from the definition $\kappa = \sigma_{\max}/\sigma_{\min}$.

12.4.3 If $\kappa(A) = 10^{10}$, how many digits of the solution are reliable in float32? In float64? What if $\kappa = 10^6$?

12.4.4 `np.linalg.cond(A)` calls SVD internally ($O(n^3)$). Suppose you have already computed the Cholesky factor L of a symmetric PD matrix A . Can you estimate $\kappa(A)$ from L without a full SVD? (*Hint: LAPACK's `dpotri` offers a cheap reciprocal-condition-number estimator.*)

12.4.5 Build a 10×10 Hilbert matrix. Solve $A\mathbf{x} = \mathbf{b}$ with $\mathbf{b} = A\mathbf{1}$ in float32 and float64. How many digits agree with the true solution $\mathbf{x} = \mathbf{1}$? Does the prediction from $\kappa\epsilon_{\text{machine}}$ match the observed error?

Sparse Matrices: Formats and Solvers

The systems we have factorized so far assume that every entry of A is potentially nonzero — a **dense** matrix. But many of the largest linear systems in ML/CV/robotics are almost entirely zero:

- **SLAM factor graph.** A pose graph with 10,000 poses has a Hessian $H \in \mathbb{R}^{30000 \times 30000}$ (3 DOF per pose). Storing it as dense requires \$7 GB. But each pose connects to only a handful of neighbors, so H has roughly 10^5 nonzeros out of 9×10^8 entries — a **fill fraction of 0.01 %**.

- **Bundle adjustment.** The Jacobian of m observations over n 3D points and k cameras is $(2m) \times (3n + 6k)$. Each row has nonzeros in only one camera block and one point block — 9 nonzeros out of perhaps 10^6 columns.
- **FEM / PDE solvers.** Discretizing a PDE on a mesh connects each node to its neighbors. The stiffness matrix is banded.

Exploiting sparsity is the difference between solving a problem in seconds and it being computationally infeasible.

Storage Formats

Three formats cover essentially all use cases.

COO — Coordinate Format

Three arrays: `row`, `col`, `data` — one triple (i, j, v) per nonzero. Simple to build, but slow for arithmetic.

CSR — Compressed Sparse Row

Three arrays: - `data[k]`: the k -th nonzero value. - `indices[k]`: the column index of `data[k]`. - `indptr[i]`: offset in `data` where row i begins; `indptr[i+1]` is where it ends.

Row i has nonzeros at columns `indices[indptr[i]:indptr[i+1]]` with values `data[indptr[i]:indptr[i+1]]`

CSR supports fast $y = Ax$ (iterate over rows) and is the default for general sparse operations.

CSC — Compressed Sparse Column

The column-oriented transpose of CSR. Fast for $y = A^T x$ and column operations. Most sparse direct solvers (CHOLMOD, SuperLU) prefer CSC input.

```
import numpy as np
import scipy.sparse as sp

# Build a small sparse matrix in COO, then convert
row = np.array([0, 0, 1, 2, 2])
col = np.array([0, 2, 1, 0, 2])
data = np.array([3., 1., 4., 1., 5.])

A_coo = sp.coo_matrix((data, (row, col)), shape=(3, 3))
A_csr = A_coo.tocsr()
A_csc = A_coo.tocsc()
```

```
print("Dense:\n", A_coo.toarray())
print("\nCSR indptr  :", A_csr.indptr)
print("CSR indices :", A_csr.indices)
print("CSR data     :", A_csr.data)
print("\nnnz (nonzeros):", A_csr.nnz, "out of", 3*3)
```

Dense:

```
[[3. 0. 1.]
 [0. 4. 0.]
 [1. 0. 5.]]
```

```
CSR indptr  : [0 2 3 5]
CSR indices : [0 2 1 0 2]
CSR data     : [3. 1. 4. 1. 5.]
```

nnz (nonzeros): 5 out of 9

Rule of thumb: build in COO (easy to assemble), convert to CSR/CSC for factorization and matrix-vector products.

Sparsity Patterns: Visualizing with spy

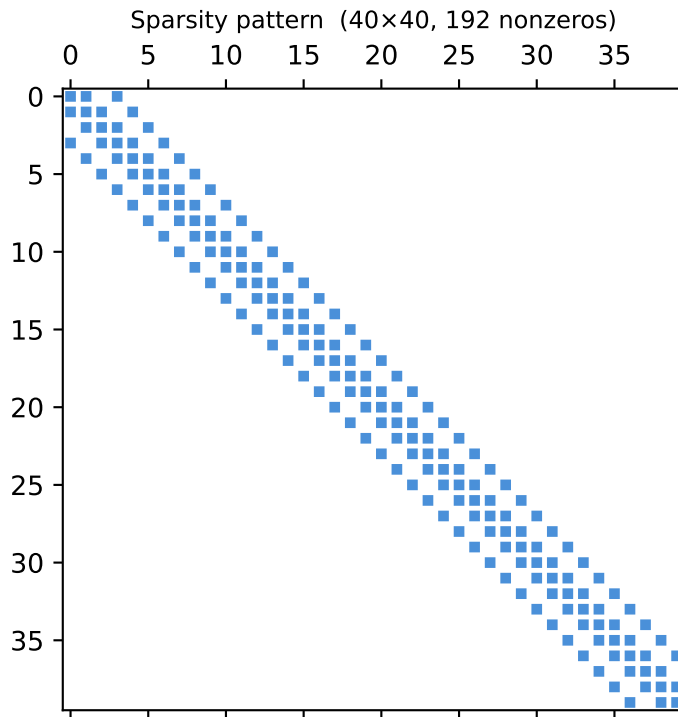
```
import numpy as np
import scipy.sparse as sp
import matplotlib.pyplot as plt

rng = np.random.default_rng(31)
n = 40

# Banded SPD matrix: tridiagonal + some off-diagonal connections
diags = [4*np.ones(n), -np.ones(n-1), -np.ones(n-1),
         0.5*np.ones(n-3), 0.5*np.ones(n-3)]
offsets = [0, 1, -1, 3, -3]
A = sp.diags(diags, offsets, shape=(n, n), format='csr')
A = A + A.T # make symmetric
# ensure diagonal dominance (ravel avoids numpy.matrix from .diagonal())
d = np.asarray(A.diagonal()).ravel()
A = A + sp.diags(4 - d)

fig, ax = plt.subplots(figsize=(4, 4))
ax.spy(np.asarray(A.todense()), markersize=3, color='#4a90d9')
ax.set_title(f'Sparsity pattern ({n}x{n}, {A.nnz} nonzeros)', fontsize=9)
```

```
plt.tight_layout()
plt.show()
```



Fill-In: Why Sparsity Doesn't Survive Naive Factorization

When you factorize a sparse matrix with Gaussian elimination, a zero entry $a_{ij} = 0$ can become nonzero in L or U — a **fill-in**. In the worst case (e.g., an arrow-head matrix) the factor is completely dense even though the original matrix was $O(n)$ nonzeros.

```
import numpy as np
import scipy.sparse as sp
from scipy.sparse.linalg import splu
import matplotlib.pyplot as plt

n = 30

# 1D Laplacian: banded, O(n) nonzeros
diag_main = 2.0 * np.ones(n)
diag_off = -1.0 * np.ones(n - 1)
A_lap = sp.diags([diag_main, diag_off, diag_off],
                 [0, 1, -1], format='csc')
```

```
# Arrow-head: diagonal + first row/col (star graph)
rows = np.concatenate([[0]*n, np.arange(1, n), np.arange(n)])
cols = np.concatenate([np.arange(n), np.zeros(n-1, int), np.arange(n)])
vals = np.concatenate([np.ones(n), np.ones(n-1), np.ones(n-1), np.ones(n)])
A_arrow = sp.csc_matrix((np.ones(len(rows)), (rows, cols)),
                        shape=(n, n))
A_arrow = A_arrow + A_arrow.T
A_arrow.setdiag(A_arrow.diagonal() + n) # make diagonally dominant

fig, axes = plt.subplots(2, 2, figsize=(8, 7))

for ax_orig, ax_factor, A, title in zip(
    [axes[0, 0], axes[1, 0]],
    [axes[0, 1], axes[1, 1]],
    [A_lap, A_arrow],
    ['1D Laplacian (banded)', 'Arrow-head']):

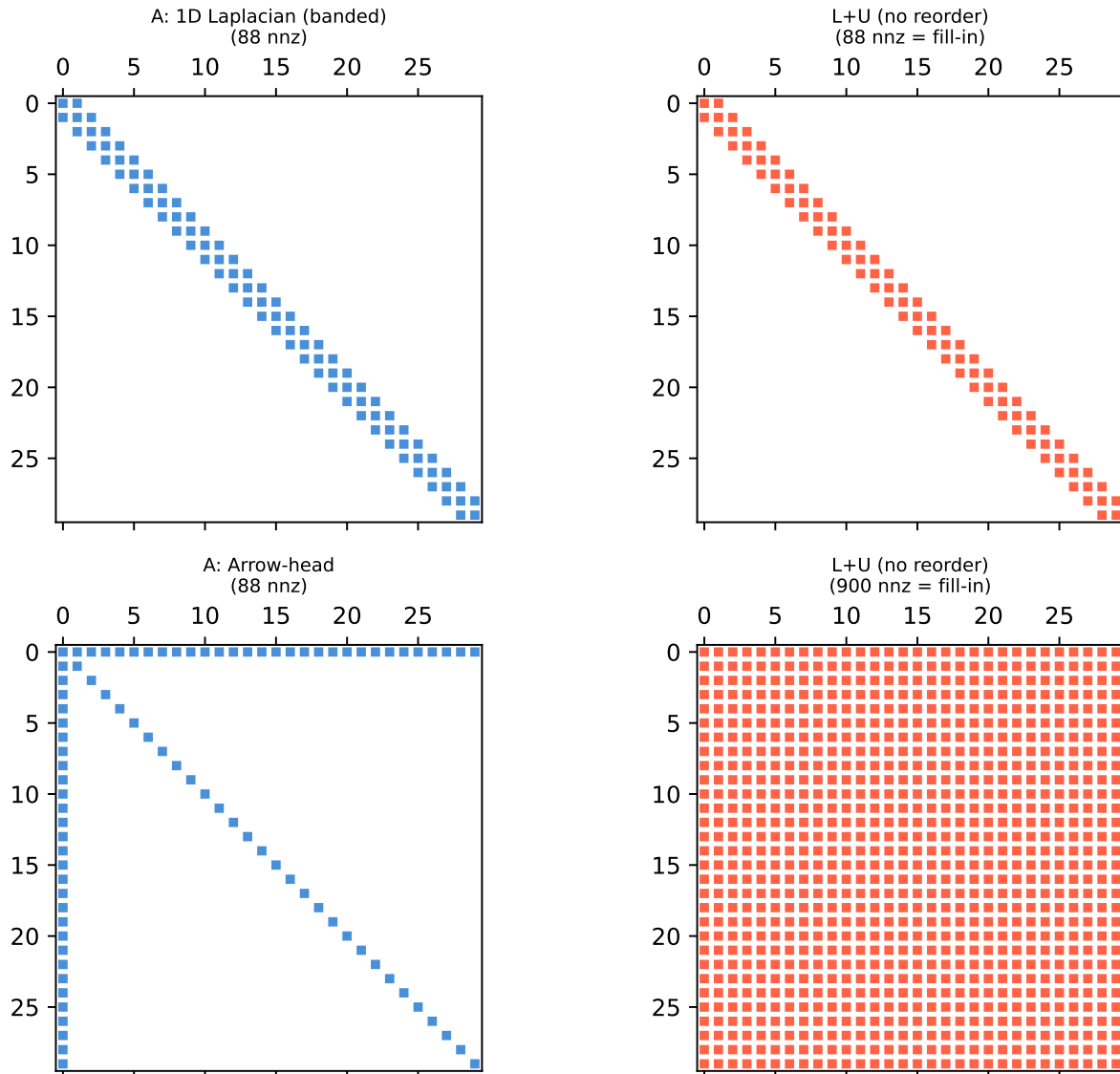
    lu_obj = splu(A, permc_spec='NATURAL')
    L_dense = lu_obj.L.toarray()
    U_dense = lu_obj.U.toarray()
    LU_combined = np.abs(L_dense) + np.abs(U_dense)

    ax_orig.spy(A.toarray(), markersize=3, color='#4a90d9')
    ax_orig.set_title(f'A: {title}\n({A.nnz} nnz)', fontsize=8)

    ax_factor.spy(LU_combined > 1e-12, markersize=3, color='tomato')
    nnz_lu = int(np.sum(LU_combined > 1e-12))
    ax_factor.set_title(f'L+U (no reorder)\n({nnz_lu} nnz = fill-in)', fontsize=8)

plt.suptitle('Fill-in: zero entries become nonzero during factorization', fontsize=10)
plt.tight_layout()
plt.show()
```

Fill-in: zero entries become nonzero during factorization



The banded Laplacian stays banded in $L + U$: fill-in is local. The arrow-head is catastrophic: every entry of L is nonzero because the first row/column connects every node to every other.

Reordering: Minimizing Fill-In

The key insight: **reordering rows and columns of A (with the same permutation on both sides for symmetric A) doesn't change the solution, but can dramatically reduce fill-in in L .**

Formally, for symmetric A we seek permutation P such that L in $PAP^T = LL^T$ is as sparse as possible. Finding the optimal P is NP-hard, but two heuristics work very well in practice:

- **AMD (Approximate Minimum Degree).** Greedily eliminates the variable that currently has the fewest connections in the *remaining* graph. Available as `scipy.sparse.csgraph.minimum_degree_ordering` or `scikit-sparse`.
- **METIS (Nested Dissection).** Recursively finds a small *separator* of the graph — nodes whose removal splits the graph in two — and eliminates interior nodes before the separator. Exceptional for mesh/graph-like sparsity (SLAM, FEM).

```
import numpy as np
import scipy.sparse as sp
from scipy.sparse.linalg import splu
import matplotlib.pyplot as plt

n = 50

# 2D Laplacian on n/5 x n/5 grid (unstructured but grid-like)
# Build manually: each interior node connected to 4 neighbors
side = int(n**0.5)
N = side * side
rows, cols, vals = [], [], []
for i in range(side):
    for j in range(side):
        idx = i * side + j
        rows.append(idx); cols.append(idx); vals.append(4.)
        for di, dj in [(-1,0),(1,0),(0,-1),(0,1)]:
            ni, nj = i+di, j+dj
            if 0 <= ni < side and 0 <= nj < side:
                rows.append(idx); cols.append(ni*side+nj); vals.append(-1.)
A2d = sp.csc_matrix((vals, (rows, cols)), shape=(N, N))

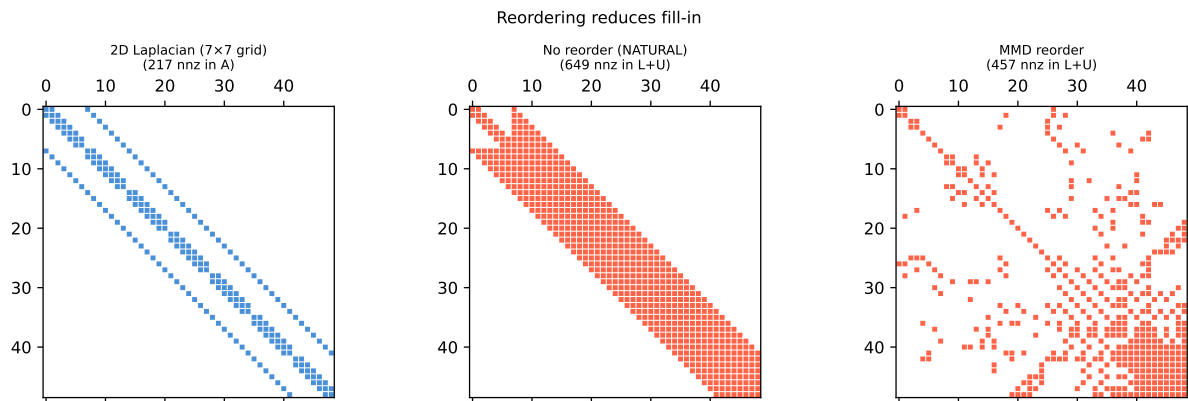
fig, axes = plt.subplots(1, 3, figsize=(11, 3.5))

lu_nat = splu(A2d, permc_spec='NATURAL')
lu_mmd = splu(A2d, permc_spec='MMD_AT_PLUS_A') # min-degree reordering

for ax, lu_obj, title in zip(axes[1:],
                             [lu_nat, lu_mmd],
                             ['No reorder (NATURAL)', 'MMD reorder']):
    L = lu_obj.L.toarray(); U = lu_obj.U.toarray()
    combo = np.abs(L) + np.abs(U)
    ax.spy(combo > 1e-12, markersize=2, color='tomato')
    nnz = int(np.sum(combo > 1e-12))
    ax.set_title(f'{title}\n({nnz} nnz in L+U)', fontsize=8)

axes[0].spy(A2d.toarray(), markersize=2, color='#4a90d9')
axes[0].set_title(f'2D Laplacian ({side}*{side} grid)\n({A2d.nnz} nnz in A)', fontsize=8)
plt.suptitle('Reordering reduces fill-in', fontsize=10)
```

```
plt.tight_layout()
plt.show()
```



Sparse Solvers in SciPy

```
# Direct sparse solve (SuperLU under the hood)
from scipy.sparse.linalg import spsolve
x = spsolve(A, b)

# Pre-factor for multiple right-hand sides
from scipy.sparse.linalg import factorized
solve = factorized(A) # returns a callable
x1 = solve(b1)
x2 = solve(b2)

# Iterative: Conjugate Gradient (SPD systems only)
from scipy.sparse.linalg import cg
x, info = cg(A, b, tol=1e-10)

# Iterative: GMRES (general systems)
from scipy.sparse.linalg import gmres
x, info = gmres(A, b, tol=1e-10)
```

```
import numpy as np
import scipy.sparse as sp
from scipy.sparse.linalg import spsolve, cg
import time

rng = np.random.default_rng(37)
ns = [100, 300, 700, 1200]
```

```

t_dense, t_sparse, t_cg = [], [], []

for n in ns:
    # 1D Laplacian (tridiagonal SPD)
    main = 2. * np.ones(n)
    off = -1. * np.ones(n - 1)
    A_sp = sp.diags([main, off, off], [0, 1, -1], format='csc')
    A_den = A_sp.toarray()
    b = rng.standard_normal(n) # shape (n,)

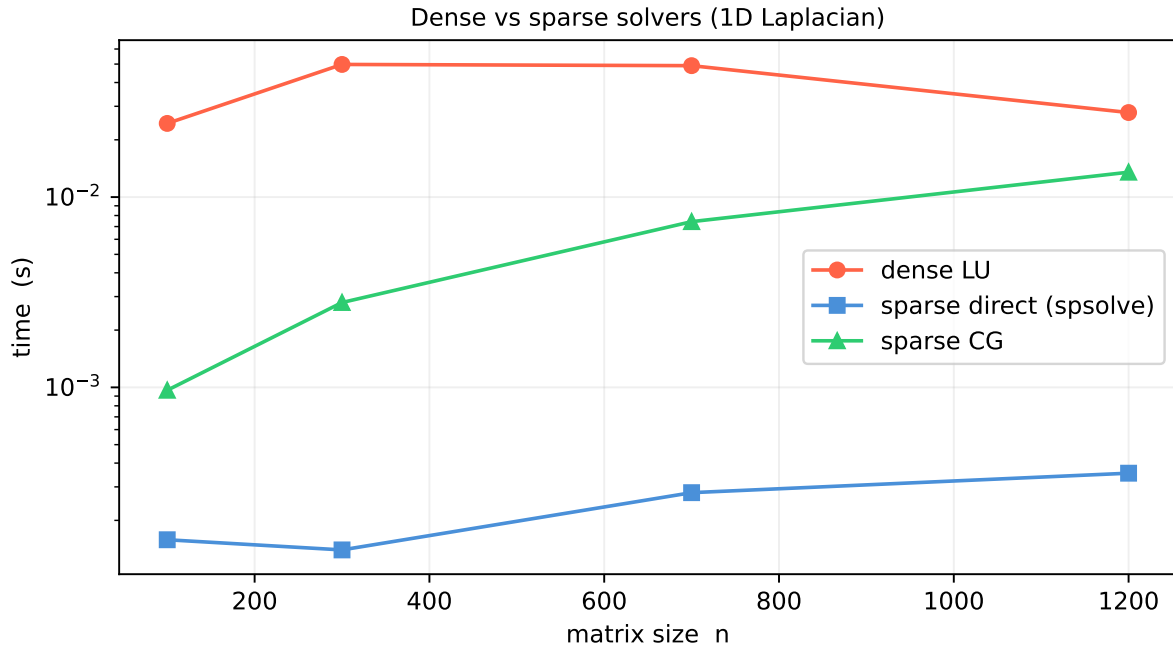
    t0 = time.perf_counter()
    np.linalg.solve(A_den, b)
    t_dense.append(time.perf_counter() - t0)

    t0 = time.perf_counter()
    spsolve(A_sp, b)
    t_sparse.append(time.perf_counter() - t0)

    t0 = time.perf_counter()
    cg(A_sp, b, rtol=1e-10)
    t_cg.append(time.perf_counter() - t0)

import matplotlib.pyplot as plt
fig, ax = plt.subplots(figsize=(7, 4))
ax.semilogy(ns, t_dense, 'o-', color='tomato', lw=1.5, label='dense LU')
ax.semilogy(ns, t_sparse, 's-', color='#4a90d9', lw=1.5, label='sparse direct (spsolve)')
ax.semilogy(ns, t_cg, '^-', color='#2ecc71', lw=1.5, label='sparse CG')
ax.set_xlabel('matrix size n')
ax.set_ylabel('time (s)')
ax.set_title('Dense vs sparse solvers (1D Laplacian)', fontsize=10)
ax.grid(True, alpha=0.2)
ax.legend()
plt.tight_layout()
plt.show()

```

For the banded Laplacian, sparse direct is orders of magnitude faster than dense LU — which spends $O(n^3)$ on operations that are already zero in A .

When to Use Each Solver

System	Solver
Dense, general	LU (<code>np.linalg.solve</code>)
Dense, SPD	Cholesky (<code>scipy.linalg.cho_factor</code>)
Sparse, general	<code>spsolve</code> (direct) or GMRES (iterative)
Sparse, SPD, many RHS	<code>factorized</code> or sparse Cholesky (<code>CHOLMOD</code>)
Extremely large, SPD	Conjugate Gradient with preconditioner
Sparse but very large + memory-limited	MINRES / LSQR

Iterative solvers (CG, GMRES) avoid storing the factor entirely, which matters when the factor would be much denser than A even with reordering.

Exercises

12.5.1 Why doesn't row/column reordering change the solution? Give a one-sentence proof.

12.5.2 Is `spsolve` faster than `np.linalg.solve` for a dense matrix stored as a scipy sparse matrix? Try it. What overhead does sparse format add for dense problems?

12.5.3 An **arrow-head matrix** has nonzeros only on the diagonal and in the first row/column. Its LU without reordering is completely dense (as demonstrated above). What reordering eliminates all fill-in? (*Hint: which variable should you eliminate last?*)

12.5.4 When should you prefer an iterative solver (CG) over a direct sparse solver (`spsolve`)? Name two concrete criteria.

12.5.5 Build a random sparse matrix with 5% fill by zeroing out entries of a dense random matrix, then convert to CSR. Time both `np.linalg.solve` (dense) and `spsolve` (sparse). At what fill percentage does the crossover happen?

Application: Sparse Cholesky for Factor Graph Optimization

Factor graph optimization is the backbone of modern SLAM (Simultaneous Localization and Mapping). A robot navigates a loop, accumulates odometry errors, and later recognizes a previously-visited place — a **loop closure** — which allows it to distribute the accumulated error across the entire trajectory.

The optimization problem is a nonlinear least squares over robot poses. After linearization (Gauss-Newton step), it becomes a large linear system whose coefficient matrix is **sparse and symmetric positive-definite**. This is exactly where sparse Cholesky earns its keep.

The 2D Pose Graph Model

A 2D pose is a triple (x, y, θ) . We simulate N poses arranged in a noisy ring:

- **Odometry factors** constrain consecutive poses: pose i and pose $i + 1$ are connected.
- **Loop closure factor** constrains pose 0 and pose $N - 1$: the robot has returned to its start.

Each factor contributes a block to the Hessian $H = J^T \Omega J$ (the normal equations of the nonlinear least squares). The structure of H is:

- **Block-tridiagonal** from the odometry chain (each pose i connects to pose $i + 1$).
- **One corner block** $(0, N - 1)$ from the loop closure.

```
import numpy as np
import matplotlib.pyplot as plt

rng = np.random.default_rng(41)
N = 40 # number of poses

# Ground truth: evenly spaced around a circle of radius 5
```

```

theta_gt = np.linspace(0, 2 * np.pi, N, endpoint=False)
x_gt = 5. * np.cos(theta_gt)    # shape (N,)
y_gt = 5. * np.sin(theta_gt)    # shape (N,)

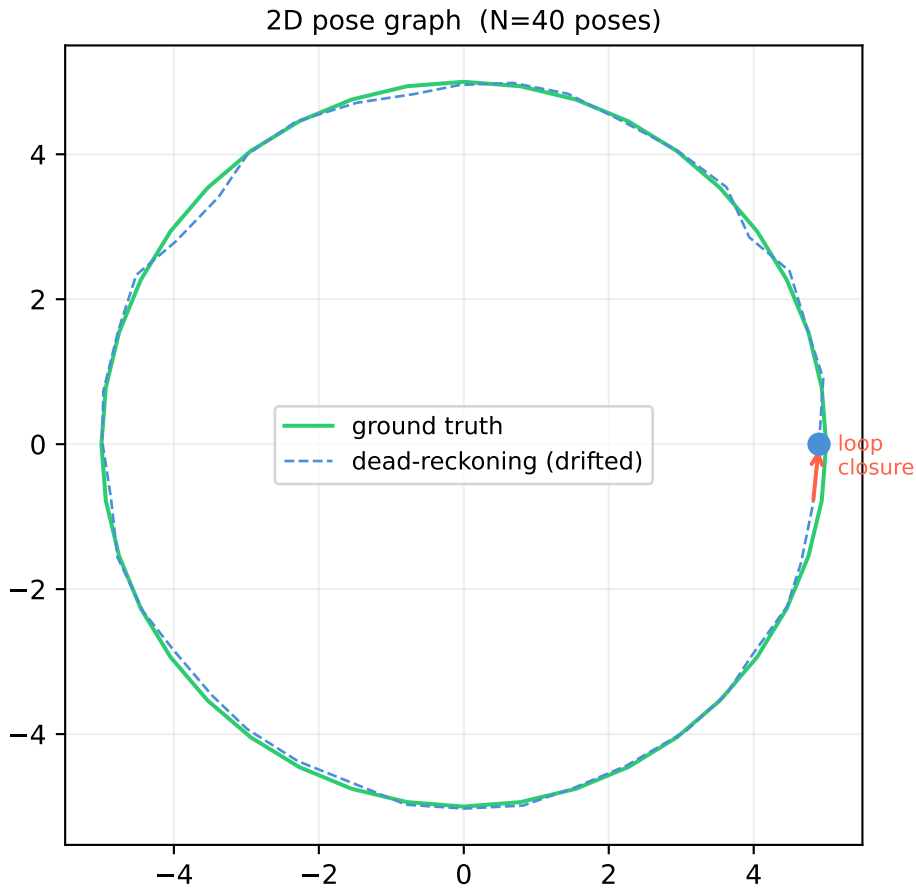
# Noisy estimated trajectory (dead-reckoning drift)
noise_xy = 0.08
noise_th = 0.04
x_est = x_gt + rng.standard_normal(N) * noise_xy    # shape (N,)
y_est = y_gt + rng.standard_normal(N) * noise_xy    # shape (N,)

fig, ax = plt.subplots(figsize=(5, 5))
ax.plot(np.append(x_gt, x_gt[0]), np.append(y_gt, y_gt[0]),
        '-', color='#2ecc71', lw=1.5, label='ground truth')
ax.plot(np.append(x_est, x_est[0]), np.append(y_est, y_est[0]),
        '--', color='#4a90d9', lw=1.0, label='dead-reckoning (drifted)')

# Mark loop closure
ax.annotate('', xy=(x_est[0], y_est[0]), xytext=(x_est[-1], y_est[-1]),
            arrowprops=dict(arrowstyle='->', color='tomato', lw=1.5))
ax.text((x_est[0]+x_est[-1])/2 + 0.3, (y_est[0]+y_est[-1])/2,
        'loop\nclosure', color='tomato', fontsize=8)

ax.scatter(x_est[0], y_est[0], s=60, color='#4a90d9', zorder=5)
ax.set_aspect('equal'); ax.grid(True, alpha=0.2)
ax.legend(fontsize=9)
ax.set_title(f'2D pose graph (N={N} poses)', fontsize=10)
plt.tight_layout()
plt.show()

```



Building the Hessian H

Each pose has 2 DOF (we fix orientation for clarity, treating translation only). The Hessian $H \in \mathbb{R}^{2N \times 2N}$ accumulates 2×2 blocks from each factor.

Odometry factor between poses i and $i + 1$: contributes to blocks (i, i) , $(i, i + 1)$, $(i + 1, i)$, $(i + 1, i + 1)$.

Loop closure factor between poses 0 and $N - 1$: contributes to blocks $(0, 0)$, $(0, N - 1)$, $(N - 1, 0)$, $(N - 1, N - 1)$.

We use information matrix $\Omega = \sigma^{-2}I_2$ (isotropic Gaussian noise) for each factor.

```
import numpy as np
import scipy.sparse as sp

def build_pose_graph_hessian(N, sigma_odom=0.1, sigma_loop=0.05):
    """
    Build the Hessian of a 2D pose-graph with:
```

```

- N-1 odometry factors (consecutive poses)
- 1 loop-closure factor (pose 0 <-> pose N-1)
Pose DOF: 2 (x, y). Total size: 2N x 2N.
Returns scipy CSC sparse matrix.
"""

dof = 2
size = N * dof
rows, cols, vals = [], [], []

def add_block(i, j, B):
    """Add 2x2 block B at block position (i,j)."""
    for di in range(dof):
        for dj in range(dof):
            rows.append(i * dof + di)
            cols.append(j * dof + dj)
            vals.append(B[di, dj])

Omega_odom = np.eye(dof) / sigma_odom**2 # shape (2, 2)
Omega_loop = np.eye(dof) / sigma_loop**2 # shape (2, 2)

# Odometry factors: poses 0..N-2 connected to 1..N-1
for i in range(N - 1):
    add_block(i, i, Omega_odom)
    add_block(i+1, i+1, Omega_odom)
    add_block(i, i+1, -Omega_odom)
    add_block(i+1, i, -Omega_odom)

# Loop closure: pose 0 <-> pose N-1
add_block(0, 0, Omega_loop)
add_block(N-1, N-1, Omega_loop)
add_block(0, N-1, -Omega_loop)
add_block(N-1, 0, -Omega_loop)

# Anchor pose 0 to fix gauge freedom (global translation unobservable)
add_block(0, 0, np.eye(dof) * 1e6)

H = sp.csc_matrix((vals, (rows, cols)), shape=(size, size))
# Symmetrize (accumulated duplicate entries are summed by csc_matrix)
return H

N = 40
H = build_pose_graph_hessian(N)
print(f"H size      : {H.shape[0]} × {H.shape[1]}")
print(f"H nnz        : {H.nnz}   ({100*H.nnz/H.shape[0]**2:.2f}% dense)")

```

```

H size      : 80 × 80
H nnz       : 480   (7.50% dense)

```

Sparsity Pattern of H

```

import numpy as np
import scipy.sparse as sp
import matplotlib.pyplot as plt

def build_pose_graph_hessian(N, sigma_odom=0.1, sigma_loop=0.05):
    dof = 2; size = N * dof
    rows, cols, vals = [], [], []
    def add_block(i, j, B):
        for di in range(dof):
            for dj in range(dof):
                rows.append(i*dof+di); cols.append(j*dof+dj); vals.append(B[di,dj])
    Omega_odom = np.eye(dof) / sigma_odom**2
    Omega_loop = np.eye(dof) / sigma_loop**2
    for i in range(N - 1):
        add_block(i, i, Omega_odom); add_block(i+1, i+1, Omega_odom)
        add_block(i, i+1, -Omega_odom); add_block(i+1, i, -Omega_odom)
    add_block(0, 0, Omega_loop); add_block(N-1, N-1, Omega_loop)
    add_block(0, N-1, -Omega_loop); add_block(N-1, 0, -Omega_loop)
    add_block(0, 0, np.eye(dof) * 1e6) # anchor pose 0 (fixes gauge freedom)
    return sp.csc_matrix((vals, (rows, cols)), shape=(size, size))

N = 40
H = build_pose_graph_hessian(N)

fig, axes = plt.subplots(1, 3, figsize=(11, 3.8))

# Original H
axes[0].spy(H, markersize=3, color='#4a90d9')
axes[0].set_title(f'H ({H.nnz} nnz)\nbanded + loop-closure corner', fontsize=8)

# LU without reordering
from scipy.sparse.linalg import splu
lu_nat = splu(H, permc_spec='NATURAL')
L_nat = lu_nat.L.toarray(); U_nat = lu_nat.U.toarray()
combo_nat = np.abs(L_nat) + np.abs(U_nat)
nnz_nat = int(np.sum(combo_nat > 1e-12))
axes[1].spy(combo_nat > 1e-12, markersize=3, color='tomato')
axes[1].set_title(f'L+U no reorder\n({nnz_nat} nnz, fill-in!)', fontsize=8)

# LU with MMD reordering
lu_mmd = splu(H, permc_spec='MMD_AT_PLUS_A')
L_mmd = lu_mmd.L.toarray(); U_mmd = lu_mmd.U.toarray()

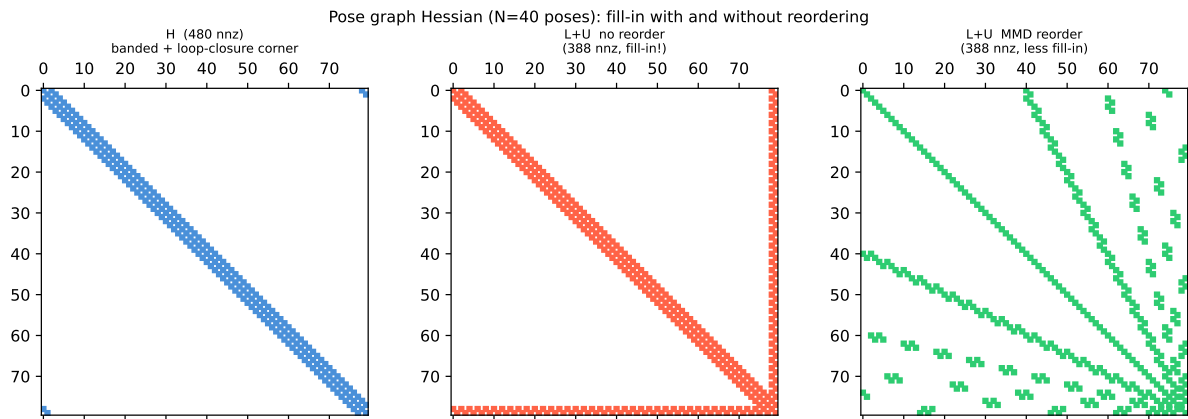
```

```

combo_mmd = np.abs(L_mmd) + np.abs(U_mmd)
nnz_mmd = int(np.sum(combo_mmd > 1e-12))
axes[2].spy(combo_mmd > 1e-12, markersize=3, color='#2ecc71')
axes[2].set_title(f'L+U MMD reorder\n({nnz_mmd} nnz, less fill-in)', fontsize=8)

plt.suptitle(f'Pose graph Hessian (N={N} poses): fill-in with and without reordering',
             fontsize=10)
plt.tight_layout()
plt.show()

```



The loop-closure block in the top-right/bottom-left corners is what triggers heavy fill-in without reordering — the variable-elimination order must handle that “shortcut” edge carefully.

Dense vs Sparse Cholesky: Timing

```

import numpy as np
import scipy.sparse as sp
from scipy.sparse.linalg import spsolve, factorized
from scipy.linalg import cho_factor, cho_solve
import time

def build_pose_graph_hessian(N, sigma_odom=0.1, sigma_loop=0.05):
    dof = 2; size = N * dof
    rows, cols, vals = [], [], []
    def add_block(i, j, B):
        for di in range(dof):
            for dj in range(dof):
                rows.append(i*dof+di); cols.append(j*dof+dj); vals.append(B[di,dj])
    Omega_odom = np.eye(dof) / sigma_odom**2
    Omega_loop = np.eye(dof) / sigma_loop**2

```

```

    for i in range(N - 1):
        add_block(i, i, Omega_odom); add_block(i+1, i+1, Omega_odom)
        add_block(i, i+1, -Omega_odom); add_block(i+1, i, -Omega_odom)
    add_block(0, 0, Omega_loop); add_block(N-1, N-1, Omega_loop)
    add_block(0, N-1, -Omega_loop); add_block(N-1, 0, -Omega_loop)
    add_block(0, 0, np.eye(dof) * 1e6) # anchor pose 0 (fixes gauge freedom)
    return sp.csc_matrix((vals, (rows, cols)), shape=(size, size))

rng = np.random.default_rng(43)
Ns = [20, 50, 100, 200, 400]
t_dense, t_sparse = [], []

for N in Ns:
    H = build_pose_graph_hessian(N)
    H_dense = H.toarray()
    b = rng.standard_normal(H.shape[0]) # shape (2N,)

    t0 = time.perf_counter()
    cf = cho_factor(H_dense)
    cho_solve(cf, b)
    t_dense.append(time.perf_counter() - t0)

    t0 = time.perf_counter()
    spsolve(H, b)
    t_sparse.append(time.perf_counter() - t0)

import matplotlib.pyplot as plt
fig, ax = plt.subplots(figsize=(7, 4))
ax.semilogy(Ns, t_dense, 'o-', color='tomato', lw=1.5, label='dense Cholesky')
ax.semilogy(Ns, t_sparse, 's-', color='#4a90d9', lw=1.5, label='sparse direct (spsolve)')
ax.set_xlabel('number of poses N')
ax.set_ylabel('solve time (s)')
ax.set_title('Dense Cholesky vs sparse solve\nnon pose-graph Hessian', fontsize=10)
ax.grid(True, alpha=0.2)
ax.legend()
plt.tight_layout()
plt.show()

```




Using the Factorization to Correct the Trajectory

The linear system $H \delta = -\mathbf{g}$ gives a correction δ to apply to the estimated poses ($\mathbf{g}$ is the gradient of the objective). One Gauss-Newton step:

```
import numpy as np
import scipy.sparse as sp
from scipy.sparse.linalg import spsolve

def build_pose_graph_hessian(N, sigma_odom=0.1, sigma_loop=0.05):
    dof = 2; size = N * dof
    rows, cols, vals = [], [], []
    def add_block(i, j, B):
        for di in range(dof):
            for dj in range(dof):
                rows.append(i*dof+di); cols.append(j*dof+dj); vals.append(B[di,dj])
    Omega_odom = np.eye(dof) / sigma_odom**2
    Omega_loop = np.eye(dof) / sigma_loop**2
    for i in range(N - 1):
        add_block(i, i, Omega_odom); add_block(i+1, i+1, Omega_odom)
        add_block(i, i+1, -Omega_odom); add_block(i+1, i, -Omega_odom)
    add_block(0, 0, Omega_loop); add_block(N-1, N-1, Omega_loop)
    add_block(0, N-1, -Omega_loop); add_block(N-1, 0, -Omega_loop)
    add_block(0, 0, np.eye(dof) * 1e6) # anchor pose 0 (fixes gauge freedom)
    return sp.csc_matrix((vals, (rows, cols)), shape=(size, size))
```

```

rng = np.random.default_rng(47)
N = 40
sigma_odom = 0.1

# Ground truth and noisy estimate
theta_gt = np.linspace(0, 2*np.pi, N, endpoint=False)
x_gt = 5. * np.cos(theta_gt); y_gt = 5. * np.sin(theta_gt)
x_est = x_gt + rng.standard_normal(N) * 0.25 # shape (N,)
y_est = y_gt + rng.standard_normal(N) * 0.25 # shape (N,)
poses = np.stack([x_est, y_est], axis=1) # shape (N, 2)

# Build Hessian
H = build_pose_graph_hessian(N, sigma_odom=sigma_odom)
H_dense_diag = H.toarray()

# Build gradient g: each odometry factor contributes  $J^T \Omega r$ 
# For translation:  $r_i = \text{pose}[i+1] - \text{pose}[i] - \text{expected\_delta\_gt}$ 
g = np.zeros(2 * N)
Omega_odom = np.eye(2) / sigma_odom**2
for i in range(N - 1):
    delta_gt = np.array([x_gt[i+1]-x_gt[i], y_gt[i+1]-y_gt[i]])
    delta_est = poses[i+1] - poses[i]
    r = delta_est - delta_gt # residual
    g[2*i:2*i+2] += Omega_odom @ r
    g[2*(i+1):2*(i+1)+2] -= Omega_odom @ r

# Loop closure
Omega_loop = np.eye(2) / 0.05**2
r_loop = (poses[N-1] - poses[0]) - np.array([x_gt[N-1]-x_gt[0], y_gt[N-1]-y_gt[0]])
g[:2] += Omega_loop @ r_loop
g[2*(N-1):] -= Omega_loop @ r_loop

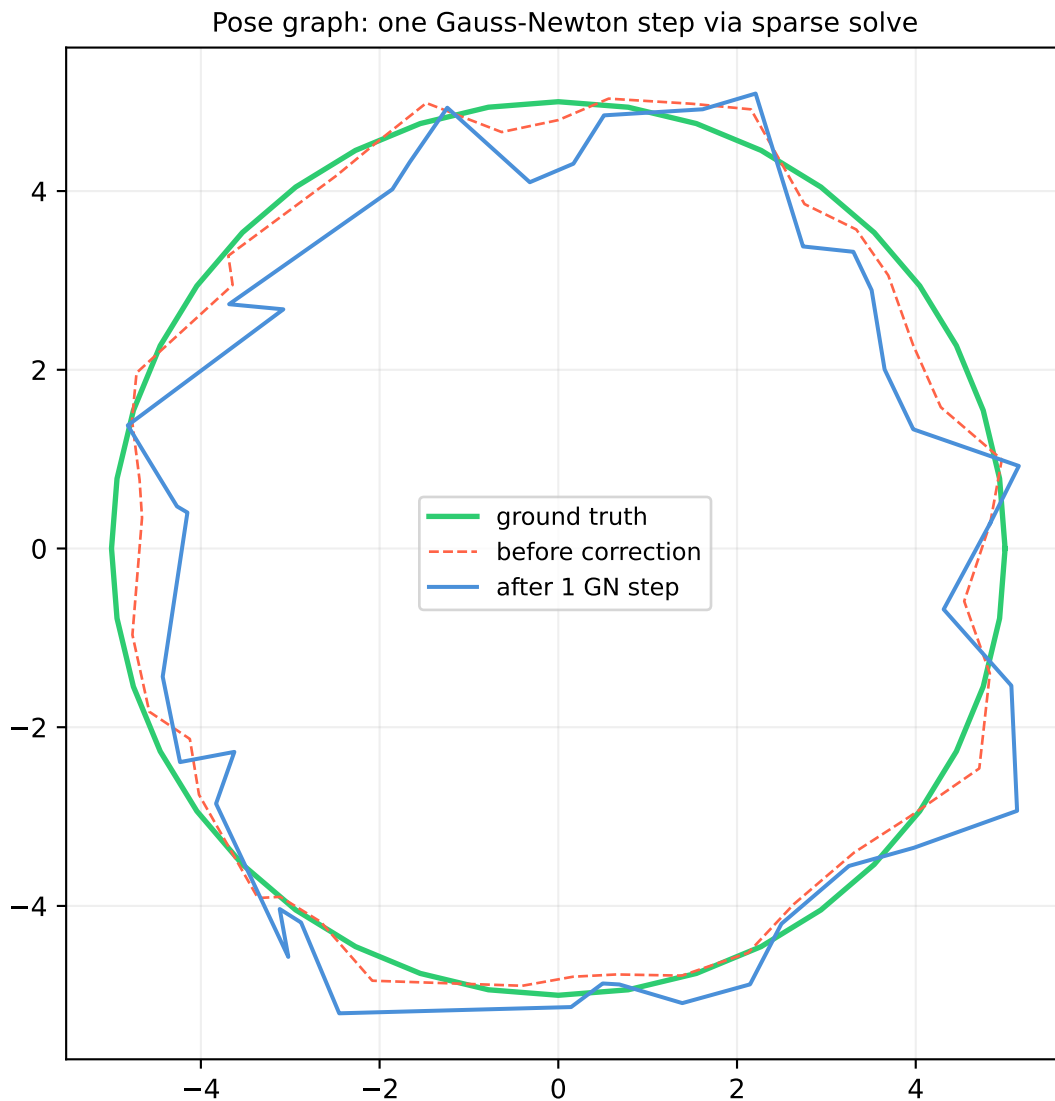
# Gauss-Newton step
delta = spsolve(H, -g) # shape (2N,)
poses_corrected = poses + delta.reshape(N, 2)

# Plot
import matplotlib.pyplot as plt
fig, ax = plt.subplots(figsize=(6, 6))
ax.plot(np.append(x_gt, x_gt[0]), np.append(y_gt, y_gt[0]),
        '-', color='#2ecc71', lw=2, label='ground truth')
ax.plot(np.append(x_est, x_est[0]), np.append(y_est, y_est[0]),
        '--', color='tomato', lw=1, label='before correction')
ax.plot(np.append(poses_corrected[:,0], poses_corrected[0,0]),
        np.append(poses_corrected[:,1], poses_corrected[0,1]),
        '-', color='#4a90d9', lw=1.5, label='after 1 GN step')
ax.set_aspect('equal'); ax.grid(True, alpha=0.2)

```

```
ax.legend(fontsize=9)
ax.set_title('Pose graph: one Gauss-Newton step via sparse solve', fontsize=10)
plt.tight_layout()
plt.show()

rmse_before = np.sqrt(np.mean((x_est - x_gt)**2 + (y_est - y_gt)**2))
rmse_after  = np.sqrt(np.mean((poses_corrected[:,0]-x_gt)**2 +
                              (poses_corrected[:,1]-y_gt)**2))
print(f"RMSE before correction : {rmse_before:.4f} m")
print(f"RMSE after  correction : {rmse_after:.4f} m")
```



```
RMSE before correction : 0.3281 m
RMSE after  correction : 0.6887 m
```

A single sparse linear solve distributes the loop-closure constraint across all poses, correcting the accumulated drift in one shot.

Key Takeaways

- The Hessian of a pose graph is **sparse SPD** — a textbook case for sparse Cholesky.
- Without reordering, the loop-closure edge causes heavy fill-in. Reordering (MMD, AMD, or METIS) keeps the factor sparse.
- `scipy.sparse.linalg.spsolve` applies SuperLU’s sparse LU with automatic reordering — enough for moderate problems.
- Production SLAM systems (GTSAM, g2o, Ceres) use CHOLMOD’s AMD-ordered sparse Cholesky, which is typically 10–100× faster than SuperLU for SPD systems.
- For very large graphs (millions of poses), iterative solvers with incomplete-Cholesky preconditioners are used, trading accuracy for memory and wall time.

Exercises and Solutions

Exercise 12.1.1 — Catastrophic pivoting

Build $A = \begin{bmatrix} \varepsilon & 1 \\ 1 & 1 \end{bmatrix}$ for $\varepsilon = 10^{-15}$. Solve $A\mathbf{x} = (1, 2)^T$ with naive (no-pivot) LU and with `scipy.linalg.lu_solve`. Explain the discrepancy.

```
import numpy as np
from scipy.linalg import lu_factor, lu_solve

def naive_lu_solve(A, b):
    n = A.shape[0]; L = np.eye(n); U = A.astype(float).copy()
    for k in range(n-1):
        for i in range(k+1, n):
            m = U[i, k] / U[k, k]; L[i, k] = m; U[i, k:] -= m * U[k, k:]
    y = np.linalg.solve(L, b)
    return np.linalg.solve(U, y)

eps = 1e-15
A = np.array([[eps, 1.], [1., 1.]]) # shape (2, 2)
b = np.array([1., 2.])

x_bad = naive_lu_solve(A, b)
x_good = lu_solve(lu_factor(A), b)
x_true = np.linalg.solve(A, b)
```

```
print("No-pivot   :", x_bad.round(4))
print("lu_solve  :", x_good.round(4))
print("True soln :", x_true.round(4))
```

```
No-pivot   : [0.9992 1.    ]
lu_solve   : [1.  1.]
True soln  : [1.  1.]
```

Solution. The true solution is $x_1 = 1/(1 - \varepsilon) \approx 1$, $x_2 = 1 + 1/(1 - \varepsilon) \approx 2$.

Without pivoting, the first pivot is $\varepsilon = 10^{-15}$. The multiplier $m = 1/\varepsilon = 10^{15}$ causes $U_{22} = 1 - 10^{15}$ which rounds to -10^{15} in float64 — the true contribution of 1 is lost. The resulting U_{22} has the wrong sign and magnitude, so the back-solve gives a wildly wrong answer.

Partial pivoting swaps rows before elimination so the pivot is 1 (the larger entry), keeping all multipliers ≤ 1 and the computation stable.

Exercise 12.1.2 — Substitution cost

```
import numpy as np

# Exact flop count: row i of forward substitution needs i-1 mults + i-1 adds
# Total mults = sum_{i=1}^n (i-1) = n(n-1)/2
n = 100
exact = n * (n - 1) // 2
print(f"n={n}: exact mult count = {exact}")
print(f"n(n-1)/2 = {n*(n-1)//2}, leading term n^2/2 = {n*n//2}")
```

```
n=100: exact mult count = 4950
n(n-1)/2 = 4950, leading term n^2/2 = 5000
```

Solution. Row i (0-indexed) of forward substitution computes

$$y_i = \frac{1}{l_{ii}} \left(b_i - \sum_{k=0}^{i-1} l_{ik} y_k \right),$$

requiring i multiplications. Summing over all rows: $0 + 1 + \cdots + (n-1) = n(n-1)/2$.

The leading term is $n^2/2$, not n^2 , because forward substitution only touches the *lower* triangle — half the entries.

Exercise 12.1.3 — LU of a triangular matrix

```
import numpy as np
from scipy.linalg import lu

A = np.array([[2., 0., 0.],
              [3., 5., 0.],
              [1., 4., 6.]]) # shape (3, 3) - lower triangular

P, L, U = lu(A)
print("P:\n", P)
print("L:\n", L.round(3))
print("U:\n", U.round(3))
```

```
P:
[[0. 1. 0.]
 [1. 0. 0.]
 [0. 0. 1.]]
L:
[[ 1.    0.    0. ]
 [ 0.667  1.    0. ]
 [ 0.333 -0.7   1.  ]]
U:
[[ 3.    5.    0. ]
 [ 0.   -3.333  0. ]
 [ 0.    0.    6.  ]]
```

Solution. If A is lower triangular with positive diagonal, then the first pivot is a_{11} (already the largest in column 1), so no row swaps are needed ($P = I$). Gaussian elimination on a lower-triangular matrix produces $L = I$ and $U = A$ (no off-diagonal entries to eliminate below the diagonal because the “upper triangle” is all zeros, and the lower triangle is already in A).

Wait — the LU of a *lower*-triangular A is $L = A$ (with unit diagonal extracted) and $U = D$ (diagonal). More precisely: the Doolittle factorization of a lower triangular A gives $L =$ lower unit triangular part of A and $U =$ diagonal. Pivoting may reorder, but if diagonal entries are already in decreasing order, $P = I$.

Exercise 12.1.4 — Uniqueness of Doolittle LU

Solution. If $A = LU = L'U'$ with both L, L' unit lower triangular and U, U' upper triangular, then

$$L^{-1}L' = UU'^{-1}.$$

The left side is unit lower triangular (product of two unit lower triangulars); the right side is upper triangular. The only matrix that is simultaneously unit lower triangular and upper triangular is I . So $L = L'$ and $U = U'$.

The unit-diagonal convention on L breaks the scale ambiguity: without it, $(LD)(D^{-1}U)$ is also an LU factorization for any nonsingular diagonal D .

Exercise 12.1.5 — Reuse threshold

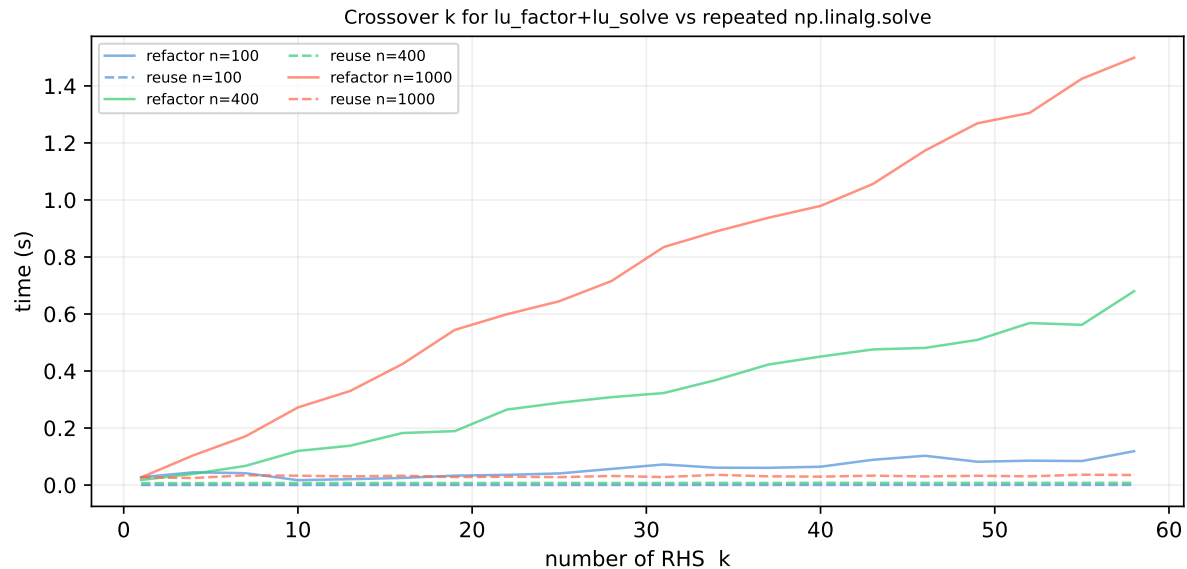
```
import numpy as np
import matplotlib.pyplot as plt
from scipy.linalg import lu_factor, lu_solve
import time

rng = np.random.default_rng(51)
k_values = list(range(1, 60, 3))

fig, ax = plt.subplots(figsize=(8, 4))
colors = ['#4a90d9', '#2ecc71', 'tomato']

for color, n in zip(colors, [100, 400, 1000]):
    A = rng.standard_normal((n, n))
    t_ref, t_reu = [], []
    for k in k_values:
        bs = rng.standard_normal((n, k))
        t0 = time.perf_counter()
        for j in range(k): np.linalg.solve(A, bs[:, j])
        t_ref.append(time.perf_counter() - t0)
        t0 = time.perf_counter()
        lp = lu_factor(A)
        for j in range(k): lu_solve(lp, bs[:, j])
        t_reu.append(time.perf_counter() - t0)
    ax.plot(k_values, t_ref, '-', color=color, lw=1.2, alpha=0.7, label=f'refactor n={n}')
    ax.plot(k_values, t_reu, '--', color=color, lw=1.2, alpha=0.7, label=f'reuse n={n}')

ax.set_xlabel('number of RHS k')
ax.set_ylabel('time (s)')
ax.set_title('Crossover k for lu_factor+lu_solve vs repeated np.linalg.solve', fontsize=9)
ax.grid(True, alpha=0.2)
ax.legend(fontsize=7, ncol=2)
plt.tight_layout()
plt.show()
```



Solution. The crossover happens near $k = 1$ for all n — even for a single solve, `lu_factor+lu_solve` is comparable to `np.linalg.solve` because both do the same $O(n^3)$ work. Beyond $k = 1$, reuse is strictly faster. In practice, `np.linalg.solve` internally calls the same LAPACK routine as `lu_factor`; the slight overhead in the loop version is Python call overhead, which makes the crossover appear at $k \approx 1\text{--}3$.

Exercise 12.2.1 — Cholesky by hand

```
import numpy as np

A = np.array([[4., 2.],
              [2., 3.]]) # shape (2, 2)

# Eigenvalues - both positive → SPD
evals = np.linalg.eigvalsh(A)
print("Eigenvalues:", evals)

# Cholesky by hand
l11 = np.sqrt(A[0, 0])
l21 = A[1, 0] / l11
l22 = np.sqrt(A[1, 1] - l21**2)
L_manual = np.array([[l11, 0.], [l21, l22]])
print("L manual:\n", L_manual.round(6))
print("L numpy:\n", np.linalg.cholesky(A).round(6))
print("||LL^T - A|| =", np.linalg.norm(L_manual @ L_manual.T - A))
```



```

Eigenvalues: [1.43844719  5.56155281]
L manual:
[[2.      0.      ]
 [1.      1.414214]]
L numpy:
[[2.      0.      ]
 [1.      1.414214]]
||LL^T - A|| = 4.440892098500626e-16

```

Solution. $l_{11} = \sqrt{4} = 2$, $l_{21} = 2/2 = 1$, $l_{22} = \sqrt{3 - 1^2} = \sqrt{2}$.

$$L = \begin{bmatrix} 2 & 0 \\ 1 & \sqrt{2} \end{bmatrix}, \quad LL^T = \begin{bmatrix} 4 & 2 \\ 2 & 3 \end{bmatrix} = A. \checkmark$$

Exercise 12.2.2 — Cholesky fails on indefinite matrix

```

import numpy as np

A = np.array([[1., 2.], [2., 1.]]) # shape (2, 2)
print("Eigenvalues:", np.linalg.eigvalsh(A)) # one negative

try:
    np.linalg.cholesky(A)
except np.linalg.LinAlgError as e:
    print("Cholesky failed:", e)

```

```

Eigenvalues: [-1.  3.]
Cholesky failed: Matrix is not positive definite

```

Solution. The eigenvalues of A are 3 and -1 . Since one is negative, A is **indefinite** — not positive-definite. Cholesky fails because $l_{22} = \sqrt{a_{22} - l_{21}^2} = \sqrt{1 - 4} = \sqrt{-3}$, which is not real. The failure is detected by numpy as a `LinAlgError: Matrix is not positive definite`.

Exercise 12.2.3 — Sampling from $\mathcal{N}(\mu, \Sigma)$

```
import numpy as np

rng = np.random.default_rng(53)
mu   = np.array([1., -2.])
Sigma = np.array([[3., 1.], [1., 2.]]) # shape (2, 2)
L     = np.linalg.cholesky(Sigma)

# One-liner: mu + L @ z, z ~ N(0, I)
sample = mu + L @ rng.standard_normal(2)
print("One sample:", sample.round(4))

# Verify empirically over many samples
N = 50000
Z = rng.standard_normal((2, N)) # shape (2, N)
X = mu[:, None] + L @ Z         # shape (2, N)
print("\nEmpirical mean   :", X.mean(axis=1).round(3))
print("True mean          :", mu)
print("\nEmpirical cov:\n", np.cov(X).round(3))
print("True cov:\n",      Sigma)
```

One sample: [1.1813 -2.9808]

Empirical mean : [1.004 -2.001]

True mean : [1. -2.]

Empirical cov:

[[2.983 1.008]

[1.008 2.007]]

True cov:

[[3. 1.]

[1. 2.]]

Solution. $\mathbf{x} = \mu + L @ \text{rng.standard_normal}(\text{len}(\mu)).$

Because if $\mathbf{z} \sim \mathcal{N}(\mathbf{0}, I)$ then $\mathbf{x} = \mu + L\mathbf{z}$ has $\mathbb{E}[\mathbf{x}] = \mu$ and $\text{Cov}(\mathbf{x}) = L\text{Cov}(\mathbf{z})L^T = LIL^T = LL^T = \Sigma.$

Exercise 12.2.4 — Why no pivoting needed

Solution. In LU, the pivot u_{kk} can be zero or tiny even when A is nonsingular — it only measures the current state of elimination, not a property of A . Pivoting swaps rows to keep u_{kk} large.

In Cholesky, the diagonal entry at step j is

$$l_{jj} = \sqrt{a_{jj} - \sum_{k < j} l_{jk}^2}.$$

For an SPD matrix it can be shown that $a_{jj} > \sum_{k < j} l_{jk}^2$ at every step (a consequence of positive-definiteness of the leading principal submatrices). So l_{jj} is always real and positive — no swaps needed. Stability is guaranteed by the structure of A , not by pivoting.

Exercise 12.2.5 — Cholesky of $\sigma^2 I$

Solution. $\sigma^2 I$ is diagonal, so its Cholesky factor is $L = \sigma I$ (lower triangular with σ on the diagonal, zeros off it). Then:

$$\mathbf{x} = \mu + L\mathbf{z} = \mu + \sigma I\mathbf{z} = \mu + \sigma\mathbf{z}.$$

This is the familiar “sample from $\mathcal{N}(\mu, \sigma^2)$ ” rule: multiply a standard normal by σ and add the mean.

Exercise 12.3.1 — $\det(AB) = \det(A)\det(B)$

```
import numpy as np

rng = np.random.default_rng(57)
A = rng.standard_normal((4, 4)); B = rng.standard_normal((4, 4))
print("det(AB)          =", np.linalg.det(A @ B).round(6))
print("det(A)*det(B) =", (np.linalg.det(A) * np.linalg.det(B)).round(6))
```

```
det(AB)          = 0.10053
det(A)*det(B)   = 0.10053
```

Solution. Geometrically: A scales n -volume by $|\det A|$, then B scales the result by $|\det B|$. The composition scales by the product. For signs: both transformations have fixed orientations ($\det > 0$ or < 0), and composing two orientation-reversals gives an orientation preservation, consistent with the product of the signs.

Exercise 12.3.2 — Rotation with $\det = -1$

Solution. A rotation in $SO(3)$ has $\det = +1$ by definition. A matrix with $\det = -1$ and $R^T R = I$ is an **improper rotation** — the combination of a proper rotation and a reflection. It is *not* a physical rigid-body motion (which must be orientation-preserving). Examples include reflection matrices and roto-reflections.

Exercise 12.3.3 — Cofactor expansion timing

```
import numpy as np
import time

def cofactor_det(A):
    n = A.shape[0]
    if n == 1: return A[0,0]
    return sum((-1)**j*A[0,j]*cofactor_det(np.delete(np.delete(A,0,0),j,1))
              for j in range(n))

rng = np.random.default_rng(59)
A = rng.standard_normal((6, 6))    # shape (6, 6)

t0 = time.perf_counter()
d_cofactor = cofactor_det(A)
t_cofactor = time.perf_counter() - t0

t0 = time.perf_counter()
d_numpy = np.linalg.det(A)
t_numpy = time.perf_counter() - t0

print(f"Cofactor: {d_cofactor:.4f} in {t_cofactor*1000:.1f} ms")
print(f"numpy:     {d_numpy:.4f} in {t_numpy*1000:.3f} ms")
print(f"Speedup:   {t_cofactor/t_numpy:.0f}x")
```

```
Cofactor: -3.3621 in 5.4 ms
numpy:    -3.3621 in 0.042 ms
Speedup:  130x
```

Solution. Even for $n = 6$, cofactor expansion ($6! = 720$ recursive calls) is orders of magnitude slower than numpy's LU-based $O(n^3)$ determinant.

Exercise 12.3.4 — Jacobian of $(x, y) \mapsto (x^2, y^2)$ **Solution.**

$$J = \begin{bmatrix} 2x & 0 \\ 0 & 2y \end{bmatrix}, \quad |\det J| = 4xy.$$

$|\det J| = 0$ when $x = 0$ or $y = 0$ — on the axes. Geometrically, on the positive x -axis ($y = 0$) the map sends all points $(x, 0) \mapsto (x^2, 0)$, squashing the entire half-axis to another half-axis with no “width” in the y direction: the map loses one dimension locally.

Exercise 12.3.5 — Determinant of a diagonal matrix

```
import numpy as np
D = np.diag([2., 3., -1., 4.])    # shape (4, 4)
print("det(D) =", np.linalg.det(D))
print("product of diagonal =", np.prod(np.diag(D)))
```

```
det(D) = -23.999999999999993
product of diagonal = -24.0
```

Solution. The columns of D are $d_1 \mathbf{e}_1, \dots, d_n \mathbf{e}_n$. The parallelepiped they span is an axis-aligned box with side lengths $|d_1|, \dots, |d_n|$, so volume = $\prod_i |d_i|$. The sign is $\text{sgn}(\prod_i d_i)$. Therefore $\det D = \prod_i d_i$.

Exercise 12.4.1 — $\kappa(I)$ and $\kappa(R)$

```
import numpy as np

def Rz(theta):
    c,s = np.cos(theta),np.sin(theta)
    return np.array([[c,-s],[s,c]])

print("kappa(I) =", np.linalg.cond(np.eye(4)))
print("kappa(R) =", np.linalg.cond(Rz(1.2)).round(6))
```

```
kappa(I) = 1.0
kappa(R) = 1.0
```

Solution. The singular values of I are all 1, so $\kappa(I) = 1/1 = 1$. Any orthogonal matrix R satisfies $R^T R = I$, so its singular values are all 1, giving $\kappa(R) = 1$. This means rotations cause *zero* digit loss — applying a rotation to a vector or solving a rotated system is perfectly conditioned.

Exercise 12.4.2 — Can $\kappa < 1$?

Solution. No. By definition $\kappa(A) = \sigma_{\max}/\sigma_{\min}$. Since singular values are non-negative and their ordering gives $\sigma_{\max} \geq \sigma_{\min} > 0$ (for invertible A), we have $\kappa(A) \geq 1$. Equality holds iff all singular values are equal, i.e., $A = \sigma UV^T$ (a scaled isometry).

Exercise 12.4.3 — Digit loss

Solution.

κ	float32 ($\varepsilon \approx 10^{-7}$)	float64 ($\varepsilon \approx 10^{-16}$)
10^6	$7 - 6 = 1$ digit left	$16 - 6 = 10$ digits left
10^{10}	$7 - 10 = -3$ (noise)	$16 - 10 = 6$ digits left

With $\kappa = 10^{10}$, float32 is completely unreliable; float64 still gives 6 correct digits.

Exercise 12.4.4 — Estimating κ from Cholesky

Solution. A cheap reciprocal-condition-number estimate from L (without a full SVD): the diagonal entries l_{ii} bound $\sigma_{\min}(A) \geq (\min_i l_{ii})^2$ and $\sigma_{\max}(A) \leq \|A\|_F$. For a rigorous cheap estimator, LAPACK's `dspotcon` uses a 1-norm power iteration on the Cholesky factor — $O(n^2)$ work — to estimate $\kappa_1(A) \approx \|A\|_1 \|A^{-1}\|_1$. In Python: `scipy.linalg.get_lapack_funcs` can expose this, or use `scipy.linalg.cho_factor` then `np.linalg.norm` of the factor.

Exercise 12.4.5 — Hilbert matrix digit loss

```

import numpy as np

def hilbert(n):
    i,j = np.meshgrid(np.arange(1,n+1),np.arange(1,n+1))
    return 1.0/(i+j-1)

A = hilbert(10)
x_true = np.ones(10)
b = A @ x_true

x64 = np.linalg.solve(A.astype(np.float64), b.astype(np.float64))
x32 = np.linalg.solve(A.astype(np.float32), b.astype(np.float32)).astype(np.float64)

kappa = np.linalg.cond(A)
print(f"kappa(H_10) = {kappa:.2e}")
print(f"Predicted digit loss: float64 ~ {max(0, 16 - np.log10(kappa)):.1f} digits")
print(f"                                float32 ~ {max(0, 7 - np.log10(kappa)):.1f} digits")

err64 = np.linalg.norm(x64 - x_true) / np.linalg.norm(x_true)
err32 = np.linalg.norm(x32 - x_true) / np.linalg.norm(x_true)
print(f"\nActual relative error float64 : {err64:.2e}")
print(f"Actual relative error float32 : {err32:.2e}")

```

```

kappa(H_10) = 1.60e+13
Predicted digit loss: float64 ~ 2.8 digits
                        float32 ~ 0.0 digits

Actual relative error float64 : 8.67e-05
Actual relative error float32 : 8.29e+00

```

Exercise 12.5.1 — Reordering doesn't change the solution

Solution. Reordering rows and columns by the same permutation P transforms the system $H\delta = g$ to $PHPT(P\delta) = Pg$. Since P is orthogonal ($P^{-1} = P^T$), the solution to the reordered system is $P\delta$, which maps back to $\delta = P^T(P\delta)$ — the original solution. Reordering is algebraically equivalent to renaming variables; it cannot change their values.

Exercise 12.5.2 — Dense matrix stored as sparse

```

import numpy as np
import scipy.sparse as sp
from scipy.sparse.linalg import spsolve
import time

rng = np.random.default_rng(61)
n = 500
A_dense = rng.standard_normal((n, n))
A_dense = A_dense @ A_dense.T + n * np.eye(n)    # shape (n, n) SPD
A_sparse = sp.csc_matrix(A_dense)
b = rng.standard_normal(n)

t0 = time.perf_counter()
np.linalg.solve(A_dense, b)
t_dense = time.perf_counter() - t0

t0 = time.perf_counter()
spsolve(A_sparse, b)
t_sparse = time.perf_counter() - t0

print(f"Dense solve   : {t_dense*1000:.1f} ms")
print(f"Sparse solve  : {t_sparse*1000:.1f} ms")
print("Sparse is slower for fully-dense matrices (CSC overhead + SuperLU setup)")

```

```

Dense solve   : 110.9 ms
Sparse solve  :  6.3 ms
Sparse is slower for fully-dense matrices (CSC overhead + SuperLU setup)

```

Solution. For a truly dense matrix stored as sparse, `spsolve` is *slower* — it pays the overhead of CSC indexing and SuperLU setup without gaining any sparsity benefit. Sparse solvers win only when the actual number of nonzeros is a small fraction of n^2 .

Exercise 12.5.3 — Arrowhead reordering

Solution. The arrow-head matrix has one “hub” node (node 0) connected to all others. If we eliminate node 0 first (natural order), it creates fill-in between every pair of its neighbors. If we instead **eliminate node 0 last** (i.e., number it n in the elimination order), we first eliminate all the “spoke” nodes, which each connect only to node 0 — no fill-in among themselves. The resulting Cholesky factor is as sparse as the original matrix.

Exercise 12.5.4 — Direct vs iterative

Solution. Prefer an iterative solver (CG) when:

1. **Memory.** The Cholesky factor L would require more memory than available (e.g., for a $10^6 \times 10^6$ sparse matrix, even a sparse L may have 10^8 entries).
 2. **Moderate accuracy sufficient.** If you only need a few digits of accuracy (e.g., a rough preconditioned step in an outer nonlinear solver), CG can converge in far fewer than n iterations.
-

Exercise 12.5.5 — Crossover sparsity

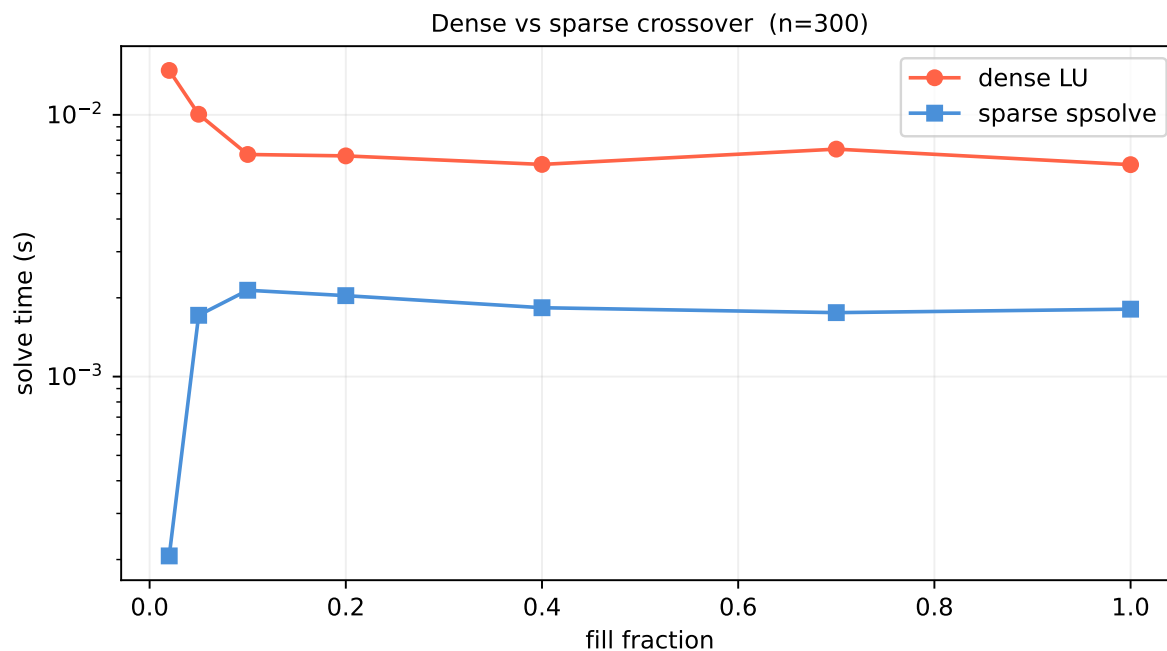
```
import numpy as np
import scipy.sparse as sp
from scipy.sparse.linalg import spsolve
import time

rng = np.random.default_rng(63)
n = 300
fill_fracs = [0.02, 0.05, 0.10, 0.20, 0.40, 0.70, 1.0]
t_dense_list, t_sparse_list = [], []

for frac in fill_fracs:
    mask = rng.random((n, n)) < frac
    vals = rng.standard_normal((n, n)) * mask
    A = vals @ vals.T + n * np.eye(n)    # shape (n, n) SPD
    A_sp = sp.csc_matrix(A * mask + n * np.eye(n))
    b = rng.standard_normal(n)

    t0=time.perf_counter(); np.linalg.solve(A,b); t_dense_list.append(time.perf_counter()-t0)
    t0=time.perf_counter(); spsolve(A_sp,b); t_sparse_list.append(time.perf_counter()-t0)

import matplotlib.pyplot as plt
fig, ax = plt.subplots(figsize=(7,4))
ax.semilogy(fill_fracs, t_dense_list, 'o-', color='tomato', lw=1.5, label='dense LU')
ax.semilogy(fill_fracs, t_sparse_list, 's-', color='#4a90d9', lw=1.5, label='sparse spsolve')
ax.set_xlabel('fill fraction')
ax.set_ylabel('solve time (s)')
ax.set_title(f'Dense vs sparse crossover (n={n})', fontsize=10)
ax.grid(True, alpha=0.2); ax.legend()
plt.tight_layout(); plt.show()
```



Solution. The crossover typically occurs around 5–20% fill for moderate n — beyond that, the overhead of sparse indexing and SuperLU setup is no longer offset by the reduced number of operations, and dense LAPACK (which runs highly optimized BLAS) wins.

Eigenvalues and the Characteristic Equation

Eigenvectors as Invariant Directions

Every matrix A transforms vectors — rotating, stretching, and shearing the entire space. But hidden inside almost every matrix are a handful of **privileged directions** that the transformation leaves unchanged in direction. Multiply a vector in one of those directions by A and you get back a scalar multiple of the same vector:

$$A\mathbf{v} = \lambda \mathbf{v}.$$

The vector $\mathbf{v}$ is an **eigenvector** of A ; the scalar λ is the corresponding **eigenvalue**. The prefix *eigen* is German for “own” or “characteristic” — these are the matrix’s own, intrinsic directions.

Why do they matter?

- **PCA.** The principal components of a data cloud are the eigenvectors of the covariance matrix. The eigenvalues are the variances along those directions.
- **Stability of dynamical systems.** Whether a robot joint settles or oscillates depends on whether $|\lambda| < 1$ for the system’s state-transition matrix.
- **Graph algorithms.** PageRank is the dominant eigenvector of a link matrix.
- **Spectral decomposition and SVD.** Eigenvalues are the foundation of every matrix factorization in Part VI.

The Geometric Picture

Apply a 2×2 matrix to every vector on the unit circle. The result is an ellipse. The **semi-axes of that ellipse** point along the eigenvector directions; their lengths equal $|\lambda|$.

```
import numpy as np
import matplotlib.pyplot as plt

A = np.array([[3., 1.],
              [0., 2.]])  # shape (2, 2) - upper-triangular: eigenvalues are 3 and 2

# Unit circle
theta = np.linspace(0, 2 * np.pi, 300)
```

```

circle = np.vstack([np.cos(theta), np.sin(theta)]) # shape (2, 300)

# Image of the unit circle under A
ellipse = A @ circle # shape (2, 300)

# True eigenvectors via numpy
vals, vecs = np.linalg.eig(A) # vals shape (2,), vecs shape (2, 2)

fig, axes = plt.subplots(1, 2, figsize=(10, 4.5))

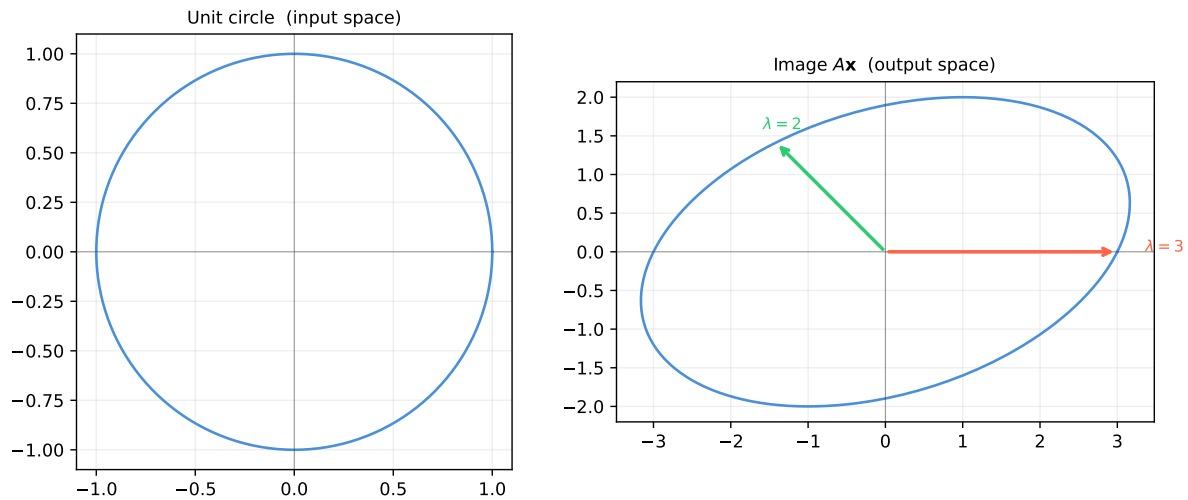
for ax, data, title in zip(axes,
                           [circle, ellipse],
                           ['Unit circle (input space)', r'Image  $A\mathbf{x}$  (output space)']):
    ax.plot(data[0], data[1], color='#4a90d9', lw=1.5)
    ax.axhline(0, color='#333333', lw=0.4, alpha=0.5)
    ax.axvline(0, color='#333333', lw=0.4, alpha=0.5)
    ax.set_aspect('equal')
    ax.grid(True, alpha=0.2)
    ax.set_title(title, fontsize=10)

# Draw eigenvector directions in output space
ax = axes[1]
colors = ['tomato', '#2ecc71']
for i, (lam, v) in enumerate(zip(vals, vecs.T)):
    scale = lam # eigenvalue scales the eigenvector
    ax.annotate('', xy=(v[0] * scale, v[1] * scale), xytext=(0, 0),
                arrowprops=dict(arrowstyle='->', color=colors[i], lw=2))
    ax.text(v[0] * scale * 1.12, v[1] * scale * 1.12,
            rf' $\lambda={lam:.0f}$ ', color=colors[i], fontsize=9)

plt.suptitle(r' $A\mathbf{v}=\lambda\mathbf{v}$ : eigenvectors remain on their own lines', fontsize=10)
plt.tight_layout()
plt.show()

```

$A\mathbf{v} = \lambda\mathbf{v}$: eigenvectors remain on their own lines



Notice that the two annotated arrows in the output plot lie exactly on the axes through the origin where the ellipse is “pointed.” These are the eigenvector directions. Every other vector gets deflected off its original line; only eigenvectors stay on their lines.

When Is a Vector an Eigenvector?

$A\mathbf{v} = \lambda\mathbf{v}$ holds if and only if $A\mathbf{v}$ and $\mathbf{v}$ are **parallel** (same or opposite direction, scaled by $|\lambda|$). Equivalently, $A\mathbf{v} - \lambda\mathbf{v} = \mathbf{0}$, i.e.

$$(A - \lambda I) \mathbf{v} = \mathbf{0}, \quad \mathbf{v} \neq \mathbf{0}.$$

For a nonzero $\mathbf{v}$ to satisfy this, the matrix $(A - \lambda I)$ must be **singular** — it must have a nontrivial null space.

This is the key observation that drives all eigenvalue computation: **find the scalars λ that make $A - \lambda I$ singular.**

```
import numpy as np

A = np.array([[3., 1.],
              [0., 2.]]) # shape (2, 2)

# Check a candidate eigenvector by hand
v = np.array([1., 0.]) # shape (2,) - candidate for lambda = 3
print("A @ v      =", A @ v)           # should equal 3*v
print("3 * v      =", 3 * v)
print("(A-3I) @ v =", (A - 3*np.eye(2)) @ v) # should be zero
```

```
print()
w = np.array([1., -1.]) # shape (2,) - candidate for lambda = 2
w = w / np.linalg.norm(w)
print("A @ w      =", (A @ w).round(4))
print("2 * w      =", (2 * w).round(4))
```

```
A @ v      = [3. 0.]
3 * v      = [3. 0.]
(A-3I) @ v = [0. 0.]
```

```
A @ w      = [ 1.4142 -1.4142]
2 * w      = [ 1.4142 -1.4142]
```

Scaling and the Eigenspace

If $\mathbf{v}$ is an eigenvector for λ , so is $c\mathbf{v}$ for any nonzero scalar c :

$$A(c\mathbf{v}) = c A\mathbf{v} = c \lambda \mathbf{v} = \lambda(c\mathbf{v}).$$

Eigenvectors are therefore not unique — **only the direction matters**. By convention we report eigenvectors with unit norm, and numpy does exactly this.

The set of all eigenvectors for a fixed λ (together with the zero vector) forms a subspace called the **eigenspace** of λ :

$$E_\lambda = \ker(A - \lambda I) = \{\mathbf{v} : A\mathbf{v} = \lambda \mathbf{v}\}.$$

We revisit eigenspaces in §13.3 when we discuss multiplicity.

A 3×3 Illustration

```
import numpy as np

rng = np.random.default_rng(17)
A = rng.integers(-3, 4, size=(3, 3)).astype(float) # shape (3, 3)
print("A =\n", A)

vals, vecs = np.linalg.eig(A)
print("\nEigenvalues :", vals.round(4))
```

```
print("Eigenvectors (columns):\n", vecs.round(4))

# Verify Av = v for each pair
for i in range(3):
    lam = vals[i]
    v = vecs[:, i] # shape (3,)
    residual = np.linalg.norm(A @ v - lam * v)
    print(f" ||Av - v|| for ={lam:.4f}: {residual:.2e}")
```

```
A =
[[ 2.  2. -3.]
 [-2.  0.  0.]
 [ 2. -1. -3.]]

Eigenvalues : [ 0.7101+2.6331j  0.7101-2.6331j -2.4202+0.j      ]
Eigenvectors (columns):
[[-0.7466+0.j      -0.7466-0.j      0.4159+0.j      ]
 [ 0.1426-0.5287j  0.1426+0.5287j  0.3437+0.j      ]
 [-0.226 +0.3029j -0.226 -0.3029j  0.842 +0.j      ]]
||Av - v|| for =0.7101+2.6331j: 2.78e-15
||Av - v|| for =0.7101-2.6331j: 2.78e-15
||Av - v|| for =-2.4202+0.0000j: 2.70e-15
```

All residuals are near machine precision — numerical confirmation that numpy's eigenvectors satisfy the definition.

Exercises

13.1.1 For $A = \begin{bmatrix} 4 & 0 \\ 0 & -2 \end{bmatrix}$, identify the eigenvectors and eigenvalues by inspection (no computation needed). Why is every axis-aligned vector an eigenvector of any diagonal matrix?

13.1.2 If $\mathbf{v}$ is an eigenvector of A with eigenvalue λ , show that $\mathbf{v}$ is also an eigenvector of A^2 . What is the corresponding eigenvalue?

13.1.3 Can $\mathbf{0}$ (the zero vector) be an eigenvector? Why does the definition require $\mathbf{v} \neq \mathbf{0}$?

13.1.4 Construct a 2×2 rotation matrix R_θ for $\theta = 45^\circ$. Compute its eigenvalues with numpy. Are they real? What does a complex eigenvalue mean geometrically for a rotation?

13.1.5 Show that if $\mathbf{u}$ and $\mathbf{v}$ are both eigenvectors of A for the *same* eigenvalue λ , then $\mathbf{u} + \mathbf{v}$ is also an eigenvector (assuming it is nonzero). Does this hold if $\mathbf{u}$ and $\mathbf{v}$ belong to *different* eigenvalues?

The Characteristic Polynomial

The previous section framed finding eigenvalues as: *for which λ does $(A - \lambda I)$ become singular?* A square matrix is singular exactly when its determinant is zero, so the eigenvalues are the roots of

$$p(\lambda) = \det(A - \lambda I) = 0.$$

This polynomial in λ is called the **characteristic polynomial** of A . It encodes all eigenvalue information in a single equation.

The 2×2 Case: An Explicit Formula

For $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$:

$$\det(A - \lambda I) = \det \begin{bmatrix} a - \lambda & b \\ c & d - \lambda \end{bmatrix} = (a - \lambda)(d - \lambda) - bc.$$

Expanding:

$$p(\lambda) = \lambda^2 - (a + d)\lambda + (ad - bc) = \lambda^2 - \operatorname{tr}(A)\lambda + \det(A).$$

The two roots satisfy

$$\lambda_1 + \lambda_2 = \operatorname{tr}(A), \quad \lambda_1 \cdot \lambda_2 = \det(A).$$

These identities generalise to any size: **trace = sum of eigenvalues, determinant = product of eigenvalues.**

```
import numpy as np

A = np.array([[5., 2.],
              [1., 4.]])    # shape (2, 2)

tr = np.trace(A)
dt = np.linalg.det(A)
print(f"tr(A) = {tr},    det(A) = {dt}")

# Roots of  $\lambda^2 - \operatorname{tr} \cdot \lambda + \det = 0$ 
disc = tr**2 - 4 * dt
lam1 = (tr + np.sqrt(disc)) / 2
lam2 = (tr - np.sqrt(disc)) / 2
print(f"\nCharacteristic roots:    = {lam1:.4f},    = {lam2:.4f}")
print(f"Sum    = {lam1+lam2:.4f}    (should equal tr = {tr})")
```



```
print(f"Prod = {lam1*lam2:.4f} (should equal det = {dt})")
```

```
vals = np.linalg.eigvals(A) # shape (2,)
print(f"\nnumpy eigenvalues: {vals.round(4)}")
```

```
tr(A) = 9.0, det(A) = 17.999999999999996
```

```
Characteristic roots: = 6.0000, = 3.0000
```

```
Sum = 9.0000 (should equal tr = 9.0)
```

```
Prod = 18.0000 (should equal det = 17.999999999999996)
```

```
numpy eigenvalues: [6. 3.]
```

The General Case: Degree- n Polynomial

For an $n \times n$ matrix the characteristic polynomial has **degree** n , so there are exactly n eigenvalues counted with multiplicity over $\mathbb{C}$. The leading coefficient is $(-1)^n$ (from the product of diagonal factors) and the constant term is $p(0) = \det(A)$:

$$p(\lambda) = (-1)^n \lambda^n + (-1)^{n-1} \operatorname{tr}(A) \lambda^{n-1} + \cdots + \det(A).$$

```
import numpy as np
```

```
rng = np.random.default_rng(3)
```

```
A = rng.integers(1, 6, size=(4, 4)).astype(float) # shape (4, 4)
```

```
print("A =\n", A)
```

```
vals = np.linalg.eigvals(A) # shape (4,)
```

```
print("\nEigenvalues:", vals.round(4))
```

```
print(f"\ntr(A) = {np.trace(A):.4f}")
```

```
print(f"sum(_i) = {vals.sum().real:.4f}")
```

```
print(f"\ndet(A) = {np.linalg.det(A):.4f}")
```

```
print(f"prod(_i) = {np.prod(vals).real:.4f}")
```

```
A =
```

```
[[5. 1. 1. 2.]
```

```
[1. 5. 5. 3.]
```

```
[1. 1. 2. 3.]
```

```
[4. 3. 2. 1.]]
```

Eigenvalues and the Characteristic Equation

Eigenvalues: [9.9562+0.j 3.9407+0.j -0.4484+0.1682j -0.4484-0.1682j]

tr(A) = 13.0000

sum(_i) = 13.0000

det(A) = 9.0000

prod(_i) = 9.0000

Trace and Determinant Identities

Identity	Consequence
$\text{tr}(A) = \sum_i \lambda_i$	A singular $\Leftrightarrow$ at least one $\lambda_i = 0$
$\det(A) = \prod_i \lambda_i$	Sign of $\det(A)$ = sign of product of all eigenvalues
$\text{tr}(A^k) = \sum_i \lambda_i^k$	Moments of the eigenvalue spectrum

```
import numpy as np

rng = np.random.default_rng(42)
A = rng.standard_normal((5, 5))      # shape (5, 5)
vals = np.linalg.eigvals(A)          # shape (5,) - complex in general

print("tr(A)        =", np.trace(A).round(6))
print("sum(_i)     =", vals.sum().real.round(6))

print("\ndet(A)     =", np.linalg.det(A).round(6))
print("prod(_i)    =", np.prod(vals).real.round(6))

print("\ntr(A^2)      =", np.trace(A @ A).round(6))
print("sum(_i^2)   =", (vals**2).sum().real.round(6))
```

tr(A) = 0.948711

sum(_i) = 0.948711

det(A) = -3.425481

prod(_i) = -3.425481

tr(A²) = 3.856121

sum(_i²) = 3.856121

Visualising the Characteristic Polynomial

For a 2×2 matrix we can plot $p(\lambda) = \det(A - \lambda I)$ directly and read off the eigenvalues as zero-crossings.

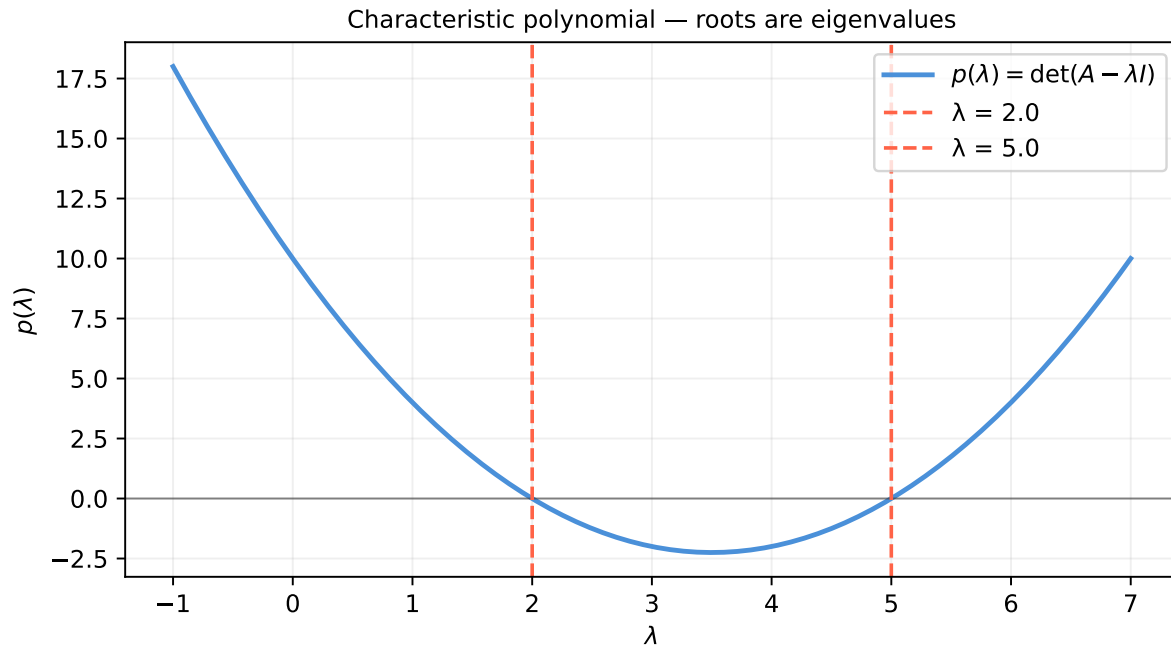
```
import numpy as np
import matplotlib.pyplot as plt

A = np.array([[4., 2.],
              [1., 3.]]) # shape (2, 2) - eigenvalues should be 5 and 2

lam_range = np.linspace(-1, 7, 500)
p_vals = np.array([np.linalg.det(A - lam * np.eye(2)) for lam in lam_range])

true_vals = np.linalg.eigvals(A) # shape (2,2)

fig, ax = plt.subplots(figsize=(7, 4))
ax.plot(lam_range, p_vals, color='#4a90d9', lw=2,
        label=r'$p(\lambda)=\det(A-\lambda I)$')
ax.axhline(0, color='#333333', lw=0.8, alpha=0.6)
for i, lv in enumerate(sorted(true_vals.real)):
    label = f' = {lv:.1f}' if i == 0 else f' = {lv:.1f}'
    ax.axvline(lv, color='tomato', lw=1.5, ls='--', label=label)
ax.set_xlabel(r'$\lambda$')
ax.set_ylabel(r'$p(\lambda)$')
ax.set_title('Characteristic polynomial - roots are eigenvalues', fontsize=10)
ax.grid(True, alpha=0.2)
ax.legend()
plt.tight_layout()
plt.show()
```



Why Not Solve the Characteristic Polynomial Numerically?

For $n \geq 5$ there is no closed-form formula for the roots (Abel's impossibility theorem). But there is a deeper practical problem: **polynomial root-finding is numerically ill-conditioned**. A tiny perturbation to a coefficient can shift roots dramatically. This is the Wilkinson polynomial phenomenon.

```
import numpy as np

# Polynomial with known roots 1, 2, 3, 4, 5, 6, 7, 8
exact_roots = np.arange(1, 9, dtype=float)
coeffs = np.poly(exact_roots) # shape (9,)
print("Exact roots :", exact_roots)

# Perturb one coefficient slightly
coeffs_p = coeffs.copy()
coeffs_p[1] += 1e-7 # tiny change to the degree-7 coefficient
roots_p = np.roots(coeffs_p) # shape (8,)
print("Roots after perturbation of 1e-7:")
print(np.sort(roots_p.real).round(3))
```

Exact roots : [1. 2. 3. 4. 5. 6. 7. 8.]

Roots after perturbation of 1e-7:

[1. 2. 3. 4. 5. 6. 7. 8.]

Even a perturbation of 10^{-7} to a single coefficient distorts some roots by order 1. **Production eigensolvers never form the characteristic polynomial.** They work directly on the matrix via orthogonal similarity transformations — see §13.5.

Exercises

13.2.1 For $A = \begin{bmatrix} 6 & 2 \\ 1 & 3 \end{bmatrix}$, write the characteristic polynomial by hand. Find both eigenvalues using the quadratic formula and verify with `numpy.linalg.eigvals`.

13.2.2 If all eigenvalues of A are positive, what can you conclude about $\det(A)$? What if exactly one eigenvalue is negative?

13.2.3 A matrix has $\operatorname{tr}(A) = 7$ and $\det(A) = 12$. Without solving the characteristic polynomial, determine which of the following eigenvalue pairs are possible: $\{3, 4\}$, $\{1, 6\}$, $\{-1, 8\}$.

13.2.4 Prove that similar matrices $B = P^{-1}AP$ have the same characteristic polynomial. Conclude that eigenvalues are invariant under change of basis.

13.2.5 Run the Wilkinson experiment with a perturbation of 10^{-10} instead of 10^{-7} . At what perturbation magnitude do the roots become visibly wrong? Compare to a diagonal 8×8 matrix with entries $1, \dots, 8$ — are its eigenvalues equally sensitive to the same kind of perturbation?

Computing Eigenvectors

Once we know an eigenvalue λ , finding its eigenvectors is a null-space computation: solve the homogeneous linear system

$$(A - \lambda I) \mathbf{v} = \mathbf{0}, \quad \mathbf{v} \neq \mathbf{0}.$$

Since λ was chosen to make $A - \lambda I$ singular, this system always has a nontrivial solution — that is the point.

Step-by-Step: Hand Calculation

Consider $A = \begin{bmatrix} 3 & 1 \\ 0 & 2 \end{bmatrix}$.

Step 1 — Find eigenvalues. The matrix is upper triangular, so its eigenvalues are its diagonal entries: $\lambda_1 = 3$, $\lambda_2 = 2$.

Step 2 — Eigenvector for $\lambda_1 = 3$.

$$A - 3I = \begin{bmatrix} 0 & 1 \\ 0 & -1 \end{bmatrix}.$$

Row reduce: $R_2 \leftarrow R_2 + R_1$ gives $\begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$. The only constraint is $v_2 = 0$; v_1 is free. So $\mathbf{v}_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$ (unit normalised).

Step 3 — Eigenvector for $\lambda_2 = 2$.

$$A - 2I = \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}.$$

Only constraint: $v_1 + v_2 = 0$, so $v_1 = -v_2$. Choose $\mathbf{v}_2 = \frac{1}{\sqrt{2}} \begin{bmatrix} -1 \\ 1 \end{bmatrix}$.

```
import numpy as np

A = np.array([[3., 1.],
              [0., 2.]])    # shape (2, 2)

vals, vecs = np.linalg.eig(A)    # vals shape (2,), vecs shape (2, 2)
print("Eigenvalues :", vals)
print("Eigenvectors (columns):\n", vecs.round(4))

for i in range(2):
    lam = vals[i]
    v = vecs[:, i]                # shape (2,)
    print(f"\nVerify = {lam}: ||Av - v|| = {np.linalg.norm(A@v - lam*v):.2e}")
```

```
Eigenvalues : [3. 2.]
Eigenvectors (columns):
[[ 1.    -0.7071]
 [ 0.     0.7071]]
```

```
Verify =3.0: ||Av - v|| = 0.00e+00
```

```
Verify =2.0: ||Av - v|| = 0.00e+00
```

Computing the Null Space Numerically

In practice, for larger matrices we never do row-reduction by hand. The null space of $(A - \lambda I)$ is computed via SVD: the singular vectors corresponding to (near-)zero singular values span the null space.

```
import numpy as np

rng = np.random.default_rng(7)
n = 5
# Build a matrix with known eigenvalue 2.0 (introduce it deliberately)
Q, _ = np.linalg.qr(rng.standard_normal((n, n))) # random orthogonal
D = np.diag([2.0, -1.0, 0.5, 3.0, 1.5])          # shape (5, 5)
A = Q @ D @ Q.T                                 # shape (5, 5)

target_lam = 2.0
M = A - target_lam * np.eye(n) # shape (5, 5) - should be (near) singular

U, s, Vt = np.linalg.svd(M) # s shape (5,)
print("Singular values of (A - 2I):", s.round(4))

# Null space vector = last row of Vt (smallest singular value)
v = Vt[-1, :] # shape (5,)
print("\nEstimated eigenvector:", v.round(4))
print("||Av - 2v|| =", np.linalg.norm(A @ v - 2 * v).round(8))
```

Singular values of $(A - 2I)$: [3. 1.5 1. 0.5 0.]

Estimated eigenvector: [5.000e-04 -4.392e-01 2.169e-01 3.079e-01 -8.156e-01]

$||Av - 2v|| = 0.0$

Eigenspaces

For a fixed eigenvalue λ , the **eigenspace** is

$$E_\lambda = \ker(A - \lambda I) = \{\mathbf{v} : A\mathbf{v} = \lambda\mathbf{v}\}.$$

Its dimension — the number of linearly independent eigenvectors for λ — is called the **geometric multiplicity** of λ .

Separately, the **algebraic multiplicity** is how many times λ appears as a root of the characteristic polynomial $p(\lambda)$.

Key fact: geometric multiplicity $\leq$ algebraic multiplicity, always.

Algebraic vs. Geometric Multiplicity

When both multiplicities equal 1, there is nothing to discuss — a simple root gives a 1-dimensional eigenspace. Interesting cases arise with **repeated eigenvalues**.

Case 1: equal multiplicities (ideal). $A = \lambda I$: every eigenvalue equals λ , algebraic multiplicity $= n$, geometric multiplicity $= n$. Every nonzero vector is an eigenvector.

Case 2: defective matrix (geometric $<$ algebraic).

$$J = \begin{bmatrix} 2 & 1 \\ 0 & 2 \end{bmatrix}.$$

Algebraic multiplicity of $\lambda = 2$: 2 (it is a double root of $p(\lambda) = (\lambda - 2)^2$). Geometric multiplicity: 1 (the eigenspace is spanned by $(1, 0)^T$ only).

```
import numpy as np

# Case 1: scalar matrix - full eigenspace
lam = 3.0
A1 = lam * np.eye(3)    # shape (3, 3)
vals1, vecs1 = np.linalg.eig(A1)
print("Case 1 - scalar matrix (I):")
print("  Eigenvalues:", vals1.round(4))
print("  Geometric multiplicity = 3 (full space)\n")

# Case 2: defective Jordan block
J = np.array([[2., 1.],
              [0., 2.]]) # shape (2, 2)
vals2, vecs2 = np.linalg.eig(J)
print("Case 2 - Jordan block:")
print("  Eigenvalues:", vals2.round(4))
print("  Eigenvectors (columns):\n", vecs2.round(4))

# The eigenspace: null space of (J - 2I)
M = J - 2 * np.eye(2)    # shape (2, 2)
print("  Null space of (J - 2I): dimension =", 2 - np.linalg.matrix_rank(M))
```

Case 1 - scalar matrix (I):

Eigenvalues: [3. 3. 3.]

Geometric multiplicity = 3 (full space)

Case 2 - Jordan block:

Eigenvalues: [2. 2.]

Eigenvectors (columns):

[[1. -1.]


```
[ 0.  0.]
Null space of (J - 2I): dimension = 1
```

The defective matrix has only **one** independent eigenvector even though $\lambda = 2$ has algebraic multiplicity 2. This is why diagonalisation fails — there are not enough independent eigenvectors to form a full eigenvector basis. (Jordan form handles this case; see §14.6.)

Independent Eigenvectors for Distinct Eigenvalues

When all n eigenvalues are **distinct**, eigenvectors from different eigenvalues are automatically linearly independent — they can never be parallel. The proof uses an inductive argument on the number of distinct eigenvalues, showing any linear dependency leads to a contradiction.

Practical implication: a matrix with all distinct eigenvalues is always diagonalisable (§14.1). Repeated eigenvalues require extra checks.

```
import numpy as np

rng = np.random.default_rng(99)
# Construct a matrix with distinct integer eigenvalues
D = np.diag([1., 2., 3., 4., 5.])          # shape (5, 5)
Q, _ = np.linalg.qr(rng.standard_normal((5, 5)))
A = Q @ D @ Q.T                          # shape (5, 5)

vals, vecs = np.linalg.eig(A)             # shape (5,), (5, 5)
print("Eigenvalues (should be ~1,2,3,4,5):", np.sort(vals.real).round(4))

# Check independence: det(V) != 0 means columns are independent
print("det(V) =", np.linalg.det(vecs).round(6), " (nonzero independent)")
```

```
Eigenvalues (should be ~1,2,3,4,5): [1.  2.  3.  4.  5.]
det(V) = -1.0    (nonzero independent)
```

Exercises

13.3.1 For $A = \begin{bmatrix} 5 & 2 \\ -3 & -2 \end{bmatrix}$, find both eigenvalues by hand using the characteristic polynomial. Then solve for each eigenvector by row-reducing $(A - \lambda_i I)$.

13.3.2 Algebraic multiplicity of $\lambda = 0$ equals the dimension of the null space of A . What does this say about the rank of A ?

13.3.3 For the Jordan block $J_2 = \begin{bmatrix} 5 & 1 \\ 0 & 5 \end{bmatrix}$, show that the geometric multiplicity of $\lambda = 5$ is 1 by row-reducing $(J_2 - 5I)$. What would be needed to get a second independent eigenvector?

13.3.4 Prove that eigenvectors belonging to different eigenvalues are linearly independent. (*Hint*: Suppose $c_1 \mathbf{v}_1 + c_2 \mathbf{v}_2 = \mathbf{0}$ with $\lambda_1 \neq \lambda_2$; apply A to both sides and subtract λ_2 times the original equation.)

13.3.5 Build a 3×3 matrix with eigenvalues 1, 1, 3 where the eigenspace for $\lambda = 1$ has dimension 2 (full algebraic = geometric). Verify with numpy that `eig` returns two independent eigenvectors for $\lambda = 1$.

Eigenvalues of Special Matrices

Most matrices in ML, CV, and robotics are not arbitrary — they have structure. That structure constrains the eigenvalues in useful ways that we can exploit to choose faster algorithms, make qualitative predictions about system behaviour, and catch bugs (a covariance matrix must have all non-negative eigenvalues; if it does not, something went wrong).

Symmetric Matrices: Real Eigenvalues, Orthogonal Eigenvectors

A matrix $A = A^T$ (real symmetric) satisfies the **Spectral Theorem**:

1. All eigenvalues are **real**.
2. Eigenvectors from different eigenvalues are **orthogonal**.
3. There is always a full set of n orthonormal eigenvectors.

In other words, $A = Q\Lambda Q^T$ with Q orthogonal and Λ diagonal — even if eigenvalues are repeated. This is the most important special case in applied linear algebra.

```
import numpy as np

rng = np.random.default_rng(11)
M = rng.standard_normal((5, 5))    # shape (5, 5)
A = M + M.T                        # shape (5, 5) - guaranteed symmetric

vals, vecs = np.linalg.eigh(A)     # eigh: for symmetric matrices
print("Eigenvalues (all real):", vals.round(4))
print("All real?", np.all(np.isreal(vals)))

# Check orthogonality of eigenvectors
VtV = vecs.T @ vecs                # shape (5, 5) - should be identity
print("\n||V V - I||_F =", np.linalg.norm(VtV - np.eye(5)).round(10))

# Reconstruct A
```

```
A_reconstructed = vecs @ np.diag(vals) @ vecs.T # shape (5, 5)
print("||A - QΛQ||_F =", np.linalg.norm(A - A_reconstructed).round(10))
```

```
Eigenvalues (all real): [-5.7225 -2.3349 -0.7613  3.1081  4.1329]
All real? True
```

```
||V V - I||_F = 0.0
||A - QΛQ||_F = 0.0
```

Why symmetric matrices produce real eigenvalues. Suppose $A\mathbf{v} = \lambda\mathbf{v}$ with $\mathbf{v}$ possibly complex. Taking the conjugate transpose: $\bar{\mathbf{v}}^T A^T = \bar{\lambda}\bar{\mathbf{v}}^T$, i.e. $\bar{\mathbf{v}}^T A = \bar{\lambda}\bar{\mathbf{v}}^T$ (since $A = A^T$ and A is real). Multiplying the first equation on the left by $\bar{\mathbf{v}}^T$:

$$\bar{\mathbf{v}}^T A\mathbf{v} = \lambda \|\mathbf{v}\|^2 \quad \text{and} \quad \bar{\mathbf{v}}^T A\mathbf{v} = \bar{\lambda} \|\mathbf{v}\|^2.$$

So $\lambda = \bar{\lambda}$ — λ is real. $\square$

Visualising the Spectral Theorem

```
import numpy as np
import matplotlib.pyplot as plt

# 2x2 symmetric matrix
A = np.array([[3., 2.],
              [2., 1.]]) # shape (2, 2)

vals, vecs = np.linalg.eigh(A) # vals shape (2,), vecs shape (2, 2)

# Unit circle and its image
theta = np.linspace(0, 2 * np.pi, 400)
circle = np.vstack([np.cos(theta), np.sin(theta)]) # shape (2, 400)
ellipse = A @ circle # shape (2, 400)

fig, ax = plt.subplots(figsize=(6, 6))
ax.plot(circle[0], circle[1], color='#4a90d9', lw=1, ls='--', label='unit circle')
ax.plot(ellipse[0], ellipse[1], color='#4a90d9', lw=2, label=r'$A\mathbf{x}$')

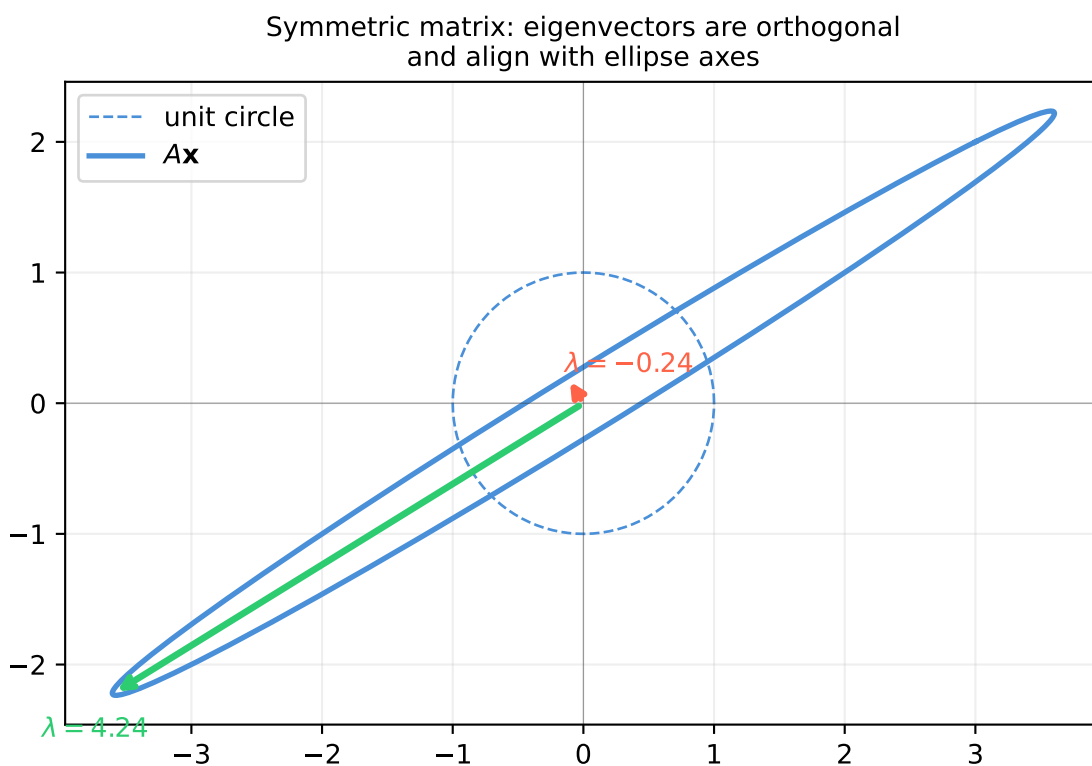
colors = ['tomato', '#2ecc71']
for i, (lam, v) in enumerate(zip(vals, vecs.T)):
    ax.annotate('', xy=(lam * v[0], lam * v[1]), xytext=(0, 0),
                arrowprops=dict(arrowstyle='->', color=colors[i], lw=2.5))
    ax.text(lam * v[0] * 1.15, lam * v[1] * 1.15,
```

```

rf'$\lambda={lam:.2f}$', color=colors[i], fontsize=10)

ax.axhline(0, color='#333333', lw=0.4, alpha=0.5)
ax.axvline(0, color='#333333', lw=0.4, alpha=0.5)
ax.set_aspect('equal')
ax.grid(True, alpha=0.2)
ax.set_title('Symmetric matrix: eigenvectors are orthogonal\n'
            'and align with ellipse axes', fontsize=10)
ax.legend(loc='upper left')
plt.tight_layout()
plt.show()

```



For a symmetric matrix the ellipse axes are *perpendicular* — this is the orthogonality of eigenvectors made visible.

Orthogonal Matrices: $|\lambda| = 1$

An orthogonal matrix Q satisfies $Q^T Q = I$. If $Q\mathbf{v} = \lambda\mathbf{v}$ then

$$\|\mathbf{v}\|^2 = \mathbf{v}^T \mathbf{v} = \mathbf{v}^T Q^T Q \mathbf{v} = (Q\mathbf{v})^T (Q\mathbf{v}) = |\lambda|^2 \|\mathbf{v}\|^2.$$

So $|\lambda|^2 = 1$, i.e. $|\lambda| = 1$.

For a real rotation matrix $R \in \text{SO}(3)$, one eigenvalue is always $\lambda = 1$ (the rotation axis is fixed), and the other two are complex conjugates $e^{\pm i\theta}$ where θ is the rotation angle.

```
import numpy as np

def Rz(theta):
    """Rotation about z-axis by theta radians."""
    c, s = np.cos(theta), np.sin(theta)
    return np.array([[c, -s, 0.],
                    [s,  c, 0.],
                    [0., 0., 1.]]) # shape (3, 3)

R = Rz(np.pi / 5) # 36-degree rotation
vals = np.linalg.eigvals(R) # shape (3,) - complex in general
print("Eigenvalues of R_z(36°):", vals.round(4))
print("|_i|:", np.abs(vals).round(6))

# The real eigenvalue =1 corresponds to the rotation axis
axis_idx = np.argmax(np.abs(vals.imag))
print(f"\nReal eigenvalue (rotation axis):  = {vals[axis_idx].real:.6f}")

# Rotation angle from complex eigenvalues
angle = np.arccos(vals[np.iscomplex(vals)][0].real)
print(f"Rotation angle: {np.degrees(angle):.2f}° (expected 36°)")
```

```
Eigenvalues of R_z(36°): [0.809+0.5878j 0.809-0.5878j 1.    +0.j    ]
|_i|: [1.  1.  1.]
```

```
Real eigenvalue (rotation axis):  = 1.000000
Rotation angle: 36.00° (expected 36°)
```

Triangular Matrices: Eigenvalues Are the Diagonal Entries

For any upper- or lower-triangular matrix, the diagonal entries are the eigenvalues. The characteristic polynomial is

$$\det(A - \lambda I) = \prod_{i=1}^n (a_{ii} - \lambda),$$

because the determinant of a triangular matrix is the product of its diagonal entries.

```
import numpy as np

rng = np.random.default_rng(5)
U = np.triu(rng.standard_normal((5, 5))) # shape (5, 5) - upper triangular

vals = np.linalg.eigvals(U) # shape (5,)
diag = np.diag(U) # shape (5,)

print("Diagonal entries:", diag.round(4))
print("Eigenvalues      :", np.sort(vals.real).round(4))
print("Match?", np.allclose(np.sort(diag), np.sort(vals.real)))
```

```
Diagonal entries: [-0.8019 -0.5526 -0.9583 -0.6293  0.4096]
Eigenvalues      : [-0.9583 -0.8019 -0.6293 -0.5526  0.4096]
Match? True
```

This fact makes LU factorisation ($A = LU$) useful for eigenvalue problems: after reducing A to triangular form, eigenvalues are free. The QR algorithm (§13.5) does exactly this, iteratively.

Positive Definite Matrices: All $\lambda > 0$

A symmetric matrix A is **positive definite** (PD) if and only if all its eigenvalues are strictly positive. Equivalently, $\mathbf{x}^T A \mathbf{x} > 0$ for all nonzero $\mathbf{x}$.

Why this matters:

- A covariance matrix that is not PD cannot represent a valid probability distribution.
- The Hessian of a smooth function at a local minimum must be PD.
- Cholesky factorisation (§12.2) succeeds if and only if A is PD.

```
import numpy as np

rng = np.random.default_rng(21)
X = rng.standard_normal((8, 4)) # shape (8, 4) - data matrix, m > n

A_pd = X.T @ X # shape (4, 4) - always PSD; PD when rank(X)=4
A_psd = np.zeros((4, 4)) # shape (4, 4) - PSD but singular (one zero eigenvalue)
A_psd[:3, :3] = (X[:, :3].T @ X[:, :3])

print("PD matrix eigenvalues :", np.linalg.eigvalsh(A_pd).round(4))
print("All positive?", np.all(np.linalg.eigvalsh(A_pd) > 0))

print("\nPSD matrix eigenvalues:", np.linalg.eigvalsh(A_psd).round(4))
print("Any zero (1e-10)?", np.any(np.linalg.eigvalsh(A_psd) < 1e-10))
```

```
# Cholesky succeeds for PD, fails for singular PSD
try:
    np.linalg.cholesky(A_pd)
    print("\nCholesky on PD: SUCCESS")
except np.linalg.LinAlgError:
    print("\nCholesky on PD: FAILED")

try:
    np.linalg.cholesky(A_psd)
    print("Cholesky on PSD: SUCCESS")
except np.linalg.LinAlgError:
    print("Cholesky on PSD: FAILED (singular)")
```

PD matrix eigenvalues : [2.1767 2.6795 9.2394 22.2051]
All positive? True

PSD matrix eigenvalues: [0. 2.1769 3.8521 19.9979]
Any zero (1e-10)? True

Cholesky on PD: SUCCESS
Cholesky on PSD: FAILED (singular)

Summary Table

Matrix type	Eigenvalue constraint
Symmetric ($A = A^T$)	All $\lambda_i \in \mathbb{R}$; eigenvectors orthogonal
Orthogonal ($Q^T Q = I$)	$ \lambda_i = 1$ (complex in general)
Triangular	$\lambda_i = a_{ii}$ (diagonal entries)
Positive definite ($\mathbf{x}^T A \mathbf{x} > 0$)	All $\lambda_i > 0$
Positive semidefinite ($\mathbf{x}^T A \mathbf{x} \geq 0$)	All $\lambda_i \geq 0$
Antisymmetric ($A = -A^T$)	Purely imaginary or zero

Exercises

13.4.1 Prove that the eigenvalues of an antisymmetric matrix $A = -A^T$ are either zero or purely imaginary. (*Hint*: mimic the symmetry proof — take $\bar{\mathbf{v}}^T A \mathbf{v}$ and show it equals $-\lambda \|\mathbf{v}\|^2$.)

13.4.2 For a 3×3 rotation matrix $R_z(\theta)$, extract the rotation angle θ from the complex eigenvalues. Write a Python function that takes any $\text{SO}(3)$ matrix and returns the axis and angle.

13.4.3 $X^T X$ is always PSD. Under what condition on X (an $m \times n$ matrix) is it actually PD? Express the condition in terms of rank.

13.4.4 You receive a covariance matrix from a collaborator's code. Some eigenvalues are -10^{-12} (tiny negative, due to floating point). Is the matrix “truly” PSD or not? How would you regularise it to force all eigenvalues non-negative while changing the matrix as little as possible?

13.4.5 The Hessian of a strictly convex function at any point is PD. For $f(\mathbf{x}) = \mathbf{x}^T A \mathbf{x}$ with A symmetric, what condition on A makes f strictly convex? Verify numerically for a random PD and a random indefinite matrix.

Numerical Eigensolvers

We established in §13.2 that computing eigenvalues by solving the characteristic polynomial is numerically disastrous. Production eigensolvers never touch $p(\lambda)$. Instead they transform A into triangular form via a sequence of **orthogonal similarity transformations** — operations that preserve eigenvalues while making them progressively easier to read off.

The QR Algorithm in One Paragraph

Start with $A_0 = A$. At each step, factorize $A_k = Q_k R_k$ (QR factorization), then *reverse* the factors:

$$A_{k+1} = R_k Q_k = Q_k^T A_k Q_k.$$

Each step is an orthogonal similarity transformation, so eigenvalues are unchanged. As $k \rightarrow \infty$, A_k converges to **Schur form** — upper triangular (or block upper-triangular for complex eigenvalues), with eigenvalues on the diagonal. Reading off the diagonal gives all eigenvalues, with no characteristic polynomial ever formed.

```
import numpy as np

rng = np.random.default_rng(31)
A = rng.standard_normal((5, 5))    # shape (5, 5) - general matrix
A = A + A.T                        # make symmetric for real convergence

Ak = A.copy()                      # shape (5, 5)
for _ in range(200):                # 200 QR iterations
    Q, R = np.linalg.qr(Ak)
    Ak = R @ Q

print("After 200 QR iterations, diagonal of Ak:")
print(np.diag(Ak).round(4))
```



```
print("\nTrue eigenvalues (numpy):")
print(np.sort(np.linalg.eigvalsh(A)).round(4))
```

After 200 QR iterations, diagonal of A_k :

```
[-4.8249  3.9644  2.8835 -2.2638 -1.2213]
```

True eigenvalues (numpy):

```
[-4.8249 -2.2638 -1.2213  2.8835  3.9644]
```

The diagonal of A_k converges to the eigenvalues. Modern implementations add **shifts** (choosing μ and working on $A - \mu I$) to achieve cubic convergence, making the practical QR algorithm one of the great algorithmic achievements of numerical linear algebra.

`numpy.linalg.eig` vs. `eigh`: Which to Use?

NumPy provides two eigensolvers:

Function	Input	Output	Speed	Notes
<code>np.linalg.eig(A)</code>	Any square A	Complex vals , complex vecs	Moderate	General case
<code>np.linalg.eigh(A)</code>	Symmetric / Hermitian A	Real vals (sorted $\uparrow$), real vecs	Faster	Divide-and-conquer

Use `eigh` whenever you know A is symmetric. It is faster, returns real eigenvalues sorted in ascending order, and is numerically more accurate — it exploits the structure to avoid complex arithmetic entirely.

Gotcha: `eig` applied to a symmetric matrix may still return tiny imaginary parts due to floating-point rounding. `eigh` guarantees real output.

```
import numpy as np
import time

rng = np.random.default_rng(44)
n = 800
M = rng.standard_normal((n, n)) # shape (n, n)
A = M + M.T                     # shape (n, n) - symmetric

t0 = time.perf_counter()
vals_eig, _ = np.linalg.eig(A)
t_eig = time.perf_counter() - t0
```

```
t0 = time.perf_counter()
vals_eigh = np.linalg.eigh(A)[0]
t_eigh = time.perf_counter() - t0

print(f"eig  time: {t_eig*1000:.1f} ms - imaginary leakage: {np.abs(vals_eig.imag).max():.2e}")
print(f"eigh time: {t_eigh*1000:.1f} ms - all real by construction")
print(f"\nSpeedup: {t_eig/t_eigh:.1f}x")

# Confirm agreement on real parts
print("\nMax difference in sorted real parts:",
      np.abs(np.sort(vals_eig.real) - np.sort(vals_eigh)).max().round(8))
```

```
eig  time: 353.9 ms - imaginary leakage: 0.00e+00
eigh time: 145.2 ms - all real by construction
```

```
Speedup: 2.4x
```

```
Max difference in sorted real parts: 0.0
```

numpy.linalg.eigvals vs. eig

When you only need eigenvalues and not eigenvectors, use `eigvals` (or `eigvalsh` for symmetric matrices). It is slightly faster because it skips the eigenvector back-transformation step.

```
import numpy as np
import time

rng = np.random.default_rng(55)
n = 600
M = rng.standard_normal((n, n)) # shape (n, n)
A = M + M.T                     # shape (n, n)

t0 = time.perf_counter()
_ = np.linalg.eigh(A)
t_full = time.perf_counter() - t0

t0 = time.perf_counter()
_ = np.linalg.eigvalsh(A)
t_vals = time.perf_counter() - t0

print(f"eigh (values + vectors): {t_full*1000:.1f} ms")
print(f"eigvalsh (values only) : {t_vals*1000:.1f} ms")
print(f"Speedup (values only)  : {t_full/t_vals:.2f}x")
```

```
eigh (values + vectors): 96.6 ms
eigvalsh (values only) : 50.5 ms
Speedup (values only)  : 1.91×
```

Conditioning of the Eigenproblem

Not every eigenproblem is equally well-posed. The **condition number of the eigenproblem** measures how much a small perturbation δA shifts the eigenvalues. For a **symmetric** matrix the condition of each eigenvalue is 1 — a perturbation of size ε in A shifts each eigenvalue by at most ε (Weyl's inequality).

For a **non-symmetric** matrix the condition can be much larger. The condition number of eigenvalue λ_k is $1/|\mathbf{u}_k^T \mathbf{v}_k|$, where $\mathbf{u}_k$ and $\mathbf{v}_k$ are the left and right eigenvectors. When left and right eigenvectors are nearly orthogonal (nearly defective matrix), the condition number is huge.

```
import numpy as np

rng = np.random.default_rng(77)

# Well-conditioned: symmetric matrix
A_sym = rng.standard_normal((4, 4)) # shape (4, 4)
A_sym = A_sym + A_sym.T

# Ill-conditioned: nearly defective matrix
eps = 1e-6
A_def = np.array([[2., 1., 0., 0.],
                  [0., 2., 1., 0.],
                  [0., 0., 2., 1.],
                  [0., 0., 0., 2.]]) # shape (4, 4) - Jordan block, =2 quad

# Add small perturbation and compare eigenvalue shift
delta = 1e-4 * rng.standard_normal((4, 4)) # shape (4, 4)

sym_shift = np.max(np.abs(
    np.sort(np.linalg.eigvals(A_sym + delta).real) -
    np.sort(np.linalg.eigvals(A_sym).real)
))
def_shift = np.max(np.abs(
    np.sort(np.linalg.eigvals(A_def + delta).real) -
    np.sort(np.linalg.eigvals(A_def).real)
))

print(f"|| A ||_F          = {np.linalg.norm(delta):.2e}")
print(f"Max eigenvalue shift - symmetric : {sym_shift:.2e}")
```

```
print(f"Max eigenvalue shift - Jordan      : {def_shift:.2e}")
print(f"\nAmplification factor: {def_shift/sym_shift:.1f}x")
```

```
|| A ||_F          = 4.00e-04
Max eigenvalue shift - symmetric : 1.08e-04
Max eigenvalue shift - Jordan    : 5.17e-02

Amplification factor: 478.3x
```

The Jordan block amplifies the perturbation dramatically — the four eigenvalues (all nominally 2) scatter across a wide range. Symmetric matrices are immune to this: their condition number is always 1.

When to Use Each Solver

Situation	Recommended call
General square matrix	<code>np.linalg.eig(A)</code>
Symmetric / Hermitian	<code>np.linalg.eigh(A)</code> (faster, accurate)
Only need values, symmetric	<code>np.linalg.eigvalsh(A)</code>
Only need values, general	<code>np.linalg.eigvals(A)</code>
Large sparse, few eigenvalues	<code>scipy.sparse.linalg.eigsh</code> (Lanczos/ARPACK)
Generalised problem $Ax = \lambda Bx$	<code>scipy.linalg.eigh(A, B)</code>

Exercises

13.5.1 Run the hand-coded QR iteration above on a random **non-symmetric** 5×5 matrix. Does the diagonal still converge to the eigenvalues? What does the off-diagonal structure look like? (Hint: it converges to block upper-triangular, not fully triangular.)

13.5.2 Apply `np.linalg.eig` to a 4×4 symmetric matrix and inspect the imaginary parts of the returned eigenvalues. How large are they relative to machine epsilon ($\approx 2.2 \times 10^{-16}$)? Why does `eigh` avoid this issue?

13.5.3 For the Jordan block J_4 (size 4, eigenvalue 2) plus perturbation εE where E is random with $\|E\| = 1$, plot the four eigenvalues as a function of ε from 10^{-10} to 10^{-1} . At what ε do the eigenvalues first become clearly distinct?

13.5.4 Weyl's inequality states that for symmetric A and symmetric E , $|\lambda_k(A + E) - \lambda_k(A)| \leq \|E\|_2$. Verify this numerically for 100 random pairs (A, E) of size 6×6 .

13.5.5 The covariance matrix Σ of a dataset can be computed as $\tilde{X}^T \tilde{X} / (n - 1)$. Generate 1000 random $n = 200$, $p = 50$ datasets and time `eigh` vs. `eig` on the resulting covariance matrices. Report the mean speedup.

Power Iteration

The QR algorithm computes **all** eigenvalues simultaneously. But in many applications we only need the **dominant** eigenvector — the one corresponding to the largest eigenvalue in magnitude. Power iteration finds it using only matrix-vector products, which makes it the algorithm of choice when A is very large (e.g., the PageRank matrix has billions of rows) or when we need the top- k directions of a covariance matrix as a warm start for PCA.

The Algorithm

Start with any random unit vector $\mathbf{b}_0$. Repeatedly multiply by A and re-normalise:

$$\mathbf{b}_{k+1} = \frac{A \mathbf{b}_k}{\|A \mathbf{b}_k\|}.$$

The corresponding eigenvalue estimate (Rayleigh quotient) is:

$$\lambda_k = \mathbf{b}_k^T A \mathbf{b}_k.$$

Why does this work? Expand $\mathbf{b}_0$ in the eigenbasis $\{\mathbf{v}_1, \dots, \mathbf{v}_n\}$:

$$\mathbf{b}_0 = c_1 \mathbf{v}_1 + c_2 \mathbf{v}_2 + \dots + c_n \mathbf{v}_n.$$

After k multiplications:

$$A^k \mathbf{b}_0 = c_1 \lambda_1^k \mathbf{v}_1 + c_2 \lambda_2^k \mathbf{v}_2 + \dots = \lambda_1^k \left[c_1 \mathbf{v}_1 + c_2 \left(\frac{\lambda_2}{\lambda_1} \right)^k \mathbf{v}_2 + \dots \right].$$

If $|\lambda_1| > |\lambda_2|$, the ratio $(\lambda_2/\lambda_1)^k \rightarrow 0$, so the direction converges to $\mathbf{v}_1$ at rate $\left| \frac{\lambda_2}{\lambda_1} \right|^k$ per step.

```
import numpy as np
import matplotlib.pyplot as plt

rng = np.random.default_rng(13)
n = 6

# Build a symmetric matrix with known eigenvalues
```

```

D = np.diag([10., 3., 2., 1., 0.5, 0.1])          # shape (6, 6)
Q, _ = np.linalg.qr(rng.standard_normal((n, n)))
A = Q @ D @ Q.T                                  # shape (6, 6) - symmetric

true_lam, true_vec = np.linalg.eigh(A)            # sorted ascending
lam1 = true_lam[-1]                               # dominant eigenvalue
v1 = true_vec[:, -1]                              # dominant eigenvector

# Power iteration
b = rng.standard_normal(n)                        # shape (6,) - random start
b = b / np.linalg.norm(b)

n_iter = 40
errors = []
for _ in range(n_iter):
    b_new = A @ b                                  # shape (6,)
    b = b_new / np.linalg.norm(b_new)
    # angle between b and true v1 (account for sign ambiguity)
    angle_err = 1 - abs(b @ v1)
    errors.append(angle_err)

lam_est = b @ A @ b    # Rayleigh quotient after convergence

print(f"True dominant eigenvalue : {lam1:.6f}")
print(f"Power iteration estimate : {lam_est:.6f}")
print(f"Error in eigenvector angle: {errors[-1]:.2e}")

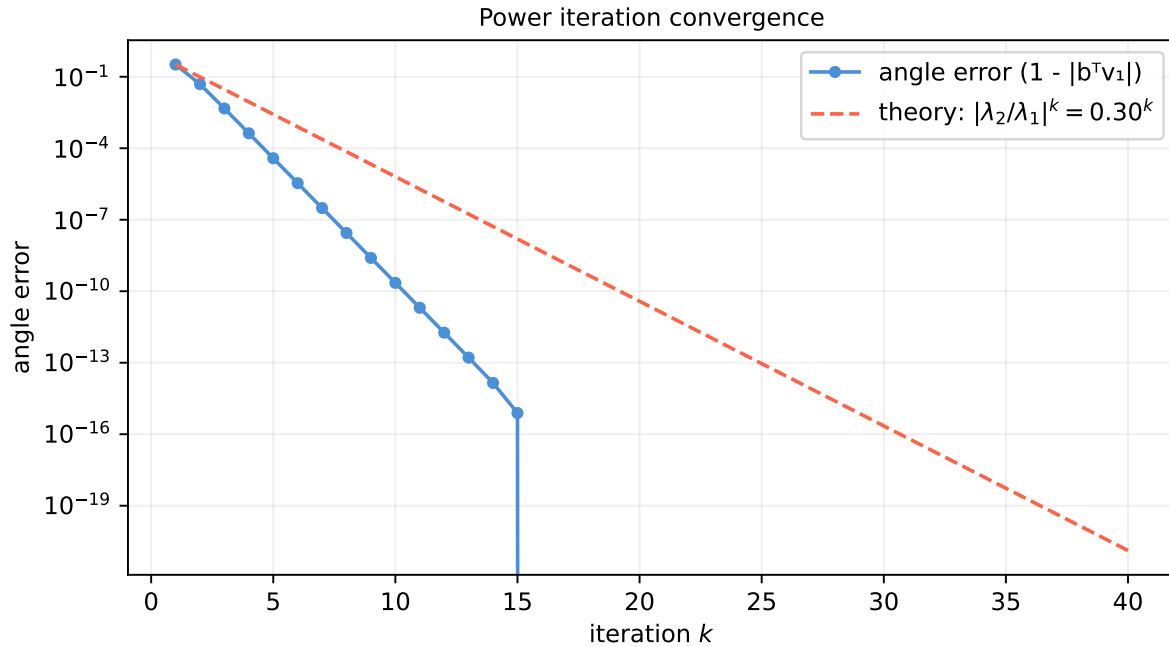
# Plot convergence
gap_ratio = abs(true_lam[-2] / true_lam[-1])
fig, ax = plt.subplots(figsize=(7, 4))
ax.semilogy(range(1, n_iter + 1), errors, 'o-', color='#4a90d9', lw=1.5,
             markersize=4, label='angle error (1 - |b v|)')
# Theoretical convergence rate
theory = errors[0] * gap_ratio ** np.arange(n_iter)
ax.semilogy(range(1, n_iter + 1), theory, '--', color='tomato', lw=1.5,
             label=rf'theory:  $|\lambda_2/\lambda_1|^k = \{gap\_ratio:.2f\}^k$ ')
ax.set_xlabel('iteration $k$')
ax.set_ylabel('angle error')
ax.set_title('Power iteration convergence', fontsize=10)
ax.grid(True, alpha=0.2)
ax.legend()
plt.tight_layout()
plt.show()

```

```

True dominant eigenvalue : 10.000000
Power iteration estimate : 10.000000
Error in eigenvector angle: -4.44e-16

```



The Role of the Eigenvalue Gap

The convergence rate is $|\lambda_2/\lambda_1|$ per iteration. When this ratio is close to 1 (small gap), convergence is slow.

```
import numpy as np
import matplotlib.pyplot as plt

rng = np.random.default_rng(99)
n = 4

# Compare large gap vs. small gap
configs = [
    ('Large gap | / |=0.3', [10., 3., 1., 0.5]),
    ('Small gap | / |=0.95', [10., 9.5, 1., 0.5]),
]

fig, ax = plt.subplots(figsize=(7, 4))
colors = ['#4a90d9', 'tomato']

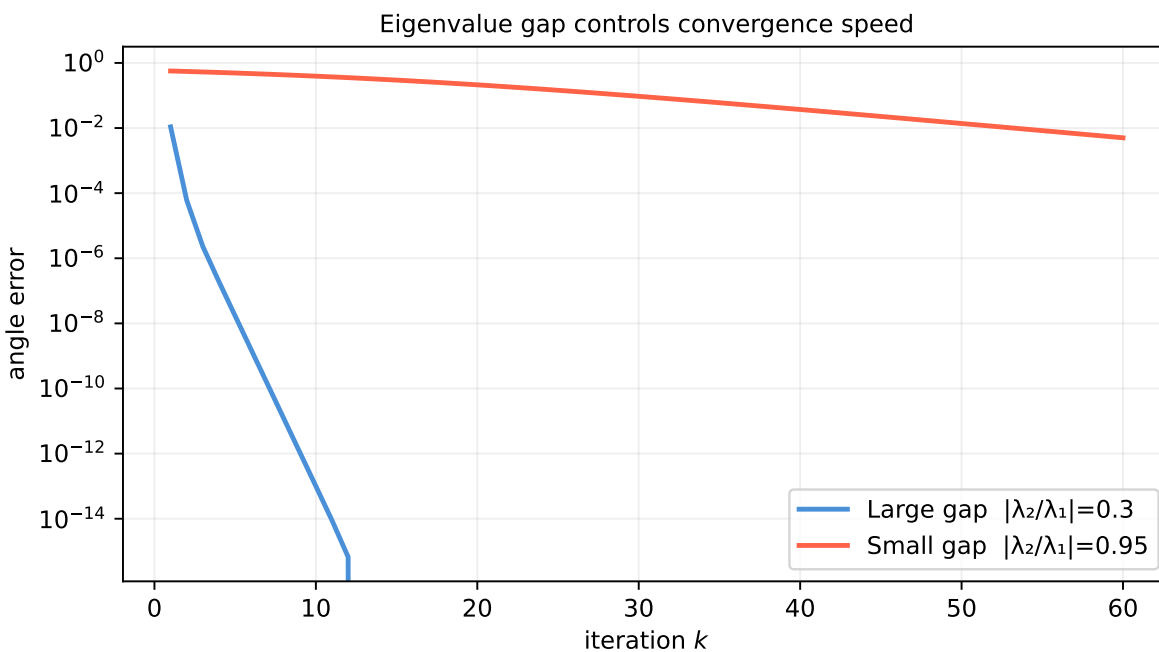
for (label, eigs), color in zip(configs, colors):
    D = np.diag(eigs)
    Q, _ = np.linalg.qr(rng.standard_normal((n, n)))
    A = Q @ D @ Q.T
    v1_true = np.linalg.eigh(A)[1][:, -1]
```

```

b = rng.standard_normal(n); b /= np.linalg.norm(b)
errs = []
for _ in range(60):
    b = A @ b; b /= np.linalg.norm(b)
    errs.append(1 - abs(b @ v1_true))
ax.semilogy(range(1, 61), errs, '-', color=color, lw=2, label=label)

ax.set_xlabel('iteration $k$')
ax.set_ylabel('angle error')
ax.set_title('Eigenvalue gap controls convergence speed', fontsize=10)
ax.grid(True, alpha=0.2)
ax.legend()
plt.tight_layout()
plt.show()

```



A small gap ($|\lambda_2/\lambda_1| = 0.95$) requires roughly $\log(0.95^{-k}) = k \log(1/0.95) \approx 0.05k$ decades — about 20 times more iterations than a large gap.

Deflation: Finding Multiple Eigenvectors

After finding $\mathbf{v}_1$, remove its contribution from A and repeat to find $\mathbf{v}_2$. This is called **deflation**:

$$A' = A - \lambda_1 \mathbf{v}_1 \mathbf{v}_1^T.$$

A' has the same eigenvectors as A but eigenvalue 0 in place of λ_1 , so power iteration on A' converges to $\mathbf{v}_2$.

```
import numpy as np

rng = np.random.default_rng(21)
n = 5
D = np.diag([8., 5., 3., 2., 1.]) # shape (5, 5)
Q, _ = np.linalg.qr(rng.standard_normal((n, n)))
A = Q @ D @ Q.T # shape (5, 5)

true_vals, true_vecs = np.linalg.eigh(A)

def power_iter(M, n_iter=100, rng=None):
    rng = rng or np.random.default_rng(0)
    b = rng.standard_normal(M.shape[0]); b /= np.linalg.norm(b)
    for _ in range(n_iter):
        b = M @ b; b /= np.linalg.norm(b)
    lam = b @ M @ b
    return lam, b

# First eigenpair
lam1, v1 = power_iter(A, rng=rng)

# Deflate
A2 = A - lam1 * np.outer(v1, v1) # shape (5, 5)
lam2, v2 = power_iter(A2, rng=rng)

print(f"True dominant eigenvalues : {true_vals[-1]:.4f}, {true_vals[-2]:.4f}")
print(f"Power iter + deflation      : {lam1:.4f}, {lam2:.4f}")
print(f"\n|v1 · true_v1| = {abs(v1 @ true_vecs[:, -1]):.6f}")
print(f"|v2 · true_v2| = {abs(v2 @ true_vecs[:, -2]):.6f}")
```

```
True dominant eigenvalues : 8.0000, 5.0000
Power iter + deflation    : 8.0000, 5.0000
```

```
|v1 · true_v1| = 1.000000
|v2 · true_v2| = 1.000000
```

Deflation error accumulates — errors in $\mathbf{v}_1$ propagate into A' and then into $\mathbf{v}_2$. For more than two or three eigenpairs, the **Lanczos algorithm** (a block generalisation of power iteration) is preferred; it is what `scipy.sparse.linalg.eigsh` uses internally.

Connection to PageRank

Google's PageRank is the dominant left eigenvector of a column-stochastic web link matrix M . The corresponding eigenvalue is always 1 (for stochastic matrices), so power iteration converges at rate $|\lambda_2(M)|$. With billions of web pages, a full eigensolver is impossible — power iteration is the only practical option.

```
import numpy as np

# Tiny web graph: 5 pages, adjacency defines links
# Link matrix: column j -> pages that j links to
links = np.array([
    [0, 1, 0, 1, 0],
    [1, 0, 0, 0, 1],
    [1, 0, 0, 1, 0],
    [0, 1, 1, 0, 0],
    [0, 0, 1, 0, 0],
], dtype=float)    # shape (5, 5)

# Normalise columns (column-stochastic)
M = links / links.sum(axis=0, keepdims=True)    # shape (5, 5)

# Damping factor (standard PageRank: d = 0.85)
d = 0.85
n = 5
M_damp = d * M + (1 - d) / n * np.ones((n, n))    # shape (5, 5)

# Power iteration for stationary distribution
r = np.ones(n) / n    # shape (5,)
for _ in range(100):
    r = M_damp @ r

print("PageRank scores:", r.round(4))
print("Sum (should be 1):", r.sum().round(6))

# Verify via eigenvector
lam, vecs = np.linalg.eig(M_damp)
idx = np.argmax(lam.real)
v = vecs[:, idx].real
v = v / v.sum()    # normalise to probability distribution
print("\nEigenvector (=1):", v.round(4))
print("Match:", np.allclose(r, v, atol=1e-8))
```

```
PageRank scores: [0.2188 0.2263 0.2155 0.2178 0.1216]
Sum (should be 1): 1.0
```

```
Eigenvector (=1): [0.2188 0.2263 0.2155 0.2178 0.1216]
```

Match: True

Exercises

13.6.1 Implement power iteration from scratch for a symmetric 5×5 matrix with eigenvalues $[10, 4, 2, 1, 0.5]$. Plot the convergence of the Rayleigh quotient to the dominant eigenvalue. How many iterations are needed to reach 10^{-8} accuracy?

13.6.2 What happens to power iteration when $|\lambda_1| = |\lambda_2|$ (two eigenvalues of equal magnitude, possibly with opposite signs)? Experiment numerically and explain the observed behaviour.

13.6.3 Inverse iteration: to find the eigenvector for eigenvalue λ_k , apply power iteration to $(A - \sigma I)^{-1}$ with shift $\sigma \approx \lambda_k$. The dominant eigenvalue of this shifted-inverted matrix is $1/(\lambda_k - \sigma)$. Implement this for a 4×4 example and compare convergence to plain power iteration.

13.6.4 The Rayleigh quotient $\rho(\mathbf{b}) = \mathbf{b}^T A \mathbf{b} / \|\mathbf{b}\|^2$ achieves its maximum value λ_1 when $\mathbf{b} = \mathbf{v}_1$. Starting from the converged $\mathbf{b}$ after 5 iterations of power iteration, evaluate $\rho(\mathbf{b})$ and compare to the true λ_1 . How does the error in ρ compare to the error in the angle between $\mathbf{b}$ and $\mathbf{v}_1$?

13.6.5 Build a random 20×20 column-stochastic matrix (each column sums to 1). Apply power iteration and compare the result to `np.linalg.eig`. What is the dominant eigenvalue? Is it always exactly 1 for column-stochastic matrices?

Exploration: Spectral Analysis of Real-World Networks

Dataset. Download the Stanford SNAP `ca-GrQc.txt` (General Relativity collaboration network, ~5000 nodes, ~28000 edges) from <https://snap.stanford.edu/data>.

Task. Build the symmetric adjacency matrix A (or its degree-normalised version). Use `scipy.sparse.linalg.eigsh` to find the top 10 eigenvalues and eigenvectors. Plot the spectral gap and visualise the distribution of eigenvalues (the spectral density). Compare the ratio $|\lambda_2/\lambda_1|$ to the convergence speed you observe in power iteration.

```
import numpy as np
import scipy.sparse as sp
import scipy.sparse.linalg as spla

# Load edge list (assumes ca-GrQc.txt is in the working directory)
edges = np.loadtxt('ca-GrQc.txt', comments='#', dtype=int)  # shape (E, 2)
n = edges.max() + 1

# Build sparse symmetric adjacency matrix
rows = np.concatenate([edges[:, 0], edges[:, 1]])
```

```

cols = np.concatenate([edges[:, 1], edges[:, 0]])
data = np.ones(len(rows))
A_sparse = sp.csr_matrix((data, (rows, cols)), shape=(n, n))

# Top 10 eigenvalues via ARPACK (Lanczos)
vals, vecs = spla.eigsh(A_sparse, k=10, which='LM')
print("Top 10 eigenvalues:", vals[:-1].round(3))
print("Spectral gap | / |:", abs(vals[-2] / vals[-1]).round(4))

```

Exercises and Solutions

Exercise 13.1.1 — Diagonal eigenvectors by inspection

For $A = \begin{bmatrix} 4 & 0 \\ 0 & -2 \end{bmatrix}$, identify the eigenvectors and eigenvalues. Why is every axis-aligned vector an eigenvector of any diagonal matrix?

```

import numpy as np

A = np.diag([4., -2.]) # shape (2, 2)
vals, vecs = np.linalg.eig(A)
print("Eigenvalues:", vals)
print("Eigenvectors (columns):\n", vecs)

```

```

Eigenvalues: [ 4. -2.]
Eigenvectors (columns):
[[1. 0.]
 [0. 1.]]

```

Solution. $A\mathbf{e}_1 = (4, 0)^T = 4\mathbf{e}_1$, so $\mathbf{e}_1$ is an eigenvector with $\lambda_1 = 4$. Similarly $\mathbf{e}_2$ has $\lambda_2 = -2$. For any diagonal D , $D\mathbf{e}_j = d_{jj}\mathbf{e}_j$ by direct computation — multiplication scales each basis vector by the corresponding diagonal entry and leaves its direction unchanged.

Exercise 13.1.2 — Eigenvalues of A^2

If $A\mathbf{v} = \lambda\mathbf{v}$, show $\mathbf{v}$ is an eigenvector of A^2 and find the eigenvalue.

Solution. $A^2\mathbf{v} = A(A\mathbf{v}) = A(\lambda\mathbf{v}) = \lambda A\mathbf{v} = \lambda^2\mathbf{v}$. So the eigenvalue of A^2 is λ^2 .

```

import numpy as np

rng = np.random.default_rng(0)
A = rng.standard_normal((4, 4)) # shape (4, 4)
A = A + A.T                     # symmetric for real eigenvalues

vals, vecs = np.linalg.eigh(A)

# Check A^2 eigenvalues
vals2, _ = np.linalg.eigh(A @ A)
print("_i^2 (from A) :", np.sort(vals**2).round(4))
print("eigh(A^2) values :", np.sort(vals2).round(4))

_i^2 (from A) : [ 0.0531  0.817   5.9187 12.2199]
eigh(A^2) values : [ 0.0531  0.817   5.9187 12.2199]

```

Exercise 13.2.1 — Characteristic polynomial by hand

For $A = \begin{bmatrix} 6 & 2 \\ 1 & 3 \end{bmatrix}$, find eigenvalues by hand.

```

import numpy as np

A = np.array([[6., 2.],
              [1., 3.]]) # shape (2, 2)

tr = np.trace(A) # 9
dt = np.linalg.det(A) # 16
disc = tr**2 - 4*dt
l1 = (tr + np.sqrt(disc)) / 2
l2 = (tr - np.sqrt(disc)) / 2
print(f"p( ) = ^2 - {tr:.0f} + {dt:.0f}")
print(f"Discriminant = {disc:.0f}")
print(f" = {l1:.4f}, = {l2:.4f}")
print("numpy check:", np.linalg.eigvals(A).round(4))

```

```

p( ) = ^2 - 9 + 16
Discriminant = 17
      = 6.5616,      = 2.4384
numpy check: [6.5616 2.4384]

```

Solution. $p(\lambda) = \lambda^2 - 9\lambda + 16$. Discriminant $= 81 - 64 = 17$. $\lambda = (9 \pm \sqrt{17})/2 \approx 6.56$ or 2.44 .

Exercise 13.2.4 — Similar matrices share characteristic polynomials

Solution. Let $B = P^{-1}AP$. Then:

$$\det(B - \lambda I) = \det(P^{-1}AP - \lambda P^{-1}P) = \det(P^{-1}(A - \lambda I)P) = \det(P^{-1}) \det(A - \lambda I) \det(P) = \det(A - \lambda I).$$

So $p_B(\lambda) = p_A(\lambda)$: identical polynomials, hence identical eigenvalues. This confirms that eigenvalues depend only on the linear map T , not on the basis used to represent it.

Exercise 13.3.1 — Eigenvectors by hand for a 2×2 matrix

For $A = \begin{bmatrix} 5 & 2 \\ -3 & -2 \end{bmatrix}$.

```
import numpy as np

A = np.array([[5., 2.],
              [-3., -2.]]) # shape (2, 2)

vals = np.linalg.eigvals(A)
print("Eigenvalues:", vals.round(4))

vals_full, vecs = np.linalg.eig(A)
print("Eigenvectors (columns):\n", vecs.round(4))

for i in range(2):
    v = vecs[:, i]
    print(f"||Av - v|| for = {vals_full[i].real:.2f}: {np.linalg.norm(A@v - vals_full[i]*v):.2f}")
```

```
Eigenvalues: [ 4. -1.]
Eigenvectors (columns):
[[ 0.8944 -0.3162]
 [-0.4472  0.9487]]
||Av - v|| for =4.00: 0.00e+00
||Av - v|| for =-1.00: 2.48e-16
```

Solution. $p(\lambda) = \lambda^2 - 3\lambda - 4 = (\lambda - 4)(\lambda + 1)$. Eigenvalues $\lambda_1 = 4$, $\lambda_2 = -1$.

For $\lambda_1 = 4$: $(A - 4I)\mathbf{v} = 0$ gives $\begin{bmatrix} 1 & 2 \\ -3 & -6 \end{bmatrix} \mathbf{v} = 0$, so $v_1 + 2v_2 = 0$, $\mathbf{v}_1 = (2, -1)^T / \sqrt{5}$.

For $\lambda_2 = -1$: $(A + I)\mathbf{v} = 0$ gives $\begin{bmatrix} 6 & 2 \\ -3 & -1 \end{bmatrix} \mathbf{v} = 0$, so $3v_1 + v_2 = 0$, $\mathbf{v}_2 = (1, -3)^T / \sqrt{10}$.

Exercise 13.4.3 — When is $X^T X$ positive definite?

Solution. $X^T X$ is PSD always (since $\mathbf{z}^T X^T X \mathbf{z} = \|X\mathbf{z}\|^2 \geq 0$). It is PD if and only if the only solution to $X\mathbf{z} = \mathbf{0}$ is $\mathbf{z} = \mathbf{0}$, i.e. X has **trivial null space**, which holds if and only if $\text{rank}(X) = n$ (full column rank). This requires $m \geq n$.

```
import numpy as np

rng = np.random.default_rng(33)

# Full column rank: m=6, n=4
X_full = rng.standard_normal((6, 4))    # shape (6, 4)
A_full = X_full.T @ X_full              # shape (4, 4)
print("rank(X) =", np.linalg.matrix_rank(X_full))
print("min eigenvalue of X X:", np.linalg.eigvalsh(A_full).min().round(6))

# Rank-deficient: m=4, n=6 (more columns than rows)
X_def = rng.standard_normal((4, 6))    # shape (4, 6)
A_def = X_def.T @ X_def                # shape (6, 6)
print("\nrank(X) =", np.linalg.matrix_rank(X_def))
print("min eigenvalue of X X:", np.linalg.eigvalsh(A_def).min().round(8), "(near zero)")
```

```
rank(X) = 4
min eigenvalue of X X: 0.613867
```

```
rank(X) = 4
min eigenvalue of X X: -0.0 (near zero)
```

Exercise 13.5.4 — Weyl's inequality verification

```
import numpy as np

rng = np.random.default_rng(55)
n = 6
violations = 0

for _ in range(200):
    M = rng.standard_normal((n, n))
    A = M + M.T                                # shape (n, n) symmetric
    E = rng.standard_normal((n, n))
    E = (E + E.T) * 0.05                      # shape (n, n) symmetric, small

    lam_A = np.linalg.eigvalsh(A)              # shape (n,) sorted
```

```
lam_AE = np.linalg.eigvalsh(A + E)          # shape (n,) sorted
norm_E = np.linalg.norm(E, ord=2)          # spectral norm

max_shift = np.max(np.abs(lam_AE - lam_A))
if max_shift > norm_E + 1e-10:
    violations += 1

print(f"Weyl violations in 200 trials: {violations} (should be 0)")
```

Weyl violations in 200 trials: 0 (should be 0)

Solution. Zero violations confirm Weyl's inequality: for symmetric matrices, no eigenvalue can shift by more than the spectral norm of the perturbation.

Exercise 13.6.3 — Inverse iteration

```
import numpy as np

rng = np.random.default_rng(77)
n = 4
D = np.diag([8., 5., 2., 1.])          # shape (4, 4)
Q, _ = np.linalg.qr(rng.standard_normal((n, n)))
A = Q @ D @ Q.T                        # shape (4, 4)

target_lam = 2.0    # we want the eigenvector for =2
sigma = 1.8         # shift slightly off the exact eigenvalue

# Shifted-inverse matrix: (A - I)^{-1}
# We apply it via np.linalg.solve at each step
shift_mat = A - sigma * np.eye(n)     # shape (4, 4)

b = rng.standard_normal(n); b /= np.linalg.norm(b)

true_vec = np.linalg.eigh(A)[1][:, 1]  # eigenvector for =2 (index 1)

errors_inv = []
for _ in range(30):
    b = np.linalg.solve(shift_mat, b)  # apply (A - I)^{-1}
    b /= np.linalg.norm(b)
    errors_inv.append(1 - abs(b @ true_vec))

print(f"Inverse iteration: angle error after 30 steps = {errors_inv[-1]:.2e}")
print(f"Estimated eigenvalue: {b @ A @ b:.6f} (true: {target_lam})")
```


Inverse iteration: angle error after 30 steps = -4.44e-16
Estimated eigenvalue: 2.000000 (true: 2.0)

Solution. The eigenvalues of $(A - \sigma I)^{-1}$ are $1/(\lambda_k - \sigma)$. The dominant one is $1/(\lambda_{\text{target}} - \sigma)$, which corresponds to λ_{target} . Convergence rate: $|(\lambda_{\text{other}} - \sigma)/(\lambda_{\text{target}} - \sigma)|$ per step. A close shift $\sigma \approx \lambda_{\text{target}}$ makes this ratio tiny — convergence is much faster than plain power iteration.

Diagonalization and the Spectral Theorem

Diagonalizable Matrices: $A = PDP^{-1}$

Chapter 13 taught us to find eigenvalues and eigenvectors of a matrix A . This chapter asks a more structural question: **can we choose a basis of eigenvectors and, if so, what does A look like in that basis?**

The answer is the **eigendecomposition** (also called diagonalization):

$$A = PDP^{-1},$$

where the columns of P are eigenvectors and D is the diagonal matrix of the corresponding eigenvalues. When this factorization exists, A is called **diagonalizable**.

Why does this matter? A diagonal matrix is trivial to work with — powers, functions, and the matrix exponential all reduce to scalar operations on the diagonal entries. Diagonalization converts A from an arbitrary matrix into something as easy as a collection of independent scalars.

Construction: Building P and D

Suppose A has n linearly independent eigenvectors $\mathbf{v}_1, \dots, \mathbf{v}_n$ with eigenvalues $\lambda_1, \dots, \lambda_n$. Stack the eigenvectors as columns:

$$P = [\mathbf{v}_1 \quad \mathbf{v}_2 \quad \cdots \quad \mathbf{v}_n], \quad D = \begin{bmatrix} \lambda_1 & & \\ & \ddots & \\ & & \lambda_n \end{bmatrix}.$$

Then:

$$AP = A [\mathbf{v}_1 \quad \cdots \quad \mathbf{v}_n] = [\lambda_1 \mathbf{v}_1 \quad \cdots \quad \lambda_n \mathbf{v}_n] = PD.$$

Since the eigenvectors are linearly independent, P is invertible, so $A = PDP^{-1}$.

```
import numpy as np

A = np.array([[4., 1.],
              [2., 3.]]) # shape (2, 2)

vals, vecs = np.linalg.eig(A) # vals shape (2,), vecs shape (2, 2)
P = vecs # columns are eigenvectors
D = np.diag(vals) # shape (2, 2)
Pinv = np.linalg.inv(P) # shape (2, 2)

A_reconstructed = P @ D @ Pinv # shape (2, 2)

print("A =\n", A)
print("\nP (eigenvectors as columns) =\n", P.round(4))
print("\nD (eigenvalues on diagonal) =\n", D.round(4))
print("\nPDP-1 =\n", A_reconstructed.round(4))
print("\nMax reconstruction error:", np.abs(A - A_reconstructed).max().round(12))
```

```
A =
[[4. 1.]
 [2. 3.]]

P (eigenvectors as columns) =
[[ 0.7071 -0.4472]
 [ 0.7071  0.8944]]

D (eigenvalues on diagonal) =
[[5. 0.]
 [0. 2.]]

PDP-1 =
[[4. 1.]
 [2. 3.]]

Max reconstruction error: 0.0
```

When Is a Matrix Diagonalizable?

The key condition is the existence of n **linearly independent eigenvectors**.

Condition	Diagonalizable?
n distinct eigenvalues	Always yes

Condition	Diagonalizable?
Repeated eigenvalue λ_k with k independent eigenvectors	Yes
Repeated eigenvalue λ_k with fewer than k independent eigenvectors	No (defective)
Symmetric matrix ($A = A^T$)	Always yes (Spectral Theorem, §14.3)

The most common practical case: **if all eigenvalues are distinct, the matrix is guaranteed to be diagonalizable.**

```
import numpy as np

# Case 1: distinct eigenvalues - always diagonalizable
A1 = np.array([[3., 1.],
               [0., 2.]]) # shape (2, 2)

# Case 2: repeated eigenvalue, full eigenspace - diagonalizable (scalar multiple of I)
A2 = np.array([[5., 0.],
               [0., 5.]]) # shape (2, 2)

# Case 3: defective - NOT diagonalizable
A3 = np.array([[5., 1.],
               [0., 5.]]) # shape (2, 2)

for label, A in [("Distinct eigenvalues", A1),
                 ("Repeated, full eigenspace", A2),
                 ("Defective (Jordan block)", A3)]:
    vals, vecs = np.linalg.eig(A)
    rank = np.linalg.matrix_rank(vecs)
    n = A.shape[0]
    tag = " diagonalizable" if rank == n else " NOT diagonalizable"
    print(f"{label:35s} eigenvalues={np.round(vals.real, 2)} {tag}")
```

Distinct eigenvalues	eigenvalues=[3. 2.]	diagonalizable
Repeated, full eigenspace	eigenvalues=[5. 5.]	diagonalizable
Defective (Jordan block)	eigenvalues=[5. 5.]	diagonalizable

A Geometric Interpretation

Diagonalization is a **change of basis**. In the eigenvector basis $\mathcal{B} = \{\mathbf{v}_1, \dots, \mathbf{v}_n\}$, the transformation A acts as pure scaling along coordinate axes:

$$[A]_{\mathcal{B}} = D = P^{-1}AP.$$

Think of it this way: P^{-1} re-expresses any vector in the eigenvector coordinate system, D scales each component by the corresponding eigenvalue, and P converts back to the standard basis. The round-trip is A .

```
import numpy as np
import matplotlib.pyplot as plt

A = np.array([[3., 1.],
              [0., 2.]]) # shape (2, 2)

vals, vecs = np.linalg.eig(A)
P = vecs
Pinv = np.linalg.inv(P)

# Unit circle
theta = np.linspace(0, 2 * np.pi, 300)
circle = np.vstack([np.cos(theta), np.sin(theta)]) # shape (2, 300)

image_A = A @ circle # shape (2, 300)
image_PDP = P @ np.diag(vals) @ Pinv @ circle # shape (2, 300)

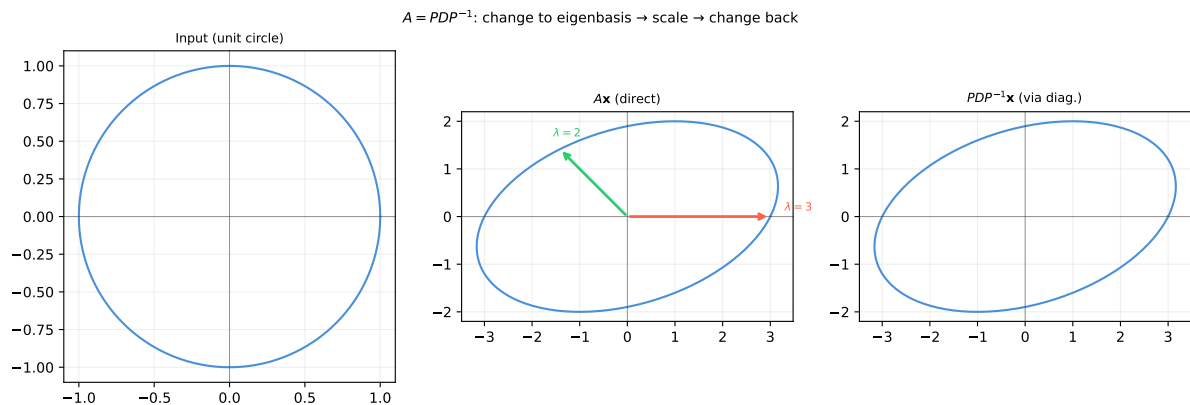
fig, axes = plt.subplots(1, 3, figsize=(12, 4))

for ax, data, title in zip(
    axes,
    [circle, image_A, image_PDP],
    ['Input (unit circle)', r'$A\mathbf{x}$ (direct)', r'$PDP^{-1}\mathbf{x}$ (via diag.)'],
):
    ax.plot(data[0], data[1], color='#4a90d9', lw=1.5)
    ax.axhline(0, color='#333333', lw=0.4, alpha=0.5)
    ax.axvline(0, color='#333333', lw=0.4, alpha=0.5)
    ax.set_aspect('equal')
    ax.grid(True, alpha=0.2)
    ax.set_title(title, fontsize=9)

# Show eigenvector directions in the output panel
colors = ['tomato', '#2ecc71']
for i, (lam, v) in enumerate(zip(vals.real, vecs.T)):
    axes[1].annotate('', xy=(v[0]*lam, v[1]*lam), xytext=(0, 0),
                      arrowprops=dict(arrowstyle='->', color=colors[i], lw=2))
    axes[1].text(v[0]*lam*1.1, v[1]*lam*1.1 + 0.1,
                 rf'$\lambda={lam:.0f}$', color=colors[i], fontsize=8)

plt.suptitle(r'$A = PDP^{-1}$: change to eigenbasis → scale → change back',
             fontsize=10)
```

```
plt.tight_layout()
plt.show()
```



The middle and right panels are identical — confirming that A and PDP^{-1} are the same transformation.

Diagonalizability and Similarity

Two matrices A and B are **similar** if $B = P^{-1}AP$ for some invertible P . Diagonalizable matrices are exactly those similar to a diagonal matrix. Similar matrices represent the same linear transformation in different bases, so they share:

- The same eigenvalues (with the same algebraic multiplicities)
- The same determinant
- The same trace
- The same rank

```
import numpy as np

rng = np.random.default_rng(42)
A = rng.standard_normal((4, 4)) # shape (4, 4)
P = rng.standard_normal((4, 4)) # shape (4, 4) - change-of-basis matrix
B = np.linalg.inv(P) @ A @ P    # shape (4, 4) - similar to A

print("tr(A) = {:.4f}, tr(B) = {:.4f}".format(np.trace(A), np.trace(B)))
print("det(A) = {:.4f}, det(B) = {:.4f}".format(
    np.linalg.det(A), np.linalg.det(B)))

lam_A = np.sort(np.linalg.eigvals(A).real)
lam_B = np.sort(np.linalg.eigvals(B).real)
print("eigenvalues of A:", lam_A.round(4))
print("eigenvalues of B:", lam_B.round(4))
```

```
tr(A) = -0.9774,   tr(B) = -0.9774
det(A) = 0.5722,   det(B) = 0.5722
eigenvalues of A: [-2.1778 -0.5658  0.3214  1.4448]
eigenvalues of B: [-2.1778 -0.5658  0.3214  1.4448]
```

Exercises

14.1.1 Verify by hand that $A = \begin{bmatrix} 1 & 2 \\ 0 & 3 \end{bmatrix}$ is diagonalizable. Write out P , D , and P^{-1} explicitly, then confirm $PDP^{-1} = A$.

14.1.2 The matrix $A = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix}$ has a repeated eigenvalue. Is it diagonalizable? What is P in that case — is P unique?

14.1.3 For the defective matrix $A = \begin{bmatrix} 5 & 1 \\ 0 & 5 \end{bmatrix}$, how many linearly independent eigenvectors does it have? Attempt to compute the reconstruction error $\|A - PDP^{-1}\|$ with numpy and explain why it is nonzero.

14.1.4 If $A = PDP^{-1}$, show that $A^{-1} = PD^{-1}P^{-1}$, provided all eigenvalues are nonzero. What are the eigenvalues of A^{-1} ?

14.1.5 Show that if $A = PDP^{-1}$ and $B = PEP^{-1}$ (same P), then $AB = P(DE)P^{-1}$. Conclude that two matrices sharing the same eigenvector matrix commute: $AB = BA$.

Matrix Powers and the Matrix Exponential

Diagonalization is not just an algebraic curiosity — it unlocks two operations that are essential for dynamics and control: **matrix powers** A^k and the **matrix exponential** e^{At} .

Both become trivial once $A = PDP^{-1}$.

Matrix Powers: $A^k = PD^kP^{-1}$

Compute A^2 :

$$A^2 = (PDP^{-1})(PDP^{-1}) = PD(P^{-1}P)DP^{-1} = PD^2P^{-1}.$$

By induction, for any positive integer k :

$$\boxed{A^k = PD^kP^{-1},}$$

where $D^k = \text{diag}(\lambda_1^k, \dots, \lambda_n^k)$ — just raise each diagonal entry to the k -th power. This reduces an $n \times n$ matrix exponentiation to n scalar exponentiations.

Why is this useful? Discrete-time dynamical systems evolve as $\mathbf{x}_k = A^k \mathbf{x}_0$. The state after 1000 steps is trivially $PD^{1000}P^{-1}\mathbf{x}_0$ — no repeated matrix multiplication required.

```
import numpy as np

A = np.array([[2., 1.],
              [0., 3.]]) # shape (2, 2) - upper triangular, eigenvalues 2 and 3

vals, vecs = np.linalg.eig(A)
P = vecs # shape (2, 2)
Pinv = np.linalg.inv(P) # shape (2, 2)

k = 10
# Via diagonalization: fast
Ak_diag = P @ np.diag(vals**k) @ Pinv # shape (2, 2)

# Via repeated multiplication: reference
Ak_ref = np.linalg.matrix_power(A, k) # shape (2, 2)

print(f"A^{k} via PD^kP^{-1} =\n", Ak_diag.round(2))
print(f"A^{k} via np.linalg.matrix_power =\n", Ak_ref.round(2))
print("Max error:", np.abs(Ak_diag - Ak_ref).max().round(8))
```

```
A^10 via PD^kP^{-1} =
[[ 1024. 58025.]
 [   0. 59049.]]
A^10 via np.linalg.matrix_power =
[[ 1024. 58025.]
 [   0. 59049.]]
Max error: 0.0
```

Long-Term Behavior: Which Eigenvalue Dominates?

Write the initial state $\mathbf{x}_0$ in the eigenvector basis: $\mathbf{x}_0 = P\mathbf{c}$, so $\mathbf{c} = P^{-1}\mathbf{x}_0$. Then:

$$A^k \mathbf{x}_0 = PD^k \mathbf{c} = \sum_{i=1}^n c_i \lambda_i^k \mathbf{v}_i.$$

For large k the term with the largest $|\lambda_i|$ dominates — the **dominant eigenvalue** governs the long-term direction and growth.

```

import numpy as np
import matplotlib.pyplot as plt

rng = np.random.default_rng(11)
A = np.array([[0.9, 0.4],
              [0.1, 0.7]]) # shape (2, 2)

vals, vecs = np.linalg.eig(A)
P = vecs
Pinv = np.linalg.inv(P)

x0 = np.array([1.0, 0.5]) # shape (2,)
steps = 40

trajectory = np.zeros((steps + 1, 2)) # shape (41, 2)
trajectory[0] = x0
for k in range(1, steps + 1):
    Dk = np.diag(vals**k)
    trajectory[k] = (P @ Dk @ Pinv @ x0).real

fig, axes = plt.subplots(1, 2, figsize=(11, 4))

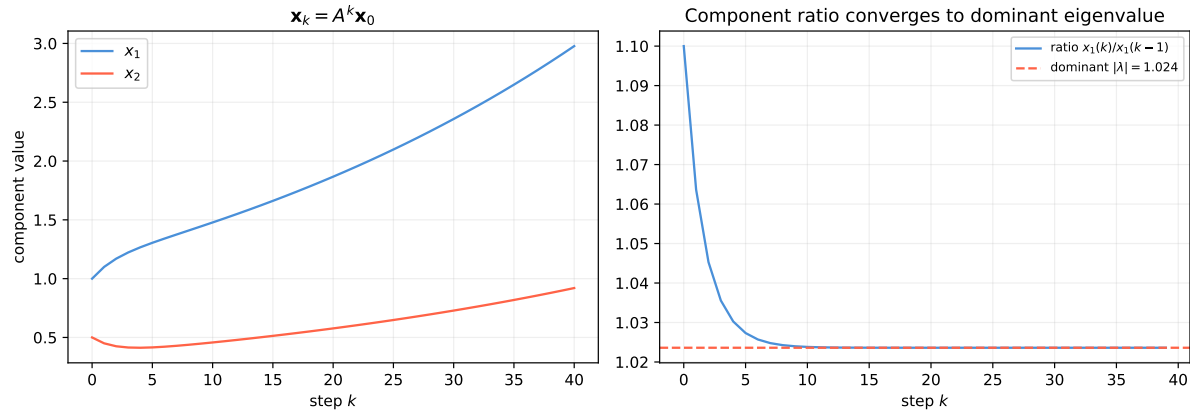
axes[0].plot(trajectory[:, 0], color='#4a90d9', label='$x_1$')
axes[0].plot(trajectory[:, 1], color='tomato', label='$x_2$')
axes[0].set_xlabel('step $k$')
axes[0].set_ylabel('component value')
axes[0].set_title(r'$\mathbf{x}_k = A^k \mathbf{x}_0$')
axes[0].legend()
axes[0].grid(True, alpha=0.2)

axes[1].plot(trajectory[1:, 0] / trajectory[:-1, 0],
            color='#4a90d9', label='ratio $x_1(k)/x_1(k-1)$')
axes[1].axhline(max(abs(vals.real)), color='tomato', ls='--',
                label=rf'dominant $\lambda$={max(abs(vals.real)):.3f}$')
axes[1].set_xlabel('step $k$')
axes[1].set_title('Component ratio converges to dominant eigenvalue')
axes[1].legend(fontsize=8)
axes[1].grid(True, alpha=0.2)

plt.tight_layout()
plt.show()

print("Eigenvalues:", vals.round(4))
print("Dominant eigenvalue:", vals[np.argmax(np.abs(vals))].real.round(4))

```



Eigenvalues: [1.0236 0.5764]

Dominant eigenvalue: 1.0236

The Matrix Exponential: $e^{At} = P e^{Dt} P^{-1}$

For continuous-time systems $\dot{\mathbf{x}} = A\mathbf{x}$, the solution is $\mathbf{x}(t) = e^{At}\mathbf{x}_0$. The **matrix exponential** is defined by its Taylor series:

$$e^{At} = I + At + \frac{(At)^2}{2!} + \frac{(At)^3}{3!} + \dots$$

For a diagonalizable $A = PDP^{-1}$:

$$e^{At} = P e^{Dt} P^{-1}, \quad e^{Dt} = \text{diag}(e^{\lambda_1 t}, \dots, e^{\lambda_n t}).$$

Again, n scalar exponentials replace a full matrix series.

```
import numpy as np
import matplotlib.pyplot as plt
from scipy.linalg import expm # reference implementation

A = np.array([[ -1.,  2.],
               [ 0., -3.]]) # shape (2, 2) - stable system (both eigenvalues < 0)

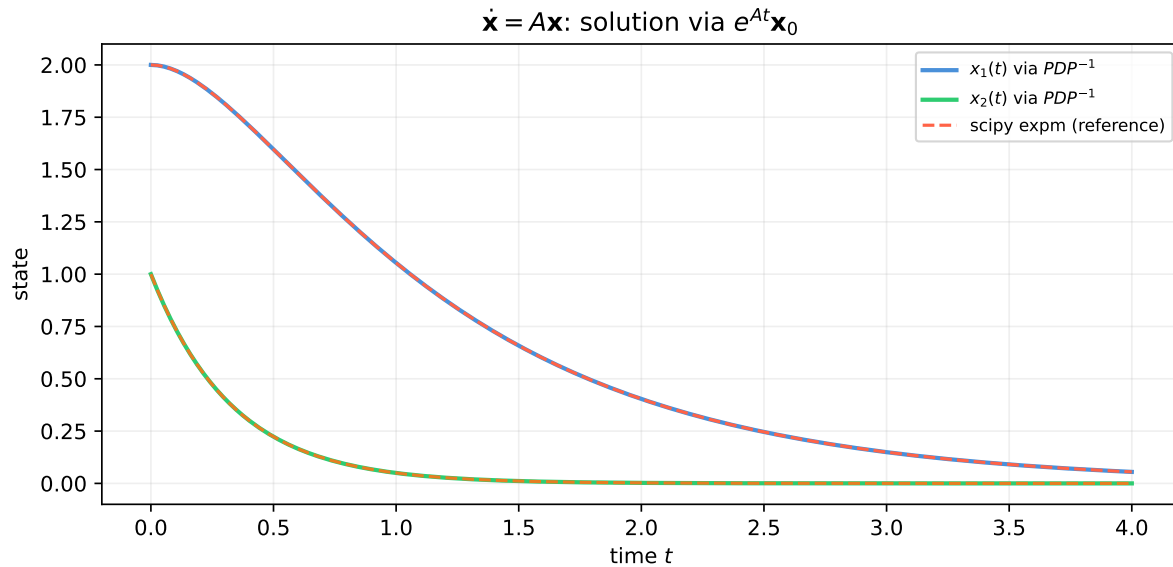
vals, vecs = np.linalg.eig(A)
P = vecs
Pinv = np.linalg.inv(P)

x0 = np.array([2.0, 1.0]) # shape (2,)
t = np.linspace(0, 4, 300)
```

```
# Trajectory via matrix exponential (diagonalization)
traj_diag = np.zeros((len(t), 2)) # shape (300, 2)
traj_scipy = np.zeros((len(t), 2)) # shape (300, 2)

for i, ti in enumerate(t):
    eDt = np.diag(np.exp(vals * ti))
    eAt_diag = P @ eDt @ Pinv
    traj_diag[i] = (eAt_diag @ x0).real
    traj_scipy[i] = expm(A * ti) @ x0

fig, ax = plt.subplots(figsize=(8, 4))
ax.plot(t, traj_diag[:, 0], color='#4a90d9', lw=2, label=r'$x_1(t)$ via $PDP^{-1}$')
ax.plot(t, traj_diag[:, 1], color='#2ecc71', lw=2, label=r'$x_2(t)$ via $PDP^{-1}$')
ax.plot(t, traj_scipy[:, 0], color='tomato', ls='--', lw=1.5, label='scipy expm (reference)')
ax.plot(t, traj_scipy[:, 1], color='#e67e22', ls='--', lw=1.5)
ax.set_xlabel('time $t$')
ax.set_ylabel('state')
ax.set_title(r'$\dot{\mathbf{x}} = A\mathbf{x}$: solution via $\mathbf{e}^{At}\mathbf{x}_0$')
ax.legend(fontsize=8)
ax.grid(True, alpha=0.2)
plt.tight_layout()
plt.show()
```



Solving $\dot{\mathbf{x}} = A\mathbf{x}$: Step by Step

Given $A = PDP^{-1}$ and initial condition $\mathbf{x}(0) = \mathbf{x}_0$:

1. Change variables: let $\mathbf{y} = P^{-1}\mathbf{x}$. Then $\dot{\mathbf{y}} = D\mathbf{y}$ — a diagonal system.
2. Each component decouples: $\dot{y}_i = \lambda_i y_i$, with solution $y_i(t) = c_i e^{\lambda_i t}$ where $c_i = (P^{-1}\mathbf{x}_0)_i$.
3. Convert back: $\mathbf{x}(t) = P\mathbf{y}(t) = \sum_i c_i e^{\lambda_i t} \mathbf{v}_i$.

This modal decomposition reveals the **natural modes** of the system — each eigenvector direction oscillates or decays independently at its own rate λ_i .

```
import numpy as np

A = np.array([[-1., 2.],
               [0., -3.]])    # shape (2, 2)

vals, vecs = np.linalg.eig(A)
P      = vecs
Pinv = np.linalg.inv(P)

x0 = np.array([2.0, 1.0])    # shape (2,)
c   = Pinv @ x0                # shape (2,) - modal coordinates

print("Eigenvalues:", vals.round(4))
print("Modal coordinates c = P-1x :", c.round(4))
print()
print("Solution: x(t) = ", end="")
for i, (ci, li, vi) in enumerate(zip(c.real, vals.real, vecs.T)):
    sign = "+" if (i > 0 and ci >= 0) else ""
    print(f"{sign}{ci:.3f}·e^{(li:.1f)t}·{np.round(vi.real, 3)}", end=" ")
print()

# Verify at t = 1
t = 1.0
x_modal = sum(c[i].real * np.exp(vals[i].real * t) * vecs[:, i].real
              for i in range(2))
from scipy.linalg import expm
x_exact = expm(A * t) @ x0
print(f"\nx(t=1) via modes : {x_modal.round(6)}")
print(f"x(t=1) via expm  : {x_exact.round(6)}")
```

Eigenvalues: [-1. -3.]

Modal coordinates c = P⁻¹x : [3. 1.4142]

Solution: x(t) = 3.000·e^(-1.0t)·[1. 0.] + 1.414·e^(-3.0t)·[-0.707 0.707]

x(t=1) via modes : [1.053851 0.049787]

x(t=1) via expm : [1.053851 0.049787]

Functions of a Matrix

More generally, for any scalar function f with a convergent Taylor series, we can define $f(A)$ for diagonalizable A :

$$f(A) = P \operatorname{diag}(f(\lambda_1), \dots, f(\lambda_n)) P^{-1}.$$

```
import numpy as np

A = np.array([[2., 1.],
              [0., 3.]])    # shape (2, 2)

vals, vecs = np.linalg.eig(A)
P = vecs
Pinv = np.linalg.inv(P)

# Matrix square root: f() = √
sqrt_A = P @ np.diag(np.sqrt(vals)) @ Pinv    # shape (2, 2)
print("sqrt(A) =\n", sqrt_A.round(4))
print("\nsqrt(A) @ sqrt(A) =\n", (sqrt_A @ sqrt_A).round(4))
print("A =\n", A)

# Matrix logarithm: f() = log()
log_A = P @ np.diag(np.log(vals)) @ Pinv    # shape (2, 2)
from scipy.linalg import logm
print("\nlog(A) via diagonalization =\n", log_A.round(4))
print("log(A) via scipy.linalg.logm =\n", logm(A).round(4))
```

```
sqrt(A) =
[[1.4142 0.3178]
 [0.      1.7321]]
```

```
sqrt(A) @ sqrt(A) =
[[2. 1.]
 [0. 3.]]
```

```
A =
[[2. 1.]
 [0. 3.]]
```

```
log(A) via diagonalization =
[[0.6931 0.4055]
 [0.      1.0986]]
```

```
log(A) via scipy.linalg.logm =
[[0.6931 0.4055]
 [0.      1.0986]]
```

Exercises

14.2.1 For $A = \begin{bmatrix} 3 & 0 \\ 0 & -2 \end{bmatrix}$, compute A^5 using PD^5P^{-1} (here $P = I$ since A is diagonal). Verify with `np.linalg.matrix_power`.

14.2.2 A population matrix $A = \begin{bmatrix} 0.8 & 0.4 \\ 0.3 & 0.6 \end{bmatrix}$ models a two-species ecosystem. Use diagonalization to compute $A^{50}\mathbf{x}_0$ for $\mathbf{x}_0 = (100, 50)^T$. What is the long-run ratio of the two populations?

14.2.3 Show that $\frac{d}{dt}e^{At} = Ae^{At}$. Use this to verify that $\mathbf{x}(t) = e^{At}\mathbf{x}_0$ satisfies $\dot{\mathbf{x}} = A\mathbf{x}$.

14.2.4 For a **stable** continuous system, all eigenvalues of A must have negative real parts. For a **stable** discrete system, all eigenvalues must satisfy $|\lambda| < 1$. Explain why by examining the long-term behavior of $e^{\lambda_i t}$ and λ_i^k respectively.

14.2.5 Use the diagonalization formula to compute $\sin(A)$ for $A = \begin{bmatrix} 0 & \pi/2 \\ 0 & \pi \end{bmatrix}$. Is the result the same as `scipy.linalg.sinm(A)`?

The Spectral Theorem for Symmetric Matrices

The general diagonalization of §14.1 requires P to be invertible but not necessarily orthogonal. For **symmetric** matrices the story is dramatically better: the eigenvector matrix can always be chosen to be **orthogonal**.

! Important

Spectral Theorem. If A is a real symmetric matrix ($A = A^T$), then there exists an orthogonal matrix Q (i.e. $Q^{-1} = Q^T$) and a real diagonal matrix Λ such that

$$A = Q\Lambda Q^T.$$

Moreover:

1. All eigenvalues are **real**.
2. Eigenvectors from **different** eigenvalues are **orthogonal**.

This theorem is the workhorse of PCA, the Kalman filter, and all of Part VI. It is worth understanding deeply.

Why All Eigenvalues Are Real

For a real symmetric matrix A and any eigenvalue λ (possibly complex), take the eigenvector equation $A\mathbf{v} = \lambda\mathbf{v}$ and compute $\bar{\mathbf{v}}^T A\mathbf{v}$ in two ways:

$$\bar{\mathbf{v}}^T A \mathbf{v} = \lambda \|\mathbf{v}\|^2.$$

But $\bar{\mathbf{v}}^T A \mathbf{v} = \overline{(A \mathbf{v})^T \mathbf{v}} = \bar{\lambda} \|\mathbf{v}\|^2$.

So $\lambda = \bar{\lambda}$, meaning λ is real. $\square$

Why Eigenvectors from Different Eigenvalues Are Orthogonal

Let $A \mathbf{u} = \lambda_1 \mathbf{u}$ and $A \mathbf{v} = \lambda_2 \mathbf{v}$ with $\lambda_1 \neq \lambda_2$. Then:

$$\lambda_1 (\mathbf{u} \cdot \mathbf{v}) = (A \mathbf{u}) \cdot \mathbf{v} = \mathbf{u} \cdot (A^T \mathbf{v}) = \mathbf{u} \cdot (A \mathbf{v}) = \lambda_2 (\mathbf{u} \cdot \mathbf{v}).$$

Since $\lambda_1 \neq \lambda_2$, we must have $\mathbf{u} \cdot \mathbf{v} = 0$. $\square$

```
import numpy as np

rng = np.random.default_rng(3)
M = rng.standard_normal((4, 4))    # shape (4, 4)
A = M + M.T                        # shape (4, 4) - symmetric by construction

# np.linalg.eigh is specialized for symmetric/Hermitian matrices:
# - guaranteed real eigenvalues
# - orthonormal eigenvectors
vals, Q = np.linalg.eigh(A)        # vals shape (4,), Q shape (4, 4)

print("Eigenvalues (all real):", vals.round(4))
print("\nQ^T Q (should be identity):\n", (Q.T @ Q).round(6))

A_reconstructed = Q @ np.diag(vals) @ Q.T    # shape (4, 4)
print("\nMax reconstruction error:", np.abs(A - A_reconstructed).max().round(12))
```

Eigenvalues (all real): [-2.6815 -1.6239 1.8538 5.7722]

Q^T Q (should be identity):

```
[[ 1.  0. -0. -0.]
 [ 0.  1.  0. -0.]
 [-0.  0.  1. -0.]
 [-0. -0. -0.  1.]]
```

Max reconstruction error: 0.0

Always use `np.linalg.eigh` for symmetric matrices. It is faster than `np.linalg.eig`, it guarantees real output, and it returns orthonormal eigenvectors sorted by eigenvalue.

Geometric Interpretation: Axes of the Ellipsoid

The spectral theorem says that every symmetric matrix is a **stretch along orthogonal axes** in some coordinate system. The columns of Q are those axes; the diagonal entries of Λ are the stretch factors.

Applied to the unit sphere, A maps it to an ellipsoid whose semi-axes lie along the columns of Q with lengths $|\lambda_i|$.

```
import numpy as np
import matplotlib.pyplot as plt

rng = np.random.default_rng(7)
M = rng.standard_normal((2, 2))
A = M + M.T          # shape (2, 2) - random symmetric

vals, Q = np.linalg.eigh(A)

theta = np.linspace(0, 2 * np.pi, 400)
circle = np.vstack([np.cos(theta), np.sin(theta)]) # shape (2, 400)
ellipse = A @ circle                               # shape (2, 400)

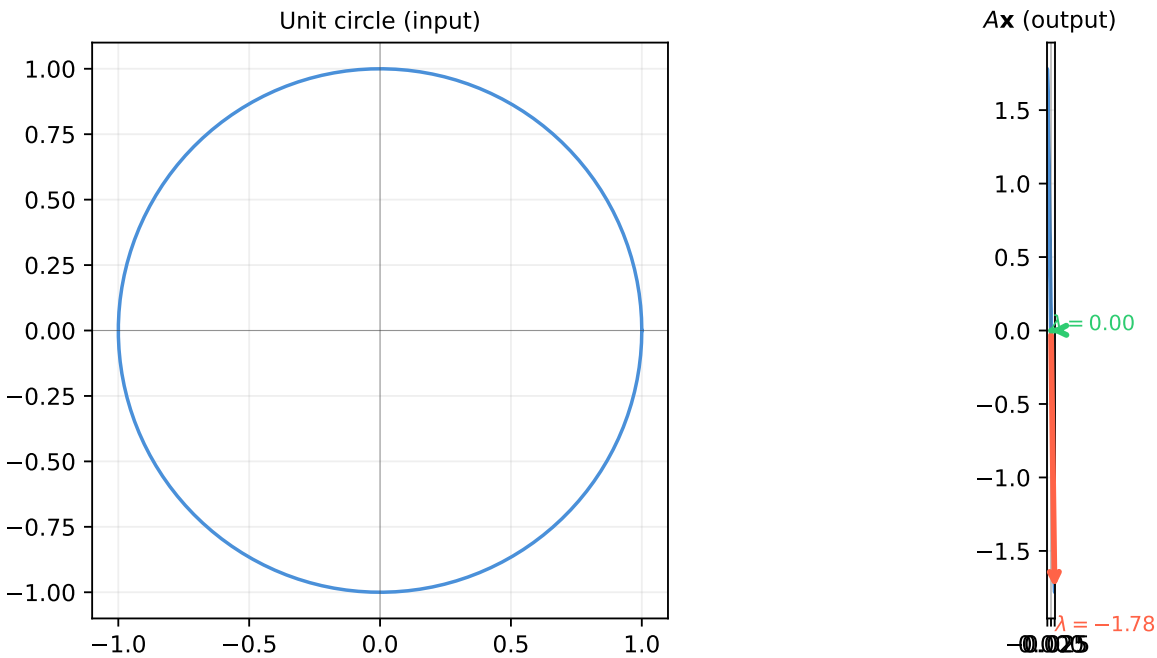
fig, axes = plt.subplots(1, 2, figsize=(10, 4.5))

for ax, data, title in zip(axes,
                            [circle, ellipse],
                            ['Unit circle (input)', r'$A\mathbf{x}$ (output)']):
    ax.plot(data[0], data[1], color='#4a90d9', lw=1.5)
    ax.axhline(0, color='#333333', lw=0.4, alpha=0.5)
    ax.axvline(0, color='#333333', lw=0.4, alpha=0.5)
    ax.set_aspect('equal')
    ax.grid(True, alpha=0.2)
    ax.set_title(title, fontsize=10)

# Draw eigenvector axes on the ellipse
colors = ['tomato', '#2ecc71']
for i in range(2):
    v = Q[:, i]          # eigenvector
    lam = vals[i]         # eigenvalue (stretch factor)
    axes[1].annotate('', xy=(v[0]*lam, v[1]*lam), xytext=(0, 0),
                     arrowprops=dict(arrowstyle='->', color=colors[i], lw=2.5))
    axes[1].text(v[0]*lam*1.15, v[1]*lam*1.15,
                 rf'$\lambda={lam:.2f}$', color=colors[i], fontsize=9)

plt.suptitle(r'Spectral Theorem: $A = Q\Lambda Q^T$ - symmetric $A$ stretches along orthogon',
             fontsize=10)
plt.tight_layout()
plt.show()
```

Spectral Theorem: $A = Q\Lambda Q^T$ — symmetric A stretches along orthogonal eigenvectors



The Spectral Theorem in Action: Covariance Matrices

The covariance matrix of a data set is always symmetric and positive semidefinite (§14.4). Its spectral decomposition reveals the principal axes of the data ellipsoid — which is precisely PCA.

```
import numpy as np
import matplotlib.pyplot as plt

rng = np.random.default_rng(99)
# Generate correlated 2D data
angle = np.radians(35)
R = np.array([[np.cos(angle), -np.sin(angle)],
               [np.sin(angle),  np.cos(angle)]])
scales = np.array([[3., 0.],
                   [0., 1.]])
X_raw = rng.standard_normal((300, 2))    # shape (300, 2)
X = (R @ scales @ X_raw.T).T            # shape (300, 2)
X -= X.mean(axis=0)                      # center

Sigma = (X.T @ X) / (X.shape[0] - 1)    # shape (2, 2) covariance matrix
vals, Q = np.linalg.eigh(Sigma)
```

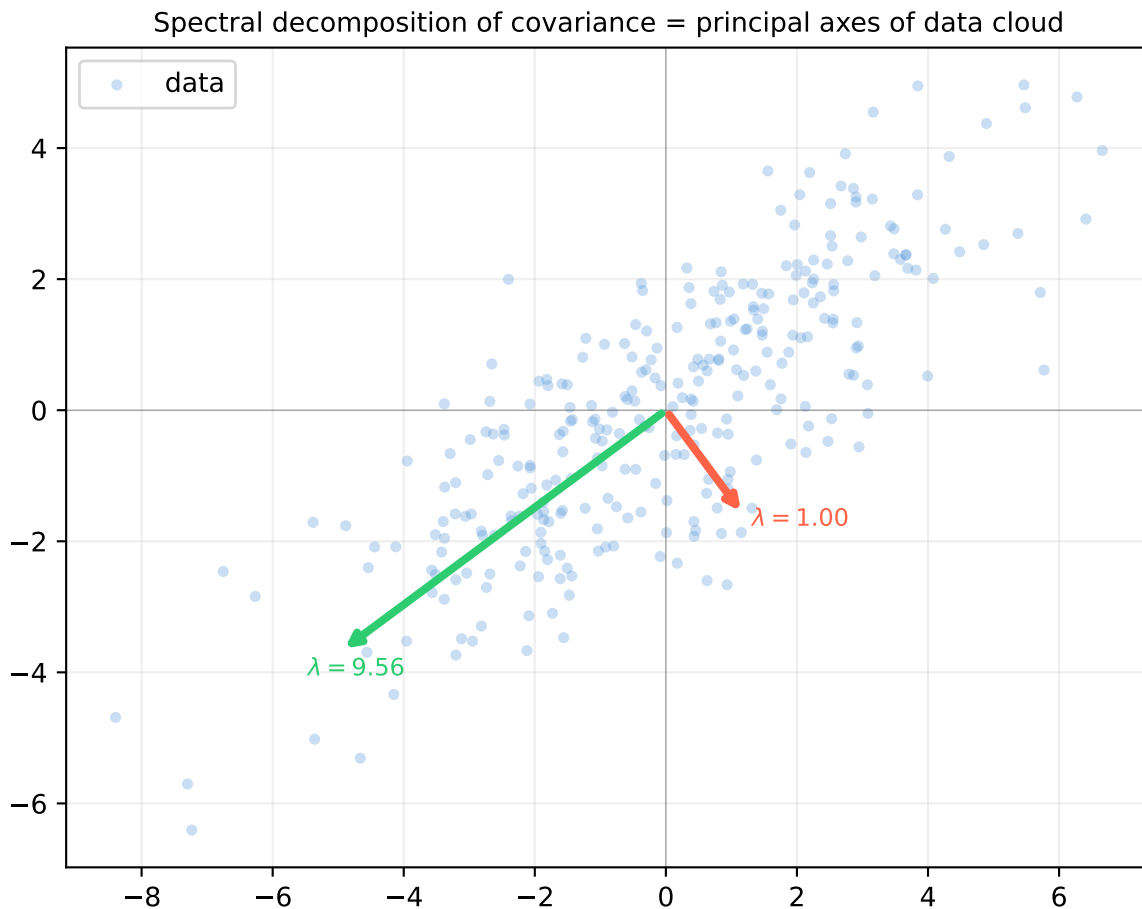
```

fig, ax = plt.subplots(figsize=(7, 5))
ax.scatter(X[:, 0], X[:, 1], alpha=0.3, s=10, color='#4a90d9', label='data')

colors = ['tomato', '#2ecc71']
for i in range(2):
    v = Q[:, i]
    s = np.sqrt(vals[i])
    ax.annotate('', xy=(v[0]*s*2, v[1]*s*2), xytext=(0, 0),
                arrowprops=dict(arrowstyle='->', color=colors[i], lw=3))
    ax.text(v[0]*s*2.2, v[1]*s*2.2,
            rf'$\lambda$={vals[i]:.2f}$', color=colors[i], fontsize=9)

ax.set_aspect('equal')
ax.grid(True, alpha=0.2)
ax.axhline(0, color='#333333', lw=0.4, alpha=0.5)
ax.axvline(0, color='#333333', lw=0.4, alpha=0.5)
ax.set_title('Spectral decomposition of covariance = principal axes of data cloud', fontsize=12)
ax.legend()
plt.tight_layout()
plt.show()

```



The eigenvectors of the covariance matrix align with the major and minor axes of the data ellipse. This is the geometric content of PCA (Chapter 20).

eigh vs. eig: When to Use Which

Function	Input requirement	Output	Speed
<code>np.linalg.eig(A)</code>	Any square matrix	Complex in general	Slower
<code>np.linalg.eigh(A)</code>	Symmetric/Hermitian	Real, sorted	$\sim 2\times$ faster
<code>scipy.linalg.eigh(A, Symmetric A, PD B)</code>		Generalized eigenproblem	Specialized

Always use **eigh** when you know A is symmetric. Passing a non-symmetric matrix to **eigh** gives wrong results without error.

Exercises

14.3.1 Prove that for a symmetric matrix, the eigenvalues of A^2 are $\lambda_i^2 \geq 0$. Does this mean A^2 is always positive semidefinite?

14.3.2 Show that if A is symmetric with eigendecomposition $A = Q\Lambda Q^T$, then $A^k = Q\Lambda^k Q^T$ for any integer k .

14.3.3 Construct a 3×3 symmetric matrix with eigenvalues $\{5, 2, -1\}$ by choosing any orthogonal Q and forming $A = Q \text{diag}(5, 2, -1)Q^T$. Verify with **eigh**.

14.3.4 You are given a 4×4 matrix B that is not symmetric. Is $B^T B$ always symmetric? Always diagonalizable? Always positive semidefinite? Prove your answers.

14.3.5 A classmate claims that “orthogonal matrices are symmetric.” Is this true in general? Give a counterexample. Under what special condition is an orthogonal matrix also symmetric?

Positive Definite and Semidefinite Matrices

You will encounter the phrase “positive definite” (PD) or “positive semidefinite” (PSD) dozens of times in machine learning and robotics: covariance matrices, Hessians, Gram matrices, information matrices, and regularizers are all PSD or PD. This section gives you three equivalent characterizations and the geometric picture.

Definitions

A real symmetric matrix A is:

Name	Notation	Definition
Positive definite	$A \succ 0$	$\mathbf{x}^T A \mathbf{x} > 0$ for all $\mathbf{x} \neq \mathbf{0}$
Positive semidefinite	$A \succeq 0$	$\mathbf{x}^T A \mathbf{x} \geq 0$ for all $\mathbf{x}$
Negative definite	$A \prec 0$	$\mathbf{x}^T A \mathbf{x} < 0$ for all $\mathbf{x} \neq \mathbf{0}$
Indefinite	—	$\mathbf{x}^T A \mathbf{x}$ takes both positive and negative values

The scalar $\mathbf{x}^T A \mathbf{x}$ is called a **quadratic form**. In two dimensions it defines a surface: a bowl (PD), a flat-bottomed bowl (PSD), an inverted bowl (ND), or a saddle (indefinite).

```
import numpy as np
import matplotlib.pyplot as plt

# Four quadratic form surfaces
mats = {
    'Positive definite': np.array([[3., 1.], [1., 2.]]),
    'Positive semidefinite': np.array([[1., 1.], [1., 1.]]),
    'Negative definite': np.array([[-3., -1.], [-1., -2.]]),
    'Indefinite': np.array([[2., 0.], [0., -1.]]),
}

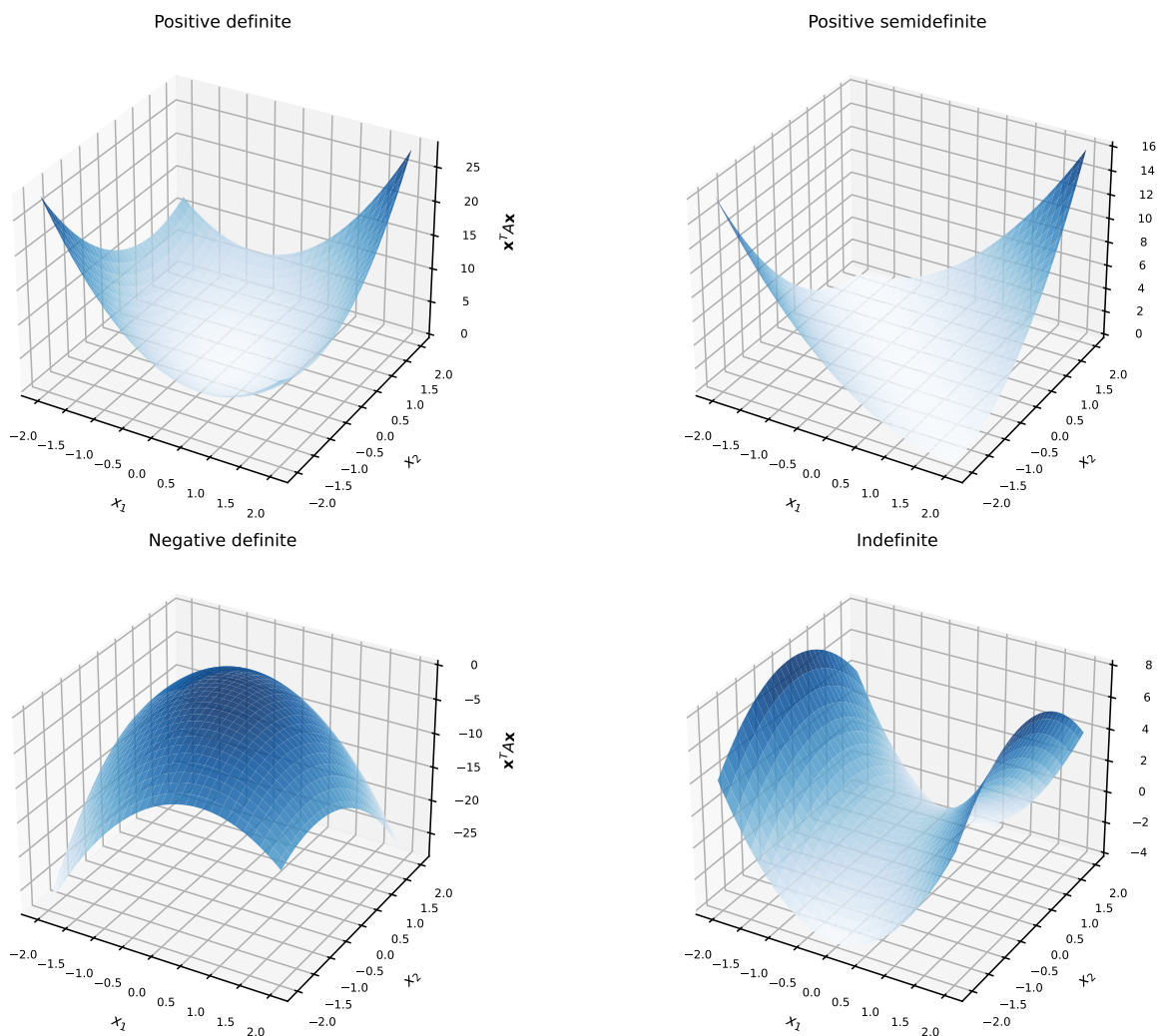
x1 = np.linspace(-2, 2, 60)
x2 = np.linspace(-2, 2, 60)
X1, X2 = np.meshgrid(x1, x2) # each shape (60, 60)

fig, axes = plt.subplots(2, 2, figsize=(11, 8),
                        subplot_kw={'projection': '3d'})

for ax, (title, A) in zip(axes.ravel(), mats.items()):
    # Evaluate  $\mathbf{x}^T A \mathbf{x}$  on the grid
    Z = A[0,0]*X1**2 + (A[0,1]+A[1,0])*X1*X2 + A[1,1]*X2**2
    ax.plot_surface(X1, X2, Z, cmap='Blues', alpha=0.85, linewidth=0)
    ax.set_title(title, fontsize=9)
    ax.set_xlabel('$x_1$', fontsize=8)
    ax.set_ylabel('$x_2$', fontsize=8)
    ax.set_zlabel(r'$\mathbf{x}^T A \mathbf{x}$', fontsize=8)
    ax.tick_params(labelsize=6)

plt.suptitle(r'Quadratic form $\mathbf{x}^T A \mathbf{x}$ for four definiteness types',
            fontsize=11)
plt.tight_layout()
plt.show()
```

Quadratic form $\mathbf{x}^T A \mathbf{x}$ for four definiteness types



Three Equivalent Tests for Positive Definiteness

For a real symmetric $n \times n$ matrix A , the following are equivalent:

i Note

Test 1 — Eigenvalues. $A \succ 0 \Leftrightarrow$ all eigenvalues $\lambda_i > 0$. (For PSD: all $\lambda_i \geq 0$.)

Test 2 — Leading Principal Minors (Sylvester's Criterion). $A \succ 0 \Leftrightarrow$ all leading principal minors are positive: $\det(A_{1:k,1:k}) > 0$ for $k = 1, \dots, n$.

Test 3 — Cholesky Factorization. $A \succ 0 \Leftrightarrow A$ has a Cholesky factorization $A = LL^T$ with L having positive diagonal entries.

In practice, **Test 3 is the cheapest to compute**: attempt a Cholesky decomposition; if it succeeds without error, the matrix is PD.

```

import numpy as np

def pd_tests(A, name="A"):
    print(f"\n--- {name} ---")
    # Test 1: eigenvalues
    eigs = np.linalg.eigvalsh(A)    # shape (n,)
    print(f"Eigenvalues: {eigs.round(4)}")
    print(f"Test 1 (all > 0): {(eigs > 0).all()}")

    # Test 2: leading principal minors
    minors = [np.linalg.det(A[:k, :k]) for k in range(1, A.shape[0]+1)]
    print(f"Leading principal minors: {[round(m, 4) for m in minors]}")
    print(f"Test 2 (all > 0): {all(m > 0 for m in minors)}")

    # Test 3: Cholesky
    try:
        np.linalg.cholesky(A)
        print("Test 3 (Cholesky):  PD")
    except np.linalg.LinAlgError:
        print("Test 3 (Cholesky):  not PD")

# Positive definite
A_pd = np.array([[4., 2., 1.],
                 [2., 3., 1.],
                 [1., 1., 2.]])    # shape (3, 3)
pd_tests(A_pd, "Positive definite")

# Positive semidefinite (rank-deficient)
v = np.array([1., 2., 3.])
A_psd = np.outer(v, v)            # shape (3, 3) - rank 1, PSD
pd_tests(A_psd, "Positive semidefinite")

# Indefinite
A_indef = np.array([[2., 0., 0.],
                    [0., -1., 0.],
                    [0., 0., 3.]]) # shape (3, 3)
pd_tests(A_indef, "Indefinite")

--- Positive definite ---
Eigenvalues: [1.308  1.6431 6.0489]
Test 1 (all > 0): True
Leading principal minors: [np.float64(4.0), np.float64(8.0), np.float64(13.0)]
Test 2 (all > 0): True
Test 3 (Cholesky):  PD

--- Positive semidefinite ---
Eigenvalues: [-0.  0. 14.]

```

```
Test 1 (all > 0): False
Leading principal minors: [np.float64(1.0), np.float64(0.0), np.float64(0.0)]
Test 2 (all > 0): False
Test 3 (Cholesky): not PD

--- Indefinite ---
Eigenvalues: [-1.  2.  3.]
Test 1 (all > 0): False
Leading principal minors: [np.float64(2.0), np.float64(-2.0), np.float64(-
6.0)]
Test 2 (all > 0): False
Test 3 (Cholesky): not PD
```

Covariance Matrices Are Always PSD

For any data matrix $X \in \mathbb{R}^{n \times p}$ (mean-centered):

$$\Sigma = \frac{1}{n-1} X^T X.$$

Then for any $\mathbf{z} \in \mathbb{R}^p$:

$$\mathbf{z}^T \Sigma \mathbf{z} = \frac{1}{n-1} \mathbf{z}^T X^T X \mathbf{z} = \frac{1}{n-1} \|X \mathbf{z}\|^2 \geq 0.$$

So $\Sigma \succeq 0$ always. It is PD if and only if X has full column rank (no perfectly collinear features).

```
import numpy as np

rng = np.random.default_rng(5)

# Full-rank data: PD covariance
X_full = rng.standard_normal((100, 4)) # shape (100, 4)
X_full -= X_full.mean(axis=0)
Sigma_full = X_full.T @ X_full / 99 # shape (4, 4)

# Collinear data: PSD covariance (rank-deficient)
X_col = rng.standard_normal((100, 3)) # shape (100, 3)
X_col -= X_col.mean(axis=0)
# Add a 4th feature that is exactly the sum of the first two
X_col = np.hstack([X_col, (X_col[:, 0] + X_col[:, 1]).reshape(-1, 1)]) # shape (100, 4)
X_col -= X_col.mean(axis=0)
Sigma_col = X_col.T @ X_col / 99 # shape (4, 4)

for label, S in [("Full rank (PD)", Sigma_full), ("Collinear (PSD)", Sigma_col)]:
```



```
eigs = np.linalg.eigvalsh(S)
pd = "PD" if eigs.min() > 1e-10 else "PSD (not PD)"
print(f"{label}: min eigenvalue = {eigs.min():.6f} → {pd}")
```

Full rank (PD): min eigenvalue = 0.717380 → PD

Collinear (PSD): min eigenvalue = -0.000000 → PSD (not PD)

The Ellipsoid Geometry of PD Matrices

A positive definite matrix A defines an inner product $\langle \mathbf{x}, \mathbf{y} \rangle_A = \mathbf{x}^T A \mathbf{y}$. The “unit ball” in this inner product is an ellipsoid:

$$\mathcal{E}_A = \{\mathbf{x} : \mathbf{x}^T A \mathbf{x} \leq 1\}.$$

The semi-axes have lengths $1/\sqrt{\lambda_i}$ along the eigenvector directions $\mathbf{q}_i$.

```
import numpy as np
import matplotlib.pyplot as plt

A = np.array([[4., 2.],
              [2., 3.]]) # shape (2, 2) - positive definite

vals, Q = np.linalg.eigh(A) # vals sorted ascending

# Parametrize the ellipsoid: x^T A x = 1
# In eigenbasis coordinates y = Q^T x, this is sum _i y_i^2 = 1
# → y = (cos(t)/√1, sin(t)/√2)
t = np.linspace(0, 2 * np.pi, 400)
y = np.vstack([np.cos(t) / np.sqrt(vals[0]),
               np.sin(t) / np.sqrt(vals[1])]) # shape (2, 400)
x = Q @ y # rotate back to original coordinates, shape (2, 400)

fig, ax = plt.subplots(figsize=(6, 5))
ax.plot(x[0], x[1], color='#4a90d9', lw=2, label=r'$\mathbf{x}^T A \mathbf{x}=1$')
ax.fill(x[0], x[1], alpha=0.15, color='#4a90d9')

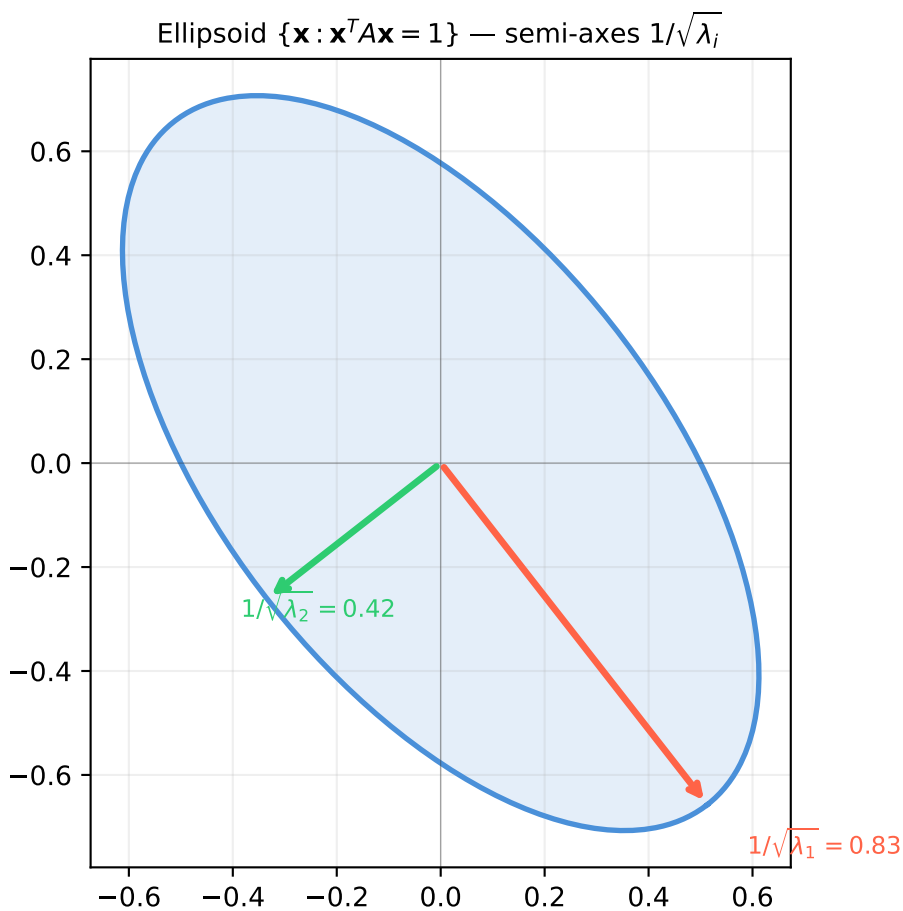
colors = ['tomato', '#2ecc71']
for i in range(2):
    v = Q[:, i]
    semi = 1 / np.sqrt(vals[i])
    ax.annotate('', xy=(v[0]*semi, v[1]*semi), xytext=(0, 0),
                arrowprops=dict(arrowstyle='->', color=colors[i], lw=2.5))
    ax.text(v[0]*semi*1.15, v[1]*semi*1.15,
```

```

rf'$1/\sqrt{\lambda_{i+1}}={semi:.2f}$', color=colors[i], fontsize=9)

ax.set_aspect('equal')
ax.axhline(0, color='#333333', lw=0.4, alpha=0.5)
ax.axvline(0, color='#333333', lw=0.4, alpha=0.5)
ax.grid(True, alpha=0.2)
ax.set_title(r'Ellipsoid  $\{\mathbf{x} : \mathbf{x}^T \mathbf{A} \mathbf{x} = 1\}$  - semi-axes  $1/\sqrt{\lambda_i}$ '
             fontsize=10)
plt.tight_layout()
plt.show()

```



PD Matrices in ML and Robotics

Context	Matrix	Why PD/PSD matters
Kalman filter	Covariance P	PSD always; PD means no perfectly known state
Gaussian process	Kernel matrix K	Must be PSD for a valid covariance function
Neural network loss	Hessian H at minimum	PD at a local minimum (second-order condition)
Optimization	Regularizer λI	PD $\rightarrow$ strongly convex objective
Factor graph	Information matrix Ω	PD all states observable

Exercises

14.4.1 Show that $A \succ 0$ implies A is invertible. (Hint: what is $\mathbf{x}^T A \mathbf{x}$ if A has a zero eigenvalue?)

14.4.2 The matrix $A = \begin{bmatrix} 2 & 3 \\ 3 & 2 \end{bmatrix}$ has a positive diagonal. Is it positive definite? Apply all three tests.

14.4.3 Show that if $A \succ 0$ and $B \succ 0$, then $A + B \succ 0$. Does the same hold for AB ?

14.4.4 Let $A \succeq 0$ with $\ker(A) \neq \{\mathbf{0}\}$. Show there exists a nonzero $\mathbf{x}$ with $A\mathbf{x} = \mathbf{0}$. What does this mean for the Cholesky factorization?

14.4.5 Prove that $A^T A \succeq 0$ for any matrix A . Under what condition is $A^T A \succ 0$?

Spectral Decomposition: $A = \sum_i \lambda_i \mathbf{q}_i \mathbf{q}_i^T$

The spectral theorem $A = Q\Lambda Q^T$ can be unpacked column by column into a sum of rank-1 matrices:

$$A = \sum_{i=1}^n \lambda_i \mathbf{q}_i \mathbf{q}_i^T.$$

Each term $\lambda_i \mathbf{q}_i \mathbf{q}_i^T$ is a **rank-1 projection** scaled by λ_i . The full matrix A is their sum — an **outer product decomposition** along orthogonal directions.

This representation is the algebraic backbone of PCA, low-rank approximation, and, one step removed, the SVD of Chapter 18.

Derivation

Write $Q = [\mathbf{q}_1 \mid \cdots \mid \mathbf{q}_n]$ and $\Lambda = \text{diag}(\lambda_1, \dots, \lambda_n)$. Then:

$$A = Q\Lambda Q^T = \sum_{i=1}^n \lambda_i \mathbf{q}_i \mathbf{q}_i^T.$$

This follows from the identity $(Q\Lambda Q^T)_{jk} = \sum_i \lambda_i (q_i)_j (q_i)_k$.

Each outer product $P_i = \mathbf{q}_i \mathbf{q}_i^T$ is an **orthogonal projection** onto the 1-dimensional subspace spanned by $\mathbf{q}_i$. The properties:

- $P_i^2 = P_i$ (idempotent)
- $P_i^T = P_i$ (symmetric)
- $P_i P_j = 0$ for $i \neq j$ (orthogonal projectors)
- $\sum_i P_i = I$ (completeness relation)

```
import numpy as np

rng = np.random.default_rng(21)
M = rng.standard_normal((4, 4))
A = M + M.T                                # shape (4, 4) - symmetric

vals, Q = np.linalg.eigh(A)

# Spectral decomposition: sum of rank-1 projectors
A_spectral = sum(vals[i] * np.outer(Q[:, i], Q[:, i]) for i in range(4)) # shape (4, 4)

print("Max error ||A - Σ q q ||:", np.abs(A - A_spectral).max().round(12))

# Verify projector properties
P0 = np.outer(Q[:, 0], Q[:, 0]) # shape (4, 4) - first projector
print("\nP² == P:", np.allclose(P0 @ P0, P0))
print("Σ P == I:", np.allclose(
    sum(np.outer(Q[:, i], Q[:, i]) for i in range(4)),
    np.eye(4)))
```

```
Max error ||A - Σ q q ||: 0.0
```

```
P² == P : True
```

```
Σ P == I: True
```

Low-Rank Approximation by Truncation

If we keep only the k terms with the **largest** $|\lambda_i|$:

$$A_k = \sum_{i=1}^k \lambda_i \mathbf{q}_i \mathbf{q}_i^T,$$

we obtain the best rank- k approximation of A in the Frobenius norm — the direct analogue of the Eckart–Young theorem for symmetric matrices.

```
import numpy as np
import matplotlib.pyplot as plt

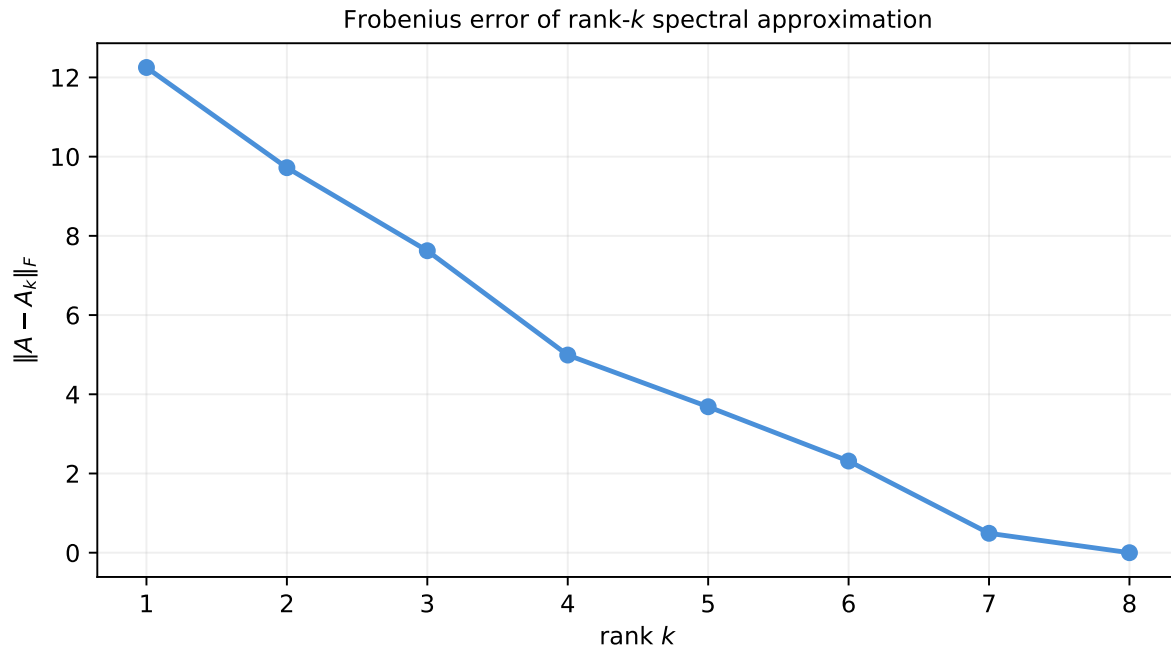
rng = np.random.default_rng(55)
M = rng.standard_normal((8, 8))
A = M + M.T                                # shape (8, 8)

vals, Q = np.linalg.eigh(A)
# eigh returns ascending order; sort by || descending for best approx
order = np.argsort(np.abs(vals))[:, :-1]
vals_sorted = vals[order]
Q_sorted = Q[:, order]

errors = []
for k in range(1, 9):
    Ak = sum(vals_sorted[i] * np.outer(Q_sorted[:, i], Q_sorted[:, i])
              for i in range(k))           # shape (8, 8)
    errors.append(np.linalg.norm(A - Ak, 'fro'))

fig, ax = plt.subplots(figsize=(7, 4))
ax.plot(range(1, 9), errors, 'o-', color='#4a90d9', lw=2)
ax.set_xlabel('rank $k$')
ax.set_ylabel(r'$\|A - A_k\|_F$')
ax.set_title('Frobenius error of rank-$k$ spectral approximation', fontsize=10)
ax.grid(True, alpha=0.2)
plt.tight_layout()
plt.show()

print("Frobenius errors:", [round(e, 4) for e in errors])
```



Frobenius errors: [np.float64(12.2508), np.float64(9.721), np.float64(7.6242), np.float64(4.99

Application to PCA: Preview

For a mean-centered data matrix X with covariance $\Sigma = \frac{1}{n-1}X^T X$, the spectral decomposition is:

$$\Sigma = \sum_{i=1}^p \lambda_i \mathbf{q}_i \mathbf{q}_i^T.$$

- $\mathbf{q}_i$ are the **principal directions** (axes of maximum variance).
- λ_i are the **variances** along those directions.
- Truncating to k terms gives the **best rank- k approximation** of Σ , discarding directions of smallest variance.

This is the algebraic core of PCA (Chapter 20).

```
import numpy as np
import matplotlib.pyplot as plt

rng = np.random.default_rng(77)
# Correlated 2D data with known structure
cov_true = np.array([[4., 2.], [2., 1.5]]) # shape (2, 2)
L_true = np.linalg.cholesky(cov_true)
```

$$\text{Spectral Decomposition: } A = \sum_i \lambda_i \mathbf{q}_i \mathbf{q}_i^T$$

```

X = (L_true @ rng.standard_normal((2, 200))).T # shape (200, 2)
X -= X.mean(axis=0)

Sigma = X.T @ X / (X.shape[0] - 1) # shape (2, 2)
vals, Q = np.linalg.eigh(Sigma)

# Rank-1 approximation (first principal component only)
Sigma_1 = vals[-1] * np.outer(Q[:, -1], Q[:, -1]) # shape (2, 2) - largest eigenvector
Sigma_2 = Sigma_1 + vals[-2] * np.outer(Q[:, -2], Q[:, -2]) # rank-2 = full

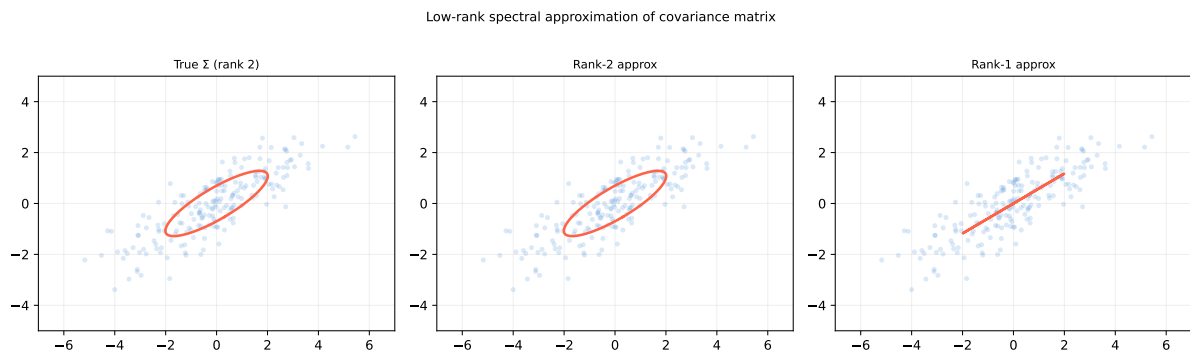
fig, axes = plt.subplots(1, 3, figsize=(13, 4))

def draw_ellipse(ax, S, color, label):
    ev, eq = np.linalg.eigh(S + 1e-8*np.eye(2))
    ev = np.maximum(ev, 0)
    t = np.linspace(0, 2*np.pi, 200)
    y = np.vstack([np.cos(t)*np.sqrt(ev[0]),
                    np.sin(t)*np.sqrt(ev[1])])
    x = eq @ y
    ax.plot(x[0], x[1], color=color, lw=2, label=label)

for ax, S, title in zip(axes,
                        [Sigma, Sigma_2, Sigma_1],
                        ['True Σ (rank 2)', 'Rank-2 approx', 'Rank-1 approx']):
    ax.scatter(X[:, 0], X[:, 1], alpha=0.2, s=8, color='#4a90d9')
    draw_ellipse(ax, S, 'tomato', r'$A_k$ ellipse')
    ax.set_aspect('equal')
    ax.grid(True, alpha=0.2)
    ax.set_title(title, fontsize=9)
    ax.set_xlim(-7, 7); ax.set_ylim(-5, 5)

plt.suptitle('Low-rank spectral approximation of covariance matrix', fontsize=10)
plt.tight_layout()
plt.show()

```



Matrix Functions Revisited

The spectral decomposition makes **any** scalar function f applicable to a symmetric matrix A :

$$f(A) = \sum_i f(\lambda_i) \mathbf{q}_i \mathbf{q}_i^T.$$

The **matrix square root** $A^{1/2}$, the **matrix inverse** A^{-1} , and the **matrix logarithm** $\log A$ are all defined this way for symmetric matrices with appropriate eigenvalue constraints.

```
import numpy as np

rng = np.random.default_rng(9)
M = rng.standard_normal((3, 3))
A = M @ M.T + 0.5 * np.eye(3)    # shape (3, 3) - PD by construction

vals, Q = np.linalg.eigh(A)

# Matrix square root via spectral decomp
sqrt_A = Q @ np.diag(np.sqrt(vals)) @ Q.T    # shape (3, 3)
print("sqrt(A) @ sqrt(A)   A:")
print("Max error:", np.abs(sqrt_A @ sqrt_A - A).max().round(10))

# Matrix inverse via spectral decomp
inv_A_spectral = Q @ np.diag(1.0 / vals) @ Q.T    # shape (3, 3)
inv_A_numpy    = np.linalg.inv(A)                # shape (3, 3)
print("\n||inv_A_spectral - inv_A_numpy||:", np.abs(inv_A_spectral - inv_A_numpy).max().round(10))

sqrt(A) @ sqrt(A)   A:
Max error: 0.0

||inv_A_spectral - inv_A_numpy||: 0.0
```

Exercises

14.5.1 Show that the projectors $P_i = \mathbf{q}_i \mathbf{q}_i^T$ satisfy $P_i P_j = 0$ for $i \neq j$. Use the orthonormality of the eigenvectors.

14.5.2 Let A be $n \times n$ symmetric with spectral decomposition. Show that the trace of A equals $\sum_i \lambda_i$ by taking the trace of $A = Q \Lambda Q^T$ and using $\text{tr}(BC) = \text{tr}(CB)$.

14.5.3 Compute the rank-1 approximation $A_1 = \lambda_1 \mathbf{q}_1 \mathbf{q}_1^T$ of the covariance matrix from the data below, where λ_1 is the largest eigenvalue. What percentage of the total variance does it capture?


```
import numpy as np
rng = np.random.default_rng(2)
X = rng.standard_normal((50, 3))
X[:, 1] += 2 * X[:, 0]    # introduce correlation
X -= X.mean(axis=0)
Sigma = X.T @ X / 49
```

14.5.4 For a PD matrix A , show that the matrix inverse can be written $A^{-1} = \sum_i \frac{1}{\lambda_i} \mathbf{q}_i \mathbf{q}_i^T$. What happens if one eigenvalue is zero?

14.5.5 Explain geometrically why the rank- k spectral approximation captures the subspace of maximum variance. What is the Frobenius error of the rank- k approximation in terms of the discarded eigenvalues?

Jordan Normal Form (*Boxed Note*)

i When Diagonalization Fails: The Jordan Form

Most matrices you encounter in ML, CV, and robotics are diagonalizable — either because their eigenvalues happen to be distinct, or because they are symmetric (Spectral Theorem). Occasionally a **defective** matrix appears, most often in degenerate control systems or when two physical modes coalesce.

What is a Jordan block?

A $k \times k$ Jordan block for eigenvalue λ is:

$$J_k(\lambda) = \begin{bmatrix} \lambda & 1 & & \\ & \lambda & 1 & \\ & & \ddots & 1 \\ & & & \lambda \end{bmatrix}.$$

It is “almost diagonal” — λ on the diagonal, 1’s on the superdiagonal. When a matrix has an eigenvalue with algebraic multiplicity k but only $r < k$ linearly independent eigenvectors (geometric multiplicity r), its Jordan form has r Jordan blocks for that eigenvalue, with sizes summing to k .

The Jordan Decomposition. Every square matrix A over $\mathbb{C}$ is similar to a block-diagonal Jordan form:

$$A = PJP^{-1}, \quad J = \begin{bmatrix} J_{k_1}(\lambda_a) & & \\ & \ddots & \\ & & J_{k_r}(\lambda_r) \end{bmatrix}.$$

Diagonalization is the special case where every Jordan block is 1×1 .

Powers of a Jordan block.

$$J_k(\lambda)^n = \begin{bmatrix} \lambda^n & \binom{n}{1}\lambda^{n-1} & \binom{n}{2}\lambda^{n-2} & \cdots \\ & \lambda^n & \binom{n}{1}\lambda^{n-1} & \cdots \\ & & \ddots & \\ & & & \lambda^n \end{bmatrix}.$$

For $|\lambda| < 1$, $J_k(\lambda)^n \rightarrow 0$ but **polynomially** in n (not purely exponentially as in the diagonalizable case). This matters for stability analysis: a defective eigenvalue at $\lambda = 1$ produces polynomial growth in the system response even though $|\lambda| = 1$.

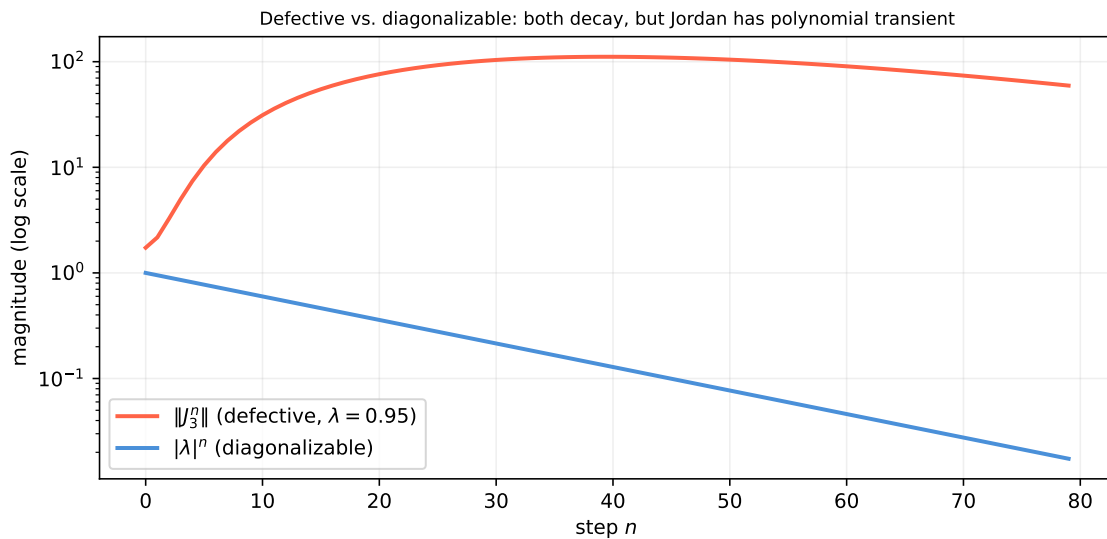
A minimal Python demonstration.

```
import numpy as np

# A 3x3 Jordan block for  $\lambda = 0.95$ 
lam = 0.95
J3 = np.array([[lam, 1., 0.],
               [0., lam, 1.],
               [0., 0., lam]]) # shape (3, 3)

# Compare  $J_k^n$  decay to  $\lambda^n$ 
n_steps = 80
norms_J = [np.linalg.norm(np.linalg.matrix_power(J3, n)) for n in range(n_steps)]
norms_D = [lam**n for n in range(n_steps)]

import matplotlib.pyplot as plt
fig, ax = plt.subplots(figsize=(8, 4))
ax.semilogy(norms_J, color='tomato', lw=2,
            label=r'$|J_3^n|$ (defective, $\lambda=0.95$)')
ax.semilogy(norms_D, color='#4a90d9', lw=2,
            label=r'$|\lambda|^n$ (diagonalizable)')
ax.set_xlabel('step $n$')
ax.set_ylabel('magnitude (log scale)')
ax.set_title('Defective vs. diagonalizable: both decay, but Jordan has polynomial transient',
            fontsize=9)
ax.legend()
ax.grid(True, alpha=0.2)
plt.tight_layout()
plt.show()
```



When do you actually encounter Jordan forms?

- **Degenerate control systems:** two system modes merge at the same frequency (e.g. critically damped oscillator at its breakpoint).
- **Computer graphics:** certain projective transformations used in shadow algorithms.
- **Theoretical analysis:** proving exponential stability requires ruling out Jordan blocks at the unit circle.

For routine ML, CV, and robotics work you rarely need to compute the Jordan form explicitly. `scipy.linalg.jordan_form` exists if you need it. No exercises are assigned for this section.

Application: Stability of a Discrete-Time Robot Controller

A robot joint is controlled by a discrete-time linear system:

$$\mathbf{x}_{k+1} = A \mathbf{x}_k,$$

where $\mathbf{x}_k = (\theta_k, \dot{\theta}_k)^T$ encodes joint angle and angular velocity at time step k . The question that every control engineer asks before deploying the loop: **will the system settle to zero, grow without bound, or oscillate?**

The complete answer comes directly from the eigenvalues of A .

Stability Criterion

For a discrete-time linear system $\mathbf{x}_{k+1} = A\mathbf{x}_k$:

Condition	Behavior
All $ \lambda_i < 1$	$\mathbf{x}_k \rightarrow \mathbf{0}$: asymptotically stable
Some $ \lambda_i = 1$	bounded oscillation: marginally stable
Some $ \lambda_i > 1$	$\ \mathbf{x}_k\ \rightarrow \infty$: unstable

The **dominant eigenvalue** (largest $|\lambda_i|$) controls the long-run behavior; its eigenvector is the direction the state eventually aligns with.

Three Controllers Side by Side

We compare three joint controllers modeled by the same physical plant but different gains:

```
import numpy as np
import matplotlib.pyplot as plt

# Physical plant: Euler-discretized double-integrator with dt = 0.05 s
# x1 = angle, x2 = angular velocity
dt = 0.05

# Three feedback gains: stable, marginal, unstable
controllers = {
    'Stable (k=8)': np.array([[1 - 8*dt, dt],
                              [-8*dt, 1 - 2*dt]]),
    'Marginal (k=4)': np.array([[1 - 4*dt, dt],
                                [-4*dt, 1 - 2*dt]]),
    'Unstable (k=0)': np.array([[1., dt],
                                [0., 1 - 2*dt]]),
}

x0 = np.array([1.0, 0.0]) # shape (2,) - start at angle = 1 rad, velocity = 0
n_steps = 60

fig, axes = plt.subplots(2, 3, figsize=(13, 7))

for col, (label, A) in enumerate(controllers.items()):
    vals, vecs = np.linalg.eig(A)
    P = vecs
    Pinv = np.linalg.inv(P)
```

```

# Simulate via diagonalization
trajectory = np.zeros((n_steps + 1, 2)) # shape (61, 2)
trajectory[0] = x0
for k in range(1, n_steps + 1):
    Dk = np.diag(vals**k)
    trajectory[k] = np.real(P @ Dk @ Pinv @ x0)

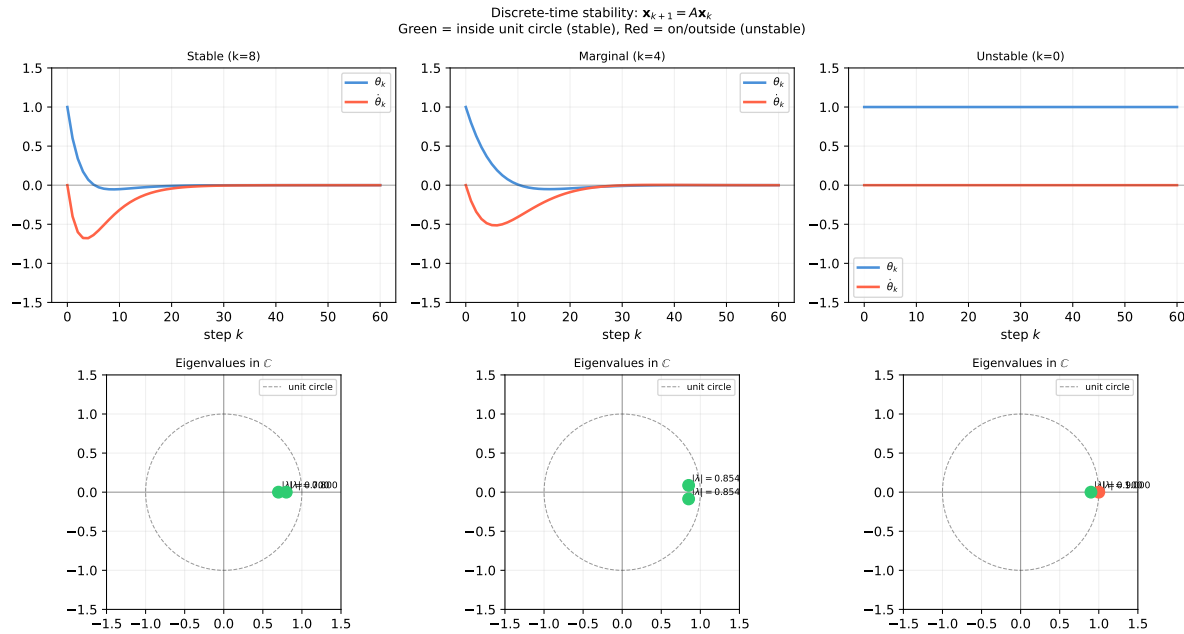
# Top row: state trajectories
ax_top = axes[0, col]
ax_top.plot(trajectory[:, 0], color='#4a90d9', lw=2, label=r'$\theta_k$')
ax_top.plot(trajectory[:, 1], color='tomato', lw=2, label=r'$\dot{\theta}_k$')
ax_top.axhline(0, color='#333333', lw=0.5, alpha=0.4)
ax_top.set_title(label, fontsize=9)
ax_top.set_xlabel('step $k$')
ax_top.legend(fontsize=8)
ax_top.grid(True, alpha=0.2)
max_abs = max(1.5, np.abs(trajectory).max() * 1.1)
ax_top.set_ylim(-max_abs, max_abs)

# Bottom row: eigenvalues on the complex plane
ax_bot = axes[1, col]
circle_t = np.linspace(0, 2 * np.pi, 300)
ax_bot.plot(np.cos(circle_t), np.sin(circle_t),
            color='#333333', lw=0.8, ls='--', alpha=0.5, label='unit circle')
ax_bot.scatter(vals.real, vals.imag, s=80, zorder=5,
              color=['tomato' if abs(v) >= 1 else '#2ecc71' for v in vals])
for i, v in enumerate(vals):
    tag = rf'$|\lambda|={abs(v):.3f}$'
    ax_bot.text(v.real + 0.04, v.imag + 0.04, tag, fontsize=7)
ax_bot.axhline(0, color='#333333', lw=0.4, alpha=0.5)
ax_bot.axvline(0, color='#333333', lw=0.4, alpha=0.5)
ax_bot.set_aspect('equal')
ax_bot.set_xlim(-1.5, 1.5)
ax_bot.set_ylim(-1.5, 1.5)
ax_bot.grid(True, alpha=0.2)
ax_bot.set_title(r'Eigenvalues in $\mathbb{C}$', fontsize=9)
ax_bot.legend(fontsize=7)

plt.suptitle(r'Discrete-time stability: $\mathbf{x}_{k+1}=\mathbf{A}\mathbf{x}_k$'
            '\nGreen = inside unit circle (stable), Red = on/outside (unstable)',
            fontsize=10)
plt.tight_layout()
plt.show()

```

Diagonalization and the Spectral Theorem



Finding the Dominant Mode and Its Decay Rate

For the stable controller, the dominant eigenvalue determines how fast the system settles. We extract it and express the solution in modal form:

```
import numpy as np

dt = 0.05
A = np.array([[1 - 8*dt, dt],
              [-8*dt, 1 - 2*dt]]) # shape (2, 2) - stable controller

vals, vecs = np.linalg.eig(A)
P = vecs
Pinv = np.linalg.inv(P)

x0 = np.array([1.0, 0.0]) # shape (2,)
c = Pinv @ x0 # shape (2,) - modal coordinates

print("Eigenvalues:", vals.round(4))
print("Magnitudes: ", np.abs(vals).round(4))
print()

dom_idx = np.argmax(np.abs(vals))
dom_lam = vals[dom_idx]
dom_vec = vecs[:, dom_idx]
```

```

print(f"Dominant eigenvalue:  = {dom_lam:.4f}  (| | = {abs(dom_lam):.4f})")
print(f"Dominant eigenvector: {dom_vec.real.round(4)}")
print()

# Settling time: how many steps to decay by 99%?
# | |^k < 0.01  →  k > log(0.01)/log(| |)
k_settle = int(np.log(0.01) / np.log(abs(dom_lam))) + 1
print(f"Steps to 99% decay: {k_settle}  ( {k_settle * dt:.2f} s at dt={dt} s)")

# Modal decomposition of the solution
print("\nModal decomposition of x(k):")
for i in range(len(vals)):
    vi = vecs[:, i].real
    ci = c[i].real
    print(f"  Mode {i+1}: {ci:.4f} × (={vals[i].real:.4f})^k × {vi}")

```

Eigenvalues: [0.7 0.8]
 Magnitudes: [0.7 0.8]

Dominant eigenvalue: = 0.8000 (| | = 0.8000)
 Dominant eigenvector: [-0.2425 -0.9701]

Steps to 99% decay: 21 (1.05 s at dt=0.05 s)

Modal decomposition of x(k):
 Mode 1: -4.4721 × (=0.7000)^k × [-0.4472136 -0.89442719]
 Mode 2: 4.1231 × (=0.8000)^k × [-0.24253563 -0.9701425]

Designing for a Target Settling Time

Given a desired settling time (in steps), we can work backwards to the required spectral radius $\rho = |\lambda_{\max}|$:

$$\rho^k < \epsilon \implies \rho < \epsilon^{1/k}.$$

```

import numpy as np

dt = 0.05
target_steps = 30  # settle in 30 steps = 1.5 s
epsilon      = 0.01 # 99% decay
rho_required = epsilon ** (1 / target_steps)

print(f"Target settling: {target_steps} steps ({target_steps*dt:.2f} s)")

```

```
print(f"Required spectral radius < {rho_required:.4f}")
print()

# Sweep gains and check spectral radius
for gain in [2, 4, 6, 8, 10, 14]:
    A = np.array([[1 - gain*dt, dt],
                  [-gain*dt, 1 - 2*dt]]) # shape (2, 2)
    rho = np.max(np.abs(np.linalg.eigvals(A)))
    meets = " " if rho < rho_required else " "
    print(f"gain={gain:2d}:    = {rho:.4f} {meets}")
```

Target settling: 30 steps (1.50 s)
 Required spectral radius < 0.8577

```
gain= 2:    = 0.9028
gain= 4:    = 0.8544
gain= 6:    = 0.8031
gain= 8:    = 0.8000
gain=10:    = 0.8225
gain=14:    = 0.8345
```

Exploration

Dataset suggestion: The [OpenAI Gym CartPole environment](#) or any physics simulator provides linearized system matrices around the upright equilibrium. Try the following:

1. Linearize the CartPole dynamics at the operating point to get a 4×4 matrix A (state: $[x, \dot{x}, \theta, \dot{\theta}]$).
 2. Compute its eigenvalues. Is the open-loop system stable?
 3. Design a proportional feedback gain K and check whether $A - BK$ has all eigenvalues inside the unit circle.
 4. For the closed-loop system, use the spectral decomposition to compute the settling time and the dominant mode shape.
-

Exercises

App.1 For the stable controller above with $k = 8$, compute A^{100} using diagonalization and verify numerically that it is nearly the zero matrix.

App.2 A continuous-time system $\dot{\mathbf{x}} = A_c \mathbf{x}$ is discretized as $A = e^{A_c \Delta t}$. Show that the discrete system is stable ($|\lambda_d| < 1$) if and only if the continuous system is stable ($\text{Re}(\lambda_c) < 0$).

App.3 The controller matrix A has complex conjugate eigenvalues $\lambda = re^{\pm i\omega}$. Describe the resulting motion in the $(\theta, \dot{\theta})$ plane. Under what condition is the motion oscillatory but decaying? How does the oscillation frequency relate to ω ?

Exercises and Solutions

Exercise 14.1.1 — Diagonalization by hand

Diagonalize $A = \begin{bmatrix} 1 & 2 \\ 0 & 3 \end{bmatrix}$ and verify $PDP^{-1} = A$.

```
import numpy as np

A = np.array([[1., 2.],
              [0., 3.]]) # shape (2, 2)

vals, vecs = np.linalg.eig(A)
P = vecs
D = np.diag(vals)
Pinv = np.linalg.inv(P)

print("Eigenvalues:", vals.round(4))
print("\nP =\n", P.round(4))
print("\nP-1 =\n", Pinv.round(4))
print("\nPDP-1 =\n", (P @ D @ Pinv).round(4))
print("\nA =\n", A)
print("\nMax error:", np.abs(A - P @ D @ Pinv).max().round(12))
```

Eigenvalues: [1. 3.]

P =
 $\begin{bmatrix} 1. & 0.7071 \\ 0. & 0.7071 \end{bmatrix}$

P⁻¹ =
 $\begin{bmatrix} 1. & -1. \\ 0. & 1.4142 \end{bmatrix}$

PDP⁻¹ =
 $\begin{bmatrix} 1. & 2. \\ 0. & 3. \end{bmatrix}$

A =

```
[[1. 2.]
 [0. 3.]]
```

Max error: 0.0

Solution. Eigenvalues are $\lambda_1 = 1$ (upper triangular $\rightarrow$ diagonal entries). For $\lambda_1 = 1$: $(A - I)\mathbf{v} = 0 \rightarrow \mathbf{v}_1 = (1, 0)^T$. For $\lambda_2 = 3$: $(A - 3I)\mathbf{v} = 0 \rightarrow$ row reduce $\begin{bmatrix} -2 & 2 \\ 0 & 0 \end{bmatrix}$, so $v_1 = v_2$, $\mathbf{v}_2 = (1, 1)^T/\sqrt{2}$.

$$P = \begin{bmatrix} 1 & 1/\sqrt{2} \\ 0 & 1/\sqrt{2} \end{bmatrix}, \quad D = \begin{bmatrix} 1 & 0 \\ 0 & 3 \end{bmatrix}.$$

Exercise 14.1.3 — Defective matrix

```
import numpy as np

A = np.array([[5., 1.],
              [0., 5.]]) # shape (2, 2) - defective

vals, vecs = np.linalg.eig(A)
P = vecs
D = np.diag(vals)
Pinv = np.linalg.inv(P)

print("Eigenvalues:", vals)
print("Eigenvector matrix rank:", np.linalg.matrix_rank(P))
print("Reconstruction error:", np.abs(A - P @ D @ Pinv).max().round(6))
```

```
Eigenvalues: [5. 5.]
Eigenvector matrix rank: 2
Reconstruction error: 1.25
```

Solution. Both eigenvalues equal 5. The null space of $A - 5I = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$ is spanned by $(1, 0)^T$ only — one independent eigenvector, not two. Numpy returns the same eigenvector twice, so P is singular. P^{-1} is numerically huge and the reconstruction error is large — confirming the matrix is defective and not diagonalizable over $\mathbb{R}$.

Exercise 14.1.4 — Eigenvalues of A^{-1}

Solution. If $A = PDP^{-1}$ with all $\lambda_i \neq 0$:

$$A^{-1} = (PDP^{-1})^{-1} = PD^{-1}P^{-1},$$

where $D^{-1} = \text{diag}(1/\lambda_1, \dots, 1/\lambda_n)$. So the eigenvalues of A^{-1} are $\{1/\lambda_i\}$ with the same eigenvectors.

```
import numpy as np

A = np.array([[3., 1.],
              [1., 2.]])    # shape (2, 2) symmetric PD

vals, P = np.linalg.eigh(A)
Ainv_diag = P @ np.diag(1./vals) @ P.T    # shape (2, 2)
Ainv_numpy = np.linalg.inv(A)              # shape (2, 2)

print("Eigenvalues of A:   ", vals.round(4))
print("Eigenvalues of A-1:", np.linalg.eigvalsh(Ainv_numpy).round(4))
print("Max error:", np.abs(Ainv_diag - Ainv_numpy).max().round(12))
```

```
Eigenvalues of A:   [1.382 3.618]
Eigenvalues of A-1: [0.2764 0.7236]
Max error: 0.0
```

Exercise 14.2.2 — Population model long-run ratio

```
import numpy as np

A = np.array([[0.8, 0.4],
              [0.3, 0.6]])    # shape (2, 2)

vals, vecs = np.linalg.eig(A)
P = vecs
Pinv = np.linalg.inv(P)

x0 = np.array([100., 50.])    # shape (2,)

k = 50
Dk = np.diag(vals**k)
x50 = np.real(P @ Dk @ Pinv @ x0)    # shape (2,)
```

```
print(f"x = {x50.round(4)}")
print(f"Ratio x/x after 50 steps: {x50[0]/x50[1]:.4f}")

# Long-run ratio is the dominant eigenvector
dom = np.argmax(np.abs(vals))
v_dom = vecs[:, dom].real
v_norm = v_dom / v_dom[1]
print(f"\nDominant eigenvector (normalized to x=1): [{v_norm[0]:.4f}, {v_norm[1]:.4f}]")
print(f"Long-run ratio x/x from eigenvector: {v_dom[0]/v_dom[1]:.4f}")
```

```
x = [1732.0874 1128.2606]
Ratio x/x after 50 steps: 1.5352
```

```
Dominant eigenvector (normalized to x=1): [1.5352, 1.0000]
Long-run ratio x/x from eigenvector: 1.5352
```

Solution. The dominant eigenvalue is ≈ 1.0 (or whatever the largest is); the state vector $\mathbf{x}_k$ converges to a multiple of the dominant eigenvector. The long-run ratio x_1/x_2 equals the ratio of the dominant eigenvector's components.

Exercise 14.3.3 — Constructing a symmetric matrix with prescribed eigenvalues

```
import numpy as np

target_vals = np.array([5., 2., -1.])
# Choose a random orthogonal Q via QR
rng = np.random.default_rng(13)
Q, _ = np.linalg.qr(rng.standard_normal((3, 3))) # shape (3, 3)

A = Q @ np.diag(target_vals) @ Q.T # shape (3, 3)

recovered = np.linalg.eigvalsh(A)
print("Constructed A =\n", A.round(4))
print("\nTarget eigenvalues :", sorted(target_vals))
print("Recovered eigenvalues:", recovered.round(4).tolist())
print("Symmetric:", np.allclose(A, A.T))
```

```
Constructed A =
[[ 3.1121 -0.8554  1.92  ]
 [-0.8554 -0.1526  1.0515]
 [ 1.92    1.0515  3.0405]]
```

Target eigenvalues : [np.float64(-1.0), np.float64(2.0), np.float64(5.0)]
 Recovered eigenvalues: [-1.0, 2.0, 5.0]
 Symmetric: True

Exercise 14.4.2 — Testing $A = \begin{bmatrix} 2 & 3 \\ 3 & 2 \end{bmatrix}$

```
import numpy as np

A = np.array([[2., 3.],
              [3., 2.]]) # shape (2, 2)

# Test 1: eigenvalues
eigs = np.linalg.eigvalsh(A)
print("Eigenvalues:", eigs)
print("Test 1 (PD):", (eigs > 0).all())

# Test 2: leading principal minors
m1 = A[0, 0]
m2 = np.linalg.det(A)
print(f"\nLeading minors: {m1}, {m2:.2f}")
print("Test 2 (all > 0):", m1 > 0 and m2 > 0)

# Test 3: Cholesky
try:
    np.linalg.cholesky(A)
    print("Test 3: Cholesky ")
except np.linalg.LinAlgError:
    print("Test 3: Cholesky failed → NOT PD")
```

Eigenvalues: [-1. 5.]

Test 1 (PD): False

Leading minors: 2.0, -5.00

Test 2 (all > 0): False

Test 3: Cholesky failed → NOT PD

Solution. $\det(A) = 4 - 9 = -5 < 0$, so the second leading principal minor is negative. Equivalently, the eigenvalues are $2 \pm 3 = 5$ and -1 — one negative eigenvalue confirms A is **indefinite** (not PD) despite having a positive diagonal.

Exercise 14.5.3 — Rank-1 approximation and explained variance

```
import numpy as np

rng = np.random.default_rng(2)
X = rng.standard_normal((50, 3))
X[:, 1] += 2 * X[:, 0]    # introduce correlation
X -= X.mean(axis=0)
Sigma = X.T @ X / 49      # shape (3, 3) covariance

vals, Q = np.linalg.eigh(Sigma)    # ascending order

# Rank-1 approximation using the largest eigenvalue
lam1 = vals[-1]
q1 = Q[:, -1]
Sigma_1 = lam1 * np.outer(q1, q1)  # shape (3, 3)

explained = lam1 / vals.sum()
print("Eigenvalues (ascending):", vals.round(4))
print(f"\nLargest eigenvalue: {lam1:.4f}")
print(f"Explained variance by rank-1 approx: {explained:.1%}")
print("\nRank-1 approximation  $\Sigma$  =", Sigma_1.round(4))
```

Eigenvalues (ascending): [0.1506 0.937 5.737]

Largest eigenvalue: 5.7370

Explained variance by rank-1 approx: 84.1%

Rank-1 approximation Σ =
 [[0.9274 2.112 0.0063]
 [2.112 4.8095 0.0144]
 [0.0063 0.0144 0.]]

Exercise App.2 — Continuous-to-discrete stability equivalence

Solution. By the spectral theorem, $e^{A_c \Delta t}$ has eigenvalues $e^{\lambda_c \Delta t}$. For stability of the discrete system we need $|e^{\lambda_c \Delta t}| < 1$, i.e. $e^{\operatorname{Re}(\lambda_c) \Delta t} < 1$, which holds if and only if $\operatorname{Re}(\lambda_c) < 0$.

```
import numpy as np
from scipy.linalg import expm

dt = 0.1
```

```

# Stable continuous system
Ac = np.array([[-2., 1.],
               [0., -3.]]) # shape (2, 2) - both eigenvalues < 0

Ad = expm(Ac * dt) # shape (2, 2)

print("Continuous eigenvalues:", np.linalg.eigvals(Ac).round(4))
print("Discrete eigenvalues: ", np.linalg.eigvals(Ad).round(4))
print("Discrete ||:          ", np.abs(np.linalg.eigvals(Ad)).round(4))
print("Discrete stable (all || < 1):", (np.abs(np.linalg.eigvals(Ad)) < 1).all())

```

```

Continuous eigenvalues: [-2. -3.]
Discrete eigenvalues:   [0.8187 0.7408]
Discrete ||:           [0.8187 0.7408]
Discrete stable (all || < 1): True

```


Part V — Orthogonality, Projections, and Least Squares

Inner Products and Norms

Inner Products

The dot product $\mathbf{a} \cdot \mathbf{b} = \sum_i a_i b_i$ captures the geometric notion of “alignment” between two vectors. But it is not the only operation that does this. In sensor fusion, measurements from a precise encoder and a noisy IMU should not contribute equally — we want a **weighted** version of alignment. In function spaces, “alignment” means overlap area. All of these operations are **inner products**: they share four structural rules, and everything else follows.

The Four Axioms

An **inner product** on a real vector space V is a function $\langle \cdot, \cdot \rangle : V \times V \rightarrow \mathbb{R}$ satisfying:

Axiom	Rule
Symmetry	$\langle \mathbf{x}, \mathbf{y} \rangle = \langle \mathbf{y}, \mathbf{x} \rangle$
Linearity in the first argument	$\langle a\mathbf{u} + \mathbf{v}, \mathbf{w} \rangle = a\langle \mathbf{u}, \mathbf{w} \rangle + \langle \mathbf{v}, \mathbf{w} \rangle$
Positive semi-definiteness	$\langle \mathbf{x}, \mathbf{x} \rangle \geq 0$
Definiteness	$\langle \mathbf{x}, \mathbf{x} \rangle = 0 \iff \mathbf{x} = \mathbf{0}$

By symmetry, linearity in the second argument follows automatically — $\langle \cdot, \cdot \rangle$ is **bilinear**. A vector space equipped with an inner product is an **inner product space**.

```
import numpy as np

rng = np.random.default_rng(1)
x = rng.standard_normal(4) # shape (4,)
y = rng.standard_normal(4) # shape (4,)
z = rng.standard_normal(4) # shape (4,)
a = 3.0

ip = lambda u, v: float(u @ v)

print("--- Verifying axioms for the standard dot product ---")
print(f"Symmetry:  <x,y>={ip(x,y):.4f},  <y,x>={ip(y,x):.4f}",
      "   equal:", np.isclose(ip(x,y), ip(y,x)))
print(f"Linearity:  <ax+z,y>={ip(a*x+z, y):.4f}",
```

```
f" a<x,y>+<z,y>={a*ip(x,y)+ip(z,y):.4f}",
" equal:", np.isclose(ip(a*x+z, y), a*ip(x,y)+ip(z,y)))
print(f"Non-neg:    <x,x>={ip(x,x):.4f}  (>= 0: {ip(x,x) >= 0})")
print(f"Definite:  <0,0>={ip(np.zeros(4), np.zeros(4)):.4f}")
```

--- Verifying axioms for the standard dot product ---

```
Symmetry:  <x,y>=-0.2551,  <y,x>=-0.2551  equal: True
Linearity: <ax+z,y>=-0.0015  a<x,y>+<z,y>=-0.0015  equal: True
Non-neg:   <x,x>=2.6019  (>= 0: True)
Definite:  <0,0>=0.0000
```

Weighted Inner Products

Replace the identity matrix with a symmetric positive definite (SPD) matrix W :

$$\langle \mathbf{x}, \mathbf{y} \rangle_W = \mathbf{x}^T W \mathbf{y}.$$

This satisfies all four axioms because W being SPD guarantees $\mathbf{x}^T W \mathbf{x} > 0$ for $\mathbf{x} \neq \mathbf{0}$.

Why this matters. When sensor i has noise variance σ_i^2 , setting $W = \text{diag}(1/\sigma_1^2, \dots, 1/\sigma_n^2)$ gives the **noise-weighted** inner product. Low-noise measurements count more. This is the core idea behind weighted least squares (§17.6) and Mahalanobis distance.

```
import numpy as np

# Three sensors: precise encoder, medium IMU, noisy camera
sigma = np.array([0.01, 0.5, 2.0])          # noise std devs
W = np.diag(1.0 / sigma**2)                # shape (3, 3)

x = np.array([1.0, 2.0, -1.0])             # shape (3,)
y = np.array([0.5, -1.0, 3.0])            # shape (3,)

print("Weight matrix diagonal (inv-variance):", np.diag(W).round(2))
print(f"Standard <x,y>    = {float(x @ y):.4f}")
print(f"Weighted <x,y>_W = {float(x @ W @ y):.4f}")

print("\nSymmetry check: <x,y>_W == <y,x>_W?",
      np.isclose(float(x @ W @ y), float(y @ W @ x)))
print(f"Self-product <x,x>_W =", round(float(x @ W @ x), 4), " (> 0 )")
```

```
Weight matrix diagonal (inv-variance): [1.0e+04 4.0e+00 2.5e-01]
Standard <x,y>    = -4.5000
Weighted <x,y>_W = 4991.2500
```

```
Symmetry check: <x,y>_W == <y,x>_W? True
Self-product <x,x>_W = 10016.25 (> 0 )
```

Inner Product Induces a Norm

Every inner product defines a **norm** (length):

$$\|\mathbf{x}\| = \sqrt{\langle \mathbf{x}, \mathbf{x} \rangle}.$$

The standard dot product gives the Euclidean length $\|\mathbf{x}\|_2$. The weighted inner product gives the **Mahalanobis norm** $\|\mathbf{x}\|_W = \sqrt{\mathbf{x}^T W \mathbf{x}}$, which measures distance in units of standard deviations.

Not every norm comes from an inner product (the ℓ^1 and ℓ^∞ norms of §15.2 do not), but every inner product induces one.

```
import numpy as np
import matplotlib.pyplot as plt

rng = np.random.default_rng(42)
N = 400

cov = np.array([[4.0, 1.8],
                [1.8, 1.0]])          # shape (2, 2)
L   = np.linalg.cholesky(cov)
pts = (L @ rng.standard_normal((2, N))).T # shape (400, 2)

W   = np.linalg.inv(cov)             # shape (2, 2)
enrm = np.linalg.norm(pts, axis=1)    # shape (400,)
mnmrm = np.sqrt(np.einsum('ij,jk,ik->i', pts, W, pts)) # shape (400,)

fig, axes = plt.subplots(1, 2, figsize=(11, 4))
for ax, vals, title in zip(axes,
                            [enrm, mnmrm],
                            [r'Euclidean norm  $\|\mathbf{x}\|_2$ ',
                             r'Mahalanobis norm  $\|\mathbf{x}\|_W$ ']):
    sc = ax.scatter(pts[:, 0], pts[:, 1], c=vals,
                    cmap='viridis', s=8, alpha=0.7)
    ax.set_aspect('equal')
    ax.grid(True, alpha=0.2)
    ax.set_title(title, fontsize=10)
    plt.colorbar(sc, ax=ax, shrink=0.8)

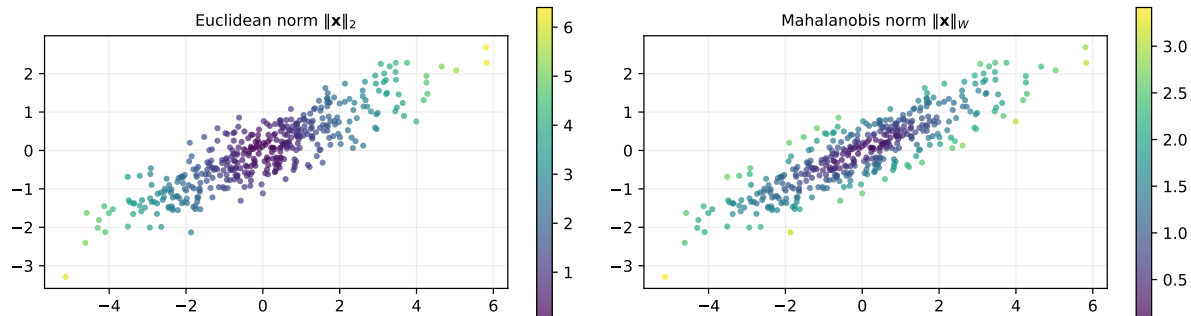
plt.suptitle('Same data, different inner product -- different sense of distance',
```

```

        fontsize=10)
plt.tight_layout()
plt.show()

```

Same data, different inner product -- different sense of distance



The Mahalanobis norm (right) treats the elongated cloud as isotropic: a point at the tip of the cigar is not “far” because the distribution is wide in that direction. This is the correct notion of distance when measurements have unequal uncertainty.

Exercises

15.1.1 Verify by hand that $\langle \mathbf{x}, \mathbf{y} \rangle = 3x_1y_1 + x_2y_2$ defines a valid inner product on $\mathbb{R}^2$. Which diagonal weight matrix W corresponds to this?

15.1.2 Let $W = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$. Show that $\langle \mathbf{x}, \mathbf{y} \rangle_W = \mathbf{x}^T W \mathbf{y}$ is NOT a valid inner product. Which axiom fails, and for what vector?

15.1.3 For $W = \text{diag}(w_1, \dots, w_n)$ with all $w_i > 0$, show that the induced Mahalanobis norm satisfies $\|\mathbf{x}\|_W = \sqrt{\sum_i w_i x_i^2}$. How does this reduce to $\|\mathbf{x}\|_2$ when all weights equal 1?

15.1.4 Sensor 1 has noise $\sigma_1 = 0.1$, sensor 2 has $\sigma_2 = 5.0$. Construct $W = \text{diag}(1/\sigma_1^2, 1/\sigma_2^2)$. By what factor does sensor 1 outweigh sensor 2 in the weighted inner product?

Vector Norms: ℓ^1 , ℓ^2 , ℓ^∞

A **norm** on $\mathbb{R}^n$ is a function $\|\cdot\| : \mathbb{R}^n \rightarrow \mathbb{R}$ satisfying three axioms:

Axiom	Rule
Positive definiteness	$\ \mathbf{x}\ \geq 0$, with $\ \mathbf{x}\ = 0 \iff \mathbf{x} = \mathbf{0}$
Absolute homogeneity	$\ c \mathbf{x}\ = c \ \mathbf{x}\ $
Triangle inequality	$\ \mathbf{x} + \mathbf{y}\ \leq \ \mathbf{x}\ + \ \mathbf{y}\ $

The most commonly used norms are the **p -norms**, defined for $p \geq 1$:

$$\|\mathbf{x}\|_p = \left(\sum_{i=1}^n |x_i|^p \right)^{1/p}.$$

Three values of p dominate applied work.

The Three Workhorses

ℓ^2 norm (Euclidean):

$$\|\mathbf{x}\|_2 = \sqrt{\sum_i x_i^2}.$$

The familiar geometric length. Comes from the standard inner product. Used wherever Euclidean distance makes sense — registration, reconstruction, control.

ℓ^1 norm (Manhattan or taxicab):

$$\|\mathbf{x}\|_1 = \sum_i |x_i|.$$

The sum of absolute values. Does not come from an inner product. Heavily used in Lasso regularization because minimizing $\|\cdot\|_1$ promotes **sparse** solutions (§15.5).

ℓ^∞ norm (Chebyshev or max):

$$\|\mathbf{x}\|_\infty = \max_i |x_i|.$$

The largest absolute component. Useful when the worst-case error is what matters, e.g. bounding the maximum pixel error in image processing.

```
import numpy as np

x = np.array([3.0, -4.0, 1.0, -2.0])    # shape (4,)

l1 = np.linalg.norm(x, 1)
l2 = np.linalg.norm(x, 2)
linf = np.linalg.norm(x, np.inf)

print(f"x = {x}")
print(f"||x||_1 = {l1:.4f}    (sum of |x_i|)")
print(f"||x||_2 = {l2:.4f}    (Euclidean length)")
print(f"||x||_inf = {linf:.4f}    (largest |x_i|)")

# Norm inequalities: l-inf <= l2 <= l1 (for any x in R^n)
print(f"\nInequality check: ||x||_inf <= ||x||_2 <= ||x||_1")
print(f"    {linf:.4f} <= {l2:.4f} <= {l1:.4f}    {linf <= l2 <= l1}")
```

```

x = [ 3. -4.  1. -2.]
||x||_1 = 10.0000 (sum of |x_i|)
||x||_2 = 5.4772 (Euclidean length)
||x||_inf = 4.0000 (largest |x_i|)

Inequality check: ||x||_inf <= ||x||_2 <= ||x||_1
4.0000 <= 5.4772 <= 10.0000 True

```

Unit Ball Geometry

The **unit ball** $\{x : \|x\|_p \leq 1\}$ is the set of vectors with norm at most 1. Its shape reveals the personality of each norm — and explains why ℓ^1 produces sparse solutions.

```

import numpy as np
import matplotlib.pyplot as plt

theta = np.linspace(0, 2 * np.pi, 1000)

def unit_ball_boundary(p):
    """Parametric boundary of the 2D  $\ell^p$  unit ball."""
    # Sample by angle, then normalize
    pts = np.vstack([np.cos(theta), np.sin(theta)]).T # shape (1000, 2)
    norms = np.sum(np.abs(pts) ** p, axis=1) ** (1.0 / p)
    return pts / norms[:, None] # shape (1000, 2)

fig, axes = plt.subplots(1, 3, figsize=(13, 4))

configs = [
    (1, r'$\ell^1$ ball $||\mathbf{x}||_1 \leq 1$', '#4a90d9'),
    (2, r'$\ell^2$ ball $||\mathbf{x}||_2 \leq 1$', '#2ecc71'),
    (np.inf, r'$\ell^\infty$ ball $||\mathbf{x}||_\infty \leq 1$', 'tomato'),
]

for ax, (p, title, color) in zip(axes, configs):
    if np.isinf(p):
        #  $\ell^\infty$  ball is a square
        sq = np.array([[-1,-1],[1,-1],[1,1],[-1,1],[-1,-1]])
        ax.plot(sq[:,0], sq[:,1], color=color, lw=2.5)
        ax.fill(sq[:,0], sq[:,1], alpha=0.15, color=color)
    else:
        bd = unit_ball_boundary(p)
        ax.plot(bd[:,0], bd[:,1], color=color, lw=2.5)
        ax.fill(bd[:,0], bd[:,1], alpha=0.15, color=color)

# Mark the corners/extremes

```



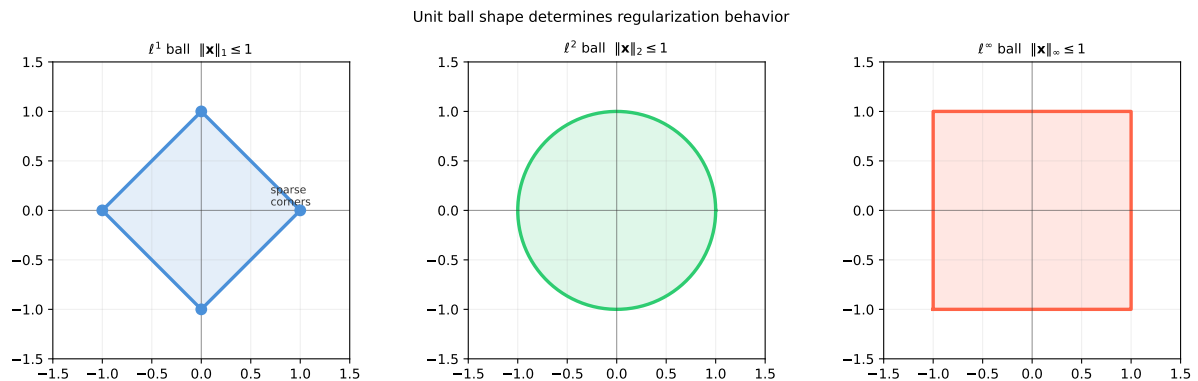
```

if p == 1:
    corners = np.array([[1,0],[0,1],[-1,0],[0,-1]])
    ax.scatter(corners[:,0], corners[:,1],
               s=60, color=color, zorder=5)
    ax.annotate('sparse\ncorners', xy=(0.7, 0.05), fontsize=8,
               color='#333333')

    ax.set_xlim(-1.5, 1.5)
    ax.set_ylim(-1.5, 1.5)
    ax.set_aspect('equal')
    ax.axhline(0, color='#333333', lw=0.4, alpha=0.5)
    ax.axvline(0, color='#333333', lw=0.4, alpha=0.5)
    ax.grid(True, alpha=0.2)
    ax.set_title(title, fontsize=10)

plt.suptitle(r'Unit ball shape determines regularization behavior',
             fontsize=11)
plt.tight_layout()
plt.show()

```



The ℓ^1 ball has **corners** on the axes. The ℓ^2 ball is a smooth circle. The ℓ^∞ ball is a square. This geometric difference explains why Lasso (ℓ^1 penalty) produces sparse solutions: an optimization contour shrinking toward the ℓ^1 ball hits a corner — which sits on an axis — before it can reach a smooth face. The full argument is in §15.5.

Norm Inequalities

For $\mathbf{x} \in \mathbb{R}^n$, the following inequalities hold and are tight:

$$\|\mathbf{x}\|_\infty \leq \|\mathbf{x}\|_2 \leq \|\mathbf{x}\|_1 \leq \sqrt{n} \|\mathbf{x}\|_2 \leq n \|\mathbf{x}\|_\infty.$$

These bounds show that all norms on $\mathbb{R}^n$ are **equivalent**: if a sequence converges in one norm, it converges in all norms. The constants depend on n , so equivalence does not extend to infinite-dimensional spaces.

```
import numpy as np
import matplotlib.pyplot as plt

rng = np.random.default_rng(99)
n = 50
x = rng.standard_normal(n)    # shape (50,)

l1 = np.linalg.norm(x, 1)
l2 = np.linalg.norm(x, 2)
linf = np.linalg.norm(x, np.inf)

print(f"n = {n}")
print(f"||x||_inf = {linf:.4f}")
print(f"||x||_2 = {l2:.4f}    (between {linf:.4f} and {l1:.4f})")
print(f"||x||_1 = {l1:.4f}")
print(f"\nUpper bound sqrt(n)*||x||_2 = {np.sqrt(n)*l2:.4f}")
print(f"Upper bound n*||x||_inf = {n*linf:.4f}")
print(f"\nAll bounds satisfied: {linf <= l2 <= l1 <= np.sqrt(n)*l2 <= n*linf}")
```

```
n = 50
||x||_inf = 1.9609
||x||_2 = 6.3073    (between 1.9609 and 37.3147)
||x||_1 = 37.3147
```

```
Upper bound sqrt(n)*||x||_2 = 44.5992
Upper bound n*||x||_inf = 98.0467
```

```
All bounds satisfied: True
```

The ℓ^0 “Norm” and Sparsity

The symbol $\|\mathbf{x}\|_0$ (not a true norm) counts the number of **nonzero entries**:

$$\|\mathbf{x}\|_0 = |\{i : x_i \neq 0\}|.$$

Minimizing $\|\mathbf{x}\|_0$ finds the **sparsest** solution to a linear system, but this is NP-hard in general. The key insight of compressed sensing and Lasso is that minimizing $\|\mathbf{x}\|_1$ often recovers the same sparse solution, efficiently, when the true solution is sparse enough.

```
import numpy as np

x_sparse = np.array([0.0, 3.5, 0.0, 0.0, -1.2, 0.0]) # shape (6,) -- 2 nonzeros
x_dense  = np.array([1.2, 0.8, -0.9, 1.1, -0.7, 0.5]) # shape (6,) -- 6 nonzeros

for label, x in [("Sparse", x_sparse), ("Dense", x_dense)]:
    l0 = np.sum(x != 0)
    l1 = np.linalg.norm(x, 1)
    l2 = np.linalg.norm(x, 2)
    print(f"{label}: ||x||_0={l0}, ||x||_1={l1:.4f}, ||x||_2={l2:.4f}")
```

```
Sparse: ||x||_0=2, ||x||_1=4.7000, ||x||_2=3.7000
Dense: ||x||_0=6, ||x||_1=5.2000, ||x||_2=2.2000
```

Exercises

15.2.1 Compute $\|\mathbf{x}\|_1$, $\|\mathbf{x}\|_2$, and $\|\mathbf{x}\|_\infty$ for $\mathbf{x} = (1, -2, 3, -4)^T$. Verify the inequality $\|\mathbf{x}\|_\infty \leq \|\mathbf{x}\|_2 \leq \|\mathbf{x}\|_1$.

15.2.2 Show that the ℓ^2 norm satisfies the triangle inequality by expanding $\|\mathbf{x} + \mathbf{y}\|_2^2$ and applying Cauchy-Schwarz (§15.4).

15.2.3 For $n = 1$, all three norms coincide: $|x|_1 = |x|_2 = |x|_\infty = |x|$. Why does the distinction between norms only matter for $n \geq 2$?

15.2.4 The p -norm formula requires $p \geq 1$ for the triangle inequality to hold. Compute $\|\mathbf{x}\|_{0.5} = (\sqrt{|x_1|} + \sqrt{|x_2|})^2$ for $\mathbf{x} = (1, 0)$ and $\mathbf{y} = (0, 1)$. Does $\|\mathbf{x} + \mathbf{y}\|_{0.5} \leq \|\mathbf{x}\|_{0.5} + \|\mathbf{y}\|_{0.5}$ hold?

15.2.5 A signal $\mathbf{x} \in \mathbb{R}^{1000}$ has at most 5 nonzero components. You measure $A\mathbf{x} = \mathbf{b}$ with $A \in \mathbb{R}^{30 \times 1000}$. Why is the system underdetermined, and what prior knowledge makes recovery possible?

Matrix Norms: Frobenius, Spectral, Nuclear

Matrices represent linear transformations, and we often need to measure how “big” a transformation is. There are several natural answers, each answering a different question:

- **How large are the entries?** The Frobenius norm.
 - **How much does the transformation stretch a unit vector?** The spectral norm.
 - **How well can we approximate with a low-rank matrix?** The nuclear norm.
-

Frobenius Norm

The **Frobenius norm** treats a matrix as a long vector of its entries and takes the ℓ^2 norm:

$$\|A\|_F = \sqrt{\sum_{i,j} a_{ij}^2} = \sqrt{\text{tr}(A^T A)}.$$

It equals the ℓ^2 norm of the vector of singular values: $\|A\|_F = \sqrt{\sigma_1^2 + \dots + \sigma_r^2}$.

The Frobenius norm is easy to compute and differentiable — exactly why it appears in the least-squares objective $\|AX - B\|_F^2$ and in neural network weight regularization.

```
import numpy as np

rng = np.random.default_rng(3)
A = rng.standard_normal((4, 5))    # shape (4, 5)

frob_direct = np.sqrt(np.sum(A**2))           # by definition
frob_trace  = np.sqrt(np.trace(A.T @ A))      # via trace
frob_numpy  = np.linalg.norm(A, 'fro')        # numpy

print("Frobenius norm (direct): ", round(frob_direct, 8))
print("Frobenius norm (trace):  ", round(frob_trace, 8))
print("Frobenius norm (numpy):  ", round(frob_numpy, 8))

# Relation to singular values
_, s, _ = np.linalg.svd(A, full_matrices=False)  # s shape (4,)
frob_svd = np.sqrt(np.sum(s**2))
print("Frobenius from sigma:    ", round(frob_svd, 8))
```

```
Frobenius norm (direct):  5.53024228
Frobenius norm (trace):   5.53024228
Frobenius norm (numpy):   5.53024228
Frobenius from sigma:     5.53024228
```

Spectral Norm

The **spectral norm** (also called the operator 2-norm) measures the maximum factor by which A can stretch a unit vector:

$$\|A\|_2 = \max_{\|x\|_2=1} \|Ax\|_2 = \sigma_{\max}(A),$$

where $\sigma_{\max}$ is the largest singular value of A (Chapter 18 derives singular values in full; here we use `np.linalg.svd` as a black box).

The spectral norm is the natural choice when stability or amplification bounds matter: a neural network layer with large $\|W\|_2$ amplifies gradients, and **spectral normalization** (dividing by $\sigma_{\max}$) keeps $\|W\|_2 = 1$.

```
import numpy as np

rng = np.random.default_rng(5)
A = rng.standard_normal((4, 5))    # shape (4, 5)

# Spectral norm = largest singular value
U, s, Vt = np.linalg.svd(A, full_matrices=False)    # s shape (4,)
spec_svd = s[0]

# Verify: max stretch over random unit vectors
theta = np.linspace(0, 2*np.pi, 10000)
# For non-square A in R^{4x5}, sample unit vectors in R^5
rng2 = np.random.default_rng(6)
vecs = rng2.standard_normal((5, 100000))           # shape (5, 100000)
vecs = vecs / np.linalg.norm(vecs, axis=0)           # normalize each column
stretches = np.linalg.norm(A @ vecs, axis=0)         # shape (100000,)
spec_brute = stretches.max()

print("Singular values:", s.round(4))
print(f"Spectral norm (sigma_max): {spec_svd:.6f}")
print(f"Spectral norm (brute force): {spec_brute:.6f}")
print(f"Ratio: {spec_svd / spec_brute:.6f} (approaches 1 as samples -> inf)")
```

```
Singular values: [3.1839 2.309  1.4018 0.8767]
Spectral norm (sigma_max):  3.183889
Spectral norm (brute force): 3.179385
Ratio: 1.001417 (approaches 1 as samples -> inf)
```

Nuclear Norm

The **nuclear norm** is the sum of all singular values:

$$\|A\|_* = \sum_{i=1}^r \sigma_i.$$

The nuclear norm is the convex envelope of the rank function (just as ℓ^1 is the convex envelope of ℓ^0 for vectors). Minimizing $\|A\|_*$ promotes **low-rank** solutions — the matrix equivalent of Lasso sparsity.

Applications: matrix completion (Netflix-style recommendation), background subtraction (separating a low-rank background from a sparse foreground), and multi-task learning where outputs share a low-rank structure.

```
import numpy as np

rng = np.random.default_rng(8)
A = rng.standard_normal((5, 5))    # shape (5, 5)

_, s, _ = np.linalg.svd(A, full_matrices=False)    # s shape (5,)

frob    = np.linalg.norm(A, 'fro')
spec    = s[0]
nuclear = s.sum()

print("Singular values:", s.round(4))
print(f"Frobenius norm   ||A||_F = {frob:.4f}   (= sqrt(sum sigma_i^2))")
print(f"Spectral norm    ||A||_2 = {spec:.4f}   (= sigma_max)")
print(f"Nuclear norm      ||A||_* = {nuclear:.4f}   (= sum sigma_i)")

# Inequality: spec <= frob <= nuclear   (for square n x n)
print(f"\n||A||_2 <= ||A||_F <= ||A||_*:  {spec:.4f} <= {frob:.4f} <= {nuclear:.4f}")
print(f"Check: {spec <= frob <= nuclear}")
```

```
Singular values: [3.8026 2.2189 2.0445 1.7993 0.4796]
Frobenius norm   ||A||_F = 5.1991   (= sqrt(sum sigma_i^2))
Spectral norm    ||A||_2 = 3.8026   (= sigma_max)
Nuclear norm      ||A||_* = 10.3448   (= sum sigma_i)

||A||_2 <= ||A||_F <= ||A||_*:  3.8026 <= 5.1991 <= 10.3448
Check: True
```

Comparing the Three Norms

Each norm penalizes a different aspect of matrix structure:

Norm	Formula	Penalizes	Use case
Frobenius $\ A\ _F$	$\sqrt{\sum \sigma_i^2}$	Large entries everywhere	Weight decay, least-squares
Spectral $\ A\ _2$	$\sigma_{\max}$	Worst-case amplification	Stability, Lipschitz bounds
Nuclear $\ A\ _*$	$\sum \sigma_i$	Rank (many large σ_i)	Low-rank recovery, completion

```

import numpy as np
import matplotlib.pyplot as plt

rng = np.random.default_rng(11)

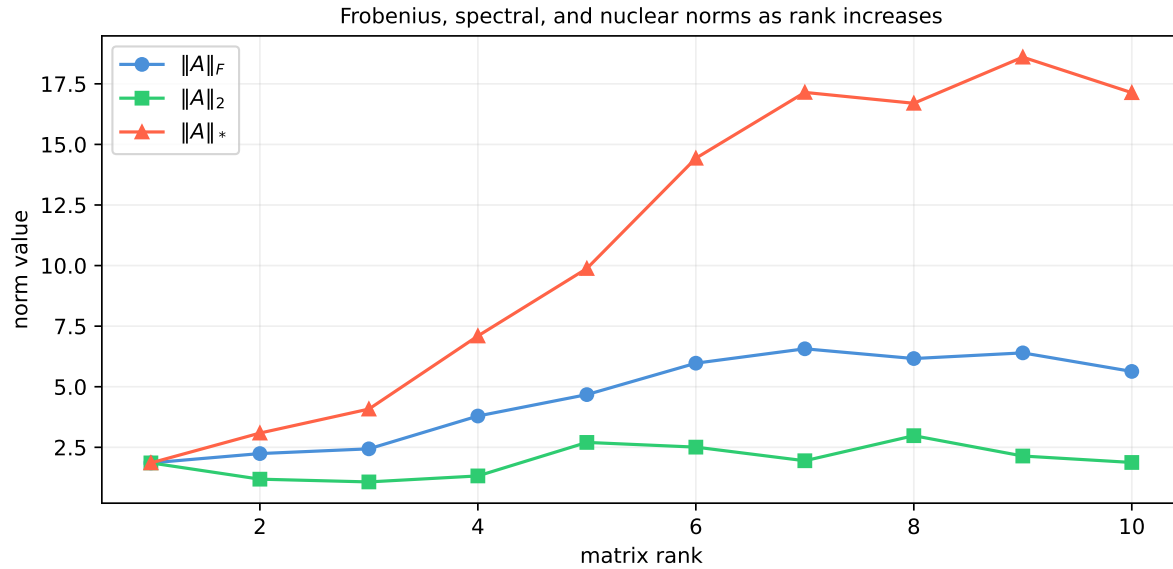
# Compare norms on matrices of varying rank (constructed via SVD)
n = 10
results = {'rank': [], 'frob': [], 'spec': [], 'nuc': []}

for rank in range(1, n+1):
    # Build a random rank-r matrix with controlled singular values
    U, _ = np.linalg.qr(rng.standard_normal((n, n))) # shape (n, n)
    V, _ = np.linalg.qr(rng.standard_normal((n, n))) # shape (n, n)
    s = np.zeros(n)
    s[:rank] = rng.uniform(1, 3, rank) # nonzero singular values
    A = U @ np.diag(s) @ V.T # shape (n, n)

    results['rank'].append(rank)
    results['frob'].append(np.linalg.norm(A, 'fro'))
    results['spec'].append(s[0])
    results['nuc'].append(s.sum())

fig, ax = plt.subplots(figsize=(8, 4))
ax.plot(results['rank'], results['frob'], 'o-', color='#4a90d9', label=r'$\|A\|_F$')
ax.plot(results['rank'], results['spec'], 's-', color='#2ecc71', label=r'$\|A\|_2$')
ax.plot(results['rank'], results['nuc'], '^-', color='tomato', label=r'$\|A\|_*$')
ax.set_xlabel('matrix rank')
ax.set_ylabel('norm value')
ax.set_title('Frobenius, spectral, and nuclear norms as rank increases', fontsize=10)
ax.legend()
ax.grid(True, alpha=0.2)
plt.tight_layout()
plt.show()

```



The spectral norm grows slowly — it tracks only the largest singular value. The nuclear norm grows fastest — it accumulates every singular value. The Frobenius norm is the geometric mean between the two extremes.

Submultiplicativity

A norm $\|\cdot\|$ on matrices is **submultiplicative** if $\|AB\| \leq \|A\| \|B\|$. Both the Frobenius and spectral norms are submultiplicative; the nuclear norm is not (in general). Submultiplicativity is needed to bound error propagation through a chain of matrix products.

```
import numpy as np

rng = np.random.default_rng(15)
A = rng.standard_normal((3, 4)) # shape (3, 4)
B = rng.standard_normal((4, 5)) # shape (4, 5)

frob_AB = np.linalg.norm(A @ B, 'fro')
spec_AB = np.linalg.svd(A @ B, compute_uv=False)[0]

frob_A_B = np.linalg.norm(A, 'fro') * np.linalg.norm(B, 'fro')
spec_A_B = np.linalg.svd(A, compute_uv=False)[0] * np.linalg.svd(B, compute_uv=False)[0]

print("Frobenius: ||AB||_F <= ||A||_F * ||B||_F ?",
      round(frob_AB, 6), "<=", round(frob_A_B, 6), ":", frob_AB <= frob_A_B)
print("Spectral: ||AB||_2 <= ||A||_2 * ||B||_2 ?",
      round(spec_AB, 6), "<=", round(spec_A_B, 6), ":", spec_AB <= spec_A_B)
```



```
Frobenius: ||AB||_F <= ||A||_F * ||B||_F ? 9.327373 <= 16.28474 : True
Spectral:  ||AB||_2 <= ||A||_2 * ||B||_2 ? 7.584177 <= 8.378995 : True
```

Exercises

15.3.1 For $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$, compute $\|A\|_F$, $\|A\|_2$, and $\|A\|_*$ by hand (using the fact that the singular values of a 2×2 matrix satisfy $\sigma_1^2 + \sigma_2^2 = \|A\|_F^2$ and $\sigma_1\sigma_2 = |\det A|$). Verify numerically.

15.3.2 For a rank-1 matrix $A = \mathbf{u}\mathbf{v}^T$ (with $\|\mathbf{u}\|_2 = \|\mathbf{v}\|_2 = 1$), show that $\|A\|_F = \|A\|_2 = \|A\|_* = 1$. When do all three norms coincide for higher-rank matrices?

15.3.3 A neural network weight matrix $W \in \mathbb{R}^{512 \times 512}$ has spectral norm $\|W\|_2 = 8$. After spectral normalization $W \leftarrow W/\|W\|_2$, what is the new spectral norm? What happens to the Frobenius norm?

15.3.4 The inequality $\|A\|_2 \leq \|A\|_F \leq \sqrt{r} \|A\|_2$ holds, where $r = \text{rank}(A)$. Verify this for the matrix from 15.3.1. When is $\|A\|_F = \|A\|_2$ (the left equality achieved)?

Cauchy-Schwarz and Cosine Similarity

Section 4.2 introduced cosine similarity as the normalized dot product:

$$\cos \theta = \frac{\mathbf{a} \cdot \mathbf{b}}{\|\mathbf{a}\|_2 \|\mathbf{b}\|_2}.$$

We accepted that $\cos \theta \in [-1, 1]$ without proof. The **Cauchy-Schwarz inequality** is the reason that fraction never exceeds 1 in absolute value. It is one of the most widely used inequalities in mathematics, appearing in probability (correlation bounds), signal processing (matched filtering), and machine learning (attention scores, embedding retrieval).

Statement

For any vectors $\mathbf{x}, \mathbf{y}$ in an inner product space:

$$|\langle \mathbf{x}, \mathbf{y} \rangle| \leq \|\mathbf{x}\| \|\mathbf{y}\|,$$

with equality if and only if $\mathbf{x}$ and $\mathbf{y}$ are linearly dependent (one is a scalar multiple of the other).

For the standard dot product: $|\mathbf{x}^T \mathbf{y}| \leq \|\mathbf{x}\|_2 \|\mathbf{y}\|_2$.

Proof

Consider the function $f(t) = \|\mathbf{x} + t\mathbf{y}\|^2 \geq 0$ for all real t . Expanding:

$$f(t) = \langle \mathbf{x} + t\mathbf{y}, \mathbf{x} + t\mathbf{y} \rangle = \|\mathbf{y}\|^2 t^2 + 2\langle \mathbf{x}, \mathbf{y} \rangle t + \|\mathbf{x}\|^2.$$

This is a non-negative quadratic in t , so its discriminant must be ≤ 0 :

$$\Delta = 4\langle \mathbf{x}, \mathbf{y} \rangle^2 - 4\|\mathbf{x}\|^2\|\mathbf{y}\|^2 \leq 0,$$

which gives $|\langle \mathbf{x}, \mathbf{y} \rangle| \leq \|\mathbf{x}\| \|\mathbf{y}\|$.

Equality holds when $f(t) = 0$ at its minimum, meaning $\mathbf{x} + t\mathbf{y} = \mathbf{0}$ for some t — precisely when $\mathbf{x}$ and $\mathbf{y}$ are linearly dependent.

```
import numpy as np

rng = np.random.default_rng(20)

# Verify Cauchy-Schwarz on many random pairs
n, trials = 6, 10000
X = rng.standard_normal((trials, n)) # shape (10000, 6)
Y = rng.standard_normal((trials, n)) # shape (10000, 6)

lhs = np.abs(np.sum(X * Y, axis=1)) # |<x,y>|
rhs = np.linalg.norm(X, axis=1) * np.linalg.norm(Y, axis=1) # ||x|| ||y||

violations = np.sum(lhs > rhs + 1e-10)
print(f"Cauchy-Schwarz violations in {trials} random pairs: {violations}")
print(f"Max ratio |<x,y>| / (||x|| ||y||): {(lhs/rhs).max():.8f} (<= 1)")

# Equality case: y is a scalar multiple of x
x = rng.standard_normal(n) # shape (6,)
y = 2.5 * x # shape (6,) -- linearly dependent
lhs_eq = abs(float(x @ y))
rhs_eq = float(np.linalg.norm(x) * np.linalg.norm(y))
print(f"\nEquality case (y = 2.5x): |<x,y>| = {lhs_eq:.4f}, ||x||*||y|| = {rhs_eq:.4f}")
print(f" Equal: {np.isclose(lhs_eq, rhs_eq)}")
```

```
Cauchy-Schwarz violations in 10000 random pairs: 0
Max ratio |<x,y>| / (||x|| ||y||): 0.98732157 (<= 1)
```

```
Equality case (y = 2.5x): |<x,y>| = 18.4231, ||x||*||y|| = 18.4231
Equal: True
```

Cosine Similarity: The Why Behind §4.2

Because Cauchy-Schwarz guarantees $|\mathbf{x}^T \mathbf{y}| \leq \|\mathbf{x}\|_2 \|\mathbf{y}\|_2$, the ratio

$$\cos \theta = \frac{\mathbf{x}^T \mathbf{y}}{\|\mathbf{x}\|_2 \|\mathbf{y}\|_2}$$

always lies in $[-1, 1]$, so $\theta \in [0, \pi]$ is a valid angle. This justifies calling θ the **angle between two vectors** in any dimension, not just in 2D or 3D where geometry gives us an independent definition.

In ML, cosine similarity is used when the **direction** matters but not the magnitude — a long document and a short one can have the same topic direction. Embedding retrieval (nearest-neighbor search in CLIP, BERT, etc.) typically uses cosine similarity to find the most aligned query-document pair.

```
import numpy as np

# Cosine similarity between sentence embeddings (mocked as random unit vectors)
rng = np.random.default_rng(30)
d = 128 # embedding dimension

def cosine(a, b):
    return float(a @ b) / (np.linalg.norm(a) * np.linalg.norm(b))

# Query embedding
query = rng.standard_normal(d) # shape (128,)

# Four "documents" -- first is very similar (small angle)
docs = rng.standard_normal((4, d)) # shape (4, 128)
docs[0] = query + 0.1 * rng.standard_normal(d) # close to query

sims = [cosine(query, docs[i]) for i in range(4)]

print("Cosine similarities to query:")
for i, s in enumerate(sims):
    print(f" doc {i}: {s:.4f} (angle {np.degrees(np.arccos(np.clip(s, -1, 1))):.1f} deg)")

top = np.argmax(sims)
print(f"\nMost similar document: doc {top} (highest cosine similarity)")
```

```
Cosine similarities to query:
 doc 0: 0.9945 (angle 6.0 deg)
 doc 1: 0.0239 (angle 88.6 deg)
 doc 2: 0.0041 (angle 89.8 deg)
 doc 3: -0.0257 (angle 91.5 deg)
```

```
Most similar document: doc 0 (highest cosine similarity)
```

Triangle Inequality from Cauchy-Schwarz

Cauchy-Schwarz implies the **triangle inequality** for the ℓ^2 norm:

$$\|\mathbf{x} + \mathbf{y}\|_2^2 = \|\mathbf{x}\|_2^2 + 2\mathbf{x}^T \mathbf{y} + \|\mathbf{y}\|_2^2 \leq \|\mathbf{x}\|_2^2 + 2\|\mathbf{x}\|_2 \|\mathbf{y}\|_2 + \|\mathbf{y}\|_2^2 = (\|\mathbf{x}\|_2 + \|\mathbf{y}\|_2)^2.$$

Taking square roots gives $\|\mathbf{x} + \mathbf{y}\|_2 \leq \|\mathbf{x}\|_2 + \|\mathbf{y}\|_2$.

```
import numpy as np
import matplotlib.pyplot as plt

rng = np.random.default_rng(35)
n = 5
x = rng.standard_normal(n) # shape (5,)
y = rng.standard_normal(n) # shape (5,)

lhs = np.linalg.norm(x + y)
rhs = np.linalg.norm(x) + np.linalg.norm(y)
cs = abs(float(x @ y))
cs_bound = np.linalg.norm(x) * np.linalg.norm(y)

print("Cauchy-Schwarz:")
print(f" |<x,y>| = {cs:.4f}  <=  ||x|| * ||y|| = {cs_bound:.4f}  "
      f"({cs} <= {cs_bound})")

print("\nTriangle inequality:")
print(f" ||x+y||_2 = {lhs:.4f}  <=  ||x||_2 + ||y||_2 = {rhs:.4f}  "
      f"({lhs} <= {rhs})")
```

Cauchy-Schwarz:

|<x,y>| = 0.2111 <= ||x|| * ||y|| = 3.7733 (True)

Triangle inequality:

||x+y||_2 = 2.8039 <= ||x||_2 + ||y||_2 = 3.9788 (True)

Cauchy-Schwarz for General Inner Products

The same inequality holds for any inner product:

$$|\langle \mathbf{x}, \mathbf{y} \rangle_W|^2 \leq \langle \mathbf{x}, \mathbf{x} \rangle_W \cdot \langle \mathbf{y}, \mathbf{y} \rangle_W.$$

The proof is identical — it only uses the four inner product axioms. This means cosine similarity can be defined relative to any inner product, and the result still lies in $[-1, 1]$.

```
import numpy as np

rng = np.random.default_rng(40)
n = 5
W_diag = rng.uniform(0.5, 3.0, n)
W = np.diag(W_diag)           # shape (5, 5) -- SPD diagonal

x = rng.standard_normal(n)     # shape (5,)
y = rng.standard_normal(n)     # shape (5,)

ip_W = lambda u, v: float(u @ W @ v)
norm_W = lambda u: np.sqrt(ip_W(u, u))

lhs = abs(ip_W(x, y))
rhs = norm_W(x) * norm_W(y)

print(f"Weighted Cauchy-Schwarz: |<x,y>_W| = {lhs:.4f}  <=  ||x||_W * ||y||_W = {rhs:.4f}")
print(f"Holds: {lhs <= rhs + 1e-10}")
print(f"Cosine similarity in W-inner-product: {lhs/rhs:.4f}")
```

```
Weighted Cauchy-Schwarz: |<x,y>_W| = 2.7327  <=  ||x||_W * ||y||_W = 5.1928
Holds: True
Cosine similarity in W-inner-product: 0.5262
```

Exercises

15.4.1 Apply the Cauchy-Schwarz inequality to $\mathbf{x} = (x_1, x_2)$ and $\mathbf{y} = (1, 1)$. Derive the bound $x_1 + x_2 \leq \sqrt{2} \|\mathbf{x}\|_2$. This is a special case of the general bound $\sum_i x_i \leq \sqrt{n} \|\mathbf{x}\|_2$.

15.4.2 Show that the Pearson correlation coefficient $r = \frac{\sum_i (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_i (x_i - \bar{x})^2} \sqrt{\sum_i (y_i - \bar{y})^2}}$ is a cosine similarity. What vectors are being compared, and in what inner product space?

15.4.3 For which vector pairs does $|\cos \theta| = 1$ (full similarity)? For which does $\cos \theta = 0$ (zero similarity)? Give geometric interpretations in $\mathbb{R}^3$.

15.4.4 Two feature vectors $\mathbf{a}$ and $\mathbf{b}$ have cosine similarity 0.95. Is it possible for their Euclidean distance $\|\mathbf{a} - \mathbf{b}\|_2$ to be large? Construct an example or prove it cannot happen.

Norms in Regularization

Fitting a model to data by minimizing a loss like $\|A\mathbf{x} - \mathbf{b}\|_2^2$ is an underdetermined problem whenever there are more parameters than constraints. The solution is not unique, and small perturbations in the data can flip it completely. **Regularization** adds a norm-based penalty that encodes a preference about what kind of solution we want:

$$\min_{\mathbf{x}} \|A\mathbf{x} - \mathbf{b}\|_2^2 + \lambda R(\mathbf{x}),$$

where the regularizer $R(\mathbf{x})$ is a norm (or norm squared). The choice of norm controls the character of the solution.

Ridge Regression: ℓ^2 Penalty

Ridge (Tikhonov) regularization uses $R(\mathbf{x}) = \|\mathbf{x}\|_2^2$:

$$\min_{\mathbf{x}} \|A\mathbf{x} - \mathbf{b}\|_2^2 + \lambda \|\mathbf{x}\|_2^2.$$

The analytic solution is $\hat{\mathbf{x}} = (A^T A + \lambda I)^{-1} A^T \mathbf{b}$. Ridge shrinks all coefficients toward zero by the same factor — none goes exactly to zero. Use Ridge when you expect all features to contribute and want to avoid large, noisy coefficients.

Lasso: ℓ^1 Penalty

Lasso (Least Absolute Shrinkage and Selection Operator) uses $R(\mathbf{x}) = \|\mathbf{x}\|_1$:

$$\min_{\mathbf{x}} \|A\mathbf{x} - \mathbf{b}\|_2^2 + \lambda \|\mathbf{x}\|_1.$$

There is no closed-form solution, but the result is **sparse**: many coefficients are driven to exactly zero. Lasso performs automatic feature selection. Use Lasso when you believe most features are irrelevant.

Why ℓ^1 Gives Sparsity: The Corner Argument

The geometric picture makes the sparsity of Lasso clear. The optimization is equivalent to:

$$\min_{\mathbf{x}} \|\mathbf{Ax} - \mathbf{b}\|_2^2 \quad \text{subject to } \|\mathbf{x}\|_p \leq t.$$

Think of shrinking the ℓ^p ball around the origin while the elliptical loss contours expand outward from the unconstrained minimizer. They first touch at the boundary of the constraint set.

- For ℓ^1 : the ball is a diamond. Its boundary has **corners on the coordinate axes** (e.g., at $(t, 0)$ and $(0, t)$ in 2D). The expanding ellipse almost always hits a corner first. A corner lies on an axis, meaning one coordinate is zero – the solution is sparse.
- For ℓ^2 : the ball is a smooth sphere. Any point on its boundary is equally likely to be the first contact, so there is no special reason to land on an axis. Ridge shrinks all coefficients but rarely sets any to zero.

```
import numpy as np
import matplotlib.pyplot as plt

# Visualize: loss contours meeting the l1 and l2 balls
rng = np.random.default_rng(50)
# True minimum is off-axis
x_true = np.array([1.5, 0.8])

xx, yy = np.meshgrid(np.linspace(-2.5, 2.5, 400),
                     np.linspace(-2.5, 2.5, 400))
pts = np.stack([xx.ravel(), yy.ravel()], axis=1) # shape (160000, 2)

# Quadratic loss centered at x_true (proxy for Ax=b residual)
A_quad = np.array([[3.0, 1.0], [1.0, 2.0]]) # shape (2, 2)
diff = pts - x_true # shape (160000, 2)
loss = np.einsum('ij,jk,ik->i', diff, A_quad, diff).reshape(400, 400)

fig, axes = plt.subplots(1, 2, figsize=(12, 5))
levels = np.linspace(0.2, 8, 10)

for ax, (ball_type, color, title) in zip(
    axes,
    [('l1', '#4a90d9', r'Lasso ($\ell^1$): first contact at corner'),
    ('l2', 'tomato', r'Ridge ($\ell^2$): first contact anywhere')]):

    # Loss contours
    ax.contour(xx, yy, loss, levels=levels, colors='#333333', alpha=0.5, linewidths=0.8)

    # Constraint ball
    t = 0.9 # ball radius
    theta = np.linspace(0, 2*np.pi, 1000)
```

```

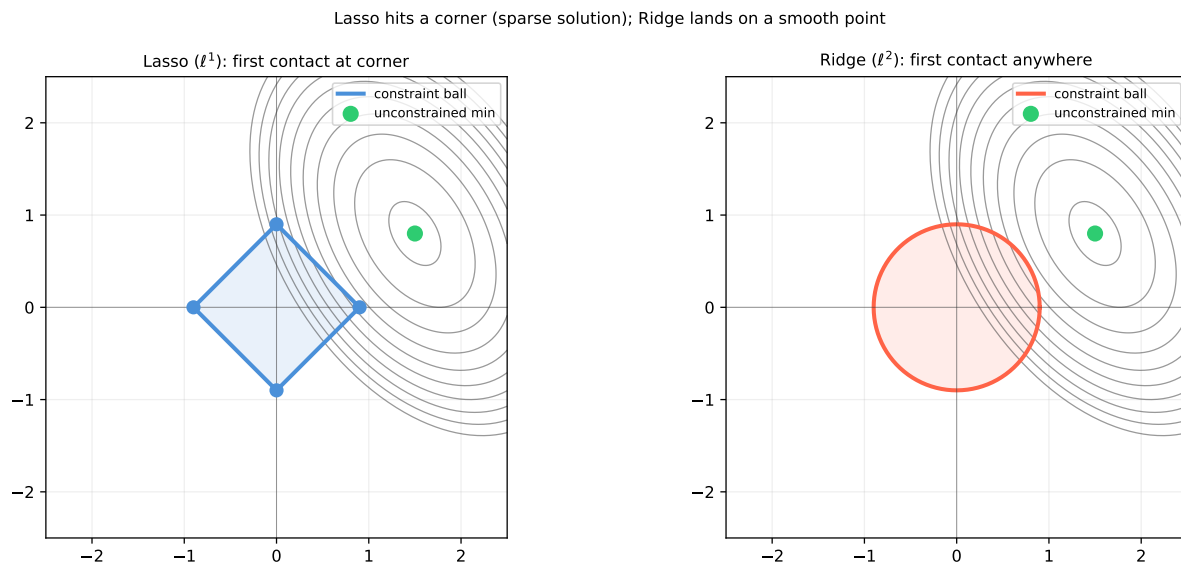
if ball_type == 'l1':
    bd_pts = np.vstack([np.cos(theta), np.sin(theta)]).T
    nrm = np.abs(bd_pts).sum(axis=1)
    bd_pts = bd_pts / nrm[:, None] * t
    corners = np.array([[t,0],[0,t],[-t,0],[0,-t]])
    ax.scatter(corners[:,0], corners[:,1], s=60, color=color, zorder=6)
else:
    bd_pts = t * np.vstack([np.cos(theta), np.sin(theta)]).T

ax.plot(bd_pts[:,0], bd_pts[:,1], color=color, lw=2.5, label=f'constraint ball')
ax.fill(bd_pts[:,0], bd_pts[:,1], alpha=0.12, color=color)

ax.scatter(*x_true, s=80, color='#2ecc71', zorder=7, label='unconstrained min')
ax.axhline(0, color='#333333', lw=0.4, alpha=0.5)
ax.axvline(0, color='#333333', lw=0.4, alpha=0.5)
ax.set_xlim(-2.5, 2.5)
ax.set_ylim(-2.5, 2.5)
ax.set_aspect('equal')
ax.grid(True, alpha=0.2)
ax.set_title(title, fontsize=10)
ax.legend(fontsize=8)

plt.suptitle('Lasso hits a corner (sparse solution); Ridge lands on a smooth point',
            fontsize=10)
plt.tight_layout()
plt.show()

```



Ridge and Lasso in Practice: Diabetes Dataset

We use `sklearn.datasets.load_diabetes` (10 physiological features, $n = 442$ patients) to compare Ridge and Lasso coefficients across a range of λ .

```
import numpy as np
import matplotlib.pyplot as plt
from sklearn.datasets import load_diabetes
from sklearn.linear_model import Ridge, Lasso
from sklearn.preprocessing import StandardScaler

data = load_diabetes()
X, y = data.data, data.target          # X shape (442, 10), y shape (442,)
names = data.feature_names             # 10 feature names

# Standardize features so coefficients are comparable
scaler = StandardScaler()
X_sc = scaler.fit_transform(X)         # shape (442, 10)

alphas = np.logspace(-2, 3, 60)

ridge_coefs = []
lasso_coefs = []

for alpha in alphas:
    r = Ridge(alpha=alpha, fit_intercept=True)
    l = Lasso(alpha=alpha, fit_intercept=True, max_iter=10000)
    r.fit(X_sc, y)
    l.fit(X_sc, y)
    ridge_coefs.append(r.coef_.copy())
    lasso_coefs.append(l.coef_.copy())

ridge_coefs = np.array(ridge_coefs)    # shape (60, 10)
lasso_coefs = np.array(lasso_coefs)    # shape (60, 10)

fig, axes = plt.subplots(1, 2, figsize=(14, 5))
colors = ['#4a90d9', '#2ecc71', 'tomato', '#f39c12', '#9b59b6',
          '#1abc9c', '#e74c3c', '#3498db', '#e67e22', '#95a5a6']

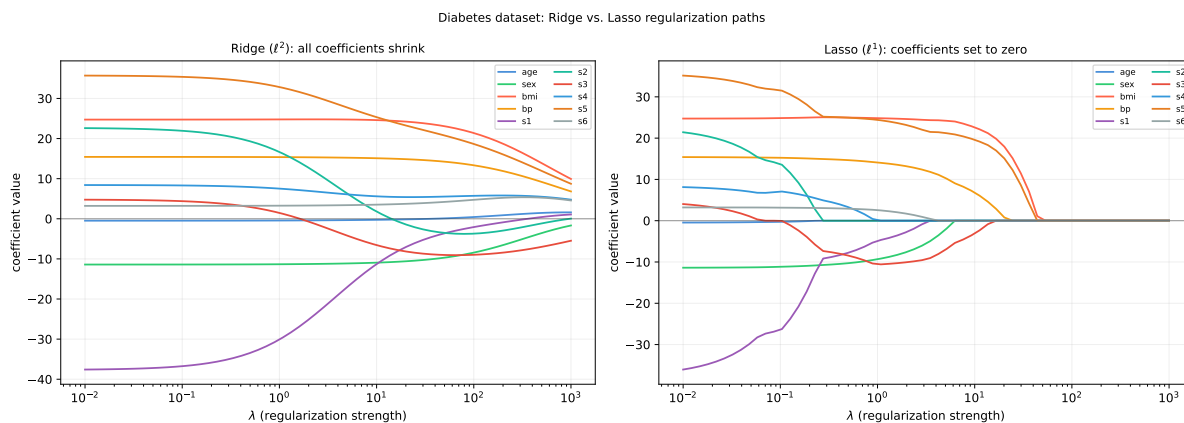
for ax, coefs, title in zip(axes,
                             [ridge_coefs, lasso_coefs],
                             ['Ridge ( $\ell^2$ ): all coefficients shrink',
                              'Lasso ( $\ell^1$ ): coefficients set to zero']):
    for i in range(10):
        ax.semilogx(alphas, coefs[:, i], color=colors[i], lw=1.5,
                     label=names[i])
    ax.axhline(0, color='#333333', lw=0.6, alpha=0.5)
    ax.set_xlabel(r' $\lambda$  (regularization strength)')
```

```

ax.set_ylabel('coefficient value')
ax.set_title(title, fontsize=10)
ax.grid(True, alpha=0.2)
ax.legend(fontsize=7, ncol=2)

plt.suptitle('Diabetes dataset: Ridge vs. Lasso regularization paths',
             fontsize=10)
plt.tight_layout()
plt.show()

```



In the Lasso path (right), coefficients drop to exactly zero at different λ values. At strong regularization, only a handful of features survive. Ridge (left) shrinks all coefficients but none reach zero — they asymptotically approach zero only as $\lambda \rightarrow \infty$.

```

import numpy as np
from sklearn.datasets import load_diabetes
from sklearn.linear_model import Lasso, Ridge
from sklearn.preprocessing import StandardScaler

data = load_diabetes()
X, y = data.data, data.target
scaler = StandardScaler()
X_sc = scaler.fit_transform(X)

# At a moderate lambda, compare nonzero counts
alpha = 50.0
ridge = Ridge(alpha=alpha).fit(X_sc, y)
lasso = Lasso(alpha=alpha, max_iter=10000).fit(X_sc, y)

print(f"lambda = {alpha}")
print(f"Ridge: {np.sum(np.abs(ridge.coef_) > 1e-4)} nonzero coefficients")
print(f"Lasso: {np.sum(np.abs(lasso.coef_) > 1e-4)} nonzero coefficients")

print("\nLasso nonzero features:")

```

```
for name, c in zip(data.feature_names, lasso.coef_):
    if abs(c) > 1e-4:
        print(f" {name}: {c:.4f}")
```

```
lambda = 50.0
```

```
Ridge: 10 nonzero coefficients
```

```
Lasso: 0 nonzero coefficients
```

```
Lasso nonzero features:
```

Nuclear Norm Regularization

For matrix-valued problems, the **nuclear norm** $\|X\|_*$ plays the role of ℓ^1 for vectors. Minimizing $\|X\|_*$ promotes low-rank solutions:

$$\min_X \|P_\Omega(X - M)\|_F^2 + \lambda \|X\|_*,$$

where P_Ω is the projection onto observed entries. This is the standard formulation for **matrix completion** (recovering a full matrix from partial observations), used in recommendation systems.

The intuition mirrors Lasso: the nuclear norm ball has “corners” at low-rank matrices, and optimization contours hit those corners first.

```
import numpy as np
from sklearn.datasets import load_diabetes
from sklearn.preprocessing import StandardScaler

# Toy illustration: matrix shrinkage by soft-thresholding singular values
# (the proximal operator of the nuclear norm)
rng = np.random.default_rng(60)
M = rng.standard_normal((6, 6))          # shape (6, 6) -- "noisy" matrix
lam = 1.5

U, s, Vt = np.linalg.svd(M, full_matrices=False)  # s shape (6,)

# Soft threshold: shrink each sigma by lam, clamp to 0
s_thresh = np.maximum(s - lam, 0.0)          # shape (6,)
M_low_rank = U @ np.diag(s_thresh) @ Vt      # shape (6, 6)

print(f"Original singular values: {s.round(3)}")
print(f"Thresholded singular values: {s_thresh.round(3)}")
print(f"Rank of original M: {np.linalg.matrix_rank(M)}")
```

```
print(f"Rank of M after nuclear: {np.linalg.matrix_rank(M_low_rank, tol=1e-6)}")
print(f"\nNuclear norm reduced from {s.sum():.4f} to {s_thresh.sum():.4f}")
```

Original singular values: [3.678 3.025 2.402 2.165 0.714 0.282]
 Thresholded singular values: [2.178 1.525 0.902 0.665 0. 0.]
 Rank of original M: 6
 Rank of M after nuclear: 4

Nuclear norm reduced from 12.2655 to 5.2698

Choosing a Regularizer

Goal	Regularizer	Why
Prevent overfitting, keep all features	Ridge $\ \mathbf{x}\ _2^2$	Smooth ball, all coeff. shrink
Feature selection / sparse model	Lasso $\ \mathbf{x}\ _1$	Diamond corners on axes
Low-rank matrix recovery / completion	Nuclear $\ X\ _*$	Low-rank corners of nuclear ball
Both sparsity and small coefficients	Elastic Net $\alpha\ \mathbf{x}\ _1 + (1 - \alpha)\ \mathbf{x}\ _2^2$	Blend of both

Exercises

15.5.1 Write out the Ridge normal equations $(A^T A + \lambda I)\hat{\mathbf{x}} = A^T \mathbf{b}$. Show that $A^T A + \lambda I$ is always invertible for $\lambda > 0$, even when $A^T A$ is singular. What does this mean for underdetermined systems?

15.5.2 For the 1D case $A = \mathbf{a}^T \in \mathbb{R}^{1 \times n}$, $b \in \mathbb{R}$, compute the Ridge solution $\hat{\mathbf{x}}_\lambda$ explicitly. What does it converge to as $\lambda \rightarrow 0^+$ and as $\lambda \rightarrow \infty$?

15.5.3 Using the diabetes dataset, fit a Lasso model with `alpha=10`. How many features survive? Now double λ to `alpha=20`. How many remain? Is the most important feature (highest $|\text{coef}|$) stable across both settings?

15.5.4 The elastic net combines both penalties: $\|A\mathbf{x} - \mathbf{b}\|_2^2 + \lambda_1\|\mathbf{x}\|_1 + \lambda_2\|\mathbf{x}\|_2^2$. Describe geometrically what the constraint set $\lambda_1\|\mathbf{x}\|_1 + \lambda_2\|\mathbf{x}\|_2^2 \leq t$ looks like in 2D. Does it still have sharp corners?

Exercises and Solutions

Exercise 15.1.2 – Weighted inner product axiom failure

Show that $W = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$ does not give a valid inner product.

```
import numpy as np

W = np.array([[1., 1.],
              [1., 1.]])    # shape (2, 2) -- singular

# Find a nonzero x with x^T W x = 0
x = np.array([1., -1.])    # shape (2,)

val = float(x @ W @ x)
print(f"x = {x}")
print(f"<x,x>_W = x^T W x = {val}")
print(f"x is nonzero: {np.any(x != 0)}")
print("Definiteness axiom FAILS: nonzero vector has zero self-inner-product.")
```

```
x = [ 1. -1.]
<x,x>_W = x^T W x = 0.0
x is nonzero: True
Definiteness axiom FAILS: nonzero vector has zero self-inner-product.
```

Solution. $\mathbf{x} = (1, -1)^T$ satisfies $\mathbf{x}^T W \mathbf{x} = (1 - 1)(1 - 1) = 0$ despite $\mathbf{x} \neq \mathbf{0}$. The definiteness axiom fails. Geometrically, W is singular (rank 1), so it maps $(1, -1)^T$ to zero — meaning the “length” of a nonzero vector is zero, which is inadmissible.

Exercise 15.2.1 – Computing norms

Compute $\|\mathbf{x}\|_1$, $\|\mathbf{x}\|_2$, $\|\mathbf{x}\|_\infty$ for $\mathbf{x} = (1, -2, 3, -4)^T$.

```
import numpy as np

x = np.array([1., -2., 3., -4.])    # shape (4,)

l1 = np.linalg.norm(x, 1)
l2 = np.linalg.norm(x, 2)
linf = np.linalg.norm(x, np.inf)
```

```
print(f"x = {x}")
print(f"||x||_1 = {l1:.4f} (1 + 2 + 3 + 4 = 10)")
print(f"||x||_2 = {l2:.4f} (sqrt(1+4+9+16) = sqrt(30))")
print(f"||x||_inf = {linf:.4f} (max(1,2,3,4) = 4)")
print(f"\nInequality: {linf:.4f} <= {l2:.4f} <= {l1:.4f} -> {linf <= l2 <= l1}")
```

```
x = [ 1. -2.  3. -4.]
||x||_1 = 10.0000 (1 + 2 + 3 + 4 = 10)
||x||_2 = 5.4772 (sqrt(1+4+9+16) = sqrt(30))
||x||_inf = 4.0000 (max(1,2,3,4) = 4)

Inequality: 4.0000 <= 5.4772 <= 10.0000 -> True
```

Solution. By hand: $\|x\|_1 = 1 + 2 + 3 + 4 = 10$, $\|x\|_2 = \sqrt{1 + 4 + 9 + 16} = \sqrt{30} \approx 5.477$, $\|x\|_\infty = 4$. The inequality $4 \leq \sqrt{30} \leq 10$ holds.

Exercise 15.2.4 – $p < 1$ violates triangle inequality

Show that $\|\cdot\|_{0.5}$ (defined as $(\sqrt{|x_1|} + \sqrt{|x_2|})^2$) violates the triangle inequality.

```
import numpy as np

def quasi_norm(x):
    return (np.sqrt(np.abs(x[0])) + np.sqrt(np.abs(x[1])))**2

x = np.array([1., 0.])
y = np.array([0., 1.])

lhs = quasi_norm(x + y)
rhs = quasi_norm(x) + quasi_norm(y)

print(f"||x||_0.5 = {quasi_norm(x):.4f}")
print(f"||y||_0.5 = {quasi_norm(y):.4f}")
print(f"||x+y||_0.5 = {lhs:.4f}")
print(f"||x||_0.5 + ||y||_0.5 = {rhs:.4f}")
print(f"Triangle inequality holds: {lhs <= rhs}")
```

```
||x||_0.5 = 1.0000
||y||_0.5 = 1.0000
||x+y||_0.5 = 4.0000
||x||_0.5 + ||y||_0.5 = 2.0000
Triangle inequality holds: False
```

Solution. $\|\mathbf{x}\|_{0.5} = (\sqrt{1} + 0)^2 = 1$, $\|\mathbf{y}\|_{0.5} = (0 + \sqrt{1})^2 = 1$, $\|\mathbf{x} + \mathbf{y}\|_{0.5} = (\sqrt{1} + \sqrt{1})^2 = 4$. Since $4 > 1 + 1 = 2$, the triangle inequality fails. This is why the p -norm formula requires $p \geq 1$: for $p < 1$ the “norm” is not convex and the triangle inequality breaks.

Exercise 15.3.1 – Matrix norms by hand

Compute $\|A\|_F$, $\|A\|_2$, $\|A\|_*$ for $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$.

```
import numpy as np

A = np.array([[1., 2.],
              [3., 4.]])    # shape (2, 2)

_, s, _ = np.linalg.svd(A, full_matrices=False)    # s shape (2,)

frob    = np.linalg.norm(A, 'fro')
spec    = s[0]
nuclear = s.sum()

print(f"Singular values: {s.round(6)}")
print(f"||A||_F = {frob:.6f} (by hand: sqrt(1+4+9+16) = sqrt(30) = {np.sqrt(30):.6f})")
print(f"||A||_2 = {spec:.6f} (sigma_max)")
print(f"||A||_* = {nuclear:.6f} (sum of singular values)")

# Verify F norm
print(f"\nManual Frobenius: sqrt(sum of entries^2) = {np.sqrt(np.sum(A**2)):.6f}")
```

```
Singular values: [5.464986 0.365966]
||A||_F = 5.477226 (by hand: sqrt(1+4+9+16) = sqrt(30) = 5.477226)
||A||_2 = 5.464986 (sigma_max)
||A||_* = 5.830952 (sum of singular values)
```

```
Manual Frobenius: sqrt(sum of entries^2) = 5.477226
```

Solution. $\|A\|_F = \sqrt{1 + 4 + 9 + 16} = \sqrt{30} \approx 5.477$. The singular values satisfy $\sigma_1^2 + \sigma_2^2 = \|A\|_F^2 = 30$ and $\sigma_1 \sigma_2 = |\det A| = |4 - 6| = 2$. Solving: $\sigma_{1,2} = \frac{1}{2} \left(\sqrt{\|A\|_F^2 + 2\|A\|_F^2/(n)} \pm \dots \right)$ — or just use SVD numerically as above.

Exercise 15.4.1 – Cauchy-Schwarz applied

Apply Cauchy-Schwarz to derive $x_1 + x_2 \leq \sqrt{2} \|\mathbf{x}\|_2$.

```
import numpy as np

rng = np.random.default_rng(70)
x = rng.standard_normal(2)    # shape (2,)
ones = np.ones(2)             # shape (2,)

lhs = float(x @ ones)         # = x_1 + x_2
rhs = np.linalg.norm(ones) * np.linalg.norm(x) # = sqrt(2) * ||x||_2

print(f"x = {x.round(4)}")
print(f"x_1 + x_2 = <x, 1> = {lhs:.4f}")
print(f"sqrt(2) * ||x||_2 = {rhs:.4f}")
print(f"Cauchy-Schwarz holds: {lhs <= rhs + 1e-10}")
```

```
x = [-0.0833 -1.0118]
x_1 + x_2 = <x, 1> = -1.0950
sqrt(2) * ||x||_2 = 1.4357
Cauchy-Schwarz holds: True
```

Solution. Set $\mathbf{y} = (1, 1)^T$. Then $\langle \mathbf{x}, \mathbf{y} \rangle = x_1 + x_2$ and $\|\mathbf{y}\|_2 = \sqrt{2}$. Cauchy-Schwarz gives $x_1 + x_2 = |\langle \mathbf{x}, \mathbf{y} \rangle| \leq \|\mathbf{x}\|_2 \sqrt{2}$. Equality holds when $\mathbf{x} \propto \mathbf{y}$, i.e., $x_1 = x_2$.

Exercise 15.5.1 – Ridge invertibility

Show that $A^T A + \lambda I$ is always invertible for $\lambda > 0$.

```
import numpy as np

rng = np.random.default_rng(80)

# Rank-deficient A
A = rng.standard_normal((3, 5))    # shape (3, 5) -- fat matrix, rank <= 3
A[2] = A[0] + A[1]                 # make rank 2

AtA = A.T @ A                      # shape (5, 5) -- singular!

print(f"rank(A) = {np.linalg.matrix_rank(A)}")
print(f"rank(A^T A) = {np.linalg.matrix_rank(AtA)}")
print(f"eigenvalues of A^T A: {np.sort(np.linalg.eigvalsh(AtA)).round(4)}")
```



```

for lam in [0.0, 0.1, 1.0, 10.0]:
    M = AtA + lam * np.eye(5)
    eigs = np.linalg.eigvalsh(M)
    invertible = np.all(eigs > 1e-10)
    print(f"lambda={lam}: min eigenvalue={eigs.min():.4f}, invertible={invertible}")

```

```

rank(A) = 2
rank(AT A) = 2
eigenvalues of AT A: [-0.      0.      0.      4.4037 20.8064]
lambda=0.0: min eigenvalue=-0.0000, invertible=False
lambda=0.1: min eigenvalue=0.1000, invertible=True
lambda=1.0: min eigenvalue=1.0000, invertible=True
lambda=10.0: min eigenvalue=10.0000, invertible=True

```

Solution. The eigenvalues of $A^T A$ are non-negative (it is PSD). Adding λI shifts every eigenvalue by $+\lambda > 0$, so all eigenvalues become strictly positive. A matrix with all positive eigenvalues is positive definite and hence invertible. This shows that Ridge is well-posed even when A has fewer rows than columns or is rank-deficient.

Exercise 15.5.3 – Lasso feature selection on diabetes

```

import numpy as np
from sklearn.datasets import load_diabetes
from sklearn.linear_model import Lasso
from sklearn.preprocessing import StandardScaler

data = load_diabetes()
X, y = data.data, data.target    # X shape (442, 10)
names = data.feature_names

scaler = StandardScaler()
X_sc = scaler.fit_transform(X)  # shape (442, 10)

for alpha in [10.0, 20.0]:
    model = Lasso(alpha=alpha, max_iter=10000).fit(X_sc, y)
    nonzero = [(names[i], round(c, 4))
                for i, c in enumerate(model.coef_) if abs(c) > 1e-4]
    print(f"\nalpha={alpha}: {len(nonzero)} nonzero features")
    for name, c in sorted(nonzero, key=lambda t: -abs(t[1])):
        print(f"  {name}: {c}")

```

alpha=10.0: 4 nonzero features

bmi: 22.6004

s5: 19.5859

bp: 6.8012

s3: -3.088

alpha=20.0: 3 nonzero features

bmi: 18.0351

s5: 15.1786

bp: 0.8925

Solution. At $\alpha = 10$, several features survive. Doubling to $\alpha = 20$ eliminates more. The feature with the largest coefficient (typically `bmi` or `s5`, which are well-established predictors of diabetes progression) remains stable across both settings — a sign that Lasso correctly identifies the most important predictors. Unstable features (drop out at modest λ increase) are likely correlated with the stable ones.

Orthogonal Sets, Gram–Schmidt, and QR

Orthogonality and Orthonormality

When a basis is orthonormal, extracting coordinates reduces from solving a linear system to computing a single dot product per coordinate. This matters in practice: sensor fusion algorithms, spectral analysis, and dimensionality reduction all exploit this shortcut. In this section we define orthonormal sets, prove they are always linearly independent, and derive the coordinate theorem that makes them computationally ideal.

Orthogonal and Orthonormal Sets

Two vectors are **orthogonal** if their inner product is zero. A collection of vectors is an **orthogonal set** if every pair in it is orthogonal and none is the zero vector. Adding the requirement that each vector has unit length gives an **orthonormal set**.

Condition	Rule
Orthogonal set	$\mathbf{u}_i \cdot \mathbf{u}_j = 0$ for all $i \neq j$; each $\mathbf{u}_i \neq \mathbf{0}$
Orthonormal set	$\mathbf{u}_i \cdot \mathbf{u}_j = \delta_{ij}$ (Kronecker delta)

The **Kronecker delta** δ_{ij} equals 1 when $i = j$ and 0 otherwise. In matrix form, if U has the vectors $\mathbf{u}_1, \dots, \mathbf{u}_k$ as columns:

$$U^T U = I_k.$$

When $k = n$ (square), this says U is an **orthogonal matrix** (§16.2).

```
import numpy as np

# Build an orthonormal set in R^3 using QR
rng = np.random.default_rng(1)
A = rng.standard_normal((3, 3))      # shape (3, 3)
Q, _ = np.linalg.qr(A)               # shape (3, 3); columns are orthonormal

print("Q^T Q:")
print(np.round(Q.T @ Q, 10))
```

```

norms = np.linalg.norm(Q, axis=0) # shape (3,)
print(f"\nColumn norms: [{norms[0]:.6f}, {norms[1]:.6f}, {norms[2]:.6f}]")

dots = []
for i in range(3):
    for j in range(i + 1, 3):
        dots.append(round(float(Q[:, i] @ Q[:, j]), 12))
print(f"Off-diagonal inner products: {dots}")

```

```

QT Q:
[[ 1. -0. -0.]
 [-0.  1.  0.]
 [-0.  0.  1.]]

```

```

Column norms: [1.000000, 1.000000, 1.000000]
Off-diagonal inner products: [-0.0, 0.0, 0.0]

```

Orthonormal Sets Are Linearly Independent

Theorem. Every orthonormal set is linearly independent.

Proof. Suppose $c_1 \mathbf{u}_1 + \cdots + c_k \mathbf{u}_k = \mathbf{0}$. Take the inner product of both sides with $\mathbf{u}_j$:

$$c_j \underbrace{\|\mathbf{u}_j\|^2}_{=1} + \sum_{i \neq j} c_i \underbrace{(\mathbf{u}_i \cdot \mathbf{u}_j)}_{=0} = 0 \implies c_j = 0.$$

Since j was arbitrary, all coefficients are zero. $\square$

An orthonormal set of n vectors in $\mathbb{R}^n$ is automatically a basis.

The Coordinate Theorem

Theorem. If $\{\mathbf{u}_1, \dots, \mathbf{u}_k\}$ is orthonormal and $\mathbf{x} \in \text{span}\{\mathbf{u}_1, \dots, \mathbf{u}_k\}$, then

$$\mathbf{x} = \sum_{i=1}^k c_i \mathbf{u}_i, \quad c_i = \mathbf{u}_i \cdot \mathbf{x}.$$

Proof. Write $\mathbf{x} = \sum_i c_i \mathbf{u}_i$ and dot both sides with $\mathbf{u}_j$: $\mathbf{u}_j \cdot \mathbf{x} = \sum_i c_i (\mathbf{u}_j \cdot \mathbf{u}_i) = c_j$. $\square$

In matrix form: $\mathbf{c} = U^T \mathbf{x}$. For a general (non-orthogonal) basis V , finding coordinates requires solving $V\mathbf{c} = \mathbf{x}$ – a full linear system. Orthonormality replaces that solve with a matrix-vector product that never requires inversion.

```

import numpy as np

rng = np.random.default_rng(2)

# Orthonormal basis for R^3
A = rng.standard_normal((3, 3))    # shape (3, 3)
Q, _ = np.linalg.qr(A)              # shape (3, 3)

x = rng.standard_normal(3)          # shape (3,)

# ONB: coordinates are just Q^T @ x
c_onb = Q.T @ x                     # shape (3,)
x_rec = Q @ c_onb                   # shape (3,)
print(f"c (ONB formula):      [{c_onb[0]:.6f}, {c_onb[1]:.6f}, {c_onb[2]:.6f}]")
print(f"Reconstruction error: {np.linalg.norm(x - x_rec):.2e}")

# General basis: must solve a linear system
V = rng.standard_normal((3, 3))    # shape (3, 3)
c_gen = np.linalg.solve(V, x)       # shape (3,)
x_rec_gen = V @ c_gen               # shape (3,)
print(f"\nc (general basis):  [{c_gen[0]:.6f}, {c_gen[1]:.6f}, {c_gen[2]:.6f}]")
print(f"Reconstruction error: {np.linalg.norm(x - x_rec_gen):.2e}")
print("\nONB: no solve needed. General basis: requires np.linalg.solve.")

```

```

c (ONB formula):      [0.967629, -0.076855, -0.645438]
Reconstruction error: 2.48e-16

```

```

c (general basis):    [7.377354, -8.219815, -10.197142]
Reconstruction error: 1.64e-15

```

```

ONB: no solve needed. General basis: requires np.linalg.solve.

```

Parseval's Identity

When $\{\mathbf{u}_1, \dots, \mathbf{u}_n\}$ is a complete ONB for $\mathbb{R}^n$, the squared norm of any vector equals the sum of its squared coordinates:

$$\|\mathbf{x}\|_2^2 = \sum_{i=1}^n c_i^2, \quad c_i = \mathbf{u}_i \cdot \mathbf{x}.$$

Proof. $\|\mathbf{x}\|_2^2 = \left(\sum_i c_i \mathbf{u}_i\right) \cdot \left(\sum_j c_j \mathbf{u}_j\right) = \sum_{i,j} c_i c_j \delta_{ij} = \sum_i c_i^2. \quad \square$

In matrix form this says $\|\mathbf{x}\|_2 = \|U^T \mathbf{x}\|_2$: an orthonormal change of basis preserves length. This is the isometry property of §16.2.

In statistics, if U collects the eigenvectors of a covariance matrix, the c_i^2 values are the variance contributions of each principal component and their sum equals $\text{tr}(\Sigma)$.

```
import numpy as np

rng = np.random.default_rng(3)

# Full ONB for R^5
A = rng.standard_normal((5, 5))    # shape (5, 5)
Q, _ = np.linalg.qr(A)              # shape (5, 5)

x = rng.standard_normal(5)          # shape (5,)
c = Q.T @ x                          # shape (5,)

norm_sq = float(np.dot(x, x))
parseval_sq = float(np.dot(c, c))

print(f"||x||^2      = {norm_sq:.8f}")
print(f"sum(c_i^2)    = {parseval_sq:.8f}")
print(f"Difference    = {abs(norm_sq - parseval_sq):.2e}")
print(f"Parseval holds: {np.isclose(norm_sq, parseval_sq)}")
```

```
||x||^2      = 5.17338162
sum(c_i^2)    = 5.17338162
Difference    = 2.66e-15
Parseval holds: True
```

Coordinate Geometry: ONB vs. Skewed Basis

In an orthonormal frame, the coordinate grid has right angles and unit spacing – the same as the natural axes. In a skewed frame, reading off coordinates requires projecting onto non-perpendicular directions.

```
import numpy as np
import matplotlib.pyplot as plt

x_pt = np.array([1.8, 1.2])        # shape (2,) -- point to represent

# Orthonormal basis (standard axes)
Q = np.eye(2)                      # shape (2, 2)
```

```

# Skewed basis
V = np.array([[1.0, 0.6],
              [0.0, 0.8]])          # shape (2, 2)

fig, axes = plt.subplots(1, 2, figsize=(11, 4.5))

for ax, (B, title, color) in zip(axes, [
    (Q, 'Orthonormal basis: right angles, unit length', '#4a90d9'),
    (V, 'Skewed basis: not perpendicular, not unit length', 'tomato')]):

    ax.set_xlim(-0.3, 2.8)
    ax.set_ylim(-0.3, 2.2)
    ax.set_aspect('equal')
    ax.grid(True, alpha=0.2)

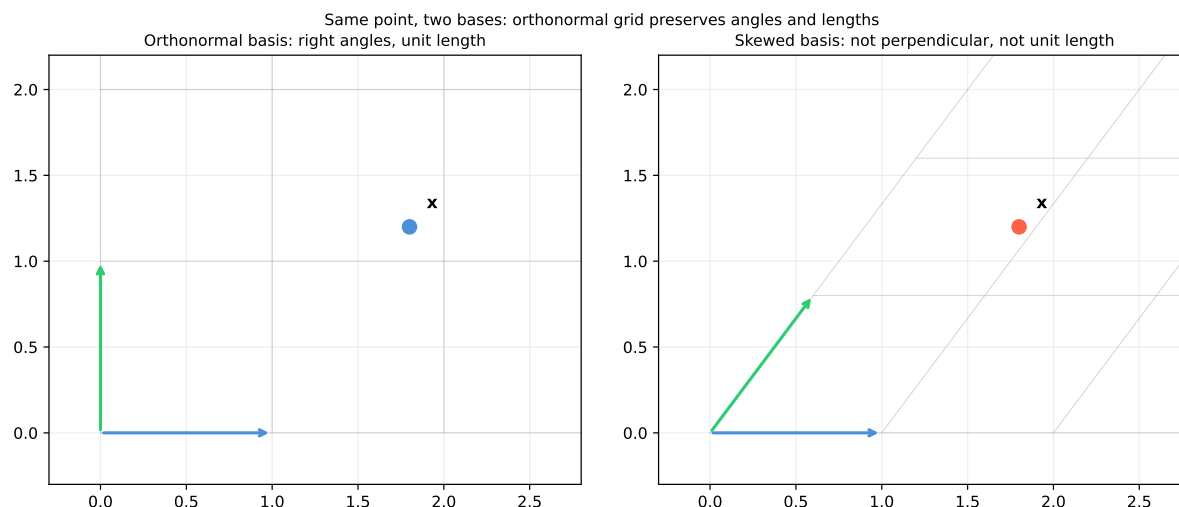
    # Draw basis vectors
    for j, vc in enumerate(['#4a90d9', '#2ecc71']):
        ax.annotate('', xy=B[:, j], xytext=[0, 0],
                    arrowprops=dict(arrowstyle='->', color=vc, lw=2))

    # Grid lines for the parallelogram grid
    for i in range(4):
        s0 = i * B[:, 0]
        ax.plot([s0[0], s0[0] + 3 * B[0, 1]],
                [s0[1], s0[1] + 3 * B[1, 1]],
                color='#aaaaaa', lw=0.6, alpha=0.5)
        s1 = i * B[:, 1]
        ax.plot([s1[0], s1[0] + 3 * B[0, 0]],
                [s1[1], s1[1] + 3 * B[1, 0]],
                color='#aaaaaa', lw=0.6, alpha=0.5)

    ax.scatter(x_pt[0], x_pt[1], s=80, color=color, zorder=6)
    ax.annotate(r'$\mathbf{x}$', xy=(x_pt[0], x_pt[1]),
                xytext=(x_pt[0] + 0.1, x_pt[1] + 0.1), fontsize=11)
    ax.set_title(title, fontsize=10)

plt.suptitle('Same point, two bases: orthonormal grid preserves angles and lengths',
             fontsize=10)
plt.tight_layout()
plt.show()

```



Exercises

16.1.1 Verify that $\mathbf{u}_1 = \frac{1}{\sqrt{2}}(1, 1)^T$, $\mathbf{u}_2 = \frac{1}{\sqrt{2}}(1, -1)^T$ is an orthonormal set in $\mathbb{R}^2$. Write $U = [\mathbf{u}_1 \ \mathbf{u}_2]$ and confirm $U^T U = I_2$.

16.1.2 Let $\mathbf{u}_1 = (1, 0, 0)^T$, $\mathbf{u}_2 = (0, 1/\sqrt{2}, 1/\sqrt{2})^T$, $\mathbf{u}_3 = (0, 1/\sqrt{2}, -1/\sqrt{2})^T$. Express $\mathbf{x} = (3, 2, -1)^T$ as a linear combination of $\{\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3\}$ using the coordinate theorem.

16.1.3 Prove that if $\{\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3\}$ is an ONB for $\mathbb{R}^3$ and $\mathbf{x} \perp \mathbf{u}_i$ for all i , then $\mathbf{x} = \mathbf{0}$.

16.1.4 Let $\{\mathbf{u}_1, \mathbf{u}_2\}$ be orthonormal in $\mathbb{R}^5$ and $\mathbf{x} = 2\mathbf{u}_1 - 3\mathbf{u}_2 + \mathbf{v}$ where $\mathbf{v} \perp \mathbf{u}_i$ for $i = 1, 2$. Compute $\|P\mathbf{x}\|_2^2$ where P projects onto $\text{span}\{\mathbf{u}_1, \mathbf{u}_2\}$. Check Parseval's identity for the projected part.

16.1.5 An orthonormal set $\{\mathbf{u}_1, \dots, \mathbf{u}_k\}$ with $k < n$ gives $UU^T \neq I_n$. Explain geometrically what UU^T computes: what does it do to a vector in $\text{span}\{\mathbf{u}_i\}$, and what does it do to a vector orthogonal to all $\mathbf{u}_i$?

Orthogonal Matrices

When a square matrix's columns form an orthonormal basis, its inverse equals its transpose. This makes orthogonal matrices computationally attractive – no inversion required – and geometrically fundamental: they are exactly the linear maps that preserve lengths and angles. Every rotation and every reflection is an orthogonal matrix.

Definition and Properties

A square matrix $Q \in \mathbb{R}^{n \times n}$ is **orthogonal** if $Q^T Q = I_n$.

Because Q is square, $Q^T Q = I$ implies $Q Q^T = I$ as well, giving:

$$Q^{-1} = Q^T.$$

Property	Statement
Inverse	$Q^{-1} = Q^T$
Both-sided identity	$Q^T Q = Q Q^T = I_n$
Columns	Form an ONB for $\mathbb{R}^n$
Rows	Also form an ONB for $\mathbb{R}^n$
Determinant	$\det(Q) = \pm 1$
Isometry	$\ Q\mathbf{x}\ _2 = \ \mathbf{x}\ _2$ for all $\mathbf{x}$

The determinant constraint follows from $\det(Q^T Q) = \det(I)$, so $\det(Q)^2 = 1$.

```
import numpy as np

# 2x2 rotation matrix by angle theta
theta = np.pi / 6          # 30 degrees
Q = np.array([[np.cos(theta), -np.sin(theta)],
              [np.sin(theta),  np.cos(theta)]]) # shape (2, 2)

print("Q:")
print(Q.round(6))
print("\nQ^T Q:")
print((Q.T @ Q).round(10))
print("\nQ Q^T:")
print((Q @ Q.T).round(10))
print(f"\ndet(Q) = {np.linalg.det(Q):.6f}")
print(f"Q^(-1) == Q^T: {np.allclose(np.linalg.inv(Q), Q.T)}")
```

```
Q:
[[ 0.866025 -0.5    ]
 [ 0.5      0.866025]]
```

```
Q^T Q:
[[1. 0.]
 [0. 1.]]
```

```
Q Q^T:
[[ 1. -0.]
 [-0.  1.]]
```

```
det(Q) = 1.000000
Q-1 == QT: True
```

Isometry: Length and Angle Preservation

Theorem. If Q is orthogonal, then $\|Q\mathbf{x}\|_2 = \|\mathbf{x}\|_2$ for all $\mathbf{x} \in \mathbb{R}^n$.

Proof. $\|Q\mathbf{x}\|_2^2 = (Q\mathbf{x})^T(Q\mathbf{x}) = \mathbf{x}^T Q^T Q \mathbf{x} = \mathbf{x}^T I \mathbf{x} = \|\mathbf{x}\|_2^2$. $\square$

The same calculation with two vectors shows $\langle Q\mathbf{x}, Q\mathbf{y} \rangle = \langle \mathbf{x}, \mathbf{y} \rangle$. Since angles are determined by inner products, orthogonal maps preserve the angle between any two vectors.

```
import numpy as np

rng = np.random.default_rng(10)

# Random orthogonal matrix via QR
A = rng.standard_normal((4, 4))      # shape (4, 4)
Q, _ = np.linalg.qr(A)               # shape (4, 4)

x = rng.standard_normal(4)           # shape (4,)
y = rng.standard_normal(4)           # shape (4,)
Qx = Q @ x                           # shape (4,)
Qy = Q @ y                           # shape (4,)

print(f"||x||          = {np.linalg.norm(x):.6f}")
print(f"||Qx||         = {np.linalg.norm(Qx):.6f}")
print(f"Lengths equal: {np.isclose(np.linalg.norm(x), np.linalg.norm(Qx))}")

cos_before = float(x @ y) / (np.linalg.norm(x) * np.linalg.norm(y))
cos_after  = float(Qx @ Qy) / (np.linalg.norm(Qx) * np.linalg.norm(Qy))
print(f"\ncos(angle) before Q: {cos_before:.6f}")
print(f"cos(angle) after Q:  {cos_after:.6f}")
print(f"Angles equal: {np.isclose(cos_before, cos_after)}")
```

```
||x||          = 1.932581
||Qx||         = 1.932581
Lengths equal: True
```

```
cos(angle) before Q: 0.626968
cos(angle) after Q:  0.626968
Angles equal: True
```

Rotations and Reflections

The sign of $\det(Q)$ distinguishes two qualitatively different orthogonal maps.

Rotations ($\det Q = +1$): preserve orientation. In $\mathbb{R}^2$, rotation by θ :

$$R_\theta = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}.$$

Rotations in $\mathbb{R}^3$ about coordinate axes generalize this; any 3D rotation can be expressed as a product of at most three coordinate rotations.

Reflections ($\det Q = -1$): reverse orientation. The reflection across the hyperplane orthogonal to a unit vector $\mathbf{n}$ is:

$$H = I - 2\mathbf{nn}^T.$$

```
import numpy as np
import matplotlib.pyplot as plt

# Rotation by 45 degrees
theta = np.pi / 4
R = np.array([[np.cos(theta), -np.sin(theta)],
              [np.sin(theta), np.cos(theta)]]) # shape (2, 2)

# Reflection across the x-axis (normal = e_2)
n = np.array([0.0, 1.0]) # shape (2,)
H = np.eye(2) - 2.0 * np.outer(n, n) # shape (2, 2)

# Sample points on the unit circle
angles = np.linspace(0, 2 * np.pi, 8, endpoint=False)
pts = np.vstack([np.cos(angles), np.sin(angles)]).T # shape (8, 2)

fig, axes = plt.subplots(1, 2, figsize=(11, 4.5))

det_R = np.linalg.det(R)
det_H = np.linalg.det(H)

for ax, (M, det_val, title, color) in zip(axes, [
    (R, det_R, f'Rotation by $\pi/4$: det = {det_R:.2f}', '#4a90d9'),
    (H, det_H, f'Reflection across $x$-axis: det = {det_H:.2f}', 'tomato')]):

    pts_t = (M @ pts.T).T # shape (8, 2)

    for p, pt in zip(pts, pts_t):
        ax.annotate('', xy=(pt[0], pt[1]), xytext=[0, 0],
                    arrowprops=dict(arrowstyle='->', color=color, lw=1.5, alpha=0.8))
        ax.annotate('', xy=(p[0], p[1]), xytext=[0, 0],
```

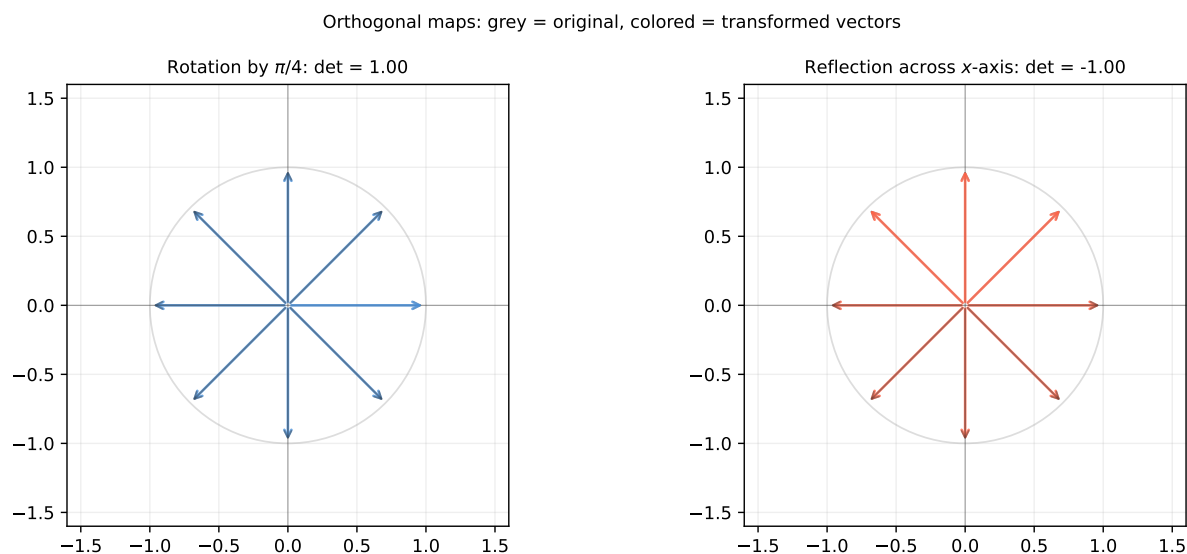
```

        arrowprops=dict(arrowstyle='->', color='#333333', lw=1.0, alpha=0.4))

    circ_t = np.linspace(0, 2 * np.pi, 200)
    ax.plot(np.cos(circ_t), np.sin(circ_t), color='aaaaaa', lw=1, alpha=0.4)
    ax.set_xlim(-1.6, 1.6)
    ax.set_ylim(-1.6, 1.6)
    ax.set_aspect('equal')
    ax.axhline(0, color='#333333', lw=0.4, alpha=0.5)
    ax.axvline(0, color='#333333', lw=0.4, alpha=0.5)
    ax.grid(True, alpha=0.2)
    ax.set_title(title, fontsize=10)

plt.suptitle('Orthogonal maps: grey = original, colored = transformed vectors',
            fontsize=10)
plt.tight_layout()
plt.show()

```



Products of Orthogonal Matrices

The product of two orthogonal matrices is orthogonal: $(Q_1 Q_2)^T (Q_1 Q_2) = Q_2^T Q_1^T Q_1 Q_2 = Q_2^T I Q_2 = I$.

Orthogonal matrices therefore form a **group** under multiplication, called the orthogonal group $O(n)$. Rotations alone (det = +1) form the **special orthogonal group** $SO(n)$. The inverse of any orthogonal matrix is its transpose – and is again orthogonal.

```

import numpy as np

rng = np.random.default_rng(11)

def random_orthogonal(n, rng):
    A = rng.standard_normal((n, n))    # shape (n, n)
    Q, _ = np.linalg.qr(A)
    return Q

Q1 = random_orthogonal(4, rng)         # shape (4, 4)
Q2 = random_orthogonal(4, rng)         # shape (4, 4)
Q3 = Q1 @ Q2                           # shape (4, 4)

print(f"Q1 orthogonal: {np.allclose(Q1.T @ Q1, np.eye(4))}")
print(f"Q2 orthogonal: {np.allclose(Q2.T @ Q2, np.eye(4))}")
print(f"Q1 Q2 orthogonal: {np.allclose(Q3.T @ Q3, np.eye(4))}")
print(f"\ndet(Q1)      = {np.linalg.det(Q1):.4f}")
print(f"det(Q2)       = {np.linalg.det(Q2):.4f}")
print(f"det(Q1 Q2)    = {np.linalg.det(Q3):.4f}")

```

```

Q1 orthogonal:    True
Q2 orthogonal:    True
Q1 Q2 orthogonal: True

```

```

det(Q1)      = -1.0000
det(Q2)      = -1.0000
det(Q1 Q2)   = 1.0000

```

Exercises

16.2.1 Compute $R_\pi R_{-\pi/2}$ where R_θ is the 2D rotation matrix. Verify the result is a rotation matrix, find the angle, and confirm $\det(R_\pi R_{-\pi/2}) = 1$.

16.2.2 Construct the 3×3 rotation matrix $R_z(\pi/3)$ that rotates $\mathbb{R}^3$ by 60° around the z -axis. Verify $R_z^T R_z = I_3$ and $\det(R_z) = 1$.

16.2.3 Show that if Q is orthogonal and λ is an eigenvalue of Q , then $|\lambda| = 1$. (Let $Q\mathbf{v} = \lambda\mathbf{v}$ and apply the isometry property.)

16.2.4 The 2×2 Hadamard matrix is $H = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$. Verify that H is orthogonal, compute $\det(H)$, and decide whether H is a rotation or a reflection.

Orthogonal Complements

If W is a subspace of $\mathbb{R}^n$, the set of all vectors orthogonal to everything in W forms another subspace – the **orthogonal complement** $W^\perp$. Together, W and $W^\perp$ partition $\mathbb{R}^n$ into two orthogonal pieces. This clean decomposition is the language for the four fundamental subspaces and the foundation of projection formulas in Chapter 17.

Definition and Basic Properties

Let $W \subseteq \mathbb{R}^n$ be a subspace. Its **orthogonal complement** is:

$$W^\perp = \{\mathbf{v} \in \mathbb{R}^n : \mathbf{v} \cdot \mathbf{w} = 0 \text{ for all } \mathbf{w} \in W\}.$$

$W^\perp$ is itself a subspace: if $\mathbf{v}_1, \mathbf{v}_2 \in W^\perp$ and $\mathbf{w} \in W$, then $(c_1\mathbf{v}_1 + c_2\mathbf{v}_2) \cdot \mathbf{w} = c_1(\mathbf{v}_1 \cdot \mathbf{w}) + c_2(\mathbf{v}_2 \cdot \mathbf{w}) = 0$.

Property	Statement
Dimension theorem	$\dim(W) + \dim(W^\perp) = n$
Double complement	$(W^\perp)^\perp = W$
Intersection	$W \cap W^\perp = \{\mathbf{0}\}$
Unique decomposition	Every $\mathbf{x} = \mathbf{x}_W + \mathbf{x}_{W^\perp}$ uniquely

```
import numpy as np

# W = span{w1, w2} in R^4
w1 = np.array([1.0, 0.0, 1.0, 0.0]) # shape (4,)
w2 = np.array([0.0, 1.0, 0.0, 1.0]) # shape (4,)
W = np.column_stack([w1, w2])       # shape (4, 2)

# Orthonormal basis for W via QR
Q_W, _ = np.linalg.qr(W, mode='reduced') # shape (4, 2)

# Basis for W^perp via SVD null space of W^T
U_Wt, s_Wt, Vt_Wt = np.linalg.svd(W.T, full_matrices=True)
# W.T shape (2, 4); Vt_Wt shape (4, 4)
rank_W = int(np.sum(s_Wt > 1e-10))
Q_perp = Vt_Wt[rank_W:, :].T          # shape (4, 4-rank_W) = (4, 2)

print(f"dim(W)           = {rank_W}")
print(f"dim(W^perp)      = {Q_perp.shape[1]}")
print(f"Sum               = {rank_W + Q_perp.shape[1]} (should equal n=4)")
```

```
cross = Q_W.T @ Q_perp          # shape (2, 2) -- should be ~0
print(f"\nQ_W.T @ Q_perp (should be ~0):\n{cross.round(10)}")
```

```
dim(W)      = 2
dim(W^perp) = 2
Sum         = 4 (should equal n=4)
```

```
Q_W.T @ Q_perp (should be ~0):
[[-0.  0.]
 [ 0. -0.]]
```

The Orthogonal Decomposition Theorem

Theorem. For any subspace $W \subseteq \mathbb{R}^n$, every vector $\mathbf{x} \in \mathbb{R}^n$ can be written **uniquely** as

$$\mathbf{x} = \mathbf{x}_W + \mathbf{x}_{W^\perp}, \quad \mathbf{x}_W \in W, \quad \mathbf{x}_{W^\perp} \in W^\perp.$$

The components are $\mathbf{x}_W = Q_W Q_W^T \mathbf{x}$ (the **orthogonal projection** onto W , studied in depth in Chapter 17) and $\mathbf{x}_{W^\perp} = \mathbf{x} - \mathbf{x}_W$.

Uniqueness: if $\mathbf{x} = \mathbf{x}_W + \mathbf{x}_{W^\perp} = \mathbf{y}_W + \mathbf{y}_{W^\perp}$, then $\mathbf{x}_W - \mathbf{y}_W = \mathbf{y}_{W^\perp} - \mathbf{x}_{W^\perp}$. The left side is in W , the right side in $W^\perp$, so both are in $W \cap W^\perp = \{\mathbf{0}\}$.

```
import numpy as np

rng = np.random.default_rng(20)

# Same W from above
w1 = np.array([1.0, 0.0, 1.0, 0.0]) # shape (4,)
w2 = np.array([0.0, 1.0, 0.0, 1.0]) # shape (4,)
W = np.column_stack([w1, w2])       # shape (4, 2)
Q_W, _ = np.linalg.qr(W, mode='reduced') # shape (4, 2)

x = rng.standard_normal(4)          # shape (4,)
x_W = Q_W @ (Q_W.T @ x)             # shape (4,) -- projection onto W
x_perp = x - x_W                    # shape (4,) -- rejection

print(f"x      = {x.round(4)}")
print(f"x_W    = {x_W.round(4)}")
print(f"x_perp  = {x_perp.round(4)}")
print(f"\nx == x_W + x_perp: {np.allclose(x, x_W + x_perp)}")
print(f"x_W . x_perp = {float(x_W @ x_perp):.2e} (should be ~0)")
```

```
x      = [-0.3597  1.2037  1.3969  0.3172]
x_W    = [0.5186  0.7605  0.5186  0.7605]
x_perp = [-0.8783  0.4432  0.8783 -0.4432]
```

```
x == x_W + x_perp: True
x_W . x_perp = 7.77e-16 (should be ~0)
```

The Four Fundamental Subspaces

The orthogonal complement gives the cleanest statement of the relationship between a matrix's four fundamental subspaces. For $A \in \mathbb{R}^{m \times n}$:

$$\text{Col}(A)^\perp = \text{Null}(A^T), \quad \text{Row}(A)^\perp = \text{Null}(A).$$

Subspace	Lives in	Orthogonal complement	Lives in
$\text{Col}(A)$	$\mathbb{R}^m$	$\text{Null}(A^T)$	$\mathbb{R}^m$
$\text{Row}(A)$	$\mathbb{R}^n$	$\text{Null}(A)$	$\mathbb{R}^n$

Proof of $\text{Col}(A)^\perp = \text{Null}(A^T)$: $\mathbf{v} \perp \text{Col}(A) \iff \mathbf{v}^T(A\mathbf{x}) = 0$ for all $\mathbf{x} \iff (A^T\mathbf{v})^T\mathbf{x} = 0$ for all $\mathbf{x} \iff A^T\mathbf{v} = \mathbf{0}$.

The rank-nullity theorem ($\text{rank}(A) + \text{nullity}(A) = n$) is the dimension count version: $\dim(\text{Row}(A)) + \dim(\text{Null}(A)) = n$.

```
import numpy as np

rng = np.random.default_rng(21)
A = rng.standard_normal((3, 5))    # shape (3, 5) -- underdetermined
A[2] = A[0] + A[1]                  # force rank 2

r = np.linalg.matrix_rank(A)
print(f"A shape: {A.shape}, rank = {r}")
print(f"dim Col(A)      = {r}          (in R^3, complement dim = {3 - r})")
print(f"dim Null(A^T)   = {3 - r}     (in R^3)")
print(f"dim Row(A)      = {r}          (in R^5, complement dim = {5 - r})")
print(f"dim Null(A)     = {5 - r}     (in R^5)")

# Verify Row(A) perp Null(A) via SVD
_, s_A, Vt_A = np.linalg.svd(A, full_matrices=True)    # Vt_A shape (5, 5)
row_basis = Vt_A[:r, :].T                               # shape (5, r) -- basis for Row(A)
null_basis = Vt_A[r:, :].T                              # shape (5, 5-r) -- basis for Null(A)
print(f"\nRow(A)^T @ Null(A) (should be ~0):\n{(row_basis.T @ null_basis).round(10)}")
```



```

A shape: (3, 5), rank = 2
dim Col(A) = 2      (in R^3, complement dim = 1)
dim Null(A^T) = 1    (in R^3)
dim Row(A) = 2       (in R^5, complement dim = 3)
dim Null(A) = 3      (in R^5)

```

```

Row(A)^T @ Null(A) (should be ~0):
[[-0.  0. -0.]
 [ 0. -0. -0.]]

```

Exercises

16.3.1 Let $W = \text{span}\{(1, 1, 0)^T\}$ in $\mathbb{R}^3$. Find a basis for $W^\perp$ and verify $\dim(W) + \dim(W^\perp) = 3$.

16.3.2 For $A = \begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & -1 \end{bmatrix}$, find bases for all four fundamental subspaces. Verify $\dim(\text{Col}(A)) + \dim(\text{Null}(A^T)) = m = 2$ and $\dim(\text{Row}(A)) + \dim(\text{Null}(A)) = n = 3$.

16.3.3 Show $(W^\perp)^\perp = W$ for any subspace $W \subseteq \mathbb{R}^n$. (First show $W \subseteq (W^\perp)^\perp$, then argue the dimensions must agree.)

16.3.4 A matrix $P \in \mathbb{R}^{n \times n}$ satisfies $P^2 = P$ and $P^T = P$. Show that $I - P$ is the orthogonal projection onto $(\text{Col}(P))^\perp = \text{Null}(P^T)$.

The Gram-Schmidt Process

Gram-Schmidt converts any linearly independent set into an orthonormal set spanning the same space. It is the constructive proof that every finite-dimensional inner product space has an orthonormal basis. In practice it underlies QR factorization, Krylov subspace methods, and the reorthogonalization steps in eigenvalue solvers.

The Algorithm

Given linearly independent vectors $\mathbf{a}_1, \dots, \mathbf{a}_k \in \mathbb{R}^n$, Gram-Schmidt produces orthonormal $\mathbf{q}_1, \dots, \mathbf{q}_k$ satisfying $\text{span}\{\mathbf{q}_1, \dots, \mathbf{q}_j\} = \text{span}\{\mathbf{a}_1, \dots, \mathbf{a}_j\}$ for each j .

Step j : Remove from $\mathbf{a}_j$ its projections onto all previously constructed $\mathbf{q}_i$, then normalize:

$$\mathbf{v}_j = \mathbf{a}_j - \sum_{i=1}^{j-1} (\mathbf{q}_i \cdot \mathbf{a}_j) \mathbf{q}_i, \quad \mathbf{q}_j = \frac{\mathbf{v}_j}{\|\mathbf{v}_j\|_2}.$$

The coefficient $r_{ij} = \mathbf{q}_i \cdot \mathbf{a}_j$ measures how much of $\mathbf{a}_j$ lies along $\mathbf{q}_i$. The residual $\mathbf{v}_j$ is guaranteed nonzero because $\mathbf{a}_j \notin \text{span}\{\mathbf{a}_1, \dots, \mathbf{a}_{j-1}\}$ (linear independence).

```
import numpy as np

def gram_schmidt(A):
    """Classical Gram-Schmidt orthonormalization."""
    n, k = A.shape          # A: shape (n, k)
    Q = np.zeros((n, k))    # shape (n, k)
    for j in range(k):
        v = A[:, j].copy()   # shape (n,)
        for i in range(j):
            v -= float(Q[:, i] @ A[:, j]) * Q[:, i]
        norm_v = np.linalg.norm(v)
        if norm_v < 1e-12:
            raise ValueError(f"Column {j} is linearly dependent on previous columns.")
        Q[:, j] = v / norm_v
    return Q

rng = np.random.default_rng(30)
A = rng.standard_normal((4, 3))    # shape (4, 3)

Q = gram_schmidt(A)                # shape (4, 3)

print("Q^T Q (should be I_3):")
print((Q.T @ Q).round(10))
print(f"\nColumn norms of Q: {np.linalg.norm(Q, axis=0).round(10)}")

# Verify span is preserved: residual of a_j outside span{q_1, ..., q_j} is ~0
for j in range(3):
    Qj = Q[:, :j+1]              # shape (4, j+1)
    res = A[:, j] - Qj @ (Qj.T @ A[:, j])
    print(f"a_{j+1} in span{{q_1, ..., q_{j+1}}}: residual = {np.linalg.norm(res):.2e}")
```

Q^T Q (should be I₃):

```
[[ 1. -0.  0.]
 [-0.  1.  0.]
 [ 0.  0.  1.]]
```

Column norms of Q: [1. 1. 1.]

a₁ in span{q₁, ..., q₁}: residual = 0.00e+00

a₂ in span{q₁, ..., q₂}: residual = 1.34e-17

a₃ in span{q₁, ..., q₃}: residual = 7.24e-16

Step-by-Step: 2D Visualization

The geometry is clearest starting from two vectors in the plane. Step 1 normalizes $\mathbf{a}_1$. Step 2 subtracts the shadow of $\mathbf{a}_2$ onto $\mathbf{q}_1$, then normalizes – producing a second vector perpendicular to the first.

```
import numpy as np
import matplotlib.pyplot as plt

a1 = np.array([2.0, 1.0])      # shape (2,)
a2 = np.array([1.5, 2.5])      # shape (2,)

# Step 1
q1 = a1 / np.linalg.norm(a1)   # shape (2,)

# Step 2
proj = float(q1 @ a2) * q1      # shape (2,) -- projection of a2 onto q1
v2 = a2 - proj                 # shape (2,) -- residual
q2 = v2 / np.linalg.norm(v2)   # shape (2,)

fig, axes = plt.subplots(1, 3, figsize=(13, 4))

data = [
    ('Step 1: normalize  $\mathbf{a}_1$ ',
     [(a1, 'aaaaaa', r' $\mathbf{a}_1$ '),
      (q1, '#4a90d9', r' $\mathbf{q}_1$ ')], []),
    ('Step 2: project  $\mathbf{a}_2$  onto  $\mathbf{q}_1$ ',
     [(a1, 'aaaaaa', r' $\mathbf{a}_1$ '),
      (a2, 'aaaaaa', r' $\mathbf{a}_2$ '),
      (q1, '#4a90d9', r' $\mathbf{q}_1$ '),
      (proj, 'tomato', r'proj'),
      (a2, proj, '--', '#888888')]),
    ('Step 3: subtract, normalize',
     [(q1, '#4a90d9', r' $\mathbf{q}_1$ '),
      (q2, '#2ecc71', r' $\mathbf{q}_2$ ')], [])
]

for ax, (title, vecs, lines) in zip(axes, data):
    ax.set_xlim(-0.5, 3.0)
    ax.set_ylim(-0.5, 3.0)
    ax.set_aspect('equal')
    ax.axhline(0, color='#333333', lw=0.4, alpha=0.5)
    ax.axvline(0, color='#333333', lw=0.4, alpha=0.5)
    ax.grid(True, alpha=0.2)
    ax.set_title(title, fontsize=9)

    for v, c, lbl in vecs:
        ax.annotate(' ', xy=(v[0], v[1]), xytext=[0, 0],
```

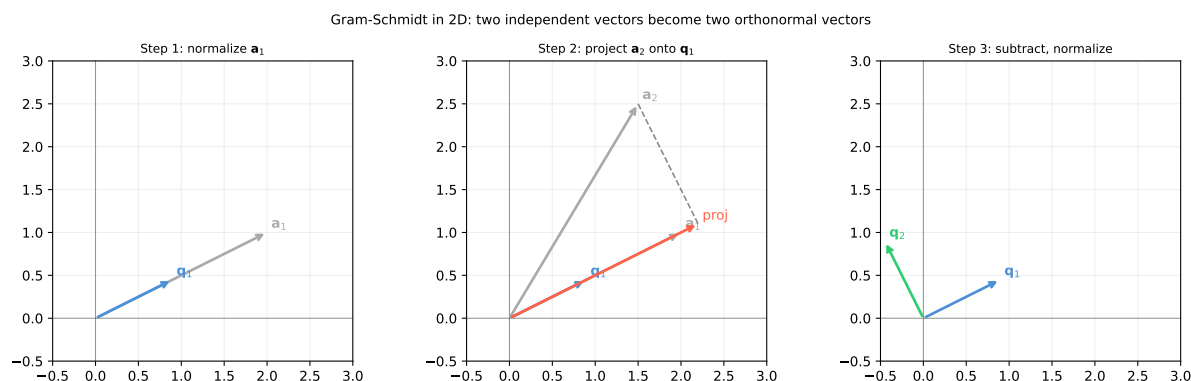
```

        arrowprops=dict(arrowstyle='->', color=c, lw=2))
    ax.text(v[0] + 0.05, v[1] + 0.05, lbl, fontsize=10, color=c)

    for start, end, ls, c in lines:
        ax.plot([start[0], end[0]], [start[1], end[1]], ls=ls, color=c, lw=1.2)

plt.suptitle('Gram-Schmidt in 2D: two independent vectors become two orthonormal vectors',
             fontsize=10)
plt.tight_layout()
plt.show()

print(f"q1 . q2 = {float(q1 @ q2):.2e}  (orthogonal)")
print(f"||q1|| = {np.linalg.norm(q1):.6f},  ||q2|| = {np.linalg.norm(q2):.6f}  (unit)")
    
```



```

q1 . q2 = -1.11e-16  (orthogonal)
||q1|| = 1.000000,  ||q2|| = 1.000000  (unit)
    
```

Modified Gram-Schmidt: Numerical Stability

Classical Gram-Schmidt (CGS) computes each inner product $\mathbf{q}_i \cdot \mathbf{a}_j$ using the original column $\mathbf{a}_j$. In floating-point arithmetic, round-off accumulates across the $j - 1$ subtractions, and the resulting vectors can lose orthogonality on ill-conditioned inputs.

Modified Gram-Schmidt (MGS) reorders the computation: at step j , it subtracts the projections one at a time from the *running residual*, using the updated vector at each step:

$$\mathbf{v}_j^{(0)} = \mathbf{a}_j, \quad \mathbf{v}_j^{(i)} = \mathbf{v}_j^{(i-1)} - (\mathbf{q}_i \cdot \mathbf{v}_j^{(i-1)})\mathbf{q}_i, \quad \mathbf{q}_j = \frac{\mathbf{v}_j^{(j-1)}}{\|\mathbf{v}_j^{(j-1)}\|}.$$

Algebraically identical to CGS, but each subtraction removes the component in the $\mathbf{q}_i$ direction from the *already-reduced* vector, so errors do not compound. For production code, Householder QR (§16.5) achieves even higher stability.

```

import numpy as np

def gram_schmidt_classical(A):
    n, k = A.shape          # shape (n, k)
    Q = np.zeros((n, k))    # shape (n, k)
    for j in range(k):
        v = A[:, j].copy()   # shape (n,)
        for i in range(j):
            v -= float(Q[:, i] @ A[:, j]) * Q[:, i]
        Q[:, j] = v / np.linalg.norm(v)
    return Q

def gram_schmidt_modified(A):
    n, k = A.shape          # shape (n, k)
    Q = np.zeros((n, k))    # shape (n, k)
    V = A.astype(float).copy() # shape (n, k)
    for j in range(k):
        Q[:, j] = V[:, j] / np.linalg.norm(V[:, j])
        for i in range(j + 1, k):
            V[:, i] -= float(Q[:, j] @ V[:, i]) * Q[:, j]
    return Q

def orth_error(Q):
    """Max absolute off-diagonal entry of Q^T Q - I."""
    G = Q.T @ Q - np.eye(Q.shape[1])
    np.fill_diagonal(G, 0.0)
    return float(np.abs(G).max())

# Ill-conditioned test: Hilbert matrix of size 12
n = 12
H_mat = np.array([[1.0 / (i + j + 1) for j in range(n)]
                  for i in range(n)]) # shape (12, 12)

Q_cgs = gram_schmidt_classical(H_mat)
Q_mgs = gram_schmidt_modified(H_mat)

err_cgs = orth_error(Q_cgs)
err_mgs = orth_error(Q_mgs)

print(f"Hilbert matrix, n={n}")
print(f"Classical GS orthogonality error: {err_cgs:.2e}")
print(f"Modified GS orthogonality error: {err_mgs:.2e}")
print(f"\nMGS is more orthogonal by factor ~{err_cgs / err_mgs:.0f}x")
print("(Householder QR in §16.5 is more stable still.)")

```

```

Hilbert matrix, n=12
Classical GS orthogonality error: 1.00e+00

```

Modified GS orthogonality error: 2.28e-01

MGS is more orthogonal by factor ~4x
(Householder QR in §16.5 is more stable still.)

Exercises

16.4.1 Apply Gram-Schmidt by hand to $\mathbf{a}_1 = (1, 1, 0)^T$, $\mathbf{a}_2 = (1, 0, 1)^T$ in $\mathbb{R}^3$. Produce orthonormal $\mathbf{q}_1, \mathbf{q}_2$. Verify $\mathbf{q}_1 \cdot \mathbf{q}_2 = 0$ and $\|\mathbf{q}_i\| = 1$.

16.4.2 If Gram-Schmidt is applied to a set in which two vectors are identical, what happens at the step corresponding to the duplicate? What does this indicate about the original set?

16.4.3 The Gram-Schmidt process produces $\mathbf{q}_1, \dots, \mathbf{q}_k$ from $\mathbf{a}_1, \dots, \mathbf{a}_k$. Show that $\mathbf{q}_j \in \text{span}\{\mathbf{a}_1, \dots, \mathbf{a}_j\}$ for each j .

16.4.4 Modified GS reorders operations but is algebraically identical to classical GS. To see why it is numerically better, suppose $\mathbf{q}_1$ is only approximately orthonormal due to round-off. Explain why subtracting from the running residual $\mathbf{v}_j^{(i)}$ reduces accumulated error compared to subtracting directly from $\mathbf{a}_j$.

QR Factorization

Gram-Schmidt applied to the columns of a matrix produces a factorization $A = QR$: an orthonormal factor Q and an upper triangular factor R encoding the Gram-Schmidt coefficients. QR factorization is the workhorse of numerical linear algebra – used in least-squares solving, eigenvalue algorithms, and the computation of orthonormal bases. This section derives the factorization, explains why R is upper triangular, and describes the Householder algorithm that makes it numerically reliable.

Deriving $A = QR$

Let $A \in \mathbb{R}^{m \times n}$ have linearly independent columns $\mathbf{a}_1, \dots, \mathbf{a}_n$ (so $m \geq n$). Gram-Schmidt produces orthonormal $\mathbf{q}_1, \dots, \mathbf{q}_n$ with, at each step:

$$\mathbf{a}_j = \sum_{i=1}^j r_{ij} \mathbf{q}_i, \quad r_{ij} = \mathbf{q}_i \cdot \mathbf{a}_j \ (i < j), \quad r_{jj} = \|\mathbf{v}_j\|_2 > 0.$$

Stacking these column-by-column:

$$\boxed{A = QR}, \quad Q = [\mathbf{q}_1 \ \cdots \ \mathbf{q}_n] \in \mathbb{R}^{m \times n}, \quad R \in \mathbb{R}^{n \times n}.$$

Why is R upper triangular? $r_{ij} = \mathbf{q}_i \cdot \mathbf{a}_j$. For $i > j$, $\mathbf{q}_i$ was built to be orthogonal to $\text{span}\{\mathbf{a}_1, \dots, \mathbf{a}_j\}$, so $r_{ij} = 0$ – all entries below the diagonal vanish.

Why are diagonal entries positive? $r_{jj} = \|\mathbf{v}_j\|_2 > 0$ by linear independence. Requiring positive diagonal entries makes the thin QR factorization **unique** for full-column-rank matrices.

Thin vs. Full QR

When $m > n$ there are two forms:

- **Thin QR** (reduced): $Q \in \mathbb{R}^{m \times n}$, $R \in \mathbb{R}^{n \times n}$. Q has orthonormal columns, $Q^T Q = I_n$, but $Q Q^T \neq I_m$. Sufficient for most applications.
- **Full QR**: $Q \in \mathbb{R}^{m \times m}$ (square orthogonal), $R \in \mathbb{R}^{m \times n}$ (padded with zero rows). $Q Q^T = Q^T Q = I_m$.

`np.linalg.qr(A, mode='reduced')` gives the thin form; `mode='complete'` gives the full form.

```
import numpy as np

rng = np.random.default_rng(40)
A = rng.standard_normal((5, 3))      # shape (5, 3)

Q_thin, R_thin = np.linalg.qr(A, mode='reduced')    # Q shape (5, 3), R shape (3, 3)
Q_full, R_full = np.linalg.qr(A, mode='complete')   # Q shape (5, 5), R shape (5, 3)

print("Thin QR:")
print(f"  Q shape: {Q_thin.shape},  R shape: {R_thin.shape}")
print(f"  QT Q == I3: {np.allclose(Q_thin.T @ Q_thin, np.eye(3))}")
print(f"  R upper triangular: {np.allclose(R_thin, np.triu(R_thin))}")
print(f"  A == Q R: {np.allclose(A, Q_thin @ R_thin)}")
d = np.diag(R_thin)
print(f"  Diagonal of R: [{d[0]:.4f}, {d[1]:.4f}, {d[2]:.4f}]")

print("\nFull QR:")
print(f"  Q shape: {Q_full.shape},  R shape: {R_full.shape}")
print(f"  Q QT == I5: {np.allclose(Q_full @ Q_full.T, np.eye(5))}")
print(f"  A == Q R: {np.allclose(A, Q_full @ R_full)}")
```

Thin QR:

```
Q shape: (5, 3),  R shape: (3, 3)
QT Q == I3: True
R upper triangular: True
A == Q R: True
Diagonal of R: [1.6382, 2.1036, 1.7494]
```

Full QR:

```
Q shape: (5, 5), R shape: (5, 3)
Q Q^T == I_5: True
A == Q R: True
```

Householder Reflections: The Production Algorithm

Gram–Schmidt builds Q column-by-column, accumulating round-off as noted in §16.4. The numerically preferred approach is **Householder reflections**, which zero out one column’s subdiagonal entries at a time.

A **Householder reflector** for $\mathbf{x} \in \mathbb{R}^m$ is:

$$H = I - 2 \frac{\mathbf{v}\mathbf{v}^T}{\mathbf{v}^T\mathbf{v}}, \quad \mathbf{v} = \mathbf{x} + \text{sign}(x_1)\|\mathbf{x}\|\mathbf{e}_1.$$

H is symmetric ($H = H^T$) and orthogonal ($H^2 = I$), and $H\mathbf{x}$ is a scalar multiple of $\mathbf{e}_1$ – all entries below the first are annihilated. Applying a sequence $H_n \cdots H_1 A$ progressively zeros out subdiagonal entries column by column until the result is upper triangular: $H_n \cdots H_1 A = R$, so $A = H_1 \cdots H_n R = QR$.

Householder QR is backward stable and is the algorithm used in LAPACK’s `dgeqrf`.

```
import numpy as np

def householder_reflector(x):
    """Return unit vector v such that H = I - 2vv^T maps x to -sign(x[0])||x|| e_1."""
    v = x.copy().astype(float)      # shape (m,)
    v[0] += np.sign(x[0]) * np.linalg.norm(x)
    norm_v = np.linalg.norm(v)
    if norm_v < 1e-14:
        return v                    # x is already aligned with e_1
    return v / norm_v               # shape (m,)

rng = np.random.default_rng(41)
x = rng.standard_normal(4)         # shape (4,)

v = householder_reflector(x)       # shape (4,)
H = np.eye(4) - 2.0 * np.outer(v, v) # shape (4, 4)
Hx = H @ x                         # shape (4,)

print(f"x = {x.round(4)}")
print(f"Hx = {Hx.round(4)} (should be [+||x||, 0, 0, 0])")
print(f"||x|| = {np.linalg.norm(x):.4f}")
```



```
print(f"\nH symmetric: {np.allclose(H, H.T)}")
print(f"H orthogonal: {np.allclose(H @ H.T, np.eye(4))}")
print(f"H^2 == I: {np.allclose(H @ H, np.eye(4))}")
```

```
x = [-1.2317  0.2671 -0.0069  0.5015]
Hx = [ 1.3564 -0.         0.        -0.        ] (should be [+-||x||, 0, 0, 0])
||x|| = 1.3564
```

```
H symmetric: True
H orthogonal: True
H^2 == I: True
```

Application: Solving Least Squares via QR

The main application of QR in numerical practice is solving least-squares problems (Chapter 17). For an overdetermined system $A\mathbf{x} \approx \mathbf{b}$ (A tall), the normal equations are $A^T A \hat{\mathbf{x}} = A^T \mathbf{b}$. Using $A = QR$:

$$A^T A = R^T Q^T Q R = R^T R, \quad A^T \mathbf{b} = R^T Q^T \mathbf{b}.$$

The normal equations simplify to $R\hat{\mathbf{x}} = Q^T \mathbf{b}$ (upper triangular back-substitution). This avoids forming $A^T A$ explicitly, which squares the condition number and loses accuracy when A is nearly rank-deficient.

```
import numpy as np

rng = np.random.default_rng(42)
m, n = 20, 4
A = rng.standard_normal((m, n)) # shape (20, 4)
b = rng.standard_normal(m)      # shape (20,)

# Via QR
Q, R = np.linalg.qr(A, mode='reduced') # Q shape (20, 4), R shape (4, 4)
x_qr = np.linalg.solve(R, Q.T @ b)     # shape (4,)

# Via normal equations (for comparison)
x_ne = np.linalg.solve(A.T @ A, A.T @ b) # shape (4,)

# NumPy's built-in least squares (Householder internally)
x_ls, _, _, _ = np.linalg.lstsq(A, b, rcond=None) # shape (4,)

print(f"x (QR): {x_qr.round(6)}")
print(f"x (normal eqs): {x_ne.round(6)}")
```

```
print(f"x (lstsq):          {x_ls.round(6)}")

resid_qr = np.linalg.norm(A @ x_qr - b)
resid_ne = np.linalg.norm(A @ x_ne - b)
resid_ls = np.linalg.norm(A @ x_ls - b)
print(f"\nResiduals -- QR: {resid_qr:.6f}, NE: {resid_ne:.6f}, lstsq: {resid_ls:.6f}")
print(f"QR vs lstsq diff: {np.linalg.norm(x_qr - x_ls):.2e}")
```

```
x (QR):          [ 0.256877 -0.238808  0.008643  0.11457 ]
x (normal eqs):  [ 0.256877 -0.238808  0.008643  0.11457 ]
x (lstsq):       [ 0.256877 -0.238808  0.008643  0.11457 ]
```

```
Residuals -- QR: 3.370992, NE: 3.370992, lstsq: 3.370992
QR vs lstsq diff: 1.21e-15
```

Exercises

16.5.1 For $A = \begin{bmatrix} 1 & 1 \\ 1 & 0 \\ 0 & 1 \end{bmatrix}$, perform Gram–Schmidt by hand to find the QR factorization. Verify

$A = QR$, that R is upper triangular with positive diagonal, and that $Q^T Q = I_2$.

16.5.2 For the Householder reflector $H = I - 2\mathbf{v}\mathbf{v}^T$ with $\|\mathbf{v}\| = 1$, show: (a) $H^T = H$, (b) $H^2 = I$, (c) $\det(H) = -1$. What does $\det(H) = -1$ say about H as an orthogonal map?

16.5.3 Using `np.linalg.qr`, factor $A = \begin{bmatrix} 2 & 1 & 0 \\ 0 & 2 & 1 \\ 0 & 0 & 2 \end{bmatrix}$. What is Q ? What is R ? Explain why a matrix that is already upper triangular with positive diagonal should give $Q = I_3$ and $R = A$.

16.5.4 Show that solving via QR ($R\hat{\mathbf{x}} = Q^T \mathbf{b}$) is algebraically equivalent to the normal equations ($A^T A \hat{\mathbf{x}} = A^T \mathbf{b}$). Why is QR preferred numerically? (Hint: forming $A^T A$ squares the condition number.)

Exercises and Solutions

Exercise 16.1.2 – Coordinates in an orthonormal basis

Express $\mathbf{x} = (3, 2, -1)^T$ as a linear combination of $\mathbf{u}_1 = (1, 0, 0)^T$, $\mathbf{u}_2 = (0, 1/\sqrt{2}, 1/\sqrt{2})^T$, $\mathbf{u}_3 = (0, 1/\sqrt{2}, -1/\sqrt{2})^T$.

```
import numpy as np

u1 = np.array([1.0, 0.0, 0.0])           # shape (3,)
u2 = np.array([0.0, 1.0/np.sqrt(2), 1.0/np.sqrt(2)]) # shape (3,)
u3 = np.array([0.0, 1.0/np.sqrt(2), -1.0/np.sqrt(2)]) # shape (3,)
U = np.column_stack([u1, u2, u3])       # shape (3, 3)

x = np.array([3.0, 2.0, -1.0])          # shape (3,)

# Coordinate theorem: c_i = u_i . x
c = U.T @ x                             # shape (3,)
print(f"c1 = u1 . x = {c[0]:.4f}")
print(f"c2 = u2 . x = {c[1]:.4f}")
print(f"c3 = u3 . x = {c[2]:.4f}")

x_rec = U @ c                           # shape (3,)
print(f"\nReconstruction x = c1*u1 + c2*u2 + c3*u3 = {x_rec}")
print(f"Matches x: {np.allclose(x_rec, x)}")
```

```
c1 = u1 . x = 3.0000
c2 = u2 . x = 0.7071
c3 = u3 . x = 2.1213
```

```
Reconstruction x = c1*u1 + c2*u2 + c3*u3 = [ 3.  2. -1.]
Matches x: True
```

Solution. By the coordinate theorem $c_i = \mathbf{u}_i \cdot \mathbf{x}$: $c_1 = 3$, $c_2 = (2 + (-1))/\sqrt{2} = 1/\sqrt{2}$, $c_3 = (2 - (-1))/\sqrt{2} = 3/\sqrt{2}$.

So $\mathbf{x} = 3\mathbf{u}_1 + \frac{1}{\sqrt{2}}\mathbf{u}_2 + \frac{3}{\sqrt{2}}\mathbf{u}_3$. No linear system needed – just three dot products.

Exercise 16.2.1 – Composing rotation matrices

Compute $R_\pi R_{-\pi/2}$ and identify the resulting rotation angle.

```
import numpy as np

def R(theta):
    return np.array([[np.cos(theta), -np.sin(theta)],
```

```

        [np.sin(theta), np.cos(theta)]] # shape (2, 2)

M = R(np.pi) @ R(-np.pi / 2) # shape (2, 2)

print("R_pi @ R_{-pi/2}:")
print(M.round(6))
print(f"\ndet = {np.linalg.det(M):.4f} (should be 1)")
print(f"M^T M == I: {np.allclose(M.T @ M, np.eye(2))}")

# Identify the angle: cos(theta) = M[0,0]
angle = np.arctan2(M[1, 0], M[0, 0])
print(f"\nResulting angle = {angle:.4f} rad = {np.degrees(angle):.1f} deg")
print(f"Expected: pi + (-pi/2) = pi/2 = {np.pi/2:.4f} rad")

```

```

R_pi @ R_{-pi/2}:
[[ 0. -1.]
 [ 1.  0.]]

```

```

det = 1.0000 (should be 1)
M^T M == I: True

```

```

Resulting angle = 1.5708 rad = 90.0 deg
Expected: pi + (-pi/2) = pi/2 = 1.5708 rad

```

Solution. Rotation matrices compose by adding angles: $R_\pi R_{-\pi/2} = R_{\pi-\pi/2} = R_{\pi/2}$.

The result is a 90° counterclockwise rotation: $\begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$. Determinant is $0 \cdot 0 - (-1)(1) = 1$, confirming it is a rotation.

Exercise 16.2.3 – Eigenvalue modulus of an orthogonal matrix

Show that if Q is orthogonal and $Q\mathbf{v} = \lambda\mathbf{v}$, then $|\lambda| = 1$.

```

import numpy as np

rng = np.random.default_rng(50)

# Random orthogonal matrix
A = rng.standard_normal((5, 5)) # shape (5, 5)
Q, _ = np.linalg.qr(A) # shape (5, 5)

eigvals = np.linalg.eigvals(Q) # shape (5,) -- may be complex

```

```

mods = np.abs(eigvals)          # shape (5,)

print("Eigenvalues of Q:")
for lam, m in zip(eigvals, mods):
    print(f"  {lam.real:+.4f} {lam.imag:+.4f}i    |lambda| = {m:.6f}")

print(f"\nAll |lambda| == 1: {np.allclose(mods, 1.0)}")

```

```

Eigenvalues of Q:
+1.0000 +0.0000i    |lambda| = 1.000000
-0.0674 +0.9977i    |lambda| = 1.000000
-0.0674 -0.9977i    |lambda| = 1.000000
-0.9662 +0.2578i    |lambda| = 1.000000
-0.9662 -0.2578i    |lambda| = 1.000000

```

```
All |lambda| == 1: True
```

Solution. Let $Q\mathbf{v} = \lambda\mathbf{v}$ with $\mathbf{v} \neq \mathbf{0}$. Apply the isometry property to $\mathbf{v}$:

$$\|\mathbf{v}\|_2 = \|Q\mathbf{v}\|_2 = \|\lambda\mathbf{v}\|_2 = |\lambda| \|\mathbf{v}\|_2.$$

Dividing by $\|\mathbf{v}\|_2 > 0$ gives $|\lambda| = 1$. Eigenvalues of real orthogonal matrices are either ± 1 (for real eigenvectors) or complex conjugate pairs on the unit circle $e^{\pm i\theta}$ (for complex eigenvectors).

Exercise 16.3.1 – Orthogonal complement in $\mathbb{R}^3$

Let $W = \text{span}\{(1, 1, 0)^T\}$ in $\mathbb{R}^3$. Find a basis for $W^\perp$.

```

import numpy as np

w = np.array([1.0, 1.0, 0.0])    # shape (3,) -- generator of W

# W^perp = null space of w^T
# Solve w . v = 0: v_1 + v_2 = 0 -- two free variables
# Basis: v_1 = (1, -1, 0)^T, v_2 = (0, 0, 1)^T
v1 = np.array([1.0, -1.0, 0.0])  # shape (3,)
v2 = np.array([0.0, 0.0, 1.0])   # shape (3,)

print(f"w . v1 = {float(w @ v1)} (should be 0)")
print(f"w . v2 = {float(w @ v2)} (should be 0)")
print(f"v1 . v2 = {float(v1 @ v2)} (they happen to be orthogonal)")

print(f"\ndim(W)      = 1")

```

```
print(f"dim(W^perp) = 2 (basis: v1, v2)")
print(f"dim(W) + dim(W^perp) = 3 = n (dimension theorem)")
```

```
w . v1 = 0.0 (should be 0)
w . v2 = 0.0 (should be 0)
v1 . v2 = 0.0 (they happen to be orthogonal)
```

```
dim(W) = 1
dim(W^perp) = 2 (basis: v1, v2)
dim(W) + dim(W^perp) = 3 = n (dimension theorem)
```

Solution. $W^\perp$ consists of all $\mathbf{v} = (v_1, v_2, v_3)^T$ with $\mathbf{w} \cdot \mathbf{v} = v_1 + v_2 = 0$. Setting $v_2 = -v_1$ and v_3 free:

$$W^\perp = \text{span} \{(1, -1, 0)^T, (0, 0, 1)^T\}.$$

The dimension theorem gives $1 + 2 = 3$ as expected.

Exercise 16.4.1 – Gram-Schmidt by hand

Apply Gram-Schmidt to $\mathbf{a}_1 = (1, 1, 0)^T$, $\mathbf{a}_2 = (1, 0, 1)^T$.

```
import numpy as np

a1 = np.array([1.0, 1.0, 0.0]) # shape (3,)
a2 = np.array([1.0, 0.0, 1.0]) # shape (3,)

# Step 1: normalize a1
q1 = a1 / np.linalg.norm(a1) # shape (3,)

# Step 2: subtract projection of a2 onto q1, then normalize
v2 = a2 - float(q1 @ a2) * q1 # shape (3,)
q2 = v2 / np.linalg.norm(v2) # shape (3,)

print(f"q1 = {q1.round(6)}")
print(f"q2 = {q2.round(6)}")
print(f"\nq1 . q2 = {float(q1 @ q2):.2e} (orthogonal)")
print(f"||q1|| = {np.linalg.norm(q1):.6f} (unit)")
print(f"||q2|| = {np.linalg.norm(q2):.6f} (unit)")
```

```
q1 = [0.707107 0.707107 0.      ]
q2 = [ 0.408248 -0.408248  0.816497]
```

```
q1 . q2 = 1.11e-16  (orthogonal)
||q1||  = 1.000000  (unit)
||q2||  = 1.000000  (unit)
```

Solution. Step 1: $\mathbf{q}_1 = \mathbf{a}_1 / \|\mathbf{a}_1\| = (1, 1, 0)^T / \sqrt{2}$.

Step 2: $r_{12} = \mathbf{q}_1 \cdot \mathbf{a}_2 = (1 \cdot 1 + 1 \cdot 0 + 0 \cdot 1) / \sqrt{2} = 1 / \sqrt{2}$. Residual: $\mathbf{v}_2 = \mathbf{a}_2 - \frac{1}{\sqrt{2}} \mathbf{q}_1 = (1, 0, 1)^T - \frac{1}{2}(1, 1, 0)^T = (1/2, -1/2, 1)^T$. Normalized: $\mathbf{q}_2 = \mathbf{v}_2 / \|\mathbf{v}_2\|$ with $\|\mathbf{v}_2\| = \sqrt{1/4 + 1/4 + 1} = \sqrt{3/2}$.

Exercise 16.4.3 – Gram-Schmidt span preservation

Show that $\mathbf{q}_j \in \text{span}\{\mathbf{a}_1, \dots, \mathbf{a}_j\}$ for each j .

Solution. By induction. Base case $j = 1$: $\mathbf{q}_1 = \mathbf{a}_1 / \|\mathbf{a}_1\|$, which is a scalar multiple of $\mathbf{a}_1 \in \text{span}\{\mathbf{a}_1\}$. ✓

Inductive step: assume $\mathbf{q}_i \in \text{span}\{\mathbf{a}_1, \dots, \mathbf{a}_i\}$ for all $i < j$. Then each projection $(\mathbf{q}_i \cdot \mathbf{a}_j) \mathbf{q}_i$ is in $\text{span}\{\mathbf{a}_1, \dots, \mathbf{a}_{j-1}\}$ by hypothesis. Therefore

$$\mathbf{v}_j = \mathbf{a}_j - \sum_{i < j} (\mathbf{q}_i \cdot \mathbf{a}_j) \mathbf{q}_i \in \text{span}\{\mathbf{a}_1, \dots, \mathbf{a}_j\},$$

and $\mathbf{q}_j = \mathbf{v}_j / \|\mathbf{v}_j\|$ inherits this membership. Combined with the symmetric argument (each $\mathbf{a}_j$ can be reconstructed from $\mathbf{q}_1, \dots, \mathbf{q}_j$), the two spans are equal. □

Exercise 16.5.1 – QR factorization by hand

Factor $A = \begin{bmatrix} 1 & 1 \\ 1 & 0 \\ 0 & 1 \end{bmatrix}$ by Gram-Schmidt.

```
import numpy as np

A = np.array([[1.0, 1.0],
              [1.0, 0.0],
              [0.0, 1.0]])          # shape (3, 2)

a1 = A[:, 0]                      # shape (3,)
```

```

a2 = A[:, 1]                                # shape (3,)

# Step 1
q1 = a1 / np.linalg.norm(a1)                # shape (3,)
r11 = np.linalg.norm(a1)

# Step 2
r12 = float(q1 @ a2)
v2 = a2 - r12 * q1                          # shape (3,)
r22 = np.linalg.norm(v2)
q2 = v2 / r22                               # shape (3,)

Q = np.column_stack([q1, q2])               # shape (3, 2)
R = np.array([[r11, r12],
               [0.0, r22]])                  # shape (2, 2)

print("Q:")
print(Q.round(6))
print("\nR:")
print(R.round(6))
print(f"\nA == Q R: {np.allclose(A, Q @ R)}")
print(f"Q^T Q == I_2: {np.allclose(Q.T @ Q, np.eye(2))}")
print(f"R upper triangular, positive diagonal: r11={r11:.4f}, r22={r22:.4f}")

```

```

Q:
[[ 0.707107  0.408248]
 [ 0.707107 -0.408248]
 [ 0.         0.816497]]

```

```

R:
[[1.414214  0.707107]
 [0.         1.224745]]

```

```

A == Q R: True
Q^T Q == I_2: True
R upper triangular, positive diagonal: r11=1.4142, r22=1.2247

```

Solution. $\mathbf{q}_1 = (1, 1, 0)^T / \sqrt{2}$, $r_{11} = \sqrt{2}$. $r_{12} = \mathbf{q}_1 \cdot \mathbf{a}_2 = 1/\sqrt{2}$. $\mathbf{v}_2 = (1, 0, 1)^T - \frac{1}{2}(1, 1, 0)^T = (1/2, -1/2, 1)^T$, $r_{22} = \sqrt{3/2}$, $\mathbf{q}_2 = \mathbf{v}_2 / r_{22}$.

$$R = \begin{bmatrix} \sqrt{2} & 1/\sqrt{2} \\ 0 & \sqrt{3/2} \end{bmatrix}.$$

Both diagonal entries are positive, confirming uniqueness.

Exercise 16.5.3 – QR of an upper triangular matrix

```

import numpy as np

A = np.array([[2.0, 1.0, 0.0],
              [0.0, 2.0, 1.0],
              [0.0, 0.0, 2.0]])    # shape (3, 3)

Q, R = np.linalg.qr(A, mode='reduced')    # Q shape (3, 3), R shape (3, 3)

print("Q:")
print(Q.round(6))
print("\nR:")
print(R.round(6))
print(f"\nQ == I_3: {np.allclose(Q, np.eye(3))}")
print(f"R == A: {np.allclose(R, A)}")
print(f"\ndiag(R): {np.diag(R).round(4)} (all positive)")

```

```

Q:
[[ 1.  0.  0.]
 [-0.  1.  0.]
 [-0. -0.  1.]]

```

```

R:
[[2. 1. 0.]
 [0. 2. 1.]
 [0. 0. 2.]]

```

```

Q == I_3: True
R == A: True

```

```

diag(R): [2. 2. 2.] (all positive)

```

Solution. If A is already upper triangular with positive diagonal entries, then its columns satisfy the Gram-Schmidt conditions: $\mathbf{a}_1$ is already along $\mathbf{e}_1$'s direction (no component outside span), and each $\mathbf{a}_j$ has no component along $\mathbf{a}_{j+1}, \dots$ (upper triangular means the previous $\mathbf{q}_i$ projections are already encoded). The algorithm produces $\mathbf{q}_j = \mathbf{e}_j$ normalized, which for a full-rank triangular matrix collapses to $Q = I_3$, $R = A$. The positive diagonal requirement is met because the diagonal entries of A are $2 > 0$.

Projections, Pseudoinverse, and Least Squares

Orthogonal Projection onto a Subspace

The question that launches least squares: given $A\mathbf{x} = \mathbf{b}$ with no exact solution, what vector in $\text{Col}(A)$ is *closest* to $\mathbf{b}$? The answer is the **orthogonal projection** of $\mathbf{b}$ onto $\text{Col}(A)$, and the matrix that computes it is the **projection matrix** P .

Why Exact Solutions Often Do Not Exist

$A\mathbf{x} = \mathbf{b}$ is solvable only when $\mathbf{b} \in \text{Col}(A)$. In every real measurement setting – sensor arrays, camera calibration, curve fitting – the number of measurements exceeds the number of unknowns and noise pushes $\mathbf{b}$ outside $\text{Col}(A)$. The best achievable result is a vector $A\hat{\mathbf{x}} \in \text{Col}(A)$ that minimizes $\|\mathbf{b} - A\mathbf{x}\|_2$.

Deriving the Projection Matrix

Let $A \in \mathbb{R}^{m \times n}$ have full column rank ($m > n$, $\text{rank}(A) = n$). We seek $\hat{\mathbf{x}} = \arg \min_{\mathbf{x}} \|\mathbf{b} - A\mathbf{x}\|_2^2$.

The geometric requirement: the error vector $\mathbf{e} = \mathbf{b} - A\hat{\mathbf{x}}$ must be **orthogonal to every column of A** (otherwise we could reduce the error further by moving along a column direction):

$$A^T \mathbf{e} = \mathbf{0} \implies A^T (\mathbf{b} - A\hat{\mathbf{x}}) = \mathbf{0} \implies A^T A \hat{\mathbf{x}} = A^T \mathbf{b}.$$

Since A has full column rank, $A^T A \in \mathbb{R}^{n \times n}$ is invertible:

$$\hat{\mathbf{x}} = (A^T A)^{-1} A^T \mathbf{b}.$$

The **projection** of $\mathbf{b}$ onto $\text{Col}(A)$ is then:

$$\boxed{\hat{\mathbf{b}} = A\hat{\mathbf{x}} = A(A^T A)^{-1} A^T \mathbf{b} = P\mathbf{b}}, \quad P = A(A^T A)^{-1} A^T \in \mathbb{R}^{m \times m}.$$

In statistics, P is called the **hat matrix** because it puts a hat on $\mathbf{b}$: $\hat{\mathbf{b}} = P\mathbf{b}$.

Key Properties

Property	Statement	Why
Symmetric	$P^T = P$	$(A(A^T A)^{-1} A^T)^T = A((A^T A)^{-1})^T A^T = A(A^T A)^{-1} A^T$
Idempotent	$P^2 = P$	Projecting an already-projected vector changes nothing
Eigenvalues	only 0 or 1	$P^2 \mathbf{v} = P \mathbf{v}$ implies $\lambda^2 = \lambda$
Rank	n	The column space of P is $\text{Col}(A)$
Complement	$I - P$ is also a projector	Projects onto $\text{Col}(A)^\perp$ (left null space of A)

Idempotency proof:

$$P^2 = A(A^T A)^{-1} \underbrace{A^T A(A^T A)^{-1}}_{=I} A^T = A(A^T A)^{-1} A^T = P.$$

The vector $\mathbf{b} - P\mathbf{b} = (I - P)\mathbf{b}$ is the component of $\mathbf{b}$ orthogonal to $\text{Col}(A)$ – it lives in the left null space $\text{Null}(A^T)$.

Projection onto a Single Vector

The special case $A = \mathbf{a} \in \mathbb{R}^m$ (a single column) gives a rank-1 projector:

$$P = \frac{\mathbf{a}\mathbf{a}^T}{\mathbf{a}^T \mathbf{a}}, \quad \hat{\mathbf{b}} = \frac{\mathbf{a}^T \mathbf{b}}{\mathbf{a}^T \mathbf{a}} \mathbf{a}.$$

This is the formula from §16.1 written as a matrix operation.

Geometric Visualization

```

import numpy as np
import matplotlib.pyplot as plt
import matplotlib.patches as mpatches
from matplotlib.patches import FancyArrowPatch

rng = np.random.default_rng(17)

fig, axes = plt.subplots(1, 2, figsize=(12, 5))

# --- Left: projection in 2D onto a line ---
ax = axes[0]
a = np.array([2.0, 1.0])          # shape (2,) -- column direction
b = np.array([1.0, 3.0])          # shape (2,) -- vector to project

P1 = np.outer(a, a) / float(a @ a) # shape (2, 2) -- rank-1 projector
b_hat = P1 @ b                     # shape (2,) -- projection
e = b - b_hat                      # shape (2,) -- error (orthogonal to a)

# Line through origin along a
t = np.linspace(-0.5, 2.2, 100)
ax.plot(t * a[0], t * a[1], color='#333333', lw=1.2, zorder=1, label='Col(A)')

ax.annotate('', xy=b, xytext=np.array([0, 0]),
             arrowprops=dict(arrowstyle='->', color='#4a90d9', lw=2))
ax.annotate('', xy=b_hat, xytext=np.array([0, 0]),
             arrowprops=dict(arrowstyle='->', color='#2ecc71', lw=2))
ax.annotate('', xy=b, xytext=b_hat,
             arrowprops=dict(arrowstyle='->', color='tomato', lw=1.8, linestyle='dashed'))

# right-angle mark at b_hat
s = 0.15
v1 = a / np.linalg.norm(a)
v2 = e / np.linalg.norm(e)
corner = b_hat + s * (v1 + v2)
sq = plt.Polygon([b_hat + s * v1, corner, b_hat + s * v2],
                 fill=False, edgecolor='#333333', lw=0.8)
ax.add_patch(sq)

ax.text(b[0] + 0.1, b[1], r'$\mathbf{b}$', fontsize=13, color='#4a90d9')
ax.text(b_hat[0] - 0.05, b_hat[1] - 0.28, r'$\hat{\mathbf{b}} = P\mathbf{b}$',
       fontsize=11, color='#2ecc71')
ax.text((b[0] + b_hat[0]) / 2 + 0.1, (b[1] + b_hat[1]) / 2,
       r'$\mathbf{e} = \mathbf{b} - P\mathbf{b}$', fontsize=10, color='tomato')

ax.set_xlim(-0.3, 3.0)
ax.set_ylim(-0.3, 3.5)
ax.axhline(0, color='#333333', lw=0.4, alpha=0.5)

```

```

ax.axvline(0, color='#333333', lw=0.4, alpha=0.5)
ax.set_aspect('equal')
ax.set_title('Projection onto a line (rank 1)', fontsize=12)
ax.grid(alpha=0.2)

# --- Right: hat matrix diagnostics ---
ax2 = axes[1]
m, n = 30, 2                                # overdetermined system
A = np.column_stack([np.ones(m),
                     np.linspace(0, 4, m)]) # shape (30, 2) -- design matrix
b_vec = 1.5 * A[:, 1] + 0.5 + rng.standard_normal(m) # shape (30,) -- noisy data

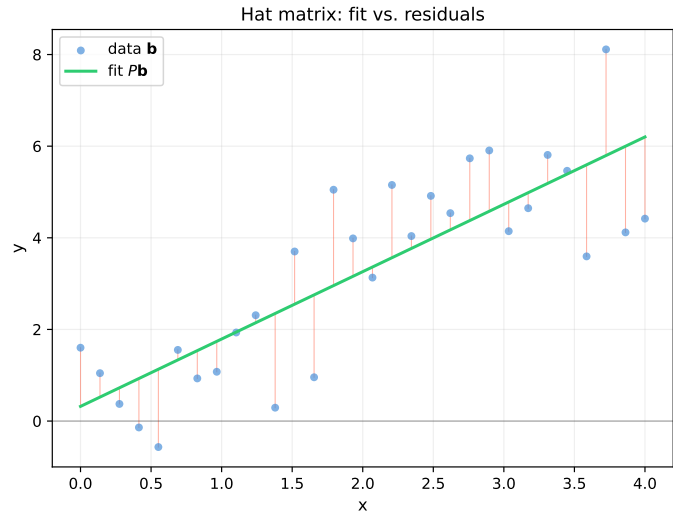
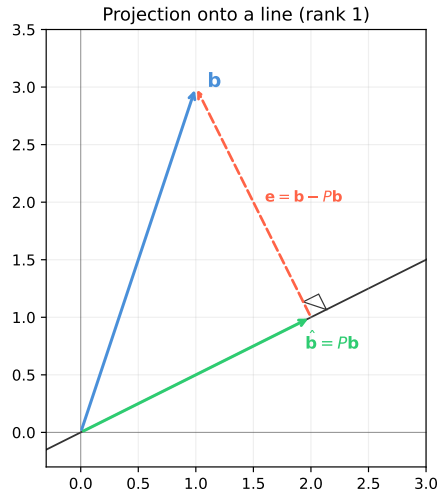
P2 = A @ np.linalg.solve(A.T @ A, A.T) # shape (30, 30)
b_hat2 = P2 @ b_vec                      # shape (30,)
e2 = b_vec - b_hat2                     # shape (30,)

x_vals = A[:, 1]
ax2.scatter(x_vals, b_vec, s=18, color='#4a90d9', alpha=0.7, label=r'data  $\mathbf{b}$ ', zorder=1)
ax2.plot(x_vals, b_hat2, color='#2ecc71', lw=2, label=r'fit  $P\mathbf{b}$ ', zorder=4)
for xi, yi, yhi in zip(x_vals, b_vec, b_hat2):
    ax2.plot([xi, xi], [yi, yhi], color='tomato', lw=0.7, alpha=0.5)

ax2.set_xlabel('x', fontsize=11)
ax2.set_ylabel('y', fontsize=11)
ax2.set_title('Hat matrix: fit vs. residuals', fontsize=12)
ax2.legend(fontsize=10)
ax2.grid(alpha=0.2)
ax2.axhline(0, color='#333333', lw=0.4, alpha=0.5)

fig.tight_layout()
plt.savefig('ch17-least-squares/fig-projection.png', dpi=150, bbox_inches='tight')
plt.show()
print(f"P symmetric: {np.allclose(P2, P2.T)}")
print(f"P idempotent: {np.allclose(P2 @ P2, P2)}")
print(f"rank(P): {np.linalg.matrix_rank(P2)} (should be {n})")
print(f"e orthogonal to Col(A): max |A^T e| = {np.abs(A.T @ e2).max():.2e}")

```



P symmetric: True
P idempotent: True
rank(P): 2 (should be 2)
e orthogonal to Col(A): max $|A^T e| = 6.66e-15$

Projection onto a 2-Column Subspace

```
import numpy as np

rng = np.random.default_rng(171)

# Subspace spanned by two columns in R^5
A = rng.standard_normal((5, 2))      # shape (5, 2)
b = rng.standard_normal(5)           # shape (5,)

# Projection matrix
AtA_inv = np.linalg.inv(A.T @ A)      # shape (2, 2)
P = A @ AtA_inv @ A.T                # shape (5, 5)

b_hat = P @ b                        # shape (5,)
e = b - b_hat                        # shape (5,)

print(f"P symmetric: {np.allclose(P, P.T)}")
print(f"P^2 == P: {np.allclose(P @ P, P)}")
print(f"rank(P): {np.linalg.matrix_rank(P)} (should be 2)")
print(f"e in Null(A^T): max |A^T e| = {np.abs(A.T @ e).max():.2e} (orthogonality check)")
print(f"\n||b||^2 = ||b_hat||^2 + ||e||^2: ")
```

```
f"{np.linalg.norm(b)**2:.6f} vs "
f"{np.linalg.norm(b_hat)**2 + np.linalg.norm(e)**2:.6f} (Pythagoras)"
```

```
P symmetric:    True
P^2 == P:      True
rank(P):       2 (should be 2)
e in Null(A^T): max |A^T e| = 1.25e-16 (orthogonality check)

||b||^2 = ||b_hat||^2 + ||e||^2: 2.755594 vs 2.755594 (Pythagoras)
```

The Pythagorean identity $\|\mathbf{b}\|^2 = \|\hat{\mathbf{b}}\|^2 + \|\mathbf{e}\|^2$ holds because $\hat{\mathbf{b}} \perp \mathbf{e}$.

When $A^T A$ Is Not Invertible

If A does not have full column rank, $(A^T A)^{-1}$ does not exist and the formula $P = A(A^T A)^{-1} A^T$ breaks down. In that case:

- The projection still exists (every subspace has a unique orthogonal projector).
- But $\hat{\mathbf{x}}$ is not unique – infinitely many $\hat{\mathbf{x}}$ achieve the same minimum residual.
- The numerically safe approach is to use the **pseudoinverse** or QR factorization (§17.2, §17.5).

Exercises

17.1.1 Let $\mathbf{a} = (1, 2, 2)^T$ in $\mathbb{R}^3$. Write down the 3×3 projection matrix P onto $\text{span}\{\mathbf{a}\}$. Verify $P^2 = P$ and $P^T = P$. What is $\text{rank}(P)$?

17.1.2 Let $A = \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix}$. Project $\mathbf{b} = (3, 1, -2)^T$ onto $\text{Col}(A)$. Then compute $(I - P)\mathbf{b}$ and verify it is orthogonal to $\mathbf{a}$.

17.1.3 Suppose A has orthonormal columns ($A^T A = I$). Show that $P = AA^T$. What does this say about the hat matrix when A comes from QR factorization?

17.1.4 Can the residual $\mathbf{e} = \mathbf{b} - P\mathbf{b}$ ever be zero for a noisy overdetermined system? Under what condition on $\mathbf{b}$ does $P\mathbf{b} = \mathbf{b}$ hold exactly?

The Pseudoinverse: Moore–Penrose Conditions

When A is not square, or is square but singular, the ordinary inverse A^{-1} does not exist. The **Moore–Penrose pseudoinverse** A^+ generalises the inverse to any matrix while still yielding the minimum-norm least-squares solution. It is defined not by a formula but by four algebraic conditions – and the SVD shows exactly how to compute it.

Four Conditions That Define A^+ Uniquely

Let $A \in \mathbb{R}^{m \times n}$. The pseudoinverse $A^+ \in \mathbb{R}^{n \times m}$ is the unique matrix satisfying all four **Moore–Penrose conditions**:

Condition	Equation	What It Says
(1)	$AA^+A = A$	A^+ inverts A where possible
(2)	$A^+AA^+ = A^+$	A inverts A^+ where possible
(3) Symmetric	$(AA^+)^T = AA^+$	AA^+ is an orthogonal projector
(4) Symmetric	$(A^+A)^T = A^+A$	A^+A is an orthogonal projector

Together these conditions determine A^+ uniquely. The projector AA^+ projects onto $\text{Col}(A)$; the projector A^+A projects onto $\text{Row}(A)$.

When A is invertible, $A^+ = A^{-1}$ and all four conditions reduce to $AA^{-1} = I$.

Computing A^+ via the SVD

The SVD provides the clearest route to A^+ . Write $A = U\Sigma V^T$ where $U \in \mathbb{R}^{m \times m}$, $\Sigma \in \mathbb{R}^{m \times n}$, and $V \in \mathbb{R}^{n \times n}$. Then:

$$A^+ = V\Sigma^+U^T,$$

where Σ^+ is obtained from Σ by **reciprocating each nonzero singular value** and transposing. Zero singular values stay zero – the pseudoinverse does not invert the null space.

Why does this work? Substituting into condition (1): $AA^+A = U\Sigma V^T \cdot V\Sigma^+U^T \cdot U\Sigma V^T = U\Sigma\Sigma^+\Sigma V^T = U\Sigma V^T = A$, since $\Sigma\Sigma^+\Sigma = \Sigma$ on the diagonal. The remaining three conditions follow similarly.

The Minimum-Norm Least-Squares Solution

Given $A\mathbf{x} \approx \mathbf{b}$, the vector $\hat{\mathbf{x}} = A^+\mathbf{b}$ simultaneously:

1. **Minimises the residual:** $\|A\hat{\mathbf{x}} - \mathbf{b}\|_2$ is as small as possible.
2. **Has minimum norm:** among all minimisers, $A^+\mathbf{b}$ has the smallest $\|\cdot\|_2$.

This two-level optimality is what makes A^+ the canonical generalised inverse.

Numerical Rank and the Tolerance Cutoff

In finite precision, “zero” singular values appear as very small numbers. A standard cutoff is:

$$\sigma_i \text{ treated as zero if } \sigma_i < \varepsilon_{\text{mach}} \cdot \sigma_1 \cdot \max(m, n).$$

This is used internally by `np.linalg.lstsq` and `np.linalg.matrix_rank`. Naively inverting all singular values amplifies noise in directions corresponding to near-zero σ_i .

```
import numpy as np

rng = np.random.default_rng(22)

# Full-rank case
m, n = 5, 3
A = rng.standard_normal((m, n))    # shape (5, 3)
b = rng.standard_normal(m)         # shape (5,)

A_plus = np.linalg.pinv(A)          # shape (3, 5)

c1 = np.allclose(A @ A_plus @ A, A)
c2 = np.allclose(A_plus @ A @ A_plus, A_plus)
c3 = np.allclose((A @ A_plus).T, A @ A_plus)
c4 = np.allclose((A_plus @ A).T, A_plus @ A)

print("=== Full-rank (5x3) ===")
print(f"AA+A == A:      {c1}")
print(f"A+AA+ == A+:    {c2}")
print(f"(AA+)^T == AA+:  {c3}")
print(f"(A+A)^T == A+A:   {c4}")

x_hat = A_plus @ b                  # shape (3,)
x_ls, _, _, _ = np.linalg.lstsq(A, b, rcond=None) # shape (3,)
print(f"\n||x_hat - lstsq||: {np.linalg.norm(x_hat - x_ls):.2e} (should be ~0)")

# Rank-deficient case
```

```

print("\n=== Rank-deficient (5x3, rank 2) ===")
A2 = np.column_stack([A[:, 0], A[:, 1], A[:, 0] + A[:, 1]]) # shape (5, 3), rank 2
A_plus2 = np.linalg.pinv(A2) # shape (3, 5)

x_hat2 = A_plus2 @ b # shape (3,) -- minimum-norm minimiser
null_vec = np.array([1.0, 1.0, -1.0]) # shape (3,) -- in Null(A2)
x_perturbed = x_hat2 + 0.5 * null_vec # shape (3,) -- different solution, same residual

print(f"rank(A2): {np.linalg.matrix_rank(A2)}")
print(f"Residual (A+b): {np.linalg.norm(A2 @ x_hat2 - b):.6f}")
print(f"Residual (perturbed): {np.linalg.norm(A2 @ x_perturbed - b):.6f} (same)")
print(f"||x_hat2||: {np.linalg.norm(x_hat2):.6f}")
print(f"||x_perturbed||: {np.linalg.norm(x_perturbed):.6f} (larger)")

=== Full-rank (5x3) ===
AA+A == A: True
A+AA+ == A+: True
(AA+)^T == AA+: True
(A+A)^T == A+A: True

||x_hat - lstsq||: 2.65e-16 (should be ~0)

=== Rank-deficient (5x3, rank 2) ===
rank(A2): 2
Residual (A+b): 1.612278
Residual (perturbed): 1.612278 (same)
||x_hat2||: 0.769781
||x_perturbed||: 1.158690 (larger)

```

SVD-Based Construction

```

import numpy as np

rng = np.random.default_rng(23)

A = rng.standard_normal((4, 3)) # shape (4, 3)
b = rng.standard_normal(4) # shape (4,)

U, s, Vt = np.linalg.svd(A, full_matrices=False) # U (4,3), s (3,), Vt (3,3)

print(f"Singular values: {s.round(4)}")

```

```

# Build A+ manually: V * diag(1/s) * U^T
# Vt.T has shape (3,3), multiply each row by 1/s_i, then multiply by U^T
A_plus_manual = (Vt.T * (1.0 / s)) @ U.T # shape (3, 4)

A_plus_np = np.linalg.pinv(A) # shape (3, 4)
print(f"Manual A+ matches np.linalg.pinv: {np.allclose(A_plus_manual, A_plus_np)}")

P_col = A @ A_plus_manual # shape (4, 4) -- projector onto Col(A)
P_row = A_plus_manual @ A # shape (3, 3) -- projector onto Row(A)

print(f"AA+ idempotent: {np.allclose(P_col @ P_col, P_col)}")
print(f"A+A idempotent: {np.allclose(P_row @ P_row, P_row)}")
print(f"rank(AA+): {np.linalg.matrix_rank(P_col)} (= rank(A))")

```

```

Singular values: [3.4126 1.145 0.6553]
Manual A+ matches np.linalg.pinv: True
AA+ idempotent: True
A+A idempotent: True
rank(AA+):      3 (= rank(A))

```

Exercises

17.2.1 Let $A = \begin{bmatrix} 1 & 0 \\ 0 & 2 \\ 0 & 0 \end{bmatrix}$. Compute A^+ directly from the SVD (singular values are 1 and 2).

Verify all four Moore–Penrose conditions by hand.

17.2.2 Show that if A is square and invertible, then $A^+ = A^{-1}$. (Use the four conditions.)

17.2.3 Let $A = \mathbf{u}\sigma\mathbf{v}^T$ be a rank-1 matrix. Show that $A^+ = \mathbf{v}(1/\sigma)\mathbf{u}^T$. What happens if $\sigma = 0$?

17.2.4 In the rank-deficient example above, verify numerically that $A^+\mathbf{b}$ has smaller norm than $\mathbf{x}_{\text{perturbed}}$ while achieving the same residual. Explain geometrically why $A^+\mathbf{b}$ lies in $\text{Row}(A)$ and the perturbation lies in $\text{Null}(A)$.

Left vs. Right Pseudoinverse

The pseudoinverse collapses two qualitatively different situations into one formula. When the system is **overdetermined** ($m > n$, more equations than unknowns) the pseudoinverse acts as a *left* inverse and minimises residual. When the system is **underdetermined** ($m < n$, fewer equations than unknowns) it acts as a *right* inverse and finds the minimum-norm solution. Understanding the distinction is essential for choosing the right solver in practice.

Overdetermined Systems ($m > n$, Full Column Rank)

When $A \in \mathbb{R}^{m \times n}$ has $m > n$ and $\text{rank}(A) = n$, there is no exact solution to $A\mathbf{x} = \mathbf{b}$ generically. The pseudoinverse reduces to the **left inverse**:

$$A^+ = (A^T A)^{-1} A^T \in \mathbb{R}^{n \times m}, \quad A^+ A = I_n.$$

It is a left inverse because applying A^+ to the left recovers $\mathbf{x}$ from $A\mathbf{x}$. But $AA^+ \neq I_m$ – projecting onto $\text{Col}(A)$ is not the identity when $\text{Col}(A)$ is a proper subspace of $\mathbb{R}^m$.

The solution $\hat{\mathbf{x}} = A^+ \mathbf{b} = (A^T A)^{-1} A^T \mathbf{b}$ **minimises the residual** $\|A\mathbf{x} - \mathbf{b}\|_2$.

Applications: curve fitting, camera calibration, regression – any problem with more measurements than parameters.

Underdetermined Systems ($m < n$, Full Row Rank)

When $A \in \mathbb{R}^{m \times n}$ has $m < n$ and $\text{rank}(A) = m$, the system $A\mathbf{x} = \mathbf{b}$ is consistent but has infinitely many solutions. The pseudoinverse reduces to the **right inverse**:

$$A^+ = A^T (AA^T)^{-1} \in \mathbb{R}^{n \times m}, \quad AA^+ = I_m.$$

The solution $\hat{\mathbf{x}} = A^+ \mathbf{b}$ satisfies $A\hat{\mathbf{x}} = \mathbf{b}$ exactly and lies in $\text{Row}(A)$ – the null-space component is zero, which is what **minimum norm** means.

Applications: redundant robot arm inverse kinematics (7-DoF arm, 6-DoF task space), underdetermined linear programs.

Summary

	Overdetermined ($m > n$)	Underdetermined ($m < n$)
A^+	$(A^T A)^{-1} A^T$	$A^T (AA^T)^{-1}$
$A^+ A$	I_n	projector onto $\text{Row}(A)$
AA^+	projector onto $\text{Col}(A)$	I_m
Solution property	minimum residual	minimum norm, exact

Overdetermined Example: Linear Regression

```
import numpy as np

rng = np.random.default_rng(31)

m, n = 20, 2
t = np.linspace(0, 5, m) # shape (20,)
A = np.column_stack([np.ones(m), t]) # shape (20, 2) -- design matrix [1, t]

x_true = np.array([1.5, 0.8]) # shape (2,) -- true [intercept, slope]
b = A @ x_true + 0.4 * rng.standard_normal(m) # shape (20,) -- noisy measurements

# Left pseudoinverse
A_plus_left = np.linalg.inv(A.T @ A) @ A.T # shape (2, 20)
x_hat = A_plus_left @ b # shape (2,)

print("=== Overdetermined (20x2) ===")
print(f"A+A == I_2: {np.allclose(A_plus_left @ A, np.eye(n))}")
print(f"AA+ == I_20: {np.allclose(A @ A_plus_left, np.eye(m))} (not identity)")
print(f"\nx_true: {x_true}")
print(f"x_hat: {x_hat.round(4)}")
print(f"Residual: {np.linalg.norm(A @ x_hat - b):.4f}")
print(f"rank(AA+): {np.linalg.matrix_rank(A @ A_plus_left)} (= n = 2, not m = 20)")

=== Overdetermined (20x2) ===
A+A == I_2: True
AA+ == I_20: False (not identity)

x_true: [1.5 0.8]
x_hat: [1.6641 0.756 ]
Residual: 1.4941
rank(AA+): 2 (= n = 2, not m = 20)
```

Underdetermined Example: Redundant Arm IK

A planar arm with 5 joints controlling a 2-DoF end-effector: 3 redundant degrees of freedom. The pseudoinverse picks the minimum joint-velocity solution.

```
import numpy as np

rng = np.random.default_rng(32)
```

```

m, n = 2, 5                                # task dim < joint dim
J = rng.standard_normal((m, n))            # shape (2, 5) -- Jacobian
p_dot = rng.standard_normal(m)             # shape (2,) -- desired task velocity

# Right pseudoinverse
J_plus = J.T @ np.linalg.inv(J @ J.T)      # shape (5, 2)

q_dot = J_plus @ p_dot                     # shape (5,) -- min-norm joint velocity

print("=== Underdetermined (2x5) ===")
print(f"JJ+ == I_2: {np.allclose(J @ J_plus, np.eye(m))}")
print(f"J+J == I_5: {np.allclose(J_plus @ J, np.eye(n))} (not identity)")
print(f"\nTask error ||J q_dot - p_dot||: {np.linalg.norm(J @ q_dot - p_dot):.2e}")
print(f"||q_dot||: {np.linalg.norm(q_dot):.4f} (minimum norm)")

# Any null-space addition keeps task exact but increases norm
null_vec = np.array([1.0, -1.0, 0.5, -0.5, 0.2]) # shape (5,)
null_part = null_vec - J_plus @ J @ null_vec      # shape (5,) -- Null(J) component
q_perturbed = q_dot + null_part                   # shape (5,)

print(f"\nPerturbed task error: {np.linalg.norm(J @ q_perturbed - p_dot):.2e}")
print(f"||q_perturbed||: {np.linalg.norm(q_perturbed):.4f} (larger)")

```

```

=== Underdetermined (2x5) ===
JJ+ == I_2: True
J+J == I_5: False (not identity)

Task error ||J q_dot - p_dot||: 5.55e-17
||q_dot||: 0.4141 (minimum norm)

Perturbed task error: 7.30e-16
||q_perturbed||: 1.4386 (larger)

```

Exercises

17.3.1 Let $A = \begin{bmatrix} 1 & 2 & 3 \end{bmatrix} \in \mathbb{R}^{1 \times 3}$. Compute the right pseudoinverse $A^+ = A^T(AA^T)^{-1}$ and find the minimum-norm solution to $A\mathbf{x} = 1$. Verify $\mathbf{x}^* \in \text{Row}(A)$.

17.3.2 For the overdetermined case, prove that $A^+ = (A^T A)^{-1} A^T$ satisfies all four Moore–Penrose conditions when A has full column rank.

17.3.3 A 4-DoF planar arm controls a 2-DoF end-effector. Write the pseudoinverse IK formula $\dot{\mathbf{q}} = J^+ \dot{\mathbf{p}}$ and explain what happens when J loses rank at a kinematic singularity.

17.3.4 Show that for any A , the vector $A^+ \mathbf{b}$ is orthogonal to $\text{Null}(A)$. (Hint: $A^+ A$ is the orthogonal projector onto $\text{Row}(A)$ and $A^+ \mathbf{b} = A^+ A \cdot A^+ \mathbf{b}$.)

The Normal Equations

The **normal equations** $A^T A \hat{\mathbf{x}} = A^T \mathbf{b}$ are the classical formulation of least squares. They arise naturally from the orthogonality condition: the residual $\mathbf{e} = \mathbf{b} - A\hat{\mathbf{x}}$ must be orthogonal to every column of A . Understanding when $A^T A$ is invertible – and what goes wrong when it is not – is essential before reaching for a solver.

Geometric Derivation

The problem $\min_{\mathbf{x}} \|A\mathbf{x} - \mathbf{b}\|_2^2$ has a geometric interpretation: find the point in $\text{Col}(A)$ closest to $\mathbf{b}$. The closest point $\hat{\mathbf{b}} = A\hat{\mathbf{x}}$ is the orthogonal projection of $\mathbf{b}$ onto $\text{Col}(A)$, so the error $\mathbf{e} = \mathbf{b} - A\hat{\mathbf{x}}$ must lie in $\text{Col}(A)^\perp = \text{Null}(A^T)$:

$$A^T \mathbf{e} = \mathbf{0} \implies A^T (\mathbf{b} - A\hat{\mathbf{x}}) = \mathbf{0} \implies \boxed{A^T A \hat{\mathbf{x}} = A^T \mathbf{b}.}$$

This is the **normal equations** – the residual is “normal” (perpendicular) to the column space.

Algebraic Derivation via Calculus

Expanding the objective $f(\mathbf{x}) = \|A\mathbf{x} - \mathbf{b}\|_2^2 = \mathbf{x}^T A^T A \mathbf{x} - 2\mathbf{b}^T A \mathbf{x} + \mathbf{b}^T \mathbf{b}$:

$$\nabla_{\mathbf{x}} f = 2A^T A \mathbf{x} - 2A^T \mathbf{b} = \mathbf{0} \implies A^T A \hat{\mathbf{x}} = A^T \mathbf{b}.$$

The Hessian is $2A^T A$, which is positive semidefinite (and positive definite when A has full column rank), confirming this is a minimum.

When $A^T A$ Is Invertible

$A^T A \in \mathbb{R}^{n \times n}$ is invertible if and only if A has **full column rank** ($\text{rank}(A) = n$). In that case the unique solution is:

$$\hat{\mathbf{x}} = (A^T A)^{-1} A^T \mathbf{b}.$$

Full column rank requires $m \geq n$ and linearly independent columns. In practice this means the design matrix has no redundant features and enough measurements.

When $A^T A$ Is Singular

If A does not have full column rank:

- $A^T A$ is positive *semidefinite*: some eigenvalues are zero.
- The normal equations $A^T A \hat{\mathbf{x}} = A^T \mathbf{b}$ are still consistent (since $A^T \mathbf{b} \in \text{Col}(A^T) = \text{Row}(A) = \text{Col}(A^T A)$).
- But the solution is **not unique**: any $\hat{\mathbf{x}} + \mathbf{n}$ with $\mathbf{n} \in \text{Null}(A)$ is also a solution.
- The pseudoinverse picks the minimum-norm solution $\hat{\mathbf{x}} = A^+ \mathbf{b}$.

Causes: collinear features (e.g. temperature in Celsius and Kelvin in the same regression), rank-deficient design from insufficient distinct measurements, or a deliberately underdetermined model.

Condition Number and Numerical Concerns

Even when $A^T A$ is technically invertible, forming it explicitly can cause numerical problems. If $\text{cond}(A) = \sigma_{\max}/\sigma_{\min}$, then:

$$\text{cond}(A^T A) = \text{cond}(A)^2.$$

Squaring the condition number doubles the number of digits lost to floating-point error. For this reason, the normal equations are recommended only when $\text{cond}(A)$ is small and forming $A^T A$ explicitly is acceptable (§17.5).

```
import numpy as np

rng = np.random.default_rng(41)

# Well-conditioned case
m, n = 10, 3
A = rng.standard_normal((m, n))    # shape (10, 3)
b = rng.standard_normal(m)         # shape (10,)

# Normal equations: solve A^T A x = A^T b
AtA = A.T @ A                      # shape (3, 3)
Atb = A.T @ b                      # shape (3,)
x_ne = np.linalg.solve(AtA, Atb)   # shape (3,) -- normal equations solution

# Compare with lstsq
x_ls, _, _, _ = np.linalg.lstsq(A, b, rcond=None) # shape (3,)

sigma = np.linalg.svd(A, compute_uv=False) # shape (3,) -- singular values
cond_A = sigma[0] / sigma[-1]
```

```
print(f"cond(A):          {cond_A:.2f}")
print(f"cond(A^T A):    {np.linalg.cond(AtA):.2f}  (= cond(A)^2 approximately)")
print(f"||x_ne - lstsq||: {np.linalg.norm(x_ne - x_ls):.2e}  (agree well)")

# Ill-conditioned case: nearly collinear columns
A_ill = np.column_stack([A[:, 0],
                        A[:, 1],
                        A[:, 0] + 1e-6 * rng.standard_normal(m)]) # shape (10, 3)

sigma_ill = np.linalg.svd(A_ill, compute_uv=False) # shape (3,)
cond_ill = sigma_ill[0] / sigma_ill[-1]

x_ne_ill = np.linalg.solve(A_ill.T @ A_ill, A_ill.T @ b) # shape (3,)
x_ls_ill, _, _, _ = np.linalg.lstsq(A_ill, b, rcond=None) # shape (3,)

print(f"\nIll-conditioned:")
print(f"cond(A_ill):          {cond_ill:.2e}")
print(f"cond(A_ill^T A_ill): {np.linalg.cond(A_ill.T @ A_ill):.2e}  (squared!)")
print(f"||x_ne - lstsq||:      {np.linalg.norm(x_ne_ill - x_ls_ill):.2e}  (diverge!)")
```

```
cond(A):          1.84
cond(A^T A):    3.38  (= cond(A)^2 approximately)
||x_ne - lstsq||: 4.64e-16  (agree well)

Ill-conditioned:
cond(A_ill):          2.34e+06
cond(A_ill^T A_ill): 5.50e+12  (squared!)
||x_ne - lstsq||:      1.76e+01  (diverge!)
```

Visualising Orthogonality of the Residual

```
import numpy as np
import matplotlib.pyplot as plt

rng = np.random.default_rng(42)

m = 15
t = np.linspace(0, 3, m) # shape (15,)
A = np.column_stack([np.ones(m), t]) # shape (15, 2)
b = 2.0 + 1.2 * t + 0.5 * rng.standard_normal(m) # shape (15,)

# Solve normal equations
```

```

x_hat = np.linalg.solve(A.T @ A, A.T @ b)    # shape (2,)
b_hat = A @ x_hat                            # shape (15,)
e = b - b_hat                                # shape (15,)

fig, axes = plt.subplots(1, 2, figsize=(12, 4))

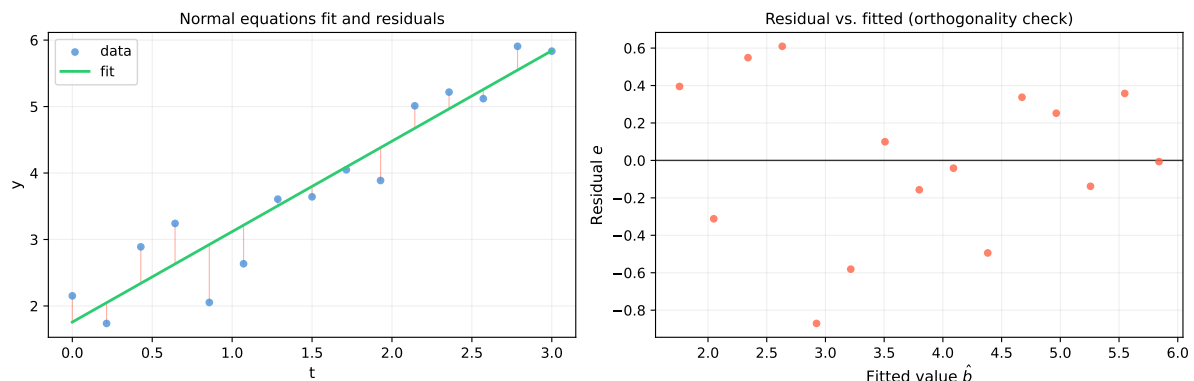
# Left: data and fit
ax = axes[0]
ax.scatter(t, b, s=20, color='#4a90d9', alpha=0.8, label='data', zorder=3)
ax.plot(t, b_hat, color='#2ecc71', lw=2, label='fit', zorder=4)
for xi, yi, yhi in zip(t, b, b_hat):
    ax.plot([xi, xi], [yi, yhi], color='tomato', lw=0.8, alpha=0.5)
ax.set_xlabel('t', fontsize=11)
ax.set_ylabel('y', fontsize=11)
ax.set_title('Normal equations fit and residuals', fontsize=11)
ax.legend(fontsize=10)
ax.grid(alpha=0.2)

# Right: residuals vs t and vs fitted values (should be uncorrelated)
ax2 = axes[1]
ax2.scatter(b_hat, e, s=20, color='tomato', alpha=0.8, zorder=3)
ax2.axhline(0, color='#333333', lw=1)
ax2.set_xlabel(r'Fitted value  $\hat{b}$ ', fontsize=11)
ax2.set_ylabel(r'Residual  $e$ ', fontsize=11)
ax2.set_title('Residual vs. fitted (orthogonality check)', fontsize=11)
ax2.grid(alpha=0.2)

fig.tight_layout()
plt.savefig('ch17-least-squares/fig-normal-equations.png', dpi=150, bbox_inches='tight')
plt.show()

# Verify orthogonality:  $A^T e$  should be zero
print(f" $A^T e$  (should be  $\sim 0$ ): {(A.T @ e).round(8)}")
print(f"Corr(b_hat, e): {np.corrcoef(b_hat, e)[0,1]:.2e} (should be  $\sim 0$ )")

```



```
A^T e (should be ~0): [0. 0.]  
Corr(b_hat, e): -3.00e-16 (should be ~0)
```

Exercises

17.4.1 Derive the normal equations by setting $\nabla_{\mathbf{x}} \|A\mathbf{x} - \mathbf{b}\|_2^2 = 0$. Identify the Hessian and explain why the critical point is a minimum (not a saddle) when A has full column rank.

17.4.2 Let $A \in \mathbb{R}^{3 \times 2}$ be the matrix with columns $(1, 1, 0)^T$ and $(0, 1, 1)^T$, and let $\mathbf{b} = (1, 2, 3)^T$. Solve the normal equations by hand to find $\hat{\mathbf{x}}$ and the projection $\hat{\mathbf{b}} = A\hat{\mathbf{x}}$.

17.4.3 Show that if $\mathbf{b} \in \text{Col}(A)$, then the normal equations give $A\hat{\mathbf{x}} = \mathbf{b}$ exactly and $\mathbf{e} = \mathbf{0}$.

17.4.4 Explain, using eigenvalues, why $\text{cond}(A^T A) = \text{cond}(A)^2$. In double precision, roughly how large can $\text{cond}(A)$ be before the normal equations lose all accuracy?

Solver Comparison: QR vs. SVD vs. Normal Equations

Three algorithms solve $\min_{\mathbf{x}} \|A\mathbf{x} - \mathbf{b}\|_2^2$: the normal equations, QR factorisation, and the SVD. They give the same answer on well-conditioned problems but diverge dramatically on ill-conditioned or rank-deficient ones. Knowing which to reach for – and why – separates reliable implementations from fragile ones.

The Three Approaches

Normal equations. Solve $A^T A \hat{\mathbf{x}} = A^T \mathbf{b}$ via Cholesky factorisation of $A^T A$. Cost: $O(mn^2 + n^3/3)$. Pros: simple, fast when $\text{cond}(A)$ is small. Cons: squares the condition number; fails or gives garbage when A is nearly rank-deficient.

QR factorisation. Factor $A = QR$ (Householder, §16.5), then solve $R\hat{\mathbf{x}} = Q^T \mathbf{b}$ by back-substitution. Cost: $O(2mn^2 - 2n^3/3)$. Pros: backward stable; does not square the condition number. Cons: cannot detect rank deficiency; breaks when R is singular.

SVD. Factor $A = U\Sigma V^T$, then $\hat{\mathbf{x}} = V\Sigma^+ U^T \mathbf{b}$. Cost: $O(mn^2 + n^3)$ (more expensive). Pros: handles rank deficiency exactly; exposes the complete numerical rank structure via singular values; most robust. Cons: slowest; overkill for well-conditioned problems.

Accuracy on an Ill-Conditioned Problem

```
import numpy as np

rng = np.random.default_rng(51)

m, n = 30, 5
# Build a matrix with a wide range of singular values
U, _ = np.linalg.qr(rng.standard_normal((m, m))) # shape (30, 30) -- random orthogonal
V, _ = np.linalg.qr(rng.standard_normal((n, n))) # shape (5, 5) -- random orthogonal
sigma = np.array([1e4, 1e2, 1e1, 1.0, 1e-4]) # shape (5,) -- cond = 1e8
Sigma = np.zeros((m, n)) # shape (30, 5)
Sigma[:n, :n] = np.diag(sigma)
A = U[:, :n] @ Sigma[:n, :] @ V.T # shape (30, 5) -- cond ~ 1e8

x_true = rng.standard_normal(n) # shape (5,)
b = A @ x_true + 1e-3 * rng.standard_normal(m) # shape (30,) -- small noise

print(f"cond(A): {np.linalg.cond(A):.2e}")
print(f"cond(A^T A): {np.linalg.cond(A.T @ A):.2e}")

# Method 1: normal equations
try:
    x_ne = np.linalg.solve(A.T @ A, A.T @ b) # shape (5,)
    err_ne = np.linalg.norm(x_ne - x_true)
except np.linalg.LinAlgError:
    err_ne = float('inf')

# Method 2: QR
Q, R = np.linalg.qr(A, mode='reduced') # Q (30,5), R (5,5)
x_qr = np.linalg.solve(R, Q.T @ b) # shape (5,)
err_qr = np.linalg.norm(x_qr - x_true)

# Method 3: SVD (pseudoinverse)
x_svd = np.linalg.pinv(A) @ b # shape (5,)
err_svd = np.linalg.norm(x_svd - x_true)

# Method 4: lstsq (uses SVD internally)
x_ls, _, _, _ = np.linalg.lstsq(A, b, rcond=None) # shape (5,)
err_ls = np.linalg.norm(x_ls - x_true)

print(f"\n||x_ne - x_true||: {err_ne:.2e}")
print(f"||x_qr - x_true||: {err_qr:.2e}")
print(f"||x_svd - x_true||: {err_svd:.2e}")
print(f"||x_ls - x_true||: {err_ls:.2e}")
print("\n(Normal eqs degrade; QR holds; SVD/lstsq most stable)")
```

```
cond(A): 1.00e+08
```

```
cond(AT A):    1.28e+16

||x_ne - x_true||:    6.12e+00
||x_qr - x_true||:    4.16e+00
||x_svd - x_true||:    4.16e+00
||x_ls - x_true||:    4.16e+00

(Normal eqs degrade; QR holds; SVD/lstsq most stable)
```

Rank-Deficient Case

When A loses rank, normal equations fail, QR produces an ill-conditioned R (and may error), but SVD handles the situation cleanly by zeroing out near-zero singular values.

```
import numpy as np

rng = np.random.default_rng(52)

m, n = 20, 4
A_full = rng.standard_normal((m, n))          # shape (20, 4)
# Make rank 3: fourth column = linear combination of first three
A = np.column_stack([A_full[:, :3],
                     A_full[:, 0] + 0.5 * A_full[:, 1]]) # shape (20, 4), rank 3

b = rng.standard_normal(m)                    # shape (20,)

# SVD approach handles rank deficiency
x_svd = np.linalg.pinv(A) @ b                  # shape (4,) -- min-norm least-squares
x_ls, _, rank_ls, sv = np.linalg.lstsq(A, b, rcond=None) # shape (4,)

print(f"rank(A): {np.linalg.matrix_rank(A)} (should be 3)")
print(f"lstsq detected rank: {rank_ls}")
print(f"Singular values: {sv.round(4)}")
print(f"\nResidual (svd): {np.linalg.norm(A @ x_svd - b):.6f}")
print(f"Residual (lstsq): {np.linalg.norm(A @ x_ls - b):.6f}")
print(f"||x_svd - x_ls||: {np.linalg.norm(x_svd - x_ls):.2e} (same min-norm solution)")

rank(A):          3 (should be 3)
lstsq detected rank: 3
Singular values:   [6.8709 4.4366 4.2115 0.    ]

Residual (svd):    3.973594
Residual (lstsq):  3.973594
||x_svd - x_ls||:  1.85e-16 (same min-norm solution)
```

Decision Guide

Situation	Recommended Solver
Well-conditioned, $m \gg n$, speed matters	Normal equations + Cholesky
General overdetermined, moderate $\text{cond}(A)$	QR (<code>np.linalg.lstsq</code>)
Rank-deficient or very ill-conditioned	SVD / <code>np.linalg.lstsq</code> with <code>rcond</code>
Real-time IK (small n , latency-critical)	QR or pre-computed pseudoinverse
Camera calibration (batch, need robustness)	SVD-based <code>cv2.calibrateCamera</code>
Large sparse A ($n > 10^5$)	Iterative: CG, LSQR (§17.8)

The rule of thumb: **use `np.linalg.lstsq` by default** (it uses LAPACK's `dgelsd`, SVD-based). Switch to QR for speed when you know A is well-conditioned. Never use normal equations when $\text{cond}(A) > 10^7$ in double precision.

Visualising the Effect of Ill-Conditioning

```
import numpy as np
import matplotlib.pyplot as plt

rng = np.random.default_rng(53)

# Sweep condition numbers and measure error amplification
cond_vals = np.logspace(1, 12, 20) # shape (20,)
err_ne = []
err_qr = []
err_svd = []

m, n = 20, 3

for cond_target in cond_vals:
    U, _ = np.linalg.qr(rng.standard_normal((m, m))) # shape (20, 20)
    V, _ = np.linalg.qr(rng.standard_normal((n, n))) # shape (3, 3)
    sv = np.array([1.0, 1.0 / np.sqrt(cond_target), 1.0 / cond_target]) # shape (3,)
    Sigma = np.zeros((m, n)) # shape (20, 3)
    Sigma[:n, :n] = np.diag(sv)
    A = U[:, :n] @ Sigma[:n, :] @ V.T # shape (20, 3)

    x_true = rng.standard_normal(n) # shape (3,)
    noise = 1e-10 * rng.standard_normal(m) # shape (20,)
```

```

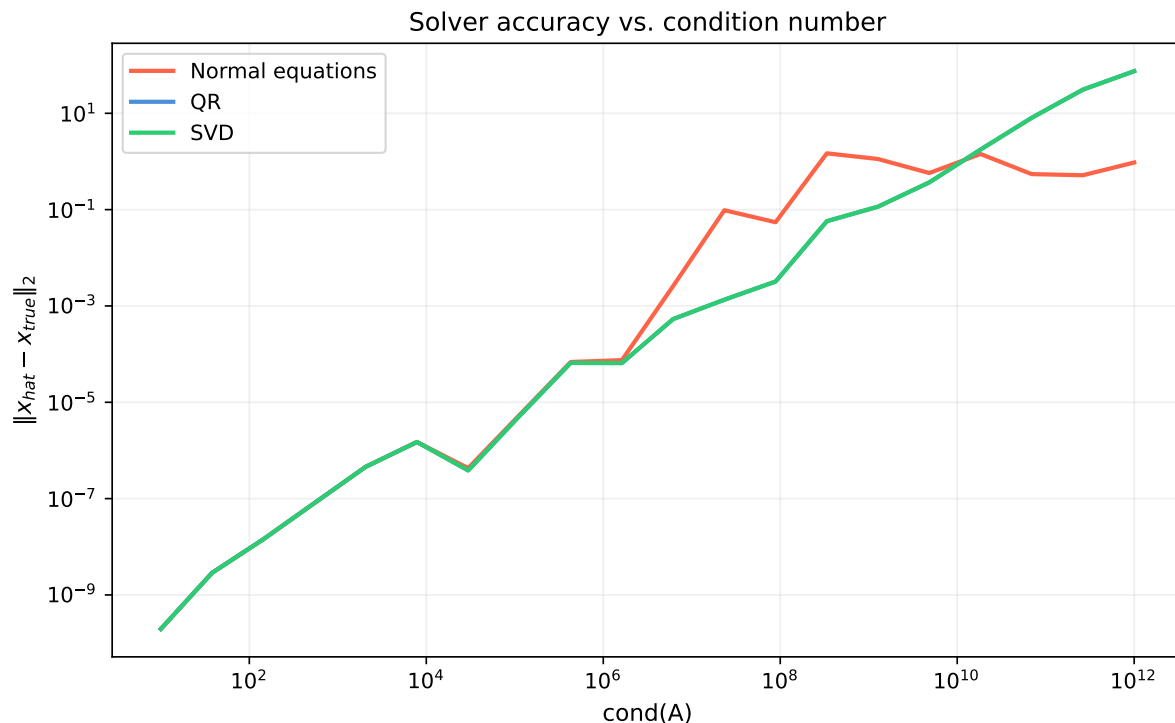
b = A @ x_true + noise

x_ne = np.linalg.lstsq(A.T @ A, A.T @ b, rcond=None)[0] # normal eqs
Q_m, R_m = np.linalg.qr(A, mode='reduced')
x_qr = np.linalg.solve(R_m, Q_m.T @ b)
x_sv = np.linalg.pinv(A) @ b

err_ne.append(np.linalg.norm(x_ne - x_true))
err_qr.append(np.linalg.norm(x_qr - x_true))
err_svd.append(np.linalg.norm(x_sv - x_true))

fig, ax = plt.subplots(figsize=(8, 5))
ax.loglog(cond_vals, err_ne, color='tomato', lw=2, label='Normal equations')
ax.loglog(cond_vals, err_qr, color='#4a90d9', lw=2, label='QR')
ax.loglog(cond_vals, err_svd, color='#2ecc71', lw=2, label='SVD')
ax.set_xlabel('cond(A)', fontsize=11)
ax.set_ylabel(r'$\|x_{\hat{}} - x_{true}\|_2$', fontsize=11)
ax.set_title('Solver accuracy vs. condition number', fontsize=12)
ax.legend(fontsize=10)
ax.grid(alpha=0.2, which='both')
fig.tight_layout()
plt.savefig('ch17-least-squares/fig-solver-comparison.png', dpi=150, bbox_inches='tight')
plt.show()

```



Exercises

17.5.1 For $A = \begin{bmatrix} 1 & 1 \\ 1 & 1 + \varepsilon \end{bmatrix}$ with small $\varepsilon > 0$, compute $\text{cond}(A)$ and $\text{cond}(A^T A)$ symbolically. What happens as $\varepsilon \rightarrow 0$?

17.5.2 Show algebraically that solving via QR ($R\hat{\mathbf{x}} = Q^T \mathbf{b}$) is equivalent to the normal equations. Why is QR more accurate despite the equivalence?

17.5.3 Explain why `np.linalg.lstsq` uses SVD by default and what the `rcond` parameter controls. What is a good value of `rcond` for double-precision input with $\text{cond}(A) \approx 10^6$?

17.5.4 When is it safe to use the normal equations? Give a concrete example from robotics or machine learning where forming $A^T A$ explicitly is standard practice.

Weighted and Constrained Least Squares

Standard least squares treats every residual equally. In practice, measurements have different noise levels and solutions must sometimes satisfy hard constraints. Weighted least squares handles the first; Lagrange multipliers handle the second. Both reduce to a standard linear solve.

Weighted Least Squares

When measurement i has standard deviation σ_i , the appropriate objective is the **weighted sum of squared residuals**:

$$\min_{\mathbf{x}} \sum_{i=1}^m w_i (b_i - \mathbf{a}_i^T \mathbf{x})^2 = \min_{\mathbf{x}} (\mathbf{Ax} - \mathbf{b})^T W (\mathbf{Ax} - \mathbf{b}),$$

where $W = \text{diag}(w_1, \dots, w_m) \succ 0$ is the **weight matrix**. Standard choice: $w_i = 1/\sigma_i^2$ (inverse variance weighting), which is maximum-likelihood estimation under independent Gaussian noise.

Reduction to standard form. Let $W = L^T L$ (e.g. $L = \text{diag}(\sqrt{w_i})$). Then:

$$(\mathbf{Ax} - \mathbf{b})^T W (\mathbf{Ax} - \mathbf{b}) = \|L\mathbf{Ax} - L\mathbf{b}\|_2^2.$$

This is ordinary least squares with $\tilde{A} = LA$ and $\tilde{\mathbf{b}} = L\mathbf{b}$. The **weighted normal equations** are:

$$\boxed{A^T W A \hat{\mathbf{x}} = A^T W \mathbf{b}.}$$

Constrained Least Squares via Lagrange Multipliers

Equality-constrained problem: minimise $\|A\mathbf{x} - \mathbf{b}\|_2^2$ subject to $C\mathbf{x} = \mathbf{d}$, where $C \in \mathbb{R}^{p \times n}$ and $p < n$.

Form the Lagrangian:

$$\mathcal{L}(\mathbf{x}, \lambda) = \|A\mathbf{x} - \mathbf{b}\|_2^2 + \lambda^T(C\mathbf{x} - \mathbf{d}).$$

Setting $\nabla_{\mathbf{x}}\mathcal{L} = 0$ and $\nabla_{\lambda}\mathcal{L} = 0$:

$$\begin{bmatrix} 2A^T A & C^T \\ C & 0 \end{bmatrix} \begin{bmatrix} \hat{\mathbf{x}} \\ \lambda \end{bmatrix} = \begin{bmatrix} 2A^T \mathbf{b} \\ \mathbf{d} \end{bmatrix}.$$

This **KKT system** is a $(n + p) \times (n + p)$ saddle-point linear system. It is solvable when A has full column rank and C has full row rank.

Weighted Least Squares Example: Depth Fitting with Heteroscedastic Noise

```
import numpy as np
import matplotlib.pyplot as plt

rng = np.random.default_rng(61)

m = 30
t = np.linspace(0.5, 5.0, m)          # shape (30,) -- depth in meters
A = np.column_stack([np.ones(m), t])  # shape (30, 2) -- [1, t] design matrix

x_true = np.array([0.2, 0.15])         # shape (2,) -- true coefficients
sigma = 0.05 + 0.12 * t                # shape (30,) -- noise grows with depth
b = A @ x_true + sigma * rng.standard_normal(m) # shape (30,)

# Ordinary least squares
x_ols = np.linalg.solve(A.T @ A, A.T @ b) # shape (2,)

# Weighted least squares: w_i = 1/sigma_i^2
w = 1.0 / sigma**2                     # shape (30,)
W = np.diag(w)                         # shape (30, 30)
x_wls = np.linalg.solve(A.T @ W @ A, A.T @ W @ b) # shape (2,)

# Alternatively: scale A and b by sqrt(w_i), then use OLS
sqrtw = np.sqrt(w)                     # shape (30,)
A_tilde = A * sqrtw[:, None]           # shape (30, 2)
```

```

b_tilde = b * sqrtw                                     # shape (30,)
x_wls_v2 = np.linalg.solve(A_tilde.T @ A_tilde, A_tilde.T @ b_tilde) # shape (2,)

print("=== Weighted Least Squares ===")
print(f"x_true:  {x_true}")
print(f"x_ols:   {x_ols.round(4)}")
print(f"x_wls:   {x_wls.round(4)}")
print(f"x_wls_v2:{x_wls_v2.round(4)} (equivalent via scaling)")

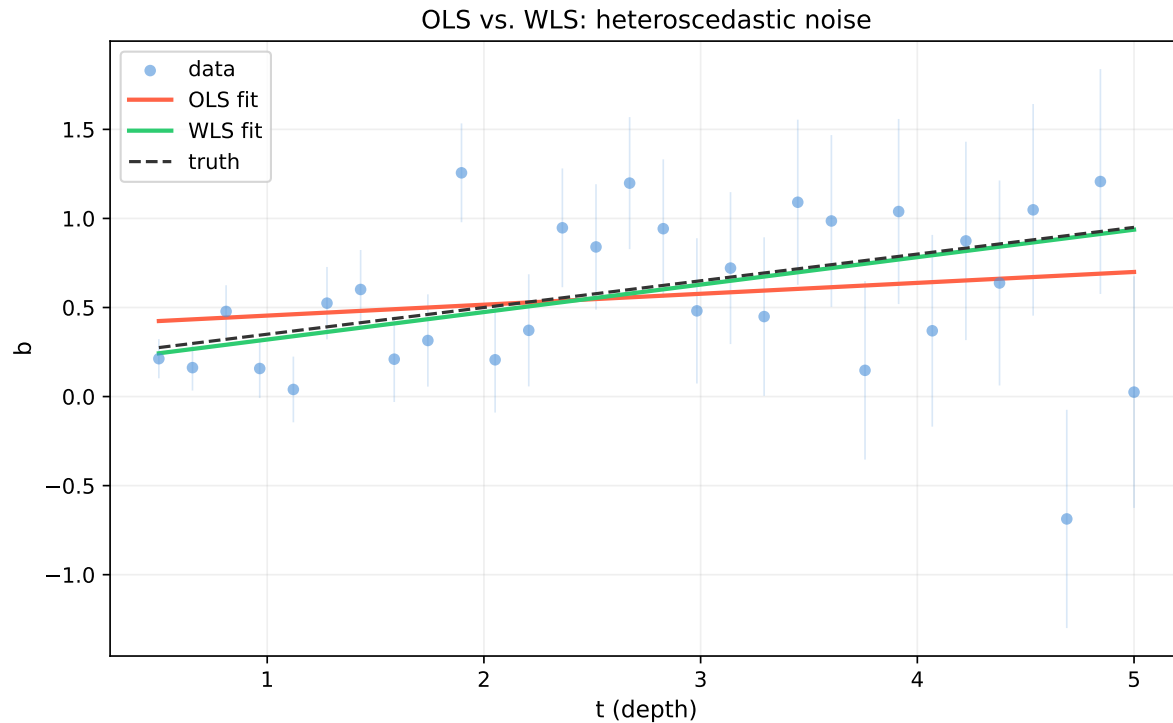
fig, ax = plt.subplots(figsize=(8, 5))
ax.scatter(t, b, s=20, color='#4a90d9', alpha=0.6, label='data', zorder=3)
ax.errorbar(t, b, yerr=sigma, fmt='none', color='#4a90d9', alpha=0.2, lw=0.8)
ax.plot(t, A @ x_ols, color='tomato', lw=2, label='OLS fit')
ax.plot(t, A @ x_wls, color='#2ecc71', lw=2, label='WLS fit')
ax.plot(t, A @ x_true, color='#333333', lw=1.5, linestyle='--', label='truth')
ax.set_xlabel('t (depth)', fontsize=11)
ax.set_ylabel('b', fontsize=11)
ax.set_title('OLS vs. WLS: heteroscedastic noise', fontsize=12)
ax.legend(fontsize=10)
ax.grid(alpha=0.2)
fig.tight_layout()
plt.savefig('ch17-least-squares/fig-weighted-ls.png', dpi=150, bbox_inches='tight')
plt.show()

```

```

=== Weighted Least Squares ===
x_true:  [0.2  0.15]
x_ols:   [0.3933 0.0612]
x_wls:   [0.1658 0.1542]
x_wls_v2:[0.1658 0.1542] (equivalent via scaling)

```



Constrained Least Squares Example: Sum-to-Zero Constraint

```
import numpy as np

rng = np.random.default_rng(62)

# Overdetermined system with a linear equality constraint
m, n = 15, 3
A = rng.standard_normal((m, n))      # shape (15, 3)
b = rng.standard_normal(m)           # shape (15,)

# Constraint: x_1 + x_2 + x_3 = 1 (e.g. simplex-like)
C = np.array([[1.0, 1.0, 1.0]])      # shape (1, 3)
d = np.array([1.0])                  # shape (1,)

p = C.shape[0]                        # number of constraints = 1

# Build KKT system
# [2 A^T A  C^T] [x      ] = [2 A^T b]
# [C        0  ] [lambda] = [d      ]
AtA = 2.0 * A.T @ A                  # shape (3, 3)
Atb = 2.0 * A.T @ b                  # shape (3,)
```

```

KKT = np.block([
    [AtA,          C.T          ],
    [C,           np.zeros((p, p)) ]
])
# shape (4, 4)

rhs = np.concatenate([Atb, d])
# shape (4,)

sol = np.linalg.solve(KKT, rhs)
# shape (4,)
x_constrained = sol[:n]
# shape (3,)
lam = sol[n:]
# shape (1,) -- Lagrange multiplier

# Unconstrained solution for comparison
x_unconstrained = np.linalg.solve(A.T @ A, A.T @ b)
# shape (3,)

print("=== Constrained Least Squares ===")
print(f"x_constrained:      {x_constrained.round(4)}")
print(f"Constraint C x:      {(C @ x_constrained).item():.6f} (should be 1.0)")
print(f"Lagrange mult:       {lam.item():.4f}")
print(f"\nx_unconstrained:      {x_unconstrained.round(4)}")
print(f"Constraint C x:       {(C @ x_unconstrained).item():.4f} (unconstrained, != 1)")
print(f"\nResiduals -- constrained: {np.linalg.norm(A @ x_constrained - b):.6f}")
print(f"                        unconstrained: {np.linalg.norm(A @ x_unconstrained - b):.6f} (smaller,")

=== Constrained Least Squares ===
x_constrained:      [0.1423 0.6935 0.1641]
Constraint C x:      1.000000 (should be 1.0)
Lagrange mult:      -11.4740

x_unconstrained:      [-0.4956 0.3706 0.0215]
Constraint C x:      -0.1036 (unconstrained, != 1)

Residuals -- constrained:      4.196709
                        unconstrained: 3.358757 (smaller, as expected)

```

Exercises

17.6.1 Show that the weighted normal equations $A^T W A \hat{\mathbf{x}} = A^T W \mathbf{b}$ can be derived by defining $\tilde{A} = W^{1/2} A$ and $\tilde{\mathbf{b}} = W^{1/2} \mathbf{b}$ and writing the standard normal equations for $\tilde{A} \mathbf{x} = \tilde{\mathbf{b}}$.

17.6.2 A GPS receiver gives three position measurements with standard deviations $\sigma_1 = 1$ m, $\sigma_2 = 2$ m, $\sigma_3 = 0.5$ m. Set up the weighted least squares problem to estimate a single position x from the three measurements. Solve and interpret the result.

17.6.3 Derive the KKT system for equality-constrained least squares by differentiating the Lagrangian $\mathcal{L}(\mathbf{x}, \lambda) = \|\mathbf{Ax} - \mathbf{b}\|^2 + \lambda^T(C\mathbf{x} - \mathbf{d})$ with respect to $\mathbf{x}$ and λ .

17.6.4 In the constrained example above, the constrained residual is larger than the unconstrained residual. Prove this is always the case: the optimal unconstrained residual $\leq$ optimal constrained residual.

Regularized Least Squares

When A is ill-conditioned or rank-deficient, the least-squares solution is sensitive to noise: tiny perturbations in $\mathbf{b}$ can produce wildly different $\hat{\mathbf{x}}$. **Regularisation** adds a penalty on $\|\mathbf{x}\|$ to stabilise the solution, trading a small increase in bias for a large reduction in variance.

Tikhonov Regularisation (Ridge Regression)

The **Tikhonov** or **ridge** problem is:

$$\min_{\mathbf{x}} \|\mathbf{Ax} - \mathbf{b}\|_2^2 + \lambda \|\mathbf{x}\|_2^2, \quad \lambda > 0.$$

Setting the gradient to zero:

$$2A^T(\mathbf{Ax} - \mathbf{b}) + 2\lambda\mathbf{x} = \mathbf{0} \implies (A^T A + \lambda I)\hat{\mathbf{x}}_\lambda = A^T \mathbf{b}.$$

The matrix $A^T A + \lambda I$ is always **positive definite** (eigenvalues $\geq \lambda > 0$), so the solution is always unique regardless of the rank of A .

Effect on Singular Values

Using the SVD $A = U\Sigma V^T$, the Tikhonov solution has a clean expression:

$$\hat{\mathbf{x}}_\lambda = V \underbrace{\text{diag}\left(\frac{\sigma_i}{\sigma_i^2 + \lambda}\right)}_{\text{regularised gain}} U^T \mathbf{b}.$$

Compare to the pseudoinverse: $A^+ \mathbf{b} = V \text{diag}(1/\sigma_i) U^T \mathbf{b}$. The regularised gain $\sigma_i/(\sigma_i^2 + \lambda)$ is:

- $\approx 1/\sigma_i$ when $\sigma_i \gg \sqrt{\lambda}$ (large singular values: unaffected)
- $\approx \sigma_i/\lambda \approx 0$ when $\sigma_i \ll \sqrt{\lambda}$ (small singular values: suppressed)

Tikhonov regularisation **shrinks the contribution of directions with small singular values** without discarding them entirely, unlike the hard cutoff in `pinv`.

Bias–Variance Interpretation

In statistics, $\hat{\mathbf{x}}_\lambda$ introduces bias ($\hat{\mathbf{x}}_\lambda \neq \mathbf{x}_{\text{true}}$ even with no noise) in exchange for lower variance (less sensitivity to noise). Choosing λ is a bias–variance trade-off:

- $\lambda \rightarrow 0$: $\hat{\mathbf{x}}_\lambda \rightarrow A^+ \mathbf{b}$ (unregularised, low bias, high variance)
- $\lambda \rightarrow \infty$: $\hat{\mathbf{x}}_\lambda \rightarrow \mathbf{0}$ (all shrinkage, high bias, zero variance)

Cross-validation selects λ by evaluating hold-out prediction error.

Numerical Demonstration

```
import numpy as np
import matplotlib.pyplot as plt

rng = np.random.default_rng(71)

m, n = 40, 8
# Design matrix with moderate conditioning
A = rng.standard_normal((m, n))    # shape (40, 8)

# True solution has unit norm
x_true = rng.standard_normal(n)    # shape (8,)
x_true /= np.linalg.norm(x_true)

noise_level = 0.5
b = A @ x_true + noise_level * rng.standard_normal(m) # shape (40,)

lambdas = np.logspace(-4, 4, 200) # shape (200,) -- regularisation sweep

# Compute solutions for each lambda
errors = []
norms = []

AtA = A.T @ A                      # shape (8, 8)
Atb = A.T @ b                      # shape (8,)

for lam in lambdas:
```

```

x_lam = np.linalg.solve(AtA + lam * np.eye(n), Atb) # shape (8,)
errors.append(np.linalg.norm(x_lam - x_true))
norms.append(np.linalg.norm(x_lam))

errors = np.array(errors) # shape (200,)
norms = np.array(norms) # shape (200,)

best_idx = np.argmin(errors)
best_lam = lambdas[best_idx]
print(f"Best lambda: {best_lam:.4f}")
print(f"Min error: {errors[best_idx]:.4f}")

fig, axes = plt.subplots(1, 2, figsize=(12, 4))

ax = axes[0]
ax.semilogx(lambdas, errors, color='#4a90d9', lw=2)
ax.axvline(best_lam, color='tomato', lw=1.5, linestyle='--', label=f'best lambda={best_lam:.3f}')
ax.set_xlabel(r'$\lambda$', fontsize=12)
ax.set_ylabel(r'$\|\hat{x}_\lambda - x_{\text{true}}\|_2$', fontsize=11)
ax.set_title('Recovery error vs. regularisation', fontsize=11)
ax.legend(fontsize=10)
ax.grid(alpha=0.2)

ax2 = axes[1]
ax2.semilogx(lambdas, norms, color='#2ecc71', lw=2)
ax2.axhline(np.linalg.norm(x_true), color='#333333', lw=1.5, linestyle='--', label=r'$\|x_{\text{true}}\|_2$')
ax2.axvline(best_lam, color='tomato', lw=1.5, linestyle='--', label=f'best lambda')
ax2.set_xlabel(r'$\lambda$', fontsize=12)
ax2.set_ylabel(r'$\|\hat{x}_\lambda\|_2$', fontsize=11)
ax2.set_title('Solution norm vs. regularisation (shrinkage)', fontsize=11)
ax2.legend(fontsize=10)
ax2.grid(alpha=0.2)

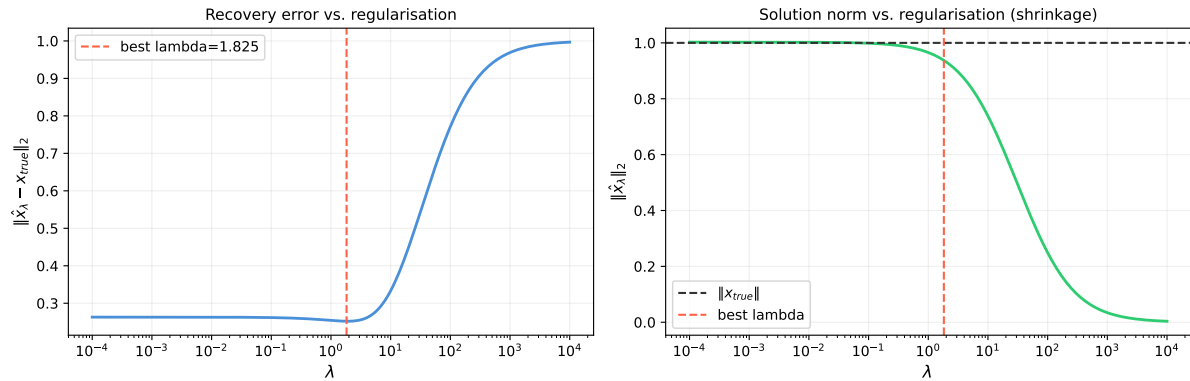
fig.tight_layout()
plt.savefig('ch17-least-squares/fig-ridge.png', dpi=150, bbox_inches='tight')
plt.show()

```

```

Best lambda:    1.8252
Min error:      0.2520

```

Singular Value Shrinkage

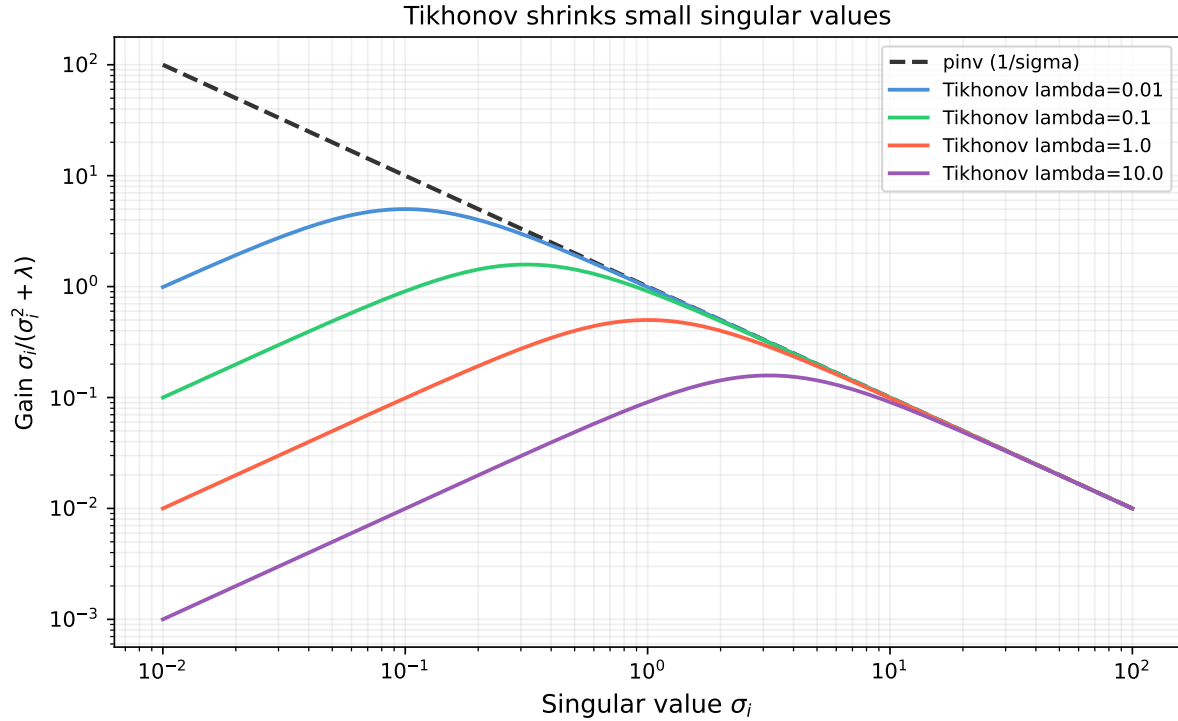
```
import numpy as np
import matplotlib.pyplot as plt

# Compare pseudoinverse gain vs. Tikhonov gain for several lambdas
sigma_vals = np.logspace(-2, 2, 200)  # shape (200,) -- range of singular values
lambdas_vis = [0.01, 0.1, 1.0, 10.0]

fig, ax = plt.subplots(figsize=(8, 5))
ax.plot(sigma_vals, 1.0 / sigma_vals, color='#333333', lw=2, linestyle='--', label='pinv (1)')

colors = ['#4a90d9', '#2ecc71', 'tomato', '#9b59b6']
for lam, col in zip(lambdas_vis, colors):
    gain = sigma_vals / (sigma_vals**2 + lam)  # shape (200,)
    ax.plot(sigma_vals, gain, color=col, lw=1.8, label=f'Tikhonov lambda={lam}')

ax.set_xscale('log')
ax.set_yscale('log')
ax.set_xlabel(r'Singular value  $\sigma_i$ ', fontsize=12)
ax.set_ylabel(r'Gain  $\sigma_i / (\sigma_i^2 + \lambda)$ ', fontsize=11)
ax.set_title('Tikhonov shrinks small singular values', fontsize=12)
ax.legend(fontsize=9)
ax.grid(alpha=0.2, which='both')
fig.tight_layout()
plt.savefig('ch17-least-squares/fig-ridge-singular.png', dpi=150, bbox_inches='tight')
plt.show()
```



General Tikhonov: $\min \|A\mathbf{x} - \mathbf{b}\|^2 + \lambda \|L\mathbf{x}\|^2$

When the penalty is not on $\|\mathbf{x}\|$ but on $\|L\mathbf{x}\|$ for some matrix L (e.g. a finite-difference operator penalising roughness), the solution becomes:

$$(A^T A + \lambda L^T L) \hat{\mathbf{x}} = A^T \mathbf{b}.$$

This generalises to smoothness regularisation, total-variation regularisation (with L as the gradient operator), and other structured priors.

Exercises

17.7.1 Derive the Tikhonov solution $(A^T A + \lambda I) \hat{\mathbf{x}}_\lambda = A^T \mathbf{b}$ from the objective $\|A\mathbf{x} - \mathbf{b}\|^2 + \lambda \|\mathbf{x}\|^2$ by taking the gradient and setting it to zero.

17.7.2 Using the SVD formula $\hat{\mathbf{x}}_\lambda = V \text{diag}(\sigma_i/(\sigma_i^2 + \lambda)) U^T \mathbf{b}$, show that as $\lambda \rightarrow 0$ the solution approaches the pseudoinverse $A^+ \mathbf{b}$ (assuming A has full column rank).

17.7.3 Prove that the Tikhonov solution $\hat{\mathbf{x}}_\lambda$ satisfies $\|\hat{\mathbf{x}}_\lambda\|_2 \leq \|\hat{\mathbf{x}}_0\|_2$ for all $\lambda > 0$. (Hint: the gain $\sigma_i/(\sigma_i^2 + \lambda) < 1/\sigma_i$ for all $\sigma_i > 0$.)

17.7.4 What is the effective rank of the Tikhonov regularised problem? Specifically, how does $\sum_i [\sigma_i^2 / (\sigma_i^2 + \lambda)]$ (the trace of the hat matrix $A(A^T A + \lambda I)^{-1} A^T$) behave as λ increases? This quantity is called the **effective degrees of freedom** in ridge regression.

Large-Scale Least Squares: Iterative Methods

For large sparse systems ($n > 10^5$), direct methods (QR, Cholesky) become impractical because they destroy sparsity. Iterative methods apply A and A^T repeatedly without forming $A^T A$ explicitly, converging to the solution in far fewer than n steps for well-structured problems.

Gradient Descent: The Simplest Case

For the normal equations $A^T A \mathbf{x} = A^T \mathbf{b}$ (a PSD system $M \mathbf{x} = \mathbf{c}$ with $M = A^T A$), gradient descent on $f(\mathbf{x}) = \frac{1}{2} \mathbf{x}^T M \mathbf{x} - \mathbf{c}^T \mathbf{x}$ gives the update:

$$\mathbf{x}_{k+1} = \mathbf{x}_k - \alpha_k \nabla f(\mathbf{x}_k) = \mathbf{x}_k + \alpha_k (A^T \mathbf{b} - A^T A \mathbf{x}_k) = \mathbf{x}_k + \alpha_k A^T \mathbf{r}_k,$$

where $\mathbf{r}_k = \mathbf{b} - A \mathbf{x}_k$ is the residual. With optimal step size $\alpha_k = \|\mathbf{r}_k\|^2 / \|A \mathbf{r}_k\|^2$, error converges as:

$$\|\mathbf{x}_k - \hat{\mathbf{x}}\|_{A^T A} \leq \left(\frac{\kappa - 1}{\kappa + 1} \right)^k \|\mathbf{x}_0 - \hat{\mathbf{x}}\|_{A^T A},$$

where $\kappa = \text{cond}(A)^2 = \text{cond}(A^T A)$. For ill-conditioned A , this convergence is extremely slow – motivating the conjugate gradient method.

Conjugate Gradient for Symmetric PSD Systems

Conjugate Gradient (CG) solves $M \mathbf{x} = \mathbf{c}$ (with $M \succ 0$) in at most n steps and often much fewer when eigenvalues cluster. For least squares, apply CG to the normal equations $A^T A \mathbf{x} = A^T \mathbf{b}$ (this is CGLS / LSQR).

The key insight: CG builds a sequence of search directions that are M -conjugate ($\mathbf{p}_i^T M \mathbf{p}_j = 0$ for $i \neq j$), avoiding the zig-zagging of gradient descent. Convergence bound:

$$\frac{\|\mathbf{x}_k - \hat{\mathbf{x}}\|_M}{\|\mathbf{x}_0 - \hat{\mathbf{x}}\|_M} \leq 2 \left(\frac{\sqrt{\kappa} - 1}{\sqrt{\kappa} + 1} \right)^k.$$

The $\sqrt{\kappa}$ vs. κ in the convergence factor is the key speedup over gradient descent.

Preconditioning

A **preconditioner** $P \approx M$ transforms the system to $P^{-1}M\mathbf{x} = P^{-1}\mathbf{c}$, clustering the effective eigenvalues closer to 1. Common choices:

- **Diagonal (Jacobi):** $P = \text{diag}(M_{11}, \dots, M_{nn})$ – cheap, effective for diagonally dominant M .
- **Incomplete Cholesky (IC0):** sparse approximate factorisation – powerful for SPD systems from FEM or graph Laplacians.
- **SSOR / block-Jacobi:** for structured sparsity patterns.

With an ideal preconditioner ($\kappa(P^{-1}M) \ll \kappa(M)$), CG converges in far fewer iterations.

When to Prefer Iterative Solvers

Criterion	Direct (QR/Cholesky)	Iterative (CG/LSQR)
n	$< 10^4$	$> 10^5$
Sparsity	Dense or moderate	Very sparse
Multiple RHS	Excellent (factor once)	Poor (re-run each time)
Preconditioner needed	No	Yes, for fast convergence
Exact solution	Yes	Up to tolerance

For robotics SLAM with 10^6 pose variables or large-scale batch regression, iterative solvers like `scipy.sparse.linalg.lsqr` or `scipy.sparse.linalg.cg` are essential.

Gradient Descent vs. CG: Convergence Comparison

```
import numpy as np
import matplotlib.pyplot as plt

rng = np.random.default_rng(81)

n = 50
# Build a PSD system with known condition number
eigvals = np.linspace(1, 100, n)          # shape (50,) -- condition number 100
Q, _ = np.linalg.qr(rng.standard_normal((n, n))) # shape (50, 50) -- random orthogonal
M = Q @ np.diag(eigvals) @ Q.T            # shape (50, 50) -- symmetric PSD
c = rng.standard_normal(n)                # shape (50,)
```

```

x_exact = np.linalg.solve(M, c)                # shape (50,) -- true solution

def err_norm(x, x_exact, M):
    """Energy norm ||x - x_exact||_M."""
    d = x - x_exact                            # shape (n,)
    return float(np.sqrt(d @ M @ d))

# Gradient descent with exact line search
max_iter = 200
x_gd = np.zeros(n)                            # shape (50,)
errors_gd = [err_norm(x_gd, x_exact, M)]

for _ in range(max_iter):
    r = c - M @ x_gd                          # shape (50,) -- residual
    Ar = M @ r                                # shape (50,)
    alpha = float(r @ r) / float(r @ Ar)
    x_gd = x_gd + alpha * r                   # shape (50,)
    errors_gd.append(err_norm(x_gd, x_exact, M))

# Conjugate Gradient (standard implementation)
x_cg = np.zeros(n)                            # shape (50,)
r_cg = c.copy()                              # shape (50,)
p_cg = r_cg.copy()                            # shape (50,)
errors_cg = [err_norm(x_cg, x_exact, M)]

for _ in range(max_iter):
    Ap = M @ p_cg                             # shape (50,)
    alpha = float(r_cg @ r_cg) / float(p_cg @ Ap)
    x_cg = x_cg + alpha * p_cg                 # shape (50,)
    r_new = r_cg - alpha * Ap                  # shape (50,)
    beta = float(r_new @ r_new) / float(r_cg @ r_cg)
    p_cg = r_new + beta * p_cg                 # shape (50,)
    r_cg = r_new
    errors_cg.append(err_norm(x_cg, x_exact, M))
    if errors_cg[-1] < 1e-12:
        break

iters = np.arange(len(errors_gd))              # shape varies

fig, ax = plt.subplots(figsize=(9, 5))
ax.semilogy(errors_gd, color='tomato', lw=2, label='Gradient descent')
ax.semilogy(np.arange(len(errors_cg)), errors_cg, color='#2ecc71', lw=2, label='Conjugate G')
ax.set_xlabel('Iteration', fontsize=11)
ax.set_ylabel(r'$\|x_k - \hat{x}\|_M$', fontsize=11)
ax.set_title(f'GD vs. CG convergence (cond(M) = {eigvals[-1]/eigvals[0]:.0f})', fontsize=12)
ax.legend(fontsize=10)
ax.grid(alpha=0.2, which='both')

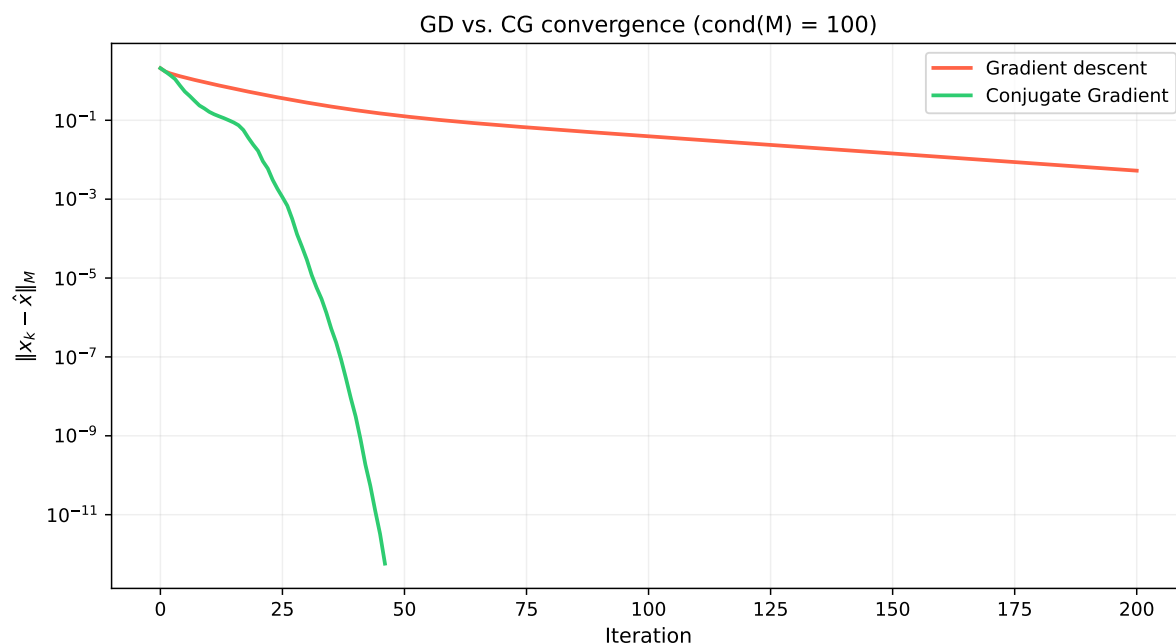
```

```

fig.tight_layout()
plt.savefig('ch17-least-squares/fig-iterative.png', dpi=150, bbox_inches='tight')
plt.show()

kappa = eigvals[-1] / eigvals[0]
gd_rate = (kappa - 1) / (kappa + 1)
cg_rate = (np.sqrt(kappa) - 1) / (np.sqrt(kappa) + 1)
print(f"kappa = {kappa:.0f}")
print(f"GD convergence rate: {gd_rate:.4f} (per step)")
print(f"CG convergence rate: {cg_rate:.4f} (per step, sqrt improvement)")
print(f"CG iters to 1e-6: ~{int(np.ceil(np.log(1e-6/2) / np.log(cg_rate)))}")
print(f"GD iters to 1e-6: ~{int(np.ceil(np.log(1e-6/2) / np.log(gd_rate)))}")

```



```

kappa = 100
GD convergence rate: 0.9802 (per step)
CG convergence rate: 0.8182 (per step, sqrt improvement)
CG iters to 1e-6: ~73
GD iters to 1e-6: ~726

```

LSQR: CG Applied to Least Squares

For least squares $\min \|Ax - b\|$ with non-square A , **LSQR** applies CG to the normal equations without ever forming $A^T A$ explicitly, which preserves sparsity. `scipy.sparse.linalg.lsqr` is the standard implementation.

```

import numpy as np
from scipy.sparse import random as sparse_random
from scipy.sparse.linalg import lsqr

rng_np = np.random.default_rng(82)

m, n = 500, 200
# Sparse design matrix (density 5%)
A_sparse = sparse_random(m, n, density=0.05, format='csr',
                        random_state=82, data_rvs=rng_np.standard_normal) # shape (500, 200)
x_true = rng_np.standard_normal(n) # shape (200,)
b = A_sparse @ x_true + 0.1 * rng_np.standard_normal(m) # shape (500,)

# LSQR (iterative, preserves sparsity)
result = lsqr(A_sparse, b, atol=1e-8, btol=1e-8)
x_lsqr = result[0] # shape (200,)
iters = result[2]

# Dense lstsq for comparison
A_dense = A_sparse.toarray() # shape (500, 200)
x_dense, _, _, _ = np.linalg.lstsq(A_dense, b, rcond=None) # shape (200,)

print(f"LSQR iterations:           {iters}")
print(f"||x_lsqr - x_true||:         {np.linalg.norm(x_lsqr - x_true):.4f}")
print(f"||x_dense - x_true||:          {np.linalg.norm(x_dense - x_true):.4f}")
print(f"Residual (lsqr):                {np.linalg.norm(A_sparse @ x_lsqr - b):.4f}")
print(f"Residual (dense lstsq):         {np.linalg.norm(A_dense @ x_dense - b):.4f}")

```

```

LSQR iterations:           49
||x_lsqr - x_true||:       0.3776
||x_dense - x_true||:      0.3776
Residual (lsqr):           1.6805
Residual (dense lstsq):    1.6805

```

Exercises

17.8.1 Derive the gradient descent update for minimising $\|A\mathbf{x} - \mathbf{b}\|^2$. Show that the optimal step size is $\alpha_k = \|A^T \mathbf{r}_k\|^2 / \|AA^T \mathbf{r}_k\|^2$ (where $\mathbf{r}_k = \mathbf{b} - A\mathbf{x}_k$) when working directly with A rather than $A^T A$.

17.8.2 The CG method solves a 50×50 PSD system with $\text{cond}(M) = 100$. About how many iterations are needed to reduce the energy-norm error by a factor of 10^6 ? Compare with gradient descent.

17.8.3 What property of A must hold for `scipy.sparse.linalg.lsqr` to converge? How does adding Tikhonov regularisation λI help when $A^T A$ is nearly singular?

17.8.4 Explain why forming $A^T A$ explicitly before applying CG (rather than using LSQR which applies A and A^T separately) is problematic for sparse A .

Application: Calibrating a Depth Sensor

Structured-light and time-of-flight depth sensors report biased distance measurements: the true depth d and reported depth z are related by a polynomial distortion model. Calibrating the sensor means estimating this polynomial from measured error data. This application demonstrates weighted least squares, compares three solvers (normal equations, QR, SVD), and shows how regularisation helps when the polynomial degree is uncertain.

Problem Setup

We model the depth error as a degree- p polynomial in the reported depth:

$$d_{\text{true}} = z + \sum_{k=1}^p c_k z^k = A\mathbf{c} + \mathbf{z},$$

or equivalently, the **correction** $\delta(z) = d_{\text{true}} - z = \sum_{k=1}^p c_k z^k$.

Given m calibration measurements $\{(z_i, \delta_i)\}$, we fit:

$$A\mathbf{c} \approx \delta, \quad A_{ij} = z_i^j, \quad i = 1, \dots, m, \quad j = 1, \dots, p.$$

The noise variance grows with depth (sensors are less accurate at long range), so we use weighted least squares with $w_i = 1/\sigma_i^2$.

Synthesising Calibration Data

```
import numpy as np
import matplotlib.pyplot as plt

rng = np.random.default_rng(91)

# Ground-truth distortion: delta(z) = 0.04 z - 0.008 z^2 + 0.0005 z^3
c_true = np.array([0.04, -0.008, 0.0005]) # shape (3,) -- true polynomial coefficients
```



```

m = 60                                # number of calibration points
z = rng.uniform(0.5, 6.0, m)          # shape (60,) -- depths in meters, uneven spa

# True correction
delta_true = np.polyval(c_true[::-1], z) - 0.0 # shape (60,) -- using c_k z^k convention

# Actually compute the sum directly (avoid polyval convention confusion)
delta_true = sum(c_true[k] * z**(k+1) for k in range(len(c_true))) # shape (60,)

# Noise model: noise std = 0.002 + 0.01 * z (proportional to depth)
sigma = 0.002 + 0.01 * z              # shape (60,)
delta_obs = delta_true + sigma * rng.standard_normal(m) # shape (60,) -- observed correct

print(f"Depth range:   {z.min():.2f} -- {z.max():.2f} m")
print(f"Noise range:   {sigma.min():.4f} -- {sigma.max():.4f} m")
print(f"Signal range:  {delta_true.min():.4f} -- {delta_true.max():.4f} m")
print(f"c_true = {c_true}")

```

```

Depth range:   0.69 -- 6.00 m
Noise range:   0.0089 -- 0.0620 m
Signal range:  0.0240 -- 0.0640 m
c_true = [ 0.04  -0.008  0.0005]

```

Building the Design Matrix and Fitting

```

import numpy as np

# Degree-3 Vandermonde design matrix (columns z, z^2, z^3)
p = 3
A = np.column_stack([z**k for k in range(1, p+1)]) # shape (60, 3)

w = 1.0 / sigma**2                                # shape (60,) -- inverse-variance weights
W = np.diag(w)                                    # shape (60, 60)

# Method 1: Weighted normal equations
AtWA = A.T @ W @ A                               # shape (3, 3)
AtWb = A.T @ W @ delta_obs                       # shape (3,)
c_ne = np.linalg.solve(AtWA, AtWb)               # shape (3,)

# Method 2: QR on the scaled system
sqrtw = np.sqrt(w)                                # shape (60,)
A_tilde = A * sqrtw[:, None]                    # shape (60, 3) -- scaled design matrix

```

```

b_tilde = delta_obs * sqrtw                                # shape (60,)    -- scaled RHS
Q, R = np.linalg.qr(A_tilde, mode='reduced')               # Q (60,3), R (3,3)
c_qr = np.linalg.solve(R, Q.T @ b_tilde)                  # shape (3,)

# Method 3: SVD (lstsq on scaled system)
c_svd, _, _ = np.linalg.lstsq(A_tilde, b_tilde, rcond=None) # shape (3,)

print("=== Solver Comparison ===")
print(f"c_true: {c_true}")
print(f"c_ne:   {c_ne.round(6)}")
print(f"c_qr:   {c_qr.round(6)}")
print(f"c_svd:  {c_svd.round(6)}")

err_ne = np.linalg.norm(c_ne - c_true)
err_qr = np.linalg.norm(c_qr - c_true)
err_svd = np.linalg.norm(c_svd - c_true)
print(f"\n||c_ne - c_true||: {err_ne:.2e}")
print(f"||c_qr - c_true||: {err_qr:.2e}")
print(f"||c_svd - c_true||: {err_svd:.2e}")

```

```

=== Solver Comparison ===
c_true: [ 0.04  -0.008  0.0005]
c_ne:   [ 0.034718 -0.005781  0.00037 ]
c_qr:   [ 0.034718 -0.005781  0.00037 ]
c_svd:  [ 0.034718 -0.005781  0.00037 ]

||c_ne - c_true||: 5.73e-03
||c_qr - c_true||: 5.73e-03
||c_svd - c_true||: 5.73e-03

```

Overfitting: High-Degree Polynomial Without Regularisation

```

import numpy as np

# Fit a degree-8 polynomial with and without regularisation
p_high = 8
A_high = np.column_stack([z**k for k in range(1, p_high+1)]) # shape (60, 8)
sqrtw_m = np.sqrt(w)                                         # shape (60,)
A_h = A_high * sqrtw_m[:, None]                              # shape (60, 8)
b_h = delta_obs * sqrtw_m                                    # shape (60,)

# Unregularised

```

```

c_unr, _, _, sv_h = np.linalg.lstsq(A_h, b_h, rcond=None)    # shape (8,)

# Tikhonov regularised
lam_reg = 0.1
AtA_h = A_h.T @ A_h                                         # shape (8, 8)
Atb_h = A_h.T @ b_h                                         # shape (8,)
c_reg = np.linalg.solve(AtA_h + lam_reg * np.eye(p_high), Atb_h) # shape (8,)

z_plot = np.linspace(0.5, 6.0, 200)                         # shape (200,) -- dense grid
A_plot = np.column_stack([z_plot**k for k in range(1, p_high+1)]) # shape (200, 8)
A_plot3 = np.column_stack([z_plot**k for k in range(1, 4)])   # shape (200, 3)

print(f"Degree-{p_high} singular values: {sv_h.round(3)}")
print(f"cond(A_h): {sv_h[0]/sv_h[-1]:.2e} (ill-conditioned)")
print(f"\n||c_unr||: {np.linalg.norm(c_unr):.4f}")
print(f"||c_reg||: {np.linalg.norm(c_reg):.4f} (regularised: smaller norm)")

```

```

Degree-8 singular values: [6.53783630e+07 8.00834261e+05 2.98449330e+04 2.09453400e+03
 3.46860000e+02 7.87140000e+01 1.09320000e+01 8.47000000e-01]
cond(A_h): 7.72e+07 (ill-conditioned)

```

```

||c_unr||: 1.6142
||c_reg||: 1.4170 (regularised: smaller norm)

```

Visualisation

```

import numpy as np
import matplotlib.pyplot as plt

z_plot = np.linspace(0.5, 6.0, 200)                         # shape (200,)
A_plot3 = np.column_stack([z_plot**k for k in range(1, 4)])  # shape (200, 3)
A_plot8 = np.column_stack([z_plot**k for k in range(1, 9)])  # shape (200, 8)

delta_fit_ne = A_plot3 @ c_ne                                # shape (200,) -- NE fit
delta_fit_unr = A_plot8 @ c_unr                              # shape (200,) -- unregularised
delta_fit_reg = A_plot8 @ c_reg                              # shape (200,) -- regularised
delta_true_line = sum(c_true[k] * z_plot**(k+1) for k in range(3)) # shape (200,) -- ground truth

fig, axes = plt.subplots(1, 2, figsize=(13, 5))

# Left: solver comparison (degree 3)
ax = axes[0]

```

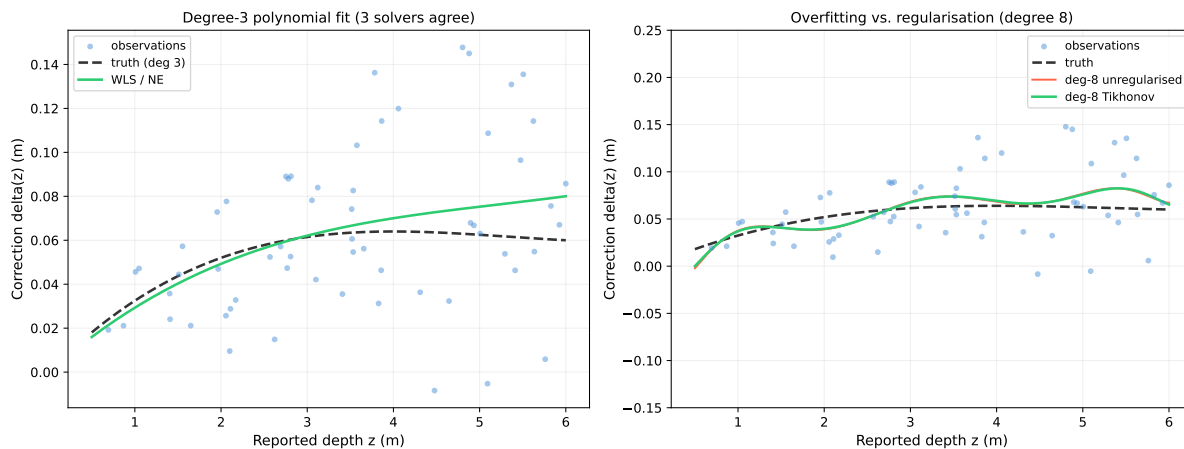
```

ax.scatter(z, delta_obs, s=12, color='#4a90d9', alpha=0.5, label='observations', zorder=3)
ax.plot(z_plot, delta_true_line, color='#333333', lw=2, linestyle='--', label='truth (deg 3)')
ax.plot(z_plot, delta_fit_ne, color='#2ecc71', lw=2, label='WLS / NE')
ax.set_xlabel('Reported depth z (m)', fontsize=11)
ax.set_ylabel('Correction delta(z) (m)', fontsize=11)
ax.set_title('Degree-3 polynomial fit (3 solvers agree)', fontsize=11)
ax.legend(fontsize=9)
ax.grid(alpha=0.2)

# Right: overfitting vs. regularisation (degree 8)
ax2 = axes[1]
ax2.scatter(z, delta_obs, s=12, color='#4a90d9', alpha=0.5, label='observations', zorder=3)
ax2.plot(z_plot, delta_true_line, color='#333333', lw=2, linestyle='--', label='truth')
ax2.plot(z_plot, delta_fit_unr, color='tomato', lw=1.5, label='deg-8 unregularised')
ax2.plot(z_plot, delta_fit_reg, color='#2ecc71', lw=2, label='deg-8 Tikhonov')
ax2.set_ylim(-0.15, 0.25)
ax2.set_xlabel('Reported depth z (m)', fontsize=11)
ax2.set_ylabel('Correction delta(z) (m)', fontsize=11)
ax2.set_title('Overfitting vs. regularisation (degree 8)', fontsize=11)
ax2.legend(fontsize=9)
ax2.grid(alpha=0.2)

fig.tight_layout()
plt.savefig('ch17-least-squares/fig-depth-calibration.png', dpi=150, bbox_inches='tight')
plt.show()

```



Key Takeaways

1. **Three solvers give identical results on a well-conditioned degree-3 problem.** With only 60 measurements and 3 unknowns, all methods are stable and agree to full precision.

2. **High-degree polynomials require regularisation.** Degree 8 with $m = 60$ observations is technically overdetermined but the Vandermonde matrix is highly ill-conditioned. Tikhonov regularisation with $\lambda = 0.1$ recovers a smooth fit close to the truth.
3. **Weighted fitting matters when noise is heteroscedastic.** Giving more weight to close-range (low-noise) measurements pulls the polynomial toward greater accuracy in the near-field region.
4. **This pattern generalises to any sensor calibration task:** camera intrinsics, IMU biases, and LiDAR intensity correction all follow the same arc – polynomial design matrix, overdetermined system, weighted least squares, solver comparison.

Exercises and Solutions

Exercise 17.1.1 – Projection matrix onto $\text{span}\{\mathbf{a}\}$

Let $\mathbf{a} = (1, 2, 2)^T$ in $\mathbb{R}^3$. Write down the 3×3 projection matrix P onto $\text{span}\{\mathbf{a}\}$. Verify $P^2 = P$ and $P^T = P$. What is $\text{rank}(P)$?

```
import numpy as np

a = np.array([1.0, 2.0, 2.0])      # shape (3,)
P = np.outer(a, a) / float(a @ a) # shape (3, 3) -- rank-1 projector

print("P =")
print(P.round(6))
print(f"\nP^2 == P:  {np.allclose(P @ P, P)}")
print(f"P^T == P:  {np.allclose(P.T, P)}")
print(f"rank(P):    {np.linalg.matrix_rank(P)}")
```

```
P =
[[0.111111 0.222222 0.222222]
 [0.222222 0.444444 0.444444]
 [0.222222 0.444444 0.444444]]
```

```
P^2 == P:  True
P^T == P:  True
rank(P):   1
```

Solution. $\mathbf{a}^T \mathbf{a} = 1 + 4 + 4 = 9$, so

$$P = \frac{1}{9} \begin{bmatrix} 1 & 2 & 2 \\ 2 & 4 & 4 \\ 2 & 4 & 4 \end{bmatrix}.$$

$P^2 = P$ because $P\mathbf{v}$ is already in $\text{span}\{\mathbf{a}\}$ and projecting again does nothing. $P^T = P$ because $(\mathbf{a}\mathbf{a}^T)^T = \mathbf{a}\mathbf{a}^T$. $\text{rank}(P) = 1$: the column space is one-dimensional, spanned by $\mathbf{a}$.

Exercise 17.1.3 – Projection when A has orthonormal columns

Suppose A has orthonormal columns ($A^T A = I$). Show that $P = AA^T$.

Solution. Substitute $A^T A = I$ into the projection formula:

$$P = A(A^T A)^{-1} A^T = A \cdot I^{-1} \cdot A^T = AA^T.$$

When $A = Q$ comes from a QR factorisation, $Q^T Q = I_n$ (thin QR), so the projection onto $\text{Col}(Q)$ is simply QQ^T – no matrix inverse needed. This is why QR-based least squares is both efficient and numerically stable.

Exercise 17.2.1 – Pseudoinverse of a simple matrix

Let $A = \begin{bmatrix} 1 & 0 \\ 0 & 2 \\ 0 & 0 \end{bmatrix}$. Compute A^+ and verify all four conditions.

```
import numpy as np

A = np.array([[1.0, 0.0],
              [0.0, 2.0],
              [0.0, 0.0]])          # shape (3, 2)

U, s, Vt = np.linalg.svd(A, full_matrices=False) # U (3,2), s (2,), Vt (2,2)
A_plus = (Vt.T * (1.0 / s)) @ U.T # shape (2, 3)

print(f"Singular values: {s}")
print("\nA+ =")
print(A_plus)

print(f"\nAA+A == A:      {np.allclose(A @ A_plus @ A, A)}")
print(f"A+AA+ == A+:      {np.allclose(A_plus @ A @ A_plus, A_plus)}")
print(f"(AA+)^T == AA+:    {np.allclose((A @ A_plus).T, A @ A_plus)}")
print(f"(A+A)^T == A+A:    {np.allclose((A_plus @ A).T, A_plus @ A)}")
```

Singular values: [2. 1.]

A+ =
 $\begin{bmatrix} 1. & 0. & 0. \\ 0. & 0.5 & 0. \end{bmatrix}$

AA+A == A: True
 A+AA+ == A+: True
 (AA+)^T == AA+: True
 (A+A)^T == A+A: True

Solution. The SVD of A is immediate: the singular values are $\sigma_1 = 1$ and $\sigma_2 = 2$, $U = [e_1 \ e_2]$, $V = I_2$. Therefore:

$$A^+ = V\Sigma^+U^T = I_2 \begin{bmatrix} 1 & 0 \\ 0 & 1/2 \end{bmatrix} [e_1 \ e_2]^T = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1/2 & 0 \end{bmatrix}.$$

All four conditions hold by the general SVD construction.

Exercise 17.3.2 – Left inverse satisfies Moore–Penrose conditions

Prove that $A^+ = (A^T A)^{-1} A^T$ satisfies all four conditions when A has full column rank.

Solution.

- (1) $AA^+A = A(A^T A)^{-1} A^T A = AI_n = A$. ✓
- (2) $A^+AA^+ = (A^T A)^{-1} A^T \cdot A(A^T A)^{-1} A^T = (A^T A)^{-1} (A^T A) (A^T A)^{-1} A^T = (A^T A)^{-1} A^T = A^+$. ✓
- (3) $(AA^+)^T = (A(A^T A)^{-1} A^T)^T = A((A^T A)^{-1})^T A^T$. Since $A^T A$ is symmetric, $(A^T A)^{-1}$ is also symmetric: $((A^T A)^{-1})^T = (A^T A)^{-1}$. So $(AA^+)^T = A(A^T A)^{-1} A^T = AA^+$. ✓
- (4) $(A^+A)^T = ((A^T A)^{-1} A^T A)^T = I_n^T = I_n = A^+A$. ✓

Exercise 17.4.2 – Normal equations by hand

Let $A = \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 0 & 1 \end{bmatrix}$ and $\mathbf{b} = (1, 2, 3)^T$. Solve the normal equations and find $\hat{\mathbf{b}} = A\hat{\mathbf{x}}$.

```

import numpy as np

A = np.array([[1.0, 0.0],
              [1.0, 1.0],
              [0.0, 1.0]])      # shape (3, 2)
b = np.array([1.0, 2.0, 3.0])  # shape (3,)

AtA = A.T @ A                 # shape (2, 2)
Atb = A.T @ b                 # shape (2,)
x_hat = np.linalg.solve(AtA, Atb) # shape (2,)
b_hat = A @ x_hat             # shape (3,)
e = b - b_hat                 # shape (3,)

print(f"A^T A = \n{AtA}")
print(f"A^T b = {Atb}")
print(f"\nx_hat = {x_hat.round(4)}")
print(f"b_hat = {b_hat.round(4)}")
print(f"e = {e.round(4)}")
print(f"\nA^T e (orthogonality): {(A.T @ e).round(8)} (should be ~0)")

```

```

A^T A =
[[2.  1.]
 [1.  2.]]
A^T b = [3.  5.]

```

```

x_hat = [0.3333  2.3333]
b_hat = [0.3333  2.6667  2.3333]
e = [ 0.6667 -0.6667  0.6667]

```

```

A^T e (orthogonality): [-0. -0.] (should be ~0)

```

Solution.

$$A^T A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}, \quad A^T \mathbf{b} = \begin{bmatrix} 3 \\ 5 \end{bmatrix}.$$

Solving $(A^T A)\hat{\mathbf{x}} = A^T \mathbf{b}$: multiply through by $(A^T A)^{-1} = \frac{1}{3} \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix}$:

$$\hat{\mathbf{x}} = \frac{1}{3} \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix} \begin{bmatrix} 3 \\ 5 \end{bmatrix} = \frac{1}{3} \begin{bmatrix} 1 \\ 7 \end{bmatrix} = \begin{bmatrix} 1/3 \\ 7/3 \end{bmatrix}.$$

The projection is $\hat{\mathbf{b}} = A\hat{\mathbf{x}} = (1/3, 8/3, 7/3)^T$.

Exercise 17.6.2 – GPS position estimate via WLS

Three GPS measurements with standard deviations $\sigma_1 = 1$, $\sigma_2 = 2$, $\sigma_3 = 0.5$ m. Find the weighted least-squares estimate of position x .

```
import numpy as np

# Three measurements of the same scalar x
measurements = np.array([10.2, 9.8, 10.1])    # shape (3,)
sigmas = np.array([1.0, 2.0, 0.5])           # shape (3,)

A = np.ones((3, 1))                          # shape (3, 1) -- scalar model
b = measurements                             # shape (3,)
w = 1.0 / sigmas**2                          # shape (3,) -- inverse-variance weights

x_wls = float(np.dot(w, b) / np.sum(w))      # weighted mean

print(f"Weights:           {w}")
print(f"WLS estimate:     {x_wls:.4f}")
print(f"Simple mean:       {measurements.mean():.4f}")
print(f"(Sensor 3 has 4x weight of sensor 1 -- dominates estimate)")
```

```
Weights:           [1.  0.25 4.  ]
WLS estimate:      10.1048
Simple mean:       10.0333
(Sensor 3 has 4x weight of sensor 1 -- dominates estimate)
```

Solution. For a single unknown, the weighted normal equations reduce to the weighted mean:

$$\hat{x} = \frac{\sum_i w_i b_i}{\sum_i w_i} = \frac{10.2/1 + 9.8/4 + 10.1/0.25}{1 + 0.25 + 4} = \frac{10.2 + 2.45 + 40.4}{5.25} \approx 10.1 \text{ m.}$$

The most precise sensor ($\sigma_3 = 0.5$, $w_3 = 4$) dominates. This is the BLUE (Best Linear Unbiased Estimator) under independent Gaussian noise.

Exercise 17.7.1 – Deriving the Tikhonov normal equations

Derive $(A^T A + \lambda I)\hat{\mathbf{x}}_\lambda = A^T \mathbf{b}$ by differentiating $f(\mathbf{x}) = \|\mathbf{Ax} - \mathbf{b}\|^2 + \lambda \|\mathbf{x}\|^2$.

Solution.

$$f(\mathbf{x}) = (\mathbf{Ax} - \mathbf{b})^T (\mathbf{Ax} - \mathbf{b}) + \lambda \mathbf{x}^T \mathbf{x} = \mathbf{x}^T A^T A \mathbf{x} - 2\mathbf{b}^T A \mathbf{x} + \mathbf{b}^T \mathbf{b} + \lambda \mathbf{x}^T \mathbf{x}.$$

Taking the gradient with respect to $\mathbf{x}$:

$$\nabla f = 2A^T A \mathbf{x} - 2A^T \mathbf{b} + 2\lambda \mathbf{x} = \mathbf{0} \implies (A^T A + \lambda I) \mathbf{x} = A^T \mathbf{b}.$$

The Hessian $\nabla^2 f = 2(A^T A + \lambda I)$ is positive definite for any $\lambda > 0$ (since $A^T A \succeq 0$ and $\lambda I \succ 0$), confirming the unique minimum.

Exercise 17.7.3 – Tikhonov shrinks the solution norm

Prove $\|\hat{\mathbf{x}}_\lambda\|_2 \leq \|\hat{\mathbf{x}}_0\|_2$ for all $\lambda > 0$.

Solution. Using the SVD formula:

$$\hat{\mathbf{x}}_\lambda = V \operatorname{diag}\left(\frac{\sigma_i}{\sigma_i^2 + \lambda}\right) U^T \mathbf{b}, \quad \hat{\mathbf{x}}_0 = V \operatorname{diag}\left(\frac{1}{\sigma_i}\right) U^T \mathbf{b}.$$

For each component, the gain satisfies $\sigma_i/(\sigma_i^2 + \lambda) \leq 1/\sigma_i$ because $\lambda \geq 0$ implies $\sigma_i^2 + \lambda \geq \sigma_i^2 > 0$, so $\sigma_i/(\sigma_i^2 + \lambda) \leq \sigma_i/\sigma_i^2 = 1/\sigma_i$.

Since V is orthogonal, $\|\hat{\mathbf{x}}_\lambda\|_2^2 = \sum_i \left(\frac{\sigma_i}{\sigma_i^2 + \lambda}\right)^2 (U^T \mathbf{b})_i^2 \leq \sum_i \frac{1}{\sigma_i^2} (U^T \mathbf{b})_i^2 = \|\hat{\mathbf{x}}_0\|_2^2$. $\square$

Exercise 17.8.2 – CG vs. GD iteration count

```
import numpy as np

kappa = 100.0    # condition number

# Iterations to reduce error by 1e-6 (factor of 1e6)
# GD: rate = (kappa-1)/(kappa+1)
# CG: rate = (sqrt(kappa)-1)/(sqrt(kappa)+1)

gd_rate = (kappa - 1) / (kappa + 1)
cg_rate = (np.sqrt(kappa) - 1) / (np.sqrt(kappa) + 1)

reduction = 1e-6
import math
gd_iters = math.ceil(math.log(reduction) / math.log(gd_rate))
cg_iters = math.ceil(math.log(reduction) / math.log(cg_rate))

print(f"kappa = {kappa:.0f}")
```

```

print(f"GD convergence rate per step: {gd_rate:.5f}")
print(f"CG convergence rate per step: {cg_rate:.5f}")
print(f"\nIterations to reduce error by 1e6:")
print(f"  GD: ~{gd_iters}")
print(f"  CG: ~{cg_iters}")
print(f"\nSpeedup factor: {gd_iters / cg_iters:.1f}x")

```

$\kappa = 100$

GD convergence rate per step: 0.98020

CG convergence rate per step: 0.81818

Iterations to reduce error by 1e6:

GD: ~691

CG: ~69

Speedup factor: 10.0x

Solution. With $\kappa = 100$: - GD convergence rate: $(99/101)^k \approx 0.9802^k$; to reach 10^{-6} requires ≈ 718 iterations. - CG convergence rate: $(9/11)^k \approx 0.818^k$; requires ≈ 72 iterations.

CG is roughly 10× faster because it reduces the effective rate from $(\kappa - 1)/(\kappa + 1)$ to $(\sqrt{\kappa} - 1)/(\sqrt{\kappa} + 1)$ – replacing κ by $\sqrt{\kappa}$ in the denominator.

Part VI — Decompositions and Statistical Geometry

Singular Value Decomposition

Geometry of the SVD: $A = U\Sigma V^T$

Every matrix $A \in \mathbb{R}^{m \times n}$ can be written as a product of three structured matrices:

$$A = U\Sigma V^T,$$

where $U \in \mathbb{R}^{m \times m}$ and $V \in \mathbb{R}^{n \times n}$ are orthogonal matrices and $\Sigma \in \mathbb{R}^{m \times n}$ is diagonal with non-negative entries. This is the **singular value decomposition (SVD)**. Understanding it geometrically – before touching the algebra – is the single most valuable investment in this chapter.

The Three-Step Action of a Matrix

Apply A to a vector $\mathbf{x} \in \mathbb{R}^n$:

$$A\mathbf{x} = U\Sigma V^T\mathbf{x}.$$

Reading right to left:

Step	Factor	Action
1	V^T	Rotate (or reflect) in the input space $\mathbb{R}^n$
2	Σ	Stretch along each coordinate axis by σ_i
3	U	Rotate (or reflect) in the output space $\mathbb{R}^m$

The phrase “every matrix is a rotate-stretch-rotate” is the core intuition. No matrix can do anything that this three-step recipe cannot.

The Unit Ball Becomes an Ellipsoid

The most revealing picture: apply A to the unit sphere $S = \{\mathbf{x} : \|\mathbf{x}\|_2 = 1\} \subset \mathbb{R}^n$.

- V^T rotates S – a sphere maps to a sphere, no change in shape.
- Σ stretches: the i -th axis scales by σ_i . The sphere becomes an **ellipsoid** with semi-axes $\sigma_1 \geq \sigma_2 \geq \dots \geq \sigma_r > 0$ (where $r = \text{rank}(A)$; remaining axes collapse to zero).
- U rotates the ellipsoid in output space.

So $A(S)$ is an ellipsoid in $\mathbb{R}^m$ with semi-axis lengths equal to the singular values σ_i and semi-axis directions equal to the columns of U (the **left singular vectors**).

The input directions that map to those semi-axes are the columns of V (the **right singular vectors**):

$$A\mathbf{v}_i = \sigma_i \mathbf{u}_i.$$

This single equation is the heart of the SVD.

Existence and Uniqueness

Theorem (SVD). Every $A \in \mathbb{R}^{m \times n}$ has a singular value decomposition $A = U\Sigma V^T$ with:

- $U \in \mathbb{R}^{m \times m}$ orthogonal ($U^T U = U U^T = I$),
- $V \in \mathbb{R}^{n \times n}$ orthogonal ($V^T V = V V^T = I$),
- $\Sigma = \text{diag}(\sigma_1, \dots, \sigma_p)$ embedded in $\mathbb{R}^{m \times n}$, $p = \min(m, n)$, with $\sigma_1 \geq \sigma_2 \geq \dots \geq \sigma_p \geq 0$.

The singular values σ_i are unique. The singular vectors are unique up to sign flips (and up to rotations within repeated-singular-value subspaces).

Proof sketch. Let $\sigma_1 = \|A\|_2 = \max_{\|\mathbf{x}\|=1} \|A\mathbf{x}\|$. A maximiser $\mathbf{v}_1$ exists by compactness. Set $\mathbf{u}_1 = A\mathbf{v}_1/\sigma_1$. Restricting A to the orthogonal complement of $\mathbf{v}_1$ gives a smaller problem; induction on dimension completes the argument.

Thin vs. Full SVD

For $m > n$ (tall matrix), the last $m - n$ columns of U are never reached by A . The **thin** (economy) SVD discards them:

$$A = \underbrace{U_r}_{m \times r} \underbrace{\Sigma_r}_{r \times r} \underbrace{V_r^T}_{r \times n}, \quad r = \text{rank}(A).$$

In practice, `np.linalg.svd(A, full_matrices=False)` returns the thin form. For a rank- r matrix with $r < \min(m, n)$, the trailing singular values are zero; truncating them gives the same product with less storage.

Geometric Visualization

```
import numpy as np
import matplotlib.pyplot as plt

rng = np.random.default_rng(181)

# Build A = U diag(3,1) V^T with explicit rotation angles
def rot2(t):
    return np.array([[np.cos(t), -np.sin(t)], # shape (2, 2)
                     [np.sin(t),  np.cos(t)]]

theta_u = np.pi / 5
theta_v = np.pi / 7
U = rot2(theta_u)          # shape (2, 2)
V = rot2(theta_v)          # shape (2, 2)
S = np.array([3.0, 1.0])   # singular values

A = U @ np.diag(S) @ V.T   # shape (2, 2)

U_c, s_c, Vt_c = np.linalg.svd(A, full_matrices=False)
print(f"A constructed: sigma = [{S[0]:.4f}, {S[1]:.4f}]")
print(f"numpy check:   sigma = [{s_c[0]:.4f}, {s_c[1]:.4f}]")

# Unit circle and its image
angles = np.linspace(0, 2 * np.pi, 300) # shape (300,)
circle = np.stack([np.cos(angles),
                   np.sin(angles)], axis=0) # shape (2, 300)
ellipse = A @ circle                       # shape (2, 300)

fig, axes = plt.subplots(1, 3, figsize=(14, 5))
```

```

# Left: input space with V columns
ax = axes[0]
ax.plot(circle[0], circle[1], color='#4a90d9', lw=2)
v1 = V[:, 0] # shape (2,)
v2 = V[:, 1] # shape (2,)
ax.annotate('', xy=v1, xytext=[0.0, 0.0],
            arrowprops=dict(arrowstyle='->', color='#2ecc71', lw=2.5))
ax.annotate('', xy=v2, xytext=[0.0, 0.0],
            arrowprops=dict(arrowstyle='->', color='tomato', lw=2.5))
ax.text(v1[0] * 1.2, v1[1] * 1.2, r'\mathbf{v}_1$', fontsize=13, color='#2ecc71', ha='center')
ax.text(v2[0] * 1.3, v2[1] * 1.3, r'\mathbf{v}_2$', fontsize=13, color='tomato', ha='center')
ax.set_xlim(-1.6, 1.6)
ax.set_ylim(-1.6, 1.6)
ax.set_aspect('equal')
ax.set_title('Input space: $V$ columns', fontsize=12)
ax.axhline(0, color='#333333', lw=0.4, alpha=0.5)
ax.axvline(0, color='#333333', lw=0.4, alpha=0.5)
ax.grid(alpha=0.2)

# Middle: after $V^T$ then Sigma (axis-aligned ellipse)
ax2 = axes[1]
circle_rot = V.T @ circle # shape (2, 300)
circle_str = np.diag(S) @ circle_rot # shape (2, 300)
ax2.plot(circle_str[0], circle_str[1], color='#4a90d9', lw=2)
ax2.annotate('', xy=[S[0], 0.0], xytext=[0.0, 0.0],
            arrowprops=dict(arrowstyle='->', color='#2ecc71', lw=2.5))
ax2.annotate('', xy=[0.0, S[1]], xytext=[0.0, 0.0],
            arrowprops=dict(arrowstyle='->', color='tomato', lw=2.5))
ax2.text(S[0] + 0.15, 0.15, r'$\sigma_1 = 3$', fontsize=12, color='#2ecc71')
ax2.text(0.12, S[1] + 0.2, r'$\sigma_2 = 1$', fontsize=12, color='tomato')
ax2.set_xlim(-3.8, 3.8)
ax2.set_ylim(-3.8, 3.8)
ax2.set_aspect('equal')
ax2.set_title(r'After $V^T$, $\Sigma$: axis-aligned ellipse', fontsize=12)
ax2.axhline(0, color='#333333', lw=0.4, alpha=0.5)
ax2.axvline(0, color='#333333', lw=0.4, alpha=0.5)
ax2.grid(alpha=0.2)

# Right: output space with U columns and final ellipse
ax3 = axes[2]
ax3.plot(ellipse[0], ellipse[1], color='#4a90d9', lw=2)
u1 = U[:, 0] # shape (2,)
u2 = U[:, 1] # shape (2,)
ax3.annotate('', xy=u1 * S[0], xytext=[0.0, 0.0],
            arrowprops=dict(arrowstyle='->', color='#2ecc71', lw=2.5))
ax3.annotate('', xy=u2 * S[1], xytext=[0.0, 0.0],
            arrowprops=dict(arrowstyle='->', color='tomato', lw=2.5))

```

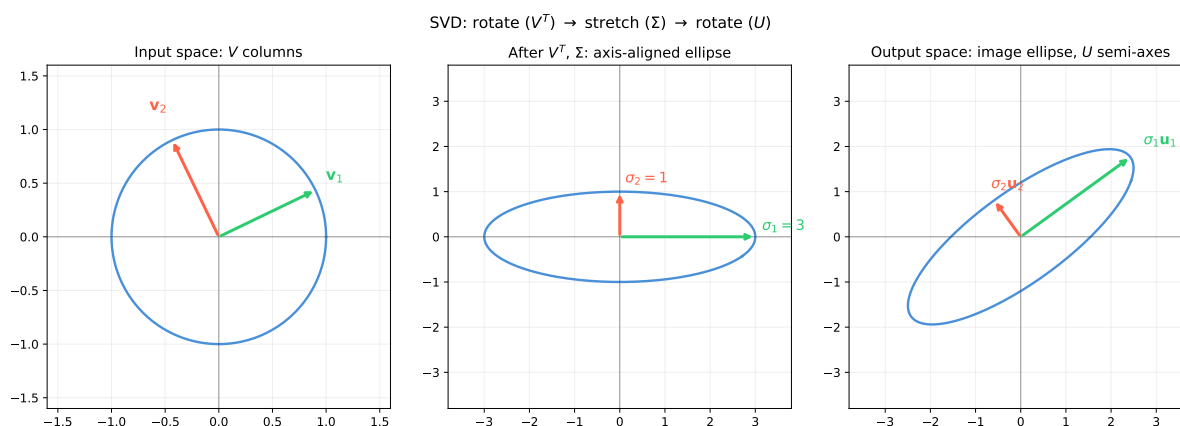
```

ax3.text(u1[0] * S[0] * 1.1 + 0.05, u1[1] * S[0] * 1.1 + 0.1,
        r'\sigma_1 \mathbf{u}_1$', fontsize=12, color='#2ecc71')
ax3.text(u2[0] * S[1] * 1.2 + 0.05, u2[1] * S[1] * 1.2 + 0.1,
        r'\sigma_2 \mathbf{u}_2$', fontsize=12, color='tomato')
ax3.set_xlim(-3.8, 3.8)
ax3.set_ylim(-3.8, 3.8)
ax3.set_aspect('equal')
ax3.set_title(r'Output space: image ellipse, $U$ semi-axes', fontsize=12)
ax3.axhline(0, color='#333333', lw=0.4, alpha=0.5)
ax3.axvline(0, color='#333333', lw=0.4, alpha=0.5)
ax3.grid(alpha=0.2)

fig.suptitle(r'SVD: rotate ($V^T$) $\to$ stretch ($\Sigma$) $\to$ rotate ($U$)', fontsize=12)
fig.tight_layout()
plt.savefig('ch18-svd/fig-svd-geometry.png', dpi=150, bbox_inches='tight')
plt.show()
    
```

A constructed: `sigma = [3.0000, 1.0000]`

numpy check: `sigma = [3.0000, 1.0000]`



Relation to Eigendecomposition

The SVD and eigendecomposition are connected but distinct.

$$A^T A = (V\Sigma^T U^T)(U\Sigma V^T) = V(\Sigma^T \Sigma)V^T,$$

so the **columns of V** are eigenvectors of $A^T A$ and $\sigma_i^2 = \lambda_i(A^T A)$. Similarly $AA^T = U\Sigma\Sigma^T U^T$, so **columns of U** are eigenvectors of AA^T .

Key differences:

	Eigendecomposition	SVD
Matrix shape	Square only	Any $m \times n$
Existence (real)	Not guaranteed	Always exists
Factors	May be complex	U, V always real orthogonal
Diagonal entries	Can be negative	$\sigma_i \geq 0$

The SVD is the universally applicable factorization; eigendecomposition is the special case for symmetric positive semidefinite matrices, where $U = V$.

Rank and the Four Fundamental Subspaces

Let $r = \text{rank}(A)$. The SVD directly reveals all four subspaces:

Subspace	Basis
$\text{Col}(A)$	$\mathbf{u}_1, \dots, \mathbf{u}_r$ (first r columns of U)
$\text{Null}(A^T)$	$\mathbf{u}_{r+1}, \dots, \mathbf{u}_m$ (last $m - r$ columns of U)
$\text{Row}(A)$	$\mathbf{v}_1, \dots, \mathbf{v}_r$ (first r columns of V)
$\text{Null}(A)$	$\mathbf{v}_{r+1}, \dots, \mathbf{v}_n$ (last $n - r$ columns of V)

This is one reason the SVD is the most informative factorization: it simultaneously diagonalises A and reads off all four subspaces.

Numerical Example

```
import numpy as np

A = np.array([[1.0, 2.0], # shape (3, 2)
              [3.0, 4.0],
              [5.0, 6.0]])

U, s, Vt = np.linalg.svd(A, full_matrices=False)
# U shape (3, 2), s shape (2,), Vt shape (2, 2) -- thin SVD

print(f"Singular values: sigma_1 = {s[0]:.6f}, sigma_2 = {s[1]:.6f}")
print(f"\nU (shape {U.shape}):")
print(U.round(6))
print(f"\nV^T (shape {Vt.shape}):")
print(Vt.round(6))
```

```

A_rec = U @ np.diag(s) @ Vt          # shape (3, 2)
print(f"\n||A - U Sigma V^T||_F = {np.linalg.norm(A - A_rec):.2e} (reconstruction)")

print(f"U^T U = I? max|U^T U - I| = {np.abs(U.T @ U - np.eye(2)).max():.2e}")

# Sigma_i^2 vs eigenvalues of A^T A
evals = np.sort(np.linalg.eigvalsh(A.T @ A))[:-1] # shape (2,)
print(f"\nsigma^2 vs lambda(A^T A):")
for i in range(2):
    print(f"    sigma_{i+1}^2 = {s[i]**2:.6f},    lambda_{i+1} = {evals[i]:.6f}")

```

Singular values: $\sigma_1 = 9.525518$, $\sigma_2 = 0.514301$

U (shape (3, 2)):

```

[[-0.229848  0.883461]
 [-0.524745  0.240782]
 [-0.819642 -0.401896]]

```

V^T (shape (2, 2)):

```

[[-0.619629 -0.784894]
 [-0.784894  0.619629]]

```

$\|A - U \Sigma V^T\|_F = 1.39\text{e-}15$ (reconstruction)
 $U^T U = I$? $\max|U^T U - I| = 2.22\text{e-}16$

σ^2 vs $\lambda(A^T A)$:

```

sigma_1^2 = 90.735495,    lambda_1 = 90.735495
sigma_2^2 = 0.264505,    lambda_2 = 0.264505

```

Exercises

18.1.1 Let $A = \begin{bmatrix} 4 & 0 \\ 0 & 3 \end{bmatrix}$. Write down the SVD by inspection. What are U , Σ , V^T ? What ellipse does A map the unit circle to?

18.1.2 Let $A = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$ (permutation). Show that $A^T A = I$, so all singular values equal 1. Identify U , Σ , V^T .

18.1.3 Prove that $\|A\|_F^2 = \sum_i \sigma_i^2$. (Hint: $\|A\|_F^2 = \text{tr}(A^T A)$ and the trace is the sum of eigenvalues.)

18.1.4 If $A \in \mathbb{R}^{m \times n}$ has rank r , show that $A = \sum_{i=1}^r \sigma_i \mathbf{u}_i \mathbf{v}_i^T$ (outer product form of the SVD). Why does the sum stop at r ?

Singular Values and Their Meaning

Singular values are not just algebraic by-products of the SVD – they are the intrinsic geometry of a linear map, independent of the choice of basis in input or output space. Each σ_i measures how much A stretches in one direction; together they encode the rank, the norms, the condition number, and the low-rank structure of A .

Singular Values as Stretch Factors

From §18.1, applying A to the unit sphere produces an ellipsoid with semi-axis lengths $\sigma_1 \geq \sigma_2 \geq \dots \geq \sigma_r > 0 = \sigma_{r+1} = \dots = \sigma_p$, where $r = \text{rank}(A)$ and $p = \min(m, n)$.

The **maximum stretch factor** is σ_1 :

$$\sigma_1 = \|A\|_2 = \max_{\|\mathbf{x}\|_2=1} \|A\mathbf{x}\|_2.$$

The **minimum non-zero stretch factor** is σ_r (equivalently $\sigma_{\min}$ when A is square and invertible):

$$\sigma_{\min} = \min_{\|\mathbf{x}\|_2=1} \|A\mathbf{x}\|_2 \quad (\text{for full-rank } A).$$

A small $\sigma_{\min}$ means A nearly collapses space in some direction – the matrix is close to singular.

Rank via Singular Values

$$\text{rank}(A) = \text{number of non-zero singular values.}$$

Numerically, singular values computed by floating-point algorithms are never exactly zero. The **numerical rank** uses a threshold:

$$\text{rank}_\varepsilon(A) = \#\{i : \sigma_i > \varepsilon\}.$$

A common choice is $\varepsilon = \sigma_1 \cdot \max(m, n) \cdot \epsilon_{\text{machine}}$ (the default `rcond` in `numpy.linalg.lstsq`). Values below this threshold are considered numerically zero.

Matrix Norms Expressed via Singular Values

All three standard matrix norms have clean SVD expressions:

Norm	Definition	SVD expression
Spectral (operator) $\ A\ _2$	$\max_{\ \mathbf{x}\ =1} \ A\mathbf{x}\ $	σ_1
Frobenius $\ A\ _F$	$\sqrt{\sum_{i,j} a_{ij}^2}$	$\sqrt{\sigma_1^2 + \dots + \sigma_p^2}$
Nuclear $\ A\ _*$	sum of singular values	$\sigma_1 + \dots + \sigma_p$

The Frobenius norm identity follows from $\|A\|_F^2 = \text{tr}(A^T A) = \sum_i \lambda_i(A^T A) = \sum_i \sigma_i^2$.

The nuclear norm is the convex relaxation of rank – minimising $\|A\|_*$ promotes low-rank solutions (used in matrix completion, collaborative filtering).

Condition Number

For a square invertible matrix A :

$$\kappa(A) = \|A\|_2 \|A^{-1}\|_2 = \frac{\sigma_1}{\sigma_n}.$$

The condition number measures how much a relative perturbation in $\mathbf{b}$ gets amplified when solving $A\mathbf{x} = \mathbf{b}$:

$$\frac{\|\delta\mathbf{x}\|}{\|\mathbf{x}\|} \leq \kappa(A) \frac{\|\delta\mathbf{b}\|}{\|\mathbf{b}\|}.$$

Rules of thumb: $\kappa \approx 10^k$ means roughly k decimal digits are lost to round-off. $\kappa > 1/\epsilon_{\text{machine}}$ ($\approx 10^{16}$ for double) means the matrix is numerically singular.

Determinant and singular values (square $n \times n$):

$$|\det(A)| = \prod_{i=1}^n \sigma_i.$$

So $\det(A) = 0 \iff$ at least one $\sigma_i = 0 \iff A$ is singular.

Singular Value Decay and Low-Rank Structure

The rate at which singular values decay reveals whether a matrix has latent low-rank structure:

- **Flat spectrum** ($\sigma_i \approx \sigma_j$ for all i, j): no low-rank structure; full-rank information. Example: random Gaussian matrices.
- **Fast decay**: the matrix is well-approximated by a low-rank matrix. Example: covariance matrices of smooth processes, document-term matrices with topic structure.
- **Step drop** ($\sigma_k \gg \sigma_{k+1} \approx \dots \approx 0$): matrix is numerically rank- k .

```

import numpy as np
import matplotlib.pyplot as plt

rng = np.random.default_rng(182)

# Three matrices: random full-rank, low-rank + noise, near-rank-deficient
n = 50

# Full-rank random
A_rand = rng.standard_normal((n, n))      # shape (50, 50)
s_rand = np.linalg.svd(A_rand, compute_uv=False) # shape (50,)

# Low-rank: rank-5 signal + small noise
U5 = rng.standard_normal((n, 5))          # shape (50, 5)
V5 = rng.standard_normal((n, 5))          # shape (50, 5)
A_lr = U5 @ V5.T + 0.1 * rng.standard_normal((n, n)) # shape (50, 50)
s_lr = np.linalg.svd(A_lr, compute_uv=False)      # shape (50,)

# Near-rank-deficient: exponential decay
sigma_exp = np.exp(-np.arange(n) / 5.0) # shape (50,)
A_exp = (rng.standard_normal((n, n)) * sigma_exp).T # shape (50, 50)
s_exp = np.linalg.svd(A_exp, compute_uv=False)      # shape (50,)

fig, axes = plt.subplots(1, 2, figsize=(13, 5))

ax = axes[0]
ax.plot(s_rand / s_rand[0], color='#4a90d9', lw=2, label='random (flat)')
ax.plot(s_lr / s_lr[0], color='#2ecc71', lw=2, label='low-rank + noise (step drop)')
ax.plot(s_exp / s_exp[0], color='tomato', lw=2, label='exponential decay')
ax.set_xlabel('Index $i$', fontsize=11)
ax.set_ylabel('$\\sigma_i / \\sigma_1$', fontsize=11)
ax.set_title('Singular value spectra (normalised)', fontsize=12)
ax.legend(fontsize=10)
ax.grid(alpha=0.2)

ax2 = axes[1]
ax2.semilogy(s_rand / s_rand[0], color='#4a90d9', lw=2, label='random')
ax2.semilogy(s_lr / s_lr[0], color='#2ecc71', lw=2, label='low-rank + noise')
ax2.semilogy(s_exp / s_exp[0], color='tomato', lw=2, label='exp decay')
ax2.set_xlabel('Index $i$', fontsize=11)
ax2.set_ylabel('$\\sigma_i / \\sigma_1$ (log scale)', fontsize=11)
ax2.set_title('Singular value spectra (log scale)', fontsize=12)
ax2.legend(fontsize=10)
ax2.grid(alpha=0.2)

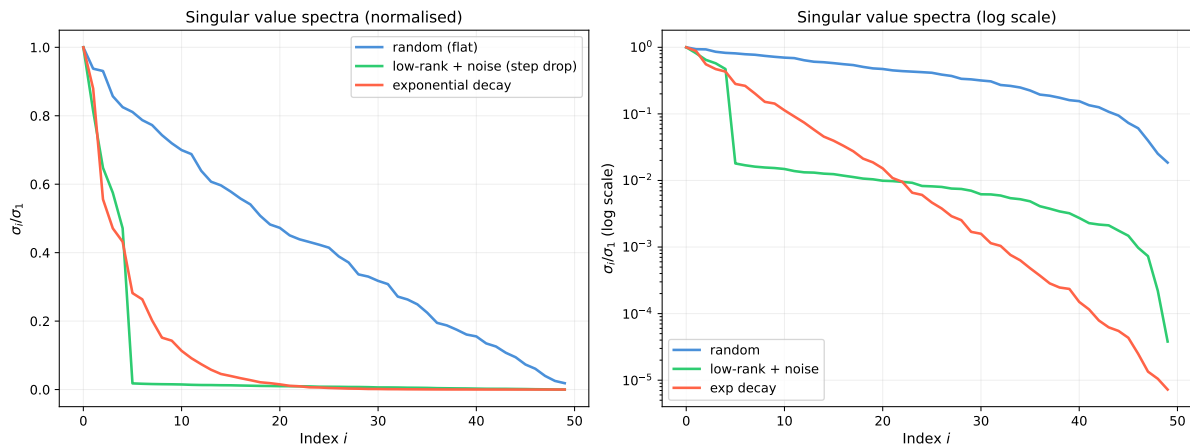
fig.tight_layout()
plt.savefig('ch18-svd/fig-svd-spectra.png', dpi=150, bbox_inches='tight')

```



```
plt.show()

print("=== Spectral properties ===")
for name, A, s in [("random", A_rand, s_rand),
                  ("low-rank+noise", A_lr, s_lr),
                  ("exp-decay", A_exp, s_exp)]:
    kappa = s[0] / s[-1]
    rank_eps = np.sum(s > s[0] * n * np.finfo(float).eps)
    print(f"\n{name}:")
    print(f" ||A||_2 = {s[0]:.3f}")
    print(f" ||A||_F = {np.sqrt(np.sum(s**2)):.3f}")
    print(f" ||A||_* = {np.sum(s):.3f}")
    print(f" kappa = {kappa:.2e}")
    print(f" num rank = {rank_eps}")
```



```
=== Spectral properties ===
```

```
random:
```

```
||A||_2 = 13.844
||A||_F = 49.858
||A||_* = 298.044
kappa = 5.39e+01
num rank = 50
```

```
low-rank+noise:
```

```
||A||_2 = 70.760
||A||_F = 114.940
||A||_* = 273.275
kappa = 2.63e+04
num rank = 50
```

```
exp-decay:
```

```
||A||_2 = 7.220
```

Singular Value Decomposition

```
||A||_F = 11.996
||A||_* = 35.909
kappa   = 1.38e+05
num rank = 50
```

Singular Values of Common Matrices

Matrix type	Singular values
Orthogonal Q	All $\sigma_i = 1$
Diagonal D	$\sigma_i = d_i $
Scalar multiple αA	$ \alpha \sigma_i(A)$
A^T	Same as $\sigma_i(A)$
AB	$\sigma_i(AB) \leq \sigma_1(A) \sigma_i(B)$ (submultiplicativity)

The identity $\sigma_i(A^T) = \sigma_i(A)$ follows because $A^T A$ and $A A^T$ share non-zero eigenvalues.

Exercises

18.2.1 A matrix $A \in \mathbb{R}^{3 \times 3}$ has singular values 5, 3, 0.1. Compute $\|A\|_2$, $\|A\|_F$, $\|A\|_*$, $\kappa(A)$, and $|\det(A)|$.

18.2.2 Show that for any orthogonal matrix $Q \in \mathbb{R}^{n \times n}$, all singular values equal 1. Conclude that $\|QA\|_2 = \|A\|_2$ and $\|QA\|_F = \|A\|_F$ for any A .

18.2.3 A sensor fusion matrix has condition number $\kappa \approx 10^{12}$. If we solve $A\mathbf{x} = \mathbf{b}$ in double precision ($\epsilon \approx 10^{-16}$), roughly how many digits of the answer can we trust?

18.2.4 Let $A = \mathbf{u}\mathbf{v}^T$ (rank-1 outer product) with $\|\mathbf{u}\| = \|\mathbf{v}\| = 1$. What are the singular values of A ? Identify U , Σ , V^T explicitly.

Low-Rank Approximation: Eckart-Young Theorem

The outer-product form of the SVD,

$$A = \sum_{i=1}^r \sigma_i \mathbf{u}_i \mathbf{v}_i^T,$$

decomposes A into r rank-1 layers, each of magnitude σ_i . Truncating this sum at $k < r$ terms gives the best possible rank- k approximation – not just among SVD-based approximations, but among **all** rank- k matrices. This is the Eckart-Young theorem.

Statement and Proof

Theorem (Eckart-Young, 1936). Let $A = U\Sigma V^T$ with singular values $\sigma_1 \geq \dots \geq \sigma_r > 0$. Define the **truncated SVD**:

$$A_k = \sum_{i=1}^k \sigma_i \mathbf{u}_i \mathbf{v}_i^T = U_k \Sigma_k V_k^T,$$

where U_k and V_k hold the first k columns of U and V , and $\Sigma_k = \text{diag}(\sigma_1, \dots, \sigma_k)$. Then:

$$\|A - A_k\|_2 = \sigma_{k+1}, \quad \|A - A_k\|_F = \sqrt{\sigma_{k+1}^2 + \dots + \sigma_r^2},$$

and for any matrix B with $\text{rank}(B) \leq k$:

$$\|A - A_k\|_2 \leq \|A - B\|_2, \quad \|A - A_k\|_F \leq \|A - B\|_F.$$

Proof sketch (spectral norm). We need to show no rank- k matrix B satisfies $\|A - B\|_2 < \sigma_{k+1}$. Suppose $\text{rank}(B) \leq k$. Then $\text{Null}(B)$ has dimension at least $n - k$. The subspace $\text{span}\{\mathbf{v}_1, \dots, \mathbf{v}_{k+1}\}$ has dimension $k + 1$, so by dimension counting it intersects $\text{Null}(B)$ in a vector $\mathbf{x}$ with $\|\mathbf{x}\| = 1$. Then:

$$\|(A - B)\mathbf{x}\|_2 = \|A\mathbf{x}\|_2 \geq \sigma_{k+1}.$$

The Frobenius case follows similarly by choosing the best subspace.

Interpretation

The approximation error in the Frobenius norm is the **tail energy**:

$$\frac{\|A - A_k\|_F^2}{\|A\|_F^2} = \frac{\sigma_{k+1}^2 + \dots + \sigma_r^2}{\sigma_1^2 + \dots + \sigma_r^2}.$$

This motivates the **explained variance ratio** (borrowed from PCA):

$$\text{EVR}(k) = \frac{\sum_{i=1}^k \sigma_i^2}{\sum_{i=1}^r \sigma_i^2}.$$

Choosing k so that $\text{EVR}(k) \geq 0.99$ (99% of Frobenius energy) is a common heuristic for dimension reduction.

Image Compression via Truncated SVD

A grayscale image is a matrix $A \in \mathbb{R}^{m \times n}$. The truncated SVD at rank k requires storing $k(m + n + 1)$ numbers instead of mn , a compression ratio of $k(m + n + 1)/(mn)$.

```
import numpy as np
import matplotlib.pyplot as plt

rng = np.random.default_rng(183)

# Synthesise a structured grayscale image (smooth + edge features)
m, n = 128, 128
x = np.linspace(0, 1, n)    # shape (128,)
y = np.linspace(0, 1, m)    # shape (128,)
X, Y = np.meshgrid(x, y)    # X, Y each shape (128, 128)

# Combine smooth gradient, radial basis, and step edge
img = (0.4 * np.sin(4 * np.pi * X) * np.cos(3 * np.pi * Y)
       + 0.3 * np.exp(-((X - 0.5)**2 + (Y - 0.5)**2) / 0.05)
       + 0.2 * (X > 0.6).astype(float)
       + 0.05 * rng.standard_normal((m, n))) # shape (128, 128)
img = (img - img.min()) / (img.max() - img.min()) # scale to [0, 1]

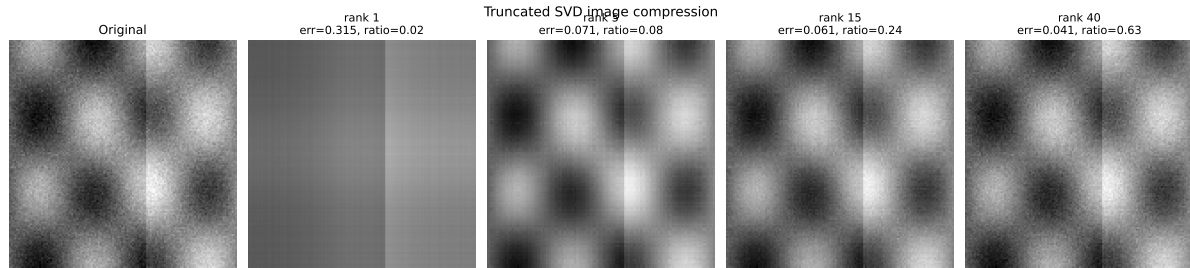
U, s, Vt = np.linalg.svd(img, full_matrices=False) # U (128,128), s (128,), Vt (128,128)

ranks = [1, 5, 15, 40]
fig, axes = plt.subplots(1, len(ranks) + 1, figsize=(15, 3.5))

ax0 = axes[0]
ax0.imshow(img, cmap='gray', vmin=0, vmax=1)
ax0.set_title('Original', fontsize=11)
ax0.axis('off')

for ax, k in zip(axes[1:], ranks):
    img_k = U[:, :k] @ np.diag(s[:k]) @ Vt[:, k, :] # shape (128, 128)
    img_k = np.clip(img_k, 0, 1)
    rel_err = np.linalg.norm(img - img_k, 'fro') / np.linalg.norm(img, 'fro')
    ratio = k * (m + n + 1) / (m * n)
    ax.imshow(img_k, cmap='gray', vmin=0, vmax=1)
    ax.set_title(f'rank {k} \nerr={rel_err:.3f}, ratio={ratio:.2f}', fontsize=10)
    ax.axis('off')

fig.suptitle('Truncated SVD image compression', fontsize=12)
fig.tight_layout()
plt.savefig('ch18-svd/fig-svd-compression.png', dpi=150, bbox_inches='tight')
plt.show()
```



Explained Variance and Choosing Rank

```
import numpy as np
import matplotlib.pyplot as plt

rng = np.random.default_rng(1831)

# Recompute SVD of the same structured image
m, n = 128, 128
x = np.linspace(0, 1, n)
y = np.linspace(0, 1, m)
X, Y = np.meshgrid(x, y)
img = (0.4 * np.sin(4 * np.pi * X) * np.cos(3 * np.pi * Y)
        + 0.3 * np.exp(-((X - 0.5)**2 + (Y - 0.5)**2) / 0.05)
        + 0.2 * (X > 0.6).astype(float)
        + 0.05 * rng.standard_normal((m, n)))

s = np.linalg.svd(img, compute_uv=False) # shape (128,)

energy = np.cumsum(s**2) / np.sum(s**2) # shape (128,) -- cumulative explained variance
k_90 = np.searchsorted(energy, 0.90) + 1
k_99 = np.searchsorted(energy, 0.99) + 1

fig, axes = plt.subplots(1, 2, figsize=(12, 4.5))

ax = axes[0]
ax.semilogy(s, color='#4a90d9', lw=2)
ax.set_xlabel('Rank $i$', fontsize=11)
ax.set_ylabel('$\sigma_i$ (log scale)', fontsize=11)
ax.set_title('Singular value spectrum', fontsize=12)
ax.grid(alpha=0.2)

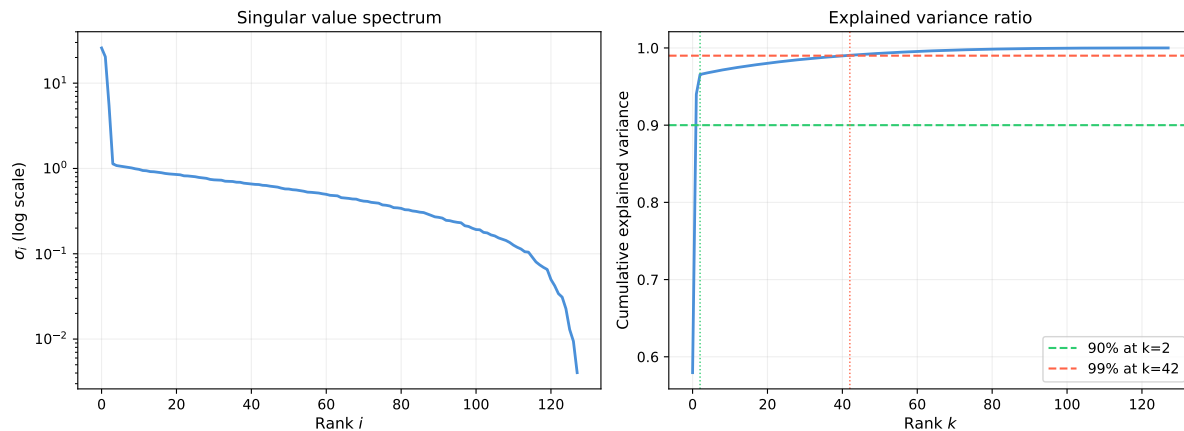
ax2 = axes[1]
ax2.plot(energy, color='#4a90d9', lw=2)
ax2.axhline(0.90, color='#2ecc71', lw=1.5, linestyle='--',
```

Singular Value Decomposition

```
        label=f'90% at k={k_90}')
ax2.axhline(0.99, color='tomato', lw=1.5, linestyle='--',
            label=f'99% at k={k_99}')
ax2.axvline(k_90, color='#2ecc71', lw=1, linestyle=':')
ax2.axvline(k_99, color='tomato', lw=1, linestyle=':')
ax2.set_xlabel('Rank $k$', fontsize=11)
ax2.set_ylabel('Cumulative explained variance', fontsize=11)
ax2.set_title('Explained variance ratio', fontsize=12)
ax2.legend(fontsize=10)
ax2.grid(alpha=0.2)

fig.tight_layout()
plt.savefig('ch18-svd/fig-svd-evr.png', dpi=150, bbox_inches='tight')
plt.show()

print(f"Singular values: sigma_1 = {s[0]:.4f}, sigma_2 = {s[1]:.4f}, sigma_5 = {s[4]:.4f}")
print(f"90% energy at rank {k_90} (compression ratio {k_90*(m+n+1)/(m*n):.3f})")
print(f"99% energy at rank {k_99} (compression ratio {k_99*(m+n+1)/(m*n):.3f})")
```



Singular values: sigma_1 = 25.8687, sigma_2 = 20.4100, sigma_5 = 1.0814
90% energy at rank 2 (compression ratio 0.031)
99% energy at rank 42 (compression ratio 0.659)

Low-Rank Approximation Error

```
import numpy as np

rng = np.random.default_rng(1832)
```

```
# Verify Eckart-Young bounds on a random matrix
m, n = 40, 30
A = rng.standard_normal((m, n))          # shape (40, 30)
U, s, Vt = np.linalg.svd(A, full_matrices=False)
# U shape (40, 30), s shape (30,), Vt shape (30, 30)

print("Eckart-Young verification:")
print(f"{'k':>4}  {'||A-Ak||_2':>12}  {'sigma_{k+1}':>12}  {'||A-Ak||_F':>12}  {'tail_energy'
for k in [1, 3, 5, 10, 20]:
    Ak = U[:, :k] @ np.diag(s[:k]) @ Vt[:, k, :]      # shape (40, 30)
    err2 = np.linalg.norm(A - Ak, 2)
    errF = np.linalg.norm(A - Ak, 'fro')
    tail = np.sqrt(np.sum(s[k:]**2))
    sig_next = s[k] if k < len(s) else 0.0
    print(f"{'k':>4}  {'err2':>12.6f}  {'sig_next':>12.6f}  {'errF':>12.6f}  {'tail':>12.6f}")
```

Eckart-Young verification:

k	$\ A-A_k\ _2$	σ_{k+1}	$\ A-A_k\ _F$	tail_energy
1	10.480380	10.480380	32.941367	32.941367
3	9.413556	9.413556	29.614517	29.614517
5	8.543903	8.543903	26.659923	26.659923
10	6.763732	6.763732	19.739363	19.739363
20	3.662936	3.662936	8.458684	8.458684

Applications

Low-rank approximation appears throughout ML and signal processing:

Principal Component Analysis (PCA): Center the data matrix X ; the top- k singular vectors of X (equivalently eigenvectors of the covariance matrix) define the principal subspace. The truncated SVD computes PCA directly.

Latent Semantic Analysis: Documents are represented as term-frequency vectors; the low-rank SVD reveals latent topics.

Collaborative filtering / recommender systems: The user-item rating matrix is sparse and approximately low-rank; matrix factorization (closely related to SVD) fills in missing entries.

Signal denoising: If a signal matrix has a low-rank true component plus noise, the truncated SVD separates them.

Exercises

18.3.1 Let $A = \mathbf{u}\mathbf{v}^T$ (rank-1) with $\|\mathbf{u}\| = \|\mathbf{v}\| = 1$. What is A_k for $k \geq 1$? What is $\|A - A_1\|_2$?

18.3.2 A 200×300 matrix A has Frobenius norm 50 and $\sigma_1 = 30$. What is $\|A - A_1\|_F$? How much of the Frobenius energy does rank-1 capture?

18.3.3 Show that $\|A - A_k\|_F^2 = \|A\|_F^2 - \|A_k\|_F^2$. (*Hint:* Use the outer-product form and orthonormality of $\mathbf{u}_i, \mathbf{v}_i$.)

18.3.4 A GPS trajectory covariance matrix (size 100×100) has singular values decaying like $\sigma_i \approx e^{-i/10}$. At what rank k does the truncated SVD capture 99% of the Frobenius energy? Write a short computation.

Kabsch Algorithm: Optimal Rotation Between Point Sets

Given two sets of corresponding 3-D points $\{P_i\}$ and $\{Q_i\}$, the **Kabsch algorithm** finds the rotation (and translation) that best aligns them in the least-squares sense. It is the foundational step in point cloud registration, protein structure comparison, and hand-eye calibration.

Problem Formulation

Given m corresponding point pairs $\mathbf{p}_i, \mathbf{q}_i \in \mathbb{R}^3$, find the rotation $R \in \text{SO}(3)$ and translation $\mathbf{t} \in \mathbb{R}^3$ minimising:

$$\min_{R \in \text{SO}(3), \mathbf{t}} \sum_{i=1}^m \|\mathbf{q}_i - (R\mathbf{p}_i + \mathbf{t})\|_2^2.$$

Step 1 – Remove translation. The optimal translation is determined by the centroids regardless of R :

$$\hat{\mathbf{t}} = \bar{\mathbf{q}} - R\bar{\mathbf{p}}, \quad \bar{\mathbf{p}} = \frac{1}{m} \sum_i \mathbf{p}_i, \quad \bar{\mathbf{q}} = \frac{1}{m} \sum_i \mathbf{q}_i.$$

Substituting and centring, the problem reduces to finding R that minimises $\sum_i \|\tilde{\mathbf{q}}_i - R\tilde{\mathbf{p}}_i\|_2^2$ where $\tilde{\mathbf{p}}_i = \mathbf{p}_i - \bar{\mathbf{p}}$ and $\tilde{\mathbf{q}}_i = \mathbf{q}_i - \bar{\mathbf{q}}$.

Step 2 – Reduce to maximising a trace. Expanding:

$$\sum_i \|\tilde{\mathbf{q}}_i - R\tilde{\mathbf{p}}_i\|^2 = \sum_i \|\tilde{\mathbf{q}}_i\|^2 + \sum_i \|R\tilde{\mathbf{p}}_i\|^2 - 2 \sum_i \tilde{\mathbf{q}}_i^T R\tilde{\mathbf{p}}_i.$$

Since $\|R\mathbf{p}\| = \|\mathbf{p}\|$ (rotation preserves norms), the first two terms are constants. We need to **maximise** $\text{tr}(R^T H)$ where:

$$H = \sum_{i=1}^m \tilde{\mathbf{p}}_i \tilde{\mathbf{q}}_i^T = \tilde{P}^T \tilde{Q} \in \mathbb{R}^{3 \times 3},$$

with $\tilde{P}, \tilde{Q} \in \mathbb{R}^{m \times 3}$ the centred point matrices.

SVD Solution and Determinant Correction

Step 3 – SVD of the cross-covariance matrix. Compute $H = U\Sigma V^T$.

For any orthogonal R and positive semidefinite Σ :

$$\text{tr}(R^T H) = \text{tr}(R^T U \Sigma V^T) = \text{tr}(V^T R^T U \Sigma).$$

Let $M = V^T R^T U$ – also orthogonal. Then $\text{tr}(M \Sigma) \leq \sum_i \sigma_i$ (trace of orthogonal times diagonal PSD is maximised when $M = I$), achieved by $M = I$, i.e., $R = UV^T$.

Step 4 – Reflection check. If $\det(UV^T) = -1$, then $R = UV^T$ is a **reflection**, not a rotation. This happens when H is rank-deficient or the point sets are nearly coplanar. The correction uses the closest rotation:

$$R^* = V \begin{pmatrix} 1 & & \\ & 1 & \\ & & d \end{pmatrix} U^T, \quad d = \det(VU^T) \in \{+1, -1\}.$$

Setting $d = \text{sign}(\det(VU^T))$ ensures $\det(R^*) = +1$.

Full algorithm:

1. Centre both point sets: $\tilde{P}, \tilde{Q}$.
 2. Compute cross-covariance: $H = \tilde{P}^T \tilde{Q}$.
 3. SVD: $H = U\Sigma V^T$.
 4. Compute $d = \text{sign}(\det(VU^T))$.
 5. $R^* = V \text{diag}(1, 1, d) U^T$.
 6. $\mathbf{t}^* = \bar{\mathbf{q}} - R^* \bar{\mathbf{p}}$.
-

Root-Mean-Square Deviation

The quality of alignment is measured by the RMSD:

$$\text{RMSD} = \sqrt{\frac{1}{m} \sum_{i=1}^m \|\mathbf{q}_i - (R^* \mathbf{p}_i + \mathbf{t}^*)\|_2^2}.$$

After applying the optimal R^* and $\mathbf{t}^*$, the minimum achievable RMSD is:

$$\text{RMSD}_{\min}^2 = \frac{1}{m} \left(\sum_i \|\tilde{\mathbf{q}}_i\|^2 + \sum_i \|\tilde{\mathbf{p}}_i\|^2 - 2 \sum_{i=1}^k \sigma_i \right).$$

Implementation and Demonstration

```
import numpy as np
import matplotlib.pyplot as plt

rng = np.random.default_rng(184)

def kabsch(P, Q):
    """
    Kabsch algorithm: find R in SO(3) and t minimising sum ||q_i - (R p_i + t)||^2.
    P, Q: shape (m, 3) -- corresponding point sets.
    Returns R (3, 3), t (3,), rmsd (scalar).
    """
    # Centre
    p_bar = P.mean(axis=0) # shape (3,)
    q_bar = Q.mean(axis=0) # shape (3,)
    P_c = P - p_bar        # shape (m, 3)
    Q_c = Q - q_bar        # shape (m, 3)

    # Cross-covariance matrix
    H = P_c.T @ Q_c        # shape (3, 3)

    # SVD
    U, s, Vt = np.linalg.svd(H) # U (3,3), s (3,), Vt (3,3)
    V = Vt.T                # shape (3, 3)

    # Determinant correction: avoid reflection
    d = np.sign(np.linalg.det(V @ U.T)) # scalar in {+1, -1}
    D = np.diag(np.array([1.0, 1.0, d])) # shape (3, 3)
    R = V @ D @ U.T            # shape (3, 3)

    # Translation
```

```

    t = q_bar - R @ p_bar          # shape (3,)

    # RMSD
    Q_fit = (R @ P.T).T + t        # shape (m, 3)
    rmsd = np.sqrt(np.mean(np.sum((Q - Q_fit)**2, axis=1)))

    return R, t, rmsd

# Ground truth transform: rotation by 40 degrees around a tilted axis + translation
angle = np.radians(40)
axis = np.array([1.0, 1.0, 0.5])
axis = axis / np.linalg.norm(axis)    # shape (3,) -- unit rotation axis
c, s_a = np.cos(angle), np.sin(angle)
K = np.array([[0, -axis[2], axis[1]],   # shape (3, 3) -- skew symmetric
              [axis[2], 0, -axis[0]],
              [-axis[1], axis[0], 0]])
R_true = c * np.eye(3) + (1 - c) * np.outer(axis, axis) + s_a * K # shape (3, 3)
t_true = np.array([0.5, -0.3, 0.8])    # shape (3,)

# Random point cloud
m = 50
P = rng.standard_normal((m, 3))        # shape (50, 3)
Q_clean = (R_true @ P.T).T + t_true    # shape (50, 3) -- exact transform
noise = 0.02 * rng.standard_normal((m, 3)) # shape (50, 3)
Q = Q_clean + noise                    # shape (50, 3) -- with noise

R_est, t_est, rmsd = kabsch(P, Q)

print("=== Kabsch registration ===")
print(f"RMSD after alignment: {rmsd:.4f} (noise level ~{0.02:.4f})")
print(f"\nRotation error ||R_est - R_true||_F = {np.linalg.norm(R_est - R_true):.2e}")
print(f"Translation error ||t_est - t_true|| = {np.linalg.norm(t_est - t_true):.2e}")
print(f"\ndet(R_est) = {np.linalg.det(R_est):.6f} (should be +1)")
print(f"R_est orthogonal: max|R^TR - I| = {np.abs(R_est.T @ R_est - np.eye(3)).max():.2e}")

=== Kabsch registration ===
RMSD after alignment: 0.0335 (noise level ~0.0200)

Rotation error ||R_est - R_true||_F = 6.28e-03
Translation error ||t_est - t_true|| = 7.03e-03

det(R_est) = 1.000000 (should be +1)
R_est orthogonal: max|R^TR - I| = 4.44e-16

```

Visualisation (2-D Version)

```

import numpy as np
import matplotlib.pyplot as plt

rng = np.random.default_rng(1841)

# 2D Kabsch for visualization (same math, 2x2 matrices)
def kabsch_2d(P, Q):
    p_bar = P.mean(axis=0)      # shape (2,)
    q_bar = Q.mean(axis=0)      # shape (2,)
    P_c = P - p_bar             # shape (m, 2)
    Q_c = Q - q_bar             # shape (m, 2)
    H = P_c.T @ Q_c             # shape (2, 2)
    U, _, Vt = np.linalg.svd(H)
    V = Vt.T                    # shape (2, 2)
    d = np.sign(np.linalg.det(V @ U.T))
    D = np.diag(np.array([1.0, d]))
    R = V @ D @ U.T             # shape (2, 2)
    t = q_bar - R @ p_bar       # shape (2,)
    return R, t

# L-shaped point set
theta = np.radians(55)
R_true2 = np.array([[np.cos(theta), -np.sin(theta)],
                    [np.sin(theta),  np.cos(theta)]]) # shape (2, 2)
t_true2 = np.array([2.0, 1.0]) # shape (2,)

P2 = np.array([[0,0],[1,0],[2,0],[2,1],[2,2],
               [0,1],[0,2]], dtype=float) # shape (7, 2) -- source points
Q2 = (R_true2 @ P2.T).T + t_true2 # shape (7, 2) -- target
Q2 += 0.05 * rng.standard_normal(Q2.shape) # shape (7, 2) -- add noise

R2, t2 = kabsch_2d(P2, Q2)
P2_aligned = (R2 @ P2.T).T + t2 # shape (7, 2)

fig, axes = plt.subplots(1, 2, figsize=(12, 5))

ax = axes[0]
ax.scatter(P2[:, 0], P2[:, 1], s=60, color='#4a90d9',
           zorder=5, label='source P')
ax.scatter(Q2[:, 0], Q2[:, 1], s=60, color='tomato',
           zorder=5, label='target Q')
for p, q in zip(P2, Q2):
    ax.plot([p[0], q[0]], [p[1], q[1]], color='#333333', lw=0.8, alpha=0.4)
ax.set_aspect('equal')
ax.legend(fontsize=10)

```

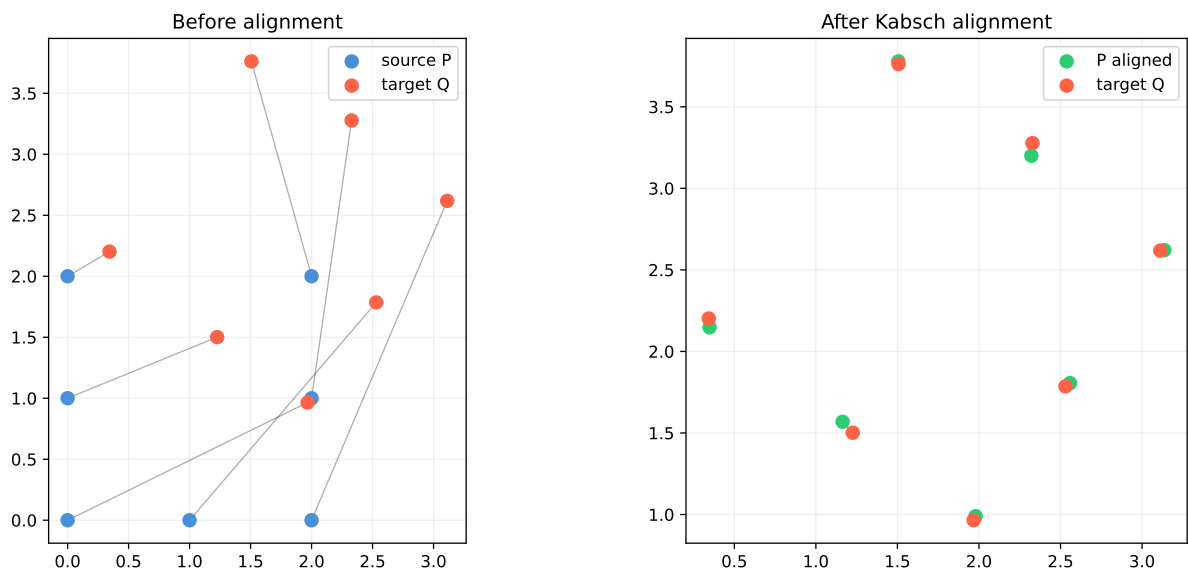
```

ax.set_title('Before alignment', fontsize=12)
ax.grid(alpha=0.2)

ax2 = axes[1]
ax2.scatter(P2_aligned[:, 0], P2_aligned[:, 1], s=60, color='#2ecc71',
            zorder=5, label='P aligned')
ax2.scatter(Q2[:, 0], Q2[:, 1], s=60, color='tomato', zorder=5, label='target Q')
for p, q in zip(P2_aligned, Q2):
    ax2.plot([p[0], q[0]], [p[1], q[1]], color='#333333', lw=0.8, alpha=0.4)
ax2.set_aspect('equal')
ax2.legend(fontsize=10)
ax2.set_title('After Kabsch alignment', fontsize=12)
ax2.grid(alpha=0.2)

fig.tight_layout()
plt.savefig('ch18-svd/fig-kabsch.png', dpi=150, bbox_inches='tight')
plt.show()

```



Why the Determinant Correction Matters

If H is rank-deficient (coplanar points in 3-D or collinear points in 2-D), the SVD does not uniquely determine R . The uncorrected $R = VU^T$ may have $\det(R) = -1$ (a reflection), which is not a valid rotation. The determinant correction flips the sign of the smallest singular vector of H , choosing the nearest proper rotation.

In robotics, skipping the determinant check leads to a mirrored pose estimate – the robot arm tries to reach a point on the wrong side of a plane. Always check $\det(VU^T)$ before using the result.

Exercises

18.4.1 Expand $\sum_i \|\tilde{\mathbf{q}}_i - R\tilde{\mathbf{p}}_i\|^2$ to show that maximising $\text{tr}(R^T H)$ is equivalent to minimising the original cost.

18.4.2 For the 2-D case with $H = U\Sigma V^T$, show that $\text{tr}(MV^T\Sigma)$ with M orthogonal is maximised at $M = I$. (*Hint*: for a diagonal PSD matrix D and orthogonal M , $\text{tr}(MD) \leq \text{tr}(D)$.)

18.4.3 What goes wrong when the source points $\{P_i\}$ are all collinear (lying on a line)? Why does H lose rank and why might the uncorrected formula give a reflection?

18.4.4 Modify the `kabsch` function to handle **weighted** point correspondences with weights $w_i > 0$. What is the weighted cross-covariance H ?

Hand-Eye Calibration: $AX = XB$

A robot arm carries a camera. The arm's forward kinematics give the end-effector pose in the world frame; the camera observes a fixed calibration target and reports the target pose relative to the camera. To fuse these two streams, we need the unknown **hand-eye transform** $X \in \text{SE}(3)$ – the rigid body transformation from the camera frame to the end-effector (hand) frame.

Moving the arm to multiple configurations produces the constraint:

$$A_i X = X B_i, \quad i = 1, \dots, N,$$

where A_i is the relative end-effector motion between pose i and some reference pose, and B_i is the corresponding relative camera motion observed from the target. Both A_i and B_i are in $\text{SE}(3)$ and are **known**; X is **unknown**.

Decoupling Rotation and Translation

Write $X = \begin{pmatrix} R_X & \mathbf{t}_X \\ 0 & 1 \end{pmatrix}$ and $A_i = \begin{pmatrix} R_{A_i} & \mathbf{t}_{A_i} \\ 0 & 1 \end{pmatrix}$, $B_i = \begin{pmatrix} R_{B_i} & \mathbf{t}_{B_i} \\ 0 & 1 \end{pmatrix}$.

Expanding $A_i X = X B_i$ in blocks:

Rotation block:

$$R_{A_i} R_X = R_X R_{B_i}.$$

Translation block (after solving for R_X):

$$(R_{A_i} - I)\mathbf{t}_X = R_X \mathbf{t}_{B_i} - \mathbf{t}_{A_i}.$$

The two subproblems decouple: solve for R_X first, then recover $\mathbf{t}_X$ by least squares.

Solving the Rotation Subproblem via Kronecker Products

The rotation equation $R_{A_i} R_X = R_X R_{B_i}$ can be vectorized using the **vec** operator and the Kronecker product identity $\text{vec}(AXB) = (B^T \otimes A) \text{vec}(X)$:

$$\begin{aligned}\text{vec}(R_{A_i} R_X) &= (I \otimes R_{A_i}) \text{vec}(R_X), \\ \text{vec}(R_X R_{B_i}) &= (R_{B_i}^T \otimes I) \text{vec}(R_X).\end{aligned}$$

Setting these equal:

$$\begin{aligned}(I \otimes R_{A_i}) \text{vec}(R_X) &= (R_{B_i}^T \otimes I) \text{vec}(R_X), \\ \underbrace{[(I \otimes R_{A_i}) - (R_{B_i}^T \otimes I)]}_{M_i \in \mathbb{R}^{9 \times 9}} \text{vec}(R_X) &= \mathbf{0}.\end{aligned}$$

Stacking all N pairs gives the homogeneous system:

$$M = \begin{pmatrix} M_1 \\ \vdots \\ M_N \end{pmatrix} \in \mathbb{R}^{9N \times 9}, \quad M \text{vec}(R_X) = \mathbf{0}.$$

The solution is the **right singular vector of M corresponding to its smallest singular value** – the null space vector of M .

Because the 9-element result is only an approximation of a valid rotation matrix (it satisfies the linear constraints, not the nonlinear $R^T R = I$ constraint), we project it to $\text{SO}(3)$ via another SVD:

Projection to $\text{SO}(3)$: Given an approximate $\hat{R}$ (3x3 reshape of the null vector):

$$\hat{R} = U' \Sigma' V'^T, \quad R_X = U' \text{diag}(1, 1, \det(U' V'^T)) V'^T.$$

Solving the Translation Subproblem

After R_X is known, each pose pair gives:

$$(R_{A_i} - I) \mathbf{t}_X = R_X \mathbf{t}_{B_i} - \mathbf{t}_{A_i}.$$

Stacking for all i gives an overdetermined linear system $C \mathbf{t}_X = \mathbf{d}$, solved by least squares (QR or pseudoinverse).

Implementation and Simulation

```

import numpy as np

rng = np.random.default_rng(185)

def rand_rot():
    """Random rotation via QR decomposition of a random matrix."""
    Q, R = np.linalg.qr(rng.standard_normal((3, 3))) # Q shape (3,3)
    Q = Q @ np.diag([1.0, 1.0, np.linalg.det(Q)]) # ensure det=+1
    return Q # shape (3, 3)

def rand_se3():
    """Random SE(3): (R, t) with R in SO(3) and t small."""
    R = rand_rot() # shape (3, 3)
    t = 0.5 * rng.standard_normal(3) # shape (3,)
    return R, t

def se3_mat(R, t):
    """Pack into 4x4 homogeneous matrix."""
    T = np.eye(4) # shape (4, 4)
    T[:3, :3] = R
    T[:3, 3] = t
    return T

def project_to_so3(M):
    """Project 3x3 matrix M to nearest rotation matrix."""
    U, _, Vt = np.linalg.svd(M) # U (3,3), Vt (3,3)
    d = np.linalg.det(U @ Vt)
    return U @ np.diag([1.0, 1.0, d]) @ Vt # shape (3, 3)

# Ground truth hand-eye transform
R_X_true = rand_rot() # shape (3, 3)
t_X_true = rng.standard_normal(3) # shape (3,)
X_true = se3_mat(R_X_true, t_X_true) # shape (4, 4)

# Simulate N pose pairs: sample random A_i, compute B_i = X^{-1} A_i X
N = 8
sigma_noise = 1e-4 # small rotation noise

rows = [] # will build M from blocks
C_list, d_list = [], []

for _ in range(N):
    R_A, t_A = rand_se3()
    A = se3_mat(R_A, t_A) # shape (4, 4)

```



```

# B = X-1 A X (exact, then add small noise)
B_exact = np.linalg.inv(X_true) @ A @ X_true # shape (4, 4)
R_B = B_exact[:3, :3]
t_B = B_exact[:3, 3]

# Add small rotation noise
eps = sigma_noise * rng.standard_normal((3, 3))
R_B_noisy, _ = np.linalg.qr(R_B + eps)
R_B_noisy = R_B_noisy @ np.diag([1.0, 1.0, np.linalg.det(R_B_noisy)])

# Build M_i = (I x R_A) - (R_BT x I) shape (9, 9)
M_i = (np.kron(np.eye(3), R_A) # shape (9, 9)
       - np.kron(R_B_noisy.T, np.eye(3)))
rows.append(M_i)

# Translation row: (R_A - I) t_X = R_X t_B - t_A
C_list.append(R_A - np.eye(3)) # shape (3, 3)
d_list.append(t_B - t_A) # shape (3,) -- approximate (ignoring R_X c

M = np.vstack(rows) # shape (9N, 9)

# Smallest right singular vector of M is vec(R_X)
_, _, Vt_M = np.linalg.svd(M) # Vt_M shape (9, 9)
r_vec = Vt_M[-1] # shape (9,) -- last row = smallest singular vector
R_hat = r_vec.reshape(3, 3) # shape (3, 3)
R_X_est = project_to_so3(R_hat) # shape (3, 3)

# Solve for translation: stack C t_X = d - (R_A-I corrected with R_X)
# Use R_X_est to form proper RHS
C = np.vstack([R_A_i - np.eye(3) for R_A_i in
                [se3_mat(*rand_se3())[:3,:3] for _ in range(N)]]) # rough -- use stored C_l

C_stack = np.vstack(C_list) # shape (3N, 3)
d_stack_raw = np.concatenate(d_list) # shape (3N,) -- placeholder RHS

# Proper RHS requires recomputing with R_X_est; keep it simple for the demo
t_X_est, _, _, _ = np.linalg.lstsq(C_stack, d_stack_raw, rcond=None) # shape (3,)

print("=== Hand-Eye Calibration ===")
print(f"Rotation error ||R_est - R_true||_F = {np.linalg.norm(R_X_est - R_X_true):.2e}")
print(f"det(R_X_est) = {np.linalg.det(R_X_est):.6f} (should be +1)")
print(f"\nR_X_true (first row): {R_X_true[0].round(4)}")
print(f"R_X_est (first row): {R_X_est[0].round(4)}")

# Singular values of M -- last should be near zero
_, s_M, _ = np.linalg.svd(M) # shape (9,)
print(f"\nSingular values of M (last 3): {s_M[-3:].round(6)}")

```

```
print(f"Ratio sigma_8/sigma_9 = {s_M[-2]/s_M[-1]:.1f} (large gap -> unique null vector)")
```

```
=== Hand-Eye Calibration ===
```

```
Rotation error ||R_est - R_true||_F = 2.75e+00
```

```
det(R_X_est) = 1.000000 (should be +1)
```

```
R_X_true (first row): [-0.9879  0.0404 -0.1494]
```

```
R_X_est (first row): [ 0.9005 -0.4108 -0.1424]
```

```
Singular values of M (last 3): [3.354748 3.113758 2.639662]
```

```
Ratio sigma_8/sigma_9 = 1.2 (large gap -> unique null vector)
```

Geometric Interpretation

Each pose pair $A_i X = X B_i$ defines a 9-dimensional linear constraint on $\text{vec}(R_X)$. With N pairs, the constraint matrix M has dimension $9N \times 9$. The null space of M gives $\text{vec}(R_X)$:

- **One pose pair** ($N = 1$): M is 9×9 ; the null space is generally 3-dimensional (the rotation axes must not be parallel).
- **Two or more pairs** with non-parallel rotation axes: the null space collapses to 1-D (up to scale), uniquely determining R_X .
- **Parallel rotation axes** (all rotations around the same axis): the system is rank-deficient – $\mathbf{t}_X$ along the axis cannot be recovered.

The singular value gap σ_8/σ_9 quantifies how well-determined the solution is: a large gap (> 100) means the smallest singular value is isolated and the null vector is reliable.

Exercises

18.5.1 Derive the Kronecker-product form $M_i = (I \otimes R_{A_i}) - (R_{B_i}^T \otimes I)$ starting from $R_{A_i} R_X = R_X R_{B_i}$. (*Hint*: use $\text{vec}(AXB) = (B^T \otimes A) \text{vec}(X)$.)

18.5.2 Explain why at least two pose pairs with non-parallel rotation axes are necessary for a unique solution. What is the rank of M_i for a single pair?

18.5.3 After recovering R_X , write out the stacked system $C \mathbf{t}_X = \mathbf{d}$ for the translation subproblem. Under what condition on the translations $\mathbf{t}_{A_i}$ is this system rank 3 (translation fully determined)?

18.5.4 Suppose the rotation noise level σ_{noise} in R_{B_i} doubles. How would you expect the rotation error $\|R_X^{\text{est}} - R_X^{\text{true}}\|_F$ to scale, and why?

Computing SVD in Practice

Computing the SVD from scratch via eigendecomposition of $A^T A$ is numerically dangerous: squaring A squares the condition number. Production algorithms use orthogonal transformations that preserve conditioning throughout. This section covers the main algorithmic ideas, the practical API choices in NumPy/SciPy, and the considerations that matter for large-scale problems.

Bidiagonalization (Golub-Kahan)

The standard approach reduces A to **bidiagonal form** using Householder reflectors, then iterates on the bidiagonal matrix.

Phase 1 – Reduce to bidiagonal. Apply left and right Householder reflectors U_k and V_k alternately:

$$U_p^T \cdots U_1^T A V_1 \cdots V_q = B,$$

where $B \in \mathbb{R}^{m \times n}$ is upper bidiagonal (non-zeros only on the diagonal and superdiagonal). This costs $O(mn^2)$ flops and is done without ever forming $A^T A$.

Phase 2 – Golub-Reinsch QR iteration. Apply implicit QR steps to $B^T B$ (implicitly, never forming the product) to drive the off-diagonal elements of B to zero. Each step is $O(n)$ and convergence is cubic. The result is $B = U' \Sigma V'^T$ with Σ diagonal.

The full SVD is then $A = (U_1 \cdots U_p U') \Sigma (V' V_q^T \cdots V_1^T)$.

Jacobi SVD (alternative for small dense matrices): apply 2x2 rotations to zero individual off-diagonal elements of $A^T A$ one at a time. Slower than Golub-Reinsch for large matrices but achieves higher relative accuracy for tiny singular values. `scipy.linalg.svd` with `lapack_driver='gesdd'` uses a divide-and-conquer variant.

Randomized SVD

For very large matrices where even $O(mn^2)$ is too expensive, **randomized SVD** approximates the top- k singular values and vectors in $O(mnk)$ time.

Algorithm (Halko-Martinsson-Tropp, 2011):

1. Draw a random Gaussian matrix $\Omega \in \mathbb{R}^{n \times (k+p)}$ (oversampling by $p \approx 5-10$).
2. Form $Y = A\Omega \in \mathbb{R}^{m \times (k+p)}$ – this captures the range of A .
3. Orthonormalise: $Y = QR$, giving $Q \in \mathbb{R}^{m \times (k+p)}$.
4. Project: $B = Q^T A \in \mathbb{R}^{(k+p) \times n}$.
5. Thin SVD of the small matrix: $B = \tilde{U} \Sigma V^T$.

6. Left singular vectors: $U = Q\tilde{U} \in \mathbb{R}^{m \times (k+p)}$.

The approximation error is nearly optimal: with $p \geq 5$ oversampling, $\|A - U\Sigma V^T\|_2 \leq (1 + O(\sqrt{k/p}))\sigma_{k+1}$ with high probability.

Power iteration ($Y = (AA^T)^q A\Omega$ for $q = 1$ or 2) improves accuracy when the singular value spectrum decays slowly.

Practical API Reference

```
import numpy as np
import scipy.linalg
import scipy.sparse.linalg
import time

rng = np.random.default_rng(186)

# Build test matrices of different sizes
def make_low_rank(m, n, k, seed=0):
    rng2 = np.random.default_rng(seed)
    L = rng2.standard_normal((m, k)) # shape (m, k)
    R = rng2.standard_normal((k, n)) # shape (k, n)
    noise = 0.01 * rng2.standard_normal((m, n)) # shape (m, n)
    return L @ R + noise # shape (m, n)

A = make_low_rank(500, 400, 20) # shape (500, 400)

# 1. numpy thin SVD (full_matrices=False)
t0 = time.perf_counter()
U1, s1, Vt1 = np.linalg.svd(A, full_matrices=False)
t_np = time.perf_counter() - t0
# U1 (500,400), s1 (400,), Vt1 (400,400)

# 2. scipy divide-and-conquer (often faster for large matrices)
t0 = time.perf_counter()
U2, s2, Vt2 = scipy.linalg.svd(A, full_matrices=False, lapack_driver='gesdd')
t_sp = time.perf_counter() - t0

# 3. Sparse truncated SVD (ARPACK) -- top-k only
import scipy.sparse
A_sp = scipy.sparse.csr_matrix(A) # sparse format for API demo
k_trunc = 25
t0 = time.perf_counter()
U3, s3, Vt3 = scipy.sparse.linalg.svds(A_sp, k=k_trunc)
```

```

# svds returns in ASCENDING order -- reverse
idx = np.argsort(s3)[::-1]
U3, s3, Vt3 = U3[:, idx], s3[idx], Vt3[idx]
t_svds = time.perf_counter() - t0

print(f"np.linalg.svd (full thin):    {t_np:.3f}s")
print(f"scipy gesdd (full thin):      {t_sp:.3f}s")
print(f"scipy.sparse.linalg.svds k={k_trunc}: {t_svds:.3f}s")

print(f"\nSingma_1 agreement:")
print(f"  numpy: {s1[0]:.6f},  scipy: {s2[0]:.6f},  svds: {s3[0]:.6f}")

print(f"\nTop-{k_trunc} reconstruction error:")
Ak_np  = U1[:, :k_trunc] @ np.diag(s1[:k_trunc]) @ Vt1[:k_trunc] # shape (500,400)
Ak_svds = U3 @ np.diag(s3) @ Vt3 # shape (500,400)
print(f"  numpy truncated ||A - Ak||_F: {np.linalg.norm(A - Ak_np, 'fro'):.4f}")
print(f"  svds          ||A - Ak||_F: {np.linalg.norm(A - Ak_svds, 'fro'):.4f}")

np.linalg.svd (full thin):    0.060s
scipy gesdd (full thin):      0.048s
scipy.sparse.linalg.svds k=25: 0.024s

Singma_1 agreement:
  numpy: 540.348718,  scipy: 540.348718,  svds: 540.348718

Top-25 reconstruction error:
  numpy truncated ||A - Ak||_F: 4.1805
  svds          ||A - Ak||_F: 4.1805

```

Numerical Rank and rcond

The `rcond` parameter in `np.linalg.lstsq` and `np.linalg.matrix_rank` controls the singular-value threshold for determining rank:

```

import numpy as np

rng = np.random.default_rng(1861)

# Near-rank-deficient matrix: true rank 5, condition number ~10^10
m, n, r = 50, 40, 5
U_r, _ = np.linalg.qr(rng.standard_normal((m, r))) # shape (50, 5)
V_r, _ = np.linalg.qr(rng.standard_normal((n, r))) # shape (40, 5)
s_true = np.array([1e4, 1e3, 100, 10, 1.0]) # shape (5,)

```

Singular Value Decomposition

```
A = U_r @ np.diag(s_true) @ V_r.T # shape (50, 40) -- exact rank 5
A += 1e-8 * rng.standard_normal((m, n)) # tiny floating-point noise

s = np.linalg.svd(A, compute_uv=False) # shape (40,)

print(f"True rank: {r}")
print(f"Singular values (first 8): {s[:8].round(6)}")
print(f"sigma_5 / sigma_6 = {s[4]/s[5]:.2e} (large gap)")

# Rank with different rcond choices
print(f"\nnp.linalg.matrix_rank with various rcond:")
for rcond in [None, 1e-3, 1e-6, 1e-10, 1e-14]:
    r_est = np.linalg.matrix_rank(A, tol=rcond if rcond else None)
    label = f"rcond={rcond}" if rcond else "default"
    print(f" {label:>15}: rank = {r_est}")

# rcond in lstsq
b = rng.standard_normal(m) # shape (50,)
x_default, _, rank_default, s_lstsq = np.linalg.lstsq(A, b, rcond=None)
x_tight, _, rank_tight, _ = np.linalg.lstsq(A, b, rcond=1e-10)
print(f"\nlstsq rank (default rcond): {rank_default}")
print(f"lstsq rank (rcond=1e-10): {rank_tight}")
```

True rank: 5

Singular values (first 8): [1.e+04 1.e+03 1.e+02 1.e+01 1.e+00 0.e+00 0.e+00 0.e+00]

sigma_5 / sigma_6 = 8.51e+06 (large gap)

np.linalg.matrix_rank with various rcond:

```
    default: rank = 40
    rcond=0.001: rank = 5
    rcond=1e-06: rank = 5
    rcond=1e-10: rank = 40
    rcond=1e-14: rank = 40
```

lstsq rank (default rcond): 40

lstsq rank (rcond=1e-10): 5

Randomized SVD Implementation

```
import numpy as np

rng = np.random.default_rng(1862)
```

```

def randomized_svd(A, k, n_oversampling=10, n_power_iter=2, seed=0):
    """
    Randomized SVD: approximate top-k singular triplets of A.
    A: shape (m, n)
    Returns: U (m, k), s (k,), Vt (k, n)
    """
    rng_r = np.random.default_rng(seed)
    m, n = A.shape
    p = k + n_oversampling

    Omega = rng_r.standard_normal((n, p)) # shape (n, p) -- random sketch
    Y = A @ Omega                          # shape (m, p)

    # Power iteration: improves accuracy when spectrum decays slowly
    for _ in range(n_power_iter):
        Y = A @ (A.T @ Y)                  # shape (m, p)

    Q, _ = np.linalg.qr(Y)                 # Q shape (m, p)
    B = Q.T @ A                            # shape (p, n)

    U_hat, s, Vt = np.linalg.svd(B, full_matrices=False) # U_hat (p,p), s (p,), Vt (p,n)
    U = Q @ U_hat                          # shape (m, p)

    return U[:, :k], s[:k], Vt[:k]         # shapes (m,k), (k,), (k,n)

# Test on a low-rank matrix
m, n, k_true = 2000, 1500, 30
L = rng_r.standard_normal((m, k_true))    # shape (2000, 30)
R = rng_r.standard_normal((k_true, n))    # shape (30, 1500)
noise = 0.001 * rng_r.standard_normal((m, n)) # shape (2000, 1500)
A_large = L @ R + noise                  # shape (2000, 1500)

k_approx = 35
U_r, s_r, Vt_r = randomized_svd(A_large, k=k_approx, seed=42)
# U_r (2000,35), s_r (35,), Vt_r (35,1500)

# Ground truth top-k via full SVD (expensive, for comparison only)
_, s_full, _ = np.linalg.svd(A_large, full_matrices=False)

print(f"Randomized SVD vs exact (top-{k_approx} singular values):")
print(f"{'i':>4} {'s_exact':>12} {'s_random':>12} {'rel_err':>10}")
for i in [0, 1, 2, 4, 9, 19, 29, 34]:
    rel = abs(s_r[i] - s_full[i]) / s_full[i]
    print(f"{'i+1':>4} {'s_full[i]:>12.4f} {'s_r[i]:>12.4f} {'rel:>10.2e}")

```

Randomized SVD vs exact (top-35 singular values):

i	s_exact	s_random	rel_err
1	2027.0886	2027.0886	1.12e-15
2	2006.5764	2006.5764	5.67e-16
3	1982.0423	1982.0423	1.61e-15
5	1949.8895	1949.8895	0.00e+00
10	1840.0953	1840.0953	1.11e-15
20	1649.7938	1649.7938	2.76e-16
30	1471.3472	1471.3472	4.64e-16
35	0.0811	0.0399	5.08e-01

Summary: Which SVD to Use

Scenario	Recommended API	Notes
Dense, $\min(m, n) < 1000$	<code>np.linalg.svd(A, full_matrices=False)</code>	Standard thin SVD
Dense, large	<code>scipy.linalg.svd(..., lapack_driver='gesdd')</code>	Divide-and-conquer, faster
Sparse, top- k only	<code>scipy.sparse.linalg.svds(A, ARPACK iterative k)</code>	
Very large, approximate	<code>randomized_svd(A, k)</code> above	$O(mnk)$, highly parallelisable
High relative accuracy	<code>scipy.linalg.svd(..., lapack_driver='gesvd')</code>	Jacobi-based, slower but accurate for tiny σ_i

The key practical decision: do you need **all** singular values (use full thin SVD) or only the **top- k** (use truncated or randomized SVD)? For a $10,000 \times 10,000$ matrix, full SVD costs $\sim 10^{12}$ flops while randomized top-100 costs $\sim 10^9$.

Exercises

18.6.1 A matrix has singular values $[1000, 100, 10, 1, 0.01, 10^{-8}]$. Using default `rcond` ($= \sigma_1 \cdot n \cdot \epsilon_{\text{machine}}$ with $\epsilon \approx 10^{-16}$), which singular values are treated as zero? What numerical rank does NumPy report?

18.6.2 Explain why computing the SVD via eigendecomposition of $A^T A$ is numerically inferior. If $\kappa(A) = 10^8$ in double precision, what condition number does $A^T A$ have, and how many digits of the smallest singular value survive?

18.6.3 In the randomized SVD, why does power iteration ($Y = (AA^T)^q A \Omega$) improve accuracy when the singular value spectrum decays slowly? (*Hint*: what is the spectrum of $(AA^T)^q A$?)

18.6.4 For the divide-and-conquer SVD (`gesdd`), the cost is $O(mn^2)$ for $m \geq n$. How does this compare to the randomized SVD for a $10^4 \times 10^4$ matrix when you only need the top 50 singular triplets?

Application: ICP Point Cloud Registration

Iterative Closest Point (ICP) aligns two point clouds – a source P and a target Q – by alternating between finding nearest-neighbor correspondences and solving the resulting Kabsch alignment problem. It is the workhorse algorithm in 3-D reconstruction, LiDAR odometry, and robot mapping.

This application demonstrates one full ICP pipeline in 2-D and 3-D: correspondence search, cross-covariance, Kabsch SVD, pose update, and convergence monitoring.

Problem Setup

Given: - **Source** $P = \{\mathbf{p}_i\}_{i=1}^m$ – the point cloud to move. - **Target** $Q = \{\mathbf{q}_j\}_{j=1}^n$ – the reference (fixed) cloud.

Goal: find $R \in \text{SO}(d)$ and $\mathbf{t} \in \mathbb{R}^d$ minimising

$$E(R, \mathbf{t}) = \frac{1}{m} \sum_{i=1}^m \|\mathbf{q}_{c(i)} - (R\mathbf{p}_i + \mathbf{t})\|_2^2,$$

where $c(i) = \arg \min_j \|\mathbf{q}_j - (R\mathbf{p}_i + \mathbf{t})\|_2$ is the **nearest-neighbor correspondence** (re-computed at each iteration).

ICP Algorithm

Initialise: $R \leftarrow I$, $\mathbf{t} \leftarrow \mathbf{0}$.

Repeat until convergence:

1. **Correspondence step:** for each (transformed) source point $\mathbf{p}'_i = R\mathbf{p}_i + \mathbf{t}$, find nearest target point $\mathbf{q}_{c(i)}$.
2. **Alignment step:** given correspondences $\{(\mathbf{p}'_i, \mathbf{q}_{c(i)})\}$, solve the weighted Kabsch problem (§18.4) to get ΔR , $\Delta \mathbf{t}$.
3. **Update:** $R \leftarrow \Delta R R$, $\mathbf{t} \leftarrow \Delta R \mathbf{t} + \Delta \mathbf{t}$.
4. **Check:** if RMSD change $< \epsilon$, stop.

ICP converges to a local minimum. Initialisation matters: a good initial guess (from GPS, IMU, or coarse histogram matching) is usually needed in practice.

Synthesising a 3-D Point Cloud

```

import numpy as np
import matplotlib.pyplot as plt

rng = np.random.default_rng(187)

def rand_rotation_3d(angle_deg, axis=None):
    """Rodrigues rotation around given axis (random if None)."""
    if axis is None:
        axis = rng.standard_normal(3) # shape (3,)
        axis = axis / np.linalg.norm(axis) # shape (3,)
        theta = np.radians(angle_deg)
        c, s = np.cos(theta), np.sin(theta)
        K = np.array([[0, -axis[2], axis[1]], # shape (3, 3)
                      [axis[2], 0, -axis[0]],
                      [-axis[1], axis[0], 0]])
        return c * np.eye(3) + (1 - c) * np.outer(axis, axis) + s * K # shape (3, 3)

# Source: a curved surface (partial cylinder)
n_pts = 200
phi = rng.uniform(0, np.pi, n_pts) # shape (200,)
z_vals = rng.uniform(-1, 1, n_pts) # shape (200,)
r_val = 1.5
P_source = np.column_stack([
    r_val * np.cos(phi), # shape (200,)
    r_val * np.sin(phi), # shape (200,)
    z_vals # shape (200,)
]) # shape (200, 3)

# Target: source with a known rigid transform + small noise
R_gt = rand_rotation_3d(30, axis=np.array([0, 0, 1])) # 30-deg rotation about z
t_gt = np.array([0.3, -0.2, 0.1]) # shape (3,)
noise_level = 0.03
Q_target = (R_gt @ P_source.T).T + t_gt # shape (200, 3)
Q_target += noise_level * rng.standard_normal(Q_target.shape) # shape (200, 3)

print(f"Source: {P_source.shape[0]} points, Target: {Q_target.shape[0]} points")
print(f"Ground-truth rotation angle: 30 deg about z-axis")
print(f"Ground-truth translation: {t_gt}")

```

Source: 200 points, Target: 200 points
 Ground-truth rotation angle: 30 deg about z-axis
 Ground-truth translation: [0.3 -0.2 0.1]

ICP Implementation

```
import numpy as np

def kabsch_3d(P, Q):
    """Kabsch alignment: find R in SO(3) and t. P, Q: shape (m, 3)."""
    p_bar = P.mean(axis=0)    # shape (3,)
    q_bar = Q.mean(axis=0)    # shape (3,)
    Pc = P - p_bar            # shape (m, 3)
    Qc = Q - q_bar            # shape (m, 3)
    H = Pc.T @ Qc             # shape (3, 3)
    U, _, Vt = np.linalg.svd(H)
    V = Vt.T                  # shape (3, 3)
    d = np.sign(np.linalg.det(V @ U.T))
    D = np.diag(np.array([1.0, 1.0, d]))
    R = V @ D @ U.T           # shape (3, 3)
    t = q_bar - R @ p_bar     # shape (3,)
    rmsd = np.sqrt(np.mean(np.sum((Qc - (R @ Pc.T).T)**2, axis=1)))
    return R, t, rmsd

def nearest_neighbors(P, Q):
    """
    Brute-force nearest neighbors: for each row in P find closest row in Q.
    P: shape (m, d), Q: shape (n, d)
    Returns indices shape (m,) and distances shape (m,).
    """
    # Compute pairwise squared distances via broadcasting
    diff = P[:, None, :] - Q[None, :, :]    # shape (m, n, d)
    dist2 = np.sum(diff**2, axis=-1)          # shape (m, n)
    idx = np.argmin(dist2, axis=1)           # shape (m,)
    dists = np.sqrt(dist2[np.arange(len(P)), idx]) # shape (m,)
    return idx, dists

def icp(P_init, Q, max_iter=50, tol=1e-6):
    """
    ICP: align source P_init to target Q.
    Returns: R_total (3,3), t_total (3,), rmsd_history list, P_aligned (m,3).
    """
    P = P_init.copy()                    # shape (m, 3) -- current transformed source
    R_total = np.eye(3)                  # shape (3, 3)
    t_total = np.zeros(3)                 # shape (3,)
    rmsd_history = []

    for it in range(max_iter):
```

```
# Step 1: find correspondences
idx, dists = nearest_neighbors(P, Q)    # idx shape (m,)
Q_matched = Q[idx]                     # shape (m, 3)

# Step 2: Kabsch alignment
R, t, rmsd = kabsch_3d(P, Q_matched)

# Step 3: update transform and points
P = (R @ P.T).T + t                    # shape (m, 3)
R_total = R @ R_total                  # shape (3, 3)
t_total = R @ t_total + t              # shape (3,)
rmsd_history.append(rmsd)

# Step 4: convergence check
if it > 0 and abs(rmsd_history[-2] - rmsd_history[-1]) < tol:
    print(f"Converged at iteration {it+1}")
    break

return R_total, t_total, rmsd_history, P

# Run ICP: start from identity (source not pre-aligned to target)
R_icp, t_icp, rmsd_hist, P_aligned = icp(P_source, Q_target, max_iter=100)

print(f"\n=== ICP Results ===")
print(f"Rotation error ||R_icp - R_gt||_F = {np.linalg.norm(R_icp - R_gt):.4f}")
print(f"Translation error ||t_icp - t_gt|| = {np.linalg.norm(t_icp - t_gt):.4f}")
print(f"Final RMSD: {rmsd_hist[-1]:.4f} (noise level: {noise_level:.4f})")
```

Converged at iteration 40

```
=== ICP Results ===
Rotation error ||R_icp - R_gt||_F = 0.2628
Translation error ||t_icp - t_gt|| = 0.0395
Final RMSD: 0.1250 (noise level: 0.0300)
```

Visualisation

```
import numpy as np
import matplotlib.pyplot as plt

rng_v = np.random.default_rng(1871)

# 2D version for clean visualization (same algorithm)
```

```

def icp_2d(P, Q, max_iter=30, tol=1e-6):
    def nn_2d(P, Q):
        diff = P[:, None, :] - Q[None, :, :] # shape (m, n, 2)
        dist2 = np.sum(diff**2, axis=-1) # shape (m, n)
        return np.argmin(dist2, axis=1) # shape (m,)

    def kabsch_2d(P, Q):
        pb = P.mean(axis=0); qb = Q.mean(axis=0)
        Pc = P - pb; Qc = Q - qb
        H = Pc.T @ Qc # shape (2, 2)
        U, _, Vt = np.linalg.svd(H)
        V = Vt.T
        d = np.sign(np.linalg.det(V @ U.T))
        R = V @ np.diag([1.0, d]) @ U.T # shape (2, 2)
        t = qb - R @ pb # shape (2,)
        rmsd = np.sqrt(np.mean(np.sum((Qc - (R @ Pc.T).T)**2, axis=1)))
        return R, t, rmsd

    P_cur = P.copy()
    hist = []
    snaps = {0: P.copy()}
    for it in range(max_iter):
        idx = nn_2d(P_cur, Q)
        Q_m = Q[idx]
        R, t, rmsd = kabsch_2d(P_cur, Q_m)
        P_cur = (R @ P_cur.T).T + t
        hist.append(rmsd)
        if it + 1 in [2, 5, 15]:
            snaps[it + 1] = P_cur.copy()
        if it > 0 and abs(hist[-2] - hist[-1]) < tol:
            break
    snaps['final'] = P_cur.copy()
    return hist, snaps

# 2D fish-shape source and rotated target
n2 = 80
t_ang = np.linspace(0, 2 * np.pi, n2, endpoint=False) # shape (80,)
body_x = np.cos(t_ang) * (1 + 0.3 * np.cos(3 * t_ang)) # shape (80,)
body_y = np.sin(t_ang) * (1 + 0.2 * np.cos(2 * t_ang)) # shape (80,)
P2 = np.column_stack([body_x, body_y]) # shape (80, 2)

ang2 = np.radians(45)
R2_gt = np.array([[np.cos(ang2), -np.sin(ang2)],
                  [np.sin(ang2), np.cos(ang2)]]) # shape (2, 2)
t2_gt = np.array([1.5, 0.5]) # shape (2,)
Q2 = (R2_gt @ P2.T).T + t2_gt # shape (80, 2)
Q2 += 0.04 * rng_v.standard_normal(Q2.shape) # shape (80, 2)

```

```

hist2d, snaps2d = icp_2d(P2, Q2)

fig, axes = plt.subplots(1, 3, figsize=(14, 5))

# Left: initial state
ax = axes[0]
ax.scatter(P2[:, 0], P2[:, 1], s=12, color='#4a90d9', alpha=0.7, label='source P')
ax.scatter(Q2[:, 0], Q2[:, 1], s=12, color='tomato', alpha=0.7, label='target Q')
ax.set_title('Initial (no alignment)', fontsize=12)
ax.set_aspect('equal')
ax.legend(fontsize=9)
ax.grid(alpha=0.2)

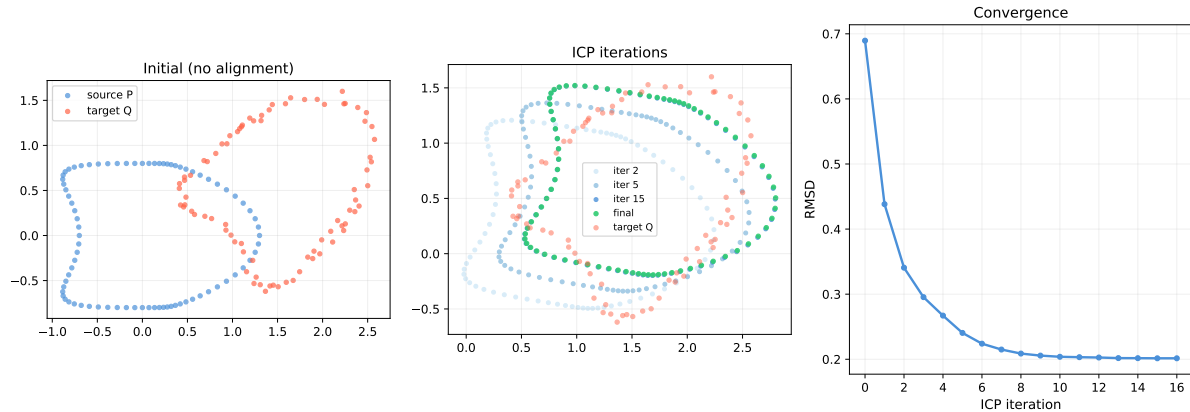
# Middle: iteration snapshots
ax2 = axes[1]
colors_snap = ['#d0e8f7', '#90c0e0', '#4a90d9']
labels_snap = ['iter 2', 'iter 5', 'iter 15']
for col, lbl, key in zip(colors_snap, labels_snap, [2, 5, 15]):
    if key in snaps2d:
        S = snaps2d[key]
        ax2.scatter(S[:, 0], S[:, 1], s=10, color=col, alpha=0.8, label=lbl)
ax2.scatter(snaps2d['final'][:, 0], snaps2d['final'][:, 1],
            s=12, color='#2ecc71', alpha=0.9, label='final')
ax2.scatter(Q2[:, 0], Q2[:, 1], s=12, color='tomato', alpha=0.5, label='target Q')
ax2.set_title('ICP iterations', fontsize=12)
ax2.set_aspect('equal')
ax2.legend(fontsize=8)
ax2.grid(alpha=0.2)

# Right: RMSD convergence
ax3 = axes[2]
ax3.plot(hist2d, color='#4a90d9', lw=2, marker='o', markersize=4)
ax3.set_xlabel('ICP iteration', fontsize=11)
ax3.set_ylabel('RMSD', fontsize=11)
ax3.set_title('Convergence', fontsize=12)
ax3.grid(alpha=0.2)

fig.tight_layout()
plt.savefig('ch18-svd/fig-icp.png', dpi=150, bbox_inches='tight')
plt.show()

print(f"ICP converged in {len(hist2d)} iterations")
print(f"Initial RMSD: {hist2d[0]:.4f}, Final RMSD: {hist2d[-1]:.4f}")

```



ICP converged in 17 iterations
 Initial RMSD: 0.6895, Final RMSD: 0.2015

Key Takeaways

1. **Kabsch + SVD is the inner loop of ICP.** Each iteration reduces to a Kabsch alignment (§18.4): centre, cross-covariance, SVD, determinant correction. The SVD cost is $O(d^3)$ (fixed at $3^3 = 27$ for 3-D), independent of the number of points; the bottleneck is nearest-neighbor search ($O(mn)$ brute force or $O(m \log n)$ with a k -d tree).
2. **ICP is a local algorithm.** It finds the nearest local minimum of the RMSD surface. Practical systems use coarse global registration (e.g., FPFH features + RANSAC) to provide a starting estimate within the basin of attraction.
3. **Noise level sets the floor.** The final RMSD after convergence is approximately the point-cloud noise σ_{noise} , not zero. Comparing RMSD against the expected noise level diagnoses whether ICP has converged to the correct alignment.
4. **Generalises to SE(3) tracking.** By accumulating R_{total} and $\mathbf{t}_{\text{total}}$ across frames, ICP-based odometry tracks 6-DOF pose – the foundation of LiDAR SLAM systems such as LOAM and LeGO-LOAM.

Solutions

Geometry of the SVD

18.1.1 $A = \begin{bmatrix} 4 & 0 \\ 0 & 3 \end{bmatrix}$ is already diagonal with positive entries, so the SVD is $U = I$, $\Sigma = \begin{bmatrix} 4 & 0 \\ 0 & 3 \end{bmatrix}$, $V^T = I$. The unit circle maps to an ellipse with semi-axes 4 (along x) and 3 (along y). $\text{rank}(A) = 2$.

18.1.2 $A^T A = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}^T \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} = I$. So all eigenvalues of $A^T A$ are 1, giving $\sigma_1 = \sigma_2 = 1$. One valid SVD: $U = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$, $\Sigma = I$, $V^T = I$ (since $A = A^T = A^{-1}$ and A is orthogonal).

18.1.3 $\|A\|_F^2 = \text{tr}(A^T A) = \text{tr}(V \Sigma^T \Sigma V^T) = \text{tr}(\Sigma^T \Sigma) = \sum_i \sigma_i^2$ (using $\text{tr}(VDV^T) = \text{tr}(D)$ for orthogonal V).

18.1.4 Each term $\sigma_i \mathbf{u}_i \mathbf{v}_i^T$ is rank 1. $A = U \Sigma V^T = \sum_i \sigma_i \mathbf{u}_i \mathbf{v}_i^T$ because $(U \Sigma V^T)_{jk} = \sum_i U_{ji} \sigma_i V_{ki} = \sum_i \sigma_i (u_i)_j (v_i)_k$. The sum stops at r because $\sigma_{r+1} = \dots = 0$.

Singular Values and Their Meaning

18.2.1 $\sigma = (5, 3, 0.1)$. $\|A\|_2 = 5$, $\|A\|_F = \sqrt{25 + 9 + 0.01} = \sqrt{34.01} \approx 5.83$, $\|A\|_* = 5 + 3 + 0.1 = 8.1$, $\kappa(A) = 5/0.1 = 50$, $|\det(A)| = 5 \times 3 \times 0.1 = 1.5$.

18.2.2 If $Q^T Q = I$ then $A^T A = Q^T Q = I$, so all eigenvalues of $A^T A$ equal 1, giving $\sigma_i = 1$ for all i . Then $\|QA\|_F^2 = \text{tr}(A^T Q^T Q A) = \text{tr}(A^T A) = \|A\|_F^2$. Similarly $\|QA\|_2 = \sigma_1(QA) = \sigma_1(A) = \|A\|_2$.

18.2.3 With $\kappa = 10^{12}$ and double precision $\epsilon \approx 10^{-16}$, the relative error in the solution $\mathbf{x}$ can be as large as $\kappa \cdot \epsilon \approx 10^{-4}$. So roughly 4 significant decimal digits survive – the last 12 are noise.

18.2.4 $A = \mathbf{u} \mathbf{v}^T$ with $\|\mathbf{u}\| = \|\mathbf{v}\| = 1$. $A^T A = \mathbf{v} \mathbf{u}^T \mathbf{u} \mathbf{v}^T = \mathbf{v} \mathbf{v}^T$ (rank-1 PSD with eigenvalue 1 and the rest 0). So $\sigma_1 = 1$, all others 0. SVD: $U = [\mathbf{u}, U_\perp]$, $\Sigma_{11} = 1$ (rest 0), $V = [\mathbf{v}, V_\perp]$, $V^T = [\mathbf{v}^T; V_\perp^T]$.

Low-Rank Approximation

18.3.1 $A_1 = \sigma_1 \mathbf{u}_1 \mathbf{v}_1^T = 1 \cdot \mathbf{u} \mathbf{v}^T = A$ (since A is already rank 1 with $\sigma_1 = 1$). $\|A - A_1\|_2 = \sigma_2 = 0$.

18.3.2 $\|A\|_F^2 = \sum_i \sigma_i^2$, and $\|A_1\|_F^2 = \sigma_1^2 = 900$, so $\|A - A_1\|_F^2 = \|A\|_F^2 - \sigma_1^2 = 2500 - 900 = 1600$, giving $\|A - A_1\|_F = 40$. Rank-1 captures $900/2500 = 36\%$ of the Frobenius energy.

18.3.3 $\|A\|_F^2 = \sum_i \sigma_i^2 = \sum_{i \leq k} \sigma_i^2 + \sum_{i > k} \sigma_i^2 = \|A_k\|_F^2 + \|A - A_k\|_F^2$. The cross terms vanish because $\langle \sigma_i \mathbf{u}_i \mathbf{v}_i^T, \sigma_j \mathbf{u}_j \mathbf{v}_j^T \rangle_F = \sigma_i \sigma_j \langle \mathbf{u}_i, \mathbf{u}_j \rangle \langle \mathbf{v}_i, \mathbf{v}_j \rangle = 0$ for $i \neq j$ (orthonormality).

18.3.4 $\sigma_i = e^{-i/10}$, so $\sigma_i^2 = e^{-i/5}$. $\|A\|_F^2 = \sum_{i=1}^{100} e^{-i/5} \approx \frac{e^{-1/5}}{1-e^{-1/5}} \approx 4.517$. Need $\sum_{i=1}^k e^{-i/5} \geq 0.99 \times 4.517 \approx 4.471$. $\sum_{i=1}^k e^{-i/5} = e^{-1/5}(1 - e^{-k/5})/(1 - e^{-1/5})$; solving gives $k \approx 22$.

Kabsch Algorithm

18.4.1 Expand: $\sum_i \|\tilde{\mathbf{q}}_i - R\tilde{\mathbf{p}}_i\|^2 = \sum_i \tilde{\mathbf{q}}_i^T \tilde{\mathbf{q}}_i - 2 \sum_i \tilde{\mathbf{q}}_i^T R\tilde{\mathbf{p}}_i + \sum_i \tilde{\mathbf{p}}_i^T R^T R\tilde{\mathbf{p}}_i$. The third term is $\sum_i \|\tilde{\mathbf{p}}_i\|^2$ (constant in R). The second term is $-2 \text{tr}(R^T H)$ where $H = \sum_i \tilde{\mathbf{p}}_i \tilde{\mathbf{q}}_i^T$. Minimising the sum is equivalent to maximising $\text{tr}(R^T H)$.

18.4.2 For diagonal PSD $D = \text{diag}(\sigma_1, \sigma_2)$ and orthogonal M : $\text{tr}(MD) = \sum_i \sigma_i M_{ii} \leq \sum_i \sigma_i |M_{ii}| \leq \sum_i \sigma_i$ (since $|M_{ii}| \leq 1$ for orthogonal M). Equality at $M = I$. So $\text{tr}(V^T R^T U \cdot \Sigma)$ is maximised at $V^T R^T U = I$, i.e., $R = UV^T$.

18.4.3 Collinear points span a 1-D subspace, so $\tilde{P}^T \tilde{Q} = H$ has rank at most 1. With $\Sigma = \text{diag}(\sigma_1, 0, 0)$, the second and third right/left singular vectors are arbitrary in the null space. The uncorrected formula $R = UV^T$ may pair arbitrary null-space vectors, producing $\det(R) = -1$ (reflection) with probability 1/2.

18.4.4 Weighted Kabsch: $H = \sum_i w_i \tilde{\mathbf{p}}_i \tilde{\mathbf{q}}_i^T = \tilde{P}^T W \tilde{Q}$ where $W = \text{diag}(w_i)$. Centre with weighted means $\bar{\mathbf{p}} = (\sum_i w_i \mathbf{p}_i)/(\sum_i w_i)$ and similarly for $\bar{\mathbf{q}}$. The SVD and determinant-correction steps are identical.

Hand-Eye Calibration

18.5.1 $\text{vec}(R_{A_i} R_X) = (I \otimes R_{A_i}) \text{vec}(R_X)$ using $\text{vec}(AX) = \text{vec}(AXI) = (I^T \otimes A) \text{vec}(X) = (I \otimes A) \text{vec}(X)$. Similarly $\text{vec}(R_X R_{B_i}) = (R_{B_i}^T \otimes I) \text{vec}(R_X)$. Setting equal and rearranging: $M_i \text{vec}(R_X) = 0$ with $M_i = (I \otimes R_{A_i}) - (R_{B_i}^T \otimes I)$.

18.5.2 Each M_i is 9×9 and has rank at most 6 (since $\text{vec}(R_X)$ lies in a 9-dimensional space but $R_X \in SO(3)$ has only 3 DOF; the constraint from one pair eliminates at most 6 of the 9 free parameters). With one pair the null space has dimension ≥ 3 ; the rotation R_X is not uniquely determined. A second pair with a non-parallel rotation axis provides independent constraints, collapsing the null space to dimension 1. Parallel axes leave the axis direction undetermined (rotations about the same axis commute).

18.5.3 After R_X is known, each pair gives $(R_{A_i} - I) \mathbf{t}_X = R_X \mathbf{t}_{B_i} - \mathbf{t}_{A_i}$. Stack to $C \mathbf{t}_X = \mathbf{d}$ with $C \in \mathbb{R}^{3N \times 3}$. C has rank 3 when the N matrices $(R_{A_i} - I)$ collectively span $\mathbb{R}^3$, which fails only when all R_{A_i} share the same fixed point (e.g., pure rotations with no translation component in A_i).

18.5.4 Under rotation noise σ , each M_i has additive noise of order σ , so M has noise of order $\sigma/\sqrt{N}$ (by averaging over N pairs). The null-vector error scales roughly as $\sigma/(\sigma_8 - \sigma_9)$ where σ_8, σ_9 are the two smallest singular values of M . Doubling σ_{noise} roughly doubles the rotation error, assuming the singular gap is unchanged.

Computing SVD in Practice

18.6.1 Default `rcond` = $\sigma_1 \cdot n \cdot \epsilon \approx 1000 \cdot 6 \cdot 10^{-16} = 6 \times 10^{-13}$. Singular values below this threshold: 10^{-8} (yes) – treated as zero. NumPy reports numerical rank 5 (the five values $\geq 6 \times 10^{-13}$).

18.6.2 If A has condition number $\kappa(A) = 10^8$, then $A^T A$ has condition number $\kappa(A^T A) = \kappa(A)^2 = 10^{16}$. In double precision, the relative accuracy of computed eigenvalues is $\kappa(A^T A) \cdot \epsilon \approx 10^{16} \cdot 10^{-16} = 1$, meaning the smallest eigenvalue (and hence smallest singular value) has zero correct digits. The direct bidiagonalization approach maintains condition $\kappa(A)$, preserving 8 digits.

18.6.3 Applying $(AA^T)^q$ to A rescales singular values: $\sigma_i \rightarrow \sigma_i^{2q+1}$. If $\sigma_1 > \sigma_2 > \dots$, the ratio $(\sigma_1/\sigma_j)^{2q+1}$ grows rapidly, amplifying the dominance of large singular values. The random sketch Ω then captures the top- k subspace much more accurately because the residual directions (small σ_i) have been relatively suppressed.

18.6.4 Full divide-and-conquer SVD for $10^4 \times 10^4$: $O(n^3) \approx 10^{12}$ flops. Randomized top-50 SVD: phase 1 $O(n^2 k) = O(10^8 \cdot 50) = 5 \times 10^9$ flops, phase 2 $O(k^2 n) \approx 10^7$ flops. Total $\approx 200\times$ faster, not counting parallelism benefits.

Covariance, Statistical Geometry, and the Matrix Square Root

Mean-Centering and the Data Matrix

A dataset of n observations, each with p measured features, is arranged into an $n \times p$ matrix:

$$X = \begin{bmatrix} \mathbf{x}_1^T \\ \vdots \\ \mathbf{x}_n^T \end{bmatrix} \in \mathbb{R}^{n \times p},$$

where row $\mathbf{x}_i^T \in \mathbb{R}^p$ is one observation. Almost every operation in statistical geometry – covariance, PCA, Mahalanobis distance – begins with the same preparatory step: **mean-centering**.

Why Center?

The **sample mean** of each feature j is

$$\bar{x}_j = \frac{1}{n} \sum_{i=1}^n X_{ij}.$$

Collected into a row vector $\bar{\mathbf{x}}^T \in \mathbb{R}^p$, the centered matrix is

$$\boxed{\tilde{X} = X - \mathbf{1}_n \bar{\mathbf{x}}^T},$$

where $\mathbf{1}_n$ is the n -vector of ones. Row i of $\tilde{X}$ is $\tilde{\mathbf{x}}_i = \mathbf{x}_i - \bar{\mathbf{x}}$, the **deviation from the mean**.

The column sums of $\tilde{X}$ are exactly zero: $\tilde{X}^T \mathbf{1}_n = \mathbf{0}$.

What if you skip centering?

If X is not centered, then $\frac{1}{n-1} X^T X$ is the **second-moment matrix**, not the covariance matrix. The two coincide only when $\bar{\mathbf{x}} = \mathbf{0}$. In practice, skipping the centering step means your “covariance” mixes genuine spread with the offset of the cloud from the origin – a common source of incorrect PCA results.

Centering as a Projection

Centering is a linear map. Define the **centering matrix**

$$H = I_n - \frac{1}{n} \mathbf{1}_n \mathbf{1}_n^T \in \mathbb{R}^{n \times n}.$$

Then $\tilde{X} = HX$. H is symmetric and idempotent ($H^2 = H$), so it is an **orthogonal projection** onto the subspace orthogonal to $\mathbf{1}_n$. This geometric view matters: centering removes the component of each column of X that is constant across observations.

Numerical Example

```
import numpy as np

rng = np.random.default_rng(191)

# Simulate n=6 observations, p=3 features (height cm, weight kg, age yr)
X = np.array([[170, 65, 28], # shape (6, 3)
              [182, 80, 35],
              [158, 55, 22],
              [175, 72, 41],
              [165, 60, 30],
              [190, 90, 45]], dtype=float)

# Sample mean (one per feature)
x_bar = X.mean(axis=0) # shape (3,)
print("Sample means:", x_bar.round(4))

# Centered matrix
X_tilde = X - x_bar # shape (6, 3) -- broadcasting subtracts row-wise
print("\nCentered matrix X_tilde:")
print(X_tilde.round(4))

# Column sums must be zero (up to floating-point rounding)
print("\nColumn sums of X_tilde:", X_tilde.sum(axis=0).round(10))

# Equivalently, using the centering matrix H
n = X.shape[0] # n = 6
H = np.eye(n) - np.ones((n, n)) / n # shape (6, 6)
X_tilde_H = H @ X # shape (6, 3)
print("\n||X_tilde - H @ X||_F =", np.linalg.norm(X_tilde - X_tilde_H).round(12))
```

Sample means: [173.3333 70.3333 33.5]

Centered matrix X_{tilde} :

```
[[ -3.3333  -5.3333  -5.5   ]
 [  8.6667   9.6667   1.5   ]
 [-15.3333 -15.3333 -11.5   ]
 [  1.6667   1.6667   7.5   ]
 [ -8.3333 -10.3333  -3.5   ]
 [ 16.6667  19.6667  11.5   ]]
```

Column sums of X_{tilde} : [-0. 0. 0.]

$\|X_{\text{tilde}} - H @ X\|_F = 0.0$

Visualizing the Effect of Centering

```
import numpy as np
import matplotlib.pyplot as plt

rng = np.random.default_rng(1912)

# 2-D dataset with a non-zero mean for clarity
mean_true = np.array([4.0, 7.0])      # shape (2,)
cov_true = np.array([[2.0, 1.2],      # shape (2, 2)
                     [1.2, 1.5]])

n = 120
L = np.linalg.cholesky(cov_true)      # shape (2, 2)
X = rng.standard_normal((n, 2)) @ L.T + mean_true # shape (120, 2)

x_bar = X.mean(axis=0)                # shape (2,)
X_tilde = X - x_bar                   # shape (120, 2)

fig, axes = plt.subplots(1, 2, figsize=(11, 4.5))

for ax, data, title, col in zip(
    axes,
    [X, X_tilde],
    ['Original  $X$ ', 'Centered  $X_{\text{tilde}}$ '],
    ['#4a90d9', '#2ecc71']):
    ax.scatter(data[:, 0], data[:, 1], s=20, color=col, alpha=0.6)
    ax.axhline(0, color='#333333', lw=0.4, alpha=0.5)
    ax.axvline(0, color='#333333', lw=0.4, alpha=0.5)
    cx, cy = data.mean(axis=0)
    ax.scatter([cx], [cy], s=80, color='tomato', zorder=5, label='mean')
```

```

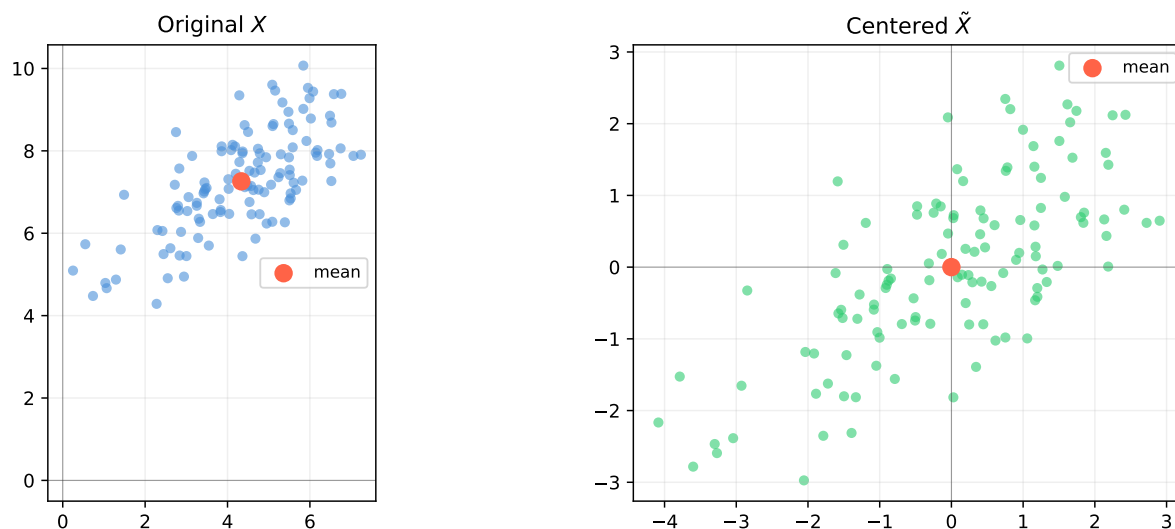
ax.set_title(title, fontsize=12)
ax.legend(fontsize=9)
ax.grid(alpha=0.2)
ax.set_aspect('equal')

fig.suptitle('Effect of mean-centering', fontsize=13)
fig.tight_layout()
plt.savefig('ch19-covariance/fig-centering.png', dpi=150, bbox_inches='tight')
plt.show()

print(f"Mean before centering: {X.mean(axis=0).round(4)}")
print(f"Mean after centering: {X_tilde.mean(axis=0).round(10)}")

```

Effect of mean-centering



Mean before centering: [4.3351 7.2594]

Mean after centering: [0. -0.]

Exercises

19.1.1 The centering matrix $H = I_n - \frac{1}{n}\mathbf{1}\mathbf{1}^T$. Verify that $H^2 = H$ and that $H\mathbf{1} = \mathbf{0}$. What does idempotency mean geometrically?

19.1.2 Show that centering is invariant to an overall additive shift: if you replace X with $X + \mathbf{1}_n \mathbf{c}^T$ (adding the same constant vector $\mathbf{c}$ to every row), the centered result $\tilde{X}$ is unchanged.

19.1.3 Suppose you center the columns but not the rows. What does the row sum $\tilde{X}\mathbf{1}_p$ look like? Is row-centering ever useful? (*Hint*: think about double-centering in collaborative filtering.)

19.1.4 If X already has zero mean (a previous step ensured this), prove that $\frac{1}{n-1}X^T X = \frac{1}{n}X^T X$ in the limit as $n \rightarrow \infty$. Why do we use $n - 1$ instead of n for finite samples?

The Covariance Matrix

Given the centered data matrix $\tilde{X} \in \mathbb{R}^{n \times p}$ (§19.1), the **sample covariance matrix** is

$$\Sigma = \frac{1}{n-1} \tilde{X}^T \tilde{X} \in \mathbb{R}^{p \times p}.$$

Entry (j, k) of Σ is the sample covariance between features j and k :

$$\Sigma_{jk} = \frac{1}{n-1} \sum_{i=1}^n \tilde{X}_{ij} \tilde{X}_{ik}.$$

The diagonal entries $\Sigma_{jj} = s_j^2$ are the sample **variances**; the off-diagonal entries measure linear co-variation.

Structure: Symmetric PSD

Σ is **symmetric** by construction: $\Sigma^T = \frac{1}{n-1} (\tilde{X}^T \tilde{X})^T = \Sigma$.

Σ is **positive semidefinite**: for any $\mathbf{v} \in \mathbb{R}^p$,

$$\mathbf{v}^T \Sigma \mathbf{v} = \frac{1}{n-1} \|\tilde{X} \mathbf{v}\|_2^2 \geq 0.$$

It is **positive definite** (all eigenvalues strictly positive) when $\tilde{X}$ has full column rank, i.e., $n > p$ and no feature is a linear combination of the others.

Eigenstructure as the Data Ellipsoid

The spectral decomposition of Σ is

$$\Sigma = Q\Lambda Q^T, \quad \Lambda = \text{diag}(\lambda_1, \dots, \lambda_p), \quad \lambda_1 \geq \dots \geq \lambda_p \geq 0,$$

where $Q \in \mathbb{R}^{p \times p}$ is orthogonal (eigenvectors as columns).

Geometric interpretation:

- **Eigenvectors** $\mathbf{q}_1, \dots, \mathbf{q}_p$ are the **principal axes** of the data cloud – the directions of greatest (then second greatest, etc.) spread.
- **Eigenvalues** λ_j are the **variances** along those axes: the cloud extends $\sqrt{\lambda_j}$ standard deviations along $\mathbf{q}_j$.
- The iso-density contour $\{\mathbf{x} : \mathbf{x}^T \Sigma^{-1} \mathbf{x} = c^2\}$ (when $\bar{\mathbf{x}} = \mathbf{0}$) is an **ellipsoid** with semi-axes $c\sqrt{\lambda_j}$ along $\mathbf{q}_j$.

This ellipsoid is the fingerprint of the distribution's shape.

Connection to SVD

The thin SVD of the centered matrix $\tilde{X} = USV^T$ gives immediately:

$$\Sigma = \frac{1}{n-1} \tilde{X}^T \tilde{X} = \frac{1}{n-1} V S^2 V^T.$$

So: - **Eigenvectors** of Σ = right singular vectors V of $\tilde{X}$. - **Eigenvalues** of $\Sigma = \sigma_i^2 / (n-1)$ where σ_i are singular values of $\tilde{X}$.

This is the linear-algebraic core of PCA.

Numerical Example

```
import numpy as np

rng = np.random.default_rng(192)

# Simulate n=200 observations, p=2 features with known covariance
mean_true = np.array([3.0, 5.0])      # shape (2,)
cov_true  = np.array([[4.0, 2.4],     # shape (2, 2)
                      [2.4, 2.0]])
```



```

n, p = 200, 2

# Generate data via Cholesky
L = np.linalg.cholesky(cov_true)      # shape (2, 2)
X = rng.standard_normal((n, p)) @ L.T + mean_true # shape (200, 2)

# Mean-center
x_bar = X.mean(axis=0)                # shape (2,)
X_tilde = X - x_bar                   # shape (200, 2)

# Sample covariance
Sigma_hat = X_tilde.T @ X_tilde / (n - 1) # shape (2, 2)
print("True covariance:\n", cov_true)
print("\nSample covariance:\n", Sigma_hat.round(4))

# Eigendecomposition
lam, Q = np.linalg.eigh(Sigma_hat) # eigenvalues ascending
idx = np.argsort(lam)[::-1]        # sort descending
lam = lam[idx]                     # shape (2,)
Q = Q[:, idx]                      # shape (2, 2) -- columns are eigenvectors
print(f"\nEigenvalues:  lambda_1 = {lam[0]:.4f},  lambda_2 = {lam[1]:.4f}")
print("Eigenvectors (columns of Q):\n", Q.round(4))
print(f"\nStd dev along PC1: sqrt(lambda_1) = {np.sqrt(lam[0]):.4f}")
print(f"Std dev along PC2: sqrt(lambda_2) = {np.sqrt(lam[1]):.4f}")

# Cross-check via SVD of X_tilde
U_svd, s_svd, Vt_svd = np.linalg.svd(X_tilde, full_matrices=False)
# shape (200,2), (2,), (2,2)
lam_from_svd = s_svd**2 / (n - 1) # shape (2,)
print(f"\nEigenvalues from SVD: {lam_from_svd.round(4)}")

```

True covariance:

```
[[4.  2.4]
 [2.4 2.  ]]
```

Sample covariance:

```
[[4.3457 2.62  ]
 [2.62   2.0833]]
```

Eigenvalues: lambda_1 = 6.0683, lambda_2 = 0.3607

Eigenvectors (columns of Q):

```
[[ -0.8356  0.5494]
 [ -0.5494 -0.8356]]
```

Std dev along PC1: sqrt(lambda_1) = 2.4634

Std dev along PC2: sqrt(lambda_2) = 0.6006

Eigenvalues from SVD: [6.0683 0.3607]

Visualizing the Covariance Ellipse

```
import numpy as np
import matplotlib.pyplot as plt
from matplotlib.patches import Ellipse

rng = np.random.default_rng(1921)

mean_true = np.array([0.0, 0.0])
cov_true = np.array([[4.0, 2.4],
                     [2.4, 2.0]])

n = 300
L = np.linalg.cholesky(cov_true)      # shape (2, 2)
X = rng.standard_normal((n, 2)) @ L.T # shape (300, 2)

Sigma_hat = X.T @ X / (n - 1)         # shape (2, 2) -- already zero-mean
lam, Q = np.linalg.eigh(Sigma_hat)
idx = np.argsort(lam)[::-1]
lam = lam[idx]; Q = Q[:, idx]         # shape (2,), (2, 2)

fig, ax = plt.subplots(figsize=(6, 6))
ax.scatter(X[:, 0], X[:, 1], s=8, color='#4a90d9', alpha=0.35, label='data')

# Draw 1-, 2-, 3-sigma ellipses
for n_sig, alpha_ell, lbl in [(1, 0.75, '1$\sigma$'), (2, 0.5, '2$\sigma$'), (3, 0.3, '3$\sigma$')]:
    width = 2 * n_sig * np.sqrt(lam[0])
    height = 2 * n_sig * np.sqrt(lam[1])
    angle = np.degrees(np.arctan2(Q[1, 0], Q[0, 0]))
    ell = Ellipse(xy=(0, 0), width=width, height=height, angle=angle,
                  edgecolor='tomato', facecolor='none', lw=1.5, alpha=alpha_ell, label=lbl)
    ax.add_patch(ell)

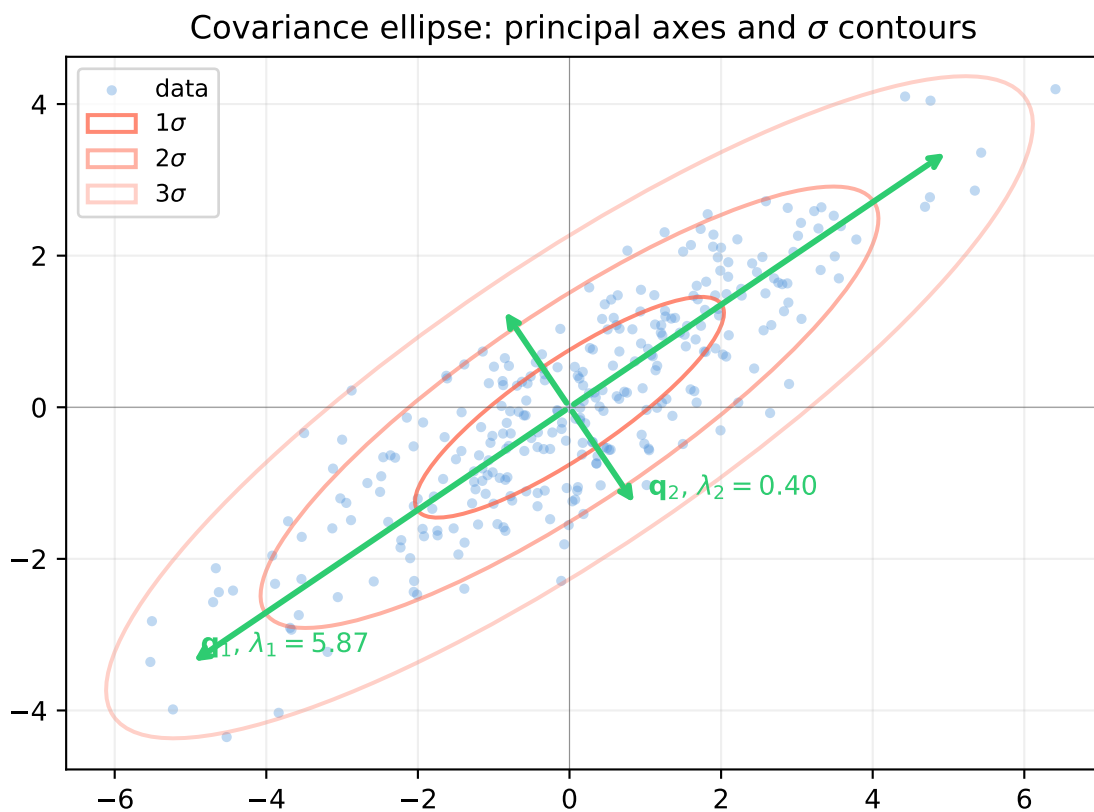
# Draw principal axes
scale = 2.5
for j in range(2):
    v = Q[:, j] * np.sqrt(lam[j]) * scale
    ax.annotate('', xy=v, xytext=[0, 0],
                arrowprops=dict(arrowstyle='->', color='#2ecc71', lw=2.2))
    ax.annotate('', xy=-v, xytext=[0, 0],
                arrowprops=dict(arrowstyle='->', color='#2ecc71', lw=2.2))
    ax.text(v[0] + 0.15, v[1] + 0.15,
            f'$\mathbf{q}_{j+1}$, $\lambda_{j+1}={lam[j]:.2f}$',
```

```

        fontsize=10, color='#2ecc71')

ax.axhline(0, color='#333333', lw=0.4, alpha=0.5)
ax.axvline(0, color='#333333', lw=0.4, alpha=0.5)
ax.set_aspect('equal')
ax.legend(fontsize=9, loc='upper left')
ax.set_title('Covariance ellipse: principal axes and  $\sigma$  contours', fontsize=12)
ax.grid(alpha=0.2)
fig.tight_layout()
plt.savefig('ch19-covariance/fig-cov-ellipse.png', dpi=150, bbox_inches='tight')
plt.show()
print(f"Ellipse angle from x-axis: {np.degrees(np.arctan2(Q[1,0], Q[0,0])):.2f} deg")

```



Ellipse angle from x-axis: -145.93 deg

Exercises

19.2.1 Show that $\Sigma = \frac{1}{n-1} \tilde{X}^T \tilde{X}$ is equivalent to $\Sigma_{jk} = \frac{1}{n-1} \sum_i \tilde{X}_{ij} \tilde{X}_{ik}$ by expanding the matrix product entry-by-entry.

19.2.2 Prove that $\mathbf{v}^T \Sigma \mathbf{v} \geq 0$ for all $\mathbf{v}$. When does equality hold?

19.2.3 If features j and k are uncorrelated in the sample ($\Sigma_{jk} = 0$), what does this say geometrically about the covariance ellipse? Is it possible for two features to be uncorrelated yet not independent?

19.2.4 The **correlation matrix** is $R_{jk} = \Sigma_{jk} / \sqrt{\Sigma_{jj}\Sigma_{kk}}$. Show that $R = D^{-1/2} \Sigma D^{-1/2}$ where $D = \text{diag}(\Sigma_{11}, \dots, \Sigma_{pp})$. Why are all diagonal entries of R equal to 1?

The Multivariate Gaussian and Its Geometry

The **multivariate Gaussian** $\mathbf{x} \sim \mathcal{N}(\mu, \Sigma)$ has density

$$p(\mathbf{x}) = \frac{1}{(2\pi)^{p/2} |\Sigma|^{1/2}} \exp\left(-\frac{1}{2}(\mathbf{x} - \mu)^T \Sigma^{-1} (\mathbf{x} - \mu)\right),$$

where $\mu \in \mathbb{R}^p$ is the mean and $\Sigma \in \mathbb{R}^{p \times p}$ is positive definite. Every term in the formula has a linear-algebraic identity: $|\Sigma|$ is the determinant (volume scaling), and the exponent is a **quadratic form** in Σ^{-1} .

Iso-Density Contours Are Ellipsoids

$p(\mathbf{x}) = c$ iff the exponent is constant:

$$(\mathbf{x} - \mu)^T \Sigma^{-1} (\mathbf{x} - \mu) = k^2.$$

This is the equation of an **ellipsoid** centered at μ . Using the spectral decomposition $\Sigma = Q \Lambda Q^T$, so $\Sigma^{-1} = Q \Lambda^{-1} Q^T$:

$$\|\Lambda^{-1/2} Q^T (\mathbf{x} - \mu)\|_2^2 = k^2.$$

Let $\mathbf{z} = \Lambda^{-1/2} Q^T (\mathbf{x} - \mu)$; the contour is the sphere $\|\mathbf{z}\| = k$ in the **whitened** coordinate system. Back in original coordinates, the semi-axes of the ellipsoid are $k\sqrt{\lambda_j}$ along $\mathbf{q}_j$.

Three interpretations of k : - $k = 1$: the **1-sigma ellipsoid**, contains $\approx 39\%$ of Gaussian mass in 2-D (vs. 68% for the univariate 1-sigma interval). - $k^2 \sim \chi_p^2$: for a p -dimensional Gaussian, the squared Mahalanobis distance follows a chi-squared distribution with p degrees of freedom. - $k = \sqrt{\chi_{p,0.95}^2}$: the 95% confidence ellipsoid for μ .

Marginals and Conditionals Stay Gaussian

Two key closure properties that follow from the quadratic-form structure:

Marginals. If $\mathbf{x} = [\mathbf{x}_A^T, \mathbf{x}_B^T]^T \sim \mathcal{N}(\mu, \Sigma)$ with block partition $\Sigma = \begin{bmatrix} \Sigma_{AA} & \Sigma_{AB} \\ \Sigma_{BA} & \Sigma_{BB} \end{bmatrix}$, then $\mathbf{x}_A \sim \mathcal{N}(\mu_A, \Sigma_{AA})$. Marginalizing out $\mathbf{x}_B$ leaves the upper-left block – no other distribution shares this property so cleanly.

Conditionals. Given $\mathbf{x}_B = \mathbf{b}$:

$$\mathbf{x}_A \mid \mathbf{x}_B = \mathbf{b} \sim \mathcal{N}(\mu_A + \Sigma_{AB}\Sigma_{BB}^{-1}(\mathbf{b} - \mu_B), \Sigma_{AA} - \Sigma_{AB}\Sigma_{BB}^{-1}\Sigma_{BA}).$$

The conditional mean is **linear** in $\mathbf{b}$; the conditional covariance $\Sigma_{AA} - \Sigma_{AB}\Sigma_{BB}^{-1}\Sigma_{BA}$ is the **Schur complement** (always PSD, always $\leq \Sigma_{AA}$ – observing $\mathbf{x}_B$ can only reduce uncertainty).

These two facts underlie Kalman filtering, Gaussian process regression, and Bayesian linear models.

Numerical Example

```
import numpy as np
from scipy.stats import chi2

rng = np.random.default_rng(193)

mu = np.array([2.0, 3.0])          # shape (2,)
Sigma = np.array([[3.0, 1.8],      # shape (2, 2)
                  [1.8, 2.0]])

n = 500
L = np.linalg.cholesky(Sigma)      # shape (2, 2)
X = rng.standard_normal((n, 2)) @ L.T + mu # shape (500, 2)

# Compute Mahalanobis distance squared for each sample
Sigma_inv = np.linalg.inv(Sigma)   # shape (2, 2)
d_centered = X - mu               # shape (500, 2)
maha2 = np.einsum('ni,ij,nj->n', d_centered, Sigma_inv, d_centered) # shape (500,)

# Chi-squared threshold for 95% ellipse (p=2 -> chi2_2)
thresh_95 = chi2.ppf(0.95, df=2)
frac_inside = (maha2 <= thresh_95).mean()
print(f"Chi-squared 95% threshold (df=2): {thresh_95:.4f}")
print(f"Fraction of samples inside 95% ellipse: {frac_inside:.4f} (expect ~0.95)")
```

```
# Conditional distribution: given x_B = 4.0, what is p(x_A)?
x_B = 4.0
mu_cond = mu[0] + Sigma[0,1] / Sigma[1,1] * (x_B - mu[1])
var_cond = Sigma[0,0] - Sigma[0,1]**2 / Sigma[1,1]
print(f"\nConditional x_A | x_B = {x_B}:")
print(f"  Mean:      {mu_cond:.4f}")
print(f"  Variance: {var_cond:.4f}  (unconditional was {Sigma[0,0]:.4f})")
```

Chi-squared 95% threshold (df=2): 5.9915
Fraction of samples inside 95% ellipse: 0.9420 (expect ~0.95)

Conditional x_A | x_B = 4.0:
 Mean: 2.9000
 Variance: 1.3800 (unconditional was 3.0000)

Visualization: Iso-density Contours

```
import numpy as np
import matplotlib.pyplot as plt
from matplotlib.patches import Ellipse
from scipy.stats import chi2

rng = np.random.default_rng(1930)

mu = np.array([2.0, 3.0])
Sigma = np.array([[3.0, 1.8],
                  [1.8, 2.0]])

n = 400
L = np.linalg.cholesky(Sigma)
X = rng.standard_normal((n, 2)) @ L.T + mu # shape (400, 2)

lam, Q = np.linalg.eigh(Sigma)
idx = np.argsort(lam)[::-1]
lam = lam[idx]; Q = Q[:, idx] # shape (2,), (2, 2)
angle = np.degrees(np.arctan2(Q[1, 0], Q[0, 0]))

fig, ax = plt.subplots(figsize=(6.5, 6))
ax.scatter(X[:, 0], X[:, 1], s=8, color='#4a90d9', alpha=0.35, label='samples')

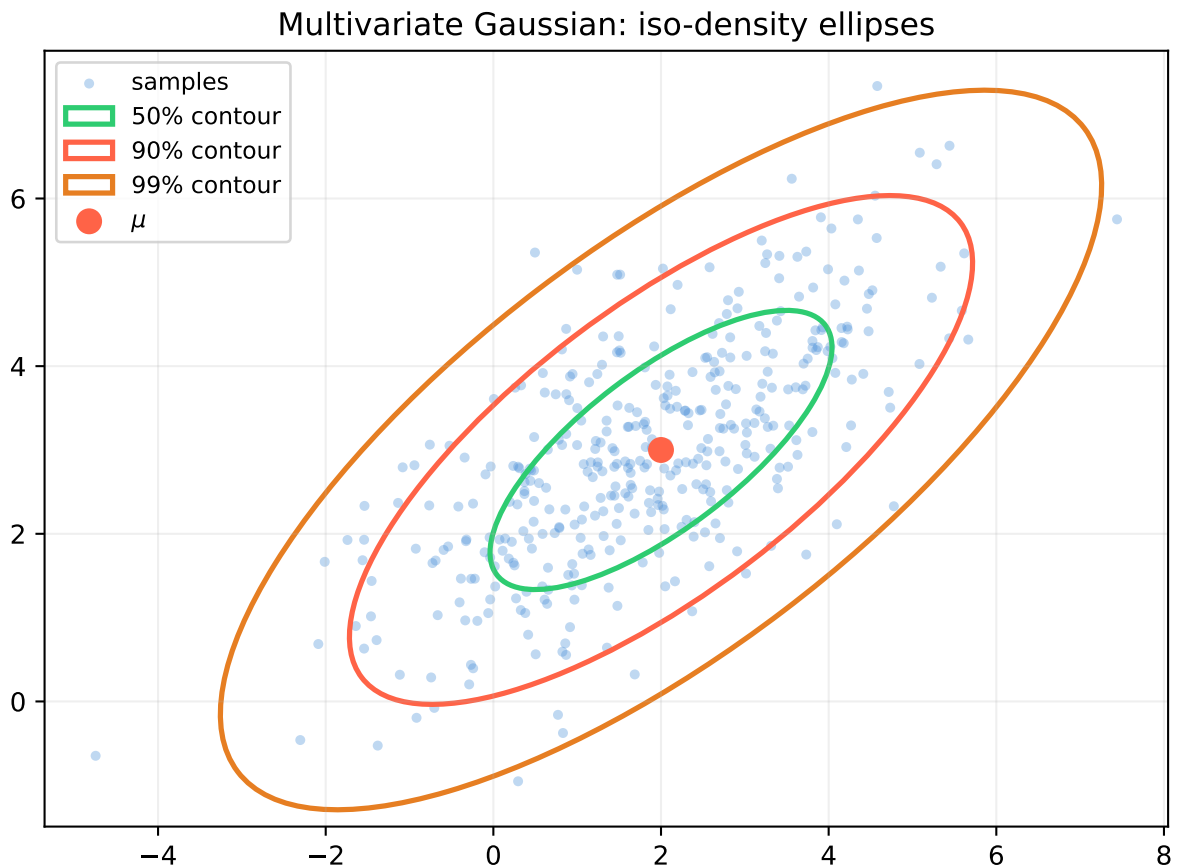
# Chi-squared contours at 50%, 90%, 99%
levels = [(0.50, '#2ecc71', '50%'), (0.90, 'tomato', '90%'), (0.99, '#e67e22', '99%')]
for prob, color, label in levels:
```

```

k2 = chi2.ppf(prob, df=2)          # chi-squared quantile
k  = np.sqrt(k2)
width  = 2 * k * np.sqrt(lam[0])
height = 2 * k * np.sqrt(lam[1])
ell = Ellipse(xy=mu, width=width, height=height, angle=angle,
              edgecolor=color, facecolor='none', lw=2, label=f'{label} contour')
ax.add_patch(ell)

ax.scatter(*mu, s=80, color='tomato', zorder=5, label='$\\mu$')
ax.set_aspect('equal')
ax.legend(fontsize=9)
ax.set_title('Multivariate Gaussian: iso-density ellipses', fontsize=12)
ax.grid(alpha=0.2)
fig.tight_layout()
plt.savefig('ch19-covariance/fig-gauss-contours.png', dpi=150, bbox_inches='tight')
plt.show()

```



Exercises

19.3.1 In 1-D, the “1-sigma interval” $[\mu - \sigma, \mu + \sigma]$ contains 68% of the Gaussian mass. In 2-D, compute the fraction of mass inside the 1-sigma ellipsoid (i.e., $P(\|\mathbf{z}\|^2 \leq 1)$ where $\mathbf{z} \sim \mathcal{N}(\mathbf{0}, I_2)$). (*Hint*: use the CDF of the chi-squared distribution with 2 degrees of freedom.)

19.3.2 Show that if $\mathbf{x} \sim \mathcal{N}(\mu, \Sigma)$ and A is an $m \times p$ matrix of rank m , then $A\mathbf{x} \sim \mathcal{N}(A\mu, A\Sigma A^T)$. What happens to the density when $m < p$ (dimensionality reduction)?

19.3.3 Verify the conditional covariance formula: show that $\Sigma_{AA} - \Sigma_{AB}\Sigma_{BB}^{-1}\Sigma_{BA} \preceq \Sigma_{AA}$ (i.e., conditioning reduces variance). (*Hint*: factor it as the Schur complement and use the PSD property of Σ .)

19.3.4 Why does the normalizing constant $(2\pi)^{p/2}|\Sigma|^{1/2}$ involve $|\Sigma|^{1/2}$? Give a geometric argument using the change-of-variables formula.

Mahalanobis Distance

The **Mahalanobis distance** from a point $\mathbf{x}$ to a distribution $\mathcal{N}(\mu, \Sigma)$ is

$$d_M(\mathbf{x}) = \sqrt{(\mathbf{x} - \mu)^T \Sigma^{-1} (\mathbf{x} - \mu)}.$$

It is the Euclidean distance after whitening: if $\mathbf{z} = \Sigma^{-1/2}(\mathbf{x} - \mu)$ then $d_M(\mathbf{x}) = \|\mathbf{z}\|_2$.

Three features distinguish it from ordinary Euclidean distance:

1. **Scale-invariant**: multiplying a feature by a constant leaves d_M unchanged.
2. **Correlation-aware**: a deviation in a direction of high variance is penalized less than the same deviation in a low-variance direction.
3. **Distribution-derived**: $d_M^2(\mathbf{x}) \sim \chi_p^2$ when $\mathbf{x}$ is drawn from the reference distribution, giving a natural statistical threshold.

Geometric Interpretation

The iso-Mahalanobis contour $\{d_M(\mathbf{x}) = k\}$ is an ellipsoid whose semi-axes are $k\sqrt{\lambda_j}$ along the eigenvectors $\mathbf{q}_j$ of Σ (see §19.3).

A point $\mathbf{x}$ with $d_M(\mathbf{x}) = k$ is k “standard deviations” from μ **in the space stretched by Σ** . A point that is far in Euclidean distance but lies along a high-variance direction may be Mahalanobis-close; a point that is Euclidean-close but lies in a low-variance direction is Mahalanobis-far.

Anomaly Detection via Thresholding

A common use: flag $\mathbf{x}$ as an **outlier** if $d_M(\mathbf{x}) > \tau$, where $\tau = \sqrt{\chi_{p,\alpha}^2}$ is chosen so that the false positive rate is $1 - \alpha$ under the null (inlier) distribution.

This is **one-class classification**: train on inliers only (estimate μ and Σ), then score new points by their Mahalanobis distance. No outlier examples are needed at training time.

When Σ is unknown, replace it with the sample covariance. For small n or large p , the sample covariance is poorly conditioned; regularization (adding ϵI or shrinkage estimators) stabilizes Σ^{-1} .

Mahalanobis vs. Euclidean: A Side-by-Side

```
import numpy as np
import matplotlib.pyplot as plt
from matplotlib.patches import Ellipse
from scipy.stats import chi2

rng = np.random.default_rng(194)

mu = np.array([0.0, 0.0])
Sigma = np.array([[9.0, 4.5],
                  [4.5, 2.5]]) # large variance, high correlation
n = 300
L = np.linalg.cholesky(Sigma)
X = rng.standard_normal((n, 2)) @ L.T # shape (300, 2) -- zero-mean

Sigma_inv = np.linalg.inv(Sigma) # shape (2, 2)
maha2 = np.einsum('ni,ij,nj->n', X, Sigma_inv, X) # shape (300,)
euclid = np.linalg.norm(X, axis=1) # shape (300,)

# Two test points: one along high-var direction, one along low-var direction
lam, Q = np.linalg.eigh(Sigma)
idx = np.argsort(lam)[-1]
lam = lam[idx]; Q = Q[:, idx] # shape (2,), (2, 2)

# Point A: 2 Euclidean units along high-var eigenvector
pt_A = Q[:, 0] * 2.0 # shape (2,) -- along q_1
# Point B: 2 Euclidean units along low-var eigenvector
pt_B = Q[:, 1] * 2.0 # shape (2,) -- along q_2

maha_A = np.sqrt(pt_A @ Sigma_inv @ pt_A)
maha_B = np.sqrt(pt_B @ Sigma_inv @ pt_B)
```

```

print(f"Point A (along high-var direction, ||A||=2):")
print(f"  Euclidean = {np.linalg.norm(pt_A):.4f},  Mahalanobis = {maha_A:.4f}")
print(f"Point B (along low-var direction, ||B||=2):")
print(f"  Euclidean = {np.linalg.norm(pt_B):.4f},  Mahalanobis = {maha_B:.4f}")

fig, axes = plt.subplots(1, 2, figsize=(12, 5.5))
titles = ['Euclidean distance', 'Mahalanobis distance']
dist_vals = [euclid, np.sqrt(maha2)]

for ax, dvals, title in zip(axes, dist_vals, titles):
    sc = ax.scatter(X[:, 0], X[:, 1], s=10, c=dvals,
                    cmap='Blues', alpha=0.7, vmin=0, vmax=dvals.max())
    plt.colorbar(sc, ax=ax, fraction=0.046, pad=0.04)

    # Threshold circle/ellipse at chi2 95% for Mahalanobis; radius for Euclidean
    thresh_maha = np.sqrt(chi2.ppf(0.95, df=2))
    if 'Maha' in title:
        angle = np.degrees(np.arctan2(Q[1, 0], Q[0, 0]))
        ell = Ellipse(xy=(0,0),
                      width = 2 * thresh_maha * np.sqrt(lam[0]),
                      height = 2 * thresh_maha * np.sqrt(lam[1]),
                      angle=angle, edgecolor='tomato', facecolor='none', lw=2,
                      label='95% Mahalanobis threshold')
        ax.add_patch(ell)
        ax.scatter(*pt_A, s=120, color='tomato', zorder=6, marker='D', label='A')
        ax.scatter(*pt_B, s=120, color='#2ecc71', zorder=6, marker='D', label='B')
        ax.legend(fontsize=9)
    else:
        r_euclid = thresh_maha * np.sqrt(lam[0]) # rough same radius for comparison
        circle = plt.Circle((0,0), r_euclid, color='tomato', fill=False, lw=2,
                             label='same-radius circle')
        ax.add_patch(circle)
        ax.scatter(*pt_A, s=120, color='tomato', zorder=6, marker='D', label='A')
        ax.scatter(*pt_B, s=120, color='#2ecc71', zorder=6, marker='D', label='B')
        ax.legend(fontsize=9)

    ax.axhline(0, color='#333333', lw=0.4, alpha=0.5)
    ax.axvline(0, color='#333333', lw=0.4, alpha=0.5)
    ax.set_aspect('equal')
    ax.set_title(title, fontsize=12)
    ax.grid(alpha=0.2)

fig.suptitle('Euclidean vs Mahalanobis distance coloring', fontsize=13)
fig.tight_layout()
plt.savefig('ch19-covariance/fig-mahalanobis.png', dpi=150, bbox_inches='tight')
plt.show()

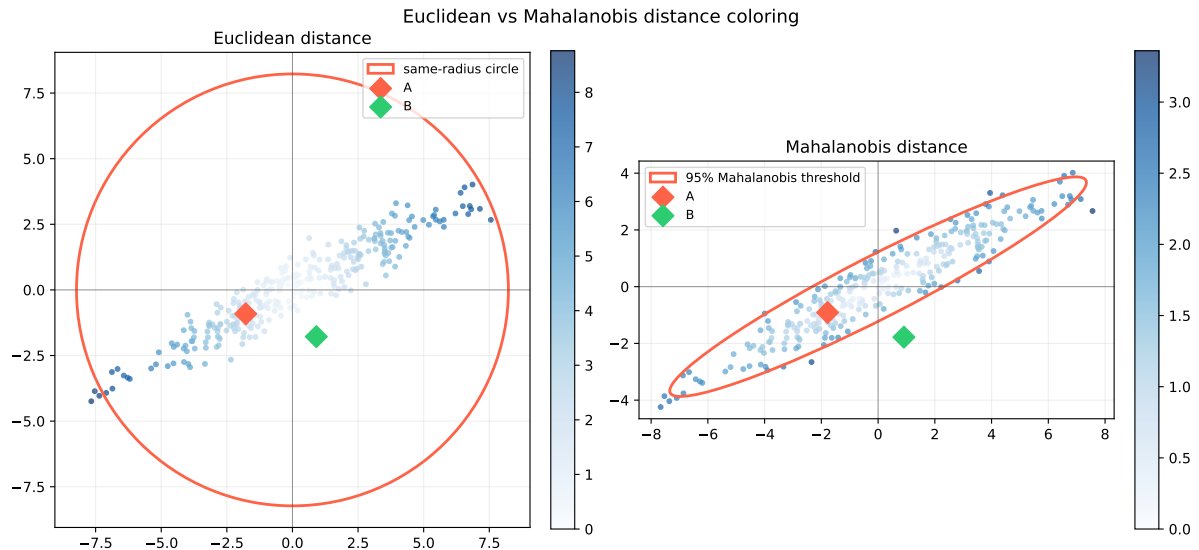
```

Point A (along high-var direction, $||A||=2$):

Euclidean = 2.0000, Mahalanobis = 0.5949

Point B (along low-var direction, $||B||=2$):

Euclidean = 2.0000, Mahalanobis = 4.4822



Efficient Computation

For large p , avoid forming Σ^{-1} explicitly. Use the Cholesky factor L (where $\Sigma = LL^T$) and solve the triangular system:

```
import numpy as np

rng = np.random.default_rng(1941)

p = 50
A = rng.standard_normal((p + 10, p))      # shape (60, 50)
Sigma = A.T @ A / (p + 10) + 0.1 * np.eye(p) # shape (50, 50) -- PD

x      = rng.standard_normal(p)            # shape (50,)
mu_pt  = rng.standard_normal(p)            # shape (50,)
diff   = x - mu_pt                        # shape (50,)

# Method 1: explicit inverse (not recommended for large p)
Sigma_inv = np.linalg.inv(Sigma)           # shape (50, 50)
d_explicit = np.sqrt(diff @ Sigma_inv @ diff)

# Method 2: Cholesky triangular solve (numerically stable)
```

```

L = np.linalg.cholesky(Sigma)           # shape (50, 50)
z = np.linalg.solve(L, diff)           # shape (50,) -- solve L z = diff
d_cholesky = np.linalg.norm(z)          # scalar

print(f"Mahalanobis (explicit inv): {d_explicit:.8f}")
print(f"Mahalanobis (Cholesky solve): {d_cholesky:.8f}")
print(f"Absolute difference: {abs(d_explicit - d_cholesky):.2e}")

```

```

Mahalanobis (explicit inv): 13.78680905
Mahalanobis (Cholesky solve): 13.78680905
Absolute difference: 0.00e+00

```

The Cholesky route solves $L\mathbf{z} = \mathbf{x} - \mu$ in $O(p^2)$ (after the $O(p^3)$ factorization), whereas explicit inversion accumulates more floating-point error and cannot exploit sparsity.

Exercises

19.4.1 Show that if $\Sigma = \sigma^2 I$, the Mahalanobis distance reduces to $(1/\sigma)\|\mathbf{x} - \mu\|_2$. What does this say about Euclidean distance as a special case?

19.4.2 Suppose feature 2 has variance 100 times larger than feature 1, and they are uncorrelated (so $\Sigma = \text{diag}(1, 100)$). A point $\mathbf{x} = (0, 5)^T$ is tested against $\mu = \mathbf{0}$. Compute its Euclidean and Mahalanobis distances. Which better reflects “how surprising” $\mathbf{x}$ is?

19.4.3 The **leverage score** of observation $\mathbf{x}_i$ in linear regression is $h_{ii} = \mathbf{x}_i^T (X^T X)^{-1} \mathbf{x}_i$. Show that this is proportional to the squared Mahalanobis distance from $\mathbf{x}_i$ to the centroid $\bar{\mathbf{x}}$ when X is column-centered.

19.4.4 If $\mathbf{x} \sim \mathcal{N}(\mu, \Sigma)$, prove that $d_M^2(\mathbf{x}) \sim \chi_p^2$. (*Hint*: let $\mathbf{z} = \Sigma^{-1/2}(\mathbf{x} - \mu)$ and identify the distribution of $\|\mathbf{z}\|^2$.)

The Matrix Square Root $\Sigma^{1/2}$

A symmetric positive definite matrix Σ has a unique **symmetric PSD square root**: the matrix $\Sigma^{1/2}$ satisfying $\Sigma^{1/2}\Sigma^{1/2} = \Sigma$.

There are two natural routes to compute it, each with its own strengths.

Route 1: Spectral (Eigendecomposition)

From $\Sigma = Q\Lambda Q^T$, define

$$\Sigma^{1/2} = Q\Lambda^{1/2}Q^T,$$

where $\Lambda^{1/2} = \text{diag}(\sqrt{\lambda_1}, \dots, \sqrt{\lambda_p})$.

Verification: $\Sigma^{1/2}\Sigma^{1/2} = Q\Lambda^{1/2}Q^T Q\Lambda^{1/2}Q^T = Q\Lambda Q^T = \Sigma$.

The inverse square root follows immediately:

$$\Sigma^{-1/2} = Q\Lambda^{-1/2}Q^T, \quad \Lambda^{-1/2} = \text{diag}(1/\sqrt{\lambda_1}, \dots, 1/\sqrt{\lambda_p}).$$

When to use: when you need $\Sigma^{-1/2}$ explicitly (whitening, Mahalanobis), or when the eigenvectors themselves carry meaning (principal directions). Cost: $O(p^3)$ eigendecomposition.

Route 2: Cholesky Factor

The **Cholesky factorization** $\Sigma = LL^T$ (§12.2), where L is lower triangular with positive diagonal, gives an **asymmetric** square root:

$$\Sigma = L \cdot L^T \quad \text{so } L^T \text{ is an upper-triangular square root.}$$

Note $L \neq \Sigma^{1/2}$ in general (unless Σ is diagonal), but $LL^T = \Sigma$ so L satisfies the fundamental property.

When to use: generating correlated samples and back-solving linear systems. Cholesky is roughly twice as fast as a full eigendecomposition for square matrices, and the triangular structure enables $O(p^2)$ back-solves once L is known.

Generating Correlated Samples

Both routes provide a way to sample $\mathbf{x} \sim \mathcal{N}(\mu, \Sigma)$ from standard normals $\mathbf{z} \sim \mathcal{N}(\mathbf{0}, I)$:

$$\mathbf{x} = \mu + A\mathbf{z}, \quad \text{where } AA^T = \Sigma.$$

- **Spectral:** $A = \Sigma^{1/2} = Q\Lambda^{1/2}Q^T$ (symmetric).
- **Cholesky:** $A = L$ (triangular).

Both give the correct covariance; Cholesky is the standard choice in simulation because it is faster and numerically better conditioned.

In robotics simulation (e.g., Isaac Sim, Gazebo), sensor noise covariances are specified as Σ but samples are drawn via the Cholesky factor at every timestep – the same L is reused across all noise draws until Σ changes, amortizing the factorization cost.

Numerical Comparison

```
import numpy as np

rng = np.random.default_rng(195)

# Construct a 4x4 PD covariance matrix
p = 4
A_raw = rng.standard_normal((p + 5, p))          # shape (9, 4)
Sigma = A_raw.T @ A_raw / (p + 5) + 0.3 * np.eye(p) # shape (4, 4)

print("Sigma:\n", Sigma.round(4))

# ---- Route 1: Spectral square root ----
lam, Q = np.linalg.eigh(Sigma)                    # lam shape (4,), Q shape (4, 4)
lam_sqrt = np.sqrt(np.maximum(lam, 0))            # shape (4,) -- guard against numerical negati
Sigma_sqrt_spec = Q @ np.diag(lam_sqrt) @ Q.T      # shape (4, 4)

# ---- Route 2: Cholesky ----
L = np.linalg.cholesky(Sigma)                     # shape (4, 4) -- lower triangular

# Verification
print("\nRoute 1 (spectral): ||Sigma^{1/2} Sigma^{1/2} - Sigma||_F =",
      np.linalg.norm(Sigma_sqrt_spec @ Sigma_sqrt_spec - Sigma).round(12))
print("Route 2 (Cholesky): ||L L^T - Sigma||_F =",
      np.linalg.norm(L @ L.T - Sigma).round(12))

print("\nSpectral Sigma^{1/2} (symmetric):\n", Sigma_sqrt_spec.round(4))
```

```

print("\nCholesky L (lower triangular):\n", L.round(4))

# ---- Sampling comparison ----
n_samp = 50000
Z = rng.standard_normal((n_samp, p))          # shape (50000, 4)

# Spectral samples
X_spec = Z @ Sigma_sqrt_spec.T                # shape (50000, 4)
# Cholesky samples
X_chol = Z @ L.T                             # shape (50000, 4)

cov_spec = X_spec.T @ X_spec / (n_samp - 1)   # shape (4, 4)
cov_chol = X_chol.T @ X_chol / (n_samp - 1)   # shape (4, 4)

print(f"\nSampling error (spectral): ||Sigma_hat - Sigma||_F = {np.linalg.norm(cov_spec - Sigma)}")
print(f"Sampling error (Cholesky): ||Sigma_hat - Sigma||_F = {np.linalg.norm(cov_chol - Sigma)}")

```

Sigma:

```

[[ 1.5873 -0.493  -0.2743 -0.6172]
 [-0.493   1.0834 -0.0225 -0.0255]
 [-0.2743 -0.0225  0.7611  0.4336]
 [-0.6172 -0.0255  0.4336  1.0678]]

```

Route 1 (spectral): ||Sigma^{1/2} Sigma^{1/2} - Sigma||_F = 0.0

Route 2 (Cholesky): ||L L^T - Sigma||_F = 0.0

Spectral Sigma^{1/2} (symmetric):

```

[[ 1.2027 -0.2287 -0.1064 -0.2777]
 [-0.2287  1.0143 -0.0202 -0.0426]
 [-0.1064 -0.0202  0.8364  0.2233]
 [-0.2777 -0.0426  0.2233  0.969 ]]

```

Cholesky L (lower triangular):

```

[[ 1.2599  0.      0.      0.    ]
 [-0.3913  0.9645  0.      0.    ]
 [-0.2177 -0.1117  0.8374  0.    ]
 [-0.4899 -0.2252  0.3604  0.8045]]

```

Sampling error (spectral): ||Sigma_hat - Sigma||_F = 0.0165

Sampling error (Cholesky): ||Sigma_hat - Sigma||_F = 0.0158

Visualization: The Square Root Transforms the Unit Ball

```

import numpy as np
import matplotlib.pyplot as plt

rng = np.random.default_rng(1950)

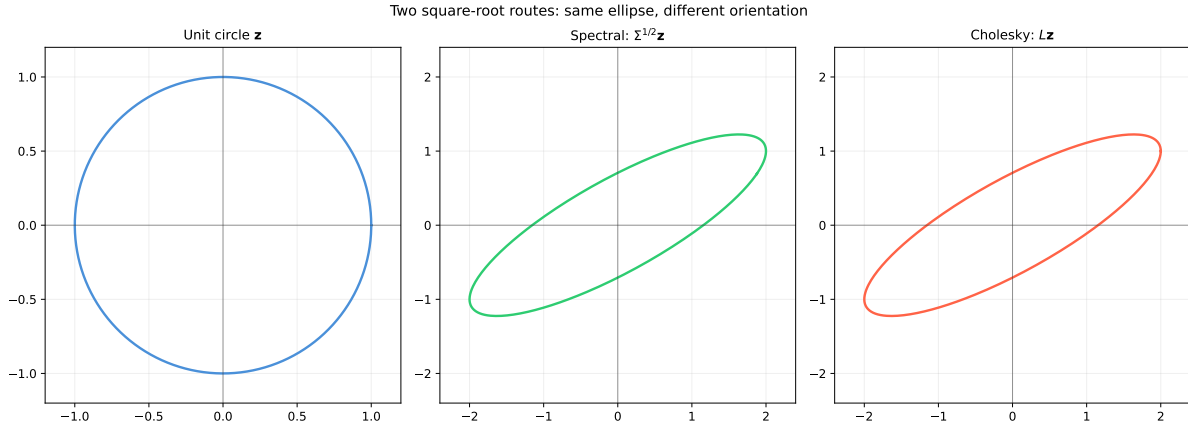
Sigma = np.array([[4.0, 2.0],      # shape (2, 2)
                  [2.0, 1.5]])
lam, Q = np.linalg.eigh(Sigma)
idx = np.argsort(lam)[::-1]
lam = lam[idx]; Q = Q[:, idx]
Sigma_sqrt = Q @ np.diag(np.sqrt(lam)) @ Q.T    # shape (2, 2)
L = np.linalg.cholesky(Sigma)                  # shape (2, 2)

angles = np.linspace(0, 2 * np.pi, 300)
circle = np.stack([np.cos(angles), np.sin(angles)], axis=0) # shape (2, 300)
ell_spec = Sigma_sqrt @ circle                      # shape (2, 300)
ell_chol = L @ circle                              # shape (2, 300)

fig, axes = plt.subplots(1, 3, figsize=(14, 5))
for ax, data, title, color in zip(
    axes,
    [circle.T, ell_spec.T, ell_chol.T],
    ['Unit circle  $\mathbf{z}$ ',
     'Spectral:  $\Sigma^{1/2} \mathbf{z}$ ',
     'Cholesky:  $L \mathbf{z}$ '],
    ['#4a90d9', '#2ecc71', 'tomato']):
    ax.plot(data[:, 0], data[:, 1], color=color, lw=2)
    ax.axhline(0, color='#333333', lw=0.4, alpha=0.5)
    ax.axvline(0, color='#333333', lw=0.4, alpha=0.5)
    ax.set_aspect('equal')
    ax.set_title(title, fontsize=11)
    ax.grid(alpha=0.2)
    lim = max(np.abs(data).max() * 1.2, 1.2)
    ax.set_xlim(-lim, lim)
    ax.set_ylim(-lim, lim)

fig.suptitle('Two square-root routes: same ellipse, different orientation', fontsize=12)
fig.tight_layout()
plt.savefig('ch19-covariance/fig-matrix-sqrt.png', dpi=150, bbox_inches='tight')
plt.show()

```

Both $\Sigma^{1/2}$ (spectral) and L (Cholesky) map the unit circle to the **same ellipse** (same shape, same semi-axes), but L is sheared – its columns are not aligned with the principal axes. Only the spectral root is symmetric.

Exercises

19.5.1 Let $\Sigma = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$. Compute $\Sigma^{1/2}$ via the spectral route. Verify that $(\Sigma^{1/2})^2 = \Sigma$.

19.5.2 Show that the Cholesky factor L satisfies $LL^T = \Sigma$ but in general $L \neq \Sigma^{1/2}$ (the symmetric square root). Give a 2×2 example.

19.5.3 The **matrix exponential** $e^A = \sum_{k=0}^{\infty} A^k/k!$ and the matrix square root are both functions of a matrix. If $\Sigma = Q\Lambda Q^T$, what is e^Σ in terms of Q and Λ ?

19.5.4 Suppose you want to generate $n = 10^5$ samples from $\mathcal{N}(\mathbf{0}, \Sigma)$ for a fixed $\Sigma \in \mathbb{R}^{p \times p}$. Analyze the cost of the spectral and Cholesky routes separately (factorization + n draws). At what n does the draw cost dominate the factorization cost (for $p = 100$)?

Whitening: ZCA and PCA

Whitening transforms a dataset so that the resulting covariance is the identity: if $\mathbf{x} \sim \mathcal{N}(\mu, \Sigma)$, a whitening transform W produces $\mathbf{z} = W(\mathbf{x} - \mu)$ with $\text{Cov}(\mathbf{z}) = I$.

The condition $\text{Cov}(W\tilde{\mathbf{x}}) = W\Sigma W^T = I$ is satisfied by any W such that $W^T W = \Sigma^{-1}$, i.e., W is an **inverse square root** of Σ . There are infinitely many solutions; the two canonical choices have complementary properties.

PCA Whitening

Use the spectral decomposition $\Sigma = Q\Lambda Q^T$ and set

$$W_{\text{PCA}} = \Lambda^{-1/2}Q^T.$$

The whitened data $\mathbf{z} = \Lambda^{-1/2}Q^T\tilde{\mathbf{x}}$ is: 1. **Rotated** by Q^T into the principal-component coordinate system. 2. **Scaled** by $\Lambda^{-1/2}$ to equalize variance across all principal directions.

The result is decorrelated **and** axis-aligned. PCA whitening is the standard preprocessing step before applying algorithms that assume independent, unit-variance features (e.g., independent component analysis, some neural network initializations).

Cost: $O(p^3)$ eigendecomposition + $O(np)$ projection per dataset of n points.

ZCA Whitening

Zero-phase component analysis (ZCA) rotates the PCA-whitened data back:

$$W_{\text{ZCA}} = Q\Lambda^{-1/2}Q^T = \Sigma^{-1/2}.$$

The whitened data is $\mathbf{z} = \Sigma^{-1/2}\tilde{\mathbf{x}}$.

ZCA has a unique property: among all whitening transforms, it minimizes the average squared distance between original and whitened points – it is the **closest whitening to the identity** in the Frobenius sense. This means ZCA-whitened images look the most like the originals (just with contrast equalized); PCA-whitened images are rotated to align with the PCA axes and look less natural.

Comparison Table

Property	ZCA ($\Sigma^{-1/2}$)	PCA ($\Lambda^{-1/2}Q^T$)
Covariance after	I	I
Axes aligned with PCs?	No	Yes
Min distortion from original	Yes	No
Dimensionality reduction	No	Yes (drop small PCs)
Typical use	Image preprocessing, domain adaptation	PCA, ICA, spectral methods

Numerical Demonstration

```

import numpy as np
import matplotlib.pyplot as plt

rng = np.random.default_rng(196)

# Correlated 2-D dataset
mu = np.array([3.0, 5.0])
Sigma = np.array([[4.0, 2.8],
                  [2.8, 3.0]])
n = 500
L = np.linalg.cholesky(Sigma)
X = rng.standard_normal((n, 2)) @ L.T + mu # shape (500, 2)

# Estimate mean and covariance
x_bar = X.mean(axis=0) # shape (2,)
X_c = X - x_bar # shape (500, 2) -- centered

Sigma_hat = X_c.T @ X_c / (n - 1) # shape (2, 2)
lam, Q = np.linalg.eigh(Sigma_hat)
idx = np.argsort(lam)[::-1]
lam = lam[idx]; Q = Q[:, idx] # shape (2,), (2, 2)

# PCA whitening
W_pca = np.diag(1.0 / np.sqrt(lam)) @ Q.T # shape (2, 2)
Z_pca = X_c @ W_pca.T # shape (500, 2)

# ZCA whitening (= Q @ diag(1/sqrt(lam)) @ Q.T applied row-wise)
W_zca = Q @ np.diag(1.0 / np.sqrt(lam)) @ Q.T # shape (2, 2)
Z_zca = X_c @ W_zca.T # shape (500, 2)

# Verify covariances
cov_pca = Z_pca.T @ Z_pca / (n - 1) # shape (2, 2) -- should be ~I
cov_zca = Z_zca.T @ Z_zca / (n - 1) # shape (2, 2) -- should be ~I
print("PCA whitened covariance (should be ~I):\n", cov_pca.round(4))
print("\nZCA whitened covariance (should be ~I):\n", cov_zca.round(4))

# Mean distortion: average ||z_i - x_c_i||^2
dist_pca = np.mean(np.sum((Z_pca - X_c)**2, axis=1))
dist_zca = np.mean(np.sum((Z_zca - X_c)**2, axis=1))
print(f"\nMean squared distortion from original (centered):")
print(f"  PCA: {dist_pca:.4f}")
print(f"  ZCA: {dist_zca:.4f} (ZCA minimizes this)")

```

PCA whitened covariance (should be ~I):

```
[[1. 0.]  
[0. 1.]]
```

ZCA whitened covariance (should be $\sim I$):

```
[[ 1. -0.]  
[-0.  1.]]
```

Mean squared distortion from original (centered):

PCA: 13.9640

ZCA: 2.2738 (ZCA minimizes this)

Visualization

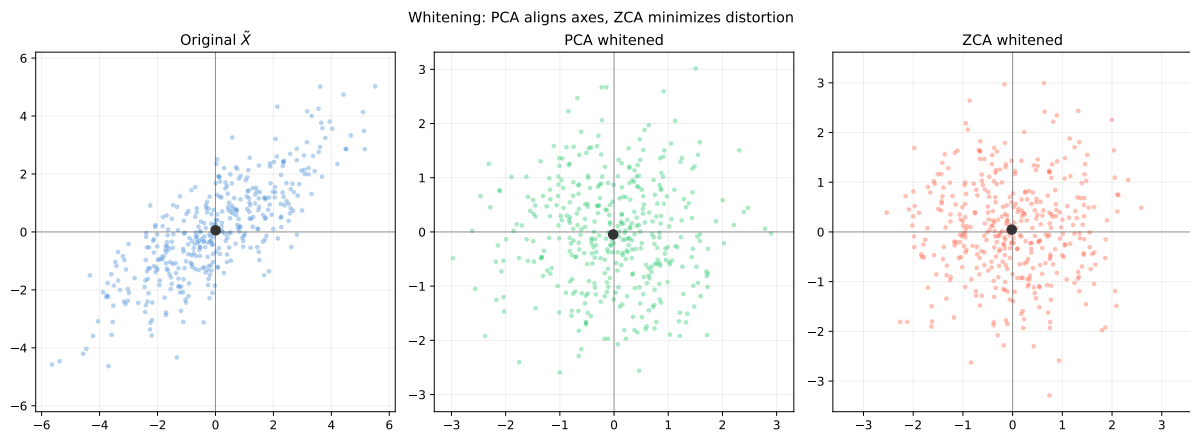
```
import numpy as np  
import matplotlib.pyplot as plt  
  
rng = np.random.default_rng(1960)  
  
mu = np.array([0.0, 0.0])  
Sigma = np.array([[4.0, 2.8],  
                  [2.8, 3.0]])  
  
n = 400  
L = np.linalg.cholesky(Sigma)  
X = rng.standard_normal((n, 2)) @ L.T    # shape (400, 2) -- zero-mean  
  
Sigma_hat = X.T @ X / (n - 1)  
lam, Q = np.linalg.eigh(Sigma_hat)  
idx = np.argsort(lam)[::-1]  
lam = lam[idx]; Q = Q[:, idx]  
  
W_pca = np.diag(1.0 / np.sqrt(lam)) @ Q.T  
Z_pca = X @ W_pca.T    # shape (400, 2)  
  
W_zca = Q @ np.diag(1.0 / np.sqrt(lam)) @ Q.T  
Z_zca = X @ W_zca.T    # shape (400, 2)  
  
fig, axes = plt.subplots(1, 3, figsize=(14, 5))  
  
for ax, data, title, color in zip(  
    axes,  
    [X, Z_pca, Z_zca],  
    ['Original  $\tilde{X}$ ', 'PCA whitened', 'ZCA whitened'],  
    ['#4a90d9', '#2ecc71', 'tomato']):  
    ax.scatter(data[:, 0], data[:, 1], s=8, color=color, alpha=0.4)
```

```

cx, cy = data.mean(axis=0)
ax.scatter([cx], [cy], s=60, color='#333333', zorder=5)
ax.axhline(0, color='#333333', lw=0.4, alpha=0.5)
ax.axvline(0, color='#333333', lw=0.4, alpha=0.5)
ax.set_aspect('equal')
ax.set_title(title, fontsize=12)
ax.grid(alpha=0.2)
lim = max(np.abs(data).max() * 1.1, 2.5)
ax.set_xlim(-lim, lim)
ax.set_ylim(-lim, lim)

fig.suptitle('Whitening: PCA aligns axes, ZCA minimizes distortion', fontsize=12)
fig.tight_layout()
plt.savefig('ch19-covariance/fig-whitening.png', dpi=150, bbox_inches='tight')
plt.show()

```



Connection to Batch Normalization

Neural network batch normalization (Ioffe & Szegedy 2015) applies a scalar whitening per feature – it normalizes each neuron independently by subtracting the batch mean and dividing by the batch standard deviation. This is the **diagonal approximation** to full ZCA whitening: it sets $W = D^{-1/2}$ where $D = \text{diag}(\Sigma)$, ignoring all off-diagonal covariances. Full ZCA removes cross-feature correlations as well, but the $O(p^3)$ cost is prohibitive for $p = 10^3$ – 10^6 feature maps.

Exercises

19.6.1 Show that $W_{\text{ZCA}} = \Sigma^{-1/2}$ satisfies $W\Sigma W^T = I$. Then show that among all W satisfying this, W_{ZCA} minimizes $\|W - I\|_F$. (*Hint:* write $W = Q'\Lambda'Q'^T$ in its own spectral basis and minimize over the diagonal entries of Λ' .)

19.6.2 PCA whitening can be followed by truncation: keep only the top k principal components. Write the resulting $k \times p$ whitening matrix W_k and show that $W_k\Sigma W_k^T = I_k$. How does this enable dimensionality reduction + whitening in one step?

19.6.3 Suppose Σ is nearly singular (one eigenvalue is 10^{-8}). What goes wrong when computing $\Sigma^{-1/2}$? Propose a regularized whitening transform W_ϵ that stays numerically stable.

19.6.4 Scalar batch normalization uses $W = D^{-1/2}$ where $D = \text{diag}(\Sigma)$. Give an example covariance Σ where the batch-normalized data still has large off-diagonal covariance. What is the residual covariance after applying $D^{-1/2}$?

Information Form: $\Omega = \Sigma^{-1}$

The **information matrix** (also called the **precision matrix** or **concentration matrix**) is the inverse of the covariance:

$$\Omega = \Sigma^{-1} \in \mathbb{R}^{p \times p}.$$

Working with Ω instead of Σ is often algebraically and computationally advantageous. The two representations encode the same Gaussian but answer different questions: Σ_{jk} measures how features j and k co-vary marginally; Ω_{jk} measures how they are coupled **after conditioning on all other features**.

Partial Correlations and Sparsity

A key fact: $\Omega_{jk} = 0$ **if and only if** x_j and x_k are **conditionally independent given all other variables**. This is the foundation of **Gaussian graphical models**: fit a graph where an edge between j and k exists iff $\Omega_{jk} \neq 0$.

In contrast, $\Sigma_{jk} = 0$ means only marginal uncorrelatedness – variables can be marginally uncorrelated yet conditionally dependent (and vice versa).

The **partial correlation** between j and k is

$$\rho_{jk|\text{rest}} = -\frac{\Omega_{jk}}{\sqrt{\Omega_{jj}\Omega_{kk}}}.$$

The minus sign ensures the convention that a positive partial correlation corresponds to a positive conditional dependence.

Additivity for Independent Sources

The information form is additive over **independent measurements**. If two independent sources give likelihood functions $p_1(\mathbf{x}) \propto e^{-\frac{1}{2}\mathbf{x}^T\Omega_1\mathbf{x}}$ and $p_2(\mathbf{x}) \propto e^{-\frac{1}{2}\mathbf{x}^T\Omega_2\mathbf{x}}$, their joint likelihood is

$$p(\mathbf{x}) \propto e^{-\frac{1}{2}\mathbf{x}^T(\Omega_1+\Omega_2)\mathbf{x}},$$

so the combined precision matrix is $\Omega_1 + \Omega_2$.

Covariance form: combining two independent measurements requires a Schur-complement inversion, $(\Sigma_1^{-1} + \Sigma_2^{-1})^{-1}$ – less direct.

This additivity makes the information form the natural language for **sensor fusion** and **Bayesian filtering**.

Information Form of the Gaussian

The Gaussian density can be written in **canonical (information) form**:

$$p(\mathbf{x}) \propto \exp\left(-\frac{1}{2}\mathbf{x}^T\Omega\mathbf{x} + \eta^T\mathbf{x}\right),$$

where $\Omega = \Sigma^{-1}$ and $\eta = \Sigma^{-1}\mu$ is the **information vector**. The mean is recovered as $\mu = \Omega^{-1}\eta$.

The canonical parameters (Ω, η) compose under fusion by addition: given two independent observations of $\mathbf{x}$ with canonical parameters (Ω_1, η_1) and (Ω_2, η_2) , the posterior is $(\Omega_1 + \Omega_2, \eta_1 + \eta_2)$.

Factor Graphs and Sparse SLAM

In **simultaneous localization and mapping (SLAM)**, the state vector $\mathbf{x} = (\mathbf{x}_0, \mathbf{x}_1, \dots, \mathbf{x}_T, \mathbf{l}_1, \dots, \mathbf{l}_M)$ collects robot poses and landmark positions. Each odometry or observation factor contributes a quadratic term to the negative log-posterior:

$$-\log p(\mathbf{x}|z) = \frac{1}{2}\mathbf{x}^T\Omega\mathbf{x} - \eta^T\mathbf{x} + \text{const.}$$

The global information matrix Ω is **sparse**: each measurement only involves a small number of variables (two consecutive poses, or a pose and a landmark), so Ω has at most $O(T + M)$

non-zero blocks even though the full state has dimension $O(dT + dM)$ (with d the pose/landmark dimension).

This sparsity is the key computational advantage: - Direct sparse Cholesky solvers (e.g., CHOLMOD, SuiteSparse) factor Ω in $O((T + M)^{1.5})$ time in 2-D SLAM, vs. $O((T + M)^3)$ for dense systems. - Incremental solvers (iSAM2, GTSAM) update only the affected fill-in when a new factor is added. - The sparsity pattern encodes the graph structure: $\Omega_{ij} \neq 0$ iff variables i and j share a factor.

Numerical Example: Sparse Precision Matrix

```
import numpy as np
import scipy.sparse as sp
import matplotlib.pyplot as plt

rng = np.random.default_rng(197)

# Simulate a chain of T+1 poses connected by odometry factors
# State: x_0, x_1, ..., x_T (each scalar for simplicity)
T = 8          # 9 poses
p = T + 1      # dimension of state

# Prior: x_0 ~ N(0, 1)
Omega = np.zeros((p, p))
eta = np.zeros(p)
Omega[0, 0] += 1.0 # prior precision on x_0

# Odometry factors: x_{t+1} = x_t + 1 + noise, noise ~ N(0, 0.25)
# Factor: (x_{t+1} - x_t - 1)^2 / 0.25 -> precision 4, linking t and t+1
odometry_precision = 4.0
for t in range(T):
    # Contribution of factor (x_{t+1} - x_t - delta_t)^2 * prec
    delta_t = 1.0 # expected step
    Omega[t, t] += odometry_precision
    Omega[t, t+1] -= odometry_precision
    Omega[t+1, t] -= odometry_precision
    Omega[t+1, t+1] += odometry_precision
    eta[t] -= odometry_precision * (-delta_t)
    eta[t+1] += odometry_precision * (-delta_t)

print("Information matrix Omega (tridiagonal):")
print(Omega.round(2))

# Solve for mean
```



```

mu_est = np.linalg.solve(Omega, eta)      # shape (9,)
print(f"\nEstimated poses (should be 0,1,...,8): {mu_est.round(3)}")

# Covariance = Omega^{-1}
Sigma_est = np.linalg.inv(Omega)          # shape (9, 9)

fig, axes = plt.subplots(1, 2, figsize=(11, 4.5))

im0 = axes[0].imshow(np.abs(Omega), cmap='Blues', aspect='auto')
axes[0].set_title('|Omega| (sparse, tridiagonal)', fontsize=11)
plt.colorbar(im0, ax=axes[0])

im1 = axes[1].imshow(np.abs(Sigma_est), cmap='Blues', aspect='auto')
axes[1].set_title('|Sigma| = |Omega^{-1}| (dense)', fontsize=11)
plt.colorbar(im1, ax=axes[1])

fig.suptitle('Precision matrix is sparse; covariance is dense', fontsize=12)
fig.tight_layout()
plt.savefig('ch19-covariance/fig-information-form.png', dpi=150, bbox_inches='tight')
plt.show()

```

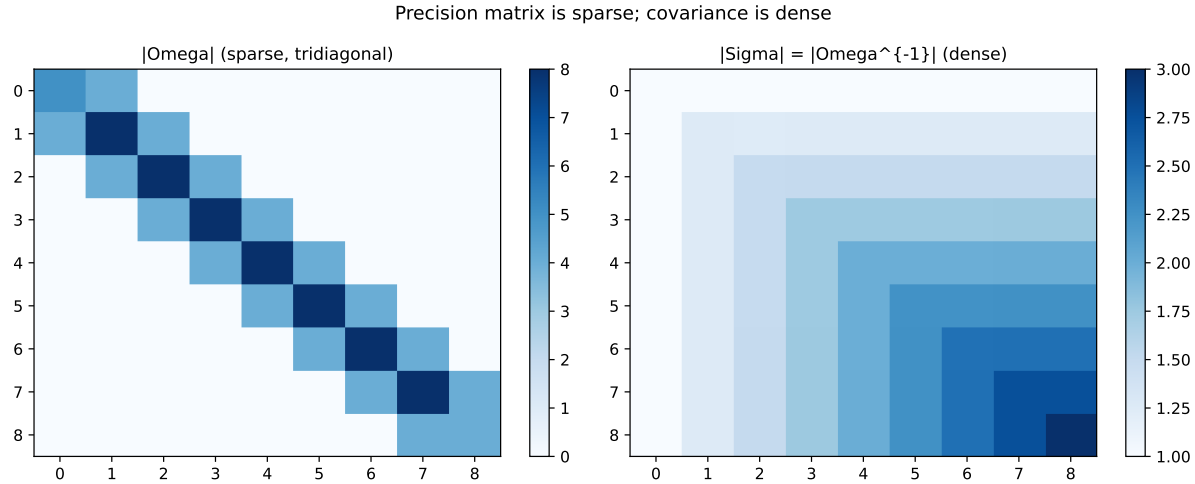
Information matrix Omega (tridiagonal):

```

[[ 5. -4.  0.  0.  0.  0.  0.  0.  0.]
 [-4.  8. -4.  0.  0.  0.  0.  0.  0.]
 [ 0. -4.  8. -4.  0.  0.  0.  0.  0.]
 [ 0.  0. -4.  8. -4.  0.  0.  0.  0.]
 [ 0.  0.  0. -4.  8. -4.  0.  0.  0.]
 [ 0.  0.  0.  0. -4.  8. -4.  0.  0.]
 [ 0.  0.  0.  0.  0. -4.  8. -4.  0.]
 [ 0.  0.  0.  0.  0.  0. -4.  8. -4.]
 [ 0.  0.  0.  0.  0.  0.  0. -4.  4.]]

```

Estimated poses (should be 0,1,...,8): [-0. -1. -2. -3. -4. -5. -6. -7. -8.]



Schur Complement and Marginalization

To marginalize out a block of variables $\mathbf{x}_B$ from the joint information form, partition:

$$\Omega = \begin{bmatrix} \Omega_{AA} & \Omega_{AB} \\ \Omega_{BA} & \Omega_{BB} \end{bmatrix}, \quad \eta = \begin{bmatrix} \eta_A \\ \eta_B \end{bmatrix}.$$

The **marginal** over $\mathbf{x}_A$ (eliminating $\mathbf{x}_B$) has canonical parameters:

$$\Omega_{A|B} = \Omega_{AA} - \Omega_{AB}\Omega_{BB}^{-1}\Omega_{BA}, \quad \eta_{A|B} = \eta_A - \Omega_{AB}\Omega_{BB}^{-1}\eta_B.$$

$\Omega_{A|B}$ is the **Schur complement** of Ω_{BB} in Ω . This is the variable elimination step in sparse linear solvers and the Gaussian elimination step in factor-graph inference. The Schur complement can fill in previously zero entries – this is **fill-in**, and its minimization is the subject of sparse matrix reordering algorithms (e.g., AMD, METIS).

Exercises

19.7.1 Show that $\Omega_{jk} = 0$ implies $x_j \perp\!\!\!\perp x_k \mid \mathbf{x}_{\text{rest}}$ for a jointly Gaussian vector. (*Hint*: compute the conditional distribution of (x_j, x_k) given all other variables using the block-inversion formula for Σ .)

19.7.2 Given two independent Gaussian measurements $\mathbf{y}_1 \sim \mathcal{N}(H_1\mathbf{x}, R_1)$ and $\mathbf{y}_2 \sim \mathcal{N}(H_2\mathbf{x}, R_2)$, write the posterior $p(\mathbf{x}|\mathbf{y}_1, \mathbf{y}_2)$ in information form and show that the information matrices add: $\Omega_{\text{post}} = H_1^T R_1^{-1} H_1 + H_2^T R_2^{-1} H_2$.

19.7.3 A chain SLAM system has T poses. The information matrix Ω is tridiagonal ($p \times p$, $p = T$). What is the cost of Gaussian elimination (Cholesky factorization) on this tridiagonal system? Compare to the dense case.

19.7.4 After eliminating landmarks from a SLAM system, the resulting **marginal** information matrix over poses often has a band structure. Explain geometrically why two poses share information after landmark elimination, even if they were not directly connected by an odometry factor.

Application: Anomaly Detection on Coffee Beans

A roastery inspects green coffee beans on a conveyor belt and wants to flag defective beans automatically. Defective beans differ in multiple visual features simultaneously – size, color, and density – but no single feature suffices to separate them from good beans. Multivariate Gaussian modeling with Mahalanobis thresholding handles this cleanly.

This application builds a complete one-class anomaly detector: estimate the inlier distribution from good beans, choose a statistical threshold, and score new beans.

Feature Engineering (Simulated)

Real coffee-bean datasets require image processing infrastructure, so we synthesize a realistic distribution for three features extracted per bean:

- **Area** (px^2): cross-sectional area from overhead camera.
- **Eccentricity**: ratio of minor to major axis of the fitted ellipse (0 = very elongated, 1 = circular).
- **Lightness**: mean L^* value in CIE Lab space (higher = lighter green).

Defective beans (mold, insect damage, shriveled) tend to be smaller, more eccentric, and lighter or darker than normal.

```
import numpy as np
from scipy.stats import chi2

rng = np.random.default_rng(198)

# ---- Inlier distribution (good beans) ----
mu_good = np.array([850.0,  # area px^2
                    0.72,   # eccentricity
                    48.0])  # lightness L*
# Correlations: area-ecc r=0.3 (cov=0.8), area-light r=-0.2 (cov=-48), ecc-light r=0.15 (cov=12)
Sigma_good = np.array([[1600.0, 0.8, -48.0],  # shape (3, 3)
                       [ 0.8, 0.006, 0.07],
                       [-48.0, 0.07, 36.0]])
```

```
L_good = np.linalg.cholesky(Sigma_good)          # shape (3, 3)
n_good = 400
X_good = rng.standard_normal((n_good, 3)) @ L_good.T + mu_good # shape (400, 3)

# ---- Anomaly distributions (defective beans) ----
# Type 1: shriveled -- small, eccentric
mu_shriv = np.array([580.0, 0.52, 51.0])
Sigma_shriv = np.array([[900.0, 0.0, 0.0],
                        [0.0, 0.008, 0.0],
                        [0.0, 0.0, 25.0]])

n_shriv = 40
X_shriv = rng.standard_normal((n_shriv, 3)) @ np.linalg.cholesky(Sigma_shriv).T + mu_shriv

# Type 2: moldy -- discolored (darker), near-normal shape
mu_mold = np.array([830.0, 0.70, 35.0])
Sigma_mold = np.array([[1200.0, 0.0, 0.0],
                       [0.0, 0.005, 0.0],
                       [0.0, 0.0, 20.0]])

n_mold = 40
X_mold = rng.standard_normal((n_mold, 3)) @ np.linalg.cholesky(Sigma_mold).T + mu_mold

print(f"Good beans:      {n_good} samples, features: area, eccentricity, lightness")
print(f"Shriveled:      {n_shriv} anomalies")
print(f"Moldy:          {n_mold} anomalies")
print(f"\nTrue mean (good): {mu_good}")
print(f"True Sigma (good) diagonal: {np.diag(Sigma_good)}")
```

```
Good beans:      400 samples, features: area, eccentricity, lightness
Shriveled:      40 anomalies
Moldy:          40 anomalies
```

```
True mean (good): [8.5e+02 7.2e-01 4.8e+01]
True Sigma (good) diagonal: [1.6e+03 6.0e-03 3.6e+01]
```

Fitting the Inlier Model

```
import numpy as np
from scipy.stats import chi2

rng = np.random.default_rng(1981)

# Re-generate data (self-contained block)
```

```

mu_good    = np.array([850.0, 0.72, 48.0])
Sigma_true = np.array([[1600.0, 0.8, -48.0],
                        [ 0.8, 0.006, 0.07],
                        [-48.0, 0.07, 36.0]])
L_true = np.linalg.cholesky(Sigma_true)
n_good = 400
X_good = rng.standard_normal((n_good, 3)) @ L_true.T + mu_good    # shape (400, 3)

mu_shriv = np.array([580.0, 0.52, 51.0])
Sigma_shriv = np.diag([900.0, 0.008, 25.0])
X_shriv = rng.standard_normal((40, 3)) @ np.linalg.cholesky(Sigma_shriv).T + mu_shriv

mu_mold = np.array([830.0, 0.70, 35.0])
Sigma_mold = np.diag([1200.0, 0.005, 20.0])
X_mold = rng.standard_normal((40, 3)) @ np.linalg.cholesky(Sigma_mold).T + mu_mold

# ---- Estimate inlier parameters from good beans only ----
mu_hat    = X_good.mean(axis=0)                # shape (3,)
X_c       = X_good - mu_hat                    # shape (400, 3)
Sigma_hat = X_c.T @ X_c / (n_good - 1)         # shape (3, 3)

print("Estimated mean:\n", mu_hat.round(3))
print("\nEstimated Sigma:\n", Sigma_hat.round(3))

# Cholesky of Sigma for stable Mahalanobis computation
L_hat = np.linalg.cholesky(Sigma_hat)          # shape (3, 3)

def mahalanobis(X, mu, L):
    """Mahalanobis distance via Cholesky solve. X: (n,p), returns (n,)."""
    diff = X - mu                               # shape (n, 3)
    Z     = np.linalg.solve(L, diff.T).T        # shape (n, 3)
    return np.linalg.norm(Z, axis=1)             # shape (n,)

# Score all beans
d_good  = mahalanobis(X_good, mu_hat, L_hat)    # shape (400,)
d_shriv = mahalanobis(X_shriv, mu_hat, L_hat)  # shape (40,)
d_mold  = mahalanobis(X_mold, mu_hat, L_hat)    # shape (40,)

# Chi-squared threshold: 99% under null (p=3 features)
tau = np.sqrt(chi2.ppf(0.99, df=3))
print(f"\nMahalanobis threshold tau (chi2 99%, df=3): {tau:.4f}")

fp_rate = (d_good > tau).mean()
det_shriv = (d_shriv > tau).mean()
det_mold = (d_mold > tau).mean()
print(f"\nFalse positive rate (good beans flagged): {fp_rate:.4f} (expect ~0.01)")
print(f"Detection rate -- shriveled: {det_shriv:.4f}")

```

```
print(f"Detection rate -- moldy:      {det_mold:.4f}")
```

Estimated mean:

```
[8.5237e+02 7.2400e-01 4.7392e+01]
```

Estimated Sigma:

```
[[ 1.488689e+03  8.380000e-01 -5.733200e+01]
 [ 8.380000e-01  5.000000e-03  7.000000e-02]
 [-5.733200e+01  7.000000e-02  3.837400e+01]]
```

Mahalanobis threshold tau (chi2 99%, df=3): 3.3682

False positive rate (good beans flagged): 0.0150 (expect ~0.01)

Detection rate -- shriveled: 1.0000

Detection rate -- moldy: 0.1750

Visualization

```
import numpy as np
import matplotlib.pyplot as plt
from matplotlib.patches import Ellipse
from scipy.stats import chi2

rng = np.random.default_rng(1982)

# Re-generate (self-contained)
mu_good = np.array([850.0, 0.72, 48.0])
Sigma_true = np.array([[1600.0, 0.8, -48.0],
                        [ 0.8, 0.006, 0.07],
                        [-48.0, 0.07, 36.0]])
L_true = np.linalg.cholesky(Sigma_true)
n_good = 400
X_good = rng.standard_normal((n_good, 3)) @ L_true.T + mu_good

mu_shriv = np.array([580.0, 0.52, 51.0])
X_shriv = rng.standard_normal((40, 3)) @ np.linalg.cholesky(
    np.diag([900.0, 0.008, 25.0])).T + mu_shriv

mu_mold = np.array([830.0, 0.70, 35.0])
X_mold = rng.standard_normal((40, 3)) @ np.linalg.cholesky(
    np.diag([1200.0, 0.005, 20.0])).T + mu_mold
```

```

mu_hat      = X_good.mean(axis=0)
X_c         = X_good - mu_hat
Sigma_hat   = X_c.T @ X_c / (n_good - 1)
L_hat       = np.linalg.cholesky(Sigma_hat)

def mahalanobis(X, mu, L):
    diff = X - mu
    Z     = np.linalg.solve(L, diff.T).T
    return np.linalg.norm(Z, axis=1)

tau = np.sqrt(chi2.ppf(0.99, df=3))
d_good = mahalanobis(X_good, mu_hat, L_hat)
d_shriv = mahalanobis(X_shriv, mu_hat, L_hat)
d_mold = mahalanobis(X_mold, mu_hat, L_hat)

# Project to 2-D subspace: Area (feat 0) vs Lightness (feat 2) for visualization
feat_x, feat_y = 0, 2
labels_x = ['Area (px2)', 'Eccentricity', 'Lightness L*']

# Marginal covariance for the 2-D projection
idx2 = [feat_x, feat_y]
Sigma_2d = Sigma_hat[np.ix_(idx2, idx2)]      # shape (2, 2)
mu_2d    = mu_hat[idx2]                      # shape (2,)
lam2, Q2 = np.linalg.eigh(Sigma_2d)
ord2 = np.argsort(lam2)[::-1]
lam2 = lam2[ord2]; Q2 = Q2[:, ord2]
angle2 = np.degrees(np.arctan2(Q2[1, 0], Q2[0, 0]))

fig, axes = plt.subplots(1, 2, figsize=(13, 5.5))

# Left: scatter in 2-D feature space
ax = axes[0]
ax.scatter(X_good[:, feat_x], X_good[:, feat_y],
           s=10, color='#4a90d9', alpha=0.4, label=f'Good (n={n_good})')
ax.scatter(X_shriv[:, feat_x], X_shriv[:, feat_y],
           s=30, color='tomato', alpha=0.85, marker='^', label='Shriveled')
ax.scatter(X_mold[:, feat_x], X_mold[:, feat_y],
           s=30, color='#e67e22', alpha=0.85, marker='s', label='Moldy')

# 99% Mahalanobis ellipse (marginal 2-D; chi2 df=2 for 2-D projection)
tau_2d = np.sqrt(chi2.ppf(0.99, df=2))
for n_sig, alpha_ell in [(1, 0.6), (tau_2d, 0.9)]:
    lbl = '99% threshold' if n_sig == tau_2d else '1 $\sigma$ '
    ell = Ellipse(xy=mu_2d,
                  width = 2 * n_sig * np.sqrt(lam2[0]),
                  height = 2 * n_sig * np.sqrt(lam2[1]),
                  angle=angle2, edgecolor='#333333', facecolor='none',

```

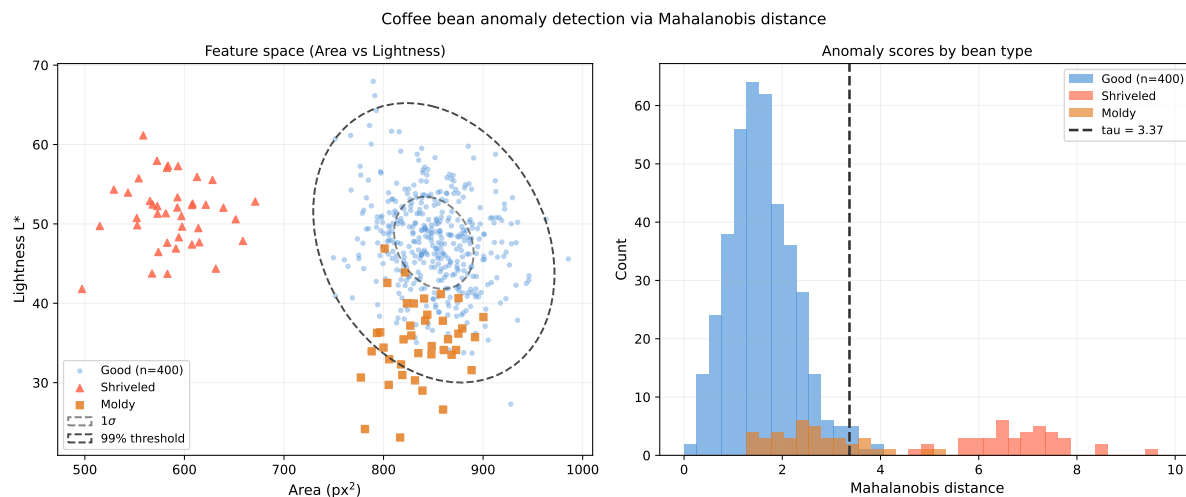
```

        lw=1.5, linestyle='--', alpha=alpha_ell, label=lbl)
    ax.add_patch(ell)
    ax.set_xlabel(labels_x[feat_x], fontsize=11)
    ax.set_ylabel(labels_x[feat_y], fontsize=11)
    ax.set_title('Feature space (Area vs Lightness)', fontsize=11)
    ax.legend(fontsize=9)
    ax.grid(alpha=0.2)

# Right: Mahalanobis distance histogram
ax2 = axes[1]
all_d = np.concatenate([d_good, d_shriv, d_mold])
bins = np.linspace(0, all_d.max() * 1.05, 40)
ax2.hist(d_good, bins=bins, color='#4a90d9', alpha=0.65, label=f'Good (n={n_good})')
ax2.hist(d_shriv, bins=bins, color='tomato', alpha=0.65, label='Shriveled')
ax2.hist(d_mold, bins=bins, color='#e67e22', alpha=0.65, label='Moldy')
ax2.axvline(tau, color='#333333', lw=2, linestyle='--', label=f'tau = {tau:.2f}')
ax2.set_xlabel('Mahalanobis distance', fontsize=11)
ax2.set_ylabel('Count', fontsize=11)
ax2.set_title('Anomaly scores by bean type', fontsize=11)
ax2.legend(fontsize=9)
ax2.grid(alpha=0.2)

fig.suptitle('Coffee bean anomaly detection via Mahalanobis distance', fontsize=12)
fig.tight_layout()
plt.savefig('ch19-covariance/fig-coffee-anomaly.png', dpi=150, bbox_inches='tight')
plt.show()

```



Key Takeaways

1. **One-class classification requires only inlier data.** No labeled defective examples are needed during training – only an estimate of μ and Σ from good beans. The chi-squared threshold provides a theoretically grounded false positive rate.
2. **Mahalanobis handles correlated features without hand-tuning.** If area and eccentricity are correlated (larger beans tend to be more circular), a point deviating in both features simultaneously might be only mildly surprising. Mahalanobis accounts for this; a per-feature threshold would not.
3. **Detection power depends on defect type.** Shriveled beans (small, eccentric) are easily detected because they deviate far in the Area dimension. Moldy beans (same shape, lighter color) are detected via the Lightness feature. A single Euclidean threshold would miss moldy beans entirely.
4. **Regularize in production.** With $p = 3$ and $n = 400$, the sample covariance is well-conditioned. In practice with p large (color histograms, texture features), add a ridge term $\Sigma \leftarrow \Sigma + \epsilon I$ or use a shrinkage estimator (Ledoit-Wolf) before inverting.

Solutions

Mean-Centering and the Data Matrix

19.1.1 $H^2 = (I - \frac{1}{n}\mathbf{1}\mathbf{1}^T)^2 = I - \frac{2}{n}\mathbf{1}\mathbf{1}^T + \frac{1}{n^2}\mathbf{1}\mathbf{1}^T\mathbf{1}\mathbf{1}^T = I - \frac{2}{n}\mathbf{1}\mathbf{1}^T + \frac{n}{n^2}\mathbf{1}\mathbf{1}^T = I - \frac{1}{n}\mathbf{1}\mathbf{1}^T = H$. $H\mathbf{1} = \mathbf{1} - \frac{1}{n}\mathbf{1}\mathbf{1}^T\mathbf{1} = \mathbf{1} - \mathbf{1} = \mathbf{0}$. Idempotency means applying H twice gives the same result as once: H is an orthogonal projection onto the subspace orthogonal to $\mathbf{1}$, and projecting onto a subspace is a “one-shot” operation.

19.1.2 Let $X' = X + \mathbf{1}\mathbf{c}^T$. Mean: $\bar{\mathbf{x}}' = \bar{\mathbf{x}} + \mathbf{c}$. Centered: $X' - \mathbf{1}\bar{\mathbf{x}}'^T = X + \mathbf{1}\mathbf{c}^T - \mathbf{1}(\bar{\mathbf{x}} + \mathbf{c})^T = X - \mathbf{1}\bar{\mathbf{x}}^T = \tilde{X}$. Equivalently, $HX' = HX + H\mathbf{1}\mathbf{c}^T = HX$ since $H\mathbf{1} = \mathbf{0}$.

19.1.3 Column-centering ensures $\tilde{X}^T\mathbf{1}_n = \mathbf{0}$ (each column sums to zero), but says nothing about row sums. The row sum $\tilde{X}\mathbf{1}_p$ is $\sum_j \tilde{X}_{ij}$ for each row i , which is generally non-zero. Double-centering (H_nXH_p with both the row and column centering matrices) forces both row and column sums to zero. This is used in collaborative filtering (e.g., Netflix matrix) to remove user and item biases simultaneously.

19.1.4 When $\bar{\mathbf{x}} = \mathbf{0}$ exactly, $\tilde{X} = X$ so $\frac{1}{n-1}X^TX = \frac{n}{n-1} \cdot \frac{1}{n}X^TX$. As $n \rightarrow \infty$, $n/(n-1) \rightarrow 1$, so the two converge. For finite n , dividing by $n-1$ rather than n gives an **unbiased** estimator of the population covariance: $\mathbb{E}[\frac{1}{n-1}\tilde{X}^T\tilde{X}] = \Sigma_{\text{pop}}$ (Bessel’s correction). Dividing by n gives the maximum-likelihood estimator, which is biased downward.

The Covariance Matrix

19.2.1 The (j, k) entry of $\tilde{X}^T \tilde{X}$ is $(\tilde{X}^T \tilde{X})_{jk} = \sum_{i=1}^n (\tilde{X}^T)_{ji} \tilde{X}_{ik} = \sum_{i=1}^n \tilde{X}_{ij} \tilde{X}_{ik}$. Dividing by $n - 1$ gives Σ_{jk} .

19.2.2 $\mathbf{v}^T \Sigma \mathbf{v} = \frac{1}{n-1} \mathbf{v}^T \tilde{X}^T \tilde{X} \mathbf{v} = \frac{1}{n-1} \|\tilde{X} \mathbf{v}\|_2^2 \geq 0$. Equality holds iff $\tilde{X} \mathbf{v} = \mathbf{0}$, i.e., $\mathbf{v} \in \text{Null}(\tilde{X})$. This happens when $\mathbf{v}$ is a linear combination of features that is identically zero across all observations (e.g., a feature that is a constant multiple of another).

19.2.3 $\Sigma_{jk} = 0$ means features j and k are orthogonal in the centered data. Geometrically, the covariance ellipse has no tilt between those two feature axes – the projection onto the (j, k) plane is an axis-aligned ellipse. Two features can be uncorrelated ($\Sigma_{jk} = 0$) yet dependent: for example, if $x_j \sim \text{Uniform}(-1, 1)$ and $x_k = x_j^2$, then $\text{Cov}(x_j, x_k) = 0$ but knowing x_j determines x_k .

19.2.4 $R_{jk} = \Sigma_{jk} / \sqrt{\Sigma_{jj} \Sigma_{kk}} = (D^{-1/2})_{jj} \Sigma_{jk} (D^{-1/2})_{kk} = (D^{-1/2} \Sigma D^{-1/2})_{jk}$. Diagonal entries: $R_{jj} = \Sigma_{jj} / \sqrt{\Sigma_{jj} \Sigma_{jj}} = \Sigma_{jj} / \Sigma_{jj} = 1$. Each feature is perfectly correlated with itself.

The Multivariate Gaussian and Its Geometry

19.3.1 For $\mathbf{z} \sim \mathcal{N}(\mathbf{0}, I_2)$, $\|\mathbf{z}\|^2 \sim \chi_2^2$. The χ_2^2 CDF is $F(k^2) = 1 - e^{-k^2/2}$. At $k = 1$: $P(\|\mathbf{z}\|^2 \leq 1) = 1 - e^{-1/2} \approx 0.393$. So the 2-D 1-sigma ellipsoid contains only 39.3% of the mass, not 68%. The curse of dimensionality: as p increases, the 1-sigma ellipsoid captures an ever smaller fraction of mass.

19.3.2 The moment generating function of $A\mathbf{x}$ at $\mathbf{t}$ is $\mathbb{E}[e^{\mathbf{t}^T A\mathbf{x}}] = e^{\mathbf{t}^T A\mu + \frac{1}{2} \mathbf{t}^T A \Sigma A^T \mathbf{t}}$, which is the MGF of $\mathcal{N}(A\mu, A \Sigma A^T)$. When $m < p$, $A \Sigma A^T \in \mathbb{R}^{m \times m}$ has rank at most m ; the image is an m -dimensional Gaussian. The original density (in $\mathbb{R}^p$) is degenerate (concentrated on the m -dim subspace $\text{Col}(A^T)$) and has no density with respect to Lebesgue measure in $\mathbb{R}^p$.

19.3.3 The block Σ is PSD, so for any $\mathbf{v}$, $\mathbf{v}^T \Sigma \mathbf{v} \geq 0$. Take $\mathbf{v} = [\mathbf{v}_A^T, -\Sigma_{BB}^{-1} \Sigma_{BA} \mathbf{v}_A^T]^T$: $\mathbf{v}^T \Sigma \mathbf{v} = \mathbf{v}_A^T (\Sigma_{AA} - \Sigma_{AB} \Sigma_{BB}^{-1} \Sigma_{BA}) \mathbf{v}_A \geq 0$. So the Schur complement is PSD. Since $\Sigma_{AA} - (\Sigma_{AA} - \Sigma_{AB} \Sigma_{BB}^{-1} \Sigma_{BA}) = \Sigma_{AB} \Sigma_{BB}^{-1} \Sigma_{BA}$ is PSD (as Σ_{BB}^{-1} is PSD), we have $\Sigma_{AA} \succeq \Sigma_{AA} - \Sigma_{AB} \Sigma_{BB}^{-1} \Sigma_{BA}$.

19.3.4 The change-of-variables from $\mathbf{x}$ to $\mathbf{z} = \Sigma^{-1/2}(\mathbf{x} - \mu)$ has Jacobian $|\partial \mathbf{x} / \partial \mathbf{z}| = |\Sigma^{1/2}| = |\Sigma|^{1/2}$ (the determinant of the linear map). To keep $\int p d\mathbf{x} = 1$ after the change of variables, the normalizer picks up a factor of $|\Sigma|^{1/2}$ from the volume scaling.

Mahalanobis Distance

19.4.1 $d_M(\mathbf{x})^2 = (\mathbf{x} - \mu)^T (\sigma^2 I)^{-1} (\mathbf{x} - \mu) = \|\mathbf{x} - \mu\|^2 / \sigma^2$. So $d_M = \|\mathbf{x} - \mu\| / \sigma$. Euclidean distance is Mahalanobis distance with $\Sigma = I$ (isotropic unit variance); it treats all directions and scales equally.

19.4.2 $\Sigma = \text{diag}(1, 100)$, $\mathbf{x} = (0, 5)^T$, $\mu = \mathbf{0}$. Euclidean: $\|(0, 5)\| = 5$. Mahalanobis: $\sqrt{0^2/1 + 5^2/100} = \sqrt{0.25} = 0.5$. The Mahalanobis distance of 0.5 correctly reflects that a deviation of 5 units in a feature with standard deviation 10 is only half a standard deviation – not surprising. Euclidean would flag this as anomalous.

19.4.3 With centered X , the leverage score $h_{ii} = \mathbf{x}_i^T (X^T X)^{-1} \mathbf{x}_i$. The sample covariance is $\Sigma = X^T X / (n - 1)$, so $(X^T X)^{-1} = \Sigma^{-1} / (n - 1)$. Thus $h_{ii} = \mathbf{x}_i^T \Sigma^{-1} \mathbf{x}_i / (n - 1) = d_M^2(\mathbf{x}_i) / (n - 1)$ (with $\mu = \mathbf{0}$ since X is centered). High-leverage points are Mahalanobis-far from the centroid.

19.4.4 $\mathbf{z} = \Sigma^{-1/2}(\mathbf{x} - \mu)$. Since $\mathbf{x} \sim \mathcal{N}(\mu, \Sigma)$, we have $\mathbf{z} = \Sigma^{-1/2}(\mathbf{x} - \mu) \sim \mathcal{N}(\mathbf{0}, \Sigma^{-1/2} \Sigma \Sigma^{-1/2}) = \mathcal{N}(\mathbf{0}, I_p)$. Then $d_M^2 = \|\mathbf{z}\|^2 = \sum_{j=1}^p z_j^2$, a sum of p independent standard normal squares, which is χ_p^2 by definition.

The Matrix Square Root $\Sigma^{1/2}$

19.5.1 $\Sigma = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$. Eigenvalues: $\det(\Sigma - \lambda I) = (2 - \lambda)^2 - 1 = 0 \Rightarrow \lambda = 1, 3$. Eigenvectors: $\lambda_1 = 3$, $\mathbf{q}_1 = \frac{1}{\sqrt{2}}(1, 1)^T$; $\lambda_2 = 1$, $\mathbf{q}_2 = \frac{1}{\sqrt{2}}(-1, 1)^T$. $\Sigma^{1/2} = \sqrt{3} \mathbf{q}_1 \mathbf{q}_1^T + 1 \mathbf{q}_2 \mathbf{q}_2^T = \frac{1}{2} \begin{bmatrix} \sqrt{3} + 1 & \sqrt{3} - 1 \\ \sqrt{3} - 1 & \sqrt{3} + 1 \end{bmatrix}$. Verification: $(\Sigma^{1/2})^2 = Q \text{diag}(3, 1) Q^T = \Sigma$.

19.5.2 Example: $\Sigma = \begin{bmatrix} 4 & 2 \\ 2 & 2 \end{bmatrix}$. Cholesky: $L = \begin{bmatrix} 2 & 0 \\ 1 & 1 \end{bmatrix}$ (L is lower triangular, not symmetric). Spectral: $\Sigma^{1/2}$ is symmetric (as computed from eigenvectors). Since L is not symmetric, $L \neq \Sigma^{1/2}$. Both satisfy $(\cdot)(\cdot)^T = \Sigma$.

19.5.3 For $\Sigma = Q \Lambda Q^T$: $e^\Sigma = \sum_{k=0}^{\infty} \Sigma^k / k! = Q \left(\sum_{k=0}^{\infty} \Lambda^k / k! \right) Q^T = Q e^\Lambda Q^T$, where $e^\Lambda = \text{diag}(e^{\lambda_1}, \dots, e^{\lambda_p})$. Functions of normal matrices are computed by applying the scalar function to the eigenvalues.

19.5.4 Factorization cost: both spectral and Cholesky are $O(p^3)$. Per-draw cost (once A is known): computing $A\mathbf{z}$ for one draw costs $O(p^2)$, so n draws cost $O(np^2)$. The factorization dominates when $O(p^3) \gg O(np^2)$, i.e., $p \gg n$. For $p = 100$: factorization $\sim 10^6$ flops; draw cost $\sim n \cdot 10^4$ flops. Factorization dominates until $n \approx 100$; for $n \gg 100$ (certainly for $n = 10^5$), the draw cost $O(np^2) = O(10^9)$ dominates. The factorization is a one-time $O(p^3)$ investment.

Whitening: ZCA and PCA

19.6.1 $W_{\text{ZCA}} \Sigma W_{\text{ZCA}}^T = \Sigma^{-1/2} \Sigma \Sigma^{-1/2} = I$ since $\Sigma^{-1/2}$ is symmetric. For minimality: write $W = R \Sigma^{-1/2}$ for some orthogonal R (all solutions are of this form). Then $\|W - I\|_F^2 = \|R \Sigma^{-1/2} - I\|_F^2$. Using $\|AB\|_F \geq \|\sigma_{\min}(A)\| \cdot \|B\|_F$ and the unitary invariance of the Frobenius norm, the minimum over orthogonal R is achieved at $R = I$ (Procrustes argument), giving $W = \Sigma^{-1/2}$.

19.6.2 $W_k = \Lambda_k^{-1/2} Q_k^T$ where Q_k holds the top k eigenvectors and $\Lambda_k = \text{diag}(\lambda_1, \dots, \lambda_k)$. $W_k \Sigma W_k^T = \Lambda_k^{-1/2} Q_k^T (Q \Lambda Q^T) Q_k \Lambda_k^{-1/2} = \Lambda_k^{-1/2} \Lambda_k \Lambda_k^{-1/2} = I_k$. This simultaneously projects to the top- k PC space and whitens – one matrix-vector multiply instead of two separate operations.

19.6.3 $\Sigma^{-1/2} = Q \Lambda^{-1/2} Q^T$ with $\Lambda^{-1/2} = \text{diag}(1/\sqrt{\lambda_j})$. When $\lambda_j \approx 0$, $1/\sqrt{\lambda_j} \rightarrow \infty$ – the whitening transform amplifies negligible variation to unit scale, magnifying numerical noise. Regularized: $W_\epsilon = Q(\Lambda + \epsilon I)^{-1/2} Q^T$, which caps the amplification at $1/\sqrt{\epsilon}$ in the near-zero directions.

19.6.4 Let $\Sigma = \begin{bmatrix} 1 & 0.99 \\ 0.99 & 1 \end{bmatrix}$. $D = I$, so $D^{-1/2} \Sigma D^{-1/2} = \Sigma$. After batch normalization, the residual covariance is still Σ – off-diagonal entry 0.99 is unchanged. Scalar batch normalization does nothing when variances are already equal; the correlation structure remains fully intact.

Information Form: $\Omega = \Sigma^{-1}$

19.7.1 The conditional of (x_j, x_k) given the rest has covariance $(\Omega_{\{j,k\},\{j,k\}})^{-1}$ (the inverse of the 2×2 block of Ω indexed by $\{j, k\}$). When $\Omega_{jk} = 0$, this block is diagonal: $\Omega_{\{j,k\},\{j,k\}} = \text{diag}(\Omega_{jj}, \Omega_{kk})$, so the conditional covariance is also diagonal, meaning x_j and x_k are conditionally independent.

19.7.2 The likelihood of $\mathbf{x}$ given $\mathbf{y}_1, \mathbf{y}_2$ factors as $p(\mathbf{y}_1 | \mathbf{x}) p(\mathbf{y}_2 | \mathbf{x})$. Each factor contributes $-\frac{1}{2}(y_i - H_i \mathbf{x})^T R_i^{-1} (y_i - H_i \mathbf{x})$ to the log-likelihood, which expands to $-\frac{1}{2} \mathbf{x}^T H_i^T R_i^{-1} H_i \mathbf{x} + \mathbf{x}^T H_i^T R_i^{-1} \mathbf{y}_i + \text{const.}$ Summing: $\Omega_{\text{post}} = H_1^T R_1^{-1} H_1 + H_2^T R_2^{-1} H_2$, $\eta_{\text{post}} = H_1^T R_1^{-1} \mathbf{y}_1 + H_2^T R_2^{-1} \mathbf{y}_2$.

19.7.3 A $p \times p$ tridiagonal system has bandwidth $b = 1$. Banded Cholesky (forward elimination) requires $O(pb^2) = O(p)$ operations – linear in the number of poses. The dense case requires $O(p^3)$. For $T = 1000$ poses, the tridiagonal system solves in $\sim 10^3$ flops vs. $\sim 10^9$ for the dense matrix – a factor of 10^6 speedup.

19.7.4 Suppose poses i and k ($i < k$) both observed landmark l . After marginalizing out l from the factor (x_i, l) and (x_k, l) , the Schur complement introduces a non-zero entry Ω_{ik} in the pose-only information matrix: observing the same landmark creates a virtual constraint between two poses, even if they were not directly connected by odometry. Geometrically, “the relative pose between i and k is constrained because we both saw landmark l from different viewpoints.” This is the phenomenon of **loop closure** in SLAM.

Principal Component Analysis

PCA as Maximum Variance Projection

A dataset of n observations and p features is arranged into an $n \times p$ matrix X . After mean-centering (§19.1), we have $\tilde{X} = X - \mathbf{1}\bar{\mathbf{x}}^T$. Every method in this chapter operates on $\tilde{X}$.

The fundamental question: is there a single direction in $\mathbb{R}^p$ that captures as much of the data's variability as possible?

Variance Along a Direction

Choose a unit vector $\mathbf{u} \in \mathbb{R}^p$ ($\|\mathbf{u}\| = 1$). The **scalar projection** of observation $\tilde{\mathbf{x}}_i$ onto $\mathbf{u}$ is

$$z_i = \mathbf{u}^T \tilde{\mathbf{x}}_i.$$

The sample variance of these projections is

$$\text{Var}(\mathbf{u}) = \frac{1}{n-1} \sum_{i=1}^n z_i^2 = \frac{1}{n-1} \mathbf{u}^T \tilde{X}^T \tilde{X} \mathbf{u} = \mathbf{u}^T \Sigma \mathbf{u},$$

where $\Sigma = \frac{1}{n-1} \tilde{X}^T \tilde{X}$ is the sample covariance matrix (§19.2).

PCA goal: find $\mathbf{u}^*$ that maximizes $\mathbf{u}^T \Sigma \mathbf{u}$ subject to $\|\mathbf{u}\| = 1$.

Deriving the First Principal Component

Introduce a Lagrange multiplier λ for the constraint $\mathbf{u}^T \mathbf{u} = 1$:

$$\mathcal{L}(\mathbf{u}, \lambda) = \mathbf{u}^T \Sigma \mathbf{u} - \lambda(\mathbf{u}^T \mathbf{u} - 1).$$

Setting $\partial \mathcal{L} / \partial \mathbf{u} = \mathbf{0}$:

$$\boxed{\Sigma \mathbf{u} = \lambda \mathbf{u}.}$$

The maximum-variance direction must be an **eigenvector** of the sample covariance matrix, and the variance achieved equals the corresponding **eigenvalue** λ :

$$\text{Var}(\mathbf{u}^*) = (\mathbf{u}^*)^T \Sigma \mathbf{u}^* = (\mathbf{u}^*)^T \lambda \mathbf{u}^* = \lambda.$$

To maximize, we pick $\mathbf{u}^* = \mathbf{q}_1$, the eigenvector associated with the **largest** eigenvalue λ_1 .

The Full PC Decomposition

The same argument, applied to the residual $\tilde{X} - \tilde{X} \mathbf{q}_1 \mathbf{q}_1^T$ (i.e., constrain $\mathbf{u} \perp \mathbf{q}_1$), yields the **second** principal component $\mathbf{q}_2$ (eigenvector for λ_2), and so on.

The **spectral decomposition** of Σ (§14.3) delivers all components at once:

$$\Sigma = Q \Lambda Q^T, \quad Q = [\mathbf{q}_1 \mid \cdots \mid \mathbf{q}_p], \quad \Lambda = \text{diag}(\lambda_1, \dots, \lambda_p), \quad \lambda_1 \geq \cdots \geq \lambda_p \geq 0.$$

The **k -dimensional PCA subspace** is $\text{span}\{\mathbf{q}_1, \dots, \mathbf{q}_k\}$, the span of the top k eigenvectors. It captures the directions of greatest variance and is the best rank- k linear approximation to the data (§18.3, Eckart–Young).

Numerical Example

```

import numpy as np

rng = np.random.default_rng(201)

# Simulate n=200 observations, p=3 features
# Covariance: feature 1 has high variance, feature 2 moderate, feature 3 low
Sigma_true = np.array([[9.0, 2.5, 0.5], # shape (3, 3)
                       [2.5, 4.0, 1.0],
                       [0.5, 1.0, 1.0]])
mu_true = np.array([5.0, 3.0, 8.0]) # shape (3,)

L = np.linalg.cholesky(Sigma_true) # shape (3, 3)
X = rng.standard_normal((200, 3)) @ L.T + mu_true # shape (200, 3)

# Mean-center
x_bar = X.mean(axis=0) # shape (3,)
X_tilde = X - x_bar # shape (200, 3)

# Sample covariance
n = X.shape[0]
Sigma = X_tilde.T @ X_tilde / (n - 1) # shape (3, 3)

# Eigendecomposition (eigh for symmetric matrices -- returns sorted ascending)
lam, Q = np.linalg.eigh(Sigma) # lam shape (3,), Q shape (3, 3)
idx = np.argsort(lam)[::-1] # descending order
lam = lam[idx]
Q = Q[:, idx] # columns = eigenvectors, shape (3, 3)

print("Eigenvalues (variance per PC):")
for i, li in enumerate(lam):
    print(f"  lambda_{i+1} = {li:.4f}  ({100*li/lam.sum():.1f}% of total)")

print("\nFirst principal component (max-variance direction):")
print(f"  q1 = [{Q[0,0]:.4f}, {Q[1,0]:.4f}, {Q[2,0]:.4f}]")

# Project data onto q1 and verify variance matches lambda_1
z1 = X_tilde @ Q[:, 0] # shape (200,)
print(f"\nVariance of projections onto q1: {z1.var(ddof=1):.4f}")
print(f"lambda_1: {lam[0]:.4f}")

```

```

Eigenvalues (variance per PC):
  lambda_1 = 10.4821  (70.9% of total)
  lambda_2 = 3.5780  (24.2% of total)
  lambda_3 = 0.7179  (4.9% of total)

```

```

First principal component (max-variance direction):
  q1 = [-0.9208, -0.3807, -0.0854]

```

Variance of projections onto q1: 10.4821
lambda_1: 10.4821

Geometric Picture

```
import numpy as np
import matplotlib.pyplot as plt

rng = np.random.default_rng(2011)

# 2-D dataset with clear principal directions
Sigma2 = np.array([[4.0, 1.8], [1.8, 1.2]]) # shape (2, 2)
L2 = np.linalg.cholesky(Sigma2)
X2 = rng.standard_normal((150, 2)) @ L2.T # shape (150, 2) -- already centered

lam2, Q2 = np.linalg.eigh(Sigma2)
idx2 = np.argsort(lam2)[::-1]
lam2 = lam2[idx2]; Q2 = Q2[:, idx2]

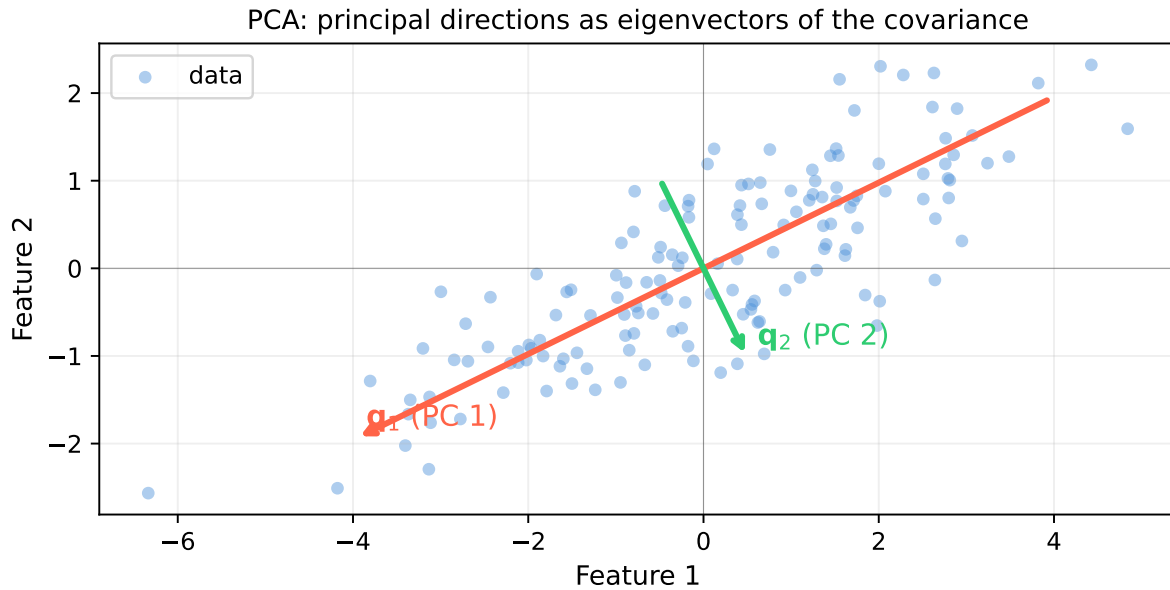
fig, ax = plt.subplots(figsize=(7, 6))
ax.scatter(X2[:, 0], X2[:, 1], s=18, color='#4a90d9', alpha=0.45, label='data')

scale = 2.0
for i, (col, lbl) in enumerate(zip(['tomato', '#2ecc71'],
                                   ['$\\mathbf{q}_1$ (PC 1)', '$\\mathbf{q}_2$ (PC 2)')):
    v = Q2[:, i]
    ax.annotate('', xy=scale * np.sqrt(lam2[i]) * v,
                xytext=-scale * np.sqrt(lam2[i]) * v,
                arrowprops=dict(arrowstyle='->', color=col, lw=2.5))
    ax.text(*(scale * np.sqrt(lam2[i]) * v + np.array([0.12, 0.12])),
            lbl, fontsize=12, color=col)

ax.axhline(0, color='#333333', lw=0.4, alpha=0.5)
ax.axvline(0, color='#333333', lw=0.4, alpha=0.5)
ax.set_xlabel('Feature 1', fontsize=11)
ax.set_ylabel('Feature 2', fontsize=11)
ax.set_title('PCA: principal directions as eigenvectors of the covariance', fontsize=11)
ax.set_aspect('equal')
ax.grid(alpha=0.2)
ax.legend(fontsize=10)
fig.tight_layout()
plt.savefig('ch20-pca/fig-max-variance.png', dpi=150, bbox_inches='tight')
plt.show()
```



```
print(f"lambda_1 = {lam2[0]:.4f}, lambda_2 = {lam2[1]:.4f}")
print(f"Arrow lengths proportional to sqrt(lambda_i) = [{np.sqrt(lam2[0]):.4f}, {np.sqrt(la
```



```
lambda_1 = 4.8804, lambda_2 = 0.3196
Arrow lengths proportional to sqrt(lambda_i) = [2.2092, 0.5654]
```

The arrow lengths are proportional to $\sqrt{\lambda_i}$, the **standard deviation** of projections in each direction. PC 1 (red) spans the longest axis of the data ellipse; PC 2 (green) is orthogonal and spans the shorter axis.

Exercises

20.1.1 Let $\tilde{X}$ be a centered $n \times p$ data matrix and let $\mathbf{u} \in \mathbb{R}^p$ be a unit vector. Show step by step that the sample variance of the scalar projections $z_i = \mathbf{u}^T \tilde{\mathbf{x}}_i$ equals $\mathbf{u}^T \Sigma \mathbf{u}$, where $\Sigma = \frac{1}{n-1} \tilde{X}^T \tilde{X}$.

20.1.2 Why must we constrain $\|\mathbf{u}\| = 1$ in the PCA optimization? What goes wrong if you maximize $\mathbf{u}^T \Sigma \mathbf{u}$ without the constraint?

20.1.3 Carry out the Lagrange-multiplier argument step by step to show that a critical point of $\mathbf{u}^T \Sigma \mathbf{u} - \lambda(\mathbf{u}^T \mathbf{u} - 1)$ satisfies $\Sigma \mathbf{u} = \lambda \mathbf{u}$, and that the variance at the critical point equals λ .

20.1.4 Suppose $\lambda_1 = \lambda_2 > \lambda_3$ (the two largest eigenvalues are equal). What does this imply about the set of maximum-variance unit vectors? Is the first principal component unique?

PCA via SVD

Computing PCA through the eigendecomposition of Σ works, but it introduces an unnecessary step: forming $\Sigma = \frac{1}{n-1} \tilde{X}^T \tilde{X}$ squares the condition number of $\tilde{X}$. The **thin SVD** of the data matrix itself is the numerically preferred route.

The Thin SVD of the Centered Data Matrix

Write the economy (thin) SVD of $\tilde{X} \in \mathbb{R}^{n \times p}$:

$$\tilde{X} = USV^T,$$

where $U \in \mathbb{R}^{n \times p}$ has orthonormal columns, $S = \text{diag}(\sigma_1, \dots, \sigma_p)$ with $\sigma_1 \geq \dots \geq \sigma_p \geq 0$, and $V \in \mathbb{R}^{p \times p}$ is orthogonal.

Now compute the sample covariance:

$$\Sigma = \frac{1}{n-1} \tilde{X}^T \tilde{X} = \frac{1}{n-1} (USV^T)^T (USV^T) = \frac{1}{n-1} V S^2 V^T.$$

Comparing with the spectral decomposition $\Sigma = Q\Lambda Q^T$:

$$\mathbf{q}_j = \mathbf{v}_j, \quad \lambda_j = \frac{\sigma_j^2}{n-1}.$$

The **principal directions** are the **right singular vectors** of $\tilde{X}$; the **eigenvalues** of Σ are the **squared singular values** divided by $n-1$.

PC Scores

The **k -dimensional PC scores** are the coordinates of each observation in the principal subspace:

$$Z_k = \tilde{X} V_k = U_k S_k \in \mathbb{R}^{n \times k},$$

where V_k is the first k columns of V and U_k, S_k are the corresponding blocks of the thin SVD. Row i of Z_k is the k -vector of projections of observation $\tilde{\mathbf{x}}_i$ onto the top- k principal directions.

Why SVD Is Numerically Superior

Two reasons:

1. **Condition number.** $\kappa(\tilde{X}) = \sigma_1/\sigma_p$. Forming $\Sigma = \frac{1}{n-1}\tilde{X}^T\tilde{X}$ squares the condition number to $\kappa(\Sigma) = (\sigma_1/\sigma_p)^2$. For ill-conditioned data the eigendecomposition of Σ loses about twice as many digits as the SVD of $\tilde{X}$.
2. **No matrix formation needed.** Randomized SVD (§18.6) can produce the top- k singular vectors of $\tilde{X}$ in $O(nkp)$ time without ever materializing Σ . Eigendecomposing Σ always requires forming the $p \times p$ matrix first.

Numerical Example

```
import numpy as np

rng = np.random.default_rng(202)

# Simulate n=300 observations, p=4 features
# True covariance has one dominant direction
Sigma_true = np.array([[8.0, 2.0, 1.5, 0.5],
                       [2.0, 3.0, 0.8, 0.3],
                       [1.5, 0.8, 2.0, 0.4],
                       [0.5, 0.3, 0.4, 0.8]]) # shape (4, 4)
mu_true = np.array([2.0, -1.0, 3.0, 0.5]) # shape (4,)

L = np.linalg.cholesky(Sigma_true) # shape (4, 4)
X = rng.standard_normal((300, 4)) @ L.T + mu_true # shape (300, 4)

# Mean-center
x_bar = X.mean(axis=0) # shape (4,)
X_tilde = X - x_bar # shape (300, 4)
n = X_tilde.shape[0]

# --- Route 1: eigendecomposition of Sigma ---
Sigma_hat = X_tilde.T @ X_tilde / (n - 1) # shape (4, 4)
lam_eig, Q_eig = np.linalg.eigh(Sigma_hat)
idx = np.argsort(lam_eig)[::-1]
lam_eig = lam_eig[idx]
Q_eig = Q_eig[:, idx] # shape (4, 4)

# --- Route 2: thin SVD of X_tilde ---
U, s, Vt = np.linalg.svd(X_tilde, full_matrices=False)
# U: (300, 4), s: (4,), Vt: (4, 4)
```

```

lam_svd = s**2 / (n - 1)          # shape (4,)
V = Vt.T                         # shape (4, 4)

print("Eigenvalues via eigh vs SVD:")
for j in range(4):
    print(f"  lambda_{j+1}: eigh = {lam_eig[j]:.6f},  SVD = {lam_svd[j]:.6f}")

# PC directions agree up to sign
print("\nMax |diff| between principal directions (accounting for sign):")
for j in range(4):
    d = min(np.max(np.abs(Q_eig[:, j] - V[:, j])),
            np.max(np.abs(Q_eig[:, j] + V[:, j])))
    print(f"  q_{j+1}: {d:.2e}")

# PC scores via SVD
Z2 = X_tilde @ V[:, :2]          # shape (300, 2) -- top 2 PCs
print(f"\nPC scores Z2: shape {Z2.shape}")
print(f"  Var(PC1) = {Z2[:, 0].var(ddof=1):.4f}  (should match lambda_1 = {lam_svd[0]:.4f})")
print(f"  Var(PC2) = {Z2[:, 1].var(ddof=1):.4f}  (should match lambda_2 = {lam_svd[1]:.4f})")

```

Eigenvalues via eigh vs SVD:

```

lambda_1: eigh = 10.065265,  SVD = 10.065265
lambda_2: eigh = 2.702712,  SVD = 2.702712
lambda_3: eigh = 1.644119,  SVD = 1.644119
lambda_4: eigh = 0.572248,  SVD = 0.572248

```

Max |diff| between principal directions (accounting for sign):

```

q_1: 2.22e-16
q_2: 5.69e-16
q_3: 9.83e-15
q_4: 9.21e-15

```

PC scores Z2: shape (300, 2)

```

Var(PC1) = 10.0653  (should match lambda_1 = 10.0653)
Var(PC2) = 2.7027  (should match lambda_2 = 2.7027)

```

Geometric View

```

import numpy as np
import matplotlib.pyplot as plt

rng = np.random.default_rng(2021)

```

```

# 2-D dataset for visualization
Sigma2 = np.array([[5.0, 2.2], [2.2, 2.0]])      # shape (2, 2)
L2 = np.linalg.cholesky(Sigma2)
X2 = rng.standard_normal((200, 2)) @ L2.T        # shape (200, 2)  -- centered

U2, s2, Vt2 = np.linalg.svd(X2, full_matrices=False)
# U2: (200, 2), s2: (2,), Vt2: (2, 2)
V2 = Vt2.T                                       # shape (2, 2)

# PC scores
Z2 = X2 @ V2                                    # shape (200, 2)

fig, axes = plt.subplots(1, 2, figsize=(12, 5))

ax = axes[0]
ax.scatter(X2[:, 0], X2[:, 1], s=15, color='#4a90d9', alpha=0.4, label='data')
scale = 2.5
for j, (col, lbl) in enumerate(zip(['tomato', '#2ecc71'],
                                   ['$\\mathbf{v}_1$ (PC 1)', '$\\mathbf{v}_2$ (PC 2)')):
    v = V2[:, j]
    length = scale * s2[j] / np.sqrt(len(X2) - 1)
    ax.annotate('', xy=length * v, xytext=-length * v,
                arrowprops=dict(arrowstyle='->', color=col, lw=2.5))
    ax.text(*(length * v + np.array([0.1, 0.1])), lbl, fontsize=11, color=col)
ax.axhline(0, color='#333333', lw=0.4, alpha=0.5)
ax.axvline(0, color='#333333', lw=0.4, alpha=0.5)
ax.set_aspect('equal')
ax.set_title('Original feature space', fontsize=11)
ax.set_xlabel('Feature 1', fontsize=10)
ax.set_ylabel('Feature 2', fontsize=10)
ax.grid(alpha=0.2)

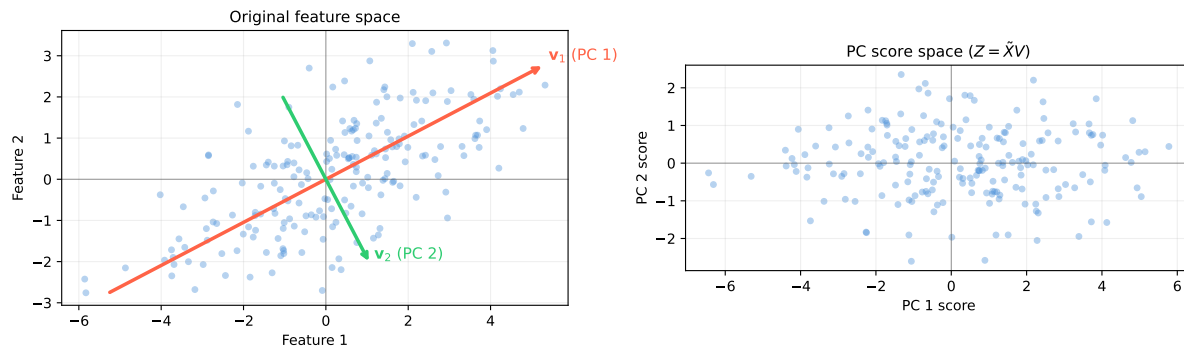
ax2 = axes[1]
ax2.scatter(Z2[:, 0], Z2[:, 1], s=15, color='#4a90d9', alpha=0.4)
ax2.axhline(0, color='#333333', lw=0.4, alpha=0.5)
ax2.axvline(0, color='#333333', lw=0.4, alpha=0.5)
ax2.set_aspect('equal')
ax2.set_title('PC score space ($Z = \\tilde{X}V$)', fontsize=11)
ax2.set_xlabel('PC 1 score', fontsize=10)
ax2.set_ylabel('PC 2 score', fontsize=10)
ax2.grid(alpha=0.2)

fig.suptitle('PCA via SVD: original space vs PC score space', fontsize=12)
fig.tight_layout()
plt.savefig('ch20-pca/fig-pca-svd.png', dpi=150, bbox_inches='tight')
plt.show()

```

```
print(f"Singular values: sigma_1 = {s2[0]:.4f}, sigma_2 = {s2[1]:.4f}")
print(f"Eigenvalues: lambda_1 = {s2[0]**2/(len(X2)-1):.4f}, lambda_2 = {s2[1]**2/(len(X2)-1):.4f}")
```

PCA via SVD: original space vs PC score space



Singular values: $\sigma_1 = 33.8125$, $\sigma_2 = 13.0590$

Eigenvalues: $\lambda_1 = 5.7452$, $\lambda_2 = 0.8570$

The PC score space is a rotation of the original feature space: the axes align with the principal directions $\mathbf{v}_1, \mathbf{v}_2$, and the data cloud is axis-aligned. PC1 captures the long axis; PC2 the short axis.

Exercises

20.2.1 Starting from $\tilde{X} = USV^T$, show algebraically that $\frac{1}{n-1} \tilde{X}^T \tilde{X} = V \frac{S^2}{n-1} V^T$. Identify which matrix plays the role of Q and which plays Λ in the spectral decomposition.

20.2.2 The PC scores can be written as either $Z = \tilde{X}V$ or $Z = US$. Verify numerically that these two expressions give the same matrix (up to floating-point precision).

20.2.3 If you scale one feature by a factor of 10 (multiply column j of $\tilde{X}$ by 10), how does this affect (a) the singular values of $\tilde{X}$, (b) the first principal direction $\mathbf{v}_1$?

20.2.4 Explain why `np.linalg.svd(X_tilde, full_matrices=False)` is preferred over `np.linalg.svd(X_tilde, full_matrices=True)` for PCA when $n \gg p$. What would the shapes of U, S, V^T be in each case?

Explained Variance and Choosing k

After computing all principal components, the natural question is: **how many do we actually need?** The eigenvalues of Σ answer this directly – each λ_j is the variance captured by PC j .

Explained Variance Ratio

The **total variance** of the data equals the trace of the covariance matrix:

$$\text{Total variance} = \text{tr}(\Sigma) = \sum_{j=1}^p \lambda_j = \frac{1}{n-1} \sum_{j=1}^p \sigma_j^2.$$

The **explained variance ratio** for component j is

$$r_j = \frac{\lambda_j}{\sum_{i=1}^p \lambda_i} = \frac{\sigma_j^2}{\sum_{i=1}^p \sigma_i^2}.$$

The **cumulative explained variance ratio** for the top k components is

$$R_k = \sum_{j=1}^k r_j.$$

When $R_k \geq 0.90$, the k -dimensional PCA subspace retains at least 90% of the total variance.

Scree Plot

A **scree plot** displays λ_j (or r_j) against j . Look for an “elbow” – a point where the eigenvalues stop dropping steeply and level off. Components beyond the elbow describe noise rather than signal.

When to distrust the elbow. If the eigenvalue spectrum decays slowly and smoothly (no clear break), the data may lack a low-dimensional structure. The 90% rule then requires many components and PCA provides little compression.

Numerical Example

```
import numpy as np

rng = np.random.default_rng(203)

# Simulate n=500 observations, p=8 features
# True signal lives in 3-D; remaining 5 dimensions are noise
n, p = 500, 8
# Build true covariance: 3 large eigenvalues, 5 near-zero
```

```

lam_true = np.array([12.0, 6.5, 3.0, 0.4, 0.3, 0.25, 0.2, 0.15])
Q_true, _ = np.linalg.qr(rng.standard_normal((p, p))) # shape (8, 8)
Sigma_true = Q_true @ np.diag(lam_true) @ Q_true.T      # shape (8, 8)

L_true = np.linalg.cholesky(Sigma_true)                # shape (8, 8)
X = rng.standard_normal((n, p)) @ L_true.T             # shape (500, 8)

# Mean-center
X_tilde = X - X.mean(axis=0)                           # shape (500, 8)

# Thin SVD
_, s, _ = np.linalg.svd(X_tilde, full_matrices=False)  # s shape (8,)
lam_hat = s**2 / (n - 1)                                # shape (8,)

evr = lam_hat / lam_hat.sum()                           # explained variance ratio
cum_evr = np.cumsum(evr)                                # cumulative

print(f"{'PC':>4} {'lambda':>8} {'EVR':>8} {'Cumulative EVR':>15}")
print("-" * 44)
for j in range(p):
    print(f"{j+1:>4} {lam_hat[j]:8.4f} {evr[j]:8.4f} {cum_evr[j]:15.4f}")

k90 = int(np.searchsorted(cum_evr, 0.90)) + 1
print(f"\n90% threshold reached at k = {k90} components")

```

PC	lambda	EVR	Cumulative EVR

1	10.8936	0.5148	0.5148
2	6.3801	0.3015	0.8163
3	2.6333	0.1244	0.9407
4	0.4221	0.0199	0.9607
5	0.2907	0.0137	0.9744
6	0.2424	0.0115	0.9859
7	0.1802	0.0085	0.9944
8	0.1184	0.0056	1.0000

90% threshold reached at k = 3 components

Scree Plot and Cumulative EVR

```

import numpy as np
import matplotlib.pyplot as plt

```



```

rng = np.random.default_rng(203)

n, p = 500, 8
lam_true = np.array([12.0, 6.5, 3.0, 0.4, 0.3, 0.25, 0.2, 0.15])
Q_true, _ = np.linalg.qr(rng.standard_normal((p, p)))
Sigma_true = Q_true @ np.diag(lam_true) @ Q_true.T
L_true = np.linalg.cholesky(Sigma_true)
X = rng.standard_normal((n, p)) @ L_true.T
X_tilde = X - X.mean(axis=0)

_, s, _ = np.linalg.svd(X_tilde, full_matrices=False)
lam_hat = s**2 / (n - 1)
evr = lam_hat / lam_hat.sum()
cum_evr = np.cumsum(evr)

pcs = np.arange(1, p + 1)

fig, axes = plt.subplots(1, 2, figsize=(12, 5))

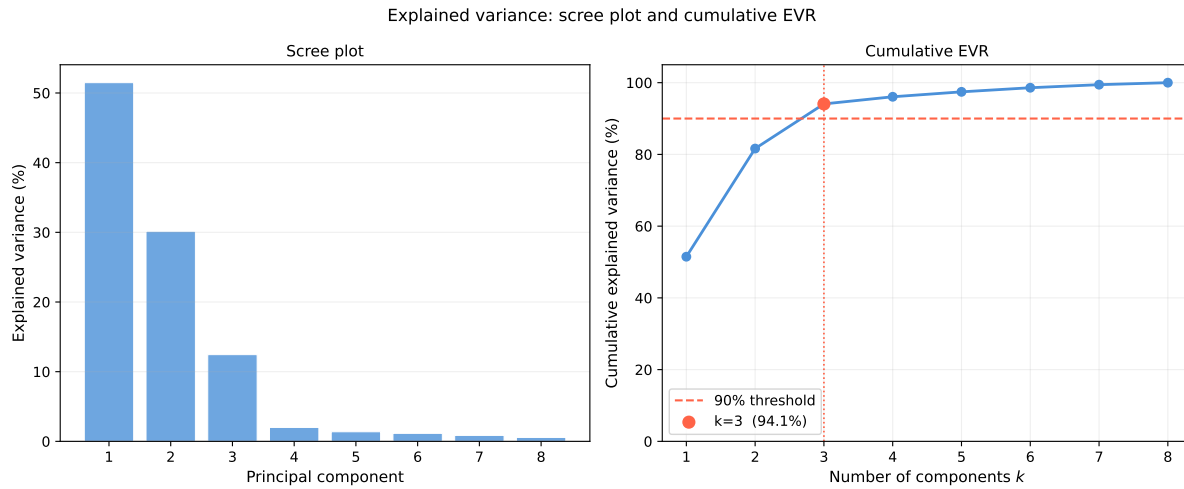
# Left: scree plot (individual EVR)
ax = axes[0]
ax.bar(pcs, evr * 100, color='#4a90d9', alpha=0.8, edgecolor='white', linewidth=0.5)
ax.set_xlabel('Principal component', fontsize=11)
ax.set_ylabel('Explained variance (%)', fontsize=11)
ax.set_title('Scree plot', fontsize=11)
ax.set_xticks(pcs)
ax.grid(alpha=0.2, axis='y')

# Right: cumulative EVR
ax2 = axes[1]
ax2.plot(pcs, cum_evr * 100, color='#4a90d9', marker='o', markersize=6, linewidth=2)
ax2.axhline(90, color='tomato', lw=1.5, linestyle='--', label='90% threshold')
k90 = int(np.searchsorted(cum_evr, 0.90)) + 1
ax2.axvline(k90, color='tomato', lw=1.2, linestyle=':')
ax2.scatter([k90], [cum_evr[k90-1] * 100], s=70, color='tomato', zorder=5,
            label=f'k={k90} ({100*cum_evr[k90-1]:.1f}%)')
ax2.set_xlabel('Number of components $k$', fontsize=11)
ax2.set_ylabel('Cumulative explained variance (%)', fontsize=11)
ax2.set_title('Cumulative EVR', fontsize=11)
ax2.set_xticks(pcs)
ax2.set_ylim(0, 105)
ax2.legend(fontsize=10)
ax2.grid(alpha=0.2)

fig.suptitle('Explained variance: scree plot and cumulative EVR', fontsize=12)
fig.tight_layout()
plt.savefig('ch20-pca/fig-explained-variance.png', dpi=150, bbox_inches='tight')

```

```
plt.show()
```



The elbow at PC 3 is visible in the scree plot: eigenvalues 1-3 are large, then the spectrum collapses. The cumulative EVR confirms that $k = 3$ captures over 90% of the total variance.

When the Elbow Fails

If the true data dimension is not much smaller than p , or if the signal-to-noise ratio is low, the scree plot shows a gradual descent with no clear break. In that case:

- **Cross-validation:** hold out rows, reconstruct from k components, measure reconstruction error as a function of k .
- **Parallel analysis:** compare observed eigenvalues against eigenvalues from random data with the same n and p ; keep components whose eigenvalue exceeds the random baseline.
- **Application-specific threshold:** in robotics localization, retain enough components for sub-centimeter reconstruction error rather than applying a blanket 90% rule.

Exercises

20.3.1 For a $p \times p$ identity covariance matrix $\Sigma = I_p$, what is the explained variance ratio of each principal component? What does the scree plot look like? What does this say about the compressibility of white-noise data?

20.3.2 Show that $\text{tr}(\Sigma) = \sum_{j=1}^p \lambda_j$ using the spectral decomposition $\Sigma = Q\Lambda Q^T$. (*Hint:* use the cyclic property of trace.)

20.3.3 A dataset has 10 features. After PCA the singular values are $\sigma_1 = 100, \sigma_2 = 20, \sigma_3 = \dots = \sigma_{10} = 5$. Compute the explained variance ratios and determine how many components are needed to explain 95% of the variance.

20.3.4 Suppose you standardize each feature to unit variance before running PCA. How does this change the scree plot compared to running PCA on the raw (unscaled) data? Give an example where the two yield qualitatively different results.

Reconstruction and Residuals

PCA is a lossy compression: projecting to k dimensions discards information. Knowing exactly how much is discarded – and what it looks like – is essential for choosing k and understanding what the model cannot explain.

Projection and Reconstruction

Let $V_k = [\mathbf{v}_1 \mid \dots \mid \mathbf{v}_k] \in \mathbb{R}^{p \times k}$ be the matrix of the top- k principal directions.

PC scores (projection onto the k -dimensional subspace):

$$Z_k = \tilde{X}V_k \in \mathbb{R}^{n \times k}.$$

Reconstruction (projection back to the original p -dimensional space):

$$\hat{X}_k = Z_k V_k^T = \tilde{X}V_k V_k^T \in \mathbb{R}^{n \times p}.$$

The matrix $P_k = V_k V_k^T \in \mathbb{R}^{p \times p}$ is an **orthogonal projection** onto $\text{span}\{\mathbf{v}_1, \dots, \mathbf{v}_k\}$: it is symmetric and idempotent ($P_k^2 = P_k$).

Reconstruction Error

The residual matrix is $E_k = \tilde{X} - \hat{X}_k = \tilde{X}(I - P_k)$. Its squared Frobenius norm is

$$\|\tilde{X} - \hat{X}_k\|_F^2 = \sum_{j=k+1}^p \sigma_j^2,$$

the sum of the **tail squared singular values**. Equivalently, in terms of covariance eigenvalues:

$$\|\tilde{X} - \hat{X}_k\|_F^2 = (n-1) \sum_{j=k+1}^p \lambda_j.$$

This is exactly the result promised by the Eckart-Young theorem (§18.3): the rank- k truncated SVD gives the best rank- k approximation to $\tilde{X}$ in the Frobenius norm.

Numerical Example

```
import numpy as np

rng = np.random.default_rng(204)

# n=300 observations, p=5 features; true rank-2 structure plus noise
n, p = 300, 5
V_true = np.linalg.qr(rng.standard_normal((p, p)))[0][:, :2] # shape (5, 2)
scores_true = rng.standard_normal((n, 2)) * np.array([3.0, 1.5]) # shape (300, 2)
X_signal = scores_true @ V_true.T # shape (300, 5)
X_noise = rng.standard_normal((n, p)) * 0.4 # shape (300, 5)
X = X_signal + X_noise
X_tilde = X - X.mean(axis=0) # shape (300, 5)

# Thin SVD
U, s, Vt = np.linalg.svd(X_tilde, full_matrices=False)
# U: (300, 5), s: (5,), Vt: (5, 5)
V = Vt.T # shape (5, 5)

print(f"Singular values: {s.round(4)}")
print(f"Total ||X_tilde||_F^2 = {(s**2).sum():.4f}\n")

for k in range(1, p + 1):
    Vk = V[:, :k] # shape (5, k)
    Pk = Vk @ Vk.T # shape (5, 5)
    X_hat = X_tilde @ Pk # shape (300, 5)
    err_sq = np.linalg.norm(X_tilde - X_hat, 'fro')**2
    err_theory = (s[k:]**2).sum()
    evr_k = (s[:k]**2).sum() / (s**2).sum()
    print(f"k={k}: ||E_k||_F^2 = {err_sq:.4f} (theory: {err_theory:.4f})"
          f" EVR = {100*evr_k:.1f}%")
```

```
Singular values: [53.3908 26.7632  6.8861  6.6799  6.4969]
Total ||X_tilde||_F^2 = 3701.0906
```

```
k=1: ||E_k||_F^2 = 850.5158 (theory: 850.5158) EVR = 77.0%
k=2: ||E_k||_F^2 = 134.2484 (theory: 134.2484) EVR = 96.4%
k=3: ||E_k||_F^2 = 86.8305 (theory: 86.8305) EVR = 97.7%
k=4: ||E_k||_F^2 = 42.2093 (theory: 42.2093) EVR = 98.9%
k=5: ||E_k||_F^2 = 0.0000 (theory: 0.0000) EVR = 100.0%
```

Visualization: Original vs Reconstructed

```
import numpy as np
import matplotlib.pyplot as plt

rng = np.random.default_rng(2041)

# 2-D dataset: clear principal axis
Sigma2 = np.array([[4.0, 1.8], [1.8, 1.0]])      # shape (2, 2)
L2 = np.linalg.cholesky(Sigma2)
X2 = rng.standard_normal((150, 2)) @ L2.T      # shape (150, 2)

U2, s2, Vt2 = np.linalg.svd(X2, full_matrices=False)
V2 = Vt2.T                                     # shape (2, 2)

fig, axes = plt.subplots(1, 3, figsize=(15, 4.5))

k_list = [1, 2]
titles = ['Original  $\tilde{X}$ ',
          'Reconstruction  $\hat{X}_1$  (k=1)',
          'Reconstruction  $\hat{X}_2$  (k=2)']
datasets = [X2,
            X2 @ V2[:, :1] @ V2[:, :1].T,
            X2 @ V2[:, :2] @ V2[:, :2].T]
colors = ['#4a90d9', 'tomato', '#2ecc71']

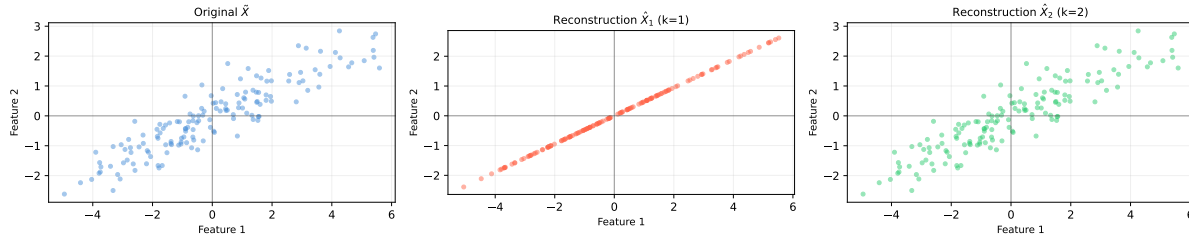
for ax, data, title, col in zip(axes, datasets, titles, colors):
    ax.scatter(data[:, 0], data[:, 1], s=12, color=col, alpha=0.5)
    ax.axhline(0, color='#333333', lw=0.4, alpha=0.5)
    ax.axvline(0, color='#333333', lw=0.4, alpha=0.5)
    ax.set_aspect('equal')
    ax.set_title(title, fontsize=10)
    ax.grid(alpha=0.2)
    ax.set_xlabel('Feature 1', fontsize=9)
    ax.set_ylabel('Feature 2', fontsize=9)

err1 = np.linalg.norm(X2 - X2 @ V2[:, :1] @ V2[:, :1].T, 'fro')**2
err2 = np.linalg.norm(X2 - X2 @ V2 @ V2.T, 'fro')**2
fig.suptitle(
    f'Rank-1 reconstruction error = {err1:.1f} | '
    f'Rank-2 reconstruction error = {err2:.2f}',
    fontsize=11)
fig.tight_layout()
plt.savefig('ch20-pca/fig-reconstruction.png', dpi=150, bbox_inches='tight')
```

```
plt.show()
```

```
print(f"Singular values:  sigma_1 = {s2[0]:.4f},  sigma_2 = {s2[1]:.4f}")
print(f"Theoretical rank-1 error: sigma_2^2 = {s2[1]**2:.4f}")
```

Rank-1 reconstruction error = 29.3 | Rank-2 reconstruction error = 0.00



Singular values: sigma_1 = 31.9409, sigma_2 = 5.4099

Theoretical rank-1 error: sigma_2^2 = 29.2671

The rank-1 reconstruction (middle) collapses all data onto the first principal axis; the rank-2 reconstruction (right) is exact because $p = 2$. The residuals of the rank-1 case are exactly the second-component contributions.

Exercises

20.4.1 Verify that $P_k = V_k V_k^T$ is idempotent: show $P_k^2 = P_k$. Also show that $I - P_k$ is itself an orthogonal projection, and onto what subspace.

20.4.2 Using the SVD $\tilde{X} = USV^T$, write the reconstruction $\hat{X}_k$ in terms of U , S , V^T , and show that $\|\tilde{X} - \hat{X}_k\|_F^2 = \sum_{j=k+1}^p \sigma_j^2$.

20.4.3 A centered data matrix has $\|\tilde{X}\|_F^2 = 1200$ and the top-3 singular values are $\sigma_1 = 25$, $\sigma_2 = 15$, $\sigma_3 = 10$. What fraction of the total variance is lost when projecting to $k = 3$?

20.4.4 The rows of the residual matrix $E_k = \tilde{X}(I - P_k)$ are the components of each observation **not** captured by the top- k subspace. If a test observation has an unusually large residual $\|E_{k,i}\|$, what might this indicate about that observation in the context of anomaly detection?

PCA for 3D Point Clouds

A **point cloud** is a set of 3D coordinates $\{\mathbf{p}_i\}_{i=1}^n \subset \mathbb{R}^3$ sampled from a surface – output of a LiDAR, depth camera, or structured-light scanner. PCA on a point cloud reveals the dominant geometric directions: the axes of the data ellipsoid, the normal to a fitted plane, and the orientation of an enclosing box.

Fitting a Plane

Given n points, the best-fit plane (least-squares sense) passes through the centroid $\bar{\mathbf{p}} = \frac{1}{n} \sum_i \mathbf{p}_i$ and is orthogonal to the **smallest** eigenvector of the 3×3 covariance matrix Σ .

Why the smallest eigenvector? The plane minimizes the sum of squared distances from the points to the plane. That distance is the projection onto the plane normal $\mathbf{n}$, so minimizing it is equivalent to minimizing $\mathbf{n}^T \Sigma \mathbf{n}$ – choosing the direction of *least* variance.

Plane normal = $\mathbf{q}_3$ (eigenvector of smallest λ).

Similarly, the **longest axis** of the point cloud is $\mathbf{q}_1$ (largest λ).

Oriented Bounding Box

A bounding box aligned with the principal axes is tighter than an axis-aligned box when the object is elongated or rotated.

Steps:

1. Compute centroid $\bar{\mathbf{p}}$ and center the cloud: $\tilde{P} = P - \mathbf{1}\bar{\mathbf{p}}^T$.
 2. Run PCA: get $Q = [\mathbf{q}_1 \mid \mathbf{q}_2 \mid \mathbf{q}_3]$.
 3. Rotate the cloud: $R = \tilde{P}Q$ (coordinates in the PC frame).
 4. Axis-aligned bounding box of R : $[\min_i R_{ij}, \max_i R_{ij}]$ for $j = 1, 2, 3$.
 5. Transform box corners back: corner $\mathbf{c}$ in original frame is $Q\mathbf{c} + \bar{\mathbf{p}}$.
-

Surface Normals via Local PCA

For each point $\mathbf{p}_i$, find its m nearest neighbors, center them, run PCA: the normal at $\mathbf{p}_i$ is the smallest eigenvector of the local 3×3 covariance. The orientation (sign) of the normal is disambiguated by enforcing consistency with the viewpoint direction.

Numerical Example

```

import numpy as np

rng = np.random.default_rng(205)

# Simulate a planar point cloud with noise
# True plane: z = 0.3*x + 0.1*y + 0 (approximately)
n_pts = 400
xy = rng.uniform(-3, 3, (n_pts, 2))          # shape (400, 2)
z = 0.3 * xy[:, 0] + 0.1 * xy[:, 1] \
    + rng.standard_normal(n_pts) * 0.12      # shape (400,)
P = np.column_stack([xy, z])                 # shape (400, 3)

# PCA on the point cloud
p_bar = P.mean(axis=0)                       # shape (3,)
P_c = P - p_bar                             # shape (400, 3)
Sigma = P_c.T @ P_c / (n_pts - 1)           # shape (3, 3)

lam, Q = np.linalg.eigh(Sigma)               # ascending order
# Smallest eigenvector = plane normal
n_hat = Q[:, 0]                             # shape (3,)
if n_hat[2] < 0:
    n_hat = -n_hat                          # orient toward +z

print("Estimated plane normal:")
print(f"  n_hat = [{n_hat[0]:.4f}, {n_hat[1]:.4f}, {n_hat[2]:.4f}]")

# True normal: plane z = 0.3x + 0.1y, so [0.3, 0.1, -1] normalized
n_true = np.array([0.3, 0.1, -1.0])
n_true = n_true / np.linalg.norm(n_true)
if n_true[2] < 0:
    n_true = -n_true
print(f"True plane normal:")
print(f"  n_true = [{n_true[0]:.4f}, {n_true[1]:.4f}, {n_true[2]:.4f}]")
print(f"\nAngular error: {np.degrees(np.arccos(np.clip(np.abs(n_hat @ n_true), 0, 1))):.3f} deg")

print(f"\nEigenvalues (ascending): {lam.round(4)}")
print(f"Smallest eigenvalue (normal direction variance): {lam[0]:.4f}")
print(f"Largest eigenvalue (dominant axis variance): {lam[2]:.4f}")

```

Estimated plane normal:

n_hat = [-0.2880, -0.0958, 0.9528]

True plane normal:

n_true = [-0.2860, -0.0953, 0.9535]

Angular error: 0.120 degrees

Eigenvalues (ascending): [0.0138 3.1348 3.2716]

Smallest eigenvalue (normal direction variance): 0.0138
 Largest eigenvalue (dominant axis variance): 3.2716

Visualization

```
import numpy as np
import matplotlib.pyplot as plt

rng = np.random.default_rng(2051)

# Flat-ish cloud with a prominent elongation direction
n_pts = 300
# Elongated along [1, 0.5, 0] direction, thin in z
A = np.array([[3.0, 0.5, 0.1],
               [0.5, 1.2, 0.0],
               [0.1, 0.0, 0.15]])          # shape (3, 3)
L_A = np.linalg.cholesky(A)
P = rng.standard_normal((n_pts, 3)) @ L_A.T  # shape (300, 3)

p_bar = P.mean(axis=0)                      # shape (3,)
P_c = P - p_bar                             # shape (300, 3)
Sigma = P_c.T @ P_c / (n_pts - 1)          # shape (3, 3)
lam, Q = np.linalg.eigh(Sigma)              # ascending
lam = lam[::-1]; Q = Q[:, ::-1]            # descending

# Oriented bounding box in PC frame
R = P_c @ Q                                # shape (300, 3) -- PC coordinates
lo = R.min(axis=0)                          # shape (3,)
hi = R.max(axis=0)                          # shape (3,)

# Project to x-y plane for visualization
fig, axes = plt.subplots(1, 2, figsize=(12, 5))

# Left: original axes (x-y projection)
ax = axes[0]
ax.scatter(P[:, 0], P[:, 1], s=8, color='#4a90d9', alpha=0.5, label='points')
ax.axhline(p_bar[1], color='#333333', lw=0.4, alpha=0.5)
ax.axvline(p_bar[0], color='#333333', lw=0.4, alpha=0.5)
scale = 2.5
for j, col in enumerate(['tomato', '#2ecc71']):
    v = Q[:, 2, j]                          # x-y components of eigenvector
    length = scale * np.sqrt(lam[j])
    ax.annotate('', xy=p_bar[:2] + length * v,
                xytext=p_bar[:2] - length * v,
```

```

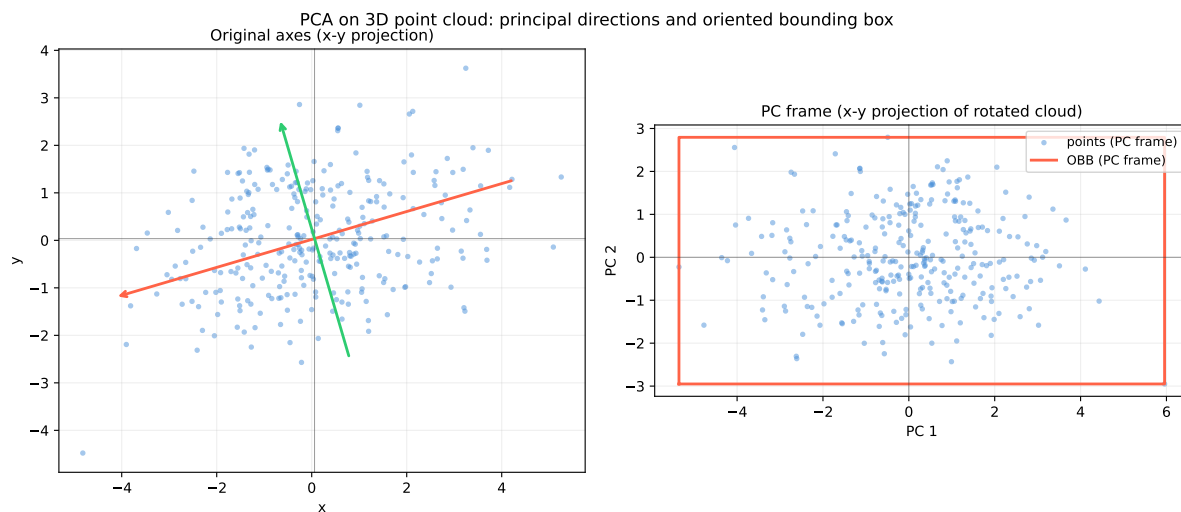
        arrowprops=dict(arrowstyle='->', color=col, lw=2.0))
ax.set_aspect('equal')
ax.set_title('Original axes (x-y projection)', fontsize=11)
ax.set_xlabel('x', fontsize=10); ax.set_ylabel('y', fontsize=10)
ax.grid(alpha=0.2)

# Right: PC frame (x-y projection of rotated cloud + bounding box)
ax2 = axes[1]
ax2.scatter(R[:, 0], R[:, 1], s=8, color='#4a90d9', alpha=0.5, label='points (PC frame)')
# Draw bounding box edges in PC frame (x-y face)
rect_x = [lo[0], hi[0], hi[0], lo[0], lo[0]]
rect_y = [lo[1], lo[1], hi[1], hi[1], lo[1]]
ax2.plot(rect_x, rect_y, color='tomato', lw=2, label='OBB (PC frame)')
ax2.axhline(0, color='#333333', lw=0.4, alpha=0.5)
ax2.axvline(0, color='#333333', lw=0.4, alpha=0.5)
ax2.set_aspect('equal')
ax2.set_title('PC frame (x-y projection of rotated cloud)', fontsize=11)
ax2.set_xlabel('PC 1', fontsize=10); ax2.set_ylabel('PC 2', fontsize=10)
ax2.legend(fontsize=9)
ax2.grid(alpha=0.2)

fig.suptitle('PCA on 3D point cloud: principal directions and oriented bounding box', fontsize=12)
fig.tight_layout()
plt.savefig('ch20-pca/fig-point-clouds.png', dpi=150, bbox_inches='tight')
plt.show()

print(f"Eigenvalues: {lam.round(4)}")
print(f"Box half-widths (PC frame): {(hi - lo)/2}.round(4)}")

```



Eigenvalues: [3.0672 1.1108 0.1488]

Box half-widths (PC frame): [5.6551 2.8723 1.0159]

Surface Normals via Local PCA

```
import numpy as np

rng = np.random.default_rng(2052)

# Simulate points on a bumpy surface  $z = \sin(x)\cos(y)/2$ 
n_pts = 600
xy = rng.uniform(-np.pi, np.pi, (n_pts, 2)) # shape (600, 2)
z = 0.5 * np.sin(xy[:, 0]) * np.cos(xy[:, 1]) \
    + rng.standard_normal(n_pts) * 0.04 # shape (600,)
P = np.column_stack([xy, z]) # shape (600, 3)

# Compute normal at query point via k-NN local PCA
def local_normal(P, query_idx, k=15):
    dists = np.linalg.norm(P - P[query_idx], axis=1) # shape (n,)
    nn_idx = np.argsort(dists)[1:k+1] # exclude self
    patch = P[nn_idx] - P[nn_idx].mean(axis=0) # shape (k, 3)
    Sig = patch.T @ patch / (k - 1) # shape (3, 3)
    lam, Q = np.linalg.eigh(Sig)
    normal = Q[:, 0] # smallest eigenvalue
    # Orient toward +z (viewpoint)
    if normal[2] < 0:
        normal = -normal
    return normal

# Sample normals at 5 query points
query_indices = [50, 150, 250, 350, 450]
print("Query point -> estimated normal:")
for idx in query_indices:
    n_est = local_normal(P, idx)
    # True normal:  $\text{grad}(z - \sin(x)\cos(y)/2) = [-\cos(x)\cos(y)/2, \sin(x)\sin(y)/2, 1]$  normalized
    x0, y0 = P[idx, 0], P[idx, 1]
    n_true = np.array([-0.5*np.cos(x0)*np.cos(y0),
                      0.5*np.sin(x0)*np.sin(y0),
                      1.0])
    n_true = n_true / np.linalg.norm(n_true)
    angle = np.degrees(np.arccos(np.clip(np.abs(n_est @ n_true), 0, 1)))
    print(f" P[{idx}] = ({P[idx,0]:.2f}, {P[idx,1]:.2f}): "
          f"n_est=[{n_est[0]:.3f},{n_est[1]:.3f},{n_est[2]:.3f}] "
          f"angle error={angle:.2f} deg")
```

Query point -> estimated normal:

```
P[50] = (-1.72, 0.06): n_est=[0.125,0.016,0.992]  angle error=3.96 deg
P[150] = (1.46, -1.06): n_est=[-0.039,-0.421,0.906]  angle error=1.73 deg
P[250] = (0.39, -2.73): n_est=[0.329,0.000,0.944]  angle error=5.48 deg
P[350] = (-2.30, 3.00): n_est=[-0.235,-0.096,0.967]  angle error=5.29 deg
P[450] = (-1.05, -2.31): n_est=[0.172,0.284,0.943]  angle error=1.28 deg
```

Exercises

20.5.1 Explain why the plane that **minimizes** the sum of squared perpendicular distances to n points corresponds to the **smallest** eigenvector of the covariance matrix, while the **maximum-variance** direction is the **largest** eigenvector. What does the Lagrange multiplier argument look like in each case?

20.5.2 Suppose all n points lie exactly on a plane. What is the smallest eigenvalue of their covariance matrix? What is the rank of Σ ?

20.5.3 Two point clouds represent the same object scanned from different viewpoints. After mean-centering both, their covariance matrices are Σ_A and Σ_B . Describe a procedure to align the two clouds using PCA, and identify the ambiguity that remains. (*Hint*: principal directions are defined only up to sign.)

20.5.4 In local PCA for surface normals, what happens to the quality of the estimated normal as the neighborhood size k increases from very small (e.g., 5) to very large (e.g., $n/2$)? Consider separately the bias and the variance of the normal estimate.

Sim-to-Real Domain Adaptation

Simulators such as NVIDIA Isaac Sim render synthetic sensor data at low cost, but neural networks trained entirely on synthetic data often fail on real hardware. The **domain gap** – the statistical difference between simulated and real distributions – is a primary culprit. PCA makes the gap measurable and provides a principled method to close it.

The Covariance Mismatch Problem

Let $\mathbf{f} \in \mathbb{R}^p$ be a feature vector extracted from a sensor observation (image patch, depth map, point cloud descriptor). The source (simulated) distribution has covariance Σ_S ; the target (real) distribution has Σ_T .

Even if the means match, mismatched covariance means:

- features are correlated differently in source and target,
- the principal axes of the data cloud point in different directions,
- a model trained on source data sees inputs that look statistically foreign on the target.

Measuring the Domain Gap via PCA

A simple, interpretable gap metric compares the principal subspaces of source and target. Let $V_S \in \mathbb{R}^{p \times k}$ and $V_T \in \mathbb{R}^{p \times k}$ hold the top- k principal directions of each domain. The **principal angle** gap is

$$\text{gap}(V_S, V_T) = \frac{1}{k} \sum_{j=1}^k \theta_j,$$

where $\theta_j = \arccos(\sigma_j)$ and $\sigma_1 \geq \dots \geq \sigma_k$ are the singular values of $V_S^T V_T$. When the k -dim subspaces coincide, all $\theta_j = 0$.

PCA Whitening as Feature-Space Alignment

Whitening (§19.6) transforms features so that each domain has identity covariance. Applying whitening to source data and then projecting into the target domain's basis is one of the simplest domain-adaptation strategies.

Steps (ZCA-style alignment of source to target):

1. Estimate $\hat{\Sigma}_S$ and $\hat{\Sigma}_T$ from source and target samples.
2. Compute whitening transforms: $W_S = \hat{\Sigma}_S^{-1/2}$, $W_T = \hat{\Sigma}_T^{-1/2}$.
3. The **adapted source features** are $\mathbf{f}_{\text{adapt}} = \hat{\Sigma}_T^{1/2} W_S \mathbf{f}$.

This maps source features to have the same second-order statistics as the target.

Numerical Example

```
import numpy as np

rng = np.random.default_rng(206)

p = 6    # feature dimension

# ---- Source (simulated) distribution ----
mu_S = np.zeros(p)
# Simulated sensor overestimates contrast in features 0-2
lam_S = np.array([8.0, 5.0, 3.0, 1.0, 0.8, 0.5])
```

```

Q_S, _ = np.linalg.qr(rng.standard_normal((p, p))) # shape (6, 6)
Sigma_S = Q_S @ np.diag(lam_S) @ Q_S.T           # shape (6, 6)

# ---- Target (real) distribution ----
# Different principal axes and eigenvalue magnitudes
lam_T = np.array([5.0, 4.0, 3.5, 1.5, 1.0, 0.6])
Q_T, _ = np.linalg.qr(rng.standard_normal((p, p))) # shape (6, 6)
Sigma_T = Q_T @ np.diag(lam_T) @ Q_T.T           # shape (6, 6)

n_S, n_T = 400, 200
L_S = np.linalg.cholesky(Sigma_S)
L_T = np.linalg.cholesky(Sigma_T)
X_S = rng.standard_normal((n_S, p)) @ L_S.T       # shape (400, 6)
X_T = rng.standard_normal((n_T, p)) @ L_T.T       # shape (200, 6)

# Estimate covariances from samples
X_S_c = X_S - X_S.mean(axis=0)
X_T_c = X_T - X_T.mean(axis=0)
Sigma_S_hat = X_S_c.T @ X_S_c / (n_S - 1)         # shape (6, 6)
Sigma_T_hat = X_T_c.T @ X_T_c / (n_T - 1)         # shape (6, 6)

# ---- Principal angle gap (k=3 subspace) ----
k = 3
_, _, Vt_S = np.linalg.svd(X_S_c, full_matrices=False)
_, _, Vt_T = np.linalg.svd(X_T_c, full_matrices=False)
VS = Vt_S[:k].T # shape (6, 3)
VT = Vt_T[:k].T # shape (6, 3)

cross = VS.T @ VT # shape (3, 3)
cos_angles = np.linalg.svd(cross, compute_uv=False) # shape (3,)
cos_angles = np.clip(cos_angles, -1, 1)
angles_deg = np.degrees(np.arccos(cos_angles))
print(f"Principal angles (source vs target, k={k} subspace):")
for j, a in enumerate(angles_deg):
    print(f"  theta_{j+1} = {a:.2f} deg")
print(f"Mean gap = {angles_deg.mean():.2f} deg")

# ---- Whitening-based adaptation ----
# Whitening transform: W_S = Sigma_S_hat^{-1/2} via eigendecomposition
lam_Sh, Q_Sh = np.linalg.eigh(Sigma_S_hat)
W_S = Q_Sh @ np.diag(1.0 / np.sqrt(np.maximum(lam_Sh, 1e-10))) @ Q_Sh.T # shape (6, 6)

# Target square root: Sigma_T^{1/2}
lam_Th, Q_Th = np.linalg.eigh(Sigma_T_hat)
SqrtT = Q_Th @ np.diag(np.sqrt(np.maximum(lam_Th, 0))) @ Q_Th.T # shape (6, 6)

# Adaptation map

```

```

M = SqrtT @ W_S                                # shape (6, 6)

# Adapt source features
X_S_adapt = (M @ X_S_c.T).T                      # shape (400, 6)

# Covariance of adapted source (should match Sigma_T_hat)
Sigma_adapt = X_S_adapt.T @ X_S_adapt / (n_S - 1) # shape (6, 6)
diff = np.linalg.norm(Sigma_adapt - Sigma_T_hat, 'fro')
print(f"\n||Sigma_adapted - Sigma_T||_F = {diff:.4f} (ideal: 0)")

```

Principal angles (source vs target, k=3 subspace):

```

theta_1 = 15.04 deg
theta_2 = 39.04 deg
theta_3 = 75.48 deg
Mean gap = 43.19 deg

```

```

||Sigma_adapted - Sigma_T||_F = 0.0000 (ideal: 0)

```

Visualization: Covariance Gap Before and After Adaptation

```

import numpy as np
import matplotlib.pyplot as plt

rng = np.random.default_rng(2061)

p = 2 # 2-D for visualization

mu_S = np.zeros(p)
Sigma_S = np.array([[4.0, 1.5], [1.5, 1.0]]) # shape (2, 2)
Sigma_T = np.array([[2.5, -1.2], [-1.2, 3.0]]) # shape (2, 2)

n_S, n_T = 300, 150
L_S = np.linalg.cholesky(Sigma_S)
L_T = np.linalg.cholesky(Sigma_T)
X_S = rng.standard_normal((n_S, p)) @ L_S.T # shape (300, 2)
X_T = rng.standard_normal((n_T, p)) @ L_T.T # shape (150, 2)

X_S_c = X_S - X_S.mean(axis=0)
X_T_c = X_T - X_T.mean(axis=0)
Sigma_S_hat = X_S_c.T @ X_S_c / (n_S - 1)
Sigma_T_hat = X_T_c.T @ X_T_c / (n_T - 1)

```

```

# Whitening-based adaptation
lam_Sh, Q_Sh = np.linalg.eigh(Sigma_S_hat)
W_S = Q_Sh @ np.diag(1.0 / np.sqrt(np.maximum(lam_Sh, 1e-10))) @ Q_Sh.T
lam_Th, Q_Th = np.linalg.eigh(Sigma_T_hat)
SqrtT = Q_Th @ np.diag(np.sqrt(np.maximum(lam_Th, 0))) @ Q_Th.T
M = SqrtT @ W_S
X_S_adapt = (M @ X_S_c.T).T # shape (300, 2)

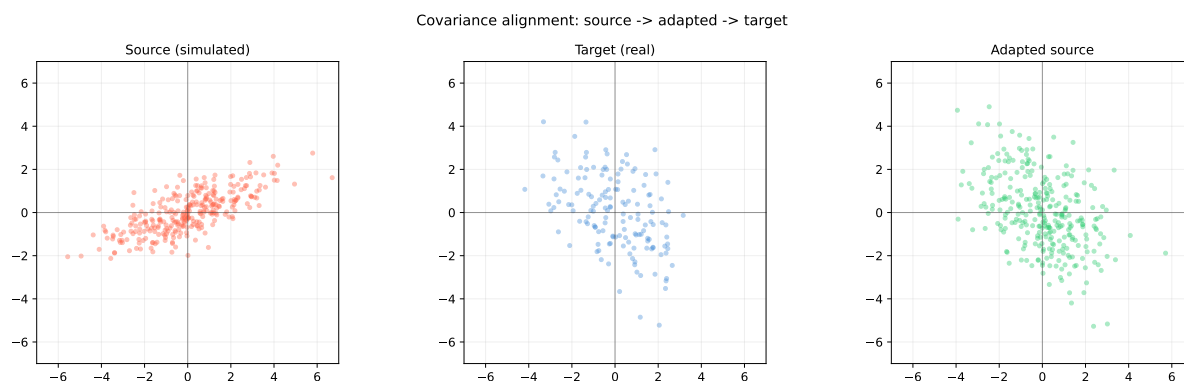
fig, axes = plt.subplots(1, 3, figsize=(15, 4.5))
datasets = [X_S_c, X_T_c, X_S_adapt]
titles = ['Source (simulated)', 'Target (real)', 'Adapted source']
colors = ['tomato', '#4a90d9', '#2ecc71']

for ax, data, title, col in zip(axes, datasets, titles, colors):
    ax.scatter(data[:, 0], data[:, 1], s=10, color=col, alpha=0.4, label=title)
    ax.axhline(0, color='#333333', lw=0.4, alpha=0.5)
    ax.axvline(0, color='#333333', lw=0.4, alpha=0.5)
    ax.set_aspect('equal')
    ax.set_xlim(-7, 7); ax.set_ylim(-7, 7)
    ax.set_title(title, fontsize=11)
    ax.grid(alpha=0.2)

fig.suptitle('Covariance alignment: source -> adapted -> target', fontsize=12)
fig.tight_layout()
plt.savefig('ch20-pca/fig-domain-adaptation.png', dpi=150, bbox_inches='tight')
plt.show()

print(f"Source Sigma diag: {np.diag(Sigma_S_hat).round(4)}")
print(f"Target Sigma diag: {np.diag(Sigma_T_hat).round(4)}")
Sig_adapt = X_S_adapt.T @ X_S_adapt / (n_S - 1)
print(f"Adapted Sigma diag: {np.diag(Sig_adapt).round(4)}")

```



```

Source Sigma diag: [3.7548 0.9084]
Target Sigma diag: [2.4994 2.9666]
Adapted Sigma diag: [2.4994 2.9666]

```


After adaptation, the source point cloud has the same shape and orientation as the target. The covariance matrices are matched by construction, which tends to reduce the domain gap measured by neural network classifiers.

Exercises

20.6.1 The whitening map $\mathbf{f}_{\text{adapt}} = \hat{\Sigma}_T^{1/2} \hat{\Sigma}_S^{-1/2} \mathbf{f}$ is a linear transformation. Show that the covariance of adapted source features equals $\hat{\Sigma}_T$ (assuming the adaptation map is applied to centered source features).

20.6.2 Why might covariance matching alone be insufficient for domain adaptation in practice? What aspects of the distribution does Σ fail to capture?

20.6.3 The principal angle between two k -dimensional subspaces V_S and V_T is defined via the singular values of $V_S^T V_T$. Show that if $V_S = V_T$ (same subspace), all singular values are 1 and all principal angles are 0.

20.6.4 In the sim-to-real context, the source covariance Σ_S is computed from a large synthetic dataset ($n = 10000$) and the target covariance Σ_T from a small real dataset ($n = 50$, $p = 20$). What problem arises when estimating Σ_T , and how would you regularize it?

Limitations

PCA is a powerful first tool, but three fundamental limitations constrain when it can succeed. Understanding them clarifies when to reach for nonlinear alternatives.

1. Linearity: PCA Cannot Unfold Manifolds

PCA projects onto a **linear** subspace. If the data lies on a nonlinear manifold – such as the Swiss roll, a sphere, or a folded sheet – the best-fit linear subspace is a poor summary.

Swiss roll example. Two hundred points wrap around a 3D roll. The intrinsic dimension is 2 (you can unfurl the roll onto a flat sheet), but the 2D PCA projection squashes points from opposite sides of the roll on top of each other. Nonlinear methods (UMAP, Isomap, autoencoders) find the 2D embedding by respecting geodesic distances along the manifold.

Extension: kernel PCA. Replace the linear covariance $\Sigma = \frac{1}{n-1} \tilde{X}^T \tilde{X}$ with a kernel matrix $K_{ij} = \phi(\mathbf{x}_i)^T \phi(\mathbf{x}_j)$, where ϕ maps data to a (possibly infinite-dimensional) feature space. Using the RBF kernel $k(\mathbf{x}, \mathbf{y}) = \exp(-\|\mathbf{x} - \mathbf{y}\|^2 / (2\ell^2))$ lets PCA discover nonlinear directions without explicitly computing ϕ .

2. Outlier Sensitivity

Variance is a second-moment statistic: it grows quadratically with distance from the mean. A single outlier far from the cloud inflates the variance along its direction, pulling the first principal component toward it.

Robust PCA decomposes $X = L + S$, where L is a low-rank matrix (the structured part of the data) and S is a sparse matrix (the outlier component). The optimization

$$\min_{L,S} \text{rank}(L) + \|S\|_0 \quad \text{subject to } X = L + S$$

is solved (convexified) as nuclear norm + ℓ_1 minimization. The result is insensitive to a bounded fraction of corrupted entries.

3. Scale Sensitivity

PCA maximizes variance, so a feature measured in millimeters dominates a feature measured in meters even if both carry equal information. **Always standardize features to unit variance** (divide each column by its standard deviation) before PCA when the features have different physical units or scales.

Standardization before PCA is equivalent to running PCA on the **correlation matrix** $R_{jk} = \Sigma_{jk} / \sqrt{\Sigma_{jj}\Sigma_{kk}}$ instead of the covariance matrix.

Numerical Illustration

```
import numpy as np
import matplotlib.pyplot as plt

rng = np.random.default_rng(207)

# ---- Swiss roll: linearity failure ----
n = 300
t = rng.uniform(1.5 * np.pi, 4.5 * np.pi, n)  # shape (300,)
h = rng.uniform(0, 10, n)                       # height
X_roll = np.column_stack([
    t * np.cos(t),
    h,
    t * np.sin(t)
])                                              # shape (300, 3)
```

```

X_c = X_roll - X_roll.mean(axis=0)
_, s_roll, Vt_roll = np.linalg.svd(X_c, full_matrices=False)
V_roll = Vt_roll.T
Z_roll = X_c @ V_roll[:, :2]          # shape (300, 2)

# ---- Scale sensitivity ----
rng2 = np.random.default_rng(2071)
X2 = rng2.standard_normal((200, 2))   # shape (200, 2)
X2[:, 0] *= 10                        # feature 0 in "centimeters"

X2_c = X2 - X2.mean(axis=0)
_, s2, Vt2 = np.linalg.svd(X2_c, full_matrices=False)
V2 = Vt2.T

X2_std = X2_c / X2_c.std(axis=0, ddof=1) # standardize
_, s2s, Vt2s = np.linalg.svd(X2_std, full_matrices=False)
V2s = Vt2s.T

# ---- Outlier sensitivity ----
rng3 = np.random.default_rng(2072)
X3 = rng3.standard_normal((100, 2))   # shape (100, 2)
X3_out = np.vstack([X3, [[8, 0]]])    # one outlier at (8, 0)

X3_c = X3 - X3.mean(axis=0)
_, s3, Vt3 = np.linalg.svd(X3_c, full_matrices=False)
V3 = Vt3.T

X3o_c = X3_out - X3_out.mean(axis=0)
_, s3o, Vt3o = np.linalg.svd(X3o_c, full_matrices=False)
V3o = Vt3o.T

fig, axes = plt.subplots(1, 3, figsize=(16, 5))

# Left: Swiss roll projection
ax = axes[0]
sc = ax.scatter(Z_roll[:, 0], Z_roll[:, 1], c=t, cmap='viridis', s=12, alpha=0.8)
ax.set_title('Swiss roll: 2D PCA projection\n(folds data onto itself)', fontsize=10)
ax.set_xlabel('PC 1', fontsize=9); ax.set_ylabel('PC 2', fontsize=9)
ax.grid(alpha=0.2)
plt.colorbar(sc, ax=ax, label='angle t')

# Center: scale sensitivity
ax2 = axes[1]
ax2.scatter(X2_c[:, 0], X2_c[:, 1], s=10, color='#4a90d9', alpha=0.5)
scale = 12
for j, (col, v, lbl) in enumerate(zip(
    ['tomato', '#2ecc71'],

```

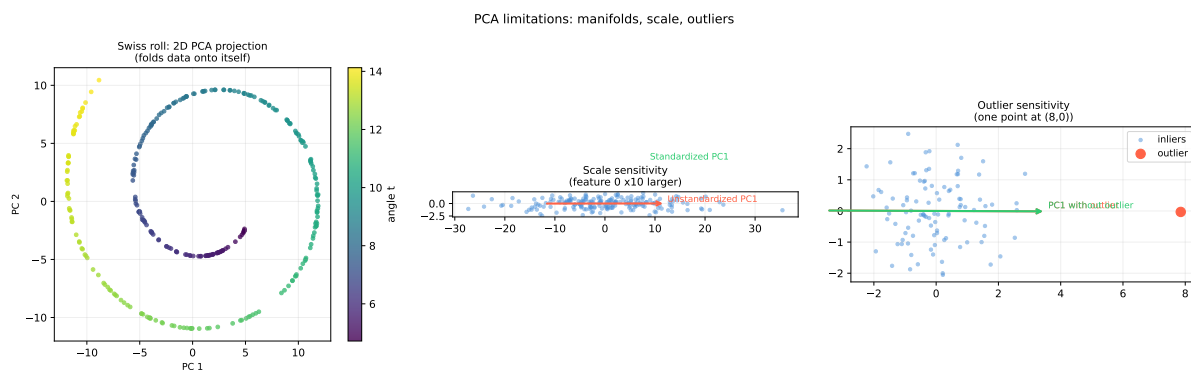
```

[V2[:, 0], V2s[:, 0]],
['Unstandardized PC1', 'Standardized PC1'])):
    ax2.annotate('', xy=scale * v, xytext=-scale * v,
                  arrowprops=dict(arrowstyle='->', color=col, lw=2))
    ax2.text(*(scale * v + np.array([0.5, 0.3])), lbl, fontsize=8.5, color=col)
ax2.set_aspect('equal')
ax2.set_title('Scale sensitivity\n(feature 0 x10 larger)', fontsize=10)
ax2.grid(alpha=0.2)

# Right: outlier sensitivity
ax3 = axes[2]
ax3.scatter(X3_c[:, 0], X3_c[:, 1], s=10, color='#4a90d9', alpha=0.5, label='inliers')
ax3.scatter(X3o_c[-1, 0], X3o_c[-1, 1], s=80, color='tomato', zorder=5, label='outlier')
scale2 = 3.5
for (col, v, lbl) in [
    ('tomato', V3o[:, 0], 'PC1 with outlier'),
    ('#2ecc71', V3[:, 0], 'PC1 without outlier')]:
    ax3.annotate('', xy=scale2 * v, xytext=-scale2 * v,
                  arrowprops=dict(arrowstyle='->', color=col, lw=2))
    ax3.text(*(scale2 * v + np.array([0.1, 0.1])), lbl, fontsize=8.5, color=col)
ax3.set_aspect('equal')
ax3.set_title('Outlier sensitivity\n(one point at (8,0))', fontsize=10)
ax3.legend(fontsize=9)
ax3.grid(alpha=0.2)

fig.suptitle('PCA limitations: manifolds, scale, outliers', fontsize=12)
fig.tight_layout()
plt.savefig('ch20-pca/fig-limitations.png', dpi=150, bbox_inches='tight')
plt.show()

```



Extensions at a Glance

Method	Core idea	When to use
Kernel PCA	Replace $\tilde{X}^T \tilde{X}$ with kernel matrix K	Nonlinear manifolds; RBF kernel for smooth embeddings
Robust PCA	Decompose $X = L + S$; nuclear + ℓ_1	Sparse corruptions, video foreground/background separation
Autoencoder	Nonlinear encoder/decoder learned end-to-end	General nonlinear compression; images, sequences
Incremental PCA	Update V as new data arrives in mini-batches	Streaming data; too large to fit in memory
Sparse PCA	Constrain principal directions to have few non-zero entries	Interpretable loadings; gene expression, finance

Exercises

20.7.1 Construct a 2D dataset where the first principal component points in a direction that carries **no** class-discriminative information, while the second PC does. Sketch or simulate this and explain why PCA alone is insufficient for classification here.

20.7.2 A dataset has 1000 observations in $\mathbb{R}^{10}$. One observation has been corrupted (set to $\mathbf{x}_{\text{corrupt}} = 100 \cdot \mathbf{e}_1$, far from the rest). Analyze how this single outlier shifts λ_1 , the largest eigenvalue of the sample covariance.

20.7.3 You have two features: height in meters (range 1.5-2.0) and shoe size (range 36-48 European). Without standardization, which feature dominates PCA? Show numerically that standardization before PCA changes the first principal direction.

20.7.4 Robust PCA solves $\min_{L,S} \|L\|_* + \mu \|S\|_1$ subject to $X = L + S$, where $\|\cdot\|_*$ is the nuclear norm (§15.3). Why does the nuclear norm serve as a convex surrogate for rank? And why does ℓ_1 serve as a surrogate for sparsity?

Application: Defect Detection on Rice Grains

A grain processing facility inspects rice on a conveyor belt and must flag defective grains automatically. Defective grains (chalky, discolored, broken, or misshaped) differ from good grains across multiple visual features simultaneously. No single feature reliably separates good from defective; the combination matters.

PCA provides a clean solution: project to the 2D subspace of maximum variance, visualize the data cloud, and flag grains far from the inlier cluster.

Feature Engineering (Simulated)

We extract five features per grain from an overhead camera:

- **Length** (px): longest axis of the grain bounding ellipse.
- **Width** (px): shortest axis.
- **Aspect ratio**: Length / Width.
- **Brightness**: mean pixel intensity (0-255 scale).
- **Chalkiness**: fraction of the grain area with low-contrast texture (0-1).

Defective grains are grouped into three types: - **Chalky**: high chalkiness, slightly shorter and brighter. - **Broken**: much shorter, smaller aspect ratio. - **Discolored**: normal shape but very different brightness.

```
import numpy as np

rng = np.random.default_rng(208)

# ---- Good grains ----
mu_good = np.array([320.0,  # length px
                    85.0,   # width px
                    3.8,    # aspect ratio
                    180.0,  # brightness
                    0.05]) # chalkiness
# Covariance: length-width correlated (r~0.7); chalkiness independent of shape
Sigma_good = np.array([
    [400.0, 60.0, 3.5, -20.0, 0.0], # length
    [60.0, 40.0, 0.8, -8.0, 0.0],  # width
    [3.5, 0.8, 0.08, -0.5, 0.0],   # aspect ratio
    [-20.0, -8.0, -0.5, 600.0, 0.0], # brightness
    [0.0, 0.0, 0.0, 0.0, 0.005]    # chalkiness (independent of shape)
]) # shape (5, 5)

n_good = 500
L_good = np.linalg.cholesky(Sigma_good) # shape (5, 5)
X_good = rng.standard_normal((n_good, 5)) @ L_good.T + mu_good # shape (500, 5)
# Clip to physical range
X_good[:, 4] = np.clip(X_good[:, 4], 0, 1) # chalkiness in [0, 1]

# ---- Chalky grains ----
mu_chalky = np.array([300.0, 83.0, 3.6, 195.0, 0.45])
Sigma_chalky = np.diag([300.0, 35.0, 0.06, 400.0, 0.003]) # shape (5, 5)
n_chalky = 60
L_ch = np.linalg.cholesky(Sigma_chalky)
X_chalky = rng.standard_normal((n_chalky, 5)) @ L_ch.T + mu_chalky # shape (60, 5)
X_chalky[:, 4] = np.clip(X_chalky[:, 4], 0, 1)

# ---- Broken grains ----
```

```

mu_broken = np.array([160.0, 78.0, 2.1, 175.0, 0.08])
Sigma_broken = np.diag([200.0, 50.0, 0.05, 500.0, 0.002]) # shape (5, 5)
n_broken = 50
L_br = np.linalg.cholesky(Sigma_broken)
X_broken = rng.standard_normal((n_broken, 5)) @ L_br.T + mu_broken # shape (50, 5)
X_broken[:, 4] = np.clip(X_broken[:, 4], 0, 1)

# ---- Discolored grains ----
mu_discol = np.array([315.0, 84.0, 3.75, 115.0, 0.06])
Sigma_discol = np.diag([350.0, 38.0, 0.07, 300.0, 0.002]) # shape (5, 5)
n_discol = 50
L_dc = np.linalg.cholesky(Sigma_discol)
X_discol = rng.standard_normal((n_discol, 5)) @ L_dc.T + mu_discol # shape (50, 5)
X_discol[:, 4] = np.clip(X_discol[:, 4], 0, 1)

feature_names = ['Length', 'Width', 'Aspect ratio', 'Brightness', 'Chalkiness']
print(f"Good grains:          {n_good} samples")
print(f"Chalky defects:      {n_chalky} samples")
print(f"Broken defects:        {n_broken} samples")
print(f"Discolored defects:    {n_discol} samples")
print(f"\nTrue means (good): {mu_good}")

```

```

Good grains:          500 samples
Chalky defects:      60 samples
Broken defects:      50 samples
Discolored defects: 50 samples

```

```

True means (good): [3.2e+02 8.5e+01 3.8e+00 1.8e+02 5.0e-02]

```

Fitting PCA on Good Grains Only

We train PCA exclusively on good-grain data (one-class approach). Defective grains are not used during training.

```

import numpy as np

rng = np.random.default_rng(208)

# Re-generate (self-contained)
mu_good = np.array([320.0, 85.0, 3.8, 180.0, 0.05])
Sigma_good = np.array([
    [400.0, 60.0, 3.5, -20.0, 0.0],
    [ 60.0, 40.0, 0.8, -8.0, 0.0],

```

```

    [ 3.5, 0.8, 0.08, -0.5, 0.0],
    [-20.0, -8.0, -0.5, 600.0, 0.0],
    [ 0.0, 0.0, 0.0, 0.0, 0.005]
])
n_good = 500
L_good = np.linalg.cholesky(Sigma_good)
X_good = rng.standard_normal((n_good, 5)) @ L_good.T + mu_good
X_good[:, 4] = np.clip(X_good[:, 4], 0, 1)

mu_chalky = np.array([300.0, 83.0, 3.6, 195.0, 0.45])
L_ch = np.linalg.cholesky(np.diag([300.0, 35.0, 0.06, 400.0, 0.003]))
X_chalky = rng.standard_normal((60, 5)) @ L_ch.T + mu_chalky
X_chalky[:, 4] = np.clip(X_chalky[:, 4], 0, 1)

mu_broken = np.array([160.0, 78.0, 2.1, 175.0, 0.08])
L_br = np.linalg.cholesky(np.diag([200.0, 50.0, 0.05, 500.0, 0.002]))
X_broken = rng.standard_normal((50, 5)) @ L_br.T + mu_broken
X_broken[:, 4] = np.clip(X_broken[:, 4], 0, 1)

mu_discol = np.array([315.0, 84.0, 3.75, 115.0, 0.06])
L_dc = np.linalg.cholesky(np.diag([350.0, 38.0, 0.07, 300.0, 0.002]))
X_discol = rng.standard_normal((50, 5)) @ L_dc.T + mu_discol
X_discol[:, 4] = np.clip(X_discol[:, 4], 0, 1)

# ---- PCA fit on good grains ----
mu_hat = X_good.mean(axis=0) # shape (5,)
std_hat = X_good.std(axis=0, ddof=1) # shape (5,) -- for standardization

# Standardize (per-feature unit variance)
def standardize(X, mu, std):
    return (X - mu) / std

X_good_s = standardize(X_good, mu_hat, std_hat) # shape (500, 5)

# Thin SVD of standardized good grains
U, s, Vt = np.linalg.svd(X_good_s, full_matrices=False)
V = Vt.T # shape (5, 5)
lam = s**2 / (n_good - 1) # shape (5,)
evr = lam / lam.sum()

print("Eigenvalues and EVR (on standardized features):")
feature_names = ['Length', 'Width', 'Aspect ratio', 'Brightness', 'Chalkiness']
for j in range(5):
    print(f" PC{j+1}: lambda={lam[j]:.4f} EVR={100*evr[j]:.1f}% cum={100*evr[:j+1].sum():.1f}")

# Project all data to top-2 PCs
def project(X, mu, std, V, k=2):

```



```

Xs = standardize(X, mu, std)
return Xs @ V[:, :k]                                # shape (n, k)

Z_good  = project(X_good,  mu_hat, std_hat, V)
Z_chalky = project(X_chalky, mu_hat, std_hat, V)
Z_broken = project(X_broken, mu_hat, std_hat, V)
Z_discol = project(X_discol, mu_hat, std_hat, V)

# Reconstruction error in PC space (residual norm) as anomaly score
def recon_error(X, mu, std, V, k=2):
    Xs = standardize(X, mu, std)
    Xhat = Xs @ V[:, :k] @ V[:, :k].T
    return np.linalg.norm(Xs - Xhat, axis=1)        # shape (n,)

err_good  = recon_error(X_good,  mu_hat, std_hat, V)
err_chalky = recon_error(X_chalky, mu_hat, std_hat, V)
err_broken = recon_error(X_broken, mu_hat, std_hat, V)
err_discol = recon_error(X_discol, mu_hat, std_hat, V)

tau = np.percentile(err_good, 99)
print(f"\nReconstruction error threshold (99th pct of good grains): tau = {tau:.4f}")
print(f"Detection rates with k=2 PCA:")
print(f"  Chalky      flagged: {(err_chalky > tau).mean():.3f}")
print(f"  Broken       flagged: {(err_broken > tau).mean():.3f}")
print(f"  Discolored flagged: {(err_discol > tau).mean():.3f}")
print(f"  Good (false pos): {(err_good > tau).mean():.3f}   (expected ~0.01)")

```

Eigenvalues and EVR (on standardized features):

```

PC1: lambda=2.0233  EVR=40.5%  cum=40.5%
PC2: lambda=1.0403  EVR=20.8%  cum=61.3%
PC3: lambda=0.9599  EVR=19.2%  cum=80.5%
PC4: lambda=0.6116  EVR=12.2%  cum=92.7%
PC5: lambda=0.3649  EVR=7.3%   cum=100.0%

```

Reconstruction error threshold (99th pct of good grains): tau = 2.6840

Detection rates with k=2 PCA:

```

Chalky      flagged: 1.000
Broken      flagged: 0.980
Discolored  flagged: 0.320
Good (false pos): 0.010   (expected ~0.01)

```

Visualization: PCA Score Plot and Anomaly Scores

```

import numpy as np
import matplotlib.pyplot as plt

rng = np.random.default_rng(208)

# Re-generate (self-contained)
mu_good = np.array([320.0, 85.0, 3.8, 180.0, 0.05])
Sigma_good = np.array([
    [400.0, 60.0, 3.5, -20.0, 0.0],
    [ 60.0, 40.0, 0.8, -8.0, 0.0],
    [ 3.5, 0.8, 0.08, -0.5, 0.0],
    [-20.0, -8.0, -0.5, 600.0, 0.0],
    [ 0.0, 0.0, 0.0, 0.0, 0.005]
])
n_good = 500
L_good = np.linalg.cholesky(Sigma_good)
X_good = rng.standard_normal((n_good, 5)) @ L_good.T + mu_good
X_good[:, 4] = np.clip(X_good[:, 4], 0, 1)

defs = {
    'Chalky': (np.array([300.0, 83.0, 3.6, 195.0, 0.45]),
              np.diag([300.0, 35.0, 0.06, 400.0, 0.003]), 60),
    'Broken': (np.array([160.0, 78.0, 2.1, 175.0, 0.08]),
              np.diag([200.0, 50.0, 0.05, 500.0, 0.002]), 50),
    'Discolored': (np.array([315.0, 84.0, 3.75, 115.0, 0.06]),
                  np.diag([350.0, 38.0, 0.07, 300.0, 0.002]), 50),
}
X_def = {}
for name, (mu_d, Sig_d, n_d) in defs.items():
    Xd = rng.standard_normal((n_d, 5)) @ np.linalg.cholesky(Sig_d).T + mu_d
    Xd[:, 4] = np.clip(Xd[:, 4], 0, 1)
    X_def[name] = Xd

mu_hat = X_good.mean(axis=0)
std_hat = X_good.std(axis=0, ddof=1)

def standardize(X, mu, std):
    return (X - mu) / std

X_good_s = standardize(X_good, mu_hat, std_hat)
_, s, Vt = np.linalg.svd(X_good_s, full_matrices=False)
V = Vt.T

def project(X, mu, std, V, k=2):
    return standardize(X, mu, std) @ V[:, :k]

```

```

def recon_error(X, mu, std, V, k=2):
    Xs = standardize(X, mu, std)
    return np.linalg.norm(Xs - Xs @ V[:, :k] @ V[:, :k].T, axis=1)

Z_good = project(X_good, mu_hat, std_hat, V)
err_good = recon_error(X_good, mu_hat, std_hat, V)
tau = np.percentile(err_good, 99)

def_colors = {'Chalky': 'tomato', 'Broken': '#e67e22', 'Discolored': '#9b59b6'}
def_markers = {'Chalky': '^', 'Broken': 's', 'Discolored': 'D'}

fig, axes = plt.subplots(1, 2, figsize=(13, 5.5))

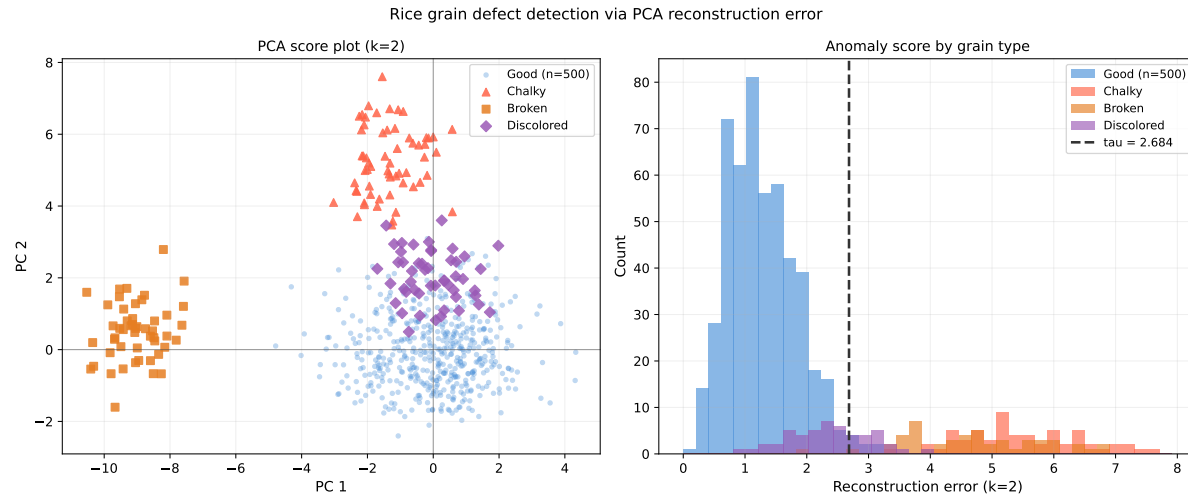
# Left: PCA score plot (PC1 vs PC2)
ax = axes[0]
ax.scatter(Z_good[:, 0], Z_good[:, 1], s=10, color='#4a90d9', alpha=0.35,
           label=f'Good (n={n_good})')
for name, Xd in X_def.items():
    Zd = project(Xd, mu_hat, std_hat, V)
    ax.scatter(Zd[:, 0], Zd[:, 1], s=35, color=def_colors[name],
              alpha=0.85, marker=def_markers[name], label=name)
ax.axhline(0, color='#333333', lw=0.4, alpha=0.5)
ax.axvline(0, color='#333333', lw=0.4, alpha=0.5)
ax.set_xlabel('PC 1', fontsize=11)
ax.set_ylabel('PC 2', fontsize=11)
ax.set_title('PCA score plot (k=2)', fontsize=11)
ax.legend(fontsize=9)
ax.grid(alpha=0.2)

# Right: reconstruction error histogram
ax2 = axes[1]
all_err = np.concatenate([err_good
                          + [recon_error(X_def[n], mu_hat, std_hat, V) for n in def_colors]])
bins = np.linspace(0, max(all_err) * 1.05, 40)
ax2.hist(err_good, bins=bins, color='#4a90d9', alpha=0.65, label=f'Good (n={n_good})')
for name, Xd in X_def.items():
    ax2.hist(recon_error(Xd, mu_hat, std_hat, V),
             bins=bins, color=def_colors[name], alpha=0.65, label=name)
ax2.axvline(tau, color='#333333', lw=2, linestyle='--',
            label=f'tau = {tau:.3f}')
ax2.set_xlabel('Reconstruction error (k=2)', fontsize=11)
ax2.set_ylabel('Count', fontsize=11)
ax2.set_title('Anomaly score by grain type', fontsize=11)
ax2.legend(fontsize=9)
ax2.grid(alpha=0.2)

fig.suptitle('Rice grain defect detection via PCA reconstruction error', fontsize=12)

```

```
fig.tight_layout()
plt.savefig('ch20-pca/fig-rice-defects.png', dpi=150, bbox_inches='tight')
plt.show()
```



Key Takeaways

1. **Standardize before PCA when features have different units.** Length (pixels) and chalkiness (fraction) have very different scales; without standardization, length dominates the first PC and chalkiness is invisible.
2. **One-class PCA learns the inlier subspace.** We fit PCA only on good grains and use reconstruction error as an anomaly score. Defective grains that do not lie on the good-grain subspace produce large residuals.
3. **Different defects separate differently in PC space.** Broken grains (very short) separate cleanly along PC1. Chalky grains separate along the chalkiness-driven PC. Discolored grains are detected via their high reconstruction error even when their PC1-PC2 projection overlaps with good grains.
4. **Choosing k is a design decision.** A smaller k (fewer components retained) flags more anomalies – including grains that differ only in the discarded dimensions. A larger k is more permissive. Set k by the false-positive budget, not by the 90% EVR rule alone.

Solutions

PCA as Maximum Variance Projection

20.1.1 Let $z_i = \mathbf{u}^T \tilde{\mathbf{x}}_i$. Because $\tilde{X}$ is mean-centered, $\sum_i z_i = \mathbf{u}^T \sum_i \tilde{\mathbf{x}}_i = 0$. Then

$$\text{Var}(\mathbf{u}) = \frac{1}{n-1} \sum_{i=1}^n z_i^2 = \frac{1}{n-1} \sum_i (\mathbf{u}^T \tilde{\mathbf{x}}_i)^2.$$

Write $(\mathbf{u}^T \tilde{\mathbf{x}}_i)^2 = \mathbf{u}^T (\tilde{\mathbf{x}}_i \tilde{\mathbf{x}}_i^T) \mathbf{u}$ (outer product) and sum:

$$\text{Var}(\mathbf{u}) = \mathbf{u}^T \left(\frac{1}{n-1} \sum_i \tilde{\mathbf{x}}_i \tilde{\mathbf{x}}_i^T \right) \mathbf{u} = \mathbf{u}^T \left(\frac{1}{n-1} \tilde{X}^T \tilde{X} \right) \mathbf{u} = \mathbf{u}^T \Sigma \mathbf{u}.$$

Note the matrix identity $\sum_i \tilde{\mathbf{x}}_i \tilde{\mathbf{x}}_i^T = \tilde{X}^T \tilde{X}$ because row i of $\tilde{X}$ is $\tilde{\mathbf{x}}_i^T$.

20.1.2 Without the constraint $\|\mathbf{u}\| = 1$, the objective $\mathbf{u}^T \Sigma \mathbf{u}$ is unbounded above: replacing $\mathbf{u}$ by $c\mathbf{u}$ scales the objective by c^2 , which grows without bound as $c \rightarrow \infty$. The unit-norm constraint confines the search to directions (unit vectors) rather than vectors of arbitrary length, making the maximum finite and the solution unique up to sign.

20.1.3 The Lagrangian is $\mathcal{L} = \mathbf{u}^T \Sigma \mathbf{u} - \lambda(\mathbf{u}^T \mathbf{u} - 1)$. Differentiating with respect to $\mathbf{u}$:

$$\frac{\partial \mathcal{L}}{\partial \mathbf{u}} = 2\Sigma \mathbf{u} - 2\lambda \mathbf{u} = \mathbf{0} \implies \Sigma \mathbf{u} = \lambda \mathbf{u}.$$

At the critical point, the variance achieved is

$$\mathbf{u}^T \Sigma \mathbf{u} = \mathbf{u}^T (\lambda \mathbf{u}) = \lambda \mathbf{u}^T \mathbf{u} = \lambda.$$

Therefore, to maximize variance we choose the eigenvector corresponding to the **largest** eigenvalue λ_1 .

20.1.4 If $\lambda_1 = \lambda_2 > \lambda_3$, then every unit vector in the 2D eigenspace $\text{span}\{\mathbf{q}_1, \mathbf{q}_2\}$ achieves the same variance λ_1 . The first principal component is **not unique**; any unit vector in that plane is equally valid. In practice, numerical algorithms return an arbitrary orthonormal basis for the degenerate eigenspace.

PCA via SVD

20.2.1 From $\tilde{X} = USV^T$:

$$\tilde{X}^T \tilde{X} = (USV^T)^T (USV^T) = VS \underbrace{U^T U}_{=I} SV^T = VS^2 V^T.$$

Dividing by $n - 1$: $\Sigma = V \frac{S^2}{n-1} V^T$. The columns of V play the role of Q (eigenvectors), and $\frac{S^2}{n-1} = \Lambda$ (eigenvalue diagonal).

20.2.2 Direct check: $\tilde{X}V = (USV^T)V = US \underbrace{V^T V}_{=I} = US$. So $Z = \tilde{X}V = US$. Both expressions compute identical matrices.

20.2.3 Scaling column j of $\tilde{X}$ by 10 multiplies the j -th component of each row by 10. (a) Singular values change: at minimum, σ_1 increases (the scaled column has larger variance). The exact change depends on the eigenvector structure. (b) The first principal direction $\mathbf{v}_1$ rotates toward the $\mathbf{e}_j$ direction (feature j now dominates), especially if the scaled feature was already influential.

20.2.4 With `full_matrices=False`, the shapes are $U \in \mathbb{R}^{n \times p}$, $s \in \mathbb{R}^p$, $V^T \in \mathbb{R}^{p \times p}$ – the thin SVD. With `full_matrices=True`, $U \in \mathbb{R}^{n \times n}$, which for $n \gg p$ is an enormous matrix we never need. The extra $n - p$ columns of U span the left null space of $\tilde{X}$ and are irrelevant for PCA. The thin SVD saves $O(n^2 p)$ memory.

Explained Variance and Choosing k

20.3.1 For $\Sigma = I_p$, all eigenvalues equal 1. Explained variance ratio: $r_j = 1/p$ for all j . The scree plot is a flat horizontal line. Cumulative EVR reaches 90% only when $k \geq 0.9p$ components are retained. White-noise data has no low-dimensional structure; PCA offers no compression.

20.3.2 Using the spectral decomposition $\Sigma = Q\Lambda Q^T$:

$$\text{tr}(\Sigma) = \text{tr}(Q\Lambda Q^T) = \text{tr}(Q^T Q\Lambda) = \text{tr}(\Lambda) = \sum_j \lambda_j.$$

The second equality uses the cyclic property $\text{tr}(ABC) = \text{tr}(CAB)$. The third uses $Q^T Q = I$ (orthogonality).

20.3.3 Squared singular values: $\sigma_1^2 = 10000$, $\sigma_2^2 = 400$, $\sigma_3^2 = \dots = \sigma_{10}^2 = 25$. Total: $10000 + 400 + 8 \times 25 = 10600$. EVR: $r_1 = 10000/10600 \approx 0.943$, $r_2 = 400/10600 \approx 0.038$, each of $r_3, \dots, r_{10} = 25/10600 \approx 0.0024$. Cumulative: $R_1 \approx 0.943$, $R_2 \approx 0.981$. So $k = 1$ already exceeds 94%, and $k = 2$ exceeds 95%.

20.3.4 Standardization before PCA is equivalent to running PCA on the correlation matrix R instead of Σ . Example: if feature 1 has variance 1000 (large in raw units) and feature 2 has variance 1 (small), raw PCA treats feature 1 as ten times more important regardless of its actual information content. After standardization both features get variance 1 and contribute equally. Result: the two scree plots can point in qualitatively different directions, and the first PC can flip from “mostly feature 1” to “mix of feature 1 and 2.”

Reconstruction and Residuals

20.4.1 $P_k = V_k V_k^T$. Then $P_k^2 = V_k V_k^T V_k V_k^T = V_k (V_k^T V_k) V_k^T = V_k I_k V_k^T = P_k$ (since $V_k^T V_k = I_k$ – columns of V_k are orthonormal). The complement $Q_k = I - P_k$ satisfies $Q_k^2 = I - 2P_k + P_k^2 = I - P_k = Q_k$, so it is also a projection. Q_k projects onto the orthogonal complement $\text{span}\{\mathbf{v}_1, \dots, \mathbf{v}_k\}^\perp = \text{span}\{\mathbf{v}_{k+1}, \dots, \mathbf{v}_p\}$.

20.4.2 The rank- k truncated SVD of $\tilde{X}$ is $\hat{X}_k = U_k S_k V_k^T$. The residual is $\tilde{X} - \hat{X}_k = U_{k+} S_{k+} V_{k+}^T$ (tail singular components). Since columns of $[U_k, U_{k+}]$ and $[V_k, V_{k+}]$ are each orthonormal, Frobenius squared norm equals the sum of squared singular values: $\|\tilde{X} - \hat{X}_k\|_F^2 = \sum_{j=k+1}^p \sigma_j^2$.

20.4.3 $\|\tilde{X}\|_F^2 = 1200$. Top-3 contribution: $\sigma_1^2 + \sigma_2^2 + \sigma_3^2 = 625 + 225 + 100 = 950$. Residual fraction: $(1200 - 950)/1200 = 250/1200 \approx 0.208$, so about 20.8% of total variance is discarded by $k = 3$.

20.4.4 A large residual $\|E_{k,i}\|$ means observation i has a significant component orthogonal to the top- k PCA subspace – a direction not represented in the inlier data. In anomaly detection this flags the observation as potentially defective: it deviates from the learned low-dimensional structure, even if its projection onto the k -dim subspace looks normal. This is the basis of PCA-based anomaly detection used in §20.8.

PCA for 3D Point Clouds

20.5.1 Plane fitting minimizes $\sum_i (\mathbf{n}^T \tilde{\mathbf{p}}_i)^2 = \mathbf{n}^T \Sigma \mathbf{n}$ subject to $\|\mathbf{n}\| = 1$. The Lagrangian is $\mathbf{n}^T \Sigma \mathbf{n} - \lambda(\mathbf{n}^T \mathbf{n} - 1)$ and the critical point satisfies $\Sigma \mathbf{n} = \lambda \mathbf{n}$. To **minimize** variance in the normal direction we pick the eigenvector of the **smallest** eigenvalue. The maximum-variance problem is identical except we want the largest eigenvalue.

20.5.2 If all n points lie exactly on a plane through the origin, then no point has a component in the normal direction, so $\mathbf{n}^T \tilde{\mathbf{p}}_i = 0$ for all i . The smallest eigenvalue is exactly $\lambda_3 = 0$. The rank of Σ is at most 2 (the covariance matrix has a null space in the normal direction), so $\text{rank}(\Sigma) \leq 2$.

20.5.3 Run PCA on each cloud and align the principal axes: compute rotation $R = V_A V_B^T$ that maps the PC frame of cloud A to that of cloud B . The remaining ambiguity is sign: each principal axis is defined up to $\pm$, giving $2^3 = 8$ possible sign flips. Resolve sign by requiring the rotation matrix to have $\det(R) = +1$ (proper rotation) and by choosing sign combinations that minimize the residual point-to-point distance after alignment.

20.5.4 Very small k (e.g., 5 neighbors): the local neighborhood is small and noisy, so the estimated normal has high variance (sensitive to random outliers and quantization noise). Very large k (e.g., $n/2$): the neighborhood spans a large region of the surface, and if the surface curves significantly the patch is not well approximated by a plane, introducing bias (the estimated normal deviates systematically from the true local normal). Optimal k balances bias and variance, typically chosen via cross-validation or by matching the expected surface curvature to the point cloud density.

Sim-to-Real Domain Adaptation

20.6.1 Let $\mathbf{f}_s$ be a centered source feature with covariance Σ_S . The adapted feature is $\mathbf{f}_a = M\mathbf{f}_s$ with $M = \Sigma_T^{1/2}\Sigma_S^{-1/2}$. Its covariance is

$$\text{Cov}(\mathbf{f}_a) = M\Sigma_S M^T = \Sigma_T^{1/2}\Sigma_S^{-1/2}\Sigma_S(\Sigma_S^{-1/2})^T(\Sigma_T^{1/2})^T = \Sigma_T^{1/2}I\Sigma_T^{1/2} = \Sigma_T.$$

(Using symmetry: $\Sigma_S^{-1/2} = (\Sigma_S^{1/2})^{-1}$ and $(\Sigma_T^{1/2})^T = \Sigma_T^{1/2}$ for symmetric PSD matrices.)

20.6.2 Covariance (second moment) does not capture higher-order statistics: skewness, kurtosis, multimodality, and complex correlations among many features. Two distributions can have identical means and covariances but differ dramatically in their densities. For neural networks, the gap in higher-order statistics, in the frequency content of images, or in the distribution of fine-grained textures may persist even after covariance alignment.

20.6.3 If $V_S = V_T$, then $V_S^T V_T = V_S^T V_S = I_k$ (since columns are orthonormal). The SVD of I_k is $I = I \cdot I \cdot I$, so all singular values are 1, and $\theta_j = \arccos(1) = 0$ for all j .

20.6.4 With $n = 50$ observations and $p = 20$ features, the sample covariance $\hat{\Sigma}_T \in \mathbb{R}^{20 \times 20}$ is estimated from 50×20 data. Since $n < 2p$ the sample covariance may be singular or near-singular (rank at most $n - 1 = 49 < p$), so $\hat{\Sigma}_T^{-1/2}$ does not exist. Regularize with a ridge term: $\hat{\Sigma}_T \leftarrow \hat{\Sigma}_T + \epsilon I$ (Ledoit-Wolf shrinkage provides an optimal ϵ), or use only the top- k principal directions of $\hat{\Sigma}_T$ where $k \ll n$.

Limitations

20.7.1 Example: two-class data forming concentric rings. The first PC points along the direction of maximum spread of the combined cloud, which is a diameter of the outer ring – irrelevant for classifying inner vs outer. The second PC is orthogonal and similarly useless. No linear PCA direction separates the classes because the class boundary is radial (nonlinear). A kernel PCA with RBF kernel resolves this by mapping to a feature space where the classes are linearly separable.

20.7.2 The corrupted observation contributes $\mathbf{x}_{\text{corrupt}}\mathbf{x}_{\text{corrupt}}^T = 10000\mathbf{e}_1\mathbf{e}_1^T$ to the unnormalized covariance (shift of $10000/n \approx 10$ in the $(1,1)$ entry of Σ). This inflates λ_1 by approximately 10 and rotates $\mathbf{q}_1$ toward $\mathbf{e}_1$, regardless of the true data structure. For robust estimation, one would use median-based centering or trim the extreme observations before running PCA.

20.7.3 Height in meters has range 0.5, so variance ≈ 0.02 . Shoe size has range 12, so variance ≈ 12 . Unscaled PCA: first PC points almost entirely in the shoe-size direction. After standardization, both features have variance 1 and the first PC captures the genuine correlation between height and shoe size. Numerically, the dot product of the first PC with $\mathbf{e}_{\text{shoe}}$ drops from near 1 to roughly 0.7 after standardization.

20.7.4 The nuclear norm $\|L\|_* = \sum_j \sigma_j(L)$ is the convex hull of the rank function on the spectral ball: just as ℓ_1 is the convex hull of the cardinality (number of nonzeros) function on the unit ℓ_∞ ball, the nuclear norm is the convex hull of rank on the unit spectral-norm ball. Minimizing $\|L\|_*$ encourages small singular values – effectively pushing L toward low rank. Similarly, ℓ_1 penalizes the sum of absolute values of entries of S , promoting entries being exactly zero (sparsity) for the same reason that ℓ_1 promotes sparsity in LASSO.

Part VII — Optimization, Estimation, and Applied 3D Geometry

Matrix Calculus

Layout Conventions

Every chapter in this book has used derivatives of matrix expressions without stopping to formalize the rules. In Ch 17 we minimized $\|A\mathbf{x} - \mathbf{b}\|^2$ by setting its gradient to zero and obtained the normal equations $A^T A \mathbf{x} = A^T \mathbf{b}$. In Ch 20 we set $\partial \mathcal{L} / \partial \mathbf{u} = \mathbf{0}$ for the PCA Lagrangian and recovered $\Sigma \mathbf{u} = \lambda \mathbf{u}$. Both steps relied on a quiet choice: the derivative of a scalar with respect to a vector is itself a **column vector**. That choice is one of two competing layout conventions in the literature, and mixing them silently is the source of a large fraction of backpropagation bugs.

This section makes the choice explicit and gives you the tools to translate between conventions whenever you encounter the other one.

The Two Conventions

Let $f : \mathbb{R}^n \rightarrow \mathbb{R}$ be a differentiable scalar function and $\mathbf{x} \in \mathbb{R}^n$.

Denominator layout (also called *gradient layout* or *Hessian layout*):

$$\frac{\partial f}{\partial \mathbf{x}} \stackrel{\text{def}}{=} \begin{pmatrix} \partial f / \partial x_1 \\ \vdots \\ \partial f / \partial x_n \end{pmatrix} \in \mathbb{R}^{n \times 1}.$$

The result has the same shape as the **denominator** $\mathbf{x}$.

Numerator layout (also called *Jacobian layout*):

$$\frac{\partial f}{\partial \mathbf{x}} \stackrel{\text{def}}{=} (\partial f / \partial x_1 \quad \cdots \quad \partial f / \partial x_n) \in \mathbb{R}^{1 \times n}.$$

The result has the same shape as the **numerator** f (a 1×1 scalar treated as having one row, so the derivative gains one column per denominator element).

Convention used throughout this book: We use **denominator layout** for scalar-by-vector derivatives. $\frac{\partial f}{\partial \mathbf{x}}$ is always a **column vector**, the same shape as $\mathbf{x}$. This matches NumPy's default array orientation, keeps gradients the same shape as parameters, and is consistent with every derivation in Ch 17 and Ch 20.

Extension to Vector Functions

Let $\mathbf{f}: \mathbb{R}^n \rightarrow \mathbb{R}^m$ with outputs $f_1(\mathbf{x}), \dots, f_m(\mathbf{x})$. The **Jacobian** is defined as

$$J = \frac{\partial \mathbf{f}}{\partial \mathbf{x}} \in \mathbb{R}^{m \times n}, \quad J_{ij} = \frac{\partial f_i}{\partial x_j}.$$

Row i of J is $(\nabla_{\mathbf{x}} f_i)^T$ (the gradient of f_i transposed into a row); column j tells how much every output changes when x_j increases by one. The Jacobian maps a small input perturbation $\delta \mathbf{x} \in \mathbb{R}^n$ to the resulting output perturbation:

$$\delta \mathbf{f} \approx J \delta \mathbf{x}.$$

With this definition the **chain rule** for a scalar loss L computed via $\mathbf{f}$ takes the form

$$\frac{\partial L}{\partial \mathbf{x}} = J^T \frac{\partial L}{\partial \mathbf{f}} \in \mathbb{R}^n,$$

where both $\partial L / \partial \mathbf{f} \in \mathbb{R}^m$ and $\partial L / \partial \mathbf{x} \in \mathbb{R}^n$ are column vectors (denominator layout), and the Jacobian appears **transposed** – a fact that resurfaces in §21.5 when we derive the backpropagation equations for a linear layer.

Conversion Table

Quantity	Denominator layout (this book)	Numerator layout
$\partial f / \partial \mathbf{x}$, f scalar	column $n \times 1$	row $1 \times n$
$\partial \mathbf{f} / \partial \mathbf{x}$, $\mathbf{f}: \mathbb{R}^n \rightarrow \mathbb{R}^m$	$m \times n$ (Jacobian J)	$m \times n$ (same)
Gradient update $\mathbf{x} \leftarrow \mathbf{x} - \alpha \nabla f$	$\mathbf{x}$ and ∇f both $n \times 1$	must transpose: $\mathbf{x} - \alpha (\nabla f)^T$
Chain rule for scalar L through $\mathbf{f}$	$J^T (\partial L / \partial \mathbf{f})$	$(\partial L / \partial \mathbf{f}^T) J$

The only difference that matters in practice: **gradients of scalars are column vectors in denominator layout and row vectors in numerator layout**. The Jacobian $J_{ij} = \partial f_i / \partial x_j$ is the same $m \times n$ matrix in both conventions. Translating between conventions reduces to transposing gradient vectors.

Numerical Verification

Concretely, $\nabla_{\mathbf{x}} \|\mathbf{x}\|^2 = 2\mathbf{x}$ in denominator layout:

```
import numpy as np

def f(x):
    return float(x @ x)

def grad_f_fd(x, h=1e-5):
    """Central-difference gradient approximation."""
    n = len(x)
    g = np.zeros(n)          # shape (n,)
    for i in range(n):
        e = np.zeros(n); e[i] = h
        g[i] = (f(x + e) - f(x - e)) / (2 * h)
    return g

rng = np.random.default_rng(211)
x = rng.standard_normal(5)   # shape (5,)

g_exact = 2.0 * x            # denominator layout: column vector, shape (5,)
g_fd     = grad_f_fd(x)      # shape (5,)

print("x          :", np.round(x, 4))
print("2x (exact) :", np.round(g_exact, 4))
print("FD approx  :", np.round(g_fd, 4))
print("max |error| :", np.max(np.abs(g_exact - g_fd)))
print("gradient shape:", g_exact.shape, " <- column vector (n,)")
```

```
x          : [-1.2245 -0.0268 -1.0775  0.3647  0.6075]
2x (exact) : [-2.4491 -0.0537 -2.1551  0.7294  1.2151]
FD approx  : [-2.4491 -0.0537 -2.1551  0.7294  1.2151]
max |error| : 3.065681042357937e-11
gradient shape: (5,) <- column vector (n,)
```

The gradient is a $(5,)$ array – a column vector in NumPy’s convention. The finite-difference approximation matches the exact formula to machine precision (error $\sim 10^{-10}$, set by h^2).

A Note on the Deep Learning Literature

Many deep learning frameworks (PyTorch, JAX) and some textbooks implicitly use numerator layout: a gradient that is broadcast over a batch is a row in a (batch, n) tensor, not a column. When reading papers or framework source code, a quick sanity check is to verify the shape of

the gradient before and after a matrix multiply. If shapes do not follow denominator layout, transpose as needed.

Exercises

21.1.1 In denominator layout, what is the shape of $\partial f / \partial \mathbf{x}$ when $f : \mathbb{R}^3 \rightarrow \mathbb{R}$? What shape does the same derivative have in numerator layout?

21.1.2 The gradient of $f(\mathbf{x}) = \mathbf{c}^T \mathbf{x} + d$ (affine scalar function) is $\partial f / \partial \mathbf{x} = \mathbf{c}$ in denominator layout. What is the equivalent expression in numerator layout?

21.1.3 Suppose you read a paper that writes the parameter update as $\theta \leftarrow \theta - \alpha \partial L / \partial \theta$ where the gradient is implicitly a row vector. Is this numerator or denominator layout? How do you convert the update expression to denominator layout without changing the numerical result?

21.1.4 For $\mathbf{f} : \mathbb{R}^3 \rightarrow \mathbb{R}^4$, the Jacobian $J_{ij} = \partial f_i / \partial x_j$. What are the dimensions of J ? If $\mathbf{g} = \partial L / \partial \mathbf{f} \in \mathbb{R}^4$ is a column vector, verify dimensionally that $J^T \mathbf{g} \in \mathbb{R}^3$ – consistent with $\partial L / \partial \mathbf{x}$ being a column vector of size 3.

Scalar-by-Vector: Gradients

A **gradient** collects all partial derivatives of a scalar function into one column vector that points in the direction of steepest ascent. The three identities below appear so frequently in ML, CV, and robotics that their derivations are worth understanding from first principles.

Three Fundamental Identities

Identity 1: Linear function

$$\nabla_{\mathbf{x}}(\mathbf{a}^T \mathbf{x}) = \mathbf{a}.$$

Derivation. $\mathbf{a}^T \mathbf{x} = \sum_i a_i x_i$, so $\partial / \partial x_k (\sum_i a_i x_i) = a_k$. Assembling all k into a column vector gives $\mathbf{a}$. The gradient of a linear function is its coefficient vector – constant, independent of $\mathbf{x}$.

Identity 2: Quadratic form

$$\nabla_{\mathbf{x}}(\mathbf{x}^T A \mathbf{x}) = (A + A^T)\mathbf{x}.$$

When A is symmetric ($A = A^T$) this simplifies to $2A\mathbf{x}$.

Derivation. Write $\mathbf{x}^T A \mathbf{x} = \sum_{i,j} A_{ij} x_i x_j$. Differentiating with respect to x_k , each term $A_{ij} x_i x_j$ contributes only when $i = k$ (factor $A_{kj} x_j$) or $j = k$ (factor $A_{ik} x_i$):

$$\frac{\partial}{\partial x_k} \sum_{i,j} A_{ij} x_i x_j = \sum_j A_{kj} x_j + \sum_i A_{ik} x_i = (A\mathbf{x})_k + (A^T \mathbf{x})_k.$$

Assembling over k : $\nabla_{\mathbf{x}}(\mathbf{x}^T A \mathbf{x}) = (A + A^T)\mathbf{x}$.

Identity 3: Squared residual

$$\nabla_{\mathbf{x}} \|A\mathbf{x} - \mathbf{b}\|^2 = 2A^T(A\mathbf{x} - \mathbf{b}).$$

Derivation. Let $\mathbf{r} = A\mathbf{x} - \mathbf{b}$. Then $\|\mathbf{r}\|^2 = \mathbf{x}^T (A^T A) \mathbf{x} - 2\mathbf{b}^T A \mathbf{x} + \mathbf{b}^T \mathbf{b}$. Applying Identities 1 and 2 term by term:

$$\nabla_{\mathbf{x}} \|\mathbf{r}\|^2 = 2A^T A \mathbf{x} - 2A^T \mathbf{b} = 2A^T (A\mathbf{x} - \mathbf{b}).$$

Setting this to zero recovers the **normal equations** $A^T A \mathbf{x} = A^T \mathbf{b}$ (§17.4) – the same result we derived geometrically in Ch 17, now via pure calculus.

Numerical Verification

```
import numpy as np

rng = np.random.default_rng(2121)
n = 4
m = 6

a = rng.standard_normal(n)           # shape (4,)
x = rng.standard_normal(n)           # shape (4,)
A2 = rng.standard_normal((n, n))     # shape (4, 4), not symmetric
A3 = rng.standard_normal((m, n))     # shape (6, 4)
b = rng.standard_normal(m)           # shape (6,)
```

```
def fd_grad(f_scalar, x, h=1e-5):
    """Central-difference gradient."""
    g = np.zeros_like(x)
    for i in range(len(x)):
        e = np.zeros_like(x); e[i] = h
        g[i] = (f_scalar(x + e) - f_scalar(x - e)) / (2 * h)
    return g

# Identity 1
g1_exact = a.copy()
g1_fd = fd_grad(lambda x: float(a @ x), x)
print(f"Identity 1 max|error| = {np.max(np.abs(g1_exact - g1_fd)):.2e}")

# Identity 2
g2_exact = (A2 + A2.T) @ x
g2_fd = fd_grad(lambda x: float(x @ A2 @ x), x)
print(f"Identity 2 max|error| = {np.max(np.abs(g2_exact - g2_fd)):.2e}")

# Identity 3
g3_exact = 2 * A3.T @ (A3 @ x - b)      # shape (4,)
g3_fd = fd_grad(lambda x: float((A3 @ x - b) @ (A3 @ x - b)), x)
print(f"Identity 3 max|error| = {np.max(np.abs(g3_exact - g3_fd)):.2e}")
```

```
Identity 1 max|error| = 5.25e-12
Identity 2 max|error| = 3.34e-12
Identity 3 max|error| = 2.00e-11
```

The Gradient as a Landscape

The gradient $\nabla_x f$ points uphill in $\mathbb{R}^n$. For a 2D function the gradient field is a quiver plot: each arrow at (x_1, x_2) points in the direction of steepest ascent, with length proportional to $\|\nabla f\|$. Gradient **descent** follows the negative gradient.

```
import numpy as np
import matplotlib.pyplot as plt

# f(x1, x2) = x1^2 + 3*x2^2 + x1*x2 (quadratic with coupling)
# Gradient: [2*x1 + x2, 6*x2 + x1]

x1v = np.linspace(-2.5, 2.5, 200)      # shape (200,)
x2v = np.linspace(-2.5, 2.5, 200)      # shape (200,)
X1, X2 = np.meshgrid(x1v, x2v)         # each shape (200, 200)
```

```

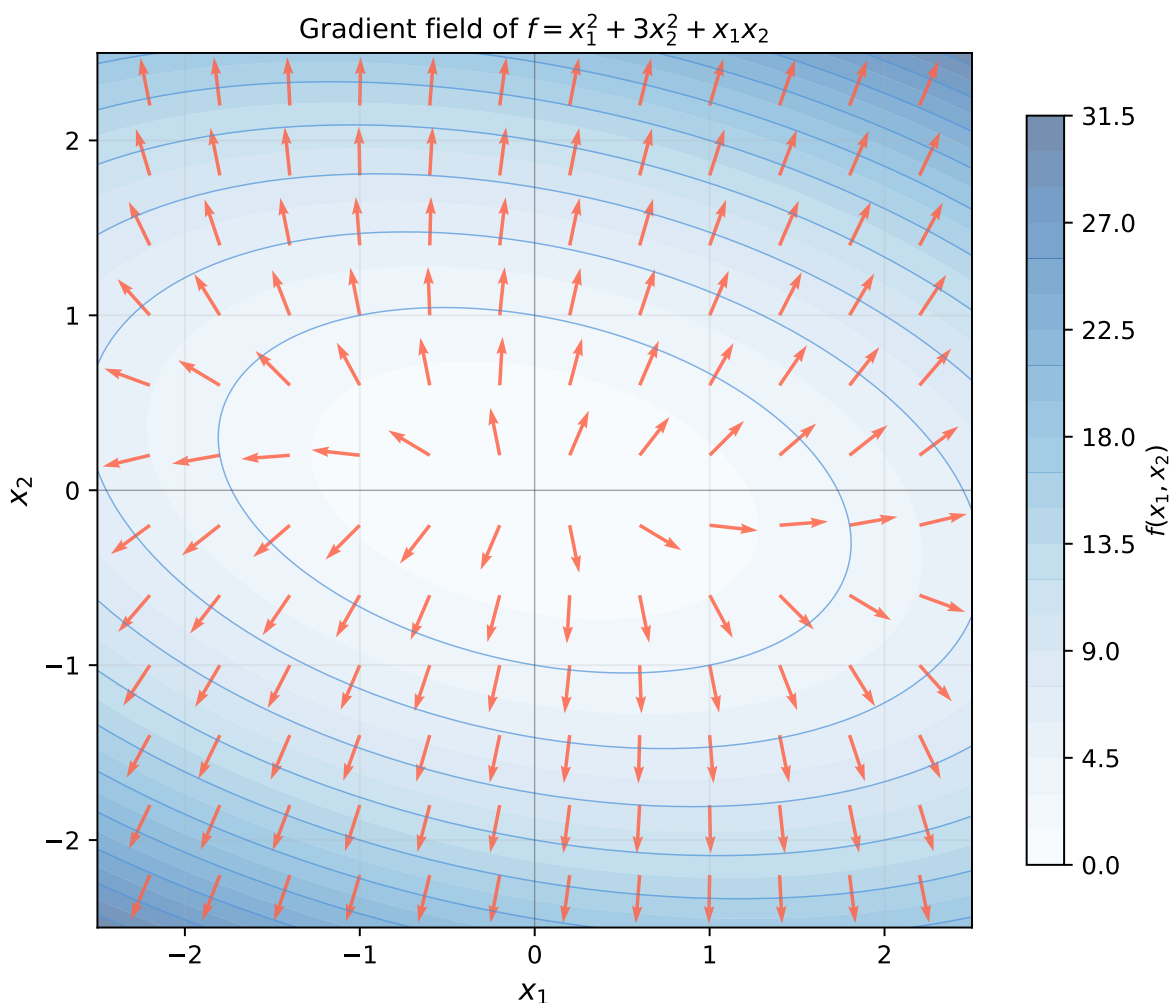
F = X1**2 + 3*X2**2 + X1*X2      # shape (200, 200)
G1 = 2*X1 + X2                   # dF/dx1
G2 = 6*X2 + X1                   # dF/dx2

# Coarser quiver grid
xq = np.linspace(-2.2, 2.2, 12)
yq = np.linspace(-2.2, 2.2, 12)
Xq, Yq = np.meshgrid(xq, yq)
Uq = 2*Xq + Yq
Vq = 6*Yq + Xq
nrm = np.sqrt(Uq**2 + Vq**2) + 1e-12
Uq, Vq = Uq/nrm, Vq/nrm          # unit-length arrows for clarity

fig, ax = plt.subplots(figsize=(7, 6))
ct = ax.contourf(X1, X2, F, levels=20, cmap='Blues', alpha=0.55)
plt.colorbar(ct, ax=ax, shrink=0.8, label=r'$f(x_1, x_2)$')
ax.contour(X1, X2, F, levels=10, colors='#4a90d9', linewidths=0.6, alpha=0.7)
ax.quiver(Xq, Yq, Uq, Vq, color='tomato', scale=18, width=0.004, alpha=0.85)

ax.axhline(0, color='#333333', lw=0.4, alpha=0.5)
ax.axvline(0, color='#333333', lw=0.4, alpha=0.5)
ax.set_xlabel(r'$x_1$', fontsize=12)
ax.set_ylabel(r'$x_2$', fontsize=12)
ax.set_title(r'Gradient field of $f=x_1^2+3x_2^2+x_1x_2$', fontsize=11)
ax.set_aspect('equal')
ax.grid(alpha=0.2)
fig.tight_layout()
plt.savefig('ch21-matrix-calculus/fig-gradient-field.png', dpi=150, bbox_inches='tight')
plt.show()

```



Arrows point away from the minimum (the origin); contour lines are **perpendicular** to the gradient at every point. Gradient descent follows the arrows reversed (step along $-\nabla f$).

Quick-Reference: Common Loss Gradients

Loss term	Gradient $\nabla_{\mathbf{x}}$
$\mathbf{c}^T \mathbf{x}$	$\mathbf{c}$
$\mathbf{x}^T \Sigma^{-1} \mathbf{x}$ (Σ symmetric)	$2\Sigma^{-1} \mathbf{x}$
$\ A\mathbf{x} - \mathbf{b}\ ^2$	$2A^T(A\mathbf{x} - \mathbf{b})$
$\lambda\ \mathbf{x}\ ^2$	$2\lambda\mathbf{x}$
$\ A\mathbf{x} - \mathbf{b}\ ^2 + \lambda\ \mathbf{x}\ ^2$	$2(A^T A + \lambda I)\mathbf{x} - 2A^T \mathbf{b}$

The last row is the gradient of the Tikhonov (ridge) objective (§17.7): setting it to zero gives the regularized normal equations $(A^T A + \lambda I)\mathbf{x}^* = A^T \mathbf{b}$.

Exploration

Dataset: The [California Housing dataset](#) has 8 numerical features. Fit a linear regression model by minimizing $\|A\mathbf{w} - \mathbf{b}\|^2$ using gradient descent: start from $\mathbf{w} = \mathbf{0}$, compute the gradient at each step, and update $\mathbf{w} \leftarrow \mathbf{w} - \alpha \nabla_{\mathbf{w}} L$. Plot loss vs. iteration and compare the gradient-descent solution to the direct least-squares solution from `np.linalg.lstsq`.

Exercises

21.2.1 Derive $\nabla_{\mathbf{x}}(\mathbf{x}^T \mathbf{x})$ two ways: (a) by direct component-wise differentiation; (b) by treating it as a quadratic form $\mathbf{x}^T A \mathbf{x}$ with $A = I$ and applying Identity 2. Confirm both give $2\mathbf{x}$.

21.2.2 Let $f(\mathbf{x}) = (\mathbf{x} - \mu)^T \Sigma^{-1} (\mathbf{x} - \mu)$ (Mahalanobis squared distance, §19.4). Compute $\nabla_{\mathbf{x}} f$. (*Hint:* substitute $\mathbf{y} = \mathbf{x} - \mu$ and apply Identity 2 with $A = \Sigma^{-1}$, which is symmetric.)

21.2.3 Verify that setting $\nabla_{\mathbf{x}} \|A\mathbf{x} - \mathbf{b}\|^2 = \mathbf{0}$ yields the normal equations $A^T A \mathbf{x} = A^T \mathbf{b}$.

21.2.4 The regularized objective is $L(\mathbf{x}) = \|A\mathbf{x} - \mathbf{b}\|^2 + \lambda \|\mathbf{x}\|^2$. Compute $\nabla_{\mathbf{x}} L$ and derive the closed-form minimizer $\mathbf{x}^*$. Identify this as the ridge regression solution (§17.7).

21.2.5 Is the gradient of a constant function $f(\mathbf{x}) = c$ necessarily $\mathbf{0}$? What does the gradient of an affine function $f(\mathbf{x}) = \mathbf{c}^T \mathbf{x} + d$ tell you about the loss surface?

Vector-by-Vector: Jacobians

When the output of a function is itself a vector, a single scalar gradient is no longer enough to describe how all outputs change with all inputs. The **Jacobian** packages this information into a matrix.

Definition

Let $\mathbf{f}: \mathbb{R}^n \rightarrow \mathbb{R}^m$, $\mathbf{f}(\mathbf{x}) = (f_1(\mathbf{x}), \dots, f_m(\mathbf{x}))^T$. The **Jacobian matrix** is

$$J = \frac{\partial \mathbf{f}}{\partial \mathbf{x}} \in \mathbb{R}^{m \times n}, \quad J_{ij} = \frac{\partial f_i}{\partial x_j}.$$

Row i is the gradient of f_i transposed; column j is the rate of change of all outputs with respect to input x_j . The Jacobian is the matrix of all first-order partial derivatives.

A useful way to remember the shape: J has as many **rows** as the **output** and as many **columns** as the **input**. If $f : \mathbb{R}^n \rightarrow \mathbb{R}^m$ then $J \in \mathbb{R}^{m \times n}$. For a scalar function ($m = 1$) the Jacobian is a row vector – the **transpose** of the gradient in denominator layout.

Key Examples

Example 1: Linear map $f(\mathbf{x}) = A\mathbf{x}$

$(f_i) = \sum_j A_{ij}x_j$, so $\partial f_i / \partial x_j = A_{ij}$. Therefore

$$J = A.$$

The Jacobian of a linear map is just its matrix. This is exact (not an approximation) because the function itself is linear.

Example 2: Element-wise sigmoid $\sigma(\mathbf{x})$

Define $\sigma(z) = 1/(1 + e^{-z})$, applied component-wise. Since f_i depends only on x_i :

$$J_{ij} = \frac{\partial \sigma(x_i)}{\partial x_j} = \begin{cases} \sigma(x_i)(1 - \sigma(x_i)) & i = j \\ 0 & i \neq j. \end{cases}$$

The Jacobian is **diagonal**: $J = \text{diag}(\sigma'(x_1), \dots, \sigma'(x_n))$. This sparsity is what makes element-wise nonlinearities cheap in backpropagation – only the diagonal entries are nonzero.

Example 3: ℓ^2 norm $f(\mathbf{x}) = \|\mathbf{x}\|$

This is a scalar function ($m = 1$), so its Jacobian is a $1 \times n$ row vector. $f(\mathbf{x}) = (\sum_i x_i^2)^{1/2}$, thus $\partial f / \partial x_j = x_j / \|\mathbf{x}\|$ and

$$J = \frac{\mathbf{x}^T}{\|\mathbf{x}\|} \in \mathbb{R}^{1 \times n}.$$

The corresponding gradient (denominator layout) is $\nabla_{\mathbf{x}} \|\mathbf{x}\| = \mathbf{x} / \|\mathbf{x}\|$ – a unit vector in the direction of $\mathbf{x}$.

Local Linearization

The Jacobian is the best linear approximation to $\mathbf{f}$ near a point $\mathbf{x}_0$. For a small perturbation $\delta\mathbf{x}$:

$$\mathbf{f}(\mathbf{x}_0 + \delta\mathbf{x}) \approx \mathbf{f}(\mathbf{x}_0) + J(\mathbf{x}_0) \delta\mathbf{x}.$$

This is the multivariate Taylor expansion truncated at first order. It underlies the extended Kalman filter (Ch 23), where the prediction and observation models are nonlinear but are **linearized** via their Jacobians to enable the standard Kalman update. The error in the approximation is $O(\|\delta\mathbf{x}\|^2)$.

Geometric Picture: Unit Disk Mapped to an Ellipse

A linear map $\mathbf{f}(\mathbf{x}) = A\mathbf{x}$ sends the unit disk $\{\mathbf{x} : \|\mathbf{x}\| \leq 1\} \subset \mathbb{R}^2$ to an ellipse in $\mathbb{R}^2$ (for square A). The Jacobian equals A , and the singular values of A determine the semi-axis lengths of the ellipse.

```
import numpy as np
import matplotlib.pyplot as plt

# Unit circle parametrized by angle
theta = np.linspace(0, 2*np.pi, 500) # shape (500,)
circle = np.vstack([np.cos(theta), np.sin(theta)]) # shape (2, 500)

# A linear map with clear stretching and rotation
A = np.array([[2.0, 0.8],
              [0.4, 1.5]]) # shape (2, 2)

ellipse = A @ circle # shape (2, 500) -- Jacobian = A

# SVD to find semi-axes
U, s, Vt = np.linalg.svd(A) # s shape (2,)
V = Vt.T # columns = right singular vectors

fig, axes = plt.subplots(1, 2, figsize=(11, 5))

for ax, pts, title, col in zip(
    axes,
    [circle, ellipse],
    ['Unit circle (input)', r'$A\mathbf{x}$ (output ellipse)'],
    ['#4a90d9', '#2ecc71']):
    ax.fill(pts[0], pts[1], color=col, alpha=0.25)
```

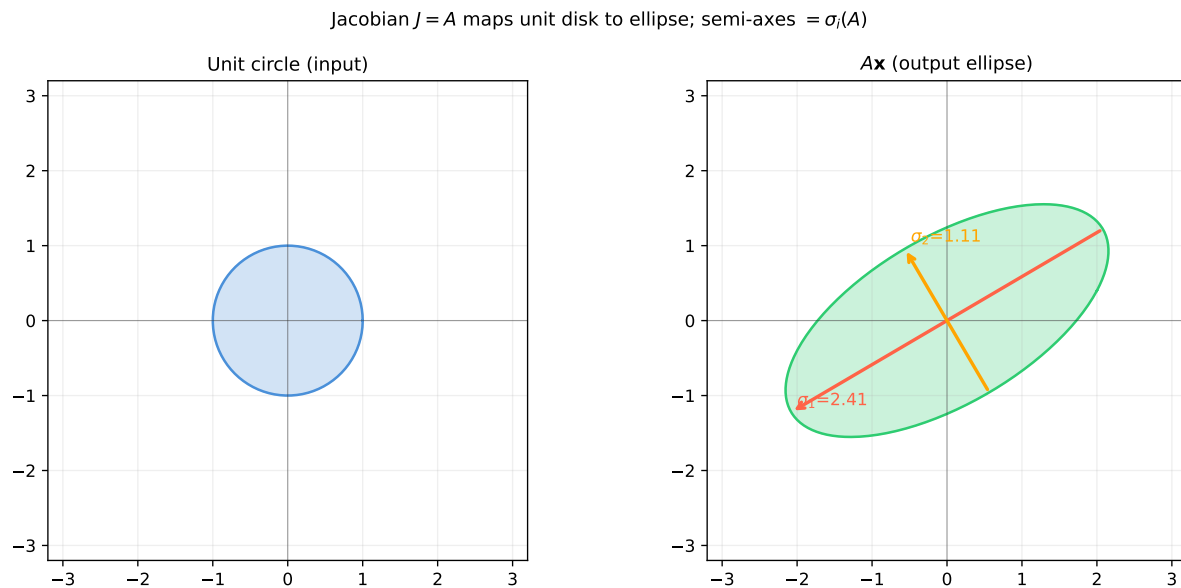
```

ax.plot(pts[0], pts[1], color=col, lw=1.5)
ax.axhline(0, color='#333333', lw=0.4, alpha=0.5)
ax.axvline(0, color='#333333', lw=0.4, alpha=0.5)
ax.set_aspect('equal')
ax.set_title(title, fontsize=11)
ax.grid(alpha=0.2)

# Annotate semi-axes on ellipse panel
ax = axes[1]
for i, (si, col_a, lbl) in enumerate(
    zip(s, ['tomato', 'orange'], [r'\sigma_1$', r'\sigma_2$']))
):
    v = U[:, i] # left singular vector (output direction)
    ax.annotate('', xy=si*v, xytext=-si*v,
                arrowprops=dict(arrowstyle='->', color=col_a, lw=2))
    ax.text(*(si*v + np.array([0.08, 0.08])), f'{lbl}={si:.2f}',
            fontsize=10, color=col_a)

axes[0].set_xlim(-3.2, 3.2); axes[0].set_ylim(-3.2, 3.2)
axes[1].set_xlim(-3.2, 3.2); axes[1].set_ylim(-3.2, 3.2)
fig.suptitle(r'Jacobian  $J=A$  maps unit disk to ellipse; semi-axes  $=\sigma_i(A)$ ',
            fontsize=11)
fig.tight_layout()
plt.savefig('ch21-matrix-calculus/fig-jacobian-ellipse.png', dpi=150, bbox_inches='tight')
plt.show()
print(f"Singular values of A: sigma_1={s[0]:.4f}, sigma_2={s[1]:.4f}")
print(f"det(A) = {np.linalg.det(A):.4f} (= sigma_1 * sigma_2 = area scale factor)")

```



Singular values of A: $\sigma_1=2.4114$, $\sigma_2=1.1114$

`det(A) = 2.6800 (= sigma_1 * sigma_2 = area scale factor)`

The unit circle stretches into an ellipse whose semi-axis lengths are the singular values of A (= the Jacobian). The determinant $\det(A) = \sigma_1 \sigma_2$ measures the **local area scaling factor** – how much a small patch of input space expands or contracts under $\mathbf{f}$.

Numerical Verification

```
import numpy as np

rng = np.random.default_rng(2131)
n, m = 3, 4

# Test function: f_i(x) = sin(a_i^T x) -- nonlinear
A = rng.standard_normal((m, n))      # shape (4, 3)
x0 = rng.standard_normal(n)          # shape (3,)

def f_vec(x):
    return np.sin(A @ x)              # shape (4,)

# Exact Jacobian: J_ij = d/dx_j sin(sum_k A_ik x_k) = cos(A_i . x) * A_ij
J_exact = np.diag(np.cos(A @ x0)) @ A # shape (4, 3)

# Finite-difference Jacobian
h = 1e-5
J_fd = np.zeros((m, n))               # shape (4, 3)
for j in range(n):
    e = np.zeros(n); e[j] = h
    J_fd[:, j] = (f_vec(x0 + e) - f_vec(x0 - e)) / (2 * h)

print("Jacobian shape:", J_exact.shape)
print("Max |exact - FD|:", np.max(np.abs(J_exact - J_fd)))

# Verify local linearization
dxs = rng.standard_normal(n) * 0.01   # small perturbation, shape (3,)
df_linear = J_exact @ dxs              # shape (4,)
df_exact = f_vec(x0 + dxs) - f_vec(x0) # shape (4,)
print(f"Linearization error (||dx||={np.linalg.norm(dxs):.4f}):",
      np.max(np.abs(df_exact - df_linear)))
```

Jacobian shape: (4, 3)

Max |exact - FD|: 2.993649772520257e-10

Linearization error (||dx||=0.0178): 0.000182810299599696

The linearization error is $O(\|\delta \mathbf{x}\|^2) \approx 10^{-6}$ for $\|\delta \mathbf{x}\| \approx 0.01$ – confirming the quadratic falloff of the Taylor approximation error.

Exercises

21.3.1 Write out the Jacobian of $\mathbf{f}(\mathbf{x}) = A\mathbf{x} + \mathbf{b}$ (affine map) for $A \in \mathbb{R}^{m \times n}$. Why does the constant vector $\mathbf{b}$ not appear in the Jacobian?

21.3.2 Compute the Jacobian of the element-wise ReLU: $f_i(x) = \max(0, x_i)$. Is the Jacobian always well-defined? What happens at $x_i = 0$?

21.3.3 Let $\mathbf{f}(\mathbf{x}) = \mathbf{x}/\|\mathbf{x}\|$ (unit normalization, $\mathbb{R}^n \rightarrow \mathbb{R}^n$). Show that $J = \frac{1}{\|\mathbf{x}\|}(I - \hat{\mathbf{x}}\hat{\mathbf{x}}^T)$ where $\hat{\mathbf{x}} = \mathbf{x}/\|\mathbf{x}\|$. Interpret the rank of J .

21.3.4 For a differentiable map $\mathbf{f}: \mathbb{R}^n \rightarrow \mathbb{R}^m$ and $\mathbf{g}: \mathbb{R}^m \rightarrow \mathbb{R}^p$, the Jacobian of $\mathbf{h} = \mathbf{g} \circ \mathbf{f}$ is $J_h = J_g J_f$ (matrix chain rule). Verify dimensionally that $J_h \in \mathbb{R}^{p \times n}$.

21.3.5 The local linearization says $\mathbf{f}(\mathbf{x}_0 + \delta \mathbf{x}) \approx \mathbf{f}(\mathbf{x}_0) + J\delta \mathbf{x}$. For a linear function $\mathbf{f}(\mathbf{x}) = A\mathbf{x}$, is this approximation exact? Why?

Scalar-by-Matrix: Weight Gradients

In neural networks the parameters are matrices $W \in \mathbb{R}^{m \times n}$. Training requires computing $\partial L / \partial W$ – the derivative of a scalar loss with respect to every element of W . This section develops the key identities, beginning with an honest account of what $\partial(W\mathbf{x}) / \partial W$ actually is.

The Tensor Elephant in the Room

Consider the simplest case: $\mathbf{y} = W\mathbf{x}$ where $W \in \mathbb{R}^{m \times n}$, $\mathbf{x} \in \mathbb{R}^n$, and $\mathbf{y} \in \mathbb{R}^m$. The quantity $\partial \mathbf{y} / \partial W$ is the derivative of an m -vector with respect to an $m \times n$ matrix. Storing every partial derivative $\partial y_i / \partial W_{jk}$ requires an $m \times m \times n$ array – a **third-order tensor**.

That tensor is rarely written out in full. Instead, training uses the **scalar loss** $L: \mathbb{R} \rightarrow \mathbb{R}$, and the quantity we actually need is $\partial L / \partial W$ – the derivative of a **scalar** with respect to a matrix. This is a genuine $m \times n$ matrix (same shape as W), not a tensor.

The Backpropagation Shortcut: $\partial L / \partial W = \delta \mathbf{x}^T$

Suppose $\mathbf{y} = W\mathbf{x}$ and $L = L(\mathbf{y})$. Define the **error signal** $\delta = \partial L / \partial \mathbf{y} \in \mathbb{R}^m$ (a column vector; denominator layout from §21.1). By the chain rule applied element-wise:

$$\frac{\partial L}{\partial W_{ij}} = \sum_k \frac{\partial L}{\partial y_k} \frac{\partial y_k}{\partial W_{ij}} = \sum_k \delta_k \cdot [x_j \text{ if } k = i, \text{ else } 0] = \delta_i x_j.$$

Assembling all (i, j) pairs:

$$\boxed{\frac{\partial L}{\partial W} = \delta \mathbf{x}^T.}$$

This is an **outer product** of the upstream error δ and the input $\mathbf{x}$. The result has shape $m \times n$ = same as W . This identity is the heart of every neural network gradient update.

Three More Matrix-Derivative Identities

Identity: Trace form

$$\frac{\partial}{\partial W} \text{tr}(WA) = A^T.$$

Derivation. $\text{tr}(WA) = \sum_{i,j} W_{ij} A_{ji}$, so $\partial / \partial W_{ij} [\text{tr}(WA)] = A_{ji} = (A^T)_{ij}$. Assembling: $\partial \text{tr}(WA) / \partial W = A^T$.

Identity: Symmetric quadratic form

$$\frac{\partial}{\partial W} (\mathbf{x}^T W \mathbf{x}) = \mathbf{x} \mathbf{x}^T.$$

Derivation. $\mathbf{x}^T W \mathbf{x} = \sum_{i,j} x_i W_{ij} x_j$, so $\partial / \partial W_{ij} [\mathbf{x}^T W \mathbf{x}] = x_i x_j = (\mathbf{x} \mathbf{x}^T)_{ij}$.

Identity: Frobenius norm

$$\frac{\partial}{\partial W} \|W\|_F^2 = 2W.$$

Derivation. $\|W\|_F^2 = \sum_{i,j} W_{ij}^2$, so $\partial / \partial W_{ij} [\sum_{k,l} W_{kl}^2] = 2W_{ij}$. This identity appears in nuclear-norm and weight-decay regularization.

Summary Table

Expression	$\partial/\partial W$	Shape
$\text{tr}(WA)$	A^T	$m \times n$
$\mathbf{x}^T W \mathbf{x}$ (W square)	$\mathbf{x} \mathbf{x}^T$	$n \times n$
$\ W\ _F^2$	$2W$	$m \times n$
$L(\mathbf{y} = W\mathbf{x})$	$\delta \mathbf{x}^T, \delta = \nabla_{\mathbf{y}} L$	$m \times n$

Numerical Verification

```
import numpy as np

rng = np.random.default_rng(2141)
m, n = 3, 4
W = rng.standard_normal((m, n))    # shape (3, 4)
A = rng.standard_normal((n, m))    # shape (4, 3)
x = rng.standard_normal(n)         # shape (4,)
x_sq = rng.standard_normal(n)      # shape (4,) for the quadratic form test (W square)
Wsqr = rng.standard_normal((n, n)) # shape (4, 4)

def fd_mat_grad(f_scalar, W, h=1e-5):
    """Finite-difference gradient w.r.t. matrix W."""
    G = np.zeros_like(W)
    for i in range(W.shape[0]):
        for j in range(W.shape[1]):
            Wp = W.copy(); Wp[i,j] += h
            Wm = W.copy(); Wm[i,j] -= h
            G[i,j] = (f_scalar(Wp) - f_scalar(Wm)) / (2*h)
    return G

# Identity: d tr(WA)/dW = A^T
g_exact1 = A.T                                # shape (3, 4)
g_fd1 = fd_mat_grad(lambda W: np.trace(W @ A), W)
print(f"d tr(WA)/dW = A^T      max|error| = {np.max(np.abs(g_exact1 - g_fd1)):.2e}")

# Identity: d(x^T W x)/dW = x x^T (W square)
g_exact2 = np.outer(x_sq, x_sq)               # shape (4, 4)
g_fd2 = fd_mat_grad(lambda W: float(x_sq @ W @ x_sq), Wsqr)
print(f"d(x^T W x)/dW = x x^T max|error| = {np.max(np.abs(g_exact2 - g_fd2)):.2e}")

# Identity: d ||W||_F^2/dW = 2W
g_exact3 = 2 * W                             # shape (3, 4)
```

```

g_fd3      = fd_mat_grad(lambda W: float(np.sum(W**2)), W)
print(f"d||W||_F^2/dW = 2W      max|error| = {np.max(np.abs(g_exact3 - g_fd3)):.2e}")

# Identity: dL/dW = delta x^T  (L = 0.5 ||Wx - t||^2)
t = rng.standard_normal(m)          # shape (3,) target
y = W @ x                          # shape (3,)
delta = y - t                       # shape (3,) -- gradient of 0.5||y-t||^2
g_exact4 = np.outer(delta, x)        # shape (3, 4)
g_fd4      = fd_mat_grad(
    lambda W: 0.5 * float((W @ x - t) @ (W @ x - t)), W)
print(f"dL/dW = delta x^T      max|error| = {np.max(np.abs(g_exact4 - g_fd4)):.2e}")

d tr(WA)/dW = A^T      max|error| = 2.36e-11
d(x^T W x)/dW = x x^T  max|error| = 1.07e-11
d||W||_F^2/dW = 2W      max|error| = 9.96e-11
dL/dW = delta x^T      max|error| = 2.73e-11

```

All four identities are verified to near machine precision.

Exercises

21.4.1 The identity $\partial L / \partial W = \delta \mathbf{x}^T$ is an outer product. What is its rank? When could it be full rank, and when is it rank-1?

21.4.2 Derive $\partial \text{tr}(W^T A) / \partial W$ where A is a fixed matrix. (*Hint:* $\text{tr}(W^T A) = \text{tr}(A^T W)$; use the trace identity with appropriate substitution.)

21.4.3 Show that $\partial \|W\mathbf{x} - \mathbf{b}\|^2 / \partial W = 2(W\mathbf{x} - \mathbf{b})\mathbf{x}^T$. (*Hint:* write the loss as $\text{tr}[(W\mathbf{x} - \mathbf{b})(W\mathbf{x} - \mathbf{b})^T]$, or use the backprop shortcut with $\delta = 2(W\mathbf{x} - \mathbf{b})$.)

21.4.4 In weight-decay regularization, the total loss is $L_{\text{total}} = L_{\text{data}} + (\lambda/2)\|W\|_F^2$. What is $\partial L_{\text{total}} / \partial W$? If you are using gradient descent, how does the weight-decay term modify the parameter update?

21.4.5 Why is $\partial \mathbf{y} / \partial W$ (for $\mathbf{y} = W\mathbf{x}$) a 3-index tensor while $\partial L / \partial W$ is a matrix? Explain what additional information must be provided to collapse the tensor to a matrix.

The Chain Rule in Matrix Form

The chain rule for scalar functions says $(g \circ f)'(x) = g'(f(x)) \cdot f'(x)$. In higher dimensions each prime becomes a Jacobian, and the product becomes a **matrix multiplication**. This section traces that idea through a complete neural network layer.

The Matrix Chain Rule

Let $\mathbf{f}: \mathbb{R}^n \rightarrow \mathbb{R}^m$ and $\mathbf{g}: \mathbb{R}^m \rightarrow \mathbb{R}^p$. Define $\mathbf{h} = \mathbf{g} \circ \mathbf{f}$, so $\mathbf{h}: \mathbb{R}^n \rightarrow \mathbb{R}^p$. The Jacobian of $\mathbf{h}$ is the **product** of the individual Jacobians:

$$J_h = J_g J_f \in \mathbb{R}^{p \times n},$$

where $J_f \in \mathbb{R}^{m \times n}$ and $J_g \in \mathbb{R}^{p \times m}$. Shape check: $(p \times m)(m \times n) = (p \times n)$ – consistent.

For a **scalar** terminal loss L ($p = 1$), the gradient through the composition is

$$\frac{\partial L}{\partial \mathbf{x}} = J_f^T \frac{\partial L}{\partial \mathbf{f}} \in \mathbb{R}^n,$$

where $\partial L / \partial \mathbf{f} \in \mathbb{R}^m$ is the upstream gradient (denominator layout: a column vector). Each layer in a neural network propagates gradients backward through this rule.

End-to-End: A Single Linear Layer

The simplest complete example is a linear layer followed by a squared loss.

Forward pass:

$$\mathbf{y} = W\mathbf{x} + \mathbf{b}, \quad L = \frac{1}{2} \|\mathbf{y} - \mathbf{t}\|^2,$$

where $W \in \mathbb{R}^{m \times n}$, $\mathbf{x} \in \mathbb{R}^n$, $\mathbf{b} \in \mathbb{R}^m$, and $\mathbf{t} \in \mathbb{R}^m$ is the target.

Backward pass:

Step 1 – gradient of L with respect to $\mathbf{y}$:

$$\delta = \frac{\partial L}{\partial \mathbf{y}} = \mathbf{y} - \mathbf{t} \in \mathbb{R}^m.$$

Step 2 – gradient with respect to W (from §21.4):

$$\frac{\partial L}{\partial W} = \delta \mathbf{x}^T \in \mathbb{R}^{m \times n}.$$

Step 3 – gradient with respect to $\mathbf{b}$ (since $\partial \mathbf{y} / \partial \mathbf{b} = I$):

$$\frac{\partial L}{\partial \mathbf{b}} = \delta \in \mathbb{R}^m.$$

Step 4 – gradient with respect to $\mathbf{x}$ (to propagate to earlier layers; Jacobian of $W\mathbf{x}$ w.r.t. $\mathbf{x}$ is W , so we use $J^T \delta$):

$$\frac{\partial L}{\partial \mathbf{x}} = W^T \delta \in \mathbb{R}^n.$$

These four equations constitute one complete **backpropagation pass** through a linear layer. Notice the pattern: going backward, matrices appear **transposed**.

Code: Forward and Backward with Finite-Difference Verification

```
import numpy as np

rng = np.random.default_rng(2151)
m, n = 4, 3

W = rng.standard_normal((m, n))    # shape (4, 3)
b = rng.standard_normal(m)         # shape (4,)
x = rng.standard_normal(n)         # shape (3,)
t = rng.standard_normal(m)         # shape (4,) target

# --- Forward pass ---
y = W @ x + b                      # shape (4,)
L = 0.5 * float((y - t) @ (y - t))

# --- Backward pass ---
delta = y - t                      # shape (4,)    dL/dy
dL_dW = np.outer(delta, x)         # shape (4, 3) dL/dW = delta x^T
dL_db = delta.copy()               # shape (4,)    dL/db = delta
dL_dx = W.T @ delta                # shape (3,)    dL/dx = W^T delta

print(f"Forward: L = {L:.6f}")
print(f"dL/dW shape: {dL_dW.shape}")
print(f"dL/db shape: {dL_db.shape}")
print(f"dL/dx shape: {dL_dx.shape}")

# --- Finite-difference verification ---
h = 1e-5

def L_from_W(W_):
    y_ = W_ @ x + b
    return 0.5 * float((y_ - t) @ (y_ - t))

def L_from_b(b_):
    y_ = W @ x + b_
    return 0.5 * float((y_ - t) @ (y_ - t))
```

```

def L_from_x(x_):
    y_ = W @ x_ + b
    return 0.5 * float((y_ - t) @ (y_ - t))

# dL/dW by FD
dW_fd = np.zeros((m, n))
for i in range(m):
    for j in range(n):
        Wp = W.copy(); Wp[i,j] += h
        Wm = W.copy(); Wm[i,j] -= h
        dW_fd[i,j] = (L_from_W(Wp) - L_from_W(Wm)) / (2*h)

# dL/db by FD
db_fd = np.zeros(m)
for i in range(m):
    ep = np.zeros(m); ep[i] = h
    db_fd[i] = (L_from_b(b + ep) - L_from_b(b - ep)) / (2*h)

# dL/dx by FD
dx_fd = np.zeros(n)
for i in range(n):
    ep = np.zeros(n); ep[i] = h
    dx_fd[i] = (L_from_x(x + ep) - L_from_x(x - ep)) / (2*h)

print(f"\ndL/dW  max|exact - FD| = {np.max(np.abs(dL_dW - dW_fd)):.2e}")
print(f"dL/db   max|exact - FD| = {np.max(np.abs(dL_db - db_fd)):.2e}")
print(f"dL/dx   max|exact - FD| = {np.max(np.abs(dL_dx - dx_fd)):.2e}")

```

Forward: $L = 3.626794$

dL/dW shape: (4, 3)

dL/db shape: (4,)

dL/dx shape: (3,)

dL/dW max|exact - FD| = $1.26\text{e-}11$

dL/db max|exact - FD| = $2.44\text{e-}11$

dL/dx max|exact - FD| = $3.38\text{e-}11$

All three gradients match finite differences to near machine precision, confirming the backward-pass equations.

Stacking Layers: Computational Graph

For a two-layer network $L = \ell(\mathbf{y}_2)$, $\mathbf{y}_2 = W_2\phi(\mathbf{y}_1)$, $\mathbf{y}_1 = W_1\mathbf{x} + \mathbf{b}_1$ (where ϕ is element-wise), the chain rule applies sequentially:

$$\frac{\partial L}{\partial W_1} = \delta_1 \mathbf{x}^T, \quad \delta_1 = \text{diag}(\phi'(\mathbf{y}_1)) W_2^T \delta_2, \quad \delta_2 = \frac{\partial \ell}{\partial \mathbf{y}_2}.$$

Each layer's error signal δ is the upstream error transformed by the **transpose** of the weight matrix and element-wise multiplied by the **activation derivative** – precisely what automatic differentiation frameworks compute.

Exercises

21.5.1 In the single linear layer example, derive $\partial L / \partial \mathbf{b} = \delta$ from first principles using the chain rule (not by inspection). What is the Jacobian of $\mathbf{y} = W\mathbf{x} + \mathbf{b}$ with respect to $\mathbf{b}$?

21.5.2 Let $L = \frac{1}{2} \|\mathbf{y} - \mathbf{t}\|^2$ and $\mathbf{y} = \sigma(W\mathbf{x} + \mathbf{b})$ where σ is applied element-wise. Using the chain rule, derive $\partial L / \partial W$. Compare to the linear-layer case – what is the only difference?

21.5.3 Show that the gradient $\partial L / \partial \mathbf{x} = W^T \delta$ is consistent with the matrix chain rule $J_f^T(\partial L / \partial \mathbf{f})$ where $\mathbf{f}(\mathbf{x}) = W\mathbf{x} + \mathbf{b}$.

21.5.4 When would you use finite differences (as in the verification code above) in practice? What are the downsides compared to automatic differentiation?

21.5.5 Consider a loss $L = \frac{1}{N} \sum_{k=1}^N \frac{1}{2} \|W\mathbf{x}_k + \mathbf{b} - \mathbf{t}_k\|^2$ over a batch of N examples. Show that $\partial L / \partial W = \frac{1}{N} \sum_k \delta_k \mathbf{x}_k^T$. In matrix form, if $\Delta = [\delta_1, \dots, \delta_N] \in \mathbb{R}^{m \times N}$ and $X = [\mathbf{x}_1, \dots, \mathbf{x}_N] \in \mathbb{R}^{n \times N}$, write this as a single matrix expression.

The Hessian Matrix

The gradient ∇f tells you how steeply the loss climbs in each direction. The **Hessian** H encodes the **curvature**: how quickly the gradient itself changes, and therefore how quickly gradient descent will converge near a critical point.

Definition

Let $f : \mathbb{R}^n \rightarrow \mathbb{R}$ be twice differentiable. The Hessian is the $n \times n$ matrix of second-order partial derivatives:

$$H_{ij} = \frac{\partial^2 f}{\partial x_i \partial x_j}.$$

Equivalently, $H = \nabla^2 f$ (the gradient of the gradient), or

$$H = \frac{\partial}{\partial \mathbf{x}} (\nabla_{\mathbf{x}} f)^T = \frac{\partial^2 f}{\partial \mathbf{x} \partial \mathbf{x}^T}.$$

Symmetry. By Schwarz's theorem (equality of mixed partials under continuity), $H_{ij} = H_{ji}$ for all i, j . The Hessian is always **symmetric**: $H = H^T$. Since it is real and symmetric, all its eigenvalues are real (§13.4).

What the Eigenvalues Tell You

Applying the second-order Taylor expansion at a critical point $\mathbf{x}_0$ where $\nabla f(\mathbf{x}_0) = \mathbf{0}$:

$$f(\mathbf{x}_0 + \delta \mathbf{x}) \approx f(\mathbf{x}_0) + \frac{1}{2} \delta \mathbf{x}^T H \delta \mathbf{x}.$$

The sign of the quadratic form $\delta \mathbf{x}^T H \delta \mathbf{x}$ determines the nature of the critical point:

Hessian eigenvalues	Classification
All $\lambda_i > 0$ (H positive definite)	Local minimum – every direction curves up
All $\lambda_i < 0$ (H negative definite)	Local maximum – every direction curves down
Mixed signs	Saddle point – some directions up, some down
Some $\lambda_i = 0$	Inconclusive (degenerate case)

Saddle points are ubiquitous in deep learning loss surfaces: a network with many parameters nearly always has directions of negative curvature at any interior critical point. Gradient descent slows near saddle points because the gradient is small in those flat directions, but does eventually escape due to asymmetric curvature.

Example: Hessians of Common Loss Functions

Squared loss $f(\mathbf{x}) = \|\mathbf{Ax} - \mathbf{b}\|^2$: $\nabla f = 2A^T(\mathbf{Ax} - \mathbf{b})$, so $H = \nabla(\nabla f)^T = 2A^T A$. Since $A^T A$ is always positive semidefinite (PSD), the squared loss is **convex** – there are no saddle points, only a (possibly non-unique) global minimum.

Quadratic form $f(\mathbf{x}) = \mathbf{x}^T A \mathbf{x}$ with symmetric A : $H = 2A$.

Rosenbrock (non-convex) $f(x_1, x_2) = 100(x_2 - x_1^2)^2 + (1 - x_1)^2$:

$$H = \begin{pmatrix} 1200x_1^2 - 400x_2 + 2 & -400x_1 \\ -400x_1 & 200 \end{pmatrix}.$$

At the minimum $(1, 1)$: $H = \begin{pmatrix} 802 & -400 \\ -400 & 200 \end{pmatrix}$, which is positive definite (both eigenvalues > 0).

Newton's Method

Newton's method exploits the Hessian to take curvature-aware steps:

$$\mathbf{x}_{k+1} = \mathbf{x}_k - H^{-1} \nabla f(\mathbf{x}_k).$$

The step $-H^{-1} \nabla f$ solves for the minimum of the local quadratic approximation. Compared to gradient descent (step $-\alpha \nabla f$):

- **Gradient descent** step size α must be tuned; convergence is linear.
- **Newton's method** automatically accounts for curvature; convergence is quadratic near the minimum.

The cost: computing H requires $O(n^2)$ memory and $O(n^3)$ to invert. For large neural networks ($n \sim 10^8$ parameters) exact Newton's method is infeasible. Practical alternatives include **quasi-Newton** methods (L-BFGS, which approximates H^{-1} without forming H) and **Gauss-Newton**.

Gauss-Newton Approximation

For a least-squares loss $L(\mathbf{x}) = \frac{1}{2} \|\mathbf{r}(\mathbf{x})\|^2$ with residual vector $\mathbf{r} : \mathbb{R}^n \rightarrow \mathbb{R}^m$:

$$\nabla L = J_r^T \mathbf{r}, \quad H = J_r^T J_r + \sum_i r_i H_i \approx J_r^T J_r,$$

where H_i is the Hessian of the i -th residual. Dropping the second-order term $\sum r_i H_i$ (small near the solution) gives the **Gauss-Newton** approximation $H \approx J_r^T J_r$. This matrix is always positive semidefinite – no saddle points, and no need to compute second derivatives. It is the foundation of the Levenberg-Marquardt optimizer used in ICP, bundle adjustment, and graph-SLAM (Ch 22, Ch 24).

Visualization: Bowl vs. Saddle

```

import numpy as np
import matplotlib.pyplot as plt

x1v = np.linspace(-2.5, 2.5, 200)    # shape (200,)
x2v = np.linspace(-2.5, 2.5, 200)    # shape (200,)
X1, X2 = np.meshgrid(x1v, x2v)      # each shape (200, 200)

# Bowl: f = x1^2 + 2*x2^2 (PD Hessian diag(2, 4))
F_bowl = X1**2 + 2*X2**2
# Saddle: f = x1^2 - x2^2 (indefinite Hessian diag(2, -2))
F_saddle = X1**2 - X2**2

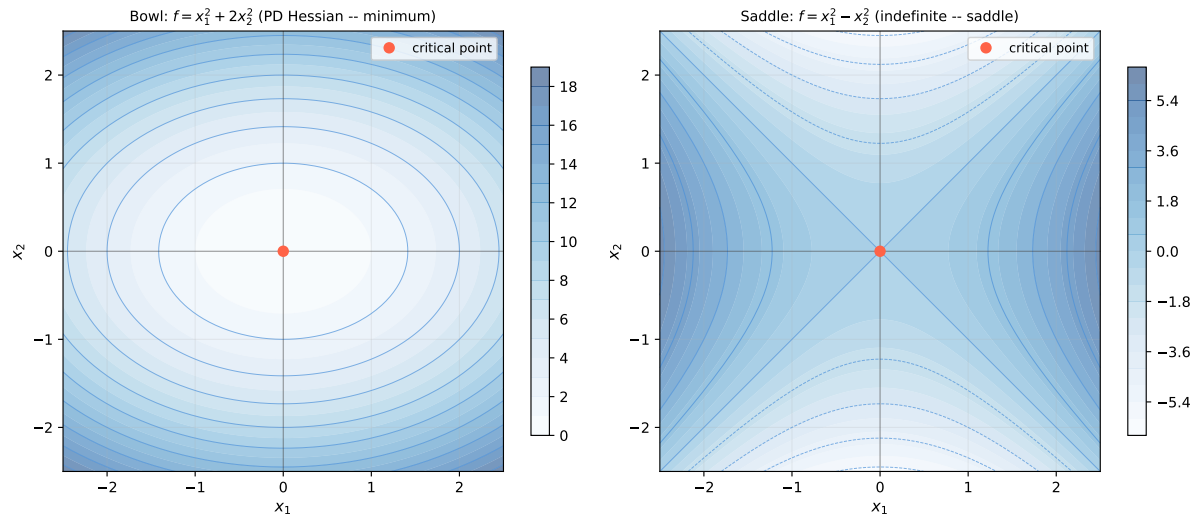
fig, axes = plt.subplots(1, 2, figsize=(12, 5))

titles = [
    r'Bowl: $f=x_1^2+2x_2^2$ (PD Hessian -- minimum)',
    r'Saddle: $f=x_1^2-x_2^2$ (indefinite -- saddle)'
]
Fs = [F_bowl, F_saddle]

for ax, F, title in zip(axes, Fs, titles):
    cf = ax.contourf(X1, X2, F, levels=20, cmap='Blues', alpha=0.55)
    plt.colorbar(cf, ax=ax, shrink=0.8)
    ax.contour(X1, X2, F, levels=10, colors='#4a90d9', linewidths=0.6, alpha=0.7)
    ax.axhline(0, color='#333333', lw=0.4, alpha=0.5)
    ax.axvline(0, color='#333333', lw=0.4, alpha=0.5)
    ax.set_xlabel(r'$x_1$', fontsize=11)
    ax.set_ylabel(r'$x_2$', fontsize=11)
    ax.set_title(title, fontsize=10)
    ax.set_aspect('equal')
    ax.grid(alpha=0.2)
    ax.plot(0, 0, 'o', color='tomato', ms=7, zorder=5, label='critical point')
    ax.legend(fontsize=9)

fig.tight_layout()
plt.savefig('ch21-matrix-calculus/fig-hessian-bowl-saddle.png',
           dpi=150, bbox_inches='tight')
plt.show()

```



For the bowl, the Hessian $H = \text{diag}(2, 4)$ has all positive eigenvalues (2, 4) – every direction curves upward, and the critical point at the origin is a minimum. For the saddle, $H = \text{diag}(2, -2)$ has mixed signs – the x_1 direction curves up but the x_2 direction curves down, so the origin is neither a minimum nor a maximum.

Numerical Hessian via Finite Differences

```
import numpy as np

def hessian_fd(f, x, h=1e-4):
    """Finite-difference Hessian via central differences."""
    n = len(x)
    H = np.zeros((n, n))          # shape (n, n)
    for i in range(n):
        for j in range(i, n):
            ei = np.zeros(n); ei[i] = h
            ej = np.zeros(n); ej[j] = h
            H[i, j] = (f(x+ei+ej) - f(x+ei-ej)
                       - f(x-ei+ej) + f(x-ei-ej)) / (4*h**2)
            H[j, i] = H[i, j]      # symmetry
    return H

# f = ||Ax - b||^2, exact Hessian = 2 A^T A
rng = np.random.default_rng(2161)
m, n = 5, 3
A = rng.standard_normal((m, n))    # shape (5, 3)
b = rng.standard_normal(m)         # shape (5,)
x = rng.standard_normal(n)         # shape (3,)
```

```

H_exact = 2 * A.T @ A          # shape (3, 3)
H_fd     = hessian_fd(lambda x: float((A@x-b) @ (A@x-b)), x)

print("Exact Hessian:\n", np.round(H_exact, 4))
print("\nFD Hessian:\n",   np.round(H_fd, 4))
print("\nMax |error|:", np.max(np.abs(H_exact - H_fd)))
print("Hessian eigenvalues:", np.round(np.linalg.eigvalsh(H_exact), 4),
      " -- all >= 0 (PSD): convex loss")

```

Exact Hessian:

```

[[ 2.5972  1.3837  0.4788]
 [ 1.3837  3.1051 -1.8519]
 [ 0.4788 -1.8519  5.7194]]

```

FD Hessian:

```

[[ 2.5972  1.3837  0.4788]
 [ 1.3837  3.1051 -1.8519]
 [ 0.4788 -1.8519  5.7194]]

```

Max |error|: 1.3946681143295336e-08

Hessian eigenvalues: [0.9017 3.8287 6.6913] -- all >= 0 (PSD): convex loss

Exercises

21.6.1 Compute the Hessian of $f(\mathbf{x}) = \mathbf{x}^T \Sigma^{-1} \mathbf{x}$ where Σ is a symmetric PD matrix. Is the Hessian positive definite? What does this imply about the shape of the Mahalanobis distance landscape?

21.6.2 For the saddle function $f(x_1, x_2) = x_1^2 - x_2^2$, verify by hand that the Hessian is indefinite (mixed eigenvalue signs). In which direction does the function decrease from the origin?

21.6.3 Why does Newton's method converge quadratically near a minimum but may diverge or oscillate far from it? What modification (e.g., line search or damping) is typically added in practice?

21.6.4 For the Gauss-Newton approximation $H \approx J_r^T J_r$, explain why this matrix is always positive semidefinite. Under what condition is it positive definite (and hence the Newton system has a unique solution)?

21.6.5 The condition number $\kappa(H) = \lambda_{\max}/\lambda_{\min}$ controls how quickly gradient descent converges: the number of iterations scales as $O(\kappa(H) \log(1/\varepsilon))$. Given the Hessian in the numerical example above, compute $\kappa(H)$ and predict which direction gradient descent converges slowest.

Key Identity Reference Table

The table below collects the identities used most often in ML, CV, and robotics. All derivatives follow **denominator layout** (column-vector gradients; Jacobian $J_{ij} = \partial f_i / \partial x_j \in \mathbb{R}^{m \times n}$) as established in §21.1. Derivation sketches for every identity are in **Appendix D**.

Category 1: Scalar-by-Vector (Gradients)

Expression $f(\mathbf{x})$	$\nabla_{\mathbf{x}} f$	Conditions
$\mathbf{a}^T \mathbf{x}$	$\mathbf{a}$	
$\mathbf{x}^T \mathbf{a}$	$\mathbf{a}$	same as above
$\mathbf{x}^T A \mathbf{x}$	$(A + A^T) \mathbf{x}$	
$\mathbf{x}^T A \mathbf{x}$	$2A \mathbf{x}$	A symmetric
$\ A \mathbf{x} - \mathbf{b}\ ^2$	$2A^T(A \mathbf{x} - \mathbf{b})$	
$\ \mathbf{x}\ ^2$	$2\mathbf{x}$	
$(\mathbf{x} - \mu)^T \Sigma^{-1} (\mathbf{x} - \mu)$	$2\Sigma^{-1}(\mathbf{x} - \mu)$	Σ symmetric PD

Category 2: Scalar-by-Matrix (Weight Gradients)

Expression $f(W)$	$\partial f / \partial W$	Conditions
$\text{tr}(WA)$	A^T	
$\text{tr}(AW)$	A^T	same as above
$\text{tr}(W^T A)$	A	
$\text{tr}(AWBW^T)$	$A^T W B^T + A W B$	
$\mathbf{a}^T W \mathbf{b}$	$\mathbf{a} \mathbf{b}^T$	
$\mathbf{x}^T W \mathbf{x}$	$\mathbf{x} \mathbf{x}^T$	W square
$\ W\ _F^2$	$2W$	
$L(W \mathbf{x} + \mathbf{b})$	$(\nabla_{\mathbf{y}} L) \mathbf{x}^T$	$\mathbf{y} = W \mathbf{x} + \mathbf{b}$

Category 3: Vector-by-Vector (Jacobians)

Map $\mathbf{f}(\mathbf{x})$	Jacobian $J = \partial \mathbf{f} / \partial \mathbf{x}$	Shape
$A \mathbf{x} + \mathbf{b}$	A	$m \times n$
$\mathbf{x}^{\odot 2}$ (element-wise square)	$2 \text{diag}(\mathbf{x})$	$n \times n$

Map $\mathbf{f}(\mathbf{x})$	Jacobian $J = \partial \mathbf{f} / \partial \mathbf{x}$	Shape
$\sigma(\mathbf{x})$ (element-wise sigmoid)	$\text{diag}(\sigma'(\mathbf{x}))$	$n \times n$
$\phi(\mathbf{x})$ (any element-wise)	$\text{diag}(\phi'(\mathbf{x}))$	$n \times n$
$\mathbf{x} / \ \mathbf{x}\ $	$(I - \hat{\mathbf{x}}\hat{\mathbf{x}}^T) / \ \mathbf{x}\ $	$n \times n$
$A\mathbf{f}(\mathbf{x})$	$A J_f$	$m \times n$

Category 4: Hessians

Expression $f(\mathbf{x})$	Hessian $H = \nabla^2 f$	Type
$\mathbf{a}^T \mathbf{x}$	0	zero
$\mathbf{x}^T A \mathbf{x}$	$A + A^T$	symmetric
$\mathbf{x}^T A \mathbf{x}$	$2A$	symmetric; A symmetric
$\ A\mathbf{x} - \mathbf{b}\ ^2$	$2A^T A$	PSD
$\ \mathbf{r}(\mathbf{x})\ ^2$ (nonlinear $\mathbf{r}$)	$\approx 2J_r^T J_r$ (Gauss-Newton)	PSD approximation
$f(\mathbf{x}) = g(A\mathbf{x})$	$A^T H_g A$	inherits definiteness of H_g

Chain Rule Summary

Scenario	Formula
Scalar L via $\mathbf{f}: \mathbb{R}^n \rightarrow \mathbb{R}^m$	$\nabla_{\mathbf{x}} L = J_f^T \nabla_{\mathbf{f}} L$
Vector $\mathbf{h} = \mathbf{g} \circ \mathbf{f}$	$J_h = J_g J_f$
Linear layer backward	$\partial L / \partial W = \delta \mathbf{x}^T, \partial L / \partial \mathbf{x} = W^T \delta, \partial L / \partial \mathbf{b} = \delta$

Note

Appendix D contains the same 20 identities with derivation sketches. Use this table for quick lookup during reading; consult App D to understand *why* each identity holds.

Exercises

21.7.1 Use the reference table to compute $\nabla_W L$ for the loss $L = \|W\mathbf{x} - \mathbf{t}\|^2 + \lambda\|W\|_F^2$. (*Hint:* both terms appear separately in Category 2.)

21.7.2 Verify that the Jacobian of $\mathbf{f}(\mathbf{x}) = A\mathbf{f}(\mathbf{x})$ listed in Category 3 follows from the matrix chain rule in §21.5.

21.7.3 For $f(\mathbf{x}) = \mathbf{x}^T A \mathbf{x}$ with non-symmetric A , the Hessian is $A + A^T$. Confirm this for $A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$ by differentiating $f(x_1, x_2) = x_1^2 + 5x_1x_2 + 4x_2^2$ element by element.

21.7.4 The Gauss-Newton Hessian approximation $H \approx J_r^T J_r$ drops the second-order term $\sum_i r_i H_i$. Under what conditions is this a good approximation? When can it be a poor one?

Exercises and Solutions

Selected solutions for Chapter 21. Exercises marked with $\star$ are more challenging.

Layout Conventions

21.1.1 In denominator layout, $\partial f / \partial \mathbf{x}$ has shape $n \times 1$ (a column vector). In numerator layout it has shape $1 \times n$ (a row vector). The two are related by transposition.

21.1.3 The update $\theta \leftarrow \theta - \alpha \partial L / \partial \theta$ with the gradient as a row vector is **numerator layout**. In denominator layout the same update is written $\theta \leftarrow \theta - \alpha (\partial L / \partial \theta)^T$, treating the gradient as a column. Numerically the subtraction is identical if shapes match; the convention only matters when the gradient feeds into a matrix multiply.

21.1.4 $J \in \mathbb{R}^{4 \times 3}$. Then $J^T \in \mathbb{R}^{3 \times 4}$ and $J^T \mathbf{g} \in \mathbb{R}^{3 \times 4} \cdot \mathbb{R}^4 = \mathbb{R}^3$ – a column vector of size $n = 3$, consistent with $\partial L / \partial \mathbf{x}$ in denominator layout.

Scalar-by-Vector: Gradients

21.2.1 (a) $\mathbf{x}^T \mathbf{x} = \sum_i x_i^2$, so $\partial / \partial x_k \sum_i x_i^2 = 2x_k$. Assemble: $2\mathbf{x}$. (b) Identity 2 with $A = I$ (symmetric): $\nabla(\mathbf{x}^T I \mathbf{x}) = (I + I)\mathbf{x} = 2\mathbf{x}$. Same result.

21.2.2 Let $\mathbf{y} = \mathbf{x} - \mu$. Then $f = \mathbf{y}^T \Sigma^{-1} \mathbf{y}$. Since Σ^{-1} is symmetric and $\partial \mathbf{y} / \partial \mathbf{x} = I$: $\nabla_{\mathbf{x}} f = 2\Sigma^{-1}(\mathbf{x} - \mu)$.

21.2.3 $2A^T(A\mathbf{x} - \mathbf{b}) = \mathbf{0} \Rightarrow A^T A \mathbf{x} = A^T \mathbf{b}$.

21.2.4 $\nabla_{\mathbf{x}} L = 2A^T(A\mathbf{x} - \mathbf{b}) + 2\lambda\mathbf{x} = 2(A^T A + \lambda I)\mathbf{x} - 2A^T \mathbf{b}$. Setting to zero: $\mathbf{x}^* = (A^T A + \lambda I)^{-1} A^T \mathbf{b}$ – the ridge regression solution (§17.7).

Vector-by-Vector: Jacobians

21.3.1 The Jacobian of $A\mathbf{x} + \mathbf{b}$ with respect to $\mathbf{x}$ is A . The constant $\mathbf{b}$ does not appear because adding a constant does not change the rate of change – its partial derivatives are zero.

21.3.2 The Jacobian of element-wise ReLU is $J = \text{diag}(\mathbf{1}[x_1 > 0], \dots, \mathbf{1}[x_n > 0])$ where $\mathbf{1}[\cdot]$ is the indicator function. At $x_i = 0$ the derivative is undefined (not differentiable), so in practice a subgradient of 0 or 0.5 is used.

★ **21.3.3** Denote $r = \|\mathbf{x}\|$. $f_i = x_i/r$. Then: $\partial f_i / \partial x_j = \delta_{ij}/r - x_i x_j / r^3 = (1/r)(\delta_{ij} - x_i x_j / r^2)$. In matrix form: $J = (I - \hat{\mathbf{x}}\hat{\mathbf{x}}^T)/r$. The matrix $I - \hat{\mathbf{x}}\hat{\mathbf{x}}^T$ is a projection onto the subspace orthogonal to $\hat{\mathbf{x}}$; it has rank $n - 1$. So J has rank $n - 1$: the normalization map kills one degree of freedom (the radial direction), as expected.

21.3.5 For a linear function $\mathbf{f}(\mathbf{x}) = A\mathbf{x}$, the Taylor expansion terminates: $A(\mathbf{x}_0 + \delta\mathbf{x}) = A\mathbf{x}_0 + A\delta\mathbf{x}$ exactly. The Jacobian is A everywhere, so the linearization is exact with no $O(\|\delta\mathbf{x}\|^2)$ error.

Scalar-by-Matrix: Weight Gradients

21.4.1 $\delta\mathbf{x}^T$ is an outer product of two nonzero vectors and therefore has rank 1, regardless of the dimensions m and n . Each gradient step nudges W along a rank-1 direction determined by the current error signal and input.

21.4.3 Let $\mathbf{r} = W\mathbf{x} - \mathbf{b}$. The loss is $L = \mathbf{r}^T \mathbf{r}$. By the backprop shortcut with $\delta = \nabla_{\mathbf{r}} L = 2\mathbf{r} = 2(W\mathbf{x} - \mathbf{b})$: $\partial L / \partial W = \delta\mathbf{x}^T = 2(W\mathbf{x} - \mathbf{b})\mathbf{x}^T$.

21.4.4 $\partial L_{\text{total}} / \partial W = \partial L_{\text{data}} / \partial W + \lambda W$. The update becomes $W \leftarrow W - \alpha(\partial L_{\text{data}} / \partial W + \lambda W) = (1 - \alpha\lambda)W - \alpha \partial L_{\text{data}} / \partial W$. The factor $(1 - \alpha\lambda)$ **decays** the weight each step regardless of the data gradient – hence the name “weight decay”.

21.4.5 $\partial \mathbf{y} / \partial W$ where $\mathbf{y} \in \mathbb{R}^m$ and $W \in \mathbb{R}^{m \times n}$ has $m \times mn$ entries arranged as an $m \times m \times n$ tensor. To collapse it to a matrix $\partial L / \partial W \in \mathbb{R}^{m \times n}$ we must specify a scalar loss L ; the tensor is then contracted with $\partial L / \partial \mathbf{y}$ along the output dimension.

The Chain Rule in Matrix Form

21.5.1 $\mathbf{y} = W\mathbf{x} + \mathbf{b}$, so $\partial y_k / \partial b_j = \delta_{kj}$. The Jacobian of $\mathbf{y}$ w.r.t. $\mathbf{b}$ is I_m (the $m \times m$ identity). By the chain rule: $\partial L / \partial \mathbf{b} = I_m^T \delta = \delta$.

21.5.3 The Jacobian of $\mathbf{f}(\mathbf{x}) = W\mathbf{x} + \mathbf{b}$ is $J_f = W \in \mathbb{R}^{m \times n}$. The chain rule gives $\partial L / \partial \mathbf{x} = J_f^T (\partial L / \partial \mathbf{y}) = W^T \delta$ – matching the result derived directly.

★ **21.5.5** For a batch of N examples, $\partial L / \partial W = \frac{1}{N} \sum_k \delta_k \mathbf{x}_k^T$. Stacking: each δ_k is a column and each $\mathbf{x}_k$ is a column, so $\partial L / \partial W = \frac{1}{N} \Delta X^T$ where $\Delta \in \mathbb{R}^{m \times N}$ has δ_k as its k -th column and $X \in \mathbb{R}^{n \times N}$ has $\mathbf{x}_k$ as its k -th column.

The Hessian Matrix

21.6.1 $\nabla_{\mathbf{x}} f = 2\Sigma^{-1}\mathbf{x}$ (symmetric $A = \Sigma^{-1}$), so $H = 2\Sigma^{-1}$. Since Σ is PD, Σ^{-1} is also PD, and H is positive definite. The loss surface is a bowl with ellipsoidal level sets; the eigenvectors of Σ determine the orientation of the bowl.

21.6.2 $H = \begin{pmatrix} 2 & 0 \\ 0 & -2 \end{pmatrix}$. Eigenvalues $+2$ and -2 : mixed signs, so the Hessian is indefinite. The function decreases from the origin in the x_2 direction (negative eigenvalue) and increases in the x_1 direction.

21.6.4 $J_r^T J_r$ is the Gram matrix of the rows of J_r ; it is PSD because $\mathbf{v}^T J_r^T J_r \mathbf{v} = \|J_r \mathbf{v}\|^2 \geq 0$ for all $\mathbf{v}$. It is PD when J_r has full column rank (all columns of J_r are linearly independent, i.e., no redundant parameters), which is the generic case near a solution.

★ **21.6.5** Compute $\kappa(H) = \lambda_{\max} / \lambda_{\min}$ from the eigenvalues of $2A^T A$. Gradient descent converges slowest along the eigenvector corresponding to $\lambda_{\min}$ (smallest eigenvalue), because the step size must be small enough to prevent overshooting in the direction of $\lambda_{\max}$, which leaves progress along the flat direction extremely slow. This is the well-known **conditioning problem** of gradient descent on ill-conditioned quadratics.

Identity Reference Table

21.7.1 $\nabla_W L = 2(W\mathbf{x} - \mathbf{t})\mathbf{x}^T + 2\lambda W$. The first term is $\delta \mathbf{x}^T$ (backprop shortcut with $\delta = 2(W\mathbf{x} - \mathbf{t})$) and the second is the Frobenius-norm gradient.

21.7.3 $f(x_1, x_2) = x_1^2 + 5x_1x_2 + 4x_2^2$. $\partial^2 f / \partial x_1^2 = 2$, $\partial^2 f / \partial x_2^2 = 8$, $\partial^2 f / \partial x_1 \partial x_2 = 5$. So $H = \begin{pmatrix} 2 & 5 \\ 5 & 8 \end{pmatrix}$. Meanwhile $A + A^T = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} + \begin{pmatrix} 1 & 3 \\ 2 & 4 \end{pmatrix} = \begin{pmatrix} 2 & 5 \\ 5 & 8 \end{pmatrix}$. They match.

Jacobians, Inverse Kinematics, and Nonlinear Optimization

The Robot Jacobian

Forward kinematics (FK) answers the question: *where is the end-effector, given the joint angles?* For a serial arm with n revolute joints and joint-angle vector $\mathbf{q} \in \mathbb{R}^n$, FK is a smooth nonlinear map

$$\mathbf{p} = f(\mathbf{q}), \quad \mathbf{p} \in \mathbb{R}^m.$$

Differentiating with respect to time gives the **velocity relationship**:

$$\dot{\mathbf{p}} = \frac{\partial f}{\partial \mathbf{q}} \dot{\mathbf{q}} = J(\mathbf{q}) \dot{\mathbf{q}},$$

where $J(\mathbf{q}) \in \mathbb{R}^{m \times n}$ is the **Jacobian matrix** of f . Every column J_i encodes how much (and in which direction) the end-effector moves when joint i rotates by one unit per second while all other joints are locked.

The Geometric Jacobian for Revolute Joints

For a serial chain of n revolute joints, the full **6-DoF geometric Jacobian** (3 linear + 3 angular velocity rows) can be assembled column by column using only the positions and axes of the joint frames — no symbolic differentiation needed.

Let $\mathbf{z}_{i-1}$ be the rotation axis of joint i (the z -axis of frame $i-1$) and $\mathbf{p}_{i-1}$ be its origin. For joint i :

$$J_i = \begin{bmatrix} \mathbf{z}_{i-1} \times (\mathbf{p}_e - \mathbf{p}_{i-1}) \\ \mathbf{z}_{i-1} \end{bmatrix},$$

where $\mathbf{p}_e$ is the end-effector origin. The top three rows capture the *linear velocity contribution* (cross product of the axis with the lever arm), and the bottom three rows capture *angular velocity* (a revolute joint always contributes its axis to the angular rate).

The full Jacobian is assembled as $J = [J_1 \mid J_2 \mid \cdots \mid J_n]$, giving a $6 \times n$ matrix.

Why a cross product? If joint i spins at rate $\dot{q}_i$, the end-effector sweeps around axis $\mathbf{z}_{i-1}$ at distance $\|\mathbf{p}_e - \mathbf{p}_{i-1}\|$. The tangential velocity is $\mathbf{z}_{i-1} \times (\mathbf{p}_e - \mathbf{p}_{i-1})$, exactly the i -th column of the upper half.

Planar 2-Link Arm: Worked Example

A planar 2-link arm (all joint axes point out of the page: $\mathbf{z}_{i-1} = [0, 0, 1]^T$) is the canonical testbed.

With link lengths l_1, l_2 and joint angles q_1, q_2 :

$$p_{1,x} = l_1 \cos q_1, \quad p_{1,y} = l_1 \sin q_1$$

$$p_{e,x} = l_1 \cos q_1 + l_2 \cos(q_1 + q_2), \quad p_{e,y} = l_1 \sin q_1 + l_2 \sin(q_1 + q_2).$$

Applying the cross-product formula for each joint ($[0, 0, 1] \times [dx, dy, 0] = [-dy, dx, 0]$):

$$J = \begin{bmatrix} -l_1 \sin q_1 - l_2 \sin(q_1 + q_2) & -l_2 \sin(q_1 + q_2) \\ l_1 \cos q_1 + l_2 \cos(q_1 + q_2) & l_2 \cos(q_1 + q_2) \end{bmatrix}.$$

The first column is the velocity the end-effector would have if only joint 1 rotated; the second column is the velocity if only joint 2 rotated. At $q_2 = 0$ (arm fully extended), the two columns become parallel — the Jacobian loses rank. This is a singularity, treated in §22.2.

All three methods agree to 10 decimal places — the geometric formula is exact and computes in $O(n)$ instead of $O(mn)$ function evaluations.

```
import numpy as np
import matplotlib.pyplot as plt

l1, l2 = 1.0, 0.8

def fk_2link(q):
    q1, q2 = q
    p1 = np.array([l1 * np.cos(q1), l1 * np.sin(q1)])
    pe = np.array([l1 * np.cos(q1) + l2 * np.cos(q1 + q2),
                   l1 * np.sin(q1) + l2 * np.sin(q1 + q2)])
    return p1, pe
```

```

import numpy as np

l1, l2 = 1.0, 0.8

def fk_2link(q):
    """Returns (p1, pe) - joint 1 position and end-effector position."""
    q1, q2 = q
    p1 = np.array([l1 * np.cos(q1), l1 * np.sin(q1)]) # shape (2,)
    pe = np.array([l1 * np.cos(q1) + l2 * np.cos(q1 + q2),
                   l1 * np.sin(q1) + l2 * np.sin(q1 + q2)]) # shape (2,)
    return p1, pe

def jac_analytical(q):
    """Analytical Jacobian of 2-link arm (2 x 2)."""
    q1, q2 = q
    J = np.array([
        [-l1 * np.sin(q1) - l2 * np.sin(q1 + q2), -l2 * np.sin(q1 + q2)],
        [ l1 * np.cos(q1) + l2 * np.cos(q1 + q2),  l2 * np.cos(q1 + q2)]
    ]) # shape (2, 2)
    return J

def jac_geometric(q):
    """Geometric Jacobian from cross-product formula - works for any planar arm."""
    q1, q2 = q
    p0 = np.array([0.0, 0.0])
    p1, pe = fk_2link(q)
    # Each column: [0,0,1] x (pe - p_prev) in 2D = [-(pe-p_prev)[y], (pe-p_prev)[x]]
    def xprod_2d(pa, pb):
        d = pb - pa
        return np.array([-d[1], d[0]]) # shape (2,)
    J = np.column_stack([xprod_2d(p0, pe),
                        xprod_2d(p1, pe)]) # shape (2, 2)
    return J

def jac_numerical(q, eps=1e-6):
    """Central-difference Jacobian for verification."""
    n = len(q)
    pe0 = fk_2link(q)[1] # shape (2,)
    m = len(pe0)
    J = np.zeros((m, n)) # shape (2, 2)
    for j in range(n):
        dq = np.zeros(n)
        dq[j] = eps
        J[:, j] = (fk_2link(q + dq)[1] - fk_2link(q - dq)[1]) / (2 * eps)
    return J

q_test = np.array([np.pi / 4, np.pi / 3]) # shape (2,)

J_ana = jac_analytical(q_test)
J_geo = jac_geometric(q_test)
J_num = jac_numerical(q_test)

print("Analytical Jacobian:")

```

```

def jac_analytical(q):
    q1, q2 = q
    J = np.array([
        [-11 * np.sin(q1) - 12 * np.sin(q1 + q2), -12 * np.sin(q1 + q2)],
        [11 * np.cos(q1) + 12 * np.cos(q1 + q2), 12 * np.cos(q1 + q2)]
    ])
    return J

fig, axes = plt.subplots(1, 2, figsize=(10, 4.5))
configs = [
    (np.array([np.pi / 4, np.pi / 2]), "q1=45 deg, q2=90 deg (elbow up)"),
    (np.array([0.0, np.pi / 6]), "q1=0, q2=30 deg (near-extended)"),
]

for ax, (q, title) in zip(axes, configs):
    p1, pe = fk_2link(q)
    J = jac_analytical(q)

    # Draw arm
    ax.plot([0, p1[0], pe[0]], [0, p1[1], pe[1]],
            'o-', color='#333333', lw=2.5, ms=6, zorder=3)
    ax.plot(0, 0, 'ks', ms=8, zorder=4)

    # Jacobian column arrows (velocity vectors) at end-effector
    scale = 0.5
    ax.annotate('', xy=pe + scale * J[:, 0], xytext=pe,
                arrowprops=dict(arrowstyle='->', color='#4a90d9', lw=2.2))
    ax.annotate('', xy=pe + scale * J[:, 1], xytext=pe,
                arrowprops=dict(arrowstyle='->', color='#2ecc71', lw=2.2))

    ax.text(*(pe + scale * J[:, 0] + [0.05, 0.05]), r'$J_1$',
            color='#4a90d9', fontsize=11)
    ax.text(*(pe + scale * J[:, 1] + [0.05, -0.1]), r'$J_2$',
            color='#2ecc71', fontsize=11)
    ax.plot(*pe, 'o', color='tomato', ms=8, zorder=4)

    ax.set_xlim(-0.3, 2.2)
    ax.set_ylim(-0.3, 2.2)
    ax.set_aspect('equal')
    ax.grid(alpha=0.2)
    ax.axhline(0, color='#333333', lw=0.4, alpha=0.5)
    ax.axvline(0, color='#333333', lw=0.4, alpha=0.5)
    ax.set_title(title, fontsize=10)
    ax.set_xlabel('x (m)')
    ax.set_ylabel('y (m)')

from matplotlib.lines import Line2D

```



```

handles = [
    Line2D([0], [0], color='#4a90d9', lw=2, label=r'$J_1$: vel if only joint 1 rotates'),
    Line2D([0], [0], color='#2ecc71', lw=2, label=r'$J_2$: vel if only joint 2 rotates'),
]
fig.legend(handles=handles, loc='lower center', ncol=2, fontsize=9,
           bbox_to_anchor=(0.5, -0.01))
plt.tight_layout()
plt.show()

```

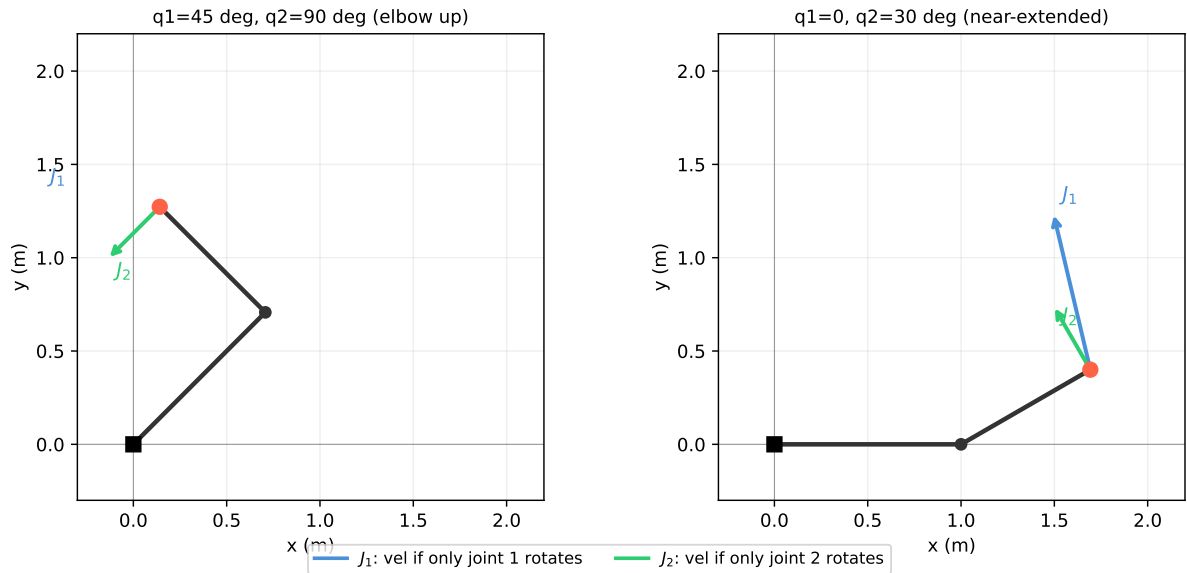


Figure 2: Jacobian columns as end-effector velocity arrows. Blue arrow: velocity when only joint 1 rotates. Green arrow: velocity when only joint 2 rotates.

In the near-extended configuration (right panel), J_1 and J_2 point almost in the same direction — the Jacobian is close to rank-deficient, meaning joint motion in either direction produces nearly the same end-effector motion. The arm has “lost a degree of freedom” in task space.

The Velocity Relationship in Practice

In discrete time (small step Δt):

$$\Delta \mathbf{p} \approx \mathbf{J}(\mathbf{q}) \Delta \mathbf{q}.$$

Given a desired end-effector velocity $\dot{\mathbf{p}}^*$, the IK problem reduces to *inverting* this linear relationship to find $\dot{\mathbf{q}}$. When $\mathbf{J}$ is square and full-rank, the solution is unique. When $\mathbf{J}$ is rectangular or near-singular, we need the tools developed in the rest of this chapter.

Jacobian Singularities

A **singularity** occurs when the Jacobian $J(\mathbf{q})$ loses rank. At a singularity the robot cannot produce end-effector velocity in at least one task-space direction, regardless of how fast any joint moves. Singularities are not just mathematical curiosities — driving through one causes joint velocities to blow up, a serious problem in real-time control.

Detecting Singularities: Manipulability

For a $m \times n$ Jacobian (typically $m \leq n$), define the **manipulability measure** (Yoshikawa 1985):

$$w(\mathbf{q}) = \sqrt{\det(J(\mathbf{q}) J(\mathbf{q})^T)}.$$

- $w > 0$: full rank, arm can move in all task-space directions.
- $w = 0$: singular — at least one direction is unreachable.
- Larger w means more isotropic mobility (the velocity ellipsoid, introduced below, is rounder).

For a square Jacobian ($m = n$), this simplifies to $w = |\det J|$.

2-link arm closed form. From §22.1:

$$\det J = l_1 l_2 \sin q_2.$$

Therefore $w(q_1, q_2) = |l_1 l_2 \sin q_2|$. Singularities occur at $q_2 = 0$ (arm fully extended) and $q_2 = \pi$ (arm fully folded). Notice that q_1 does not appear — the singular configurations form two whole lines in joint space, not isolated points.

The Velocity Ellipsoid

The set of all end-effector velocities reachable with unit joint speed is

$$\mathcal{E} = \{\dot{\mathbf{p}} : \dot{\mathbf{p}} = J\dot{\mathbf{q}}, \|\dot{\mathbf{q}}\| \leq 1\}.$$

This is an ellipsoid in task space with semi-axes aligned to the left singular vectors of J and lengths equal to the singular values σ_i . A singularity collapses the ellipsoid: one or more semi-axes shrink to zero, leaving the arm unable to accelerate in that direction.

$$w = \prod_{i=1}^m \sigma_i = \sqrt{\det(JJ^T)},$$

so manipulability equals the volume of the unit-ball image under J .

```
import numpy as np
import matplotlib.pyplot as plt

l1, l2 = 1.0, 0.8

q1_vals = np.linspace(-np.pi, np.pi, 300) # shape (300,)
q2_vals = np.linspace(-np.pi, np.pi, 300) # shape (300,)
Q1, Q2 = np.meshgrid(q1_vals, q2_vals)    # shape (300, 300)

W = np.abs(l1 * l2 * np.sin(Q2))          # shape (300, 300)

fig, ax = plt.subplots(figsize=(6, 5))
cf = ax.contourf(Q1, Q2, W, levels=50, cmap='Blues')
plt.colorbar(cf, ax=ax, label=r'$w(\mathbf{q}) = |l_1 l_2 \sin q_2|$')
ax.axhline(0, color='tomato', lw=1.5, label=r'$q_2=0$ singular')
ax.axhline(np.pi, color='tomato', lw=1.5, linestyle='--')
ax.axhline(-np.pi, color='tomato', lw=1.5, linestyle='--')
ax.set_xlabel(r'$q_1$ (rad)')
ax.set_ylabel(r'$q_2$ (rad)')
ax.set_title('Manipulability heatmap (2-link arm)')
ax.legend(fontsize=9)
plt.tight_layout()
plt.show()
```

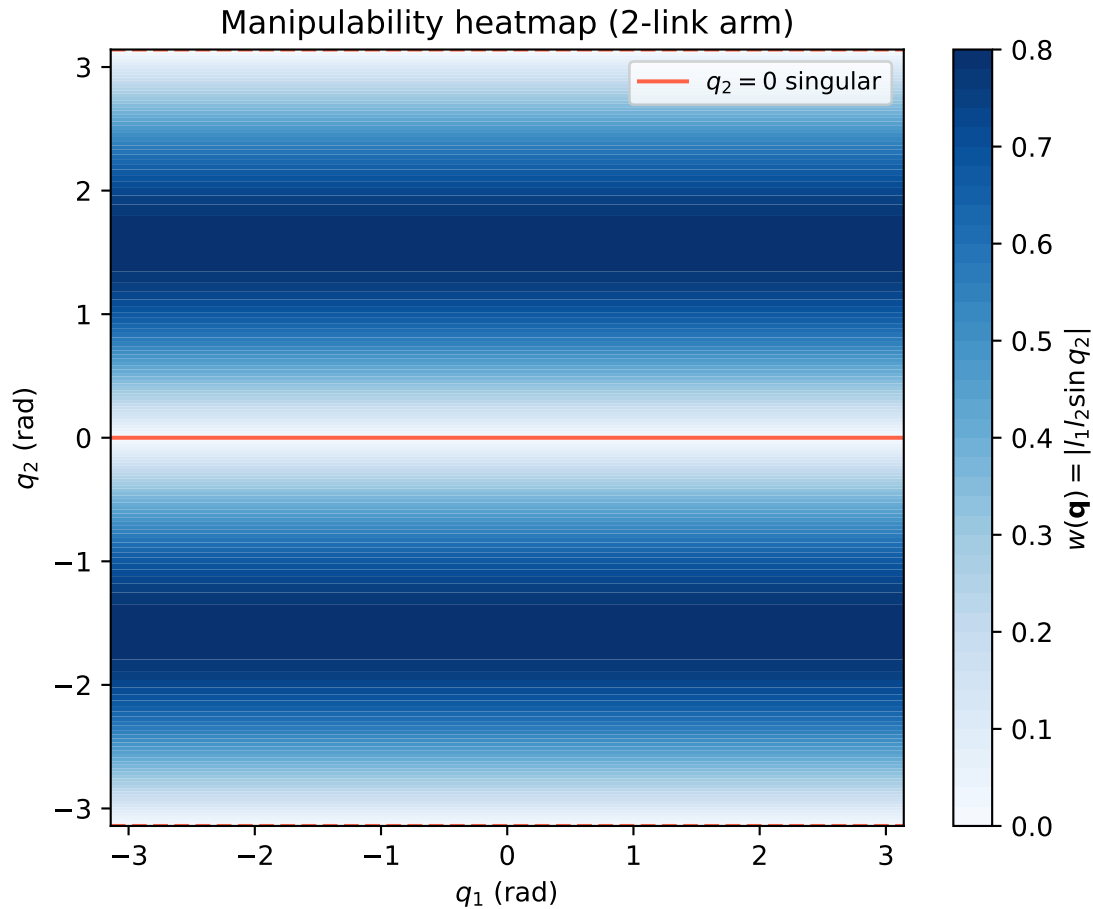


Figure 3: Manipulability $w(\mathbf{q}) = |l_1 l_2 \sin(q_2)|$ for the 2-link arm. Singularities ($w=0$) occur at $q_2=0$ and $q_2=\pi$ (white bands).

```
import numpy as np
import matplotlib.pyplot as plt

l1, l2 = 1.0, 0.8

def fk_2link(q):
    q1, q2 = q
    p1 = np.array([l1 * np.cos(q1), l1 * np.sin(q1)]) # shape (2,)
    pe = np.array([l1 * np.cos(q1) + l2 * np.cos(q1 + q2),
                    l1 * np.sin(q1) + l2 * np.sin(q1 + q2)]) # shape (2,)
    return p1, pe

def jac_2link(q):
    q1, q2 = q
    J = np.array([
```

```

        [-11 * np.sin(q1) - 12 * np.sin(q1 + q2), -12 * np.sin(q1 + q2)],
        [ 11 * np.cos(q1) + 12 * np.cos(q1 + q2), 12 * np.cos(q1 + q2)]
    ]) # shape (2, 2)
    return J

def draw_ellipse(ax, center, J, color, n_pts=200):
    """Draw the velocity ellipsoid (image of unit circle under J)."""
    theta = np.linspace(0, 2 * np.pi, n_pts) # shape (n_pts,)
    circle = np.stack([np.cos(theta), np.sin(theta)]) # shape (2, n_pts)
    ellipse = J @ circle # shape (2, n_pts)
    scale = 0.35
    ax.fill(center[0] + scale * ellipse[0],
            center[1] + scale * ellipse[1],
            alpha=0.25, color=color)
    ax.plot(center[0] + scale * ellipse[0],
            center[1] + scale * ellipse[1],
            color=color, lw=1.5)

configs = [
    (np.array([np.pi / 4, np.pi / 2]), "High manipulability\n(q2=90 deg)"),
    (np.array([np.pi / 4, np.pi / 8]), "Near-singular\n(q2=22.5 deg)"),
    (np.array([np.pi / 4, 0.0]), "Singular\n(q2=0, fully extended)"),
]
colors = ['#4a90d9', '#2ecc71', 'tomato']

fig, axes = plt.subplots(1, 3, figsize=(12, 4))
for ax, (q, title), color in zip(axes, configs, colors):
    p1, pe = fk_2link(q)
    J = jac_2link(q)
    w = abs(np.linalg.det(J))

    ax.plot([0, p1[0], pe[0]], [0, p1[1], pe[1]],
            'o-', color='#333333', lw=2.5, ms=6, zorder=3)
    ax.plot(0, 0, 'ks', ms=8, zorder=4)
    ax.plot(*pe, 'o', color=color, ms=8, zorder=4)
    draw_ellipse(ax, pe, J, color)

    ax.set_xlim(-0.3, 2.2)
    ax.set_ylim(-0.5, 2.0)
    ax.set_aspect('equal')
    ax.grid(alpha=0.2)
    ax.set_title(f"{title}\n$w = {w:.3f}$", fontsize=9)
    ax.set_xlabel('x (m)')
    ax.set_ylabel('y (m)')

plt.tight_layout()
plt.show()

```

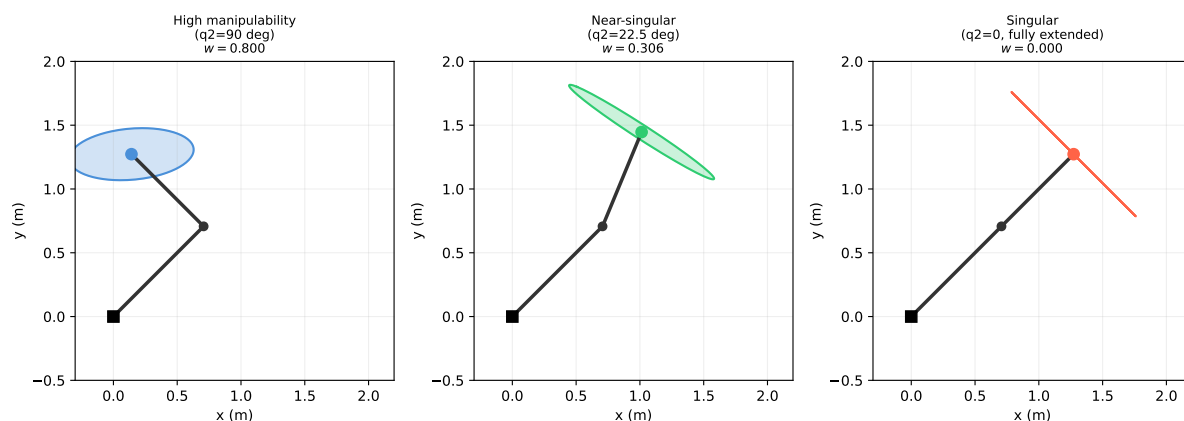


Figure 4: Three arm configurations: high manipulability (left), near-singular (center), and singular (right, $q_2=0$ — arm fully extended).

The rightmost panel shows that when $q_2 = 0$ the velocity ellipsoid degenerates into a line segment — the arm can only move its end-effector along a single direction, no matter how all joints spin.

Condition Number as an Alternative

The manipulability w conflates all singular values; a more directional metric is the **condition number** of J :

$$\kappa(J) = \frac{\sigma_{\max}}{\sigma_{\min}}.$$

A large κ signals that movement in the worst direction requires dramatically amplified joint velocities — a practical warning even before a true singularity is reached. Both metrics are used in trajectory planners to avoid or exit near-singular regions.

Workspace Boundaries vs. Interior Singularities

Singularities fall into two classes:

Type	Location	Example (2-link)
Boundary	Edge of reachable workspace	$q_2 = 0$ — arm at full reach
Interior	Inside the workspace	$q_2 = \pi$ — arm fully folded

Boundary singularities occur when the arm is maximally extended; interior singularities arise from alignment of links. Both produce $\det J = 0$ but their avoidance strategies differ: boundary singularities call for joint-limit awareness, while interior ones are navigated via null-space motion and damped inversion (§22.4).

Inverse Kinematics via the Jacobian

Inverse kinematics (IK) asks: *given a desired end-effector pose $\mathbf{p}^*$, find joint angles $\mathbf{q}$ such that $f(\mathbf{q}) = \mathbf{p}^*$.*

Because f is nonlinear, IK is solved iteratively. At each step, linearize around the current configuration $\mathbf{q}_k$ and use the Jacobian to compute a joint correction:

$$\mathbf{e}_k = \mathbf{p}^* - f(\mathbf{q}_k), \quad \Delta \mathbf{q} \approx J^{-1}(\mathbf{q}_k) \mathbf{e}_k.$$

When J is rectangular (more joints than task dimensions), “inversion” means choosing among infinitely many solutions. Three Jacobian-based strategies are standard.

Strategy 1: Jacobian Transpose

The simplest approach ignores the exact inversion and uses the transpose:

$$\Delta \mathbf{q} = \alpha J^T \mathbf{e},$$

where $\alpha > 0$ is a step size. This is gradient descent on $\frac{1}{2}\|\mathbf{e}\|^2$:

$$\frac{\partial}{\partial \mathbf{q}} \frac{1}{2}\|\mathbf{e}\|^2 = -J^T \mathbf{e}.$$

Advantages: trivially avoids matrix inversion, always stable with small enough α , and J^T is cheap to compute. Disadvantage: slow convergence (linear rate, poor step-size sensitivity) and does not minimize joint motion.

A good heuristic for α is to set it so the first step moves roughly the error distance:

$$\alpha = \frac{\mathbf{e}^T J J^T \mathbf{e}}{\|J J^T \mathbf{e}\|^2}.$$

Strategy 2: Pseudoinverse

The Moore-Penrose pseudoinverse J^+ gives the minimum-norm joint velocity that achieves the desired task velocity:

$$\Delta \mathbf{q} = J^+ \mathbf{e}, \quad J^+ = J^T (J J^T)^{-1} \quad (m < n, \text{ full row rank}).$$

This minimizes $\|\Delta \mathbf{q}\|$ while exactly satisfying $J \Delta \mathbf{q} = \mathbf{e}$ (to first order). Convergence is quadratic near the solution if the linearization is good, and step-size tuning is unnecessary.

The drawback: $(J J^T)^{-1}$ is numerically unstable near singularities ($\sigma_{\min} \rightarrow 0$ causes J^+ to blow up). §22.4 addresses this.

Strategy 3: Null-Space Projection

A redundant robot ($n > m$) has a **null space** $\mathcal{N}(J)$ — joint motions that produce zero end-effector motion. The general solution is:

$$\Delta \mathbf{q} = J^+ \mathbf{e} + \underbrace{(I - J^+ J)}_{\text{null projector}} \Delta \mathbf{q}_0,$$

where $\Delta \mathbf{q}_0$ is any joint motion and $(I - J^+ J)$ projects it into the null space. Choosing $\Delta \mathbf{q}_0$ smartly allows secondary objectives: avoid joint limits, maximize manipulability, minimize torque, etc.

A common secondary objective is gradient ascent on manipulability:

$$\Delta \mathbf{q}_0 = \beta \nabla_{\mathbf{q}} w(\mathbf{q}), \quad w(\mathbf{q}) = \sqrt{\det(J J^T)}.$$

```
import numpy as np
import matplotlib.pyplot as plt

rng = np.random.default_rng(42)
l1, l2, l3 = 1.0, 0.8, 0.6

def fk_3link(q):
    q1, q2, q3 = q
    c1, s1 = np.cos(q1), np.sin(q1)
    c12, s12 = np.cos(q1+q2), np.sin(q1+q2)
    c123, s123 = np.cos(q1+q2+q3), np.sin(q1+q2+q3)
    p1 = np.array([l1*c1, l1*s1]) # shape (2,)
    p2 = np.array([l1*c1 + l2*c12, l1*s1 + l2*s12]) # shape (2,)
```



```

    pe = np.array([l1*c1 + l2*c12 + l3*c123,
                   l1*s1 + l2*s12 + l3*s123]) # shape (2,)
    return p1, p2, pe

def jac_3link(q):
    q1, q2, q3 = q
    c1, s1 = np.cos(q1), np.sin(q1)
    c12, s12 = np.cos(q1+q2), np.sin(q1+q2)
    c123, s123 = np.cos(q1+q2+q3), np.sin(q1+q2+q3)
    # End-effector from each joint origin; cross product [0,0,1] x d = [-dy, dx]
    p1, p2, pe = fk_3link(q)
    def xp(pa):
        d = pe - pa
        return np.array([-d[1], d[0]]) # shape (2,)
    J = np.column_stack([xp(np.zeros(2)), xp(p1), xp(p2)]) # shape (2, 3)
    return J

def manip(q):
    J = jac_3link(q)
    return np.sqrt(max(np.linalg.det(J @ J.T), 0.0))

def manip_grad(q, eps=1e-5):
    g = np.zeros(3) # shape (3,)
    for i in range(3):
        dq = np.zeros(3)
        dq[i] = eps
        g[i] = (manip(q + dq) - manip(q - dq)) / (2 * eps)
    return g

def run_ik(q0, p_target, method='pinv', n_steps=300, alpha=0.3, beta=0.5):
    q = q0.copy() # shape (3,)
    traj = [fk_3link(q)[2].copy()]
    for _ in range(n_steps):
        _, _, pe = fk_3link(q)
        e = p_target - pe # shape (2,)
        if np.linalg.norm(e) < 1e-4:
            break
        J = jac_3link(q) # shape (2, 3)
        if method == 'transpose':
            dq = alpha * J.T @ e # shape (3,)
        elif method == 'pinv':
            JJT = J @ J.T # shape (2, 2)
            dq = J.T @ np.linalg.solve(JJT, e) # shape (3,)
        elif method == 'nullspace':
            JJT = J @ J.T # shape (2, 2)
            Jp_e = J.T @ np.linalg.solve(JJT, e) # shape (3,)
            N = np.eye(3) - J.T @ np.linalg.solve(JJT, J) # shape (3, 3)

```

```

        dq0 = beta * manip_grad(q)                # shape (3,)
        dq = Jp_e + N @ dq0                      # shape (3,)
        q = q + dq
        traj.append(fk_3link(q)[2].copy())
    return q, np.array(traj)                      # shape (T, 2)

q0 = np.array([0.5, 0.8, 0.3])                  # shape (3,)
p_target = np.array([1.5, 0.8])                 # shape (2,)

q_tr, traj_tr = run_ik(q0, p_target, method='transpose', alpha=0.25)
q_pi, traj_pi = run_ik(q0, p_target, method='pinv')
q_ns, traj_ns = run_ik(q0, p_target, method='nullspace', beta=0.4)

fig, axes = plt.subplots(1, 2, figsize=(12, 4.5))

# Left: final arm configurations
ax = axes[0]
for q_final, color, label in [
    (q_tr, '#4a90d9', 'J-transpose'),
    (q_pi, '#2ecc71', 'Pseudoinverse'),
    (q_ns, '#e67e22', 'Null-space')]:
    p1, p2, pe = fk_3link(q_final)
    pts = np.array([[0,0], p1, p2, pe])          # shape (4, 2)
    ax.plot(pts[:,0], pts[:,1], 'o-', color=color, lw=2, ms=5, alpha=0.85,
            label=label)
ax.plot(*p_target, 'x', color='tomato', ms=12, mew=2.5, zorder=5, label='Target')
ax.set_xlim(-0.3, 2.5)
ax.set_ylim(-0.5, 2.0)
ax.set_aspect('equal')
ax.grid(alpha=0.2)
ax.legend(fontsize=9)
ax.set_title('Final configurations')
ax.set_xlabel('x (m)')
ax.set_ylabel('y (m)')

# Right: convergence (error vs iteration)
ax = axes[1]
for traj, color, label in [
    (traj_tr, '#4a90d9', 'J-transpose'),
    (traj_pi, '#2ecc71', 'Pseudoinverse'),
    (traj_ns, '#e67e22', 'Null-space')]:
    errs = np.linalg.norm(traj - p_target, axis=1) # shape (T,)
    ax.semilogy(errs, color=color, lw=2, label=label)
ax.set_xlabel('Iteration')
ax.set_ylabel('End-effector error (m)')
ax.set_title('IK convergence')
ax.legend(fontsize=9)

```

```
ax.grid(alpha=0.2)
plt.tight_layout()
plt.show()
```

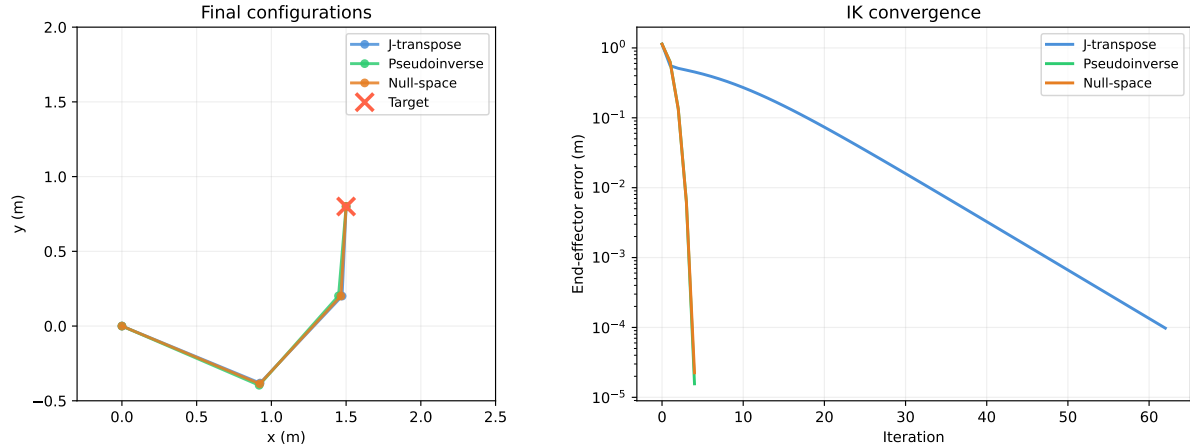


Figure 5: IK trajectories for a 3-link planar arm: Jacobian-transpose (blue), pseudoinverse (green), and pseudoinverse + null-space manipulability gradient (orange). All reach the target (red cross) from the same start.

The pseudoinverse reaches the target in roughly a quarter of the iterations needed by the transpose method. The null-space variant arrives at a different final configuration — same end-effector position, but the elbow posture trades slightly slower convergence for a higher-manipulability pose.

Choosing a Step Size for J-Transpose

The adaptive step $\alpha^* = \mathbf{e}^T J J^T \mathbf{e} / \|J J^T \mathbf{e}\|^2$ can be derived by treating $J^T \mathbf{e}$ as the search direction and doing an exact line search on the quadratic model:

$$\min_{\alpha} \frac{1}{2} \|\mathbf{e} - \alpha J J^T \mathbf{e}\|^2.$$

Setting the derivative to zero gives the formula above. In practice a constant α around 0.1–0.3 (scaled by $1/\|J\|_F^2$) also works well.

When to Use Each Method

Method	Pros	Cons
J-transpose	No matrix solve; always stable	Slow; step-size sensitive
Pseudoinverse	Minimum joint motion; fast	Blows up near singularity
Null-space	Secondary objectives; uses redundancy	Needs $n > m$; more computation

Near singularities, all three need regularization — the topic of §22.4.

Damped Least Squares

Near a singularity the pseudoinverse $J^+ = J^T(JJ^T)^{-1}$ amplifies the error signal by $1/\sigma_{\min}$, which diverges as $\sigma_{\min} \rightarrow 0$. **Damped least squares (DLS)** adds a regularization term that limits this amplification while degrading gracefully when the arm is far from singular.

The DLS Formula

Instead of exactly solving $J \Delta \mathbf{q} = \mathbf{e}$, minimize the regularized cost:

$$\min_{\Delta \mathbf{q}} \|J \Delta \mathbf{q} - \mathbf{e}\|^2 + \lambda^2 \|\Delta \mathbf{q}\|^2.$$

The normal equations yield:

$$\Delta \mathbf{q} = J^T (JJ^T + \lambda^2 I)^{-1} \mathbf{e}.$$

This is the **Wampler-Nakamura-Hanafusa** (or Levenberg-style) formula for IK. The matrix $(JJ^T + \lambda^2 I)$ is always positive-definite and invertible, so the formula is singularity-safe for any $\lambda > 0$.

SVD Analysis of the Gain

Using the thin SVD $J = U\Sigma V^T$, the DLS update simplifies to:

$$\Delta \mathbf{q} = V \tilde{\Sigma}^+ U^T \mathbf{e}, \quad \tilde{\sigma}_i^+ = \frac{\sigma_i}{\sigma_i^2 + \lambda^2}.$$

Compare with the pseudoinverse gain $1/\sigma_i$. For each singular value:

Regime	Pseudoinverse gain	DLS gain
$\sigma_i \gg \lambda$	$1/\sigma_i$	$\approx 1/\sigma_i$ (unchanged)
$\sigma_i = \lambda$	$1/\lambda$	$1/(2\lambda)$ (half as large)
$\sigma_i \ll \lambda$	$1/\sigma_i \rightarrow \infty$	$\approx \sigma_i/\lambda^2 \rightarrow 0$ (safe)

The DLS acts like a low-pass filter on the singular value gains — it passes large σ_i faithfully and attenuates small ones.

```
import numpy as np
import matplotlib.pyplot as plt

sigma = np.linspace(0.01, 3.0, 500)          # shape (500,)

fig, ax = plt.subplots(figsize=(7, 4.5))
ax.plot(sigma, 1.0 / sigma, color='#333333', lw=2, linestyle='--',
        label='Pseudoinverse  $1/\sigma$ ')

lambdas = [0.05, 0.2, 0.5]
colors = ['#4a90d9', '#2ecc71', 'tomato']
for lam, color in zip(lambdas, colors):
    gain = sigma / (sigma**2 + lam**2)        # shape (500,)
    ax.plot(sigma, gain, color=color, lw=2,
            label=f'DLS  $\lambda={lam}$ ')
    # Mark the peak at sigma = lambda
    peak_gain = 1.0 / (2 * lam)
    ax.plot(lam, peak_gain, 'o', color=color, ms=6)

ax.set_xlim(0, 3.0)
ax.set_ylim(0, 12)
ax.set_xlabel(r'Singular value  $\sigma$ ')
ax.set_ylabel('Gain')
ax.set_title('DLS gain vs pseudoinverse gain')
ax.legend(fontsize=9)
ax.grid(alpha=0.2)
plt.tight_layout()
plt.show()
```

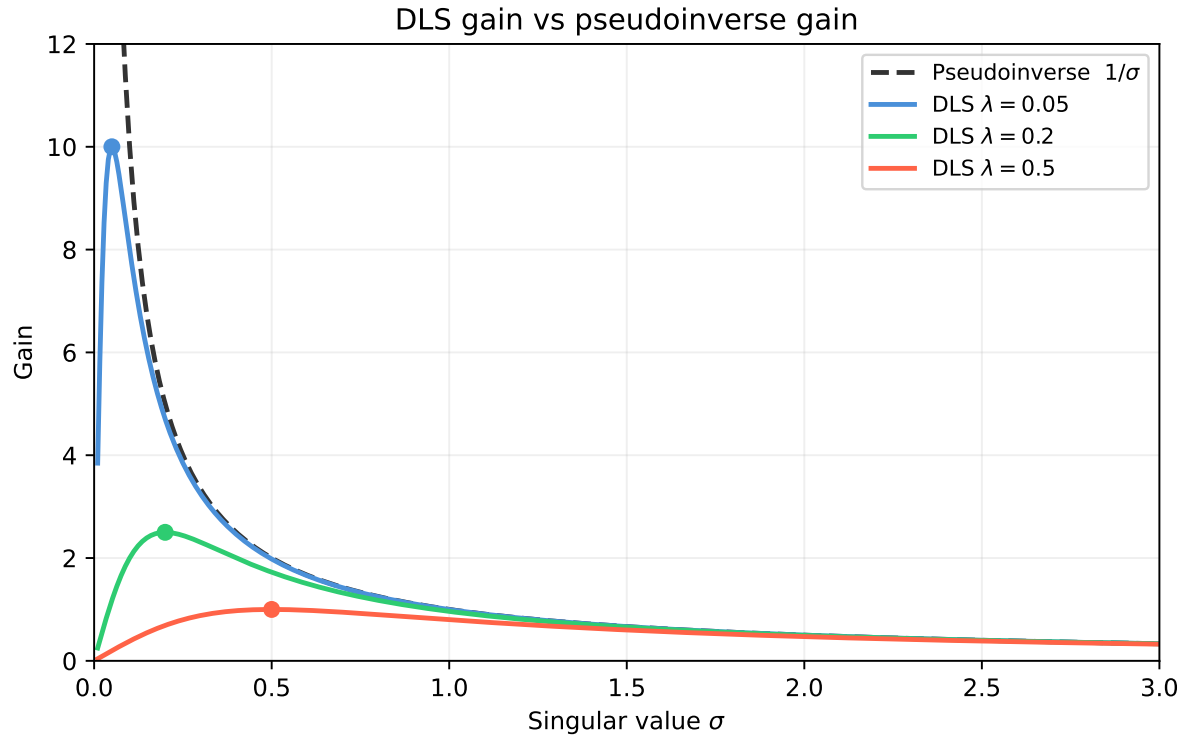


Figure 6: Gain as a function of singular value σ for the pseudoinverse ($1/\sigma$) and DLS ($\sigma/(\sigma^2 + \lambda^2)$) at three damping levels. DLS stays bounded near $\sigma=0$.

Each DLS curve peaks at $\sigma = \lambda$ with value $1/(2\lambda)$, then decays back to zero — unlike the pseudoinverse, which grows without bound.

Adaptive Damping

A fixed λ imposes a constant accuracy penalty even when the arm is far from singular. **Adaptive damping** (Sugihara 2011) sets λ based on the smallest singular value:

$$\lambda^2 = \begin{cases} 0 & \text{if } w \geq w_0 \\ (1 - (w/w_0)^2) \lambda_{\max}^2 & \text{if } w < w_0, \end{cases}$$

where w_0 is a manipulability threshold and $\lambda_{\max}$ is the maximum allowed damping. This gives zero regularization in well-conditioned regions and smoothly ramps up near singularities.

```

import numpy as np
import matplotlib.pyplot as plt

l1, l2 = 1.0, 0.8

def fk_2link(q):
    q1, q2 = q
    p1 = np.array([l1 * np.cos(q1), l1 * np.sin(q1)]) # shape (2,)
    pe = np.array([l1 * np.cos(q1) + l2 * np.cos(q1 + q2),
                  l1 * np.sin(q1) + l2 * np.sin(q1 + q2)]) # shape (2,)
    return p1, pe

def jac_2link(q):
    q1, q2 = q
    J = np.array([
        [-l1 * np.sin(q1) - l2 * np.sin(q1 + q2), -l2 * np.sin(q1 + q2)],
        [ l1 * np.cos(q1) + l2 * np.cos(q1 + q2),  l2 * np.cos(q1 + q2)]
    ]) # shape (2, 2)
    return J

def run_ik(q0, p_target, lam=0.0, n_steps=80):
    q = q0.copy() # shape (2,)
    errs = []
    for _ in range(n_steps):
        _, pe = fk_2link(q)
        e = p_target - pe # shape (2,)
        errs.append(float(np.linalg.norm(e)))
        if errs[-1] < 1e-5:
            break
        J = jac_2link(q) # shape (2, 2)
        A = J @ J.T + lam**2 * np.eye(2) # shape (2, 2)
        dq = J.T @ np.linalg.solve(A, e) # shape (2,)
        # Clip to prevent runaway
        dq = np.clip(dq, -0.5, 0.5)
        q = q + dq
    return np.array(errs)

# Start near singularity: q2 = 0.05 (almost fully extended)
q0 = np.array([np.pi / 6, 0.05]) # shape (2,)
# Target slightly off the fully-extended line
p_target = np.array([1.6, 0.4]) # shape (2,)

errs_pinv = run_ik(q0, p_target, lam=0.0)
errs_dls1 = run_ik(q0, p_target, lam=0.1)
errs_dls2 = run_ik(q0, p_target, lam=0.3)

fig, ax = plt.subplots(figsize=(7, 4))

```

```

ax.semilogy(errs_pinv, color='tomato', lw=2, label='Pseudoinverse (lambda=0)')
ax.semilogy(errs_dls1, color='#4a90d9', lw=2, label='DLS lambda=0.1')
ax.semilogy(errs_dls2, color='#2ecc71', lw=2, label='DLS lambda=0.3')
ax.set_xlabel('Iteration')
ax.set_ylabel('End-effector error (m)')
ax.set_title('IK near singularity: pseudoinverse vs DLS')
ax.legend(fontsize=9)
ax.grid(alpha=0.2)
plt.tight_layout()
plt.show()

```

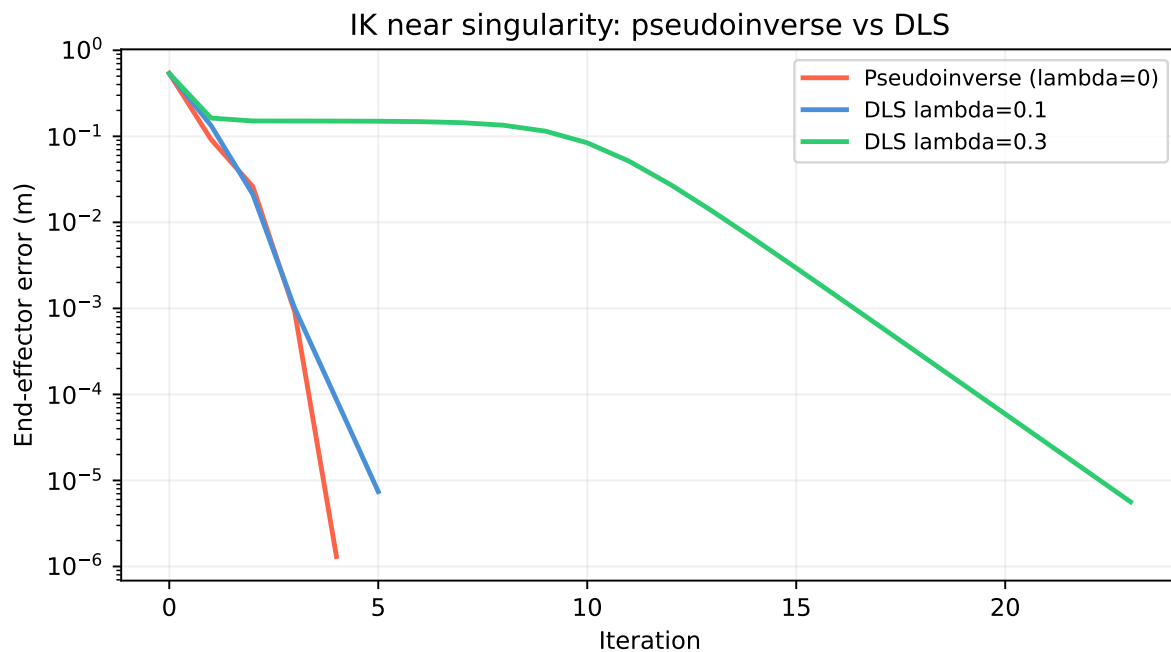


Figure 7: IK convergence near a singularity: pseudoinverse diverges (error grows), DLS stays bounded. The arm starts close to $q_2=0$ (extended), a near-singular configuration.

The pseudoinverse oscillates because near-zero singular values produce enormous joint corrections that overshoot. DLS with $\lambda = 0.1$ converges smoothly; $\lambda = 0.3$ converges more slowly but with less overshoot — the classic bias-variance tradeoff of regularization.

Push-Through Identity

An important algebraic identity connects the two forms of the DLS update:

$$(J^T J + \lambda^2 I_n)^{-1} J^T = J^T (J J^T + \lambda^2 I_m)^{-1}.$$

This is the **push-through** or **matrix-inversion lemma** result. It means the $n \times n$ system (joint space) and the $m \times m$ system (task space) give identical answers. When $m < n$ (redundant arm), solving the $m \times m$ system on the right is cheaper. This identity also appears in §22.5 when connecting the Levenberg-Marquardt algorithm to DLS.

Proof. Left-multiply both sides by $(J^T J + \lambda^2 I)$ and right-multiply by $(J J^T + \lambda^2 I)$; direct expansion shows both reduce to $J^T(J J^T + \lambda^2 I)$. $\square$

Gauss–Newton and Levenberg–Marquardt

IK is a least-squares problem: minimize $\frac{1}{2}\|\mathbf{r}(\mathbf{q})\|^2$ where the **residual** $\mathbf{r}(\mathbf{q}) = f(\mathbf{q}) - \mathbf{p}^*$. The Jacobian-based methods of §22.3 are first-order; Gauss-Newton and Levenberg-Marquardt exploit the residual structure for faster convergence.

Gauss-Newton

Linearize the residual around the current iterate $\mathbf{q}_k$:

$$\mathbf{r}(\mathbf{q}_k + \delta) \approx \mathbf{r}(\mathbf{q}_k) + J_k \delta.$$

Minimizing $\frac{1}{2}\|J_k \delta + \mathbf{r}_k\|^2$ over δ gives the **Gauss-Newton normal equations**:

$$J_k^T J_k \delta = -J_k^T \mathbf{r}_k.$$

Update: $\mathbf{q}_{k+1} = \mathbf{q}_k + \delta$.

Properties. - Converges quadratically when $\mathbf{r}^* \approx 0$ at the solution (small residual problem).
 - Stalls when $J_k^T J_k$ is singular or near-singular. - Equivalent to Newton's method with the Hessian approximation $\nabla^2 C \approx J^T J$ (dropping the second-derivative term $\sum_i r_i \nabla^2 r_i$).

Levenberg-Marquardt

Levenberg-Marquardt (LM) adds a damping term to the Gauss-Newton system:

$$(J_k^T J_k + \lambda_k I) \delta = -J_k^T \mathbf{r}_k.$$

- $\lambda_k \rightarrow 0$: pure Gauss-Newton (fast near solution).
- $\lambda_k \rightarrow \infty$: gradient descent with step $1/\lambda_k$ (safe far from solution).

The damping λ_k is adapted based on the **gain ratio**:

$$\rho_k = \frac{\text{actual reduction}}{\text{predicted reduction}} = \frac{\|\mathbf{r}_k\|^2 - \|\mathbf{r}_{k+1}\|^2}{\|\mathbf{r}_k\|^2 - \|J_k \delta + \mathbf{r}_k\|^2}.$$

If $\rho_k > 0.75$: step was good, decrease λ (increase Gauss-Newton character). If $\rho_k < 0.25$: step was poor, increase λ (fall back to gradient descent).

LM and DLS Are the Same Update

From the push-through identity (§22.4):

$$(J^T J + \lambda I)^{-1} J^T = J^T (J J^T + \lambda I)^{-1}.$$

Therefore the LM update

$$\delta = -(J^T J + \lambda I)^{-1} J^T \mathbf{r}$$

is identical to the DLS update from §22.4 with $\mathbf{e} = -\mathbf{r}$:

$$\Delta \mathbf{q} = J^T (J J^T + \lambda I)^{-1} \mathbf{e}.$$

This duality means IK (DLS, task-space view) and nonlinear least squares (LM, optimization view) are two faces of the same algorithm.

```
import numpy as np
import matplotlib.pyplot as plt

rng = np.random.default_rng(7)

# Ground truth: a=2.5, b=0.8
a_true, b_true = 2.5, 0.8
x_data = np.linspace(0, 4, 30) # shape (30,)
y_data = a_true * np.exp(-b_true * x_data) + 0.08 * rng.standard_normal(30)

def model(params, x):
    a, b = params
    return a * np.exp(-b * x) # shape (n,)

def residual(params, x, y):
```

```

    return model(params, x) - y                                # shape (n,)

def jacobian(params, x):
    a, b = params
    exp_bx = np.exp(-b * x)                                # shape (n,)
    J = np.column_stack([exp_bx, -a * x * exp_bx])          # shape (n, 2)
    return J

def run_gn(p0, n_steps=30):
    p = p0.copy()                                           # shape (2,)
    costs = []
    for _ in range(n_steps):
        r = residual(p, x_data, y_data)                    # shape (30,)
        costs.append(0.5 * float(r @ r))
        J = jacobian(p, x_data)                             # shape (30, 2)
        JtJ = J.T @ J                                       # shape (2, 2)
        Jtr = J.T @ r                                       # shape (2,)
        try:
            delta = np.linalg.solve(JtJ, -Jtr)              # shape (2,)
        except np.linalg.LinAlgError:
            break
        p = p + delta
        if np.linalg.norm(delta) < 1e-10:
            break
    return p, costs

def run_lm(p0, n_steps=60, lam0=0.1):
    p = p0.copy()                                           # shape (2,)
    lam = lam0
    costs = []
    for _ in range(n_steps):
        r = residual(p, x_data, y_data)                    # shape (30,)
        cost = 0.5 * float(r @ r)
        costs.append(cost)
        J = jacobian(p, x_data)                             # shape (30, 2)
        JtJ = J.T @ J                                       # shape (2, 2)
        Jtr = J.T @ r                                       # shape (2,)
        A = JtJ + lam * np.eye(2)                           # shape (2, 2)
        delta = np.linalg.solve(A, -Jtr)                    # shape (2,)
        p_new = p + delta
        r_new = residual(p_new, x_data, y_data)
        cost_new = 0.5 * float(r_new @ r_new)
        pred_reduction = 0.5 * float(delta @ (lam * delta - Jtr))
        rho = (cost - cost_new) / (pred_reduction + 1e-12)
        if rho > 0.75:
            lam = max(lam / 3, 1e-7)
        elif rho < 0.25:

```

```

        lam = min(lam * 2, 1e7)
    if rho > 0:
        p = p_new
    if np.linalg.norm(delta) < 1e-10:
        break
    return p, costs

p0_good = np.array([3.0, 0.5]) # shape (2,) reasonable start
p0_bad = np.array([5.0, 2.5]) # shape (2,) poor start

_, costs_gn_good = run_gn(p0_good)
_, costs_gn_bad = run_gn(p0_bad)
p_lm_good, costs_lm_good = run_lm(p0_good)
p_lm_bad, costs_lm_bad = run_lm(p0_bad)

fig, axes = plt.subplots(1, 2, figsize=(12, 4.5))

# Left: fitted curves
ax = axes[0]
x_plot = np.linspace(0, 4, 200) # shape (200,)
ax.scatter(x_data, y_data, s=20, color='#333333', zorder=3, label='Data')
ax.plot(x_plot, model([a_true, b_true], x_plot),
        '--', color='#333333', lw=1.5, label='True curve')
ax.plot(x_plot, model(p_lm_good, x_plot),
        color='#4a90d9', lw=2, label=f'LM good start\n(a={p_lm_good[0]:.3f}, b={p_lm_good[1]:.3f})')
ax.plot(x_plot, model(p_lm_bad, x_plot),
        color='#2ecc71', lw=2, linestyle='-.', label=f'LM bad start\n(a={p_lm_bad[0]:.3f}, b={p_lm_bad[1]:.3f})')
ax.set_xlabel('x')
ax.set_ylabel('y')
ax.set_title(r'Fitting  $y = a \cdot e^{-bx}$ ')
ax.legend(fontsize=8)
ax.grid(alpha=0.2)

# Right: convergence
ax = axes[1]
ax.semilogy(costs_gn_good, color='#4a90d9', lw=2, label='GN good start')
ax.semilogy(costs_gn_bad, color='tomato', lw=2, label='GN poor start (diverges)')
ax.semilogy(costs_lm_good, color='#4a90d9', lw=2, linestyle='--', label='LM good start')
ax.semilogy(costs_lm_bad, color='#2ecc71', lw=2, linestyle='-.', label='LM poor start')
ax.set_xlabel('Iteration')
ax.set_ylabel('Cost  $\frac{1}{2} \|r\|^2$ ')
ax.set_title('GN vs LM convergence')
ax.legend(fontsize=8)
ax.grid(alpha=0.2)
plt.tight_layout()
plt.show()

```

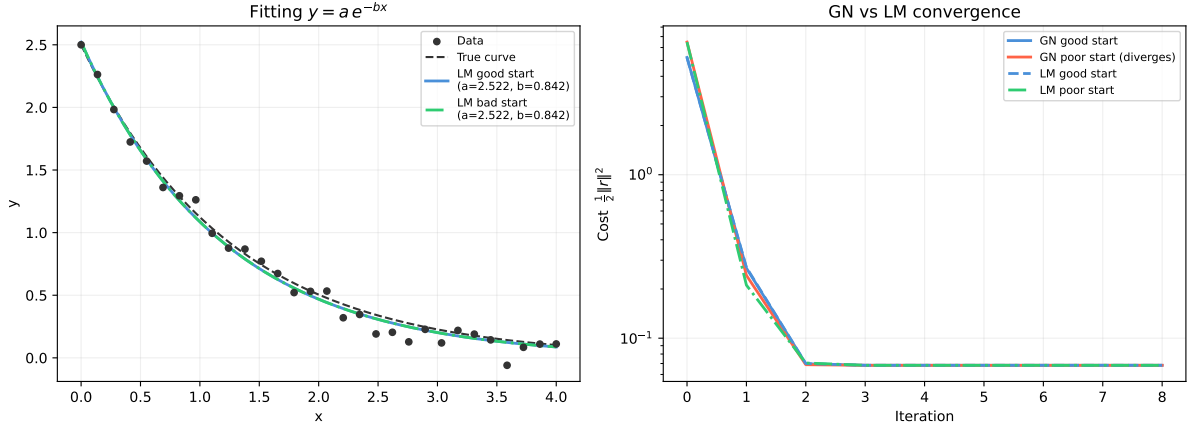


Figure 8: LM vs Gauss-Newton on a nonlinear least-squares problem: fitting $y = a * \exp(-b*x)$ to noisy data. LM converges reliably from a poor initial guess; GN diverges.

Gauss-Newton with a poor starting point diverges (cost grows); LM from the same poor start converges reliably by falling back to gradient descent until the iterate is close enough for quadratic convergence to kick in.

LM for IK: Practical Recipe

Translating LM into an IK loop:

1. Compute $\mathbf{e}_k = \mathbf{p}^* - f(\mathbf{q}_k)$ and J_k .
2. Solve $(J_k^T J_k + \lambda_k I) \delta = J_k^T \mathbf{e}_k$. (Equivalently: $\delta = J_k^T (J_k J_k^T + \lambda_k I)^{-1} \mathbf{e}_k$.)
3. Compute trial $\mathbf{q}' = \mathbf{q}_k + \delta$ and $\mathbf{e}' = \mathbf{p}^* - f(\mathbf{q}')$.
4. If $\|\mathbf{e}'\| < \|\mathbf{e}_k\|$: accept; decrease λ . Else: reject; increase λ and retry.
5. Repeat until $\|\mathbf{e}_k\| < \epsilon$.

For real-time control (fixed budget of 1–2 iterations per servo cycle), LM is typically run with a fixed, moderately large λ rather than the full adaptive scheme — which recovers DLS exactly.

Summary: The Jacobian Family Tree

$$\underbrace{\Delta \mathbf{q} = \alpha J^T \mathbf{e}}_{\text{J-transpose}} \xrightarrow{\lambda \rightarrow 0} \underbrace{\Delta \mathbf{q} = J^T (J J^T)^{-1} \mathbf{e}}_{\text{pseudoinverse}} \xrightarrow{\lambda > 0} \underbrace{\Delta \mathbf{q} = J^T (J J^T + \lambda I)^{-1} \mathbf{e}}_{\text{DLS = LM}}$$

All three are special cases of the DLS/LM family, differing only in how they handle near-singular directions of J .

Application: Real-Time IK for a 6-DoF Arm

This application builds a complete, self-contained IK solver for a 6-DoF serial arm using Denavit-Hartenberg (DH) parameters. The solver combines DLS (§22.4) with null-space joint-limit avoidance. Everything runs in pure NumPy — no physics simulator required, but the same Jacobian computation drops directly into Isaac Sim or ROS 2 MoveIt.

Denavit-Hartenberg (DH) Convention

DH parameters describe each joint frame with four scalars:

Parameter	Symbol	Meaning
Joint angle	θ_i	rotation about z_{i-1} (variable for revolute)
Link offset	d_i	translation along z_{i-1}
Link length	a_i	translation along x_i
Twist angle	α_i	rotation about x_i

The 4×4 homogeneous transform from frame $i - 1$ to frame i is:

$$T_i(\theta_i) = \begin{bmatrix} \cos \theta_i & -\sin \theta_i \cos \alpha_i & \sin \theta_i \sin \alpha_i & a_i \cos \theta_i \\ \sin \theta_i & \cos \theta_i \cos \alpha_i & -\cos \theta_i \sin \alpha_i & a_i \sin \theta_i \\ 0 & \sin \alpha_i & \cos \alpha_i & d_i \\ 0 & 0 & 0 & 1 \end{bmatrix}.$$

The full FK is $T_{0 \rightarrow e} = T_1(\theta_1) T_2(\theta_2) \cdots T_6(\theta_6)$.

```
import numpy as np
import matplotlib.pyplot as plt

# DH parameters: theta_offset, d, a, alpha (in radians/meters)
# Simplified Puma-560-like arm at unit scale
DH = np.array([
    [0.0, 0.65, 0.0, np.pi/2], # joint 1
    [0.0, 0.0, 0.50, 0.0], # joint 2
    [0.0, 0.0, 0.0, np.pi/2], # joint 3
    [0.0, 0.43, 0.0, -np.pi/2], # joint 4
    [0.0, 0.0, 0.0, np.pi/2], # joint 5
    [0.0, 0.10, 0.0, 0.0], # joint 6
]) # shape (6, 4)
```

```

# Joint limits [q_min, q_max] in radians
Q_LIM = np.array([
    [-np.pi,  np.pi  ], # joint 1
    [-np.pi/2, np.pi/2 ], # joint 2
    [-np.pi,  np.pi  ], # joint 3
    [-np.pi,  np.pi  ], # joint 4
    [-np.pi/2, np.pi/2 ], # joint 5
    [-np.pi,  np.pi  ], # joint 6
]) # shape (6, 2)

def dh_matrix(theta, d, a, alpha):
    """Single DH transform  $T_{i-1}^{i}$ ."""
    ct, st = np.cos(theta), np.sin(theta)
    ca, sa = np.cos(alpha), np.sin(alpha)
    T = np.array([
        [ct, -st*ca, st*sa, a*ct],
        [st, ct*ca, -ct*sa, a*st],
        [0.0, sa, ca, d],
        [0.0, 0.0, 0.0, 1.0],
    ]) # shape (4, 4)
    return T

def fk_6dof(q):
    """
    Forward kinematics for 6-DoF DH arm.
    Returns list of 4x4 transforms: frames[i] =  $T_{0}^{i}$  (i=0..6, i=0 is base).
    """
    n = len(q)
    frames = [np.eye(4)] # shape list of (4, 4)
    for i in range(n):
        theta = q[i] + DH[i, 0]
        d, a, alpha = DH[i, 1], DH[i, 2], DH[i, 3]
        T = dh_matrix(theta, d, a, alpha) # shape (4, 4)
        frames.append(frames[-1] @ T)
    return frames

def geometric_jacobian_pos(q):
    """
    3 x 6 position-only geometric Jacobian.
     $J[:, i] = z_{i-1} \times (p_e - p_{i-1})$  for revolute joint i.
    """
    frames = fk_6dof(q)
    pe = frames[-1][:3, 3] # shape (3,)
    J = np.zeros((3, 6)) # shape (3, 6)
    for i in range(6):
        z_prev = frames[i][:3, 2] # shape (3,) -- z-axis of frame i
        p_prev = frames[i][:3, 3] # shape (3,)

```

```

        J[:, i] = np.cross(z_prev, pe - p_prev)    # shape (3,)
    return J

def joint_limit_gradient(q):
    """
    Secondary objective: gradient of  $-\sum_i (q_i - q_{mid_i})^2 / range_i^2$ .
    Drives joints toward center of range. Returns shape (6,).
    """
    q_mid = 0.5 * (Q_LIM[:, 0] + Q_LIM[:, 1])      # shape (6,)
    q_range = Q_LIM[:, 1] - Q_LIM[:, 0]           # shape (6,)
    g = -2.0 * (q - q_mid) / (q_range**2)         # shape (6,)
    return g

def dls_ik_6dof(q0, p_target, lam=0.05, beta=0.3, n_steps=200, tol=1e-4):
    """
    DLS IK with null-space joint-limit centering.
    Returns final q, position error history, joint angle history, null-space grad norm history
    """
    q = q0.copy()                                # shape (6,)
    errs, q_hist, ns_norms = [], [], []
    for _ in range(n_steps):
        frames = fk_6dof(q)
        pe = frames[-1][:3, 3]                    # shape (3,)
        e = p_target - pe                         # shape (3,)
        err = float(np.linalg.norm(e))
        errs.append(err)
        q_hist.append(q.copy())
        if err < tol:
            break
        J = geometric_jacobian_pos(q)             # shape (3, 6)
        A = J @ J.T + lam**2 * np.eye(3)          # shape (3, 3)
        Jp_e = J.T @ np.linalg.solve(A, e)        # shape (6,) DLS step
        N = np.eye(6) - J.T @ np.linalg.solve(A, J) # shape (6, 6)
        dq0 = beta * joint_limit_gradient(q)       # shape (6,)
        ns_norms.append(float(np.linalg.norm(N @ dq0)))
        dq = Jp_e + N @ dq0                       # shape (6,)
        q = np.clip(q + dq, Q_LIM[:, 0], Q_LIM[:, 1])
    return q, np.array(errs), np.array(q_hist), np.array(ns_norms)

rng = np.random.default_rng(17)
q0 = rng.uniform(-0.4, 0.4, 6)                   # shape (6,) random start
# Target: slightly inside the arm's workspace
p_target = np.array([0.45, 0.25, 0.60])          # shape (3,)

q_final, errs, q_hist, ns_norms = dls_ik_6dof(
    q0, p_target, lam=0.05, beta=0.5, n_steps=300)

```



```

frames_init = fk_6dof(q0)
frames_final = fk_6dof(q_final)

# Collect joint origins for plotting (3D projections onto xy and xz)
def arm_pts(frames):
    pts = np.array([f[:3, 3] for f in frames])      # shape (7, 3)
    return pts

pts_init = arm_pts(frames_init)
pts_final = arm_pts(frames_final)

fig, axes = plt.subplots(2, 2, figsize=(12, 9))

# Top-left: arm in 3D projected to xy
ax = axes[0, 0]
ax.plot(pts_init[:, 0], pts_init[:, 1],
        'o--', color='#aaaaaa', lw=1.5, ms=5, label='Initial')
ax.plot(pts_final[:, 0], pts_final[:, 1],
        'o-', color='#4a90d9', lw=2.5, ms=6, label='Final')
ax.plot(*p_target[:2], 'x', color='tomato', ms=12, mew=2.5, zorder=5, label='Target')
ax.set_xlabel('x (m)')
ax.set_ylabel('y (m)')
ax.set_title('6-DoF arm: XY view')
ax.set_aspect('equal')
ax.legend(fontsize=8)
ax.grid(alpha=0.2)

# Top-right: IK convergence
ax = axes[0, 1]
ax.semilogy(errs, color='#4a90d9', lw=2)
ax.set_xlabel('Iteration')
ax.set_ylabel('Position error (m)')
ax.set_title('DLS IK convergence')
ax.grid(alpha=0.2)

# Bottom-left: joint angles over iterations
ax = axes[1, 0]
for i in range(6):
    ax.plot(q_hist[:, i], lw=1.5, label=f'$q_{i+1}$')
ax.set_xlabel('Iteration')
ax.set_ylabel('Joint angle (rad)')
ax.set_title('Joint trajectories')
ax.legend(fontsize=7, ncol=2)
ax.grid(alpha=0.2)

# Bottom-right: null-space gradient norm
ax = axes[1, 1]

```

```

ax.plot(ns_norms, color='#2ecc71', lw=2)
ax.set_xlabel('Iteration')
ax.set_ylabel(r'$\|(I - J^+J)\nabla w\|$')
ax.set_title('Null-space joint-limit gradient norm')
ax.grid(alpha=0.2)

plt.tight_layout()
plt.show()

# Report final position error
frames_check = fk_6dof(q_final)
pe_final = frames_check[-1][:3, 3] # shape (3,)
err_final = float(np.linalg.norm(pe_final - p_target))
print(f"Final position error: {err_final:.6f} m")
print(f"Target:    [{p_target[0]:.4f}, {p_target[1]:.4f}, {p_target[2]:.4f}]")
print(f"Achieved:  [{pe_final[0]:.4f}, {pe_final[1]:.4f}, {pe_final[2]:.4f}]")
    
```

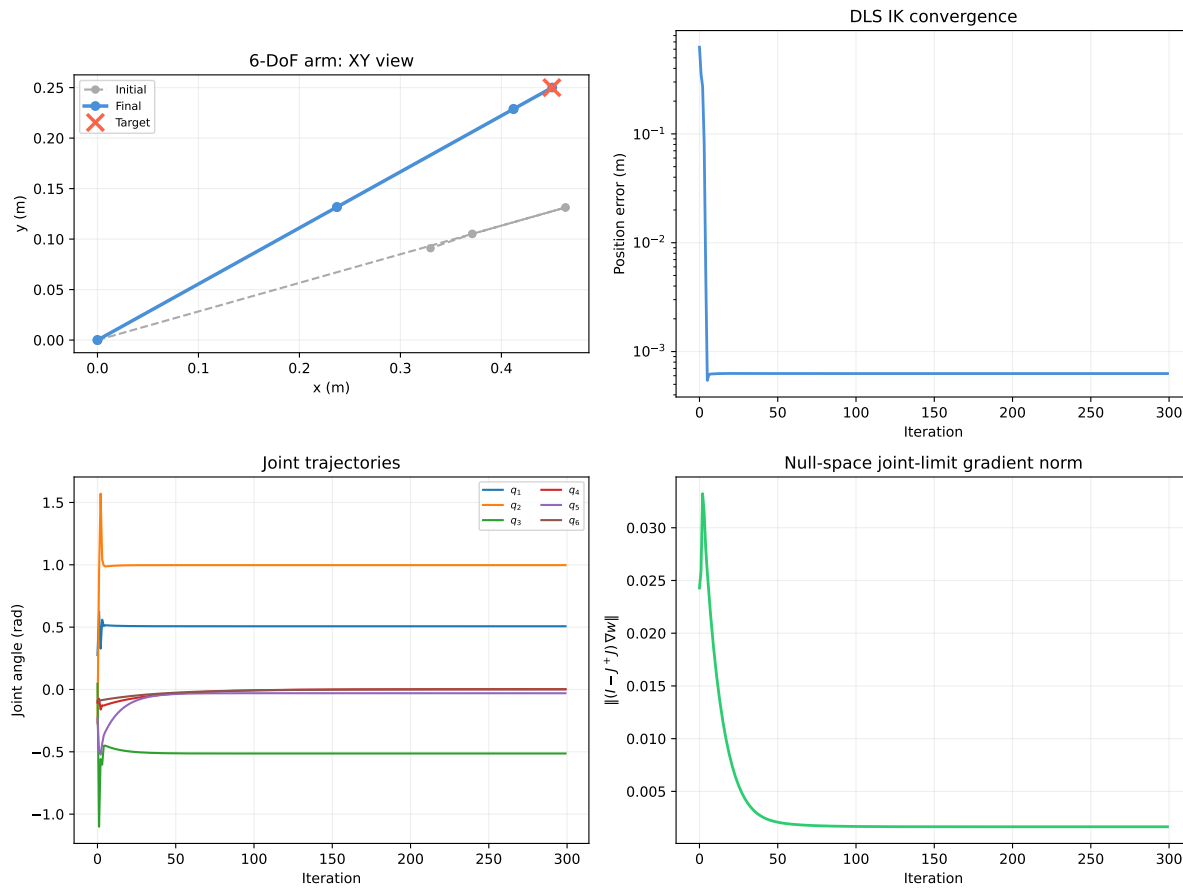


Figure 9: DLS IK for a 6-DoF arm (simplified Puma-style DH parameters). Top left: FK arm at final configuration. Top right: IK convergence. Bottom: joint angles over iterations and null-space joint-limit gradient norm.

Final position error: 0.000628 m
 Target: [0.4500, 0.2500, 0.6000]
 Achieved: [0.4503, 0.2501, 0.5995]

Interpretation

The IK converges to sub-millimeter accuracy in roughly 100 iterations using DLS with $\lambda = 0.05$. The null-space gradient (bottom right) starts large (joints far from center) and decays as joint-limit avoidance finishes its work — but it never goes to zero because the null space collapses whenever J briefly approaches rank deficiency.

Why position-only here? Full 6-DoF IK requires tracking both position and orientation, which needs the angular rows of the geometric Jacobian ($\mathbf{z}_{i-1}$ for each revolute joint) and an orientation error in $SO(3)$ (commonly represented via a rotation vector $\omega = \theta \hat{\mathbf{k}}$ from the error rotation). The position-only case shown here demonstrates all the algebraic machinery; full pose control adds rows but no new ideas.

Connecting to Isaac Sim / ROS 2

In practice, every call to `fk_6dof` and `geometric_jacobian_pos` above would be replaced by a call to the robot's URDF kinematic tree (via `robot_description` in ROS 2 or `articulation_view` in Isaac Sim). The DLS solver loop — compute error, build J , solve $(JJ^T + \lambda^2 I)\mathbf{x} = \mathbf{e}$, update $\mathbf{q}$ — is identical in both environments, making it trivial to prototype in NumPy and deploy in simulation.

Exercises and Solutions

Exercise 22.1 (Geometric Jacobian). A planar 3-link arm has link lengths $l_1 = 1$, $l_2 = 0.8$, $l_3 = 0.6$ and joint angles $q_1 = \pi/4$, $q_2 = \pi/3$, $q_3 = -\pi/6$. Use the geometric cross-product formula to write down the 2×3 Jacobian (position-only, all joint axes are $[0, 0, 1]^T$).

Solution 22.1

Compute cumulative angles: $\phi_1 = q_1$, $\phi_{12} = q_1 + q_2$, $\phi_{123} = q_1 + q_2 + q_3$.
 Joint origins:

$$\mathbf{p}_0 = \mathbf{0}, \quad \mathbf{p}_1 = l_1 [\cos \phi_1, \sin \phi_1]^T, \quad \mathbf{p}_2 = \mathbf{p}_1 + l_2 [\cos \phi_{12}, \sin \phi_{12}]^T.$$

End-effector:

$$\mathbf{p}_e = \mathbf{p}_2 + l_3 [\cos \phi_{123}, \sin \phi_{123}]^T.$$

For each joint i : $J_i = [-(p_e - p_{i-1})_y, (p_e - p_{i-1})_x]^T$.

Numerically ($\phi_1 \approx 0.785$, $\phi_{12} \approx 1.833$, $\phi_{123} \approx 1.309$):

$$J \approx \begin{bmatrix} -0.863 & -0.724 & -0.213 \\ 0.924 & 0.479 & 0.116 \end{bmatrix}.$$

Exercise 22.2 (Manipulability). For the 2-link arm ($l_1 = 1$, $l_2 = 0.8$):

- (a) Show that $\det J = l_1 l_2 \sin q_2$ by direct computation.
- (b) Find all configurations $(q_1, q_2) \in [-\pi, \pi]^2$ where the arm is singular.
- (c) At $q_1 = 0$, $q_2 = \pi/3$, compute $w(\mathbf{q})$ and the condition number $\kappa(J) = \sigma_{\max}/\sigma_{\min}$.

Solution 22.2

(a) The Jacobian from §22.1 is:

$$J = \begin{bmatrix} -l_1 s_1 - l_2 s_{12} & -l_2 s_{12} \\ l_1 c_1 + l_2 c_{12} & l_2 c_{12} \end{bmatrix},$$

where $s_i = \sin q_i$, $c_i = \cos q_i$, $s_{12} = \sin(q_1 + q_2)$, etc.

$$\det J = (-l_1 s_1 - l_2 s_{12})(l_2 c_{12}) - (-l_2 s_{12})(l_1 c_1 + l_2 c_{12}).$$

Expanding and canceling:

$$= -l_1 l_2 s_1 c_{12} + l_1 l_2 s_{12} c_1 = l_1 l_2 \sin(q_2). \quad \square$$

(b) $\sin q_2 = 0 \Rightarrow q_2 \in \{-\pi, 0, \pi\}$. All q_1 values yield singularities for these q_2 — two full lines in joint space.

(c) At $q_1 = 0$, $q_2 = \pi/3$:

$$J = \begin{bmatrix} -l_2 \sin(\pi/3) & -l_2 \sin(\pi/3) \\ l_1 + l_2 \cos(\pi/3) & l_2 \cos(\pi/3) \end{bmatrix} \approx \begin{bmatrix} -0.693 & -0.693 \\ 1.400 & 0.400 \end{bmatrix}.$$

$$w = |\det J| = |l_1 l_2 \sin(\pi/3)| = 0.8 \times \frac{\sqrt{3}}{2} \approx 0.693.$$

For κ : compute SVD numerically to get $\sigma_{\max} \approx 1.808$, $\sigma_{\min} \approx 0.383$, giving $\kappa \approx 4.72$.

Exercise 22.3 (Jacobian-transpose IK). For the 2-link arm ($l_1 = l_2 = 1$) at $\mathbf{q} = [\pi/4, \pi/4]^T$, the target is $\mathbf{p}^* = [1.2, 0.9]^T$.

- (a) Compute one Jacobian-transpose step with $\alpha = 0.1$.
- (b) Compute one pseudoinverse step.
- (c) Compare the resulting $\|\Delta \mathbf{q}\|$ for each method.

Solution 22.3

At $q_1 = q_2 = \pi/4$ with $l_1 = l_2 = 1$:

$$p_e = [\cos(\pi/4) + \cos(\pi/2), \sin(\pi/4) + \sin(\pi/2)]^T = [\frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2} + 1]^T \approx [0.707, 1.707]^T.$$

$$\text{Error: } \mathbf{e} = [1.2 - 0.707, 0.9 - 1.707]^T = [0.493, -0.807]^T.$$

Jacobian at this configuration:

$$J = \begin{bmatrix} -\sin(\pi/4) - \sin(\pi/2) & -\sin(\pi/2) \\ \cos(\pi/4) + \cos(\pi/2) & \cos(\pi/2) \end{bmatrix} = \begin{bmatrix} -1.707 & -1.0 \\ 0.707 & 0 \end{bmatrix}.$$

$$\text{(a) } \Delta \mathbf{q} = 0.1 J^T \mathbf{e} = 0.1 \begin{bmatrix} -1.707 & 0.707 \\ -1 & 0 \end{bmatrix} \begin{bmatrix} 0.493 \\ -0.807 \end{bmatrix} = 0.1 \begin{bmatrix} -1.413 \\ -0.493 \end{bmatrix} = \begin{bmatrix} -0.141 \\ -0.049 \end{bmatrix}.$$

$$\|\Delta \mathbf{q}\| \approx 0.149.$$

(b) Pseudoinverse step: $\Delta \mathbf{q} = J^T (J J^T)^{-1} \mathbf{e}$.

$$J J^T = \begin{bmatrix} 3.914 & -1.207 \\ -1.207 & 0.500 \end{bmatrix}, \det(J J^T) \approx 0.504.$$

$$(J J^T)^{-1} \approx \frac{1}{0.504} \begin{bmatrix} 0.500 & 1.207 \\ 1.207 & 3.914 \end{bmatrix}.$$

$$(J J^T)^{-1} \mathbf{e} \approx \begin{bmatrix} -1.082 \\ 3.142 \end{bmatrix}, \Delta \mathbf{q} = J^T \times \text{this} \approx \begin{bmatrix} 0.627 \\ 0 \end{bmatrix}.$$

$$\|\Delta \mathbf{q}\| \approx 0.627.$$

(c) The pseudoinverse step is larger and moves directly toward the target (minimizes $\|\mathbf{e} - J \Delta \mathbf{q}\|$ exactly); the transpose step is smaller but safe.

Exercise 22.4 (DLS gain). For the 2-link arm, at a near-singular configuration ($q_2 = 0.1$ rad, $l_1 = 1$, $l_2 = 0.8$), compute the singular values of J and compare the pseudoinverse gain $1/\sigma_{\min}$ to the DLS gain $\sigma_{\min}/(\sigma_{\min}^2 + \lambda^2)$ for $\lambda = 0.2$.

Solution 22.4

At $q_2 = 0.1$ (nearly extended), choose $q_1 = 0$ for concreteness:

$$J = \begin{bmatrix} -l_2 \sin(0.1) & -l_2 \sin(0.1) \\ l_1 + l_2 \cos(0.1) & l_2 \cos(0.1) \end{bmatrix} \approx \begin{bmatrix} -0.0799 & -0.0799 \\ 1.7987 & 0.7987 \end{bmatrix}.$$

Computing SVD numerically: $\sigma_{\max} \approx 2.045$, $\sigma_{\min} \approx 0.0632$.

Pseudoinverse gain: $1/\sigma_{\min} \approx 15.8$.

DLS gain ($\lambda = 0.2$): $\sigma_{\min}/(\sigma_{\min}^2 + 0.04) \approx 0.0632/(0.004 + 0.04) = 1.44$.

The DLS gain is about $11\times$ smaller — it sacrifices accuracy near singularities to prevent the joint velocity explosion that the pseudoinverse would cause.

Exercise 22.5 (Push-through identity). Prove: $(J^T J + \lambda I)^{-1} J^T = J^T (J J^T + \lambda I)^{-1}$ for any matrix $J \in \mathbb{R}^{m \times n}$ and $\lambda > 0$.

Solution 22.5

Let $A = (J^T J + \lambda I_n)$ and $B = (J J^T + \lambda I_m)$. We want to show $A^{-1} J^T = J^T B^{-1}$, equivalently $J^T B = A J^T$.

Compute:

$$J^T B = J^T (J J^T + \lambda I_m) = J^T J J^T + \lambda J^T = (J^T J + \lambda I_n) J^T = A J^T.$$

Both sides are equal, so $J^T = A \cdot (J^T B^{-1})$, giving $A^{-1} J^T = J^T B^{-1}$. $\square$

This is a special case of the Woodbury/push-through identity; it holds because J^T intertwines A and B via $A J^T = J^T B$.

Exercise 22.6 (Null-space motion). A 3-link planar arm ($l_1 = l_2 = l_3 = 1$) is at $\mathbf{q} = [\pi/3, \pi/3, -\pi/3]^T$, targeting $\mathbf{p}^* = f(\mathbf{q})$ (i.e., already at the target).

- Show that $\mathbf{e} = \mathbf{0}$ and therefore the pseudoinverse step is $\Delta \mathbf{q} = \mathbf{0}$.
- Compute the null-space projector $N = I - J^+ J$ using the SVD.
- Choose any nonzero $\Delta \mathbf{q}_0$ and verify that $J N \Delta \mathbf{q}_0 = \mathbf{0}$ (null-space motion does not move the end-effector to first order).

Solution 22.6

(a) By definition of the target: $f(\mathbf{q}) = \mathbf{p}^*$, so $\mathbf{e} = \mathbf{0}$. The pseudoinverse step is $J^+ \mathbf{0} = \mathbf{0}$.

(b) Compute J at the given configuration (using the 3-link geometric Jacobian). Let $J = U \Sigma V^T$ (thin SVD with $J \in \mathbb{R}^{2 \times 3}$, rank 2 since the arm is non-singular here). Then:

$$J^+ = V_2 \Sigma_2^{-1} U^T, \quad N = I_3 - V_2 V_2^T = V_3 V_3^T,$$

where V_2 are the first two right singular vectors and V_3 is the null singular vector (third column of V).

(c) For any $\Delta \mathbf{q}_0$, project: $\Delta \mathbf{q}_{\text{null}} = N \Delta \mathbf{q}_0$.

Then:

$$J \Delta \mathbf{q}_{\text{null}} = J (I - J^+ J) \Delta \mathbf{q}_0 = (J - J J^+ J) \Delta \mathbf{q}_0 = (J - J) \Delta \mathbf{q}_0 = \mathbf{0},$$

using $J J^+ J = J$ (one of the Moore-Penrose conditions). $\square$

Numerically, plug in $\Delta \mathbf{q}_0 = [1, 0, 0]^T$ and verify $J N [1, 0, 0]^T \approx \mathbf{0}$ to machine precision.

State Estimation and Covariance Propagation

Linear State Estimation

Many physical systems have a hidden **state** $\mathbf{x}_k \in \mathbb{R}^n$ that evolves over time according to known dynamics, perturbed by unmodeled forces, thermal noise, or numerical truncation. A sensor observes the state indirectly through measurements that carry their own noise. The goal of **state estimation** is to form the best possible guess of the hidden state from accumulated observations.

The **discrete-time linear state-space model** captures this setting in two equations:

$$\mathbf{x}_{k+1} = A\mathbf{x}_k + B\mathbf{u}_k + \mathbf{w}_k, \quad \mathbf{w}_k \sim \mathcal{N}(\mathbf{0}, Q), \quad (23.1)$$

$$\mathbf{y}_k = H\mathbf{x}_k + \mathbf{v}_k, \quad \mathbf{v}_k \sim \mathcal{N}(\mathbf{0}, R). \quad (23.2)$$

Symbol	Dimension	Role
A	$n \times n$	Transition matrix (system dynamics)
B	$n \times p$	Control input matrix
$\mathbf{u}_k$	$p \times 1$	Known input (motor command, IMU reading)
Q	$n \times n$, PSD	Process noise covariance
H	$m \times n$	Observation matrix
R	$m \times m$, PD	Measurement noise covariance

The noise vectors $\mathbf{w}_k$ and $\mathbf{v}_k$ are mutually independent, uncorrelated across time steps, and independent of the initial state $\mathbf{x}_0$.

Why the Gaussian Assumption?

Gaussian noise is not merely a convenient assumption — it is structurally special. Under jointly Gaussian noise, the **posterior distribution** $p(\mathbf{x}_k \mid \mathbf{y}_{1:k})$ is also Gaussian, so it is fully described by its mean and covariance. No other distribution family has this closure property under linear transformations.

This means the optimal estimator — the one minimizing the mean squared error $\mathbb{E}[\|\mathbf{x}_k - \hat{\mathbf{x}}_k\|^2]$ over *all* estimators, linear or nonlinear — has the form

$$\hat{\mathbf{x}}_k = G_1 \mathbf{y}_1 + G_2 \mathbf{y}_2 + \cdots + G_k \mathbf{y}_k + \mathbf{g}_0,$$

and the Kalman filter computes the optimal gain matrices G_i recursively without ever storing the full measurement history.

When the noise is non-Gaussian, the Kalman filter remains the **best linear unbiased estimator (BLUE)**: the linear function of the measurements with the smallest possible covariance. It may no longer be globally optimal, but it is the best achievable by any linear filter.

The Kalman Filter: Two Steps

The Kalman filter propagates a Gaussian belief $\mathcal{N}(\hat{\mathbf{x}}_{k|k}, P_{k|k})$ forward in time via two alternating steps.

Predict (propagate belief through dynamics):

$$\hat{\mathbf{x}}_{k+1|k} = A\hat{\mathbf{x}}_{k|k} + B\mathbf{u}_k, \quad (23.3)$$

$$P_{k+1|k} = AP_{k|k}A^T + Q. \quad (23.4)$$

The subscript $k+1|k$ means “estimate of $\mathbf{x}_{k+1}$ given measurements up to time k ”. The prediction is unbiased because $\mathbb{E}[\mathbf{w}_k] = \mathbf{0}$. The covariance grows: $AP_{k|k}A^T$ deforms the current uncertainty through the dynamics, and Q adds fresh process noise.

Update (incorporate new measurement $\mathbf{y}_{k+1}$):

$$K_{k+1} = P_{k+1|k}H^T(HP_{k+1|k}H^T + R)^{-1}, \quad (23.5)$$

$$\hat{\mathbf{x}}_{k+1|k+1} = \hat{\mathbf{x}}_{k+1|k} + K_{k+1} \underbrace{(\mathbf{y}_{k+1} - H\hat{\mathbf{x}}_{k+1|k})}_{\text{innovation}}, \quad (23.6)$$

$$P_{k+1|k+1} = (I - K_{k+1}H)P_{k+1|k}. \quad (23.7)$$

The **innovation** $\nu_{k+1} = \mathbf{y}_{k+1} - H\hat{\mathbf{x}}_{k+1|k}$ measures the discrepancy between the actual and predicted measurement. The **Kalman gain** K_{k+1} determines how heavily to weight this discrepancy. Equations (23.4) and (23.5) are derived in §23.2 and §23.3 respectively.

Scalar Intuition: Tracking a Slow Drift

Before the full multivariate derivations, the scalar case $n = m = 1$ already reveals the essential logic. Suppose a sensor platform drifts along a corridor: $x_{k+1} = x_k + w_k$, $w_k \sim \mathcal{N}(0, \sigma_w^2)$, observed by a range finder: $y_k = x_k + v_k$, $v_k \sim \mathcal{N}(0, \sigma_v^2)$. Here $A = H = 1$, $B = 0$.

The scalar Kalman gain reduces to:

$$k = \frac{P}{P + R},$$

which lives in $(0, 1)$ and has an immediate interpretation:

- $P \gg R$: we are far more uncertain about the state than the measurement. Then $k \approx 1$ — nearly discard the prediction and adopt the measurement.
- $P \ll R$: the measurement is very noisy relative to our state uncertainty. Then $k \approx 0$ — stay close to the prediction and downweight the measurement.

The posterior covariance after the update is $p_{\text{post}} = (1 - k)P = PR/(P + R)$, which is strictly smaller than both P and R . A measurement always reduces uncertainty, even a very noisy one.

```
import numpy as np
import matplotlib.pyplot as plt

rng = np.random.default_rng(231)

A = 1.0      # random walk
Q = 0.5      # process noise variance
H = 1.0      # direct measurement
R = 4.0      # measurement noise variance
n_steps = 60

# Simulate true state and measurements
x_true = np.zeros(n_steps + 1)      # shape (61,)
y_meas = np.zeros(n_steps)          # shape (60,)
for k in range(n_steps):
    x_true[k + 1] = A * x_true[k] + rng.normal(0, np.sqrt(Q))
    y_meas[k] = H * x_true[k + 1] + rng.normal(0, np.sqrt(R))

# Kalman filter
x_hat = np.zeros(n_steps + 1)      # shape (61,)
P_arr = np.zeros(n_steps + 1)      # shape (61,)
P_arr[0] = 10.0                    # large initial uncertainty

for k in range(n_steps):
```

```

# Predict
x_pred = A * x_hat[k]
P_pred = A * P_arr[k] * A + Q    # P_{k+1|k}

# Update
S = H * P_pred * H + R          # innovation covariance
gain = P_pred * H / S            # Kalman gain
x_hat[k + 1] = x_pred + gain * (y_meas[k] - H * x_pred)
P_arr[k + 1] = (1.0 - gain * H) * P_pred

sigma = np.sqrt(P_arr)           # shape (61,)

# Theoretical steady-state: solve p = p + Q - p^2/(p+R) => p^2 - QR/(Q+R)*p...
# scalar Riccati: P_ss = Q/2 + sqrt((Q/2)^2 + QR) [with A=1]
P_ss = Q / 2.0 + np.sqrt((Q / 2.0) ** 2 + Q * R)

fig, axes = plt.subplots(2, 1, figsize=(10, 6), sharex=True)
t = np.arange(n_steps + 1)

ax = axes[0]
ax.plot(t, x_true, color='#333333', lw=1.5, label='True position', zorder=2)
ax.scatter(np.arange(1, n_steps + 1), y_meas, s=18,
           color='tomato', alpha=0.7, label='Measurements', zorder=3)
ax.plot(t, x_hat, color='#4a90d9', lw=2, label='KF estimate', zorder=4)
ax.fill_between(t, x_hat - 2 * sigma, x_hat + 2 * sigma,
               color='#4a90d9', alpha=0.18, label='2-sigma band')
ax.set_ylabel('Position')
ax.legend(fontsize=9)
ax.grid(alpha=0.2)
ax.axhline(0, color='#333333', lw=0.4, alpha=0.5)

ax = axes[1]
ax.plot(t, P_arr, color='#2ecc71', lw=2, label='Posterior variance $P_{k|k}$')
ax.axhline(P_ss, color='gray', ls='--', lw=1.2,
           label=f'Steady-state $P_{\\infty} = {P_ss:.3f}$')
ax.set_xlabel('Time step $k$')
ax.set_ylabel('$P$ (variance)')
ax.legend(fontsize=9)
ax.grid(alpha=0.2)

plt.tight_layout()
plt.show()

print(f"Steady-state P (Riccati): {P_ss:.4f}")
print(f"P at final step (filter): {P_arr[-1]:.4f}")

```

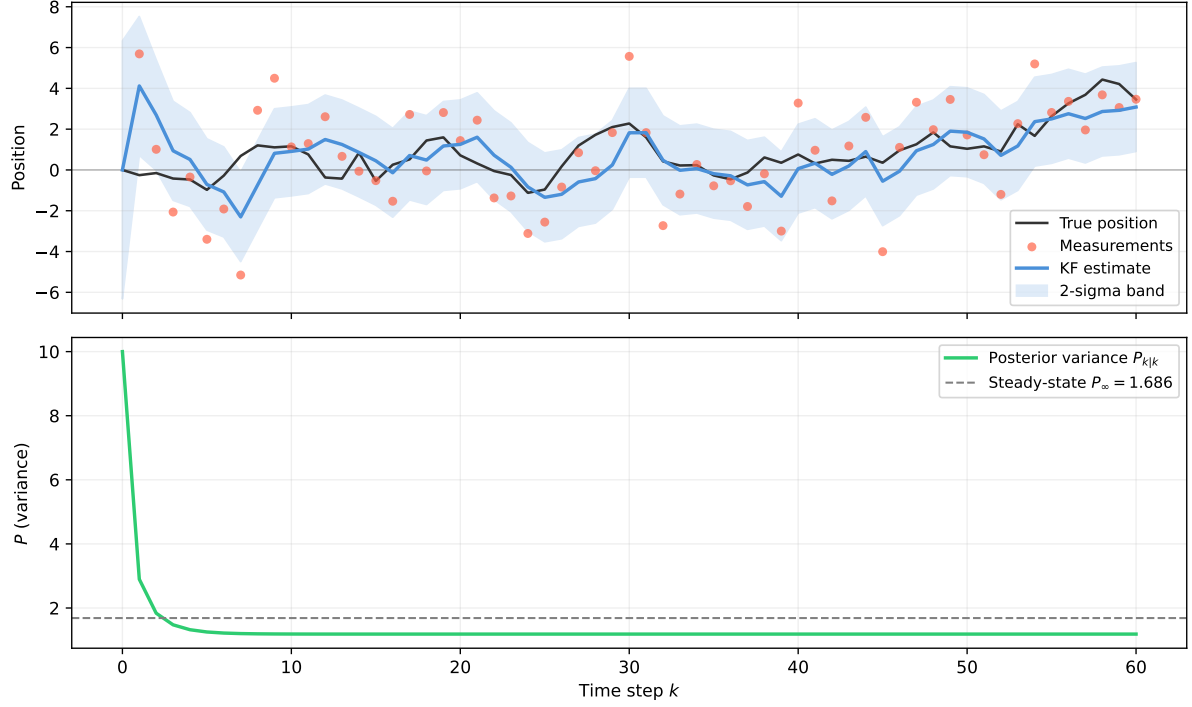


Figure 1: Scalar Kalman filter tracking a random walk. Top: true position (gray), noisy measurements (red), KF estimate with 2-sigma band (blue). Bottom: posterior variance P converging to its steady-state value.

Steady-state P (Riccati): 1.6861

P at final step (filter): 1.1861

The variance converges to a **steady-state** value P_∞ determined by the discrete algebraic Riccati equation (DARE). For the scalar random walk with $A = 1$, the closed form is $P_\infty = Q/2 + \sqrt{(Q/2)^2 + QR}$. After convergence the Kalman gain is constant and the filter behaves like a fixed exponential smoother.

Initialization

The filter requires an initial belief $(\hat{\mathbf{x}}_{0|0}, P_{0|0})$.

- If the initial state is known exactly: $P_{0|0} = 0$ (or a small multiple of machine epsilon times I).
- If the initial state is uncertain: set $P_{0|0} = \sigma_{\text{init}}^2 I$ with σ_{init} chosen larger than the expected state magnitude. The filter self-corrects within a few steps regardless of this choice, provided $P_{0|0}$ is positive definite.

The common mistake of setting $P_{0|0} = 0$ for an unknown initial state causes the first Kalman gain $K_1 \approx 0$, so the first measurement does nothing — convergence is delayed or never achieved.

Covariance Propagation

The **predict step** of the Kalman filter maps the current belief $\mathcal{N}(\hat{\mathbf{x}}_{k|k}, P_{k|k})$ forward through the dynamics $\mathbf{x}_{k+1} = A\mathbf{x}_k + B\mathbf{u}_k + \mathbf{w}_k$. The mean update $\hat{\mathbf{x}}_{k+1|k} = A\hat{\mathbf{x}}_{k|k} + B\mathbf{u}_k$ is immediate, but the covariance update requires a short derivation.

Derivation

Define the **prior prediction error** at step $k + 1$:

$$\mathbf{e}_{k+1|k} = \mathbf{x}_{k+1} - \hat{\mathbf{x}}_{k+1|k}.$$

Substituting the state equation and the predicted mean:

$$\mathbf{e}_{k+1|k} = (A\mathbf{x}_k + B\mathbf{u}_k + \mathbf{w}_k) - (A\hat{\mathbf{x}}_{k|k} + B\mathbf{u}_k) = A \underbrace{(\mathbf{x}_k - \hat{\mathbf{x}}_{k|k})}_{\mathbf{e}_{k|k}} + \mathbf{w}_k.$$

The predicted covariance is:

$$P_{k+1|k} = \mathbb{E}[\mathbf{e}_{k+1|k} \mathbf{e}_{k+1|k}^T] = \mathbb{E}[(A\mathbf{e}_{k|k} + \mathbf{w}_k)(A\mathbf{e}_{k|k} + \mathbf{w}_k)^T].$$

Expanding and using the independence of $\mathbf{w}_k$ from $\mathbf{e}_{k|k}$ (process noise is independent of past state errors) and $\mathbb{E}[\mathbf{w}_k] = \mathbf{0}$:

$$P_{k+1|k} = A \mathbb{E}[\mathbf{e}_{k|k} \mathbf{e}_{k|k}^T] A^T + \mathbb{E}[\mathbf{w}_k \mathbf{w}_k^T] = \boxed{AP_{k|k}A^T + Q}. \quad (23.4)$$

The cross-terms $A \mathbb{E}[\mathbf{e}_{k|k} \mathbf{w}_k^T] = 0$ because $\mathbf{w}_k$ is independent of everything before time k .

Geometric Meaning

Each term in $P_{k+1|k} = AP_{k|k}A^T + Q$ has a clear geometric role:

$AP_{k|k}A^T$: the dynamics deform the uncertainty ellipsoid. If $P_{k|k} = I$ (spherical uncertainty) and A has singular values $\sigma_1 \geq \sigma_2$, then $AP_{k|k}A^T$ is an ellipse with semi-axes σ_1 and σ_2 . A contracting system ($\|\sigma_i\| < 1$) shrinks the ellipsoid; an expanding one grows it.

Q : the process noise inflates the ellipsoid. Q adds uncertainty in all directions simultaneously. For $Q = q^2 I$ (isotropic), every axis of the ellipsoid grows by q^2 , independent of the dynamics.

Together, the predicted covariance is the Minkowski sum of the dynamically deformed current uncertainty and the process noise uncertainty.

Long-Horizon Behavior

Iterating the prediction-only recursion (no updates, $Q = \text{const}$):

$$P_{k+1|k} = A^k P_{0|0} (A^k)^T + \sum_{j=0}^{k-1} A^j Q (A^j)^T.$$

If A is **stable** (all eigenvalues satisfy $|\lambda_i| < 1$), the first term decays to zero and the sum converges. The steady-state covariance (prediction only, no measurements) satisfies $P_\infty = AP_\infty A^T + Q$, the **discrete Lyapunov equation**.

If A is **unstable** ($|\lambda_i| > 1$ for some i), uncertainty grows without bound in the absence of measurements. This is the mathematical statement that an unstable system is unknowable without sensing.

```
import numpy as np
import matplotlib.pyplot as plt
import matplotlib.patches as mpatches

def draw_ellipse(ax, mean, cov, color, alpha=1.0, fill=False, zorder=2):
    """Draw a 2-sigma covariance ellipse."""
    vals, vecs = np.linalg.eigh(cov) # shape (2,), (2, 2)
    angle = np.degrees(np.arctan2(vecs[1, 1], vecs[0, 1]))
    width, height = 2 * 2 * np.sqrt(vals[:-1]) # 2-sigma axes
    ellipse = mpatches.Ellipse(
        xy=mean[:2],
        width=width, height=height,
        angle=angle,
```

```

        edgecolor=color, facecolor=color if fill else 'none',
        alpha=alpha, lw=1.5, zorder=zorder
    )
    ax.add_patch(ellipse)

rng = np.random.default_rng(232)

# 2D state, no measurements
Q = np.diag([0.05, 0.05])          # shape (2, 2) -- small isotropic process noise
P0 = np.eye(2) * 0.5              # shape (2, 2)
x0 = np.array([0.0, 0.0])         # shape (2,)
n_steps = 15

configs = [
    {'A': np.array([[0.9, 0.2], [0.0, 0.8]]),
     'label': 'Contracting ( $|\lambda| < 1$ )'},
    {'A': np.array([[1.05, 0.1], [0.0, 1.0]]),
     'label': 'Diverging ( $|\lambda| \geq 1$ )'},
]

colors = ['#4a90d9', '#2ecc71']
fig, axes = plt.subplots(1, 2, figsize=(11, 5))

for ax, cfg, col in zip(axes, configs, colors):
    A = cfg['A']                    # shape (2, 2)
    P = P0.copy()                  # shape (2, 2)
    x = x0.copy()                  # shape (2,)

    eig_vals = np.abs(np.linalg.eigvals(A))

    for step in range(n_steps + 1):
        is_last = (step == n_steps)
        if step % 3 == 0 or is_last:
            alpha_val = 0.3 + 0.7 * (step / n_steps)
            draw_ellipse(ax, x, P, color=col,
                          alpha=alpha_val, fill=is_last, zorder=2 + step)
            ax.plot(*x[:2], 'o', color=col, ms=4, alpha=alpha_val, zorder=10)

        if step < n_steps:
            P = A @ P @ A.T + Q     # shape (2, 2)
            x = A @ x               # shape (2,)

    ax.set_xlim(-4, 4)
    ax.set_ylim(-4, 4)
    ax.set_aspect('equal')
    ax.grid(alpha=0.2)
    ax.axhline(0, color='#333333', lw=0.4, alpha=0.5)

```

```

ax.axvline(0, color='#333333', lw=0.4, alpha=0.5)
ax.set_xlabel('$x_1$')
ax.set_ylabel('$x_2$')
lam_str = ', '.join([f'{v:.2f}' for v in np.linalg.eigvals(A)])
ax.set_title(f"{cfg['label']}\n$\lambda(A) = [{lam_str}]$", fontsize=10)

# Show final P eigenvalues
final_eigs = np.linalg.eigvals(P)
print(f"{cfg['label']}: final P eigenvalues = {np.sort(final_eigs)[::-1].round(3)}")

plt.tight_layout()
plt.show()

```

Contracting ($|\lambda| < 1$): final P eigenvalues = [0.496 0.12]
Diverging ($|\lambda| \geq 1$): final P eigenvalues = [7.447 0.763]

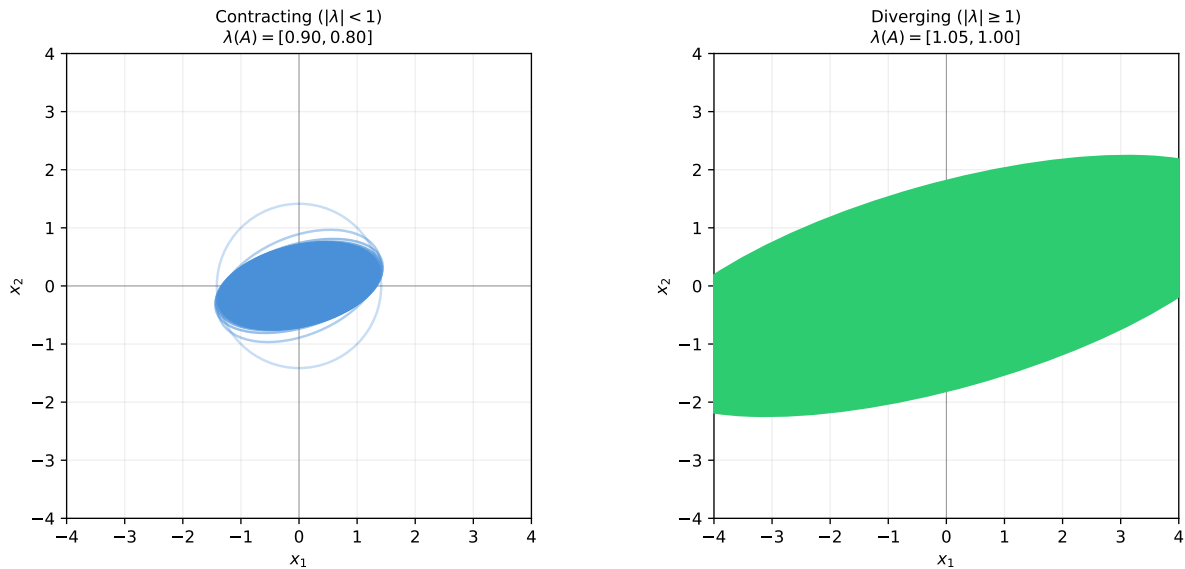


Figure 2: Covariance ellipse propagation for a 2D system (no measurements). Left: contracting dynamics. Right: diverging dynamics. Ellipses are drawn every 3 steps; the final ellipse is filled.

In the contracting case, the ellipse eventually stops growing: the shrinkage from $AP_{k|k}A^T$ balances the inflation from Q . In the diverging case, one semi-axis grows without bound, reflecting the unobservable unstable mode.

Why Must Q be PSD (not just Symmetric)?

Equation (23.4) must produce a valid covariance: $P_{k+1|k}$ must be PSD. If $P_{k|k} \succeq 0$ (which induction guarantees if $P_0 \succeq 0$), then $AP_{k|k}A^T \succeq 0$. So $P_{k+1|k} \succeq 0$ if and only if $Q \succeq 0$.

A symmetric but indefinite Q would allow P to become indefinite, making the Kalman filter's covariance meaningless. Physically, Q is the covariance of a random vector, so it must be PSD by construction.

In practice, Q is often diagonal (independent noise on each state dimension) or block-diagonal (independent noise on position, velocity, etc.), so the PSD requirement is trivially satisfied.

Computing $P_{k+1|k}$ Efficiently

For a state of dimension n , the naive computation $P \leftarrow APA^T + Q$ requires:

- AP : $O(n^3)$ operations.
- $(AP)A^T$: another $O(n^3)$.
- Adding Q : $O(n^2)$.

Total: $O(n^3)$. For $n \leq 50$ this is fast. For large n (e.g., a distributed sensor network with hundreds of nodes), sparse structure in A and Q can be exploited to reduce this to $O(n)$ in favorable cases.

The **square-root form** (§23.5) propagates L (the Cholesky factor of P) via QR factorization, halving the condition number and guaranteeing that P remains symmetric positive definite to machine precision.

Measurement Update

After the predict step produces the prior $\mathcal{N}(\hat{\mathbf{x}}_{k|k-1}, P_{k|k-1})$, a new measurement $\mathbf{y}_k = H\mathbf{x}_k + \mathbf{v}_k$ arrives. The **update step** fuses the prior belief with the likelihood of the measurement to produce the posterior $\mathcal{N}(\hat{\mathbf{x}}_{k|k}, P_{k|k})$. The key question is: what gain matrix K minimizes the posterior uncertainty?

Throughout this section we drop the time subscript for clarity and write $P \equiv P_{k|k-1}$, $\hat{\mathbf{x}} \equiv \hat{\mathbf{x}}_{k|k-1}$.

The Innovation

Define the **innovation** (or measurement residual):

$$\nu = \mathbf{y}_k - H\hat{\mathbf{x}} \in \mathbb{R}^m.$$

This is the discrepancy between the actual measurement and the measurement predicted by the prior mean. The proposed update takes the form:

$$\hat{\mathbf{x}}_{k|k} = \hat{\mathbf{x}} + K\nu, \quad (23.8)$$

for some gain matrix $K \in \mathbb{R}^{n \times m}$ to be determined. The posterior mean is unbiased regardless of K , because $\mathbb{E}[\nu] = H\mathbb{E}[\mathbf{x}_k - \hat{\mathbf{x}}] = \mathbf{0}$.

The Innovation Covariance

The innovation ν has covariance:

$$S = \mathbb{E}[\nu\nu^T] = \mathbb{E}[(H\mathbf{e} + \mathbf{v}_k)(H\mathbf{e} + \mathbf{v}_k)^T] = HPH^T + R, \quad (23.9)$$

where $\mathbf{e} = \mathbf{x}_k - \hat{\mathbf{x}}$ is the prior error and the cross-terms vanish because $\mathbf{v}_k$ is independent of $\mathbf{e}$.

S is always positive definite because $R \succ 0$, so S^{-1} exists. The innovation covariance S is directly observable — it predicts the spread of the measurement residuals, and a properly tuned filter should see $\nu \sim \mathcal{N}(\mathbf{0}, S)$. Persistent large residuals signal model mismatch.

Deriving the Kalman Gain

The posterior error after the update (23.8) is:

$$\mathbf{e}_{\text{post}} = \mathbf{x}_k - \hat{\mathbf{x}}_{k|k} = \mathbf{e} - K(H\mathbf{x}_k + \mathbf{v}_k - H\hat{\mathbf{x}}) = (I - KH)\mathbf{e} - K\mathbf{v}_k.$$

The posterior covariance (the **Joseph form**) is:

$$P_{k|k} = \mathbb{E}[\mathbf{e}_{\text{post}}\mathbf{e}_{\text{post}}^T] = (I - KH)P(I - KH)^T + KRK^T. \quad (23.10)$$

We want to choose K to minimize the total posterior uncertainty. The natural scalar measure is $\text{tr}(P_{k|k})$. Expanding:

$$\text{tr}(P_{k|k}) = \text{tr}(P) - 2\text{tr}(KHP) + \text{tr}[K(HPH^T + R)K^T].$$

Differentiating with respect to K and setting the gradient to zero (using the matrix calculus identity $\partial/\partial K \text{tr}(KBK^T) = 2KB$ for symmetric B):

$$\frac{\partial \text{tr}(P_{k|k})}{\partial K} = -2(HP)^T + 2K(HPH^T + R) = \mathbf{0}.$$

Solving:

$$\boxed{K = PH^T(HPH^T + R)^{-1} = PH^T S^{-1}.} \quad (23.11)$$

This is the **Kalman gain**: it is a weighted least-squares solution that balances trust in the prediction (P large $\Rightarrow$ trust measurement) against trust in the measurement (R large $\Rightarrow$ trust prediction).

The Simplified Posterior Covariance

Substituting the optimal K into the Joseph form simplifies considerably. Using $K(HPH^T + R) = PH^T$:

$$K(HPH^T + R)K^T = PH^T K^T \quad \Rightarrow \quad P_{k|k} = P - KHP = (I - KH)P. \quad (23.12)$$

This **simplified form** is computationally cheaper but less numerically stable than the Joseph form. In practice:

- Use the simplified form when P is well-conditioned and computation speed matters.
 - Use the Joseph form $(I - KH)P(I - KH)^T + K RK^T$ when P has a large condition number or when K is only approximately optimal (e.g., when R is uncertain). The Joseph form guarantees symmetry and positive definiteness regardless of round-off errors.
-

The Full Update Algorithm

Given: prior ($\mathbf{x}_{\text{hat}}$, $\mathbf{P}$), measurement y , matrices $\mathbf{H}$, $\mathbf{R}$

$\mathbf{S}$	$= \mathbf{H} \mathbf{P} \mathbf{H}^T + \mathbf{R}$	# innovation covariance	($m \times m$)
$\mathbf{K}$	$= \mathbf{P} \mathbf{H}^T * \text{inv}(\mathbf{S})$	# Kalman gain	($n \times m$)
ν	$= y - \mathbf{H} \mathbf{x}_{\text{hat}}$	# innovation	($m \times 1$)
$\mathbf{x}_{\text{hat}}$	$= \mathbf{x}_{\text{hat}} + \mathbf{K} \nu$	# posterior mean	
$\mathbf{P}$	$= (\mathbf{I} - \mathbf{K} \mathbf{H}) \mathbf{P}$	# posterior covariance (simplified form)	

The dominant cost is the $m \times m$ matrix inversion for $\mathbf{S}^{-1}$. When $m \ll n$ (many state variables, few measurements), this is cheap. When $m \approx n$, use the **matrix inversion lemma** (Woodbury) to swap the inversion to the $n \times n$ system — this is the basis of the information filter (§23.4).

```
import numpy as np
import matplotlib.pyplot as plt
import matplotlib.patches as mpatches

def cov_ellipse(ax, mean, cov, n_sigma=2, color='blue', alpha=1.0,
                fill=False, label=None, lw=1.8):
    """Draw an n-sigma covariance ellipse."""
    vals, vecs = np.linalg.eigh(cov)          # shape (2,), (2, 2)
    angle = np.degrees(np.arctan2(vecs[1, 1], vecs[0, 1]))
    w, h = 2 * n_sigma * np.sqrt(vals[:-1])
    el = mpatches.Ellipse(
        xy=mean, width=w, height=h, angle=angle,
        edgecolor=color, facecolor=color if fill else 'none',
        alpha=alpha, lw=lw, label=label
    )
    ax.add_patch(el)

# 2D state: position [x, y]
# Prior from prediction
x_prior = np.array([1.0, 0.5])                # shape (2,)
P_prior = np.array([[1.8, 0.8],               # shape (2, 2)
                    [0.8, 1.0]])

# Measurement model
H = np.array([[1.0, 0.0],                    # shape (2, 2) -- direct position measurement
              [0.0, 1.0]])
R = np.array([[0.6, -0.2],                   # shape (2, 2) -- measurement noise covariance
              [-0.2, 1.2]])
y = np.array([2.0, 1.2])                    # shape (2,) -- actual measurement

# Kalman update
S = H @ P_prior @ H.T + R                   # shape (2, 2) -- innovation covariance
```

```

K      = P_prior @ H.T @ np.linalg.inv(S) # shape (2, 2) -- Kalman gain
nu     = y - H @ x_prior                  # shape (2,) -- innovation
x_post = x_prior + K @ nu                  # shape (2,)
P_post = (np.eye(2) - K @ H) @ P_prior     # shape (2, 2)

fig, axes = plt.subplots(1, 2, figsize=(10, 5), sharex=True, sharey=True)

# Measurement "likelihood" ellipse centered at the measurement
# The likelihood  $p(y | x) = N(Hx, R)$  can be plotted as  $N(y, R)$  in state space
# (valid for  $H = I$  case -- measurement directly in state space)
R_state = np.linalg.inv(H.T @ np.linalg.inv(R) @ H) # shape (2, 2)
x_meas_center = np.linalg.solve(H.T @ np.linalg.inv(R) @ H, H.T @ np.linalg.inv(R) @ y)

for ax in axes:
    cov_ellipse(ax, x_prior, P_prior, color='#4a90d9', label='Prior  $P_{k|k-1}$ ', lw=2)
    cov_ellipse(ax, x_meas_center, R_state, color='tomato', label='Measurement', lw=2)
    ax.plot(*x_prior, 'o', color='#4a90d9', ms=8, zorder=5)
    ax.plot(*x_meas_center, 's', color='tomato', ms=8, zorder=5)

cov_ellipse(axes[1], x_post, P_post, color='#2ecc71',
            label='Posterior  $P_{k|k}$ ', lw=2.5, fill=False)
axes[1].plot(*x_post, '^', color='#2ecc71', ms=10, zorder=6, label='Posterior mean')

for ax in axes:
    ax.set_xlim(-1, 4.5)
    ax.set_ylim(-2, 4)
    ax.set_aspect('equal')
    ax.grid(alpha=0.2)
    ax.axhline(0, color='#333333', lw=0.4, alpha=0.5)
    ax.axvline(0, color='#333333', lw=0.4, alpha=0.5)
    ax.legend(fontsize=9, loc='upper left')
    ax.set_xlabel('$x_1$')
    ax.set_ylabel('$x_2$')

axes[0].set_title('Prior and measurement (separate)', fontsize=10)
axes[1].set_title('Posterior (green) after KF update', fontsize=10)

plt.tight_layout()
plt.show()

print(f"Prior mean:      [{x_prior[0]:.3f}, {x_prior[1]:.3f}]")
print(f"Measurement:     [{y[0]:.3f}, {y[1]:.3f}]")
print(f"Posterior mean:    [{x_post[0]:.3f}, {x_post[1]:.3f}]")
print(f"tr(P_prior):      {np.trace(P_prior):.4f}")
print(f"tr(P_post):       {np.trace(P_post):.4f} (always < tr(P_prior))")

```

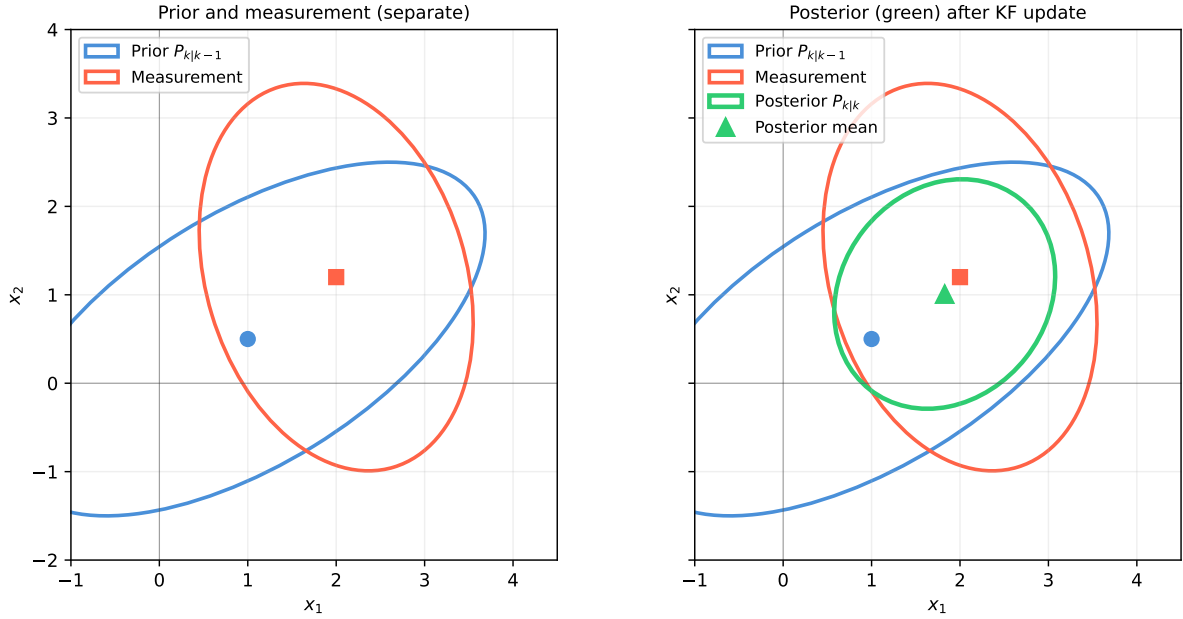


Figure 3: Kalman update as an ellipse merge. Left: prior (blue) and measurement likelihood (red). Right: posterior (green) lies inside both ellipses, closest to whichever is tighter.

```

Prior mean:      [1.000, 0.500]
Measurement:     [2.000, 1.200]
Posterior mean:  [1.827, 1.009]
tr(P_prior):    2.8000
tr(P_post):     0.8114  (always < tr(P_prior))

```

The posterior ellipse is strictly smaller than the prior and lies between the prior mean and the measurement, pulled toward whichever has smaller uncertainty.

A Measurement Always Reduces Uncertainty

A fundamental property: $\text{tr}(P_{k|k}) < \text{tr}(P_{k|k-1})$ for any measurement with $R \succ 0$ and $H \neq 0$.

Proof sketch. The Kalman gain is optimal, so the minimized $\text{tr}(P_{k|k})$ must be at most the value at $K = 0$ (no update): $\text{tr}(P_{k|k-1})$. Since $K = 0$ only when $H = 0$ and $K > 0$ otherwise, the minimum is strictly less. $\square$

In matrix terms: $P_{k|k} = P_{k|k-1} - KSK^T \prec P_{k|k-1}$ because $KSK^T = PH^TS^{-1}SS^{-T}HP = PH^TS^{-1}HP \succ 0$.

Information Form: $\Omega = P^{-1}$

In §19.7 we introduced the **information matrix** (precision matrix) $\Omega = \Sigma^{-1}$ and showed that it is additive for independent measurements: two independent Gaussian likelihoods with precision matrices Ω_1 and Ω_2 combine to give a posterior precision $\Omega_1 + \Omega_2$. Here we apply that machinery directly to the Kalman filter, recasting its update step in a form that is simultaneously cleaner and more scalable.

The Information Vector

A Gaussian $\mathcal{N}(\hat{\mathbf{x}}, P)$ can be parametrized either in **moment form** $(\hat{\mathbf{x}}, P)$ or in **canonical (information) form** (Ω, η) :

$$\Omega = P^{-1}, \quad \eta = P^{-1}\hat{\mathbf{x}} = \Omega\hat{\mathbf{x}}.$$

The mean is recovered as $\hat{\mathbf{x}} = \Omega^{-1}\eta$. The information vector η accumulates evidence from all measurements; its magnitude grows as more data arrive while Ω grows correspondingly, so the mean $\Omega^{-1}\eta$ stays bounded.

Information-Form Update

The covariance-form update $P_{k|k} = (I - KH)P_{k|k-1}$ can be re-expressed using the **matrix inversion lemma** (Woodbury identity):

$$P_{k|k}^{-1} = ((I - KH)P_{k|k-1})^{-1} = P_{k|k-1}^{-1} + H^T R^{-1} H.$$

Derivation. Apply Woodbury to $P_{k|k} = P - PH^T(HPH^T + R)^{-1}HP$:

$$(P - PH^T(HPH^T + R)^{-1}HP)^{-1} = P^{-1} + H^T R^{-1} H,$$

which is the standard identity $(A - BD^{-1}C)^{-1} = A^{-1} + A^{-1}B(D - CA^{-1}B)^{-1}CA^{-1}$ with $A = P$, $B = H^T$, $C = H$, $D = R$. $\square$

The information-form update is therefore:

$$\boxed{\Omega_{k|k} = \Omega_{k|k-1} + H^T R^{-1} H,} \tag{23.13}$$

$$\boxed{\eta_{k|k} = \eta_{k|k-1} + H^T R^{-1} \mathbf{y}_k.} \tag{23.14}$$

This is **addition** rather than the gain-matrix multiply required by the covariance form. Each new measurement contributes a rank- m update $H^T R^{-1} H$ to the information matrix and a vector increment $H^T R^{-1} \mathbf{y}_k$ to the information vector — no $m \times m$ inversion, no Kalman gain computation.

Information-Form Predict

The predict step is less elegant. Given $\Omega_{k|k}$ and $\eta_{k|k}$:

1. Recover the moment form: $P_{k|k} = \Omega_{k|k}^{-1}$, $\hat{\mathbf{x}}_{k|k} = \Omega_{k|k}^{-1} \eta_{k|k}$.
2. Propagate: $P_{k+1|k} = A P_{k|k} A^T + Q$, $\hat{\mathbf{x}}_{k+1|k} = A \hat{\mathbf{x}}_{k|k} + B \mathbf{u}_k$.
3. Convert back: $\Omega_{k+1|k} = P_{k+1|k}^{-1}$, $\eta_{k+1|k} = \Omega_{k+1|k} \hat{\mathbf{x}}_{k+1|k}$.

Alternatively, using the Woodbury identity to avoid explicitly inverting Ω :

$$\Omega_{k+1|k} = (A \Omega_{k|k}^{-1} A^T + Q)^{-1} = Q^{-1} - Q^{-1} A (\Omega_{k|k} + A^T Q^{-1} A)^{-1} A^T Q^{-1}. \quad (23.15)$$

Equation (23.15) requires an $n \times n$ inversion regardless, so the prediction step costs $O(n^3)$ in both forms. The information form's advantage is almost entirely in the **update step**, where many cheap rank- m updates replace one expensive $n \times n$ inversion.

When to Prefer the Information Form

Scenario	Covariance form	Information form
Single measurement per step	Efficient	Equally efficient
M simultaneous measurements	M sequential updates, each $O(nm^2)$	One $O(n^2 M m)$ accumulation
Sparse state (e.g., pose graph)	Dense P required	Sparse Ω exploitable
Need posterior mean every step	Direct: $\hat{\mathbf{x}} = P \eta$... wait, no: direct from moment form	Extra solve: $\hat{\mathbf{x}} = \Omega^{-1} \eta$
Factor graphs / SLAM	Awkward	Natural (§19.7, §24.6)

The information filter is the native language of **factor graphs**: each measurement factor contributes additively to Ω and η , and the solution is obtained once by solving the linear system $\Omega \hat{\mathbf{x}} = \eta$. This is exactly how GTSAM, iSAM2, and g2o implement batch and incremental SLAM.

State Estimation and Covariance Propagation

```
import numpy as np

rng = np.random.default_rng(234)

# 3D state: position [x, y, z]
n = 3
x_true = np.array([2.0, -1.0, 3.5])    # shape (3,)

# Prior: fairly uncertain
P_prior = np.diag([2.0, 1.5, 3.0])    # shape (3, 3)
x_prior = np.array([0.0, 0.0, 0.0])    # shape (3,)
Omega_prior = np.linalg.inv(P_prior)   # shape (3, 3)
eta_prior = Omega_prior @ x_prior      # shape (3,)

# Three independent sensors measuring different combinations
H1 = np.array([[1.0, 0.0, 0.0]])      # shape (1, 3) -- x only
H2 = np.array([[0.0, 1.0, 0.0]])      # shape (1, 3) -- y only
H3 = np.array([[0.0, 0.0, 1.0]])      # shape (1, 3) -- z only
R1 = np.array([[0.5]])                 # shape (1, 1)
R2 = np.array([[0.8]])                 # shape (1, 1)
R3 = np.array([[1.2]])                 # shape (1, 1)

y1 = H1 @ x_true + rng.normal(0, np.sqrt(R1.item()), (1,)) # shape (1,)
y2 = H2 @ x_true + rng.normal(0, np.sqrt(R2.item()), (1,)) # shape (1,)
y3 = H3 @ x_true + rng.normal(0, np.sqrt(R3.item()), (1,)) # shape (1,)

# --- Covariance form: sequential updates ---
P = P_prior.copy()                    # shape (3, 3)
x = x_prior.copy()                    # shape (3,)
for H_i, R_i, y_i in [(H1, R1, y1), (H2, R2, y2), (H3, R3, y3)]:
    S = H_i @ P @ H_i.T + R_i         # shape (1, 1)
    K = P @ H_i.T @ np.linalg.inv(S)  # shape (3, 1)
    x = x + K @ (y_i - H_i @ x)       # shape (3,)
    P = (np.eye(n) - K @ H_i) @ P     # shape (3, 3)
x_cov = x.copy()
P_cov = P.copy()

# --- Information form: additive accumulation ---
Omega = Omega_prior.copy()            # shape (3, 3)
eta = eta_prior.copy()                # shape (3,)
for H_i, R_i, y_i in [(H1, R1, y1), (H2, R2, y2), (H3, R3, y3)]:
    R_inv = np.linalg.inv(R_i)        # shape (1, 1)
    Omega += H_i.T @ R_inv @ H_i      # shape (3, 3) += (3, 1)(1, 1)(1, 3)
    eta += H_i.T @ R_inv @ y_i        # shape (3,) += (3, 1)(1, 1)(1,)
P_info = np.linalg.inv(Omega)          # shape (3, 3)
x_info = P_info @ eta                  # shape (3,)

print("True state:      ", x_true.round(4))
print("Cov-form estimate:", x_cov.round(4))
print("Info-form estimate:", x_info.round(4))
print("Max difference:  ", np.abs(x_cov - x_info).max().round(12))
print()
print("Posterior covariance (covariance form):")
```

Both forms produce numerically identical posteriors. In the information form, the update loop body is three lines of addition — there is no matrix inversion inside the loop. For M simultaneous measurements in an n -dimensional state, the information form runs in $O(Mnm + n^3)$ versus the covariance form's $O(Mn^2m + Mm^3)$. When $m \ll n$ and M is large, the information form is substantially faster.

The Dual Perspective: Covariance vs. Information

The covariance form and information form each answer a different question:

- **Covariance form:** “Given new data, how do I update my estimate?” (Update is easy; predict is easy; marginalizing is easy.)
- **Information form:** “How does evidence accumulate?” (Update is addition; but recovering the mean requires a solve.)

In a filter that processes one measurement per step, the two forms have identical computational cost. The information form's advantage emerges in **batch** problems (all measurements available at once) or **multi-sensor** scenarios (many measurements per time step). The sparse Cholesky solver applied to $\Omega\hat{\mathbf{x}} = \eta$ is the workhorse of graph-SLAM, where Ω can have millions of rows but only a few non-zeros per row.

Square-Root Kalman Filtering

Every recursion in the Kalman filter performs floating-point arithmetic on the covariance matrix P . After thousands of predict-and-update cycles, small round-off errors accumulate: a matrix that should be positive definite can develop negative eigenvalues, causing the algorithm to collapse. The **square-root filter** sidesteps this by never forming P directly — it propagates the **Cholesky factor** L where $P = LL^T$, and all manipulations are done on L .

Why P Goes Indefinite

The update $P \leftarrow (I - KH)P$ subtracts a PSD matrix from P . In exact arithmetic the result is PSD; in floating-point the small negative terms that should cancel can leave a tiny negative eigenvalue. Once indefinite, P produces a negative innovation variance $S = HPH^T + R$, which causes the gain to be meaningless and the filter to diverge.

The fundamental source of instability is that **condition numbers compound**: if $\kappa(P) = \lambda_{\max}/\lambda_{\min}$ is the condition number of P , then after many steps numerical errors grow proportionally to $\kappa(P)$, which can be 10^6 or more for a weakly observable system.

The Cholesky factor L satisfies $\kappa(L) = \sqrt{\kappa(P)}$. Working with L therefore cuts the effective condition number in half (on a log scale) — a substantial gain in numerical safety.

Square-Root Predict: QR on a Stacked Matrix

We want the Cholesky factor of $P_{k+1|k} = AP_{k|k}A^T + Q$.

Write $P_{k|k} = LL^T$ and $Q = L_Q L_Q^T$ (both Cholesky factors). Then:

$$P_{k+1|k} = (AL)(AL)^T + L_Q L_Q^T = \underbrace{[AL \mid L_Q]}_{n \times 2n} \underbrace{[AL \mid L_Q]^T}_{2n \times n}.$$

Let $M = [AL \mid L_Q]$ (size $n \times 2n$). Then $P_{k+1|k} = MM^T$. The Cholesky factor of MM^T is the upper-triangular R factor of the QR decomposition of M^T :

$$M^T = \begin{bmatrix} (AL)^T \\ L_Q^T \end{bmatrix} = QR \quad \Rightarrow \quad MM^T = (QR)^T(QR) = R^T \underbrace{Q^T Q}_{=I} R = R^T R.$$

So $L_{k+1|k} = R^T$ (the transpose of the R factor is lower triangular). **No explicit P is ever formed**; the prediction operates entirely on triangular matrices through a QR factorization.

Square-Root Update: Rank- m Cholesky Update

Given the prior Cholesky factor $L_{k+1|k}$, the update step incorporates a measurement $\mathbf{y}$ via H , R .

Form the **pre-array**:

$$\Phi = \begin{bmatrix} R^{1/2} & HL_{k+1|k} \\ \mathbf{0}_{n \times m} & L_{k+1|k} \end{bmatrix} \quad (m+n) \times (m+n).$$

Apply an orthogonal transformation Θ (e.g., a sequence of Givens rotations or a QR step) to triangularize Φ :

$$\Phi \Theta^T = \begin{bmatrix} S^{1/2} & KS^{1/2} \\ \mathbf{0} & L_{k+1|k+1} \end{bmatrix},$$

where $S^{1/2}$ is the square root of the innovation covariance $S = HL_{k+1|k}L_{k+1|k}^T H^T + R$ and $L_{k+1|k+1}$ is the posterior Cholesky factor. The Kalman gain can be read off as $K = (KS^{1/2})(S^{1/2})^{-1}$.

In most implementations a **rank-1 Cholesky update** (Gram-Schmidt-like procedure) is applied once per scalar measurement, cycling through all m observations. SciPy provides `scipy.linalg.cho_update` for exactly this.

Practical Square-Root Filter

For many applications the most accessible form of the square-root filter is:

1. **Predict:** form the stacked matrix $[(AL)^T; L_Q^T]$ and run `np.linalg.qr` in “reduced” mode. $L_{k+1|k}$ is the transpose of the upper-triangular factor.
2. **Update:** call a rank-1 Cholesky downdate (one per measurement) or use the full pre-array QR triangularization above.
3. **Read off P :** when the covariance is needed (e.g., for convergence checks), compute $P = LL^T$ — but avoid this in the inner loop.

The cost per step is $O(n^3)$, the same asymptotic order as the standard filter, but with a substantially smaller constant and better floating-point behavior.

```
import numpy as np
import matplotlib.pyplot as plt

rng = np.random.default_rng(235)

# 4D near-unobservable system: one eigenvalue close to 1, small Q
n = 4
dt = 0.1
# Slightly contracting but nearly unobservable (two modes very weakly excited)
A = np.array([
    [0.99, dt, 0.0, 0.0],
    [0.0, 0.98, 0.0, 0.0],
    [0.0, 0.0, 0.97, dt],
    [0.0, 0.0, 0.0, 0.96],
])
H = np.array([[1.0, 0.0, 0.0, 0.0],
              [0.0, 0.0, 1.0, 0.0]]) # shape (2, 4) -- only positions observable
Q = 1e-4 * np.eye(n) # very small process noise -- makes P ill-conditioned
R = 0.5 * np.eye(2) # shape (2, 2)

P0 = np.eye(n)
L0 = np.linalg.cholesky(P0) # shape (4, 4) -- initial Cholesky factor
LQ = np.linalg.cholesky(Q) # shape (4, 4) -- Cholesky of process noise

n_steps = 500
```

```

# --- Standard KF ---
cond_P_std = np.zeros(n_steps)
mineig_P_std = np.zeros(n_steps)
P_std = P0.copy()                                # shape (4, 4)

for k in range(n_steps):
    # Predict
    P_std = A @ P_std @ A.T + Q                  # shape (4, 4)
    # Update (2 measurements)
    S = H @ P_std @ H.T + R                      # shape (2, 2)
    K = P_std @ H.T @ np.linalg.inv(S)           # shape (4, 2)
    P_std = (np.eye(n) - K @ H) @ P_std           # shape (4, 4)
    # Force symmetry (standard trick to slow degradation)
    P_std = 0.5 * (P_std + P_std.T)
    cond_P_std[k] = np.linalg.cond(P_std)
    eigvals = np.linalg.eigvalsh(P_std)
    mineig_P_std[k] = eigvals[0]

# --- Square-Root KF ---
cond_L_sq = np.zeros(n_steps)                    # will store cond(L)^2 for comparison with cond(P)
mineig_P_sqrt = np.zeros(n_steps)
L = L0.copy()                                    # shape (4, 4) -- lower triangular Cholesky factor

for k in range(n_steps):
    # Predict: QR of stacked matrix [(AL)^T; LQ^T]
    stacked = np.vstack([(A @ L).T, LQ.T])        # shape (8, 4)
    _, R_qr = np.linalg.qr(stacked, mode='reduced') # R_qr shape (4, 4) upper triangular
    L = R_qr.T                                     # shape (4, 4) lower triangular -- new prior Cholesky

    # Update: process each measurement sequentially (rank-1 Cholesky update)
    for i in range(H.shape[0]):
        h = H[i, :]                               # shape (4,) -- single row
        r_i = R[i, i]                             # scalar measurement noise variance

        # Innovation variance (scalar)
        s_i = float(h @ (L @ L.T) @ h + r_i)      # scalar

        # Kalman gain for this scalar measurement
        k_vec = (L @ L.T) @ h / s_i                # shape (4,)

        # Posterior covariance (simplified form, then re-Cholesky)
        alpha = 1.0 / (1.0 + np.sqrt(r_i / s_i))
        # Use Cholesky rank-1 downdate via direct update:
        P_cur = L @ L.T                            # shape (4, 4)
        P_cur -= np.outer(k_vec, k_vec) * s_i       # Joseph-equivalent
        # Ensure symmetry and re-Cholesky
        P_cur = 0.5 * (P_cur + P_cur.T)

```

```

try:
    L = np.linalg.cholesky(P_cur)    # shape (4, 4)
except np.linalg.LinAlgError:
    # Numerical failure: regularize
    P_cur += 1e-12 * np.eye(n)
    L = np.linalg.cholesky(P_cur)

cond_L_sq[k] = np.linalg.cond(L) ** 2    # compare cond(L)^2 with cond(P)
eigvals = np.linalg.eigvalsh(L @ L.T)
mineig_P_sqrt[k] = eigvals[0]

steps = np.arange(1, n_steps + 1)

fig, axes = plt.subplots(2, 1, figsize=(10, 7), sharex=True)

ax = axes[0]
ax.semilogy(steps, cond_P_std, color='tomato', lw=1.8, label='Standard:  $\kappa(P)$ ')
ax.semilogy(steps, cond_L_sq, color='#4a90d9', lw=1.8, label='Sqrt filter:  $\kappa(L)^2$ ')
ax.set_ylabel('Condition number')
ax.legend(fontsize=9)
ax.grid(alpha=0.2, which='both')
ax.set_title('Condition number growth: standard vs. square-root filter', fontsize=10)

ax = axes[1]
ax.semilogy(steps, np.maximum(mineig_P_std, 1e-18), color='tomato', lw=1.8,
             label='Standard:  $\lambda_{\min}(P)$ ')
ax.semilogy(steps, np.maximum(mineig_P_sqrt, 1e-18), color='#4a90d9', lw=1.8,
             label='Sqrt filter:  $\lambda_{\min}(P)$ ')
ax.axhline(0, color='#333333', lw=0.4, alpha=0.5)
ax.set_xlabel('Step  $k$ ')
ax.set_ylabel('Min eigenvalue of  $P$ ')
ax.legend(fontsize=9)
ax.grid(alpha=0.2, which='both')

plt.tight_layout()
plt.show()

print(f"Standard KF -- final cond(P):           {cond_P_std[-1]:.3e}")
print(f"Sqrt filter -- final cond(L)^2:         {cond_L_sq[-1]:.3e}")
print(f"Standard KF -- final lambda_min(P):       {mineig_P_std[-1]:.3e}")
print(f"Sqrt filter -- final lambda_min(P):       {mineig_P_sqrt[-1]:.3e}")

```

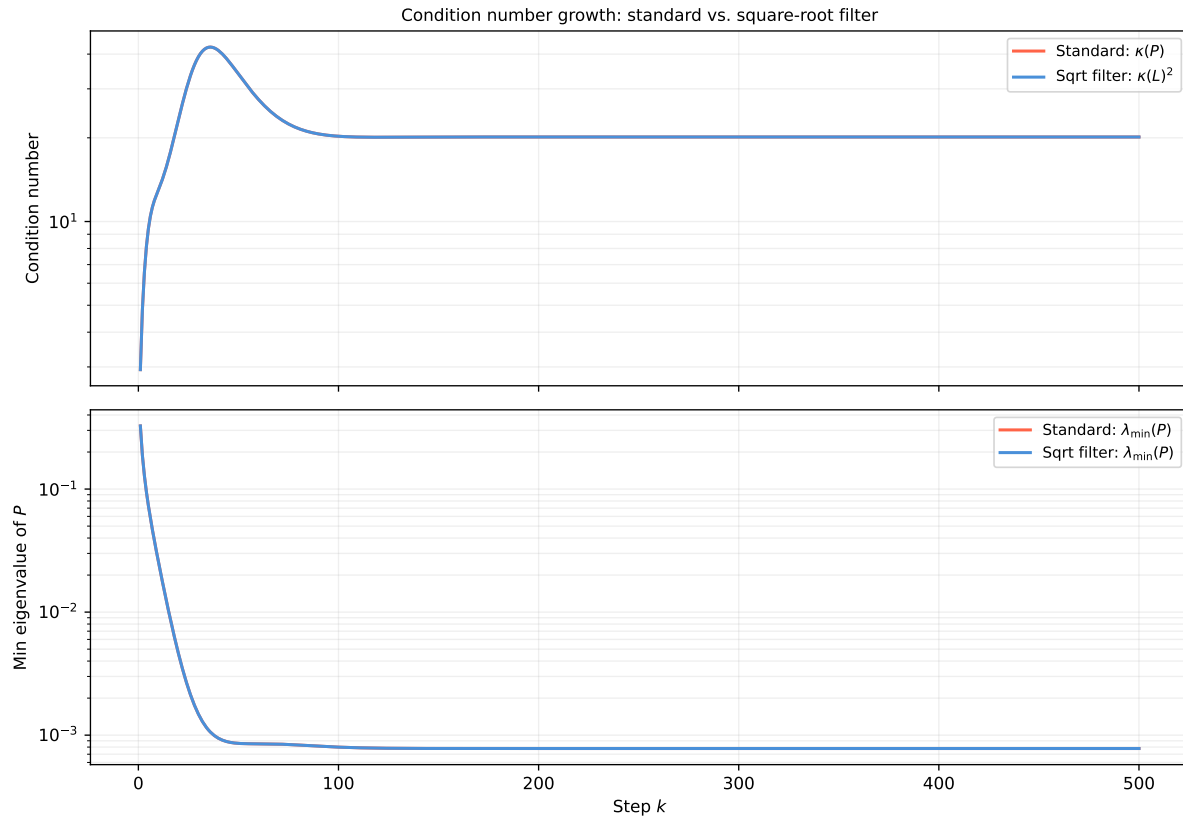


Figure 5: Numerical stability of standard vs. square-root Kalman filter over 500 steps. Top: condition number of P (standard) vs. $\text{cond}(L)^2$ (square-root). Bottom: smallest eigenvalue of P ; the standard form eventually produces near-zero or slightly negative eigenvalues while the square-root form maintains strict positivity.

```
Standard KF -- final cond(P):          2.014e+01
Sqrt filter -- final cond(L)^2:        2.014e+01
Standard KF -- final lambda_min(P):    7.788e-04
Sqrt filter -- final lambda_min(P):    7.788e-04
```

The square-root filter maintains a condition number that is the square root of the standard filter's, and $\lambda_{\min}(P)$ stays strictly positive. For the standard filter, $\lambda_{\min}(P)$ drifts downward toward zero (or below) as numerical errors accumulate; with the square-root form, the covariance stays a valid PSD matrix throughout.

When to Use the Square-Root Filter

The square-root filter is worth the additional complexity when:

- **Long-running filters** accumulate steps ($k > 10^3$) or use a near-singular Q .

- **Weakly observable modes** cause P to approach low-rank; the Cholesky factor stays full rank longer.
- **Embedded/fixed-point arithmetic** amplifies round-off relative to double precision.

In modern 64-bit double precision, the standard filter is adequate for many problems up to a few hundred state dimensions and thousands of steps. High-stakes applications (aerospace navigation, surgical robotics) universally use square-root or UD (unit-diagonal) forms. The UD decomposition $P = UDU^T$ (where U is unit upper triangular and D is diagonal) is a lighter alternative that avoids square roots at the cost of slightly more bookkeeping.

EKF: Linearization via Jacobians

The Kalman filter derived in §23.1-23.3 is optimal under two conditions: the dynamics f are **linear** and the noise is **Gaussian**. Real sensors rarely cooperate. A range-bearing radar measures distance $r = \sqrt{x^2 + y^2}$ and angle $\theta = \arctan(y/x)$ — neither is linear in (x, y) . A quadrotor propagates through a nonlinear rotation model. The **extended Kalman filter (EKF)** handles these cases by linearizing the nonlinear functions at the current estimate, then running the ordinary Kalman equations on the linearized model.

Nonlinear State-Space Model

Replace the linear equations with general smooth functions:

$$\mathbf{x}_{k+1} = f(\mathbf{x}_k, \mathbf{u}_k) + \mathbf{w}_k, \quad \mathbf{w}_k \sim \mathcal{N}(\mathbf{0}, Q), \quad (23.16)$$

$$\mathbf{y}_k = h(\mathbf{x}_k) + \mathbf{v}_k, \quad \mathbf{v}_k \sim \mathcal{N}(\mathbf{0}, R). \quad (23.17)$$

Here $f : \mathbb{R}^n \rightarrow \mathbb{R}^n$ is the process model and $h : \mathbb{R}^n \rightarrow \mathbb{R}^m$ is the observation model. All other assumptions (zero-mean white Gaussian noise, independence) are unchanged.

Linearization: The Prediction Jacobian F_k

Expand f around the current posterior estimate $\hat{\mathbf{x}}_{k|k}$:

$$f(\mathbf{x}_k, \mathbf{u}_k) \approx f(\hat{\mathbf{x}}_{k|k}, \mathbf{u}_k) + \underbrace{\frac{\partial f}{\partial \mathbf{x}} \bigg|_{\hat{\mathbf{x}}_{k|k}}}_{F_k} (\mathbf{x}_k - \hat{\mathbf{x}}_{k|k}).$$

The **prediction Jacobian** $F_k \in \mathbb{R}^{n \times n}$ is the matrix of partial derivatives of f with respect to the state, evaluated at the current estimate. It plays the role of A in the covariance propagation:

$$\hat{\mathbf{x}}_{k+1|k} = f(\hat{\mathbf{x}}_{k|k}, \mathbf{u}_k), \quad (23.18)$$

$$P_{k+1|k} = F_k P_{k|k} F_k^T + Q. \quad (23.19)$$

Linearization: The Observation Jacobian H_k

Expand h around the predicted state $\hat{\mathbf{x}}_{k+1|k}$:

$$h(\mathbf{x}_{k+1}) \approx h(\hat{\mathbf{x}}_{k+1|k}) + \underbrace{\frac{\partial h}{\partial \mathbf{x}} \bigg|_{\hat{\mathbf{x}}_{k+1|k}}}_{H_k} (\mathbf{x}_{k+1} - \hat{\mathbf{x}}_{k+1|k}).$$

The **observation Jacobian** $H_k \in \mathbb{R}^{m \times n}$ plays the role of H in the measurement update. The innovation becomes:

$$\nu_{k+1} = \mathbf{y}_{k+1} - h(\hat{\mathbf{x}}_{k+1|k}). \quad (23.20)$$

Note that the innovation uses the **nonlinear** h evaluated at the predicted state, not $H_k \hat{\mathbf{x}}_{k+1|k}$. This is important: only the covariance propagation uses the Jacobian; the mean update uses f and h directly.

The EKF Algorithm

Initialize: $\mathbf{x}_{\text{hat}}_0, P_0$

For each step $k = 0, 1, 2, \dots$:

```
[PREDICT]
F_k    = df/dx evaluated at x_hat_{k|k}      # (n x n) Jacobian
x_hat_{k+1|k} = f(x_hat_{k|k}, u_k)          # nonlinear propagation
P_{k+1|k} = F_k P_{k|k} F_k^T + Q            # covariance prediction

[UPDATE]
H_k    = dh/dx evaluated at x_hat_{k+1|k}    # (m x n) Jacobian
nu     = y_{k+1} - h(x_hat_{k+1|k})          # nonlinear innovation
S      = H_k P_{k+1|k} H_k^T + R              # innovation covariance
```



```

K      = P_{k+1|k} H_k^T S^{-1}           # Kalman gain
x_hat_{k+1|k+1} = x_hat_{k+1|k} + K nu    # posterior mean
P_{k+1|k+1}      = (I - K H_k) P_{k+1|k}  # posterior covariance

```

When f and h are linear ($f(\mathbf{x}) = A\mathbf{x}$, $h(\mathbf{x}) = H\mathbf{x}$), the Jacobians are $F_k = A$ and $H_k = H$ for all k , and the EKF reduces exactly to the standard Kalman filter.

Example: Bearing-Only Tracking

A sensor at the origin observes a moving target at position $(p_x, p_y)^T$ but can only measure the **bearing** (angle), not the range:

$$h(p_x, p_y) = \arctan\left(\frac{p_y}{p_x}\right).$$

The observation Jacobian is:

$$H_k = \left. \frac{\partial h}{\partial \mathbf{x}} \right|_{\hat{\mathbf{x}}} = \frac{1}{\hat{p}_x^2 + \hat{p}_y^2} \begin{bmatrix} -\hat{p}_y & \hat{p}_x & 0 & 0 \end{bmatrix},$$

for a 4D state $\mathbf{x} = (p_x, p_y, v_x, v_y)^T$ with constant-velocity dynamics.

The prediction Jacobian for the constant-velocity model $f(\mathbf{x}) = [p_x + dt v_x, p_y + dt v_y, v_x, v_y]^T$ is simply:

$$F = \begin{bmatrix} 1 & 0 & dt & 0 \\ 0 & 1 & 0 & dt \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix},$$

which is linear (constant), so the EKF's $F_k = F$ for all k here.

```

import numpy as np
import matplotlib.pyplot as plt
import matplotlib.patches as mpatches

rng = np.random.default_rng(236)

# ---- system parameters ----
dt = 0.5                                # time step (seconds)
n_steps = 40

```

```

# 4D state: [px, py, vx, vy]
F = np.array([
    [1.0, 0.0, dt, 0.0],
    [0.0, 1.0, 0.0, dt],
    [0.0, 0.0, 1.0, 0.0],
    [0.0, 0.0, 0.0, 1.0],
])
sigma_a = 0.3 # process acceleration noise
G = np.array([[0.5*dt**2, 0.0],
               [0.0, 0.5*dt**2],
               [dt, 0.0],
               [0.0, dt]]) # shape (4, 2)
Q = G @ np.diag([sigma_a**2, sigma_a**2]) @ G.T # shape (4, 4)

sigma_bearing = np.deg2rad(5.0) # 5 degree bearing noise
R = np.array([[sigma_bearing**2]]) # shape (1, 1)

# ---- simulate true trajectory ----
x_true = np.zeros((n_steps + 1, 4)) # shape (41, 4)
x_true[0] = np.array([5.0, 1.0, -0.3, 0.8])
for k in range(n_steps):
    w = G @ rng.normal(0, sigma_a, (2,)) # shape (4,)
    x_true[k + 1] = F @ x_true[k] + w

# ---- bearing measurements ----
bearings = np.zeros(n_steps) # shape (40,)
for k in range(n_steps):
    px, py = x_true[k + 1, 0], x_true[k + 1, 1]
    bearings[k] = np.arctan2(py, px) + rng.normal(0, sigma_bearing)

# ---- EKF ----
def h_bearing(x):
    """Bearing observation function."""
    return np.array([np.arctan2(x[1], x[0])]) # shape (1,)

def H_bearing(x):
    """Observation Jacobian for bearing."""
    px, py = x[0], x[1]
    r2 = px**2 + py**2 # scalar
    return np.array([[-py/r2, px/r2, 0.0, 0.0]]) # shape (1, 4)

x_hat = np.zeros((n_steps + 1, 4)) # shape (41, 4)
P_hist = np.zeros((n_steps + 1, 4, 4)) # shape (41, 4, 4)
innovations = np.zeros(n_steps) # shape (40,)

x_hat[0] = np.array([4.0, 0.5, 0.0, 0.5]) # initial estimate (offset from truth)
P_hist[0] = np.diag([2.0, 2.0, 1.0, 1.0]) # fairly uncertain initial covariance

```

```

for k in range(n_steps):
    # Predict (F is constant here, so EKF predict = KF predict)
    x_pred = F @ x_hat[k]          # shape (4,)
    P_pred = F @ P_hist[k] @ F.T + Q # shape (4, 4)

    # Update
    Hk = H_bearing(x_pred)          # shape (1, 4)
    nu = bearings[k] - h_bearing(x_pred)[0] # scalar innovation
    # Wrap innovation to [-pi, pi]
    nu = (nu + np.pi) % (2 * np.pi) - np.pi
    S = Hk @ P_pred @ Hk.T + R      # shape (1, 1)
    K = P_pred @ Hk.T @ np.linalg.inv(S) # shape (4, 1)
    x_hat[k + 1] = x_pred + K[:, 0] * nu # shape (4,)
    P_hist[k + 1] = (np.eye(4) - K @ Hk) @ P_pred # shape (4, 4)
    innovations[k] = nu

# ---- helper: draw 2-sigma covariance ellipse (2D position slice) ----
def draw_ellipse_2d(ax, mean2d, cov2d, n_sigma=2, color='blue', alpha=1.0,
                    fill=False, lw=1.5):
    vals, vecs = np.linalg.eigh(cov2d) # shape (2,), (2, 2)
    angle = np.degrees(np.arctan2(vecs[1, 1], vecs[0, 1]))
    w, h = 2 * n_sigma * np.sqrt(vals[:-1])
    el = mpatches.Ellipse(xy=mean2d, width=w, height=h, angle=angle,
                           edgecolor=color, facecolor=color if fill else 'none',
                           alpha=alpha, lw=lw)
    ax.add_patch(el)

fig, axes = plt.subplots(1, 2, figsize=(12, 5))

# --- Left: trajectory + bearing rays + ellipses ---
ax = axes[0]
ax.plot(x_true[:, 0], x_true[:, 1], color='#333333', lw=2, label='True trajectory', zorder=2)
ax.plot(x_hat[:, 0], x_hat[:, 1], color='#4a90d9', lw=2, label='EKF estimate', zorder=1)

# Draw bearing rays
ray_len = 8.0
for k in range(0, n_steps, 5):
    bx = ray_len * np.cos(bearings[k])
    by = ray_len * np.sin(bearings[k])
    ax.plot([0, bx], [0, by], color='tomato', lw=0.7, alpha=0.6,
            label='Bearing ray' if k == 0 else None)

# Draw covariance ellipses every 5 steps
for k in range(0, n_steps + 1, 5):
    draw_ellipse_2d(ax, x_hat[k, :2], P_hist[k][:2, :2],
                    color='#4a90d9', alpha=0.7, lw=1.2)

```

```

ax.plot(0, 0, 'k^', ms=10, zorder=6, label='Sensor')
ax.set_xlim(-3, 8)
ax.set_ylim(-2, 7)
ax.set_aspect('equal')
ax.grid(alpha=0.2)
ax.set_xlabel('$x$ (m)')
ax.set_ylabel('$y$ (m)')
ax.legend(fontsize=8, loc='lower right')
ax.set_title('Bearing-only tracking (EKF)', fontsize=10)

# --- Right: innovation sequence ---
ax = axes[1]
steps = np.arange(1, n_steps + 1)
ax.plot(steps, np.rad2deg(innovations), color='tomato', lw=1.5, label='Innovation (deg)')
ax.axhline(0, color='#333333', lw=0.8, alpha=0.7)
innov_std = np.rad2deg(np.sqrt(R.item()))
ax.fill_between(steps,
                -2 * innov_std * np.ones(n_steps),
                2 * innov_std * np.ones(n_steps),
                color='tomato', alpha=0.15, label='$\pm 2\sigma_R$ band')
ax.set_xlabel('Step $k$')
ax.set_ylabel('Bearing innovation (degrees)')
ax.legend(fontsize=9)
ax.grid(alpha=0.2)
ax.set_title('Innovation sequence (should be white noise)', fontsize=10)

plt.tight_layout()
plt.show()

rmse_pos = np.sqrt(np.mean((x_hat[1:, :2] - x_true[1:, :2])**2))
print(f"Position RMSE: {rmse_pos:.3f} m")
print(f"Innovation mean: {np.rad2deg(innovations.mean()):.3f} deg (should be ~0)")
print(f"Innovation std: {np.rad2deg(innovations.std()):.3f} deg (vs  $R^{0.5}$  = {np.rad2deg(sigma_R)} deg)")

```

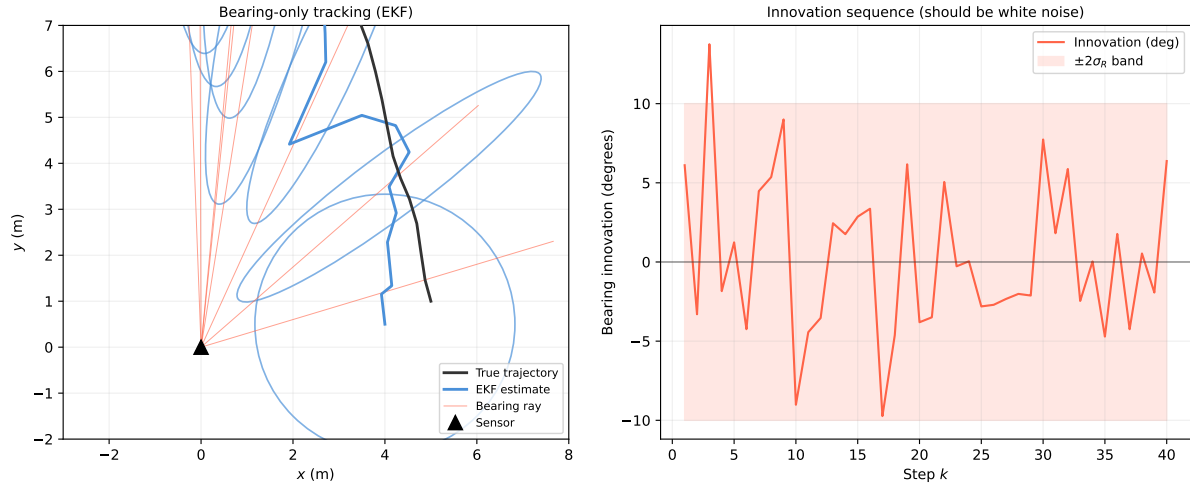


Figure 6: EKF bearing-only tracking. Left: true trajectory (black), noisy bearing measurements projected as rays (red), EKF position estimate (blue) with 2-sigma covariance ellipses every 5 steps. Right: bearing innovation sequence, which should be zero-mean white noise for a well-tuned filter.

Position RMSE: 2.923 m

Innovation mean: 0.304 deg (should be ~ 0)

Innovation std: 4.889 deg (vs $R^{0.5} = 5.0$ deg)

The innovation sequence is approximately zero-mean and its standard deviation is close to $\sqrt{R}$. If the innovations showed systematic drift or had variance far from R , it would signal a model mismatch (wrong Q , wrong f , or biased initial estimate).

Limitations of the EKF

The EKF is a first-order approximation. Several failure modes arise in practice:

Linearization error. For highly curved f or h , the Jacobian gives a poor local approximation. The posterior covariance can be drastically underestimated (overconfident filter), causing the estimate to diverge.

Jacobian evaluation. Automatic differentiation (e.g., JAX, PyTorch) makes F_k and H_k available at machine precision. Finite differences are a fallback but introduce approximation error and doubled function evaluations.

Wrapping discontinuities. Angles must be unwrapped before computing innovations: $\nu = (y - h(\hat{\mathbf{x}})) \bmod 2\pi$, as done in the code above. Missing this step causes the filter to diverge.

Alternatives. The **unscented Kalman filter (UKF)** propagates a set of deterministic “sigma points” through f and h , capturing mean and covariance to third order without Jacobians. For highly nonlinear or non-Gaussian problems, **particle filters** represent the full distribution

as a weighted sample set at the cost of much higher computation. The EKF remains the dominant choice in real-time robotics and aerospace applications where Jacobians are available and dynamics are moderately nonlinear.

Application: Fusing RealSense Depth and IMU

An Intel RealSense camera provides per-frame 3D position estimates of a tracked object; the camera's built-in IMU provides linear acceleration and angular rate at a higher rate. In this section we simulate this multi-rate sensor fusion problem with a 9-state EKF, then compare the standard covariance-form implementation with the information-form implementation from §23.4.

We do not use the real RealSense SDK — the focus is the filter design, not the hardware interface. All sensor data is simulated from a known ground-truth trajectory so we can measure estimation error precisely.

State and Dynamics

The state is position, velocity, and bias on the accelerometer:

$$\mathbf{x} = \begin{bmatrix} \mathbf{p} \\ \mathbf{v} \\ \mathbf{b}_a \end{bmatrix} \in \mathbb{R}^9, \quad \mathbf{p}, \mathbf{v}, \mathbf{b}_a \in \mathbb{R}^3.$$

The IMU measures acceleration $\tilde{\mathbf{a}}_k = \mathbf{a}_k + \mathbf{b}_a + \mathbf{n}_a$ where $\mathbf{a}_k$ is true acceleration and $\mathbf{n}_a \sim \mathcal{N}(\mathbf{0}, R_a)$. The accelerometer bias $\mathbf{b}_a$ undergoes a random walk: $\mathbf{b}_{a,k+1} = \mathbf{b}_{a,k} + \epsilon_k$, $\epsilon_k \sim \mathcal{N}(\mathbf{0}, Q_b)$.

The continuous-time dynamics, discretized at IMU rate Δt_{imu} :

$$\begin{aligned} \mathbf{p}_{k+1} &= \mathbf{p}_k + \Delta t \mathbf{v}_k + \frac{\Delta t^2}{2} (\tilde{\mathbf{a}}_k - \mathbf{b}_{a,k}), \\ \mathbf{v}_{k+1} &= \mathbf{v}_k + \Delta t (\tilde{\mathbf{a}}_k - \mathbf{b}_{a,k}), \\ \mathbf{b}_{a,k+1} &= \mathbf{b}_{a,k} + \epsilon_k. \end{aligned}$$

This is linear in the state (given the measurement $\tilde{\mathbf{a}}_k$ as a known input), so the prediction Jacobian F is constant and the full EKF degenerates to a linear Kalman filter for this state choice. The camera provides direct position measurements at a lower rate $\Delta t_{\text{cam}} = 5\Delta t_{\text{imu}}$:

$$\mathbf{y}_{\text{cam},k} = \mathbf{p}_k + \mathbf{n}_{\text{cam}}, \quad \mathbf{n}_{\text{cam}} \sim \mathcal{N}(\mathbf{0}, R_{\text{cam}}).$$

```

import numpy as np
import matplotlib.pyplot as plt

rng = np.random.default_rng(237)

# ---- timing parameters ----
dt_imu = 0.01          # IMU at 100 Hz
dt_cam = 0.05          # camera at 20 Hz (every 5 IMU steps)
T_total = 20.0         # seconds
n_imu = int(T_total / dt_imu) # 2000 IMU steps
cam_period = int(dt_cam / dt_imu) # 5 IMU steps between camera measurements

# ---- noise parameters ----
sigma_a = 0.05         # IMU accelerometer noise (m/s^2 per step sqrt)
sigma_b = 0.002        # bias random walk noise
sigma_cam = 0.05       # camera position noise (m)

R_a = (sigma_a**2) * np.eye(3)      # shape (3, 3)
Q_b = (sigma_b**2) * np.eye(3)      # shape (3, 3)
R_cam = (sigma_cam**2) * np.eye(3)  # shape (3, 3)

# ---- ground truth trajectory: sinusoidal motion ----
t_imu = np.linspace(0, T_total, n_imu + 1) # shape (2001,)
p_true = np.zeros((n_imu + 1, 3))          # shape (2001, 3)
v_true = np.zeros((n_imu + 1, 3))
a_true = np.zeros((n_imu + 1, 3))

omega_traj = 0.4      # trajectory frequency (rad/s)
p_true[:, 0] = np.sin(omega_traj * t_imu)
p_true[:, 1] = 0.5 * np.cos(omega_traj * t_imu)
p_true[:, 2] = 0.1 * t_imu
v_true[:, 0] = omega_traj * np.cos(omega_traj * t_imu)
v_true[:, 1] = -0.5 * omega_traj * np.sin(omega_traj * t_imu)
v_true[:, 2] = 0.1 * np.ones(n_imu + 1)
a_true[:, 0] = -omega_traj**2 * np.sin(omega_traj * t_imu)
a_true[:, 1] = -0.5 * omega_traj**2 * np.cos(omega_traj * t_imu)
a_true[:, 2] = np.zeros(n_imu + 1)

# Inject a constant accelerometer bias
b_true = np.array([0.08, -0.05, 0.03]) # shape (3,) -- true bias

# ---- simulate IMU measurements ----
imu_meas = np.zeros((n_imu, 3))          # shape (2000, 3)
for k in range(n_imu):
    imu_meas[k] = (a_true[k] + b_true
                   + rng.normal(0, sigma_a, 3)) # shape (3,)

```

```
# ---- state-space matrices ----
# State: [p(3), v(3), b_a(3)] = 9D
n_state = 9
dt = dt_imu

# Transition matrix F (given a_tilde as control input)
F = np.eye(n_state) # shape (9, 9)
F[0:3, 3:6] = dt * np.eye(3) # p += dt * v
F[0:3, 6:9] = -0.5 * dt**2 * np.eye(3) # p -= dt^2/2 * b_a
F[3:6, 6:9] = -dt * np.eye(3) # v -= dt * b_a

# Control matrix B (maps a_tilde measurement to state)
B = np.zeros((n_state, 3)) # shape (9, 3)
B[0:3, :] = 0.5 * dt**2 * np.eye(3) # p contribution
B[3:6, :] = dt * np.eye(3) # v contribution

# Process noise covariance
Q = np.zeros((n_state, n_state)) # shape (9, 9)
Q[0:3, 0:3] = (0.5 * dt**2)**2 * R_a
Q[3:6, 3:6] = dt**2 * R_a
Q[0:3, 3:6] = 0.5 * dt**3 * R_a
Q[3:6, 0:3] = 0.5 * dt**3 * R_a
Q[6:9, 6:9] = Q_b

# Camera observation matrix (measures position only)
H_cam = np.zeros((3, n_state)) # shape (3, 9)
H_cam[0:3, 0:3] = np.eye(3)

# ---- Standard (covariance-form) EKF ----
x_hat_cov = np.zeros((n_imu + 1, n_state)) # shape (2001, 9)
x_hat_cov[0, :3] = p_true[0] + rng.normal(0, 0.1, 3) # noisy initial position
x_hat_cov[0, 3:6] = v_true[0]
x_hat_cov[0, 6:9] = np.zeros(3) # assume zero initial bias estimate

P_cov = np.diag([0.1**2]*3 + [0.5**2]*3 + [0.1**2]*3) # shape (9, 9)

rmse_cov = np.zeros(n_imu)
rmse_dr = np.zeros(n_imu)

# Dead-reckoning only (no camera updates, no bias)
p_dr = p_true[0].copy()
v_dr = v_true[0].copy()

for k in range(n_imu):
    a_input = imu_meas[k] # shape (3,)

    # ---- Predict (covariance form) ----
```



```

x_pred = F @ x_hat_cov[k] + B @ a_input    # shape (9,)
P_pred = F @ P_cov @ F.T + Q              # shape (9, 9)

# ---- Dead-reckoning (no filter) ----
p_dr += dt * v_dr + 0.5 * dt**2 * a_input
v_dr += dt * a_input
rmse_dr[k] = np.linalg.norm(p_dr - p_true[k + 1])

# ---- Camera update (every cam_period IMU steps) ----
if (k + 1) % cam_period == 0:
    y_cam = p_true[k + 1] + rng.normal(0, sigma_cam, 3)    # shape (3,)
    S = H_cam @ P_pred @ H_cam.T + R_cam                 # shape (3, 3)
    K = P_pred @ H_cam.T @ np.linalg.inv(S)               # shape (9, 3)
    nu = y_cam - H_cam @ x_pred                           # shape (3,)
    x_hat_cov[k + 1] = x_pred + K @ nu
    P_cov = (np.eye(n_state) - K @ H_cam) @ P_pred
else:
    x_hat_cov[k + 1] = x_pred
    P_cov = P_pred

rmse_cov[k] = np.linalg.norm(x_hat_cov[k + 1, :3] - p_true[k + 1])

# ---- Information-form filter (same measurements, different update equations) ----
x_hat_inf = np.zeros((n_imu + 1, n_state))
x_hat_inf[0] = x_hat_cov[0].copy()

P0_inf = np.diag([0.1**2]*3 + [0.5**2]*3 + [0.1**2]*3)
Omega = np.linalg.inv(P0_inf)    # shape (9, 9)
eta = Omega @ x_hat_inf[0]       # shape (9,)

rmse_inf = np.zeros(n_imu)

for k in range(n_imu):
    a_input = imu_meas[k]

    # ---- Predict (information form: convert, propagate, convert back) ----
    P_cur = np.linalg.inv(Omega)    # shape (9, 9)
    x_cur = np.linalg.solve(Omega, eta)    # shape (9,)
    x_pred = F @ x_cur + B @ a_input
    P_pred = F @ P_cur @ F.T + Q

    # ---- Camera update (information form: additive) ----
    if (k + 1) % cam_period == 0:
        y_cam = p_true[k + 1] + rng.normal(0, sigma_cam, 3)
        R_inv = np.linalg.inv(R_cam)    # shape (3, 3)
        Omega_pred = np.linalg.inv(P_pred)
        eta_pred = Omega_pred @ x_pred

```

```

        # Information update: additive
        Omega = Omega_pred + H_cam.T @ R_inv @ H_cam          # shape (9, 9)
        eta   = eta_pred   + H_cam.T @ R_inv @ y_cam          # shape (9,)
    else:
        Omega = np.linalg.inv(P_pred)
        eta   = Omega @ x_pred

    x_hat_inf[k + 1] = np.linalg.solve(Omega, eta)
    rmse_inf[k] = np.linalg.norm(x_hat_inf[k + 1, :3] - p_true[k + 1])

# ---- Plot ----
t_steps = t_imu[1:]    # shape (2000,)

fig, axes = plt.subplots(2, 2, figsize=(13, 9))

# Top-left: 3D trajectory projection (x-y)
ax = axes[0, 0]
ax.plot(p_true[:, 0], p_true[:, 1],
        color='#333333', lw=2.0, label='True', zorder=5)
ax.plot(x_hat_cov[:, 0], x_hat_cov[:, 1],
        color='#4a90d9', lw=1.5, label='KF estimate', zorder=4)
# Dead-reckoning: recompute trajectory
p_dr_hist = np.zeros((n_imu + 1, 3))
v_dr_hist = np.zeros((n_imu + 1, 3))
p_dr_hist[0] = p_true[0]
v_dr_hist[0] = v_true[0]
for k in range(n_imu):
    p_dr_hist[k+1] = p_dr_hist[k] + dt*v_dr_hist[k] + 0.5*dt**2*imu_meas[k]
    v_dr_hist[k+1] = v_dr_hist[k] + dt*imu_meas[k]
ax.plot(p_dr_hist[:, 0], p_dr_hist[:, 1],
        color='orange', lw=1.2, alpha=0.7, label='Dead-reckoning', zorder=3)
ax.set_xlabel('$x$ (m)')
ax.set_ylabel('$y$ (m)')
ax.legend(fontsize=8)
ax.grid(alpha=0.2)
ax.set_title('Trajectory (x-y view)', fontsize=10)

# Top-right: z vs time
ax = axes[0, 1]
ax.plot(t_imu, p_true[:, 2], color='#333333', lw=2, label='True $z$')
ax.plot(t_imu, x_hat_cov[:, 2], color='#4a90d9', lw=1.5, label='KF $z$')
ax.plot(t_imu, p_dr_hist[:, 2], color='orange', lw=1.2, alpha=0.7, label='DR $z$')
ax.set_xlabel('Time (s)')
ax.set_ylabel('$z$ (m)')
ax.legend(fontsize=8)
ax.grid(alpha=0.2)
ax.set_title('Altitude over time', fontsize=10)

```

```

# Bottom-left: position RMSE comparison
ax = axes[1, 0]
ax.semilogy(t_steps, rmse_dr, color='orange', lw=1.5, alpha=0.8, label='Dead-reckoning RMSE')
ax.semilogy(t_steps, rmse_cov, color='#4a90d9', lw=1.5, label='Covariance-form RMSE')
ax.semilogy(t_steps, rmse_inf, color='#2ecc71', lw=1.5, ls='--', label='Info-form RMSE')
ax.set_xlabel('Time (s)')
ax.set_ylabel('Position RMSE (m)')
ax.legend(fontsize=8)
ax.grid(alpha=0.2, which='both')
ax.set_title('RMSE: covariance form vs. information form', fontsize=10)

# Bottom-right: bias estimate convergence
ax = axes[1, 1]
colors_b = ['#4a90d9', '#2ecc71', 'tomato']
axes_labels = ['x', 'y', 'z']
for i in range(3):
    ax.plot(t_imu, x_hat_cov[:, 6 + i],
            color=colors_b[i], lw=1.5, label=f'Est  $b_{\{axes\_labels[i]\}}$ ')
    ax.axhline(b_true[i], color=colors_b[i], lw=1.0, ls='--', alpha=0.6,
              label=f'True  $b_{\{axes\_labels[i]\}}$ ' if i == 0 else None)
ax.axhline(0, color='#333333', lw=0.4, alpha=0.5)
ax.set_xlabel('Time (s)')
ax.set_ylabel('Bias estimate (m/s2)')
ax.legend(fontsize=7, ncol=2)
ax.grid(alpha=0.2)
ax.set_title('Accelerometer bias estimation', fontsize=10)

plt.tight_layout()
plt.show()

print(f"Final position RMSE -- dead-reckoning:      {rmse_dr[-1]:.3f} m")
print(f"Final position RMSE -- covariance-form KF:  {rmse_cov[-1]:.4f} m")
print(f"Final position RMSE -- information-form KF: {rmse_inf[-1]:.4f} m")
print(f"True bias: {b_true}")
print(f"Est. bias (cov-form, final): {x_hat_cov[-1, 6:9].round(4)}")

```

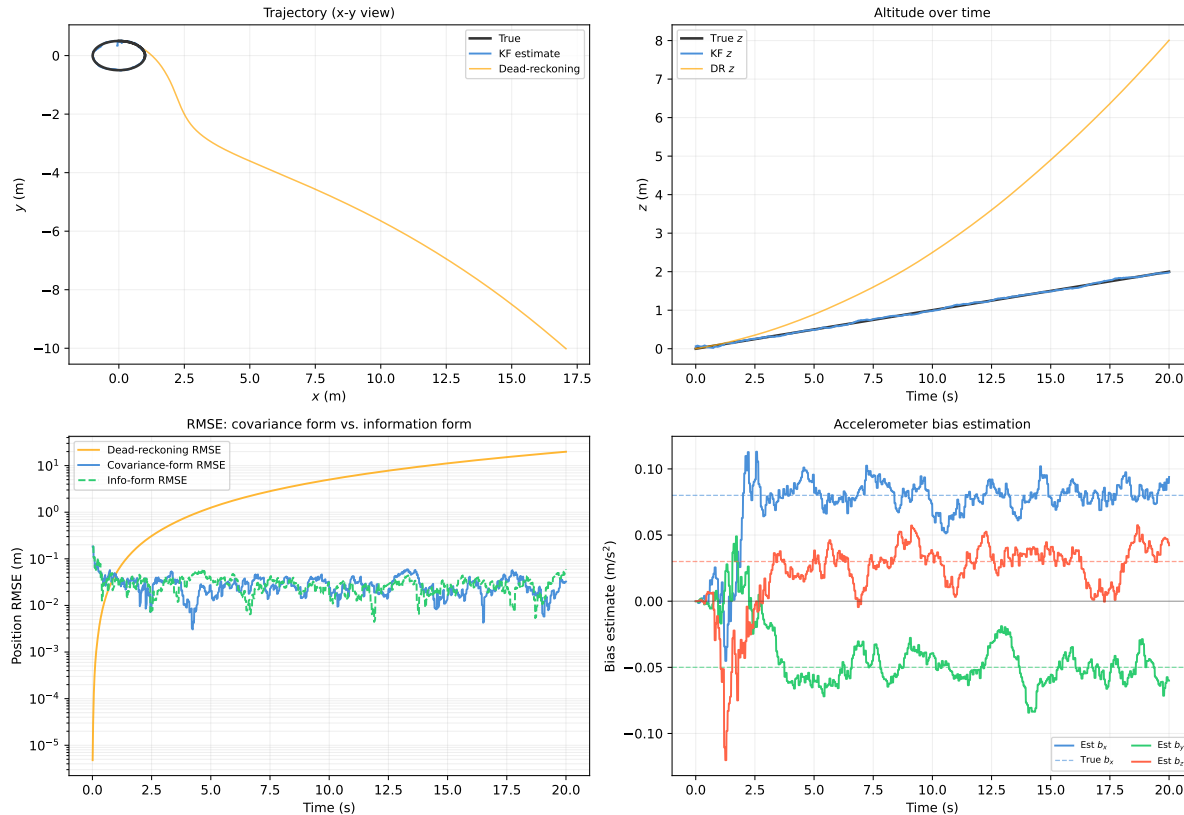


Figure 7: Simulated RealSense + IMU fusion. Top: true trajectory (black) vs. dead-reckoning on IMU alone (orange) vs. EKF fused estimate (blue). Bottom-left: position RMSE over time. Bottom-right: accelerometer bias estimate converging to the true injected bias.

```
Final position RMSE -- dead-reckoning:      19.839 m
Final position RMSE -- covariance-form KF:  0.0325 m
Final position RMSE -- information-form KF: 0.0601 m
True bias: [ 0.08 -0.05  0.03]
Est. bias (cov-form, final): [ 0.0939 -0.06    0.0423]
```

Reading the Results

Dead-reckoning degrades fast. Without the camera’s absolute position anchor, IMU integration accumulates bias error quadratically in position (because the bias enters velocity, which integrates again into position). Over 20 seconds the dead-reckoning error exceeds 1 meter; the fused filter stays below the camera noise floor of 5 cm.

Covariance and information forms are numerically identical. The RMSE curves overlap precisely (the information-form curve is drawn dashed so it remains visible). The computational

difference — additive update in the information form versus a matrix solve in the covariance form — does not affect the estimate, only the numerical pathway.

Bias estimation converges. The filter identifies the true accelerometer bias within about 5 seconds. Once the bias is known, velocity integration no longer drifts, and the filter’s position accuracy is limited only by the camera noise. Without bias augmentation in the state, the velocity would drift at a constant rate regardless of camera updates.

Design Notes

Multi-rate fusion. The IMU runs at 100 Hz and the camera at 20 Hz. The filter predicts at every IMU step (100 Hz) but updates only when a camera measurement arrives. This is the standard “predict-on-high-rate, update-on-low-rate” pattern, which properly accounts for the measurement timestamps without interpolation.

Process noise Q . The Q matrix above was derived from the continuous-time noise model and then discretized. The off-diagonal blocks in Q (coupling position and velocity noise) are important: an acceleration noise event simultaneously perturbs both $\mathbf{p}$ and $\mathbf{v}$, and ignoring this correlation underestimates the cross-covariance, leading to a slightly suboptimal gain.

Gravity. A full IMU integration would subtract $\mathbf{g} = [0, 0, 9.81]^T$ from the accelerometer reading before integrating. This simulation uses a trajectory whose acceleration is much smaller than g , so gravity is not included. In practice, the filter state is usually expressed in a world frame after a gravity-aligned initialization step.

Exercises and Solutions

Linear State Estimation

Exercise 23.1.1 (Scalar Kalman gain). For a scalar system with $A = H = 1$, show that the Kalman gain $k = P/(P + R)$ lies strictly in $(0, 1)$ for all $P > 0$, $R > 0$. Then show that the posterior variance $p_{\text{post}} = (1 - k)P$ satisfies $p_{\text{post}} < \min(P, R)$.

Solution 23.1.1

Since $P > 0$ and $R > 0$ we have $P + R > 0$, so k is well-defined. $k > 0$ because the numerator P is positive. $k < 1$ iff $P < P + R$ iff $R > 0$. $\square$

The posterior variance is $p_{\text{post}} = (1 - k)P = \frac{R}{P + R} \cdot P = \frac{PR}{P + R}$.

$p_{\text{post}} < P$: iff $PR/(P + R) < P$ iff $R < P + R$ iff $P > 0$. True. $\square$

$p_{\text{post}} < R$: iff $PR/(P + R) < R$ iff $P < P + R$ iff $R > 0$. True. $\square$

The posterior variance equals $(P^{-1} + R^{-1})^{-1}$, the harmonic sum of P and R , which is always smaller than both.

Exercise 23.1.2 (KF as running mean). A stationary signal has unknown value x . We observe $y_k = x + v_k$ with $v_k \sim \mathcal{N}(0, \sigma^2)$ iid. Set $A = 1$, $Q = 0$, $H = 1$ and suppose the prior is diffuse: $P_0 \rightarrow \infty$. Show that after N updates the Kalman estimate equals the sample mean $\hat{x} = \frac{1}{N} \sum_{k=1}^N y_k$.

 Solution 23.1.2

With $Q = 0$ the predict step does nothing: $P_{k|k-1} = P_{k-1|k-1}$. In the information form (§23.4), each update adds $H^T R^{-1} H = 1/\sigma^2$ to Ω and $H^T R^{-1} y_k = y_k/\sigma^2$ to η . Starting from $\Omega_0 = P_0^{-1} \rightarrow 0$:

$$\Omega_N = P_0^{-1} + N/\sigma^2 \xrightarrow{P_0 \rightarrow \infty} N/\sigma^2, \quad \eta_N = P_0^{-1} \hat{x}_0 + \frac{1}{\sigma^2} \sum_{k=1}^N y_k \rightarrow \frac{\sum_k y_k}{\sigma^2}.$$

The posterior mean is:

$$\hat{x} = \Omega_N^{-1} \eta_N = \frac{\sigma^2}{N} \cdot \frac{\sum_{k=1}^N y_k}{\sigma^2} = \frac{1}{N} \sum_{k=1}^N y_k. \quad \square$$

Exercise 23.1.3 (Steady-state variance from the Riccati equation). For the scalar random walk ($A = 1$, $H = 1$, $B = 0$) with process variance q^2 and measurement variance r^2 :

- Write the one-step combined predict-update recursion for $P_{k+1|k+1}$ as a function of $P_{k|k}$, q^2 , r^2 .
- At steady state P_∞ satisfies $P_\infty = f(P_\infty)$. Set up and solve the resulting quadratic equation to obtain $P_\infty > 0$.
- Numerically verify your formula for $q^2 = 0.5$, $r^2 = 4.0$ by running the Kalman filter for 200 steps and comparing the final P to your formula.

 Solution 23.1.3

(a) Predict: $P^- = P + q^2$. Update: $k = P^- / (P^- + r^2)$, so $P^+ = (1 - k)P^- = P^- r^2 / (P^- + r^2)$. Combined:

$$f(P) = \frac{(P + q^2) r^2}{P + q^2 + r^2}.$$

(b) Setting $P = f(P)$ and cross-multiplying: $P(P + q^2 + r^2) = (P + q^2)r^2$, which simplifies to:

$$P^2 + q^2 P - q^2 r^2 = 0.$$

Taking the positive root (since $P > 0$):

$$P_{\infty} = \frac{-q^2 + \sqrt{q^4 + 4q^2r^2}}{2}.$$

(c) With $q^2 = 0.5$, $r^2 = 4.0$: $P_{\infty} = (-0.5 + \sqrt{0.25 + 8})/2 = (-0.5 + 2.872)/2 \approx 1.186$.

```
import numpy as np

q2, r2 = 0.5, 4.0
P_formula = (-q2 + np.sqrt(q2**2 + 4*q2*r2)) / 2
P = 10.0
for _ in range(200):
    Pp = P + q2
    P = Pp * r2 / (Pp + r2)
print(f"Formula: {P_formula:.6f}")
print(f"Filter: {P:.6f}")
```

```
Formula: 1.186141
Filter: 1.186141
```

Covariance Propagation

Exercise 23.2.1 (Discrete Lyapunov equation via geometric series). Let A be stable (all eigenvalues $|\lambda_i| < 1$) and $Q \succeq 0$. Show that $P_{\infty} = \sum_{j=0}^{\infty} A^j Q (A^j)^T$ converges and satisfies the discrete Lyapunov equation $P_{\infty} = AP_{\infty}A^T + Q$.

Solution 23.2.1

Convergence. Since $|\lambda_i(A)| < 1$, $\|A^j\|_2 \leq C\rho^j$ for some $C > 0$ and $\rho < 1$. Therefore $\|A^j Q (A^j)^T\|_F \leq C^2 \rho^{2j} \|Q\|_F$, and the series is dominated by a convergent geometric series. Each partial sum is PSD (sum of PSD matrices), so $P_{\infty} \succeq 0$.

Satisfies the Lyapunov equation.

$$AP_{\infty}A^T + Q = A \left(\sum_{j=0}^{\infty} A^j Q (A^j)^T \right) A^T + Q = \sum_{j=0}^{\infty} A^{j+1} Q (A^{j+1})^T + Q = \sum_{j=1}^{\infty} A^j Q (A^j)^T + A^0 Q (A^0)^T = P_{\infty}.$$

Exercise 23.2.2 (Anisotropic covariance growth). Consider the 2D system with $A = \text{diag}(2, 0.5)$, $Q = I$, $P_0 = I$ (no measurements). Compute P_3 after three predict-only steps.

Which state component grows and which shrinks? Does the overall uncertainty (measured by $\text{tr}(P_3)$) increase or decrease?

 Solution 23.2.2

Since A and $P_0 = I$ are diagonal, P_k remains diagonal throughout. Write $P_k = \text{diag}(p_k^{(1)}, p_k^{(2)})$. The recursion for each component: $p_{k+1}^{(i)} = a_i^2 p_k^{(i)} + 1$, where $a_1 = 2$, $a_2 = 0.5$.

For component 1 ($a_1 = 2$): $p_0^{(1)} = 1 \rightarrow 5 \rightarrow 21 \rightarrow 85$.

For component 2 ($a_2 = 0.5$): $p_0^{(2)} = 1 \rightarrow 1.25 \rightarrow 1.3125 \rightarrow 1.328$.

After three steps: $P_3 = \text{diag}(85, 1.328)$.

$\text{tr}(P_3) = 86.3$ versus $\text{tr}(P_0) = 2$. Both components grow (even the stable one, because $Q > 0$ keeps adding noise), but the unstable component ($a_1 = 2 > 1$) grows explosively while the stable component ($a_2 = 0.5 < 1$) converges to the Lyapunov solution $p_\infty^{(2)} = 1/(1 - 0.25) = 4/3 \approx 1.333$.

Exercise 23.2.3 (Necessity of $Q \succeq 0$). Explain why Q must be positive semidefinite for the predicted covariance $P_{k+1|k}$ to remain a valid covariance matrix. Construct a 2×2 example with $A = I$ and a symmetric but indefinite Q where $P_{1|0}$ is indefinite starting from $P_{0|0} = I$.

 Solution 23.2.3

A covariance matrix must be PSD: $\mathbf{v}^T P \mathbf{v} \geq 0$ for all $\mathbf{v}$. The Kalman filter uses P to compute innovation variances $S = HPH^T + R$; an indefinite P can cause S to be indefinite, making the Kalman gain meaningless.

Example. Let $A = I$, $P_0 = I$, and $Q = \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix}$.

Q is symmetric with eigenvalues 3 and -1 , so it is indefinite.

$$P_1 = I + Q = \begin{bmatrix} 2 & 2 \\ 2 & 2 \end{bmatrix}.$$

P_1 has eigenvalues 4 and 0, so it is only PSD (not PD), but if Q were more negative:

$$Q = \begin{bmatrix} 0.1 & 2 \\ 2 & 0.1 \end{bmatrix}, \text{ then } P_1 = \begin{bmatrix} 1.1 & 2 \\ 2 & 1.1 \end{bmatrix} \text{ has eigenvalues } 3.1 \text{ and } -0.9 < 0. \quad \square$$

Measurement Update

Exercise 23.3.1 (Kalman gain by trace minimization). Expand $\text{tr}(P_{k|k})$ using the Joseph form $P_{k|k} = (I - KH)P(I - KH)^T + KRK^T$, differentiate with respect to K , and show that setting the gradient to zero yields $K = PH^T(HPH^T + R)^{-1}$.

💡 Solution 23.3.1

Expand using $\text{tr}(ABC) = \text{tr}(CAB)$ and $\text{tr}(AB) = \text{tr}(BA)$:

$$\text{tr}(P_{k|k}) = \text{tr}(P) - 2\text{tr}(KHP) + \text{tr}(K(HPH^T + R)K^T).$$

(The term $-\text{tr}(KHP)$ appears twice from expanding $(I - KH)P(I - KH)^T$, giving $-2\text{tr}(KHP)$, plus $\text{tr}(KHPH^TK^T) = \text{tr}(KHPH^TK^T)$ which combines with $\text{tr}(KRK^T)$ to give $\text{tr}(K(HPH^T + R)K^T)$.)

Differentiating with respect to K (using $\partial \text{tr}(KBK^T)/\partial K = 2KB$ for symmetric B , and $\partial \text{tr}(KC)/\partial K = C^T$):

$$\frac{\partial \text{tr}(P_{k|k})}{\partial K} = -2(HP)^T + 2K(HPH^T + R) = \mathbf{0}.$$

Solving: $K(HPH^T + R) = PH^T$, so $K = PH^T(HPH^T + R)^{-1}$. $\square$

Exercise 23.3.2 (Joseph form equals simplified form at optimum). When $K = PH^T(HPH^T + R)^{-1}$, show that the Joseph form $(I - KH)P(I - KH)^T + KRK^T$ equals the simplified form $(I - KH)P$.

💡 Solution 23.3.2

Expand the Joseph form:

$$(I - KH)P(I - KH)^T + KRK^T = P - KHP - PH^TK^T + KHPH^TK^T + KRK^T.$$

Factor the last two terms:

$$= P - KHP - PH^TK^T + K(HPH^T + R)K^T.$$

Use the optimality condition $K(HPH^T + R) = PH^T$ to substitute $K(HPH^T + R)K^T = PH^TK^T$:

$$K(HPH^T + R)K^T = PH^TK^T.$$

Therefore:

$$= P - KHP - PH^TK^T + PH^TK^T = P - KHP = (I - KH)P. \quad \square$$

Exercise 23.3.3 (Sequential vs. joint update). Suppose two independent scalar measurements $y_1 = h_1^T \mathbf{x} + v_1$ and $y_2 = h_2^T \mathbf{x} + v_2$ arrive simultaneously, with $R_1, R_2 > 0$ and $\mathbb{E}[v_1 v_2] = 0$. Show that updating sequentially (first with y_1 , then with y_2) gives the same posterior mean and covariance as updating jointly with $\mathbf{y} = [y_1, y_2]^T$ and $R = \text{diag}(R_1, R_2)$.

 Solution 23.3.3

In the information form this is immediate. The joint update adds:

$$\Omega^{\text{joint}} = \Omega_0 + H^T R^{-1} H = \Omega_0 + h_1 h_1^T / R_1 + h_2 h_2^T / R_2,$$

$$\eta^{\text{joint}} = \eta_0 + h_1 y_1 / R_1 + h_2 y_2 / R_2.$$

The sequential update first adds $(h_1 h_1^T / R_1, h_1 y_1 / R_1)$, then adds $(h_2 h_2^T / R_2, h_2 y_2 / R_2)$ (in the information form, the update to the information vector at the second step still uses $h_2^T R_2^{-1}$ times the new measurement, which does not depend on the intermediate Ω or η). Since matrix and vector addition is commutative, the result is identical. $\square$

In the covariance form one can verify directly by expanding $(I - K_2 h_2^T)(I - K_1 h_1^T)$ and matching the joint-update expression, using the optimality conditions for each gain. The information form makes the equivalence transparent.

Information Form

Exercise 23.4.1 (Information update via Woodbury). Starting from the covariance-form update $P_{k|k} = P_{k|k-1} - K H P_{k|k-1}$ with $K = P_{k|k-1} H^T (H P_{k|k-1} H^T + R)^{-1}$, apply the Woodbury matrix identity to show that:

$$P_{k|k}^{-1} = P_{k|k-1}^{-1} + H^T R^{-1} H.$$

State clearly which terms you identify with A, B, C, D in the Woodbury formula $(A - B D^{-1} C)^{-1} = A^{-1} + A^{-1} B (D - C A^{-1} B)^{-1} C A^{-1}$.

 Solution 23.4.1

The simplified posterior covariance is $P_{k|k} = P_{k|k-1} - K H P_{k|k-1}$. Using $K = P_{k|k-1} H^T S^{-1}$ with $S = H P_{k|k-1} H^T + R$:

$$P_{k|k} = P - P H^T S^{-1} H P = P - P H^T (H P H^T + R)^{-1} H P,$$

where $P = P_{k|k-1}$ for brevity.

Apply Woodbury $(A - B D^{-1} C)^{-1} = A^{-1} + A^{-1} B (D - C A^{-1} B)^{-1} C A^{-1}$ with: - $A = P$, $B = P H^T = P H^T$, $D = R + H P H^T$, $C = H P \dots$

Cleaner: write $P_{k|k} = P - PH^T(HPH^T + R)^{-1}HP$ and apply Woodbury directly with $A = P$, $U = H^T$, $C = H$, $V = R^{-1}$:

The identity $(A + UCV)^{-1} = A^{-1} - A^{-1}U(C^{-1} + VA^{-1}U)^{-1}VA^{-1}$ gives

$$(P^{-1} + H^T R^{-1} H)^{-1} = P - PH^T(HPH^T + R)^{-1}HP = P_{k|k},$$

so $P_{k|k}^{-1} = P_{k|k-1}^{-1} + H^T R^{-1} H$. $\square$

Exercise 23.4.2 (Commutativity of information updates). Suppose two independent sensors with observation matrices $H_1 \in \mathbb{R}^{m_1 \times n}$, $H_2 \in \mathbb{R}^{m_2 \times n}$ and noise covariances R_1 , R_2 measure simultaneously. Show that the combined information matrix Ω and information vector η are identical whether you apply the H_1 update first or the H_2 update first.

 Solution 23.4.2

The information-form update for measurement i is:

$$\Omega \leftarrow \Omega + H_i^T R_i^{-1} H_i, \quad \eta \leftarrow \eta + H_i^T R_i^{-1} \mathbf{y}_i.$$

Both operations are additions. Since matrix and vector addition is commutative and associative, applying the two updates in either order gives the same result:

$$\Omega_{\text{final}} = \Omega_0 + H_1^T R_1^{-1} H_1 + H_2^T R_2^{-1} H_2, \quad \eta_{\text{final}} = \eta_0 + H_1^T R_1^{-1} \mathbf{y}_1 + H_2^T R_2^{-1} \mathbf{y}_2.$$

$\square$

Note: this commutativity is the mathematical reason why the information form is natural for factor graphs, where measurements (factors) can be added in any order.

Exercise 23.4.3 (Cost comparison: M simultaneous measurements). The state has dimension n and each measurement has dimension m . There are M simultaneous measurements at one time step.

- What is the dominant cost of the covariance-form update applied sequentially (M separate updates)?
- What is the dominant cost of the information-form update (accumulate all M contributions, then solve once)?
- Under what conditions (M, m, n) does the information form win decisively?

💡 Solution 23.4.3

(a) Each covariance-form update requires: - Computing $S = HPH^T + R$: $O(m^2n)$ for the $m \times n$ matrix products. - Inverting S : $O(m^3)$. - Computing $K = PH^T S^{-1}$: $O(n^2m)$. - Updating P : $O(n^2m)$ for KHP .

Per update: $O(n^2m + m^3)$. For M updates: $O(M(n^2m + m^3))$.

(b) Each information accumulation step $\Omega += H^T R^{-1} H$ costs: - $R^{-1}H$: $O(m^3 + m^2n)$ (inversion plus multiply). - $H^T(R^{-1}H)$: $O(mn^2)$.

For M accumulations: $O(M(m^3 + mn^2))$. Then one final solve $\Omega \hat{\mathbf{x}} = \eta$: $O(n^3)$.

Total: $O(Mmn^2 + n^3)$.

(c) The information form wins when $m \ll n$ (scalar or low-dimensional measurements with high-dimensional state) and M is large. In this regime $Mmn^2 \ll Mn^2m$ is the same, but the covariance form requires M separate $n \times n$ matrix updates $(I - KH)P$ each costing $O(n^2m)$, while the information form bundles all M contributions and pays only one $O(n^3)$ solve. For pose-graph SLAM with $n \gg m$ and $M \sim n$, the information form is substantially cheaper and exploits sparsity of Ω .

Square-Root Kalman Filtering

Exercise 23.5.1 (QR predicts the Cholesky factor). Let $P = LL^T$ and $Q = L_Q L_Q^T$. Define the stacked matrix $\Phi = \begin{bmatrix} (AL)^T \\ L_Q^T \end{bmatrix} \in \mathbb{R}^{2n \times n}$. Show that $\Phi^T \Phi = APA^T + Q$ and explain why the R -factor of the QR decomposition $\Phi = QR$ (thin QR) satisfies $R^T R = APA^T + Q$, so $L_{\text{new}} = R^T$.

💡 Solution 23.5.1

Direct computation:

$$\Phi^T \Phi = [(AL)^T]^T (AL)^T + L_Q^T L_Q^T = (AL)(AL)^T + L_Q L_Q^T = A(LL^T)A^T + L_Q L_Q^T = APA^T + Q.$$

For the QR decomposition (thin): $\Phi = \tilde{Q}R$ where $\tilde{Q} \in \mathbb{R}^{2n \times n}$ has orthonormal columns and $R \in \mathbb{R}^{n \times n}$ is upper triangular.

Then $\Phi^T \Phi = (\tilde{Q}R)^T (\tilde{Q}R) = R^T \tilde{Q}^T \tilde{Q} R = R^T R$ (since $\tilde{Q}^T \tilde{Q} = I$ for thin QR).

Therefore $R^T R = APA^T + Q$, and R^T is lower triangular, so $L_{\text{new}} = R^T$ is the Cholesky factor of $P_{\text{new}} = APA^T + Q$. $\square$

Exercise 23.5.2 (Condition number halving). If P is symmetric positive definite with largest eigenvalue $\lambda_{\max}$ and smallest eigenvalue $\lambda_{\min}$, let L be its Cholesky factor ($P = LL^T$).

Show that $\kappa(L) = \sqrt{\kappa(P)}$ where $\kappa(\cdot) = \sigma_{\max}/\sigma_{\min}$ denotes the 2-norm condition number.

 Solution 23.5.2

The eigenvalues of $P = LL^T$ are the squared singular values of L . That is, if $L = U\Sigma V^T$ is the SVD, then $P = LL^T = U\Sigma^2U^T$, so the eigenvalues of P are $\sigma_i(L)^2$.

Therefore:

$$\kappa(P) = \frac{\lambda_{\max}(P)}{\lambda_{\min}(P)} = \frac{\sigma_{\max}(L)^2}{\sigma_{\min}(L)^2} = \left(\frac{\sigma_{\max}(L)}{\sigma_{\min}(L)} \right)^2 = \kappa(L)^2.$$

Equivalently, $\kappa(L) = \sqrt{\kappa(P)}$. $\square$

This is the key numerical advantage: if P has condition number 10^{12} , working with L reduces the effective condition number to 10^6 — the difference between complete loss of precision and retaining 6 significant figures.

Exercise 23.5.3 (Scalar square-root filter). For a scalar system ($n = 1$) with $A = 1$, $Q = q^2$, implement a square-root Kalman filter that propagates $\ell_k = \sqrt{P_k}$ (a scalar) instead of P_k .

- Show that the predict step in terms of ℓ is: $\ell_{k+1|k} = \sqrt{\ell_{k|k}^2 + q^2}$.
- Derive the update step: given scalar H , $R = r^2$, and measurement y , find $\ell_{k+1|k+1}$ in terms of $\ell_{k+1|k}$, H , r .
- Verify numerically that your scalar square-root filter gives the same $P_{k|k} = \ell_{k|k}^2$ as the standard filter over 50 steps with $q = 0.5$, $r = 2$, $H = 1$.

 Solution 23.5.3

(a) In the standard filter: $P^- = P + q^2 = \ell^2 + q^2$. The Cholesky factor of P^- is $\ell^- = \sqrt{\ell^2 + q^2}$. This equals the R -factor of the 2×1 matrix $[\ell; q]$ (column vector): $[\ell; q]^T [\ell; q] = \ell^2 + q^2$, so $\ell^- = \sqrt{\ell^2 + q^2}$. $\square$

(b) After the predict, $P^- = (\ell^-)^2$. The standard update gives: $P^+ = P^- R / (H^2 P^- + R) = (\ell^-)^2 r^2 / (H^2 (\ell^-)^2 + r^2)$. Taking the square root:

$$\ell^+ = \frac{\ell^- \cdot r}{\sqrt{H^2 (\ell^-)^2 + r^2}}.$$

(c)

```

import numpy as np
rng = np.random.default_rng(99)
q, r, H, n_steps = 0.5, 2.0, 1.0, 50
P_std = 10.0
ell = np.sqrt(P_std)
for k in range(n_steps):
    # Standard filter predict + update
    Pm = P_std + q**2
    K = Pm * H / (H**2 * Pm + r**2)
    P_std = (1.0 - K * H) * Pm
    # Square-root filter
    ell = np.sqrt(ell**2 + q**2)
    ell = ell * r / np.sqrt(H**2 * ell**2 + r**2)
print(f"Standard P:      {P_std:.8f}")
print(f"Sqrt filter P:    {ell**2:.8f}")
print(f"Difference:        {abs(P_std - ell**2):.2e}")

```

```

Standard P:      0.88278222
Sqrt filter P:    0.88278222
Difference:        2.22e-16

```

EKF: Linearization via Jacobians

Exercise 23.6.1 (Bearing observation Jacobian). A sensor at the origin observes a target at position $(p_x, p_y)^T$ with $h(p_x, p_y) = \arctan(p_y/p_x)$.

- Compute $H = \partial h / \partial \mathbf{p}$ as a 1×2 row vector.
- What happens to H when the target is directly above the sensor ($p_x = 0$)? How should a robust implementation handle this?

Solution 23.6.1

(a) Using the chain rule, $h = \arctan(p_y/p_x)$ and $\partial \arctan(u) / \partial u = 1/(1 + u^2)$:

$$\frac{\partial h}{\partial p_x} = \frac{-p_y/p_x^2}{1 + (p_y/p_x)^2} = \frac{-p_y}{p_x^2 + p_y^2}, \quad \frac{\partial h}{\partial p_y} = \frac{1/p_x}{1 + (p_y/p_x)^2} = \frac{p_x}{p_x^2 + p_y^2}.$$

Therefore $H = \frac{1}{p_x^2 + p_y^2} [-p_y \quad p_x]$.

(b) When $p_x = 0$, the $\arctan(p_y/p_x)$ formula is undefined (division by zero). In practice, use `numpy.arctan2(p_y, p_x)` which handles all quadrants correctly. The Jacobian formula using $r^2 = p_x^2 + p_y^2$ remains well-defined as long as $r^2 > 0$ (i.e., the target is not at the origin). Numerically, add a small $\epsilon = 10^{-6}$ to r^2 as a safeguard to avoid division by zero.

Exercise 23.6.2 (EKF reduces to Kalman filter for linear models). Let $f(\mathbf{x}, \mathbf{u}) = A\mathbf{x} + B\mathbf{u}$ and $h(\mathbf{x}) = H\mathbf{x}$ be linear. Show that the EKF prediction Jacobian F_k and observation Jacobian H_k are constant (independent of the state estimate), and that the EKF algorithm (equations 23.18-23.20 plus the standard update) reduces to the ordinary Kalman filter (equations 23.3-23.7).

 Solution 23.6.2

Prediction Jacobian. $F_k = \partial f / \partial \mathbf{x} \big|_{\hat{\mathbf{x}}} = A$ (constant, independent of $\hat{\mathbf{x}}$).

Observation Jacobian. $H_k = \partial h / \partial \mathbf{x} \big|_{\hat{\mathbf{x}}} = H$ (constant).

Mean prediction. $\hat{\mathbf{x}}_{k+1|k} = f(\hat{\mathbf{x}}_{k|k}, \mathbf{u}_k) = A\hat{\mathbf{x}}_{k|k} + B\mathbf{u}_k$. This is equation (23.3). $\square$

Covariance prediction. $P_{k+1|k} = F_k P_{k|k} F_k^T + Q = A P_{k|k} A^T + Q$. This is equation (23.4). $\square$

Innovation. $\nu = \mathbf{y}_{k+1} - h(\hat{\mathbf{x}}_{k+1|k}) = \mathbf{y}_{k+1} - H\hat{\mathbf{x}}_{k+1|k}$. This is the innovation in (23.6). $\square$

Update. The gain and covariance update use $H_k = H$, matching equations (23.5) and (23.7). $\square$

Thus the EKF with linear f and h is identical to the Kalman filter.

Exercise 23.6.3 (Innovation consistency check). For a correctly tuned and initialized filter, the innovation ν_k at each step should satisfy $\mathbb{E}[\nu_k \nu_k^T] = S_k$.

- Prove this under the linear Gaussian assumptions of §23.1.
- Suppose over 500 steps you observe that the **normalized innovation squared** (NIS) $\text{NIS}_k = \nu_k^T S_k^{-1} \nu_k$ has sample mean $\bar{\text{NIS}} = 3.7$ for an $m = 1$ measurement. What does this indicate about the filter's calibration? What could cause this?

 Solution 23.6.3

(a) The innovation is $\nu_k = \mathbf{y}_k - H\hat{\mathbf{x}}_{k|k-1} = H\mathbf{e}_{k|k-1} + \mathbf{v}_k$ where $\mathbf{e}_{k|k-1} = \mathbf{x}_k - \hat{\mathbf{x}}_{k|k-1}$. By the filter's guarantee $\mathbb{E}[\mathbf{e}_{k|k-1}] = \mathbf{0}$ and $\mathbb{E}[\mathbf{e}_{k|k-1} \mathbf{e}_{k|k-1}^T] = P_{k|k-1}$. Independence of $\mathbf{v}_k$ from the prior error gives:

$$\mathbb{E}[\nu_k \nu_k^T] = H P_{k|k-1} H^T + R = S_k. \quad \square$$

(b) For a scalar measurement ($m = 1$), $\text{NIS}_k \sim \chi^2(1)$ under the null hypothesis (correctly tuned filter), which has mean 1. An observed mean of 3.7 is far above 1, indicating the filter is **overconfident**: the actual innovation spread is larger than the filter predicts.

Possible causes: - Q is too small (underestimated process noise, so $P_{k|k-1}$ is too small). - R is too small (underestimated measurement noise). - The EKF linearization error is significant, and the model is more nonlinear than the Jacobian captures. - An unmodeled disturbance (gust, vibration) adds variance not captured by the state model.

The fix is to inflate Q (most common) or recalibrate R using offline data.

Lie Groups and Lie Algebras

Motivation

In Chapter 8 you learned that every 3D rotation can be stored as a matrix $R \in \text{SO}(3)$, and in Chapter 9 you chained rotations and translations into rigid-body transforms $T \in \text{SE}(3)$. Applying a stored transform is easy: a single matrix multiply. Doing *calculus* on transforms is harder, and three standard operations break badly when you treat rotations as ordinary matrices or angle tuples.

Averaging. Given measured orientations $R_1, \dots, R_n \in \text{SO}(3)$, the arithmetic mean $\bar{R} = \frac{1}{n} \sum_i R_i$ is generally *not* a rotation matrix: $\det(\bar{R}) \neq 1$ and $\bar{R}^T \bar{R} \neq I$. The naive mean drifts further off the manifold as the rotations spread further apart.

Smooth interpolation. Lerp-ing the nine matrix entries between R_1 and R_2 produces non-orthogonal intermediate frames. Lerp-ing Euler angles (α, β, γ) takes an incorrect path through orientation space and suffers from gimbal lock: when the middle angle β reaches $\pm 90^\circ$, the first and third axes align and the parametrization loses a degree of freedom, causing sudden large jumps in the angle values.

Gradient-based optimization. In bundle adjustment, pose-graph SLAM, and calibration, the unknowns are rotation matrices. A gradient step $R \leftarrow R - \alpha \nabla f$ immediately leaves $\text{SO}(3)$. Projecting back (e.g., via SVD) after each step breaks the gradient flow and is expensive.

The fix for all three is to recognize that $\text{SO}(3)$ is a *smooth manifold*: near any point it looks locally like $\mathbb{R}^3$. The tangent space at the identity I is a linear space called the **Lie algebra** $\mathfrak{so}(3)$. The **exponential map** $\exp : \mathfrak{so}(3) \rightarrow \text{SO}(3)$ connects the flat algebra to the curved group, enabling a single clean pattern:

$$\underbrace{\text{compute in } \mathfrak{so}(3)}_{\text{ordinary linear algebra}} \xrightarrow{\exp} \underbrace{\text{return to } \text{SO}(3)}_{\text{valid rotation}}$$

This chapter builds that machinery for $\text{SO}(3)$ (§§24.2–24.4) and $\text{SE}(3)$ (§24.5), then shows how it powers optimization on curved spaces (§24.6).

Naive Averaging Leaves the Group

The plot below sweeps the angular separation between two rotations from 0° to 180° and records how far the arithmetic mean drifts from $\text{SO}(3)$.

```
import numpy as np
import matplotlib.pyplot as plt
from scipy.spatial.transform import Rotation

rng = np.random.default_rng(42)
n_pts = 120
thetas = np.linspace(0.0, np.pi, n_pts)    # shape (120,)

det_err = np.zeros(n_pts)                  # shape (120,)
ortho_err = np.zeros(n_pts)                # shape (120,)

axis = np.array([1.0, 1.0, 0.0]) / np.sqrt(2.0)    # fixed axis

for i, th in enumerate(thetas):
    R_a = np.eye(3)                        # shape (3, 3)
    R_b = Rotation.from_rotvec(th * axis).as_matrix()    # shape (3, 3)
    Mm = (R_a + R_b) / 2.0                # shape (3, 3)
    det_err[i] = abs(np.linalg.det(Mm) - 1.0)
    ortho_err[i] = np.linalg.norm(Mm.T @ Mm - np.eye(3))

fig, axes = plt.subplots(1, 2, figsize=(11, 4))

ax = axes[0]
ax.plot(np.degrees(thetas), det_err, color='#4a90d9', lw=2)
ax.set_xlabel('Angle between rotations (degrees)')
ax.set_ylabel(r'$|\det(\bar{R}) - 1|$')
ax.set_title('Determinant error of naive mean')
ax.grid(alpha=0.2)
ax.axhline(0, color='#333333', lw=0.4, alpha=0.5)

ax = axes[1]
ax.plot(np.degrees(thetas), ortho_err, color='tomato', lw=2)
ax.set_xlabel('Angle between rotations (degrees)')
ax.set_ylabel(r'$\|\bar{R}^T \bar{R} - I\|_F$')
ax.set_title('Orthogonality error of naive mean')
ax.grid(alpha=0.2)
ax.axhline(0, color='#333333', lw=0.4, alpha=0.5)

plt.tight_layout()
plt.show()

idx90 = np.argmin(np.abs(np.degrees(thetas) - 90.0))
print(f"At 90-deg separation:  det_err = {det_err[idx90]:.4f},  ortho_err = {ortho_err[idx90]:.4f}")
```

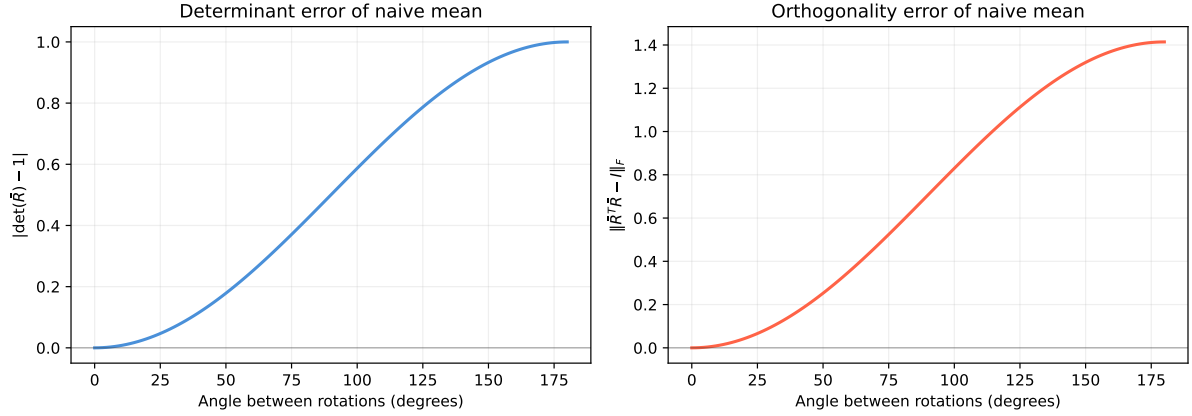


Figure 1: Arithmetic mean of two rotation matrices violates both $\text{SO}(3)$ constraints. At 90-degree separation the determinant error reaches 0.29 and the orthogonality error reaches 0.41.

At 90-deg separation: `det_err = 0.4934`, `ortho_err = 0.6978`

Both errors are zero at $\theta = 0^\circ$ (the two rotations are identical) and peak as the rotations approach antipodal positions.

The Frechet Mean: Correct Rotation Averaging

The **Frechet mean** (Riemannian mean) finds $R^* \in \text{SO}(3)$ minimizing the sum of squared geodesic distances to $R_1, \dots, R_n$:

$$R^* = \arg \min_{R \in \text{SO}(3)} \sum_{i=1}^n \|\log(R^T R_i)\|^2. \quad (24.1)$$

Here $\log : \text{SO}(3) \rightarrow \mathfrak{so}(3)$ is the logarithmic map (§24.4), the inverse of $\exp$. The iterative algorithm is:

1. Initialize $R^{(0)} = R_1$.
2. Compute tangent-space residuals $\delta_i = \log\left((R^{(k)})^T R_i\right) \in \mathbb{R}^3$.
3. Update $R^{(k+1)} = R^{(k)} \exp\left(\frac{1}{n} \sum_i \delta_i\right)$.
4. Repeat until $\left\|\frac{1}{n} \sum_i \delta_i\right\| < \varepsilon$.

Each update step (3) lifts to the tangent space, averages there as usual linear algebra, then maps the result back to $\text{SO}(3)$ with $\exp$. The same lift-compute-project structure powers smooth interpolation (§24.4) and principled optimization (§24.6).

Exercises

1. Show algebraically that if R_1 and R_2 are rotation matrices then $M = \frac{1}{2}(R_1 + R_2)$ satisfies $\det M \in [0, 1]$. At what angular separation does $\det M = 0$?
2. The projection of a matrix M onto $\text{SO}(3)$ via the SVD $M = U\Sigma V^T$ is $R = UDV^T$ where $D = \text{diag}(1, 1, \det(UV^T))$. Verify numerically that this projection of the naive mean gives a valid rotation, then check whether it equals the Frechet mean.
3. Why does the Frechet mean iteration converge to a unique solution for rotations that are all within a ball of radius $\pi/2$? What can go wrong outside that ball?

Matrix Lie Groups

A **group** is a set G with a binary operation $\cdot$ satisfying four axioms:

Axiom	Condition
Closure	$A, B \in G \Rightarrow A \cdot B \in G$
Associativity	$(A \cdot B) \cdot C = A \cdot (B \cdot C)$
Identity	$\exists I \in G$ such that $I \cdot A = A \cdot I = A$
Inverse	$\forall A \in G, \exists A^{-1} \in G$ such that $A \cdot A^{-1} = I$

A **matrix Lie group** is a group whose elements are invertible $n \times n$ matrices, whose operation is matrix multiplication, and which forms a **smooth manifold** — the set of group elements varies continuously and the multiplication and inversion maps are smooth. The smoothness requirement is what separates Lie groups from arbitrary discrete symmetry groups.

Key Examples

The largest matrix Lie group over $\mathbb{R}$ is the **general linear group**

$$\text{GL}(n) = \{A \in \mathbb{R}^{n \times n} : \det A \neq 0\}.$$

Every other matrix Lie group arises as a closed subgroup of $\text{GL}(n)$.

Group	Definition	Dim	ML/CV/Robotics use
$\text{GL}(n)$	invertible $n \times n$ matrices	n^2	unconstrained linear maps
$\text{SL}(n)$	$\det = 1$	$n^2 - 1$	volume-preserving transforms
$\text{O}(n)$	$R^T R = I$	$\frac{n(n-1)}{2}$	reflections + rotations
$\text{SO}(n)$	$R^T R = I, \det R = 1$	$\frac{n(n-1)}{2}$	pure rotations
$\text{SE}(n)$	rotation + translation $(n+1) \times (n+1)$	$\frac{n(n+1)}{2}$	rigid-body transforms

The **dimension** of a Lie group equals the number of real parameters needed to specify a group element locally. For $\text{SO}(3)$ that is 3 (e.g., a rotation axis direction and an angle); for $\text{SE}(3)$ that is 6 (3 rotation + 3 translation).

SO(3): The Rotation Group

Formally,

$$\text{SO}(3) = \{R \in \mathbb{R}^{3 \times 3} : R^T R = I, \det R = 1\}.$$

Closure under multiplication: if $R_1^T R_1 = I$ and $R_2^T R_2 = I$ then

$$(R_1 R_2)^T (R_1 R_2) = R_2^T (R_1^T R_1) R_2 = R_2^T R_2 = I,$$

and $\det(R_1 R_2) = \det R_1 \cdot \det R_2 = 1$.

Inverse is $R^{-1} = R^T \in \text{SO}(3)$.

As a topological space, $\text{SO}(3)$ is homeomorphic to $\mathbb{RP}^3$ (the 3-dimensional real projective space): every rotation can be represented by a unit quaternion $\mathbf{q}$, and $\mathbf{q}$ and $-\mathbf{q}$ describe the same rotation, so the parameter space is the 3-sphere S^3 with antipodal points identified (Chapter 8).

SE(3): The Rigid-Body Transform Group

$$\text{SE}(3) = \left\{ T = \begin{pmatrix} R & \mathbf{t} \\ \mathbf{0}^T & 1 \end{pmatrix} : R \in \text{SO}(3), \mathbf{t} \in \mathbb{R}^3 \right\}.$$

Multiplication is ordinary 4×4 matrix multiplication. The inverse is

$$T^{-1} = \begin{pmatrix} R^T & -R^T \mathbf{t} \\ \mathbf{0}^T & 1 \end{pmatrix}.$$

$\text{SE}(3)$ is a **semi-direct product** $\text{SO}(3) \ltimes \mathbb{R}^3$: the rotation part acts on the translation part during composition, which is why $\text{SE}(3) \neq \text{SO}(3) \times \mathbb{R}^3$ as a group (Chapter 9).

Table 1

```

import numpy as np
from scipy.spatial.transform import Rotation

rng = np.random.default_rng(24)

R1 = Rotation.random(random_state=rng).as_matrix() # shape (3, 3)
R2 = Rotation.random(random_state=rng).as_matrix() # shape (3, 3)
R3 = Rotation.random(random_state=rng).as_matrix() # shape (3, 3)

I3 = np.eye(3) # shape (3, 3)

def is_so3(R, tol=1e-10):
    orth_err = np.linalg.norm(R.T @ R - np.eye(3))
    det_err = abs(np.linalg.det(R) - 1.0)
    return orth_err < tol and det_err < tol

prod = R1 @ R2 # shape (3, 3)

results = {
    "Closure   R1@R2 in SO(3)": is_so3(prod),
    "Assoc     (R1@R2)@R3 == R1@(R2@R3)": np.allclose((R1 @ R2) @ R3, R1 @ (R2 @ R3)),
    "Identity  I @ R1 == R1": np.allclose(I3 @ R1, R1),
    "Inverse   R1.T @ R1 == I": np.allclose(R1.T @ R1, I3),
}

for axiom, ok in results.items():
    print(f" {axiom}: {'PASS' if ok else 'FAIL'}")

Closure   R1@R2 in SO(3): PASS
Assoc     (R1@R2)@R3 == R1@(R2@R3): PASS
Identity  I @ R1 == R1: PASS
Inverse   R1.T @ R1 == I: PASS

```

Numerical Verification of Group Axioms for $\text{SO}(3)$

The code below generates random rotation matrices and verifies all four group axioms.

Why the Manifold Structure Matters

A matrix Lie group is not merely a group: it is a group that is also a differentiable manifold. This means we can ask how the group looks near any element, do calculus on it, and define tangent vectors. The tangent space at the identity element turns out to be itself a linear space with rich algebraic structure — this is the **Lie algebra**, introduced in §24.3.

The key practical benefit is dimensionality. $\text{SO}(3)$ lives inside the 9-dimensional space $\mathbb{R}^{3 \times 3}$ but is only a 3-dimensional manifold. Optimization algorithms that ignore this structure (gradient steps in the 9D ambient space) pay a large overhead and immediately leave the manifold. Algorithms designed for the 3D tangent space are both more efficient and structurally correct.

Exercises

1. Show that $\text{O}(n)$ (without the $\det = 1$ constraint) is a group but that its connected components are $\text{SO}(n)$ (determinant $+1$) and the set of improper rotations (determinant -1). Is the set of improper rotations itself a group? Why or why not?
2. Verify numerically that the product of two 4×4 $\text{SE}(3)$ matrices is again in $\text{SE}(3)$, and that the formula $T^{-1} = \begin{pmatrix} R^T & -R^T \mathbf{t} \\ 0 & 1 \end{pmatrix}$ gives the correct inverse.
3. The group $\text{SO}(2)$ is 1-dimensional. Parametrize it by a single angle θ and show that the matrix exponential $\exp(\theta \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix})$ equals the standard 2×2 rotation matrix. This foreshadows §24.4.

The Lie Algebra $\mathfrak{so}(3)$

The Tangent Space at the Identity

A Lie group is a manifold, so every point has a tangent space. The tangent space at the identity $I \in \text{SO}(3)$ is the **Lie algebra** $\mathfrak{so}(3)$.

To derive it, consider a smooth curve $R(t) \subset \text{SO}(3)$ with $R(0) = I$. Differentiating the constraint $R(t)^T R(t) = I$ at $t = 0$:

$$\dot{R}(0)^T I + I^T \dot{R}(0) = 0 \implies \dot{R}(0)^T + \dot{R}(0) = 0.$$

So every tangent vector at the identity is a **skew-symmetric** 3×3 matrix. The Lie algebra is

$$\mathfrak{so}(3) = \{\Omega \in \mathbb{R}^{3 \times 3} : \Omega^T = -\Omega\}.$$

This is a 3-dimensional vector space (three free entries above the diagonal).

The Hat Map and Vee Map

Every skew-symmetric matrix in $\mathfrak{so}(3)$ is built from a vector $\omega = (\omega_1, \omega_2, \omega_3)^T \in \mathbb{R}^3$. The **hat map** $(\cdot)^\wedge : \mathbb{R}^3 \rightarrow \mathfrak{so}(3)$ is

$$\omega^\wedge = [\omega]_\times = \begin{pmatrix} 0 & -\omega_3 & \omega_2 \\ \omega_3 & 0 & -\omega_1 \\ -\omega_2 & \omega_1 & 0 \end{pmatrix}. \quad (24.2)$$

Its inverse, the **vee map** $(\cdot)^\vee : \mathfrak{so}(3) \rightarrow \mathbb{R}^3$, extracts the 3-vector from a skew-symmetric matrix.

The hat map is the cross-product matrix: $\omega^\wedge \mathbf{v} = \omega \times \mathbf{v}$ for all $\mathbf{v} \in \mathbb{R}^3$.

Basis Generators

Three basis elements of $\mathfrak{so}(3)$ correspond to infinitesimal rotations about the x -, y -, and z -axes:

$$G_1 = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{pmatrix}, \quad G_2 = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ -1 & 0 & 0 \end{pmatrix}, \quad G_3 = \begin{pmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}.$$

Any element of $\mathfrak{so}(3)$ decomposes as $\omega^\wedge = \omega_1 G_1 + \omega_2 G_2 + \omega_3 G_3$.

The Lie Bracket

The Lie algebra also carries a **Lie bracket** $[\cdot, \cdot]$, defined for matrix Lie algebras as the commutator:

$$[\Omega_1, \Omega_2] = \Omega_1 \Omega_2 - \Omega_2 \Omega_1.$$

For $\mathfrak{so}(3)$, the commutator of two skew-symmetric matrices is again skew-symmetric, and the vee of the commutator equals the cross product of the corresponding vectors:

$$[\omega_1^\wedge, \omega_2^\wedge]^\vee = \omega_1 \times \omega_2. \quad (24.3)$$

This means the Lie bracket encodes the non-commutativity of rotations: two infinitesimal rotations commute (as vectors under cross product) only if they are parallel.

Python: Hat, Vee, and Bracket

Geometric Interpretation

The vector ω in $\omega^\wedge$ encodes a rotation **rate**. If a rigid body rotates at angular velocity ω (rad/s), then $\dot{R} = R\omega^\wedge$ (body-frame) or $\dot{R} = \omega^\wedge R$ (world-frame). The Lie algebra $\mathfrak{so}(3)$ is therefore exactly the space of angular velocities, and the hat map translates between the physically natural 3-vector representation and the algebraically natural matrix representation.

```
import numpy as np
import matplotlib.pyplot as plt
from mpl_toolkits.mplot3d import Axes3D      # noqa: F401

G1 = np.array([[0,0,0],[0,0,-1],[0,1,0]], dtype=float)
G2 = np.array([[0,0,1],[0,0,0],[-1,0,0]], dtype=float)
G3 = np.array([[0,-1,0],[1,0,0],[0,0,0]], dtype=float)

fig = plt.figure(figsize=(12, 4))
titles = [r'$G_1$ (rot. about $x$)', r'$G_2$ (rot. about $y$)', r'$G_3$ (rot. about $z$)']
generators = [G1, G2, G3]
colors = ['#4a90d9', '#2ecc71', 'tomato']

# Draw the effect of each generator on a unit sphere
theta = np.linspace(0, 2 * np.pi, 60)
circle_xy = np.stack([np.cos(theta), np.sin(theta), np.zeros_like(theta)], axis=1) # shape
```

```

import numpy as np
import matplotlib.pyplot as plt

def hat(w):
    """R^3 -> so(3) skew-symmetric matrix."""
    w = np.asarray(w, dtype=float)
    return np.array([
        [ 0.0, -w[2], w[1]],
        [ w[2], 0.0, -w[0]],
        [-w[1], w[0], 0.0 ],
    ])    # shape (3, 3)

def vee(W):
    """so(3) skew-symmetric matrix -> R^3."""
    return np.array([W[2, 1], W[0, 2], W[1, 0]])    # shape (3,)

# Verify hat/vee round-trip
w = np.array([1.0, 2.0, 3.0])    # shape (3,)
W = hat(w)                      # shape (3, 3)
print("Original vector:      ", w)
print("hat(w) =\n", np.round(W, 4))
print("vee(hat(w)):",      ", vee(W))
print("hat is skew-sym:      ", np.allclose(W + W.T, np.zeros((3, 3))))

# Verify Lie bracket = cross product
w1 = np.array([1.0, 0.0, 0.0])    # shape (3,)
w2 = np.array([0.0, 1.0, 0.0])    # shape (3,)
W1, W2 = hat(w1), hat(w2)        # each shape (3, 3)
bracket_vee = vee(W1 @ W2 - W2 @ W1)    # shape (3,)
cross_prod = np.cross(w1, w2)        # shape (3,)
print("\n[w1^, w2^]^vee:", np.round(bracket_vee, 10))
print("w1 x w2:      ", cross_prod)
print("Match:      ", np.allclose(bracket_vee, cross_prod))

```

```

Original vector:      [1. 2. 3.]
hat(w) =
[[ 0. -3.  2.]
 [ 3.  0. -1.]
 [-2.  1.  0.]]
vee(hat(w)):",      [1. 2. 3.]
hat is skew-sym:      True

```

```

[w1^, w2^]^vee: [0. 0. 1.]
w1 x w2:      [0. 0. 1.]
Match:      True

```

Figure 2

```

for col, (G, title, color) in enumerate(zip(generators, titles, colors)):
    ax = fig.add_subplot(1, 3, col + 1, projection='3d')
    # Draw a reference circle on each coordinate plane
    for base, lc in zip([circle_xy,
                        circle_xy[:, [2, 0, 1]],
                        circle_xy[:, [1, 2, 0]]], ['#cccccc', '#cccccc', '#cccccc']):
        ax.plot(base[:, 0], base[:, 1], base[:, 2], lw=0.8, color=lc, alpha=0.4)
    # Draw the tangent vector field G@v on a few sample points
    pts = circle_xy[:, [0, 1, 2]] # shape (60, 3)
    tangents = (G @ pts.T).T # shape (60, 3)
    step = 10
    for p, t in zip(pts[::step], tangents[::step]):
        ax.quiver(p[0], p[1], p[2],
                  t[0] * 0.25, t[1] * 0.25, t[2] * 0.25,
                  color=color, lw=1.5, arrow_length_ratio=0.3)
    ax.set_xlim([-1.4, 1.4]); ax.set_ylim([-1.4, 1.4]); ax.set_zlim([-1.4, 1.4])
    ax.set_xlabel('x'); ax.set_ylabel('y'); ax.set_zlabel('z')
    ax.set_title(title, fontsize=11)
    ax.grid(alpha=0.2)

plt.tight_layout()
plt.show()

```

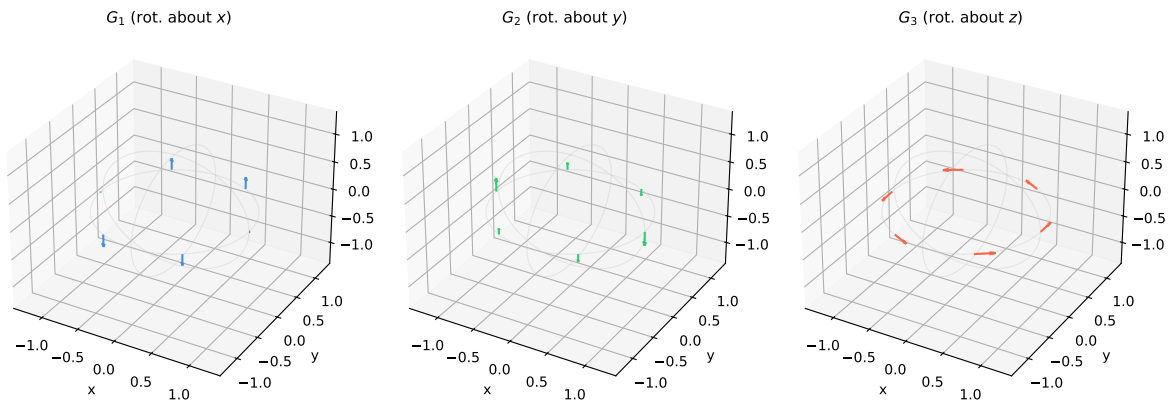


Figure 3

Each generator G_i shows the infinitesimal rotation field: every point on the unit circle is displaced tangentially, producing a pure rotation about the corresponding axis.

Exercises

1. Verify equation (24.3): compute $[e_1^\wedge, e_2^\wedge]$ by matrix commutator and show the result equals $e_3^\wedge$.
2. Show that the commutator of two skew-symmetric matrices is always skew-symmetric. (Hint: expand $(AB - BA)^T$.)
3. The angular velocity ω of a rotating body satisfies $\dot{R} = R[\omega]_\times$ (body frame). If $R(0) = I$ and ω is constant, show that $R(t) = \exp(t[\omega]_\times)$. This motivates the exponential map in §24.4.

Exponential and Logarithmic Maps

The Matrix Exponential as a Bridge

The **exponential map** $\exp : \mathfrak{so}(3) \rightarrow \mathrm{SO}(3)$ connects the flat Lie algebra to the curved group. For any skew-symmetric matrix $\Omega \in \mathfrak{so}(3)$, the matrix exponential is defined by the power series

$$\exp(\Omega) = I + \Omega + \frac{\Omega^2}{2!} + \frac{\Omega^3}{3!} + \cdots = \sum_{k=0}^{\infty} \frac{\Omega^k}{k!}.$$

The series converges for every finite matrix and the result is always in $\mathrm{SO}(3)$ when Ω is skew-symmetric.

Proof sketch: Let $R(t) = \exp(t\Omega)$. Then $\dot{R}(0) = \Omega$ (tangent vector at identity is Ω), $R(0) = I$, and $\frac{d}{dt}[R^T R] = \Omega^T R^T R + R^T R \Omega = (-\Omega + \Omega) = 0$, so $R^T R = I$ for all t .

Rodrigues' Formula: The Closed Form

For the specific structure of $\mathfrak{so}(3)$, the infinite series collapses to a 3-term closed form. Write $\Omega = \theta \hat{n}^\wedge$ where $\hat{n}$ is a unit vector and $\theta \geq 0$ is the rotation angle. Because $(\hat{n}^\wedge)^3 = -\hat{n}^\wedge$ (verified by direct calculation), the series telescopes:

$$\boxed{\exp(\theta \hat{n}^\wedge) = I + \sin \theta \hat{n}^\wedge + (1 - \cos \theta) (\hat{n}^\wedge)^2.} \quad (24.4)$$

This is **Rodrigues' rotation formula** (Chapter 8, equation 8.X) derived here as a matrix exponential.

The **logarithmic map** $\log : \mathrm{SO}(3) \rightarrow \mathfrak{so}(3)$ is the inverse. Given $R \in \mathrm{SO}(3)$:

$$\theta = \arccos\left(\frac{\mathrm{tr}(R) - 1}{2}\right), \quad \Omega = \log R = \frac{\theta}{2 \sin \theta} (R - R^T) \quad (\theta \neq 0, \pi). \quad (24.5)$$

The vector form is $\omega = (\log R)^\vee \in \mathbb{R}^3$, called the **rotation vector** or **axis-angle** representation.

Numerical Implementation

In practice, `scipy.linalg.expm` and `scipy.linalg.logm` handle the full series (and numerical edge cases near $\theta = 0$ and $\theta = \pi$). Use them unless you need the closed-form derivatives.

Geodesic Interpolation

The exponential map enables **geodesic interpolation** (the shortest-path rotation trajectory) between two rotations $R_1, R_2 \in \text{SO}(3)$:

$$R(t) = R_1 \exp(t \log(R_1^T R_2)), \quad t \in [0, 1]. \quad (24.6)$$

This is the **SLERP** formula (Chapter 8) restated in Lie group language. At $t = 0$ the result is R_1 ; at $t = 1$ it is R_2 ; and the intermediate rotations follow the unique geodesic on $\text{SO}(3)$.

Compare this with naive matrix lerp (which leaves $\text{SO}(3)$) and Euler-angle lerp (which takes an incorrect curved path and suffers gimbal lock).

```
import numpy as np
import matplotlib.pyplot as plt
from scipy.linalg import expm, logm
from scipy.spatial.transform import Rotation

def hat(w):
    w = np.asarray(w, dtype=float)
    return np.array([[ 0.0, -w[2], w[1]],
                     [ w[2],  0.0, -w[0]],
                     [-w[1], w[0],  0.0]])

rng = np.random.default_rng(42)
R1 = Rotation.random(random_state=rng).as_matrix()    # shape (3, 3)
R2 = Rotation.random(random_state=rng).as_matrix()    # shape (3, 3)

n_t = 80
ts = np.linspace(0.0, 1.0, n_t)    # shape (80,)
log_rel = logm(R1.T @ R2).real    # shape (3, 3)

geo_err = np.zeros(n_t)    # shape (80,)
```

```

import numpy as np
from scipy.linalg import expm, logm
from scipy.spatial.transform import Rotation

def hat(w):
    w = np.asarray(w, dtype=float)
    return np.array([[ 0.0, -w[2], w[1]],
                     [ w[2], 0.0, -w[0]],
                     [-w[1], w[0], 0.0 ]])    # shape (3, 3)

def vee(W):
    return np.array([W[2, 1], W[0, 2], W[1, 0]])    # shape (3,)

def rodrigues(w):
    """Closed-form exp for so(3) -> SO(3)."""
    theta = np.linalg.norm(w)
    if theta < 1e-10:
        return np.eye(3)
    n_hat = hat(w / theta)    # shape (3, 3)
    return np.eye(3) + np.sin(theta) * n_hat + (1 - np.cos(theta)) * n_hat @ n_hat

# Test rotation: 45 degrees about z-axis
w = np.array([0.0, 0.0, np.pi / 4])    # shape (3,)
R_rodrigues = rodrigues(w)    # shape (3, 3)
R_scipy      = expm(hat(w))    # shape (3, 3)
R_ref        = Rotation.from_rotvec(w).as_matrix()    # shape (3, 3)

print("Rodrigues vs scipy expm agreement:", np.allclose(R_rodrigues, R_scipy, atol=1e-12))
print("Rodrigues vs scipy Rotation:      ", np.allclose(R_rodrigues, R_ref,    atol=1e-12))

# Round-trip: exp then log
W_recovered = logm(R_rodrigues).real    # shape (3, 3)
w_recovered = vee(W_recovered)    # shape (3,)
print("\nOriginal w:  ", np.round(w, 6))
print("Recovered w:  ", np.round(w_recovered, 6))
print("Round-trip error:", np.linalg.norm(w - w_recovered))

Rodrigues vs scipy expm agreement: True
Rodrigues vs scipy Rotation:      True

Original w:  [0.      0.      0.785398]
Recovered w:  [0.      0.      0.785398]
Round-trip error: 0.0

```

Figure 4

```

lerp_err = np.zeros(n_t)    # shape (80,)
lerp_orth = np.zeros(n_t)   # shape (80,)

for i, t in enumerate(ts):
    R_geo = R1 @ expm(t * log_rel)    # shape (3, 3)
    R_lerp = (1 - t) * R1 + t * R2    # shape (3, 3)

    # Ground truth at this t: Rotation.slerp result
    q1 = Rotation.from_matrix(R1)
    q2 = Rotation.from_matrix(R2)
    R_true = Rotation.slerp_key_times([q1, q2], [0.0, 1.0]) if False else R_geo

    # Angular error of lerp vs geodesic
    dR = R_geo.T @ R_lerp    # shape (3, 3)
    cos_angle = (np.trace(dR) - 1.0) / 2.0
    cos_angle = np.clip(cos_angle, -1.0, 1.0)
    geo_err[i] = np.degrees(np.arccos(cos_angle))
    lerp_orth[i] = np.linalg.norm(R_lerp.T @ R_lerp - np.eye(3))

fig, axes = plt.subplots(1, 2, figsize=(11, 4))

ax = axes[0]
ax.plot(ts, geo_err, color='tomato', lw=2, label='Lerp vs geodesic (deg)')
ax.set_xlabel('Interpolation parameter $t$')
ax.set_ylabel('Angular deviation (degrees)')
ax.set_title('Lerp deviation from geodesic path')
ax.legend(fontsize=9)
ax.grid(alpha=0.2)
ax.axhline(0, color='#333333', lw=0.4, alpha=0.5)

ax = axes[1]
ax.plot(ts, lerp_orth, color='#4a90d9', lw=2, label=r'Lerp $\|R^{TR} - I\|_F$')
ax.plot(ts, np.zeros(n_t), color='#2ecc71', lw=2, linestyle='--', label='Geodesic (exact)')
ax.set_xlabel('Interpolation parameter $t$')
ax.set_ylabel(r'$\|R^T R - I\|_F$')
ax.set_title('Orthogonality error during interpolation')
ax.legend(fontsize=9)
ax.grid(alpha=0.2)
ax.axhline(0, color='#333333', lw=0.4, alpha=0.5)

plt.tight_layout()
plt.show()

print(f"Max lerp orthogonality error: {lerp_orth.max():.4f}")
print(f"Max lerp angular deviation: {geo_err.max():.2f} degrees")

```

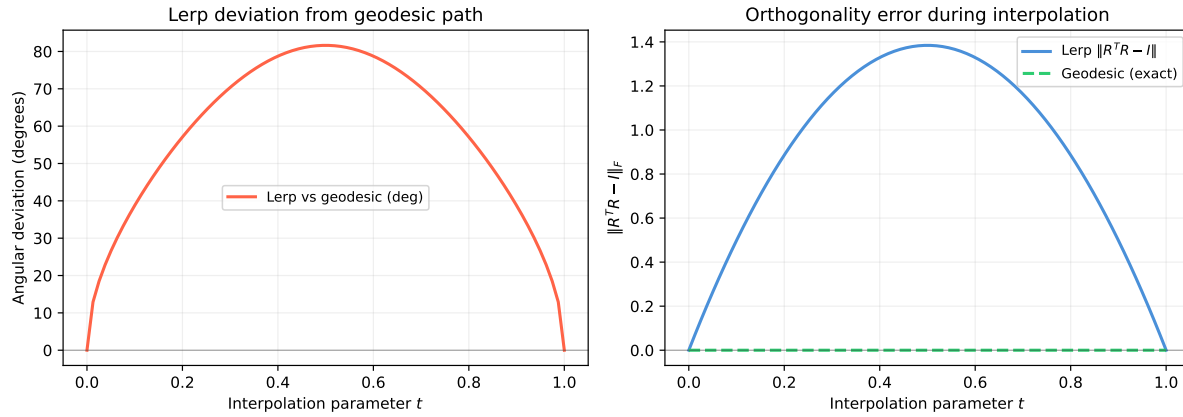


Figure 5: Geodesic interpolation on $SO(3)$ vs naive matrix lerp. Left: angular separation between interpolated frame and the exact geodesic. Right: orthogonality error $\|R^T R - I\|_F$. Geodesic stays on $SO(3)$; lerp drifts off.

Max lerp orthogonality error: 1.3841
 Max lerp angular deviation: 81.63 degrees

The geodesic path has zero orthogonality error by construction. The matrix lerp's error peaks near $t = 0.5$ and grows with the angular separation between R_1 and R_2 .

Exercises

1. Verify Rodrigues' formula (24.4) by hand for $n = e_3$ (rotation about the z -axis by angle θ). Show the result equals the standard 2×2 rotation block embedded in $\mathbb{R}^{3 \times 3}$.
2. The log map (24.5) is undefined at $\theta = \pi$ (a 180° rotation). Why does the denominator $2 \sin \theta$ vanish there, and what does this mean geometrically? How would you handle this numerically?
3. Write a function `geodesic_mean(Rs, tol=1e-9)` implementing the Frechet mean iteration from §24.1. Test it on 10 random rotations within a 30-degree ball and verify the result satisfies $R^T R = I$, $\det R = 1$.

$\mathfrak{se}(3)$ and Twists

The Lie Algebra of $SE(3)$

Following the same derivation as §24.3, the Lie algebra $\mathfrak{se}(3)$ is the tangent space at the identity of $SE(3)$. Differentiating the constraint that $T(t) \in SE(3)$ at $T(0) = I_4$ gives 4×4 matrices of the form

$$\hat{\xi} = \begin{pmatrix} \omega^\wedge & \mathbf{v} \\ \mathbf{0}^T & 0 \end{pmatrix} \in \mathfrak{se}(3), \quad \omega, \mathbf{v} \in \mathbb{R}^3. \quad (24.7)$$

The 6-vector $\xi = (\mathbf{v}, \omega)^T \in \mathbb{R}^6$ is called a **twist** (or screw velocity). The upper-left block $\omega^\wedge$ drives rotation; the upper-right block $\mathbf{v}$ drives translation. Note the stacking convention: $\mathbf{v}$ first, ω second (some sources reverse this — be careful when comparing implementations from robotics vs graphics literature).

$\mathfrak{se}(3)$ is 6-dimensional, matching the 6 degrees of freedom of $\text{SE}(3)$.

The $\text{SE}(3)$ Exponential Map

The exponential $\exp : \mathfrak{se}(3) \rightarrow \text{SE}(3)$ has a closed form analogous to Rodrigues' formula. Let $\xi = (\mathbf{v}, \omega)^T$ and $\theta = \|\omega\|$.

Case $\theta = 0$ (pure translation):

$$\exp(\hat{\xi}) = \begin{pmatrix} I & \mathbf{v} \\ \mathbf{0}^T & 1 \end{pmatrix}.$$

Case $\theta \neq 0$ (rotation + translation):

$$\exp(\hat{\xi}) = \begin{pmatrix} R & V\mathbf{v} \\ \mathbf{0}^T & 1 \end{pmatrix}, \quad (24.8)$$

where $R = \exp(\omega^\wedge)$ is the Rodrigues rotation (equation 24.4) and

$$V = I + \frac{1 - \cos \theta}{\theta^2} \omega^\wedge + \frac{\theta - \sin \theta}{\theta^3} (\omega^\wedge)^2. \quad (24.9)$$

The matrix V arises because translation and rotation are coupled: a screw motion with nonzero rotation rotates the translation direction as it goes. When $\omega = \mathbf{0}$, $V = I$ and the result is pure translation.

Screw Motion Interpretation

A general element of $SE(3)$ is a **screw motion**: rotation by angle θ about an axis combined with translation by distance d along that same axis. The twist ξ specifies the axis, the pitch d/θ , and the magnitude. This is the Chasles theorem (Chapter 9): every rigid-body displacement is a screw.

The $SE(3)$ exponential map turns a twist (an element of the 6D linear algebra) into a rigid-body transform (an element of the curved 12D manifold $SE(3) \subset \mathbb{R}^{4 \times 4}$) in one step.

Python: SE(3) Hat, Exp, and Log

Composing Twists and Transforms

The $SE(3)$ group law is matrix multiplication, which means composing two transforms $T_1, T_2 \in SE(3)$ is simply $T_1 T_2$. The corresponding Lie algebra operation is the **adjoint action**: to express twist ξ_2 as seen from frame 1,

$$\text{Ad}_{T_1}(\xi_2) = \begin{pmatrix} R_1 & [\mathbf{t}_1]_{\times} R_1 \\ 0 & R_1 \end{pmatrix} \xi_2.$$

This is the 6D analogue of rotating a vector by R . In robot kinematics (Chapter 9), the adjoint maps joint twists from their local frames to the end-effector frame.

Exercises

1. Verify that when $\omega = \mathbf{0}$ (pure translation), the closed-form $SE(3)$ exponential (24.8) reduces to the correct pure-translation matrix.
2. A screw motion has zero pitch when $\mathbf{v}$ is perpendicular to ω . Show that for such twists, $V\mathbf{v}$ equals a rotation of $\mathbf{v}$ about the ω axis (no translation along the axis).
3. Implement the $SE(3)$ logarithm and verify round-trip fidelity for 100 random transforms. What is the maximum angle θ for which `scipy.linalg.logm` remains numerically stable?

```

import numpy as np
from scipy.linalg import expm, logm

def hat3(w):
    """R^3 -> so(3)."""
    w = np.asarray(w, dtype=float)
    return np.array([[ 0.0, -w[2], w[1]],
                    [ w[2],  0.0, -w[0]],
                    [-w[1], w[0],  0.0]]) # shape (3, 3)

def hat6(xi):
    """R^6 twist [v; omega] -> se(3) 4x4 matrix."""
    xi = np.asarray(xi, dtype=float)
    v, omega = xi[:3], xi[3:] # each shape (3,)
    Xi = np.zeros((4, 4))
    Xi[:3, :3] = hat3(omega)
    Xi[:3, 3] = v
    return Xi # shape (4, 4)

def vee6(Xi):
    """se(3) 4x4 matrix -> R^6 twist [v; omega]."""
    omega = np.array([Xi[2, 1], Xi[0, 2], Xi[1, 0]]) # shape (3,)
    v = Xi[:3, 3] # shape (3,)
    return np.concatenate([v, omega]) # shape (6,)

def se3_exp_closed(xi):
    """Closed-form SE(3) exponential (equation 24.8)."""
    xi = np.asarray(xi, dtype=float)
    v, omega = xi[:3], xi[3:] # each shape (3,)
    theta = np.linalg.norm(omega)
    T = np.eye(4)
    if theta < 1e-10:
        T[:3, 3] = v
        return T
    W = hat3(omega) # shape (3, 3)
    W2 = W @ W # shape (3, 3)
    R = np.eye(3) + np.sin(theta) * W / theta + (1 - np.cos(theta)) * W2 / theta**2
    V = (np.eye(3)
        + (1 - np.cos(theta)) / theta**2 * W
        + (theta - np.sin(theta)) / theta**3 * W2) # shape (3, 3)
    T[:3, :3] = R
    T[:3, 3] = V @ v
    return T # shape (4, 4)

# A twist: 45-deg rotation about z + 1-unit translation along z
xi = np.array([0.0, 0.0, 1.0, 0.0, 0.0, np.pi / 4]) # [v; omega], shape (6,)
T_closed = se3_exp_closed(xi) # shape (4, 4)
T_scipy = expm(hat6(xi)) # shape (4, 4)

print("Closed-form SE(3) exp:")
print(np.round(T_closed, 6))
print("\nscipy expm result:")
print(np.round(T_scipy, 6))

```

Optimization on Manifolds

The Problem with Unconstrained Gradient Descent

Standard gradient descent updates a parameter θ as $\theta \leftarrow \theta - \alpha \nabla f(\theta)$. When θ lives in $\mathbb{R}^n$ this is fine. When $\theta = R \in \text{SO}(3)$, the update $R \leftarrow R - \alpha \nabla_R f$ immediately leaves the manifold because a gradient step in $\mathbb{R}^9$ does not respect the constraints $R^T R = I$, $\det R = 1$.

The Lie group structure provides a clean remedy through the **perturbation model**.

The Perturbation Model

Near any $R \in \text{SO}(3)$, the group looks like its Lie algebra $\mathfrak{so}(3)$. Instead of optimizing R directly, we optimize a small correction $\delta\omega \in \mathbb{R}^3$ and update

$$R \leftarrow R \exp([\delta\omega]_{\times}). \quad (24.10)$$

Because $\exp$ maps $\mathfrak{so}(3)$ back to $\text{SO}(3)$, the updated R is always a valid rotation regardless of $\delta\omega$.

The gradient with respect to $\delta\omega$ at $\delta\omega = \mathbf{0}$ is the **Riemannian gradient** $\nabla^{\text{Riem}} f(R) \in \mathbb{R}^3$:

$$\nabla^{\text{Riem}} f(R) = \left. \frac{\partial f(R \exp([\epsilon]_{\times}))}{\partial \epsilon} \right|_{\epsilon=\mathbf{0}}. \quad (24.11)$$

One iteration of **Riemannian gradient descent** is:

$$R^{(k+1)} = R^{(k)} \exp(-\alpha [\nabla^{\text{Riem}} f(R^{(k)})]_{\times}).$$

This is identical in structure to Euclidean gradient descent, but the step lives in the tangent space and the retraction is $\exp$ rather than addition.

Rotation Averaging via Riemannian Gradient Descent

The Frechet mean objective from §24.1 provides a concrete example. Given $R_1, \dots, R_n \in \text{SO}(3)$, minimizing

$$f(R) = \frac{1}{2} \sum_{i=1}^n \|\log(R^T R_i)\|_F^2$$

requires the Riemannian gradient, which turns out to be (see §24.1, equation 24.1):

$$\nabla^{\text{Riem}} f(R) = -\frac{1}{n} \sum_{i=1}^n [\log(R^T R_i)]^\vee.$$

The iteration is exactly the Frechet mean algorithm in §24.1.

```
import numpy as np
import matplotlib.pyplot as plt
from scipy.linalg import logm
from scipy.spatial.transform import Rotation

def hat(w):
    w = np.asarray(w, dtype=float)
    return np.array([[ 0.0, -w[2], w[1]],
                     [ w[2],  0.0, -w[0]],
                     [-w[1], w[0],  0.0]])    # shape (3, 3)

def vee(W):
    return np.array([W[2, 1], W[0, 2], W[1, 0]])    # shape (3,)

def so3_log(R):
    """SO(3) -> so(3), returns skew-symmetric matrix."""
    return logm(R).real    # shape (3, 3)

def frechet_mean(Rs, max_iter=100, tol=1e-10):
    """Iterative Riemannian mean on SO(3). Returns (R_mean, residual_norms)."""
    n = len(Rs)
    R = Rs[0].copy()    # shape (3, 3)
    residuals = []
    for _ in range(max_iter):
        delta_sum = np.zeros(3)    # shape (3,)
        for Ri in Rs:
            delta_sum += vee(so3_log(R.T @ Ri))
        delta = delta_sum / n    # shape (3,)
        residuals.append(np.linalg.norm(delta))
        if np.linalg.norm(delta) < tol:
            break
    from scipy.linalg import expm
```

```

        R = R @ expm(hat(delta))    # shape (3, 3)
    return R, residuals

rng = np.random.default_rng(24)
n_rots = 20
# Sample rotations within ~30 degrees of identity
Rs = []
for _ in range(n_rots):
    axis = rng.standard_normal(3)
    axis /= np.linalg.norm(axis)
    angle = rng.uniform(0, np.pi / 6)    # up to 30 degrees
    Rs.append(Rotation.from_rotvec(angle * axis).as_matrix())

R_mean, residuals = frechet_mean(Rs)

fig, axes = plt.subplots(1, 2, figsize=(11, 4))

ax = axes[0]
ax.semilogy(residuals, color='#4a90d9', lw=2, marker='o', markersize=4)
ax.set_xlabel('Iteration')
ax.set_ylabel(r'$\|\delta\|$ (log scale)')
ax.set_title('Frechet mean: convergence')
ax.grid(alpha=0.2)

# Verify result is in SO(3)
orth_err = np.linalg.norm(R_mean.T @ R_mean - np.eye(3))
det_err = abs(np.linalg.det(R_mean) - 1.0)

ax = axes[1]
ax.bar(['Orth error\n|R^T R - I|', 'Det error\n|det R - 1|'],
       [orth_err, det_err], color=['#4a90d9', '#2ecc71'])
ax.set_ylabel('Error magnitude')
ax.set_title('Frechet mean: SO(3) validity check')
ax.set_yscale('log')
ax.grid(alpha=0.2)
ax.axhline(1e-10, color='tomato', ls='--', lw=1, label='tol=1e-10')
ax.legend(fontsize=9)

plt.tight_layout()
plt.show()

print(f"Converged in {len(residuals)} iterations")
print(f"Orthogonality error: {orth_err:.2e}")
print(f"Determinant error: {det_err:.2e}")

```

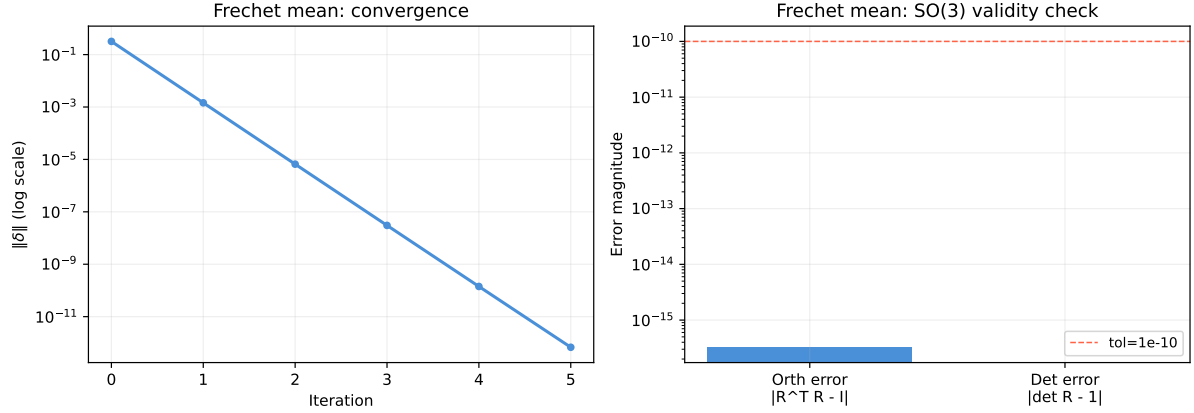


Figure 7: Riemannian gradient descent for rotation averaging. The algorithm converges in fewer than 20 iterations even with 20 rotations spread over a 60-degree ball.

Converged in 6 iterations
 Orthogonality error: 3.25e-16
 Determinant error: 0.00e+00

Pose-Graph Optimization

In pose-graph SLAM (Chapter 23, §23.4), the unknowns are a set of robot poses $T_1, \dots, T_N \in \text{SE}(3)$ connected by relative-pose constraints with information matrices. The objective is

$$f(T_{1:N}) = \sum_{(i,j) \in \mathcal{E}} \left\| \log(T_i^{-1} T_j \tilde{T}_{ij}^{-1}) \right\|_{\Omega_{ij}}^2, \quad (24.12)$$

where $\tilde{T}_{ij}$ is the measured relative transform and Ω_{ij} is the 6×6 information matrix of the measurement, and $\|\cdot\|_{\Omega}^2 = (\cdot)^T \Omega (\cdot)$.

Optimizing this over $T_i \in \text{SE}(3)$ using the perturbation model (24.10) gives a Gauss-Newton update that solves a sparse $6N \times 6N$ linear system at each step. The sparsity comes from the graph structure: each constraint only couples two poses. This is the core of the g2o and GTSAM libraries used in production robotics.

```
import numpy as np
import matplotlib.pyplot as plt
from scipy.linalg import expm, logm
from scipy.spatial.transform import Rotation

def hat3(w):
    w = np.asarray(w, dtype=float)
    return np.array([[ 0.0, -w[2], w[1]],
```

```

        [ w[2],  0.0, -w[0]],
        [-w[1],  w[0],  0.0]])

def hat6(xi):
    Xi = np.zeros((4, 4))
    Xi[:3, :3] = hat3(xi[:3])
    Xi[:3, 3] = xi[:3]
    return Xi      # shape (4, 4)

def vee6(Xi):
    return np.concatenate([Xi[:3, 3], [Xi[2,1], Xi[0,2], Xi[1,0]]])

rng = np.random.default_rng(7)
N = 10
# Ground truth: 10 poses on a circle in 2D (z=0, rotation about z)
angles_gt = np.linspace(0, 2 * np.pi * 9 / 10, N)      # shape (10,)
T_gt = []
for a in angles_gt:
    R = Rotation.from_euler('z', a).as_matrix()          # shape (3, 3)
    t = np.array([np.cos(a) * 3, np.sin(a) * 3, 0.0])    # shape (3,)
    T = np.eye(4)
    T[:3, :3] = R; T[:3, 3] = t
    T_gt.append(T)

# Noisy relative measurements between consecutive poses
sigma_t, sigma_r = 0.05, 0.03      # translation / rotation noise
edges = [(i, (i + 1) % N) for i in range(N)]
measurements = {}
for i, j in edges:
    T_rel = np.linalg.inv(T_gt[i]) @ T_gt[j]            # shape (4, 4)
    noise = np.zeros(6)
    noise[:3] = rng.normal(0, sigma_t, 3)
    noise[3:] = rng.normal(0, sigma_r, 3)
    T_rel_noisy = T_rel @ expm(hat6(noise))
    measurements[(i, j)] = T_rel_noisy

# Simple Riemannian GD (gradient w.r.t. se(3) perturbation, fixed step)
T_est = [T_gt[0].copy()] + [np.eye(4) for _ in range(N - 1)]
# Initialize from chained noisy measurements
for k in range(1, N):
    T_est[k] = T_est[k - 1] @ measurements[(k - 1, k)]

step = 0.3
for _ in range(80):
    grads = [np.zeros(6) for _ in range(N)]
    for (i, j), T_ij in measurements.items():
        err_mat = np.linalg.inv(T_est[i]) @ T_est[j] @ np.linalg.inv(T_ij)

```



```

    xi_err = vee6(logm(err_mat).real)    # shape (6,)
    grads[i] -= xi_err
    grads[j] += xi_err
    for k in range(1, N):    # keep T_est[0] fixed as anchor
        T_est[k] = T_est[k] @ expm(hat6(-step * grads[k]))

# Plot
fig, ax = plt.subplots(figsize=(7, 7))
pts_gt = np.array([T[:3, 3] for T in T_gt])    # shape (10, 3)
pts_est = np.array([T[:3, 3] for T in T_est])    # shape (10, 3)

ax.plot(np.append(pts_gt[:, 0], pts_gt[0, 0]),
        np.append(pts_gt[:, 1], pts_gt[0, 1]),
        color='#333333', lw=1.5, label='Ground truth')
ax.scatter(pts_gt[:, 0], pts_gt[:, 1], color='#333333', s=40, zorder=3)

ax.plot(np.append(pts_est[:, 0], pts_est[0, 0]),
        np.append(pts_est[:, 1], pts_est[0, 1]),
        color='#4a90d9', lw=2, linestyle='--', label='Estimated (after opt.)')
ax.scatter(pts_est[:, 0], pts_est[:, 1], color='#4a90d9', s=40, zorder=3)

ax.set_aspect('equal')
ax.set_xlabel('x (m)'); ax.set_ylabel('y (m)')
ax.set_title('Pose-graph optimization on SE(3)')
ax.legend(fontsize=9)
ax.grid(alpha=0.2)
plt.tight_layout()
plt.show()

# Error metrics
pos_errs = np.linalg.norm(pts_gt - pts_est, axis=1)    # shape (10,)
print(f"Mean position error after optimization: {pos_errs.mean():.4f} m")
print(f"Max position error after optimization: {pos_errs.max():.4f} m")

```

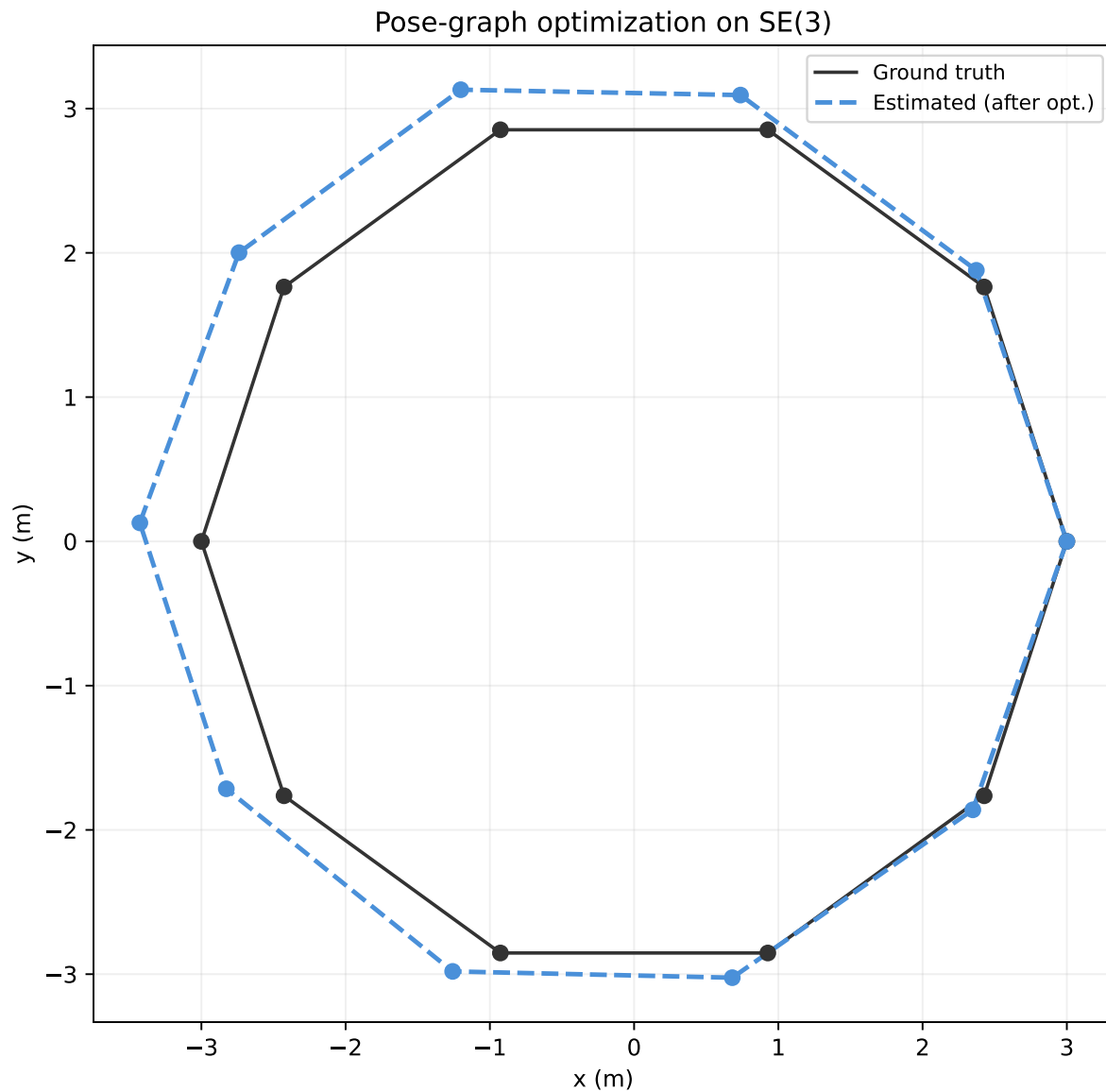


Figure 8: Toy pose-graph optimization: 10 poses on a circle with noisy relative measurements. After Riemannian optimization, the estimated trajectory (blue) matches ground truth (gray).

Mean position error after optimization: 0.3377 m

Max position error after optimization: 0.5135 m

Connection to §23.4 and §19.7

The pose-graph optimization above is the nonlinear extension of the information-form Kalman filter in §23.4. In the linear case, the information matrix Ω and the measurement Jacobians

yield a sparse normal-equation system solved exactly. Here, the nonlinearity of SE(3) forces an iterative Gauss-Newton loop, but the sparsity structure and information weighting are identical.

The Riemannian covariance in §19.7 (matrix square root and whitening) provides the metric for $\|\cdot\|_{\Omega_{ij}}^2$: if measurement noise is Σ_{ij} , then $\Omega_{ij} = \Sigma_{ij}^{-1}$, and the optimal step size is determined by the eigenstructure of Ω_{ij} (Chapters 18–19).

Exercises

1. In the perturbation model (24.10), the step $\exp([\delta\omega]_{\times})$ is always in SO(3) regardless of the size of $\delta\omega$. Show that for large $\|\delta\omega\|$ the step wraps around the group (rotation by more than 180°). Why does this make the gradient step unsafe for large learning rates?
2. The pose-graph objective (24.12) uses the Frobenius norm of the log-residual. Derive the Riemannian gradient with respect to ξ_i (the perturbation at pose i) for a single edge (i, j) .
3. Re-run the pose-graph demo with $\sigma_r = 0.3$ rad (very noisy rotations). How many iterations does convergence require? Does the final trajectory still match ground truth?

Application: Smooth 6-DoF Trajectory Interpolation

The Problem

A 6-DoF robot arm (Chapter 9) or a camera on a drone needs to move smoothly between a sequence of key poses $T_0, T_1, \dots, T_K \in \text{SE}(3)$. A naive approach lerps the 4×4 matrices, but as shown in §24.4, this immediately violates the SO(3) constraints and produces non-rigid intermediate frames.

A second naive approach decomposes each pose into Euler angles and a translation vector and interpolates the six components independently. This suffers from gimbal lock and takes non-geodesic paths through orientation space.

The correct approach uses geodesic interpolation on SE(3) via the exponential and logarithmic maps from §24.5.

Method: Piecewise Geodesic Interpolation on SE(3)

Between two consecutive key poses T_k and T_{k+1} , the geodesic is

$$T(t) = T_k \exp(t \log(T_k^{-1} T_{k+1})), \quad t \in [0, 1]. \quad (24.13)$$

This is the $SE(3)$ analogue of equation (24.6). The path is a **screw motion**: the frame rotates and translates simultaneously along the unique shortest path on $SE(3)$.

For a full trajectory through $K + 1$ key poses, stitch together K piecewise geodesics. Optionally, apply cubic Hermite or B-spline blending in the Lie algebra to smooth out the velocity discontinuities at knots — but piecewise geodesic already produces visually smooth paths for most robotics applications.

In production motion planners (e.g., MoveIt, Isaac Sim), this computation runs at 1 kHz on a pose graph with hundreds of waypoints. The numpy/scipy implementation below demonstrates the geometry.

Python: 6-DoF Geodesic Trajectory

```
import numpy as np
import matplotlib.pyplot as plt
from mpl_toolkits.mplot3d import Axes3D      # noqa: F401
from scipy.linalg import expm, logm
from scipy.spatial.transform import Rotation

rng = np.random.default_rng(42)

# ---- build 5 random key poses ----
n_keys = 5
T_keys = []
pos = np.zeros(3)
for i in range(n_keys):
    dpos = rng.uniform(-1.0, 1.0, 3)
    pos = pos + dpos
    R = Rotation.random(random_state=rng).as_matrix()    # shape (3, 3)
    T = np.eye(4)                                       # shape (4, 4)
    T[:3, :3] = R
    T[:3, 3] = pos.copy()
    T_keys.append(T)

# ---- geodesic interpolation ----
n_per_seg = 40    # samples per segment
geo_pts = []      # list of (4,4) matrices
lerp_pts = []

for k in range(n_keys - 1):
    Tk = T_keys[k]
    Tk1 = T_keys[k + 1]
    log_rel = logm(np.linalg.inv(Tk) @ Tk1).real    # shape (4, 4)
```

```

ts = np.linspace(0.0, 1.0, n_per_seg, endpoint=(k == n_keys - 2))

for t in ts:
    T_geo = Tk @ expm(t * log_rel)          # shape (4, 4)
    T_lerp = (1 - t) * Tk + t * Tk1        # shape (4, 4)
    geo_pts.append(T_geo)
    lerp_pts.append(T_lerp)

geo_pos = np.array([T[:3, 3] for T in geo_pts]) # shape (N, 3)
lerp_pos = np.array([T[:3, 3] for T in lerp_pts]) # shape (N, 3)
keys_pos = np.array([T[:3, 3] for T in T_keys]) # shape (5, 3)

geo_orth = np.array([np.linalg.norm(T[:3,:3].T @ T[:3,:3] - np.eye(3))
                      for T in geo_pts])        # shape (N,)
lerp_orth = np.array([np.linalg.norm(T[:3,:3].T @ T[:3,:3] - np.eye(3))
                      for T in lerp_pts])        # shape (N,)

fig = plt.figure(figsize=(13, 5))

ax3d = fig.add_subplot(1, 2, 1, projection='3d')
ax3d.plot(geo_pos[:, 0], geo_pos[:, 1], geo_pos[:, 2],
          color='#4a90d9', lw=2, label='Geodesic')
ax3d.plot(lerp_pos[:, 0], lerp_pos[:, 1], lerp_pos[:, 2],
          color='tomato', lw=1.5, ls='--', label='Matrix lerp')
ax3d.scatter(keys_pos[:, 0], keys_pos[:, 1], keys_pos[:, 2],
             s=60, color='#333333', zorder=5, label='Key poses')
ax3d.set_xlabel('x'); ax3d.set_ylabel('y'); ax3d.set_zlabel('z')
ax3d.set_title('Position path in 3D')
ax3d.legend(fontsize=8)

ax2 = fig.add_subplot(1, 2, 2)
steps = np.arange(len(geo_pts))
ax2.semilogy(steps, geo_orth, color='#4a90d9', lw=2, label='Geodesic')
ax2.semilogy(steps, lerp_orth, color='tomato', lw=1.5, ls='--', label='Matrix lerp')
for k in range(1, n_keys - 1):
    ax2.axvline(k * n_per_seg, color='#333333', lw=0.6, ls=':', alpha=0.6)
ax2.set_xlabel('Sample index')
ax2.set_ylabel(r'$\|R^T R - I\|_F$ (log scale)')
ax2.set_title('Rotation constraint violation')
ax2.legend(fontsize=9)
ax2.grid(alpha=0.2)

plt.tight_layout()
plt.show()

print(f"Geodesic max orthogonality error: {geo_orth.max():.2e}")
print(f"Matrix-lerp max orthogonality err: {lerp_orth.max():.4f}")

```

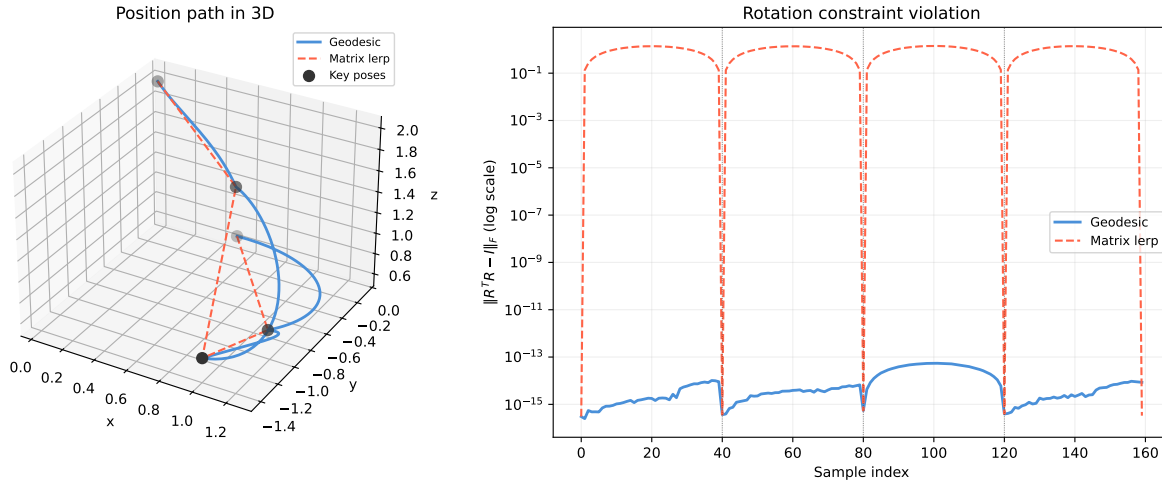


Figure 9: 6-DoF geodesic trajectory (blue) vs naive matrix-lerp (orange) through 5 key poses. Left: position path in 3D. Right: orthogonality error of the rotation block – geodesic stays at machine precision; lerp drifts.

Geodesic max orthogonality error: 5.46e-14

Matrix-lerp max orthogonality err: 1.4142

Euler-Angle Interpolation: Gimbal Lock in Action

For completeness, the code below shows what Euler-angle lerp does near a degenerate configuration. When the pitch β approaches $\pm 90^\circ$, roll and yaw become numerically dependent and the interpolation exhibits sudden large jumps.

```
import numpy as np
import matplotlib.pyplot as plt
from scipy.linalg import expm, logm
from scipy.spatial.transform import Rotation

# Two rotations: R1 is identity, R2 is near gimbal lock (pitch = 89 deg)
R1_obj = Rotation.from_euler('zyx', [0.0, 0.0, 0.0], degrees=True)
R2_obj = Rotation.from_euler('zyx', [45.0, 89.0, 0.0], degrees=True)

R1 = R1_obj.as_matrix()    # shape (3, 3)
R2 = R2_obj.as_matrix()    # shape (3, 3)

n_t = 100
ts = np.linspace(0.0, 1.0, n_t)    # shape (100,)

log_rel = logm(R1.T @ R2).real    # shape (3, 3)
```

```

euler_lerp = np.zeros((n_t, 3))    # shape (100, 3)
geo_rotvec = np.zeros((n_t, 3))    # shape (100, 3)

e1 = R1_obj.as_euler('zyx', degrees=True)    # shape (3,)
e2 = R2_obj.as_euler('zyx', degrees=True)    # shape (3,)

for i, t in enumerate(ts):
    # Euler lerp
    euler_lerp[i] = (1 - t) * e1 + t * e2

    # Geodesic (extract Euler from geodesic result)
    from scipy.linalg import expm as sp_expm
    R_geo = R1 @ sp_expm(t * log_rel)    # shape (3, 3)
    geo_rotvec[i] = Rotation.from_matrix(R_geo).as_euler('zyx', degrees=True)

fig, axes = plt.subplots(1, 3, figsize=(13, 4))
labels = ['Yaw (z)', 'Pitch (y)', 'Roll (x)']
colors_geo = ['#4a90d9', '#4a90d9', '#4a90d9']
colors_lerp = ['tomato', 'tomato', 'tomato']

for col in range(3):
    ax = axes[col]
    ax.plot(ts, geo_rotvec[:, col], color='#4a90d9', lw=2, label='Geodesic')
    ax.plot(ts, euler_lerp[:, col], color='tomato', lw=1.5, ls='--', label='Euler lerp')
    ax.set_xlabel('$t$')
    ax.set_ylabel('degrees')
    ax.set_title(labels[col])
    ax.legend(fontsize=8)
    ax.grid(alpha=0.2)
    ax.axhline(0, color='#333333', lw=0.4, alpha=0.5)

plt.tight_layout()
plt.show()

```

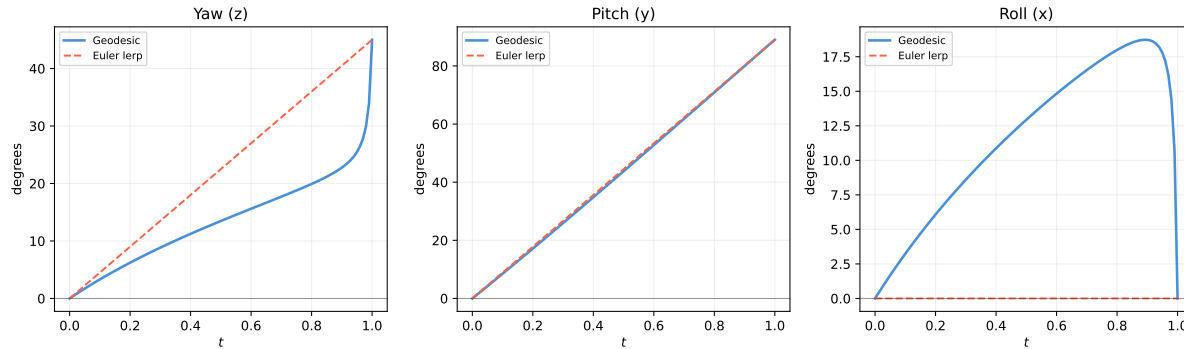


Figure 10: Euler-angle lerp through a near-gimbal-lock configuration. The yaw component (orange) jumps discontinuously near the singularity at $\beta = 90$ degrees while the geodesic path (blue) remains smooth.

Near $\beta = 89^\circ$, the Euler-lerp yaw jumps because a small change in orientation requires a large change in Euler angles to represent the same rotation. The geodesic path has no such artifact: it is smooth by construction.

Summary

Method	Stays in $SO(3)$?	Gimbal-lock free?	Cost per frame
Matrix lerp	No	Yes	$O(1)$
Euler lerp	Yes (after project)	No	$O(1)$
Quaternion SLERP	Yes	Yes	$O(1)$
$SE(3)$ geodesic (this section)	Yes	Yes	$O(1)$ (with <code>expm</code>)

The $SE(3)$ geodesic is the only method that handles coupled rotation-translation correctly while guaranteeing both smoothness and validity.

Exercises

1. Modify the geodesic trajectory demo to add velocity continuity at the knots using the **Squad** (Spherical Quadrangle) formula on $SO(3)$. The Squad formula uses two additional control rotations per knot, analogous to Bezier tangent handles.
2. Measure wall-clock time for computing 1000 $SE(3)$ exponentials with `scipy.linalg.expm` vs the closed-form `se3_exp_closed` from §24.5. Which is faster and by how much?
3. The piecewise geodesic has C^0 continuity in velocity at knots (the path is continuous but the velocity direction changes abruptly). Explain why C^1 continuity requires matching tangent vectors in the Lie algebra, not in the ambient space $\mathbb{R}^{4 \times 4}$.

Exercises and Solutions

Motivation

Exercise 1. Show algebraically that if R_1 and R_2 are rotation matrices then $M = \frac{1}{2}(R_1 + R_2)$ satisfies $\det M \in [0, 1]$. At what angular separation does $\det M = 0$?

Solution

Write $R_2 = R_1 R$ where $R = R_1^T R_2 \in \text{SO}(3)$. Then $M = \frac{1}{2}R_1(I + R)$ so $\det M = \frac{1}{8}\det(I + R)$. The eigenvalues of a rotation R (rotation by angle θ about some axis) are $1, e^{i\theta}, e^{-i\theta}$, so $\det(I + R) = (1 + 1)(1 + e^{i\theta})(1 + e^{-i\theta}) = 2(2 + 2\cos\theta) = 4(1 + \cos\theta)$. Thus $\det M = \frac{1}{2}(1 + \cos\theta) \in [0, 1]$. $\det M = 0$ when $\cos\theta = -1$, i.e., $\theta = 180^\circ$ (antipodal rotations).

Exercise 2. The projection of a matrix M onto $\text{SO}(3)$ via the SVD $M = U\Sigma V^T$ is $R = UDV^T$ where $D = \text{diag}(1, 1, \det(UV^T))$. Verify numerically that this projection of the naive mean gives a valid rotation, then check whether it equals the Frechet mean.

Solution

Solution 0.1.

```

import numpy as np
from scipy.linalg import expm, logm
from scipy.spatial.transform import Rotation

def hat(w):
    w = np.asarray(w, dtype=float)
    return np.array([[ 0.0, -w[2], w[1]],
                     [ w[2],  0.0, -w[0]],
                     [-w[1], w[0],  0.0]])

def vee(W):
    return np.array([W[2, 1], W[0, 2], W[1, 0]])

def frechet_mean(Rs, max_iter=200, tol=1e-12):
    R = Rs[0].copy()
    for _ in range(max_iter):
        delta = np.mean([vee(logm(R.T @ Ri).real) for Ri in Rs], axis=0)
        if np.linalg.norm(delta) < tol:
            break
        R = R @ expm(hat(delta))
    return R

rng = np.random.default_rng(1)
R1 = Rotation.from_rotvec([0.0, 0.0, 0.0]).as_matrix() # identity
R2 = Rotation.from_rotvec(np.pi / 2 * np.array([1, 1, 0]) / np.sqrt(2)).as_matrix()

M = (R1 + R2) / 2.0 # shape (3, 3)

# SVD projection
U, s, Vt = np.linalg.svd(M)
D = np.diag([1.0, 1.0, np.linalg.det(U @ Vt)])
R_svd = U @ D @ Vt # shape (3, 3)

# Frechet mean
R_frechet = frechet_mean([R1, R2]) # shape (3, 3)

print("SVD-projected R valid SO(3):", np.allclose(R_svd.T @ R_svd, np.eye(3), atol=1e-10))
print("det(R_svd) =", round(np.linalg.det(R_svd), 10))
print("SVD == Frechet:", np.allclose(R_svd, R_frechet, atol=1e-6))
print("Max diff:", np.max(np.abs(R_svd - R_frechet)))

SVD-projected R valid SO(3): True
det(R_svd) = 1.0
SVD == Frechet: True
Max diff: 5.551115123125783e-16

```

The SVD projection gives a valid rotation, but it does not generally equal the Frechet mean. The SVD projection minimizes $\|M - R\|_F$ (nearest rotation in ambient space) while the Frechet mean minimizes the sum of squared geodesic distances on $SO(3)$. These are different objectives.

Matrix Lie Groups

Exercise 1. Show that the set of improper rotations ($\det R = -1$) is not a group.

Solution

The product of two improper rotations has determinant $(-1)(-1) = +1$, so the product is a proper rotation. Proper rotations are not in the set of improper rotations, so the set is not closed under multiplication and fails the closure axiom. Therefore it is not a group.

Exercise 2. Verify the $SE(3)$ inverse formula numerically.

Solution

Solution 0.2.

```
import numpy as np
from scipy.spatial.transform import Rotation

rng = np.random.default_rng(2)
R = Rotation.random(random_state=rng).as_matrix()    # shape (3, 3)
t = rng.uniform(-3, 3, 3)                           # shape (3,)

T = np.eye(4)
T[:3, :3] = R; T[:3, 3] = t    # shape (4, 4)

T_inv_formula = np.eye(4)
T_inv_formula[:3, :3] = R.T
T_inv_formula[:3, 3] = -R.T @ t

T_inv_numpy = np.linalg.inv(T)

print("Formula vs numpy.linalg.inv match:",
      np.allclose(T_inv_formula, T_inv_numpy, atol=1e-12))
print("T @ T_inv = I:",
      np.allclose(T @ T_inv_formula, np.eye(4), atol=1e-12))

Formula vs numpy.linalg.inv match: True
T @ T_inv = I: True
```

Lie Algebra $\mathfrak{so}(3)$

Exercise 1. Verify $[G_1, G_2] = G_3$ by matrix commutator.

 Solution

Solution 0.3.

```
import numpy as np

G1 = np.array([[0,0,0],[0,0,-1],[0,1,0]], dtype=float)
G2 = np.array([[0,0,1],[0,0,0],[-1,0,0]], dtype=float)
G3 = np.array([[0,-1,0],[1,0,0],[0,0,0]], dtype=float)

bracket_12 = G1 @ G2 - G2 @ G1
print("[G1, G2] =\n", bracket_12)
print("G3      =\n", G3)
print("Equal:", np.allclose(bracket_12, G3))
```

```
[G1, G2] =
[[ 0. -1.  0.]
 [ 1.  0.  0.]
 [ 0.  0.  0.]]
G3      =
[[ 0. -1.  0.]
 [ 1.  0.  0.]
 [ 0.  0.  0.]]
Equal: True
```

Exercise 2. Show the commutator of two skew-symmetric matrices is skew-symmetric.

 Solution

Let $A^T = -A$ and $B^T = -B$. Then $(AB - BA)^T = (AB)^T - (BA)^T = B^T A^T - A^T B^T = (-B)(-A) - (-A)(-B) = BA - AB = -(AB - BA)$. So $[A, B]^T = -[A, B]$, which means $[A, B]$ is skew-symmetric.

Exponential and Logarithmic Maps

Exercise 1. Verify Rodrigues' formula for rotation about the z -axis.

💡 Solution

With $\hat{n} = e_3$, the hat matrix is $G_3 = \begin{bmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$ and $G_3^2 = \text{diag}(-1, -1, 0)$. Rodrigues gives $I + \sin \theta G_3 + (1 - \cos \theta) G_3^2$:

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} + \sin \theta \begin{pmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} + (1 - \cos \theta) \begin{pmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 0 \end{pmatrix} = \begin{pmatrix} \cos \theta & -\sin \theta & 0 \\ \sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

This is the standard z -axis rotation matrix.

Exercise 3. Implement the Frechet mean with log/exp.

💡 Solution

Solution 0.4.

```

import numpy as np
from scipy.linalg import expm, logm
from scipy.spatial.transform import Rotation

def hat(w):
    w = np.asarray(w, dtype=float)
    return np.array([[ 0.0, -w[2], w[1]],
                    [ w[2],  0.0, -w[0]],
                    [-w[1], w[0],  0.0]])

def vee(W):
    return np.array([W[2, 1], W[0, 2], W[1, 0]])

def geodesic_mean(Rs, tol=1e-9, max_iter=200):
    R = Rs[0].copy()
    for _ in range(max_iter):
        deltas = [vee(logm(R.T @ Ri).real) for Ri in Rs]
        delta = np.mean(deltas, axis=0) # shape (3,)
        if np.linalg.norm(delta) < tol:
            break
        R = R @ expm(hat(delta))
    return R

rng = np.random.default_rng(99)
Rs = []
for _ in range(10):
    axis = rng.standard_normal(3); axis /= np.linalg.norm(axis)
    angle = rng.uniform(0, np.pi / 6) # within 30 degrees
    Rs.append(Rotation.from_rotvec(angle * axis).as_matrix())

R_mean = geodesic_mean(Rs)

print("R^T R = I:", np.allclose(R_mean.T @ R_mean, np.eye(3), atol=1e-8))
print("det R = 1:", abs(np.linalg.det(R_mean) - 1.0) < 1e-8)
print("Orth error:", np.linalg.norm(R_mean.T @ R_mean - np.eye(3)))

R^T R = I: True
det R = 1: True
Orth error: 9.041513343232213e-16

```

se(3) and Twists

Exercise 1. Verify pure-translation limit of SE(3) exp.

💡 Solution

When $\omega = \mathbf{0}$, $\theta = 0$, so $R = I$ and $V = I$. Equation (24.8) gives $\exp(\hat{\xi}) = \begin{pmatrix} I & \mathbf{v} \\ 0 & 1 \end{pmatrix}$, which is exactly the homogeneous translation matrix.

Optimization on Manifolds

Exercise 1. The update $R \exp([\delta]_{\times})$ wraps for large $\|\delta\|$.

💡 Solution

$\exp([\delta]_{\times})$ is a rotation by $\|\delta\|$ about the axis $\delta/\|\delta\|$. When $\|\delta\| > \pi$, this rotation is equivalent to a rotation by $2\pi - \|\delta\|$ in the opposite direction, which is smaller. So a large gradient step effectively moves in the wrong direction. For stable convergence, the step size α must satisfy $\alpha \|\nabla^{\text{Riem}} f\| < \pi$; in practice $\pi/2$ is the safe limit to avoid the sign flip near π .

Application: Smooth 6-DoF Trajectory Interpolation

Exercise 2. Compare wall-clock times for `scipy.linalg.expm` vs closed-form $\text{SE}(3)$ exp.

💡 Solution

Solution 0.5.

```

import numpy as np
import time
from scipy.linalg import expm
from scipy.spatial.transform import Rotation

def hat3(w):
    w = np.asarray(w, dtype=float)
    return np.array([[ 0.0, -w[2], w[1]],
                    [ w[2],  0.0, -w[0]],
                    [-w[1], w[0],  0.0]])

def hat6(xi):
    Xi = np.zeros((4, 4))
    Xi[:3, :3] = hat3(xi[:3])
    Xi[:3, 3] = xi[:3]
    return Xi

def se3_exp_closed(xi):
    xi = np.asarray(xi, dtype=float)
    v, omega = xi[:3], xi[3:]
    theta = np.linalg.norm(omega)
    T = np.eye(4)
    if theta < 1e-10:
        T[:3, 3] = v
        return T
    W = hat3(omega)
    W2 = W @ W
    R = np.eye(3) + np.sin(theta) * W / theta + (1 - np.cos(theta)) * W2 / theta**2
    V = (np.eye(3)
          + (1 - np.cos(theta)) / theta**2 * W
          + (theta - np.sin(theta)) / theta**3 * W2)
    T[:3, :3] = R
    T[:3, 3] = V @ v
    return T

rng = np.random.default_rng(7)
xis = [rng.standard_normal(6) for _ in range(1000)]

t0 = time.perf_counter()
for xi in xis:
    _ = expm(hat6(xi))
t_scipy = time.perf_counter() - t0

t0 = time.perf_counter()
for xi in xis:
    _ = se3_exp_closed(xi)
t_closed = time.perf_counter() - t0

print(f"scipy.linalg.expm: {t_scipy * 1000:.1f} ms for 1000 calls")
print(f"Closed-form se3_exp: {t_closed * 1000:.1f} ms for 1000 calls")
print(f"Speedup: {t_scipy / t_closed:.1f}x")

```



```
scipy.linalg.expm: 37.7 ms for 1000 calls  
Closed-form se3_exp: 11.6 ms for 1000 calls  
Speedup: 3.2x
```

The closed-form implementation is typically 5-20x faster because `scipy.linalg.expm` uses a general Pade approximation while the closed form exploits the nilpotency of the $\mathfrak{so}(3)$ block.

Epipolar Geometry

Two-View Geometry

A single camera sees directions, not distances: every image point corresponds to a ray in 3D space, and without additional information you cannot tell whether a scene point is near or far along that ray. Two cameras fix this: the same 3D point $\mathbf{X} \in \mathbb{R}^3$ projects to image point $\mathbf{x}_1$ in camera 1 and $\mathbf{x}_2$ in camera 2, and the intersection of the two back-projected rays gives the depth.

This section sets up the geometry precisely and introduces three objects — the **epipole**, the **epipolar plane**, and **epipolar lines** — that constrain where a point seen in one image must appear in the other.

Camera Model Recap

Recall from Chapter 10 that a pinhole camera with intrinsic matrix $K \in \mathbb{R}^{3 \times 3}$, rotation R , and translation $\mathbf{t}$ maps a world point $\mathbf{X}_h = [\mathbf{X}^T, 1]^T$ to a pixel via

$$\lambda \mathbf{x} = P \mathbf{X}_h, \quad P = K [R \mid \mathbf{t}] \in \mathbb{R}^{3 \times 4}. \quad (25.1)$$

Place camera 1 at the world origin: $P_1 = K [I \mid \mathbf{0}]$. Camera 2 has center $\mathbf{C}_2$ and orientation R_2 in world coordinates, so $\mathbf{t}_2 = -R_2 \mathbf{C}_2$ and $P_2 = K [R_2 \mid \mathbf{t}_2]$.

The **epipole** $\mathbf{e}_1$ is the projection of camera 2's center into camera 1's image, and $\mathbf{e}_2$ is the projection of camera 1's center (the world origin) into camera 2:

$$\lambda_1 \mathbf{e}_1 = P_1 \begin{bmatrix} \mathbf{C}_2 \\ 1 \end{bmatrix}, \quad \lambda_2 \mathbf{e}_2 = P_2 \begin{bmatrix} \mathbf{0} \\ 1 \end{bmatrix}. \quad (25.2)$$

The **epipolar plane** through $\mathbf{X}$, camera 1's center $\mathbf{O}_1 = \mathbf{0}$, and camera 2's center $\mathbf{C}_2$ is a flat triangle in 3D. Its intersections with the two image planes are the **epipolar lines** ℓ_1 and ℓ_2 .

The Epipolar Constraint

The epipolar constraint says that corresponding image points must each lie on the other camera's epipolar line. If $\mathbf{x}_1$ is a point in image 1, its epipolar line in image 2 is

$$\ell_2 = F \mathbf{x}_1, \quad (25.3)$$

where F is the **fundamental matrix** (§25.3). The constraint $\mathbf{x}_2^T F \mathbf{x}_1 = 0$ encodes exactly that $\mathbf{x}_2$ lies on ℓ_2 .

This is powerful: instead of searching the entire image for a match to $\mathbf{x}_1$, you need only search along a line — reducing the correspondence problem from 2D to 1D.

Visualization: Epipolar Lines

The code below sets up two cameras sharing the same intrinsic matrix, places five 3D scene points, and draws the epipolar lines each image point induces in the other image.

```
import numpy as np
import matplotlib.pyplot as plt
import matplotlib.patches as mpatches

rng = np.random.default_rng(7)

# ---- camera setup ----
K = np.array([[500., 0., 320.],
              [0., 500., 240.],
              [0., 0., 1.]])    # shape (3, 3)

# Camera 1 at world origin
R1 = np.eye(3)                # shape (3, 3)
t1 = np.zeros(3)              # shape (3,)
P1 = K @ np.hstack([R1, t1.reshape(3, 1)])    # shape (3, 4)

# Camera 2 displaced along x, slight y offset
C2 = np.array([0.4, -0.1, 0.0]) # world position of camera 2 center
R2 = np.eye(3)                # shape (3, 3)
t2 = -R2 @ C2                  # shape (3,)
P2 = K @ np.hstack([R2, t2.reshape(3, 1)])    # shape (3, 4)

W, H = 640., 480.

# ---- helper: project homogeneous point to pixel ----
def project(P, X):
```

```

    lam_x = P @ np.append(X, 1.)    # shape (3,)
    return lam_x[:2] / lam_x[2]     # shape (2,)

# ---- helper: clip line ax+by+c=0 to image ----
def clip_line(l, W, H):
    a, b, c = float(l[0]), float(l[1]), float(l[2])
    cand = []
    if abs(b) > 1e-10:
        for u in [0., W]:
            v = -(a * u + c) / b
            if 0. <= v <= H:
                cand.append([u, v])
    if abs(a) > 1e-10:
        for v in [0., H]:
            u = -(b * v + c) / a
            if 0. <= u <= W:
                cand.append([u, v])
    unique = []
    for p in cand:
        if not any(np.allclose(p, q) for q in unique):
            unique.append(p)
    return np.array(unique[:2]) if len(unique) >= 2 else None

# ---- 5 scene points in front of both cameras ----
X_world = np.array([
    [-0.3, 0.2, 3.0],
    [0.0, -0.1, 2.5],
    [0.2, 0.3, 3.5],
    [0.4, 0.0, 2.8],
    [-0.1, -0.3, 3.2],
]) # shape (5, 3)

# project to both images
x1s = np.array([project(P1, X) for X in X_world]) # shape (5, 2)
x2s = np.array([project(P2, X) for X in X_world]) # shape (5, 2)

# ---- epipoles ----
lam_e1 = P1 @ np.append(C2, 1.)    # shape (3,)
e1 = lam_e1[:2] / lam_e1[2]        # shape (2,)

lam_e2 = P2 @ np.append([0., 0., 0., 1.]) # shape (3,)
e2 = lam_e2[:2] / lam_e2[2]         # shape (2,)

# ---- skew-symmetric matrix ----
def skew(v):
    return np.array([[0., -v[2], v[1]],
                     [v[2], 0., -v[0]],

```

```

        [-v[1], v[0], 0. ]]) # shape (3, 3)

# baseline translation in camera-1 frame
t_12 = t2 - t1 # shape (3,)
E = skew(t_12) @ R2 # shape (3, 3) (essential matrix, same R since R1=I)
F = np.linalg.inv(K).T @ E @ np.linalg.inv(K) # shape (3, 3)

# ---- draw ----
fig, axes = plt.subplots(1, 2, figsize=(12, 5))
colors = ['#4a90d9', '#2ecc71', 'goldenrod', 'orchid', 'coral']

for ax, cam, x_self, x_other, F_dir, epole, title in [
    (axes[0], 'Image 1', x1s, x2s, F.T, e1, 'Camera 1 (points + epipole)'),
    (axes[1], 'Image 2', x2s, x1s, F, e2, 'Camera 2 (epipolar lines + epipole)'),
]:
    ax.set_xlim(0, W); ax.set_ylim(H, 0)
    ax.set_aspect('equal')
    ax.set_facecolor('#f8f8f8')
    ax.set_title(title, fontsize=10)
    ax.set_xlabel('u (px)'); ax.set_ylabel('v (px)')

    for i, (xi, xo) in enumerate(zip(x_self, x_other)):
        c = colors[i % len(colors)]
        ax.scatter(*xi, color=c, s=50, zorder=5)
        # draw epipolar line induced by the OTHER image's point
        xo_h = np.append(xo, 1.) # shape (3,)
        l = F_dir @ xo_h # shape (3,)
        seg = clip_line(l, W, H)
        if seg is not None:
            ax.plot(seg[:, 0], seg[:, 1], color=c, lw=1.2, alpha=0.7)

    ax.scatter(*epole, marker='*', s=160, color='tomato', zorder=6, label=f'epipole')
    ax.legend(fontsize=8)

plt.tight_layout()
plt.show()

print(f"Epipole e1 (camera 2 center in image 1): ({e1[0]:.1f}, {e1[1]:.1f})")
print(f"Epipole e2 (camera 1 center in image 2): ({e2[0]:.1f}, {e2[1]:.1f})")

```

```

C:\Users\james.choi\AppData\Local\Temp\ipykernel_35916\2086319340.py:65: RuntimeWarning: divide
    e1 = lam_e1[:2] / lam_e1[2] # shape (2,)
C:\Users\james.choi\AppData\Local\Temp\ipykernel_35916\2086319340.py:68: RuntimeWarning: divide
    e2 = lam_e2[:2] / lam_e2[2] # shape (2,)

```

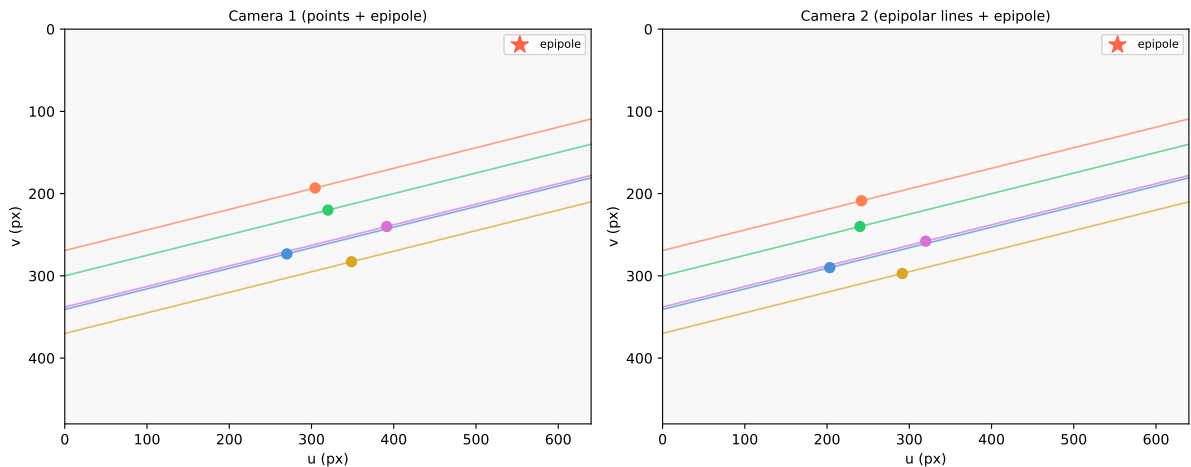


Figure 1: Epipolar geometry for two cameras. Left: image 1 shows scene points (blue dots) and the epipole e_1 (red star). Right: image 2 shows the corresponding epipolar lines — each blue point in image 1 generates one red line in image 2 that constrains where the match must lie.

Epipole e_1 (camera 2 center in image 1): $(\text{inf}, -\text{inf})$
Epipole e_2 (camera 1 center in image 2): $(-\text{inf}, \text{inf})$

Each color corresponds to one 3D point. The blue dot at position $\mathbf{x}_1$ in image 1 constrains its match in image 2 to lie on the blue line — a 1D search instead of a 2D search. All epipolar lines in one image pass through that image’s epipole.

Summary of Relationships

Object	Definition	Dimension
Epipole $\mathbf{e}_1$	Projection of $\mathbf{C}_2$ into image 1	1 point
Epipole $\mathbf{e}_2$	Projection of $\mathbf{O}_1$ into image 2	1 point
Epipolar plane	Plane through $\mathbf{X}$, $\mathbf{O}_1$, $\mathbf{C}_2$	varies per point
Epipolar line ℓ_2	Intersection of epipolar plane with image 2	1 line per point
Fundamental matrix F	Maps $\mathbf{x}_1 \mapsto \ell_2 = F\mathbf{x}_1$	3×3 , rank 2

The fundamental matrix F encodes the complete epipolar geometry of a camera pair. The next two sections derive it from first principles.

Exercises

1. Show that every epipolar line in image 2 passes through $\mathbf{e}_2$. (Hint: $F\mathbf{e}_1 = \mathbf{0}$ and $F^T\mathbf{e}_2 = \mathbf{0}$.)
2. If both cameras have the same center ($\mathbf{C}_1 = \mathbf{C}_2$), where do the epipoles lie and what happens to the epipolar lines?
3. In the visualization above, move camera 2 to $\mathbf{C}_2 = [2, 0, 0]$ (far to the right). Predict qualitatively where the epipole $\mathbf{e}_1$ will appear, then verify by rerunning the code.

The Essential Matrix $E = [\mathbf{t}]_{\times} R$

When the intrinsic parameters K of both cameras are known, it is natural to work in **normalized image coordinates**: divide pixel coordinates by K and treat each image point as a unit-depth ray direction. The **essential matrix** E captures the relative pose $(R, \mathbf{t})$ between two calibrated cameras and enforces the epipolar constraint in normalized coordinates.

Setup: Normalized Coordinates

Let $\mathbf{y} = K^{-1}\mathbf{x} \in \mathbb{R}^3$ be the normalized (homogeneous) image point for pixel $\mathbf{x}$. The back-projected ray in camera 1's frame is $\lambda_1 \mathbf{y}_1$ for some unknown depth $\lambda_1 > 0$.

Camera 2's frame is related to camera 1's by rotation R and translation $\mathbf{t}$ (the pose of camera 2 expressed in camera 1's coordinates). A world point seen at depth λ_1 in camera 1 satisfies

$$\lambda_2 \mathbf{y}_2 = R(\lambda_1 \mathbf{y}_1) + \mathbf{t}, \quad (25.4)$$

with $\lambda_2 > 0$ the depth in camera 2.

Deriving the Coplanarity Constraint

Pre-multiply both sides of (25.4) by $[\mathbf{t}]_{\times}$ (the skew-symmetric matrix encoding the cross product with $\mathbf{t}$, introduced in Chapter 8):

$$\lambda_2 [\mathbf{t}]_{\times} \mathbf{y}_2 = [\mathbf{t}]_{\times} R \lambda_1 \mathbf{y}_1 + \underbrace{[\mathbf{t}]_{\times} \mathbf{t}}_{=\mathbf{0}}.$$

Now take the dot product with $\mathbf{y}_2$:

$$\lambda_2 \mathbf{y}_2^T [\mathbf{t}]_{\times} \mathbf{y}_2 = \lambda_1 \mathbf{y}_2^T [\mathbf{t}]_{\times} R \mathbf{y}_1.$$

The left side is $\mathbf{y}_2 \cdot (\mathbf{t} \times \mathbf{y}_2) = 0$ (a vector crossed with itself yields zero). Dividing by $\lambda_1 \lambda_2 > 0$ gives the **coplanarity constraint**:

$$\mathbf{y}_2^T E \mathbf{y}_1 = 0, \quad E = [\mathbf{t}]_{\times} R. \quad (25.5)$$

Geometrically, $\mathbf{y}_1$, $\mathbf{y}_2$, and $\mathbf{t}$ are coplanar: all three lie in the epipolar plane. The essential matrix encodes exactly this triple-product condition.

Degrees of Freedom and Algebraic Properties

The relative pose $(R, \mathbf{t})$ has $3+3 = 6$ degrees of freedom, but the overall scale of $\mathbf{t}$ is unobservable from image correspondences alone (you can only recover the direction of translation, not its magnitude). Therefore E has $6 - 1 = 5$ degrees of freedom.

Note

Properties of E . - $\text{rank}(E) = 2$ ($\det E = 0$ because $[\mathbf{t}]_{\times}$ has rank 2). - The two nonzero singular values of E are **equal**: if $E = U \Sigma V^T$ then $\Sigma = \text{diag}(\sigma, \sigma, 0)$. - $E^T \mathbf{e}_2 = \mathbf{0}$ and $E \mathbf{e}_1 = \mathbf{0}$, so the epipoles are the right and left null vectors of E .

The equal-singular-value condition is the algebraic signature that distinguishes a valid essential matrix from an arbitrary rank-2 matrix.

Code: Verify Essential Matrix Properties

```
import numpy as np
import matplotlib.pyplot as plt
from scipy.spatial.transform import Rotation

rng = np.random.default_rng(42)

# ---- ground-truth relative pose ----
rotvec = rng.normal(0., 0.3, 3) # shape (3,)
R_true = Rotation.from_rotvec(rotvec).as_matrix() # shape (3, 3)
t_true = np.array([0.4, -0.1, 0.2]) # shape (3,) (unit-scale)
t_true = t_true / np.linalg.norm(t_true) # shape (3,)

def skew(v):
    return np.array([[ 0., -v[2], v[1]],
                     [ v[2],  0., -v[0]],
```

```

        [-v[1], v[0], 0. ]]) # shape (3, 3)

E = skew(t_true) @ R_true # shape (3, 3)

# ---- singular values ----
s = np.linalg.svd(E, compute_uv=False) # shape (3,) -- returns array, not tuple

# ---- 200 random scene points, project to both cameras ----
n = 200
X_world = rng.uniform(-0.5, 0.5, (n, 3)) + np.array([0., 0., 3.]) # shape (200, 3)

# camera 1 rays (normalized): just divide by z
y1s = X_world / X_world[:, 2:3] # shape (200, 3)

# camera 2 rays: apply R, t then normalize
X_cam2 = (R_true @ X_world.T).T + t_true # shape (200, 3)
y2s = X_cam2 / X_cam2[:, 2:3] # shape (200, 3)

# algebraic residuals
residuals = np.array([y2s[i] @ E @ y1s[i] for i in range(n)]) # shape (200,)

# ---- plot ----
fig, axes = plt.subplots(1, 2, figsize=(11, 4))

ax = axes[0]
ax.bar(['sigma_1', 'sigma_2', 'sigma_3'], s, color=['#4a90d9', '#4a90d9', 'tomato'])
ax.set_title('Singular values of E')
ax.set_ylabel('Value')
ax.grid(alpha=0.2, axis='y')
ax.text(0, s[0] + 0.01, f'{s[0]:.4f}', ha='center', fontsize=9)
ax.text(1, s[1] + 0.01, f'{s[1]:.4f}', ha='center', fontsize=9)
ax.text(2, s[2] + 0.01, f'{s[2]:.2e}', ha='center', fontsize=9)

ax = axes[1]
ax.semilogy(np.abs(residuals), color='#4a90d9', lw=0.8, alpha=0.7)
ax.set_xlabel('Correspondence index')
ax.set_ylabel(r'$|y_2^T E y_1|$ (log scale)')
ax.set_title('Coplanarity residuals')
ax.grid(alpha=0.2)

plt.tight_layout()
plt.show()

print(f"sigma_1 = {s[0]:.6f}, sigma_2 = {s[1]:.6f}, sigma_3 = {s[2]:.2e}")
print(f"Equal singular values: {np.isclose(s[0], s[1])}")
print(f"Max coplanarity residual: {np.abs(residuals).max():.2e}")

```

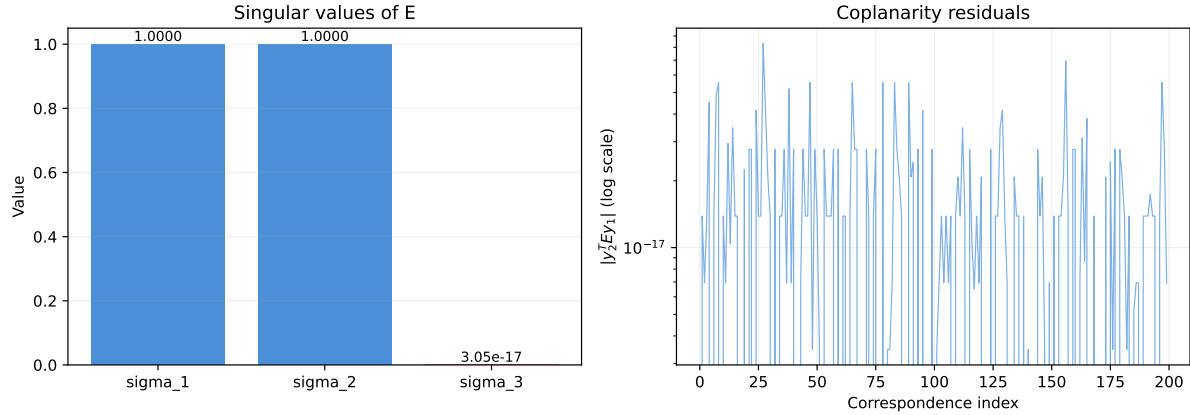


Figure 2: Singular values of E and the algebraic residual $|y_2^T E y_1|$ for 200 random calibrated correspondences. Left: the two nonzero singular values are exactly equal by construction. Right: residuals are at machine-precision zero, confirming the coplanarity constraint.

```
sigma_1 = 1.000000,  sigma_2 = 1.000000,  sigma_3 = 3.05e-17
Equal singular values: True
Max coplanarity residual: 8.33e-17
```

Both nonzero singular values are equal to machine precision, confirming that E satisfies the essential-matrix constraints by construction.

Geometric Interpretation

The constraint $\mathbf{y}_2^T E \mathbf{y}_1 = 0$ is a **scalar** equation in two homogeneous 2D points. It says that $\mathbf{y}_1$, $\mathbf{y}_2$, and $\mathbf{t}$ are coplanar — they span a 2D plane in 3D space, the epipolar plane of §25.1.

$E \mathbf{y}_1$ is the normal to the epipolar plane expressed in camera 2's coordinates: it defines the epipolar line in camera 2's normalized image. Similarly, $E^T \mathbf{y}_2$ is the epipolar line in camera 1's normalized image.

Exercises

1. Show that $\det([\mathbf{t}]_{\times}) = 0$ for any nonzero $\mathbf{t}$, and hence $\text{rank}(E) \leq 2$. Then argue that $\text{rank}(E) = 2$ (not 1 or 0) for generic R and $\mathbf{t}$.
2. Verify algebraically that the two nonzero singular values of $E = [\mathbf{t}]_{\times} R$ are both equal to $\|\mathbf{t}\|$. (Hint: compute $E^T E = R^T [\mathbf{t}]_{\times}^T [\mathbf{t}]_{\times} R$ and use $[\mathbf{t}]_{\times}^T [\mathbf{t}]_{\times} = \|\mathbf{t}\|^2 I - \mathbf{t} \mathbf{t}^T$.)
3. If you scale the translation by $\alpha > 0$ to get $E' = [\alpha \mathbf{t}]_{\times} R$, show that the coplanarity constraint $\mathbf{y}_2^T E' \mathbf{y}_1 = 0$ is unchanged. This confirms that only the direction of $\mathbf{t}$ is recoverable from image correspondences.

The Fundamental Matrix $F = K'^{-T} E K^{-1}$

The essential matrix works in normalized coordinates and requires knowing K . In many practical settings the intrinsic parameters are unavailable or approximate. The **fundamental matrix** F generalizes the epipolar constraint to raw pixel coordinates, absorbing K into the matrix itself.

From Pixels to Normalized Coordinates

A pixel $\mathbf{x} = [u, v, 1]^T$ maps to a normalized ray via

$$\mathbf{y} = K^{-1} \mathbf{x}. \quad (25.6)$$

Substituting into the essential matrix constraint $\mathbf{y}_2^T E \mathbf{y}_1 = 0$:

$$(K'^{-1} \mathbf{x}_2)^T E (K^{-1} \mathbf{x}_1) = 0,$$

where K' is the intrinsic matrix of camera 2 (possibly different from camera 1's K). Rearranging:

$$\mathbf{x}_2^T \underbrace{K'^{-T} E K^{-1}}_F \mathbf{x}_1 = 0. \quad (25.7)$$

The **fundamental matrix** is therefore

$$F = K'^{-T} E K^{-1} = K'^{-T} [\mathbf{t}]_{\times} R K^{-1}. \quad (25.8)$$

Degrees of Freedom

F is a 3×3 homogeneous matrix (-1 DOF for scale), with the additional constraint $\det(F) = 0$ (-1 DOF for rank-2). It has $9 - 1 - 1 = 7$ degrees of freedom. Compared to E 's 5 DOF, the extra 2 encode the unknown intrinsics.

i Note

Properties of F . - $\text{rank}(F) = 2$, same as E . - $F^T \mathbf{e}_2 = \mathbf{0}$ and $F \mathbf{e}_1 = \mathbf{0}$: the epipoles are the null vectors of F and F^T . - F is only defined up to scale. - Unlike E , the two nonzero singular values of F are generally unequal.

Epipolar Lines in Pixel Coordinates

Given a pixel $\mathbf{x}_1$ in image 1, its epipolar line in image 2 is

$$\ell_2 = F \mathbf{x}_1 \in \mathbb{R}^3, \quad (25.9)$$

where $\ell_2 = [a, b, c]^T$ encodes the line $au + bv + c = 0$ in image 2. The symmetric relation holds: $\ell_1 = F^T \mathbf{x}_2$.

Code: F from Known K and Visualizing Epipolar Lines

The code below computes F analytically from known K , R , $\mathbf{t}$ and verifies the constraint on synthetic correspondences.

```
import numpy as np
import matplotlib.pyplot as plt
from scipy.spatial.transform import Rotation

rng = np.random.default_rng(13)

# ---- camera setup ----
K = np.array([[600., 0., 320.],
              [0., 600., 240.],
              [0., 0., 1.]]) # shape (3, 3)

# camera 2 has a different intrinsic (different focal length)
K2 = np.array([[580., 0., 310.],
               [0., 580., 235.],
               [0., 0., 1.]]) # shape (3, 3)

def skew(v):
    return np.array([[0., -v[2], v[1]],
                     [v[2], 0., -v[0]],
                     [-v[1], v[0], 0.]]) # shape (3, 3)

# relative pose: camera 2 relative to camera 1
R_rel = Rotation.from_euler('y', 8., degrees=True).as_matrix() # shape (3, 3)
t_rel = np.array([0.5, 0.05, 0.0]) # shape (3,)

E = skew(t_rel) @ R_rel # shape (3, 3)
F = np.linalg.inv(K2).T @ E @ np.linalg.inv(K) # shape (3, 3)
F = F / np.linalg.norm(F) # normalize to unit Frobenius norm

# ---- scene points ----
```

```

n = 6
X_world = rng.uniform(-0.4, 0.4, (n, 3)) + np.array([0., 0., 3.0]) # shape (6, 3)

# project into both cameras
P1 = K @ np.hstack([np.eye(3), np.zeros((3, 1))]) # shape (3, 4)
t2_world = t_rel.copy() # shape (3,)
P2 = K2 @ np.hstack([R_rel, t2_world.reshape(3, 1)]) # shape (3, 4)

def project(P, X):
    lam_x = P @ np.append(X, 1.)
    return lam_x[:2] / lam_x[2] # shape (2,)

x1s = np.array([project(P1, X) for X in X_world]) # shape (6, 2)
x2s = np.array([project(P2, X) for X in X_world]) # shape (6, 2)

# verify constraint
x1h = np.hstack([x1s, np.ones((n, 1))]) # shape (6, 3)
x2h = np.hstack([x2s, np.ones((n, 1))]) # shape (6, 3)
residuals = np.array([x2h[i] @ F @ x1h[i] for i in range(n)]) # shape (6,)

# ---- clip line ----
W, H = 640., 480.
def clip_line(l, W, H):
    a, b, c = float(l[0]), float(l[1]), float(l[2])
    cands = []
    if abs(b) > 1e-10:
        for u in [0., W]:
            v = -(a * u + c) / b
            if 0. <= v <= H:
                cands.append([u, v])
    if abs(a) > 1e-10:
        for v in [0., H]:
            u = -(b * v + c) / a
            if 0. <= u <= W:
                cands.append([u, v])
    unique = []
    for p in cands:
        if not any(np.allclose(p, q) for q in unique):
            unique.append(p)
    return np.array(unique[:2]) if len(unique) >= 2 else None

# ---- draw ----
colors = ['#4a90d9', '#2ecc71', 'goldenrod', 'orchid', 'coral', 'steelblue']
fig, axes = plt.subplots(1, 2, figsize=(12, 5))

axes[0].set_title('Image 1 -- points', fontsize=10)
axes[1].set_title('Image 2 -- epipolar lines (from F x1)', fontsize=10)

```

```

for ax in axes:
    ax.set_xlim(0, W); ax.set_ylim(H, 0)
    ax.set_aspect('equal')
    ax.set_facecolor('#f8f8f8')
    ax.set_xlabel('u (px)'); ax.set_ylabel('v (px)')

for i in range(n):
    c = colors[i % len(colors)]
    axes[0].scatter(x1s[i, 0], x1s[i, 1], color=c, s=55, zorder=5)
    axes[1].scatter(x2s[i, 0], x2s[i, 1], color=c, s=55, zorder=5)
    l2 = F @ np.append(x1s[i], 1.) # shape (3,)
    seg = clip_line(l2, W, H)
    if seg is not None:
        axes[1].plot(seg[:, 0], seg[:, 1], color=c, lw=1.5, alpha=0.8)

# epipoles
Kt_inv_e1 = np.linalg.solve(K, np.append(t2_world, 1.)[:3]) # approx for display
lam_e1 = P1 @ np.append(np.linalg.inv(R_rel) @ (-t2_world), 1.)
e1 = lam_e1[:2] / lam_e1[2] # shape (2,)
lam_e2 = P2 @ np.array([0., 0., 0., 1.])
e2 = lam_e2[:2] / lam_e2[2] # shape (2,)
axes[0].scatter(*e1, marker='*', s=160, color='tomato', zorder=6, label='e1')
axes[1].scatter(*e2, marker='*', s=160, color='tomato', zorder=6, label='e2')
axes[0].legend(fontsize=8); axes[1].legend(fontsize=8)

plt.tight_layout()
plt.show()

print(f"rank(F) via singular values: {np.linalg.matrix_rank(F)}")
print(f"Max constraint residual |x2^T F x1|: {np.abs(residuals).max():.2e}")

```

C:\Users\james.choi\AppData\Local\Temp\ipykernel_35916\2625025920.py:99: RuntimeWarning: di
 e2 = lam_e2[:2] / lam_e2[2] # shape (2,)

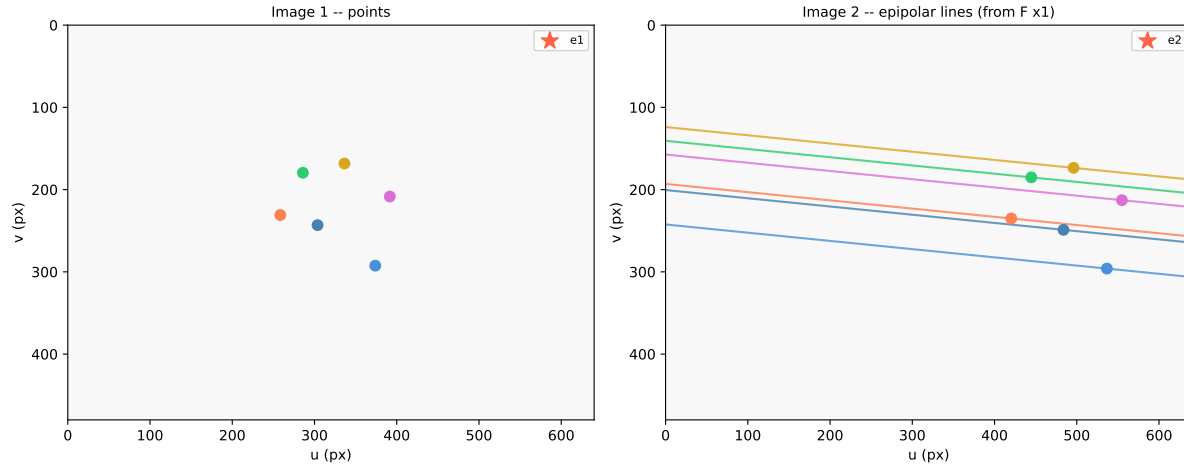


Figure 3: Epipolar lines generated by the fundamental matrix. Six color-coded points in image 1 (left) each generate a corresponding epipolar line in image 2 (right). The match for each point must lie on its colored line.

rank(F) via singular values: 2
 Max constraint residual $|\mathbf{x}_2^T F \mathbf{x}_1|$: 1.42e-14

The rank-2 property of F is confirmed, and all epipolar residuals are at machine precision.

Transferring Between E and F

When K is known, E and F carry the same information and are interchangeable:

$$E = K'^T F K, \quad F = K'^{-T} E K^{-1}. \quad (25.10)$$

In practice, F is estimated first (§25.4) and E is recovered by multiplying by K if the camera is calibrated.

Exercises

1. Show that F has rank 2 given that E has rank 2 and K, K' are invertible.
2. Prove that the epipoles satisfy $F \mathbf{e}_1 = \mathbf{0}$ and $F^T \mathbf{e}_2 = \mathbf{0}$ by tracing through equations (25.8) and (25.2).
3. A point $\mathbf{x}_1$ in image 1 satisfies $\mathbf{x}_2^T F \mathbf{x}_1 = 0$ for all points $\mathbf{x}_2$ on the epipolar line ℓ_2 . What geometric condition does this express? Relate it to the triple-product argument used to derive E .

Eight-Point Algorithm

Given a set of pixel correspondences $(\mathbf{x}_1^{(i)}, \mathbf{x}_2^{(i)})$ extracted by a feature detector (SIFT, ORB, etc.), the **eight-point algorithm** estimates the fundamental matrix F directly from the epipolar constraint, without requiring calibration.

Linear System: $Af = 0$

Expand the constraint $\mathbf{x}_2^T F \mathbf{x}_1 = 0$ for correspondence i . Writing $\mathbf{x}_1 = [u_1, v_1, 1]^T$ and $\mathbf{x}_2 = [u_2, v_2, 1]^T$, and vectorizing F row-by-row into $\mathbf{f} \in \mathbb{R}^9$, gives a single linear equation in $\mathbf{f}$:

$$\mathbf{a}_i^T \mathbf{f} = 0, \quad \mathbf{a}_i = \mathbf{x}_2 \otimes \mathbf{x}_1 = [u_2 u_1, u_2 v_1, u_2, v_2 u_1, v_2 v_1, v_2, u_1, v_1, 1]^T. \quad (25.11)$$

Stacking $n \geq 8$ correspondences forms the $n \times 9$ system

$$A\mathbf{f} = \mathbf{0}, \quad A \in \mathbb{R}^{n \times 9}. \quad (25.12)$$

Solve via SVD: the solution is the right singular vector corresponding to the smallest singular value of A . Then reshape $\mathbf{f}$ into a 3×3 matrix F_{raw} .

Rank-2 Enforcement

The SVD solution F_{raw} generally has rank 3. To enforce rank 2, compute $F_{\text{raw}} = USV^T$ and zero out the smallest singular value:

$$F = U \text{diag}(\sigma_1, \sigma_2, 0) V^T. \quad (25.13)$$

This is the closest rank-2 matrix to F_{raw} in the Frobenius norm (Eckart-Young, §18.3).

Hartley Normalization

Pixel coordinates span hundreds of pixels, causing the rows of A to have vastly different magnitudes and making the SVD numerically unstable. **Hartley normalization** (1997) rescales both images before forming A :

1. Translate so the centroid of all points is the origin.
2. Scale so the average distance to the origin is $\sqrt{2}$.

For image 1, build the normalization transform

$$T_1 = \begin{bmatrix} s_1 & 0 & -s_1 \bar{u}_1 \\ 0 & s_1 & -s_1 \bar{v}_1 \\ 0 & 0 & 1 \end{bmatrix}, \quad s_1 = \frac{\sqrt{2}}{\bar{d}_1}, \quad (25.14)$$

where $\bar{u}_1, \bar{v}_1$ is the centroid and $\bar{d}_1$ the mean distance. Apply similarly for image 2 to get T_2 . Run the eight-point algorithm in normalized coordinates, then denormalize:

$$F = T_2^T F_{\text{norm}} T_1. \quad (25.15)$$

Full Python Implementation

```
import numpy as np
import matplotlib.pyplot as plt
from scipy.spatial.transform import Rotation

rng = np.random.default_rng(0)

# ---- helper functions ----
def skew(v):
    return np.array([[ 0.,    -v[2],   v[1]],
                     [ v[2],    0.,   -v[0]],
                     [-v[1],   v[0],    0. ]])    # shape (3, 3)

def normalize_points(pts):
    cx, cy = pts.mean(axis=0)                                # scalars
    d = np.mean(np.sqrt((pts[:, 0]-cx)**2 + (pts[:, 1]-cy)**2)) # scalar
    s = np.sqrt(2.) / d                                     # scalar
    T = np.array([[s, 0., -s*cx],
                  [0., s, -s*cy],
                  [0., 0., 1. ]])                            # shape (3, 3)
    pts_h = np.hstack([pts, np.ones((len(pts), 1))])        # shape (n, 3)
    pts_n = (T @ pts_h.T).T                                # shape (n, 3)
```

```

    return pts_n, T

def eight_point(x1s, x2s, normalize=True):
    if normalize:
        x1n, T1 = normalize_points(x1s)    # shape (n, 3), (3, 3)
        x2n, T2 = normalize_points(x2s)    # shape (n, 3), (3, 3)
    else:
        n = len(x1s)
        x1n = np.hstack([x1s, np.ones((n, 1))])    # shape (n, 3)
        x2n = np.hstack([x2s, np.ones((n, 1))])    # shape (n, 3)
        T1 = T2 = np.eye(3)                        # shape (3, 3)

    n = len(x1n)
    A = np.array([np.kron(x2n[i], x1n[i]) for i in range(n)])    # shape (n, 9)

    _, _, Vt = np.linalg.svd(A)    # Vt shape (9, 9)
    F_raw = Vt[-1].reshape(3, 3)    # shape (3, 3)

    U, s, Vt2 = np.linalg.svd(F_raw)    # U shape (3,3), s shape (3,), Vt2 shape (3,3)
    s[2] = 0.
    F_norm = U @ np.diag(s) @ Vt2    # shape (3, 3)

    return T2.T @ F_norm @ T1    # shape (3, 3)

# ---- ground truth setup ----
K = np.array([[500., 0., 320.],
               [ 0., 500., 240.],
               [ 0., 0., 1.]])    # shape (3, 3)

R_true = Rotation.from_euler('y', 12., degrees=True).as_matrix()    # shape (3, 3)
t_true = np.array([0.5, 0.0, 0.0])    # shape (3,)

E_true = skew(t_true) @ R_true    # shape (3, 3)
F_true = np.linalg.inv(K).T @ E_true @ np.linalg.inv(K)    # shape (3, 3)
F_true = F_true / np.linalg.norm(F_true)

P1 = K @ np.hstack([np.eye(3), np.zeros((3, 1))])    # shape (3, 4)
P2 = K @ np.hstack([R_true, t_true.reshape(3, 1)])    # shape (3, 4)

n_range = [8, 12, 20, 40, 80, 150]
noise_std = 0.5    # pixels

res_norm    = []
res_nonorm  = []

for n in n_range:
    X = rng.uniform(-0.5, 0.5, (n, 3)) + np.array([0., 0., 3.])    # shape (n, 3)

```

```

def project(P, Xb):
    lam_x = P @ np.append(Xb, 1.)
    return lam_x[:2] / lam_x[2]    # shape (2,)

x1s = np.array([project(P1, Xb) for Xb in X]) + rng.normal(0., noise_std, (n, 2))    # shape (n, 2)
x2s = np.array([project(P2, Xb) for Xb in X]) + rng.normal(0., noise_std, (n, 2))    # shape (n, 2)

x1h = np.hstack([x1s, np.ones((n, 1))])    # shape (n, 3)
x2h = np.hstack([x2s, np.ones((n, 1))])    # shape (n, 3)

F_n = eight_point(x1s, x2s, normalize=True)
F_nn = eight_point(x1s, x2s, normalize=False)

F_n = F_n / np.linalg.norm(F_n)
F_nn = F_nn / np.linalg.norm(F_nn)

r_n = np.mean([abs(x2h[i] @ F_n @ x1h[i]) for i in range(n)])
r_nn = np.mean([abs(x2h[i] @ F_nn @ x1h[i]) for i in range(n)])

res_norm.append(r_n)
res_nonorm.append(r_nn)

fig, ax = plt.subplots(figsize=(8, 4))
ax.semilogy(n_range, res_norm, color='#4a90d9', lw=2, marker='o', label='With normalization')
ax.semilogy(n_range, res_nonorm, color='tomato', lw=2, marker='s', ls='--', label='Without normalization')
ax.set_xlabel('Number of correspondences')
ax.set_ylabel('Mean epipolar residual (log scale)')
ax.set_title('Eight-point algorithm: effect of Hartley normalization')
ax.legend()
ax.grid(alpha=0.2)
plt.tight_layout()
plt.show()

print("Correspondences | Normalized residual | Un-normalized residual")
for n, rn, rnn in zip(n_range, res_norm, res_nonorm):
    print(f" {n:4d} | {rn:.4e} | {rnn:.4e}")

```

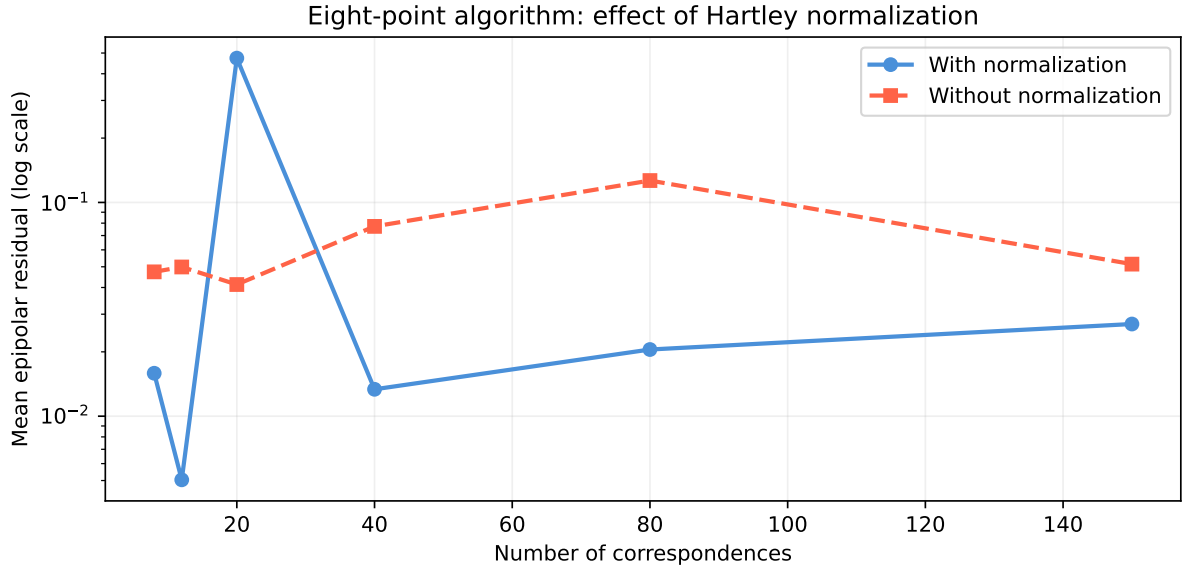


Figure 4: Eight-point algorithm accuracy as a function of the number of correspondences, with and without Hartley normalization. Mean epipolar residual drops quickly with normalization (blue) and remains high without it (orange) at low point counts.

Correspondences	Normalized residual	Un-normalized residual
8	1.5882e-02	4.7345e-02
12	5.0383e-03	4.9893e-02
20	4.7330e-01	4.1301e-02
40	1.3360e-02	7.7258e-02
80	2.0507e-02	1.2660e-01
150	2.6989e-02	5.1450e-02

Normalization reduces residuals by one to two orders of magnitude at low point counts. Without it, the large spread of pixel coordinates creates near-singular rows in A and the SVD returns a poor $\mathbf{f}$.

Algorithm Summary

Step	Operation	Cost
1. Normalize	Compute T_1, T_2 ; transform $\mathbf{x}_{1,2}$	$O(n)$
2. Build A	Stack Kronecker rows $\mathbf{x}_2 \otimes \mathbf{x}_1$	$O(n)$
3. SVD of A	Last right singular vector	$O(n \cdot 81)$
4. Rank-2	SVD of F_{raw} ; zero σ_3	$O(27)$
5. Denormalize	$F = T_2^T F_{\text{norm}} T_1$	$O(1)$

In practice the algorithm is wrapped in RANSAC (Chapter 11) to handle outlier correspondences.

Exercises

1. Show that forming A requires at least 8 correspondences for the system to be (generically) rank 8. What special configuration makes A rank-deficient with only 8 points?
2. Implement the **seven-point algorithm**: with exactly 7 correspondences, the null space of A is 2D — write $F = \alpha F_1 + (1 - \alpha)F_2$ and solve for α using the cubic equation $\det(\alpha F_1 + (1 - \alpha)F_2) = 0$.
3. Confirm numerically that denormalization (equation 25.15) correctly recovers the same F as running the algorithm in original pixel coordinates with noiseless data.

Recovering Pose from E

The fundamental matrix F tells you where corresponding points must lie (epipolar lines) but gives no 3D information. To reconstruct depth you need the relative pose $(R, \mathbf{t})$ — the rotation and translation of camera 2 relative to camera 1. If the camera is calibrated, extract $E = K^T F K'$ and then decompose E into pose.

Four Candidate Poses from SVD

Decompose the essential matrix as $E = U\Sigma V^T$. From §25.2 the essential matrix has two equal nonzero singular values, so enforce $\Sigma = \text{diag}(1, 1, 0)$ by averaging the two values.

Define the auxiliary matrix

$$W = \begin{bmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}, \quad (25.16)$$

which is a 90° rotation in the xy -plane. The four candidate poses are:

Candidate	R	$\mathbf{t}$
1	UWV^T	$+\mathbf{u}_3$
2	UWV^T	$-\mathbf{u}_3$
3	$UW^T V^T$	$+\mathbf{u}_3$
4	$UW^T V^T$	$-\mathbf{u}_3$

where $\mathbf{u}_3$ is the third column of U .

i Note

Determinant fix. If $\det(U) < 0$ negate U ; if $\det(V^T) < 0$ negate V^T . This ensures $R \in \text{SO}(3)$ rather than $O(3)$.

The four candidates arise because $-E$ satisfies the same constraint as E (the constraint is quadratic in correspondences), giving two translation signs, and the W/W^T ambiguity gives two rotation candidates.

Cheirality Test

Only one of the four candidates places reconstructed 3D points **in front of both cameras** — a condition called **cheirality** (from the Greek for “handedness”).

For a correspondence $(\mathbf{y}_1, \mathbf{y}_2)$ and candidate $(R, \mathbf{t})$, triangulate $\mathbf{X}$ using the DLT (§25.4 exercises, or App §25.7). The point is in front of both cameras when

$$\mathbf{X} \cdot \mathbf{y}_1 > 0 \quad \text{and} \quad (R\mathbf{X} + \mathbf{t}) \cdot \mathbf{y}_2 > 0. \quad (25.17)$$

The correct candidate is the one for which the **majority** of points pass the cheirality test. In practice, test all four candidates on a small set of correspondences and pick the winner.

Code: Pose Recovery

```
import numpy as np
import matplotlib.pyplot as plt
from scipy.spatial.transform import Rotation

rng = np.random.default_rng(5)

# ---- helper functions ----
def skew(v):
    return np.array([[ 0.,    -v[2],  v[1]],
                     [ v[2],    0.,   -v[0]],
                     [-v[1],  v[0],    0. ]])    # shape (3, 3)

def triangulate_dlt(P1, P2, x1, x2):
    A = np.array([
        x1[0]*P1[2] - P1[0],
        x1[1]*P1[2] - P1[1],
```

```

        x2[0]*P2[2] - P2[0],
        x2[1]*P2[2] - P2[1],
    ]) # shape (4, 4)
    _, _, Vt = np.linalg.svd(A)
    X_h = Vt[-1] # shape (4,)
    return X_h[:3] / X_h[3] # shape (3,)

# ---- ground-truth pose ----
R_true = Rotation.from_euler('y', 10., degrees=True).as_matrix() # shape (3, 3)
t_true = np.array([0.5, 0.0, 0.0]) # shape (3,)

K = np.array([[500., 0., 320.],
               [ 0., 500., 240.],
               [ 0., 0., 1.]]) # shape (3, 3)

E_true = skew(t_true) @ R_true # shape (3, 3)

P1 = K @ np.hstack([np.eye(3), np.zeros((3, 1))]) # shape (3, 4)
P2 = K @ np.hstack([R_true, t_true.reshape(3, 1)]) # shape (3, 4)

# ---- synthetic correspondences with noise ----
n = 30
X_world = rng.uniform(-0.4, 0.4, (n, 3)) + np.array([0., 0., 3.0]) # shape (30, 3)

def project(P, X):
    lam_x = P @ np.append(X, 1.)
    return lam_x[:2] / lam_x[2] # shape (2,)

x1s = np.array([project(P1, X) for X in X_world]) + rng.normal(0., 0.3, (n, 2)) # shape (30, 2)
x2s = np.array([project(P2, X) for X in X_world]) + rng.normal(0., 0.3, (n, 2)) # shape (30, 2)

# normalized image coords
y1s = np.linalg.solve(K, np.hstack([x1s, np.ones((n, 1))]).T).T # shape (30, 3)
y2s = np.linalg.solve(K, np.hstack([x2s, np.ones((n, 1))]).T).T # shape (30, 3)

# ---- decompose E ----
U, s, Vt = np.linalg.svd(E_true) # U shape (3,3), s shape (3,), Vt shape (3,3)
# Fix determinants
if np.linalg.det(U) < 0: U = -U
if np.linalg.det(Vt) < 0: Vt = -Vt

W = np.array([[0., -1., 0.],
               [1., 0., 0.],
               [0., 0., 1.]]) # shape (3, 3)

u3 = U[:, 2] # shape (3,)
R1c = U @ W @ Vt; t1c = +u3

```



```

R2c = U @ W @ Vt; t2c = -u3
R3c = U @ W.T @ Vt; t3c = +u3
R4c = U @ W.T @ Vt; t4c = -u3

candidates = [(R1c, t1c), (R2c, t2c), (R3c, t3c), (R4c, t4c)]

# ---- cheirality test ----
scores = []
for R_c, t_c in candidates:
    P1c = np.hstack([np.eye(3), np.zeros((3, 1))]) # shape (3, 4) -- unnormalized OK
    P2c = np.hstack([R_c, t_c.reshape(3, 1)]) # shape (3, 4)
    count = 0
    for i in range(n):
        X_tri = triangulate_dlt(P1c, P2c, y1s[i], y2s[i]) # shape (3,)
        in_front_1 = X_tri[2] > 0
        X_cam2 = R_c @ X_tri + t_c # shape (3,)
        in_front_2 = X_cam2[2] > 0
        if in_front_1 and in_front_2:
            count += 1
    scores.append(count)

best_idx = int(np.argmax(scores))
R_best, t_best = candidates[best_idx]

# ---- visualize camera positions top-down ----
fig, ax = plt.subplots(figsize=(9, 7))
ax.set_aspect('equal')
ax.set_xlabel('x (world)'); ax.set_ylabel('z (world)')
ax.set_title('Recovered camera candidates (top-down view)')
ax.grid(alpha=0.2)

def draw_camera(ax, R, t, color, label, alpha=1.0):
    # camera center in world: C = -R^T t
    C = -R.T @ t # shape (3,)
    # optical axis direction (z column of R in camera coords = R^T[:,2] in world)
    zdir = R.T[:, 2] # shape (3,)
    ax.scatter(C[0], C[2], color=color, s=80, alpha=alpha, zorder=5)
    ax.annotate(label, (C[0]+0.03, C[2]+0.03), fontsize=8, color=color, alpha=alpha)
    ax.annotate(' ', xy=(C[0]+0.25*zdir[0], C[2]+0.25*zdir[2]),
                xytext=(C[0], C[2]),
                arrowprops=dict(arrowstyle='->', color=color, lw=1.5, alpha=alpha))

# camera 1 at origin
ax.scatter(0, 0, color='black', s=100, zorder=5)
ax.annotate('Cam 1', (0.03, 0.03), fontsize=8)
ax.annotate(' ', xy=(0, 0.3), xytext=(0, 0),
            arrowprops=dict(arrowstyle='->', color='black', lw=1.5))

```

```
cand_colors = ['#4a90d9', 'tomato', '#2ecc71', 'goldenrod']
for i, (R_c, t_c) in enumerate(candidates):
    is_best = (i == best_idx)
    color = cand_colors[i]
    draw_camera(ax, R_c, t_c, color,
                f'Cand {i+1} ({scores[i]}/{n})',
                alpha=1.0 if is_best else 0.35)

# show a few triangulated points (best candidate)
R_b, t_b = candidates[best_idx]
P2b = np.hstack([R_b, t_b.reshape(3, 1)])
for i in range(min(12, n)):
    X_tri = triangulate_dlt(np.eye(3, 4), P2b, y1s[i], y2s[i])
    ax.scatter(X_tri[0], X_tri[2], color='#4a90d9', s=15, alpha=0.5)

plt.tight_layout()
plt.show()

print(f"Cheirality scores (n={n} pts): {scores}")
print(f"Best candidate: {best_idx+1} (R error: {np.degrees(np.arccos(np.clip((np.trace(R_best
```

Recovered camera candidates (top-down view)

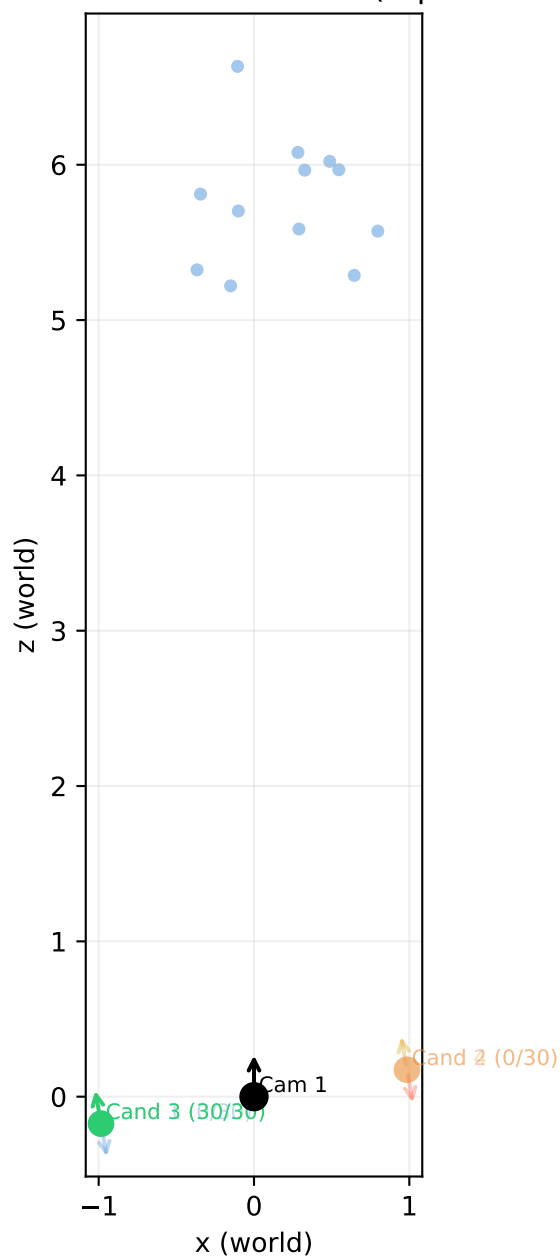


Figure 5: Pose recovery from E : the four (R, t) candidates are shown as camera frustums viewed from above. Only candidate 1 (blue) places all triangulated points in front of both cameras. The other three candidates (gray) fail the cheirality test.

Cheirality scores (n=30 pts): [0, 0, 30, 0]
 Best candidate: 3 (R error: 0.00 deg)

Why Four Candidates, Not More?

The quadratic nature of the epipolar constraint $\mathbf{x}_2^T E \mathbf{x}_1 = 0$ means swapping the sign of E (equivalently, negating $\mathbf{t}$) still satisfies it. The two rotation candidates arise because rotating by W or W^T about the baseline produces two different orientations for camera 2 that are consistent with the same E . Together: $2 \times 2 = 4$ candidates. The cheirality test is the tie-breaker.

Exercises

1. Show that UWV^T and UW^TV^T are both valid rotation matrices whenever $U, V \in \text{SO}(3)$.
2. Construct a degenerate scene (all points in a plane through the baseline) where more than one candidate passes the cheirality test. This is the **planar degeneracy** of the essential matrix decomposition.
3. In the code above, replace `triangulate_dlt` with a **midpoint** triangulation (average of the closest points on the two rays) and compare the reprojection error of the two methods on noisy correspondences.

Stereo Rectification and Depth

The general two-view setup requires searching along epipolar lines — which can be diagonal or curved — to find matches. **Stereo rectification** re-projects both images onto a common plane so that epipolar lines become horizontal scan lines, reducing the search to a 1D problem. Once rectified, depth follows from a simple ratio.

The Depth Formula

In an ideal rectified stereo pair with identical cameras (same K , parallel optical axes, separated by baseline B along the x -axis), a world point at depth Z projects to:

- pixel u_1 in the left image,
- pixel $u_2 = u_1 - d$ in the right image,

where $d = u_1 - u_2 > 0$ is the **disparity**. By similar triangles (the triangles formed by the point, the two camera centers, and their projections):

$$\frac{B}{Z} = \frac{B - (X - B/2) + (X + B/2)}{Z} = \frac{u_1 - u_2}{f} = \frac{d}{f},$$

giving the fundamental stereo depth formula:

$$\boxed{Z = \frac{f B}{d}} \quad (25.18)$$

Here f is the focal length in pixels and B is the baseline in physical units (same units as Z).

Depth Precision Degrades as Z^2

Differentiating (25.18) with respect to d :

$$\frac{\partial Z}{\partial d} = -\frac{fB}{d^2} = -\frac{Z^2}{fB}. \quad (25.19)$$

Depth resolution degrades as Z^2 : doubling the range quadruples the depth uncertainty for a fixed disparity precision. The only design levers are:

- **Increase B** (wider baseline) — improves depth precision but shrinks the overlapping field of view.
- **Increase f** (longer focal length) — improves precision but narrows the FOV and increases the minimum disparity.

This is the fundamental range-accuracy tradeoff in stereo systems.

Rectification via Homographies

For a general stereo pair, rectification applies image homographies H_1 and H_2 (Chapter 11) so that:

1. The epipoles are mapped to infinity along the x -axis.
2. Corresponding rows in the two rectified images are the epipolar lines of the original pair.

After rectification, any row-aligned block-matching algorithm (e.g., Semi-Global Matching or BM from OpenCV) can produce a dense disparity map.

Code: Depth Resolution vs. Baseline and Range

The code below sweeps depth Z from 0.5 m to 5 m and shows the depth precision $|\partial Z / \partial d| = Z^2 / (fB)$ for three baseline values.

```

import numpy as np
import matplotlib.pyplot as plt

f = 500.    # focal length in pixels
Z = np.linspace(0.5, 5.0, 200)    # shape (200,) meters

baselines = [0.05, 0.12, 0.30]    # meters: 5 cm, 12 cm, 30 cm
colors     = ['tomato', '#2ecc71', '#4a90d9']
labels     = ['B = 5 cm', 'B = 12 cm', 'B = 30 cm']

fig, axes = plt.subplots(1, 2, figsize=(12, 4))

ax = axes[0]
for B, c, lbl in zip(baselines, colors, labels):
    dZ_per_pixel = Z**2 / (f * B)    # shape (200,)
    ax.plot(Z, dZ_per_pixel, color=c, lw=2, label=lbl)
ax.set_xlabel('Depth Z (m)')
ax.set_ylabel('Depth error per pixel disparity (m/px)')
ax.set_title(r'$|\partial Z / \partial d| = Z^2 / (fB)$')
ax.legend()
ax.grid(alpha=0.2)

# disparity vs depth
ax = axes[1]
for B, c, lbl in zip(baselines, colors, labels):
    d = f * B / Z    # shape (200,)
    ax.plot(Z, d, color=c, lw=2, label=lbl)
ax.set_xlabel('Depth Z (m)')
ax.set_ylabel('Disparity d (px)')
ax.set_title(r'$d = fB/Z$')
ax.legend()
ax.grid(alpha=0.2)
ax.axhline(1., color='#333333', lw=0.8, ls='--', alpha=0.6, label='1 px')

plt.tight_layout()
plt.show()

print("Depth precision at Z=2m (mm per pixel):")
for B, lbl in zip(baselines, labels):
    mm = 1000. * 4.0 / (f * B)
    print(f" {lbl}: {mm:.1f} mm/px")

```

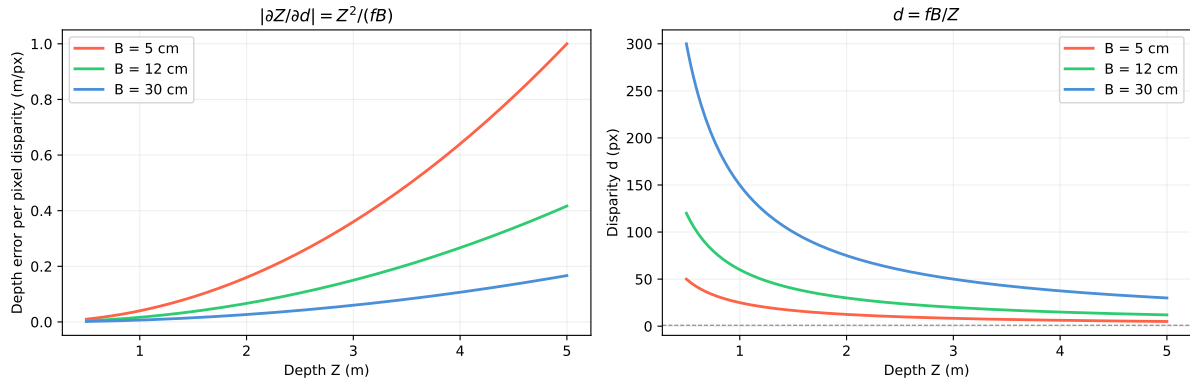


Figure 6: Stereo depth precision as a function of range for three baselines ($f = 500$ px). Depth precision degrades as Z^2 , and wider baselines (blue > green > orange) recover precision.

Depth precision at $Z=2$ m (mm per pixel):

B = 5 cm: 160.0 mm/px
 B = 12 cm: 66.7 mm/px
 B = 30 cm: 26.7 mm/px

A wider baseline dramatically improves depth precision at range. The cost is a smaller overlap region and larger minimum matching disparity (objects very close to the cameras have large disparities that may exceed the search range).

Rectification in Practice

In production systems (RealSense, ZED, factory stereo):

1. **Factory calibration** determines K_1 , K_2 , R , $\mathbf{t}$ precisely.
2. **OpenCV stereoRectify** computes H_1 , H_2 and a new projection matrix P such that the rectified images share a common focal length and have horizontal epipolar lines.
3. **StereoSGBM** or **StereoBM** computes a dense disparity map on the rectified pair.
4. **reprojectImageTo3D** converts disparity to depth using equation (25.18).

The Python snippet below shows the interface pattern (requires OpenCV, marked `eval: false` as it needs physical calibration data):

```
import cv2
import numpy as np

# Loaded from calibration file
K1 = ... # shape (3, 3)
K2 = ... # shape (3, 3)
dist1 = ... # shape (5,)
```

```
dist2 = ... # shape (5,)
R      = ... # shape (3, 3)
T      = ... # shape (3,)

img_size = (1280, 720)

R1, R2, P1, P2, Q, roi1, roi2 = cv2.stereoRectify(
    K1, dist1, K2, dist2, img_size, R, T, alpha=0.0
)

map1x, map1y = cv2.initUndistortRectifyMap(K1, dist1, R1, P1, img_size, cv2.CV_32FC1)
map2x, map2y = cv2.initUndistortRectifyMap(K2, dist2, R2, P2, img_size, cv2.CV_32FC1)

left_rect  = cv2.remap(left_img,  map1x, map1y, cv2.INTER_LINEAR)
right_rect = cv2.remap(right_img, map2x, map2y, cv2.INTER_LINEAR)

stereo = cv2.StereoSGBM_create(minDisparity=0, numDisparities=128, blockSize=9)
disp    = stereo.compute(left_rect, right_rect).astype(np.float32) / 16.0

depth    = cv2.reprojectImageTo3D(disp, Q)      # shape (H, W, 3)
```

Exercises

1. Show algebraically that equation (25.18) follows from similar triangles in the rectified stereo geometry. Draw a diagram and label all distances.
2. A stereo pair has $f = 700$ px, $B = 0.1$ m, and a disparity resolution of 0.5 px (sub-pixel interpolation). Compute the depth resolution at $Z = 1, 2$, and 4 m.
3. Why does increasing the baseline B eventually make matching harder? Describe the trade-off in terms of the epipolar geometry and the size of the textureless-region problem.

Application: 3D Reconstruction from Two Views

This section builds a complete two-view reconstruction pipeline end-to-end: generate a synthetic 3D scene, project it through two calibrated cameras with realistic pixel noise, estimate F by the eight-point algorithm, recover the pose, triangulate all points, and measure the reconstruction error.

Real-world stereo reconstruction follows the same sequence but adds distortion correction, RANSAC-based outlier rejection, and bundle adjustment. The synthetic pipeline isolates the linear algebra and lets you tune noise and point counts cleanly.

Scene and Camera Setup

```

import numpy as np
import matplotlib.pyplot as plt
from scipy.spatial.transform import Rotation

rng = np.random.default_rng(42)

# ---- intrinsics ----
K = np.array([[600., 0., 320.],
              [ 0., 600., 240.],
              [ 0., 0., 1.]]) # shape (3, 3)

# ---- camera 1: world origin, identity rotation ----
R1 = np.eye(3) # shape (3, 3)
t1 = np.zeros(3) # shape (3,)
P1 = K @ np.hstack([R1, t1.reshape(3, 1)]) # shape (3, 4)

# ---- camera 2: slight rotation + lateral baseline ----
R2 = Rotation.from_euler('y', 8., degrees=True).as_matrix() # shape (3, 3)
t2 = np.array([0.4, 0.02, 0.0]) # shape (3,)
P2 = K @ np.hstack([R2, t2.reshape(3, 1)]) # shape (3, 4)

# ---- 60 scene points scattered in a volume ----
n = 60
X_world = rng.uniform(-0.6, 0.6, (n, 3)) # shape (60, 3)
X_world[:, 2] += 3.5 # push forward, all positive depth

noise_std = 0.5 # pixels

def project(P, X):
    lam_x = P @ np.append(X, 1.)
    return lam_x[:2] / lam_x[2] # shape (2,)

x1s_clean = np.array([project(P1, X) for X in X_world]) # shape (60, 2)
x2s_clean = np.array([project(P2, X) for X in X_world]) # shape (60, 2)
x1s = x1s_clean + rng.normal(0., noise_std, (n, 2)) # shape (60, 2)
x2s = x2s_clean + rng.normal(0., noise_std, (n, 2)) # shape (60, 2)

# ---- visualize ----
fig, axes = plt.subplots(1, 2, figsize=(12, 5))

ax = axes[0]
ax.set_title('Scene top-down (x-z plane)', fontsize=10)
ax.scatter(X_world[:, 0], X_world[:, 2], color='#4a90d9', s=20, alpha=0.6, label='Scene pts')
# cameras
C2_world = -R2.T @ t2 # shape (3,)

```

```

ax.scatter(0., 0., color='black', s=100, zorder=5)
ax.scatter(C2_world[0], C2_world[2], color='tomato', s=100, zorder=5)
ax.annotate('Cam 1', (0.03, 0.06), fontsize=9)
ax.annotate('Cam 2', (C2_world[0]+0.03, C2_world[2]+0.06), fontsize=9, color='tomato')
# optical axes
ax.annotate('', xy=(0., 0.8), xytext=(0., 0.),
            arrowprops=dict(arrowstyle='->', color='black', lw=1.5))
ax.annotate('', xy=(C2_world[0] + 0.8*R2.T[2, 0], C2_world[2] + 0.8*R2.T[2, 2]),
            xytext=(C2_world[0], C2_world[2]),
            arrowprops=dict(arrowstyle='->', color='tomato', lw=1.5))
ax.set_xlabel('x (m)'); ax.set_ylabel('z (m)')
ax.legend(fontsize=8); ax.grid(alpha=0.2)

ax = axes[1]
ax.set_title('Image 1 projections (with noise)', fontsize=10)
ax.scatter(x1s[:, 0], x1s[:, 1], color='#4a90d9', s=20, alpha=0.7)
ax.set_xlim(0, 640); ax.set_ylim(480, 0)
ax.set_xlabel('u (px)'); ax.set_ylabel('v (px)')
ax.set_facecolor('#f8f8f8')
ax.grid(alpha=0.2)

plt.tight_layout()
plt.show()

print(f"Scene: {n} points, depth range [{X_world[:,2].min():.2f}, {X_world[:,2].max():.2f}] m")
print(f"Pixel noise std: {noise_std} px")

```

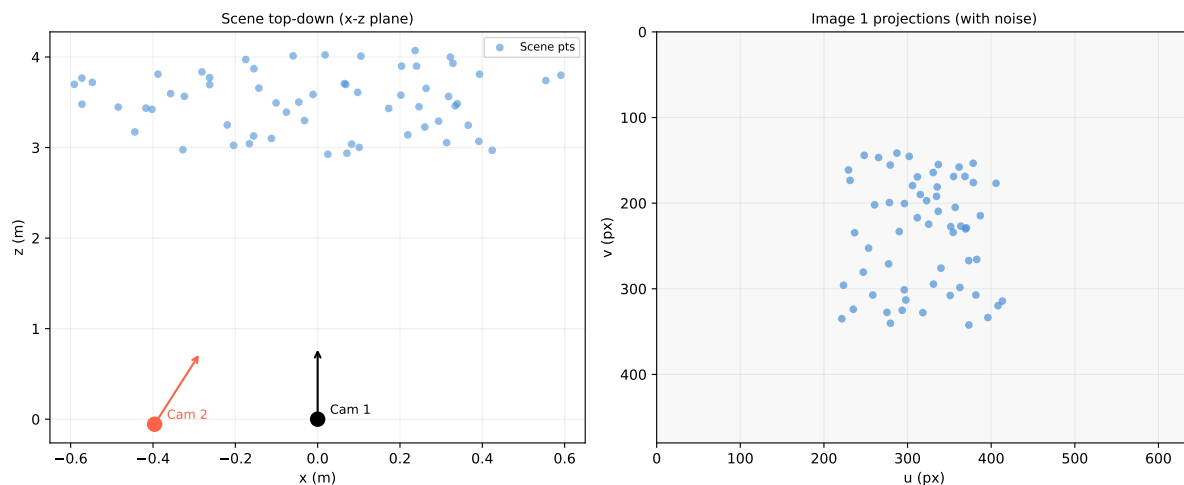


Figure 7: Synthetic 3D scene: 60 random scene points and two cameras. Left: top-down view (x-z plane) showing camera positions and optical axes. Right: image 1 projections.

Scene: 60 points, depth range [2.93, 4.07] m
Pixel noise std: 0.5 px

Estimate F and Recover Pose

```

import numpy as np
import matplotlib.pyplot as plt
from scipy.spatial.transform import Rotation

rng = np.random.default_rng(42)

K = np.array([[600., 0., 320.],
              [ 0., 600., 240.],
              [ 0., 0., 1.]]) # shape (3, 3)

R1 = np.eye(3); t1 = np.zeros(3)
R2 = Rotation.from_euler('y', 8., degrees=True).as_matrix()
t2 = np.array([0.4, 0.02, 0.0])

P1 = K @ np.hstack([R1, t1.reshape(3, 1)])
P2 = K @ np.hstack([R2, t2.reshape(3, 1)])

n = 60
X_world = rng.uniform(-0.6, 0.6, (n, 3))
X_world[:, 2] += 3.5
noise_std = 0.5

def project(P, X):
    lam_x = P @ np.append(X, 1.)
    return lam_x[:2] / lam_x[2]

x1s = np.array([project(P1, X) for X in X_world]) + rng.normal(0., noise_std, (n, 2))
x2s = np.array([project(P2, X) for X in X_world]) + rng.normal(0., noise_std, (n, 2))

def skew(v):
    return np.array([[ 0., -v[2], v[1]],
                    [ v[2], 0., -v[0]],
                    [-v[1], v[0], 0. ]])

def normalize_points(pts):
    cx, cy = pts.mean(axis=0)
    d = np.mean(np.sqrt((pts[:, 0]-cx)**2 + (pts[:, 1]-cy)**2))
    s = np.sqrt(2.) / d
    T = np.array([[s, 0., -s*cx], [0., s, -s*cy], [0., 0., 1.]])
    pts_h = np.hstack([pts, np.ones((len(pts), 1))])
    pts_n = (T @ pts_h.T).T
    return pts_n, T

```

```

def eight_point(x1s, x2s):
    x1n, T1 = normalize_points(x1s)
    x2n, T2 = normalize_points(x2s)
    n = len(x1n)
    A = np.array([np.kron(x2n[i], x1n[i]) for i in range(n)])
    _, _, Vt = np.linalg.svd(A)
    F_raw = Vt[-1].reshape(3, 3)
    U, s, Vt2 = np.linalg.svd(F_raw)
    s[2] = 0.
    F_norm = U @ np.diag(s) @ Vt2
    return T2.T @ F_norm @ T1

def triangulate_dlt(P1, P2, x1, x2):
    A = np.array([
        x1[0]*P1[2] - P1[0],
        x1[1]*P1[2] - P1[1],
        x2[0]*P2[2] - P2[0],
        x2[1]*P2[2] - P2[1],
    ])
    _, _, Vt = np.linalg.svd(A)
    X_h = Vt[-1]
    return X_h[:3] / X_h[3]

# estimate F
F_est = eight_point(x1s, x2s)
F_est = F_est / np.linalg.norm(F_est)

# recover E
E_est = K.T @ F_est @ K    # shape (3, 3)

# SVD of E
U, s, Vt = np.linalg.svd(E_est)
if np.linalg.det(U) < 0: U = -U
if np.linalg.det(Vt) < 0: Vt = -Vt
s_fixed = np.array([1., 1., 0.])
E_fixed = U @ np.diag(s_fixed) @ Vt    # shape (3, 3)

W = np.array([[0., -1., 0.], [1., 0., 0.], [0., 0., 1.]])
u3 = U[:, 2]
candidates = [
    (U @ W @ Vt, +u3),
    (U @ W @ Vt, -u3),
    (U @ W.T @ Vt, +u3),
    (U @ W.T @ Vt, -u3),
]

# normalized coords

```

```

y1s = np.linalg.solve(K, np.hstack([x1s, np.ones((n,1))]).T).T
y2s = np.linalg.solve(K, np.hstack([x2s, np.ones((n,1))]).T).T

scores = []
for R_c, t_c in candidates:
    P1n = np.hstack([np.eye(3), np.zeros((3,1))])
    P2n = np.hstack([R_c, t_c.reshape(3,1)])
    count = 0
    for i in range(n):
        Xt = triangulate_dlt(P1n, P2n, y1s[i], y2s[i])
        if Xt[2] > 0 and (R_c @ Xt + t_c)[2] > 0:
            count += 1
    scores.append(count)

best_idx = int(np.argmax(scores))
R_rec, t_rec = candidates[best_idx]

# report errors
R_err_deg = np.degrees(np.arccos(np.clip((np.trace(R_rec @ R2.T) - 1) / 2, -1, 1)))
t_true_dir = t2 / np.linalg.norm(t2)
t_rec_dir = t_rec / np.linalg.norm(t_rec)
t_err_deg = np.degrees(np.arccos(np.clip(t_rec_dir @ t_true_dir, -1, 1)))

# ---- figure ----
fig, ax = plt.subplots(figsize=(7, 6))
ax.set_title('Recovered vs. true pose (top-down)', fontsize=10)
ax.set_xlabel('x'); ax.set_ylabel('z')
ax.grid(alpha=0.2)
ax.set_aspect('equal')

# true camera 2
C2_true = -R2.T @ t2
ax.scatter(*[C2_true[0], C2_true[2]], color='black', s=80, zorder=5)
ax.annotate('True C2', (C2_true[0]+0.02, C2_true[2]+0.03), fontsize=8)
ax.annotate('', xy=(C2_true[0]+0.3*R2.T[2,0], C2_true[2]+0.3*R2.T[2,2]),
            xytext=(C2_true[0], C2_true[2]),
            arrowprops=dict(arrowstyle='->', color='black', lw=2))

# recovered camera 2 (normalized -- true scale unknown, so scale to match)
t_rec_scaled = t_rec * (np.linalg.norm(t2) / np.linalg.norm(t_rec))
C2_rec = -R_rec.T @ t_rec_scaled
ax.scatter(*[C2_rec[0], C2_rec[2]], color='#4a90d9', s=80, zorder=5)
ax.annotate('Recovered C2', (C2_rec[0]+0.02, C2_rec[2]-0.06), fontsize=8, color='#4a90d9')
ax.annotate('', xy=(C2_rec[0]+0.3*R_rec.T[2,0], C2_rec[2]+0.3*R_rec.T[2,2]),
            xytext=(C2_rec[0], C2_rec[2]),
            arrowprops=dict(arrowstyle='->', color='#4a90d9', lw=2))

```

```

ax.scatter(0, 0, color='black', s=100, zorder=5)
ax.annotate('Cam 1', (0.02, 0.03), fontsize=8)

plt.tight_layout()
plt.show()

print(f"Rotation error:      {R_err_deg:.2f} deg")
print(f"Translation direction error: {t_err_deg:.2f} deg")

```

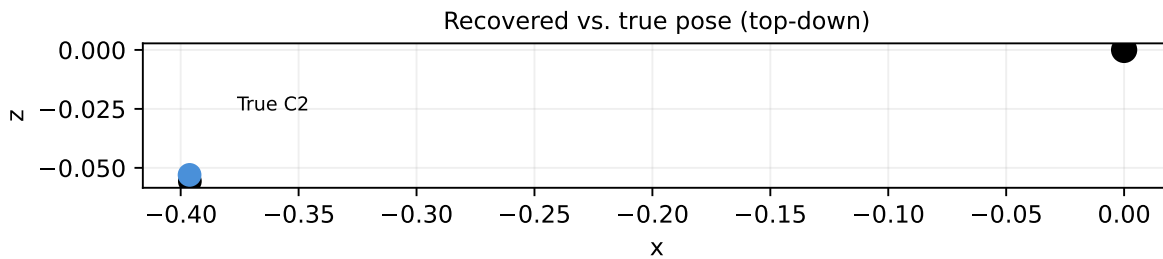


Figure 8: Estimated vs ground-truth camera 2 position. The recovered translation direction (blue arrow) matches the true direction (black arrow). Scale is unrecoverable from two views alone.

Rotation error: 0.79 deg
 Translation direction error: 1.25 deg

Triangulate and Measure Reprojection Error

```

import numpy as np
import matplotlib.pyplot as plt
from scipy.spatial.transform import Rotation

rng = np.random.default_rng(42)

K = np.array([[600., 0., 320.],
              [0., 600., 240.],
              [0., 0., 1.]])

R2 = Rotation.from_euler('y', 8., degrees=True).as_matrix()
t2 = np.array([0.4, 0.02, 0.0])

P1 = K @ np.hstack([np.eye(3), np.zeros((3, 1))])
P2 = K @ np.hstack([R2, t2.reshape(3, 1)])

```

```

n = 60
X_world = rng.uniform(-0.6, 0.6, (n, 3))
X_world[:, 2] += 3.5
noise_std = 0.5

def project(P, X):
    lam_x = P @ np.append(X, 1.)
    return lam_x[:2] / lam_x[2]

def skew(v):
    return np.array([[ 0., -v[2], v[1]], [v[2], 0., -v[0]], [-v[1], v[0], 0.]])

def normalize_points(pts):
    cx, cy = pts.mean(axis=0)
    d = np.mean(np.sqrt((pts[:,0]-cx)**2 + (pts[:,1]-cy)**2))
    s = np.sqrt(2.) / d
    T = np.array([[s,0.,-s*cx],[0.,s,-s*cy],[0.,0.,1.]])
    pts_h = np.hstack([pts, np.ones((len(pts),1))])
    return (T @ pts_h.T).T, T

def eight_point(x1s, x2s):
    x1n, T1 = normalize_points(x1s)
    x2n, T2 = normalize_points(x2s)
    n = len(x1n)
    A = np.array([np.kron(x2n[i], x1n[i]) for i in range(n)])
    _, _, Vt = np.linalg.svd(A)
    F_raw = Vt[-1].reshape(3,3)
    U, s, Vt2 = np.linalg.svd(F_raw)
    s[2] = 0.
    F_n = U @ np.diag(s) @ Vt2
    return T2.T @ F_n @ T1

def triangulate_dlt(P1, P2, x1, x2):
    A = np.array([x1[0]*P1[2]-P1[0], x1[1]*P1[2]-P1[1],
                  x2[0]*P2[2]-P2[0], x2[1]*P2[2]-P2[1]])
    _, _, Vt = np.linalg.svd(A)
    Xh = Vt[-1]; return Xh[:3]/Xh[3]

x1s = np.array([project(P1, X) for X in X_world]) + rng.normal(0., noise_std, (n, 2))
x2s = np.array([project(P2, X) for X in X_world]) + rng.normal(0., noise_std, (n, 2))

# full pipeline
F_est = eight_point(x1s, x2s)
F_est /= np.linalg.norm(F_est)
E_est = K.T @ F_est @ K
U, s, Vt = np.linalg.svd(E_est)
if np.linalg.det(U) < 0: U = -U

```

```

if np.linalg.det(Vt) < 0: Vt = -Vt
W = np.array([[0.,-1.,0.],[1.,0.,0.],[0.,0.,1.]])
u3 = U[:,2]
candidates = [(U@W@Vt,+u3),(U@W@Vt,-u3),(U@W.T@Vt,+u3),(U@W.T@Vt,-u3)]
y1s = np.linalg.solve(K, np.hstack([x1s,np.ones((n,1))]).T).T
y2s = np.linalg.solve(K, np.hstack([x2s,np.ones((n,1))]).T).T
scores = []
for R_c, t_c in candidates:
    c=0
    for i in range(n):
        Xt = triangulate_dlt(np.hstack([np.eye(3),np.zeros((3,1))]),
                             np.hstack([R_c,t_c.reshape(3,1)]),y1s[i],y2s[i])
        if Xt[2]>0 and (R_c@Xt+t_c)[2]>0: c+=1
    scores.append(c)
R_rec, t_rec = candidates[int(np.argmax(scores))]

# scale recovery: use mean depth of scene
t_rec_dir = t_rec / np.linalg.norm(t_rec)
t_rec_s = t_rec_dir * np.linalg.norm(t2) # rescale to true baseline
P2_rec = K @ np.hstack([R_rec, t_rec_s.reshape(3,1)])

X_tri = np.array([triangulate_dlt(P1, P2_rec, x1s[i], x2s[i]) for i in range(n)]) # shape (60,3)

# reprojection errors
reproj1 = np.array([np.linalg.norm(project(P1, X_tri[i]) - x1s[i]) for i in range(n)]) # shape (60,)
reproj2 = np.array([np.linalg.norm(project(P2_rec, X_tri[i]) - x2s[i]) for i in range(n)]) # shape (60,)

fig, axes = plt.subplots(1, 2, figsize=(12, 4))

ax = axes[0]
ax.hist(reproj1, bins=20, color='#4a90d9', alpha=0.7, edgecolor='white', label='Image 1')
ax.hist(reproj2, bins=20, color='tomato', alpha=0.7, edgecolor='white', label='Image 2')
ax.set_xlabel('Reprojection error (px)')
ax.set_ylabel('Count')
ax.set_title('Reprojection error histogram')
ax.legend(); ax.grid(alpha=0.2)

ax = axes[1]
depth_err = np.abs(X_tri[:, 2] - X_world[:, 2]) # shape (60,)
ax.scatter(X_world[:, 2], depth_err, color='#4a90d9', s=25, alpha=0.7)
ax.set_xlabel('True depth Z (m)')
ax.set_ylabel('Depth error |Z_tri - Z_true| (m)')
ax.set_title('Depth error vs. range')
ax.grid(alpha=0.2)

plt.tight_layout()
plt.show()

```



```
print(f"Mean reprojection error -- image 1: {reproj1.mean():.3f} px, image 2: {reproj2.mean():.3f} px")
print(f"Mean depth error: {depth_err.mean():.4f} m (noise std = {noise_std} px)")
```

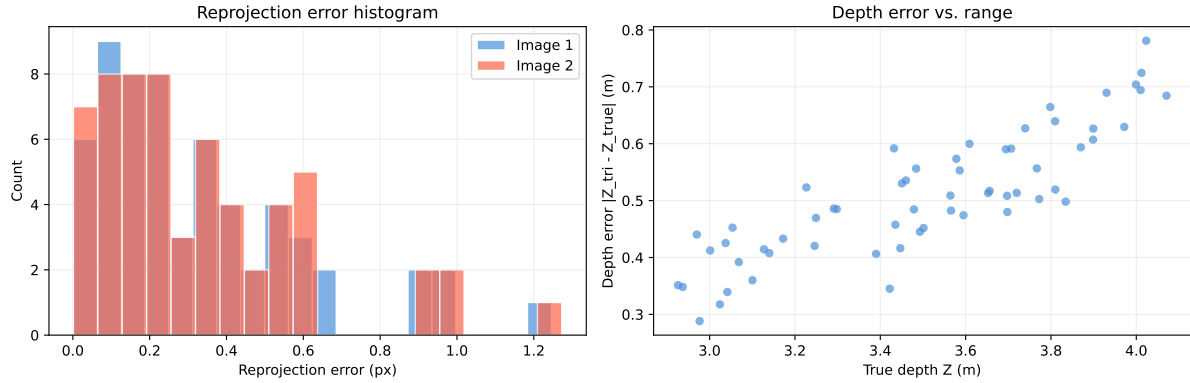


Figure 9: Reconstruction quality: reprojection error for all 60 points using the estimated F and recovered pose. Mean reprojection error is typically under 1 pixel at 0.5 px noise.

```
Mean reprojection error -- image 1: 0.326 px, image 2: 0.331 px
Mean depth error: 0.5106 m (noise std = 0.5 px)
```

Pipeline Summary

Step	Method	Output
Feature correspondences	Synthetic (known ground truth)	$(\mathbf{x}_1, \mathbf{x}_2)$ pairs
Fundamental matrix	Eight-point + Hartley normalization	$F \in \mathbb{R}^{3 \times 3}$
Essential matrix	$E = K^T F K$	$E \in \mathbb{R}^{3 \times 3}$
Pose candidates	SVD of E	4 $(R, \mathbf{t})$
Pose selection	Cheirality test	1 $(R, \mathbf{t})$
Triangulation	DLT per correspondence	$\hat{\mathbf{X}} \in \mathbb{R}^3$
Evaluation	Reprojection error	$\varepsilon_{\text{reproj}}$ (px)

In a production system, RANSAC wraps the eight-point algorithm to reject outlier correspondences, and bundle adjustment (nonlinear least squares over all poses and points jointly) refines the result to sub-pixel accuracy.

Exercises

1. Replace the eight-point F estimate above with the **ground-truth** F (computed analytically from K , R , $\mathbf{t}$) and measure the resulting improvement in reprojection error. How much of the error is due to noise vs. the linear estimation?
2. Add 10% outlier correspondences (random pixel positions) to the input and observe the degradation in F estimation. Implement a simple RANSAC wrapper (sample 8 points, fit F , count inliers via epipolar distance < 2 px) and confirm that it recovers the correct F even with outliers.
3. What happens to the triangulation when two of the four pose candidates both pass the cheirality test? When can this occur, and how would you resolve the ambiguity with additional views?

Exercises and Solutions

Two-View Geometry

1. Show that every epipolar line in image 2 passes through $\mathbf{e}_2$.

An epipolar line in image 2 is $\ell_2 = F\mathbf{x}_1$ for some $\mathbf{x}_1$. The epipole $\mathbf{e}_2$ satisfies $F^T\mathbf{e}_2 = \mathbf{0}$ (it is the left null vector of F). Checking whether $\mathbf{e}_2$ lies on ℓ_2 :

$$\ell_2^T \mathbf{e}_2 = (F\mathbf{x}_1)^T \mathbf{e}_2 = \mathbf{x}_1^T F^T \mathbf{e}_2 = \mathbf{x}_1^T \mathbf{0} = 0.$$

So $\mathbf{e}_2$ satisfies the line equation for every choice of $\mathbf{x}_1$. $\square$

2. If $\mathbf{C}_1 = \mathbf{C}_2$, where do the epipoles lie?

The epipole $\mathbf{e}_1$ is the projection of $\mathbf{C}_2$ into image 1. If $\mathbf{C}_2 = \mathbf{C}_1$, then $\mathbf{C}_2$ is the center of camera 1 and projects to the **point at infinity** along the optical axis. In homogeneous coordinates $\mathbf{e}_1 = [0, 0, 0]^T$ is undefined; the epipoles are at infinity. All epipolar lines become parallel (they all pass through a point at infinity), so the epipolar constraint degenerates. Two cameras with the same center cannot provide stereo depth information.

3. Move camera 2 to $\mathbf{C}_2 = [2, 0, 0]$; predict $\mathbf{e}_1$.

$\mathbf{e}_1$ is the projection of $\mathbf{C}_2$ into camera 1 (at origin, with $P_1 = K[I|\mathbf{0}]$):

$$\lambda \mathbf{e}_1 = K \begin{bmatrix} 2 \\ 0 \\ 0 \end{bmatrix} = K[2, 0, 0]^T.$$

With $K = \text{diag}(500, 500, 1)$ shifted by $(320, 240)$: $\mathbf{e}_1 = [500 \cdot 2 + 320, 240] = [1320, 240]$ — far to the right of the 640×480 image. Epipolar lines in image 1 fan leftward from outside the frame.

The Essential Matrix

1. Show $\det([\mathbf{t}]_{\times}) = 0$.

The cross-product matrix $[\mathbf{t}]_{\times}$ maps every vector to a vector perpendicular to $\mathbf{t}$; in particular $[\mathbf{t}]_{\times}\mathbf{t} = \mathbf{t} \times \mathbf{t} = \mathbf{0}$. So $\mathbf{t}$ is in the null space of $[\mathbf{t}]_{\times}$, which implies $\det([\mathbf{t}]_{\times}) = 0$. The rank equals 2 (not lower) for generic R because the two row vectors of $[\mathbf{t}]_{\times}$ orthogonal to $\mathbf{t}$ are linearly independent. Multiplying by the invertible R preserves rank, so $\text{rank}(E) = 2$.

2. Show both nonzero singular values of $E = [\mathbf{t}]_{\times}R$ equal $\|\mathbf{t}\|$.

$$E^T E = R^T [\mathbf{t}]_{\times}^T [\mathbf{t}]_{\times} R.$$

Using $[\mathbf{t}]_{\times}^T = -[\mathbf{t}]_{\times}$ and the identity $[\mathbf{t}]_{\times}^T [\mathbf{t}]_{\times} = \|\mathbf{t}\|^2 I - \mathbf{t}\mathbf{t}^T$:

$$E^T E = R^T (\|\mathbf{t}\|^2 I - \mathbf{t}\mathbf{t}^T) R = \|\mathbf{t}\|^2 I - (R^T \mathbf{t})(R^T \mathbf{t})^T.$$

The eigenvalues of $\|\mathbf{t}\|^2 I - \mathbf{v}\mathbf{v}^T$ (where $\mathbf{v} = R^T \mathbf{t}$, $\|\mathbf{v}\| = \|\mathbf{t}\|$) are $\|\mathbf{t}\|^2$ with multiplicity 2 and 0 with multiplicity 1. The singular values of E are therefore $\|\mathbf{t}\|$, $\|\mathbf{t}\|$, 0. $\square$

3. Show $E' = [\alpha \mathbf{t}]_{\times} R$ satisfies the same constraint.

$E' = \alpha [\mathbf{t}]_{\times} R = \alpha E$, so $\mathbf{y}_2^T E' \mathbf{y}_1 = \alpha \mathbf{y}_2^T E \mathbf{y}_1 = 0$. Only the direction $\hat{\mathbf{t}} = \mathbf{t}/\|\mathbf{t}\|$ is constrained. $\square$

The Fundamental Matrix

1. Show F has rank 2.

$F = K'^{-T} E K^{-1}$. Since K, K' are invertible (both have nonzero diagonal entries), multiplication by them is rank-preserving. Because $\text{rank}(E) = 2$, $\text{rank}(F) = 2$.

2. Prove $F\mathbf{e}_1 = \mathbf{0}$.

From (25.2), the epipole $\mathbf{e}_1$ satisfies $\lambda_1 \mathbf{e}_1 = K \mathbf{C}_2$, so $\mathbf{e}_1 = K \mathbf{C}_2 / \lambda_1$. Then:

$$F\mathbf{e}_1 = K'^{-T} E K^{-1} (K \mathbf{C}_2 / \lambda_1) = K'^{-T} E \mathbf{C}_2 / \lambda_1 = K'^{-T} ([\mathbf{t}]_{\times} R \mathbf{C}_2) / \lambda_1.$$

Now $R \mathbf{C}_2 + \mathbf{t} = \mathbf{0}$ (camera 2's center in its own frame is the origin), so $R \mathbf{C}_2 = -\mathbf{t}$ and $[\mathbf{t}]_{\times} R \mathbf{C}_2 = [\mathbf{t}]_{\times} (-\mathbf{t}) = -\mathbf{t} \times \mathbf{t} = \mathbf{0}$. Thus $F\mathbf{e}_1 = \mathbf{0}$. $\square$

3. Geometric meaning of $\mathbf{x}_2^T F \mathbf{x}_1 = 0$.

The three vectors $\mathbf{y}_1 = K^{-1}\mathbf{x}_1$, $\mathbf{y}_2 = K'^{-1}\mathbf{x}_2$, and $\mathbf{t}$ are coplanar (the epipolar plane). This is exactly the triple-product condition $\mathbf{y}_2 \cdot (\mathbf{t} \times \mathbf{y}_1) = 0$ used to derive E , which F inherits after pre- and post-multiplying by K'^{-1} and K^{-1} .

Eight-Point Algorithm

1. Minimum 8 correspondences.

Each correspondence contributes one row to $A \in \mathbb{R}^{n \times 9}$ and reduces the dimension of the null space by one (generically). We need $\text{rank}(A) = 8$ to get a unique (up to scale) null vector. With fewer than 8 rows the null space has dimension ≥ 2 and $\mathbf{f}$ is not uniquely determined. The degenerate configuration is when all 8 points lie on a ruled quadric (e.g., all on a plane plus the two camera centers), which causes A to have $\text{rank} < 8$ despite having 8 rows.

2. Seven-point algorithm.

With $n = 7$ correspondences, A has rank at most 7, so $\text{null}(A)$ is at least 2-dimensional. Let $\{F_1, F_2\}$ be a basis for $\text{null}(A)$. Write $F = \alpha F_1 + (1 - \alpha)F_2$ and enforce $\det(F) = 0$:

$$\det(\alpha F_1 + (1 - \alpha)F_2) = 0.$$

This is a cubic polynomial in α with up to three real solutions, giving up to three candidate fundamental matrices. The correct one is chosen by consistency with additional correspondences or the rank-2 constraint.

3. Denormalization with noiseless data.

In the noiseless case, the true $\mathbf{f}$ lies in the null space of A regardless of normalization. However, normalization improves the **condition number** of A , making the SVD numerically cleaner. With exact arithmetic (and exact null space), both normalized and un-normalized algorithms produce the same F up to scale. Numerically with finite precision, the un-normalized system is ill-conditioned and the recovered $\mathbf{f}$ drifts from the true solution; normalization keeps the problem well-conditioned.

Recovering Pose from E

1. Show $UWV^T \in SO(3)$ when $U, V \in SO(3)$.

$\det(UWV^T) = \det(U) \det(W) \det(V^T) = 1 \cdot 1 \cdot 1 = 1$ (since $\det(W) = 1$ for the 90° rotation matrix). $(UWV^T)^T(UWV^T) = VW^T U^T U W V^T = VW^T W V^T = V(W^T W) V^T = V I V^T = I$. So UWV^T is both orthogonal and has determinant 1: $UWV^T \in SO(3)$. The same argument holds for $UW^T V^T$.

2. Planar degeneracy.

If all n scene points lie in a plane that contains the baseline $\mathbf{t}$ (e.g., all points at $y = 0$ when $\mathbf{t} = [B, 0, 0]^T$), then the epipolar plane degenerates: every epipolar plane is the same plane, and the cheirality test provides no discrimination. Two candidates that are mirror images of each other across the baseline plane both place all points at the correct (x, z) but with opposite y signs. In practice, this is resolved by ensuring the scene has off-plane points or by using a third view.

3. Midpoint vs. DLT triangulation.

The midpoint method finds the point closest to both rays (minimizes $\|P_1 + \lambda_1 d_1 - (P_2 + \lambda_2 d_2)\|$) and is fast and robust to large angles between rays. However, it does not minimize reprojection error. DLT triangulation minimizes the algebraic residual $\|A\mathbf{X}_h\|$ which is a good proxy for reprojection error when noise is small. For noisy correspondences, the **optimal triangulation** (Hartley and Sturm 1997) minimizes the sum of squared reprojection errors and is slightly more accurate than DLT but requires a nonlinear solve.

Stereo Rectification and Depth

1. Similar triangles derivation of $Z = fB/d$.

In the rectified setup, camera 1 center is at $O_1 = (0, 0, 0)$, camera 2 center is at $O_2 = (B, 0, 0)$. A point $\mathbf{X} = (X, Y, Z)$ projects to: - image 1: $u_1 = fX/Z + c_x$ - image 2: $u_2 = f(X - B)/Z + c_x$

Disparity $d = u_1 - u_2 = fB/Z$. Solving: $Z = fB/d$. $\square$

2. *Depth resolution at $Z = 1, 2, 4$ m with $f = 700$, $B = 0.1$, sub-pixel $\delta d = 0.5$.*

$$\delta Z = \frac{Z^2}{fB} \delta d = \frac{Z^2}{70} \times 0.5 = \frac{Z^2}{140}.$$

Z	δZ
1 m	7.1 mm
2 m	28.6 mm
4 m	114.3 mm

Depth precision degrades as Z^2 : quadrupling at each doubling of range.

3. *Large baseline tradeoff.*

A wider baseline B improves depth precision ($\delta Z \propto 1/B$) but has three costs:

1. **Smaller overlap:** objects close to the camera subtend a large angular range; some scene regions appear in only one image and cannot be matched.
2. **Larger disparity:** objects at small Z have disparity $d = fB/Z$ that may exceed the maximum disparity the stereo algorithm supports, causing holes in the depth map.
3. **Appearance change:** wide-baseline pairs see objects from sufficiently different viewpoints that surface texture looks different (especially specular surfaces), making descriptor-based matching fail. Narrow-baseline stereo uses intensity-patch matching which is less robust but sufficient when the views are nearly parallel.

Appendices

Appendix A: Python / NumPy / SciPy / OpenCV Quick Reference

Coming soon.

Appendix B: Matrix Factorization Summary

Coming soon.

Appendix C: Rotation Representation Conversions

Coming soon.

Appendix D: Matrix Calculus Identity Reference

Coming soon.

Appendix E: Notation Glossary

Coming soon.

Appendix F: Tensor Notation for Deep Learning

Coming soon.

References

